

# round1\_clean

September 20, 2024

## 1 MDA Hydroxyacyle-CoA Lyase variant design round 1: Higher order variants

In Round 0, SSM over 64 residues was conducted. Here in Round 1, we use this data to inform some higher order combinatorial variants to test. These variants are design to target exploration and exploitation.

**Plate 1:** *The null hypothesis*, combine the top 12 SSM mutations at depths of 2 and 3 mutations. Each of the 12 mutations are explored evenly over the plate. The top performing variant according to an additive model of 3 mutations is included.

**Plate 2:** *Exploitation* A supervised ML model is trained on the SSM data, using ESM2 embeddings. Under the assumption that the ESM2 embeddings capture global context, learnings from the SSM data provide **better than random** (but do not promise much better than that...) predictions on higher order mutants. Two and three order mutants are selected greedily according to model predictions, enforcing that no single mutation is seen more than four times. Top 12 mutations explored additively in plate 1 are only allowed to be used once, and only alone in a variant.

**Plate 3:** *Exploration*, the plate is chosen according to the Upper Confidence Bound of the model, which incorporates approximate model uncertainty, forcing exploration of the protein landscape that the model has few information on. Mutation count is enforced as in plate 2.

### 1.1 0. Imports and tools

```
[1]: # builtins and utils
import os
import json
import hashlib
from tqdm.notebook import tqdm
import itertools

# DS stack
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats

# Downstream supervised tools
```

```

from sklearn.model_selection import KFold
from sklearn.ensemble import BaggingRegressor, RandomForestRegressor
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.pipeline import Pipeline
from sklearn.compose import TransformedTargetRegressor
from sklearn.neural_network import MLPRegressor
from sklearn.model_selection import RandomizedSearchCV, train_test_split
from sklearn.decomposition import PCA
from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.metrics import make_scorer, r2_score
import joblib

# aide predict
import aide_predict as ap
from aide_predict.utils.plotting import plot_mutation_heatmap,
    plot_protein_sequence_heatmap

# data handling
import datasets

# typing
from typing import List, Dict, Tuple, Union

```

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-packages/Bio/pairwise2.py:278: BiopythonDeprecationWarning: Bio.pairwise2 has been deprecated, and we intend to remove it in a future release of Biopython. As an alternative, please consider using Bio.Align.PairwiseAligner as a replacement, and contact the Biopython developers if you still need the Bio.pairwise2 module.

```
warnings.warn(
```

```

[61]: # globals
POSITION_OFFSET = 10
MAX_MUTATION_COUNT = 4

MUTATION_COUNTS_OBSERVED = {}

```

```

[3]: def add_observed_mutations(mutations: List[str]):
    for mutation in mutations:
        if mutation in MUTATION_COUNTS_OBSERVED:
            MUTATION_COUNTS_OBSERVED[mutation] += 1
        else:
            MUTATION_COUNTS_OBSERVED[mutation] = 1

```

```

[4]: # mutating wt to new variant
# the aide package does nto have an option for position offset
def mutate_wt(

```

```

        wild_type: ap.ProteinSequence,
        mutations: Union[str, List[str]],
        offset: int = POSITION_OFFSET,
        one_indexed: bool=True,
        return_str: bool=True
    ) -> ap.ProteinSequence:
        """
    if isinstance(mutations, str):
        mutations = [mutations]

    mutated = list(str(wild_type))
    for mutation in mutations:
        original_aa = mutation[0]
        position = int(mutation[1:-1])
        new_aa = mutation[-1]
        if one_indexed:
            position = position - 1
        if offset != 0:
            position = position - offset
        assert wild_type[position] == original_aa, f"Mutation {mutation} does_
↳not match wild type {wild_type[position]}"
        mutated[position] = new_aa
    if return_str:
        return "".join(mutated)
    else:
        return ap.ProteinSequence("".join(mutated))

```

```

[5]: def get_mutations(
    wt: ap.ProteinSequence,
    variant: ap.ProteinSequence,
    offset: int = POSITION_OFFSET,
    one_indexed: bool=True
) -> List[str]:
    """Parse mutations from two protein sequences"""
    if not isinstance(wt, ap.ProteinSequence):
        wt = ap.ProteinSequence(wt)
    if not isinstance(variant, ap.ProteinSequence):
        variant = ap.ProteinSequence(variant)
    mutations = wt.get_mutations(variant)
    mutations_ = []
    for mutation in mutations:
        position = int(mutation[1:-1])
        if not one_indexed:
            position = position - 1
        if offset != 0:
            position = position + offset
        mutations_.append(f"{mutation[0]}{position}{mutation[-1]}")

```

```
return mutations_
```

```
[6]: # hashing protein sequences
def protein_hash(protein: str) -> str:
    return hashlib.md5(str(protein).encode()).hexdigest()
```

```
[7]: # create a model class
model_no_embedder = Pipeline([
    ('scaler', StandardScaler()),
    ('pca', PCA(n_components=0.98)),
    ('regressor', BaggingRegressor(
        estimator=TransformedTargetRegressor(
            regressor=MLPRegressor(
                max_iter=1000,
                early_stopping=True,
                n_iter_no_change=10,
                activation='relu',
                hidden_layer_sizes=(100, 100),
                alpha=0.0001,
                learning_rate_init=0.001
            ),
            transformer=StandardScaler()
        ),
        n_estimators=10,
        max_samples=0.9,
        max_features=0.9,
        bootstrap=True,
        bootstrap_features=True,
    ))
])
```

```
[8]: def predict_ensemble(model, X):
    """Take average and std of predictions from all estimators in the ensemble.
    ↪ """
    all_predictions = []
    X_transformed = model[:-1].transform(X)
    for i, estimator in enumerate(model[-1].estimators_):
        feature_indexes = model[-1].estimators_features_[i]
        model_input = X_transformed[:, feature_indexes]
        y_pred_ = estimator.predict(model_input)
        all_predictions.append(y_pred_)
    all_predictions = np.array(all_predictions)
    means = np.mean(all_predictions, axis=0)
    stds = np.std(all_predictions, axis=0)
    return means, stds
```

## 1.2 1. Data preparation and initial analysis

```
[9]: alignment = ap.ProteinSequences.from_fasta('./data/mda_hacl1_msa.a2m')
     wt = alignment[0]
```

### 1.2.1 1.1 Load and preprocess SSM data

Convert the excel into a pandas dataframe, rename columns, parse positions, melt what is originally a wide format.

NOTE: The positions in the excel are forward 10 residues because of His tag and 1 indexed.

```
[10]: ssm_data = pd.read_excel('./data/ssm_compiled.xlsx', header=1)

[11]: # remove column 0, 2 and row 1
     ssm_data = ssm_data.drop(ssm_data.columns[0], axis=1)
     ssm_data = ssm_data.drop(ssm_data.columns[1], axis=1)
     ssm_data = ssm_data.drop(ssm_data.index[0])
     # format long with columns [mutation_string, original, position, mutation,
     ↪activity]
     ssm_data.rename(columns={'Unnamed: 1': 'original_position_string'},
     ↪inplace=True)

[12]: # parse positions/mutations
     ssm_data['original_aa'] = ssm_data['original_position_string'].str[0]
     ssm_data['position'] = ssm_data['original_position_string'].str[1:].astype(int)
     ssm_data.drop(columns=['original_position_string'], inplace=True)
     ssm_data = ssm_data.melt(id_vars=['original_aa', 'position'],
     ↪var_name='mutation', value_name='activity').sort_values(by='position').
     ↪reset_index(drop=True)

[13]: # compute the position on our sequence, 0 indexed
     ssm_data['position_on_wt_0_indexed'] = ssm_data['position'] - POSITION_OFFSET -
     ↪1
     for _, row in ssm_data.iterrows():
         assert wt[row['position_on_wt_0_indexed']] == row['original_aa']

[14]: # parse the mutation string
     ssm_data['mutation_string'] = ssm_data['original_aa'] + ssm_data['position'].
     ↪astype(str) + ssm_data['mutation']

[15]: # get the variant and the hash, save to file
     ssm_data['variant'] = ssm_data.apply(lambda x: mutate_wt(wt,
     ↪x['mutation_string']), axis=1)
     ssm_data['hash'] = ssm_data['variant'].apply(protein_hash)

[16]: ssm_data.dropna().to_csv('./data/ssm_data_clean.csv', index=False)
```

```
[17]: ssm_data
```

```
[17]:      original_aa  position mutation  activity  position_on_wt_0_indexed  \
0              V        35         A  0.000000                      24
1              V        35         V  1.024488                      24
2              V        35         E  0.000000                      24
3              V        35         L  0.120022                      24
4              V        35         W  0.000000                      24
...           ...      ...      ...      ...                      ...
1275           K        573         P  0.809104                      562
1276           K        573         S  1.111239                      562
1277           K        573         T  0.799904                      562
1278           K        573         H  0.739676                      562
1279           K        573         V      NaN                      562

      mutation_string                                variant  \
0          V35A  MAQLTGEQIIARSLKTQGIDKIFGAVGIPVTGIASACQAEGLTYTG...
1          V35V  MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLTYTG...
2          V35E  MAQLTGEQIIARSLKTQGIDKIFGEVGIPVTGIASACQAEGLTYTG...
3          V35L  MAQLTGEQIIARSLKTQGIDKIFGLVGIPVTGIASACQAEGLTYTG...
4          V35W  MAQLTGEQIIARSLKTQGIDKIFGWVGIPVTGIASACQAEGLTYTG...
...           ...      ...
1275       K573P  MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLTYTG...
1276       K573S  MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLTYTG...
1277       K573T  MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLTYTG...
1278       K573H  MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLTYTG...
1279       K573V  MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLTYTG...

      hash
0  92631f37c9f61a25441937b30499d155
1  354634911f28df897d00b02eb6fbf492
2  f38ee4c7d5b67593a91ae508c92950bf
3  f210c6f54a5c28b239e283e1a85f39bc
4  4818ed9923d4e7b88695e26418f8f579
...
1275  d75303ee75f5a01874ace15e62cb8c0f
1276  ae2a7556c5ccd6edccbd6c7d62e6345b
1277  debc8517ff2754b7665e29b780b80e90
1278  7e97eba15d71f9e86d5b5d7bbcf798c4
1279  9c6d3fa77cb4f9ebadb4c10c2027081
```

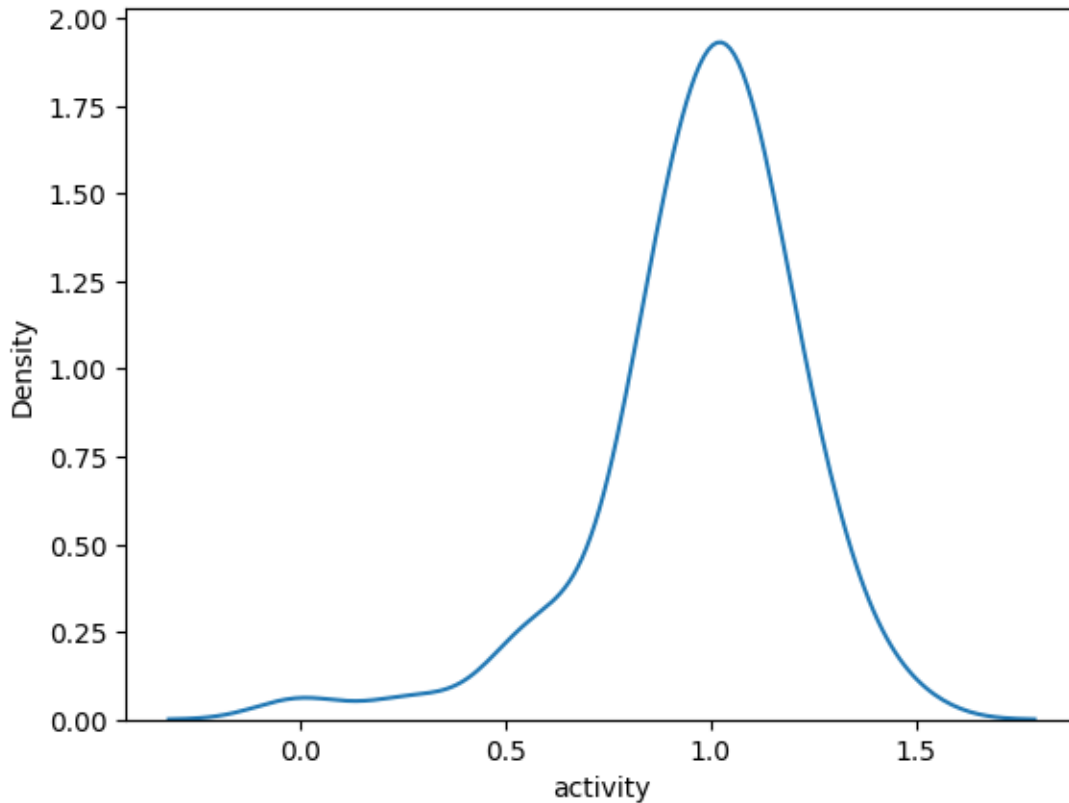
```
[1280 rows x 8 columns]
```

### 1.2.2 1.2 Visualize SSM data

**1.2.1 Statistical significance of hits** Determine how many and which single mutations are statistically significant based on the control values.

```
[18]: # mark control mutations
ssm_data['control'] = ssm_data['original_aa'] == ssm_data['mutation']
sns.kdeplot(ssm_data[ssm_data['control']]['activity'], label='control')
```

```
[18]: <AxesSubplot: xlabel='activity', ylabel='Density'>
```



```
[19]: control_data = ssm_data[ssm_data['control']]
# drop failed control
control_data = control_data[control_data['activity'] > 0.01]
# stats
control_mean, control_std = control_data['activity'].mean(),
    ↪ control_data['activity'].std()
ssm_data['significant'] = np.abs((ssm_data['activity'] - control_mean) /
    ↪ control_std) > 2
print('Frac significant: ', ssm_data['significant'].mean())
```

```
Frac significant: 0.7265625
```

### 1.2.2 SSM heatmap

```
[20]: from matplotlib.colors import Normalize, TwoSlopeNorm, LogNorm
from matplotlib.colorbar import ColorbarBase
```

```

from matplotlib.cm import ScalarMappable
sns.set_context('paper')

class CustomNorm(Normalize):
    def __init__(self, vmin, vcenter, vmax):
        super().__init__(vmin, vmax)
        self.vcenter = vcenter
        self.lower_norm = Normalize(vmin=vmin, vmax=vcenter)
        self.upper_norm = LogNorm(vmin=vcenter, vmax=vmax)

    def __call__(self, value, clip=None):
        lower = value <= self.vcenter
        upper = value > self.vcenter
        result = np.zeros_like(value, dtype=float)
        result[lower] = self.lower_norm(value[lower])/2.0
        result[upper] = 0.5 + 0.5 * self.upper_norm(value[upper])
        return result

    def inverse(self, value):
        lower = value <= 0.5
        upper = value > 0.5
        result = np.zeros_like(value, dtype=float)
        result[lower] = self.lower_norm.inverse(value[lower] * 2.0)
        result[upper] = self.upper_norm.inverse((value[upper] - 0.5) * 2.0)
        return result

def plot_mutation_heatmap(mutations, scores, all_positions=True,
    wt_activity=None, zero_minimum=False, log_scale_upper=False):
    """
    Plot a heatmap of single point mutation scores.

    Parameters:
    mutations (list): List of mutation strings (e.g., ["L1V", "A2G", ...])
    scores (list): List of corresponding scores
    all_positions (bool): If True, show all positions, even those without
    wt_activity (float): Wild-type activity score for centering the colormap
    zero_minimum (bool): If True, set the minimum of the colormap to 0.0
    log_scale_upper (bool): If True, use log scale for upper half of colormap

    Returns:
    fig, ax: The created figure and axis objects
    """
    # All possible amino acids
    all_aas = 'ACDEFGHIKLMNPQRSTVWY'

    # Extract residue positions and mutant amino acids

```



```

positions = [int(m[1:-1]) for m in mutations]
mutant_aas = [m[-1] for m in mutations]
original_aas = [m[0] for m in mutations]

# Create a DataFrame
df = pd.DataFrame({
    'Position': positions,
    'Mutant_AA': mutant_aas,
    'Original_AA': original_aas,
    'Score': scores
})
original_aa_dict = dict(zip(df['Position'], df['Original_AA']))

if all_positions:
    # Create a full matrix with all amino acids and positions
    full_matrix = pd.DataFrame(index=range(1, max(positions)+1),
    ↪columns=list(all_aas))
else:
    full_matrix = pd.DataFrame(index=np.sort(np.unique(positions)),
    ↪columns=list(all_aas))

# Fill the matrix with scores
for _, row in df.iterrows():
    full_matrix.at[row['Position'], row['Mutant_AA']] = row['Score']

# Create the heatmap
width = 8 # Increased width to accommodate colorbar
height = 6/20 * len(full_matrix)
fig, (ax, cax) = plt.subplots(1, 2, figsize=(width, height),
    gridspec_kw={'width_ratios': [20, 1]})

# Determine colormap and normalization
if wt_activity is None:
    cmap = sns.color_palette("viridis", as_cmap=True)
    vmin = 0.0 if zero_minimum else np.nanmin(full_matrix.values)
    vmax = np.nanmax(full_matrix.values)
    norm = Normalize(vmin=vmin, vmax=vmax)
else:
    cmap = sns.color_palette("vlag", as_cmap=True)
    vmin = 0.0 if zero_minimum else min(np.nanmin(full_matrix.values),
    ↪wt_activity)
    vmax = max(np.nanmax(full_matrix.values), wt_activity)
    if log_scale_upper:
        norm = CustomNorm(vmin=vmin, vcenter=wt_activity, vmax=vmax)
    else:
        norm = TwoSlopeNorm(vmin=vmin, vcenter=wt_activity, vmax=vmax)

```

```

# Plot the heatmap
sns.heatmap(full_matrix.astype(np.float64), ax=ax, cmap=cmap, norm=norm,
            cbar=False, square=True, linewidths=0.5, linecolor='black',
            mask=full_matrix.isnull())

# Create custom colorbar
sm = ScalarMappable(cmap=cmap, norm=norm)
sm.set_array([])
cb = fig.colorbar(sm, cax=cax, label='Score')

# Customize the plot
ax.set_title('Single Point Mutation Scores')
ax.set_xlabel('Mutant Amino Acid')
ax.set_ylabel('Residue Position')

# Add original amino acids to y-axis labels
ax.set_yticks(np.array(range(len(full_matrix)))+0.5)
ax.set_yticklabels([f'{original_aa_dict.get(i, "?")} {i}' for i in
    ↪full_matrix.index])

# Adjust aspect ratio to make cells square
ax.set_aspect('equal')

plt.tight_layout()

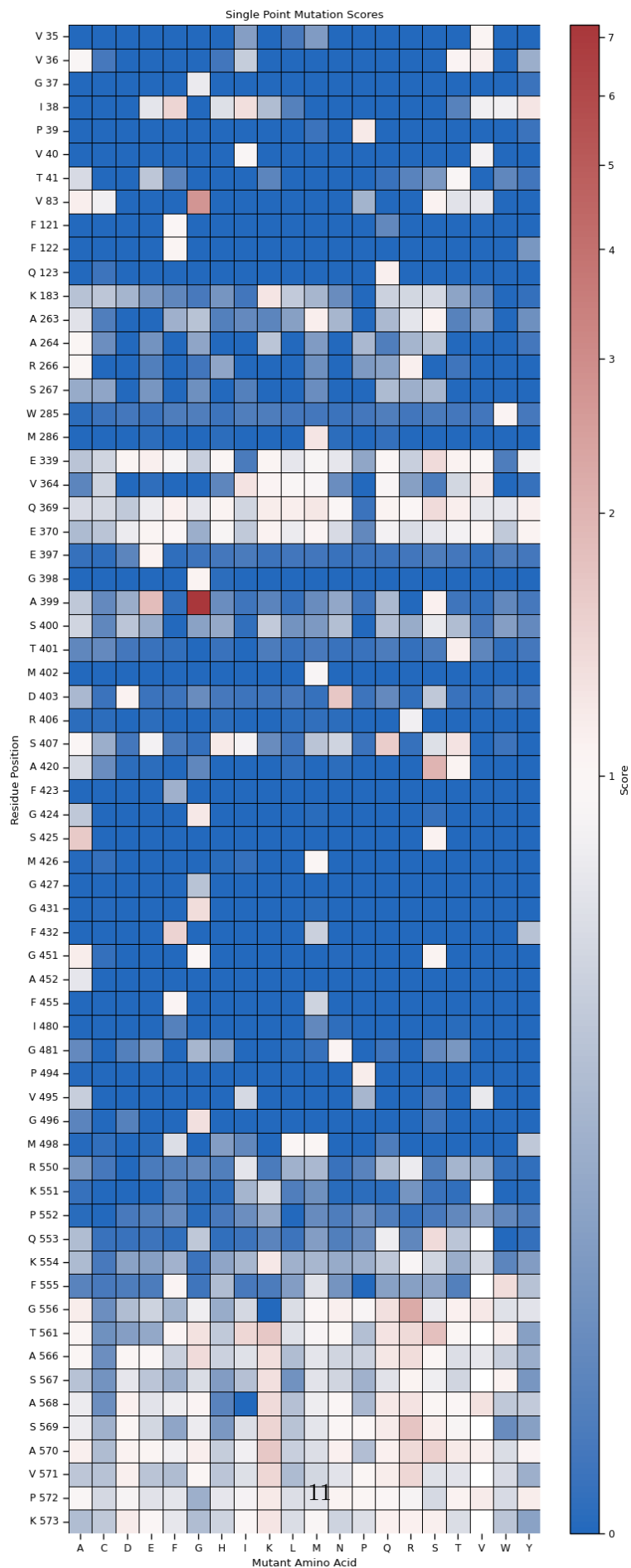
return fig, ax, full_matrix

```

```

[21]: fig, ax, matrix = plot_mutation_heatmap(mutations =
    ↪ssm_data['mutation_string'], scores=ssm_data['activity'],
    ↪all_positions=False, wt_activity=control_mean, zero_minimum=True,
    ↪log_scale_upper=True)

```



```
[22]: # highlight top 12 boxes
top_12 = ssm_data.sort_values(by='activity', ascending=False).head(12)
for _, row in top_12.iterrows():
    pos, to_aa = row['position'], row['mutation'][-1]
    row_pos = matrix.index.get_loc(pos)
    col_pos = matrix.columns.get_loc(to_aa)
    ax.add_patch(plt.Rectangle((col_pos, row_pos), 1, 1, fill=False,
    ↪edgecolor='red', lw=2))
```

```
[23]: fig.savefig('./data/ssm_heatmap.png', dpi=300)
fig
```

```
[23]:
```



```
[24]: putative_mutatable_positions = top_12['position'].unique()
print(', '.join(putative_mutatable_positions.astype(str)))
```

399,83,556,420,561,569,403,570,425,407

```
[25]: putative_mutatable_positions_wt_offset_zero_index =
↳ putative_mutatable_positions - POSITION_OFFSET - 1
```

### 1.2.3 1.3 Opportunity for Epistasis

Two methods: compute evolutionary couplings of the protein, check tcouplings between chosen sites based on top 12 SSM mutants to the overall pairwise couplings. We should expect that the couplings across our 10 sites are not on the low end compared to bulk.

Second method: look at the structure and see if some are proximal. This is qualitative. We want to see multiple nearby residue interactions.

#### 1.3.1 Evolutionary couplings

```
[42]: evc = ap.EVMutationWrapper(wt=wt, metadata_folder='./evmutation_metadata')
evc.fit(alignment)
```

/Users/ekomp/Documents/repos/EVcouplings/evcouplings/couplings/pairs.py:595:

RuntimeWarning: divide by zero encountered in log

```
z * np.log(
```

/Users/ekomp/Documents/repos/EVcouplings/evcouplings/couplings/pairs.py:595:

RuntimeWarning: invalid value encountered in multiply

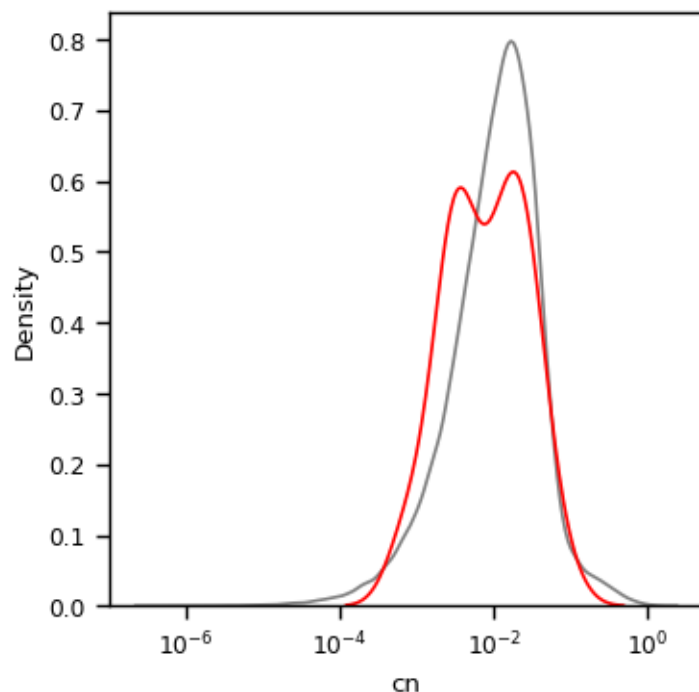
```
z * np.log(
```

```
[42]: EVMutationWrapper(metadata_folder='./evmutation_metadata',
                        wt=ProteinSequence(id='MDA_HACL/1-565',
                        seq='MAQLTGEEQIIARSLKTQGID...'))
```

```
[45]: couplings = evc.model_.ecs
target_couplings = couplings[couplings['i'].
↳ isin(putative_mutatable_positions_wt_offset_zero_index+1) & couplings['j'].
↳ isin(putative_mutatable_positions_wt_offset_zero_index+1)]
```

```
[46]: fig, ax = plt.subplots(figsize=(4, 4))
sns.kdeplot(couplings['cn'], ax=ax, color='grey', log_scale=True, label='All
↳ Couplings')
sns.kdeplot(target_couplings['cn'], ax=ax, color='red', log_scale=True,
↳ label='Target Couplings')
```

```
[46]: <AxesSubplot: xlabel='cn', ylabel='Density'>
```



```
[47]: couplings['is_target'] = couplings.index.isin(target_couplings.index)
target_cn_rankings = 1 - np.where(couplings['is_target'])[0]/len(couplings)
print('Mean CN ranks for target couplings:', target_cn_rankings.mean())
print('quartiles:', np.quantile(target_cn_rankings, [0.25, 0.5, 0.75]))
```

Mean CN ranks for target couplings: 0.6507917180973103  
quartiles: [0.56278792 0.66346576 0.81334965]

The couplings for our target sites are saturated with high coupling. Eg. 7 pairwise interactions are 90%ile couplings over whole MSA couplings. The 25%ile of target couplings is > 56% of bulk couplings. The structure also indicates that the target sites are in close proximity to each other, more opportunity for epistasis.

### 1.3.2 Structure

```
[ ]:
```

## 1.3 2. Plate 1: Additive Variant Selection (null hypothesis, no epistasis)

For the top 12 mutations, over 10 sites: 1. 9 Variants of top additive mutations 2. Remainder of the plates 2-3 mutations chosen from the top 12 mutations, each mutation is sampled equally over the plate.

```
[26]:
```

```
# drop the ssm data down to only the positions we care about eg. the top 12
↳ mutations
top_12_ssm_data = ssm_data.sort_values(by='activity', ascending=False).head(12).
↳ reset_index(drop=True)
```

### 1.3.1 2.1 Generate additive variants

```
[27]: # create all combinations of 3 mutations, and 2 mutations
```

```
from itertools import combinations
combination_indexes = list(combinations(
    top_12_ssm_data.index, 3
))

combination_indexes += list(combinations(
    top_12_ssm_data.index, 2
))
additive_df = pd.Series(name='top12_index_combinations',
↳ data=combination_indexes)
additive_df = pd.DataFrame(additive_df)
```

```
[28]: # remove any combinations that occur at the same position
```

```
def check_positions(row):
    positions = top_12_ssm_data.loc[list(row), 'position']
    return len(positions) == len(set(positions))
additive_df = additive_df[additive_df['top12_index_combinations'].
↳ apply(check_positions)]
```

```
[29]: # compute additive scores
```

```
def get_additive_score(row):
    top12_indexes = row['top12_index_combinations']
    score = (top_12_ssm_data.loc[list(top12_indexes), 'activity'].values - 1.0).
↳ sum() + 1.0
    return score
additive_df['additive_activity'] = additive_df.apply(get_additive_score, axis=1)
```

```
[30]: # sort by additive score
```

```
# chose the top plate, split into order 2 and order 3
# do not add the same mutation more than 96/12 = 8 times
# so that we explore
additive_df.sort_values(by='additive_activity', ascending=False, inplace=True)
additive_df['n_mutations'] = additive_df['top12_index_combinations'].apply(len)
```

```
[31]: # create OHE of top 12
```

```
for top12_index in top_12_ssm_data.index:
    additive_df[f'has_{top12_index}'] = additive_df['top12_index_combinations'].
↳ apply(lambda x: top12_index in x)
```



### 1.3.2 2.2 Select top variants for plate 1

**2.2.1 Top additive variants of size 1-10** Here the same mutations will be sampled many times, not balanced.

```
[32]: top_12_no_repeat_positions = top_12_ssm_data.drop_duplicates(subset='position')
```

```
[33]: top_additive = []
for i in range(2,10):
    mutations = top_12_no_repeat_positions['mutation_string'].iloc[:i].tolist()
    activities = top_12_no_repeat_positions['activity'].iloc[:i].values - 1.0
    variant = mutate_wt(wt, mutations)
    hash_ = protein_hash(variant)
    top_additive.append({
        'mutations': mutations,
        'additive_activity': 1+ sum(activities),
        'variant': variant,
        'hash': hash_,
    })
top_additive = pd.DataFrame(top_additive)
top_additive['type'] = 'TopAdditive'
```

**2.2.2 Sample from top 12 mutations max order 2-3** Here we sample mutations in order to maximize a representation score of individual mutants

```
[34]: # remove from additive_df any rows that are the same as top_additive
def check_row(row):
    top12_indexes = row['top12_index_combinations']
    mutations = top_12_ssm_data.loc[list(top12_indexes), 'mutation_string']
    for top_mutations in top_additive['mutations']:
        if set(mutations) == set(top_mutations):
            return False
    return True
additive_df = additive_df[additive_df.apply(check_row, axis=1)]
```

```
[35]: # starting with 0 representation of mutants
# add representation from top additive variants
current_mutation_representative_count = np.zeros(12)
for mutations in top_additive['mutations']:
    for mutation in mutations:
        index = top_12_ssm_data[top_12_ssm_data['mutation_string'] == mutation].
        ↪index[0]
        current_mutation_representative_count[index] += 1
```

```
[36]: additive_df
```

```
[36]:   top12_index_combinations  additive_activity  n_mutations  has_0  has_1  \
1              (0, 1, 3)              9.973371              3   True   True
```

|     |           |          |     |       |       |
|-----|-----------|----------|-----|-------|-------|
| 3   | (0, 1, 5) | 9.737164 | 3   | True  | True  |
| 4   | (0, 1, 6) | 9.674993 | 3   | True  | True  |
| 5   | (0, 1, 7) | 9.644304 | 3   | True  | True  |
| 6   | (0, 1, 8) | 9.604477 | 3   | True  | True  |
| ..  | ...       | ...      | ... | ...   | ...   |
| 281 | (8, 10)   | 2.237175 | 2   | False | False |
| 283 | (9, 10)   | 2.236267 | 2   | False | False |
| 282 | (8, 11)   | 2.199659 | 2   | False | False |
| 284 | (9, 11)   | 2.198751 | 2   | False | False |
| 285 | (10, 11)  | 2.172426 | 2   | False | False |

|     | has_2 | has_3 | has_4 | has_5 | has_6 | has_7 | has_8 | has_9 | has_10 | has_11 |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|
| 1   | False | True  | False | False | False | False | False | False | False  | False  |
| 3   | False | False | False | True  | False | False | False | False | False  | False  |
| 4   | False | False | False | False | True  | False | False | False | False  | False  |
| 5   | False | False | False | False | False | True  | False | False | False  | False  |
| 6   | False | False | False | False | False | False | True  | False | False  | False  |
| ..  | ...   | ...   | ...   | ...   | ...   | ...   | ...   | ...   | ...    | ...    |
| 281 | False | False | False | False | False | False | True  | False | True   | False  |
| 283 | False | False | False | False | False | False | False | True  | True   | False  |
| 282 | False | False | False | False | False | False | True  | False | False  | True   |
| 284 | False | False | False | False | False | False | False | True  | False  | True   |
| 285 | False | False | False | False | False | False | False | False | True   | True   |

[262 rows x 15 columns]

```
[37]: def get_representation_score(top_12_indexes):
        """Get a score that is how underrepresented the variant is"""
        # compute inverse fractions from the counts
        if sum(current_mutation_representative_count) == 0:
            return 1.0
        else:
            inverse_ratios = 1 - current_mutation_representative_count /
            ↪sum(current_mutation_representative_count)

            return sum([inverse_ratios[ind] for ind in top_12_indexes]) /
            ↪len(top_12_indexes)

        selected_additive_indexes = []

        two_or_three_bool_switch = False
        while len(selected_additive_indexes) < (96 - len(top_additive)):

            if two_or_three_bool_switch:
                n_mutations = 3
            else:
                n_mutations = 2
```

```

# switch
two_or_three_bool_switch = not two_or_three_bool_switch

# df subset to the correct mutation count and remove already chosen variants
df_subset = additive_df[additive_df['n_mutations'] == n_mutations]
df_subset = df_subset[~df_subset.index.isin(selected_additive_indexes)]

# select the mutation with the current least representation
# show only variants with that mutation
selected_mutation_indexes_sorted = np.
↳argsort(current_mutation_representative_count)
for selected_mutation_index in selected_mutation_indexes_sorted:
    if df_subset[f'has_{selected_mutation_index}'].sum() > 0:
        break

# select the variant that maximizes representation
df_subset['representation_score'] = df_subset['top12_index_combinations'].
↳apply(get_representation_score)
# sort by representation score and select the first
df_subset.sort_values(by='representation_score', ascending=False,
↳inplace=True)

selected_additive_index = df_subset.index[0]
selected_variant_12_indexes = df_subset.loc[selected_additive_index,
↳'top12_index_combinations']

# add the selected index to the list
selected_additive_indexes.append(selected_additive_index)
print(len(selected_additive_indexes))
# update the representative count
for ind in selected_variant_12_indexes:
    current_mutation_representative_count[ind] += 1

print("Selected: ", selected_additive_index, selected_variant_12_indexes,
↳"Additive score: ", df_subset.loc[selected_additive_index,
↳'additive_activity'])
print("Current representative mutation ratio: ",
↳current_mutation_representative_count/np.
↳sum(current_mutation_representative_count))
print("=====")

```

1

```

Selected: 261 (4, 8) Additive score: 2.4751760490517722
Current representative mutation ratio: [0.17391304 0.17391304 0.15217391
0.13043478 0.02173913 0.10869565
0.08695652 0.06521739 0.02173913 0.04347826 0.02173913 0.
]
=====

```

2

Selected: 181 (4, 8, 11) Additive score: 3.042631393968174  
Current representative mutation ratio: [0.16326531 0.16326531 0.14285714  
0.12244898 0.04081633 0.10204082  
0.08163265 0.06122449 0.04081633 0.04081633 0.02040816 0.02040816]  
=====

3

Selected: 285 (10, 11) Additive score: 2.172426303012551  
Current representative mutation ratio: [0.15686275 0.15686275 0.1372549  
0.11764706 0.03921569 0.09803922  
0.07843137 0.05882353 0.03921569 0.03921569 0.03921569 0.03921569]  
=====

4

Selected: 219 (9, 10, 11) Additive score: 2.80372218650148  
Current representative mutation ratio: [0.14814815 0.14814815 0.12962963  
0.11111111 0.03703704 0.09259259  
0.07407407 0.05555556 0.03703704 0.05555556 0.05555556 0.05555556]  
=====

5

Selected: 264 (4, 11) Additive score: 2.4104274615938044  
Current representative mutation ratio: [0.14285714 0.14285714 0.125  
0.10714286 0.05357143 0.08928571  
0.07142857 0.05357143 0.03571429 0.05357143 0.05357143 0.07142857]  
=====

6

Selected: 175 (4, 7, 8) Additive score: 3.147206375142583  
Current representative mutation ratio: [0.13559322 0.13559322 0.11864407  
0.10169492 0.06779661 0.08474576  
0.06779661 0.06779661 0.05084746 0.05084746 0.05084746 0.06779661]  
=====

7

Selected: 283 (9, 10) Additive score: 2.236266841585078  
Current representative mutation ratio: [0.13114754 0.13114754 0.1147541  
0.09836066 0.06557377 0.08196721  
0.06557377 0.06557377 0.04918033 0.06557377 0.06557377 0.06557377]  
=====

8

Selected: 218 (8, 10, 11) Additive score: 2.8046302353869206  
Current representative mutation ratio: [0.125 0.125 0.109375 0.09375  
0.0625 0.078125 0.0625 0.0625  
0.0625 0.0625 0.078125 0.078125]  
=====

9

Selected: 271 (6, 7) Additive score: 2.374750619162735  
Current representative mutation ratio: [0.12121212 0.12121212 0.10606061  
0.09090909 0.06060606 0.07575758  
0.07575758 0.07575758 0.06060606 0.06060606 0.07575758 0.07575758]  
=====

10  
 Selected: 179 (4, 8, 9) Additive score: 3.1064719325407015  
 Current representative mutation ratio: [0.11594203 0.11594203 0.10144928  
 0.08695652 0.07246377 0.07246377  
 0.07246377 0.07246377 0.07246377 0.07246377 0.07246377 0.07246377]  
 =====

11  
 Selected: 284 (9, 11) Additive score: 2.1987512284053308  
 Current representative mutation ratio: [0.11267606 0.11267606 0.09859155  
 0.08450704 0.07042254 0.07042254  
 0.07042254 0.07042254 0.07042254 0.08450704 0.07042254 0.08450704]  
 =====

12  
 Selected: 177 (4, 7, 10) Additive score: 3.1199734008643625  
 Current representative mutation ratio: [0.10810811 0.10810811 0.09459459  
 0.08108108 0.08108108 0.06756757  
 0.06756757 0.08108108 0.06756757 0.08108108 0.08108108 0.08108108]  
 =====

13  
 Selected: 265 (5, 6) Additive score: 2.467610958507764  
 Current representative mutation ratio: [0.10526316 0.10526316 0.09210526  
 0.07894737 0.07894737 0.07894737  
 0.07894737 0.07894737 0.06578947 0.07894737 0.07894737 0.07894737]  
 =====

14  
 Selected: 217 (8, 9, 11) Additive score: 2.8309551607797006  
 Current representative mutation ratio: [0.10126582 0.10126582 0.08860759  
 0.07594937 0.07594937 0.07594937  
 0.07594937 0.07594937 0.07594937 0.08860759 0.07594937 0.08860759]  
 =====

15  
 Selected: 258 (4, 5) Additive score: 2.6078627821132425  
 Current representative mutation ratio: [0.09876543 0.09876543 0.08641975  
 0.07407407 0.08641975 0.08641975  
 0.07407407 0.07407407 0.07407407 0.08641975 0.07407407 0.08641975]  
 =====

16  
 Selected: 156 (3, 7, 10) Additive score: 3.2780989414636723  
 Current representative mutation ratio: [0.0952381 0.0952381 0.08333333  
 0.08333333 0.08333333 0.08333333  
 0.07142857 0.08333333 0.07142857 0.08333333 0.08333333 0.08333333]  
 =====

17  
 Selected: 272 (6, 8) Additive score: 2.3349242254462936  
 Current representative mutation ratio: [0.09302326 0.09302326 0.08139535  
 0.08139535 0.08139535 0.08139535  
 0.08139535 0.08139535 0.08139535 0.08139535 0.08139535 0.08139535]  
 =====

18

Selected: 107 (2, 3, 11) Additive score: 3.7771828915841725  
Current representative mutation ratio: [0.08988764 0.08988764 0.08988764  
0.08988764 0.07865169 0.07865169  
0.07865169 0.07865169 0.07865169 0.07865169 0.07865169 0.08988764]  
=====

19

Selected: 269 (5, 10) Additive score: 2.369861623531989  
Current representative mutation ratio: [0.08791209 0.08791209 0.08791209  
0.08791209 0.07692308 0.08791209  
0.07692308 0.07692308 0.07692308 0.07692308 0.08791209 0.08791209]  
=====

20

Selected: 200 (6, 7, 8) Additive score: 3.0069545515371043  
Current representative mutation ratio: [0.08510638 0.08510638 0.08510638  
0.08510638 0.07446809 0.08510638  
0.08510638 0.08510638 0.08510638 0.07446809 0.08510638 0.08510638]  
=====

21

Selected: 262 (4, 9) Additive score: 2.4742680001663317  
Current representative mutation ratio: [0.08333333 0.08333333 0.08333333  
0.08333333 0.08333333 0.08333333  
0.08333333 0.08333333 0.08333333 0.08333333 0.08333333 0.08333333]  
=====

22

Selected: 1 (0, 1, 3) Additive score: 9.973370839698907  
Current representative mutation ratio: [0.09090909 0.09090909 0.08080808  
0.09090909 0.08080808 0.08080808  
0.08080808 0.08080808 0.08080808 0.08080808 0.08080808 0.08080808]  
=====

23

Selected: 247 (2, 9) Additive score: 2.8399257728799876  
Current representative mutation ratio: [0.08910891 0.08910891 0.08910891  
0.08910891 0.07920792 0.07920792  
0.07920792 0.07920792 0.07920792 0.08910891 0.07920792 0.07920792]  
=====

24

Selected: 215 (7, 10, 11) Additive score: 2.8444566291033615  
Current representative mutation ratio: [0.08653846 0.08653846 0.08653846  
0.08653846 0.07692308 0.07692308  
0.07692308 0.08653846 0.07692308 0.08653846 0.08653846 0.08653846]  
=====

25

Selected: 259 (4, 6) Additive score: 2.5456924097493268  
Current representative mutation ratio: [0.08490566 0.08490566 0.08490566  
0.08490566 0.08490566 0.0754717  
0.08490566 0.08490566 0.0754717 0.08490566 0.08490566 0.08490566]  
=====

26

Selected: 3 (0, 1, 5) Additive score: 9.737163847858033  
Current representative mutation ratio: [0.09174312 0.09174312 0.08256881  
0.08256881 0.08256881 0.08256881  
0.08256881 0.08256881 0.0733945 0.08256881 0.08256881 0.08256881]  
=====

27

Selected: 246 (2, 8) Additive score: 2.840833821765428  
Current representative mutation ratio: [0.09009009 0.09009009 0.09009009  
0.08108108 0.08108108 0.08108108  
0.08108108 0.08108108 0.08108108 0.08108108 0.08108108 0.08108108]  
=====

28

Selected: 216 (8, 9, 10) Additive score: 2.868470773959448  
Current representative mutation ratio: [0.0877193 0.0877193 0.0877193  
0.07894737 0.07894737 0.07894737  
0.07894737 0.07894737 0.0877193 0.0877193 0.0877193 0.07894737]  
=====

29

Selected: 250 (3, 4) Additive score: 2.844069773954115  
Current representative mutation ratio: [0.0862069 0.0862069 0.0862069  
0.0862069 0.0862069 0.07758621  
0.07758621 0.07758621 0.0862069 0.0862069 0.0862069 0.07758621]  
=====

30

Selected: 185 (5, 6, 7) Additive score: 3.1396412845985746  
Current representative mutation ratio: [0.08403361 0.08403361 0.08403361  
0.08403361 0.08403361 0.08403361  
0.08403361 0.08403361 0.08403361 0.08403361 0.08403361 0.07563025]  
=====

31

Selected: 282 (8, 11) Additive score: 2.1996592772907713  
Current representative mutation ratio: [0.08264463 0.08264463 0.08264463  
0.08264463 0.08264463 0.08264463  
0.08264463 0.08264463 0.09090909 0.08264463 0.08264463 0.08264463]  
=====

32

Selected: 4 (0, 1, 6) Additive score: 9.674993475494118  
Current representative mutation ratio: [0.08870968 0.08870968 0.08064516  
0.08064516 0.08064516 0.08064516  
0.08870968 0.08064516 0.08870968 0.08064516 0.08064516 0.08064516]  
=====

33

Selected: 245 (2, 7) Additive score: 2.880660215481869  
Current representative mutation ratio: [0.08730159 0.08730159 0.08730159  
0.07936508 0.07936508 0.07936508  
0.08730159 0.08730159 0.08730159 0.07936508 0.07936508 0.07936508]  
=====

34

Selected: 148 (3, 5, 11) Additive score: 3.333443667628954  
Current representative mutation ratio: [0.08527132 0.08527132 0.08527132  
0.08527132 0.07751938 0.08527132  
0.08527132 0.08527132 0.08527132 0.07751938 0.07751938 0.08527132]  
=====

35

Selected: 263 (4, 10) Additive score: 2.4479430747735518  
Current representative mutation ratio: [0.08396947 0.08396947 0.08396947  
0.08396947 0.08396947 0.08396947  
0.08396947 0.08396947 0.08396947 0.07633588 0.08396947 0.08396947]  
=====

36

Selected: 87 (1, 6, 9) Additive score: 4.083201157017857  
Current representative mutation ratio: [0.08208955 0.08955224 0.08208955  
0.08208955 0.08208955 0.08208955  
0.08955224 0.08208955 0.08208955 0.08208955 0.08208955 0.08208955]  
=====

37

Selected: 221 (0, 2) Additive score: 8.431718091356249  
Current representative mutation ratio: [0.08823529 0.08823529 0.08823529  
0.08088235 0.08088235 0.08088235  
0.08823529 0.08088235 0.08088235 0.08088235 0.08088235 0.08088235]  
=====

38

Selected: 214 (7, 9, 11) Additive score: 2.8707815544961415  
Current representative mutation ratio: [0.08633094 0.08633094 0.08633094  
0.07913669 0.07913669 0.07913669  
0.08633094 0.08633094 0.07913669 0.08633094 0.07913669 0.08633094]  
=====

39

Selected: 281 (8, 10) Additive score: 2.2371748904705187  
Current representative mutation ratio: [0.08510638 0.08510638 0.08510638  
0.07801418 0.07801418 0.07801418  
0.08510638 0.08510638 0.08510638 0.08510638 0.08510638 0.08510638]  
=====

40

Selected: 136 (3, 4, 5) Additive score: 3.608960439389955  
Current representative mutation ratio: [0.08333333 0.08333333 0.08333333  
0.08333333 0.08333333 0.08333333  
0.08333333 0.08333333 0.08333333 0.08333333 0.08333333 0.08333333]  
=====

41

Selected: 222 (0, 3) Additive score: 8.224185859241903  
Current representative mutation ratio: [0.0890411 0.08219178 0.08219178  
0.0890411 0.08219178 0.08219178  
0.08219178 0.08219178 0.08219178 0.08219178 0.08219178 0.08219178]  
=====



42

Selected: 108 (2, 4, 5) Additive score: 3.8164926715043013  
Current representative mutation ratio: [0.08724832 0.08053691 0.08724832  
0.08724832 0.08724832 0.08724832  
0.08053691 0.08053691 0.08053691 0.08053691 0.08053691 0.08053691]  
=====

43

Selected: 279 (7, 11) Additive score: 2.2394856710072126  
Current representative mutation ratio: [0.08609272 0.0794702 0.08609272  
0.08609272 0.08609272 0.08609272  
0.0794702 0.08609272 0.0794702 0.0794702 0.0794702 0.08609272]  
=====

44

Selected: 88 (1, 6, 10) Additive score: 4.056876231625077  
Current representative mutation ratio: [0.08441558 0.08441558 0.08441558  
0.08441558 0.08441558 0.08441558  
0.08441558 0.08441558 0.07792208 0.07792208 0.08441558 0.08441558]  
=====

45

Selected: 280 (8, 9) Additive score: 2.2634998158632986  
Current representative mutation ratio: [0.08333333 0.08333333 0.08333333  
0.08333333 0.08333333 0.08333333  
0.08333333 0.08333333 0.08333333 0.08333333 0.08333333 0.08333333]  
=====

46

Selected: 5 (0, 1, 7) Additive score: 9.644303508513005  
Current representative mutation ratio: [0.08805031 0.08805031 0.08176101  
0.08176101 0.08176101 0.08176101  
0.08176101 0.08805031 0.08176101 0.08176101 0.08176101 0.08176101]  
=====

47

Selected: 244 (2, 6) Additive score: 2.911350182462982  
Current representative mutation ratio: [0.08695652 0.08695652 0.08695652  
0.08074534 0.08074534 0.08074534  
0.08695652 0.08695652 0.08074534 0.08074534 0.08074534 0.08074534]  
=====

48

Selected: 160 (3, 8, 11) Additive score: 3.200756934567484  
Current representative mutation ratio: [0.08536585 0.08536585 0.08536585  
0.08536585 0.07926829 0.07926829  
0.08536585 0.08536585 0.08536585 0.07926829 0.07926829 0.08536585]  
=====

49

Selected: 268 (5, 9) Additive score: 2.396186548924769  
Current representative mutation ratio: [0.08433735 0.08433735 0.08433735  
0.08433735 0.07831325 0.08433735  
0.08433735 0.08433735 0.08433735 0.08433735 0.07831325 0.08433735]  
=====

50  
 Selected: 77 (1, 4, 10) Additive score: 4.197128055230556  
 Current representative mutation ratio: [0.08284024 0.0887574 0.08284024  
 0.08284024 0.08284024 0.08284024  
 0.08284024 0.08284024 0.08284024 0.08284024 0.08284024 0.08284024]  
 =====

51  
 Selected: 224 (0, 5) Additive score: 7.98797886740103  
 Current representative mutation ratio: [0.0877193 0.0877193 0.08187135  
 0.08187135 0.08187135 0.0877193  
 0.08187135 0.08187135 0.08187135 0.08187135 0.08187135 0.08187135]  
 =====

52  
 Selected: 109 (2, 4, 6) Additive score: 3.7543222991403855  
 Current representative mutation ratio: [0.0862069 0.0862069 0.0862069  
 0.08045977 0.0862069 0.0862069  
 0.0862069 0.08045977 0.08045977 0.08045977 0.08045977 0.08045977]  
 =====

53  
 Selected: 257 (3, 11) Additive score: 2.568553002193114  
 Current representative mutation ratio: [0.08522727 0.08522727 0.08522727  
 0.08522727 0.08522727 0.08522727  
 0.08522727 0.07954545 0.07954545 0.07954545 0.07954545 0.08522727]  
 =====

54  
 Selected: 213 (7, 9, 10) Additive score: 2.908297167675889  
 Current representative mutation ratio: [0.08379888 0.08379888 0.08379888  
 0.08379888 0.08379888 0.08379888  
 0.08379888 0.08379888 0.07821229 0.08379888 0.08379888 0.08379888]  
 =====

55  
 Selected: 227 (0, 8) Additive score: 7.855292134339559  
 Current representative mutation ratio: [0.08839779 0.08287293 0.08287293  
 0.08287293 0.08287293 0.08287293  
 0.08287293 0.08287293 0.08287293 0.08287293 0.08287293 0.08287293]  
 =====

56  
 Selected: 106 (2, 3, 10) Additive score: 3.8146985047639195  
 Current representative mutation ratio: [0.08695652 0.08152174 0.08695652  
 0.08695652 0.08152174 0.08152174  
 0.08152174 0.08152174 0.08152174 0.08152174 0.08695652 0.08152174]  
 =====

57  
 Selected: 275 (6, 11) Additive score: 2.2701756379883262  
 Current representative mutation ratio: [0.08602151 0.08064516 0.08602151  
 0.08602151 0.08064516 0.08064516  
 0.08602151 0.08064516 0.08064516 0.08064516 0.08602151 0.08602151]  
 =====

58

Selected: 76 (1, 4, 9) Additive score: 4.223452980623336  
Current representative mutation ratio: [0.08465608 0.08465608 0.08465608  
0.08465608 0.08465608 0.07936508  
0.08465608 0.07936508 0.07936508 0.08465608 0.08465608 0.08465608]  
=====

59

Selected: 276 (7, 8) Additive score: 2.30423425846518  
Current representative mutation ratio: [0.08376963 0.08376963 0.08376963  
0.08376963 0.08376963 0.07853403  
0.08376963 0.08376963 0.08376963 0.08376963 0.08376963 0.08376963]  
=====

60

Selected: 167 (4, 5, 9) Additive score: 3.2391586656021714  
Current representative mutation ratio: [0.08247423 0.08247423 0.08247423  
0.08247423 0.08762887 0.08247423  
0.08247423 0.08247423 0.08247423 0.08762887 0.08247423 0.08247423]  
=====

61

Selected: 225 (0, 6) Additive score: 7.925808495037114  
Current representative mutation ratio: [0.08673469 0.08163265 0.08163265  
0.08163265 0.08673469 0.08163265  
0.08673469 0.08163265 0.08163265 0.08673469 0.08163265 0.08163265]  
=====

62

Selected: 212 (7, 8, 11) Additive score: 2.871689603381582  
Current representative mutation ratio: [0.08542714 0.08040201 0.08040201  
0.08040201 0.08542714 0.08040201  
0.08542714 0.08542714 0.08542714 0.08542714 0.08040201 0.08542714]  
=====

63

Selected: 243 (2, 5) Additive score: 2.973520554826898  
Current representative mutation ratio: [0.08457711 0.07960199 0.08457711  
0.07960199 0.08457711 0.08457711  
0.08457711 0.08457711 0.08457711 0.08457711 0.07960199 0.08457711]  
=====

64

Selected: 70 (1, 3, 10) Additive score: 4.355253595829866  
Current representative mutation ratio: [0.08333333 0.08333333 0.08333333  
0.08333333 0.08333333 0.08333333  
0.08333333 0.08333333 0.08333333 0.08333333 0.08333333 0.08333333]  
=====

65

Selected: 226 (0, 7) Additive score: 7.895118528056001  
Current representative mutation ratio: [0.08737864 0.08252427 0.08252427  
0.08252427 0.08252427 0.08252427  
0.08252427 0.08737864 0.08252427 0.08252427 0.08252427 0.08252427]  
=====

66

Selected: 105 (2, 3, 9) Additive score: 3.8410234301566994  
Current representative mutation ratio: [0.0861244 0.08133971 0.0861244  
0.0861244 0.08133971 0.08133971  
0.08133971 0.0861244 0.08133971 0.0861244 0.08133971 0.08133971]  
=====

67

Selected: 274 (6, 10) Additive score: 2.307691251168073  
Current representative mutation ratio: [0.08530806 0.08056872 0.08530806  
0.08530806 0.08056872 0.08056872  
0.08530806 0.08530806 0.08056872 0.08530806 0.08530806 0.08056872]  
=====

68

Selected: 78 (1, 4, 11) Additive score: 4.159612442050809  
Current representative mutation ratio: [0.08411215 0.08411215 0.08411215  
0.08411215 0.08411215 0.07943925  
0.08411215 0.08411215 0.07943925 0.08411215 0.08411215 0.08411215]  
=====

69

Selected: 270 (5, 11) Additive score: 2.332346010352242  
Current representative mutation ratio: [0.08333333 0.08333333 0.08333333  
0.08333333 0.08333333 0.08333333  
0.08333333 0.08333333 0.0787037 0.08333333 0.08333333 0.08796296]  
=====

70

Selected: 6 (0, 1, 8) Additive score: 9.604477114796564  
Current representative mutation ratio: [0.08675799 0.08675799 0.08219178  
0.08219178 0.08219178 0.08219178  
0.08219178 0.08219178 0.08219178 0.08219178 0.08219178 0.08675799]  
=====

71

Selected: 242 (2, 4) Additive score: 3.051602006068461  
Current representative mutation ratio: [0.08597285 0.08597285 0.08597285  
0.08144796 0.08597285 0.08144796  
0.08144796 0.08144796 0.08144796 0.08144796 0.08144796 0.08597285]  
=====

72

Selected: 161 (3, 9, 10) Additive score: 3.2373644988617905  
Current representative mutation ratio: [0.08482143 0.08482143 0.08482143  
0.08482143 0.08482143 0.08035714  
0.08035714 0.08035714 0.08035714 0.08482143 0.08482143 0.08482143]  
=====

73

Selected: 266 (5, 7) Additive score: 2.4369209915266508  
Current representative mutation ratio: [0.0840708 0.0840708 0.0840708  
0.0840708 0.0840708 0.0840708  
0.07964602 0.0840708 0.07964602 0.0840708 0.0840708 0.0840708 ]  
=====

74

Selected: 171 (4, 6, 8) Additive score: 3.177896342123696  
Current representative mutation ratio: [0.08296943 0.08296943 0.08296943  
0.08296943 0.08733624 0.08296943  
0.08296943 0.08296943 0.08296943 0.08296943 0.08296943 0.08296943]  
=====

75

Selected: 228 (0, 9) Additive score: 7.854384085454119  
Current representative mutation ratio: [0.08658009 0.08225108 0.08225108  
0.08225108 0.08658009 0.08225108  
0.08225108 0.08225108 0.08225108 0.08658009 0.08225108 0.08225108]  
=====

76

Selected: 209 (6, 10, 11) Additive score: 2.875146596084475  
Current representative mutation ratio: [0.08547009 0.08119658 0.08119658  
0.08119658 0.08547009 0.08119658  
0.08547009 0.08119658 0.08119658 0.08547009 0.08547009 0.08547009]  
=====

77

Selected: 241 (2, 3) Additive score: 3.2097275466677706  
Current representative mutation ratio: [0.08474576 0.08050847 0.08474576  
0.08474576 0.08474576 0.08050847  
0.08474576 0.08050847 0.08050847 0.08474576 0.08474576 0.08474576]  
=====

78

Selected: 80 (1, 5, 7) Additive score: 4.186105971983655  
Current representative mutation ratio: [0.08368201 0.08368201 0.08368201  
0.08368201 0.08368201 0.08368201  
0.08368201 0.08368201 0.07949791 0.08368201 0.08368201 0.08368201]  
=====

79

Selected: 254 (3, 8) Additive score: 2.633301589651082  
Current representative mutation ratio: [0.08298755 0.08298755 0.08298755  
0.08713693 0.08298755 0.08298755  
0.08298755 0.08298755 0.08298755 0.08298755 0.08298755 0.08298755]  
=====

80

Selected: 7 (0, 1, 9) Additive score: 9.603569065911124  
Current representative mutation ratio: [0.08606557 0.08606557 0.08196721  
0.08606557 0.08196721 0.08196721  
0.08196721 0.08196721 0.08196721 0.08606557 0.08196721 0.08196721]  
=====

81

Selected: 278 (7, 10) Additive score: 2.2770012841869596  
Current representative mutation ratio: [0.08536585 0.08536585 0.08130081  
0.08536585 0.08130081 0.08130081  
0.08130081 0.08536585 0.08130081 0.08536585 0.08536585 0.08130081]  
=====

```

82
Selected: 174 (4, 6, 11) Additive score: 3.1131477546657287
Current representative mutation ratio: [0.08433735 0.08433735 0.08032129
0.08433735 0.08433735 0.08032129
0.08433735 0.08433735 0.08032129 0.08433735 0.08433735 0.08433735]
=====
83
Selected: 249 (2, 11) Additive score: 2.7760852343074602
Current representative mutation ratio: [0.08366534 0.08366534 0.08366534
0.08366534 0.08366534 0.07968127
0.08366534 0.08366534 0.07968127 0.08366534 0.08366534 0.0876494 ]
=====
84
Selected: 131 (2, 8, 10) Additive score: 3.445804779861577
Current representative mutation ratio: [0.08267717 0.08267717 0.08661417
0.08267717 0.08267717 0.07874016
0.08267717 0.08267717 0.08267717 0.08267717 0.08661417 0.08661417]
=====
85
Selected: 234 (1, 5) Additive score: 3.514075645892844
Current representative mutation ratio: [0.08203125 0.0859375 0.0859375
0.08203125 0.08203125 0.08203125
0.08203125 0.08203125 0.08203125 0.08203125 0.0859375 0.0859375 ]
=====
86
Selected: 154 (3, 7, 8) Additive score: 3.305331915741893
Current representative mutation ratio: [0.08108108 0.08494208 0.08494208
0.08494208 0.08108108 0.08108108
0.08108108 0.08494208 0.08494208 0.08108108 0.08494208 0.08494208]
=====
87
Selected: 273 (6, 9) Additive score: 2.334016176560853
Current representative mutation ratio: [0.08045977 0.08429119 0.08429119
0.08429119 0.08045977 0.08045977
0.08429119 0.08429119 0.08429119 0.08429119 0.08429119 0.08429119]
=====
88
Selected: 168 (4, 5, 10) Additive score: 3.2128337402093914
Current representative mutation ratio: [0.07954545 0.08333333 0.08333333
0.08333333 0.08333333 0.08333333
0.08333333 0.08333333 0.08333333 0.08333333 0.08712121 0.08333333]
=====

```

```
[38]: current_mutation_representative_count
```

```
[38]: array([21., 22., 22., 22., 22., 22., 22., 22., 22., 22., 23., 22.])
```

```
[39]: # df of the combinations of mutants to chose
selected_additive = additive_df.loc[selected_additive_indexes]

[40]: # parse mutation strings and get variant scores
# format into dataframe as expected for output
def get_variant_metadata_from_top12(row):
    top12_indexes = row['top12_index_combinations']

    mutations = []
    for ind in top12_indexes:
        top12_row = top_12_ssm_data.loc[ind]
        mutation_string = top12_row['mutation_string']
        mutations.append(mutation_string)

    variant = mutate_wt(wt, mutations)
    hash_ = protein_hash(variant)
    activity = row['additive_activity']
    return {
        'mutations': mutations,
        'additive_activity': activity,
        'variant': variant,
        'hash': hash_,
    }

plate1_df = pd.DataFrame(list(selected_additive.
    ↪apply(get_variant_metadata_from_top12, axis=1)))
plate1_df['type'] = 'Additive'

[41]: plate1_df = pd.concat([top_additive, plate1_df], ignore_index=True)

[42]: plate1_df['mutations'] = plate1_df['mutations'].apply(lambda x: '|'.join(x))

[43]: plate1_df.to_csv('./data/plate1.csv', index=False)
```

## 1.4 3. ML development

Check whether supervised models can predict the SSM data, optimize them for accuracy and error calibration.

### 1.4.1 3.1 Analysis of ESM embeddings - variance, PCA

Embed SSM data with ESM, check for signal between components and activity.

```
[42]: ssm_ml_df = ssm_data.copy().sample(frac=1.0)
ssm_ml_df = ssm_ml_df[['activity', 'mutation_string', 'variant', 'hash']]

# group by hash and average (duplicate controls)
ssm_ml_df = ssm_ml_df.groupby('variant').mean().reset_index().dropna()
```

```
[43]: X_seq = ap.ProteinSequences.from_list(ssm_ml_df['variant'].tolist())
      y = ssm_ml_df['activity'].values
```

### 3.1.1 Embed sequences

```
[44]: esm_full = ap.ESM2Embedding(
      model_checkpoint='esm2_t33_650M_UR50D',
      metadata_folder='esm2_embedder',
      pool=True,
      device='mps'
    )
    esm_full.fit([])
```

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-packages/huggingface\_hub/file\_download.py:1132: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force\_download=True`.

```
warnings.warn(
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t33_650M_UR50D and are newly initialized:
['esm.pooler.dense.bias', 'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
```

```
[44]: ESM2Embedding(device='mps', metadata_folder='esm2_embedder',
      model_checkpoint='esm2_t33_650M_UR50D', pool=True,
      use_cache=True)
```

```
[45]: X_embeddings = esm_full.transform(X_seq)
```

/Users/ekomp/Documents/repos/lab/aide\_predict/aide\_predict/bespoke\_models/base.py:794: UserWarning: Cache found but model state has changed. Clearing cache.

```
warnings.warn("Cache found but model state has changed. Clearing cache.")
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t33_650M_UR50D and are newly initialized:
['esm.pooler.dense.bias', 'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
Computing ESM2 embeddings: 100%|          | 1210/1210 [03:36<00:00, 5.59it/s]
```

### 3.1.2 PCA and variance

```
[46]: scaler = StandardScaler()
      pca = PCA(n_components=0.98)
```

```
[47]: X = scaler.fit_transform(X_embeddings)
      X = pca.fit_transform(X)
```

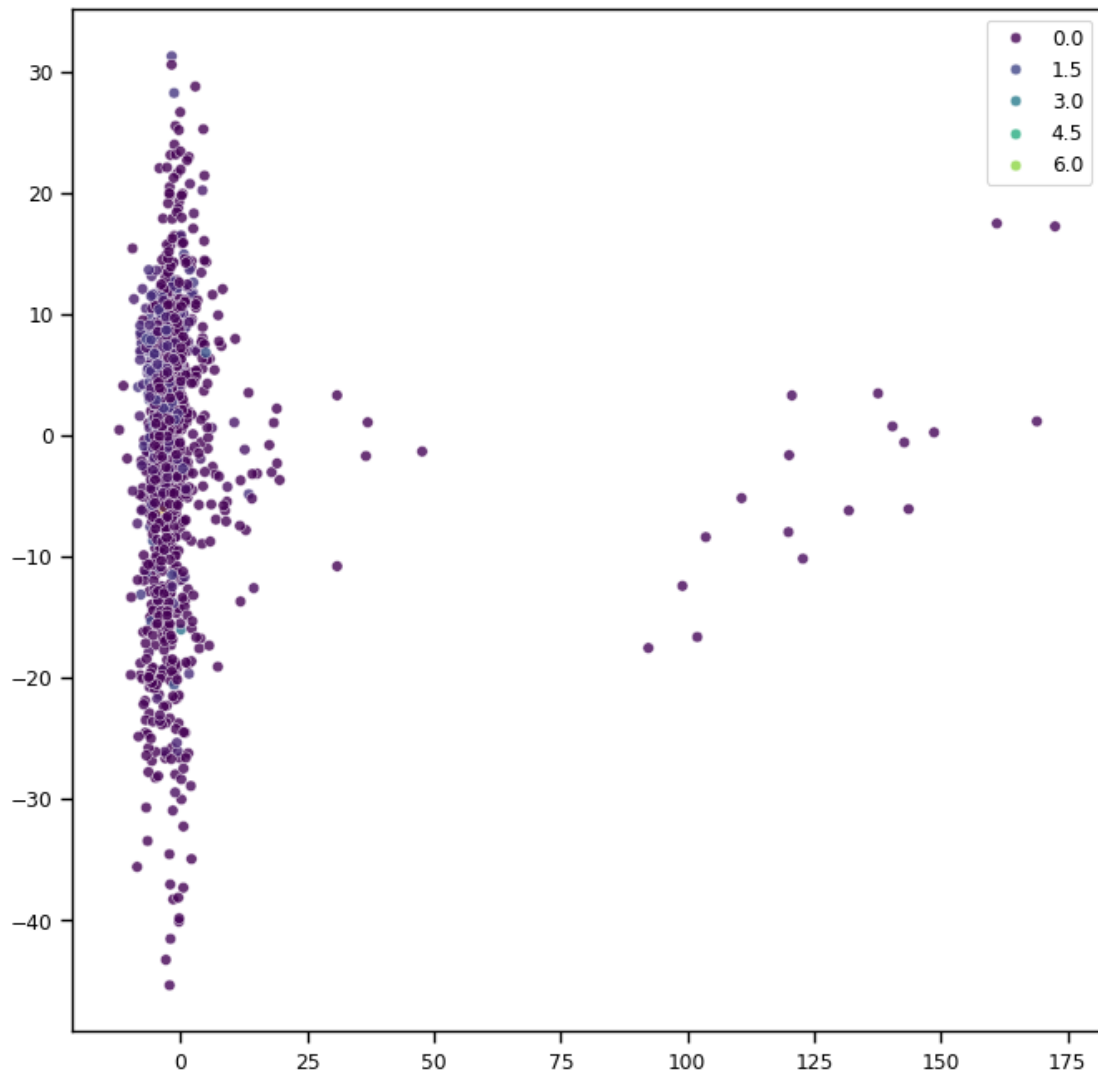
```
[48]: pca.explained_variance_ratio_
```



```
[48]: array([2.20430776e-01, 1.00171655e-01, 6.48374632e-02, 5.24517186e-02,
4.82834242e-02, 3.54132280e-02, 3.25674973e-02, 2.77404450e-02,
2.68537533e-02, 2.27018856e-02, 2.02325471e-02, 1.66699849e-02,
1.48824872e-02, 1.41711663e-02, 1.32797863e-02, 1.26281166e-02,
1.12953540e-02, 1.05874473e-02, 1.02719953e-02, 9.79800802e-03,
9.22717899e-03, 8.34317598e-03, 7.97180738e-03, 7.58118369e-03,
7.23217288e-03, 6.60497043e-03, 6.39181305e-03, 6.17097644e-03,
5.81966899e-03, 5.32503147e-03, 5.18564461e-03, 4.80412552e-03,
4.65377513e-03, 4.45784954e-03, 4.25511226e-03, 4.10102122e-03,
3.92152742e-03, 3.75464256e-03, 3.60957813e-03, 3.31092207e-03,
3.20709799e-03, 3.05719045e-03, 2.94623105e-03, 2.82893004e-03,
2.76155234e-03, 2.69750878e-03, 2.60946085e-03, 2.58828141e-03,
2.37940066e-03, 2.18662200e-03, 2.14336836e-03, 2.01146980e-03,
1.93342404e-03, 1.92660757e-03, 1.84134638e-03, 1.80536788e-03,
1.70585630e-03, 1.69525912e-03, 1.62384554e-03, 1.59703055e-03,
1.52482756e-03, 1.47623417e-03, 1.43441465e-03, 1.42456673e-03,
1.31011393e-03, 1.30071433e-03, 1.25715078e-03, 1.17763260e-03,
1.17481034e-03, 1.14777952e-03, 1.11895439e-03, 1.07421679e-03,
1.04254612e-03, 1.02899747e-03, 1.00289145e-03, 9.77655756e-04,
9.46696964e-04, 9.12887277e-04, 8.88093840e-04, 8.54960177e-04,
8.41316942e-04, 8.11704260e-04, 7.99254980e-04, 7.77122390e-04,
7.69915525e-04, 7.49942206e-04, 7.18838244e-04, 7.12969631e-04,
7.03328929e-04, 6.88933535e-04, 6.69935951e-04, 6.47423731e-04,
6.39143109e-04, 6.08829141e-04, 6.07325986e-04, 5.98396698e-04,
5.81785687e-04, 5.71659184e-04, 5.50495693e-04, 5.49532298e-04,
5.37420507e-04, 5.29502926e-04, 5.04819327e-04, 4.97289933e-04,
4.81719413e-04, 4.77640366e-04, 4.66758356e-04, 4.61997872e-04,
4.52400389e-04, 4.41174285e-04, 4.35584545e-04, 4.30428161e-04,
4.26048064e-04, 4.23744525e-04, 4.07051615e-04, 3.97450029e-04,
3.92814894e-04, 3.88218905e-04, 3.77352233e-04, 3.67625355e-04,
3.63005296e-04, 3.52105912e-04, 3.46449204e-04, 3.41924053e-04,
3.38412967e-04, 3.33162723e-04, 3.27872491e-04, 3.25990288e-04,
3.14704754e-04, 3.10601172e-04, 3.06186848e-04, 3.00744432e-04,
2.95114325e-04, 2.91654520e-04, 2.89676711e-04, 2.86087190e-04,
2.79461237e-04, 2.74361984e-04, 2.69965502e-04, 2.69522337e-04,
2.67611642e-04, 2.61527835e-04, 2.57833221e-04, 2.53948470e-04,
2.50073936e-04, 2.43692193e-04, 2.39779401e-04, 2.36836757e-04,
2.34601335e-04, 2.32159524e-04, 2.26337172e-04, 2.23069947e-04,
2.21753100e-04, 2.17200417e-04, 2.16282875e-04, 2.11949504e-04,
2.10261001e-04, 2.06676399e-04, 2.04487442e-04, 2.02323907e-04,
1.99593007e-04], dtype=float32)
```

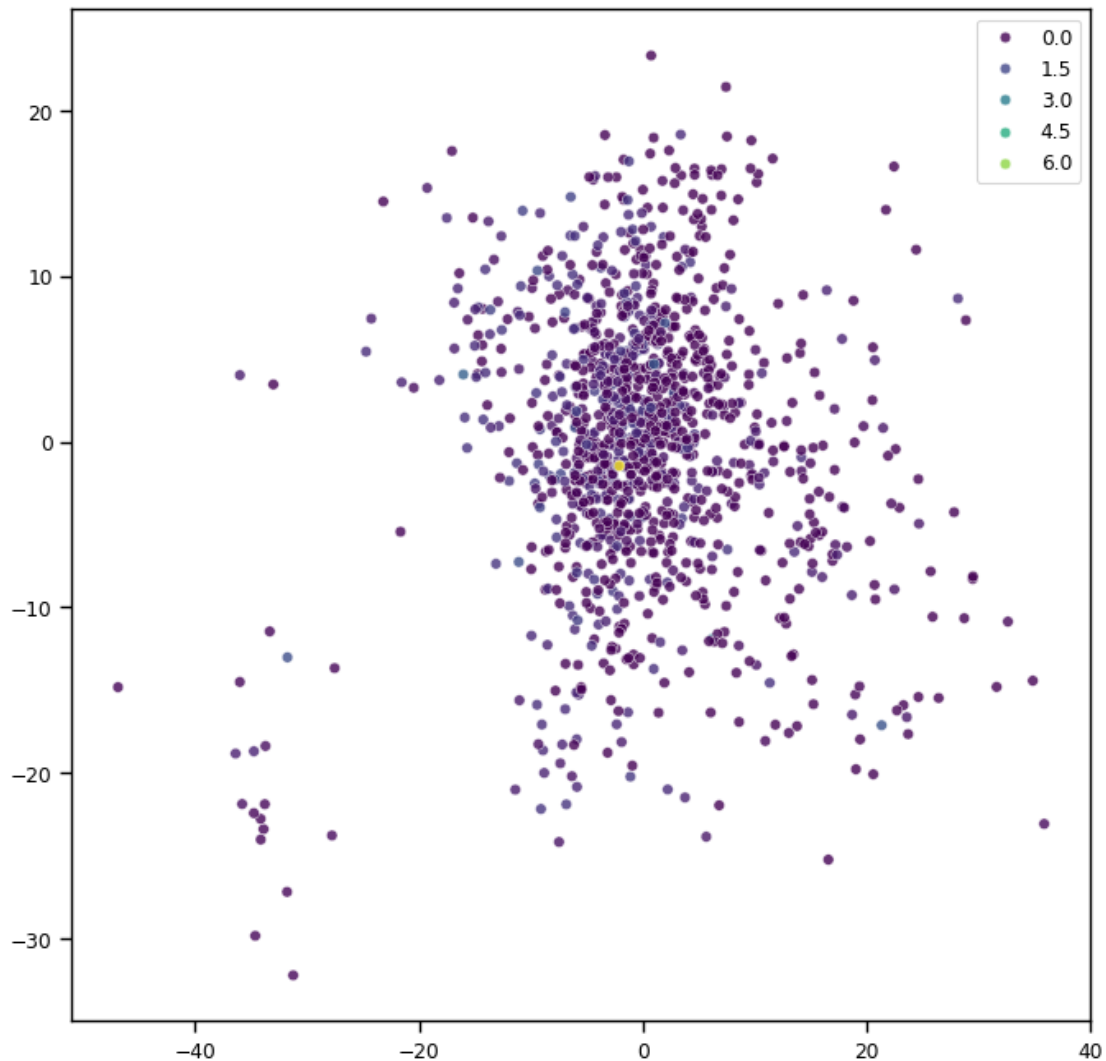
```
[49]: fig, ax = plt.subplots(figsize=(8, 8))
sns.scatterplot(x=X[:, 0], y=X[:, 1], hue=y, ax=ax, palette='viridis', alpha=0.
↪8)
```

```
[49]: <AxesSubplot: >
```



```
[50]: fig, ax = plt.subplots(figsize=(8, 8))  
sns.scatterplot(x=X[:, 2], y=X[:, 3], hue=y, ax=ax, palette='viridis', alpha=0.  
↪8)
```

```
[50]: <AxesSubplot: >
```



### 1.4.2 3.2 Model training and optimization

#### 3.2.1 Check for signal on a train test split

```
[51]: X_train, X_test, y_train, y_test = train_test_split(X_embeddings, y,
↳ test_size=0.2, random_state=42)
```

```
[52]: oracle = model_no_embedder.fit(X_train, y_train)
oracle.score(X_test, y_test)
```

```
[52]: 0.4848778823794505
```

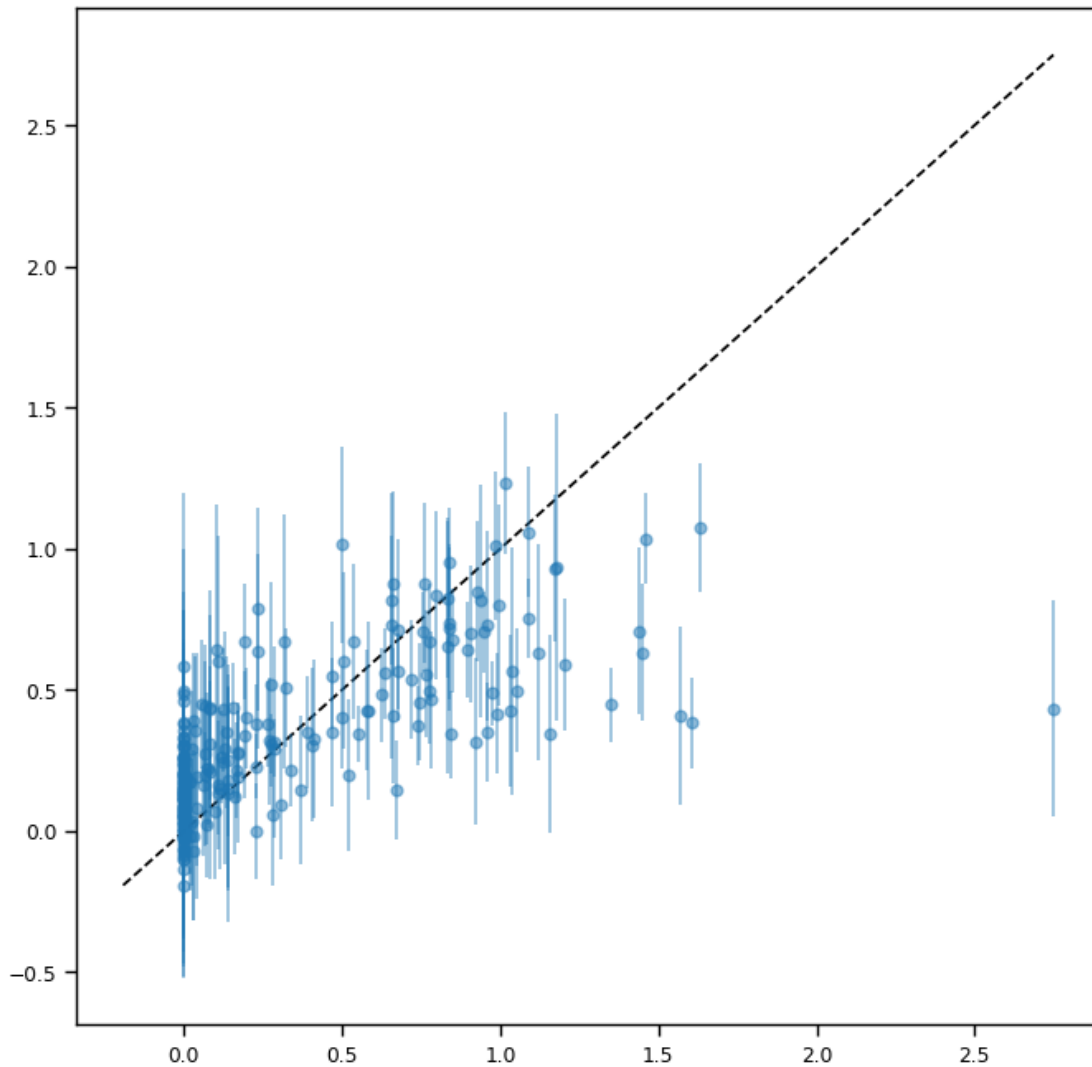
R2 of 0.5 is not too bad. Hopefully we can increase this with hyperparameter optimization. How do predictions look?

```
[53]: pred_mean, pred_std = predict_ensemble(oracle, X_test)
```

```
[58]: fig, ax = plt.subplots(figsize=(8, 8))
min_min = min(y_test.min(), pred_mean.min())
max_max = max(y_test.max(), pred_mean.max())
ax.plot([min_min, max_max], [min_min, max_max], color='black', linestyle='--')

ax.errorbar(y_test, pred_mean, yerr=pred_std, fmt='o', alpha=0.5)
```

[58]: <ErrorbarContainer object of 3 artists>



**3.2.2 Hyperparameter optimization** Create best possible predictor. Random search is used.

Optimize over R2

Eg. error is penalized less when the model is less certain.

```
[59]: # scores that balance model uncertainty and accuracy
def uncertainty_nmse(y_true, y_pred, y_std):
    """
    Compute the normalized mean squared error of the predictions.

    """
    squared_error = (y_true - y_pred)**2
    normalized_squared_error = squared_error / (y_std**2 + 1e-8) # Add small
    ↪epsilon to avoid division by zero
    return np.mean(normalized_squared_error)

[56]: # function to score the model with uncertainty calibrated score
def evaluate_model(model, X, y):
    y_pred, y_std = predict_ensemble(model, X)
    nmse = uncertainty_nmse(y, y_pred, y_std)
    r2 = r2_score(y, y_pred)
    return nmse, r2

def custom_scorer(model, X, y, **kwargs):
    nmse, r2 = evaluate_model(model, X, y)
    return r2

[57]: param_space = {
    'regressor__max_samples': stats.uniform(0.5, 0.5),
    'regressor__max_features': stats.uniform(0.5, 0.5),
    'regressor__estimator__regressor__alpha': stats.loguniform(0.0001, 0.1),
    'regressor__estimator__regressor__learning_rate_init': stats.loguniform(0.
    ↪0001, 0.1),
    'regressor__estimator__regressor__hidden_layer_sizes': [(50, 50), (50, 50,
    ↪50), (50, 50, 50, 50)],
    'regressor__estimator__regressor__activation': ['relu', 'tanh']
}

[62]: search = RandomizedSearchCV(
    model_no_embedder,
    param_space,
    n_iter=500,
    scoring=custom_scorer,
    cv=5,
    n_jobs=16,
    random_state=42,
    verbose=2
)

[63]: search.fit(X_embeddings, y)
```

Fitting 5 folds for each of 500 candidates, totalling 2500 fits

```

[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02452612631133676,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01570297088405538,
regressor__max_features=0.7993292420985183,
regressor__max_samples=0.5780093202212182; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02452612631133676,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01570297088405538,
regressor__max_features=0.7993292420985183,
regressor__max_samples=0.5780093202212182; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02452612631133676,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01570297088405538,
regressor__max_features=0.7993292420985183,
regressor__max_samples=0.5780093202212182; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02452612631133676,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01570297088405538,
regressor__max_features=0.7993292420985183,
regressor__max_samples=0.5780093202212182; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02452612631133676,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01570297088405538,
regressor__max_features=0.7993292420985183,
regressor__max_samples=0.5780093202212182; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0068384784309640425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.001976218934028007,
regressor__max_features=0.645614570099021,
regressor__max_samples=0.8059264473611898; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001994916615063393,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.039676050770529867,
regressor__max_features=0.8005575058716043,
regressor__max_samples=0.8540362888980227; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001994916615063393,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.039676050770529867,
regressor__max_features=0.8005575058716043,
regressor__max_samples=0.8540362888980227; total time= 3.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014765043713594345,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00043352817949515634,
regressor__max_features=0.5909124836035503,
regressor__max_samples=0.5917022549267169; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0068384784309640425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.001976218934028007,
regressor__max_features=0.645614570099021,
regressor__max_samples=0.8059264473611898; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014765043713594345,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00043352817949515634,
regressor__max_features=0.5909124836035503,
regressor__max_samples=0.5917022549267169; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014765043713594345,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00043352817949515634,
regressor__max_features=0.5909124836035503,
regressor__max_samples=0.5917022549267169; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0068384784309640425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.001976218934028007,
regressor__max_features=0.645614570099021,
regressor__max_samples=0.8059264473611898; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0068384784309640425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.001976218934028007,
regressor__max_features=0.645614570099021,
regressor__max_samples=0.8059264473611898; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014765043713594345,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00043352817949515634,
regressor__max_features=0.5909124836035503,
regressor__max_samples=0.5917022549267169; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014765043713594345,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00043352817949515634,
regressor__max_features=0.5909124836035503,
regressor__max_samples=0.5917022549267169; total time= 3.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001994916615063393,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.039676050770529867,
regressor__max_features=0.8005575058716043,
regressor__max_samples=0.8540362888980227; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001994916615063393,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.039676050770529867,
regressor__max_features=0.8005575058716043,
regressor__max_samples=0.8540362888980227; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001994916615063393,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.039676050770529867,
regressor__max_features=0.8005575058716043,
regressor__max_samples=0.8540362888980227; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0068384784309640425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.001976218934028007,
regressor__max_features=0.645614570099021,
regressor__max_samples=0.8059264473611898; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013803746963532806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002334586407601623,
regressor__max_features=0.8925879806965068,
regressor__max_samples=0.5998368910791798; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013803746963532806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002334586407601623,
regressor__max_features=0.8925879806965068,
regressor__max_samples=0.5998368910791798; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013803746963532806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002334586407601623,
regressor__max_features=0.8925879806965068,
regressor__max_samples=0.5998368910791798; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001095960453692584,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07886714129990491,
regressor__max_features=0.9041986740582306,
regressor__max_samples=0.6523068845866853; total time= 1.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001095960453692584,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07886714129990491,
regressor__max_features=0.9041986740582306,
regressor__max_samples=0.6523068845866853; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001095960453692584,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07886714129990491,
regressor__max_features=0.9041986740582306,
regressor__max_samples=0.6523068845866853; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013803746963532806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002334586407601623,
regressor__max_features=0.8925879806965068,
regressor__max_samples=0.5998368910791798; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001095960453692584,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07886714129990491,
regressor__max_features=0.9041986740582306,
regressor__max_samples=0.6523068845866853; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013803746963532806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002334586407601623,
regressor__max_features=0.8925879806965068,
regressor__max_samples=0.5998368910791798; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001095960453692584,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07886714129990491,
regressor__max_features=0.9041986740582306,
regressor__max_samples=0.6523068845866853; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004928122297695502,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011214774784159456,
regressor__max_features=0.8049983288913105,
regressor__max_samples=0.9165974558680822; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004928122297695502,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011214774784159456,
regressor__max_features=0.8049983288913105,
regressor__max_samples=0.9165974558680822; total time= 3.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004928122297695502,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011214774784159456,
regressor__max_features=0.8049983288913105,
regressor__max_samples=0.9165974558680822; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004928122297695502,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011214774784159456,
regressor__max_features=0.8049983288913105,
regressor__max_samples=0.9165974558680822; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.05345166110646816,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.009717775305059633,
regressor__max_features=0.6558555380447055,
regressor__max_samples=0.7600340105889054; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0050477865657106045,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08105016126411582,
regressor__max_features=0.8875664116805573,
regressor__max_samples=0.9697494707820946; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.05345166110646816,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.009717775305059633,
regressor__max_features=0.6558555380447055,
regressor__max_samples=0.7600340105889054; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.05345166110646816,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.009717775305059633,
regressor__max_features=0.6558555380447055,
regressor__max_samples=0.7600340105889054; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.05345166110646816,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.009717775305059633,
regressor__max_features=0.6558555380447055,
regressor__max_samples=0.7600340105889054; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.05345166110646816,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.009717775305059633,
regressor__max_features=0.6558555380447055,
regressor__max_samples=0.7600340105889054; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004928122297695502,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011214774784159456,
regressor__max_features=0.8049983288913105,
regressor__max_samples=0.9165974558680822; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0050477865657106045,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08105016126411582,
regressor__max_features=0.8875664116805573,
regressor__max_samples=0.9697494707820946; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0050477865657106045,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08105016126411582,
regressor__max_features=0.8875664116805573,
regressor__max_samples=0.9697494707820946; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0050477865657106045,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08105016126411582,
regressor__max_features=0.8875664116805573,
regressor__max_samples=0.9697494707820946; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.060252711710953796,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05829384542994737,
regressor__max_features=0.5442462510259598,
regressor__max_samples=0.5979914312095727; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.060252711710953796,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05829384542994737,
regressor__max_features=0.5442462510259598,
regressor__max_samples=0.5979914312095727; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0050477865657106045,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08105016126411582,
regressor__max_features=0.8875664116805573,
regressor__max_samples=0.9697494707820946; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.060252711710953796,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05829384542994737,
regressor__max_features=0.5442462510259598,
regressor__max_samples=0.5979914312095727; total time= 2.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.060252711710953796,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05829384542994737,
regressor__max_features=0.5442462510259598,
regressor__max_samples=0.5979914312095727; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.060252711710953796,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05829384542994737,
regressor__max_features=0.5442462510259598,
regressor__max_samples=0.5979914312095727; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08906204386161681,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013783237455007182,
regressor__max_features=0.8037724259507192,
regressor__max_samples=0.5852620618436457; total time= 7.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08906204386161681,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013783237455007182,
regressor__max_features=0.8037724259507192,
regressor__max_samples=0.5852620618436457; total time= 8.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08906204386161681,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013783237455007182,
regressor__max_features=0.8037724259507192,
regressor__max_samples=0.5852620618436457; total time= 8.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08906204386161681,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013783237455007182,
regressor__max_features=0.8037724259507192,
regressor__max_samples=0.5852620618436457; total time= 9.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07647439907543506,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014656553886225332,
regressor__max_features=0.6356745158869479,
regressor__max_samples=0.9143687545759647; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07866216981266842,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026471141828218156,
regressor__max_features=0.9010984903770198,
regressor__max_samples=0.5372753218398854; total time= 4.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07866216981266842,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026471141828218156,
regressor__max_features=0.9010984903770198,
regressor__max_samples=0.5372753218398854; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07866216981266842,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026471141828218156,
regressor__max_features=0.9010984903770198,
regressor__max_samples=0.5372753218398854; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07647439907543506,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014656553886225332,
regressor__max_features=0.6356745158869479,
regressor__max_samples=0.9143687545759647; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07866216981266842,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026471141828218156,
regressor__max_features=0.9010984903770198,
regressor__max_samples=0.5372753218398854; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07866216981266842,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026471141828218156,
regressor__max_features=0.9010984903770198,
regressor__max_samples=0.5372753218398854; total time= 4.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07647439907543506,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014656553886225332,
regressor__max_features=0.6356745158869479,
regressor__max_samples=0.9143687545759647; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07647439907543506,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014656553886225332,
regressor__max_features=0.6356745158869479,
regressor__max_samples=0.9143687545759647; total time= 6.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08906204386161681,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013783237455007182,
regressor__max_features=0.8037724259507192,
regressor__max_samples=0.5852620618436457; total time= 11.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07647439907543506,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014656553886225332,
regressor__max_features=0.6356745158869479,
regressor__max_samples=0.9143687545759647; total time= 6.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.013615257464006502,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.020597335357437185,
regressor__max_features=0.5370223258670452,
regressor__max_samples=0.6792328642721364; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.055574943088396424,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00741129978108324,
regressor__max_features=0.6654490124263246,
regressor__max_samples=0.5317791751430119; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.013615257464006502,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.020597335357437185,
regressor__max_features=0.5370223258670452,
regressor__max_samples=0.6792328642721364; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.013615257464006502,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.020597335357437185,
regressor__max_features=0.5370223258670452,
regressor__max_samples=0.6792328642721364; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.013615257464006502,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.020597335357437185,
regressor__max_features=0.5370223258670452,
regressor__max_samples=0.6792328642721364; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.013615257464006502,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.020597335357437185,
regressor__max_features=0.5370223258670452,
regressor__max_samples=0.6792328642721364; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.055574943088396424,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00741129978108324,
regressor__max_features=0.6654490124263246,
regressor__max_samples=0.5317791751430119; total time= 1.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.055574943088396424,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00741129978108324,
regressor__max_features=0.6654490124263246,
regressor__max_samples=0.5317791751430119; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.055574943088396424,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00741129978108324,
regressor__max_features=0.6654490124263246,
regressor__max_samples=0.5317791751430119; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.055574943088396424,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00741129978108324,
regressor__max_features=0.6654490124263246,
regressor__max_samples=0.5317791751430119; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010151348978007601,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01544608907504707,
regressor__max_features=0.8187787356776066,
regressor__max_samples=0.9436063712881633; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010151348978007601,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01544608907504707,
regressor__max_features=0.8187787356776066,
regressor__max_samples=0.9436063712881633; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018629639119246218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003945908811099999,
regressor__max_features=0.5027610585618012,
regressor__max_samples=0.9077307142274171; total time= 6.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010151348978007601,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01544608907504707,
regressor__max_features=0.8187787356776066,
regressor__max_samples=0.9436063712881633; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0014085771022963815,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013795402040204173,
regressor__max_features=0.8803925243084487,
regressor__max_samples=0.7806385987847482; total time= 2.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010151348978007601,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01544608907504707,
regressor__max_features=0.8187787356776066,
regressor__max_samples=0.9436063712881633; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0014085771022963815,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013795402040204173,
regressor__max_features=0.8803925243084487,
regressor__max_samples=0.7806385987847482; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000586414861763494,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0036999724314638075,
regressor__max_features=0.7137705091792748,
regressor__max_samples=0.5127095633720475; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010151348978007601,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01544608907504707,
regressor__max_features=0.8187787356776066,
regressor__max_samples=0.9436063712881633; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0014085771022963815,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013795402040204173,
regressor__max_features=0.8803925243084487,
regressor__max_samples=0.7806385987847482; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018629639119246218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003945908811099999,
regressor__max_features=0.5027610585618012,
regressor__max_samples=0.9077307142274171; total time= 8.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018629639119246218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003945908811099999,
regressor__max_features=0.5027610585618012,
regressor__max_samples=0.9077307142274171; total time= 9.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0014085771022963815,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013795402040204173,
regressor__max_features=0.8803925243084487,
regressor__max_samples=0.7806385987847482; total time= 2.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018629639119246218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003945908811099999,
regressor__max_features=0.5027610585618012,
regressor__max_samples=0.9077307142274171; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0014085771022963815,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013795402040204173,
regressor__max_features=0.8803925243084487,
regressor__max_samples=0.7806385987847482; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000586414861763494,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0036999724314638075,
regressor__max_features=0.7137705091792748,
regressor__max_samples=0.5127095633720475; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000586414861763494,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0036999724314638075,
regressor__max_features=0.7137705091792748,
regressor__max_samples=0.5127095633720475; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018629639119246218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003945908811099999,
regressor__max_features=0.5027610585618012,
regressor__max_samples=0.9077307142274171; total time= 8.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000586414861763494,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0036999724314638075,
regressor__max_features=0.7137705091792748,
regressor__max_samples=0.5127095633720475; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000586414861763494,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0036999724314638075,
regressor__max_features=0.7137705091792748,
regressor__max_samples=0.5127095633720475; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000559598687800608,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004164120360093466,
regressor__max_features=0.6015306123673847,
regressor__max_samples=0.9714267852789905; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000559598687800608,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004164120360093466,
regressor__max_features=0.6015306123673847,
regressor__max_samples=0.9714267852789905; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000559598687800608,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004164120360093466,
regressor__max_features=0.6015306123673847,
regressor__max_samples=0.9714267852789905; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004028632082859626,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026675383361699162,
regressor__max_features=0.7816377859881918,
regressor__max_samples=0.8477580432130638; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000559598687800608,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004164120360093466,
regressor__max_features=0.6015306123673847,
regressor__max_samples=0.9714267852789905; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004028632082859626,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026675383361699162,
regressor__max_features=0.7816377859881918,
regressor__max_samples=0.8477580432130638; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004028632082859626,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026675383361699162,
regressor__max_features=0.7816377859881918,
regressor__max_samples=0.8477580432130638; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007400385759087374,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0437928810005208,
regressor__max_features=0.8121770240668966,
regressor__max_samples=0.647816842918857; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000559598687800608,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004164120360093466,
regressor__max_features=0.6015306123673847,
regressor__max_samples=0.9714267852789905; total time= 2.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007400385759087374,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0437928810005208,
regressor__max_features=0.8121770240668966,
regressor__max_samples=0.647816842918857; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004028632082859626,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026675383361699162,
regressor__max_features=0.7816377859881918,
regressor__max_samples=0.8477580432130638; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004028632082859626,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026675383361699162,
regressor__max_features=0.7816377859881918,
regressor__max_samples=0.8477580432130638; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007400385759087374,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0437928810005208,
regressor__max_features=0.8121770240668966,
regressor__max_samples=0.647816842918857; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04115113049561089,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04465198567657794,
regressor__max_features=0.662172510502637,
regressor__max_samples=0.5610439773503366; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007400385759087374,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0437928810005208,
regressor__max_features=0.8121770240668966,
regressor__max_samples=0.647816842918857; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007400385759087374,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0437928810005208,
regressor__max_features=0.8121770240668966,
regressor__max_samples=0.647816842918857; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04115113049561089,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04465198567657794,
regressor__max_features=0.662172510502637,
regressor__max_samples=0.5610439773503366; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04115113049561089,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04465198567657794,
regressor__max_features=0.662172510502637,
regressor__max_samples=0.5610439773503366; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04115113049561089,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04465198567657794,
regressor__max_features=0.662172510502637,
regressor__max_samples=0.5610439773503366; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04115113049561089,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04465198567657794,
regressor__max_features=0.662172510502637,
regressor__max_samples=0.5610439773503366; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008995191735587165,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006552344878295669,
regressor__max_features=0.8238450602706812,
regressor__max_samples=0.5002601884976579; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008995191735587165,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006552344878295669,
regressor__max_features=0.8238450602706812,
regressor__max_samples=0.5002601884976579; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008995191735587165,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006552344878295669,
regressor__max_features=0.8238450602706812,
regressor__max_samples=0.5002601884976579; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008995191735587165,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006552344878295669,
regressor__max_features=0.8238450602706812,
regressor__max_samples=0.5002601884976579; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008995191735587165,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006552344878295669,
regressor__max_features=0.8238450602706812,
regressor__max_samples=0.5002601884976579; total time= 3.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012327891605450796,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005777583646404136,
regressor__max_features=0.6234380314193007,
regressor__max_samples=0.8481521364198942; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03821129441691226,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003118656620551979,
regressor__max_features=0.767044709687721,
regressor__max_samples=0.7424149856794916; total time= 4.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007991621920384354,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09845129999868117,
regressor__max_features=0.6333905071376424,
regressor__max_samples=0.9883074779163264; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012327891605450796,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005777583646404136,
regressor__max_features=0.6234380314193007,
regressor__max_samples=0.8481521364198942; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03821129441691226,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003118656620551979,
regressor__max_features=0.767044709687721,
regressor__max_samples=0.7424149856794916; total time= 6.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012327891605450796,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005777583646404136,
regressor__max_features=0.6234380314193007,
regressor__max_samples=0.8481521364198942; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012327891605450796,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005777583646404136,
regressor__max_features=0.6234380314193007,
regressor__max_samples=0.8481521364198942; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012327891605450796,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005777583646404136,
regressor__max_features=0.6234380314193007,
regressor__max_samples=0.8481521364198942; total time= 4.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007991621920384354,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09845129999868117,
regressor__max_features=0.6333905071376424,
regressor__max_samples=0.9883074779163264; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007991621920384354,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09845129999868117,
regressor__max_features=0.6333905071376424,
regressor__max_samples=0.9883074779163264; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00022887381144600966,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031979607306038295,
regressor__max_features=0.6093821097865351,
regressor__max_samples=0.7790510010086706; total time= 6.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03821129441691226,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003118656620551979,
regressor__max_features=0.767044709687721,
regressor__max_samples=0.7424149856794916; total time= 6.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00022887381144600966,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031979607306038295,
regressor__max_features=0.6093821097865351,
regressor__max_samples=0.7790510010086706; total time= 5.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007991621920384354,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09845129999868117,
regressor__max_features=0.6333905071376424,
regressor__max_samples=0.9883074779163264; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007991621920384354,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09845129999868117,
regressor__max_features=0.6333905071376424,
regressor__max_samples=0.9883074779163264; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03821129441691226,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003118656620551979,
regressor__max_features=0.767044709687721,
regressor__max_samples=0.7424149856794916; total time= 7.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03821129441691226,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003118656620551979,
regressor__max_features=0.767044709687721,
regressor__max_samples=0.7424149856794916; total time= 7.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014270403521460836,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.001084460515052502,
regressor__max_features=0.8171756723506819,
regressor__max_samples=0.8403527257773834; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00022887381144600966,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031979607306038295,
regressor__max_features=0.6093821097865351,
regressor__max_samples=0.7790510010086706; total time= 7.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014270403521460836,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.001084460515052502,
regressor__max_features=0.8171756723506819,
regressor__max_samples=0.8403527257773834; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014270403521460836,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.001084460515052502,
regressor__max_features=0.8171756723506819,
regressor__max_samples=0.8403527257773834; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00022887381144600966,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031979607306038295,
regressor__max_features=0.6093821097865351,
regressor__max_samples=0.7790510010086706; total time= 7.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014270403521460836,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.001084460515052502,
regressor__max_features=0.8171756723506819,
regressor__max_samples=0.8403527257773834; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00022887381144600966,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031979607306038295,
regressor__max_features=0.6093821097865351,
regressor__max_samples=0.7790510010086706; total time= 8.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014270403521460836,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.001084460515052502,
regressor__max_features=0.8171756723506819,
regressor__max_samples=0.8403527257773834; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005326964650459662,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012686721262104848,
regressor__max_features=0.8161529152967897,
regressor__max_samples=0.8167648553804474; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005326964650459662,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012686721262104848,
regressor__max_features=0.8161529152967897,
regressor__max_samples=0.8167648553804474; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002940074130903305,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005323114580928885,
regressor__max_features=0.8360677737029393,
regressor__max_samples=0.8808098076643588; total time= 5.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005326964650459662,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012686721262104848,
regressor__max_features=0.8161529152967897,
regressor__max_samples=0.8167648553804474; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005326964650459662,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012686721262104848,
regressor__max_features=0.8161529152967897,
regressor__max_samples=0.8167648553804474; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005326964650459662,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012686721262104848,
regressor__max_features=0.8161529152967897,
regressor__max_samples=0.8167648553804474; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002940074130903305,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005323114580928885,
regressor__max_features=0.8360677737029393,
regressor__max_samples=0.8808098076643588; total time= 5.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002940074130903305,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005323114580928885,
regressor__max_features=0.8360677737029393,
regressor__max_samples=0.8808098076643588; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002940074130903305,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005323114580928885,
regressor__max_features=0.8360677737029393,
regressor__max_samples=0.8808098076643588; total time= 5.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002940074130903305,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005323114580928885,
regressor__max_features=0.8360677737029393,
regressor__max_samples=0.8808098076643588; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008620220426018043,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011825328508422647,
regressor__max_features=0.6933676731502687,
regressor__max_samples=0.9683649943683672; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008620220426018043,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011825328508422647,
regressor__max_features=0.6933676731502687,
regressor__max_samples=0.9683649943683672; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008620220426018043,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011825328508422647,
regressor__max_features=0.6933676731502687,
regressor__max_samples=0.9683649943683672; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008620220426018043,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011825328508422647,
regressor__max_features=0.6933676731502687,
regressor__max_samples=0.9683649943683672; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008620220426018043,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011825328508422647,
regressor__max_features=0.6933676731502687,
regressor__max_samples=0.9683649943683672; total time= 1.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0059248700519943165,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004578499200611471,
regressor__max_features=0.855574766219009,
regressor__max_samples=0.9047505230698577; total time= 7.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028138155131170208,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028323872575338924,
regressor__max_features=0.7540993883703593,
regressor__max_samples=0.8479064033954409; total time= 8.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00048489763440102805,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08841074088520708,
regressor__max_features=0.7583179456355071,
regressor__max_samples=0.6304145874152045; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0059248700519943165,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004578499200611471,
regressor__max_features=0.855574766219009,
regressor__max_samples=0.9047505230698577; total time= 6.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0059248700519943165,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004578499200611471,
regressor__max_features=0.855574766219009,
regressor__max_samples=0.9047505230698577; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028138155131170208,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028323872575338924,
regressor__max_features=0.7540993883703593,
regressor__max_samples=0.8479064033954409; total time= 8.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00048489763440102805,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08841074088520708,
regressor__max_features=0.7583179456355071,
regressor__max_samples=0.6304145874152045; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00048489763440102805,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08841074088520708,
regressor__max_features=0.7583179456355071,
regressor__max_samples=0.6304145874152045; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00048489763440102805,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08841074088520708,
regressor__max_features=0.7583179456355071,
regressor__max_samples=0.6304145874152045; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028138155131170208,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028323872575338924,
regressor__max_features=0.7540993883703593,
regressor__max_samples=0.8479064033954409; total time= 9.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028138155131170208,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028323872575338924,
regressor__max_features=0.7540993883703593,
regressor__max_samples=0.8479064033954409; total time= 9.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00048489763440102805,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08841074088520708,
regressor__max_features=0.7583179456355071,
regressor__max_samples=0.6304145874152045; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0059248700519943165,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004578499200611471,
regressor__max_features=0.855574766219009,
regressor__max_samples=0.9047505230698577; total time= 7.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00019024283247489577,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0502636672316142,
regressor__max_features=0.816550728636634,
regressor__max_samples=0.6695148955243504; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00019024283247489577,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0502636672316142,
regressor__max_features=0.816550728636634,
regressor__max_samples=0.6695148955243504; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00019024283247489577,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0502636672316142,
regressor__max_features=0.816550728636634,
regressor__max_samples=0.6695148955243504; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028138155131170208,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028323872575338924,
regressor__max_features=0.7540993883703593,
regressor__max_samples=0.8479064033954409; total time= 11.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0059248700519943165,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004578499200611471,
regressor__max_features=0.855574766219009,
regressor__max_samples=0.9047505230698577; total time= 8.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00019024283247489577,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0502636672316142,
regressor__max_features=0.816550728636634,
regressor__max_samples=0.6695148955243504; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00019024283247489577,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0502636672316142,
regressor__max_features=0.816550728636634,
regressor__max_samples=0.6695148955243504; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001788215664787948,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.010161662443413273,
regressor__max_features=0.7903433107182274,
regressor__max_samples=0.6861413832808716; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001788215664787948,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.010161662443413273,
regressor__max_features=0.7903433107182274,
regressor__max_samples=0.6861413832808716; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001788215664787948,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.010161662443413273,
regressor__max_features=0.7903433107182274,
regressor__max_samples=0.6861413832808716; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010643154121134478,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00021898812429056914,
regressor__max_features=0.9623468091392814,
regressor__max_samples=0.9386696766904905; total time= 8.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001788215664787948,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.010161662443413273,
regressor__max_features=0.7903433107182274,
regressor__max_samples=0.6861413832808716; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001788215664787948,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.010161662443413273,
regressor__max_features=0.7903433107182274,
regressor__max_samples=0.6861413832808716; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012620410750629132,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03706575712109999,
regressor__max_features=0.702254063561095,
regressor__max_samples=0.9438850493804799; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012620410750629132,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03706575712109999,
regressor__max_features=0.702254063561095,
regressor__max_samples=0.9438850493804799; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012620410750629132,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03706575712109999,
regressor__max_features=0.702254063561095,
regressor__max_samples=0.9438850493804799; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010643154121134478,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00021898812429056914,
regressor__max_features=0.9623468091392814,
regressor__max_samples=0.9386696766904905; total time= 9.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012620410750629132,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03706575712109999,
regressor__max_features=0.702254063561095,
regressor__max_samples=0.9438850493804799; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010643154121134478,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00021898812429056914,
regressor__max_features=0.9623468091392814,
regressor__max_samples=0.9386696766904905; total time= 9.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009783749110062342,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007108253772669103,
regressor__max_features=0.6526819301721967,
regressor__max_samples=0.7428068767931133; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012620410750629132,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03706575712109999,
regressor__max_features=0.702254063561095,
regressor__max_samples=0.9438850493804799; total time= 4.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010643154121134478,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00021898812429056914,
regressor__max_features=0.9623468091392814,
regressor__max_samples=0.9386696766904905; total time= 10.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009783749110062342,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007108253772669103,
regressor__max_features=0.6526819301721967,
regressor__max_samples=0.7428068767931133; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009783749110062342,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007108253772669103,
regressor__max_features=0.6526819301721967,
regressor__max_samples=0.7428068767931133; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017356982090814235,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014548552070164159,
regressor__max_features=0.6540303959261946,
regressor__max_samples=0.7712701152774497; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010643154121134478,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00021898812429056914,
regressor__max_features=0.9623468091392814,
regressor__max_samples=0.9386696766904905; total time= 11.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009783749110062342,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007108253772669103,
regressor__max_features=0.6526819301721967,
regressor__max_samples=0.7428068767931133; total time= 4.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009783749110062342,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007108253772669103,
regressor__max_features=0.6526819301721967,
regressor__max_samples=0.7428068767931133; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017356982090814235,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014548552070164159,
regressor__max_features=0.6540303959261946,
regressor__max_samples=0.7712701152774497; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017356982090814235,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014548552070164159,
regressor__max_features=0.6540303959261946,
regressor__max_samples=0.7712701152774497; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017356982090814235,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014548552070164159,
regressor__max_features=0.6540303959261946,
regressor__max_samples=0.7712701152774497; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017356982090814235,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014548552070164159,
regressor__max_features=0.6540303959261946,
regressor__max_samples=0.7712701152774497; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009034076805537767,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003371131999605197,
regressor__max_features=0.5090376818077604,
regressor__max_samples=0.7469468575917173; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009034076805537767,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003371131999605197,
regressor__max_features=0.5090376818077604,
regressor__max_samples=0.7469468575917173; total time= 5.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04743945221058974,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002009940015359038,
regressor__max_features=0.6750392038473378,
regressor__max_samples=0.8225516810152824; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009034076805537767,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003371131999605197,
regressor__max_features=0.5090376818077604,
regressor__max_samples=0.7469468575917173; total time= 6.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009034076805537767,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003371131999605197,
regressor__max_features=0.5090376818077604,
regressor__max_samples=0.7469468575917173; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009034076805537767,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003371131999605197,
regressor__max_features=0.5090376818077604,
regressor__max_samples=0.7469468575917173; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04743945221058974,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002009940015359038,
regressor__max_features=0.6750392038473378,
regressor__max_samples=0.8225516810152824; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0012680822080429405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005641381315708953,
regressor__max_features=0.7949354237802719,
regressor__max_samples=0.9894464291375045; total time= 4.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04743945221058974,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002009940015359038,
regressor__max_features=0.6750392038473378,
regressor__max_samples=0.8225516810152824; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04743945221058974,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002009940015359038,
regressor__max_features=0.6750392038473378,
regressor__max_samples=0.8225516810152824; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0012680822080429405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005641381315708953,
regressor__max_features=0.7949354237802719,
regressor__max_samples=0.9894464291375045; total time= 5.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008638073353835192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09630559802166859,
regressor__max_features=0.7349722569954715,
regressor__max_samples=0.6397801708983792; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008638073353835192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09630559802166859,
regressor__max_features=0.7349722569954715,
regressor__max_samples=0.6397801708983792; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0012680822080429405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005641381315708953,
regressor__max_features=0.7949354237802719,
regressor__max_samples=0.9894464291375045; total time= 6.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04743945221058974,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002009940015359038,
regressor__max_features=0.6750392038473378,
regressor__max_samples=0.8225516810152824; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008638073353835192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09630559802166859,
regressor__max_features=0.7349722569954715,
regressor__max_samples=0.6397801708983792; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0012680822080429405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005641381315708953,
regressor__max_features=0.7949354237802719,
regressor__max_samples=0.9894464291375045; total time= 5.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008638073353835192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09630559802166859,
regressor__max_features=0.7349722569954715,
regressor__max_samples=0.6397801708983792; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008638073353835192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09630559802166859,
regressor__max_features=0.7349722569954715,
regressor__max_samples=0.6397801708983792; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012896625857891446,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0723128604484139,
regressor__max_features=0.6653751523352569,
regressor__max_samples=0.776382483417745; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0012680822080429405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005641381315708953,
regressor__max_features=0.7949354237802719,
regressor__max_samples=0.9894464291375045; total time= 6.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012896625857891446,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0723128604484139,
regressor__max_features=0.6653751523352569,
regressor__max_samples=0.776382483417745; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012896625857891446,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0723128604484139,
regressor__max_features=0.6653751523352569,
regressor__max_samples=0.776382483417745; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012896625857891446,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0723128604484139,
regressor__max_features=0.6653751523352569,
regressor__max_samples=0.776382483417745; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012896625857891446,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0723128604484139,
regressor__max_features=0.6653751523352569,
regressor__max_samples=0.776382483417745; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0030029844369151284,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004904060335291482,
regressor__max_features=0.7495966899423762,
regressor__max_samples=0.7860020996045916; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0030029844369151284,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004904060335291482,
regressor__max_features=0.7495966899423762,
regressor__max_samples=0.7860020996045916; total time= 5.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001626478746126889,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006998769331752383,
regressor__max_features=0.9950269250521316,
regressor__max_samples=0.570042007618262; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0030029844369151284,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004904060335291482,
regressor__max_features=0.7495966899423762,
regressor__max_samples=0.7860020996045916; total time= 5.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001626478746126889,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006998769331752383,
regressor__max_features=0.9950269250521316,
regressor__max_samples=0.570042007618262; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0030029844369151284,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004904060335291482,
regressor__max_features=0.7495966899423762,
regressor__max_samples=0.7860020996045916; total time= 5.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008928243277966445,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004681702225035901,
regressor__max_features=0.9680773870803905,
regressor__max_samples=0.8480148983374864; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001626478746126889,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006998769331752383,
regressor__max_features=0.9950269250521316,
regressor__max_samples=0.570042007618262; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0030029844369151284,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004904060335291482,
regressor__max_features=0.7495966899423762,
regressor__max_samples=0.7860020996045916; total time= 6.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008928243277966445,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004681702225035901,
regressor__max_features=0.9680773870803905,
regressor__max_samples=0.8480148983374864; total time= 2.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008928243277966445,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004681702225035901,
regressor__max_features=0.9680773870803905,
regressor__max_samples=0.8480148983374864; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008928243277966445,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004681702225035901,
regressor__max_features=0.9680773870803905,
regressor__max_samples=0.8480148983374864; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001626478746126889,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006998769331752383,
regressor__max_features=0.9950269250521316,
regressor__max_samples=0.570042007618262; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001626478746126889,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006998769331752383,
regressor__max_features=0.9950269250521316,
regressor__max_samples=0.570042007618262; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007689033715836228,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.016684217902257083,
regressor__max_features=0.848507870497634,
regressor__max_samples=0.8512420419935547; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008928243277966445,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004681702225035901,
regressor__max_features=0.9680773870803905,
regressor__max_samples=0.8480148983374864; total time= 4.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007689033715836228,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.016684217902257083,
regressor__max_features=0.848507870497634,
regressor__max_samples=0.8512420419935547; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007689033715836228,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.016684217902257083,
regressor__max_features=0.848507870497634,
regressor__max_samples=0.8512420419935547; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007689033715836228,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.016684217902257083,
regressor__max_features=0.848507870497634,
regressor__max_samples=0.8512420419935547; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004554701612208506,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.026796832798314643,
regressor__max_features=0.9050566973395904,
regressor__max_samples=0.9335361592900518; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004554701612208506,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.026796832798314643,
regressor__max_features=0.9050566973395904,
regressor__max_samples=0.9335361592900518; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007689033715836228,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.016684217902257083,
regressor__max_features=0.848507870497634,
regressor__max_samples=0.8512420419935547; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004554701612208506,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.026796832798314643,
regressor__max_features=0.9050566973395904,
regressor__max_samples=0.9335361592900518; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004554701612208506,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.026796832798314643,
regressor__max_features=0.9050566973395904,
regressor__max_samples=0.9335361592900518; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004554701612208506,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.026796832798314643,
regressor__max_features=0.9050566973395904,
regressor__max_samples=0.9335361592900518; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07777856590163645,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001682824274470241,
regressor__max_features=0.652848509643591,
regressor__max_samples=0.5954555155751731; total time= 7.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0015500706089539398,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.046775240101761456,
regressor__max_features=0.6689975784257679,
regressor__max_samples=0.687791476319972; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07777856590163645,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001682824274470241,
regressor__max_features=0.652848509643591,
regressor__max_samples=0.5954555155751731; total time= 8.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0015500706089539398,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.046775240101761456,
regressor__max_features=0.6689975784257679,
regressor__max_samples=0.687791476319972; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017570436113703507,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031955740374573207,
regressor__max_features=0.8991475894833876,
regressor__max_samples=0.8249819653888826; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0015500706089539398,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.046775240101761456,
regressor__max_features=0.6689975784257679,
regressor__max_samples=0.687791476319972; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017570436113703507,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031955740374573207,
regressor__max_features=0.8991475894833876,
regressor__max_samples=0.8249819653888826; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017570436113703507,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031955740374573207,
regressor__max_features=0.8991475894833876,
regressor__max_samples=0.8249819653888826; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0015500706089539398,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.046775240101761456,
regressor__max_features=0.6689975784257679,
regressor__max_samples=0.687791476319972; total time= 1.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017570436113703507,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031955740374573207,
regressor__max_features=0.8991475894833876,
regressor__max_samples=0.8249819653888826; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07777856590163645,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001682824274470241,
regressor__max_features=0.652848509643591,
regressor__max_samples=0.5954555155751731; total time= 9.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0015500706089539398,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.046775240101761456,
regressor__max_features=0.6689975784257679,
regressor__max_samples=0.687791476319972; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017570436113703507,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031955740374573207,
regressor__max_features=0.8991475894833876,
regressor__max_samples=0.8249819653888826; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07777856590163645,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001682824274470241,
regressor__max_features=0.652848509643591,
regressor__max_samples=0.5954555155751731; total time= 10.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004085519442448726,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.003687480905792529,
regressor__max_features=0.8849967765493054,
regressor__max_samples=0.6079105137484215; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004085519442448726,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.003687480905792529,
regressor__max_features=0.8849967765493054,
regressor__max_samples=0.6079105137484215; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004085519442448726,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.003687480905792529,
regressor__max_features=0.8849967765493054,
regressor__max_samples=0.6079105137484215; total time= 1.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004085519442448726,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.003687480905792529,
regressor__max_features=0.8849967765493054,
regressor__max_samples=0.6079105137484215; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07777856590163645,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001682824274470241,
regressor__max_features=0.652848509643591,
regressor__max_samples=0.5954555155751731; total time= 11.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004085519442448726,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.003687480905792529,
regressor__max_features=0.8849967765493054,
regressor__max_samples=0.6079105137484215; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002600514485557595,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0810234256958155,
regressor__max_features=0.857297552089976,
regressor__max_samples=0.5205337583839379; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002600514485557595,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0810234256958155,
regressor__max_features=0.857297552089976,
regressor__max_samples=0.5205337583839379; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002600514485557595,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0810234256958155,
regressor__max_features=0.857297552089976,
regressor__max_samples=0.5205337583839379; total time= 1.4s

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:690:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
the optimization hasn't converged yet.
  warnings.warn(

[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020673897991199588,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001281819344392312,
regressor__max_features=0.7327990090662301,
regressor__max_samples=0.7713223173537883; total time= 9.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002600514485557595,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0810234256958155,
regressor__max_features=0.857297552089976,
regressor__max_samples=0.5205337583839379; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020673897991199588,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001281819344392312,
regressor__max_features=0.7327990090662301,
regressor__max_samples=0.7713223173537883; total time= 9.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020673897991199588,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001281819344392312,
regressor__max_features=0.7327990090662301,
regressor__max_samples=0.7713223173537883; total time= 9.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002600514485557595,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0810234256958155,
regressor__max_features=0.857297552089976,
regressor__max_samples=0.5205337583839379; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020673897991199588,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001281819344392312,
regressor__max_features=0.7327990090662301,
regressor__max_samples=0.7713223173537883; total time= 11.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020673897991199588,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001281819344392312,
regressor__max_features=0.7327990090662301,
regressor__max_samples=0.7713223173537883; total time= 11.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0023539405203496107,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001429042560943209,
regressor__max_features=0.7656773157840739,
regressor__max_samples=0.7703175608050532; total time= 11.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0023539405203496107,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001429042560943209,
regressor__max_features=0.7656773157840739,
regressor__max_samples=0.7703175608050532; total time= 12.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03287006070977729,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012345274466024465,
regressor__max_features=0.5186740943746072,
regressor__max_samples=0.9113002803298291; total time= 15.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0023539405203496107,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001429042560943209,
regressor__max_features=0.7656773157840739,
regressor__max_samples=0.7703175608050532; total time= 13.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0023539405203496107,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001429042560943209,
regressor__max_features=0.7656773157840739,
regressor__max_samples=0.7703175608050532; total time= 14.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0023539405203496107,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001429042560943209,
regressor__max_features=0.7656773157840739,
regressor__max_samples=0.7703175608050532; total time= 14.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006493767776722162,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017193692051833037,
regressor__max_features=0.5126753717077288,
regressor__max_samples=0.9813242073389625; total time= 7.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03287006070977729,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012345274466024465,
regressor__max_features=0.5186740943746072,
regressor__max_samples=0.9113002803298291; total time= 17.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004443037656955539,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009562955396652195,
regressor__max_features=0.6399669484729714,
regressor__max_samples=0.977432640331597; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006493767776722162,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017193692051833037,
regressor__max_features=0.5126753717077288,
regressor__max_samples=0.9813242073389625; total time= 9.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004443037656955539,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009562955396652195,
regressor__max_features=0.6399669484729714,
regressor__max_samples=0.977432640331597; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03287006070977729,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012345274466024465,
regressor__max_features=0.5186740943746072,
regressor__max_samples=0.9113002803298291; total time= 18.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004443037656955539,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009562955396652195,
regressor__max_features=0.6399669484729714,
regressor__max_samples=0.977432640331597; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004443037656955539,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009562955396652195,
regressor__max_features=0.6399669484729714,
regressor__max_samples=0.977432640331597; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009361253058793687,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0068416718280138135,
regressor__max_features=0.709800031213895,
regressor__max_samples=0.6238654947505787; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006493767776722162,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017193692051833037,
regressor__max_features=0.5126753717077288,
regressor__max_samples=0.9813242073389625; total time= 11.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006493767776722162,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017193692051833037,
regressor__max_features=0.5126753717077288,
regressor__max_samples=0.9813242073389625; total time= 10.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004443037656955539,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009562955396652195,
regressor__max_features=0.6399669484729714,
regressor__max_samples=0.977432640331597; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009361253058793687,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0068416718280138135,
regressor__max_features=0.709800031213895,
regressor__max_samples=0.6238654947505787; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03287006070977729,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012345274466024465,
regressor__max_features=0.5186740943746072,
regressor__max_samples=0.9113002803298291; total time= 20.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001927273832483682,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014947920745125016,
regressor__max_features=0.9576068638132402,
regressor__max_samples=0.7211761148655522; total time= 10.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009361253058793687,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0068416718280138135,
regressor__max_features=0.709800031213895,
regressor__max_samples=0.6238654947505787; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009361253058793687,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0068416718280138135,
regressor__max_features=0.709800031213895,
regressor__max_samples=0.6238654947505787; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03287006070977729,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012345274466024465,
regressor__max_features=0.5186740943746072,
regressor__max_samples=0.9113002803298291; total time= 21.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006493767776722162,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017193692051833037,
regressor__max_features=0.5126753717077288,
regressor__max_samples=0.9813242073389625; total time= 11.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009361253058793687,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0068416718280138135,
regressor__max_features=0.709800031213895,
regressor__max_samples=0.6238654947505787; total time= 2.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008594882189642618,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012911407200549917,
regressor__max_features=0.7370869145436626,
regressor__max_samples=0.5489170803255008; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008594882189642618,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012911407200549917,
regressor__max_features=0.7370869145436626,
regressor__max_samples=0.5489170803255008; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026347470756444176,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002477544357860109,
regressor__max_features=0.8248868413213817,
regressor__max_samples=0.5240294620985169; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008594882189642618,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012911407200549917,
regressor__max_features=0.7370869145436626,
regressor__max_samples=0.5489170803255008; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008594882189642618,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012911407200549917,
regressor__max_features=0.7370869145436626,
regressor__max_samples=0.5489170803255008; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001927273832483682,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014947920745125016,
regressor__max_features=0.9576068638132402,
regressor__max_samples=0.7211761148655522; total time= 8.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008594882189642618,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012911407200549917,
regressor__max_features=0.7370869145436626,
regressor__max_samples=0.5489170803255008; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026347470756444176,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002477544357860109,
regressor__max_features=0.8248868413213817,
regressor__max_samples=0.5240294620985169; total time= 1.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026347470756444176,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002477544357860109,
regressor__max_features=0.8248868413213817,
regressor__max_samples=0.5240294620985169; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001927273832483682,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014947920745125016,
regressor__max_features=0.9576068638132402,
regressor__max_samples=0.7211761148655522; total time= 11.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026347470756444176,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002477544357860109,
regressor__max_features=0.8248868413213817,
regressor__max_samples=0.5240294620985169; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026347470756444176,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002477544357860109,
regressor__max_features=0.8248868413213817,
regressor__max_samples=0.5240294620985169; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002991846741962466,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011045375019318373,
regressor__max_features=0.5580363202534582,
regressor__max_samples=0.5230013210108764; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001927273832483682,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014947920745125016,
regressor__max_features=0.9576068638132402,
regressor__max_samples=0.7211761148655522; total time= 9.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001927273832483682,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014947920745125016,
regressor__max_features=0.9576068638132402,
regressor__max_samples=0.7211761148655522; total time= 10.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03710816880757538,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011269153853645159,
regressor__max_features=0.8079255821949569,
regressor__max_samples=0.9719458022803786; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03710816880757538,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011269153853645159,
regressor__max_features=0.8079255821949569,
regressor__max_samples=0.9719458022803786; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03710816880757538,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011269153853645159,
regressor__max_features=0.8079255821949569,
regressor__max_samples=0.9719458022803786; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002991846741962466,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011045375019318373,
regressor__max_features=0.5580363202534582,
regressor__max_samples=0.5230013210108764; total time= 6.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008040461052901444,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006062906489871084,
regressor__max_features=0.5076522701451924,
regressor__max_samples=0.9667181540397415; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002991846741962466,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011045375019318373,
regressor__max_features=0.5580363202534582,
regressor__max_samples=0.5230013210108764; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03710816880757538,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011269153853645159,
regressor__max_features=0.8079255821949569,
regressor__max_samples=0.9719458022803786; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03710816880757538,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011269153853645159,
regressor__max_features=0.8079255821949569,
regressor__max_samples=0.9719458022803786; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002991846741962466,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011045375019318373,
regressor__max_features=0.5580363202534582,
regressor__max_samples=0.5230013210108764; total time= 6.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002991846741962466,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011045375019318373,
regressor__max_features=0.5580363202534582,
regressor__max_samples=0.5230013210108764; total time= 7.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008040461052901444,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006062906489871084,
regressor__max_features=0.5076522701451924,
regressor__max_samples=0.9667181540397415; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008040461052901444,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006062906489871084,
regressor__max_features=0.5076522701451924,
regressor__max_samples=0.9667181540397415; total time= 4.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00012009749020630266,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02528412239850358,
regressor__max_features=0.8385841711914909,
regressor__max_samples=0.7866835208359666; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00012009749020630266,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02528412239850358,
regressor__max_features=0.8385841711914909,
regressor__max_samples=0.7866835208359666; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008040461052901444,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006062906489871084,
regressor__max_features=0.5076522701451924,
regressor__max_samples=0.9667181540397415; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00012009749020630266,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02528412239850358,
regressor__max_features=0.8385841711914909,
regressor__max_samples=0.7866835208359666; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008040461052901444,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006062906489871084,
regressor__max_features=0.5076522701451924,
regressor__max_samples=0.9667181540397415; total time= 5.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00012009749020630266,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02528412239850358,
regressor__max_features=0.8385841711914909,
regressor__max_samples=0.7866835208359666; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00012009749020630266,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02528412239850358,
regressor__max_features=0.8385841711914909,
regressor__max_samples=0.7866835208359666; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0023700563480372546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02896799435860002,
regressor__max_features=0.8129698350507863,
regressor__max_samples=0.910213421885634; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0023700563480372546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02896799435860002,
regressor__max_features=0.8129698350507863,
regressor__max_samples=0.910213421885634; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0023700563480372546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02896799435860002,
regressor__max_features=0.8129698350507863,
regressor__max_samples=0.910213421885634; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0023700563480372546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02896799435860002,
regressor__max_features=0.8129698350507863,
regressor__max_samples=0.910213421885634; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0023700563480372546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02896799435860002,
regressor__max_features=0.8129698350507863,
regressor__max_samples=0.910213421885634; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.052005817291141204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016146704696224918,
regressor__max_features=0.5503890006887133,
regressor__max_samples=0.5091109128257749; total time= 4.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.052005817291141204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016146704696224918,
regressor__max_features=0.5503890006887133,
regressor__max_samples=0.5091109128257749; total time= 4.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.052005817291141204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016146704696224918,
regressor__max_features=0.5503890006887133,
regressor__max_samples=0.5091109128257749; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06455716578617578,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.025730178030080857,
regressor__max_features=0.6410172862856532,
regressor__max_samples=0.5887197718898614; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010516560107591674,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007007519651093011,
regressor__max_features=0.5590824138108281,
regressor__max_samples=0.8483685826820753; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.052005817291141204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016146704696224918,
regressor__max_features=0.5503890006887133,
regressor__max_samples=0.5091109128257749; total time= 5.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.052005817291141204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016146704696224918,
regressor__max_features=0.5503890006887133,
regressor__max_samples=0.5091109128257749; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010516560107591674,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007007519651093011,
regressor__max_features=0.5590824138108281,
regressor__max_samples=0.8483685826820753; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06455716578617578,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.025730178030080857,
regressor__max_features=0.6410172862856532,
regressor__max_samples=0.5887197718898614; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010516560107591674,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007007519651093011,
regressor__max_features=0.5590824138108281,
regressor__max_samples=0.8483685826820753; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06455716578617578,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.025730178030080857,
regressor__max_features=0.6410172862856532,
regressor__max_samples=0.5887197718898614; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06455716578617578,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.025730178030080857,
regressor__max_features=0.6410172862856532,
regressor__max_samples=0.5887197718898614; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06455716578617578,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.025730178030080857,
regressor__max_features=0.6410172862856532,
regressor__max_samples=0.5887197718898614; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0050764586293791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09365164840715295,
regressor__max_features=0.7063088384557132,
regressor__max_samples=0.6860090428963916; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010516560107591674,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007007519651093011,
regressor__max_features=0.5590824138108281,
regressor__max_samples=0.8483685826820753; total time= 4.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010516560107591674,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007007519651093011,
regressor__max_features=0.5590824138108281,
regressor__max_samples=0.8483685826820753; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0050764586293791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09365164840715295,
regressor__max_features=0.7063088384557132,
regressor__max_samples=0.6860090428963916; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0050764586293791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09365164840715295,
regressor__max_features=0.7063088384557132,
regressor__max_samples=0.6860090428963916; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0050764586293791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09365164840715295,
regressor__max_features=0.7063088384557132,
regressor__max_samples=0.6860090428963916; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0055160988091309555,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06198291213894245,
regressor__max_features=0.9292063759215059,
regressor__max_samples=0.7144970136875092; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0055160988091309555,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06198291213894245,
regressor__max_features=0.9292063759215059,
regressor__max_samples=0.7144970136875092; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0050764586293791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09365164840715295,
regressor__max_features=0.7063088384557132,
regressor__max_samples=0.6860090428963916; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0055160988091309555,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06198291213894245,
regressor__max_features=0.9292063759215059,
regressor__max_samples=0.7144970136875092; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0055160988091309555,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06198291213894245,
regressor__max_features=0.9292063759215059,
regressor__max_samples=0.7144970136875092; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014752113960427669,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04859254947164246,
regressor__max_features=0.6946008393670815,
regressor__max_samples=0.5054188257401492; total time= 1.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014752113960427669,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04859254947164246,
regressor__max_features=0.6946008393670815,
regressor__max_samples=0.5054188257401492; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0055160988091309555,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06198291213894245,
regressor__max_features=0.9292063759215059,
regressor__max_samples=0.7144970136875092; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014752113960427669,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04859254947164246,
regressor__max_features=0.6946008393670815,
regressor__max_samples=0.5054188257401492; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014752113960427669,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04859254947164246,
regressor__max_features=0.6946008393670815,
regressor__max_samples=0.5054188257401492; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014752113960427669,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04859254947164246,
regressor__max_features=0.6946008393670815,
regressor__max_samples=0.5054188257401492; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007157504883579034,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016351814261589403,
regressor__max_features=0.6594878151468806,
regressor__max_samples=0.9224376554847273; total time= 10.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007157504883579034,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016351814261589403,
regressor__max_features=0.6594878151468806,
regressor__max_samples=0.9224376554847273; total time= 11.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007157504883579034,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016351814261589403,
regressor__max_features=0.6594878151468806,
regressor__max_samples=0.9224376554847273; total time= 11.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007157504883579034,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016351814261589403,
regressor__max_features=0.6594878151468806,
regressor__max_samples=0.9224376554847273; total time= 12.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03478799766775241,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009076970265355106,
regressor__max_features=0.9750309835254025,
regressor__max_samples=0.975303573468778; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.016585412649095323,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00020387858300971332,
regressor__max_features=0.9512764533397833,
regressor__max_samples=0.7526261862239285; total time= 7.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03478799766775241,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009076970265355106,
regressor__max_features=0.9750309835254025,
regressor__max_samples=0.975303573468778; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03478799766775241,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009076970265355106,
regressor__max_features=0.9750309835254025,
regressor__max_samples=0.975303573468778; total time= 4.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03478799766775241,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009076970265355106,
regressor__max_features=0.9750309835254025,
regressor__max_samples=0.975303573468778; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03478799766775241,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009076970265355106,
regressor__max_features=0.9750309835254025,
regressor__max_samples=0.975303573468778; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007157504883579034,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016351814261589403,
regressor__max_features=0.6594878151468806,
regressor__max_samples=0.9224376554847273; total time= 15.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.016585412649095323,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00020387858300971332,
regressor__max_features=0.9512764533397833,
regressor__max_samples=0.7526261862239285; total time= 8.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.016585412649095323,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00020387858300971332,
regressor__max_features=0.9512764533397833,
regressor__max_samples=0.7526261862239285; total time= 9.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.016585412649095323,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00020387858300971332,
regressor__max_features=0.9512764533397833,
regressor__max_samples=0.7526261862239285; total time= 9.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.018076884714446042,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022074361547391334,
regressor__max_features=0.556732301008724,
regressor__max_samples=0.9654645528436463; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.018076884714446042,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022074361547391334,
regressor__max_features=0.556732301008724,
regressor__max_samples=0.9654645528436463; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.016585412649095323,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00020387858300971332,
regressor__max_features=0.9512764533397833,
regressor__max_samples=0.7526261862239285; total time= 10.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.018076884714446042,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022074361547391334,
regressor__max_features=0.556732301008724,
regressor__max_samples=0.9654645528436463; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.018076884714446042,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022074361547391334,
regressor__max_features=0.556732301008724,
regressor__max_samples=0.9654645528436463; total time= 3.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.018076884714446042,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022074361547391334,
regressor__max_features=0.556732301008724,
regressor__max_samples=0.9654645528436463; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00022449209314333798,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01925448447327093,
regressor__max_features=0.8091090316581305,
regressor__max_samples=0.5505613380613952; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00022449209314333798,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01925448447327093,
regressor__max_features=0.8091090316581305,
regressor__max_samples=0.5505613380613952; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00022449209314333798,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01925448447327093,
regressor__max_features=0.8091090316581305,
regressor__max_samples=0.5505613380613952; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00022449209314333798,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01925448447327093,
regressor__max_features=0.8091090316581305,
regressor__max_samples=0.5505613380613952; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00022449209314333798,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01925448447327093,
regressor__max_features=0.8091090316581305,
regressor__max_samples=0.5505613380613952; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003013540089495475,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018841368421034364,
regressor__max_features=0.8012204629505595,
regressor__max_samples=0.7768515261729003; total time= 10.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018979048042173073,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015486554522575941,
regressor__max_features=0.9459484543883883,
regressor__max_samples=0.5733024108460127; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003013540089495475,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018841368421034364,
regressor__max_features=0.8012204629505595,
regressor__max_samples=0.7768515261729003; total time= 10.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003013540089495475,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018841368421034364,
regressor__max_features=0.8012204629505595,
regressor__max_samples=0.7768515261729003; total time= 9.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018979048042173073,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015486554522575941,
regressor__max_features=0.9459484543883883,
regressor__max_samples=0.5733024108460127; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018979048042173073,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015486554522575941,
regressor__max_features=0.9459484543883883,
regressor__max_samples=0.5733024108460127; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018979048042173073,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015486554522575941,
regressor__max_features=0.9459484543883883,
regressor__max_samples=0.5733024108460127; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018979048042173073,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015486554522575941,
regressor__max_features=0.9459484543883883,
regressor__max_samples=0.5733024108460127; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003013540089495475,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018841368421034364,
regressor__max_features=0.8012204629505595,
regressor__max_samples=0.7768515261729003; total time= 11.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014882513324386765,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014710026802320958,
regressor__max_features=0.8685177783955045,
regressor__max_samples=0.7729578850824945; total time= 7.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017968554542843702,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005545187705601485,
regressor__max_features=0.9315692609945019,
regressor__max_samples=0.9401799843192085; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014882513324386765,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014710026802320958,
regressor__max_features=0.8685177783955045,
regressor__max_samples=0.7729578850824945; total time= 8.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014882513324386765,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014710026802320958,
regressor__max_features=0.8685177783955045,
regressor__max_samples=0.7729578850824945; total time= 7.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014882513324386765,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014710026802320958,
regressor__max_features=0.8685177783955045,
regressor__max_samples=0.7729578850824945; total time= 8.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06946175184647421,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005965766024488796,
regressor__max_features=0.6751091233501567,
regressor__max_samples=0.8540905977915999; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06946175184647421,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005965766024488796,
regressor__max_features=0.6751091233501567,
regressor__max_samples=0.8540905977915999; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017968554542843702,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005545187705601485,
regressor__max_features=0.9315692609945019,
regressor__max_samples=0.9401799843192085; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06946175184647421,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005965766024488796,
regressor__max_features=0.6751091233501567,
regressor__max_samples=0.8540905977915999; total time= 2.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017968554542843702,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005545187705601485,
regressor__max_features=0.9315692609945019,
regressor__max_samples=0.9401799843192085; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017968554542843702,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005545187705601485,
regressor__max_features=0.9315692609945019,
regressor__max_samples=0.9401799843192085; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017968554542843702,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005545187705601485,
regressor__max_features=0.9315692609945019,
regressor__max_samples=0.9401799843192085; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06946175184647421,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005965766024488796,
regressor__max_features=0.6751091233501567,
regressor__max_samples=0.8540905977915999; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.021450664374153838,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013039264435721998,
regressor__max_features=0.6243620772151439,
regressor__max_samples=0.6651262645095493; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06946175184647421,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005965766024488796,
regressor__max_features=0.6751091233501567,
regressor__max_samples=0.8540905977915999; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014882513324386765,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014710026802320958,
regressor__max_features=0.8685177783955045,
regressor__max_samples=0.7729578850824945; total time= 9.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.021450664374153838,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013039264435721998,
regressor__max_features=0.6243620772151439,
regressor__max_samples=0.6651262645095493; total time= 1.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.021450664374153838,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013039264435721998,
regressor__max_features=0.6243620772151439,
regressor__max_samples=0.6651262645095493; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0030052133738230707,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016428815800640642,
regressor__max_features=0.7857365126025053,
regressor__max_samples=0.8704869304553342; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0030052133738230707,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016428815800640642,
regressor__max_features=0.7857365126025053,
regressor__max_samples=0.8704869304553342; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0030052133738230707,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016428815800640642,
regressor__max_features=0.7857365126025053,
regressor__max_samples=0.8704869304553342; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.021450664374153838,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013039264435721998,
regressor__max_features=0.6243620772151439,
regressor__max_samples=0.6651262645095493; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003013540089495475,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018841368421034364,
regressor__max_features=0.8012204629505595,
regressor__max_samples=0.7768515261729003; total time= 13.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022520536912837993,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.017085786327350676,
regressor__max_features=0.8405197135484546,
regressor__max_samples=0.618753235646235; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.021450664374153838,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013039264435721998,
regressor__max_features=0.6243620772151439,
regressor__max_samples=0.6651262645095493; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0030052133738230707,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016428815800640642,
regressor__max_features=0.7857365126025053,
regressor__max_samples=0.8704869304553342; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022520536912837993,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.017085786327350676,
regressor__max_features=0.8405197135484546,
regressor__max_samples=0.618753235646235; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022520536912837993,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.017085786327350676,
regressor__max_features=0.8405197135484546,
regressor__max_samples=0.618753235646235; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0030052133738230707,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016428815800640642,
regressor__max_features=0.7857365126025053,
regressor__max_samples=0.8704869304553342; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022520536912837993,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.017085786327350676,
regressor__max_features=0.8405197135484546,
regressor__max_samples=0.618753235646235; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022520536912837993,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.017085786327350676,
regressor__max_features=0.8405197135484546,
regressor__max_samples=0.618753235646235; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012224921159070404,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09716148079839168,
regressor__max_features=0.8484625279319074,
regressor__max_samples=0.69210091568961; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.05719356209230236,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012196448023200598,
regressor__max_features=0.5377172735448846,
regressor__max_samples=0.5831077159546265; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.05719356209230236,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012196448023200598,
regressor__max_features=0.5377172735448846,
regressor__max_samples=0.5831077159546265; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.05719356209230236,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012196448023200598,
regressor__max_features=0.5377172735448846,
regressor__max_samples=0.5831077159546265; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.05719356209230236,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012196448023200598,
regressor__max_features=0.5377172735448846,
regressor__max_samples=0.5831077159546265; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.05719356209230236,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012196448023200598,
regressor__max_features=0.5377172735448846,
regressor__max_samples=0.5831077159546265; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012224921159070404,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09716148079839168,
regressor__max_features=0.8484625279319074,
regressor__max_samples=0.69210091568961; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012224921159070404,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09716148079839168,
regressor__max_features=0.8484625279319074,
regressor__max_samples=0.69210091568961; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012224921159070404,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09716148079839168,
regressor__max_features=0.8484625279319074,
regressor__max_samples=0.69210091568961; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012224921159070404,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09716148079839168,
regressor__max_features=0.8484625279319074,
regressor__max_samples=0.69210091568961; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00013687973036288436,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07099609342475928,
regressor__max_features=0.9451318919454581,
regressor__max_samples=0.7278283763928565; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00013687973036288436,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07099609342475928,
regressor__max_features=0.9451318919454581,
regressor__max_samples=0.7278283763928565; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00013687973036288436,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07099609342475928,
regressor__max_features=0.9451318919454581,
regressor__max_samples=0.7278283763928565; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00013687973036288436,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07099609342475928,
regressor__max_features=0.9451318919454581,
regressor__max_samples=0.7278283763928565; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0013323297901105763,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017728580556091047,
regressor__max_features=0.7641850606486046,
regressor__max_samples=0.718167915803278; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00013687973036288436,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07099609342475928,
regressor__max_features=0.9451318919454581,
regressor__max_samples=0.7278283763928565; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0013323297901105763,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017728580556091047,
regressor__max_features=0.7641850606486046,
regressor__max_samples=0.718167915803278; total time= 7.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010875855381980924,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013473478567267632,
regressor__max_features=0.9455715568490355,
regressor__max_samples=0.76385055454315; total time= 8.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0013323297901105763,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017728580556091047,
regressor__max_features=0.7641850606486046,
regressor__max_samples=0.718167915803278; total time= 8.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010875855381980924,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013473478567267632,
regressor__max_features=0.9455715568490355,
regressor__max_samples=0.76385055454315; total time= 8.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010875855381980924,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013473478567267632,
regressor__max_features=0.9455715568490355,
regressor__max_samples=0.76385055454315; total time= 8.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000627789105651048,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08378850141664125,
regressor__max_features=0.9931053722398014,
regressor__max_samples=0.8490808570098726; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010875855381980924,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013473478567267632,
regressor__max_features=0.9455715568490355,
regressor__max_samples=0.76385055454315; total time= 8.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000627789105651048,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08378850141664125,
regressor__max_features=0.9931053722398014,
regressor__max_samples=0.8490808570098726; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010162023488201283,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003667443899176896,
regressor__max_features=0.7318492024699911,
regressor__max_samples=0.6766761140130264; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010162023488201283,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003667443899176896,
regressor__max_features=0.7318492024699911,
regressor__max_samples=0.6766761140130264; total time= 4.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0013323297901105763,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017728580556091047,
regressor__max_features=0.7641850606486046,
regressor__max_samples=0.718167915803278; total time= 9.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010875855381980924,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013473478567267632,
regressor__max_features=0.9455715568490355,
regressor__max_samples=0.76385055454315; total time= 9.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000627789105651048,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08378850141664125,
regressor__max_features=0.9931053722398014,
regressor__max_samples=0.8490808570098726; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010162023488201283,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003667443899176896,
regressor__max_features=0.7318492024699911,
regressor__max_samples=0.6766761140130264; total time= 4.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000627789105651048,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08378850141664125,
regressor__max_features=0.9931053722398014,
regressor__max_samples=0.8490808570098726; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000627789105651048,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08378850141664125,
regressor__max_features=0.9931053722398014,
regressor__max_samples=0.8490808570098726; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000690701694569607,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02763026556593472,
regressor__max_features=0.8423655862769397,
regressor__max_samples=0.5813084696724456; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010162023488201283,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003667443899176896,
regressor__max_features=0.7318492024699911,
regressor__max_samples=0.6766761140130264; total time= 4.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000690701694569607,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02763026556593472,
regressor__max_features=0.8423655862769397,
regressor__max_samples=0.5813084696724456; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0013323297901105763,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017728580556091047,
regressor__max_features=0.7641850606486046,
regressor__max_samples=0.718167915803278; total time= 10.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000690701694569607,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02763026556593472,
regressor__max_features=0.8423655862769397,
regressor__max_samples=0.5813084696724456; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000690701694569607,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02763026556593472,
regressor__max_features=0.8423655862769397,
regressor__max_samples=0.5813084696724456; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010162023488201283,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003667443899176896,
regressor__max_features=0.7318492024699911,
regressor__max_samples=0.6766761140130264; total time= 4.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000690701694569607,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02763026556593472,
regressor__max_features=0.8423655862769397,
regressor__max_samples=0.5813084696724456; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001771108878273569,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.039645296344066765,
regressor__max_features=0.5226093350530947,
regressor__max_samples=0.5131834872486261; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016927824148697275,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07069679730050325,
regressor__max_features=0.86285975419418,
regressor__max_samples=0.8067075979678949; total time= 2.5s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016927824148697275,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07069679730050325,
regressor__max_features=0.86285975419418,
regressor__max_samples=0.8067075979678949; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016927824148697275,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07069679730050325,
regressor__max_features=0.86285975419418,
regressor__max_samples=0.8067075979678949; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001771108878273569,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.039645296344066765,
regressor__max_features=0.5226093350530947,
regressor__max_samples=0.5131834872486261; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001771108878273569,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.039645296344066765,
regressor__max_features=0.5226093350530947,
regressor__max_samples=0.5131834872486261; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001771108878273569,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.039645296344066765,
regressor__max_features=0.5226093350530947,
regressor__max_samples=0.5131834872486261; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016927824148697275,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07069679730050325,
regressor__max_features=0.86285975419418,
regressor__max_samples=0.8067075979678949; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016927824148697275,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07069679730050325,
regressor__max_features=0.86285975419418,
regressor__max_samples=0.8067075979678949; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001771108878273569,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.039645296344066765,
regressor__max_features=0.5226093350530947,
regressor__max_samples=0.5131834872486261; total time= 1.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.021518245161453526,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09158585158712827,
regressor__max_features=0.5752084455517641,
regressor__max_samples=0.7970653576760676; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.021518245161453526,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09158585158712827,
regressor__max_features=0.5752084455517641,
regressor__max_samples=0.7970653576760676; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.021518245161453526,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09158585158712827,
regressor__max_features=0.5752084455517641,
regressor__max_samples=0.7970653576760676; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.021518245161453526,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09158585158712827,
regressor__max_features=0.5752084455517641,
regressor__max_samples=0.7970653576760676; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0015099700976862259,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.033601353329690616,
regressor__max_features=0.9191643523555689,
regressor__max_samples=0.7343465798974851; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.021518245161453526,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09158585158712827,
regressor__max_features=0.5752084455517641,
regressor__max_samples=0.7970653576760676; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0015099700976862259,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.033601353329690616,
regressor__max_features=0.9191643523555689,
regressor__max_samples=0.7343465798974851; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0015099700976862259,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.033601353329690616,
regressor__max_features=0.9191643523555689,
regressor__max_samples=0.7343465798974851; total time= 3.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0015099700976862259,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.033601353329690616,
regressor__max_features=0.9191643523555689,
regressor__max_samples=0.7343465798974851; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0015099700976862259,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.033601353329690616,
regressor__max_features=0.9191643523555689,
regressor__max_samples=0.7343465798974851; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0013600398943955081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002247151774034237,
regressor__max_features=0.5645797075757475,
regressor__max_samples=0.9770255136293612; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.017771834016496495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004637619535713969,
regressor__max_features=0.8844937075902553,
regressor__max_samples=0.9723828649412141; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0013600398943955081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002247151774034237,
regressor__max_features=0.5645797075757475,
regressor__max_samples=0.9770255136293612; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0013600398943955081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002247151774034237,
regressor__max_features=0.5645797075757475,
regressor__max_samples=0.9770255136293612; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.017771834016496495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004637619535713969,
regressor__max_features=0.8844937075902553,
regressor__max_samples=0.9723828649412141; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0013600398943955081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002247151774034237,
regressor__max_features=0.5645797075757475,
regressor__max_samples=0.9770255136293612; total time= 1.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004184879400270116,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01035385439403384,
regressor__max_features=0.8090641202289479,
regressor__max_samples=0.6790813590164202; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.017771834016496495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004637619535713969,
regressor__max_features=0.8844937075902553,
regressor__max_samples=0.9723828649412141; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.017771834016496495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004637619535713969,
regressor__max_features=0.8844937075902553,
regressor__max_samples=0.9723828649412141; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0013600398943955081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002247151774034237,
regressor__max_features=0.5645797075757475,
regressor__max_samples=0.9770255136293612; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.017771834016496495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004637619535713969,
regressor__max_features=0.8844937075902553,
regressor__max_samples=0.9723828649412141; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004184879400270116,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01035385439403384,
regressor__max_features=0.8090641202289479,
regressor__max_samples=0.6790813590164202; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004184879400270116,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01035385439403384,
regressor__max_features=0.8090641202289479,
regressor__max_samples=0.6790813590164202; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004184879400270116,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01035385439403384,
regressor__max_features=0.8090641202289479,
regressor__max_samples=0.6790813590164202; total time= 2.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004184879400270116,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01035385439403384,
regressor__max_features=0.8090641202289479,
regressor__max_samples=0.6790813590164202; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00011656125803581683,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003638506061069969,
regressor__max_features=0.8861591958678197,
regressor__max_samples=0.7600817505559967; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00011656125803581683,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003638506061069969,
regressor__max_features=0.8861591958678197,
regressor__max_samples=0.7600817505559967; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00011656125803581683,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003638506061069969,
regressor__max_features=0.8861591958678197,
regressor__max_samples=0.7600817505559967; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00011656125803581683,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003638506061069969,
regressor__max_features=0.8861591958678197,
regressor__max_samples=0.7600817505559967; total time= 2.1s

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:690:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
the optimization hasn't converged yet.
  warnings.warn(

[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00011656125803581683,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003638506061069969,
regressor__max_features=0.8861591958678197,
regressor__max_samples=0.7600817505559967; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012199668475623268,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022594607000759603,
regressor__max_features=0.8196806784871211,
regressor__max_samples=0.9025223337596809; total time= 3.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012199668475623268,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022594607000759603,
regressor__max_features=0.8196806784871211,
regressor__max_samples=0.9025223337596809; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012199668475623268,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022594607000759603,
regressor__max_features=0.8196806784871211,
regressor__max_samples=0.9025223337596809; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012199668475623268,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022594607000759603,
regressor__max_features=0.8196806784871211,
regressor__max_samples=0.9025223337596809; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012199668475623268,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022594607000759603,
regressor__max_features=0.8196806784871211,
regressor__max_samples=0.9025223337596809; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00025651598090037223,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011265816586723024,
regressor__max_features=0.7949588434273165,
regressor__max_samples=0.6961220225498661; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02812918864708231,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018417245197099389,
regressor__max_features=0.8959089197348165,
regressor__max_samples=0.794977948568818; total time= 6.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00025651598090037223,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011265816586723024,
regressor__max_features=0.7949588434273165,
regressor__max_samples=0.6961220225498661; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02812918864708231,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018417245197099389,
regressor__max_features=0.8959089197348165,
regressor__max_samples=0.794977948568818; total time= 6.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02812918864708231,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018417245197099389,
regressor__max_features=0.8959089197348165,
regressor__max_samples=0.794977948568818; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00025651598090037223,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011265816586723024,
regressor__max_features=0.7949588434273165,
regressor__max_samples=0.6961220225498661; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00025651598090037223,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011265816586723024,
regressor__max_features=0.7949588434273165,
regressor__max_samples=0.6961220225498661; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001875021591566566,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011085783711666662,
regressor__max_features=0.7569947445799055,
regressor__max_samples=0.8918265063705715; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02812918864708231,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018417245197099389,
regressor__max_features=0.8959089197348165,
regressor__max_samples=0.794977948568818; total time= 8.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007832102357729433,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03864480553672624,
regressor__max_features=0.9747603118288211,
regressor__max_samples=0.573536740464519; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00025651598090037223,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011265816586723024,
regressor__max_features=0.7949588434273165,
regressor__max_samples=0.6961220225498661; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001875021591566566,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011085783711666662,
regressor__max_features=0.7569947445799055,
regressor__max_samples=0.8918265063705715; total time= 3.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001875021591566566,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011085783711666662,
regressor__max_features=0.7569947445799055,
regressor__max_samples=0.8918265063705715; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02812918864708231,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018417245197099389,
regressor__max_features=0.8959089197348165,
regressor__max_samples=0.794977948568818; total time= 8.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007832102357729433,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03864480553672624,
regressor__max_features=0.9747603118288211,
regressor__max_samples=0.573536740464519; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007832102357729433,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03864480553672624,
regressor__max_features=0.9747603118288211,
regressor__max_samples=0.573536740464519; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002520904284595546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014761364011262927,
regressor__max_features=0.9323611881275267,
regressor__max_samples=0.9064505045650388; total time= 15.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007832102357729433,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03864480553672624,
regressor__max_features=0.9747603118288211,
regressor__max_samples=0.573536740464519; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007832102357729433,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03864480553672624,
regressor__max_features=0.9747603118288211,
regressor__max_samples=0.573536740464519; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001875021591566566,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011085783711666662,
regressor__max_features=0.7569947445799055,
regressor__max_samples=0.8918265063705715; total time= 3.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002520904284595546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014761364011262927,
regressor__max_features=0.9323611881275267,
regressor__max_samples=0.9064505045650388; total time= 15.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001875021591566566,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011085783711666662,
regressor__max_features=0.7569947445799055,
regressor__max_samples=0.8918265063705715; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001102148781071297,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00794698810466966,
regressor__max_features=0.6200728093890966,
regressor__max_samples=0.5379316640543319; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001102148781071297,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00794698810466966,
regressor__max_features=0.6200728093890966,
regressor__max_samples=0.5379316640543319; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001102148781071297,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00794698810466966,
regressor__max_features=0.6200728093890966,
regressor__max_samples=0.5379316640543319; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001102148781071297,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00794698810466966,
regressor__max_features=0.6200728093890966,
regressor__max_samples=0.5379316640543319; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001102148781071297,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00794698810466966,
regressor__max_features=0.6200728093890966,
regressor__max_samples=0.5379316640543319; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03359496902928335,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005952962748435576,
regressor__max_features=0.7295678781191306,
regressor__max_samples=0.9900162876427385; total time= 3.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002520904284595546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014761364011262927,
regressor__max_features=0.9323611881275267,
regressor__max_samples=0.9064505045650388; total time= 18.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03359496902928335,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005952962748435576,
regressor__max_features=0.7295678781191306,
regressor__max_samples=0.9900162876427385; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0024008082618669087,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015881863780290717,
regressor__max_features=0.8488337868484659,
regressor__max_samples=0.5900336361746492; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03359496902928335,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005952962748435576,
regressor__max_features=0.7295678781191306,
regressor__max_samples=0.9900162876427385; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0024008082618669087,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015881863780290717,
regressor__max_features=0.8488337868484659,
regressor__max_samples=0.5900336361746492; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03359496902928335,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005952962748435576,
regressor__max_features=0.7295678781191306,
regressor__max_samples=0.9900162876427385; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002520904284595546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014761364011262927,
regressor__max_features=0.9323611881275267,
regressor__max_samples=0.9064505045650388; total time= 18.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002520904284595546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014761364011262927,
regressor__max_features=0.9323611881275267,
regressor__max_samples=0.9064505045650388; total time= 18.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0024008082618669087,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015881863780290717,
regressor__max_features=0.8488337868484659,
regressor__max_samples=0.5900336361746492; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0024008082618669087,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015881863780290717,
regressor__max_features=0.8488337868484659,
regressor__max_samples=0.5900336361746492; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0024008082618669087,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015881863780290717,
regressor__max_features=0.8488337868484659,
regressor__max_samples=0.5900336361746492; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03359496902928335,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005952962748435576,
regressor__max_features=0.7295678781191306,
regressor__max_samples=0.9900162876427385; total time= 5.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010889347318773486,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04197133802914918,
regressor__max_features=0.7576180274321054,
regressor__max_samples=0.9865551746350522; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010889347318773486,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04197133802914918,
regressor__max_features=0.7576180274321054,
regressor__max_samples=0.9865551746350522; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003774566285211953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.029199264391537895,
regressor__max_features=0.6725413139111933,
regressor__max_samples=0.6738096072413418; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010889347318773486,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04197133802914918,
regressor__max_features=0.7576180274321054,
regressor__max_samples=0.9865551746350522; total time= 3.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003774566285211953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.029199264391537895,
regressor__max_features=0.6725413139111933,
regressor__max_samples=0.6738096072413418; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010889347318773486,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04197133802914918,
regressor__max_features=0.7576180274321054,
regressor__max_samples=0.9865551746350522; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00018454725988311613,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0040111648528343875,
regressor__max_features=0.677995742135993,
regressor__max_samples=0.9471086314479629; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00018454725988311613,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0040111648528343875,
regressor__max_features=0.677995742135993,
regressor__max_samples=0.9471086314479629; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003774566285211953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.029199264391537895,
regressor__max_features=0.6725413139111933,
regressor__max_samples=0.6738096072413418; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003774566285211953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.029199264391537895,
regressor__max_features=0.6725413139111933,
regressor__max_samples=0.6738096072413418; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003774566285211953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.029199264391537895,
regressor__max_features=0.6725413139111933,
regressor__max_samples=0.6738096072413418; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010889347318773486,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04197133802914918,
regressor__max_features=0.7576180274321054,
regressor__max_samples=0.9865551746350522; total time= 3.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00018454725988311613,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0040111648528343875,
regressor__max_features=0.677995742135993,
regressor__max_samples=0.9471086314479629; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00018454725988311613,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0040111648528343875,
regressor__max_features=0.677995742135993,
regressor__max_samples=0.9471086314479629; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00018454725988311613,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0040111648528343875,
regressor__max_features=0.677995742135993,
regressor__max_samples=0.9471086314479629; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0262182583965218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006192047037327743,
regressor__max_features=0.9931284979334076,
regressor__max_samples=0.7682953229936662; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0262182583965218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006192047037327743,
regressor__max_features=0.9931284979334076,
regressor__max_samples=0.7682953229936662; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0262182583965218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006192047037327743,
regressor__max_features=0.9931284979334076,
regressor__max_samples=0.7682953229936662; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0262182583965218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006192047037327743,
regressor__max_features=0.9931284979334076,
regressor__max_samples=0.7682953229936662; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003237813373628106,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009220368833005388,
regressor__max_features=0.5461452931143731,
regressor__max_samples=0.7405726970281465; total time= 3.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0262182583965218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006192047037327743,
regressor__max_features=0.9931284979334076,
regressor__max_samples=0.7682953229936662; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001759045591749253,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002957556487590962,
regressor__max_features=0.6886429826210363,
regressor__max_samples=0.5012975121918233; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003237813373628106,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009220368833005388,
regressor__max_features=0.5461452931143731,
regressor__max_samples=0.7405726970281465; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003237813373628106,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009220368833005388,
regressor__max_features=0.5461452931143731,
regressor__max_samples=0.7405726970281465; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001759045591749253,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002957556487590962,
regressor__max_features=0.6886429826210363,
regressor__max_samples=0.5012975121918233; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002841639333505349,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01904873880436062,
regressor__max_features=0.7656328769867671,
regressor__max_samples=0.8602580306957324; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003237813373628106,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009220368833005388,
regressor__max_features=0.5461452931143731,
regressor__max_samples=0.7405726970281465; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002841639333505349,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01904873880436062,
regressor__max_features=0.7656328769867671,
regressor__max_samples=0.8602580306957324; total time= 1.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001759045591749253,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002957556487590962,
regressor__max_features=0.6886429826210363,
regressor__max_samples=0.5012975121918233; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001759045591749253,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002957556487590962,
regressor__max_features=0.6886429826210363,
regressor__max_samples=0.5012975121918233; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001759045591749253,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002957556487590962,
regressor__max_features=0.6886429826210363,
regressor__max_samples=0.5012975121918233; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003237813373628106,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009220368833005388,
regressor__max_features=0.5461452931143731,
regressor__max_samples=0.7405726970281465; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002841639333505349,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01904873880436062,
regressor__max_features=0.7656328769867671,
regressor__max_samples=0.8602580306957324; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002841639333505349,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01904873880436062,
regressor__max_features=0.7656328769867671,
regressor__max_samples=0.8602580306957324; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002841639333505349,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01904873880436062,
regressor__max_features=0.7656328769867671,
regressor__max_samples=0.8602580306957324; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001288396714510227,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013805392471755742,
regressor__max_features=0.9476034188435997,
regressor__max_samples=0.755838721057833; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001288396714510227,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013805392471755742,
regressor__max_features=0.9476034188435997,
regressor__max_samples=0.755838721057833; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001288396714510227,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013805392471755742,
regressor__max_features=0.9476034188435997,
regressor__max_samples=0.755838721057833; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001288396714510227,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013805392471755742,
regressor__max_features=0.9476034188435997,
regressor__max_samples=0.755838721057833; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001288396714510227,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013805392471755742,
regressor__max_features=0.9476034188435997,
regressor__max_samples=0.755838721057833; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012342878621258996,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002199060108815142,
regressor__max_features=0.7663086332275115,
regressor__max_samples=0.6212352518173648; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01630105145459998,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004919209016427628,
regressor__max_features=0.9203549929389656,
regressor__max_samples=0.5446021643560282; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01630105145459998,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004919209016427628,
regressor__max_features=0.9203549929389656,
regressor__max_samples=0.5446021643560282; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012342878621258996,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002199060108815142,
regressor__max_features=0.7663086332275115,
regressor__max_samples=0.6212352518173648; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012342878621258996,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002199060108815142,
regressor__max_features=0.7663086332275115,
regressor__max_samples=0.6212352518173648; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01630105145459998,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004919209016427628,
regressor__max_features=0.9203549929389656,
regressor__max_samples=0.5446021643560282; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012342878621258996,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002199060108815142,
regressor__max_features=0.7663086332275115,
regressor__max_samples=0.6212352518173648; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01630105145459998,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004919209016427628,
regressor__max_features=0.9203549929389656,
regressor__max_samples=0.5446021643560282; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002287106538387775,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010685149251813218,
regressor__max_features=0.736984970715278,
regressor__max_samples=0.6775521516396523; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012342878621258996,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002199060108815142,
regressor__max_features=0.7663086332275115,
regressor__max_samples=0.6212352518173648; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002287106538387775,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010685149251813218,
regressor__max_features=0.736984970715278,
regressor__max_samples=0.6775521516396523; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01630105145459998,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004919209016427628,
regressor__max_features=0.9203549929389656,
regressor__max_samples=0.5446021643560282; total time= 3.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002287106538387775,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010685149251813218,
regressor__max_features=0.736984970715278,
regressor__max_samples=0.6775521516396523; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002287106538387775,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010685149251813218,
regressor__max_features=0.736984970715278,
regressor__max_samples=0.6775521516396523; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002287106538387775,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010685149251813218,
regressor__max_features=0.736984970715278,
regressor__max_samples=0.6775521516396523; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0031284311157851762,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0056571598982579074,
regressor__max_features=0.8684112376974564,
regressor__max_samples=0.7788711329752187; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0031284311157851762,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0056571598982579074,
regressor__max_features=0.8684112376974564,
regressor__max_samples=0.7788711329752187; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0031284311157851762,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0056571598982579074,
regressor__max_features=0.8684112376974564,
regressor__max_samples=0.7788711329752187; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0031284311157851762,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0056571598982579074,
regressor__max_features=0.8684112376974564,
regressor__max_samples=0.7788711329752187; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0019723257224501817,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013687302885982927,
regressor__max_features=0.668723416935959,
regressor__max_samples=0.9498236985725061; total time= 3.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.005156261256674519,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09110574133808463,
regressor__max_features=0.8028874096784435,
regressor__max_samples=0.6186133958679972; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0031284311157851762,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0056571598982579074,
regressor__max_features=0.8684112376974564,
regressor__max_samples=0.7788711329752187; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.005156261256674519,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09110574133808463,
regressor__max_features=0.8028874096784435,
regressor__max_samples=0.6186133958679972; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0019723257224501817,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013687302885982927,
regressor__max_features=0.668723416935959,
regressor__max_samples=0.9498236985725061; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.005156261256674519,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09110574133808463,
regressor__max_features=0.8028874096784435,
regressor__max_samples=0.6186133958679972; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.005156261256674519,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09110574133808463,
regressor__max_features=0.8028874096784435,
regressor__max_samples=0.6186133958679972; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0019723257224501817,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013687302885982927,
regressor__max_features=0.668723416935959,
regressor__max_samples=0.9498236985725061; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.005156261256674519,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09110574133808463,
regressor__max_features=0.8028874096784435,
regressor__max_samples=0.6186133958679972; total time= 2.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0019723257224501817,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013687302885982927,
regressor__max_features=0.668723416935959,
regressor__max_samples=0.9498236985725061; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026157405212333017,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.033646671331902085,
regressor__max_features=0.5258177389372565,
regressor__max_samples=0.5091212407141805; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00047424023059781196,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005468562564708447,
regressor__max_features=0.5803406866297778,
regressor__max_samples=0.5932835120256529; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0019723257224501817,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013687302885982927,
regressor__max_features=0.668723416935959,
regressor__max_samples=0.9498236985725061; total time= 4.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026157405212333017,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.033646671331902085,
regressor__max_features=0.5258177389372565,
regressor__max_samples=0.5091212407141805; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026157405212333017,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.033646671331902085,
regressor__max_features=0.5258177389372565,
regressor__max_samples=0.5091212407141805; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00047424023059781196,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005468562564708447,
regressor__max_features=0.5803406866297778,
regressor__max_samples=0.5932835120256529; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00047424023059781196,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005468562564708447,
regressor__max_features=0.5803406866297778,
regressor__max_samples=0.5932835120256529; total time= 3.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026157405212333017,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.033646671331902085,
regressor__max_features=0.5258177389372565,
regressor__max_samples=0.5091212407141805; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026157405212333017,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.033646671331902085,
regressor__max_features=0.5258177389372565,
regressor__max_samples=0.5091212407141805; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00047424023059781196,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005468562564708447,
regressor__max_features=0.5803406866297778,
regressor__max_samples=0.5932835120256529; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00047424023059781196,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005468562564708447,
regressor__max_features=0.5803406866297778,
regressor__max_samples=0.5932835120256529; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028262866872541346,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0029665135595118054,
regressor__max_features=0.8614480483108878,
regressor__max_samples=0.9104307365766763; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00468046151958953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006917310667555202,
regressor__max_features=0.8847464665959685,
regressor__max_samples=0.5935218742787617; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00468046151958953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006917310667555202,
regressor__max_features=0.8847464665959685,
regressor__max_samples=0.5935218742787617; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028262866872541346,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0029665135595118054,
regressor__max_features=0.8614480483108878,
regressor__max_samples=0.9104307365766763; total time= 2.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028262866872541346,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0029665135595118054,
regressor__max_features=0.8614480483108878,
regressor__max_samples=0.9104307365766763; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08853925242554317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04895880274925847,
regressor__max_features=0.78799920890283,
regressor__max_samples=0.9586978065427658; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08853925242554317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04895880274925847,
regressor__max_features=0.78799920890283,
regressor__max_samples=0.9586978065427658; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08853925242554317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04895880274925847,
regressor__max_features=0.78799920890283,
regressor__max_samples=0.9586978065427658; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.013578676335889984,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01617117403710517,
regressor__max_features=0.7649200115905583,
regressor__max_samples=0.8536150478222222; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00468046151958953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006917310667555202,
regressor__max_features=0.8847464665959685,
regressor__max_samples=0.5935218742787617; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08853925242554317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04895880274925847,
regressor__max_features=0.78799920890283,
regressor__max_samples=0.9586978065427658; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00468046151958953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006917310667555202,
regressor__max_features=0.8847464665959685,
regressor__max_samples=0.5935218742787617; total time= 3.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028262866872541346,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0029665135595118054,
regressor__max_features=0.8614480483108878,
regressor__max_samples=0.9104307365766763; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.013578676335889984,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01617117403710517,
regressor__max_features=0.7649200115905583,
regressor__max_samples=0.8536150478222222; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00468046151958953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006917310667555202,
regressor__max_features=0.8847464665959685,
regressor__max_samples=0.5935218742787617; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08853925242554317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04895880274925847,
regressor__max_features=0.78799920890283,
regressor__max_samples=0.9586978065427658; total time= 4.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.013578676335889984,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01617117403710517,
regressor__max_features=0.7649200115905583,
regressor__max_samples=0.8536150478222222; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.013578676335889984,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01617117403710517,
regressor__max_features=0.7649200115905583,
regressor__max_samples=0.8536150478222222; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028262866872541346,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0029665135595118054,
regressor__max_features=0.8614480483108878,
regressor__max_samples=0.9104307365766763; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007343306571887155,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0032984649159558425,
regressor__max_features=0.966007171128467,
regressor__max_samples=0.6603210960797572; total time= 1.5s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007343306571887155,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0032984649159558425,
regressor__max_features=0.966007171128467,
regressor__max_samples=0.6603210960797572; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.013578676335889984,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01617117403710517,
regressor__max_features=0.7649200115905583,
regressor__max_samples=0.8536150478222222; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007343306571887155,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0032984649159558425,
regressor__max_features=0.966007171128467,
regressor__max_samples=0.6603210960797572; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007343306571887155,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0032984649159558425,
regressor__max_features=0.966007171128467,
regressor__max_samples=0.6603210960797572; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007343306571887155,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0032984649159558425,
regressor__max_features=0.966007171128467,
regressor__max_samples=0.6603210960797572; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001430443176737134,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0023057059307165324,
regressor__max_features=0.7743015589690766,
regressor__max_samples=0.7744609899444886; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001430443176737134,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0023057059307165324,
regressor__max_features=0.7743015589690766,
regressor__max_samples=0.7744609899444886; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001430443176737134,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0023057059307165324,
regressor__max_features=0.7743015589690766,
regressor__max_samples=0.7744609899444886; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001430443176737134,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0023057059307165324,
regressor__max_features=0.7743015589690766,
regressor__max_samples=0.7744609899444886; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001430443176737134,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0023057059307165324,
regressor__max_features=0.7743015589690766,
regressor__max_samples=0.7744609899444886; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006122322768201859,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017199774079177854,
regressor__max_features=0.6744341332714976,
regressor__max_samples=0.9647645721239129; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009269611546439868,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018348659173827745,
regressor__max_features=0.5694123867210101,
regressor__max_samples=0.5013554497071648; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009269611546439868,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018348659173827745,
regressor__max_features=0.5694123867210101,
regressor__max_samples=0.5013554497071648; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006122322768201859,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017199774079177854,
regressor__max_features=0.6744341332714976,
regressor__max_samples=0.9647645721239129; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006122322768201859,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017199774079177854,
regressor__max_features=0.6744341332714976,
regressor__max_samples=0.9647645721239129; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009269611546439868,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018348659173827745,
regressor__max_features=0.5694123867210101,
regressor__max_samples=0.5013554497071648; total time= 4.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006122322768201859,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017199774079177854,
regressor__max_features=0.6744341332714976,
regressor__max_samples=0.9647645721239129; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006122322768201859,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017199774079177854,
regressor__max_features=0.6744341332714976,
regressor__max_samples=0.9647645721239129; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001373048110687019,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01672483908098154,
regressor__max_features=0.787236556589956,
regressor__max_samples=0.920914388379136; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001373048110687019,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01672483908098154,
regressor__max_features=0.787236556589956,
regressor__max_samples=0.920914388379136; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001373048110687019,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01672483908098154,
regressor__max_features=0.787236556589956,
regressor__max_samples=0.920914388379136; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001373048110687019,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01672483908098154,
regressor__max_features=0.787236556589956,
regressor__max_samples=0.920914388379136; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009269611546439868,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018348659173827745,
regressor__max_features=0.5694123867210101,
regressor__max_samples=0.5013554497071648; total time= 5.7s

```

```

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/joblib/externals/loky/process_executor.py:752: UserWarning: A worker
stopped while some jobs were given to the executor. This can be caused by a too
short worker timeout or by a memory leak.

```

```

warnings.warn(

```

```

[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009269611546439868,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018348659173827745,
regressor__max_features=0.5694123867210101,
regressor__max_samples=0.5013554497071648; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001373048110687019,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01672483908098154,
regressor__max_features=0.787236556589956,
regressor__max_samples=0.920914388379136; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.030466506756675733,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00040260759199206917,
regressor__max_features=0.5818279714328523,
regressor__max_samples=0.5821328989654965; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.030466506756675733,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00040260759199206917,
regressor__max_features=0.5818279714328523,
regressor__max_samples=0.5821328989654965; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005094191163759966,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0037084828445683614,
regressor__max_features=0.6794152420617512,
regressor__max_samples=0.9386002704065541; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.030466506756675733,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00040260759199206917,
regressor__max_features=0.5818279714328523,
regressor__max_samples=0.5821328989654965; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017295119880053788,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00023598895136582365,
regressor__max_features=0.8654337376018222,
regressor__max_samples=0.9691702284105189; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017295119880053788,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00023598895136582365,
regressor__max_features=0.8654337376018222,
regressor__max_samples=0.9691702284105189; total time= 6.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.030466506756675733,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00040260759199206917,
regressor__max_features=0.5818279714328523,
regressor__max_samples=0.5821328989654965; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005094191163759966,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0037084828445683614,
regressor__max_features=0.6794152420617512,
regressor__max_samples=0.9386002704065541; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005094191163759966,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0037084828445683614,
regressor__max_features=0.6794152420617512,
regressor__max_samples=0.9386002704065541; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017295119880053788,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00023598895136582365,
regressor__max_features=0.8654337376018222,
regressor__max_samples=0.9691702284105189; total time= 8.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005094191163759966,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0037084828445683614,
regressor__max_features=0.6794152420617512,
regressor__max_samples=0.9386002704065541; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.030466506756675733,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00040260759199206917,
regressor__max_features=0.5818279714328523,
regressor__max_samples=0.5821328989654965; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005094191163759966,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0037084828445683614,
regressor__max_features=0.6794152420617512,
regressor__max_samples=0.9386002704065541; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017295119880053788,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00023598895136582365,
regressor__max_features=0.8654337376018222,
regressor__max_samples=0.9691702284105189; total time= 7.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011192991919984775,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013846734485006352,
regressor__max_features=0.9498337226460866,
regressor__max_samples=0.8120507720909188; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011192991919984775,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013846734485006352,
regressor__max_features=0.9498337226460866,
regressor__max_samples=0.8120507720909188; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011192991919984775,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013846734485006352,
regressor__max_features=0.9498337226460866,
regressor__max_samples=0.8120507720909188; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004272268935624073,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004251556274263087,
regressor__max_features=0.7157640715682213,
regressor__max_samples=0.6664078929562421; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011192991919984775,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013846734485006352,
regressor__max_features=0.9498337226460866,
regressor__max_samples=0.8120507720909188; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017295119880053788,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00023598895136582365,
regressor__max_features=0.8654337376018222,
regressor__max_samples=0.9691702284105189; total time= 8.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01749154374434863,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0054008089256499135,
regressor__max_features=0.677681225909198,
regressor__max_samples=0.695741055100751; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01749154374434863,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0054008089256499135,
regressor__max_features=0.677681225909198,
regressor__max_samples=0.695741055100751; total time= 2.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01749154374434863,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0054008089256499135,
regressor__max_features=0.677681225909198,
regressor__max_samples=0.695741055100751; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004272268935624073,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004251556274263087,
regressor__max_features=0.7157640715682213,
regressor__max_samples=0.6664078929562421; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004272268935624073,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004251556274263087,
regressor__max_features=0.7157640715682213,
regressor__max_samples=0.6664078929562421; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004272268935624073,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004251556274263087,
regressor__max_features=0.7157640715682213,
regressor__max_samples=0.6664078929562421; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01749154374434863,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0054008089256499135,
regressor__max_features=0.677681225909198,
regressor__max_samples=0.695741055100751; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01749154374434863,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0054008089256499135,
regressor__max_features=0.677681225909198,
regressor__max_samples=0.695741055100751; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011192991919984775,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013846734485006352,
regressor__max_features=0.9498337226460866,
regressor__max_samples=0.8120507720909188; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004272268935624073,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004251556274263087,
regressor__max_features=0.7157640715682213,
regressor__max_samples=0.6664078929562421; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0010109201286713741,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.043240165359728217,
regressor__max_features=0.7477118132019829,
regressor__max_samples=0.8707304555695228; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0010109201286713741,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.043240165359728217,
regressor__max_features=0.7477118132019829,
regressor__max_samples=0.8707304555695228; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012606580582274818,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013873698320843767,
regressor__max_features=0.7469907435654313,
regressor__max_samples=0.8773176459300582; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.023216226582091153,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.018080455195272262,
regressor__max_features=0.8534902306217296,
regressor__max_samples=0.8892859695253755; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.023216226582091153,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.018080455195272262,
regressor__max_features=0.8534902306217296,
regressor__max_samples=0.8892859695253755; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.023216226582091153,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.018080455195272262,
regressor__max_features=0.8534902306217296,
regressor__max_samples=0.8892859695253755; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.023216226582091153,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.018080455195272262,
regressor__max_features=0.8534902306217296,
regressor__max_samples=0.8892859695253755; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.023216226582091153,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.018080455195272262,
regressor__max_features=0.8534902306217296,
regressor__max_samples=0.8892859695253755; total time= 3.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012606580582274818,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013873698320843767,
regressor__max_features=0.7469907435654313,
regressor__max_samples=0.8773176459300582; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012606580582274818,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013873698320843767,
regressor__max_features=0.7469907435654313,
regressor__max_samples=0.8773176459300582; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0010109201286713741,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.043240165359728217,
regressor__max_features=0.7477118132019829,
regressor__max_samples=0.8707304555695228; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0010109201286713741,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.043240165359728217,
regressor__max_features=0.7477118132019829,
regressor__max_samples=0.8707304555695228; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012606580582274818,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013873698320843767,
regressor__max_features=0.7469907435654313,
regressor__max_samples=0.8773176459300582; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0010109201286713741,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.043240165359728217,
regressor__max_features=0.7477118132019829,
regressor__max_samples=0.8707304555695228; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.029091888474996042,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013691930836407576,
regressor__max_features=0.7285001096408392,
regressor__max_samples=0.8019789360362563; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012606580582274818,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013873698320843767,
regressor__max_features=0.7469907435654313,
regressor__max_samples=0.8773176459300582; total time= 3.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.029091888474996042,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013691930836407576,
regressor__max_features=0.7285001096408392,
regressor__max_samples=0.8019789360362563; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.029091888474996042,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013691930836407576,
regressor__max_features=0.7285001096408392,
regressor__max_samples=0.8019789360362563; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.029091888474996042,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013691930836407576,
regressor__max_features=0.7285001096408392,
regressor__max_samples=0.8019789360362563; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.029091888474996042,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013691930836407576,
regressor__max_features=0.7285001096408392,
regressor__max_samples=0.8019789360362563; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.024653926524984868,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0028778784423756257,
regressor__max_features=0.7044775885427466,
regressor__max_samples=0.8859409902991225; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.024653926524984868,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0028778784423756257,
regressor__max_features=0.7044775885427466,
regressor__max_samples=0.8859409902991225; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008383838400775534,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00497193041678123,
regressor__max_features=0.8580895373664761,
regressor__max_samples=0.7995146823886632; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008383838400775534,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00497193041678123,
regressor__max_features=0.8580895373664761,
regressor__max_samples=0.7995146823886632; total time= 2.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003505848545428616,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010655473946580738,
regressor__max_features=0.6136754915149936,
regressor__max_samples=0.7117984312102414; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008383838400775534,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00497193041678123,
regressor__max_features=0.8580895373664761,
regressor__max_samples=0.7995146823886632; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008383838400775534,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00497193041678123,
regressor__max_features=0.8580895373664761,
regressor__max_samples=0.7995146823886632; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008383838400775534,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00497193041678123,
regressor__max_features=0.8580895373664761,
regressor__max_samples=0.7995146823886632; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.024653926524984868,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0028778784423756257,
regressor__max_features=0.7044775885427466,
regressor__max_samples=0.8859409902991225; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003505848545428616,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010655473946580738,
regressor__max_features=0.6136754915149936,
regressor__max_samples=0.7117984312102414; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003505848545428616,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010655473946580738,
regressor__max_features=0.6136754915149936,
regressor__max_samples=0.7117984312102414; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.024653926524984868,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0028778784423756257,
regressor__max_features=0.7044775885427466,
regressor__max_samples=0.8859409902991225; total time= 3.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.024653926524984868,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0028778784423756257,
regressor__max_features=0.7044775885427466,
regressor__max_samples=0.8859409902991225; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06947843362776954,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005737061375884855,
regressor__max_features=0.752944339442233,
regressor__max_samples=0.805727117717324; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003505848545428616,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010655473946580738,
regressor__max_features=0.6136754915149936,
regressor__max_samples=0.7117984312102414; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06947843362776954,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005737061375884855,
regressor__max_features=0.752944339442233,
regressor__max_samples=0.805727117717324; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06947843362776954,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005737061375884855,
regressor__max_features=0.752944339442233,
regressor__max_samples=0.805727117717324; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06947843362776954,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005737061375884855,
regressor__max_features=0.752944339442233,
regressor__max_samples=0.805727117717324; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003505848545428616,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010655473946580738,
regressor__max_features=0.6136754915149936,
regressor__max_samples=0.7117984312102414; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06947843362776954,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005737061375884855,
regressor__max_features=0.752944339442233,
regressor__max_samples=0.805727117717324; total time= 2.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015076021849198939,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06256837091475116,
regressor__max_features=0.7825665917946045,
regressor__max_samples=0.8483254119384461; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015076021849198939,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06256837091475116,
regressor__max_features=0.7825665917946045,
regressor__max_samples=0.8483254119384461; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00017059378161320057,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0635478767122729,
regressor__max_features=0.9627842564533882,
regressor__max_samples=0.725419685702066; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015076021849198939,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06256837091475116,
regressor__max_features=0.7825665917946045,
regressor__max_samples=0.8483254119384461; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00017059378161320057,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0635478767122729,
regressor__max_features=0.9627842564533882,
regressor__max_samples=0.725419685702066; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015076021849198939,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06256837091475116,
regressor__max_features=0.7825665917946045,
regressor__max_samples=0.8483254119384461; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015076021849198939,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06256837091475116,
regressor__max_features=0.7825665917946045,
regressor__max_samples=0.8483254119384461; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00017059378161320057,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0635478767122729,
regressor__max_features=0.9627842564533882,
regressor__max_samples=0.725419685702066; total time= 2.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00017059378161320057,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0635478767122729,
regressor__max_features=0.9627842564533882,
regressor__max_samples=0.725419685702066; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00017059378161320057,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0635478767122729,
regressor__max_features=0.9627842564533882,
regressor__max_samples=0.725419685702066; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005101737437929794,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03286202028082996,
regressor__max_features=0.5623313406016335,
regressor__max_samples=0.9604209413086862; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005101737437929794,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03286202028082996,
regressor__max_features=0.5623313406016335,
regressor__max_samples=0.9604209413086862; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00011723784162011009,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028682508353628937,
regressor__max_features=0.7881441800834066,
regressor__max_samples=0.803357523191428; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005101737437929794,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03286202028082996,
regressor__max_features=0.5623313406016335,
regressor__max_samples=0.9604209413086862; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005101737437929794,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03286202028082996,
regressor__max_features=0.5623313406016335,
regressor__max_samples=0.9604209413086862; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000859672020553953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005940545235233702,
regressor__max_features=0.6995013519350651,
regressor__max_samples=0.5273808194110157; total time= 2.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005101737437929794,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03286202028082996,
regressor__max_features=0.5623313406016335,
regressor__max_samples=0.9604209413086862; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000859672020553953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005940545235233702,
regressor__max_features=0.6995013519350651,
regressor__max_samples=0.5273808194110157; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000859672020553953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005940545235233702,
regressor__max_features=0.6995013519350651,
regressor__max_samples=0.5273808194110157; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00011723784162011009,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028682508353628937,
regressor__max_features=0.7881441800834066,
regressor__max_samples=0.803357523191428; total time= 5.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00011723784162011009,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028682508353628937,
regressor__max_features=0.7881441800834066,
regressor__max_samples=0.803357523191428; total time= 5.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000859672020553953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005940545235233702,
regressor__max_features=0.6995013519350651,
regressor__max_samples=0.5273808194110157; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000859672020553953,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005940545235233702,
regressor__max_features=0.6995013519350651,
regressor__max_samples=0.5273808194110157; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00011723784162011009,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028682508353628937,
regressor__max_features=0.7881441800834066,
regressor__max_samples=0.803357523191428; total time= 6.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00011723784162011009,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028682508353628937,
regressor__max_features=0.7881441800834066,
regressor__max_samples=0.803357523191428; total time= 6.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00577038343436332,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022767667922905187,
regressor__max_features=0.5704285101898999,
regressor__max_samples=0.5881934932531117; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00577038343436332,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022767667922905187,
regressor__max_features=0.5704285101898999,
regressor__max_samples=0.5881934932531117; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00577038343436332,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022767667922905187,
regressor__max_features=0.5704285101898999,
regressor__max_samples=0.5881934932531117; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00577038343436332,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022767667922905187,
regressor__max_features=0.5704285101898999,
regressor__max_samples=0.5881934932531117; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00577038343436332,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022767667922905187,
regressor__max_features=0.5704285101898999,
regressor__max_samples=0.5881934932531117; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03895421185245975,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010940520471305285,
regressor__max_features=0.6734766009481139,
regressor__max_samples=0.8687506240548741; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03895421185245975,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010940520471305285,
regressor__max_features=0.6734766009481139,
regressor__max_samples=0.8687506240548741; total time= 4.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000313718371749915,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.05553128244188592,
regressor__max_features=0.6811969495583166,
regressor__max_samples=0.7902941751390218; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03895421185245975,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010940520471305285,
regressor__max_features=0.6734766009481139,
regressor__max_samples=0.8687506240548741; total time= 4.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000313718371749915,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.05553128244188592,
regressor__max_features=0.6811969495583166,
regressor__max_samples=0.7902941751390218; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000313718371749915,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.05553128244188592,
regressor__max_features=0.6811969495583166,
regressor__max_samples=0.7902941751390218; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000313718371749915,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.05553128244188592,
regressor__max_features=0.6811969495583166,
regressor__max_samples=0.7902941751390218; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03895421185245975,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010940520471305285,
regressor__max_features=0.6734766009481139,
regressor__max_samples=0.8687506240548741; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00041375135125002913,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010325142852622373,
regressor__max_features=0.6667495858455721,
regressor__max_samples=0.6990843467954717; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004573952416273123,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.009786155583853933,
regressor__max_features=0.5890179834348757,
regressor__max_samples=0.9805351587347275; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03895421185245975,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0010940520471305285,
regressor__max_features=0.6734766009481139,
regressor__max_samples=0.8687506240548741; total time= 6.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000313718371749915,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.05553128244188592,
regressor__max_features=0.6811969495583166,
regressor__max_samples=0.7902941751390218; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004573952416273123,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.009786155583853933,
regressor__max_features=0.5890179834348757,
regressor__max_samples=0.9805351587347275; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004573952416273123,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.009786155583853933,
regressor__max_features=0.5890179834348757,
regressor__max_samples=0.9805351587347275; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004573952416273123,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.009786155583853933,
regressor__max_features=0.5890179834348757,
regressor__max_samples=0.9805351587347275; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004573952416273123,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.009786155583853933,
regressor__max_features=0.5890179834348757,
regressor__max_samples=0.9805351587347275; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00041375135125002913,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010325142852622373,
regressor__max_features=0.6667495858455721,
regressor__max_samples=0.6990843467954717; total time= 7.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00041375135125002913,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010325142852622373,
regressor__max_features=0.6667495858455721,
regressor__max_samples=0.6990843467954717; total time= 9.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008507672799208652,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.017777938979152075,
regressor__max_features=0.6049527965477929,
regressor__max_samples=0.9490271447203569; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008507672799208652,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.017777938979152075,
regressor__max_features=0.6049527965477929,
regressor__max_samples=0.9490271447203569; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008507672799208652,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.017777938979152075,
regressor__max_features=0.6049527965477929,
regressor__max_samples=0.9490271447203569; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008507672799208652,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.017777938979152075,
regressor__max_features=0.6049527965477929,
regressor__max_samples=0.9490271447203569; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008507672799208652,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.017777938979152075,
regressor__max_features=0.6049527965477929,
regressor__max_samples=0.9490271447203569; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00041375135125002913,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010325142852622373,
regressor__max_features=0.6667495858455721,
regressor__max_samples=0.6990843467954717; total time= 10.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00041375135125002913,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010325142852622373,
regressor__max_features=0.6667495858455721,
regressor__max_samples=0.6990843467954717; total time= 10.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007623142491508385,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002290163581857119,
regressor__max_features=0.7621951346637901,
regressor__max_samples=0.720381373469114; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007623142491508385,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002290163581857119,
regressor__max_features=0.7621951346637901,
regressor__max_samples=0.720381373469114; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007623142491508385,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002290163581857119,
regressor__max_features=0.7621951346637901,
regressor__max_samples=0.720381373469114; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007623142491508385,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002290163581857119,
regressor__max_features=0.7621951346637901,
regressor__max_samples=0.720381373469114; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007623142491508385,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002290163581857119,
regressor__max_features=0.7621951346637901,
regressor__max_samples=0.720381373469114; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0046980334720751705,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018032211984108181,
regressor__max_features=0.9984371259229736,
regressor__max_samples=0.7510975051656212; total time= 8.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0046980334720751705,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018032211984108181,
regressor__max_features=0.9984371259229736,
regressor__max_samples=0.7510975051656212; total time= 8.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0046980334720751705,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018032211984108181,
regressor__max_features=0.9984371259229736,
regressor__max_samples=0.7510975051656212; total time= 9.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002539191673772048,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001287208810047458,
regressor__max_features=0.7360334725549996,
regressor__max_samples=0.7824205666313082; total time= 8.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0046980334720751705,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018032211984108181,
regressor__max_features=0.9984371259229736,
regressor__max_samples=0.7510975051656212; total time= 10.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002539191673772048,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001287208810047458,
regressor__max_features=0.7360334725549996,
regressor__max_samples=0.7824205666313082; total time= 9.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0030976749693105298,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002922272679181134,
regressor__max_features=0.5909640652476356,
regressor__max_samples=0.9308928105067587; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01818606178601756,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006489840441726854,
regressor__max_features=0.8219997716195079,
regressor__max_samples=0.7043670855490483; total time= 5.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01818606178601756,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006489840441726854,
regressor__max_features=0.8219997716195079,
regressor__max_samples=0.7043670855490483; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01818606178601756,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006489840441726854,
regressor__max_features=0.8219997716195079,
regressor__max_samples=0.7043670855490483; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002539191673772048,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001287208810047458,
regressor__max_features=0.7360334725549996,
regressor__max_samples=0.7824205666313082; total time= 10.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0046980334720751705,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00018032211984108181,
regressor__max_features=0.9984371259229736,
regressor__max_samples=0.7510975051656212; total time= 12.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08645583444926509,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03990077030384057,
regressor__max_features=0.7039922476824223,
regressor__max_samples=0.7758612976296699; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002539191673772048,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001287208810047458,
regressor__max_features=0.7360334725549996,
regressor__max_samples=0.7824205666313082; total time= 10.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08645583444926509,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03990077030384057,
regressor__max_features=0.7039922476824223,
regressor__max_samples=0.7758612976296699; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0030976749693105298,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002922272679181134,
regressor__max_features=0.5909640652476356,
regressor__max_samples=0.9308928105067587; total time= 7.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08645583444926509,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03990077030384057,
regressor__max_features=0.7039922476824223,
regressor__max_samples=0.7758612976296699; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0030976749693105298,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002922272679181134,
regressor__max_features=0.5909640652476356,
regressor__max_samples=0.9308928105067587; total time= 8.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08645583444926509,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03990077030384057,
regressor__max_features=0.7039922476824223,
regressor__max_samples=0.7758612976296699; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01818606178601756,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006489840441726854,
regressor__max_features=0.8219997716195079,
regressor__max_samples=0.7043670855490483; total time= 4.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08645583444926509,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03990077030384057,
regressor__max_features=0.7039922476824223,
regressor__max_samples=0.7758612976296699; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004934286811215813,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0032849112258193686,
regressor__max_features=0.7975244898502103,
regressor__max_samples=0.6696356229480955; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01818606178601756,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006489840441726854,
regressor__max_features=0.8219997716195079,
regressor__max_samples=0.7043670855490483; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0030976749693105298,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002922272679181134,
regressor__max_features=0.5909640652476356,
regressor__max_samples=0.9308928105067587; total time= 8.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0030976749693105298,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002922272679181134,
regressor__max_features=0.5909640652476356,
regressor__max_samples=0.9308928105067587; total time= 8.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002539191673772048,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001287208810047458,
regressor__max_features=0.7360334725549996,
regressor__max_samples=0.7824205666313082; total time= 12.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004934286811215813,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0032849112258193686,
regressor__max_features=0.7975244898502103,
regressor__max_samples=0.6696356229480955; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00034326852679397653,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004679126153775273,
regressor__max_features=0.8604065481552533,
regressor__max_samples=0.9026574836789525; total time= 1.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00034326852679397653,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004679126153775273,
regressor__max_features=0.8604065481552533,
regressor__max_samples=0.9026574836789525; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00034326852679397653,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004679126153775273,
regressor__max_features=0.8604065481552533,
regressor__max_samples=0.9026574836789525; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004934286811215813,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0032849112258193686,
regressor__max_features=0.7975244898502103,
regressor__max_samples=0.6696356229480955; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00034326852679397653,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004679126153775273,
regressor__max_features=0.8604065481552533,
regressor__max_samples=0.9026574836789525; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01468252657463809,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.030949924483757867,
regressor__max_features=0.6985917648092275,
regressor__max_samples=0.8340425682853231; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004934286811215813,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0032849112258193686,
regressor__max_features=0.7975244898502103,
regressor__max_samples=0.6696356229480955; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004934286811215813,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0032849112258193686,
regressor__max_features=0.7975244898502103,
regressor__max_samples=0.6696356229480955; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01468252657463809,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.030949924483757867,
regressor__max_features=0.6985917648092275,
regressor__max_samples=0.8340425682853231; total time= 2.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01468252657463809,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.030949924483757867,
regressor__max_features=0.6985917648092275,
regressor__max_samples=0.8340425682853231; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00034326852679397653,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004679126153775273,
regressor__max_features=0.8604065481552533,
regressor__max_samples=0.9026574836789525; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01468252657463809,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.030949924483757867,
regressor__max_features=0.6985917648092275,
regressor__max_samples=0.8340425682853231; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01468252657463809,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.030949924483757867,
regressor__max_features=0.6985917648092275,
regressor__max_samples=0.8340425682853231; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0016564446358289192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04886607481436338,
regressor__max_features=0.506500961755368,
regressor__max_samples=0.542754265427234; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0016564446358289192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04886607481436338,
regressor__max_features=0.506500961755368,
regressor__max_samples=0.542754265427234; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0016564446358289192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04886607481436338,
regressor__max_features=0.506500961755368,
regressor__max_samples=0.542754265427234; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0016564446358289192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04886607481436338,
regressor__max_features=0.506500961755368,
regressor__max_samples=0.542754265427234; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0016564446358289192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04886607481436338,
regressor__max_features=0.506500961755368,
regressor__max_samples=0.542754265427234; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03502549537949661,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04259241129816898,
regressor__max_features=0.8214627818071266,
regressor__max_samples=0.8519730675692547; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03502549537949661,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04259241129816898,
regressor__max_features=0.8214627818071266,
regressor__max_samples=0.8519730675692547; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.031042988191650682,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016913557203040608,
regressor__max_features=0.7760235904259901,
regressor__max_samples=0.7180632645676583; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03502549537949661,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04259241129816898,
regressor__max_features=0.8214627818071266,
regressor__max_samples=0.8519730675692547; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.031042988191650682,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016913557203040608,
regressor__max_features=0.7760235904259901,
regressor__max_samples=0.7180632645676583; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03502549537949661,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04259241129816898,
regressor__max_features=0.8214627818071266,
regressor__max_samples=0.8519730675692547; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.031042988191650682,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016913557203040608,
regressor__max_features=0.7760235904259901,
regressor__max_samples=0.7180632645676583; total time= 1.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03502549537949661,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04259241129816898,
regressor__max_features=0.8214627818071266,
regressor__max_samples=0.8519730675692547; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.031042988191650682,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016913557203040608,
regressor__max_features=0.7760235904259901,
regressor__max_samples=0.7180632645676583; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02833548726001961,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006001644960741875,
regressor__max_features=0.6898462040836335,
regressor__max_samples=0.7951474712574038; total time= 4.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.031042988191650682,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016913557203040608,
regressor__max_features=0.7760235904259901,
regressor__max_samples=0.7180632645676583; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09038616013990274,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02509376160713166,
regressor__max_features=0.7126067522346167,
regressor__max_samples=0.5112346541600588; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09038616013990274,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02509376160713166,
regressor__max_features=0.7126067522346167,
regressor__max_samples=0.5112346541600588; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09038616013990274,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02509376160713166,
regressor__max_features=0.7126067522346167,
regressor__max_samples=0.5112346541600588; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00045554530523903063,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01953517843677422,
regressor__max_features=0.5700565878832262,
regressor__max_samples=0.9342339879489564; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00045554530523903063,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01953517843677422,
regressor__max_features=0.5700565878832262,
regressor__max_samples=0.9342339879489564; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00045554530523903063,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01953517843677422,
regressor__max_features=0.5700565878832262,
regressor__max_samples=0.9342339879489564; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00045554530523903063,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01953517843677422,
regressor__max_features=0.5700565878832262,
regressor__max_samples=0.9342339879489564; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09038616013990274,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02509376160713166,
regressor__max_features=0.7126067522346167,
regressor__max_samples=0.5112346541600588; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00045554530523903063,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01953517843677422,
regressor__max_features=0.5700565878832262,
regressor__max_samples=0.9342339879489564; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02833548726001961,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006001644960741875,
regressor__max_features=0.6898462040836335,
regressor__max_samples=0.7951474712574038; total time= 6.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02833548726001961,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006001644960741875,
regressor__max_features=0.6898462040836335,
regressor__max_samples=0.7951474712574038; total time= 6.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016790391887609266,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007951236285054583,
regressor__max_features=0.6289438427166012,
regressor__max_samples=0.5696780370364121; total time= 1.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09038616013990274,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02509376160713166,
regressor__max_features=0.7126067522346167,
regressor__max_samples=0.5112346541600588; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016790391887609266,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007951236285054583,
regressor__max_features=0.6289438427166012,
regressor__max_samples=0.5696780370364121; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016790391887609266,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007951236285054583,
regressor__max_features=0.6289438427166012,
regressor__max_samples=0.5696780370364121; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02833548726001961,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006001644960741875,
regressor__max_features=0.6898462040836335,
regressor__max_samples=0.7951474712574038; total time= 6.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016790391887609266,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007951236285054583,
regressor__max_features=0.6289438427166012,
regressor__max_samples=0.5696780370364121; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016790391887609266,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007951236285054583,
regressor__max_features=0.6289438427166012,
regressor__max_samples=0.5696780370364121; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018412459461044919,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0037763352883158235,
regressor__max_features=0.5858396429241517,
regressor__max_samples=0.6361536632596845; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02833548726001961,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006001644960741875,
regressor__max_features=0.6898462040836335,
regressor__max_samples=0.7951474712574038; total time= 6.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018412459461044919,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0037763352883158235,
regressor__max_features=0.5858396429241517,
regressor__max_samples=0.6361536632596845; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029359700955069516,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02787168886076715,
regressor__max_features=0.7490453767555029,
regressor__max_samples=0.533587699957855; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018412459461044919,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0037763352883158235,
regressor__max_features=0.5858396429241517,
regressor__max_samples=0.6361536632596845; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018412459461044919,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0037763352883158235,
regressor__max_features=0.5858396429241517,
regressor__max_samples=0.6361536632596845; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018412459461044919,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0037763352883158235,
regressor__max_features=0.5858396429241517,
regressor__max_samples=0.6361536632596845; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029359700955069516,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02787168886076715,
regressor__max_features=0.7490453767555029,
regressor__max_samples=0.533587699957855; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029359700955069516,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02787168886076715,
regressor__max_features=0.7490453767555029,
regressor__max_samples=0.533587699957855; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029359700955069516,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02787168886076715,
regressor__max_features=0.7490453767555029,
regressor__max_samples=0.533587699957855; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009000386613598924,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014474639124711983,
regressor__max_features=0.5532893650643371,
regressor__max_samples=0.5700092693302246; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029359700955069516,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02787168886076715,
regressor__max_features=0.7490453767555029,
regressor__max_samples=0.533587699957855; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009000386613598924,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014474639124711983,
regressor__max_features=0.5532893650643371,
regressor__max_samples=0.5700092693302246; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009000386613598924,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014474639124711983,
regressor__max_features=0.5532893650643371,
regressor__max_samples=0.5700092693302246; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009000386613598924,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014474639124711983,
regressor__max_features=0.5532893650643371,
regressor__max_samples=0.5700092693302246; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009000386613598924,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014474639124711983,
regressor__max_features=0.5532893650643371,
regressor__max_samples=0.5700092693302246; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0501251777329287,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01572366329751463,
regressor__max_features=0.6253653302807989,
regressor__max_samples=0.8165490316787238; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0501251777329287,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01572366329751463,
regressor__max_features=0.6253653302807989,
regressor__max_samples=0.8165490316787238; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0501251777329287,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01572366329751463,
regressor__max_features=0.6253653302807989,
regressor__max_samples=0.8165490316787238; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0501251777329287,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01572366329751463,
regressor__max_features=0.6253653302807989,
regressor__max_samples=0.8165490316787238; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001383241150170212,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.016371966573507313,
regressor__max_features=0.5413993339625632,
regressor__max_samples=0.801576054733194; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0501251777329287,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01572366329751463,
regressor__max_features=0.6253653302807989,
regressor__max_samples=0.8165490316787238; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001383241150170212,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.016371966573507313,
regressor__max_features=0.5413993339625632,
regressor__max_samples=0.801576054733194; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001383241150170212,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.016371966573507313,
regressor__max_features=0.5413993339625632,
regressor__max_samples=0.801576054733194; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00033582562823916007,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03268292290770511,
regressor__max_features=0.7021907507745122,
regressor__max_samples=0.8854556885321352; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00033582562823916007,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03268292290770511,
regressor__max_features=0.7021907507745122,
regressor__max_samples=0.8854556885321352; total time= 2.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001383241150170212,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.016371966573507313,
regressor__max_features=0.5413993339625632,
regressor__max_samples=0.801576054733194; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00033582562823916007,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03268292290770511,
regressor__max_features=0.7021907507745122,
regressor__max_samples=0.8854556885321352; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00033582562823916007,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03268292290770511,
regressor__max_features=0.7021907507745122,
regressor__max_samples=0.8854556885321352; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001383241150170212,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.016371966573507313,
regressor__max_features=0.5413993339625632,
regressor__max_samples=0.801576054733194; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00033582562823916007,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03268292290770511,
regressor__max_features=0.7021907507745122,
regressor__max_samples=0.8854556885321352; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016023194927662347,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007346512304317998,
regressor__max_features=0.6778363582324746,
regressor__max_samples=0.8595229525921227; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0067797550151820945,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026801012826931717,
regressor__max_features=0.8318355826813241,
regressor__max_samples=0.968414869662379; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0067797550151820945,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026801012826931717,
regressor__max_features=0.8318355826813241,
regressor__max_samples=0.968414869662379; total time= 3.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0067797550151820945,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026801012826931717,
regressor__max_features=0.8318355826813241,
regressor__max_samples=0.968414869662379; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016023194927662347,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007346512304317998,
regressor__max_features=0.6778363582324746,
regressor__max_samples=0.8595229525921227; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016023194927662347,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007346512304317998,
regressor__max_features=0.6778363582324746,
regressor__max_samples=0.8595229525921227; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016023194927662347,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007346512304317998,
regressor__max_features=0.6778363582324746,
regressor__max_samples=0.8595229525921227; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0067797550151820945,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026801012826931717,
regressor__max_features=0.8318355826813241,
regressor__max_samples=0.968414869662379; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016023194927662347,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007346512304317998,
regressor__max_features=0.6778363582324746,
regressor__max_samples=0.8595229525921227; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0067797550151820945,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026801012826931717,
regressor__max_features=0.8318355826813241,
regressor__max_samples=0.968414869662379; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.034611749186297204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00617345869515527,
regressor__max_features=0.6671219454084805,
regressor__max_samples=0.885456101872911; total time= 2.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.034611749186297204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00617345869515527,
regressor__max_features=0.6671219454084805,
regressor__max_samples=0.885456101872911; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0010793224142510984,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017313331788414118,
regressor__max_features=0.8524627748444765,
regressor__max_samples=0.7991430291340773; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.034611749186297204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00617345869515527,
regressor__max_features=0.6671219454084805,
regressor__max_samples=0.885456101872911; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.034611749186297204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00617345869515527,
regressor__max_features=0.6671219454084805,
regressor__max_samples=0.885456101872911; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0010793224142510984,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017313331788414118,
regressor__max_features=0.8524627748444765,
regressor__max_samples=0.7991430291340773; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.034611749186297204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00617345869515527,
regressor__max_features=0.6671219454084805,
regressor__max_samples=0.885456101872911; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0010793224142510984,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017313331788414118,
regressor__max_features=0.8524627748444765,
regressor__max_samples=0.7991430291340773; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0010793224142510984,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017313331788414118,
regressor__max_features=0.8524627748444765,
regressor__max_samples=0.7991430291340773; total time= 3.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0010793224142510984,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017313331788414118,
regressor__max_features=0.8524627748444765,
regressor__max_samples=0.7991430291340773; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.005897826610165949,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000878596907769424,
regressor__max_features=0.9037151259083807,
regressor__max_samples=0.9810493928820588; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.005897826610165949,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000878596907769424,
regressor__max_features=0.9037151259083807,
regressor__max_samples=0.9810493928820588; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.005897826610165949,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000878596907769424,
regressor__max_features=0.9037151259083807,
regressor__max_samples=0.9810493928820588; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.005897826610165949,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000878596907769424,
regressor__max_features=0.9037151259083807,
regressor__max_samples=0.9810493928820588; total time= 4.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.005897826610165949,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000878596907769424,
regressor__max_features=0.9037151259083807,
regressor__max_samples=0.9810493928820588; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07091074717067322,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003663548980044111,
regressor__max_features=0.5433526059310918,
regressor__max_samples=0.9790853965019652; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07091074717067322,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003663548980044111,
regressor__max_features=0.5433526059310918,
regressor__max_samples=0.9790853965019652; total time= 3.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07091074717067322,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003663548980044111,
regressor__max_features=0.5433526059310918,
regressor__max_samples=0.9790853965019652; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07091074717067322,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003663548980044111,
regressor__max_features=0.5433526059310918,
regressor__max_samples=0.9790853965019652; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0028266030666403864,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012403647205371554,
regressor__max_features=0.6311320221499126,
regressor__max_samples=0.7975389653501275; total time= 12.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015728041963915255,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.018928122440471713,
regressor__max_features=0.8620516646365692,
regressor__max_samples=0.8186157898947435; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015728041963915255,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.018928122440471713,
regressor__max_features=0.8620516646365692,
regressor__max_samples=0.8186157898947435; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005610531778339117,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00870953076056879,
regressor__max_features=0.8465876057284624,
regressor__max_samples=0.5257979033293549; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005610531778339117,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00870953076056879,
regressor__max_features=0.8465876057284624,
regressor__max_samples=0.5257979033293549; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015728041963915255,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.018928122440471713,
regressor__max_features=0.8620516646365692,
regressor__max_samples=0.8186157898947435; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07091074717067322,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003663548980044111,
regressor__max_features=0.5433526059310918,
regressor__max_samples=0.9790853965019652; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015728041963915255,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.018928122440471713,
regressor__max_features=0.8620516646365692,
regressor__max_samples=0.8186157898947435; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005610531778339117,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00870953076056879,
regressor__max_features=0.8465876057284624,
regressor__max_samples=0.5257979033293549; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015728041963915255,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.018928122440471713,
regressor__max_features=0.8620516646365692,
regressor__max_samples=0.8186157898947435; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005610531778339117,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00870953076056879,
regressor__max_features=0.8465876057284624,
regressor__max_samples=0.5257979033293549; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005610531778339117,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00870953076056879,
regressor__max_features=0.8465876057284624,
regressor__max_samples=0.5257979033293549; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0028266030666403864,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012403647205371554,
regressor__max_features=0.6311320221499126,
regressor__max_samples=0.7975389653501275; total time= 15.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0028266030666403864,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012403647205371554,
regressor__max_features=0.6311320221499126,
regressor__max_samples=0.7975389653501275; total time= 16.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0028266030666403864,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012403647205371554,
regressor__max_features=0.6311320221499126,
regressor__max_samples=0.7975389653501275; total time= 16.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013130474908130515,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005818235165844919,
regressor__max_features=0.9996767497829174,
regressor__max_samples=0.7779032907185899; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013130474908130515,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005818235165844919,
regressor__max_features=0.9996767497829174,
regressor__max_samples=0.7779032907185899; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013130474908130515,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005818235165844919,
regressor__max_features=0.9996767497829174,
regressor__max_samples=0.7779032907185899; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013130474908130515,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005818235165844919,
regressor__max_features=0.9996767497829174,
regressor__max_samples=0.7779032907185899; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0028266030666403864,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012403647205371554,
regressor__max_features=0.6311320221499126,
regressor__max_samples=0.7975389653501275; total time= 16.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005485372181571514,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017111813262848922,
regressor__max_features=0.5011696953128891,
regressor__max_samples=0.9838496941035433; total time= 11.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013130474908130515,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005818235165844919,
regressor__max_features=0.9996767497829174,
regressor__max_samples=0.7779032907185899; total time= 2.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005485372181571514,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017111813262848922,
regressor__max_features=0.5011696953128891,
regressor__max_samples=0.9838496941035433; total time= 14.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.08187941582538087,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.024323469000845397,
regressor__max_features=0.8725322740601102,
regressor__max_samples=0.5274413073232375; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.08187941582538087,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.024323469000845397,
regressor__max_features=0.8725322740601102,
regressor__max_samples=0.5274413073232375; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005485372181571514,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017111813262848922,
regressor__max_features=0.5011696953128891,
regressor__max_samples=0.9838496941035433; total time= 13.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.08187941582538087,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.024323469000845397,
regressor__max_features=0.8725322740601102,
regressor__max_samples=0.5274413073232375; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.08187941582538087,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.024323469000845397,
regressor__max_features=0.8725322740601102,
regressor__max_samples=0.5274413073232375; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.08187941582538087,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.024323469000845397,
regressor__max_features=0.8725322740601102,
regressor__max_samples=0.5274413073232375; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0009063614378928192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008583016588214048,
regressor__max_features=0.8250836752216141,
regressor__max_samples=0.682348624728653; total time= 2.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0009063614378928192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008583016588214048,
regressor__max_features=0.8250836752216141,
regressor__max_samples=0.682348624728653; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05577825106318865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002299152277164545,
regressor__max_features=0.8655118357291408,
regressor__max_samples=0.5962995138928152; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05577825106318865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002299152277164545,
regressor__max_features=0.8655118357291408,
regressor__max_samples=0.5962995138928152; total time= 4.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005485372181571514,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017111813262848922,
regressor__max_features=0.5011696953128891,
regressor__max_samples=0.9838496941035433; total time= 14.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0009063614378928192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008583016588214048,
regressor__max_features=0.8250836752216141,
regressor__max_samples=0.682348624728653; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005485372181571514,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017111813262848922,
regressor__max_features=0.5011696953128891,
regressor__max_samples=0.9838496941035433; total time= 15.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0009063614378928192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008583016588214048,
regressor__max_features=0.8250836752216141,
regressor__max_samples=0.682348624728653; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022998560676594382,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04518508146032548,
regressor__max_features=0.7651325663174501,
regressor__max_samples=0.7205160808740808; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022998560676594382,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04518508146032548,
regressor__max_features=0.7651325663174501,
regressor__max_samples=0.7205160808740808; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0009063614378928192,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008583016588214048,
regressor__max_features=0.8250836752216141,
regressor__max_samples=0.682348624728653; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022998560676594382,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04518508146032548,
regressor__max_features=0.7651325663174501,
regressor__max_samples=0.7205160808740808; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022998560676594382,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04518508146032548,
regressor__max_features=0.7651325663174501,
regressor__max_samples=0.7205160808740808; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05577825106318865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002299152277164545,
regressor__max_features=0.8655118357291408,
regressor__max_samples=0.5962995138928152; total time= 5.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05577825106318865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002299152277164545,
regressor__max_features=0.8655118357291408,
regressor__max_samples=0.5962995138928152; total time= 5.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007545258605068627,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.007040274045310659,
regressor__max_features=0.6164797373768169,
regressor__max_samples=0.5122003907782691; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022998560676594382,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04518508146032548,
regressor__max_features=0.7651325663174501,
regressor__max_samples=0.7205160808740808; total time= 2.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007545258605068627,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.007040274045310659,
regressor__max_features=0.6164797373768169,
regressor__max_samples=0.5122003907782691; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007545258605068627,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.007040274045310659,
regressor__max_features=0.6164797373768169,
regressor__max_samples=0.5122003907782691; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007545258605068627,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.007040274045310659,
regressor__max_features=0.6164797373768169,
regressor__max_samples=0.5122003907782691; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007545258605068627,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.007040274045310659,
regressor__max_features=0.6164797373768169,
regressor__max_samples=0.5122003907782691; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00015706124018367235,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.025818495905497583,
regressor__max_features=0.7692007451903937,
regressor__max_samples=0.8308275875402708; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00015706124018367235,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.025818495905497583,
regressor__max_features=0.7692007451903937,
regressor__max_samples=0.8308275875402708; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00015706124018367235,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.025818495905497583,
regressor__max_features=0.7692007451903937,
regressor__max_samples=0.8308275875402708; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00015706124018367235,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.025818495905497583,
regressor__max_features=0.7692007451903937,
regressor__max_samples=0.8308275875402708; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00015706124018367235,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.025818495905497583,
regressor__max_features=0.7692007451903937,
regressor__max_samples=0.8308275875402708; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09354572388201218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.042082837785532345,
regressor__max_features=0.764468567013606,
regressor__max_samples=0.969533849256448; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09354572388201218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.042082837785532345,
regressor__max_features=0.764468567013606,
regressor__max_samples=0.969533849256448; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09354572388201218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.042082837785532345,
regressor__max_features=0.764468567013606,
regressor__max_samples=0.969533849256448; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05577825106318865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002299152277164545,
regressor__max_features=0.8655118357291408,
regressor__max_samples=0.5962995138928152; total time= 8.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09354572388201218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.042082837785532345,
regressor__max_features=0.764468567013606,
regressor__max_samples=0.969533849256448; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09354572388201218,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.042082837785532345,
regressor__max_features=0.764468567013606,
regressor__max_samples=0.969533849256448; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0027500714697919594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015662538513939,
regressor__max_features=0.5458496675557151,
regressor__max_samples=0.668185231914211; total time= 2.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016221429269930602,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011275490576893408,
regressor__max_features=0.8835941444655635,
regressor__max_samples=0.7009654568046211; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0027500714697919594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015662538513939,
regressor__max_features=0.5458496675557151,
regressor__max_samples=0.668185231914211; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016221429269930602,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011275490576893408,
regressor__max_features=0.8835941444655635,
regressor__max_samples=0.7009654568046211; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016221429269930602,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011275490576893408,
regressor__max_features=0.8835941444655635,
regressor__max_samples=0.7009654568046211; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0027500714697919594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015662538513939,
regressor__max_features=0.5458496675557151,
regressor__max_samples=0.668185231914211; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0027500714697919594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015662538513939,
regressor__max_features=0.5458496675557151,
regressor__max_samples=0.668185231914211; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016221429269930602,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011275490576893408,
regressor__max_features=0.8835941444655635,
regressor__max_samples=0.7009654568046211; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016221429269930602,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011275490576893408,
regressor__max_features=0.8835941444655635,
regressor__max_samples=0.7009654568046211; total time= 3.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0027500714697919594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015662538513939,
regressor__max_features=0.5458496675557151,
regressor__max_samples=0.668185231914211; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000774326925883869,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005852286829541617,
regressor__max_features=0.6386328228483868,
regressor__max_samples=0.9551695510287672; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000774326925883869,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005852286829541617,
regressor__max_features=0.6386328228483868,
regressor__max_samples=0.9551695510287672; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000774326925883869,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005852286829541617,
regressor__max_features=0.6386328228483868,
regressor__max_samples=0.9551695510287672; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000774326925883869,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005852286829541617,
regressor__max_features=0.6386328228483868,
regressor__max_samples=0.9551695510287672; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000774326925883869,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005852286829541617,
regressor__max_features=0.6386328228483868,
regressor__max_samples=0.9551695510287672; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0019200803416557317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014958581831993768,
regressor__max_features=0.5622187786484282,
regressor__max_samples=0.9782078487797126; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0019200803416557317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014958581831993768,
regressor__max_features=0.5622187786484282,
regressor__max_samples=0.9782078487797126; total time= 2.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0019200803416557317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014958581831993768,
regressor__max_features=0.5622187786484282,
regressor__max_samples=0.9782078487797126; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0025246695017047643,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09280144303883059,
regressor__max_features=0.8716764114468121,
regressor__max_samples=0.532493204741447; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0019200803416557317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014958581831993768,
regressor__max_features=0.5622187786484282,
regressor__max_samples=0.9782078487797126; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.022857832359148136,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005829921230240514,
regressor__max_features=0.9911652575400408,
regressor__max_samples=0.9420994890657515; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0025246695017047643,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09280144303883059,
regressor__max_features=0.8716764114468121,
regressor__max_samples=0.532493204741447; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0019200803416557317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014958581831993768,
regressor__max_features=0.5622187786484282,
regressor__max_samples=0.9782078487797126; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0025246695017047643,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09280144303883059,
regressor__max_features=0.8716764114468121,
regressor__max_samples=0.532493204741447; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0025246695017047643,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09280144303883059,
regressor__max_features=0.8716764114468121,
regressor__max_samples=0.532493204741447; total time= 1.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0025246695017047643,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09280144303883059,
regressor__max_features=0.8716764114468121,
regressor__max_samples=0.532493204741447; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.022857832359148136,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005829921230240514,
regressor__max_features=0.9911652575400408,
regressor__max_samples=0.9420994890657515; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.022857832359148136,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005829921230240514,
regressor__max_features=0.9911652575400408,
regressor__max_samples=0.9420994890657515; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.022857832359148136,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005829921230240514,
regressor__max_features=0.9911652575400408,
regressor__max_samples=0.9420994890657515; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.022857832359148136,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005829921230240514,
regressor__max_features=0.9911652575400408,
regressor__max_samples=0.9420994890657515; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016321005260615848,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010233850919121932,
regressor__max_features=0.7336553832688263,
regressor__max_samples=0.6485014382274411; total time= 8.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06732793932247673,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004917083545766455,
regressor__max_features=0.9153212588753123,
regressor__max_samples=0.5601693810174979; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06732793932247673,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004917083545766455,
regressor__max_features=0.9153212588753123,
regressor__max_samples=0.5601693810174979; total time= 2.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06732793932247673,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004917083545766455,
regressor__max_features=0.9153212588753123,
regressor__max_samples=0.5601693810174979; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06732793932247673,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004917083545766455,
regressor__max_features=0.9153212588753123,
regressor__max_samples=0.5601693810174979; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06732793932247673,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004917083545766455,
regressor__max_features=0.9153212588753123,
regressor__max_samples=0.5601693810174979; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016321005260615848,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010233850919121932,
regressor__max_features=0.7336553832688263,
regressor__max_samples=0.6485014382274411; total time= 9.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016321005260615848,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010233850919121932,
regressor__max_features=0.7336553832688263,
regressor__max_samples=0.6485014382274411; total time= 9.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016321005260615848,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010233850919121932,
regressor__max_features=0.7336553832688263,
regressor__max_samples=0.6485014382274411; total time= 10.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016321005260615848,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010233850919121932,
regressor__max_features=0.7336553832688263,
regressor__max_samples=0.6485014382274411; total time= 11.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003324189303176931,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003513530864565823,
regressor__max_features=0.7455931704343133,
regressor__max_samples=0.9981678879033622; total time= 3.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003324189303176931,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003513530864565823,
regressor__max_features=0.7455931704343133,
regressor__max_samples=0.9981678879033622; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028022015887899803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007392854328849203,
regressor__max_features=0.8739358020461143,
regressor__max_samples=0.5174325801169124; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003324189303176931,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003513530864565823,
regressor__max_features=0.7455931704343133,
regressor__max_samples=0.9981678879033622; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003324189303176931,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003513530864565823,
regressor__max_features=0.7455931704343133,
regressor__max_samples=0.9981678879033622; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00014911729937656495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00044562758771665587,
regressor__max_features=0.759130183338196,
regressor__max_samples=0.7986370362151705; total time= 5.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003324189303176931,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003513530864565823,
regressor__max_features=0.7455931704343133,
regressor__max_samples=0.9981678879033622; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028022015887899803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007392854328849203,
regressor__max_features=0.8739358020461143,
regressor__max_samples=0.5174325801169124; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00014911729937656495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00044562758771665587,
regressor__max_features=0.759130183338196,
regressor__max_samples=0.7986370362151705; total time= 6.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00014911729937656495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00044562758771665587,
regressor__max_features=0.759130183338196,
regressor__max_samples=0.7986370362151705; total time= 5.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028022015887899803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007392854328849203,
regressor__max_features=0.8739358020461143,
regressor__max_samples=0.5174325801169124; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028022015887899803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007392854328849203,
regressor__max_features=0.8739358020461143,
regressor__max_samples=0.5174325801169124; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00549868224361865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002356656542301788,
regressor__max_features=0.6955848830401898,
regressor__max_samples=0.6364796874281575; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028022015887899803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007392854328849203,
regressor__max_features=0.8739358020461143,
regressor__max_samples=0.5174325801169124; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00549868224361865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002356656542301788,
regressor__max_features=0.6955848830401898,
regressor__max_samples=0.6364796874281575; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00549868224361865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002356656542301788,
regressor__max_features=0.6955848830401898,
regressor__max_samples=0.6364796874281575; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00014911729937656495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00044562758771665587,
regressor__max_features=0.759130183338196,
regressor__max_samples=0.7986370362151705; total time= 6.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00549868224361865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002356656542301788,
regressor__max_features=0.6955848830401898,
regressor__max_samples=0.6364796874281575; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00549868224361865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002356656542301788,
regressor__max_features=0.6955848830401898,
regressor__max_samples=0.6364796874281575; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.046370762746530715,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03602305988309308,
regressor__max_features=0.5177111842092725,
regressor__max_samples=0.6645256826849355; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.046370762746530715,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03602305988309308,
regressor__max_features=0.5177111842092725,
regressor__max_samples=0.6645256826849355; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.046370762746530715,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03602305988309308,
regressor__max_features=0.5177111842092725,
regressor__max_samples=0.6645256826849355; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00014911729937656495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00044562758771665587,
regressor__max_features=0.759130183338196,
regressor__max_samples=0.7986370362151705; total time= 8.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.046370762746530715,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03602305988309308,
regressor__max_features=0.5177111842092725,
regressor__max_samples=0.6645256826849355; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01479995316532707,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003961024769620464,
regressor__max_features=0.9810609862960249,
regressor__max_samples=0.554734430027227; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.046370762746530715,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03602305988309308,
regressor__max_features=0.5177111842092725,
regressor__max_samples=0.6645256826849355; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004493369280060231,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006985104082013648,
regressor__max_features=0.6263245469804426,
regressor__max_samples=0.5025410699090129; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01479995316532707,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003961024769620464,
regressor__max_features=0.9810609862960249,
regressor__max_samples=0.554734430027227; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004493369280060231,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006985104082013648,
regressor__max_features=0.6263245469804426,
regressor__max_samples=0.5025410699090129; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01479995316532707,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003961024769620464,
regressor__max_features=0.9810609862960249,
regressor__max_samples=0.554734430027227; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004493369280060231,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006985104082013648,
regressor__max_features=0.6263245469804426,
regressor__max_samples=0.5025410699090129; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01479995316532707,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003961024769620464,
regressor__max_features=0.9810609862960249,
regressor__max_samples=0.554734430027227; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004493369280060231,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006985104082013648,
regressor__max_features=0.6263245469804426,
regressor__max_samples=0.5025410699090129; total time= 3.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004493369280060231,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006985104082013648,
regressor__max_features=0.6263245469804426,
regressor__max_samples=0.5025410699090129; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.01479995316532707,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003961024769620464,
regressor__max_features=0.9810609862960249,
regressor__max_samples=0.554734430027227; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000338519815899524,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012888305214463,
regressor__max_features=0.9786351314916628,
regressor__max_samples=0.9131028478969829; total time= 13.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02978810649238247,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0042789451944459395,
regressor__max_features=0.9322464401536743,
regressor__max_samples=0.8669549594768509; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02978810649238247,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0042789451944459395,
regressor__max_features=0.9322464401536743,
regressor__max_samples=0.8669549594768509; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02978810649238247,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0042789451944459395,
regressor__max_features=0.9322464401536743,
regressor__max_samples=0.8669549594768509; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02978810649238247,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0042789451944459395,
regressor__max_features=0.9322464401536743,
regressor__max_samples=0.8669549594768509; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02978810649238247,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0042789451944459395,
regressor__max_features=0.9322464401536743,
regressor__max_samples=0.8669549594768509; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.024676187686654617,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00029649814397776793,
regressor__max_features=0.580200052665036,
regressor__max_samples=0.6670044393726189; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016919696886896933,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00015295056257293572,
regressor__max_features=0.9414531345687307,
regressor__max_samples=0.7580395102218083; total time= 7.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016919696886896933,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00015295056257293572,
regressor__max_features=0.9414531345687307,
regressor__max_samples=0.7580395102218083; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.024676187686654617,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00029649814397776793,
regressor__max_features=0.580200052665036,
regressor__max_samples=0.6670044393726189; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000338519815899524,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012888305214463,
regressor__max_features=0.9786351314916628,
regressor__max_samples=0.9131028478969829; total time= 18.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.024676187686654617,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00029649814397776793,
regressor__max_features=0.580200052665036,
regressor__max_samples=0.6670044393726189; total time= 5.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016919696886896933,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00015295056257293572,
regressor__max_features=0.9414531345687307,
regressor__max_samples=0.7580395102218083; total time= 8.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000338519815899524,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012888305214463,
regressor__max_features=0.9786351314916628,
regressor__max_samples=0.9131028478969829; total time= 20.5s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016919696886896933,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00015295056257293572,
regressor__max_features=0.9414531345687307,
regressor__max_samples=0.7580395102218083; total time= 7.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.024676187686654617,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00029649814397776793,
regressor__max_features=0.580200052665036,
regressor__max_samples=0.6670044393726189; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.024676187686654617,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00029649814397776793,
regressor__max_features=0.580200052665036,
regressor__max_samples=0.6670044393726189; total time= 5.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04391493836948081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024912082090084737,
regressor__max_features=0.9395357401140658,
regressor__max_samples=0.5025725853773574; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000338519815899524,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012888305214463,
regressor__max_features=0.9786351314916628,
regressor__max_samples=0.9131028478969829; total time= 21.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00016919696886896933,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00015295056257293572,
regressor__max_features=0.9414531345687307,
regressor__max_samples=0.7580395102218083; total time= 9.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04391493836948081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024912082090084737,
regressor__max_features=0.9395357401140658,
regressor__max_samples=0.5025725853773574; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04391493836948081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024912082090084737,
regressor__max_features=0.9395357401140658,
regressor__max_samples=0.5025725853773574; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.000338519815899524,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012888305214463,
regressor__max_features=0.9786351314916628,
regressor__max_samples=0.9131028478969829; total time= 21.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04391493836948081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024912082090084737,
regressor__max_features=0.9395357401140658,
regressor__max_samples=0.5025725853773574; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010124103183444479,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05322356094052614,
regressor__max_features=0.5005600557400547,
regressor__max_samples=0.7780908675198785; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010124103183444479,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05322356094052614,
regressor__max_features=0.5005600557400547,
regressor__max_samples=0.7780908675198785; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04391493836948081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024912082090084737,
regressor__max_features=0.9395357401140658,
regressor__max_samples=0.5025725853773574; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012371272838122673,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000207731463833222,
regressor__max_features=0.8494953690197938,
regressor__max_samples=0.5946107698925029; total time= 6.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010124103183444479,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05322356094052614,
regressor__max_features=0.5005600557400547,
regressor__max_samples=0.7780908675198785; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010124103183444479,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05322356094052614,
regressor__max_features=0.5005600557400547,
regressor__max_samples=0.7780908675198785; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010124103183444479,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05322356094052614,
regressor__max_features=0.5005600557400547,
regressor__max_samples=0.7780908675198785; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012371272838122673,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000207731463833222,
regressor__max_features=0.8494953690197938,
regressor__max_samples=0.5946107698925029; total time= 8.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012371272838122673,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000207731463833222,
regressor__max_features=0.8494953690197938,
regressor__max_samples=0.5946107698925029; total time= 6.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003329685395839853,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00037966327857701245,
regressor__max_features=0.5420491122339923,
regressor__max_samples=0.6632167882516464; total time= 4.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003329685395839853,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00037966327857701245,
regressor__max_features=0.5420491122339923,
regressor__max_samples=0.6632167882516464; total time= 4.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003329685395839853,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00037966327857701245,
regressor__max_features=0.5420491122339923,
regressor__max_samples=0.6632167882516464; total time= 5.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012371272838122673,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000207731463833222,
regressor__max_features=0.8494953690197938,
regressor__max_samples=0.5946107698925029; total time= 7.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003329685395839853,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00037966327857701245,
regressor__max_features=0.5420491122339923,
regressor__max_samples=0.6632167882516464; total time= 4.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007860326410149302,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013492303008532825,
regressor__max_features=0.5045362023263624,
regressor__max_samples=0.9093397102499209; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007860326410149302,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013492303008532825,
regressor__max_features=0.5045362023263624,
regressor__max_samples=0.9093397102499209; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007860326410149302,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013492303008532825,
regressor__max_features=0.5045362023263624,
regressor__max_samples=0.9093397102499209; total time= 4.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007860326410149302,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013492303008532825,
regressor__max_features=0.5045362023263624,
regressor__max_samples=0.9093397102499209; total time= 4.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007904341210841664,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01824301025028901,
regressor__max_features=0.8944034909353722,
regressor__max_samples=0.5868767376537833; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007860326410149302,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013492303008532825,
regressor__max_features=0.5045362023263624,
regressor__max_samples=0.9093397102499209; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003329685395839853,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00037966327857701245,
regressor__max_features=0.5420491122339923,
regressor__max_samples=0.6632167882516464; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007904341210841664,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01824301025028901,
regressor__max_features=0.8944034909353722,
regressor__max_samples=0.5868767376537833; total time= 1.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007904341210841664,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01824301025028901,
regressor__max_features=0.8944034909353722,
regressor__max_samples=0.5868767376537833; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012371272838122673,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000207731463833222,
regressor__max_features=0.8494953690197938,
regressor__max_samples=0.5946107698925029; total time= 9.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007904341210841664,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01824301025028901,
regressor__max_features=0.8944034909353722,
regressor__max_samples=0.5868767376537833; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007904341210841664,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01824301025028901,
regressor__max_features=0.8944034909353722,
regressor__max_samples=0.5868767376537833; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002093832403277015,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011357510741203337,
regressor__max_features=0.7161203919462562,
regressor__max_samples=0.9200607513706094; total time= 8.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002093832403277015,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011357510741203337,
regressor__max_features=0.7161203919462562,
regressor__max_samples=0.9200607513706094; total time= 7.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015977586495966143,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031559692337327467,
regressor__max_features=0.8854921462660678,
regressor__max_samples=0.8542860023013147; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015977586495966143,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031559692337327467,
regressor__max_features=0.8854921462660678,
regressor__max_samples=0.8542860023013147; total time= 4.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015977586495966143,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031559692337327467,
regressor__max_features=0.8854921462660678,
regressor__max_samples=0.8542860023013147; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002093832403277015,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011357510741203337,
regressor__max_features=0.7161203919462562,
regressor__max_samples=0.9200607513706094; total time= 7.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002093832403277015,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011357510741203337,
regressor__max_features=0.7161203919462562,
regressor__max_samples=0.9200607513706094; total time= 8.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08065542163437615,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002456171684576671,
regressor__max_features=0.8791315979645113,
regressor__max_samples=0.5122934582294008; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015977586495966143,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031559692337327467,
regressor__max_features=0.8854921462660678,
regressor__max_samples=0.8542860023013147; total time= 6.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015977586495966143,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031559692337327467,
regressor__max_features=0.8854921462660678,
regressor__max_samples=0.8542860023013147; total time= 6.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002093832403277015,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011357510741203337,
regressor__max_features=0.7161203919462562,
regressor__max_samples=0.9200607513706094; total time= 11.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029143307736381567,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002923678211551315,
regressor__max_features=0.8852037089038967,
regressor__max_samples=0.8416476883032527; total time= 3.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00029978841764769854,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09803327254101647,
regressor__max_features=0.7130906511179864,
regressor__max_samples=0.7256935121648378; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029143307736381567,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002923678211551315,
regressor__max_features=0.8852037089038967,
regressor__max_samples=0.8416476883032527; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010905590117399008,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016752639895789228,
regressor__max_features=0.9262073295993238,
regressor__max_samples=0.9950741863753235; total time= 8.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029143307736381567,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002923678211551315,
regressor__max_features=0.8852037089038967,
regressor__max_samples=0.8416476883032527; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010905590117399008,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016752639895789228,
regressor__max_features=0.9262073295993238,
regressor__max_samples=0.9950741863753235; total time= 8.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00029978841764769854,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09803327254101647,
regressor__max_features=0.7130906511179864,
regressor__max_samples=0.7256935121648378; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010905590117399008,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016752639895789228,
regressor__max_features=0.9262073295993238,
regressor__max_samples=0.9950741863753235; total time= 9.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029143307736381567,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002923678211551315,
regressor__max_features=0.8852037089038967,
regressor__max_samples=0.8416476883032527; total time= 3.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029143307736381567,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002923678211551315,
regressor__max_features=0.8852037089038967,
regressor__max_samples=0.8416476883032527; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08065542163437615,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002456171684576671,
regressor__max_features=0.8791315979645113,
regressor__max_samples=0.5122934582294008; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010905590117399008,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016752639895789228,
regressor__max_features=0.9262073295993238,
regressor__max_samples=0.9950741863753235; total time= 9.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08065542163437615,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002456171684576671,
regressor__max_features=0.8791315979645113,
regressor__max_samples=0.5122934582294008; total time= 5.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08065542163437615,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002456171684576671,
regressor__max_features=0.8791315979645113,
regressor__max_samples=0.5122934582294008; total time= 5.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00029978841764769854,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09803327254101647,
regressor__max_features=0.7130906511179864,
regressor__max_samples=0.7256935121648378; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00029978841764769854,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09803327254101647,
regressor__max_features=0.7130906511179864,
regressor__max_samples=0.7256935121648378; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.08065542163437615,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002456171684576671,
regressor__max_features=0.8791315979645113,
regressor__max_samples=0.5122934582294008; total time= 5.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0038112925735921226,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.012051654628567821,
regressor__max_features=0.6103848063943802,
regressor__max_samples=0.5411905228079213; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0038112925735921226,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.012051654628567821,
regressor__max_features=0.6103848063943802,
regressor__max_samples=0.5411905228079213; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0038112925735921226,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.012051654628567821,
regressor__max_features=0.6103848063943802,
regressor__max_samples=0.5411905228079213; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0038112925735921226,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.012051654628567821,
regressor__max_features=0.6103848063943802,
regressor__max_samples=0.5411905228079213; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0038112925735921226,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.012051654628567821,
regressor__max_features=0.6103848063943802,
regressor__max_samples=0.5411905228079213; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00029978841764769854,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09803327254101647,
regressor__max_features=0.7130906511179864,
regressor__max_samples=0.7256935121648378; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004899604773576175,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006603566884894323,
regressor__max_features=0.9754317811252051,
regressor__max_samples=0.5755289458904501; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004899604773576175,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006603566884894323,
regressor__max_features=0.9754317811252051,
regressor__max_samples=0.5755289458904501; total time= 2.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004899604773576175,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006603566884894323,
regressor__max_features=0.9754317811252051,
regressor__max_samples=0.5755289458904501; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015624646517788865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0018162764453973897,
regressor__max_features=0.8192629738320427,
regressor__max_samples=0.6987971989813971; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004899604773576175,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006603566884894323,
regressor__max_features=0.9754317811252051,
regressor__max_samples=0.5755289458904501; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004899604773576175,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006603566884894323,
regressor__max_features=0.9754317811252051,
regressor__max_samples=0.5755289458904501; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0021227374754566367,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016904487901714292,
regressor__max_features=0.9470496018395672,
regressor__max_samples=0.6149773029455431; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0021227374754566367,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016904487901714292,
regressor__max_features=0.9470496018395672,
regressor__max_samples=0.6149773029455431; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0021227374754566367,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016904487901714292,
regressor__max_features=0.9470496018395672,
regressor__max_samples=0.6149773029455431; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0021227374754566367,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016904487901714292,
regressor__max_features=0.9470496018395672,
regressor__max_samples=0.6149773029455431; total time= 2.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0021227374754566367,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016904487901714292,
regressor__max_features=0.9470496018395672,
regressor__max_samples=0.6149773029455431; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015624646517788865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0018162764453973897,
regressor__max_features=0.8192629738320427,
regressor__max_samples=0.6987971989813971; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015624646517788865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0018162764453973897,
regressor__max_features=0.8192629738320427,
regressor__max_samples=0.6987971989813971; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015624646517788865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0018162764453973897,
regressor__max_features=0.8192629738320427,
regressor__max_samples=0.6987971989813971; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015624646517788865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0018162764453973897,
regressor__max_features=0.8192629738320427,
regressor__max_samples=0.6987971989813971; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010905590117399008,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016752639895789228,
regressor__max_features=0.9262073295993238,
regressor__max_samples=0.9950741863753235; total time= 13.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003990960511143544,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009015721260209724,
regressor__max_features=0.6842631718618806,
regressor__max_samples=0.9321791249198186; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003990960511143544,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009015721260209724,
regressor__max_features=0.6842631718618806,
regressor__max_samples=0.9321791249198186; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003990960511143544,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009015721260209724,
regressor__max_features=0.6842631718618806,
regressor__max_samples=0.9321791249198186; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003990960511143544,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009015721260209724,
regressor__max_features=0.6842631718618806,
regressor__max_samples=0.9321791249198186; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003990960511143544,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009015721260209724,
regressor__max_features=0.6842631718618806,
regressor__max_samples=0.9321791249198186; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004262833964220881,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004084169406694094,
regressor__max_features=0.5455822452030958,
regressor__max_samples=0.5768290190805563; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.034225490771244337,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007562422985437138,
regressor__max_features=0.5656224388411938,
regressor__max_samples=0.516263089745626; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.034225490771244337,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007562422985437138,
regressor__max_features=0.5656224388411938,
regressor__max_samples=0.516263089745626; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.034225490771244337,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007562422985437138,
regressor__max_features=0.5656224388411938,
regressor__max_samples=0.516263089745626; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.034225490771244337,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007562422985437138,
regressor__max_features=0.5656224388411938,
regressor__max_samples=0.516263089745626; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07989761046135668,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024338391474407867,
regressor__max_features=0.500615917583213,
regressor__max_samples=0.7991636325612135; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.034225490771244337,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007562422985437138,
regressor__max_features=0.5656224388411938,
regressor__max_samples=0.516263089745626; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004262833964220881,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004084169406694094,
regressor__max_features=0.5455822452030958,
regressor__max_samples=0.5768290190805563; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07989761046135668,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024338391474407867,
regressor__max_features=0.500615917583213,
regressor__max_samples=0.7991636325612135; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004262833964220881,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004084169406694094,
regressor__max_features=0.5455822452030958,
regressor__max_samples=0.5768290190805563; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004262833964220881,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004084169406694094,
regressor__max_features=0.5455822452030958,
regressor__max_samples=0.5768290190805563; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07989761046135668,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024338391474407867,
regressor__max_features=0.500615917583213,
regressor__max_samples=0.7991636325612135; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004262833964220881,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004084169406694094,
regressor__max_features=0.5455822452030958,
regressor__max_samples=0.5768290190805563; total time= 3.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07989761046135668,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024338391474407867,
regressor__max_features=0.500615917583213,
regressor__max_samples=0.7991636325612135; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07989761046135668,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024338391474407867,
regressor__max_features=0.500615917583213,
regressor__max_samples=0.7991636325612135; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.011394515452184195,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003995528014939306,
regressor__max_features=0.7457972733707184,
regressor__max_samples=0.5321044685375671; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009501154753026081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02469888237742627,
regressor__max_features=0.6551809794617869,
regressor__max_samples=0.7276100745409007; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009501154753026081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02469888237742627,
regressor__max_features=0.6551809794617869,
regressor__max_samples=0.7276100745409007; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.011394515452184195,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003995528014939306,
regressor__max_features=0.7457972733707184,
regressor__max_samples=0.5321044685375671; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.011394515452184195,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003995528014939306,
regressor__max_features=0.7457972733707184,
regressor__max_samples=0.5321044685375671; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009501154753026081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02469888237742627,
regressor__max_features=0.6551809794617869,
regressor__max_samples=0.7276100745409007; total time= 2.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.011394515452184195,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003995528014939306,
regressor__max_features=0.7457972733707184,
regressor__max_samples=0.5321044685375671; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
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regressor__estimator__regressor__learning_rate_init=0.02469888237742627,
regressor__max_features=0.6551809794617869,
regressor__max_samples=0.7276100745409007; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009501154753026081,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02469888237742627,
regressor__max_features=0.6551809794617869,
regressor__max_samples=0.7276100745409007; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009353013747523849,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03805109785166522,
regressor__max_features=0.8899254944466749,
regressor__max_samples=0.5198094126741786; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002476257420079486,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001504806537220564,
regressor__max_features=0.7399694173585618,
regressor__max_samples=0.8000102740596519; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002476257420079486,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001504806537220564,
regressor__max_features=0.7399694173585618,
regressor__max_samples=0.8000102740596519; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002476257420079486,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001504806537220564,
regressor__max_features=0.7399694173585618,
regressor__max_samples=0.8000102740596519; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009353013747523849,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03805109785166522,
regressor__max_features=0.8899254944466749,
regressor__max_samples=0.5198094126741786; total time= 1.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002476257420079486,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001504806537220564,
regressor__max_features=0.7399694173585618,
regressor__max_samples=0.8000102740596519; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009353013747523849,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03805109785166522,
regressor__max_features=0.8899254944466749,
regressor__max_samples=0.5198094126741786; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.011394515452184195,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003995528014939306,
regressor__max_features=0.7457972733707184,
regressor__max_samples=0.5321044685375671; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009353013747523849,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03805109785166522,
regressor__max_features=0.8899254944466749,
regressor__max_samples=0.5198094126741786; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
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regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001504806537220564,
regressor__max_features=0.7399694173585618,
regressor__max_samples=0.8000102740596519; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009353013747523849,
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regressor__estimator__regressor__learning_rate_init=0.03805109785166522,
regressor__max_features=0.8899254944466749,
regressor__max_samples=0.5198094126741786; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0027994269849891196,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004774699570297257,
regressor__max_features=0.5048854237095919,
regressor__max_samples=0.6632306541219959; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0027994269849891196,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004774699570297257,
regressor__max_features=0.5048854237095919,
regressor__max_samples=0.6632306541219959; total time= 1.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0027994269849891196,
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regressor__estimator__regressor__learning_rate_init=0.004774699570297257,
regressor__max_features=0.5048854237095919,
regressor__max_samples=0.6632306541219959; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0027994269849891196,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004774699570297257,
regressor__max_features=0.5048854237095919,
regressor__max_samples=0.6632306541219959; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0027994269849891196,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004774699570297257,
regressor__max_features=0.5048854237095919,
regressor__max_samples=0.6632306541219959; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004006144572543498,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011268881023014168,
regressor__max_features=0.5166015543956832,
regressor__max_samples=0.5392892485775104; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017680108150822427,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.000532273748614762,
regressor__max_features=0.9933312966335732,
regressor__max_samples=0.5712477714507567; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004006144572543498,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011268881023014168,
regressor__max_features=0.5166015543956832,
regressor__max_samples=0.5392892485775104; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017680108150822427,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.000532273748614762,
regressor__max_features=0.9933312966335732,
regressor__max_samples=0.5712477714507567; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017680108150822427,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.000532273748614762,
regressor__max_features=0.9933312966335732,
regressor__max_samples=0.5712477714507567; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004006144572543498,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011268881023014168,
regressor__max_features=0.5166015543956832,
regressor__max_samples=0.5392892485775104; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004006144572543498,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011268881023014168,
regressor__max_features=0.5166015543956832,
regressor__max_samples=0.5392892485775104; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017680108150822427,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.000532273748614762,
regressor__max_features=0.9933312966335732,
regressor__max_samples=0.5712477714507567; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.017680108150822427,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.000532273748614762,
regressor__max_features=0.9933312966335732,
regressor__max_samples=0.5712477714507567; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004006144572543498,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011268881023014168,
regressor__max_features=0.5166015543956832,
regressor__max_samples=0.5392892485775104; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009253032906695801,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005042228754966153,
regressor__max_features=0.8447324845686328,
regressor__max_samples=0.9002933495545414; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009253032906695801,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005042228754966153,
regressor__max_features=0.8447324845686328,
regressor__max_samples=0.9002933495545414; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009253032906695801,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005042228754966153,
regressor__max_features=0.8447324845686328,
regressor__max_samples=0.9002933495545414; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009253032906695801,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005042228754966153,
regressor__max_features=0.8447324845686328,
regressor__max_samples=0.9002933495545414; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013290780507289545,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07155663725519353,
regressor__max_features=0.5621099833640297,
regressor__max_samples=0.9481106513425192; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0009253032906695801,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005042228754966153,
regressor__max_features=0.8447324845686328,
regressor__max_samples=0.9002933495545414; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013290780507289545,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07155663725519353,
regressor__max_features=0.5621099833640297,
regressor__max_samples=0.9481106513425192; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013290780507289545,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07155663725519353,
regressor__max_features=0.5621099833640297,
regressor__max_samples=0.9481106513425192; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013290780507289545,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07155663725519353,
regressor__max_features=0.5621099833640297,
regressor__max_samples=0.9481106513425192; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013290780507289545,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07155663725519353,
regressor__max_features=0.5621099833640297,
regressor__max_samples=0.9481106513425192; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.013875888340116141,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022401039808450252,
regressor__max_features=0.806128992940579,
regressor__max_samples=0.7421992237870203; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.013875888340116141,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022401039808450252,
regressor__max_features=0.806128992940579,
regressor__max_samples=0.7421992237870203; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.013875888340116141,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022401039808450252,
regressor__max_features=0.806128992940579,
regressor__max_samples=0.7421992237870203; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.013875888340116141,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022401039808450252,
regressor__max_features=0.806128992940579,
regressor__max_samples=0.7421992237870203; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.013875888340116141,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022401039808450252,
regressor__max_features=0.806128992940579,
regressor__max_samples=0.7421992237870203; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06436306818637239,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006899736804878476,
regressor__max_features=0.6494179431686569,
regressor__max_samples=0.9676009076497298; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06436306818637239,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006899736804878476,
regressor__max_features=0.6494179431686569,
regressor__max_samples=0.9676009076497298; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.048355955381429803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.028306965128008513,
regressor__max_features=0.72615914225915,
regressor__max_samples=0.8217888475982624; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06436306818637239,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006899736804878476,
regressor__max_features=0.6494179431686569,
regressor__max_samples=0.9676009076497298; total time= 3.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.048355955381429803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.028306965128008513,
regressor__max_features=0.72615914225915,
regressor__max_samples=0.8217888475982624; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06436306818637239,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006899736804878476,
regressor__max_features=0.6494179431686569,
regressor__max_samples=0.9676009076497298; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.06436306818637239,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006899736804878476,
regressor__max_features=0.6494179431686569,
regressor__max_samples=0.9676009076497298; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.048355955381429803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.028306965128008513,
regressor__max_features=0.72615914225915,
regressor__max_samples=0.8217888475982624; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.048355955381429803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.028306965128008513,
regressor__max_features=0.72615914225915,
regressor__max_samples=0.8217888475982624; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.048355955381429803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.028306965128008513,
regressor__max_features=0.72615914225915,
regressor__max_samples=0.8217888475982624; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.041257065722194,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009465059022669212,
regressor__max_features=0.7490978582265695,
regressor__max_samples=0.7776817754688157; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.041257065722194,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009465059022669212,
regressor__max_features=0.7490978582265695,
regressor__max_samples=0.7776817754688157; total time= 1.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.041257065722194,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009465059022669212,
regressor__max_features=0.7490978582265695,
regressor__max_samples=0.7776817754688157; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.041257065722194,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009465059022669212,
regressor__max_features=0.7490978582265695,
regressor__max_samples=0.7776817754688157; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.041257065722194,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009465059022669212,
regressor__max_features=0.7490978582265695,
regressor__max_samples=0.7776817754688157; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009115679717910939,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0974993562361509,
regressor__max_features=0.9873965810733916,
regressor__max_samples=0.8251628431734684; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009115679717910939,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0974993562361509,
regressor__max_features=0.9873965810733916,
regressor__max_samples=0.8251628431734684; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009115679717910939,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0974993562361509,
regressor__max_features=0.9873965810733916,
regressor__max_samples=0.8251628431734684; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009115679717910939,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0974993562361509,
regressor__max_features=0.9873965810733916,
regressor__max_samples=0.8251628431734684; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009115679717910939,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0974993562361509,
regressor__max_features=0.9873965810733916,
regressor__max_samples=0.8251628431734684; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0027405440415034224,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004546131167488908,
regressor__max_features=0.9879326279095657,
regressor__max_samples=0.6684478958855534; total time= 4.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0027405440415034224,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004546131167488908,
regressor__max_features=0.9879326279095657,
regressor__max_samples=0.6684478958855534; total time= 5.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003104548457688909,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017574781976312703,
regressor__max_features=0.5301760419952819,
regressor__max_samples=0.6235516170050732; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003331041241700745,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.015188231543237177,
regressor__max_features=0.8713532605999905,
regressor__max_samples=0.7127466672240377; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0027405440415034224,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004546131167488908,
regressor__max_features=0.9879326279095657,
regressor__max_samples=0.6684478958855534; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003331041241700745,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.015188231543237177,
regressor__max_features=0.8713532605999905,
regressor__max_samples=0.7127466672240377; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0027405440415034224,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004546131167488908,
regressor__max_features=0.9879326279095657,
regressor__max_samples=0.6684478958855534; total time= 5.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0027405440415034224,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004546131167488908,
regressor__max_features=0.9879326279095657,
regressor__max_samples=0.6684478958855534; total time= 6.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003104548457688909,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017574781976312703,
regressor__max_features=0.5301760419952819,
regressor__max_samples=0.6235516170050732; total time= 7.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003104548457688909,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017574781976312703,
regressor__max_features=0.5301760419952819,
regressor__max_samples=0.6235516170050732; total time= 7.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003331041241700745,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.015188231543237177,
regressor__max_features=0.8713532605999905,
regressor__max_samples=0.7127466672240377; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003331041241700745,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.015188231543237177,
regressor__max_features=0.8713532605999905,
regressor__max_samples=0.7127466672240377; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003331041241700745,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.015188231543237177,
regressor__max_features=0.8713532605999905,
regressor__max_samples=0.7127466672240377; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003104548457688909,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017574781976312703,
regressor__max_features=0.5301760419952819,
regressor__max_samples=0.6235516170050732; total time= 8.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016071295852933945,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09182241084536058,
regressor__max_features=0.5200545957062412,
regressor__max_samples=0.9335157480612243; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016071295852933945,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09182241084536058,
regressor__max_features=0.5200545957062412,
regressor__max_samples=0.9335157480612243; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0016210039882567113,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016466269795608539,
regressor__max_features=0.5153262511029031,
regressor__max_samples=0.6288414442556068; total time= 6.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016071295852933945,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09182241084536058,
regressor__max_features=0.5200545957062412,
regressor__max_samples=0.9335157480612243; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016071295852933945,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09182241084536058,
regressor__max_features=0.5200545957062412,
regressor__max_samples=0.9335157480612243; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016071295852933945,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09182241084536058,
regressor__max_features=0.5200545957062412,
regressor__max_samples=0.9335157480612243; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003104548457688909,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00017574781976312703,
regressor__max_features=0.5301760419952819,
regressor__max_samples=0.6235516170050732; total time= 9.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0058099866471453855,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01498901110104321,
regressor__max_features=0.7433344707123515,
regressor__max_samples=0.9367116190408302; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0058099866471453855,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01498901110104321,
regressor__max_features=0.7433344707123515,
regressor__max_samples=0.9367116190408302; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0058099866471453855,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01498901110104321,
regressor__max_features=0.7433344707123515,
regressor__max_samples=0.9367116190408302; total time= 2.5s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0058099866471453855,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01498901110104321,
regressor__max_features=0.7433344707123515,
regressor__max_samples=0.9367116190408302; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0016210039882567113,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016466269795608539,
regressor__max_features=0.5153262511029031,
regressor__max_samples=0.6288414442556068; total time= 7.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014175600182534812,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007846584576690928,
regressor__max_features=0.8665565112076405,
regressor__max_samples=0.5657838425636299; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0016210039882567113,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016466269795608539,
regressor__max_features=0.5153262511029031,
regressor__max_samples=0.6288414442556068; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014175600182534812,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007846584576690928,
regressor__max_features=0.8665565112076405,
regressor__max_samples=0.5657838425636299; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014175600182534812,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007846584576690928,
regressor__max_features=0.8665565112076405,
regressor__max_samples=0.5657838425636299; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0058099866471453855,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01498901110104321,
regressor__max_features=0.7433344707123515,
regressor__max_samples=0.9367116190408302; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014175600182534812,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007846584576690928,
regressor__max_features=0.8665565112076405,
regressor__max_samples=0.5657838425636299; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014175600182534812,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007846584576690928,
regressor__max_features=0.8665565112076405,
regressor__max_samples=0.5657838425636299; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0016210039882567113,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016466269795608539,
regressor__max_features=0.5153262511029031,
regressor__max_samples=0.6288414442556068; total time= 8.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02392590003188629,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006768359088577831,
regressor__max_features=0.7961751642966811,
regressor__max_samples=0.9561816728083454; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02392590003188629,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006768359088577831,
regressor__max_features=0.7961751642966811,
regressor__max_samples=0.9561816728083454; total time= 4.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02392590003188629,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006768359088577831,
regressor__max_features=0.7961751642966811,
regressor__max_samples=0.9561816728083454; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0016210039882567113,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00016466269795608539,
regressor__max_features=0.5153262511029031,
regressor__max_samples=0.6288414442556068; total time= 9.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001734177407166203,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00299801339788349,
regressor__max_features=0.6236155372015886,
regressor__max_samples=0.9353749506362556; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02392590003188629,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006768359088577831,
regressor__max_features=0.7961751642966811,
regressor__max_samples=0.9561816728083454; total time= 5.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001734177407166203,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00299801339788349,
regressor__max_features=0.6236155372015886,
regressor__max_samples=0.9353749506362556; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02392590003188629,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006768359088577831,
regressor__max_features=0.7961751642966811,
regressor__max_samples=0.9561816728083454; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001734177407166203,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00299801339788349,
regressor__max_features=0.6236155372015886,
regressor__max_samples=0.9353749506362556; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006008349669693534,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011959144356418752,
regressor__max_features=0.7964754257174638,
regressor__max_samples=0.5817619362925114; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0016441087824883795,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000345978671408765,
regressor__max_features=0.6187716624619388,
regressor__max_samples=0.9856975470208198; total time= 4.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001734177407166203,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00299801339788349,
regressor__max_features=0.6236155372015886,
regressor__max_samples=0.9353749506362556; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001734177407166203,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00299801339788349,
regressor__max_features=0.6236155372015886,
regressor__max_samples=0.9353749506362556; total time= 4.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006008349669693534,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011959144356418752,
regressor__max_features=0.7964754257174638,
regressor__max_samples=0.5817619362925114; total time= 2.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006008349669693534,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011959144356418752,
regressor__max_features=0.7964754257174638,
regressor__max_samples=0.5817619362925114; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0016441087824883795,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000345978671408765,
regressor__max_features=0.6187716624619388,
regressor__max_samples=0.9856975470208198; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006008349669693534,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011959144356418752,
regressor__max_features=0.7964754257174638,
regressor__max_samples=0.5817619362925114; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008638616014419022,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0374937398547688,
regressor__max_features=0.5045152020647752,
regressor__max_samples=0.7942063166703027; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0016441087824883795,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000345978671408765,
regressor__max_features=0.6187716624619388,
regressor__max_samples=0.9856975470208198; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008638616014419022,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0374937398547688,
regressor__max_features=0.5045152020647752,
regressor__max_samples=0.7942063166703027; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006008349669693534,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011959144356418752,
regressor__max_features=0.7964754257174638,
regressor__max_samples=0.5817619362925114; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008638616014419022,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0374937398547688,
regressor__max_features=0.5045152020647752,
regressor__max_samples=0.7942063166703027; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0016441087824883795,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000345978671408765,
regressor__max_features=0.6187716624619388,
regressor__max_samples=0.9856975470208198; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008638616014419022,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0374937398547688,
regressor__max_features=0.5045152020647752,
regressor__max_samples=0.7942063166703027; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0016441087824883795,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.000345978671408765,
regressor__max_features=0.6187716624619388,
regressor__max_samples=0.9856975470208198; total time= 6.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003230529530139748,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022198964745885775,
regressor__max_features=0.8346952007296806,
regressor__max_samples=0.5728782841223758; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008638616014419022,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0374937398547688,
regressor__max_features=0.5045152020647752,
regressor__max_samples=0.7942063166703027; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003230529530139748,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022198964745885775,
regressor__max_features=0.8346952007296806,
regressor__max_samples=0.5728782841223758; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003230529530139748,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022198964745885775,
regressor__max_features=0.8346952007296806,
regressor__max_samples=0.5728782841223758; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003230529530139748,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022198964745885775,
regressor__max_features=0.8346952007296806,
regressor__max_samples=0.5728782841223758; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003230529530139748,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022198964745885775,
regressor__max_features=0.8346952007296806,
regressor__max_samples=0.5728782841223758; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005573690096253459,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07010517286949926,
regressor__max_features=0.6262638910377096,
regressor__max_samples=0.6796253739581228; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005573690096253459,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07010517286949926,
regressor__max_features=0.6262638910377096,
regressor__max_samples=0.6796253739581228; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005573690096253459,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07010517286949926,
regressor__max_features=0.6262638910377096,
regressor__max_samples=0.6796253739581228; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002469567960854912,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005141425785650661,
regressor__max_features=0.6173370191307755,
regressor__max_samples=0.9418548348303764; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002469567960854912,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005141425785650661,
regressor__max_features=0.6173370191307755,
regressor__max_samples=0.9418548348303764; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005573690096253459,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07010517286949926,
regressor__max_features=0.6262638910377096,
regressor__max_samples=0.6796253739581228; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005573690096253459,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07010517286949926,
regressor__max_features=0.6262638910377096,
regressor__max_samples=0.6796253739581228; total time= 2.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00012882919135887,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011754283443825332,
regressor__max_features=0.7667559626498478,
regressor__max_samples=0.8908951649225147; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026122242622516133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.040254444839396324,
regressor__max_features=0.6029206050183842,
regressor__max_samples=0.5559598096988639; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00012882919135887,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011754283443825332,
regressor__max_features=0.7667559626498478,
regressor__max_samples=0.8908951649225147; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00012882919135887,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011754283443825332,
regressor__max_features=0.7667559626498478,
regressor__max_samples=0.8908951649225147; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002469567960854912,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005141425785650661,
regressor__max_features=0.6173370191307755,
regressor__max_samples=0.9418548348303764; total time= 5.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002469567960854912,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005141425785650661,
regressor__max_features=0.6173370191307755,
regressor__max_samples=0.9418548348303764; total time= 4.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026122242622516133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.040254444839396324,
regressor__max_features=0.6029206050183842,
regressor__max_samples=0.5559598096988639; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00012882919135887,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011754283443825332,
regressor__max_features=0.7667559626498478,
regressor__max_samples=0.8908951649225147; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026122242622516133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.040254444839396324,
regressor__max_features=0.6029206050183842,
regressor__max_samples=0.5559598096988639; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002469567960854912,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005141425785650661,
regressor__max_features=0.6173370191307755,
regressor__max_samples=0.9418548348303764; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09121766292968007,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.003922008982627202,
regressor__max_features=0.9683028461474807,
regressor__max_samples=0.5196717703342548; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00012882919135887,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011754283443825332,
regressor__max_features=0.7667559626498478,
regressor__max_samples=0.8908951649225147; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09121766292968007,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.003922008982627202,
regressor__max_features=0.9683028461474807,
regressor__max_samples=0.5196717703342548; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026122242622516133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.040254444839396324,
regressor__max_features=0.6029206050183842,
regressor__max_samples=0.5559598096988639; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09121766292968007,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.003922008982627202,
regressor__max_features=0.9683028461474807,
regressor__max_samples=0.5196717703342548; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026122242622516133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.040254444839396324,
regressor__max_features=0.6029206050183842,
regressor__max_samples=0.5559598096988639; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09121766292968007,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.003922008982627202,
regressor__max_features=0.9683028461474807,
regressor__max_samples=0.5196717703342548; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09121766292968007,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.003922008982627202,
regressor__max_features=0.9683028461474807,
regressor__max_samples=0.5196717703342548; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004721755891957781,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.06333226970652636,
regressor__max_features=0.6580780524883753,
regressor__max_samples=0.7536174043441962; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01160442424098791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002752271321139521,
regressor__max_features=0.842983867991492,
regressor__max_samples=0.5086898950927399; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01160442424098791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002752271321139521,
regressor__max_features=0.842983867991492,
regressor__max_samples=0.5086898950927399; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01160442424098791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002752271321139521,
regressor__max_features=0.842983867991492,
regressor__max_samples=0.5086898950927399; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004721755891957781,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.06333226970652636,
regressor__max_features=0.6580780524883753,
regressor__max_samples=0.7536174043441962; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01160442424098791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002752271321139521,
regressor__max_features=0.842983867991492,
regressor__max_samples=0.5086898950927399; total time= 1.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01160442424098791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002752271321139521,
regressor__max_features=0.842983867991492,
regressor__max_samples=0.5086898950927399; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008267226064893465,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005499670735023073,
regressor__max_features=0.6271122899487467,
regressor__max_samples=0.7608650233181066; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004721755891957781,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.06333226970652636,
regressor__max_features=0.6580780524883753,
regressor__max_samples=0.7536174043441962; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008267226064893465,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005499670735023073,
regressor__max_features=0.6271122899487467,
regressor__max_samples=0.7608650233181066; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008267226064893465,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005499670735023073,
regressor__max_features=0.6271122899487467,
regressor__max_samples=0.7608650233181066; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004721755891957781,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.06333226970652636,
regressor__max_features=0.6580780524883753,
regressor__max_samples=0.7536174043441962; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004721755891957781,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.06333226970652636,
regressor__max_features=0.6580780524883753,
regressor__max_samples=0.7536174043441962; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008267226064893465,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005499670735023073,
regressor__max_features=0.6271122899487467,
regressor__max_samples=0.7608650233181066; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008267226064893465,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005499670735023073,
regressor__max_features=0.6271122899487467,
regressor__max_samples=0.7608650233181066; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015801443223781,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003673222630346554,
regressor__max_features=0.8654147374354382,
regressor__max_samples=0.5780819223960869; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015801443223781,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003673222630346554,
regressor__max_features=0.8654147374354382,
regressor__max_samples=0.5780819223960869; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015801443223781,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003673222630346554,
regressor__max_features=0.8654147374354382,
regressor__max_samples=0.5780819223960869; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015801443223781,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003673222630346554,
regressor__max_features=0.8654147374354382,
regressor__max_samples=0.5780819223960869; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015801443223781,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003673222630346554,
regressor__max_features=0.8654147374354382,
regressor__max_samples=0.5780819223960869; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.040218582802687364,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005833370676644822,
regressor__max_features=0.8308161751855734,
regressor__max_samples=0.9893156437002653; total time= 4.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.040218582802687364,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005833370676644822,
regressor__max_features=0.8308161751855734,
regressor__max_samples=0.9893156437002653; total time= 4.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.040218582802687364,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005833370676644822,
regressor__max_features=0.8308161751855734,
regressor__max_samples=0.9893156437002653; total time= 4.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.040218582802687364,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005833370676644822,
regressor__max_features=0.8308161751855734,
regressor__max_samples=0.9893156437002653; total time= 4.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012335625585959236,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011710629825633166,
regressor__max_features=0.8382566928886233,
regressor__max_samples=0.6078375761988586; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.040218582802687364,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005833370676644822,
regressor__max_features=0.8308161751855734,
regressor__max_samples=0.9893156437002653; total time= 5.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012335625585959236,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011710629825633166,
regressor__max_features=0.8382566928886233,
regressor__max_samples=0.6078375761988586; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012335625585959236,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011710629825633166,
regressor__max_features=0.8382566928886233,
regressor__max_samples=0.6078375761988586; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012335625585959236,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011710629825633166,
regressor__max_features=0.8382566928886233,
regressor__max_samples=0.6078375761988586; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.030235464930471193,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0053895897715609325,
regressor__max_features=0.6742868336209853,
regressor__max_samples=0.5796437226214026; total time= 1.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00012335625585959236,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011710629825633166,
regressor__max_features=0.8382566928886233,
regressor__max_samples=0.6078375761988586; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.030235464930471193,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0053895897715609325,
regressor__max_features=0.6742868336209853,
regressor__max_samples=0.5796437226214026; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.030235464930471193,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0053895897715609325,
regressor__max_features=0.6742868336209853,
regressor__max_samples=0.5796437226214026; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011213574813286702,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005400598248884807,
regressor__max_features=0.9154385956708527,
regressor__max_samples=0.9323044179775184; total time= 6.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.030235464930471193,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0053895897715609325,
regressor__max_features=0.6742868336209853,
regressor__max_samples=0.5796437226214026; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.030235464930471193,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0053895897715609325,
regressor__max_features=0.6742868336209853,
regressor__max_samples=0.5796437226214026; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011213574813286702,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005400598248884807,
regressor__max_features=0.9154385956708527,
regressor__max_samples=0.9323044179775184; total time= 5.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001395198872367614,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03974806602804837,
regressor__max_features=0.9966684686793142,
regressor__max_samples=0.8762138813077862; total time= 1.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001395198872367614,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03974806602804837,
regressor__max_features=0.9966684686793142,
regressor__max_samples=0.8762138813077862; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011213574813286702,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005400598248884807,
regressor__max_features=0.9154385956708527,
regressor__max_samples=0.9323044179775184; total time= 5.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001395198872367614,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03974806602804837,
regressor__max_features=0.9966684686793142,
regressor__max_samples=0.8762138813077862; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001395198872367614,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03974806602804837,
regressor__max_features=0.9966684686793142,
regressor__max_samples=0.8762138813077862; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011213574813286702,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005400598248884807,
regressor__max_features=0.9154385956708527,
regressor__max_samples=0.9323044179775184; total time= 6.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001395198872367614,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03974806602804837,
regressor__max_features=0.9966684686793142,
regressor__max_samples=0.8762138813077862; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010064220810489967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05059725413916119,
regressor__max_features=0.9389455795206638,
regressor__max_samples=0.6585978583826992; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010064220810489967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05059725413916119,
regressor__max_features=0.9389455795206638,
regressor__max_samples=0.6585978583826992; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010064220810489967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05059725413916119,
regressor__max_features=0.9389455795206638,
regressor__max_samples=0.6585978583826992; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010064220810489967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05059725413916119,
regressor__max_features=0.9389455795206638,
regressor__max_samples=0.6585978583826992; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011213574813286702,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005400598248884807,
regressor__max_features=0.9154385956708527,
regressor__max_samples=0.9323044179775184; total time= 7.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010064220810489967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05059725413916119,
regressor__max_features=0.9389455795206638,
regressor__max_samples=0.6585978583826992; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010013658413533487,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012300443475745694,
regressor__max_features=0.6539649707955955,
regressor__max_samples=0.8523403813683234; total time= 13.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028412055118924747,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004299988497918371,
regressor__max_features=0.5028793302490608,
regressor__max_samples=0.6622929149501351; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028412055118924747,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004299988497918371,
regressor__max_features=0.5028793302490608,
regressor__max_samples=0.6622929149501351; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028412055118924747,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004299988497918371,
regressor__max_features=0.5028793302490608,
regressor__max_samples=0.6622929149501351; total time= 2.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028412055118924747,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004299988497918371,
regressor__max_features=0.5028793302490608,
regressor__max_samples=0.6622929149501351; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.028412055118924747,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004299988497918371,
regressor__max_features=0.5028793302490608,
regressor__max_samples=0.6622929149501351; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006752979269254169,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01220117398486036,
regressor__max_features=0.6942790506351728,
regressor__max_samples=0.7243468113112653; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006752979269254169,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01220117398486036,
regressor__max_features=0.6942790506351728,
regressor__max_samples=0.7243468113112653; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006752979269254169,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01220117398486036,
regressor__max_features=0.6942790506351728,
regressor__max_samples=0.7243468113112653; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006752979269254169,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01220117398486036,
regressor__max_features=0.6942790506351728,
regressor__max_samples=0.7243468113112653; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010013658413533487,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012300443475745694,
regressor__max_features=0.6539649707955955,
regressor__max_samples=0.8523403813683234; total time= 17.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006752979269254169,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01220117398486036,
regressor__max_features=0.6942790506351728,
regressor__max_samples=0.7243468113112653; total time= 2.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010013658413533487,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012300443475745694,
regressor__max_features=0.6539649707955955,
regressor__max_samples=0.8523403813683234; total time= 18.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0050239901928572405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002124853024020752,
regressor__max_features=0.8252558238958578,
regressor__max_samples=0.8825954322778924; total time= 10.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010013658413533487,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012300443475745694,
regressor__max_features=0.6539649707955955,
regressor__max_samples=0.8523403813683234; total time= 19.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00846696777455624,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024574355204864994,
regressor__max_features=0.6898928900896149,
regressor__max_samples=0.9316668247859126; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00846696777455624,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024574355204864994,
regressor__max_features=0.6898928900896149,
regressor__max_samples=0.9316668247859126; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0043357143864403164,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00048062779398098536,
regressor__max_features=0.5365979618842218,
regressor__max_samples=0.801724296690689; total time= 4.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.010013658413533487,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012300443475745694,
regressor__max_features=0.6539649707955955,
regressor__max_samples=0.8523403813683234; total time= 19.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0043357143864403164,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00048062779398098536,
regressor__max_features=0.5365979618842218,
regressor__max_samples=0.801724296690689; total time= 5.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00846696777455624,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024574355204864994,
regressor__max_features=0.6898928900896149,
regressor__max_samples=0.9316668247859126; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03479381773511331,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.027624782348050502,
regressor__max_features=0.5544591709233657,
regressor__max_samples=0.5744959279850974; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0050239901928572405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002124853024020752,
regressor__max_features=0.8252558238958578,
regressor__max_samples=0.8825954322778924; total time= 12.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03479381773511331,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.027624782348050502,
regressor__max_features=0.5544591709233657,
regressor__max_samples=0.5744959279850974; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03479381773511331,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.027624782348050502,
regressor__max_features=0.5544591709233657,
regressor__max_samples=0.5744959279850974; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03479381773511331,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.027624782348050502,
regressor__max_features=0.5544591709233657,
regressor__max_samples=0.5744959279850974; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0043357143864403164,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00048062779398098536,
regressor__max_features=0.5365979618842218,
regressor__max_samples=0.801724296690689; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.03479381773511331,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.027624782348050502,
regressor__max_features=0.5544591709233657,
regressor__max_samples=0.5744959279850974; total time= 1.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0043357143864403164,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00048062779398098536,
regressor__max_features=0.5365979618842218,
regressor__max_samples=0.801724296690689; total time= 5.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0043357143864403164,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00048062779398098536,
regressor__max_features=0.5365979618842218,
regressor__max_samples=0.801724296690689; total time= 6.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0050239901928572405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002124853024020752,
regressor__max_features=0.8252558238958578,
regressor__max_samples=0.8825954322778924; total time= 12.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00846696777455624,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024574355204864994,
regressor__max_features=0.6898928900896149,
regressor__max_samples=0.9316668247859126; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07354803733588544,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.029321099820312828,
regressor__max_features=0.7154310457690167,
regressor__max_samples=0.9483571449200612; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00846696777455624,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024574355204864994,
regressor__max_features=0.6898928900896149,
regressor__max_samples=0.9316668247859126; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07354803733588544,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.029321099820312828,
regressor__max_features=0.7154310457690167,
regressor__max_samples=0.9483571449200612; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07354803733588544,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.029321099820312828,
regressor__max_features=0.7154310457690167,
regressor__max_samples=0.9483571449200612; total time= 2.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00550195593330085,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001524389579911143,
regressor__max_features=0.7747071192152266,
regressor__max_samples=0.8734378580679114; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0050239901928572405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002124853024020752,
regressor__max_features=0.8252558238958578,
regressor__max_samples=0.8825954322778924; total time= 13.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07354803733588544,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.029321099820312828,
regressor__max_features=0.7154310457690167,
regressor__max_samples=0.9483571449200612; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00550195593330085,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001524389579911143,
regressor__max_features=0.7747071192152266,
regressor__max_samples=0.8734378580679114; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07354803733588544,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.029321099820312828,
regressor__max_features=0.7154310457690167,
regressor__max_samples=0.9483571449200612; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00550195593330085,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001524389579911143,
regressor__max_features=0.7747071192152266,
regressor__max_samples=0.8734378580679114; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00550195593330085,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001524389579911143,
regressor__max_features=0.7747071192152266,
regressor__max_samples=0.8734378580679114; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00550195593330085,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001524389579911143,
regressor__max_features=0.7747071192152266,
regressor__max_samples=0.8734378580679114; total time= 3.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04944150097436515,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009206175385005228,
regressor__max_features=0.6900372544686026,
regressor__max_samples=0.5985770632864387; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04944150097436515,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009206175385005228,
regressor__max_features=0.6900372544686026,
regressor__max_samples=0.5985770632864387; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0050239901928572405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002124853024020752,
regressor__max_features=0.8252558238958578,
regressor__max_samples=0.8825954322778924; total time= 15.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04944150097436515,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009206175385005228,
regressor__max_features=0.6900372544686026,
regressor__max_samples=0.5985770632864387; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04944150097436515,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009206175385005228,
regressor__max_features=0.6900372544686026,
regressor__max_samples=0.5985770632864387; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04944150097436515,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009206175385005228,
regressor__max_features=0.6900372544686026,
regressor__max_samples=0.5985770632864387; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0008353527271566405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003951669727793275,
regressor__max_features=0.7904660266313989,
regressor__max_samples=0.5448131695774711; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0008353527271566405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003951669727793275,
regressor__max_features=0.7904660266313989,
regressor__max_samples=0.5448131695774711; total time= 1.5s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0008353527271566405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003951669727793275,
regressor__max_features=0.7904660266313989,
regressor__max_samples=0.5448131695774711; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0008353527271566405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003951669727793275,
regressor__max_features=0.7904660266313989,
regressor__max_samples=0.5448131695774711; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0008353527271566405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003951669727793275,
regressor__max_features=0.7904660266313989,
regressor__max_samples=0.5448131695774711; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.014075117292982282,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08514824791915436,
regressor__max_features=0.9115870077978301,
regressor__max_samples=0.7612405202580592; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.014075117292982282,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08514824791915436,
regressor__max_features=0.9115870077978301,
regressor__max_samples=0.7612405202580592; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.014075117292982282,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08514824791915436,
regressor__max_features=0.9115870077978301,
regressor__max_samples=0.7612405202580592; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.014075117292982282,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08514824791915436,
regressor__max_features=0.9115870077978301,
regressor__max_samples=0.7612405202580592; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.014075117292982282,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08514824791915436,
regressor__max_features=0.9115870077978301,
regressor__max_samples=0.7612405202580592; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011049778232899942,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.025080909798182286,
regressor__max_features=0.5922925647685164,
regressor__max_samples=0.647639144610155; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011049778232899942,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.025080909798182286,
regressor__max_features=0.5922925647685164,
regressor__max_samples=0.647639144610155; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011049778232899942,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.025080909798182286,
regressor__max_features=0.5922925647685164,
regressor__max_samples=0.647639144610155; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011049778232899942,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.025080909798182286,
regressor__max_features=0.5922925647685164,
regressor__max_samples=0.647639144610155; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.011049778232899942,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.025080909798182286,
regressor__max_features=0.5922925647685164,
regressor__max_samples=0.647639144610155; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005034271794333396,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05665750783502933,
regressor__max_features=0.712263712598181,
regressor__max_samples=0.5435194812938293; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005034271794333396,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05665750783502933,
regressor__max_features=0.712263712598181,
regressor__max_samples=0.5435194812938293; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005034271794333396,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05665750783502933,
regressor__max_features=0.712263712598181,
regressor__max_samples=0.5435194812938293; total time= 1.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005034271794333396,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05665750783502933,
regressor__max_features=0.712263712598181,
regressor__max_samples=0.5435194812938293; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004473831535390865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.007933304552450494,
regressor__max_features=0.9757016711033285,
regressor__max_samples=0.8008059100731656; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004473831535390865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.007933304552450494,
regressor__max_features=0.9757016711033285,
regressor__max_samples=0.8008059100731656; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004473831535390865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.007933304552450494,
regressor__max_features=0.9757016711033285,
regressor__max_samples=0.8008059100731656; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005034271794333396,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05665750783502933,
regressor__max_features=0.712263712598181,
regressor__max_samples=0.5435194812938293; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004473831535390865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.007933304552450494,
regressor__max_features=0.9757016711033285,
regressor__max_samples=0.8008059100731656; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0011346660527581627,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003154967016086411,
regressor__max_features=0.8086758787064872,
regressor__max_samples=0.7615423622908923; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003511765524604632,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.024056450668010612,
regressor__max_features=0.8233029567114301,
regressor__max_samples=0.5873867214136885; total time= 1.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0011346660527581627,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003154967016086411,
regressor__max_features=0.8086758787064872,
regressor__max_samples=0.7615423622908923; total time= 5.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003511765524604632,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.024056450668010612,
regressor__max_features=0.8233029567114301,
regressor__max_samples=0.5873867214136885; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003511765524604632,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.024056450668010612,
regressor__max_features=0.8233029567114301,
regressor__max_samples=0.5873867214136885; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003511765524604632,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.024056450668010612,
regressor__max_features=0.8233029567114301,
regressor__max_samples=0.5873867214136885; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004473831535390865,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.007933304552450494,
regressor__max_features=0.9757016711033285,
regressor__max_samples=0.8008059100731656; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0011346660527581627,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003154967016086411,
regressor__max_features=0.8086758787064872,
regressor__max_samples=0.7615423622908923; total time= 6.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003511765524604632,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.024056450668010612,
regressor__max_features=0.8233029567114301,
regressor__max_samples=0.5873867214136885; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0011346660527581627,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003154967016086411,
regressor__max_features=0.8086758787064872,
regressor__max_samples=0.7615423622908923; total time= 7.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03577031647556194,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01006119804781128,
regressor__max_features=0.5477676569173341,
regressor__max_samples=0.8119296623382866; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03577031647556194,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01006119804781128,
regressor__max_features=0.5477676569173341,
regressor__max_samples=0.8119296623382866; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03577031647556194,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01006119804781128,
regressor__max_features=0.5477676569173341,
regressor__max_samples=0.8119296623382866; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03577031647556194,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01006119804781128,
regressor__max_features=0.5477676569173341,
regressor__max_samples=0.8119296623382866; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03577031647556194,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01006119804781128,
regressor__max_features=0.5477676569173341,
regressor__max_samples=0.8119296623382866; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0011346660527581627,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003154967016086411,
regressor__max_features=0.8086758787064872,
regressor__max_samples=0.7615423622908923; total time= 9.3s

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:690:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
the optimization hasn't converged yet.
  warnings.warn(

[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03308140213679068,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005135842264582366,
regressor__max_features=0.8902627576219309,
regressor__max_samples=0.637403017774068; total time= 6.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03308140213679068,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005135842264582366,
regressor__max_features=0.8902627576219309,
regressor__max_samples=0.637403017774068; total time= 6.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00820129398457831,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014883942641903353,
regressor__max_features=0.7972381675930865,
regressor__max_samples=0.7483834294295847; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03308140213679068,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005135842264582366,
regressor__max_features=0.8902627576219309,
regressor__max_samples=0.637403017774068; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00820129398457831,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014883942641903353,
regressor__max_features=0.7972381675930865,
regressor__max_samples=0.7483834294295847; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03308140213679068,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005135842264582366,
regressor__max_features=0.8902627576219309,
regressor__max_samples=0.637403017774068; total time= 7.9s

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:690:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
the optimization hasn't converged yet.
  warnings.warn(

[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018699691369115268,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003191851099565076,
regressor__max_features=0.868436872472091,
regressor__max_samples=0.9313985387716143; total time= 7.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00820129398457831,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014883942641903353,
regressor__max_features=0.7972381675930865,
regressor__max_samples=0.7483834294295847; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00820129398457831,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014883942641903353,
regressor__max_features=0.7972381675930865,
regressor__max_samples=0.7483834294295847; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00820129398457831,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014883942641903353,
regressor__max_features=0.7972381675930865,
regressor__max_samples=0.7483834294295847; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018699691369115268,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003191851099565076,
regressor__max_features=0.868436872472091,
regressor__max_samples=0.9313985387716143; total time= 7.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018699691369115268,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003191851099565076,
regressor__max_features=0.868436872472091,
regressor__max_samples=0.9313985387716143; total time= 8.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.014912503348486539,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09413865327592391,
regressor__max_features=0.5641971447270626,
regressor__max_samples=0.5520548246920831; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006662247491841096,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012174010223877665,
regressor__max_features=0.7021593840521461,
regressor__max_samples=0.7140998074628073; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018699691369115268,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003191851099565076,
regressor__max_features=0.868436872472091,
regressor__max_samples=0.9313985387716143; total time= 9.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.014912503348486539,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09413865327592391,
regressor__max_features=0.5641971447270626,
regressor__max_samples=0.5520548246920831; total time= 2.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006662247491841096,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012174010223877665,
regressor__max_features=0.7021593840521461,
regressor__max_samples=0.7140998074628073; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.014912503348486539,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09413865327592391,
regressor__max_features=0.5641971447270626,
regressor__max_samples=0.5520548246920831; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006662247491841096,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012174010223877665,
regressor__max_features=0.7021593840521461,
regressor__max_samples=0.7140998074628073; total time= 2.4s

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:690:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
the optimization hasn't converged yet.
  warnings.warn(

[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03308140213679068,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005135842264582366,
regressor__max_features=0.8902627576219309,
regressor__max_samples=0.637403017774068; total time= 10.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006662247491841096,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012174010223877665,
regressor__max_features=0.7021593840521461,
regressor__max_samples=0.7140998074628073; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0018699691369115268,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003191851099565076,
regressor__max_features=0.868436872472091,
regressor__max_samples=0.9313985387716143; total time= 8.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.014912503348486539,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09413865327592391,
regressor__max_features=0.5641971447270626,
regressor__max_samples=0.5520548246920831; total time= 1.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006662247491841096,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012174010223877665,
regressor__max_features=0.7021593840521461,
regressor__max_samples=0.7140998074628073; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.014912503348486539,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.09413865327592391,
regressor__max_features=0.5641971447270626,
regressor__max_samples=0.5520548246920831; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010272624141157904,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02079065550523091,
regressor__max_features=0.6324477236183668,
regressor__max_samples=0.8226449187628584; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010272624141157904,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02079065550523091,
regressor__max_features=0.6324477236183668,
regressor__max_samples=0.8226449187628584; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010272624141157904,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02079065550523091,
regressor__max_features=0.6324477236183668,
regressor__max_samples=0.8226449187628584; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010272624141157904,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02079065550523091,
regressor__max_features=0.6324477236183668,
regressor__max_samples=0.8226449187628584; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0010272624141157904,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02079065550523091,
regressor__max_features=0.6324477236183668,
regressor__max_samples=0.8226449187628584; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003763587124895471,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004785961566240636,
regressor__max_features=0.6774981529159687,
regressor__max_samples=0.5347119213730467; total time= 2.5s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003763587124895471,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004785961566240636,
regressor__max_features=0.6774981529159687,
regressor__max_samples=0.5347119213730467; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003763587124895471,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004785961566240636,
regressor__max_features=0.6774981529159687,
regressor__max_samples=0.5347119213730467; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003763587124895471,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004785961566240636,
regressor__max_features=0.6774981529159687,
regressor__max_samples=0.5347119213730467; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003763587124895471,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004785961566240636,
regressor__max_features=0.6774981529159687,
regressor__max_samples=0.5347119213730467; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004037144605041932,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011773775706697352,
regressor__max_features=0.820985750221633,
regressor__max_samples=0.8035470180443904; total time= 11.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00896668900330703,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004925922417211657,
regressor__max_features=0.7284293224401843,
regressor__max_samples=0.5951624314095338; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00896668900330703,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004925922417211657,
regressor__max_features=0.7284293224401843,
regressor__max_samples=0.5951624314095338; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00896668900330703,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004925922417211657,
regressor__max_features=0.7284293224401843,
regressor__max_samples=0.5951624314095338; total time= 2.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09603246655027282,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012269668114456866,
regressor__max_features=0.9977402453521285,
regressor__max_samples=0.7231761755988211; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00896668900330703,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004925922417211657,
regressor__max_features=0.7284293224401843,
regressor__max_samples=0.5951624314095338; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004037144605041932,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011773775706697352,
regressor__max_features=0.820985750221633,
regressor__max_samples=0.8035470180443904; total time= 14.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09603246655027282,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012269668114456866,
regressor__max_features=0.9977402453521285,
regressor__max_samples=0.7231761755988211; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00896668900330703,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004925922417211657,
regressor__max_features=0.7284293224401843,
regressor__max_samples=0.5951624314095338; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09603246655027282,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012269668114456866,
regressor__max_features=0.9977402453521285,
regressor__max_samples=0.7231761755988211; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09603246655027282,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012269668114456866,
regressor__max_features=0.9977402453521285,
regressor__max_samples=0.7231761755988211; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09603246655027282,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012269668114456866,
regressor__max_features=0.9977402453521285,
regressor__max_samples=0.7231761755988211; total time= 2.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015150502164610583,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07462354106730308,
regressor__max_features=0.5876514711788299,
regressor__max_samples=0.8450024454103329; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004037144605041932,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011773775706697352,
regressor__max_features=0.820985750221633,
regressor__max_samples=0.8035470180443904; total time= 15.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004037144605041932,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011773775706697352,
regressor__max_features=0.820985750221633,
regressor__max_samples=0.8035470180443904; total time= 15.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015150502164610583,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07462354106730308,
regressor__max_features=0.5876514711788299,
regressor__max_samples=0.8450024454103329; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015150502164610583,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07462354106730308,
regressor__max_features=0.5876514711788299,
regressor__max_samples=0.8450024454103329; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015150502164610583,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07462354106730308,
regressor__max_features=0.5876514711788299,
regressor__max_samples=0.8450024454103329; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015150502164610583,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07462354106730308,
regressor__max_features=0.5876514711788299,
regressor__max_samples=0.8450024454103329; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007990230815634968,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.060469093548503126,
regressor__max_features=0.6173210409277268,
regressor__max_samples=0.699657957904654; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007990230815634968,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.060469093548503126,
regressor__max_features=0.6173210409277268,
regressor__max_samples=0.699657957904654; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007990230815634968,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.060469093548503126,
regressor__max_features=0.6173210409277268,
regressor__max_samples=0.699657957904654; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004037144605041932,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011773775706697352,
regressor__max_features=0.820985750221633,
regressor__max_samples=0.8035470180443904; total time= 18.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001148362673702768,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03358148392170399,
regressor__max_features=0.760478988636021,
regressor__max_samples=0.8117928440013799; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03130324351826899,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001621044052495004,
regressor__max_features=0.820084111914083,
regressor__max_samples=0.6355737182500925; total time= 10.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03130324351826899,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001621044052495004,
regressor__max_features=0.820084111914083,
regressor__max_samples=0.6355737182500925; total time= 9.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007990230815634968,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.060469093548503126,
regressor__max_features=0.6173210409277268,
regressor__max_samples=0.699657957904654; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007990230815634968,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.060469093548503126,
regressor__max_features=0.6173210409277268,
regressor__max_samples=0.699657957904654; total time= 2.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03130324351826899,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001621044052495004,
regressor__max_features=0.820084111914083,
regressor__max_samples=0.6355737182500925; total time= 10.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001148362673702768,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03358148392170399,
regressor__max_features=0.760478988636021,
regressor__max_samples=0.8117928440013799; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001148362673702768,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03358148392170399,
regressor__max_features=0.760478988636021,
regressor__max_samples=0.8117928440013799; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001148362673702768,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03358148392170399,
regressor__max_features=0.760478988636021,
regressor__max_samples=0.8117928440013799; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02310499957556439,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006690510212939794,
regressor__max_features=0.7534192159102413,
regressor__max_samples=0.5796520103094902; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02310499957556439,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006690510212939794,
regressor__max_features=0.7534192159102413,
regressor__max_samples=0.5796520103094902; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02310499957556439,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006690510212939794,
regressor__max_features=0.7534192159102413,
regressor__max_samples=0.5796520103094902; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02310499957556439,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006690510212939794,
regressor__max_features=0.7534192159102413,
regressor__max_samples=0.5796520103094902; total time= 1.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02310499957556439,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006690510212939794,
regressor__max_features=0.7534192159102413,
regressor__max_samples=0.5796520103094902; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001148362673702768,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03358148392170399,
regressor__max_features=0.760478988636021,
regressor__max_samples=0.8117928440013799; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03130324351826899,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001621044052495004,
regressor__max_features=0.820084111914083,
regressor__max_samples=0.6355737182500925; total time= 13.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06469540993886787,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0033819932794782364,
regressor__max_features=0.9126430541164484,
regressor__max_samples=0.6398517692357396; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06469540993886787,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0033819932794782364,
regressor__max_features=0.9126430541164484,
regressor__max_samples=0.6398517692357396; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06469540993886787,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0033819932794782364,
regressor__max_features=0.9126430541164484,
regressor__max_samples=0.6398517692357396; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06469540993886787,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0033819932794782364,
regressor__max_features=0.9126430541164484,
regressor__max_samples=0.6398517692357396; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03130324351826899,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001621044052495004,
regressor__max_features=0.820084111914083,
regressor__max_samples=0.6355737182500925; total time= 13.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001283752024731195,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00048091896113473153,
regressor__max_features=0.6708372509484604,
regressor__max_samples=0.6904547627554936; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001283752024731195,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00048091896113473153,
regressor__max_features=0.6708372509484604,
regressor__max_samples=0.6904547627554936; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.06469540993886787,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0033819932794782364,
regressor__max_features=0.9126430541164484,
regressor__max_samples=0.6398517692357396; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001283752024731195,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00048091896113473153,
regressor__max_features=0.6708372509484604,
regressor__max_samples=0.6904547627554936; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001283752024731195,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00048091896113473153,
regressor__max_features=0.6708372509484604,
regressor__max_samples=0.6904547627554936; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001283752024731195,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00048091896113473153,
regressor__max_features=0.6708372509484604,
regressor__max_samples=0.6904547627554936; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.035832315488596625,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031875646182882296,
regressor__max_features=0.6985893887146055,
regressor__max_samples=0.7698663093799714; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.035832315488596625,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031875646182882296,
regressor__max_features=0.6985893887146055,
regressor__max_samples=0.7698663093799714; total time= 1.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.035832315488596625,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031875646182882296,
regressor__max_features=0.6985893887146055,
regressor__max_samples=0.7698663093799714; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003246637603885204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001949976371651364,
regressor__max_features=0.7251854681290926,
regressor__max_samples=0.878081664447623; total time= 10.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003246637603885204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001949976371651364,
regressor__max_features=0.7251854681290926,
regressor__max_samples=0.878081664447623; total time= 10.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003246637603885204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001949976371651364,
regressor__max_features=0.7251854681290926,
regressor__max_samples=0.878081664447623; total time= 10.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.035832315488596625,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031875646182882296,
regressor__max_features=0.6985893887146055,
regressor__max_samples=0.7698663093799714; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.035832315488596625,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031875646182882296,
regressor__max_features=0.6985893887146055,
regressor__max_samples=0.7698663093799714; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0013863404032766384,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.040194818350022164,
regressor__max_features=0.9029627000588105,
regressor__max_samples=0.8950152181535307; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006335530705377297,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003056980257985932,
regressor__max_features=0.8931379783785472,
regressor__max_samples=0.7872936911588833; total time= 5.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006335530705377297,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003056980257985932,
regressor__max_features=0.8931379783785472,
regressor__max_samples=0.7872936911588833; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003246637603885204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001949976371651364,
regressor__max_features=0.7251854681290926,
regressor__max_samples=0.878081664447623; total time= 12.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006335530705377297,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003056980257985932,
regressor__max_features=0.8931379783785472,
regressor__max_samples=0.7872936911588833; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006335530705377297,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003056980257985932,
regressor__max_features=0.8931379783785472,
regressor__max_samples=0.7872936911588833; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003246637603885204,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001949976371651364,
regressor__max_features=0.7251854681290926,
regressor__max_samples=0.878081664447623; total time= 12.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.028589398105446806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016178585049031369,
regressor__max_features=0.5867622574932543,
regressor__max_samples=0.8474755444512309; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0013863404032766384,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.040194818350022164,
regressor__max_features=0.9029627000588105,
regressor__max_samples=0.8950152181535307; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0013863404032766384,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.040194818350022164,
regressor__max_features=0.9029627000588105,
regressor__max_samples=0.8950152181535307; total time= 3.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006335530705377297,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003056980257985932,
regressor__max_features=0.8931379783785472,
regressor__max_samples=0.7872936911588833; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0013863404032766384,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.040194818350022164,
regressor__max_features=0.9029627000588105,
regressor__max_samples=0.8950152181535307; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.028589398105446806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016178585049031369,
regressor__max_features=0.5867622574932543,
regressor__max_samples=0.8474755444512309; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.028589398105446806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016178585049031369,
regressor__max_features=0.5867622574932543,
regressor__max_samples=0.8474755444512309; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07446782400555742,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00837367752657903,
regressor__max_features=0.9112402818171805,
regressor__max_samples=0.5662623363709243; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.028589398105446806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016178585049031369,
regressor__max_features=0.5867622574932543,
regressor__max_samples=0.8474755444512309; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0013863404032766384,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.040194818350022164,
regressor__max_features=0.9029627000588105,
regressor__max_samples=0.8950152181535307; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07446782400555742,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00837367752657903,
regressor__max_features=0.9112402818171805,
regressor__max_samples=0.5662623363709243; total time= 1.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07446782400555742,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00837367752657903,
regressor__max_features=0.9112402818171805,
regressor__max_samples=0.5662623363709243; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07446782400555742,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00837367752657903,
regressor__max_features=0.9112402818171805,
regressor__max_samples=0.5662623363709243; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07446782400555742,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00837367752657903,
regressor__max_features=0.9112402818171805,
regressor__max_samples=0.5662623363709243; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.028589398105446806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016178585049031369,
regressor__max_features=0.5867622574932543,
regressor__max_samples=0.8474755444512309; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0008733394404960459,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0028919165485592003,
regressor__max_features=0.8031264688754887,
regressor__max_samples=0.8824049004514978; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0008733394404960459,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0028919165485592003,
regressor__max_features=0.8031264688754887,
regressor__max_samples=0.8824049004514978; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0008733394404960459,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0028919165485592003,
regressor__max_features=0.8031264688754887,
regressor__max_samples=0.8824049004514978; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0008733394404960459,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0028919165485592003,
regressor__max_features=0.8031264688754887,
regressor__max_samples=0.8824049004514978; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0008733394404960459,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0028919165485592003,
regressor__max_features=0.8031264688754887,
regressor__max_samples=0.8824049004514978; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008965909350433643,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001570323331261061,
regressor__max_features=0.5731869960811246,
regressor__max_samples=0.6837672111457658; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001983780582382386,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.06446849348758972,
regressor__max_features=0.9380525962990213,
regressor__max_samples=0.6388763118746306; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008965909350433643,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001570323331261061,
regressor__max_features=0.5731869960811246,
regressor__max_samples=0.6837672111457658; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008965909350433643,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001570323331261061,
regressor__max_features=0.5731869960811246,
regressor__max_samples=0.6837672111457658; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008965909350433643,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001570323331261061,
regressor__max_features=0.5731869960811246,
regressor__max_samples=0.6837672111457658; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0008965909350433643,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.001570323331261061,
regressor__max_features=0.5731869960811246,
regressor__max_samples=0.6837672111457658; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001983780582382386,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.06446849348758972,
regressor__max_features=0.9380525962990213,
regressor__max_samples=0.6388763118746306; total time= 1.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001983780582382386,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.06446849348758972,
regressor__max_features=0.9380525962990213,
regressor__max_samples=0.6388763118746306; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001983780582382386,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.06446849348758972,
regressor__max_features=0.9380525962990213,
regressor__max_samples=0.6388763118746306; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001983780582382386,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.06446849348758972,
regressor__max_features=0.9380525962990213,
regressor__max_samples=0.6388763118746306; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002720834145150403,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004984872734420615,
regressor__max_features=0.539982697179651,
regressor__max_samples=0.8883618152358463; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002720834145150403,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004984872734420615,
regressor__max_features=0.539982697179651,
regressor__max_samples=0.8883618152358463; total time= 5.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014905332997952562,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08933811314963759,
regressor__max_features=0.6195900521079628,
regressor__max_samples=0.5711246843435981; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002720834145150403,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004984872734420615,
regressor__max_features=0.539982697179651,
regressor__max_samples=0.8883618152358463; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002720834145150403,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004984872734420615,
regressor__max_features=0.539982697179651,
regressor__max_samples=0.8883618152358463; total time= 5.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014905332997952562,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08933811314963759,
regressor__max_features=0.6195900521079628,
regressor__max_samples=0.5711246843435981; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002720834145150403,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004984872734420615,
regressor__max_features=0.539982697179651,
regressor__max_samples=0.8883618152358463; total time= 6.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014905332997952562,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08933811314963759,
regressor__max_features=0.6195900521079628,
regressor__max_samples=0.5711246843435981; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014905332997952562,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08933811314963759,
regressor__max_features=0.6195900521079628,
regressor__max_samples=0.5711246843435981; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001527738000708154,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00037613813491631145,
regressor__max_features=0.6058776753327392,
regressor__max_samples=0.7423007935094323; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00014905332997952562,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08933811314963759,
regressor__max_features=0.6195900521079628,
regressor__max_samples=0.5711246843435981; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001527738000708154,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00037613813491631145,
regressor__max_features=0.6058776753327392,
regressor__max_samples=0.7423007935094323; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001527738000708154,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00037613813491631145,
regressor__max_features=0.6058776753327392,
regressor__max_samples=0.7423007935094323; total time= 5.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001527738000708154,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00037613813491631145,
regressor__max_features=0.6058776753327392,
regressor__max_samples=0.7423007935094323; total time= 6.5s

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:690:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
the optimization hasn't converged yet.
  warnings.warn(

[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001527738000708154,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00037613813491631145,
regressor__max_features=0.6058776753327392,
regressor__max_samples=0.7423007935094323; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007935401003125989,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004988116816937099,
regressor__max_features=0.6104013148928196,
regressor__max_samples=0.5868762978311473; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007935401003125989,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004988116816937099,
regressor__max_features=0.6104013148928196,
regressor__max_samples=0.5868762978311473; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007935401003125989,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004988116816937099,
regressor__max_features=0.6104013148928196,
regressor__max_samples=0.5868762978311473; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02228887364559327,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00020097286931492,
regressor__max_features=0.846080670847066,
regressor__max_samples=0.5311458990650313; total time= 5.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007935401003125989,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004988116816937099,
regressor__max_features=0.6104013148928196,
regressor__max_samples=0.5868762978311473; total time= 2.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02228887364559327,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00020097286931492,
regressor__max_features=0.846080670847066,
regressor__max_samples=0.5311458990650313; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007935401003125989,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004988116816937099,
regressor__max_features=0.6104013148928196,
regressor__max_samples=0.5868762978311473; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003811749848727884,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014891978791028618,
regressor__max_features=0.8960468168802715,
regressor__max_samples=0.7023992145425999; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02228887364559327,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00020097286931492,
regressor__max_features=0.846080670847066,
regressor__max_samples=0.5311458990650313; total time= 7.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003811749848727884,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014891978791028618,
regressor__max_features=0.8960468168802715,
regressor__max_samples=0.7023992145425999; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003811749848727884,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014891978791028618,
regressor__max_features=0.8960468168802715,
regressor__max_samples=0.7023992145425999; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02228887364559327,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00020097286931492,
regressor__max_features=0.846080670847066,
regressor__max_samples=0.5311458990650313; total time= 6.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003835902284581968,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00377149786241922,
regressor__max_features=0.7408174232674312,
regressor__max_samples=0.5884435725719273; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003835902284581968,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00377149786241922,
regressor__max_features=0.7408174232674312,
regressor__max_samples=0.5884435725719273; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003835902284581968,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00377149786241922,
regressor__max_features=0.7408174232674312,
regressor__max_samples=0.5884435725719273; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003835902284581968,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00377149786241922,
regressor__max_features=0.7408174232674312,
regressor__max_samples=0.5884435725719273; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003811749848727884,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014891978791028618,
regressor__max_features=0.8960468168802715,
regressor__max_samples=0.7023992145425999; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003835902284581968,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00377149786241922,
regressor__max_features=0.7408174232674312,
regressor__max_samples=0.5884435725719273; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003811749848727884,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014891978791028618,
regressor__max_features=0.8960468168802715,
regressor__max_samples=0.7023992145425999; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02228887364559327,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00020097286931492,
regressor__max_features=0.846080670847066,
regressor__max_samples=0.5311458990650313; total time= 7.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00836525037471417,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011299609337563688,
regressor__max_features=0.8012765819214325,
regressor__max_samples=0.6823991095722648; total time= 13.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006017825998364116,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011038866513671945,
regressor__max_features=0.5948474240873386,
regressor__max_samples=0.8094740965038607; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006017825998364116,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011038866513671945,
regressor__max_features=0.5948474240873386,
regressor__max_samples=0.8094740965038607; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006017825998364116,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011038866513671945,
regressor__max_features=0.5948474240873386,
regressor__max_samples=0.8094740965038607; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0001087332386811596,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007724963178874974,
regressor__max_features=0.5994281388876479,
regressor__max_samples=0.5323083399220431; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006017825998364116,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011038866513671945,
regressor__max_features=0.5948474240873386,
regressor__max_samples=0.8094740965038607; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006017825998364116,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011038866513671945,
regressor__max_features=0.5948474240873386,
regressor__max_samples=0.8094740965038607; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02031957784609174,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005568532685131212,
regressor__max_features=0.5669018681397444,
regressor__max_samples=0.7092903108119677; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00836525037471417,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011299609337563688,
regressor__max_features=0.8012765819214325,
regressor__max_samples=0.6823991095722648; total time= 14.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02031957784609174,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005568532685131212,
regressor__max_features=0.5669018681397444,
regressor__max_samples=0.7092903108119677; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02031957784609174,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005568532685131212,
regressor__max_features=0.5669018681397444,
regressor__max_samples=0.7092903108119677; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00010873323868111596,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007724963178874974,
regressor__max_features=0.5994281388876479,
regressor__max_samples=0.5323083399220431; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00010873323868111596,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007724963178874974,
regressor__max_features=0.5994281388876479,
regressor__max_samples=0.5323083399220431; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00836525037471417,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011299609337563688,
regressor__max_features=0.8012765819214325,
regressor__max_samples=0.6823991095722648; total time= 14.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02031957784609174,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005568532685131212,
regressor__max_features=0.5669018681397444,
regressor__max_samples=0.7092903108119677; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00010873323868111596,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007724963178874974,
regressor__max_features=0.5994281388876479,
regressor__max_samples=0.5323083399220431; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02031957784609174,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005568532685131212,
regressor__max_features=0.5669018681397444,
regressor__max_samples=0.7092903108119677; total time= 4.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00010873323868111596,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007724963178874974,
regressor__max_features=0.5994281388876479,
regressor__max_samples=0.5323083399220431; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00836525037471417,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011299609337563688,
regressor__max_features=0.8012765819214325,
regressor__max_samples=0.6823991095722648; total time= 16.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002679503866766425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006954148069488681,
regressor__max_features=0.9297826158234729,
regressor__max_samples=0.6961968607642373; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002679503866766425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006954148069488681,
regressor__max_features=0.9297826158234729,
regressor__max_samples=0.6961968607642373; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002907783928040276,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006274128510897404,
regressor__max_features=0.823045061731254,
regressor__max_samples=0.7496669174893091; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002679503866766425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006954148069488681,
regressor__max_features=0.9297826158234729,
regressor__max_samples=0.6961968607642373; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002907783928040276,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006274128510897404,
regressor__max_features=0.823045061731254,
regressor__max_samples=0.7496669174893091; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002907783928040276,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006274128510897404,
regressor__max_features=0.823045061731254,
regressor__max_samples=0.7496669174893091; total time= 3.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002679503866766425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006954148069488681,
regressor__max_features=0.9297826158234729,
regressor__max_samples=0.6961968607642373; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002907783928040276,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006274128510897404,
regressor__max_features=0.823045061731254,
regressor__max_samples=0.7496669174893091; total time= 4.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002907783928040276,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006274128510897404,
regressor__max_features=0.823045061731254,
regressor__max_samples=0.7496669174893091; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002679503866766425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.006954148069488681,
regressor__max_features=0.9297826158234729,
regressor__max_samples=0.6961968607642373; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00836525037471417,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011299609337563688,
regressor__max_features=0.8012765819214325,
regressor__max_samples=0.6823991095722648; total time= 19.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00019690091864530893,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00024076828987018033,
regressor__max_features=0.9323301960702088,
regressor__max_samples=0.6217389842024337; total time= 4.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00019690091864530893,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00024076828987018033,
regressor__max_features=0.9323301960702088,
regressor__max_samples=0.6217389842024337; total time= 4.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00019690091864530893,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00024076828987018033,
regressor__max_features=0.9323301960702088,
regressor__max_samples=0.6217389842024337; total time= 5.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020549883879480133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006888158657155729,
regressor__max_features=0.9651594424189118,
regressor__max_samples=0.6265292164003923; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00019690091864530893,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00024076828987018033,
regressor__max_features=0.9323301960702088,
regressor__max_samples=0.6217389842024337; total time= 5.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020549883879480133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006888158657155729,
regressor__max_features=0.9651594424189118,
regressor__max_samples=0.6265292164003923; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020549883879480133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006888158657155729,
regressor__max_features=0.9651594424189118,
regressor__max_samples=0.6265292164003923; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020549883879480133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006888158657155729,
regressor__max_features=0.9651594424189118,
regressor__max_samples=0.6265292164003923; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00019690091864530893,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00024076828987018033,
regressor__max_features=0.9323301960702088,
regressor__max_samples=0.6217389842024337; total time= 6.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020549883879480133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0006888158657155729,
regressor__max_features=0.9651594424189118,
regressor__max_samples=0.6265292164003923; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002423440680947825,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0030644766063940955,
regressor__max_features=0.6931744549241577,
regressor__max_samples=0.7250622162098799; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002423440680947825,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0030644766063940955,
regressor__max_features=0.6931744549241577,
regressor__max_samples=0.7250622162098799; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002423440680947825,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0030644766063940955,
regressor__max_features=0.6931744549241577,
regressor__max_samples=0.7250622162098799; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002423440680947825,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0030644766063940955,
regressor__max_features=0.6931744549241577,
regressor__max_samples=0.7250622162098799; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002423440680947825,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0030644766063940955,
regressor__max_features=0.6931744549241577,
regressor__max_samples=0.7250622162098799; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006947055273094172,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003591071612510307,
regressor__max_features=0.5750845004085208,
regressor__max_samples=0.8687168843607428; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006947055273094172,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003591071612510307,
regressor__max_features=0.5750845004085208,
regressor__max_samples=0.8687168843607428; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006947055273094172,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003591071612510307,
regressor__max_features=0.5750845004085208,
regressor__max_samples=0.8687168843607428; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006947055273094172,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003591071612510307,
regressor__max_features=0.5750845004085208,
regressor__max_samples=0.8687168843607428; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016359505922420298,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07404644196440942,
regressor__max_features=0.8687541797295035,
regressor__max_samples=0.6766257042510321; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.04745195029562856,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002025109863642414,
regressor__max_features=0.8293460731037218,
regressor__max_samples=0.9714155707066309; total time= 7.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006947055273094172,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003591071612510307,
regressor__max_features=0.5750845004085208,
regressor__max_samples=0.8687168843607428; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.04745195029562856,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002025109863642414,
regressor__max_features=0.8293460731037218,
regressor__max_samples=0.9714155707066309; total time= 7.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016359505922420298,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07404644196440942,
regressor__max_features=0.8687541797295035,
regressor__max_samples=0.6766257042510321; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016359505922420298,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07404644196440942,
regressor__max_features=0.8687541797295035,
regressor__max_samples=0.6766257042510321; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016359505922420298,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07404644196440942,
regressor__max_features=0.8687541797295035,
regressor__max_samples=0.6766257042510321; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016359505922420298,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07404644196440942,
regressor__max_features=0.8687541797295035,
regressor__max_samples=0.6766257042510321; total time= 1.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.04745195029562856,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002025109863642414,
regressor__max_features=0.8293460731037218,
regressor__max_samples=0.9714155707066309; total time= 8.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09435506683884383,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06713456341998962,
regressor__max_features=0.8919213383838369,
regressor__max_samples=0.6774206801284453; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.04745195029562856,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002025109863642414,
regressor__max_features=0.8293460731037218,
regressor__max_samples=0.9714155707066309; total time= 9.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09435506683884383,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06713456341998962,
regressor__max_features=0.8919213383838369,
regressor__max_samples=0.6774206801284453; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00019732719804111334,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031017255622707934,
regressor__max_features=0.5343564162591339,
regressor__max_samples=0.7121484429424441; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09435506683884383,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06713456341998962,
regressor__max_features=0.8919213383838369,
regressor__max_samples=0.6774206801284453; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09435506683884383,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06713456341998962,
regressor__max_features=0.8919213383838369,
regressor__max_samples=0.6774206801284453; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.04745195029562856,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002025109863642414,
regressor__max_features=0.8293460731037218,
regressor__max_samples=0.9714155707066309; total time= 9.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.09435506683884383,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06713456341998962,
regressor__max_features=0.8919213383838369,
regressor__max_samples=0.6774206801284453; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00019732719804111334,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031017255622707934,
regressor__max_features=0.5343564162591339,
regressor__max_samples=0.7121484429424441; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00019732719804111334,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031017255622707934,
regressor__max_features=0.5343564162591339,
regressor__max_samples=0.7121484429424441; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00019732719804111334,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031017255622707934,
regressor__max_features=0.5343564162591339,
regressor__max_samples=0.7121484429424441; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00019732719804111334,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031017255622707934,
regressor__max_features=0.5343564162591339,
regressor__max_samples=0.7121484429424441; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017640667019967838,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007697227670737768,
regressor__max_features=0.8846115739190668,
regressor__max_samples=0.8123317844271318; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017640667019967838,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007697227670737768,
regressor__max_features=0.8846115739190668,
regressor__max_samples=0.8123317844271318; total time= 4.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017640667019967838,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007697227670737768,
regressor__max_features=0.8846115739190668,
regressor__max_samples=0.8123317844271318; total time= 4.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017640667019967838,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007697227670737768,
regressor__max_features=0.8846115739190668,
regressor__max_samples=0.8123317844271318; total time= 4.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017640667019967838,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007697227670737768,
regressor__max_features=0.8846115739190668,
regressor__max_samples=0.8123317844271318; total time= 4.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0026485509370245113,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006392117784388237,
regressor__max_features=0.7912113291036232,
regressor__max_samples=0.874036589714766; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010047342202245233,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00023129127728359916,
regressor__max_features=0.8075064840194333,
regressor__max_samples=0.8873168903188939; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010047342202245233,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00023129127728359916,
regressor__max_features=0.8075064840194333,
regressor__max_samples=0.8873168903188939; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0021370725320543943,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001300365817395367,
regressor__max_features=0.8731036287409193,
regressor__max_samples=0.8295177671018403; total time= 14.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010047342202245233,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00023129127728359916,
regressor__max_features=0.8075064840194333,
regressor__max_samples=0.8873168903188939; total time= 8.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010047342202245233,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00023129127728359916,
regressor__max_features=0.8075064840194333,
regressor__max_samples=0.8873168903188939; total time= 7.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010047342202245233,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00023129127728359916,
regressor__max_features=0.8075064840194333,
regressor__max_samples=0.8873168903188939; total time= 7.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0026485509370245113,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006392117784388237,
regressor__max_features=0.7912113291036232,
regressor__max_samples=0.874036589714766; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0021370725320543943,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001300365817395367,
regressor__max_features=0.8731036287409193,
regressor__max_samples=0.8295177671018403; total time= 15.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0026485509370245113,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006392117784388237,
regressor__max_features=0.7912113291036232,
regressor__max_samples=0.874036589714766; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0026485509370245113,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006392117784388237,
regressor__max_features=0.7912113291036232,
regressor__max_samples=0.874036589714766; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0021370725320543943,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001300365817395367,
regressor__max_features=0.8731036287409193,
regressor__max_samples=0.8295177671018403; total time= 14.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0021370725320543943,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001300365817395367,
regressor__max_features=0.8731036287409193,
regressor__max_samples=0.8295177671018403; total time= 15.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0021370725320543943,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001300365817395367,
regressor__max_features=0.8731036287409193,
regressor__max_samples=0.8295177671018403; total time= 15.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0026485509370245113,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006392117784388237,
regressor__max_features=0.7912113291036232,
regressor__max_samples=0.874036589714766; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00019276806900515918,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004469673719814785,
regressor__max_features=0.6943263993149236,
regressor__max_samples=0.5818419007326507; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00019276806900515918,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004469673719814785,
regressor__max_features=0.6943263993149236,
regressor__max_samples=0.5818419007326507; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00019276806900515918,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004469673719814785,
regressor__max_features=0.6943263993149236,
regressor__max_samples=0.5818419007326507; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00019276806900515918,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004469673719814785,
regressor__max_features=0.6943263993149236,
regressor__max_samples=0.5818419007326507; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00019276806900515918,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004469673719814785,
regressor__max_features=0.6943263993149236,
regressor__max_samples=0.5818419007326507; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005273247257954318,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.010664655395556106,
regressor__max_features=0.7480239193399433,
regressor__max_samples=0.5063574449934467; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005273247257954318,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.010664655395556106,
regressor__max_features=0.7480239193399433,
regressor__max_samples=0.5063574449934467; total time= 1.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005273247257954318,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.010664655395556106,
regressor__max_features=0.7480239193399433,
regressor__max_samples=0.5063574449934467; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005273247257954318,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.010664655395556106,
regressor__max_features=0.7480239193399433,
regressor__max_samples=0.5063574449934467; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005273247257954318,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.010664655395556106,
regressor__max_features=0.7480239193399433,
regressor__max_samples=0.5063574449934467; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008094914969249167,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024077893646803203,
regressor__max_features=0.9209992764450308,
regressor__max_samples=0.8642453385808067; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007206433240176898,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03838044352041299,
regressor__max_features=0.923257169345139,
regressor__max_samples=0.9594632668197772; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007206433240176898,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03838044352041299,
regressor__max_features=0.923257169345139,
regressor__max_samples=0.9594632668197772; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007071298666499002,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00024226308267532445,
regressor__max_features=0.6691337529972492,
regressor__max_samples=0.9640418201630249; total time= 7.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007071298666499002,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00024226308267532445,
regressor__max_features=0.6691337529972492,
regressor__max_samples=0.9640418201630249; total time= 7.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007206433240176898,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03838044352041299,
regressor__max_features=0.923257169345139,
regressor__max_samples=0.9594632668197772; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007206433240176898,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03838044352041299,
regressor__max_features=0.923257169345139,
regressor__max_samples=0.9594632668197772; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0007206433240176898,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03838044352041299,
regressor__max_features=0.923257169345139,
regressor__max_samples=0.9594632668197772; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007071298666499002,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00024226308267532445,
regressor__max_features=0.6691337529972492,
regressor__max_samples=0.9640418201630249; total time= 7.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008094914969249167,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024077893646803203,
regressor__max_features=0.9209992764450308,
regressor__max_samples=0.8642453385808067; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007071298666499002,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00024226308267532445,
regressor__max_features=0.6691337529972492,
regressor__max_samples=0.9640418201630249; total time= 8.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07576690623833664,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013361452444615547,
regressor__max_features=0.9842443887971919,
regressor__max_samples=0.8993570958272873; total time= 13.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008094914969249167,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024077893646803203,
regressor__max_features=0.9209992764450308,
regressor__max_samples=0.8642453385808067; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008094914969249167,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024077893646803203,
regressor__max_features=0.9209992764450308,
regressor__max_samples=0.8642453385808067; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008094914969249167,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0024077893646803203,
regressor__max_features=0.9209992764450308,
regressor__max_samples=0.8642453385808067; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07576690623833664,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013361452444615547,
regressor__max_features=0.9842443887971919,
regressor__max_samples=0.8993570958272873; total time= 17.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007071298666499002,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00024226308267532445,
regressor__max_features=0.6691337529972492,
regressor__max_samples=0.9640418201630249; total time= 9.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00034816394490551296,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009123878590156009,
regressor__max_features=0.8765936679567015,
regressor__max_samples=0.6346656491510125; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00034816394490551296,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009123878590156009,
regressor__max_features=0.8765936679567015,
regressor__max_samples=0.6346656491510125; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00034816394490551296,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009123878590156009,
regressor__max_features=0.8765936679567015,
regressor__max_samples=0.6346656491510125; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00034816394490551296,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009123878590156009,
regressor__max_features=0.8765936679567015,
regressor__max_samples=0.6346656491510125; total time= 3.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00034816394490551296,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0009123878590156009,
regressor__max_features=0.8765936679567015,
regressor__max_samples=0.6346656491510125; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07576690623833664,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013361452444615547,
regressor__max_features=0.9842443887971919,
regressor__max_samples=0.8993570958272873; total time= 15.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001347476523472232,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00021078084631804208,
regressor__max_features=0.8426490080579367,
regressor__max_samples=0.5459354438550224; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07576690623833664,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013361452444615547,
regressor__max_features=0.9842443887971919,
regressor__max_samples=0.8993570958272873; total time= 18.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001347476523472232,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00021078084631804208,
regressor__max_features=0.8426490080579367,
regressor__max_samples=0.5459354438550224; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001347476523472232,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00021078084631804208,
regressor__max_features=0.8426490080579367,
regressor__max_samples=0.5459354438550224; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001347476523472232,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00021078084631804208,
regressor__max_features=0.8426490080579367,
regressor__max_samples=0.5459354438550224; total time= 4.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004370529932419393,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028315766550591993,
regressor__max_features=0.6711918180167733,
regressor__max_samples=0.7067082246476307; total time= 4.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001347476523472232,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00021078084631804208,
regressor__max_features=0.8426490080579367,
regressor__max_samples=0.5459354438550224; total time= 5.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005902596165670503,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007809897857574717,
regressor__max_features=0.782510996429508,
regressor__max_samples=0.8444426510250453; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004370529932419393,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028315766550591993,
regressor__max_features=0.6711918180167733,
regressor__max_samples=0.7067082246476307; total time= 4.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005389126150928375,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.019202802318323553,
regressor__max_features=0.5800358174772865,
regressor__max_samples=0.730778737209895; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005389126150928375,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.019202802318323553,
regressor__max_features=0.5800358174772865,
regressor__max_samples=0.730778737209895; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005389126150928375,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.019202802318323553,
regressor__max_features=0.5800358174772865,
regressor__max_samples=0.730778737209895; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005389126150928375,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.019202802318323553,
regressor__max_features=0.5800358174772865,
regressor__max_samples=0.730778737209895; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005902596165670503,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007809897857574717,
regressor__max_features=0.782510996429508,
regressor__max_samples=0.8444426510250453; total time= 4.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.027243107907277023,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04409124265444333,
regressor__max_features=0.6389510486244339,
regressor__max_samples=0.5969769336338875; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.027243107907277023,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04409124265444333,
regressor__max_features=0.6389510486244339,
regressor__max_samples=0.5969769336338875; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.027243107907277023,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04409124265444333,
regressor__max_features=0.6389510486244339,
regressor__max_samples=0.5969769336338875; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004370529932419393,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028315766550591993,
regressor__max_features=0.6711918180167733,
regressor__max_samples=0.7067082246476307; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.07576690623833664,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013361452444615547,
regressor__max_features=0.9842443887971919,
regressor__max_samples=0.8993570958272873; total time= 20.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004370529932419393,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028315766550591993,
regressor__max_features=0.6711918180167733,
regressor__max_samples=0.7067082246476307; total time= 6.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.027243107907277023,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04409124265444333,
regressor__max_features=0.6389510486244339,
regressor__max_samples=0.5969769336338875; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005389126150928375,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.019202802318323553,
regressor__max_features=0.5800358174772865,
regressor__max_samples=0.730778737209895; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.027243107907277023,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.04409124265444333,
regressor__max_features=0.6389510486244339,
regressor__max_samples=0.5969769336338875; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004370529932419393,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00028315766550591993,
regressor__max_features=0.6711918180167733,
regressor__max_samples=0.7067082246476307; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005902596165670503,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007809897857574717,
regressor__max_features=0.782510996429508,
regressor__max_samples=0.8444426510250453; total time= 5.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005902596165670503,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007809897857574717,
regressor__max_features=0.782510996429508,
regressor__max_samples=0.8444426510250453; total time= 5.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005902596165670503,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007809897857574717,
regressor__max_features=0.782510996429508,
regressor__max_samples=0.8444426510250453; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019185462808171422,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031480484521667386,
regressor__max_features=0.5055158071312329,
regressor__max_samples=0.8747727422344441; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019185462808171422,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031480484521667386,
regressor__max_features=0.5055158071312329,
regressor__max_samples=0.8747727422344441; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02513118146552806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008566171993770244,
regressor__max_features=0.8711740252862545,
regressor__max_samples=0.907559192240184; total time= 3.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02513118146552806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008566171993770244,
regressor__max_features=0.8711740252862545,
regressor__max_samples=0.907559192240184; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003665001897665076,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013359177287082529,
regressor__max_features=0.7693478336715196,
regressor__max_samples=0.7620795853630811; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02513118146552806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008566171993770244,
regressor__max_features=0.8711740252862545,
regressor__max_samples=0.907559192240184; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02513118146552806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008566171993770244,
regressor__max_features=0.8711740252862545,
regressor__max_samples=0.907559192240184; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003665001897665076,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013359177287082529,
regressor__max_features=0.7693478336715196,
regressor__max_samples=0.7620795853630811; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003665001897665076,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013359177287082529,
regressor__max_features=0.7693478336715196,
regressor__max_samples=0.7620795853630811; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019185462808171422,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031480484521667386,
regressor__max_features=0.5055158071312329,
regressor__max_samples=0.8747727422344441; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003665001897665076,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013359177287082529,
regressor__max_features=0.7693478336715196,
regressor__max_samples=0.7620795853630811; total time= 3.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.02513118146552806,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008566171993770244,
regressor__max_features=0.8711740252862545,
regressor__max_samples=0.907559192240184; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003665001897665076,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0013359177287082529,
regressor__max_features=0.7693478336715196,
regressor__max_samples=0.7620795853630811; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019185462808171422,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031480484521667386,
regressor__max_features=0.5055158071312329,
regressor__max_samples=0.8747727422344441; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019185462808171422,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031480484521667386,
regressor__max_features=0.5055158071312329,
regressor__max_samples=0.8747727422344441; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0012700541967491338,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01856923334406577,
regressor__max_features=0.9209851857483544,
regressor__max_samples=0.7351595270508928; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0012700541967491338,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01856923334406577,
regressor__max_features=0.9209851857483544,
regressor__max_samples=0.7351595270508928; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00025910076632872976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017292808117964627,
regressor__max_features=0.8478090732484034,
regressor__max_samples=0.5016091318021394; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00025910076632872976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017292808117964627,
regressor__max_features=0.8478090732484034,
regressor__max_samples=0.5016091318021394; total time= 1.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0012700541967491338,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01856923334406577,
regressor__max_features=0.9209851857483544,
regressor__max_samples=0.7351595270508928; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00025910076632872976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017292808117964627,
regressor__max_features=0.8478090732484034,
regressor__max_samples=0.5016091318021394; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00025910076632872976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017292808117964627,
regressor__max_features=0.8478090732484034,
regressor__max_samples=0.5016091318021394; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0012700541967491338,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01856923334406577,
regressor__max_features=0.9209851857483544,
regressor__max_samples=0.7351595270508928; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0012700541967491338,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01856923334406577,
regressor__max_features=0.9209851857483544,
regressor__max_samples=0.7351595270508928; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00025910076632872976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0017292808117964627,
regressor__max_features=0.8478090732484034,
regressor__max_samples=0.5016091318021394; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0974996401650133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022173379642989955,
regressor__max_features=0.8610354691013685,
regressor__max_samples=0.9430978902325373; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0974996401650133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022173379642989955,
regressor__max_features=0.8610354691013685,
regressor__max_samples=0.9430978902325373; total time= 3.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00845233258977658,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005494793328590266,
regressor__max_features=0.7807422939797619,
regressor__max_samples=0.7803300472080854; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0974996401650133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022173379642989955,
regressor__max_features=0.8610354691013685,
regressor__max_samples=0.9430978902325373; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00845233258977658,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005494793328590266,
regressor__max_features=0.7807422939797619,
regressor__max_samples=0.7803300472080854; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003167053162623584,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014555287368917943,
regressor__max_features=0.7011635623432779,
regressor__max_samples=0.9371312353051189; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00845233258977658,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005494793328590266,
regressor__max_features=0.7807422939797619,
regressor__max_samples=0.7803300472080854; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00845233258977658,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005494793328590266,
regressor__max_features=0.7807422939797619,
regressor__max_samples=0.7803300472080854; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003167053162623584,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014555287368917943,
regressor__max_features=0.7011635623432779,
regressor__max_samples=0.9371312353051189; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002882657111916091,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02599961991012661,
regressor__max_features=0.6349103600641072,
regressor__max_samples=0.912524705090031; total time= 1.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003167053162623584,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014555287368917943,
regressor__max_features=0.7011635623432779,
regressor__max_samples=0.9371312353051189; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003167053162623584,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014555287368917943,
regressor__max_features=0.7011635623432779,
regressor__max_samples=0.9371312353051189; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003167053162623584,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014555287368917943,
regressor__max_features=0.7011635623432779,
regressor__max_samples=0.9371312353051189; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0974996401650133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022173379642989955,
regressor__max_features=0.8610354691013685,
regressor__max_samples=0.9430978902325373; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00845233258977658,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.005494793328590266,
regressor__max_features=0.7807422939797619,
regressor__max_samples=0.7803300472080854; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0974996401650133,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0022173379642989955,
regressor__max_features=0.8610354691013685,
regressor__max_samples=0.9430978902325373; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002882657111916091,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02599961991012661,
regressor__max_features=0.6349103600641072,
regressor__max_samples=0.912524705090031; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002882657111916091,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02599961991012661,
regressor__max_features=0.6349103600641072,
regressor__max_samples=0.912524705090031; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002882657111916091,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02599961991012661,
regressor__max_features=0.6349103600641072,
regressor__max_samples=0.912524705090031; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002882657111916091,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.02599961991012661,
regressor__max_features=0.6349103600641072,
regressor__max_samples=0.912524705090031; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00042611417794798074,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0020921701131534514,
regressor__max_features=0.5737262633531419,
regressor__max_samples=0.6640963768046557; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00042611417794798074,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0020921701131534514,
regressor__max_features=0.5737262633531419,
regressor__max_samples=0.6640963768046557; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00042611417794798074,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0020921701131534514,
regressor__max_features=0.5737262633531419,
regressor__max_samples=0.6640963768046557; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00042611417794798074,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0020921701131534514,
regressor__max_features=0.5737262633531419,
regressor__max_samples=0.6640963768046557; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00042611417794798074,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0020921701131534514,
regressor__max_features=0.5737262633531419,
regressor__max_samples=0.6640963768046557; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015879890014017867,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008466337919401422,
regressor__max_features=0.7106240266470361,
regressor__max_samples=0.8180886823541164; total time= 2.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015879890014017867,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008466337919401422,
regressor__max_features=0.7106240266470361,
regressor__max_samples=0.8180886823541164; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015879890014017867,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008466337919401422,
regressor__max_features=0.7106240266470361,
regressor__max_samples=0.8180886823541164; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015879890014017867,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008466337919401422,
regressor__max_features=0.7106240266470361,
regressor__max_samples=0.8180886823541164; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.015879890014017867,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008466337919401422,
regressor__max_features=0.7106240266470361,
regressor__max_samples=0.8180886823541164; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003425763680444612,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016971282750263474,
regressor__max_features=0.9199011428178085,
regressor__max_samples=0.691916475966522; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003425763680444612,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016971282750263474,
regressor__max_features=0.9199011428178085,
regressor__max_samples=0.691916475966522; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005100030782890044,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00045902449145900087,
regressor__max_features=0.7991126470318695,
regressor__max_samples=0.8678315571195949; total time= 6.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005100030782890044,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00045902449145900087,
regressor__max_features=0.7991126470318695,
regressor__max_samples=0.8678315571195949; total time= 6.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003425763680444612,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016971282750263474,
regressor__max_features=0.9199011428178085,
regressor__max_samples=0.691916475966522; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005100030782890044,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00045902449145900087,
regressor__max_features=0.7991126470318695,
regressor__max_samples=0.8678315571195949; total time= 6.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003425763680444612,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016971282750263474,
regressor__max_features=0.9199011428178085,
regressor__max_samples=0.691916475966522; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005100030782890044,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00045902449145900087,
regressor__max_features=0.7991126470318695,
regressor__max_samples=0.8678315571195949; total time= 7.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003425763680444612,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016971282750263474,
regressor__max_features=0.9199011428178085,
regressor__max_samples=0.691916475966522; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0005100030782890044,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00045902449145900087,
regressor__max_features=0.7991126470318695,
regressor__max_samples=0.8678315571195949; total time= 7.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012611184468785609,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03126735402896098,
regressor__max_features=0.6365354049864063,
regressor__max_samples=0.7590393831194078; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012611184468785609,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03126735402896098,
regressor__max_features=0.6365354049864063,
regressor__max_samples=0.7590393831194078; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012611184468785609,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03126735402896098,
regressor__max_features=0.6365354049864063,
regressor__max_samples=0.7590393831194078; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008118428182187172,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014984863379378192,
regressor__max_features=0.6671191593480303,
regressor__max_samples=0.8924484885557338; total time= 10.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012611184468785609,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03126735402896098,
regressor__max_features=0.6365354049864063,
regressor__max_samples=0.7590393831194078; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.012611184468785609,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.03126735402896098,
regressor__max_features=0.6365354049864063,
regressor__max_samples=0.7590393831194078; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0709476898446109,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003576257317188646,
regressor__max_features=0.6811177208261442,
regressor__max_samples=0.6672556443709661; total time= 5.0s

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:690:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
the optimization hasn't converged yet.
  warnings.warn(
/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:690:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
the optimization hasn't converged yet.
  warnings.warn(

[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008118428182187172,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014984863379378192,
regressor__max_features=0.6671191593480303,
regressor__max_samples=0.8924484885557338; total time= 13.1s

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/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-  
packages/sklearn/neural_network/multilayer_perceptron.py:690:  
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and  
the optimization hasn't converged yet.
```

```
warnings.warn(  

```

```
[CV] END regressor__estimator__regressor__activation=relu,  
regressor__estimator__regressor__alpha=0.0709476898446109,  
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),  
regressor__estimator__regressor__learning_rate_init=0.0003576257317188646,  
regressor__max_features=0.6811177208261442,  
regressor__max_samples=0.6672556443709661; total time= 6.6s  
[CV] END regressor__estimator__regressor__activation=relu,  
regressor__estimator__regressor__alpha=0.002427719131898409,  
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),  
regressor__estimator__regressor__learning_rate_init=0.0020208831266257787,  
regressor__max_features=0.9686766471079322,  
regressor__max_samples=0.8530095038320122; total time= 3.5s  
[CV] END regressor__estimator__regressor__activation=relu,  
regressor__estimator__regressor__alpha=0.0709476898446109,  
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),  
regressor__estimator__regressor__learning_rate_init=0.0003576257317188646,  
regressor__max_features=0.6811177208261442,  
regressor__max_samples=0.6672556443709661; total time= 9.0s  
[CV] END regressor__estimator__regressor__activation=relu,  
regressor__estimator__regressor__alpha=0.002427719131898409,  
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),  
regressor__estimator__regressor__learning_rate_init=0.0020208831266257787,  
regressor__max_features=0.9686766471079322,  
regressor__max_samples=0.8530095038320122; total time= 2.5s  
[CV] END regressor__estimator__regressor__activation=relu,  
regressor__estimator__regressor__alpha=0.002427719131898409,  
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),  
regressor__estimator__regressor__learning_rate_init=0.0020208831266257787,  
regressor__max_features=0.9686766471079322,  
regressor__max_samples=0.8530095038320122; total time= 3.1s  
[CV] END regressor__estimator__regressor__activation=relu,  
regressor__estimator__regressor__alpha=0.02227131289931191,  
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),  
regressor__estimator__regressor__learning_rate_init=0.0005996395596976923,  
regressor__max_features=0.7148284063694675,  
regressor__max_samples=0.9363651252423009; total time= 6.4s  
[CV] END regressor__estimator__regressor__activation=relu,  
regressor__estimator__regressor__alpha=0.0709476898446109,  
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),  
regressor__estimator__regressor__learning_rate_init=0.0003576257317188646,  
regressor__max_features=0.6811177208261442,  
regressor__max_samples=0.6672556443709661; total time= 9.8s
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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002427719131898409,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0020208831266257787,
regressor__max_features=0.9686766471079322,
regressor__max_samples=0.8530095038320122; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02227131289931191,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005996395596976923,
regressor__max_features=0.7148284063694675,
regressor__max_samples=0.9363651252423009; total time= 6.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02227131289931191,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005996395596976923,
regressor__max_features=0.7148284063694675,
regressor__max_samples=0.9363651252423009; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02227131289931191,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005996395596976923,
regressor__max_features=0.7148284063694675,
regressor__max_samples=0.9363651252423009; total time= 6.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008118428182187172,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014984863379378192,
regressor__max_features=0.6671191593480303,
regressor__max_samples=0.8924484885557338; total time= 17.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002427719131898409,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0020208831266257787,
regressor__max_features=0.9686766471079322,
regressor__max_samples=0.8530095038320122; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00017450366043682804,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05149202729370194,
regressor__max_features=0.9546950960041981,
regressor__max_samples=0.8901065317150512; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00017450366043682804,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05149202729370194,
regressor__max_features=0.9546950960041981,
regressor__max_samples=0.8901065317150512; total time= 2.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02227131289931191,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005996395596976923,
regressor__max_features=0.7148284063694675,
regressor__max_samples=0.9363651252423009; total time= 7.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008118428182187172,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014984863379378192,
regressor__max_features=0.6671191593480303,
regressor__max_samples=0.8924484885557338; total time= 17.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.008118428182187172,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00014984863379378192,
regressor__max_features=0.6671191593480303,
regressor__max_samples=0.8924484885557338; total time= 17.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00017450366043682804,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05149202729370194,
regressor__max_features=0.9546950960041981,
regressor__max_samples=0.8901065317150512; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00017450366043682804,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05149202729370194,
regressor__max_features=0.9546950960041981,
regressor__max_samples=0.8901065317150512; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00017450366043682804,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05149202729370194,
regressor__max_features=0.9546950960041981,
regressor__max_samples=0.8901065317150512; total time= 3.5s

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/neural_network/_multilayer_perceptron.py:690:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
the optimization hasn't converged yet.
  warnings.warn(

[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019794233860884507,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00047388297559597197,
regressor__max_features=0.5656747040724533,
regressor__max_samples=0.6994180148518647; total time= 3.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0709476898446109,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003576257317188646,
regressor__max_features=0.6811177208261442,
regressor__max_samples=0.6672556443709661; total time= 12.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006716755804251612,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.052331567271268205,
regressor__max_features=0.624773099924753,
regressor__max_samples=0.6359748630643155; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006716755804251612,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.052331567271268205,
regressor__max_features=0.624773099924753,
regressor__max_samples=0.6359748630643155; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006716755804251612,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.052331567271268205,
regressor__max_features=0.624773099924753,
regressor__max_samples=0.6359748630643155; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019794233860884507,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00047388297559597197,
regressor__max_features=0.5656747040724533,
regressor__max_samples=0.6994180148518647; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019794233860884507,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00047388297559597197,
regressor__max_features=0.5656747040724533,
regressor__max_samples=0.6994180148518647; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019794233860884507,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00047388297559597197,
regressor__max_features=0.5656747040724533,
regressor__max_samples=0.6994180148518647; total time= 4.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006716755804251612,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.052331567271268205,
regressor__max_features=0.624773099924753,
regressor__max_samples=0.6359748630643155; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0019349094094120137,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.021386103823289624,
regressor__max_features=0.5326830787821926,
regressor__max_samples=0.7437855968366922; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0019349094094120137,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.021386103823289624,
regressor__max_features=0.5326830787821926,
regressor__max_samples=0.7437855968366922; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0019349094094120137,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.021386103823289624,
regressor__max_features=0.5326830787821926,
regressor__max_samples=0.7437855968366922; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0019349094094120137,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.021386103823289624,
regressor__max_features=0.5326830787821926,
regressor__max_samples=0.7437855968366922; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0019349094094120137,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.021386103823289624,
regressor__max_features=0.5326830787821926,
regressor__max_samples=0.7437855968366922; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006716755804251612,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.052331567271268205,
regressor__max_features=0.624773099924753,
regressor__max_samples=0.6359748630643155; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019794233860884507,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00047388297559597197,
regressor__max_features=0.5656747040724533,
regressor__max_samples=0.6994180148518647; total time= 5.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013788015384611475,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05239771640986394,
regressor__max_features=0.5696226855587976,
regressor__max_samples=0.7662103411376086; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013788015384611475,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05239771640986394,
regressor__max_features=0.5696226855587976,
regressor__max_samples=0.7662103411376086; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013788015384611475,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05239771640986394,
regressor__max_features=0.5696226855587976,
regressor__max_samples=0.7662103411376086; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013788015384611475,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05239771640986394,
regressor__max_features=0.5696226855587976,
regressor__max_samples=0.7662103411376086; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02068107828372359,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05006105952499202,
regressor__max_features=0.5109116983877448,
regressor__max_samples=0.8318948430877946; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00013788015384611475,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05239771640986394,
regressor__max_features=0.5696226855587976,
regressor__max_samples=0.7662103411376086; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02068107828372359,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05006105952499202,
regressor__max_features=0.5109116983877448,
regressor__max_samples=0.8318948430877946; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009351159588440742,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001610178595158641,
regressor__max_features=0.5285273605756271,
regressor__max_samples=0.6410935373466; total time= 6.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02068107828372359,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05006105952499202,
regressor__max_features=0.5109116983877448,
regressor__max_samples=0.8318948430877946; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009351159588440742,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001610178595158641,
regressor__max_features=0.5285273605756271,
regressor__max_samples=0.6410935373466; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009351159588440742,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001610178595158641,
regressor__max_features=0.5285273605756271,
regressor__max_samples=0.6410935373466; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02068107828372359,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05006105952499202,
regressor__max_features=0.5109116983877448,
regressor__max_samples=0.8318948430877946; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009351159588440742,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001610178595158641,
regressor__max_features=0.5285273605756271,
regressor__max_samples=0.6410935373466; total time= 7.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002420831855742883,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015147589935015783,
regressor__max_features=0.9139341876610865,
regressor__max_samples=0.8938565077638543; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.02068107828372359,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05006105952499202,
regressor__max_features=0.5109116983877448,
regressor__max_samples=0.8318948430877946; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002420831855742883,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015147589935015783,
regressor__max_features=0.9139341876610865,
regressor__max_samples=0.8938565077638543; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002420831855742883,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015147589935015783,
regressor__max_features=0.9139341876610865,
regressor__max_samples=0.8938565077638543; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01557624949482281,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08682654524298877,
regressor__max_features=0.5724825361899085,
regressor__max_samples=0.8625778637755575; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01557624949482281,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08682654524298877,
regressor__max_features=0.5724825361899085,
regressor__max_samples=0.8625778637755575; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009351159588440742,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0001610178595158641,
regressor__max_features=0.5285273605756271,
regressor__max_samples=0.6410935373466; total time= 8.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002420831855742883,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015147589935015783,
regressor__max_features=0.9139341876610865,
regressor__max_samples=0.8938565077638543; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01557624949482281,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08682654524298877,
regressor__max_features=0.5724825361899085,
regressor__max_samples=0.8625778637755575; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01557624949482281,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08682654524298877,
regressor__max_features=0.5724825361899085,
regressor__max_samples=0.8625778637755575; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002420831855742883,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0015147589935015783,
regressor__max_features=0.9139341876610865,
regressor__max_samples=0.8938565077638543; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01557624949482281,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08682654524298877,
regressor__max_features=0.5724825361899085,
regressor__max_samples=0.8625778637755575; total time= 2.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00021993145868038364,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014455200469547113,
regressor__max_features=0.7089949669352422,
regressor__max_samples=0.5444201225437821; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00021993145868038364,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014455200469547113,
regressor__max_features=0.7089949669352422,
regressor__max_samples=0.5444201225437821; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00021993145868038364,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014455200469547113,
regressor__max_features=0.7089949669352422,
regressor__max_samples=0.5444201225437821; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004826900703700316,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08740785565992541,
regressor__max_features=0.6137047479306894,
regressor__max_samples=0.7019889381632741; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00021993145868038364,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014455200469547113,
regressor__max_features=0.7089949669352422,
regressor__max_samples=0.5444201225437821; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00021993145868038364,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014455200469547113,
regressor__max_features=0.7089949669352422,
regressor__max_samples=0.5444201225437821; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004826900703700316,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08740785565992541,
regressor__max_features=0.6137047479306894,
regressor__max_samples=0.7019889381632741; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004826900703700316,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08740785565992541,
regressor__max_features=0.6137047479306894,
regressor__max_samples=0.7019889381632741; total time= 1.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004826900703700316,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08740785565992541,
regressor__max_features=0.6137047479306894,
regressor__max_samples=0.7019889381632741; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.004826900703700316,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08740785565992541,
regressor__max_features=0.6137047479306894,
regressor__max_samples=0.7019889381632741; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00015958978355011392,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004431145086856649,
regressor__max_features=0.8064500804309747,
regressor__max_samples=0.6130735257949704; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00015958978355011392,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004431145086856649,
regressor__max_features=0.8064500804309747,
regressor__max_samples=0.6130735257949704; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00015958978355011392,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004431145086856649,
regressor__max_features=0.8064500804309747,
regressor__max_samples=0.6130735257949704; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00015958978355011392,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004431145086856649,
regressor__max_features=0.8064500804309747,
regressor__max_samples=0.6130735257949704; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00015958978355011392,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004431145086856649,
regressor__max_features=0.8064500804309747,
regressor__max_samples=0.6130735257949704; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004965817603685986,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008435216429300589,
regressor__max_features=0.9735179938693614,
regressor__max_samples=0.9836780888941377; total time= 5.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004965817603685986,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008435216429300589,
regressor__max_features=0.9735179938693614,
regressor__max_samples=0.9836780888941377; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0024371748227955824,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008972643266447733,
regressor__max_features=0.7727159347034527,
regressor__max_samples=0.5311365529409429; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0024371748227955824,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008972643266447733,
regressor__max_features=0.7727159347034527,
regressor__max_samples=0.5311365529409429; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0024371748227955824,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008972643266447733,
regressor__max_features=0.7727159347034527,
regressor__max_samples=0.5311365529409429; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004965817603685986,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008435216429300589,
regressor__max_features=0.9735179938693614,
regressor__max_samples=0.9836780888941377; total time= 5.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004965817603685986,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008435216429300589,
regressor__max_features=0.9735179938693614,
regressor__max_samples=0.9836780888941377; total time= 6.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004965817603685986,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008435216429300589,
regressor__max_features=0.9735179938693614,
regressor__max_samples=0.9836780888941377; total time= 5.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0024371748227955824,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008972643266447733,
regressor__max_features=0.7727159347034527,
regressor__max_samples=0.5311365529409429; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0024371748227955824,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008972643266447733,
regressor__max_features=0.7727159347034527,
regressor__max_samples=0.5311365529409429; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002845043174776692,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026477616193989285,
regressor__max_features=0.8320939572527086,
regressor__max_samples=0.6080452265926537; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00021667448902860774,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0023862702825851213,
regressor__max_features=0.5259782887771343,
regressor__max_samples=0.8931390996920165; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002845043174776692,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026477616193989285,
regressor__max_features=0.8320939572527086,
regressor__max_samples=0.6080452265926537; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002845043174776692,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026477616193989285,
regressor__max_features=0.8320939572527086,
regressor__max_samples=0.6080452265926537; total time= 4.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00021667448902860774,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0023862702825851213,
regressor__max_features=0.5259782887771343,
regressor__max_samples=0.8931390996920165; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00021667448902860774,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0023862702825851213,
regressor__max_features=0.5259782887771343,
regressor__max_samples=0.8931390996920165; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00021667448902860774,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0023862702825851213,
regressor__max_features=0.5259782887771343,
regressor__max_samples=0.8931390996920165; total time= 2.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001131975521993357,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009525506970930383,
regressor__max_features=0.5808099056429357,
regressor__max_samples=0.6277466723278023; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002845043174776692,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026477616193989285,
regressor__max_features=0.8320939572527086,
regressor__max_samples=0.6080452265926537; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00021667448902860774,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0023862702825851213,
regressor__max_features=0.5259782887771343,
regressor__max_samples=0.8931390996920165; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001131975521993357,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009525506970930383,
regressor__max_features=0.5808099056429357,
regressor__max_samples=0.6277466723278023; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001131975521993357,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009525506970930383,
regressor__max_features=0.5808099056429357,
regressor__max_samples=0.6277466723278023; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001131975521993357,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009525506970930383,
regressor__max_features=0.5808099056429357,
regressor__max_samples=0.6277466723278023; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001131975521993357,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.009525506970930383,
regressor__max_features=0.5808099056429357,
regressor__max_samples=0.6277466723278023; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002845043174776692,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026477616193989285,
regressor__max_features=0.8320939572527086,
regressor__max_samples=0.6080452265926537; total time= 5.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002257969298474326,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01787263231792577,
regressor__max_features=0.7789903693913199,
regressor__max_samples=0.7762568073913626; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002257969298474326,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01787263231792577,
regressor__max_features=0.7789903693913199,
regressor__max_samples=0.7762568073913626; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002257969298474326,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01787263231792577,
regressor__max_features=0.7789903693913199,
regressor__max_samples=0.7762568073913626; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0024648706322381825,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014865690848576284,
regressor__max_features=0.8991720941121641,
regressor__max_samples=0.6859663339584324; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002257969298474326,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01787263231792577,
regressor__max_features=0.7789903693913199,
regressor__max_samples=0.7762568073913626; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0024648706322381825,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014865690848576284,
regressor__max_features=0.8991720941121641,
regressor__max_samples=0.6859663339584324; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.002257969298474326,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01787263231792577,
regressor__max_features=0.7789903693913199,
regressor__max_samples=0.7762568073913626; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0024648706322381825,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014865690848576284,
regressor__max_features=0.8991720941121641,
regressor__max_samples=0.6859663339584324; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0024648706322381825,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014865690848576284,
regressor__max_features=0.8991720941121641,
regressor__max_samples=0.6859663339584324; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0024648706322381825,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014865690848576284,
regressor__max_features=0.8991720941121641,
regressor__max_samples=0.6859663339584324; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003715257277745937,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00540480584607934,
regressor__max_features=0.8172912087751518,
regressor__max_samples=0.898957079807947; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003715257277745937,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00540480584607934,
regressor__max_features=0.8172912087751518,
regressor__max_samples=0.898957079807947; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003715257277745937,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00540480584607934,
regressor__max_features=0.8172912087751518,
regressor__max_samples=0.898957079807947; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001045520167868836,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031197447338602457,
regressor__max_features=0.6651075324357487,
regressor__max_samples=0.8783757528443175; total time= 6.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003715257277745937,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00540480584607934,
regressor__max_features=0.8172912087751518,
regressor__max_samples=0.898957079807947; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003715257277745937,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00540480584607934,
regressor__max_features=0.8172912087751518,
regressor__max_samples=0.898957079807947; total time= 1.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001045520167868836,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031197447338602457,
regressor__max_features=0.6651075324357487,
regressor__max_samples=0.8783757528443175; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001045520167868836,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031197447338602457,
regressor__max_features=0.6651075324357487,
regressor__max_samples=0.8783757528443175; total time= 6.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001045520167868836,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031197447338602457,
regressor__max_features=0.6651075324357487,
regressor__max_samples=0.8783757528443175; total time= 7.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.001045520167868836,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00031197447338602457,
regressor__max_features=0.6651075324357487,
regressor__max_samples=0.8783757528443175; total time= 6.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003868320397374223,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003972707604884497,
regressor__max_features=0.578977411417674,
regressor__max_samples=0.8479495590844013; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003868320397374223,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003972707604884497,
regressor__max_features=0.578977411417674,
regressor__max_samples=0.8479495590844013; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000314849450916637,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012282936591988124,
regressor__max_features=0.9485931704823094,
regressor__max_samples=0.5844271033388431; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003868320397374223,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003972707604884497,
regressor__max_features=0.578977411417674,
regressor__max_samples=0.8479495590844013; total time= 2.5s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003868320397374223,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003972707604884497,
regressor__max_features=0.578977411417674,
regressor__max_samples=0.8479495590844013; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0003868320397374223,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003972707604884497,
regressor__max_features=0.578977411417674,
regressor__max_samples=0.8479495590844013; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000314849450916637,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012282936591988124,
regressor__max_features=0.9485931704823094,
regressor__max_samples=0.5844271033388431; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000314849450916637,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012282936591988124,
regressor__max_features=0.9485931704823094,
regressor__max_samples=0.5844271033388431; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000314849450916637,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012282936591988124,
regressor__max_features=0.9485931704823094,
regressor__max_samples=0.5844271033388431; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029262368953835792,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00019064433991351597,
regressor__max_features=0.7205360791253919,
regressor__max_samples=0.5978046365222576; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029262368953835792,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00019064433991351597,
regressor__max_features=0.7205360791253919,
regressor__max_samples=0.5978046365222576; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029262368953835792,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00019064433991351597,
regressor__max_features=0.7205360791253919,
regressor__max_samples=0.5978046365222576; total time= 7.0s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000314849450916637,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0012282936591988124,
regressor__max_features=0.9485931704823094,
regressor__max_samples=0.5844271033388431; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029262368953835792,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00019064433991351597,
regressor__max_features=0.7205360791253919,
regressor__max_samples=0.5978046365222576; total time= 7.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002859649787424546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007012842337430562,
regressor__max_features=0.5417320424946916,
regressor__max_samples=0.9412081842036694; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002859649787424546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007012842337430562,
regressor__max_features=0.5417320424946916,
regressor__max_samples=0.9412081842036694; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002859649787424546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007012842337430562,
regressor__max_features=0.5417320424946916,
regressor__max_samples=0.9412081842036694; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002859649787424546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007012842337430562,
regressor__max_features=0.5417320424946916,
regressor__max_samples=0.9412081842036694; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002859649787424546,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.007012842337430562,
regressor__max_features=0.5417320424946916,
regressor__max_samples=0.9412081842036694; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029262368953835792,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00019064433991351597,
regressor__max_features=0.7205360791253919,
regressor__max_samples=0.5978046365222576; total time= 8.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016564983066291596,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07968559976694965,
regressor__max_features=0.7088804896375498,
regressor__max_samples=0.992055159518249; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016564983066291596,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07968559976694965,
regressor__max_features=0.7088804896375498,
regressor__max_samples=0.992055159518249; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016564983066291596,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07968559976694965,
regressor__max_features=0.7088804896375498,
regressor__max_samples=0.992055159518249; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00024517678556881594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004778230002667454,
regressor__max_features=0.7636998110613382,
regressor__max_samples=0.8596768111845793; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016564983066291596,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07968559976694965,
regressor__max_features=0.7088804896375498,
regressor__max_samples=0.992055159518249; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.016564983066291596,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07968559976694965,
regressor__max_features=0.7088804896375498,
regressor__max_samples=0.992055159518249; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00024517678556881594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004778230002667454,
regressor__max_features=0.7636998110613382,
regressor__max_samples=0.8596768111845793; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00024517678556881594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004778230002667454,
regressor__max_features=0.7636998110613382,
regressor__max_samples=0.8596768111845793; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00024517678556881594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004778230002667454,
regressor__max_features=0.7636998110613382,
regressor__max_samples=0.8596768111845793; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.014981546413452848,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003146767715820979,
regressor__max_features=0.940963879152322,
regressor__max_samples=0.7137448439603076; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.014981546413452848,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003146767715820979,
regressor__max_features=0.940963879152322,
regressor__max_samples=0.7137448439603076; total time= 5.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00024517678556881594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.004778230002667454,
regressor__max_features=0.7636998110613382,
regressor__max_samples=0.8596768111845793; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00042585214886369145,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.015649103792813424,
regressor__max_features=0.5937059720487912,
regressor__max_samples=0.929088526578582; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.014981546413452848,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003146767715820979,
regressor__max_features=0.940963879152322,
regressor__max_samples=0.7137448439603076; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00042585214886369145,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.015649103792813424,
regressor__max_features=0.5937059720487912,
regressor__max_samples=0.929088526578582; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00042585214886369145,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.015649103792813424,
regressor__max_features=0.5937059720487912,
regressor__max_samples=0.929088526578582; total time= 2.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000264322774409246,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012593626396960184,
regressor__max_features=0.6818465623780701,
regressor__max_samples=0.8663176173504163; total time= 9.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.014981546413452848,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003146767715820979,
regressor__max_features=0.940963879152322,
regressor__max_samples=0.7137448439603076; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000264322774409246,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012593626396960184,
regressor__max_features=0.6818465623780701,
regressor__max_samples=0.8663176173504163; total time= 9.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05683606455595568,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013512272526947604,
regressor__max_features=0.6571750565918741,
regressor__max_samples=0.7355839494664074; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05683606455595568,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013512272526947604,
regressor__max_features=0.6571750565918741,
regressor__max_samples=0.7355839494664074; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.014981546413452848,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003146767715820979,
regressor__max_features=0.940963879152322,
regressor__max_samples=0.7137448439603076; total time= 6.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00042585214886369145,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.015649103792813424,
regressor__max_features=0.5937059720487912,
regressor__max_samples=0.929088526578582; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05683606455595568,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013512272526947604,
regressor__max_features=0.6571750565918741,
regressor__max_samples=0.7355839494664074; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05683606455595568,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013512272526947604,
regressor__max_features=0.6571750565918741,
regressor__max_samples=0.7355839494664074; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05683606455595568,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.013512272526947604,
regressor__max_features=0.6571750565918741,
regressor__max_samples=0.7355839494664074; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00042585214886369145,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.015649103792813424,
regressor__max_features=0.5937059720487912,
regressor__max_samples=0.929088526578582; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000264322774409246,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012593626396960184,
regressor__max_features=0.6818465623780701,
regressor__max_samples=0.8663176173504163; total time= 11.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009015789351339376,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011841174901904119,
regressor__max_features=0.74710622362515,
regressor__max_samples=0.9141246262186091; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009015789351339376,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011841174901904119,
regressor__max_features=0.74710622362515,
regressor__max_samples=0.9141246262186091; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000264322774409246,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012593626396960184,
regressor__max_features=0.6818465623780701,
regressor__max_samples=0.8663176173504163; total time= 12.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009015789351339376,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011841174901904119,
regressor__max_features=0.74710622362515,
regressor__max_samples=0.9141246262186091; total time= 3.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00026663427096190485,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.016558229670925143,
regressor__max_features=0.5956302089572934,
regressor__max_samples=0.8068887094078495; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00026663427096190485,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.016558229670925143,
regressor__max_features=0.5956302089572934,
regressor__max_samples=0.8068887094078495; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000264322774409246,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012593626396960184,
regressor__max_features=0.6818465623780701,
regressor__max_samples=0.8663176173504163; total time= 13.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00026663427096190485,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.016558229670925143,
regressor__max_features=0.5956302089572934,
regressor__max_samples=0.8068887094078495; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009015789351339376,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011841174901904119,
regressor__max_features=0.74710622362515,
regressor__max_samples=0.9141246262186091; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.009015789351339376,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011841174901904119,
regressor__max_features=0.74710622362515,
regressor__max_samples=0.9141246262186091; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00026663427096190485,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.016558229670925143,
regressor__max_features=0.5956302089572934,
regressor__max_samples=0.8068887094078495; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00026663427096190485,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.016558229670925143,
regressor__max_features=0.5956302089572934,
regressor__max_samples=0.8068887094078495; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007987785883649299,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01067905163065924,
regressor__max_features=0.6625520077158495,
regressor__max_samples=0.8431635983430488; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0072909765671399455,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0163759826683439,
regressor__max_features=0.9030979405018954,
regressor__max_samples=0.9218686632033621; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007987785883649299,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01067905163065924,
regressor__max_features=0.6625520077158495,
regressor__max_samples=0.8431635983430488; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0072909765671399455,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0163759826683439,
regressor__max_features=0.9030979405018954,
regressor__max_samples=0.9218686632033621; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007987785883649299,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01067905163065924,
regressor__max_features=0.6625520077158495,
regressor__max_samples=0.8431635983430488; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0072909765671399455,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0163759826683439,
regressor__max_features=0.9030979405018954,
regressor__max_samples=0.9218686632033621; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0072909765671399455,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0163759826683439,
regressor__max_features=0.9030979405018954,
regressor__max_samples=0.9218686632033621; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0072909765671399455,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0163759826683439,
regressor__max_features=0.9030979405018954,
regressor__max_samples=0.9218686632033621; total time= 3.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007987785883649299,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01067905163065924,
regressor__max_features=0.6625520077158495,
regressor__max_samples=0.8431635983430488; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003769682188994372,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002181873789605835,
regressor__max_features=0.5435897339159156,
regressor__max_samples=0.8595387236769279; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.011091963893749005,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.036915756843628036,
regressor__max_features=0.6135897785117574,
regressor__max_samples=0.9185205626214701; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.011091963893749005,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.036915756843628036,
regressor__max_features=0.6135897785117574,
regressor__max_samples=0.9185205626214701; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007987785883649299,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01067905163065924,
regressor__max_features=0.6625520077158495,
regressor__max_samples=0.8431635983430488; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.011091963893749005,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.036915756843628036,
regressor__max_features=0.6135897785117574,
regressor__max_samples=0.9185205626214701; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003769682188994372,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002181873789605835,
regressor__max_features=0.5435897339159156,
regressor__max_samples=0.8595387236769279; total time= 7.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003769682188994372,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002181873789605835,
regressor__max_features=0.5435897339159156,
regressor__max_samples=0.8595387236769279; total time= 7.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.011091963893749005,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.036915756843628036,
regressor__max_features=0.6135897785117574,
regressor__max_samples=0.9185205626214701; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017030760274838383,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012090707659898519,
regressor__max_features=0.7563273765725265,
regressor__max_samples=0.6526553383483656; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.011091963893749005,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.036915756843628036,
regressor__max_features=0.6135897785117574,
regressor__max_samples=0.9185205626214701; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003769682188994372,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002181873789605835,
regressor__max_features=0.5435897339159156,
regressor__max_samples=0.8595387236769279; total time= 8.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017030760274838383,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012090707659898519,
regressor__max_features=0.7563273765725265,
regressor__max_samples=0.6526553383483656; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017030760274838383,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012090707659898519,
regressor__max_features=0.7563273765725265,
regressor__max_samples=0.6526553383483656; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017030760274838383,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012090707659898519,
regressor__max_features=0.7563273765725265,
regressor__max_samples=0.6526553383483656; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00017030760274838383,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.012090707659898519,
regressor__max_features=0.7563273765725265,
regressor__max_samples=0.6526553383483656; total time= 2.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004992170864478967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02319220439069824,
regressor__max_features=0.8324924517379977,
regressor__max_samples=0.6301434287830014; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004992170864478967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02319220439069824,
regressor__max_features=0.8324924517379977,
regressor__max_samples=0.6301434287830014; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004992170864478967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02319220439069824,
regressor__max_features=0.8324924517379977,
regressor__max_samples=0.6301434287830014; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004992170864478967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02319220439069824,
regressor__max_features=0.8324924517379977,
regressor__max_samples=0.6301434287830014; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0004992170864478967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.02319220439069824,
regressor__max_features=0.8324924517379977,
regressor__max_samples=0.6301434287830014; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.003769682188994372,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0002181873789605835,
regressor__max_features=0.5435897339159156,
regressor__max_samples=0.8595387236769279; total time= 9.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000478840801172029,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00436460213294378,
regressor__max_features=0.6902002744852048,
regressor__max_samples=0.9884001642553297; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04194499745820249,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004800891521038383,
regressor__max_features=0.5554947346067405,
regressor__max_samples=0.7235277719032127; total time= 2.5s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000478840801172029,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00436460213294378,
regressor__max_features=0.6902002744852048,
regressor__max_samples=0.9884001642553297; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016051522056499332,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008162800900564543,
regressor__max_features=0.8265813952656416,
regressor__max_samples=0.9691524092242414; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000478840801172029,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00436460213294378,
regressor__max_features=0.6902002744852048,
regressor__max_samples=0.9884001642553297; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000478840801172029,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00436460213294378,
regressor__max_features=0.6902002744852048,
regressor__max_samples=0.9884001642553297; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04194499745820249,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004800891521038383,
regressor__max_features=0.5554947346067405,
regressor__max_samples=0.7235277719032127; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04194499745820249,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004800891521038383,
regressor__max_features=0.5554947346067405,
regressor__max_samples=0.7235277719032127; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000478840801172029,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00436460213294378,
regressor__max_features=0.6902002744852048,
regressor__max_samples=0.9884001642553297; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04194499745820249,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004800891521038383,
regressor__max_features=0.5554947346067405,
regressor__max_samples=0.7235277719032127; total time= 2.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.04194499745820249,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.004800891521038383,
regressor__max_features=0.5554947346067405,
regressor__max_samples=0.7235277719032127; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016051522056499332,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008162800900564543,
regressor__max_features=0.8265813952656416,
regressor__max_samples=0.9691524092242414; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016051522056499332,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008162800900564543,
regressor__max_features=0.8265813952656416,
regressor__max_samples=0.9691524092242414; total time= 4.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016051522056499332,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008162800900564543,
regressor__max_features=0.8265813952656416,
regressor__max_samples=0.9691524092242414; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016051522056499332,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008162800900564543,
regressor__max_features=0.8265813952656416,
regressor__max_samples=0.9691524092242414; total time= 5.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03299003136916782,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07734479117296668,
regressor__max_features=0.8633354960351616,
regressor__max_samples=0.6523755476930666; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03299003136916782,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07734479117296668,
regressor__max_features=0.8633354960351616,
regressor__max_samples=0.6523755476930666; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03299003136916782,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07734479117296668,
regressor__max_features=0.8633354960351616,
regressor__max_samples=0.6523755476930666; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03299003136916782,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07734479117296668,
regressor__max_features=0.8633354960351616,
regressor__max_samples=0.6523755476930666; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03299003136916782,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.07734479117296668,
regressor__max_features=0.8633354960351616,
regressor__max_samples=0.6523755476930666; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00024119143325046998,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007527568069339216,
regressor__max_features=0.9553850808219576,
regressor__max_samples=0.876980951798392; total time= 4.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003963032740702592,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04152039903665235,
regressor__max_features=0.5562901952059685,
regressor__max_samples=0.8518429887303979; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003963032740702592,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04152039903665235,
regressor__max_features=0.5562901952059685,
regressor__max_samples=0.8518429887303979; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003963032740702592,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04152039903665235,
regressor__max_features=0.5562901952059685,
regressor__max_samples=0.8518429887303979; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003963032740702592,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04152039903665235,
regressor__max_features=0.5562901952059685,
regressor__max_samples=0.8518429887303979; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00024119143325046998,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007527568069339216,
regressor__max_features=0.9553850808219576,
regressor__max_samples=0.876980951798392; total time= 5.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003963032740702592,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.04152039903665235,
regressor__max_features=0.5562901952059685,
regressor__max_samples=0.8518429887303979; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00024119143325046998,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007527568069339216,
regressor__max_features=0.9553850808219576,
regressor__max_samples=0.876980951798392; total time= 5.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00024119143325046998,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007527568069339216,
regressor__max_features=0.9553850808219576,
regressor__max_samples=0.876980951798392; total time= 6.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00024119143325046998,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007527568069339216,
regressor__max_features=0.9553850808219576,
regressor__max_samples=0.876980951798392; total time= 6.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016033295859372426,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.000531718149669136,
regressor__max_features=0.5062020181386574,
regressor__max_samples=0.7343839827322529; total time= 4.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016033295859372426,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.000531718149669136,
regressor__max_features=0.5062020181386574,
regressor__max_samples=0.7343839827322529; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016033295859372426,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.000531718149669136,
regressor__max_features=0.5062020181386574,
regressor__max_samples=0.7343839827322529; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015612548975896913,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.010078921197842261,
regressor__max_features=0.7966107850536921,
regressor__max_samples=0.9461512490698261; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016033295859372426,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.000531718149669136,
regressor__max_features=0.5062020181386574,
regressor__max_samples=0.7343839827322529; total time= 5.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015612548975896913,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.010078921197842261,
regressor__max_features=0.7966107850536921,
regressor__max_samples=0.9461512490698261; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029758046924479954,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007793156956114083,
regressor__max_features=0.6499595557534923,
regressor__max_samples=0.8715965993658017; total time= 4.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015612548975896913,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.010078921197842261,
regressor__max_features=0.7966107850536921,
regressor__max_samples=0.9461512490698261; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002736592858706199,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013368233587304028,
regressor__max_features=0.8699520603834652,
regressor__max_samples=0.9590382760995753; total time= 11.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029758046924479954,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007793156956114083,
regressor__max_features=0.6499595557534923,
regressor__max_samples=0.8715965993658017; total time= 5.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029758046924479954,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007793156956114083,
regressor__max_features=0.6499595557534923,
regressor__max_samples=0.8715965993658017; total time= 5.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029758046924479954,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007793156956114083,
regressor__max_features=0.6499595557534923,
regressor__max_samples=0.8715965993658017; total time= 5.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.029758046924479954,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007793156956114083,
regressor__max_features=0.6499595557534923,
regressor__max_samples=0.8715965993658017; total time= 4.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015612548975896913,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.010078921197842261,
regressor__max_features=0.7966107850536921,
regressor__max_samples=0.9461512490698261; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002736592858706199,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013368233587304028,
regressor__max_features=0.8699520603834652,
regressor__max_samples=0.9590382760995753; total time= 12.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00015612548975896913,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.010078921197842261,
regressor__max_features=0.7966107850536921,
regressor__max_samples=0.9461512490698261; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016033295859372426,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.000531718149669136,
regressor__max_features=0.5062020181386574,
regressor__max_samples=0.7343839827322529; total time= 7.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002736592858706199,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013368233587304028,
regressor__max_features=0.8699520603834652,
regressor__max_samples=0.9590382760995753; total time= 12.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007560116967031211,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.032693133708690536,
regressor__max_features=0.6426451691786582,
regressor__max_samples=0.7145912966981784; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007560116967031211,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.032693133708690536,
regressor__max_features=0.6426451691786582,
regressor__max_samples=0.7145912966981784; total time= 2.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007560116967031211,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.032693133708690536,
regressor__max_features=0.6426451691786582,
regressor__max_samples=0.7145912966981784; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.09689749472907107,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0052486551192160065,
regressor__max_features=0.8294125350700352,
regressor__max_samples=0.6261536249197033; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.09689749472907107,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0052486551192160065,
regressor__max_features=0.8294125350700352,
regressor__max_samples=0.6261536249197033; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.09689749472907107,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0052486551192160065,
regressor__max_features=0.8294125350700352,
regressor__max_samples=0.6261536249197033; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007560116967031211,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.032693133708690536,
regressor__max_features=0.6426451691786582,
regressor__max_samples=0.7145912966981784; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.09689749472907107,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0052486551192160065,
regressor__max_features=0.8294125350700352,
regressor__max_samples=0.6261536249197033; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.09689749472907107,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0052486551192160065,
regressor__max_features=0.8294125350700352,
regressor__max_samples=0.6261536249197033; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006265208443249563,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01631913921389563,
regressor__max_features=0.7626198787600801,
regressor__max_samples=0.5163228387728342; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002736592858706199,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013368233587304028,
regressor__max_features=0.8699520603834652,
regressor__max_samples=0.9590382760995753; total time= 13.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.007560116967031211,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.032693133708690536,
regressor__max_features=0.6426451691786582,
regressor__max_samples=0.7145912966981784; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006265208443249563,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01631913921389563,
regressor__max_features=0.7626198787600801,
regressor__max_samples=0.5163228387728342; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.002736592858706199,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013368233587304028,
regressor__max_features=0.8699520603834652,
regressor__max_samples=0.9590382760995753; total time= 13.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006265208443249563,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01631913921389563,
regressor__max_features=0.7626198787600801,
regressor__max_samples=0.5163228387728342; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006265208443249563,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01631913921389563,
regressor__max_features=0.7626198787600801,
regressor__max_samples=0.5163228387728342; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.006265208443249563,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.01631913921389563,
regressor__max_features=0.7626198787600801,
regressor__max_samples=0.5163228387728342; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006626264020467989,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0021663984528695764,
regressor__max_features=0.8944011319526798,
regressor__max_samples=0.8198027805888409; total time= 2.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006626264020467989,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0021663984528695764,
regressor__max_features=0.8944011319526798,
regressor__max_samples=0.8198027805888409; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006626264020467989,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0021663984528695764,
regressor__max_features=0.8944011319526798,
regressor__max_samples=0.8198027805888409; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006626264020467989,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0021663984528695764,
regressor__max_features=0.8944011319526798,
regressor__max_samples=0.8198027805888409; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006626264020467989,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0021663984528695764,
regressor__max_features=0.8944011319526798,
regressor__max_samples=0.8198027805888409; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003444223555716761,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022921920686245958,
regressor__max_features=0.9853831032373914,
regressor__max_samples=0.6313378212993259; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020127409500000993,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00025856594491429003,
regressor__max_features=0.6704078498949233,
regressor__max_samples=0.582412337673381; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020127409500000993,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00025856594491429003,
regressor__max_features=0.6704078498949233,
regressor__max_samples=0.582412337673381; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020127409500000993,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00025856594491429003,
regressor__max_features=0.6704078498949233,
regressor__max_samples=0.582412337673381; total time= 4.5s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020127409500000993,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00025856594491429003,
regressor__max_features=0.6704078498949233,
regressor__max_samples=0.582412337673381; total time= 4.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003444223555716761,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022921920686245958,
regressor__max_features=0.9853831032373914,
regressor__max_samples=0.6313378212993259; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003444223555716761,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022921920686245958,
regressor__max_features=0.9853831032373914,
regressor__max_samples=0.6313378212993259; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003444223555716761,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022921920686245958,
regressor__max_features=0.9853831032373914,
regressor__max_samples=0.6313378212993259; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006338014079964283,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08943776513713486,
regressor__max_features=0.6344794729132355,
regressor__max_samples=0.7034859548171319; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006338014079964283,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08943776513713486,
regressor__max_features=0.6344794729132355,
regressor__max_samples=0.7034859548171319; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0020127409500000993,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00025856594491429003,
regressor__max_features=0.6704078498949233,
regressor__max_samples=0.582412337673381; total time= 5.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.003444223555716761,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.022921920686245958,
regressor__max_features=0.9853831032373914,
regressor__max_samples=0.6313378212993259; total time= 2.4s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006338014079964283,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08943776513713486,
regressor__max_features=0.6344794729132355,
regressor__max_samples=0.7034859548171319; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006338014079964283,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08943776513713486,
regressor__max_features=0.6344794729132355,
regressor__max_samples=0.7034859548171319; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006338014079964283,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.08943776513713486,
regressor__max_features=0.6344794729132355,
regressor__max_samples=0.7034859548171319; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0006767728430239078,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002245928390159994,
regressor__max_features=0.7364856730477551,
regressor__max_samples=0.8112598478648726; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03075108175863855,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011896682598382425,
regressor__max_features=0.8472252280653544,
regressor__max_samples=0.5411035318558908; total time= 9.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0006767728430239078,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002245928390159994,
regressor__max_features=0.7364856730477551,
regressor__max_samples=0.8112598478648726; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0006767728430239078,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002245928390159994,
regressor__max_features=0.7364856730477551,
regressor__max_samples=0.8112598478648726; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006797359087601082,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026476785759436787,
regressor__max_features=0.7954337206355208,
regressor__max_samples=0.7685037441756792; total time= 1.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03075108175863855,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011896682598382425,
regressor__max_features=0.8472252280653544,
regressor__max_samples=0.5411035318558908; total time= 9.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03075108175863855,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011896682598382425,
regressor__max_features=0.8472252280653544,
regressor__max_samples=0.5411035318558908; total time= 9.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0006767728430239078,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002245928390159994,
regressor__max_features=0.7364856730477551,
regressor__max_samples=0.8112598478648726; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0006767728430239078,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.002245928390159994,
regressor__max_features=0.7364856730477551,
regressor__max_samples=0.8112598478648726; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006797359087601082,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026476785759436787,
regressor__max_features=0.7954337206355208,
regressor__max_samples=0.7685037441756792; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006797359087601082,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026476785759436787,
regressor__max_features=0.7954337206355208,
regressor__max_samples=0.7685037441756792; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006797359087601082,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026476785759436787,
regressor__max_features=0.7954337206355208,
regressor__max_samples=0.7685037441756792; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03075108175863855,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011896682598382425,
regressor__max_features=0.8472252280653544,
regressor__max_samples=0.5411035318558908; total time= 10.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.006797359087601082,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0026476785759436787,
regressor__max_features=0.7954337206355208,
regressor__max_samples=0.7685037441756792; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00010543677118402842,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031619224521365785,
regressor__max_features=0.921962306305862,
regressor__max_samples=0.6424540107654804; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00010543677118402842,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031619224521365785,
regressor__max_features=0.921962306305862,
regressor__max_samples=0.6424540107654804; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03075108175863855,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011896682598382425,
regressor__max_features=0.8472252280653544,
regressor__max_samples=0.5411035318558908; total time= 11.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00010543677118402842,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031619224521365785,
regressor__max_features=0.921962306305862,
regressor__max_samples=0.6424540107654804; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00010543677118402842,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031619224521365785,
regressor__max_features=0.921962306305862,
regressor__max_samples=0.6424540107654804; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00010543677118402842,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.031619224521365785,
regressor__max_features=0.921962306305862,
regressor__max_samples=0.6424540107654804; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.025571549145433428,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00045240719480757,
regressor__max_features=0.6983984235431582,
regressor__max_samples=0.5221938860854272; total time= 3.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.025571549145433428,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00045240719480757,
regressor__max_features=0.6983984235431582,
regressor__max_samples=0.5221938860854272; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.025571549145433428,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00045240719480757,
regressor__max_features=0.6983984235431582,
regressor__max_samples=0.5221938860854272; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.025571549145433428,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00045240719480757,
regressor__max_features=0.6983984235431582,
regressor__max_samples=0.5221938860854272; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.025571549145433428,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00045240719480757,
regressor__max_features=0.6983984235431582,
regressor__max_samples=0.5221938860854272; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0015525483131886009,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007139814118592002,
regressor__max_features=0.9091033343741237,
regressor__max_samples=0.7657997323022154; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005796618819964793,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013161272918742,
regressor__max_features=0.8020358385356097,
regressor__max_samples=0.9536900823564235; total time= 10.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0015525483131886009,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007139814118592002,
regressor__max_features=0.9091033343741237,
regressor__max_samples=0.7657997323022154; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0015525483131886009,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007139814118592002,
regressor__max_features=0.9091033343741237,
regressor__max_samples=0.7657997323022154; total time= 4.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0015525483131886009,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007139814118592002,
regressor__max_features=0.9091033343741237,
regressor__max_samples=0.7657997323022154; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0015525483131886009,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0007139814118592002,
regressor__max_features=0.9091033343741237,
regressor__max_samples=0.7657997323022154; total time= 4.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005796618819964793,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013161272918742,
regressor__max_features=0.8020358385356097,
regressor__max_samples=0.9536900823564235; total time= 11.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005796618819964793,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013161272918742,
regressor__max_features=0.8020358385356097,
regressor__max_samples=0.9536900823564235; total time= 11.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005796618819964793,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013161272918742,
regressor__max_features=0.8020358385356097,
regressor__max_samples=0.9536900823564235; total time= 12.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05282420962115538,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005950038317965755,
regressor__max_features=0.8373039264581081,
regressor__max_samples=0.648363820552045; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05282420962115538,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005950038317965755,
regressor__max_features=0.8373039264581081,
regressor__max_samples=0.648363820552045; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05282420962115538,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005950038317965755,
regressor__max_features=0.8373039264581081,
regressor__max_samples=0.648363820552045; total time= 2.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005796618819964793,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013161272918742,
regressor__max_features=0.8020358385356097,
regressor__max_samples=0.9536900823564235; total time= 14.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07331309167458976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014594202008118394,
regressor__max_features=0.9314402514562061,
regressor__max_samples=0.5010519215879417; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07331309167458976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014594202008118394,
regressor__max_features=0.9314402514562061,
regressor__max_samples=0.5010519215879417; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0033284023207554064,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012274524276948825,
regressor__max_features=0.8377101037735446,
regressor__max_samples=0.5224267840117843; total time= 8.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05282420962115538,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005950038317965755,
regressor__max_features=0.8373039264581081,
regressor__max_samples=0.648363820552045; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.05282420962115538,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0005950038317965755,
regressor__max_features=0.8373039264581081,
regressor__max_samples=0.648363820552045; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07331309167458976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014594202008118394,
regressor__max_features=0.9314402514562061,
regressor__max_samples=0.5010519215879417; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002480089534395398,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003130486375675573,
regressor__max_features=0.694936176470772,
regressor__max_samples=0.8649301544242392; total time= 6.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07331309167458976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014594202008118394,
regressor__max_features=0.9314402514562061,
regressor__max_samples=0.5010519215879417; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0033284023207554064,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012274524276948825,
regressor__max_features=0.8377101037735446,
regressor__max_samples=0.5224267840117843; total time= 7.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.07331309167458976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.014594202008118394,
regressor__max_features=0.9314402514562061,
regressor__max_samples=0.5010519215879417; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0033284023207554064,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012274524276948825,
regressor__max_features=0.8377101037735446,
regressor__max_samples=0.5224267840117843; total time= 8.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002480089534395398,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003130486375675573,
regressor__max_features=0.694936176470772,
regressor__max_samples=0.8649301544242392; total time= 6.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0033284023207554064,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012274524276948825,
regressor__max_features=0.8377101037735446,
regressor__max_samples=0.5224267840117843; total time= 8.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0033284023207554064,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012274524276948825,
regressor__max_features=0.8377101037735446,
regressor__max_samples=0.5224267840117843; total time= 8.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002480089534395398,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003130486375675573,
regressor__max_features=0.694936176470772,
regressor__max_samples=0.8649301544242392; total time= 8.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014174790509488791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00972732752720773,
regressor__max_features=0.5461423960366286,
regressor__max_samples=0.7981607594301842; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002480089534395398,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003130486375675573,
regressor__max_features=0.694936176470772,
regressor__max_samples=0.8649301544242392; total time= 8.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014174790509488791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00972732752720773,
regressor__max_features=0.5461423960366286,
regressor__max_samples=0.7981607594301842; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014174790509488791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00972732752720773,
regressor__max_features=0.5461423960366286,
regressor__max_samples=0.7981607594301842; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014174790509488791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00972732752720773,
regressor__max_features=0.5461423960366286,
regressor__max_samples=0.7981607594301842; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0014174790509488791,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00972732752720773,
regressor__max_features=0.5461423960366286,
regressor__max_samples=0.7981607594301842; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01792809494176582,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016319963718854724,
regressor__max_features=0.6605405485435251,
regressor__max_samples=0.8749253822182331; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01792809494176582,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016319963718854724,
regressor__max_features=0.6605405485435251,
regressor__max_samples=0.8749253822182331; total time= 2.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01792809494176582,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016319963718854724,
regressor__max_features=0.6605405485435251,
regressor__max_samples=0.8749253822182331; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010132530977824449,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.008696615523304559,
regressor__max_features=0.9440195321800589,
regressor__max_samples=0.7993462011595487; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010132530977824449,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.008696615523304559,
regressor__max_features=0.9440195321800589,
regressor__max_samples=0.7993462011595487; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010132530977824449,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.008696615523304559,
regressor__max_features=0.9440195321800589,
regressor__max_samples=0.7993462011595487; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0002480089534395398,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003130486375675573,
regressor__max_features=0.694936176470772,
regressor__max_samples=0.8649301544242392; total time= 9.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010132530977824449,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.008696615523304559,
regressor__max_features=0.9440195321800589,
regressor__max_samples=0.7993462011595487; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01792809494176582,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016319963718854724,
regressor__max_features=0.6605405485435251,
regressor__max_samples=0.8749253822182331; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.010132530977824449,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.008696615523304559,
regressor__max_features=0.9440195321800589,
regressor__max_samples=0.7993462011595487; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.01792809494176582,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016319963718854724,
regressor__max_features=0.6605405485435251,
regressor__max_samples=0.8749253822182331; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022997215199876915,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0038902236413083766,
regressor__max_features=0.9145088662727455,
regressor__max_samples=0.9956316133668166; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022997215199876915,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0038902236413083766,
regressor__max_features=0.9145088662727455,
regressor__max_samples=0.9956316133668166; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022997215199876915,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0038902236413083766,
regressor__max_features=0.9145088662727455,
regressor__max_samples=0.9956316133668166; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010842260365421879,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003958021038337783,
regressor__max_features=0.7244265358373541,
regressor__max_samples=0.5421961902722712; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000592681895282092,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013233545430326254,
regressor__max_features=0.9357988636441152,
regressor__max_samples=0.628347114155023; total time= 7.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000592681895282092,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013233545430326254,
regressor__max_features=0.9357988636441152,
regressor__max_samples=0.628347114155023; total time= 8.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022997215199876915,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0038902236413083766,
regressor__max_features=0.9145088662727455,
regressor__max_samples=0.9956316133668166; total time= 2.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008096328841797104,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00022243996600679152,
regressor__max_features=0.5771607826415397,
regressor__max_samples=0.5859242406635472; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008096328841797104,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00022243996600679152,
regressor__max_features=0.5771607826415397,
regressor__max_samples=0.5859242406635472; total time= 4.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010842260365421879,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003958021038337783,
regressor__max_features=0.7244265358373541,
regressor__max_samples=0.5421961902722712; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00022997215199876915,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0038902236413083766,
regressor__max_features=0.9145088662727455,
regressor__max_samples=0.9956316133668166; total time= 4.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010842260365421879,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003958021038337783,
regressor__max_features=0.7244265358373541,
regressor__max_samples=0.5421961902722712; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000592681895282092,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013233545430326254,
regressor__max_features=0.9357988636441152,
regressor__max_samples=0.628347114155023; total time= 9.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008096328841797104,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00022243996600679152,
regressor__max_features=0.5771607826415397,
regressor__max_samples=0.5859242406635472; total time= 5.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000592681895282092,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013233545430326254,
regressor__max_features=0.9357988636441152,
regressor__max_samples=0.628347114155023; total time= 8.8s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010842260365421879,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003958021038337783,
regressor__max_features=0.7244265358373541,
regressor__max_samples=0.5421961902722712; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010842260365421879,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003958021038337783,
regressor__max_features=0.7244265358373541,
regressor__max_samples=0.5421961902722712; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008096328841797104,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00022243996600679152,
regressor__max_features=0.5771607826415397,
regressor__max_samples=0.5859242406635472; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009054718345594803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011953042949497715,
regressor__max_features=0.7149655386950741,
regressor__max_samples=0.8364833111972162; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009054718345594803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011953042949497715,
regressor__max_features=0.7149655386950741,
regressor__max_samples=0.8364833111972162; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009054718345594803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011953042949497715,
regressor__max_features=0.7149655386950741,
regressor__max_samples=0.8364833111972162; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00013527489855720798,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.026550205672814953,
regressor__max_features=0.6080256825058437,
regressor__max_samples=0.7333038923486079; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00013527489855720798,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.026550205672814953,
regressor__max_features=0.6080256825058437,
regressor__max_samples=0.7333038923486079; total time= 2.2s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00013527489855720798,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.026550205672814953,
regressor__max_features=0.6080256825058437,
regressor__max_samples=0.7333038923486079; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00013527489855720798,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.026550205672814953,
regressor__max_features=0.6080256825058437,
regressor__max_samples=0.7333038923486079; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00013527489855720798,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.026550205672814953,
regressor__max_features=0.6080256825058437,
regressor__max_samples=0.7333038923486079; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009054718345594803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011953042949497715,
regressor__max_features=0.7149655386950741,
regressor__max_samples=0.8364833111972162; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.009054718345594803,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.011953042949497715,
regressor__max_features=0.7149655386950741,
regressor__max_samples=0.8364833111972162; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.008096328841797104,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00022243996600679152,
regressor__max_features=0.5771607826415397,
regressor__max_samples=0.5859242406635472; total time= 6.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.000592681895282092,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00013233545430326254,
regressor__max_features=0.9357988636441152,
regressor__max_samples=0.628347114155023; total time= 11.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004242316885508213,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.023278526134547502,
regressor__max_features=0.723209758200791,
regressor__max_samples=0.8991901702591762; total time= 2.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004242316885508213,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.023278526134547502,
regressor__max_features=0.723209758200791,
regressor__max_samples=0.8991901702591762; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004242316885508213,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.023278526134547502,
regressor__max_features=0.723209758200791,
regressor__max_samples=0.8991901702591762; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004242316885508213,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.023278526134547502,
regressor__max_features=0.723209758200791,
regressor__max_samples=0.8991901702591762; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.004242316885508213,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.023278526134547502,
regressor__max_features=0.723209758200791,
regressor__max_samples=0.8991901702591762; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003166508113097114,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.008654092448309982,
regressor__max_features=0.5223892252523341,
regressor__max_samples=0.8809534100662078; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0022406296752880317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01896937741106349,
regressor__max_features=0.5494368088203176,
regressor__max_samples=0.9448260165087832; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0022406296752880317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01896937741106349,
regressor__max_features=0.5494368088203176,
regressor__max_samples=0.9448260165087832; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0022406296752880317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01896937741106349,
regressor__max_features=0.5494368088203176,
regressor__max_samples=0.9448260165087832; total time= 3.3s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003166508113097114,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.008654092448309982,
regressor__max_features=0.5223892252523341,
regressor__max_samples=0.8809534100662078; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0022406296752880317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01896937741106349,
regressor__max_features=0.5494368088203176,
regressor__max_samples=0.9448260165087832; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0022406296752880317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.01896937741106349,
regressor__max_features=0.5494368088203176,
regressor__max_samples=0.9448260165087832; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003166508113097114,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.008654092448309982,
regressor__max_features=0.5223892252523341,
regressor__max_samples=0.8809534100662078; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003166508113097114,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.008654092448309982,
regressor__max_features=0.5223892252523341,
regressor__max_samples=0.8809534100662078; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0003166508113097114,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.008654092448309982,
regressor__max_features=0.5223892252523341,
regressor__max_samples=0.8809534100662078; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00029065596597594134,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0035621987354834852,
regressor__max_features=0.6894551390294072,
regressor__max_samples=0.50335519515203; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00029065596597594134,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0035621987354834852,
regressor__max_features=0.6894551390294072,
regressor__max_samples=0.50335519515203; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00029065596597594134,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0035621987354834852,
regressor__max_features=0.6894551390294072,
regressor__max_samples=0.50335519515203; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00029065596597594134,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0035621987354834852,
regressor__max_features=0.6894551390294072,
regressor__max_samples=0.50335519515203; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00029065596597594134,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0035621987354834852,
regressor__max_features=0.6894551390294072,
regressor__max_samples=0.50335519515203; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016411023468562347,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006137147009952793,
regressor__max_features=0.5506299573075795,
regressor__max_samples=0.9235697448468925; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012410268697078425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002301467590999335,
regressor__max_features=0.8441371178901185,
regressor__max_samples=0.5702764837891804; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012410268697078425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002301467590999335,
regressor__max_features=0.8441371178901185,
regressor__max_samples=0.5702764837891804; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012410268697078425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002301467590999335,
regressor__max_features=0.8441371178901185,
regressor__max_samples=0.5702764837891804; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012410268697078425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002301467590999335,
regressor__max_features=0.8441371178901185,
regressor__max_samples=0.5702764837891804; total time= 1.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016411023468562347,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006137147009952793,
regressor__max_features=0.5506299573075795,
regressor__max_samples=0.9235697448468925; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012410268697078425,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.002301467590999335,
regressor__max_features=0.8441371178901185,
regressor__max_samples=0.5702764837891804; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016411023468562347,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006137147009952793,
regressor__max_features=0.5506299573075795,
regressor__max_samples=0.9235697448468925; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016411023468562347,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006137147009952793,
regressor__max_features=0.5506299573075795,
regressor__max_samples=0.9235697448468925; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016411023468562347,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006137147009952793,
regressor__max_features=0.5506299573075795,
regressor__max_samples=0.9235697448468925; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00048468573752918,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003942195152460661,
regressor__max_features=0.5678945235319686,
regressor__max_samples=0.5496511013768465; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00048468573752918,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003942195152460661,
regressor__max_features=0.5678945235319686,
regressor__max_samples=0.5496511013768465; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00048468573752918,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003942195152460661,
regressor__max_features=0.5678945235319686,
regressor__max_samples=0.5496511013768465; total time= 1.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00048468573752918,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003942195152460661,
regressor__max_features=0.5678945235319686,
regressor__max_samples=0.5496511013768465; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00048468573752918,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.003942195152460661,
regressor__max_features=0.5678945235319686,
regressor__max_samples=0.5496511013768465; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012987252083829638,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012085397238104457,
regressor__max_features=0.6964871126726456,
regressor__max_samples=0.7629062994366619; total time= 7.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012987252083829638,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012085397238104457,
regressor__max_features=0.6964871126726456,
regressor__max_samples=0.7629062994366619; total time= 7.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012987252083829638,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012085397238104457,
regressor__max_features=0.6964871126726456,
regressor__max_samples=0.7629062994366619; total time= 8.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0029077267483647216,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05014753025371397,
regressor__max_features=0.5580694801002628,
regressor__max_samples=0.9779594144776595; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0029077267483647216,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05014753025371397,
regressor__max_features=0.5580694801002628,
regressor__max_samples=0.9779594144776595; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0029077267483647216,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05014753025371397,
regressor__max_features=0.5580694801002628,
regressor__max_samples=0.9779594144776595; total time= 3.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0026425260997570412,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006439151294939814,
regressor__max_features=0.9133536798707438,
regressor__max_samples=0.9920065816052321; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012987252083829638,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012085397238104457,
regressor__max_features=0.6964871126726456,
regressor__max_samples=0.7629062994366619; total time= 10.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001238418333762051,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003020471966947368,
regressor__max_features=0.511676048164093,
regressor__max_samples=0.8781068069750884; total time= 4.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0029077267483647216,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05014753025371397,
regressor__max_features=0.5580694801002628,
regressor__max_samples=0.9779594144776595; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0026425260997570412,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006439151294939814,
regressor__max_features=0.9133536798707438,
regressor__max_samples=0.9920065816052321; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0029077267483647216,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.05014753025371397,
regressor__max_features=0.5580694801002628,
regressor__max_samples=0.9779594144776595; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0012987252083829638,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00012085397238104457,
regressor__max_features=0.6964871126726456,
regressor__max_samples=0.7629062994366619; total time= 10.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0026425260997570412,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006439151294939814,
regressor__max_features=0.9133536798707438,
regressor__max_samples=0.9920065816052321; total time= 3.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0026425260997570412,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006439151294939814,
regressor__max_features=0.9133536798707438,
regressor__max_samples=0.9920065816052321; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001238418333762051,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003020471966947368,
regressor__max_features=0.511676048164093,
regressor__max_samples=0.8781068069750884; total time= 6.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.08716195791917405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00458283747821586,
regressor__max_features=0.770852406205002,
regressor__max_samples=0.880950906302435; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0066165564902486125,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014738242047630224,
regressor__max_features=0.6927559523684612,
regressor__max_samples=0.6701935521272528; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0026425260997570412,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006439151294939814,
regressor__max_features=0.9133536798707438,
regressor__max_samples=0.9920065816052321; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0066165564902486125,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014738242047630224,
regressor__max_features=0.6927559523684612,
regressor__max_samples=0.6701935521272528; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001238418333762051,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003020471966947368,
regressor__max_features=0.511676048164093,
regressor__max_samples=0.8781068069750884; total time= 7.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.08716195791917405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00458283747821586,
regressor__max_features=0.770852406205002,
regressor__max_samples=0.880950906302435; total time= 1.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0066165564902486125,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014738242047630224,
regressor__max_features=0.6927559523684612,
regressor__max_samples=0.6701935521272528; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001238418333762051,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003020471966947368,
regressor__max_features=0.511676048164093,
regressor__max_samples=0.8781068069750884; total time= 7.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.08716195791917405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00458283747821586,
regressor__max_features=0.770852406205002,
regressor__max_samples=0.880950906302435; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.08716195791917405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00458283747821586,
regressor__max_features=0.770852406205002,
regressor__max_samples=0.880950906302435; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0066165564902486125,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014738242047630224,
regressor__max_features=0.6927559523684612,
regressor__max_samples=0.6701935521272528; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0066165564902486125,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014738242047630224,
regressor__max_features=0.6927559523684612,
regressor__max_samples=0.6701935521272528; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.08716195791917405,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.00458283747821586,
regressor__max_features=0.770852406205002,
regressor__max_samples=0.880950906302435; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001238418333762051,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0003020471966947368,
regressor__max_features=0.511676048164093,
regressor__max_samples=0.8781068069750884; total time= 8.8s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03278292081571619,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008072345619509482,
regressor__max_features=0.6296674486585472,
regressor__max_samples=0.597311918352436; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03278292081571619,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008072345619509482,
regressor__max_features=0.6296674486585472,
regressor__max_samples=0.597311918352436; total time= 3.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03278292081571619,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008072345619509482,
regressor__max_features=0.6296674486585472,
regressor__max_samples=0.597311918352436; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03278292081571619,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008072345619509482,
regressor__max_features=0.6296674486585472,
regressor__max_samples=0.597311918352436; total time= 4.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.03278292081571619,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0008072345619509482,
regressor__max_features=0.6296674486585472,
regressor__max_samples=0.597311918352436; total time= 3.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00041872318578279027,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09583963845085632,
regressor__max_features=0.9401573810744634,
regressor__max_samples=0.8117029790797201; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00041872318578279027,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09583963845085632,
regressor__max_features=0.9401573810744634,
regressor__max_samples=0.8117029790797201; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0033291939624350317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00064675988685946,
regressor__max_features=0.9832086929332037,
regressor__max_samples=0.7788544131250219; total time= 4.4s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0033291939624350317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00064675988685946,
regressor__max_features=0.9832086929332037,
regressor__max_samples=0.7788544131250219; total time= 4.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0033291939624350317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00064675988685946,
regressor__max_features=0.9832086929332037,
regressor__max_samples=0.7788544131250219; total time= 4.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0033291939624350317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00064675988685946,
regressor__max_features=0.9832086929332037,
regressor__max_samples=0.7788544131250219; total time= 4.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00041872318578279027,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09583963845085632,
regressor__max_features=0.9401573810744634,
regressor__max_samples=0.8117029790797201; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00041872318578279027,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09583963845085632,
regressor__max_features=0.9401573810744634,
regressor__max_samples=0.8117029790797201; total time= 1.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0033291939624350317,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00064675988685946,
regressor__max_features=0.9832086929332037,
regressor__max_samples=0.7788544131250219; total time= 4.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00041872318578279027,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.09583963845085632,
regressor__max_features=0.9401573810744634,
regressor__max_samples=0.8117029790797201; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0238541626344453,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004017319043436887,
regressor__max_features=0.6975688716321593,
regressor__max_samples=0.5197297664454497; total time= 3.6s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0238541626344453,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004017319043436887,
regressor__max_features=0.6975688716321593,
regressor__max_samples=0.5197297664454497; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0238541626344453,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004017319043436887,
regressor__max_features=0.6975688716321593,
regressor__max_samples=0.5197297664454497; total time= 3.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0238541626344453,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004017319043436887,
regressor__max_features=0.6975688716321593,
regressor__max_samples=0.5197297664454497; total time= 3.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00031313106228508846,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026157130796257666,
regressor__max_features=0.7220218746011015,
regressor__max_samples=0.8131171607105319; total time= 7.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0011491645538265378,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0407293499728615,
regressor__max_features=0.5653261719117315,
regressor__max_samples=0.8952548971374426; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0238541626344453,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004017319043436887,
regressor__max_features=0.6975688716321593,
regressor__max_samples=0.5197297664454497; total time= 4.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00031313106228508846,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026157130796257666,
regressor__max_features=0.7220218746011015,
regressor__max_samples=0.8131171607105319; total time= 8.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00031313106228508846,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026157130796257666,
regressor__max_features=0.7220218746011015,
regressor__max_samples=0.8131171607105319; total time= 8.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001490126559874452,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.042783050088531144,
regressor__max_features=0.9720210469126989,
regressor__max_samples=0.5743979847025578; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.013572321856928637,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004822244298809203,
regressor__max_features=0.9820144235827568,
regressor__max_samples=0.954727305468621; total time= 5.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001490126559874452,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.042783050088531144,
regressor__max_features=0.9720210469126989,
regressor__max_samples=0.5743979847025578; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001490126559874452,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.042783050088531144,
regressor__max_features=0.9720210469126989,
regressor__max_samples=0.5743979847025578; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001490126559874452,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.042783050088531144,
regressor__max_features=0.9720210469126989,
regressor__max_samples=0.5743979847025578; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0001490126559874452,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.042783050088531144,
regressor__max_features=0.9720210469126989,
regressor__max_samples=0.5743979847025578; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.013572321856928637,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004822244298809203,
regressor__max_features=0.9820144235827568,
regressor__max_samples=0.954727305468621; total time= 5.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0011491645538265378,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0407293499728615,
regressor__max_features=0.5653261719117315,
regressor__max_samples=0.8952548971374426; total time= 3.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.013572321856928637,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004822244298809203,
regressor__max_features=0.9820144235827568,
regressor__max_samples=0.954727305468621; total time= 6.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.013572321856928637,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004822244298809203,
regressor__max_features=0.9820144235827568,
regressor__max_samples=0.954727305468621; total time= 6.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0011491645538265378,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0407293499728615,
regressor__max_features=0.5653261719117315,
regressor__max_samples=0.8952548971374426; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0011491645538265378,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0407293499728615,
regressor__max_features=0.5653261719117315,
regressor__max_samples=0.8952548971374426; total time= 4.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00031313106228508846,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026157130796257666,
regressor__max_features=0.7220218746011015,
regressor__max_samples=0.8131171607105319; total time= 11.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00031313106228508846,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00026157130796257666,
regressor__max_features=0.7220218746011015,
regressor__max_samples=0.8131171607105319; total time= 10.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0034570626372934996,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005837065495535647,
regressor__max_features=0.687664953959801,
regressor__max_samples=0.642891777762935; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0034570626372934996,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005837065495535647,
regressor__max_features=0.687664953959801,
regressor__max_samples=0.642891777762935; total time= 1.9s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0011491645538265378,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0407293499728615,
regressor__max_features=0.5653261719117315,
regressor__max_samples=0.8952548971374426; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.013572321856928637,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0004822244298809203,
regressor__max_features=0.9820144235827568,
regressor__max_samples=0.954727305468621; total time= 7.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0034570626372934996,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005837065495535647,
regressor__max_features=0.687664953959801,
regressor__max_samples=0.642891777762935; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0034570626372934996,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005837065495535647,
regressor__max_features=0.687664953959801,
regressor__max_samples=0.642891777762935; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0034570626372934996,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.005837065495535647,
regressor__max_features=0.687664953959801,
regressor__max_samples=0.642891777762935; total time= 2.5s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015036591094157914,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08754231574262933,
regressor__max_features=0.7037921042869084,
regressor__max_samples=0.9137594610531515; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015036591094157914,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08754231574262933,
regressor__max_features=0.7037921042869084,
regressor__max_samples=0.9137594610531515; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0048768017039658735,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014441806457096897,
regressor__max_features=0.7556377311003756,
regressor__max_samples=0.746162740000591; total time= 3.0s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00594593665674591,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07381445725956805,
regressor__max_features=0.6002372577642014,
regressor__max_samples=0.5546321051548107; total time= 1.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00594593665674591,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07381445725956805,
regressor__max_features=0.6002372577642014,
regressor__max_samples=0.5546321051548107; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015036591094157914,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08754231574262933,
regressor__max_features=0.7037921042869084,
regressor__max_samples=0.9137594610531515; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015036591094157914,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08754231574262933,
regressor__max_features=0.7037921042869084,
regressor__max_samples=0.9137594610531515; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0048768017039658735,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014441806457096897,
regressor__max_features=0.7556377311003756,
regressor__max_samples=0.746162740000591; total time= 3.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0048768017039658735,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014441806457096897,
regressor__max_features=0.7556377311003756,
regressor__max_samples=0.746162740000591; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.015036591094157914,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.08754231574262933,
regressor__max_features=0.7037921042869084,
regressor__max_samples=0.9137594610531515; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0048768017039658735,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014441806457096897,
regressor__max_features=0.7556377311003756,
regressor__max_samples=0.746162740000591; total time= 3.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00594593665674591,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07381445725956805,
regressor__max_features=0.6002372577642014,
regressor__max_samples=0.5546321051548107; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010221239055667372,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03474613979443704,
regressor__max_features=0.9465451025977074,
regressor__max_samples=0.5312287968095357; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00594593665674591,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07381445725956805,
regressor__max_features=0.6002372577642014,
regressor__max_samples=0.5546321051548107; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010221239055667372,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03474613979443704,
regressor__max_features=0.9465451025977074,
regressor__max_samples=0.5312287968095357; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010221239055667372,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03474613979443704,
regressor__max_features=0.9465451025977074,
regressor__max_samples=0.5312287968095357; total time= 1.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00594593665674591,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.07381445725956805,
regressor__max_features=0.6002372577642014,
regressor__max_samples=0.5546321051548107; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010221239055667372,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03474613979443704,
regressor__max_features=0.9465451025977074,
regressor__max_samples=0.5312287968095357; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0048768017039658735,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0014441806457096897,
regressor__max_features=0.7556377311003756,
regressor__max_samples=0.746162740000591; total time= 4.1s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005863307824622438,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0033985797626583476,
regressor__max_features=0.8132962555466051,
regressor__max_samples=0.9631913033650434; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.00010221239055667372,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.03474613979443704,
regressor__max_features=0.9465451025977074,
regressor__max_samples=0.5312287968095357; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005863307824622438,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0033985797626583476,
regressor__max_features=0.8132962555466051,
regressor__max_samples=0.9631913033650434; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005863307824622438,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0033985797626583476,
regressor__max_features=0.8132962555466051,
regressor__max_samples=0.9631913033650434; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005863307824622438,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0033985797626583476,
regressor__max_features=0.8132962555466051,
regressor__max_samples=0.9631913033650434; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019399439583185935,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011565545654135936,
regressor__max_features=0.8613534788491142,
regressor__max_samples=0.846266215768554; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019399439583185935,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011565545654135936,
regressor__max_features=0.8613534788491142,
regressor__max_samples=0.846266215768554; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007900095293048748,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011916441251705794,
regressor__max_features=0.9022186038397525,
regressor__max_samples=0.6393799362072765; total time= 1.5s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019399439583185935,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011565545654135936,
regressor__max_features=0.8613534788491142,
regressor__max_samples=0.846266215768554; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007900095293048748,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011916441251705794,
regressor__max_features=0.9022186038397525,
regressor__max_samples=0.6393799362072765; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0005863307824622438,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0033985797626583476,
regressor__max_features=0.8132962555466051,
regressor__max_samples=0.9631913033650434; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019399439583185935,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011565545654135936,
regressor__max_features=0.8613534788491142,
regressor__max_samples=0.846266215768554; total time= 2.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007900095293048748,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011916441251705794,
regressor__max_features=0.9022186038397525,
regressor__max_samples=0.6393799362072765; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.019399439583185935,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.011565545654135936,
regressor__max_features=0.8613534788491142,
regressor__max_samples=0.846266215768554; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007900095293048748,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011916441251705794,
regressor__max_features=0.9022186038397525,
regressor__max_samples=0.6393799362072765; total time= 1.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.007900095293048748,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0011916441251705794,
regressor__max_features=0.9022186038397525,
regressor__max_samples=0.6393799362072765; total time= 2.1s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016109743390347128,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.018208902344655022,
regressor__max_features=0.8844613910538205,
regressor__max_samples=0.8292702578350275; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016109743390347128,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.018208902344655022,
regressor__max_features=0.8844613910538205,
regressor__max_samples=0.8292702578350275; total time= 2.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016109743390347128,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.018208902344655022,
regressor__max_features=0.8844613910538205,
regressor__max_samples=0.8292702578350275; total time= 2.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016109743390347128,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.018208902344655022,
regressor__max_features=0.8844613910538205,
regressor__max_samples=0.8292702578350275; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00016109743390347128,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.018208902344655022,
regressor__max_features=0.8844613910538205,
regressor__max_samples=0.8292702578350275; total time= 2.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.045006249817708885,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006931262436187275,
regressor__max_features=0.5443023569848395,
regressor__max_samples=0.7438139048336054; total time= 2.8s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.045006249817708885,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006931262436187275,
regressor__max_features=0.5443023569848395,
regressor__max_samples=0.7438139048336054; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.045006249817708885,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006931262436187275,
regressor__max_features=0.5443023569848395,
regressor__max_samples=0.7438139048336054; total time= 2.7s

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[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.045006249817708885,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006931262436187275,
regressor__max_features=0.5443023569848395,
regressor__max_samples=0.7438139048336054; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.045006249817708885,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.006931262436187275,
regressor__max_features=0.5443023569848395,
regressor__max_samples=0.7438139048336054; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026589221997327595,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.012395636388129811,
regressor__max_features=0.5030443741199195,
regressor__max_samples=0.5504705999887607; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026589221997327595,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.012395636388129811,
regressor__max_features=0.5030443741199195,
regressor__max_samples=0.5504705999887607; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0038524262121627495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010631385234226691,
regressor__max_features=0.998910427840989,
regressor__max_samples=0.8384106494286978; total time= 11.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026589221997327595,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.012395636388129811,
regressor__max_features=0.5030443741199195,
regressor__max_samples=0.5504705999887607; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0038524262121627495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010631385234226691,
regressor__max_features=0.998910427840989,
regressor__max_samples=0.8384106494286978; total time= 12.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0038524262121627495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010631385234226691,
regressor__max_features=0.998910427840989,
regressor__max_samples=0.8384106494286978; total time= 12.7s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0038524262121627495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010631385234226691,
regressor__max_features=0.998910427840989,
regressor__max_samples=0.8384106494286978; total time= 12.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026589221997327595,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.012395636388129811,
regressor__max_features=0.5030443741199195,
regressor__max_samples=0.5504705999887607; total time= 1.3s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.026589221997327595,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.012395636388129811,
regressor__max_features=0.5030443741199195,
regressor__max_samples=0.5504705999887607; total time= 1.1s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007919486701328569,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031759437667851233,
regressor__max_features=0.8972577722453743,
regressor__max_samples=0.853543238633349; total time= 1.4s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007919486701328569,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031759437667851233,
regressor__max_features=0.8972577722453743,
regressor__max_samples=0.853543238633349; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007919486701328569,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031759437667851233,
regressor__max_features=0.8972577722453743,
regressor__max_samples=0.853543238633349; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006934312065801004,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011843916714498214,
regressor__max_features=0.6530686704397966,
regressor__max_samples=0.7745651681385863; total time= 9.9s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007919486701328569,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031759437667851233,
regressor__max_features=0.8972577722453743,
regressor__max_samples=0.853543238633349; total time= 1.6s

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[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002726399783839976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016166642992073996,
regressor__max_features=0.6476452470239716,
regressor__max_samples=0.6161921615280239; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0038524262121627495,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00010631385234226691,
regressor__max_features=0.998910427840989,
regressor__max_samples=0.8384106494286978; total time= 14.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002726399783839976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016166642992073996,
regressor__max_features=0.6476452470239716,
regressor__max_samples=0.6161921615280239; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006934312065801004,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011843916714498214,
regressor__max_features=0.6530686704397966,
regressor__max_samples=0.7745651681385863; total time= 10.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002726399783839976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016166642992073996,
regressor__max_features=0.6476452470239716,
regressor__max_samples=0.6161921615280239; total time= 1.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0007919486701328569,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50),
regressor__estimator__regressor__learning_rate_init=0.0031759437667851233,
regressor__max_features=0.8972577722453743,
regressor__max_samples=0.853543238633349; total time= 1.7s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002726399783839976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016166642992073996,
regressor__max_features=0.6476452470239716,
regressor__max_samples=0.6161921615280239; total time= 2.6s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006934312065801004,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011843916714498214,
regressor__max_features=0.6530686704397966,
regressor__max_samples=0.7745651681385863; total time= 10.6s

```

```

[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006934312065801004,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011843916714498214,
regressor__max_features=0.6530686704397966,
regressor__max_samples=0.7745651681385863; total time= 12.2s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0002726399783839976,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.0016166642992073996,
regressor__max_features=0.6476452470239716,
regressor__max_samples=0.6161921615280239; total time= 2.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00018596724737978982,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011039409232505874,
regressor__max_features=0.868934759280765,
regressor__max_samples=0.9170722985003379; total time= 16.8s
[CV] END regressor__estimator__regressor__activation=tanh,
regressor__estimator__regressor__alpha=0.0006934312065801004,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011843916714498214,
regressor__max_features=0.6530686704397966,
regressor__max_samples=0.7745651681385863; total time= 11.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001599256308971964,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06133158112138356,
regressor__max_features=0.7025513472637832,
regressor__max_samples=0.9530555010662485; total time= 2.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001599256308971964,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06133158112138356,
regressor__max_features=0.7025513472637832,
regressor__max_samples=0.9530555010662485; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001599256308971964,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06133158112138356,
regressor__max_features=0.7025513472637832,
regressor__max_samples=0.9530555010662485; total time= 3.4s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00018596724737978982,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011039409232505874,
regressor__max_features=0.868934759280765,
regressor__max_samples=0.9170722985003379; total time= 18.8s

```

```

[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001599256308971964,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06133158112138356,
regressor__max_features=0.7025513472637832,
regressor__max_samples=0.9530555010662485; total time= 4.2s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.001599256308971964,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.06133158112138356,
regressor__max_features=0.7025513472637832,
regressor__max_samples=0.9530555010662485; total time= 3.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004404374299581594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008779147342877831,
regressor__max_features=0.5614603392326938,
regressor__max_samples=0.9443295401326993; total time= 3.1s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004404374299581594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008779147342877831,
regressor__max_features=0.5614603392326938,
regressor__max_samples=0.9443295401326993; total time= 3.3s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00018596724737978982,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011039409232505874,
regressor__max_features=0.868934759280765,
regressor__max_samples=0.9170722985003379; total time= 20.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00018596724737978982,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011039409232505874,
regressor__max_features=0.868934759280765,
regressor__max_samples=0.9170722985003379; total time= 20.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004404374299581594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008779147342877831,
regressor__max_features=0.5614603392326938,
regressor__max_samples=0.9443295401326993; total time= 2.9s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004404374299581594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008779147342877831,
regressor__max_features=0.5614603392326938,
regressor__max_samples=0.9443295401326993; total time= 3.0s

```

```

[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005620467972266967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00047652643016778405,
regressor__max_features=0.820238021087922,
regressor__max_samples=0.9894905575874045; total time= 5.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.00018596724737978982,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00011039409232505874,
regressor__max_features=0.868934759280765,
regressor__max_samples=0.9170722985003379; total time= 21.5s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.0004404374299581594,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.008779147342877831,
regressor__max_features=0.5614603392326938,
regressor__max_samples=0.9443295401326993; total time= 3.0s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005620467972266967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00047652643016778405,
regressor__max_features=0.820238021087922,
regressor__max_samples=0.9894905575874045; total time= 5.7s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005620467972266967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00047652643016778405,
regressor__max_features=0.820238021087922,
regressor__max_samples=0.9894905575874045; total time= 6.4s

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/model_selection/_search.py:412: VisibleDeprecationWarning:
Creating an ndarray from ragged nested sequences (which is a list-or-tuple of
lists-or-tuples-or ndarrays with different lengths or shapes) is deprecated. If
you meant to do this, you must specify 'dtype=object' when creating the ndarray.
    arr = np.array(param_list)

[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005620467972266967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00047652643016778405,
regressor__max_features=0.820238021087922,
regressor__max_samples=0.9894905575874045; total time= 6.6s
[CV] END regressor__estimator__regressor__activation=relu,
regressor__estimator__regressor__alpha=0.005620467972266967,
regressor__estimator__regressor__hidden_layer_sizes=(50, 50, 50, 50),
regressor__estimator__regressor__learning_rate_init=0.00047652643016778405,
regressor__max_features=0.820238021087922,

```

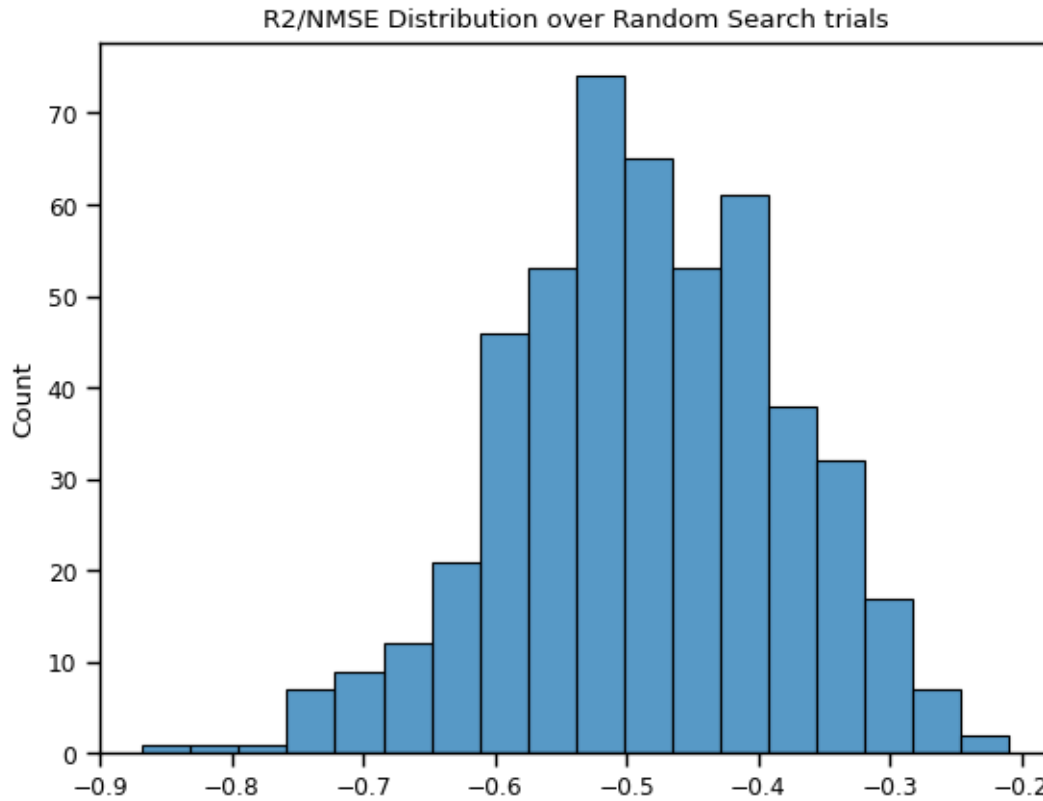
```
regressor__max_samples=0.9894905575874045; total time= 6.7s
```

```
[63]: RandomizedSearchCV(cv=5,
                        estimator=Pipeline(steps=[('scaler', StandardScaler()),
                                                  ('pca', PCA(n_components=0.98)),
                                                  ('regressor',
BaggingRegressor(bootstrap_features=True,
estimator=TransformedTargetRegressor(regressor=MLPRegressor(early_stopping=True,
                                                              hidden_layer_sizes=(100,
                                                                      100),
                                                              max_iter=1000),
transformer=StandardScaler()),
                                                              max_features=0.9,
max_samples=0.9)...
'regressor__estimator__regressor__learning_rate_init':
<scipy.stats._distn_infrastructure.rv_continuous_frozen object at
0x7f82cb20f460>,
'regressor__max_features':
<scipy.stats._distn_infrastructure.rv_continuous_frozen object at
0x7f82cb20f2e0>,
'regressor__max_samples':
<scipy.stats._distn_infrastructure.rv_continuous_frozen object at
0x7f821ff08c40>},
random_state=42,
scoring=<function custom_scorer at 0x7f82cb206310>,
verbose=2)
```

```
[64]: sns.histplot(search.cv_results_['mean_test_score'])
plt.title('R2/NMSE Distribution over Random Search trials')
```

```
[64]: Text(0.5, 1.0, 'R2/NMSE Distribution over Random Search trials')
```





```
[65]: # check the CV predictions of the best model
# worst_params = search.cv_results_['params'][np.argmin(search.
#     ↪cv_results_['mean_test_score'])]
# best_model = model_no_embedder.set_params(**worst_params)
best_model = model_no_embedder.set_params(**search.best_params_)
fold_predictions = []
fold_targets = []
fold_stds = []

kfold = KFold(n_splits=5, shuffle=True, random_state=42)
for train_index, test_index in kfold.split(X_embeddings):
    X_train, X_test = X_embeddings[train_index], X_embeddings[test_index]
    y_train, y_test = y[train_index], y[test_index]

    model = best_model.fit(X_train, y_train)
    preds, stds = predict_ensemble(model, X_test)

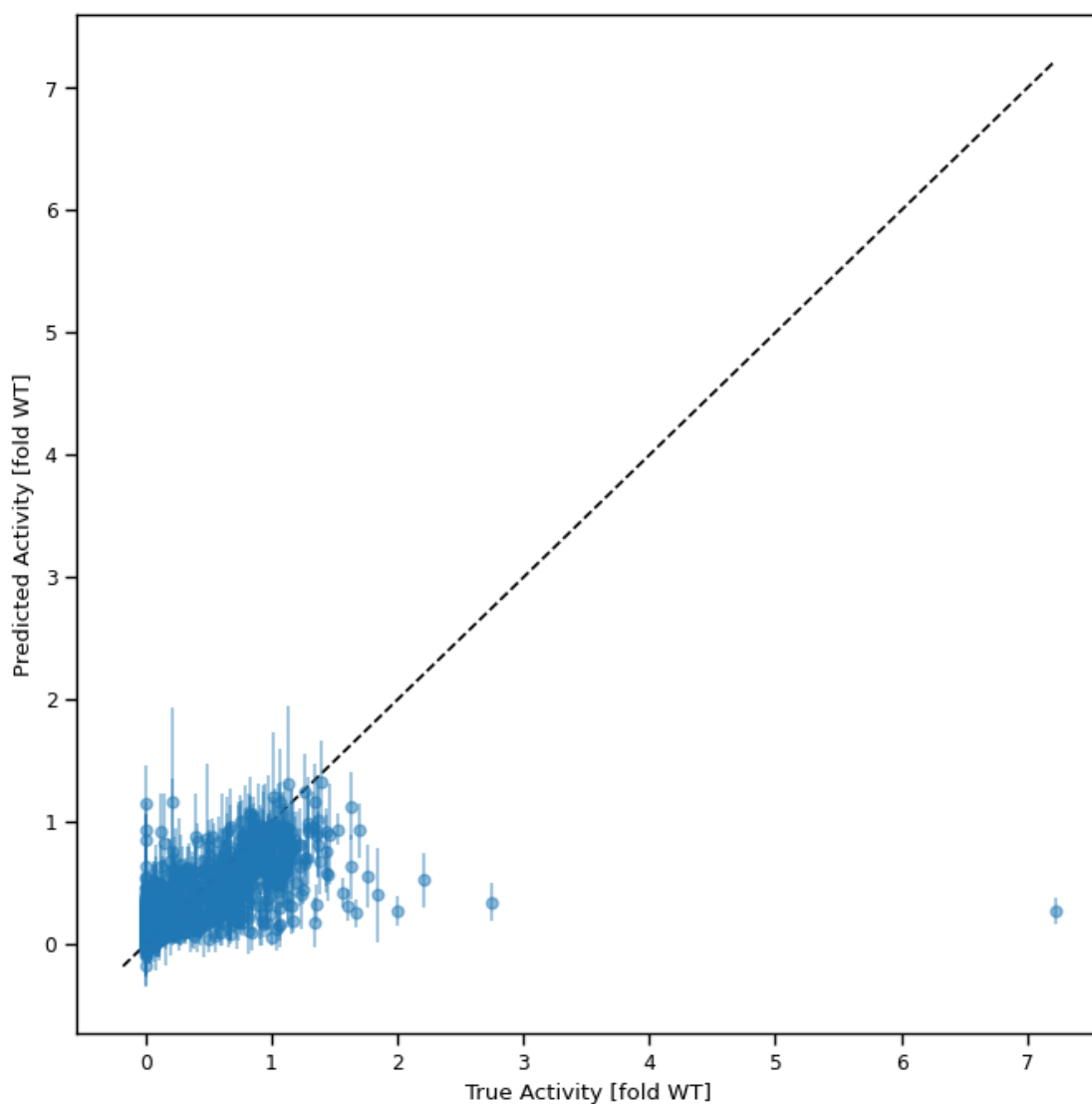
    fold_predictions.append(preds)
    fold_targets.append(y_test)
    fold_stds.append(stds)
```

```

# combine all predictions and plot
fold_predictions = np.concatenate(fold_predictions)
fold_targets = np.concatenate(fold_targets)
fold_stds = np.concatenate(fold_stds)
fig, ax = plt.subplots(figsize=(8, 8))
min_min = min(fold_targets.min(), fold_predictions.min())
max_max = max(fold_targets.max(), fold_predictions.max())
ax.plot([min_min, max_max], [min_min, max_max], color='black', linestyle='--')
ax.errorbar(fold_targets, fold_predictions, yerr=fold_stds, fmt='o', alpha=0.5)
ax.set_xlabel('True Activity [fold WT]')
ax.set_ylabel('Predicted Activity [fold WT]')

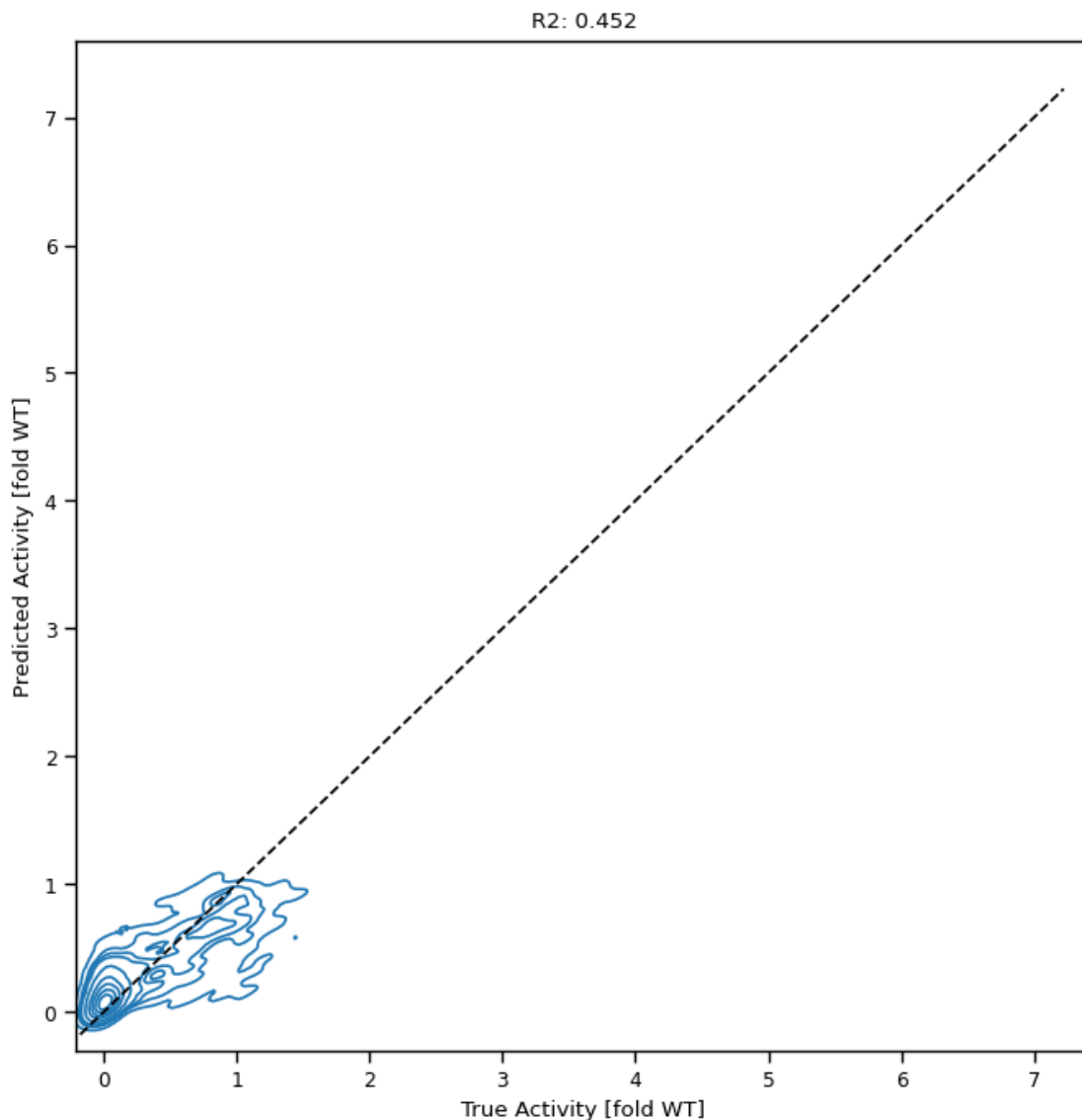
```

[65]: Text(0, 0.5, 'Predicted Activity [fold WT]')

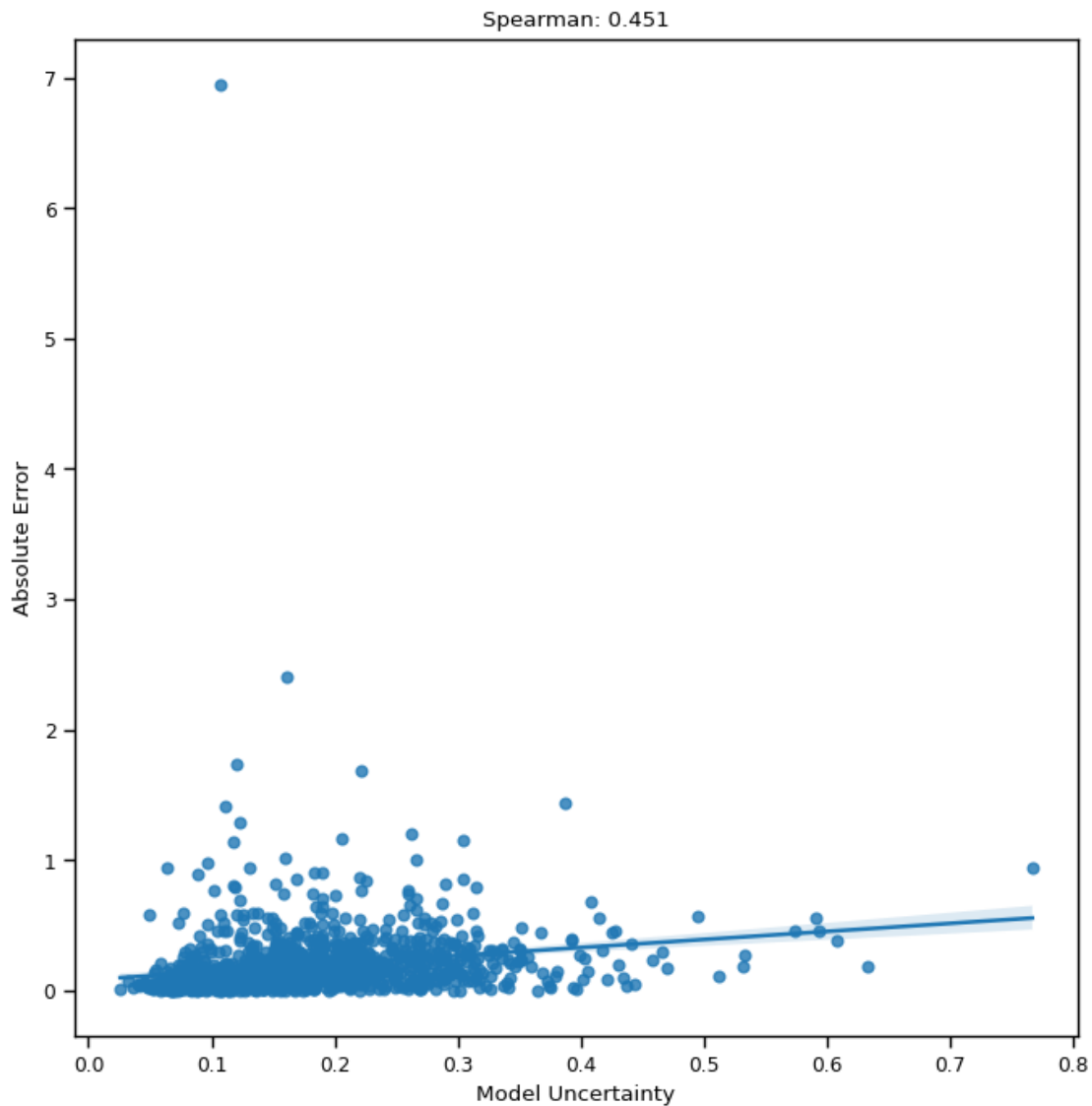


```
[66]: # use kdeplot because of too many points
r2 = r2_score(fold_targets, fold_predictions)
fig, ax = plt.subplots(figsize=(8, 8))
sns.kdeplot(x=fold_targets, y=fold_predictions, ax=ax, fill=False,
            palette='Blues', levels=10, bw_adjust=0.5)
ax.plot([min_min, max_max], [min_min, max_max], color='black', linestyle='--')
ax.set_xlabel('True Activity [fold WT]')
ax.set_ylabel('Predicted Activity [fold WT]')
ax.set_title(f'R2: {r2:.3f}')
plt.savefig('./data/r1_5fcv_model_parity.png', dpi=300, bbox_inches='tight')
```

/var/folders/4b/6yg7r65x0152cnhzpgzdhmvrjcvswz/T/ipykernel\_35182/572736163.py:4:  
 UserWarning: Ignoring `palette` because no `hue` variable has been assigned.  
 sns.kdeplot(x=fold\_targets, y=fold\_predictions, ax=ax, fill=False,  
 palette='Blues', levels=10, bw\_adjust=0.5)



```
[67]: # let's look at actual error vs model uncertainty
spearman = stats.spearmanr(fold_stds, np.abs(fold_targets - fold_predictions))
fig, ax = plt.subplots(figsize=(8, 8))
sns.regplot(x=fold_stds, y=np.abs(fold_targets - fold_predictions), ax=ax)
ax.set_xlabel('Model Uncertainty')
ax.set_ylabel('Absolute Error')
ax.set_title(f'Spearman: {spearman.correlation:.3f}')
plt.savefig('./data/r1_5fcv_model_calibration.png', dpi=300,
            bbox_inches='tight')
```



```
[68]: with open('./data/best_params.json', 'w') as f:
      json.dump(search.best_params_, f)
```

```
[69]: search.best_params_
```

```
[69]: {'regressor__estimator__regressor__activation': 'relu',
      'regressor__estimator__regressor__alpha': 0.022857832359148136,
      'regressor__estimator__regressor__hidden_layer_sizes': (50, 50, 50, 50),
      'regressor__estimator__regressor__learning_rate_init': 0.005829921230240514,
      'regressor__max_features': 0.9911652575400408,
      'regressor__max_samples': 0.9420994890657515}
```

```
[70]: predict_df = pd.DataFrame({
      'predictions': fold_predictions,
      'targets': fold_targets})
```

```
[71]: predict_df['is deleterious'] = predict_df['targets'] < 0.9
      (predict_df['predictions'][predict_df['is deleterious']]>1.0).mean()
```

```
[71]: 0.00471253534401508
```

```
[72]: predict_df['is beneficial'] = predict_df['targets'] > 1.1
      (predict_df['predictions'][predict_df['is beneficial']]<0.5).mean()
```

```
[72]: 0.234375
```

```
[73]: # kendal tau
      stats.kendalltau(fold_targets, fold_predictions)
```

```
[73]: SignificanceResult(statistic=0.5764297786782238, pvalue=2.5183837954493978e-173)
```

**3.2.3 Compare ESM sizes for speed performance** This assumes that the hyperopt from the first model size applies to other sizes.

```
[175]: model_sizes = [
      'esm2_t33_650M_UR50D',
      'esm2_t30_150M_UR50D',
      'esm2_t12_35M_UR50D',
      'esm2_t6_8M_UR50D',
      ]

      model_size_scores_dict = {}
      for model_size in model_sizes:
          esm_full = ap.ESM2Embedding(
              model_checkpoint=model_size,
              metadata_folder='esm2_embedder',
              pool=True,
              device='mps')
```

```

)
esm_full.fit([])
X_embeddings = esm_full.transform(X_seq)

kfold = KFold(n_splits=5, shuffle=True, random_state=42)
scores = []
for train_index, test_index in kfold.split(X_embeddings):
    X_train, X_test = X_embeddings[train_index], X_embeddings[test_index]
    y_train, y_test = y[train_index], y[test_index]

    oracle = model_no_embedder.set_params(**search.best_params_)
    oracle.fit(X_train, y_train)
    score = oracle.score(X_test, y_test)
    scores.append(score)
model_size_scores_dict[model_size] = np.mean(scores), np.std(scores)

```

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-packages/huggingface\_hub/file\_download.py:1132: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force\_download=True`.

warnings.warn(

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t33\_650M\_UR50D and are newly initialized:

['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-packages/huggingface\_hub/file\_download.py:1132: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force\_download=True`.

warnings.warn(

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t30\_150M\_UR50D and are newly initialized:

['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

/Users/ekomp/Documents/repos/lab/aide\_predict/aide\_predict/bespoke\_models/base.py:794: UserWarning: Cache found but model state has changed. Clearing cache.

warnings.warn("Cache found but model state has changed. Clearing cache.")

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t30\_150M\_UR50D and are newly initialized:

['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 100% | 1210/1210 [01:31<00:00, 13.24it/s]

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-packages/huggingface\_hub/file\_download.py:1132: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when

```

possible. If you want to force a new download, use `force_download=True`.
warnings.warn(
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
/Users/ekomp/Documents/repos/lab/aide_predict/aide_predict/bespoke_models/base.p
y:794: UserWarning: Cache found but model state has changed. Clearing cache.
warnings.warn("Cache found but model state has changed. Clearing cache.")
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
Computing ESM2 embeddings: 100%|      | 1210/1210 [00:31<00:00, 38.48it/s]
/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/huggingface_hub/file_download.py:1132: FutureWarning: `resume_download`
is deprecated and will be removed in version 1.0.0. Downloads always resume when
possible. If you want to force a new download, use `force_download=True`.
warnings.warn(
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t6_8M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
/Users/ekomp/Documents/repos/lab/aide_predict/aide_predict/bespoke_models/base.p
y:794: UserWarning: Cache found but model state has changed. Clearing cache.
warnings.warn("Cache found but model state has changed. Clearing cache.")
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t6_8M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
Computing ESM2 embeddings: 100%|      | 1210/1210 [00:14<00:00, 85.81it/s]

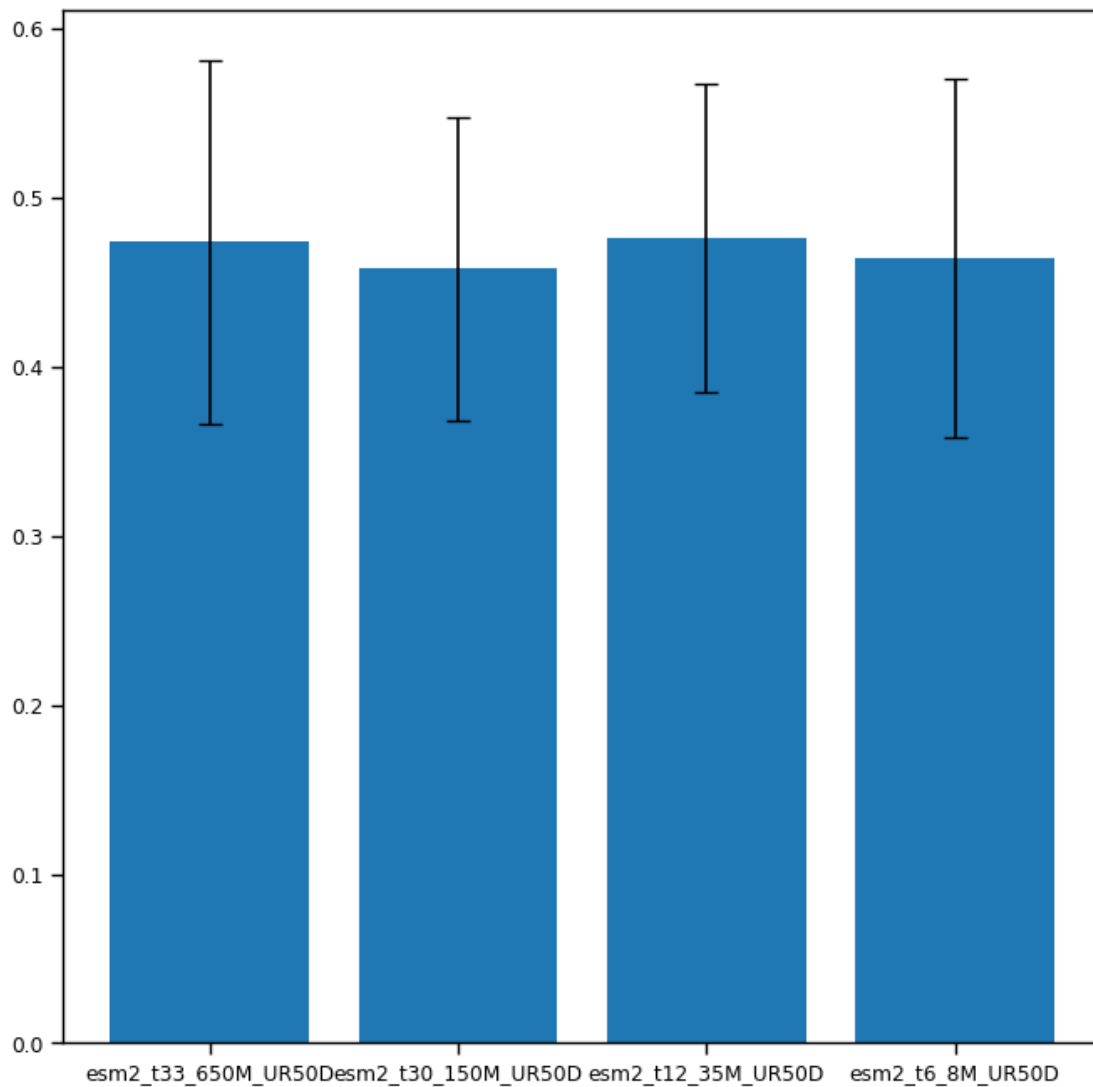
```

```
[186]: model_size_scores_dict
```

```
[186]: {'esm2_t33_650M_UR50D': (0.4737074900653944, 0.10731933340444119),
'esm2_t30_150M_UR50D': (0.45774848000856494, 0.08984262577902609),
'esm2_t12_35M_UR50D': (0.4758897433666832, 0.09149050401815977),
'esm2_t6_8M_UR50D': (0.4643347797815247, 0.10609726210803963)}
```

```
[191]: fig, ax = plt.subplots(figsize=(8, 8))
scores = np.array([v[0] for v in model_size_scores_dict.values()])
errors = np.array([v[1] for v in model_size_scores_dict.values()])
sizes = np.array(list(model_size_scores_dict.keys()))
ax.bar(x=sizes, height=scores, yerr=errors, capsize=5)
```

[191]: <BarContainer object of 4 artists>



Model size is near negligible, save maybe the smallest. Go with 35M param model.

### 1.4.3 3.3 Embedding all variants

Create datasets of schema: - **variant**: the variant sequence - **hash**: the hash of the variant - **embedding**: the embedding of the variant - **activity**: (optional) the activity of the variant - **mutations**: list of mutations compared to WT.

1. Embed the SSM data and wild type.
2. Embed plate 1
3. Embed the 2 and 3 order variants over 10 sites.



```
[74]: embedder = ap.ESM2Embedding(
        model_checkpoint='esm2_t12_35M_UR50D',
        metadata_folder='esm2_embedder',
        pool=True,
        device='mps'
    )
    embedder.fit([])
```

```
/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/huggingface_hub/file_download.py:1132: FutureWarning: `resume_download`
is deprecated and will be removed in version 1.0.0. Downloads always resume when
possible. If you want to force a new download, use `force_download=True`.
    warnings.warn(
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
```

```
[74]: ESM2Embedding(device='mps', metadata_folder='esm2_embedder',
                    model_checkpoint='esm2_t12_35M_UR50D', pool=True, use_cache=True)
```

### 3.3.1 Embed SSM, plate 1 into HF datasets

```
[75]: # add hash, mutations strings back in
    ssm_dataset = ssm_ml_df.copy()
    ssm_dataset['hash'] = ssm_dataset['variant'].apply(protein_hash)
    ssm_dataset['mutations'] = ssm_dataset['variant'].apply(lambda x:
        ↪get_mutations(wt, x))
```

```
[76]: ssm_seqs = ap.ProteinSequences.from_list(ssm_dataset['variant'].tolist())
    X_ssm = embedder.transform(ssm_seqs)
```

```
/Users/ekomp/Documents/repos/lab/aide_predict/aide_predict/bespoke_models/base.p
y:794: UserWarning: Cache found but model state has changed. Clearing cache.
    warnings.warn("Cache found but model state has changed. Clearing cache.")
/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/huggingface_hub/file_download.py:1132: FutureWarning: `resume_download`
is deprecated and will be removed in version 1.0.0. Downloads always resume when
possible. If you want to force a new download, use `force_download=True`.
    warnings.warn(
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
Computing ESM2 embeddings: 100%|          | 1210/1210 [00:31<00:00, 38.76it/s]
```

```
[77]: dataset_dict = {
        'hash': ssm_dataset['hash'].to_list(),
        'variant': ssm_dataset['variant'].to_list(),
        'activity': ssm_dataset['activity'].to_list(),
        'embedding': X_ssm,
        'mutations': ssm_dataset['mutations'].to_list(),
    }
    ssm_dataset = datasets.Dataset.from_dict(dataset_dict)
    del X_ssm
```

```
[78]: ssm_dataset.save_to_disk('./data/hf_ssm_dataset')
```

Saving the dataset (0/1 shards): 0% | 0/1210 [00:00<?, ? examples/s]

```
[79]: plate1_seqs = ap.ProteinSequences.from_list(plate1_df['variant'].tolist())
    X_plate1 = embedder.transform(plate1_seqs)
```

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 100% | 96/96 [00:02<00:00, 36.13it/s]

```
[80]: dataset_dict = {
        'hash': plate1_df['hash'].to_list(),
        'variant': plate1_df['variant'].to_list(),
        'activity': plate1_df['additive_activity'].to_list(),
        'embedding': X_plate1,
        'mutations': plate1_df['mutations'].to_list(),
        'type': plate1_df['type'].to_list()
    }
    plate1_dataset = datasets.Dataset.from_dict(dataset_dict)
    del X_plate1
```

```
[81]: plate1_dataset.save_to_disk('./data/hf_plate1_dataset')
```

Saving the dataset (0/1 shards): 0% | 0/96 [00:00<?, ? examples/s]

**3.3.2 Produce all possible 2 and 3 order mutants and embed them** We define some iterators to iterate over combinations of mutations at 2 and 3 depth. We use these to generate variants and embeddings, skipping those that are already in the SSM data or plate 1. We lastly filter for redundant variants due to selecting WT AA.

```
[82]: putative_mutatable_positions_wt_offset_zero_index
```

```
[82]: array([388, 72, 545, 409, 550, 558, 392, 559, 414, 396])
```

```
[83]: def mutation_combinations(n_positions, n_options, max_order=2):
    """Generate tuples of ints representing mutations at n_positions with
    ↪n_options options.

    Ints are from 0. Map back to actual positions as post processing.
    """

    to_yield = [None for _ in range(n_positions)]

    for positions in itertools.combinations(range(n_positions), max_order):
        for options in itertools.product(range(n_options),
        ↪repeat=len(positions)):
            for i, pos in enumerate(positions):
                to_yield[pos] = options[i]
            yield tuple(to_yield)
            to_yield = [None for _ in range(n_positions)]

[84]: def sequence_string_from_mutation_combination_iterator(
    wt_sequence, positions_mutating, max_order=2
):
    """Generate a sequence strings by iterating over mutation combinations

    Parameters
    -----
    wt_sequence : str
        The wild-type sequence
    positions_mutating : list
        List of positions to mutate, 0 indexed
    max_order : int
        Maximum order of mutations to consider
    """
    AA_LIST = 'ACDEFGHIKLMNPQRSTVWY'
    n_positions = len(positions_mutating)
    n_options = len(AA_LIST)

    for mutation_combination in mutation_combinations(n_positions, n_options,
    ↪max_order):
        sequence = list(wt_sequence)
        mutation_strings = []
        for sequence_index, mutation_index in zip(positions_mutating,
        ↪mutation_combination):
            if mutation_index is None:
                continue
            if sequence[sequence_index] == AA_LIST[mutation_index]:
                continue
            sequence[sequence_index] = AA_LIST[mutation_index]
```

```

        mutation_strings.
    ↪append(f'{wt_sequence[sequence_index]}{sequence_index+POSITION_OFFSET+1}-{AA_LIST[mutation_i
        hash_ = hashlib.md5('').join(sequence).encode()).hexdigest()
    ↪yield ''.join(sequence), hash_, mutation_strings

```

```

[85]: def multiple_mutation_orders_sequence_string_combination_iterator(
        wt_sequence, positions_mutating, max_order=3
    ):
        """Runs sequence_string_from_mutation_combination_iterator over multiple_
        ↪mutation depths"""
        for order in range(2, max_order+1):
            for sequence, hash_, mutation_strings in_
    ↪sequence_string_from_mutation_combination_iterator(wt_sequence,
    ↪positions_mutating, order):
                yield sequence, hash_, mutation_strings

```

```

[87]: # embed them
block_size = 1000
# get the hashes already in the dataset
existing_hashes = set(ssm_dataset['hash']).union(set(plate1_dataset['hash']))

def dataset_generator():
    """Generate a dataset of all possible 2-3 mutations not already sampled
    """

    bar = tqdm()

    block = {
        'mutations': [],
        'hash': [],
        'variant': [],
        'embedding': None
    }
    for i, (sequence, hash_, mutation_strings) in_
    ↪enumerate(multiple_mutation_orders_sequence_string_combination_iterator(wt,
    ↪putative_mutatable_positions_wt_offset_zero_index, max_order=3)):
        if hash_ in existing_hashes:
            continue

        block['hash'].append(hash_)
        block['mutations'].append(mutation_strings)
        block['variant'].append(sequence)
        if len(block['hash']) == block_size:
            block['embedding'] = embedder.transform(block['variant'])

            for i in range(len(block['hash'])):
                example = {

```

```

        'mutations': block['mutations'][i],
        'hash': block['hash'][i],
        'variant': block['variant'][i],
        'embedding': block['embedding'][i]
    }
    yield example

    block = {
        'mutations': [],
        'hash': [],
        'variant': [],
        'embedding': None
    }
    bar.update(block_size)

```

```
[88]: variants_dataset = datasets.Dataset.from_generator(dataset_generator)
```

Generating train split: 0 examples [00:00, ? examples/s]

0it [00:00, ?it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```

Computing ESM2 embeddings:  0%|          | 0/1000 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|          | 32/1000 [00:00<00:25, 38.31it/s]
Computing ESM2 embeddings:  6%|          | 64/1000 [00:01<00:24, 38.60it/s]
Computing ESM2 embeddings: 10%|         | 96/1000 [00:02<00:23, 39.07it/s]
Computing ESM2 embeddings: 13%|         | 128/1000 [00:03<00:22, 39.34it/s]
Computing ESM2 embeddings: 16%|         | 160/1000 [00:04<00:21, 39.50it/s]
Computing ESM2 embeddings: 19%|         | 192/1000 [00:04<00:20, 39.58it/s]
Computing ESM2 embeddings: 22%|         | 224/1000 [00:05<00:19, 39.67it/s]
Computing ESM2 embeddings: 26%|         | 256/1000 [00:06<00:18, 39.71it/s]
Computing ESM2 embeddings: 29%|         | 288/1000 [00:07<00:18, 39.49it/s]
Computing ESM2 embeddings: 32%|         | 320/1000 [00:08<00:17, 39.47it/s]

```

```

Computing ESM2 embeddings: 35%|          | 352/1000 [00:08<00:16, 39.53it/s]
Computing ESM2 embeddings: 38%|          | 384/1000 [00:09<00:15, 39.41it/s]
Computing ESM2 embeddings: 42%|          | 416/1000 [00:10<00:14, 39.33it/s]
Computing ESM2 embeddings: 45%|          | 448/1000 [00:11<00:14, 39.27it/s]
Computing ESM2 embeddings: 48%|          | 480/1000 [00:12<00:13, 39.18it/s]
Computing ESM2 embeddings: 51%|          | 512/1000 [00:13<00:12, 39.18it/s]
Computing ESM2 embeddings: 54%|          | 544/1000 [00:13<00:11, 39.13it/s]
Computing ESM2 embeddings: 58%|          | 576/1000 [00:14<00:10, 39.12it/s]
Computing ESM2 embeddings: 61%|          | 608/1000 [00:15<00:10, 39.13it/s]
Computing ESM2 embeddings: 64%|          | 640/1000 [00:16<00:09, 39.10it/s]
Computing ESM2 embeddings: 67%|          | 672/1000 [00:17<00:08, 39.03it/s]
Computing ESM2 embeddings: 70%|          | 704/1000 [00:17<00:07, 39.09it/s]
Computing ESM2 embeddings: 74%|          | 736/1000 [00:18<00:06, 39.06it/s]
Computing ESM2 embeddings: 77%|          | 768/1000 [00:19<00:06, 38.55it/s]
Computing ESM2 embeddings: 80%|          | 800/1000 [00:20<00:05, 38.76it/s]
Computing ESM2 embeddings: 83%|          | 832/1000 [00:21<00:04, 38.90it/s]
Computing ESM2 embeddings: 86%|          | 864/1000 [00:22<00:03, 38.86it/s]
Computing ESM2 embeddings: 90%|          | 896/1000 [00:22<00:02, 38.83it/s]
Computing ESM2 embeddings: 93%|          | 928/1000 [00:23<00:01, 38.98it/s]
Computing ESM2 embeddings: 96%|          | 960/1000 [00:24<00:01, 39.17it/s]
Computing ESM2 embeddings: 99%|          | 992/1000 [00:25<00:00, 39.34it/s]

```

```

Computing ESM2 embeddings: 100%|         | 1000/1000 [00:25<00:00, 38.89it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```

for predictions and inference.

|                            |     |                                   |
|----------------------------|-----|-----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/1000 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/1000 [00:00<00:26, 37.10it/s]  |
| Computing ESM2 embeddings: | 6%  | 64/1000 [00:01<00:24, 37.81it/s]  |
| Computing ESM2 embeddings: | 10% | 96/1000 [00:02<00:23, 38.65it/s]  |
| Computing ESM2 embeddings: | 13% | 128/1000 [00:03<00:22, 38.90it/s] |
| Computing ESM2 embeddings: | 16% | 160/1000 [00:04<00:21, 39.08it/s] |
| Computing ESM2 embeddings: | 19% | 192/1000 [00:04<00:20, 39.34it/s] |
| Computing ESM2 embeddings: | 22% | 224/1000 [00:05<00:19, 39.45it/s] |
| Computing ESM2 embeddings: | 26% | 256/1000 [00:06<00:18, 39.56it/s] |
| Computing ESM2 embeddings: | 29% | 288/1000 [00:07<00:17, 39.62it/s] |
| Computing ESM2 embeddings: | 32% | 320/1000 [00:08<00:17, 39.61it/s] |
| Computing ESM2 embeddings: | 35% | 352/1000 [00:08<00:16, 39.65it/s] |
| Computing ESM2 embeddings: | 38% | 384/1000 [00:09<00:15, 39.69it/s] |
| Computing ESM2 embeddings: | 42% | 416/1000 [00:10<00:14, 39.72it/s] |
| Computing ESM2 embeddings: | 45% | 448/1000 [00:11<00:13, 39.67it/s] |
| Computing ESM2 embeddings: | 48% | 480/1000 [00:12<00:13, 39.56it/s] |
| Computing ESM2 embeddings: | 51% | 512/1000 [00:13<00:12, 39.46it/s] |
| Computing ESM2 embeddings: | 54% | 544/1000 [00:13<00:11, 39.41it/s] |
| Computing ESM2 embeddings: | 58% | 576/1000 [00:14<00:10, 39.37it/s] |
| Computing ESM2 embeddings: | 61% | 608/1000 [00:15<00:09, 39.32it/s] |
| Computing ESM2 embeddings: | 64% | 640/1000 [00:16<00:09, 39.15it/s] |
| Computing ESM2 embeddings: | 67% | 672/1000 [00:17<00:08, 39.13it/s] |
| Computing ESM2 embeddings: | 70% | 704/1000 [00:17<00:07, 39.11it/s] |

```

Computing ESM2 embeddings: 74%|      | 736/1000 [00:18<00:06, 39.00it/s]
Computing ESM2 embeddings: 77%|      | 768/1000 [00:19<00:05, 38.70it/s]
Computing ESM2 embeddings: 80%|      | 800/1000 [00:20<00:05, 38.83it/s]
Computing ESM2 embeddings: 83%|      | 832/1000 [00:21<00:04, 38.85it/s]
Computing ESM2 embeddings: 86%|      | 864/1000 [00:22<00:03, 38.71it/s]
Computing ESM2 embeddings: 90%|      | 896/1000 [00:22<00:02, 38.59it/s]
Computing ESM2 embeddings: 93%|      | 928/1000 [00:23<00:01, 38.48it/s]
Computing ESM2 embeddings: 96%|      | 960/1000 [00:24<00:01, 38.66it/s]
Computing ESM2 embeddings: 99%|      | 992/1000 [00:25<00:00, 38.90it/s]

Computing ESM2 embeddings: 100%|     | 1000/1000 [00:25<00:00, 39.05it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/1000 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/1000 [00:00<00:25, 38.20it/s]
Computing ESM2 embeddings:  6%|      | 64/1000 [00:01<00:24, 38.75it/s]
Computing ESM2 embeddings: 10%|      | 96/1000 [00:02<00:23, 38.67it/s]
Computing ESM2 embeddings: 13%|      | 128/1000 [00:03<00:22, 38.78it/s]
Computing ESM2 embeddings: 16%|      | 160/1000 [00:04<00:21, 38.87it/s]
Computing ESM2 embeddings: 19%|      | 192/1000 [00:04<00:20, 38.97it/s]
Computing ESM2 embeddings: 22%|      | 224/1000 [00:05<00:19, 38.93it/s]
Computing ESM2 embeddings: 26%|      | 256/1000 [00:06<00:19, 38.89it/s]
Computing ESM2 embeddings: 29%|      | 288/1000 [00:07<00:18, 38.22it/s]
Computing ESM2 embeddings: 32%|      | 320/1000 [00:08<00:17, 37.99it/s]

```



```

Computing ESM2 embeddings: 35%|          | 352/1000 [00:09<00:16, 38.24it/s]
Computing ESM2 embeddings: 38%|          | 384/1000 [00:10<00:16, 37.48it/s]
Computing ESM2 embeddings: 42%|          | 416/1000 [00:10<00:15, 37.86it/s]
Computing ESM2 embeddings: 45%|          | 448/1000 [00:11<00:14, 38.29it/s]
Computing ESM2 embeddings: 48%|          | 480/1000 [00:12<00:13, 38.26it/s]
Computing ESM2 embeddings: 51%|          | 512/1000 [00:13<00:12, 38.13it/s]
Computing ESM2 embeddings: 54%|          | 544/1000 [00:14<00:11, 38.06it/s]
Computing ESM2 embeddings: 58%|          | 576/1000 [00:15<00:11, 38.40it/s]
Computing ESM2 embeddings: 61%|          | 608/1000 [00:15<00:10, 38.68it/s]
Computing ESM2 embeddings: 64%|          | 640/1000 [00:16<00:09, 38.70it/s]
Computing ESM2 embeddings: 67%|          | 672/1000 [00:17<00:08, 38.58it/s]
Computing ESM2 embeddings: 70%|          | 704/1000 [00:18<00:07, 38.74it/s]
Computing ESM2 embeddings: 74%|          | 736/1000 [00:19<00:06, 38.90it/s]
Computing ESM2 embeddings: 77%|          | 768/1000 [00:19<00:05, 39.06it/s]
Computing ESM2 embeddings: 80%|          | 800/1000 [00:20<00:05, 38.93it/s]
Computing ESM2 embeddings: 83%|          | 832/1000 [00:21<00:04, 38.84it/s]
Computing ESM2 embeddings: 86%|          | 864/1000 [00:22<00:03, 38.72it/s]
Computing ESM2 embeddings: 90%|          | 896/1000 [00:23<00:02, 38.84it/s]
Computing ESM2 embeddings: 93%|          | 928/1000 [00:24<00:01, 38.89it/s]
Computing ESM2 embeddings: 96%|          | 960/1000 [00:24<00:01, 38.70it/s]
Computing ESM2 embeddings: 99%|          | 992/1000 [00:25<00:00, 38.62it/s]

```

```

Computing ESM2 embeddings: 100%|         | 1000/1000 [00:25<00:00, 38.47it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```

for predictions and inference.

|                            |     |                                   |
|----------------------------|-----|-----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/1000 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/1000 [00:01<00:34, 27.73it/s]  |
| Computing ESM2 embeddings: | 6%  | 64/1000 [00:02<00:31, 29.28it/s]  |
| Computing ESM2 embeddings: | 10% | 96/1000 [00:03<00:27, 32.98it/s]  |
| Computing ESM2 embeddings: | 13% | 128/1000 [00:03<00:24, 35.01it/s] |
| Computing ESM2 embeddings: | 16% | 160/1000 [00:04<00:23, 36.19it/s] |
| Computing ESM2 embeddings: | 19% | 192/1000 [00:05<00:22, 36.70it/s] |
| Computing ESM2 embeddings: | 22% | 224/1000 [00:06<00:21, 35.75it/s] |
| Computing ESM2 embeddings: | 26% | 256/1000 [00:07<00:20, 36.19it/s] |
| Computing ESM2 embeddings: | 29% | 288/1000 [00:08<00:19, 36.62it/s] |
| Computing ESM2 embeddings: | 32% | 320/1000 [00:09<00:18, 37.00it/s] |
| Computing ESM2 embeddings: | 35% | 352/1000 [00:09<00:17, 37.36it/s] |
| Computing ESM2 embeddings: | 38% | 384/1000 [00:10<00:16, 37.26it/s] |
| Computing ESM2 embeddings: | 42% | 416/1000 [00:11<00:15, 37.38it/s] |
| Computing ESM2 embeddings: | 45% | 448/1000 [00:12<00:14, 37.53it/s] |
| Computing ESM2 embeddings: | 48% | 480/1000 [00:13<00:13, 37.54it/s] |
| Computing ESM2 embeddings: | 51% | 512/1000 [00:14<00:12, 37.59it/s] |
| Computing ESM2 embeddings: | 54% | 544/1000 [00:14<00:12, 37.58it/s] |
| Computing ESM2 embeddings: | 58% | 576/1000 [00:15<00:11, 37.50it/s] |
| Computing ESM2 embeddings: | 61% | 608/1000 [00:16<00:10, 37.50it/s] |
| Computing ESM2 embeddings: | 64% | 640/1000 [00:17<00:09, 37.74it/s] |
| Computing ESM2 embeddings: | 67% | 672/1000 [00:18<00:08, 37.88it/s] |
| Computing ESM2 embeddings: | 70% | 704/1000 [00:19<00:07, 37.80it/s] |

```

Computing ESM2 embeddings: 74%|      | 736/1000 [00:20<00:06, 37.85it/s]
Computing ESM2 embeddings: 77%|      | 768/1000 [00:20<00:06, 37.92it/s]
Computing ESM2 embeddings: 80%|      | 800/1000 [00:21<00:05, 38.08it/s]
Computing ESM2 embeddings: 83%|      | 832/1000 [00:22<00:04, 38.06it/s]
Computing ESM2 embeddings: 86%|      | 864/1000 [00:23<00:03, 38.16it/s]
Computing ESM2 embeddings: 90%|      | 896/1000 [00:24<00:02, 37.85it/s]
Computing ESM2 embeddings: 93%|      | 928/1000 [00:25<00:01, 38.08it/s]
Computing ESM2 embeddings: 96%|      | 960/1000 [00:25<00:01, 38.01it/s]
Computing ESM2 embeddings: 99%|      | 992/1000 [00:26<00:00, 37.92it/s]

Computing ESM2 embeddings: 100%|     | 1000/1000 [00:27<00:00, 36.98it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/998 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/998 [00:00<00:28, 33.57it/s]
Computing ESM2 embeddings:  6%|      | 64/998 [00:01<00:26, 35.90it/s]
Computing ESM2 embeddings: 10%|      | 96/998 [00:02<00:24, 36.68it/s]
Computing ESM2 embeddings: 13%|      | 128/998 [00:03<00:23, 37.11it/s]
Computing ESM2 embeddings: 16%|      | 160/998 [00:04<00:22, 37.29it/s]
Computing ESM2 embeddings: 19%|      | 192/998 [00:05<00:21, 36.99it/s]
Computing ESM2 embeddings: 22%|      | 224/998 [00:06<00:20, 36.97it/s]
Computing ESM2 embeddings: 26%|      | 256/998 [00:06<00:19, 37.32it/s]
Computing ESM2 embeddings: 29%|      | 288/998 [00:07<00:18, 37.58it/s]
Computing ESM2 embeddings: 32%|      | 320/998 [00:08<00:18, 37.51it/s]

```

Computing ESM2 embeddings: 35%| | 352/998 [00:09<00:17, 37.64it/s]

Computing ESM2 embeddings: 38%| | 384/998 [00:10<00:16, 37.70it/s]

Computing ESM2 embeddings: 42%| | 416/998 [00:11<00:15, 37.74it/s]

Computing ESM2 embeddings: 45%| | 448/998 [00:12<00:14, 37.76it/s]

Computing ESM2 embeddings: 48%| | 480/998 [00:12<00:13, 37.75it/s]

Computing ESM2 embeddings: 51%| | 512/998 [00:13<00:12, 37.67it/s]

Computing ESM2 embeddings: 55%| | 544/998 [00:14<00:12, 37.71it/s]

Computing ESM2 embeddings: 58%| | 576/998 [00:15<00:11, 37.69it/s]

Computing ESM2 embeddings: 61%| | 608/998 [00:16<00:10, 37.63it/s]

Computing ESM2 embeddings: 64%| | 640/998 [00:17<00:09, 37.84it/s]

Computing ESM2 embeddings: 67%| | 672/998 [00:17<00:08, 37.67it/s]

Computing ESM2 embeddings: 71%| | 704/998 [00:18<00:07, 37.43it/s]

Computing ESM2 embeddings: 74%| | 736/998 [00:19<00:07, 37.32it/s]

Computing ESM2 embeddings: 77%| | 768/998 [00:20<00:06, 37.39it/s]

Computing ESM2 embeddings: 80%| | 800/998 [00:21<00:05, 37.41it/s]

Computing ESM2 embeddings: 83%| | 832/998 [00:22<00:04, 37.52it/s]

Computing ESM2 embeddings: 87%| | 864/998 [00:23<00:03, 37.43it/s]

Computing ESM2 embeddings: 90%| | 896/998 [00:23<00:02, 37.48it/s]

Computing ESM2 embeddings: 93%| | 928/998 [00:24<00:01, 37.34it/s]

Computing ESM2 embeddings: 96%| | 960/998 [00:25<00:01, 37.42it/s]

Computing ESM2 embeddings: 99%| | 992/998 [00:26<00:00, 37.36it/s]

Computing ESM2 embeddings: 100%| | 998/998 [00:27<00:00, 36.70it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/1000 [00:00<?, ?it/s]
Computing ESM2 embeddings: 3%|          | 32/1000 [00:00<00:28, 34.54it/s]
Computing ESM2 embeddings: 6%|          | 64/1000 [00:01<00:26, 35.68it/s]
Computing ESM2 embeddings: 10%|         | 96/1000 [00:02<00:24, 37.08it/s]
Computing ESM2 embeddings: 13%|         | 128/1000 [00:03<00:23, 37.78it/s]
Computing ESM2 embeddings: 16%|         | 160/1000 [00:04<00:22, 38.11it/s]
Computing ESM2 embeddings: 19%|         | 192/1000 [00:05<00:21, 38.17it/s]
Computing ESM2 embeddings: 22%|         | 224/1000 [00:05<00:20, 38.09it/s]
Computing ESM2 embeddings: 26%|         | 256/1000 [00:06<00:19, 38.22it/s]
Computing ESM2 embeddings: 29%|         | 288/1000 [00:07<00:18, 38.25it/s]
Computing ESM2 embeddings: 32%|         | 320/1000 [00:08<00:17, 38.42it/s]
Computing ESM2 embeddings: 35%|         | 352/1000 [00:09<00:17, 37.70it/s]
Computing ESM2 embeddings: 38%|         | 384/1000 [00:10<00:16, 37.88it/s]
Computing ESM2 embeddings: 42%|         | 416/1000 [00:10<00:15, 38.08it/s]
Computing ESM2 embeddings: 45%|         | 448/1000 [00:11<00:14, 38.13it/s]
Computing ESM2 embeddings: 48%|         | 480/1000 [00:12<00:13, 38.24it/s]
Computing ESM2 embeddings: 51%|         | 512/1000 [00:13<00:12, 38.31it/s]
Computing ESM2 embeddings: 54%|         | 544/1000 [00:14<00:11, 38.44it/s]
Computing ESM2 embeddings: 58%|         | 576/1000 [00:15<00:11, 38.47it/s]
Computing ESM2 embeddings: 61%|         | 608/1000 [00:15<00:10, 38.46it/s]
Computing ESM2 embeddings: 64%|         | 640/1000 [00:16<00:09, 38.67it/s]
Computing ESM2 embeddings: 67%|         | 672/1000 [00:17<00:08, 38.78it/s]
Computing ESM2 embeddings: 70%|         | 704/1000 [00:18<00:07, 38.74it/s]
```

```

Computing ESM2 embeddings: 74%|      | 736/1000 [00:19<00:06, 38.55it/s]
Computing ESM2 embeddings: 77%|      | 768/1000 [00:20<00:06, 38.47it/s]
Computing ESM2 embeddings: 80%|      | 800/1000 [00:20<00:05, 38.53it/s]
Computing ESM2 embeddings: 83%|      | 832/1000 [00:21<00:04, 38.24it/s]
Computing ESM2 embeddings: 86%|      | 864/1000 [00:22<00:03, 37.80it/s]
Computing ESM2 embeddings: 90%|      | 896/1000 [00:23<00:02, 37.85it/s]
Computing ESM2 embeddings: 93%|      | 928/1000 [00:24<00:01, 37.96it/s]
Computing ESM2 embeddings: 96%|      | 960/1000 [00:25<00:01, 35.86it/s]
Computing ESM2 embeddings: 99%|      | 992/1000 [00:26<00:00, 36.25it/s]

Computing ESM2 embeddings: 100%|     | 1000/1000 [00:26<00:00, 37.76it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/998 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/998 [00:00<00:27, 35.15it/s]
Computing ESM2 embeddings:  6%|      | 64/998 [00:01<00:25, 36.78it/s]
Computing ESM2 embeddings: 10%|      | 96/998 [00:02<00:24, 37.24it/s]
Computing ESM2 embeddings: 13%|      | 128/998 [00:03<00:22, 37.90it/s]
Computing ESM2 embeddings: 16%|      | 160/998 [00:04<00:21, 38.14it/s]
Computing ESM2 embeddings: 19%|      | 192/998 [00:05<00:21, 38.27it/s]
Computing ESM2 embeddings: 22%|      | 224/998 [00:05<00:20, 38.13it/s]
Computing ESM2 embeddings: 26%|      | 256/998 [00:06<00:19, 38.33it/s]
Computing ESM2 embeddings: 29%|      | 288/998 [00:07<00:18, 38.29it/s]
Computing ESM2 embeddings: 32%|      | 320/998 [00:08<00:17, 38.51it/s]

```

Computing ESM2 embeddings: 35%| | 352/998 [00:09<00:16, 38.77it/s]

Computing ESM2 embeddings: 38%| | 384/998 [00:10<00:15, 38.98it/s]

Computing ESM2 embeddings: 42%| | 416/998 [00:10<00:14, 39.08it/s]

Computing ESM2 embeddings: 45%| | 448/998 [00:11<00:14, 38.98it/s]

Computing ESM2 embeddings: 48%| | 480/998 [00:12<00:13, 38.98it/s]

Computing ESM2 embeddings: 51%| | 512/998 [00:13<00:12, 38.81it/s]

Computing ESM2 embeddings: 55%| | 544/998 [00:14<00:11, 38.69it/s]

Computing ESM2 embeddings: 58%| | 576/998 [00:14<00:10, 38.77it/s]

Computing ESM2 embeddings: 61%| | 608/998 [00:15<00:10, 38.86it/s]

Computing ESM2 embeddings: 64%| | 640/998 [00:16<00:09, 38.88it/s]

Computing ESM2 embeddings: 67%| | 672/998 [00:17<00:08, 38.87it/s]

Computing ESM2 embeddings: 71%| | 704/998 [00:18<00:07, 38.84it/s]

Computing ESM2 embeddings: 74%| | 736/998 [00:19<00:06, 38.46it/s]

Computing ESM2 embeddings: 77%| | 768/998 [00:19<00:05, 38.47it/s]

Computing ESM2 embeddings: 80%| | 800/998 [00:20<00:05, 37.98it/s]

Computing ESM2 embeddings: 83%| | 832/998 [00:21<00:04, 37.87it/s]

Computing ESM2 embeddings: 87%| | 864/998 [00:22<00:03, 38.02it/s]

Computing ESM2 embeddings: 90%| | 896/998 [00:23<00:02, 37.94it/s]

Computing ESM2 embeddings: 93%| | 928/998 [00:24<00:01, 37.76it/s]

Computing ESM2 embeddings: 96%| | 960/998 [00:25<00:01, 37.76it/s]

Computing ESM2 embeddings: 99%| | 992/998 [00:25<00:00, 37.75it/s]

Computing ESM2 embeddings: 100%| | 998/998 [00:26<00:00, 38.22it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                   |
|----------------------------|-----|-----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/1000 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/1000 [00:00<00:27, 34.61it/s]  |
| Computing ESM2 embeddings: | 6%  | 64/1000 [00:01<00:26, 35.78it/s]  |
| Computing ESM2 embeddings: | 10% | 96/1000 [00:02<00:25, 35.28it/s]  |
| Computing ESM2 embeddings: | 13% | 128/1000 [00:03<00:24, 35.22it/s] |
| Computing ESM2 embeddings: | 16% | 160/1000 [00:04<00:23, 35.17it/s] |
| Computing ESM2 embeddings: | 19% | 192/1000 [00:05<00:23, 34.95it/s] |
| Computing ESM2 embeddings: | 22% | 224/1000 [00:06<00:22, 35.15it/s] |
| Computing ESM2 embeddings: | 26% | 256/1000 [00:07<00:21, 35.04it/s] |
| Computing ESM2 embeddings: | 29% | 288/1000 [00:08<00:20, 35.30it/s] |
| Computing ESM2 embeddings: | 32% | 320/1000 [00:09<00:19, 35.49it/s] |
| Computing ESM2 embeddings: | 35% | 352/1000 [00:09<00:18, 35.64it/s] |
| Computing ESM2 embeddings: | 38% | 384/1000 [00:10<00:17, 35.76it/s] |
| Computing ESM2 embeddings: | 42% | 416/1000 [00:11<00:16, 35.26it/s] |
| Computing ESM2 embeddings: | 45% | 448/1000 [00:12<00:15, 35.36it/s] |
| Computing ESM2 embeddings: | 48% | 480/1000 [00:13<00:14, 35.65it/s] |
| Computing ESM2 embeddings: | 51% | 512/1000 [00:14<00:13, 35.37it/s] |
| Computing ESM2 embeddings: | 54% | 544/1000 [00:15<00:12, 35.40it/s] |
| Computing ESM2 embeddings: | 58% | 576/1000 [00:16<00:12, 35.16it/s] |
| Computing ESM2 embeddings: | 61% | 608/1000 [00:17<00:11, 35.30it/s] |
| Computing ESM2 embeddings: | 64% | 640/1000 [00:18<00:10, 35.04it/s] |
| Computing ESM2 embeddings: | 67% | 672/1000 [00:19<00:09, 35.29it/s] |
| Computing ESM2 embeddings: | 70% | 704/1000 [00:19<00:08, 35.45it/s] |



Computing ESM2 embeddings: 74%| | 736/1000 [00:20<00:07, 35.50it/s]

Computing ESM2 embeddings: 77%| | 768/1000 [00:21<00:06, 35.54it/s]

Computing ESM2 embeddings: 80%| | 800/1000 [00:22<00:05, 35.46it/s]

Computing ESM2 embeddings: 83%| | 832/1000 [00:23<00:04, 35.31it/s]

Computing ESM2 embeddings: 86%| | 864/1000 [00:24<00:03, 35.06it/s]

Computing ESM2 embeddings: 90%| | 896/1000 [00:25<00:02, 34.75it/s]

Computing ESM2 embeddings: 93%| | 928/1000 [00:26<00:02, 34.54it/s]

Computing ESM2 embeddings: 96%| | 960/1000 [00:27<00:01, 34.45it/s]

Computing ESM2 embeddings: 99%| | 992/1000 [00:28<00:00, 34.46it/s]

Computing ESM2 embeddings: 100%| | 1000/1000 [00:28<00:00, 35.11it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/999 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/999 [00:00<00:29, 32.27it/s]

Computing ESM2 embeddings: 6%| | 64/999 [00:01<00:27, 33.42it/s]

Computing ESM2 embeddings: 10%| | 96/999 [00:02<00:26, 34.06it/s]

Computing ESM2 embeddings: 13%| | 128/999 [00:03<00:25, 34.32it/s]

Computing ESM2 embeddings: 16%| | 160/999 [00:04<00:24, 34.47it/s]

Computing ESM2 embeddings: 19%| | 192/999 [00:05<00:23, 34.80it/s]

Computing ESM2 embeddings: 22%| | 224/999 [00:06<00:22, 34.88it/s]

Computing ESM2 embeddings: 26%| | 256/999 [00:07<00:21, 34.87it/s]

Computing ESM2 embeddings: 29%| | 288/999 [00:08<00:20, 34.94it/s]

Computing ESM2 embeddings: 32%| | 320/999 [00:09<00:19, 34.26it/s]

Computing ESM2 embeddings: 35%| | 352/999 [00:10<00:18, 34.64it/s]

Computing ESM2 embeddings: 38%| | 384/999 [00:11<00:17, 34.69it/s]

Computing ESM2 embeddings: 42%| | 416/999 [00:12<00:16, 34.91it/s]

Computing ESM2 embeddings: 45%| | 448/999 [00:12<00:15, 34.79it/s]

Computing ESM2 embeddings: 48%| | 480/999 [00:13<00:14, 34.81it/s]

Computing ESM2 embeddings: 51%| | 512/999 [00:14<00:14, 34.77it/s]

Computing ESM2 embeddings: 54%| | 544/999 [00:15<00:13, 34.85it/s]

Computing ESM2 embeddings: 58%| | 576/999 [00:16<00:12, 34.93it/s]

Computing ESM2 embeddings: 61%| | 608/999 [00:17<00:11, 34.97it/s]

Computing ESM2 embeddings: 64%| | 640/999 [00:18<00:10, 35.06it/s]

Computing ESM2 embeddings: 67%| | 672/999 [00:19<00:09, 35.07it/s]

Computing ESM2 embeddings: 70%| | 704/999 [00:20<00:08, 35.25it/s]

Computing ESM2 embeddings: 74%| | 736/999 [00:21<00:07, 35.43it/s]

Computing ESM2 embeddings: 77%| | 768/999 [00:22<00:06, 35.32it/s]

Computing ESM2 embeddings: 80%| | 800/999 [00:22<00:05, 35.44it/s]

Computing ESM2 embeddings: 83%| | 832/999 [00:23<00:04, 35.56it/s]

Computing ESM2 embeddings: 86%| | 864/999 [00:24<00:03, 35.58it/s]

Computing ESM2 embeddings: 90%| | 896/999 [00:25<00:02, 35.80it/s]

Computing ESM2 embeddings: 93%| | 928/999 [00:26<00:01, 35.85it/s]

Computing ESM2 embeddings: 96%| | 960/999 [00:27<00:01, 35.80it/s]

Computing ESM2 embeddings: 99%| | 992/999 [00:28<00:00, 35.78it/s]

Computing ESM2 embeddings: 100%| | 999/999 [00:28<00:00, 34.85it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/999 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/999 [00:00<00:27, 34.80it/s]  |
| Computing ESM2 embeddings: | 6%  | 64/999 [00:01<00:26, 35.64it/s]  |
| Computing ESM2 embeddings: | 10% | 96/999 [00:02<00:25, 35.77it/s]  |
| Computing ESM2 embeddings: | 13% | 128/999 [00:03<00:24, 36.00it/s] |
| Computing ESM2 embeddings: | 16% | 160/999 [00:04<00:23, 35.82it/s] |
| Computing ESM2 embeddings: | 19% | 192/999 [00:05<00:22, 36.01it/s] |
| Computing ESM2 embeddings: | 22% | 224/999 [00:06<00:21, 36.15it/s] |
| Computing ESM2 embeddings: | 26% | 256/999 [00:07<00:20, 36.12it/s] |
| Computing ESM2 embeddings: | 29% | 288/999 [00:08<00:19, 36.17it/s] |
| Computing ESM2 embeddings: | 32% | 320/999 [00:08<00:18, 36.03it/s] |
| Computing ESM2 embeddings: | 35% | 352/999 [00:09<00:17, 36.12it/s] |
| Computing ESM2 embeddings: | 38% | 384/999 [00:10<00:16, 36.18it/s] |
| Computing ESM2 embeddings: | 42% | 416/999 [00:11<00:16, 36.20it/s] |
| Computing ESM2 embeddings: | 45% | 448/999 [00:12<00:15, 35.68it/s] |
| Computing ESM2 embeddings: | 48% | 480/999 [00:13<00:14, 35.76it/s] |
| Computing ESM2 embeddings: | 51% | 512/999 [00:14<00:13, 35.68it/s] |
| Computing ESM2 embeddings: | 54% | 544/999 [00:15<00:12, 35.90it/s] |
| Computing ESM2 embeddings: | 58% | 576/999 [00:16<00:11, 36.18it/s] |
| Computing ESM2 embeddings: | 61% | 608/999 [00:16<00:10, 36.30it/s] |
| Computing ESM2 embeddings: | 64% | 640/999 [00:17<00:09, 36.43it/s] |
| Computing ESM2 embeddings: | 67% | 672/999 [00:18<00:08, 36.44it/s] |
| Computing ESM2 embeddings: | 70% | 704/999 [00:19<00:08, 36.52it/s] |

```

Computing ESM2 embeddings: 74%|          | 736/999 [00:20<00:07, 36.51it/s]
Computing ESM2 embeddings: 77%|          | 768/999 [00:21<00:06, 36.65it/s]
Computing ESM2 embeddings: 80%|          | 800/999 [00:22<00:05, 36.85it/s]
Computing ESM2 embeddings: 83%|          | 832/999 [00:22<00:04, 36.75it/s]
Computing ESM2 embeddings: 86%|          | 864/999 [00:23<00:03, 36.81it/s]
Computing ESM2 embeddings: 90%|          | 896/999 [00:24<00:02, 36.83it/s]
Computing ESM2 embeddings: 93%|          | 928/999 [00:25<00:01, 36.71it/s]
Computing ESM2 embeddings: 96%|          | 960/999 [00:26<00:01, 36.75it/s]
Computing ESM2 embeddings: 99%|          | 992/999 [00:27<00:00, 36.70it/s]

Computing ESM2 embeddings: 100%|         | 999/999 [00:27<00:00, 36.24it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/1000 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|          | 32/1000 [00:00<00:27, 35.05it/s]
Computing ESM2 embeddings:  6%|          | 64/1000 [00:01<00:25, 36.45it/s]
Computing ESM2 embeddings: 10%|          | 96/1000 [00:02<00:24, 36.36it/s]
Computing ESM2 embeddings: 13%|          | 128/1000 [00:03<00:23, 36.69it/s]
Computing ESM2 embeddings: 16%|          | 160/1000 [00:04<00:22, 36.93it/s]
Computing ESM2 embeddings: 19%|          | 192/1000 [00:05<00:21, 36.93it/s]
Computing ESM2 embeddings: 22%|          | 224/1000 [00:06<00:20, 37.24it/s]
Computing ESM2 embeddings: 26%|          | 256/1000 [00:06<00:19, 37.71it/s]
Computing ESM2 embeddings: 29%|          | 288/1000 [00:07<00:18, 37.99it/s]
Computing ESM2 embeddings: 32%|          | 320/1000 [00:08<00:17, 38.18it/s]

```

Computing ESM2 embeddings: 35%| | 352/1000 [00:09<00:17, 37.68it/s]

Computing ESM2 embeddings: 38%| | 384/1000 [00:10<00:16, 36.57it/s]

Computing ESM2 embeddings: 42%| | 416/1000 [00:11<00:15, 37.22it/s]

Computing ESM2 embeddings: 45%| | 448/1000 [00:12<00:14, 37.65it/s]

Computing ESM2 embeddings: 48%| | 480/1000 [00:12<00:13, 37.92it/s]

Computing ESM2 embeddings: 51%| | 512/1000 [00:13<00:12, 38.13it/s]

Computing ESM2 embeddings: 54%| | 544/1000 [00:14<00:11, 38.24it/s]

Computing ESM2 embeddings: 58%| | 576/1000 [00:15<00:11, 38.34it/s]

Computing ESM2 embeddings: 61%| | 608/1000 [00:16<00:10, 38.44it/s]

Computing ESM2 embeddings: 64%| | 640/1000 [00:16<00:09, 38.52it/s]

Computing ESM2 embeddings: 67%| | 672/1000 [00:17<00:08, 38.57it/s]

Computing ESM2 embeddings: 70%| | 704/1000 [00:18<00:07, 38.55it/s]

Computing ESM2 embeddings: 74%| | 736/1000 [00:19<00:06, 38.61it/s]

Computing ESM2 embeddings: 77%| | 768/1000 [00:20<00:06, 38.65it/s]

Computing ESM2 embeddings: 80%| | 800/1000 [00:21<00:05, 38.68it/s]

Computing ESM2 embeddings: 83%| | 832/1000 [00:21<00:04, 38.69it/s]

Computing ESM2 embeddings: 86%| | 864/1000 [00:22<00:03, 38.71it/s]

Computing ESM2 embeddings: 90%| | 896/1000 [00:23<00:02, 38.76it/s]

Computing ESM2 embeddings: 93%| | 928/1000 [00:24<00:01, 38.75it/s]

Computing ESM2 embeddings: 96%| | 960/1000 [00:25<00:01, 38.74it/s]

Computing ESM2 embeddings: 99%| | 992/1000 [00:26<00:00, 38.78it/s]

Computing ESM2 embeddings: 100%| | 1000/1000 [00:26<00:00, 38.01it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/996 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/996 [00:00<00:26, 36.73it/s]  |
| Computing ESM2 embeddings: | 6%  | 64/996 [00:01<00:24, 38.03it/s]  |
| Computing ESM2 embeddings: | 10% | 96/996 [00:02<00:23, 38.46it/s]  |
| Computing ESM2 embeddings: | 13% | 128/996 [00:03<00:22, 38.31it/s] |
| Computing ESM2 embeddings: | 16% | 160/996 [00:04<00:21, 38.54it/s] |
| Computing ESM2 embeddings: | 19% | 192/996 [00:05<00:20, 38.57it/s] |
| Computing ESM2 embeddings: | 22% | 224/996 [00:05<00:20, 38.55it/s] |
| Computing ESM2 embeddings: | 26% | 256/996 [00:06<00:19, 38.58it/s] |
| Computing ESM2 embeddings: | 29% | 288/996 [00:07<00:18, 38.71it/s] |
| Computing ESM2 embeddings: | 32% | 320/996 [00:08<00:17, 38.81it/s] |
| Computing ESM2 embeddings: | 35% | 352/996 [00:09<00:16, 38.88it/s] |
| Computing ESM2 embeddings: | 39% | 384/996 [00:09<00:15, 38.90it/s] |
| Computing ESM2 embeddings: | 42% | 416/996 [00:10<00:14, 38.83it/s] |
| Computing ESM2 embeddings: | 45% | 448/996 [00:11<00:14, 38.79it/s] |
| Computing ESM2 embeddings: | 48% | 480/996 [00:12<00:13, 38.89it/s] |
| Computing ESM2 embeddings: | 51% | 512/996 [00:13<00:12, 38.86it/s] |
| Computing ESM2 embeddings: | 55% | 544/996 [00:14<00:11, 38.87it/s] |
| Computing ESM2 embeddings: | 58% | 576/996 [00:14<00:10, 38.87it/s] |
| Computing ESM2 embeddings: | 61% | 608/996 [00:15<00:10, 38.74it/s] |
| Computing ESM2 embeddings: | 64% | 640/996 [00:16<00:09, 38.08it/s] |
| Computing ESM2 embeddings: | 67% | 672/996 [00:17<00:08, 38.33it/s] |
| Computing ESM2 embeddings: | 71% | 704/996 [00:18<00:07, 38.57it/s] |

```

Computing ESM2 embeddings: 74%|      | 736/996 [00:19<00:06, 38.70it/s]
Computing ESM2 embeddings: 77%|      | 768/996 [00:19<00:05, 38.77it/s]
Computing ESM2 embeddings: 80%|      | 800/996 [00:20<00:05, 38.81it/s]
Computing ESM2 embeddings: 84%|      | 832/996 [00:21<00:04, 38.85it/s]
Computing ESM2 embeddings: 87%|      | 864/996 [00:22<00:03, 38.90it/s]
Computing ESM2 embeddings: 90%|      | 896/996 [00:23<00:02, 38.97it/s]
Computing ESM2 embeddings: 93%|      | 928/996 [00:23<00:01, 39.06it/s]
Computing ESM2 embeddings: 96%|      | 960/996 [00:24<00:00, 39.08it/s]
Computing ESM2 embeddings: 100%|     | 992/996 [00:25<00:00, 39.10it/s]

Computing ESM2 embeddings: 100%|     | 996/996 [00:25<00:00, 38.46it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/997 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/997 [00:00<00:25, 37.67it/s]
Computing ESM2 embeddings:  6%|      | 64/997 [00:01<00:24, 38.02it/s]
Computing ESM2 embeddings: 10%|      | 96/997 [00:02<00:23, 38.46it/s]
Computing ESM2 embeddings: 13%|      | 128/997 [00:03<00:22, 38.68it/s]
Computing ESM2 embeddings: 16%|      | 160/997 [00:04<00:21, 38.47it/s]
Computing ESM2 embeddings: 19%|      | 192/997 [00:04<00:20, 38.64it/s]
Computing ESM2 embeddings: 22%|      | 224/997 [00:05<00:19, 38.83it/s]
Computing ESM2 embeddings: 26%|      | 256/997 [00:06<00:19, 38.70it/s]
Computing ESM2 embeddings: 29%|      | 288/997 [00:07<00:18, 38.53it/s]
Computing ESM2 embeddings: 32%|      | 320/997 [00:08<00:17, 38.65it/s]

```

Computing ESM2 embeddings: 35%| | 352/997 [00:09<00:16, 38.62it/s]

Computing ESM2 embeddings: 39%| | 384/997 [00:09<00:15, 38.73it/s]

Computing ESM2 embeddings: 42%| | 416/997 [00:10<00:14, 38.87it/s]

Computing ESM2 embeddings: 45%| | 448/997 [00:11<00:14, 38.88it/s]

Computing ESM2 embeddings: 48%| | 480/997 [00:12<00:13, 38.79it/s]

Computing ESM2 embeddings: 51%| | 512/997 [00:13<00:12, 38.95it/s]

Computing ESM2 embeddings: 55%| | 544/997 [00:14<00:11, 38.69it/s]

Computing ESM2 embeddings: 58%| | 576/997 [00:14<00:10, 38.59it/s]

Computing ESM2 embeddings: 61%| | 608/997 [00:15<00:10, 38.65it/s]

Computing ESM2 embeddings: 64%| | 640/997 [00:16<00:09, 38.81it/s]

Computing ESM2 embeddings: 67%| | 672/997 [00:17<00:08, 38.90it/s]

Computing ESM2 embeddings: 71%| | 704/997 [00:18<00:07, 38.98it/s]

Computing ESM2 embeddings: 74%| | 736/997 [00:18<00:06, 39.03it/s]

Computing ESM2 embeddings: 77%| | 768/997 [00:19<00:05, 39.07it/s]

Computing ESM2 embeddings: 80%| | 800/997 [00:20<00:05, 39.08it/s]

Computing ESM2 embeddings: 83%| | 832/997 [00:21<00:04, 39.11it/s]

Computing ESM2 embeddings: 87%| | 864/997 [00:22<00:03, 39.11it/s]

Computing ESM2 embeddings: 90%| | 896/997 [00:23<00:02, 39.13it/s]

Computing ESM2 embeddings: 93%| | 928/997 [00:23<00:01, 39.12it/s]

Computing ESM2 embeddings: 96%| | 960/997 [00:24<00:00, 39.17it/s]

Computing ESM2 embeddings: 99%| | 992/997 [00:25<00:00, 39.18it/s]

Computing ESM2 embeddings: 100%| | 997/997 [00:25<00:00, 38.54it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it



for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/998 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/998 [00:00<00:26, 36.83it/s]  |
| Computing ESM2 embeddings: | 6%  | 64/998 [00:01<00:24, 38.07it/s]  |
| Computing ESM2 embeddings: | 10% | 96/998 [00:02<00:23, 38.62it/s]  |
| Computing ESM2 embeddings: | 13% | 128/998 [00:03<00:22, 38.89it/s] |
| Computing ESM2 embeddings: | 16% | 160/998 [00:04<00:21, 39.03it/s] |
| Computing ESM2 embeddings: | 19% | 192/998 [00:04<00:20, 39.14it/s] |
| Computing ESM2 embeddings: | 22% | 224/998 [00:05<00:19, 39.21it/s] |
| Computing ESM2 embeddings: | 26% | 256/998 [00:06<00:18, 39.29it/s] |
| Computing ESM2 embeddings: | 29% | 288/998 [00:07<00:18, 39.12it/s] |
| Computing ESM2 embeddings: | 32% | 320/998 [00:08<00:17, 38.97it/s] |
| Computing ESM2 embeddings: | 35% | 352/998 [00:09<00:16, 38.99it/s] |
| Computing ESM2 embeddings: | 38% | 384/998 [00:09<00:15, 39.08it/s] |
| Computing ESM2 embeddings: | 42% | 416/998 [00:10<00:14, 39.13it/s] |
| Computing ESM2 embeddings: | 45% | 448/998 [00:11<00:14, 39.19it/s] |
| Computing ESM2 embeddings: | 48% | 480/998 [00:12<00:13, 39.22it/s] |
| Computing ESM2 embeddings: | 51% | 512/998 [00:13<00:12, 39.23it/s] |
| Computing ESM2 embeddings: | 55% | 544/998 [00:13<00:11, 38.88it/s] |
| Computing ESM2 embeddings: | 58% | 576/998 [00:14<00:11, 37.99it/s] |
| Computing ESM2 embeddings: | 61% | 608/998 [00:15<00:10, 37.39it/s] |
| Computing ESM2 embeddings: | 64% | 640/998 [00:16<00:09, 37.06it/s] |
| Computing ESM2 embeddings: | 67% | 672/998 [00:17<00:08, 37.53it/s] |
| Computing ESM2 embeddings: | 71% | 704/998 [00:18<00:07, 37.98it/s] |

```

Computing ESM2 embeddings: 74%|      | 736/998 [00:19<00:06, 38.11it/s]
Computing ESM2 embeddings: 77%|      | 768/998 [00:19<00:06, 38.28it/s]
Computing ESM2 embeddings: 80%|      | 800/998 [00:20<00:05, 38.44it/s]
Computing ESM2 embeddings: 83%|      | 832/998 [00:21<00:04, 38.45it/s]
Computing ESM2 embeddings: 87%|      | 864/998 [00:22<00:03, 38.40it/s]
Computing ESM2 embeddings: 90%|      | 896/998 [00:23<00:02, 38.48it/s]
Computing ESM2 embeddings: 93%|      | 928/998 [00:24<00:01, 38.58it/s]
Computing ESM2 embeddings: 96%|      | 960/998 [00:24<00:00, 38.46it/s]
Computing ESM2 embeddings: 99%|      | 992/998 [00:25<00:00, 38.29it/s]
Computing ESM2 embeddings: 100%|     | 998/998 [00:25<00:00, 38.45it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/999 [00:00<?, ?it/s]
Computing ESM2 embeddings: 3%|      | 32/999 [00:00<00:25, 37.37it/s]
Computing ESM2 embeddings: 6%|      | 64/999 [00:01<00:24, 38.47it/s]
Computing ESM2 embeddings: 10%|     | 96/999 [00:02<00:23, 38.81it/s]
Computing ESM2 embeddings: 13%|     | 128/999 [00:03<00:22, 39.02it/s]
Computing ESM2 embeddings: 16%|     | 160/999 [00:04<00:21, 39.16it/s]
Computing ESM2 embeddings: 19%|     | 192/999 [00:04<00:20, 39.26it/s]
Computing ESM2 embeddings: 22%|     | 224/999 [00:05<00:19, 39.36it/s]
Computing ESM2 embeddings: 26%|     | 256/999 [00:06<00:18, 39.45it/s]
Computing ESM2 embeddings: 29%|     | 288/999 [00:07<00:18, 39.48it/s]
Computing ESM2 embeddings: 32%|     | 320/999 [00:08<00:17, 39.12it/s]

```

```

Computing ESM2 embeddings: 35%|          | 352/999 [00:09<00:16, 39.12it/s]
Computing ESM2 embeddings: 38%|          | 384/999 [00:09<00:15, 39.21it/s]
Computing ESM2 embeddings: 42%|          | 416/999 [00:10<00:14, 39.21it/s]
Computing ESM2 embeddings: 45%|          | 448/999 [00:11<00:14, 38.99it/s]
Computing ESM2 embeddings: 48%|          | 480/999 [00:12<00:13, 38.86it/s]
Computing ESM2 embeddings: 51%|          | 512/999 [00:13<00:12, 38.71it/s]
Computing ESM2 embeddings: 54%|          | 544/999 [00:13<00:11, 38.80it/s]
Computing ESM2 embeddings: 58%|          | 576/999 [00:14<00:10, 38.97it/s]
Computing ESM2 embeddings: 61%|          | 608/999 [00:15<00:09, 39.14it/s]
Computing ESM2 embeddings: 64%|          | 640/999 [00:16<00:09, 39.25it/s]
Computing ESM2 embeddings: 67%|          | 672/999 [00:17<00:08, 39.33it/s]
Computing ESM2 embeddings: 70%|          | 704/999 [00:18<00:07, 39.24it/s]
Computing ESM2 embeddings: 74%|          | 736/999 [00:18<00:06, 38.96it/s]
Computing ESM2 embeddings: 77%|          | 768/999 [00:19<00:05, 39.03it/s]
Computing ESM2 embeddings: 80%|          | 800/999 [00:20<00:05, 39.17it/s]
Computing ESM2 embeddings: 83%|          | 832/999 [00:21<00:04, 39.23it/s]
Computing ESM2 embeddings: 86%|          | 864/999 [00:22<00:03, 39.31it/s]
Computing ESM2 embeddings: 90%|          | 896/999 [00:22<00:02, 39.38it/s]
Computing ESM2 embeddings: 93%|          | 928/999 [00:23<00:01, 39.45it/s]
Computing ESM2 embeddings: 96%|          | 960/999 [00:24<00:00, 39.50it/s]
Computing ESM2 embeddings: 99%|          | 992/999 [00:25<00:00, 39.53it/s]

```

```

Computing ESM2 embeddings: 100%|         | 999/999 [00:25<00:00, 39.13it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```

for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/1000 [00:00<?, ?it/s]
Computing ESM2 embeddings: 3%|          | 32/1000 [00:00<00:26, 36.92it/s]
Computing ESM2 embeddings: 6%|          | 64/1000 [00:01<00:24, 38.12it/s]
Computing ESM2 embeddings: 10%|         | 96/1000 [00:02<00:23, 38.70it/s]
Computing ESM2 embeddings: 13%|         | 128/1000 [00:03<00:22, 38.77it/s]
Computing ESM2 embeddings: 16%|         | 160/1000 [00:04<00:21, 38.77it/s]
Computing ESM2 embeddings: 19%|         | 192/1000 [00:04<00:20, 38.73it/s]
Computing ESM2 embeddings: 22%|         | 224/1000 [00:05<00:20, 38.69it/s]
Computing ESM2 embeddings: 26%|         | 256/1000 [00:06<00:19, 38.31it/s]
Computing ESM2 embeddings: 29%|         | 288/1000 [00:07<00:18, 38.14it/s]
Computing ESM2 embeddings: 32%|         | 320/1000 [00:08<00:17, 38.10it/s]
Computing ESM2 embeddings: 35%|         | 352/1000 [00:09<00:17, 38.03it/s]
Computing ESM2 embeddings: 38%|         | 384/1000 [00:10<00:16, 38.06it/s]
Computing ESM2 embeddings: 42%|         | 416/1000 [00:10<00:15, 38.20it/s]
Computing ESM2 embeddings: 45%|         | 448/1000 [00:11<00:14, 38.24it/s]
Computing ESM2 embeddings: 48%|         | 480/1000 [00:12<00:13, 38.41it/s]
Computing ESM2 embeddings: 51%|         | 512/1000 [00:13<00:12, 38.42it/s]
Computing ESM2 embeddings: 54%|         | 544/1000 [00:14<00:11, 38.29it/s]
Computing ESM2 embeddings: 58%|         | 576/1000 [00:15<00:11, 38.54it/s]
Computing ESM2 embeddings: 61%|         | 608/1000 [00:15<00:10, 38.80it/s]
Computing ESM2 embeddings: 64%|         | 640/1000 [00:16<00:09, 39.01it/s]
Computing ESM2 embeddings: 67%|         | 672/1000 [00:17<00:08, 39.20it/s]
Computing ESM2 embeddings: 70%|         | 704/1000 [00:18<00:07, 39.32it/s]
```

```

Computing ESM2 embeddings: 74%|          | 736/1000 [00:19<00:06, 39.38it/s]
Computing ESM2 embeddings: 77%|          | 768/1000 [00:19<00:05, 39.08it/s]
Computing ESM2 embeddings: 80%|          | 800/1000 [00:20<00:05, 38.93it/s]
Computing ESM2 embeddings: 83%|          | 832/1000 [00:21<00:04, 38.99it/s]
Computing ESM2 embeddings: 86%|          | 864/1000 [00:22<00:03, 39.17it/s]
Computing ESM2 embeddings: 90%|          | 896/1000 [00:23<00:02, 39.29it/s]
Computing ESM2 embeddings: 93%|          | 928/1000 [00:23<00:01, 39.35it/s]
Computing ESM2 embeddings: 96%|          | 960/1000 [00:24<00:01, 39.36it/s]
Computing ESM2 embeddings: 99%|          | 992/1000 [00:25<00:00, 39.37it/s]

Computing ESM2 embeddings: 100%|         | 1000/1000 [00:25<00:00, 38.71it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/600 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|          | 32/600 [00:00<00:15, 36.83it/s]
Computing ESM2 embeddings: 11%|          | 64/600 [00:01<00:14, 38.05it/s]
Computing ESM2 embeddings: 16%|          | 96/600 [00:02<00:13, 38.65it/s]
Computing ESM2 embeddings: 21%|          | 128/600 [00:03<00:12, 39.00it/s]
Computing ESM2 embeddings: 27%|          | 160/600 [00:04<00:11, 39.22it/s]
Computing ESM2 embeddings: 32%|          | 192/600 [00:04<00:10, 39.34it/s]
Computing ESM2 embeddings: 37%|          | 224/600 [00:05<00:09, 39.41it/s]
Computing ESM2 embeddings: 43%|          | 256/600 [00:06<00:08, 39.39it/s]
Computing ESM2 embeddings: 48%|          | 288/600 [00:07<00:07, 39.45it/s]
Computing ESM2 embeddings: 53%|          | 320/600 [00:08<00:07, 39.46it/s]

```

Computing ESM2 embeddings: 59%| | 352/600 [00:08<00:06, 39.46it/s]  
 Computing ESM2 embeddings: 64%| | 384/600 [00:09<00:05, 39.49it/s]  
 Computing ESM2 embeddings: 69%| | 416/600 [00:10<00:04, 39.49it/s]  
 Computing ESM2 embeddings: 75%| | 448/600 [00:11<00:03, 39.52it/s]  
 Computing ESM2 embeddings: 80%| | 480/600 [00:12<00:03, 39.57it/s]  
 Computing ESM2 embeddings: 85%| | 512/600 [00:13<00:02, 39.19it/s]  
 Computing ESM2 embeddings: 91%| | 544/600 [00:13<00:01, 38.69it/s]  
 Computing ESM2 embeddings: 96%| | 576/600 [00:14<00:00, 38.67it/s]

Computing ESM2 embeddings: 100%| | 600/600 [00:15<00:00, 38.11it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:23, 36.92it/s]  
 Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:22, 38.07it/s]  
 Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:21, 38.69it/s]  
 Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:20, 39.00it/s]  
 Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:19, 39.21it/s]  
 Computing ESM2 embeddings: 21%| | 192/915 [00:04<00:18, 39.32it/s]  
 Computing ESM2 embeddings: 24%| | 224/915 [00:05<00:17, 39.11it/s]  
 Computing ESM2 embeddings: 28%| | 256/915 [00:06<00:16, 39.18it/s]  
 Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:15, 39.30it/s]  
 Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:15, 39.35it/s]  
 Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:14, 39.11it/s]

```

Computing ESM2 embeddings: 42%|          | 384/915 [00:09<00:13, 39.09it/s]
Computing ESM2 embeddings: 45%|          | 416/915 [00:10<00:12, 39.24it/s]
Computing ESM2 embeddings: 49%|          | 448/915 [00:11<00:11, 39.35it/s]
Computing ESM2 embeddings: 52%|          | 480/915 [00:12<00:11, 39.42it/s]
Computing ESM2 embeddings: 56%|          | 512/915 [00:13<00:10, 39.46it/s]
Computing ESM2 embeddings: 59%|          | 544/915 [00:13<00:09, 39.47it/s]
Computing ESM2 embeddings: 63%|          | 576/915 [00:14<00:08, 39.43it/s]
Computing ESM2 embeddings: 66%|          | 608/915 [00:15<00:07, 39.43it/s]
Computing ESM2 embeddings: 70%|          | 640/915 [00:16<00:06, 39.48it/s]
Computing ESM2 embeddings: 73%|          | 672/915 [00:17<00:06, 39.51it/s]
Computing ESM2 embeddings: 77%|          | 704/915 [00:17<00:05, 39.55it/s]
Computing ESM2 embeddings: 80%|          | 736/915 [00:18<00:04, 39.56it/s]
Computing ESM2 embeddings: 84%|          | 768/915 [00:19<00:03, 39.49it/s]
Computing ESM2 embeddings: 87%|          | 800/915 [00:20<00:02, 39.29it/s]
Computing ESM2 embeddings: 91%|          | 832/915 [00:21<00:02, 39.28it/s]
Computing ESM2 embeddings: 94%|          | 864/915 [00:22<00:01, 39.38it/s]
Computing ESM2 embeddings: 98%|          | 896/915 [00:22<00:00, 39.19it/s]

Computing ESM2 embeddings: 100%|         | 915/915 [00:23<00:00, 38.97it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/897 [00:00<00:23, 37.24it/s]
Computing ESM2 embeddings:  7%|          | 64/897 [00:01<00:21, 38.04it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/897 [00:02<00:20, 38.46it/s]  |
| Computing ESM2 embeddings: 14% | 128/897 [00:03<00:19, 38.86it/s] |
| Computing ESM2 embeddings: 18% | 160/897 [00:04<00:18, 39.06it/s] |
| Computing ESM2 embeddings: 21% | 192/897 [00:04<00:17, 39.26it/s] |
| Computing ESM2 embeddings: 25% | 224/897 [00:05<00:17, 39.38it/s] |
| Computing ESM2 embeddings: 29% | 256/897 [00:06<00:16, 39.45it/s] |
| Computing ESM2 embeddings: 32% | 288/897 [00:07<00:15, 39.44it/s] |
| Computing ESM2 embeddings: 36% | 320/897 [00:08<00:14, 39.43it/s] |
| Computing ESM2 embeddings: 39% | 352/897 [00:08<00:13, 39.48it/s] |
| Computing ESM2 embeddings: 43% | 384/897 [00:09<00:12, 39.48it/s] |
| Computing ESM2 embeddings: 46% | 416/897 [00:10<00:12, 39.52it/s] |
| Computing ESM2 embeddings: 50% | 448/897 [00:11<00:11, 39.52it/s] |
| Computing ESM2 embeddings: 54% | 480/897 [00:12<00:10, 39.31it/s] |
| Computing ESM2 embeddings: 57% | 512/897 [00:13<00:09, 39.26it/s] |
| Computing ESM2 embeddings: 61% | 544/897 [00:13<00:09, 39.16it/s] |
| Computing ESM2 embeddings: 64% | 576/897 [00:14<00:08, 39.21it/s] |
| Computing ESM2 embeddings: 68% | 608/897 [00:15<00:07, 39.23it/s] |
| Computing ESM2 embeddings: 71% | 640/897 [00:16<00:06, 39.07it/s] |
| Computing ESM2 embeddings: 75% | 672/897 [00:17<00:05, 39.11it/s] |
| Computing ESM2 embeddings: 78% | 704/897 [00:17<00:04, 39.19it/s] |
| Computing ESM2 embeddings: 82% | 736/897 [00:18<00:04, 39.32it/s] |
| Computing ESM2 embeddings: 86% | 768/897 [00:19<00:03, 39.36it/s] |
| Computing ESM2 embeddings: 89% | 800/897 [00:20<00:02, 39.45it/s] |
| Computing ESM2 embeddings: 93% | 832/897 [00:21<00:01, 39.47it/s] |



Computing ESM2 embeddings: 96%| | 864/897 [00:22<00:00, 39.48it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:23<00:00, 38.33it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 36.42it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.98it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.71it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 39.15it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.17it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.06it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.14it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.23it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 39.04it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 39.12it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 39.25it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.31it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.40it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.47it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 39.50it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 39.53it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.58it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.29it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.32it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.39it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.39it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.47it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.55it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.56it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.58it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.61it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.62it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.63it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.13it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.15it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.30it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.84it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.14it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.34it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 39.41it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 39.48it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 39.52it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 39.57it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 39.54it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:13, 39.51it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:12, 39.42it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 39.46it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.51it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.51it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 39.49it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.53it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.37it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.25it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.97it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:06, 36.68it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 36.26it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 36.65it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 36.97it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 37.23it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 37.40it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 37.54it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.43it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 36.65it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:22, 38.18it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 38.57it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 38.78it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.94it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 39.03it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 39.14it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 39.17it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:15, 39.26it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 39.12it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 39.17it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 39.31it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 39.43it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.50it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:10, 39.53it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 39.55it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.20it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.07it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.20it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:06, 39.27it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 39.08it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 39.15it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.29it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 39.31it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 39.40it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.40it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.19it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.17it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.09it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 37.63it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 38.62it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 39.08it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:19, 39.26it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:18, 39.40it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:17, 39.46it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 39.49it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 39.51it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 39.54it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 39.57it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:08<00:13, 39.57it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:09<00:12, 39.55it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 39.51it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 39.52it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 39.52it/s]
Computing ESM2 embeddings: 57%|      | 512/896 [00:12<00:09, 39.52it/s]

```

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.53it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.53it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.52it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.53it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 39.54it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 39.52it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.53it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 39.50it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 39.36it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.31it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 39.40it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.39it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.21it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 38.62it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 38.99it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 39.22it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.32it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.25it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.24it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.33it/s]

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Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:15, 39.39it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 39.44it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:08<00:14, 39.46it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:09<00:13, 39.48it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 39.51it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.30it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 39.12it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 39.21it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:13<00:09, 39.29it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 39.37it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.42it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:06, 39.46it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 39.23it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:17<00:05, 39.01it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 39.07it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 39.22it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 39.31it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.16it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 39.17it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:22<00:00, 39.13it/s]

```

```

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 39.13it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|          | 0/570 [00:00<?, ?it/s]
Computing ESM2 embeddings: 6%|          | 32/570 [00:00<00:14, 37.48it/s]
Computing ESM2 embeddings: 11%|         | 64/570 [00:01<00:13, 38.48it/s]
Computing ESM2 embeddings: 17%|         | 96/570 [00:02<00:12, 38.49it/s]
Computing ESM2 embeddings: 22%|         | 128/570 [00:03<00:11, 38.81it/s]
Computing ESM2 embeddings: 28%|         | 160/570 [00:04<00:10, 39.08it/s]
Computing ESM2 embeddings: 34%|         | 192/570 [00:04<00:09, 39.24it/s]
Computing ESM2 embeddings: 39%|         | 224/570 [00:05<00:08, 39.35it/s]
Computing ESM2 embeddings: 45%|         | 256/570 [00:06<00:07, 39.38it/s]
Computing ESM2 embeddings: 51%|         | 288/570 [00:07<00:07, 39.35it/s]
Computing ESM2 embeddings: 56%|         | 320/570 [00:08<00:06, 39.40it/s]
Computing ESM2 embeddings: 62%|         | 352/570 [00:08<00:05, 39.36it/s]
Computing ESM2 embeddings: 67%|         | 384/570 [00:09<00:04, 39.05it/s]
Computing ESM2 embeddings: 73%|         | 416/570 [00:10<00:03, 39.09it/s]
Computing ESM2 embeddings: 79%|         | 448/570 [00:11<00:03, 39.21it/s]
Computing ESM2 embeddings: 84%|         | 480/570 [00:12<00:02, 39.29it/s]
Computing ESM2 embeddings: 90%|         | 512/570 [00:13<00:01, 39.37it/s]
Computing ESM2 embeddings: 95%|         | 544/570 [00:13<00:00, 39.40it/s]

Computing ESM2 embeddings: 100%|        | 570/570 [00:14<00:00, 39.01it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings: 0%|          | 0/915 [00:00<?, ?it/s]
Computing ESM2 embeddings: 3%|          | 32/915 [00:00<00:23, 37.02it/s]

```



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/915 [00:01<00:22, 38.34it/s]  |
| Computing ESM2 embeddings: 10% | 96/915 [00:02<00:21, 38.87it/s]  |
| Computing ESM2 embeddings: 14% | 128/915 [00:03<00:20, 39.17it/s] |
| Computing ESM2 embeddings: 17% | 160/915 [00:04<00:19, 39.31it/s] |
| Computing ESM2 embeddings: 21% | 192/915 [00:04<00:18, 39.39it/s] |
| Computing ESM2 embeddings: 24% | 224/915 [00:05<00:17, 39.38it/s] |
| Computing ESM2 embeddings: 28% | 256/915 [00:06<00:16, 39.42it/s] |
| Computing ESM2 embeddings: 31% | 288/915 [00:07<00:15, 39.47it/s] |
| Computing ESM2 embeddings: 35% | 320/915 [00:08<00:15, 39.49it/s] |
| Computing ESM2 embeddings: 38% | 352/915 [00:08<00:14, 39.49it/s] |
| Computing ESM2 embeddings: 42% | 384/915 [00:09<00:13, 39.51it/s] |
| Computing ESM2 embeddings: 45% | 416/915 [00:10<00:12, 39.52it/s] |
| Computing ESM2 embeddings: 49% | 448/915 [00:11<00:11, 39.51it/s] |
| Computing ESM2 embeddings: 52% | 480/915 [00:12<00:11, 39.49it/s] |
| Computing ESM2 embeddings: 56% | 512/915 [00:13<00:10, 39.25it/s] |
| Computing ESM2 embeddings: 59% | 544/915 [00:13<00:09, 39.03it/s] |
| Computing ESM2 embeddings: 63% | 576/915 [00:14<00:08, 39.08it/s] |
| Computing ESM2 embeddings: 66% | 608/915 [00:15<00:07, 39.21it/s] |
| Computing ESM2 embeddings: 70% | 640/915 [00:16<00:06, 39.32it/s] |
| Computing ESM2 embeddings: 73% | 672/915 [00:17<00:06, 39.38it/s] |
| Computing ESM2 embeddings: 77% | 704/915 [00:17<00:05, 39.16it/s] |
| Computing ESM2 embeddings: 80% | 736/915 [00:18<00:04, 38.93it/s] |
| Computing ESM2 embeddings: 84% | 768/915 [00:19<00:03, 38.82it/s] |
| Computing ESM2 embeddings: 87% | 800/915 [00:20<00:02, 38.86it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/915 [00:21<00:02, 39.05it/s]
Computing ESM2 embeddings: 94%|      | 864/915 [00:22<00:01, 39.18it/s]
Computing ESM2 embeddings: 98%|      | 896/915 [00:22<00:00, 38.42it/s]
Computing ESM2 embeddings: 100%|     | 915/915 [00:23<00:00, 39.01it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/898 [00:00<00:23, 37.12it/s]
Computing ESM2 embeddings:  7%|      | 64/898 [00:01<00:21, 38.40it/s]
Computing ESM2 embeddings: 11%|      | 96/898 [00:02<00:20, 38.84it/s]
Computing ESM2 embeddings: 14%|      | 128/898 [00:03<00:19, 39.04it/s]
Computing ESM2 embeddings: 18%|      | 160/898 [00:04<00:18, 39.11it/s]
Computing ESM2 embeddings: 21%|      | 192/898 [00:04<00:18, 39.16it/s]
Computing ESM2 embeddings: 25%|      | 224/898 [00:05<00:17, 38.96it/s]
Computing ESM2 embeddings: 29%|      | 256/898 [00:06<00:16, 39.04it/s]
Computing ESM2 embeddings: 32%|      | 288/898 [00:07<00:15, 39.21it/s]
Computing ESM2 embeddings: 36%|      | 320/898 [00:08<00:14, 39.29it/s]
Computing ESM2 embeddings: 39%|      | 352/898 [00:09<00:13, 39.31it/s]
Computing ESM2 embeddings: 43%|      | 384/898 [00:09<00:13, 39.39it/s]
Computing ESM2 embeddings: 46%|      | 416/898 [00:10<00:12, 39.44it/s]
Computing ESM2 embeddings: 50%|      | 448/898 [00:11<00:11, 39.50it/s]
Computing ESM2 embeddings: 53%|      | 480/898 [00:12<00:10, 39.52it/s]
Computing ESM2 embeddings: 57%|      | 512/898 [00:13<00:09, 39.54it/s]

```

Computing ESM2 embeddings: 61%| | 544/898 [00:13<00:08, 39.51it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:14<00:08, 39.49it/s]

Computing ESM2 embeddings: 68%| | 608/898 [00:15<00:07, 39.51it/s]

Computing ESM2 embeddings: 71%| | 640/898 [00:16<00:06, 39.50it/s]

Computing ESM2 embeddings: 75%| | 672/898 [00:17<00:05, 39.35it/s]

Computing ESM2 embeddings: 78%| | 704/898 [00:17<00:04, 38.99it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:18<00:04, 39.04it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:19<00:03, 39.16it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:20<00:02, 39.00it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:21<00:01, 38.77it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:22<00:00, 38.64it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:23<00:00, 38.63it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 36.99it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 38.17it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.44it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.84it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.12it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.33it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.46it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.55it/s]

```

Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:15, 39.61it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 39.39it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 39.12it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:09<00:13, 39.16it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 39.29it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.42it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 39.44it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 39.52it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:13<00:09, 39.56it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 39.59it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.59it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:06, 39.59it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 39.62it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:17<00:05, 39.64it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 39.62it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 39.61it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 39.64it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.64it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:21<00:01, 39.37it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:22<00:00, 39.33it/s]

```

```

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 39.20it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.09it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.80it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.14it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.30it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.37it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:17, 39.42it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 39.46it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.50it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.51it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.53it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 39.32it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 39.27it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 39.33it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.40it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.41it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 39.34it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:08, 39.12it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.96it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.88it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.85it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 38.88it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:17<00:04, 38.86it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:18<00:04, 38.82it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 38.72it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 38.67it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 38.72it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 38.82it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.03it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.00it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:22, 37.46it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.77it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:20, 38.29it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 38.74it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:18, 38.66it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 38.84it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 39.04it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:15, 39.22it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 39.32it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 39.41it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 39.18it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 39.19it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 39.03it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 38.66it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.68it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.68it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.69it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.77it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 39.01it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.18it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 39.30it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.38it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.45it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.48it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.49it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.69it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.65it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.83it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 37.94it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:21, 38.53it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:20, 38.84it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:19, 39.08it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:18, 39.27it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:04<00:17, 39.39it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25% | 224/895 [00:05<00:17, 39.44it/s] |
| Computing ESM2 embeddings: 29% | 256/895 [00:06<00:16, 39.51it/s] |
| Computing ESM2 embeddings: 32% | 288/895 [00:07<00:15, 39.52it/s] |
| Computing ESM2 embeddings: 36% | 320/895 [00:08<00:14, 39.54it/s] |
| Computing ESM2 embeddings: 39% | 352/895 [00:08<00:13, 39.58it/s] |
| Computing ESM2 embeddings: 43% | 384/895 [00:09<00:12, 39.58it/s] |
| Computing ESM2 embeddings: 46% | 416/895 [00:10<00:12, 39.53it/s] |
| Computing ESM2 embeddings: 50% | 448/895 [00:11<00:11, 39.54it/s] |
| Computing ESM2 embeddings: 54% | 480/895 [00:12<00:10, 39.56it/s] |
| Computing ESM2 embeddings: 57% | 512/895 [00:12<00:09, 39.56it/s] |
| Computing ESM2 embeddings: 61% | 544/895 [00:13<00:08, 39.58it/s] |
| Computing ESM2 embeddings: 64% | 576/895 [00:14<00:08, 39.56it/s] |
| Computing ESM2 embeddings: 68% | 608/895 [00:15<00:07, 39.54it/s] |
| Computing ESM2 embeddings: 72% | 640/895 [00:16<00:06, 39.55it/s] |
| Computing ESM2 embeddings: 75% | 672/895 [00:17<00:05, 39.57it/s] |
| Computing ESM2 embeddings: 79% | 704/895 [00:17<00:04, 39.30it/s] |
| Computing ESM2 embeddings: 82% | 736/895 [00:18<00:04, 39.06it/s] |
| Computing ESM2 embeddings: 86% | 768/895 [00:19<00:03, 39.09it/s] |
| Computing ESM2 embeddings: 89% | 800/895 [00:20<00:02, 39.22it/s] |
| Computing ESM2 embeddings: 93% | 832/895 [00:21<00:01, 39.33it/s] |
| Computing ESM2 embeddings: 97% | 864/895 [00:21<00:00, 39.42it/s] |

Computing ESM2 embeddings: 100%| | 895/895 [00:22<00:00, 39.05it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it



for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.14it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:22, 37.89it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 38.16it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.52it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.88it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 39.11it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.26it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.38it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:15, 39.42it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 39.48it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 39.51it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.54it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.54it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.56it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 39.24it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 39.25it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 39.34it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.38it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.14it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 39.12it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 39.24it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:17<00:05, 39.34it/s] |

```

Computing ESM2 embeddings: 81%|      | 736/914 [00:18<00:04, 39.39it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:19<00:03, 39.43it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 39.47it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.50it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.46it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.10it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.08it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/572 [00:00<00:14, 38.06it/s]
Computing ESM2 embeddings: 11%|      | 64/572 [00:01<00:13, 38.73it/s]
Computing ESM2 embeddings: 17%|      | 96/572 [00:02<00:12, 39.07it/s]
Computing ESM2 embeddings: 22%|      | 128/572 [00:03<00:11, 39.31it/s]
Computing ESM2 embeddings: 28%|      | 160/572 [00:04<00:10, 39.44it/s]
Computing ESM2 embeddings: 34%|      | 192/572 [00:04<00:09, 39.50it/s]
Computing ESM2 embeddings: 39%|      | 224/572 [00:05<00:08, 39.53it/s]
Computing ESM2 embeddings: 45%|      | 256/572 [00:06<00:07, 39.55it/s]
Computing ESM2 embeddings: 50%|      | 288/572 [00:07<00:07, 39.57it/s]
Computing ESM2 embeddings: 56%|      | 320/572 [00:08<00:06, 39.58it/s]
Computing ESM2 embeddings: 62%|      | 352/572 [00:08<00:05, 39.58it/s]
Computing ESM2 embeddings: 67%|      | 384/572 [00:09<00:04, 39.59it/s]
Computing ESM2 embeddings: 73%|      | 416/572 [00:10<00:03, 39.38it/s]

```

Computing ESM2 embeddings: 78%| | 448/572 [00:11<00:03, 39.36it/s]

Computing ESM2 embeddings: 84%| | 480/572 [00:12<00:02, 39.42it/s]

Computing ESM2 embeddings: 90%| | 512/572 [00:12<00:01, 39.45it/s]

Computing ESM2 embeddings: 95%| | 544/572 [00:13<00:00, 39.49it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:14<00:00, 38.93it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/906 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/906 [00:00<00:23, 36.95it/s]

Computing ESM2 embeddings: 7%| | 64/906 [00:01<00:22, 37.84it/s]

Computing ESM2 embeddings: 11%| | 96/906 [00:02<00:21, 38.48it/s]

Computing ESM2 embeddings: 14%| | 128/906 [00:03<00:19, 38.92it/s]

Computing ESM2 embeddings: 18%| | 160/906 [00:04<00:19, 39.16it/s]

Computing ESM2 embeddings: 21%| | 192/906 [00:04<00:18, 39.31it/s]

Computing ESM2 embeddings: 25%| | 224/906 [00:05<00:17, 39.42it/s]

Computing ESM2 embeddings: 28%| | 256/906 [00:06<00:16, 39.47it/s]

Computing ESM2 embeddings: 32%| | 288/906 [00:07<00:15, 39.50it/s]

Computing ESM2 embeddings: 35%| | 320/906 [00:08<00:14, 39.52it/s]

Computing ESM2 embeddings: 39%| | 352/906 [00:08<00:14, 39.54it/s]

Computing ESM2 embeddings: 42%| | 384/906 [00:09<00:13, 39.50it/s]

Computing ESM2 embeddings: 46%| | 416/906 [00:10<00:12, 39.50it/s]

Computing ESM2 embeddings: 49%| | 448/906 [00:11<00:11, 39.48it/s]

Computing ESM2 embeddings: 53%| | 480/906 [00:12<00:10, 39.51it/s]

Computing ESM2 embeddings: 57%| | 512/906 [00:13<00:09, 39.53it/s]

Computing ESM2 embeddings: 60%| | 544/906 [00:13<00:09, 39.54it/s]

Computing ESM2 embeddings: 64%| | 576/906 [00:14<00:08, 39.57it/s]

Computing ESM2 embeddings: 67%| | 608/906 [00:15<00:07, 39.44it/s]

Computing ESM2 embeddings: 71%| | 640/906 [00:16<00:06, 39.40it/s]

Computing ESM2 embeddings: 74%| | 672/906 [00:17<00:05, 39.48it/s]

Computing ESM2 embeddings: 78%| | 704/906 [00:17<00:05, 39.52it/s]

Computing ESM2 embeddings: 81%| | 736/906 [00:18<00:04, 39.58it/s]

Computing ESM2 embeddings: 85%| | 768/906 [00:19<00:03, 39.61it/s]

Computing ESM2 embeddings: 88%| | 800/906 [00:20<00:02, 39.53it/s]

Computing ESM2 embeddings: 92%| | 832/906 [00:21<00:01, 38.75it/s]

Computing ESM2 embeddings: 95%| | 864/906 [00:22<00:01, 38.81it/s]

Computing ESM2 embeddings: 99%| | 896/906 [00:22<00:00, 39.03it/s]

Computing ESM2 embeddings: 100%| | 906/906 [00:23<00:00, 38.96it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/908 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/908 [00:00<00:23, 37.11it/s]

Computing ESM2 embeddings: 7%| | 64/908 [00:01<00:21, 38.39it/s]

Computing ESM2 embeddings: 11%| | 96/908 [00:02<00:20, 38.97it/s]

Computing ESM2 embeddings: 14%| | 128/908 [00:03<00:19, 39.21it/s]

Computing ESM2 embeddings: 18%| | 160/908 [00:04<00:19, 39.35it/s]

Computing ESM2 embeddings: 21%| | 192/908 [00:04<00:18, 39.12it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/908 [00:05<00:17, 38.94it/s] |
| Computing ESM2 embeddings: 28%  | 256/908 [00:06<00:16, 39.02it/s] |
| Computing ESM2 embeddings: 32%  | 288/908 [00:07<00:15, 39.16it/s] |
| Computing ESM2 embeddings: 35%  | 320/908 [00:08<00:15, 38.96it/s] |
| Computing ESM2 embeddings: 39%  | 352/908 [00:09<00:14, 39.02it/s] |
| Computing ESM2 embeddings: 42%  | 384/908 [00:09<00:13, 39.18it/s] |
| Computing ESM2 embeddings: 46%  | 416/908 [00:10<00:12, 39.30it/s] |
| Computing ESM2 embeddings: 49%  | 448/908 [00:11<00:11, 39.40it/s] |
| Computing ESM2 embeddings: 53%  | 480/908 [00:12<00:10, 39.45it/s] |
| Computing ESM2 embeddings: 56%  | 512/908 [00:13<00:10, 39.52it/s] |
| Computing ESM2 embeddings: 60%  | 544/908 [00:13<00:09, 39.53it/s] |
| Computing ESM2 embeddings: 63%  | 576/908 [00:14<00:08, 39.48it/s] |
| Computing ESM2 embeddings: 67%  | 608/908 [00:15<00:07, 39.19it/s] |
| Computing ESM2 embeddings: 70%  | 640/908 [00:16<00:06, 39.02it/s] |
| Computing ESM2 embeddings: 74%  | 672/908 [00:17<00:06, 39.09it/s] |
| Computing ESM2 embeddings: 78%  | 704/908 [00:17<00:05, 39.25it/s] |
| Computing ESM2 embeddings: 81%  | 736/908 [00:18<00:04, 39.33it/s] |
| Computing ESM2 embeddings: 85%  | 768/908 [00:19<00:03, 39.35it/s] |
| Computing ESM2 embeddings: 88%  | 800/908 [00:20<00:02, 39.42it/s] |
| Computing ESM2 embeddings: 92%  | 832/908 [00:21<00:01, 39.49it/s] |
| Computing ESM2 embeddings: 95%  | 864/908 [00:22<00:01, 38.71it/s] |
| Computing ESM2 embeddings: 99%  | 896/908 [00:22<00:00, 38.83it/s] |
| Computing ESM2 embeddings: 100% | 908/908 [00:23<00:00, 38.83it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.31it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.90it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.24it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.41it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.48it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:17, 39.44it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 39.51it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.53it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.57it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.40it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 39.31it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 39.37it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 39.37it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.44it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.47it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:12<00:09, 39.50it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:08, 39.52it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.54it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.55it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.48it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 39.56it/s] |

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 39.29it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 38.10it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.43it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.74it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.76it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.90it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.17it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.47it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 38.44it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.53it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.82it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.12it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.22it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.11it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.17it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:15, 39.31it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 39.39it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:14, 39.46it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.50it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.53it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.55it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 39.56it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 39.57it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.53it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.33it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.94it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 39.04it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.16it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 38.91it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 38.99it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.15it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.53it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.71it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.96it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.15it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.09it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.70it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.20it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.39it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.47it/s]



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:18, 39.51it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:17, 39.54it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:16, 39.56it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 39.56it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 39.58it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 39.57it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:08<00:13, 39.58it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:13, 39.38it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:12, 39.28it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 39.33it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 39.37it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:12<00:09, 39.30it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:13<00:08, 39.29it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 39.31it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 39.32it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 39.36it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 39.39it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:17<00:04, 39.17it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:18<00:04, 39.27it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 39.32it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 39.36it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:21<00:01, 39.32it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:21<00:00, 39.37it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:22<00:00, 39.35it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.82it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 38.69it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.07it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 39.21it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 39.31it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 39.41it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.42it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.41it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:15, 39.41it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 39.44it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 39.43it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.42it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.41it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.41it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 39.42it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 39.42it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 39.42it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.41it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.36it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.23it/s] |

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Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 39.19it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:17<00:05, 39.21it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:18<00:04, 39.32it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:19<00:03, 38.65it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 38.82it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.04it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.18it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.29it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.20it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 39.17it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 39.48it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 39.17it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:19, 39.30it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:18, 39.47it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:17, 39.57it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:16, 39.61it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 39.64it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 39.64it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 39.64it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:08<00:13, 39.40it/s]

```

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Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:13, 39.04it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 39.12it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.30it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 39.44it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:12<00:09, 39.45it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:08, 39.43it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 39.46it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 39.48it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.51it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 39.54it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 39.54it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 38.47it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.72it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.96it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.15it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 39.31it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.28it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/588 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|          | 32/588 [00:00<00:14, 37.51it/s]
Computing ESM2 embeddings: 11%|          | 64/588 [00:01<00:14, 36.45it/s]
Computing ESM2 embeddings: 16%|          | 96/588 [00:02<00:13, 37.81it/s]

```

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Computing ESM2 embeddings: 22%|          | 128/588 [00:03<00:11, 38.48it/s]
Computing ESM2 embeddings: 27%|          | 160/588 [00:04<00:11, 38.84it/s]
Computing ESM2 embeddings: 33%|          | 192/588 [00:04<00:10, 39.03it/s]
Computing ESM2 embeddings: 38%|          | 224/588 [00:05<00:09, 39.21it/s]
Computing ESM2 embeddings: 44%|          | 256/588 [00:06<00:08, 39.33it/s]
Computing ESM2 embeddings: 49%|          | 288/588 [00:07<00:07, 39.40it/s]
Computing ESM2 embeddings: 54%|          | 320/588 [00:08<00:06, 39.47it/s]
Computing ESM2 embeddings: 60%|          | 352/588 [00:09<00:05, 39.43it/s]
Computing ESM2 embeddings: 65%|          | 384/588 [00:09<00:05, 39.49it/s]
Computing ESM2 embeddings: 71%|          | 416/588 [00:10<00:04, 39.53it/s]
Computing ESM2 embeddings: 76%|          | 448/588 [00:11<00:03, 39.53it/s]
Computing ESM2 embeddings: 82%|          | 480/588 [00:12<00:02, 39.54it/s]
Computing ESM2 embeddings: 87%|          | 512/588 [00:13<00:01, 39.55it/s]
Computing ESM2 embeddings: 93%|          | 544/588 [00:13<00:01, 39.53it/s]
Computing ESM2 embeddings: 98%|          | 576/588 [00:14<00:00, 39.30it/s]

Computing ESM2 embeddings: 100%|         | 588/588 [00:15<00:00, 39.01it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/897 [00:00<00:22, 37.98it/s]
Computing ESM2 embeddings:  7%|          | 64/897 [00:01<00:21, 38.72it/s]
Computing ESM2 embeddings: 11%|          | 96/897 [00:02<00:20, 39.14it/s]
Computing ESM2 embeddings: 14%|          | 128/897 [00:03<00:19, 39.31it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/897 [00:04<00:18, 39.41it/s] |
| Computing ESM2 embeddings: 21%  | 192/897 [00:04<00:17, 39.47it/s] |
| Computing ESM2 embeddings: 25%  | 224/897 [00:05<00:17, 39.50it/s] |
| Computing ESM2 embeddings: 29%  | 256/897 [00:06<00:16, 39.53it/s] |
| Computing ESM2 embeddings: 32%  | 288/897 [00:07<00:15, 39.53it/s] |
| Computing ESM2 embeddings: 36%  | 320/897 [00:08<00:14, 39.56it/s] |
| Computing ESM2 embeddings: 39%  | 352/897 [00:08<00:13, 39.60it/s] |
| Computing ESM2 embeddings: 43%  | 384/897 [00:09<00:13, 39.43it/s] |
| Computing ESM2 embeddings: 46%  | 416/897 [00:10<00:12, 39.39it/s] |
| Computing ESM2 embeddings: 50%  | 448/897 [00:11<00:11, 39.40it/s] |
| Computing ESM2 embeddings: 54%  | 480/897 [00:12<00:10, 39.47it/s] |
| Computing ESM2 embeddings: 57%  | 512/897 [00:12<00:09, 39.48it/s] |
| Computing ESM2 embeddings: 61%  | 544/897 [00:13<00:08, 39.52it/s] |
| Computing ESM2 embeddings: 64%  | 576/897 [00:14<00:08, 39.55it/s] |
| Computing ESM2 embeddings: 68%  | 608/897 [00:15<00:07, 39.31it/s] |
| Computing ESM2 embeddings: 71%  | 640/897 [00:16<00:06, 39.27it/s] |
| Computing ESM2 embeddings: 75%  | 672/897 [00:17<00:05, 39.35it/s] |
| Computing ESM2 embeddings: 78%  | 704/897 [00:17<00:04, 39.01it/s] |
| Computing ESM2 embeddings: 82%  | 736/897 [00:18<00:04, 39.04it/s] |
| Computing ESM2 embeddings: 86%  | 768/897 [00:19<00:03, 38.34it/s] |
| Computing ESM2 embeddings: 89%  | 800/897 [00:20<00:02, 38.52it/s] |
| Computing ESM2 embeddings: 93%  | 832/897 [00:21<00:01, 38.72it/s] |
| Computing ESM2 embeddings: 96%  | 864/897 [00:22<00:00, 38.94it/s] |
| Computing ESM2 embeddings: 100% | 897/897 [00:22<00:00, 39.20it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/915 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/915 [00:00<00:22, 38.52it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/915 [00:01<00:21, 39.17it/s]  |
| Computing ESM2 embeddings: | 10% | 96/915 [00:02<00:21, 38.85it/s]  |
| Computing ESM2 embeddings: | 14% | 128/915 [00:03<00:20, 38.70it/s] |
| Computing ESM2 embeddings: | 17% | 160/915 [00:04<00:19, 38.86it/s] |
| Computing ESM2 embeddings: | 21% | 192/915 [00:04<00:18, 39.10it/s] |
| Computing ESM2 embeddings: | 24% | 224/915 [00:05<00:17, 39.18it/s] |
| Computing ESM2 embeddings: | 28% | 256/915 [00:06<00:16, 39.30it/s] |
| Computing ESM2 embeddings: | 31% | 288/915 [00:07<00:15, 39.36it/s] |
| Computing ESM2 embeddings: | 35% | 320/915 [00:08<00:15, 39.40it/s] |
| Computing ESM2 embeddings: | 38% | 352/915 [00:08<00:14, 39.45it/s] |
| Computing ESM2 embeddings: | 42% | 384/915 [00:09<00:13, 39.47it/s] |
| Computing ESM2 embeddings: | 45% | 416/915 [00:10<00:12, 39.49it/s] |
| Computing ESM2 embeddings: | 49% | 448/915 [00:11<00:11, 39.29it/s] |
| Computing ESM2 embeddings: | 52% | 480/915 [00:12<00:11, 39.17it/s] |
| Computing ESM2 embeddings: | 56% | 512/915 [00:13<00:10, 38.99it/s] |
| Computing ESM2 embeddings: | 59% | 544/915 [00:13<00:09, 39.05it/s] |
| Computing ESM2 embeddings: | 63% | 576/915 [00:14<00:08, 39.17it/s] |
| Computing ESM2 embeddings: | 66% | 608/915 [00:15<00:07, 38.87it/s] |
| Computing ESM2 embeddings: | 70% | 640/915 [00:16<00:07, 38.98it/s] |

```

Computing ESM2 embeddings: 73%|      | 672/915 [00:17<00:06, 39.12it/s]
Computing ESM2 embeddings: 77%|      | 704/915 [00:17<00:05, 39.23it/s]
Computing ESM2 embeddings: 80%|      | 736/915 [00:18<00:04, 39.32it/s]
Computing ESM2 embeddings: 84%|      | 768/915 [00:19<00:03, 39.43it/s]
Computing ESM2 embeddings: 87%|      | 800/915 [00:20<00:02, 39.47it/s]
Computing ESM2 embeddings: 91%|      | 832/915 [00:21<00:02, 39.49it/s]
Computing ESM2 embeddings: 94%|      | 864/915 [00:22<00:01, 39.48it/s]
Computing ESM2 embeddings: 98%|      | 896/915 [00:22<00:00, 39.49it/s]

Computing ESM2 embeddings: 100%|     | 915/915 [00:23<00:00, 39.22it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.12it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 38.66it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 39.07it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:19, 39.28it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:18, 39.38it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:17, 39.47it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 39.49it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 39.31it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 39.08it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 39.12it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:08<00:13, 39.20it/s]

```



Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 39.26it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 39.33it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.40it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.43it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 39.44it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.48it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.51it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.50it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.51it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 39.50it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 39.53it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.49it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 39.27it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 39.26it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.06it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 39.11it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.28it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.77it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 38.46it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.23it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 38.52it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.91it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 39.05it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 38.93it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 38.75it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.88it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 39.10it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 39.23it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 39.32it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 39.40it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.44it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 39.40it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 39.11it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.14it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.25it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.33it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:06, 39.39it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 39.44it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 39.47it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.25it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 39.33it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 39.41it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:21<00:02, 39.28it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:22<00:01, 39.19it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.29it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.99it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.56it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 37.97it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.62it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.07it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.48it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.60it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 37.87it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 38.16it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 38.40it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 38.63it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 38.78it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 38.87it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.89it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.95it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 38.99it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 39.02it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 39.04it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.05it/s]

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Computing ESM2 embeddings: 68%|      | 608/896 [00:15<00:07, 38.96it/s]
Computing ESM2 embeddings: 71%|      | 640/896 [00:16<00:06, 38.67it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 38.74it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:04, 38.88it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 38.95it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 39.01it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 39.02it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.08it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 39.08it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 38.70it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/914 [00:00<00:23, 38.12it/s]
Computing ESM2 embeddings: 7%|      | 64/914 [00:01<00:22, 38.54it/s]
Computing ESM2 embeddings: 11%|     | 96/914 [00:02<00:21, 38.74it/s]
Computing ESM2 embeddings: 14%|     | 128/914 [00:03<00:20, 38.85it/s]
Computing ESM2 embeddings: 18%|     | 160/914 [00:04<00:19, 38.92it/s]
Computing ESM2 embeddings: 21%|     | 192/914 [00:04<00:18, 38.81it/s]
Computing ESM2 embeddings: 25%|     | 224/914 [00:05<00:17, 38.82it/s]
Computing ESM2 embeddings: 28%|     | 256/914 [00:06<00:16, 38.89it/s]
Computing ESM2 embeddings: 32%|     | 288/914 [00:07<00:16, 38.89it/s]
Computing ESM2 embeddings: 35%|     | 320/914 [00:08<00:15, 38.93it/s]

```

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Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 38.94it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:09<00:13, 38.88it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 38.91it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 38.89it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.19it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.42it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 38.60it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 38.65it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 38.74it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.81it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.87it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.90it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 38.93it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.91it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.89it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 38.88it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 38.83it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.86it/s]

```

```

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.77it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/865 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/865 [00:00<00:21, 38.21it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/865 [00:01<00:20, 38.61it/s]  |
| Computing ESM2 embeddings: 11% | 96/865 [00:02<00:19, 38.70it/s]  |
| Computing ESM2 embeddings: 15% | 128/865 [00:03<00:19, 38.76it/s] |
| Computing ESM2 embeddings: 18% | 160/865 [00:04<00:18, 38.90it/s] |
| Computing ESM2 embeddings: 22% | 192/865 [00:04<00:17, 38.94it/s] |
| Computing ESM2 embeddings: 26% | 224/865 [00:05<00:16, 38.98it/s] |
| Computing ESM2 embeddings: 30% | 256/865 [00:06<00:15, 39.00it/s] |
| Computing ESM2 embeddings: 33% | 288/865 [00:07<00:14, 39.00it/s] |
| Computing ESM2 embeddings: 37% | 320/865 [00:08<00:13, 38.99it/s] |
| Computing ESM2 embeddings: 41% | 352/865 [00:09<00:13, 39.01it/s] |
| Computing ESM2 embeddings: 44% | 384/865 [00:09<00:12, 39.02it/s] |
| Computing ESM2 embeddings: 48% | 416/865 [00:10<00:11, 39.04it/s] |
| Computing ESM2 embeddings: 52% | 448/865 [00:11<00:10, 39.02it/s] |
| Computing ESM2 embeddings: 55% | 480/865 [00:12<00:09, 39.01it/s] |
| Computing ESM2 embeddings: 59% | 512/865 [00:13<00:09, 38.99it/s] |
| Computing ESM2 embeddings: 63% | 544/865 [00:13<00:08, 38.98it/s] |
| Computing ESM2 embeddings: 67% | 576/865 [00:14<00:07, 38.98it/s] |
| Computing ESM2 embeddings: 70% | 608/865 [00:15<00:06, 39.01it/s] |
| Computing ESM2 embeddings: 74% | 640/865 [00:16<00:05, 38.98it/s] |
| Computing ESM2 embeddings: 78% | 672/865 [00:17<00:04, 39.03it/s] |
| Computing ESM2 embeddings: 81% | 704/865 [00:18<00:04, 39.07it/s] |
| Computing ESM2 embeddings: 85% | 736/865 [00:18<00:03, 39.09it/s] |
| Computing ESM2 embeddings: 89% | 768/865 [00:19<00:02, 39.09it/s] |
| Computing ESM2 embeddings: 92% | 800/865 [00:20<00:01, 38.19it/s] |

Computing ESM2 embeddings: 96%| | 832/865 [00:21<00:00, 37.26it/s]

Computing ESM2 embeddings: 100%| | 865/865 [00:22<00:00, 38.70it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/619 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/619 [00:00<00:15, 37.89it/s]

Computing ESM2 embeddings: 10%| | 64/619 [00:01<00:14, 38.69it/s]

Computing ESM2 embeddings: 16%| | 96/619 [00:02<00:13, 38.92it/s]

Computing ESM2 embeddings: 21%| | 128/619 [00:03<00:12, 38.97it/s]

Computing ESM2 embeddings: 26%| | 160/619 [00:04<00:11, 39.06it/s]

Computing ESM2 embeddings: 31%| | 192/619 [00:04<00:10, 39.06it/s]

Computing ESM2 embeddings: 36%| | 224/619 [00:05<00:10, 39.11it/s]

Computing ESM2 embeddings: 41%| | 256/619 [00:06<00:09, 39.15it/s]

Computing ESM2 embeddings: 47%| | 288/619 [00:07<00:08, 39.18it/s]

Computing ESM2 embeddings: 52%| | 320/619 [00:08<00:07, 39.18it/s]

Computing ESM2 embeddings: 57%| | 352/619 [00:09<00:06, 39.17it/s]

Computing ESM2 embeddings: 62%| | 384/619 [00:09<00:05, 39.19it/s]

Computing ESM2 embeddings: 67%| | 416/619 [00:10<00:05, 39.20it/s]

Computing ESM2 embeddings: 72%| | 448/619 [00:11<00:04, 39.19it/s]

Computing ESM2 embeddings: 78%| | 480/619 [00:12<00:03, 39.20it/s]

Computing ESM2 embeddings: 83%| | 512/619 [00:13<00:02, 39.17it/s]

Computing ESM2 embeddings: 88%| | 544/619 [00:13<00:01, 39.17it/s]

Computing ESM2 embeddings: 93%| | 576/619 [00:14<00:01, 39.18it/s]

Computing ESM2 embeddings: 98%| | 608/619 [00:15<00:00, 39.15it/s]

Computing ESM2 embeddings: 100%| | 619/619 [00:16<00:00, 38.64it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.15it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.32it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.63it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 38.87it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 38.96it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:18, 39.00it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 39.08it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 39.04it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 39.04it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 39.09it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:13, 39.11it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 39.14it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 39.16it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.11it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.14it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 39.15it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.13it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.10it/s]



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Computing ESM2 embeddings: 68%|      | 608/896 [00:15<00:07, 39.06it/s]
Computing ESM2 embeddings: 71%|      | 640/896 [00:16<00:06, 39.10it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 39.14it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:04, 39.14it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:18<00:04, 39.17it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 39.15it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 38.41it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 38.63it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 38.81it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 38.92it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/916 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/916 [00:00<00:23, 38.39it/s]
Computing ESM2 embeddings:  7%|      | 64/916 [00:01<00:21, 38.91it/s]
Computing ESM2 embeddings: 10%|      | 96/916 [00:02<00:21, 39.01it/s]
Computing ESM2 embeddings: 14%|      | 128/916 [00:03<00:20, 39.06it/s]
Computing ESM2 embeddings: 17%|      | 160/916 [00:04<00:19, 39.16it/s]
Computing ESM2 embeddings: 21%|      | 192/916 [00:04<00:18, 39.19it/s]
Computing ESM2 embeddings: 24%|      | 224/916 [00:05<00:17, 39.23it/s]
Computing ESM2 embeddings: 28%|      | 256/916 [00:06<00:16, 39.17it/s]
Computing ESM2 embeddings: 31%|      | 288/916 [00:07<00:16, 39.19it/s]
Computing ESM2 embeddings: 35%|      | 320/916 [00:08<00:15, 39.12it/s]

```

Computing ESM2 embeddings: 38%| | 352/916 [00:08<00:14, 39.16it/s]

Computing ESM2 embeddings: 42%| | 384/916 [00:09<00:13, 39.17it/s]

Computing ESM2 embeddings: 45%| | 416/916 [00:10<00:12, 39.13it/s]

Computing ESM2 embeddings: 49%| | 448/916 [00:11<00:11, 39.16it/s]

Computing ESM2 embeddings: 52%| | 480/916 [00:12<00:11, 39.18it/s]

Computing ESM2 embeddings: 56%| | 512/916 [00:13<00:10, 39.17it/s]

Computing ESM2 embeddings: 59%| | 544/916 [00:13<00:09, 39.20it/s]

Computing ESM2 embeddings: 63%| | 576/916 [00:14<00:08, 39.15it/s]

Computing ESM2 embeddings: 66%| | 608/916 [00:15<00:07, 39.15it/s]

Computing ESM2 embeddings: 70%| | 640/916 [00:16<00:07, 39.17it/s]

Computing ESM2 embeddings: 73%| | 672/916 [00:17<00:06, 39.15it/s]

Computing ESM2 embeddings: 77%| | 704/916 [00:17<00:05, 39.15it/s]

Computing ESM2 embeddings: 80%| | 736/916 [00:18<00:04, 38.26it/s]

Computing ESM2 embeddings: 84%| | 768/916 [00:19<00:03, 38.50it/s]

Computing ESM2 embeddings: 87%| | 800/916 [00:20<00:02, 38.71it/s]

Computing ESM2 embeddings: 91%| | 832/916 [00:21<00:02, 38.87it/s]

Computing ESM2 embeddings: 94%| | 864/916 [00:22<00:01, 38.99it/s]

Computing ESM2 embeddings: 98%| | 896/916 [00:22<00:00, 39.00it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:23<00:00, 38.80it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 38.07it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:21, 38.67it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:20, 38.89it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:19, 38.98it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:18, 39.03it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:04<00:18, 39.00it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 39.04it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:16, 39.07it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:15, 39.10it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:14, 39.08it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:13, 39.04it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:09<00:13, 39.07it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:10<00:12, 39.08it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:11<00:11, 39.09it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 39.09it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:09, 38.96it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:13<00:09, 38.94it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:14<00:08, 38.99it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:15<00:07, 39.03it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:16<00:06, 39.05it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:17<00:05, 39.03it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:18<00:05, 38.19it/s] |
| Computing ESM2 embeddings: | 82% | 736/895 [00:18<00:04, 38.46it/s] |
| Computing ESM2 embeddings: | 86% | 768/895 [00:19<00:03, 38.65it/s] |
| Computing ESM2 embeddings: | 89% | 800/895 [00:20<00:02, 38.80it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/895 [00:21<00:01, 38.85it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:22<00:00, 38.93it/s]
Computing ESM2 embeddings: 100%|     | 895/895 [00:23<00:00, 38.90it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.41it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 38.83it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:20, 39.00it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:20, 39.08it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 39.10it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:04<00:18, 39.13it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 39.14it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 39.16it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:15, 39.17it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 39.11it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 39.12it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 39.15it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 39.17it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 39.19it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 39.13it/s]
Computing ESM2 embeddings: 56%|      | 512/914 [00:13<00:10, 39.14it/s]
Computing ESM2 embeddings: 60%|      | 544/914 [00:13<00:09, 39.15it/s]

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```

Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 39.15it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.16it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.25it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.52it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.70it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 38.84it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.91it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.93it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.01it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 39.07it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:22<00:00, 39.08it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.99it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.56it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 38.74it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 38.92it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:19, 38.84it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:18, 38.91it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:04<00:18, 38.95it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 39.00it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 39.03it/s]

```

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 39.01it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 39.03it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:13, 39.06it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 39.07it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 39.09it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.06it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.06it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 39.06it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:09, 39.09it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.06it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.01it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.14it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.42it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.54it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 38.65it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.72it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.83it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.89it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.93it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.86it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 4%  | 32/914 [00:00<00:23, 38.26it/s]  |
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:21, 38.74it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 38.87it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 38.93it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.98it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 38.98it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 38.99it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 38.96it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.96it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 38.92it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 38.97it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 39.01it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 38.98it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.02it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 39.05it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 39.06it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.09it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.03it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:08, 38.17it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 38.45it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 38.66it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 38.79it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 38.81it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 38.87it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 38.91it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 38.96it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.02it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 39.00it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.87it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/808 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/808 [00:00<00:20, 38.61it/s]
Computing ESM2 embeddings:  8%|      | 64/808 [00:01<00:19, 38.91it/s]
Computing ESM2 embeddings: 12%|      | 96/808 [00:02<00:18, 39.05it/s]
Computing ESM2 embeddings: 16%|      | 128/808 [00:03<00:17, 39.09it/s]
Computing ESM2 embeddings: 20%|      | 160/808 [00:04<00:16, 39.12it/s]
Computing ESM2 embeddings: 24%|      | 192/808 [00:04<00:15, 39.07it/s]
Computing ESM2 embeddings: 28%|      | 224/808 [00:05<00:14, 39.09it/s]
Computing ESM2 embeddings: 32%|      | 256/808 [00:06<00:14, 39.09it/s]
Computing ESM2 embeddings: 36%|      | 288/808 [00:07<00:13, 39.11it/s]
Computing ESM2 embeddings: 40%|      | 320/808 [00:08<00:12, 39.12it/s]
Computing ESM2 embeddings: 44%|      | 352/808 [00:09<00:11, 39.05it/s]
Computing ESM2 embeddings: 48%|      | 384/808 [00:09<00:10, 39.06it/s]
Computing ESM2 embeddings: 51%|      | 416/808 [00:10<00:10, 39.12it/s]
Computing ESM2 embeddings: 55%|      | 448/808 [00:11<00:09, 39.13it/s]
Computing ESM2 embeddings: 59%|      | 480/808 [00:12<00:08, 39.13it/s]

```



```

Computing ESM2 embeddings: 63%|      | 512/808 [00:13<00:07, 39.08it/s]
Computing ESM2 embeddings: 67%|      | 544/808 [00:13<00:06, 39.10it/s]
Computing ESM2 embeddings: 71%|      | 576/808 [00:14<00:05, 39.11it/s]
Computing ESM2 embeddings: 75%|      | 608/808 [00:15<00:05, 39.12it/s]
Computing ESM2 embeddings: 79%|      | 640/808 [00:16<00:04, 38.54it/s]
Computing ESM2 embeddings: 83%|      | 672/808 [00:17<00:03, 38.61it/s]
Computing ESM2 embeddings: 87%|      | 704/808 [00:18<00:02, 38.74it/s]
Computing ESM2 embeddings: 91%|      | 736/808 [00:18<00:01, 38.82it/s]
Computing ESM2 embeddings: 95%|      | 768/808 [00:19<00:01, 38.89it/s]
Computing ESM2 embeddings: 99%|      | 800/808 [00:20<00:00, 38.96it/s]

Computing ESM2 embeddings: 100%|     | 808/808 [00:20<00:00, 38.91it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/677 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/677 [00:00<00:17, 37.05it/s]
Computing ESM2 embeddings:  9%|      | 64/677 [00:01<00:16, 37.95it/s]
Computing ESM2 embeddings: 14%|      | 96/677 [00:02<00:15, 38.50it/s]
Computing ESM2 embeddings: 19%|      | 128/677 [00:03<00:14, 38.78it/s]
Computing ESM2 embeddings: 24%|      | 160/677 [00:04<00:13, 38.90it/s]
Computing ESM2 embeddings: 28%|      | 192/677 [00:04<00:12, 38.94it/s]
Computing ESM2 embeddings: 33%|      | 224/677 [00:05<00:11, 38.89it/s]
Computing ESM2 embeddings: 38%|      | 256/677 [00:06<00:10, 38.97it/s]
Computing ESM2 embeddings: 43%|      | 288/677 [00:07<00:09, 39.02it/s]

```

Computing ESM2 embeddings: 47%| | 320/677 [00:08<00:09, 39.06it/s]

Computing ESM2 embeddings: 52%| | 352/677 [00:09<00:08, 39.05it/s]

Computing ESM2 embeddings: 57%| | 384/677 [00:09<00:07, 38.97it/s]

Computing ESM2 embeddings: 61%| | 416/677 [00:10<00:06, 38.98it/s]

Computing ESM2 embeddings: 66%| | 448/677 [00:11<00:05, 39.00it/s]

Computing ESM2 embeddings: 71%| | 480/677 [00:12<00:05, 38.97it/s]

Computing ESM2 embeddings: 76%| | 512/677 [00:13<00:04, 38.99it/s]

Computing ESM2 embeddings: 80%| | 544/677 [00:13<00:03, 38.98it/s]

Computing ESM2 embeddings: 85%| | 576/677 [00:14<00:02, 38.99it/s]

Computing ESM2 embeddings: 90%| | 608/677 [00:15<00:01, 38.45it/s]

Computing ESM2 embeddings: 95%| | 640/677 [00:16<00:00, 38.62it/s]

Computing ESM2 embeddings: 99%| | 672/677 [00:17<00:00, 38.76it/s]

Computing ESM2 embeddings: 100%| | 677/677 [00:17<00:00, 38.75it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/898 [00:00<00:22, 38.34it/s]

Computing ESM2 embeddings: 7%| | 64/898 [00:01<00:21, 38.73it/s]

Computing ESM2 embeddings: 11%| | 96/898 [00:02<00:20, 38.90it/s]

Computing ESM2 embeddings: 14%| | 128/898 [00:03<00:19, 38.98it/s]

Computing ESM2 embeddings: 18%| | 160/898 [00:04<00:18, 39.04it/s]

Computing ESM2 embeddings: 21%| | 192/898 [00:04<00:18, 39.09it/s]

Computing ESM2 embeddings: 25%| | 224/898 [00:05<00:17, 39.01it/s]

```

Computing ESM2 embeddings: 29%|          | 256/898 [00:06<00:16, 39.04it/s]
Computing ESM2 embeddings: 32%|          | 288/898 [00:07<00:15, 39.06it/s]
Computing ESM2 embeddings: 36%|          | 320/898 [00:08<00:14, 39.06it/s]
Computing ESM2 embeddings: 39%|          | 352/898 [00:09<00:13, 39.06it/s]
Computing ESM2 embeddings: 43%|          | 384/898 [00:09<00:13, 38.97it/s]
Computing ESM2 embeddings: 46%|          | 416/898 [00:10<00:12, 38.98it/s]
Computing ESM2 embeddings: 50%|          | 448/898 [00:11<00:11, 39.00it/s]
Computing ESM2 embeddings: 53%|          | 480/898 [00:12<00:10, 38.99it/s]
Computing ESM2 embeddings: 57%|          | 512/898 [00:13<00:09, 38.98it/s]
Computing ESM2 embeddings: 61%|          | 544/898 [00:13<00:09, 38.92it/s]
Computing ESM2 embeddings: 64%|          | 576/898 [00:14<00:08, 38.82it/s]
Computing ESM2 embeddings: 68%|          | 608/898 [00:15<00:07, 37.88it/s]
Computing ESM2 embeddings: 71%|          | 640/898 [00:16<00:06, 38.06it/s]
Computing ESM2 embeddings: 75%|          | 672/898 [00:17<00:05, 38.30it/s]
Computing ESM2 embeddings: 78%|          | 704/898 [00:18<00:05, 38.45it/s]
Computing ESM2 embeddings: 82%|          | 736/898 [00:18<00:04, 38.58it/s]
Computing ESM2 embeddings: 86%|          | 768/898 [00:19<00:03, 38.70it/s]
Computing ESM2 embeddings: 89%|          | 800/898 [00:20<00:02, 38.78it/s]
Computing ESM2 embeddings: 93%|          | 832/898 [00:21<00:01, 38.84it/s]
Computing ESM2 embeddings: 96%|          | 864/898 [00:22<00:00, 38.83it/s]

```

```

Computing ESM2 embeddings: 100%|         | 898/898 [00:23<00:00, 38.74it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/912 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/912 [00:00<00:25, 34.82it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/912 [00:01<00:22, 37.53it/s]  |
| Computing ESM2 embeddings: | 11% | 96/912 [00:02<00:21, 38.70it/s]  |
| Computing ESM2 embeddings: | 14% | 128/912 [00:03<00:19, 39.27it/s] |
| Computing ESM2 embeddings: | 18% | 160/912 [00:04<00:19, 39.54it/s] |
| Computing ESM2 embeddings: | 21% | 192/912 [00:04<00:18, 39.69it/s] |
| Computing ESM2 embeddings: | 25% | 224/912 [00:05<00:17, 39.85it/s] |
| Computing ESM2 embeddings: | 28% | 256/912 [00:06<00:16, 39.96it/s] |
| Computing ESM2 embeddings: | 32% | 288/912 [00:07<00:15, 40.03it/s] |
| Computing ESM2 embeddings: | 35% | 320/912 [00:08<00:14, 40.07it/s] |
| Computing ESM2 embeddings: | 39% | 352/912 [00:08<00:13, 40.04it/s] |
| Computing ESM2 embeddings: | 42% | 384/912 [00:09<00:13, 40.09it/s] |
| Computing ESM2 embeddings: | 46% | 416/912 [00:10<00:12, 40.13it/s] |
| Computing ESM2 embeddings: | 49% | 448/912 [00:11<00:11, 40.15it/s] |
| Computing ESM2 embeddings: | 53% | 480/912 [00:12<00:10, 40.16it/s] |
| Computing ESM2 embeddings: | 56% | 512/912 [00:12<00:09, 40.11it/s] |
| Computing ESM2 embeddings: | 60% | 544/912 [00:13<00:09, 40.14it/s] |
| Computing ESM2 embeddings: | 63% | 576/912 [00:14<00:08, 40.16it/s] |
| Computing ESM2 embeddings: | 67% | 608/912 [00:15<00:07, 38.71it/s] |
| Computing ESM2 embeddings: | 70% | 640/912 [00:16<00:06, 39.03it/s] |
| Computing ESM2 embeddings: | 74% | 672/912 [00:16<00:06, 39.31it/s] |
| Computing ESM2 embeddings: | 77% | 704/912 [00:17<00:05, 39.57it/s] |
| Computing ESM2 embeddings: | 81% | 736/912 [00:18<00:04, 39.75it/s] |

```

Computing ESM2 embeddings: 84%|      | 768/912 [00:19<00:03, 39.89it/s]
Computing ESM2 embeddings: 88%|      | 800/912 [00:20<00:02, 39.99it/s]
Computing ESM2 embeddings: 91%|      | 832/912 [00:20<00:02, 39.93it/s]
Computing ESM2 embeddings: 95%|      | 864/912 [00:21<00:01, 39.90it/s]
Computing ESM2 embeddings: 98%|      | 896/912 [00:22<00:00, 39.99it/s]

Computing ESM2 embeddings: 100%|     | 912/912 [00:23<00:00, 39.13it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/898 [00:00<00:23, 37.26it/s]
Computing ESM2 embeddings:  7%|      | 64/898 [00:01<00:21, 38.51it/s]
Computing ESM2 embeddings: 11%|      | 96/898 [00:02<00:20, 39.27it/s]
Computing ESM2 embeddings: 14%|      | 128/898 [00:03<00:19, 39.61it/s]
Computing ESM2 embeddings: 18%|      | 160/898 [00:04<00:18, 39.87it/s]
Computing ESM2 embeddings: 21%|      | 192/898 [00:04<00:17, 40.04it/s]
Computing ESM2 embeddings: 25%|      | 224/898 [00:05<00:16, 40.14it/s]
Computing ESM2 embeddings: 29%|      | 256/898 [00:06<00:15, 40.20it/s]
Computing ESM2 embeddings: 32%|      | 288/898 [00:07<00:15, 40.21it/s]
Computing ESM2 embeddings: 36%|      | 320/898 [00:08<00:14, 40.24it/s]
Computing ESM2 embeddings: 39%|      | 352/898 [00:08<00:13, 40.30it/s]
Computing ESM2 embeddings: 43%|      | 384/898 [00:09<00:12, 40.30it/s]
Computing ESM2 embeddings: 46%|      | 416/898 [00:10<00:11, 40.31it/s]
Computing ESM2 embeddings: 50%|      | 448/898 [00:11<00:11, 40.24it/s]

```

Computing ESM2 embeddings: 53%| | 480/898 [00:11<00:10, 40.28it/s]

Computing ESM2 embeddings: 57%| | 512/898 [00:12<00:09, 40.29it/s]

Computing ESM2 embeddings: 61%| | 544/898 [00:13<00:08, 40.30it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:14<00:07, 40.32it/s]

Computing ESM2 embeddings: 68%| | 608/898 [00:15<00:07, 38.82it/s]

Computing ESM2 embeddings: 71%| | 640/898 [00:16<00:06, 38.91it/s]

Computing ESM2 embeddings: 75%| | 672/898 [00:16<00:05, 39.19it/s]

Computing ESM2 embeddings: 78%| | 704/898 [00:17<00:04, 39.52it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:18<00:04, 39.74it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:19<00:03, 39.86it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:20<00:02, 39.99it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:20<00:01, 40.09it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:21<00:00, 40.13it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:22<00:00, 39.78it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/907 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/907 [00:00<00:23, 37.30it/s]

Computing ESM2 embeddings: 7%| | 64/907 [00:01<00:21, 38.38it/s]

Computing ESM2 embeddings: 11%| | 96/907 [00:02<00:20, 39.00it/s]

Computing ESM2 embeddings: 14%| | 128/907 [00:03<00:19, 39.54it/s]

Computing ESM2 embeddings: 18%| | 160/907 [00:04<00:18, 39.84it/s]

Computing ESM2 embeddings: 21%| | 192/907 [00:04<00:17, 40.04it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/907 [00:05<00:17, 40.15it/s] |
| Computing ESM2 embeddings: 28%  | 256/907 [00:06<00:16, 40.16it/s] |
| Computing ESM2 embeddings: 32%  | 288/907 [00:07<00:15, 40.23it/s] |
| Computing ESM2 embeddings: 35%  | 320/907 [00:08<00:14, 40.29it/s] |
| Computing ESM2 embeddings: 39%  | 352/907 [00:08<00:13, 40.32it/s] |
| Computing ESM2 embeddings: 42%  | 384/907 [00:09<00:12, 40.34it/s] |
| Computing ESM2 embeddings: 46%  | 416/907 [00:10<00:12, 40.29it/s] |
| Computing ESM2 embeddings: 49%  | 448/907 [00:11<00:11, 40.29it/s] |
| Computing ESM2 embeddings: 53%  | 480/907 [00:11<00:10, 40.32it/s] |
| Computing ESM2 embeddings: 56%  | 512/907 [00:12<00:09, 40.35it/s] |
| Computing ESM2 embeddings: 60%  | 544/907 [00:13<00:09, 40.33it/s] |
| Computing ESM2 embeddings: 64%  | 576/907 [00:14<00:08, 40.29it/s] |
| Computing ESM2 embeddings: 67%  | 608/907 [00:15<00:07, 39.46it/s] |
| Computing ESM2 embeddings: 71%  | 640/907 [00:16<00:06, 39.62it/s] |
| Computing ESM2 embeddings: 74%  | 672/907 [00:16<00:05, 39.85it/s] |
| Computing ESM2 embeddings: 78%  | 704/907 [00:17<00:05, 40.00it/s] |
| Computing ESM2 embeddings: 81%  | 736/907 [00:18<00:04, 40.07it/s] |
| Computing ESM2 embeddings: 85%  | 768/907 [00:19<00:03, 40.15it/s] |
| Computing ESM2 embeddings: 88%  | 800/907 [00:19<00:02, 40.21it/s] |
| Computing ESM2 embeddings: 92%  | 832/907 [00:20<00:01, 40.25it/s] |
| Computing ESM2 embeddings: 95%  | 864/907 [00:21<00:01, 40.29it/s] |
| Computing ESM2 embeddings: 99%  | 896/907 [00:22<00:00, 40.26it/s] |
| Computing ESM2 embeddings: 100% | 907/907 [00:22<00:00, 39.96it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/903 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/903 [00:00<00:23, 37.42it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/903 [00:01<00:21, 38.47it/s]  |
| Computing ESM2 embeddings: | 11% | 96/903 [00:02<00:20, 39.13it/s]  |
| Computing ESM2 embeddings: | 14% | 128/903 [00:03<00:19, 39.57it/s] |
| Computing ESM2 embeddings: | 18% | 160/903 [00:04<00:18, 39.83it/s] |
| Computing ESM2 embeddings: | 21% | 192/903 [00:04<00:17, 39.94it/s] |
| Computing ESM2 embeddings: | 25% | 224/903 [00:05<00:16, 40.05it/s] |
| Computing ESM2 embeddings: | 28% | 256/903 [00:06<00:16, 40.15it/s] |
| Computing ESM2 embeddings: | 32% | 288/903 [00:07<00:15, 40.15it/s] |
| Computing ESM2 embeddings: | 35% | 320/903 [00:08<00:14, 40.20it/s] |
| Computing ESM2 embeddings: | 39% | 352/903 [00:08<00:13, 40.19it/s] |
| Computing ESM2 embeddings: | 43% | 384/903 [00:09<00:12, 40.21it/s] |
| Computing ESM2 embeddings: | 46% | 416/903 [00:10<00:12, 40.25it/s] |
| Computing ESM2 embeddings: | 50% | 448/903 [00:11<00:11, 40.28it/s] |
| Computing ESM2 embeddings: | 53% | 480/903 [00:12<00:10, 40.30it/s] |
| Computing ESM2 embeddings: | 57% | 512/903 [00:12<00:09, 40.27it/s] |
| Computing ESM2 embeddings: | 60% | 544/903 [00:13<00:08, 40.29it/s] |
| Computing ESM2 embeddings: | 64% | 576/903 [00:14<00:08, 40.30it/s] |
| Computing ESM2 embeddings: | 67% | 608/903 [00:15<00:07, 39.16it/s] |
| Computing ESM2 embeddings: | 71% | 640/903 [00:16<00:06, 39.37it/s] |
| Computing ESM2 embeddings: | 74% | 672/903 [00:16<00:05, 39.59it/s] |



```

Computing ESM2 embeddings: 78%|      | 704/903 [00:17<00:04, 39.82it/s]
Computing ESM2 embeddings: 82%|      | 736/903 [00:18<00:04, 39.95it/s]
Computing ESM2 embeddings: 85%|      | 768/903 [00:19<00:03, 40.06it/s]
Computing ESM2 embeddings: 89%|      | 800/903 [00:20<00:02, 40.12it/s]
Computing ESM2 embeddings: 92%|      | 832/903 [00:20<00:01, 40.12it/s]
Computing ESM2 embeddings: 96%|      | 864/903 [00:21<00:00, 40.17it/s]
Computing ESM2 embeddings: 99%|      | 896/903 [00:22<00:00, 40.22it/s]

Computing ESM2 embeddings: 100%|     | 903/903 [00:22<00:00, 39.91it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/902 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/902 [00:00<00:22, 37.84it/s]
Computing ESM2 embeddings:  7%|      | 64/902 [00:01<00:21, 39.24it/s]
Computing ESM2 embeddings: 11%|      | 96/902 [00:02<00:20, 39.73it/s]
Computing ESM2 embeddings: 14%|      | 128/902 [00:03<00:19, 39.89it/s]
Computing ESM2 embeddings: 18%|      | 160/902 [00:04<00:18, 40.06it/s]
Computing ESM2 embeddings: 21%|      | 192/902 [00:04<00:17, 40.14it/s]
Computing ESM2 embeddings: 25%|      | 224/902 [00:05<00:17, 38.47it/s]
Computing ESM2 embeddings: 28%|      | 256/902 [00:06<00:16, 38.91it/s]
Computing ESM2 embeddings: 32%|      | 288/902 [00:07<00:15, 39.26it/s]
Computing ESM2 embeddings: 35%|      | 320/902 [00:08<00:14, 39.56it/s]
Computing ESM2 embeddings: 39%|      | 352/902 [00:08<00:13, 39.77it/s]
Computing ESM2 embeddings: 43%|      | 384/902 [00:09<00:12, 39.91it/s]

```

Computing ESM2 embeddings: 46%| | 416/902 [00:10<00:12, 40.02it/s]

Computing ESM2 embeddings: 50%| | 448/902 [00:11<00:11, 40.03it/s]

Computing ESM2 embeddings: 53%| | 480/902 [00:12<00:10, 40.10it/s]

Computing ESM2 embeddings: 57%| | 512/902 [00:12<00:09, 40.15it/s]

Computing ESM2 embeddings: 60%| | 544/902 [00:13<00:08, 40.18it/s]

Computing ESM2 embeddings: 64%| | 576/902 [00:14<00:08, 40.18it/s]

Computing ESM2 embeddings: 67%| | 608/902 [00:15<00:07, 40.13it/s]

Computing ESM2 embeddings: 71%| | 640/902 [00:16<00:06, 40.16it/s]

Computing ESM2 embeddings: 75%| | 672/902 [00:16<00:05, 40.19it/s]

Computing ESM2 embeddings: 78%| | 704/902 [00:17<00:04, 40.18it/s]

Computing ESM2 embeddings: 82%| | 736/902 [00:18<00:04, 40.19it/s]

Computing ESM2 embeddings: 85%| | 768/902 [00:19<00:03, 40.13it/s]

Computing ESM2 embeddings: 89%| | 800/902 [00:20<00:02, 40.16it/s]

Computing ESM2 embeddings: 92%| | 832/902 [00:20<00:01, 39.34it/s]

Computing ESM2 embeddings: 96%| | 864/902 [00:21<00:00, 39.45it/s]

Computing ESM2 embeddings: 99%| | 896/902 [00:22<00:00, 39.66it/s]

Computing ESM2 embeddings: 100%| | 902/902 [00:22<00:00, 39.72it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/760 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/760 [00:00<00:18, 39.33it/s]

Computing ESM2 embeddings: 8%| | 64/760 [00:01<00:17, 39.81it/s]

Computing ESM2 embeddings: 13%| | 96/760 [00:02<00:16, 40.08it/s]

```

Computing ESM2 embeddings: 17%|          | 128/760 [00:03<00:15, 40.20it/s]
Computing ESM2 embeddings: 21%|          | 160/760 [00:03<00:14, 40.25it/s]
Computing ESM2 embeddings: 25%|          | 192/760 [00:04<00:14, 40.29it/s]
Computing ESM2 embeddings: 29%|          | 224/760 [00:05<00:13, 40.25it/s]
Computing ESM2 embeddings: 34%|          | 256/760 [00:06<00:12, 40.28it/s]
Computing ESM2 embeddings: 38%|          | 288/760 [00:07<00:11, 40.30it/s]
Computing ESM2 embeddings: 42%|          | 320/760 [00:07<00:10, 40.32it/s]
Computing ESM2 embeddings: 46%|          | 352/760 [00:08<00:10, 40.32it/s]
Computing ESM2 embeddings: 51%|          | 384/760 [00:09<00:09, 40.19it/s]
Computing ESM2 embeddings: 55%|          | 416/760 [00:10<00:08, 40.18it/s]
Computing ESM2 embeddings: 59%|          | 448/760 [00:11<00:07, 40.20it/s]
Computing ESM2 embeddings: 63%|          | 480/760 [00:11<00:06, 40.21it/s]
Computing ESM2 embeddings: 67%|          | 512/760 [00:12<00:06, 40.21it/s]
Computing ESM2 embeddings: 72%|          | 544/760 [00:13<00:05, 40.14it/s]
Computing ESM2 embeddings: 76%|          | 576/760 [00:14<00:04, 40.15it/s]
Computing ESM2 embeddings: 80%|          | 608/760 [00:15<00:03, 38.76it/s]
Computing ESM2 embeddings: 84%|          | 640/760 [00:16<00:03, 39.08it/s]
Computing ESM2 embeddings: 88%|          | 672/760 [00:16<00:02, 39.40it/s]
Computing ESM2 embeddings: 93%|          | 704/760 [00:17<00:01, 39.60it/s]
Computing ESM2 embeddings: 97%|          | 736/760 [00:18<00:00, 39.73it/s]

Computing ESM2 embeddings: 100%|         | 760/760 [00:19<00:00, 39.89it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |      |                                  |
|----------------------------|------|----------------------------------|
| Computing ESM2 embeddings: | 0%   | 0/719 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%   | 32/719 [00:00<00:18, 37.76it/s]  |
| Computing ESM2 embeddings: | 9%   | 64/719 [00:01<00:16, 39.08it/s]  |
| Computing ESM2 embeddings: | 13%  | 96/719 [00:02<00:15, 39.60it/s]  |
| Computing ESM2 embeddings: | 18%  | 128/719 [00:03<00:14, 39.92it/s] |
| Computing ESM2 embeddings: | 22%  | 160/719 [00:04<00:13, 40.02it/s] |
| Computing ESM2 embeddings: | 27%  | 192/719 [00:04<00:13, 40.15it/s] |
| Computing ESM2 embeddings: | 31%  | 224/719 [00:05<00:12, 40.23it/s] |
| Computing ESM2 embeddings: | 36%  | 256/719 [00:06<00:11, 40.27it/s] |
| Computing ESM2 embeddings: | 40%  | 288/719 [00:07<00:10, 40.29it/s] |
| Computing ESM2 embeddings: | 45%  | 320/719 [00:07<00:09, 40.26it/s] |
| Computing ESM2 embeddings: | 49%  | 352/719 [00:08<00:09, 40.30it/s] |
| Computing ESM2 embeddings: | 53%  | 384/719 [00:09<00:08, 40.31it/s] |
| Computing ESM2 embeddings: | 58%  | 416/719 [00:10<00:07, 40.31it/s] |
| Computing ESM2 embeddings: | 62%  | 448/719 [00:11<00:06, 40.33it/s] |
| Computing ESM2 embeddings: | 67%  | 480/719 [00:11<00:05, 40.27it/s] |
| Computing ESM2 embeddings: | 71%  | 512/719 [00:12<00:05, 40.30it/s] |
| Computing ESM2 embeddings: | 76%  | 544/719 [00:13<00:04, 40.32it/s] |
| Computing ESM2 embeddings: | 80%  | 576/719 [00:14<00:03, 40.34it/s] |
| Computing ESM2 embeddings: | 85%  | 608/719 [00:15<00:02, 39.48it/s] |
| Computing ESM2 embeddings: | 89%  | 640/719 [00:16<00:01, 39.52it/s] |
| Computing ESM2 embeddings: | 93%  | 672/719 [00:16<00:01, 39.77it/s] |
| Computing ESM2 embeddings: | 98%  | 704/719 [00:17<00:00, 39.94it/s] |
| Computing ESM2 embeddings: | 100% | 719/719 [00:18<00:00, 39.55it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/916 [00:00<?, ?it/s]
Computing ESM2 embeddings: 3%|          | 32/916 [00:00<00:23, 37.35it/s]
Computing ESM2 embeddings: 7%|          | 64/916 [00:01<00:22, 38.59it/s]
Computing ESM2 embeddings: 10%|         | 96/916 [00:02<00:20, 39.29it/s]
Computing ESM2 embeddings: 14%|         | 128/916 [00:03<00:19, 39.63it/s]
Computing ESM2 embeddings: 17%|         | 160/916 [00:04<00:18, 39.88it/s]
Computing ESM2 embeddings: 21%|         | 192/916 [00:04<00:18, 40.03it/s]
Computing ESM2 embeddings: 24%|         | 224/916 [00:05<00:17, 40.13it/s]
Computing ESM2 embeddings: 28%|         | 256/916 [00:06<00:16, 40.20it/s]
Computing ESM2 embeddings: 31%|         | 288/916 [00:07<00:15, 40.17it/s]
Computing ESM2 embeddings: 35%|         | 320/916 [00:08<00:14, 40.21it/s]
Computing ESM2 embeddings: 38%|         | 352/916 [00:08<00:14, 40.23it/s]
Computing ESM2 embeddings: 42%|         | 384/916 [00:09<00:13, 40.26it/s]
Computing ESM2 embeddings: 45%|         | 416/916 [00:10<00:12, 40.28it/s]
Computing ESM2 embeddings: 49%|         | 448/916 [00:11<00:11, 40.24it/s]
Computing ESM2 embeddings: 52%|         | 480/916 [00:11<00:10, 40.27it/s]
Computing ESM2 embeddings: 56%|         | 512/916 [00:12<00:10, 40.21it/s]
Computing ESM2 embeddings: 59%|         | 544/916 [00:13<00:09, 38.80it/s]
Computing ESM2 embeddings: 63%|         | 576/916 [00:14<00:08, 39.10it/s]
Computing ESM2 embeddings: 66%|         | 608/916 [00:15<00:07, 39.33it/s]
Computing ESM2 embeddings: 70%|         | 640/916 [00:16<00:06, 39.59it/s]
```

```

Computing ESM2 embeddings: 73%|      | 672/916 [00:16<00:06, 39.77it/s]
Computing ESM2 embeddings: 77%|      | 704/916 [00:17<00:05, 39.91it/s]
Computing ESM2 embeddings: 80%|      | 736/916 [00:18<00:04, 40.02it/s]
Computing ESM2 embeddings: 84%|      | 768/916 [00:19<00:03, 40.04it/s]
Computing ESM2 embeddings: 87%|      | 800/916 [00:20<00:02, 40.09it/s]
Computing ESM2 embeddings: 91%|      | 832/916 [00:20<00:02, 40.14it/s]
Computing ESM2 embeddings: 94%|      | 864/916 [00:21<00:01, 40.19it/s]
Computing ESM2 embeddings: 98%|      | 896/916 [00:22<00:00, 40.20it/s]

Computing ESM2 embeddings: 100%|     | 916/916 [00:22<00:00, 39.85it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/896 [00:00<00:22, 38.81it/s]
Computing ESM2 embeddings: 7%|      | 64/896 [00:01<00:21, 39.34it/s]
Computing ESM2 embeddings: 11%|     | 96/896 [00:02<00:20, 39.81it/s]
Computing ESM2 embeddings: 14%|     | 128/896 [00:03<00:19, 40.05it/s]
Computing ESM2 embeddings: 18%|     | 160/896 [00:04<00:18, 40.17it/s]
Computing ESM2 embeddings: 21%|     | 192/896 [00:04<00:17, 40.27it/s]
Computing ESM2 embeddings: 25%|     | 224/896 [00:05<00:16, 40.24it/s]
Computing ESM2 embeddings: 29%|     | 256/896 [00:06<00:15, 40.28it/s]
Computing ESM2 embeddings: 32%|     | 288/896 [00:07<00:15, 40.30it/s]
Computing ESM2 embeddings: 36%|     | 320/896 [00:07<00:14, 40.32it/s]
Computing ESM2 embeddings: 39%|     | 352/896 [00:08<00:13, 40.32it/s]

```

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:12, 40.29it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:11, 40.31it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 40.33it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:11<00:10, 40.33it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:12<00:09, 40.30it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:09, 38.85it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.16it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.50it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.76it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:16<00:05, 39.93it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 39.94it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:03, 40.06it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 39.88it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 39.90it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:20<00:01, 40.03it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 40.05it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.94it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.93it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.22it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.77it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:19, 39.95it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:18, 40.12it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:17, 40.21it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 40.26it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 40.31it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:15, 40.30it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:14, 40.05it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:08<00:14, 40.03it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 40.13it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 40.05it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 40.10it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:11<00:10, 40.14it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:12<00:10, 40.17it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 38.76it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 38.73it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.06it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:06, 39.39it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:16<00:06, 39.66it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 39.76it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.89it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 39.99it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 40.06it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:20<00:02, 40.11it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:21<00:01, 40.09it/s] |



Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 40.14it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:22<00:00, 39.82it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:24, 34.64it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 37.52it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.71it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.37it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.74it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 39.97it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:16, 40.04it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:15, 40.08it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 40.18it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 40.27it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:13, 40.32it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:12, 40.27it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:11, 40.32it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 40.34it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 40.36it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:12<00:09, 40.36it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 40.31it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:07, 40.34it/s]

```

Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 40.36it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.97it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:16<00:05, 39.20it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 39.54it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 39.80it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 39.97it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 40.10it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:20<00:01, 40.11it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 40.20it/s]
Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.85it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:22, 38.50it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 39.27it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:20, 39.68it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:19, 39.95it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:18, 40.10it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:04<00:17, 40.17it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:17, 40.23it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:16, 40.18it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:15, 40.23it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:07<00:14, 40.26it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:13, 40.29it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 40.20it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 40.24it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 40.27it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:11<00:10, 40.31it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:12<00:09, 40.24it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 40.26it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 40.26it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.43it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.53it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:16<00:06, 39.76it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.93it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 40.06it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 40.06it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:19<00:02, 40.12it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:20<00:02, 40.19it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 40.21it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 40.15it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:22<00:00, 40.01it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.95it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.96it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 38.96it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.26it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.65it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:17, 39.88it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:16, 40.04it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:15, 40.08it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 40.15it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 40.23it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 40.27it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:12, 40.19it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:11, 40.24it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 40.27it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 40.30it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:12<00:09, 40.27it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:08, 40.28it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.89it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.19it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.48it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:16<00:05, 39.72it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:17<00:04, 39.90it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:18<00:03, 40.03it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:19<00:03, 40.07it/s] |
| Computing ESM2 embeddings: | 89% | 800/896 [00:20<00:02, 40.15it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/896 [00:20<00:01, 40.21it/s]

Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 40.25it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.88it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/707 [00:00<?, ?it/s]

Computing ESM2 embeddings:  5%|      | 32/707 [00:00<00:17, 38.48it/s]

Computing ESM2 embeddings:  9%|      | 64/707 [00:01<00:16, 39.36it/s]

Computing ESM2 embeddings: 14%|      | 96/707 [00:02<00:15, 39.84it/s]

Computing ESM2 embeddings: 18%|      | 128/707 [00:03<00:14, 40.08it/s]

Computing ESM2 embeddings: 23%|      | 160/707 [00:04<00:13, 40.14it/s]

Computing ESM2 embeddings: 27%|      | 192/707 [00:04<00:12, 40.22it/s]

Computing ESM2 embeddings: 32%|      | 224/707 [00:05<00:11, 40.28it/s]

Computing ESM2 embeddings: 36%|      | 256/707 [00:06<00:11, 40.31it/s]

Computing ESM2 embeddings: 41%|      | 288/707 [00:07<00:10, 40.27it/s]

Computing ESM2 embeddings: 45%|      | 320/707 [00:07<00:09, 40.30it/s]

Computing ESM2 embeddings: 50%|      | 352/707 [00:08<00:08, 40.32it/s]

Computing ESM2 embeddings: 54%|      | 384/707 [00:09<00:08, 40.25it/s]

Computing ESM2 embeddings: 59%|      | 416/707 [00:10<00:07, 40.22it/s]

Computing ESM2 embeddings: 63%|      | 448/707 [00:11<00:06, 40.27it/s]

Computing ESM2 embeddings: 68%|      | 480/707 [00:11<00:05, 40.29it/s]

Computing ESM2 embeddings: 72%|      | 512/707 [00:12<00:04, 40.30it/s]

Computing ESM2 embeddings: 77%|      | 544/707 [00:13<00:04, 40.26it/s]

```

```

Computing ESM2 embeddings: 81%|      | 576/707 [00:14<00:03, 38.84it/s]
Computing ESM2 embeddings: 86%|      | 608/707 [00:15<00:02, 38.83it/s]
Computing ESM2 embeddings: 91%|      | 640/707 [00:16<00:01, 39.14it/s]
Computing ESM2 embeddings: 95%|      | 672/707 [00:16<00:00, 39.40it/s]
Computing ESM2 embeddings: 100%|     | 707/707 [00:17<00:00, 39.45it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/777 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/777 [00:00<00:19, 38.59it/s]
Computing ESM2 embeddings:  8%|      | 64/777 [00:01<00:18, 39.36it/s]
Computing ESM2 embeddings: 12%|      | 96/777 [00:02<00:17, 39.80it/s]
Computing ESM2 embeddings: 16%|      | 128/777 [00:03<00:16, 39.95it/s]
Computing ESM2 embeddings: 21%|      | 160/777 [00:04<00:15, 40.03it/s]
Computing ESM2 embeddings: 25%|      | 192/777 [00:04<00:14, 40.11it/s]
Computing ESM2 embeddings: 29%|      | 224/777 [00:05<00:13, 40.15it/s]
Computing ESM2 embeddings: 33%|      | 256/777 [00:06<00:12, 40.13it/s]
Computing ESM2 embeddings: 37%|      | 288/777 [00:07<00:12, 40.16it/s]
Computing ESM2 embeddings: 41%|      | 320/777 [00:07<00:11, 40.20it/s]
Computing ESM2 embeddings: 45%|      | 352/777 [00:08<00:10, 40.21it/s]
Computing ESM2 embeddings: 49%|      | 384/777 [00:09<00:09, 40.12it/s]
Computing ESM2 embeddings: 54%|      | 416/777 [00:10<00:08, 40.17it/s]
Computing ESM2 embeddings: 58%|      | 448/777 [00:11<00:08, 40.19it/s]
Computing ESM2 embeddings: 62%|      | 480/777 [00:11<00:07, 40.19it/s]

```

Computing ESM2 embeddings: 66%| | 512/777 [00:12<00:06, 40.16it/s]

Computing ESM2 embeddings: 70%| | 544/777 [00:13<00:05, 40.16it/s]

Computing ESM2 embeddings: 74%| | 576/777 [00:14<00:05, 38.83it/s]

Computing ESM2 embeddings: 78%| | 608/777 [00:15<00:04, 39.11it/s]

Computing ESM2 embeddings: 82%| | 640/777 [00:16<00:03, 39.40it/s]

Computing ESM2 embeddings: 86%| | 672/777 [00:16<00:02, 39.67it/s]

Computing ESM2 embeddings: 91%| | 704/777 [00:17<00:01, 39.84it/s]

Computing ESM2 embeddings: 95%| | 736/777 [00:18<00:01, 39.98it/s]

Computing ESM2 embeddings: 99%| | 768/777 [00:19<00:00, 40.02it/s]

Computing ESM2 embeddings: 100%| | 777/777 [00:19<00:00, 39.58it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/916 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/916 [00:00<00:23, 38.29it/s]

Computing ESM2 embeddings: 7%| | 64/916 [00:01<00:21, 39.15it/s]

Computing ESM2 embeddings: 10%| | 96/916 [00:02<00:20, 39.70it/s]

Computing ESM2 embeddings: 14%| | 128/916 [00:03<00:19, 39.82it/s]

Computing ESM2 embeddings: 17%| | 160/916 [00:04<00:18, 40.04it/s]

Computing ESM2 embeddings: 21%| | 192/916 [00:04<00:18, 40.11it/s]

Computing ESM2 embeddings: 24%| | 224/916 [00:05<00:17, 40.18it/s]

Computing ESM2 embeddings: 28%| | 256/916 [00:06<00:16, 40.16it/s]

Computing ESM2 embeddings: 31%| | 288/916 [00:07<00:15, 40.21it/s]

Computing ESM2 embeddings: 35%| | 320/916 [00:07<00:14, 40.23it/s]

Computing ESM2 embeddings: 38%| | 352/916 [00:08<00:14, 40.26it/s]

Computing ESM2 embeddings: 42%| | 384/916 [00:09<00:13, 40.19it/s]

Computing ESM2 embeddings: 45%| | 416/916 [00:10<00:12, 40.24it/s]

Computing ESM2 embeddings: 49%| | 448/916 [00:11<00:11, 40.23it/s]

Computing ESM2 embeddings: 52%| | 480/916 [00:11<00:10, 40.27it/s]

Computing ESM2 embeddings: 56%| | 512/916 [00:12<00:10, 40.20it/s]

Computing ESM2 embeddings: 59%| | 544/916 [00:13<00:09, 40.24it/s]

Computing ESM2 embeddings: 63%| | 576/916 [00:14<00:08, 38.82it/s]

Computing ESM2 embeddings: 66%| | 608/916 [00:15<00:07, 39.13it/s]

Computing ESM2 embeddings: 70%| | 640/916 [00:16<00:07, 39.40it/s]

Computing ESM2 embeddings: 73%| | 672/916 [00:16<00:06, 39.68it/s]

Computing ESM2 embeddings: 77%| | 704/916 [00:17<00:05, 39.83it/s]

Computing ESM2 embeddings: 80%| | 736/916 [00:18<00:04, 39.98it/s]

Computing ESM2 embeddings: 84%| | 768/916 [00:19<00:03, 40.00it/s]

Computing ESM2 embeddings: 87%| | 800/916 [00:20<00:02, 40.10it/s]

Computing ESM2 embeddings: 91%| | 832/916 [00:20<00:02, 40.12it/s]

Computing ESM2 embeddings: 94%| | 864/916 [00:21<00:01, 40.18it/s]

Computing ESM2 embeddings: 98%| | 896/916 [00:22<00:00, 40.13it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:22<00:00, 39.94it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.22it/s]



|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.16it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.70it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.78it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.96it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:17, 40.08it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:16, 40.16it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:15, 40.14it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 40.20it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 40.22it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 40.24it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:12, 40.18it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:11, 40.21it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 40.22it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:11<00:10, 40.25it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:12<00:09, 40.16it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:08, 40.15it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.81it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.12it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.35it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:16<00:05, 39.59it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:17<00:04, 39.76it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:18<00:04, 39.89it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:19<00:03, 39.91it/s] |
| Computing ESM2 embeddings: | 89% | 800/896 [00:20<00:02, 40.00it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/896 [00:20<00:01, 40.05it/s]

Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 40.11it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.86it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.98it/s]

Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 39.26it/s]

Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:20, 39.72it/s]

Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:19, 39.93it/s]

Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:18, 39.99it/s]

Computing ESM2 embeddings: 21%|      | 192/914 [00:04<00:18, 40.06it/s]

Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 40.11it/s]

Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 40.15it/s]

Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:15, 40.11it/s]

Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:14, 40.13it/s]

Computing ESM2 embeddings: 39%|      | 352/914 [00:08<00:13, 40.16it/s]

Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 40.16it/s]

Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 40.12it/s]

Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 40.15it/s]

Computing ESM2 embeddings: 53%|      | 480/914 [00:11<00:10, 40.16it/s]

Computing ESM2 embeddings: 56%|      | 512/914 [00:12<00:10, 40.17it/s]

Computing ESM2 embeddings: 60%|      | 544/914 [00:13<00:09, 40.13it/s]

```

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.32it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.46it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.60it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:16<00:06, 39.58it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.71it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.85it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.96it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.98it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:20<00:02, 40.01it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 40.06it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 40.11it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:22<00:00, 39.86it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.99it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.30it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.13it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.60it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.80it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 39.98it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:16, 40.09it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:15, 40.17it/s]

```

Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 40.13it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 40.18it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:08<00:13, 40.15it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:12, 40.21it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:11, 40.19it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 40.22it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 40.23it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:12<00:09, 40.26it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:08, 40.21it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 39.52it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 39.72it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.88it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:16<00:05, 39.96it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 40.07it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:03, 40.15it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 40.20it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 40.15it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:20<00:01, 40.21it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 40.23it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.95it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 4%  | 32/914 [00:00<00:22, 38.55it/s]  |
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:21, 39.28it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:20, 39.75it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:19, 40.00it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:18, 40.12it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:17, 40.13it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 40.20it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 40.24it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:15, 40.26it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:07<00:14, 40.21it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:08<00:13, 40.25it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 40.25it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 40.26it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 40.21it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:11<00:10, 40.25it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:12<00:09, 40.26it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.40it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.47it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.72it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:06, 39.89it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:16<00:06, 40.01it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 40.03it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 40.12it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 40.15it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/914 [00:19<00:02, 40.20it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:20<00:02, 40.16it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:21<00:01, 40.21it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 40.20it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:22<00:00, 40.00it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/895 [00:00<00:22, 38.15it/s]
Computing ESM2 embeddings:  7%|      | 64/895 [00:01<00:21, 39.08it/s]
Computing ESM2 embeddings: 11%|      | 96/895 [00:02<00:20, 38.38it/s]
Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:19, 38.94it/s]
Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:18, 39.44it/s]
Computing ESM2 embeddings: 21%|      | 192/895 [00:04<00:17, 39.67it/s]
Computing ESM2 embeddings: 25%|      | 224/895 [00:05<00:16, 39.88it/s]
Computing ESM2 embeddings: 29%|      | 256/895 [00:06<00:15, 40.01it/s]
Computing ESM2 embeddings: 32%|      | 288/895 [00:07<00:15, 40.11it/s]
Computing ESM2 embeddings: 36%|      | 320/895 [00:08<00:14, 40.11it/s]
Computing ESM2 embeddings: 39%|      | 352/895 [00:08<00:13, 40.20it/s]
Computing ESM2 embeddings: 43%|      | 384/895 [00:09<00:12, 40.21it/s]
Computing ESM2 embeddings: 46%|      | 416/895 [00:10<00:11, 40.26it/s]
Computing ESM2 embeddings: 50%|      | 448/895 [00:11<00:11, 40.19it/s]
Computing ESM2 embeddings: 54%|      | 480/895 [00:12<00:10, 40.23it/s]

```

Computing ESM2 embeddings: 57%| | 512/895 [00:12<00:09, 40.24it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:13<00:08, 40.25it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:14<00:07, 40.17it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:15<00:07, 39.83it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:16<00:06, 38.39it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:16<00:05, 38.84it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:17<00:04, 39.20it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:18<00:04, 39.54it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:19<00:03, 39.75it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:20<00:02, 39.92it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:20<00:01, 39.95it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:21<00:00, 40.05it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:22<00:00, 39.69it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/650 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/650 [00:00<00:16, 36.57it/s]

Computing ESM2 embeddings: 10%| | 64/650 [00:01<00:15, 38.41it/s]

Computing ESM2 embeddings: 15%| | 96/650 [00:02<00:14, 39.21it/s]

Computing ESM2 embeddings: 20%| | 128/650 [00:03<00:13, 39.69it/s]

Computing ESM2 embeddings: 25%| | 160/650 [00:04<00:12, 39.94it/s]

Computing ESM2 embeddings: 30%| | 192/650 [00:04<00:11, 40.11it/s]

Computing ESM2 embeddings: 34%| | 224/650 [00:05<00:10, 40.15it/s]

```

Computing ESM2 embeddings: 39%|          | 256/650 [00:06<00:09, 40.24it/s]
Computing ESM2 embeddings: 44%|          | 288/650 [00:07<00:08, 40.28it/s]
Computing ESM2 embeddings: 49%|          | 320/650 [00:08<00:08, 40.35it/s]
Computing ESM2 embeddings: 54%|          | 352/650 [00:08<00:07, 40.28it/s]
Computing ESM2 embeddings: 59%|          | 384/650 [00:09<00:06, 40.32it/s]
Computing ESM2 embeddings: 64%|          | 416/650 [00:10<00:05, 40.35it/s]
Computing ESM2 embeddings: 69%|          | 448/650 [00:11<00:05, 40.36it/s]
Computing ESM2 embeddings: 74%|          | 480/650 [00:12<00:04, 39.02it/s]
Computing ESM2 embeddings: 79%|          | 512/650 [00:12<00:03, 39.29it/s]
Computing ESM2 embeddings: 84%|          | 544/650 [00:13<00:02, 39.58it/s]
Computing ESM2 embeddings: 89%|          | 576/650 [00:14<00:01, 39.79it/s]
Computing ESM2 embeddings: 94%|          | 608/650 [00:15<00:01, 39.59it/s]
Computing ESM2 embeddings: 98%|          | 640/650 [00:16<00:00, 39.69it/s]

```

```

Computing ESM2 embeddings: 100%|         | 650/650 [00:16<00:00, 39.72it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/818 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/818 [00:00<00:21, 37.33it/s]
Computing ESM2 embeddings: 8%|           | 64/818 [00:01<00:19, 39.07it/s]
Computing ESM2 embeddings: 12%|          | 96/818 [00:02<00:18, 39.57it/s]
Computing ESM2 embeddings: 16%|          | 128/818 [00:03<00:17, 39.79it/s]
Computing ESM2 embeddings: 20%|          | 160/818 [00:04<00:16, 40.00it/s]
Computing ESM2 embeddings: 23%|          | 192/818 [00:04<00:15, 40.11it/s]

```



```

Computing ESM2 embeddings: 27%|          | 224/818 [00:05<00:14, 40.13it/s]
Computing ESM2 embeddings: 31%|          | 256/818 [00:06<00:13, 40.21it/s]
Computing ESM2 embeddings: 35%|          | 288/818 [00:07<00:13, 40.25it/s]
Computing ESM2 embeddings: 39%|          | 320/818 [00:08<00:12, 40.31it/s]
Computing ESM2 embeddings: 43%|          | 352/818 [00:08<00:11, 40.28it/s]
Computing ESM2 embeddings: 47%|          | 384/818 [00:09<00:10, 40.32it/s]
Computing ESM2 embeddings: 51%|          | 416/818 [00:10<00:09, 40.31it/s]
Computing ESM2 embeddings: 55%|          | 448/818 [00:11<00:09, 40.34it/s]
Computing ESM2 embeddings: 59%|          | 480/818 [00:11<00:08, 40.29it/s]
Computing ESM2 embeddings: 63%|          | 512/818 [00:12<00:07, 40.31it/s]
Computing ESM2 embeddings: 67%|          | 544/818 [00:13<00:07, 38.95it/s]
Computing ESM2 embeddings: 70%|          | 576/818 [00:14<00:06, 39.23it/s]
Computing ESM2 embeddings: 74%|          | 608/818 [00:15<00:05, 39.48it/s]
Computing ESM2 embeddings: 78%|          | 640/818 [00:16<00:04, 39.73it/s]
Computing ESM2 embeddings: 82%|          | 672/818 [00:16<00:03, 39.91it/s]
Computing ESM2 embeddings: 86%|          | 704/818 [00:17<00:02, 40.05it/s]
Computing ESM2 embeddings: 90%|          | 736/818 [00:18<00:02, 40.09it/s]
Computing ESM2 embeddings: 94%|          | 768/818 [00:19<00:01, 40.17it/s]
Computing ESM2 embeddings: 98%|          | 800/818 [00:20<00:00, 40.22it/s]

Computing ESM2 embeddings: 100%|         | 818/818 [00:20<00:00, 39.92it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          | 0/917 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 3%  | 32/917 [00:00<00:23, 38.39it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/917 [00:01<00:21, 39.18it/s]  |
| Computing ESM2 embeddings: | 10% | 96/917 [00:02<00:20, 39.75it/s]  |
| Computing ESM2 embeddings: | 14% | 128/917 [00:03<00:19, 39.98it/s] |
| Computing ESM2 embeddings: | 17% | 160/917 [00:04<00:18, 40.12it/s] |
| Computing ESM2 embeddings: | 21% | 192/917 [00:04<00:18, 40.15it/s] |
| Computing ESM2 embeddings: | 24% | 224/917 [00:05<00:17, 40.23it/s] |
| Computing ESM2 embeddings: | 28% | 256/917 [00:06<00:16, 40.25it/s] |
| Computing ESM2 embeddings: | 31% | 288/917 [00:07<00:15, 40.30it/s] |
| Computing ESM2 embeddings: | 35% | 320/917 [00:07<00:14, 40.26it/s] |
| Computing ESM2 embeddings: | 38% | 352/917 [00:08<00:14, 40.30it/s] |
| Computing ESM2 embeddings: | 42% | 384/917 [00:09<00:13, 40.31it/s] |
| Computing ESM2 embeddings: | 45% | 416/917 [00:10<00:12, 40.33it/s] |
| Computing ESM2 embeddings: | 49% | 448/917 [00:11<00:11, 40.26it/s] |
| Computing ESM2 embeddings: | 52% | 480/917 [00:11<00:10, 40.29it/s] |
| Computing ESM2 embeddings: | 56% | 512/917 [00:12<00:10, 40.30it/s] |
| Computing ESM2 embeddings: | 59% | 544/917 [00:13<00:09, 39.52it/s] |
| Computing ESM2 embeddings: | 63% | 576/917 [00:14<00:08, 39.57it/s] |
| Computing ESM2 embeddings: | 66% | 608/917 [00:15<00:07, 39.80it/s] |
| Computing ESM2 embeddings: | 70% | 640/917 [00:15<00:06, 39.95it/s] |
| Computing ESM2 embeddings: | 73% | 672/917 [00:16<00:06, 40.07it/s] |
| Computing ESM2 embeddings: | 77% | 704/917 [00:17<00:05, 40.06it/s] |
| Computing ESM2 embeddings: | 80% | 736/917 [00:18<00:04, 40.15it/s] |
| Computing ESM2 embeddings: | 84% | 768/917 [00:19<00:03, 40.16it/s] |

Computing ESM2 embeddings: 87%| | 800/917 [00:19<00:02, 40.16it/s]

Computing ESM2 embeddings: 91%| | 832/917 [00:20<00:02, 40.15it/s]

Computing ESM2 embeddings: 94%| | 864/917 [00:21<00:01, 40.21it/s]

Computing ESM2 embeddings: 98%| | 896/917 [00:22<00:00, 40.24it/s]

Computing ESM2 embeddings: 100%| | 917/917 [00:23<00:00, 39.85it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.38it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.15it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.65it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.92it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 40.06it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 40.09it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:16, 40.16it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:15, 40.22it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 40.26it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:07<00:14, 40.25it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:13, 40.27it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:12, 40.27it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:11, 40.26it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 40.18it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:11<00:10, 40.18it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:12<00:09, 38.81it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:09, 39.10it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.36it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.60it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.79it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:16<00:05, 39.90it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 39.92it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.99it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 40.05it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 40.08it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:20<00:01, 40.04it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 40.09it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.90it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.69it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.40it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.80it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 40.03it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:18, 40.18it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:17, 40.25it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 40.26it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 40.28it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:15, 40.32it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:07<00:14, 40.36it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:13, 40.31it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 40.32it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 40.35it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 40.35it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:11<00:10, 40.30it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:12<00:10, 38.92it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.23it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.56it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.74it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.90it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:16<00:06, 40.03it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 40.12it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 40.13it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 40.18it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:19<00:02, 40.24it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:20<00:02, 40.27it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 40.24it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 40.27it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:22<00:00, 40.05it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.59it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.34it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.74it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.99it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 40.15it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:17, 40.18it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:16, 40.16it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:15, 40.22it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 40.27it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:07<00:14, 40.29it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 40.25it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:12, 40.27it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:11, 40.30it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 40.32it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:11<00:10, 40.28it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:12<00:09, 39.43it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:08, 39.57it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.79it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.90it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:15<00:06, 40.03it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:16<00:05, 40.12it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:17<00:04, 40.18it/s] |

```

Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:03, 40.18it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 40.21it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:19<00:02, 40.24it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:20<00:01, 40.25it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 40.22it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 40.06it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 38.19it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 39.20it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:20, 39.71it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:19, 39.90it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:18, 40.06it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:04<00:17, 40.15it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:17, 39.69it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:16, 39.64it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:15, 39.78it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:14, 39.91it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:08<00:14, 40.00it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:09<00:13, 40.01it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 40.07it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 40.10it/s]

```

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 40.13it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:12<00:10, 38.93it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.20it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.51it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.72it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.80it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:16<00:06, 39.92it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 40.01it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 40.06it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 40.05it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 40.10it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:20<00:02, 40.13it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 40.13it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 40.08it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:22<00:00, 39.83it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.07it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.13it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.67it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.84it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 40.06it/s]



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:17, 39.98it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:16, 40.08it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:15, 40.05it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 40.11it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 40.13it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:08<00:13, 40.18it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:12, 40.13it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:11, 40.17it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 40.17it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:11<00:10, 40.17it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:12<00:09, 38.74it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:13<00:09, 39.06it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 39.39it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 39.56it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 39.67it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:16<00:05, 39.84it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:17<00:04, 39.94it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:18<00:03, 40.03it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 40.01it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 40.08it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:20<00:01, 40.11it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:21<00:00, 40.15it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:22<00:00, 39.81it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

`'esm.pooler.dense.weight']`

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/608 [00:00<?, ?it/s]
Computing ESM2 embeddings: 5%|          | 32/608 [00:00<00:15, 37.43it/s]
Computing ESM2 embeddings: 11%|         | 64/608 [00:01<00:13, 38.89it/s]
Computing ESM2 embeddings: 16%|         | 96/608 [00:02<00:12, 39.54it/s]
Computing ESM2 embeddings: 21%|         | 128/608 [00:03<00:12, 39.86it/s]
Computing ESM2 embeddings: 26%|         | 160/608 [00:04<00:11, 39.93it/s]
Computing ESM2 embeddings: 32%|         | 192/608 [00:04<00:10, 40.07it/s]
Computing ESM2 embeddings: 37%|         | 224/608 [00:05<00:09, 40.15it/s]
Computing ESM2 embeddings: 42%|         | 256/608 [00:06<00:08, 40.18it/s]
Computing ESM2 embeddings: 47%|         | 288/608 [00:07<00:07, 40.15it/s]
Computing ESM2 embeddings: 53%|         | 320/608 [00:08<00:07, 40.20it/s]
Computing ESM2 embeddings: 58%|         | 352/608 [00:08<00:06, 40.24it/s]
Computing ESM2 embeddings: 63%|         | 384/608 [00:09<00:05, 40.19it/s]
Computing ESM2 embeddings: 68%|         | 416/608 [00:10<00:04, 40.11it/s]
Computing ESM2 embeddings: 74%|         | 448/608 [00:11<00:03, 40.12it/s]
Computing ESM2 embeddings: 79%|         | 480/608 [00:12<00:03, 40.12it/s]
Computing ESM2 embeddings: 84%|         | 512/608 [00:12<00:02, 39.27it/s]
Computing ESM2 embeddings: 89%|         | 544/608 [00:13<00:01, 39.35it/s]
Computing ESM2 embeddings: 95%|         | 576/608 [00:14<00:00, 39.59it/s]
```

```
Computing ESM2 embeddings: 100%|        | 608/608 [00:15<00:00, 39.75it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/880 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/880 [00:00<00:22, 37.71it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/880 [00:01<00:21, 38.51it/s]  |
| Computing ESM2 embeddings: | 11% | 96/880 [00:02<00:19, 39.20it/s]  |
| Computing ESM2 embeddings: | 15% | 128/880 [00:03<00:18, 39.60it/s] |
| Computing ESM2 embeddings: | 18% | 160/880 [00:04<00:18, 39.81it/s] |
| Computing ESM2 embeddings: | 22% | 192/880 [00:04<00:17, 39.94it/s] |
| Computing ESM2 embeddings: | 25% | 224/880 [00:05<00:16, 40.00it/s] |
| Computing ESM2 embeddings: | 29% | 256/880 [00:06<00:15, 40.10it/s] |
| Computing ESM2 embeddings: | 33% | 288/880 [00:07<00:14, 40.17it/s] |
| Computing ESM2 embeddings: | 36% | 320/880 [00:08<00:13, 40.22it/s] |
| Computing ESM2 embeddings: | 40% | 352/880 [00:08<00:13, 40.19it/s] |
| Computing ESM2 embeddings: | 44% | 384/880 [00:09<00:12, 40.22it/s] |
| Computing ESM2 embeddings: | 47% | 416/880 [00:10<00:11, 40.25it/s] |
| Computing ESM2 embeddings: | 51% | 448/880 [00:11<00:10, 40.27it/s] |
| Computing ESM2 embeddings: | 55% | 480/880 [00:12<00:09, 40.23it/s] |
| Computing ESM2 embeddings: | 58% | 512/880 [00:12<00:09, 39.45it/s] |
| Computing ESM2 embeddings: | 62% | 544/880 [00:13<00:08, 39.54it/s] |
| Computing ESM2 embeddings: | 65% | 576/880 [00:14<00:07, 39.76it/s] |
| Computing ESM2 embeddings: | 69% | 608/880 [00:15<00:06, 39.85it/s] |
| Computing ESM2 embeddings: | 73% | 640/880 [00:16<00:06, 39.99it/s] |
| Computing ESM2 embeddings: | 76% | 672/880 [00:16<00:05, 40.08it/s] |
| Computing ESM2 embeddings: | 80% | 704/880 [00:17<00:04, 40.15it/s] |

```

Computing ESM2 embeddings: 84%|          | 736/880 [00:18<00:03, 40.13it/s]
Computing ESM2 embeddings: 87%|          | 768/880 [00:19<00:02, 40.19it/s]
Computing ESM2 embeddings: 91%|          | 800/880 [00:20<00:01, 40.23it/s]
Computing ESM2 embeddings: 95%|          | 832/880 [00:20<00:01, 40.26it/s]
Computing ESM2 embeddings: 98%|          | 864/880 [00:21<00:00, 40.21it/s]

Computing ESM2 embeddings: 100%|         | 880/880 [00:22<00:00, 39.92it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/918 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|          | 32/918 [00:00<00:23, 38.37it/s]
Computing ESM2 embeddings:  7%|          | 64/918 [00:01<00:21, 39.14it/s]
Computing ESM2 embeddings: 10%|          | 96/918 [00:02<00:20, 39.61it/s]
Computing ESM2 embeddings: 14%|          | 128/918 [00:03<00:19, 39.78it/s]
Computing ESM2 embeddings: 17%|          | 160/918 [00:04<00:18, 39.97it/s]
Computing ESM2 embeddings: 21%|          | 192/918 [00:04<00:18, 40.06it/s]
Computing ESM2 embeddings: 24%|          | 224/918 [00:05<00:17, 40.11it/s]
Computing ESM2 embeddings: 28%|          | 256/918 [00:06<00:16, 40.08it/s]
Computing ESM2 embeddings: 31%|          | 288/918 [00:07<00:15, 40.12it/s]
Computing ESM2 embeddings: 35%|          | 320/918 [00:08<00:14, 40.15it/s]
Computing ESM2 embeddings: 38%|          | 352/918 [00:08<00:14, 40.11it/s]
Computing ESM2 embeddings: 42%|          | 384/918 [00:09<00:13, 40.07it/s]
Computing ESM2 embeddings: 45%|          | 416/918 [00:10<00:12, 40.11it/s]
Computing ESM2 embeddings: 49%|          | 448/918 [00:11<00:11, 40.15it/s]

```

Computing ESM2 embeddings: 52%| | 480/918 [00:11<00:10, 40.18it/s]

Computing ESM2 embeddings: 56%| | 512/918 [00:12<00:10, 39.27it/s]

Computing ESM2 embeddings: 59%| | 544/918 [00:13<00:09, 39.40it/s]

Computing ESM2 embeddings: 63%| | 576/918 [00:14<00:08, 39.65it/s]

Computing ESM2 embeddings: 66%| | 608/918 [00:15<00:07, 39.81it/s]

Computing ESM2 embeddings: 70%| | 640/918 [00:16<00:06, 39.87it/s]

Computing ESM2 embeddings: 73%| | 672/918 [00:16<00:06, 39.98it/s]

Computing ESM2 embeddings: 77%| | 704/918 [00:17<00:05, 40.05it/s]

Computing ESM2 embeddings: 80%| | 736/918 [00:18<00:04, 40.10it/s]

Computing ESM2 embeddings: 84%| | 768/918 [00:19<00:03, 40.11it/s]

Computing ESM2 embeddings: 87%| | 800/918 [00:20<00:02, 40.16it/s]

Computing ESM2 embeddings: 91%| | 832/918 [00:20<00:02, 40.20it/s]

Computing ESM2 embeddings: 94%| | 864/918 [00:21<00:01, 40.17it/s]

Computing ESM2 embeddings: 98%| | 896/918 [00:22<00:00, 40.13it/s]

Computing ESM2 embeddings: 100%| | 918/918 [00:23<00:00, 39.69it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.53it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.05it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.63it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.84it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 40.01it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:17, 40.12it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:16, 40.18it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:15, 40.16it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 40.21it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 40.22it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:08<00:13, 40.25it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:12, 40.21it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:11, 40.24it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 40.27it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:11<00:10, 40.26it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:12<00:09, 39.00it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:13<00:08, 39.24it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 39.53it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 39.76it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 39.87it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:16<00:05, 39.99it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:17<00:04, 40.08it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:18<00:03, 40.14it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 40.14it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 40.17it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:20<00:01, 40.21it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:21<00:00, 40.24it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:22<00:00, 39.92it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/913 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/913 [00:00<00:23, 37.04it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/913 [00:01<00:21, 38.72it/s]  |
| Computing ESM2 embeddings: | 11% | 96/913 [00:02<00:20, 39.42it/s]  |
| Computing ESM2 embeddings: | 14% | 128/913 [00:03<00:19, 39.77it/s] |
| Computing ESM2 embeddings: | 18% | 160/913 [00:04<00:18, 39.89it/s] |
| Computing ESM2 embeddings: | 21% | 192/913 [00:04<00:18, 40.04it/s] |
| Computing ESM2 embeddings: | 25% | 224/913 [00:05<00:17, 40.12it/s] |
| Computing ESM2 embeddings: | 28% | 256/913 [00:06<00:16, 40.18it/s] |
| Computing ESM2 embeddings: | 32% | 288/913 [00:07<00:15, 40.15it/s] |
| Computing ESM2 embeddings: | 35% | 320/913 [00:08<00:14, 40.20it/s] |
| Computing ESM2 embeddings: | 39% | 352/913 [00:08<00:13, 40.23it/s] |
| Computing ESM2 embeddings: | 42% | 384/913 [00:09<00:13, 40.24it/s] |
| Computing ESM2 embeddings: | 46% | 416/913 [00:10<00:12, 40.19it/s] |
| Computing ESM2 embeddings: | 49% | 448/913 [00:11<00:11, 40.21it/s] |
| Computing ESM2 embeddings: | 53% | 480/913 [00:12<00:11, 38.86it/s] |
| Computing ESM2 embeddings: | 56% | 512/913 [00:12<00:10, 39.17it/s] |
| Computing ESM2 embeddings: | 60% | 544/913 [00:13<00:09, 39.43it/s] |
| Computing ESM2 embeddings: | 63% | 576/913 [00:14<00:08, 39.68it/s] |
| Computing ESM2 embeddings: | 67% | 608/913 [00:15<00:07, 39.84it/s] |
| Computing ESM2 embeddings: | 70% | 640/913 [00:16<00:06, 39.97it/s] |
| Computing ESM2 embeddings: | 74% | 672/913 [00:16<00:06, 39.72it/s] |

```

Computing ESM2 embeddings: 77%|          | 704/913 [00:17<00:05, 39.76it/s]
Computing ESM2 embeddings: 81%|          | 736/913 [00:18<00:04, 39.90it/s]
Computing ESM2 embeddings: 84%|          | 768/913 [00:19<00:03, 39.99it/s]
Computing ESM2 embeddings: 88%|          | 800/913 [00:20<00:02, 39.97it/s]
Computing ESM2 embeddings: 91%|          | 832/913 [00:20<00:02, 40.06it/s]
Computing ESM2 embeddings: 95%|          | 864/913 [00:21<00:01, 40.11it/s]
Computing ESM2 embeddings: 98%|          | 896/913 [00:22<00:00, 40.17it/s]

Computing ESM2 embeddings: 100%|         | 913/913 [00:23<00:00, 39.36it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 37.63it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 38.54it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 39.12it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:19, 39.54it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:18, 39.41it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:04<00:17, 39.55it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:16, 39.65it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 39.76it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 39.84it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 39.93it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:08<00:13, 39.99it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:12, 40.04it/s]

```



Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:11, 40.01it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 40.07it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 38.75it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:12<00:09, 39.06it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.34it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.59it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.76it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.85it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:16<00:05, 39.88it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 39.96it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:03, 40.02it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 40.05it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 40.02it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:20<00:01, 40.03it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 40.01it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.68it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.36it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 38.38it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.00it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 39.48it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:18, 39.73it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 39.84it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 39.97it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 40.06it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:15, 40.09it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:14, 40.08it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:08<00:14, 40.11it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 40.13it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 40.15it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 40.09it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 39.31it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:12<00:10, 39.46it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.67it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.76it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.88it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:06, 39.99it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:16<00:06, 40.05it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 39.99it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 40.05it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 40.09it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 40.12it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:20<00:02, 40.09it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:21<00:01, 40.10it/s] |
| Computing ESM2 embeddings: 98% | 896/914 [00:22<00:00, 40.12it/s] |

Computing ESM2 embeddings: 100%| | 914/914 [00:22<00:00, 39.82it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: 4%  | 32/896 [00:00<00:22, 38.78it/s]  |
| Computing ESM2 embeddings: 7%  | 64/896 [00:01<00:21, 39.50it/s]  |
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:20, 39.86it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:19, 40.03it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:18, 40.05it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:04<00:17, 39.98it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:16, 40.05it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:15, 40.06it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 40.11it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 40.09it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:08<00:13, 40.11it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:12, 40.10it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:11, 40.13it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 40.10it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 38.86it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:12<00:09, 39.15it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:13<00:08, 39.46it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 39.63it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 39.82it/s] |

```

Computing ESM2 embeddings: 71%|      | 640/896 [00:16<00:06, 39.93it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:16<00:05, 40.02it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:17<00:04, 40.02it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:18<00:03, 40.07it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 40.11it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 40.15it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:20<00:01, 40.11it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 40.15it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.87it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/589 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/589 [00:00<00:14, 37.83it/s]
Computing ESM2 embeddings: 11%|      | 64/589 [00:01<00:13, 39.06it/s]
Computing ESM2 embeddings: 16%|      | 96/589 [00:02<00:12, 39.51it/s]
Computing ESM2 embeddings: 22%|      | 128/589 [00:03<00:11, 39.85it/s]
Computing ESM2 embeddings: 27%|      | 160/589 [00:04<00:10, 40.06it/s]
Computing ESM2 embeddings: 33%|      | 192/589 [00:04<00:09, 40.14it/s]
Computing ESM2 embeddings: 38%|      | 224/589 [00:05<00:09, 40.15it/s]
Computing ESM2 embeddings: 43%|      | 256/589 [00:06<00:08, 40.22it/s]
Computing ESM2 embeddings: 49%|      | 288/589 [00:07<00:07, 40.28it/s]
Computing ESM2 embeddings: 54%|      | 320/589 [00:07<00:06, 40.31it/s]
Computing ESM2 embeddings: 60%|      | 352/589 [00:08<00:05, 40.29it/s]

```

Computing ESM2 embeddings: 65%| | 384/589 [00:09<00:05, 40.31it/s]

Computing ESM2 embeddings: 71%| | 416/589 [00:10<00:04, 40.34it/s]

Computing ESM2 embeddings: 76%| | 448/589 [00:11<00:03, 40.34it/s]

Computing ESM2 embeddings: 81%| | 480/589 [00:12<00:02, 39.43it/s]

Computing ESM2 embeddings: 87%| | 512/589 [00:12<00:01, 39.55it/s]

Computing ESM2 embeddings: 92%| | 544/589 [00:13<00:01, 39.76it/s]

Computing ESM2 embeddings: 98%| | 576/589 [00:14<00:00, 39.89it/s]

Computing ESM2 embeddings: 100%| | 589/589 [00:14<00:00, 39.38it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:00<00:23, 37.29it/s]

Computing ESM2 embeddings: 7%| | 64/897 [00:01<00:21, 38.77it/s]

Computing ESM2 embeddings: 11%| | 96/897 [00:02<00:20, 39.47it/s]

Computing ESM2 embeddings: 14%| | 128/897 [00:03<00:19, 39.78it/s]

Computing ESM2 embeddings: 18%| | 160/897 [00:04<00:18, 39.89it/s]

Computing ESM2 embeddings: 21%| | 192/897 [00:04<00:17, 40.02it/s]

Computing ESM2 embeddings: 25%| | 224/897 [00:05<00:16, 40.12it/s]

Computing ESM2 embeddings: 29%| | 256/897 [00:06<00:15, 40.09it/s]

Computing ESM2 embeddings: 32%| | 288/897 [00:07<00:15, 40.09it/s]

Computing ESM2 embeddings: 36%| | 320/897 [00:08<00:14, 40.14it/s]

Computing ESM2 embeddings: 39%| | 352/897 [00:08<00:13, 40.18it/s]

Computing ESM2 embeddings: 43%| | 384/897 [00:09<00:12, 40.21it/s]

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Computing ESM2 embeddings: 46%|          | 416/897 [00:10<00:11, 40.17it/s]
Computing ESM2 embeddings: 50%|          | 448/897 [00:11<00:11, 40.18it/s]
Computing ESM2 embeddings: 54%|          | 480/897 [00:12<00:10, 38.78it/s]
Computing ESM2 embeddings: 57%|          | 512/897 [00:12<00:09, 38.80it/s]
Computing ESM2 embeddings: 61%|          | 544/897 [00:13<00:09, 39.02it/s]
Computing ESM2 embeddings: 64%|          | 576/897 [00:14<00:08, 39.36it/s]
Computing ESM2 embeddings: 68%|          | 608/897 [00:15<00:07, 39.61it/s]
Computing ESM2 embeddings: 71%|          | 640/897 [00:16<00:06, 39.78it/s]
Computing ESM2 embeddings: 75%|          | 672/897 [00:16<00:05, 39.83it/s]
Computing ESM2 embeddings: 78%|          | 704/897 [00:17<00:04, 39.95it/s]
Computing ESM2 embeddings: 82%|          | 736/897 [00:18<00:04, 40.03it/s]
Computing ESM2 embeddings: 86%|          | 768/897 [00:19<00:03, 40.08it/s]
Computing ESM2 embeddings: 89%|          | 800/897 [00:20<00:02, 40.07it/s]
Computing ESM2 embeddings: 93%|          | 832/897 [00:20<00:01, 40.11it/s]
Computing ESM2 embeddings: 96%|          | 864/897 [00:21<00:00, 39.78it/s]

Computing ESM2 embeddings: 100%|         | 897/897 [00:22<00:00, 39.66it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/916 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|          | 32/916 [00:00<00:23, 38.03it/s]
Computing ESM2 embeddings:  7%|          | 64/916 [00:01<00:21, 39.00it/s]
Computing ESM2 embeddings: 10%|          | 96/916 [00:02<00:20, 39.63it/s]
Computing ESM2 embeddings: 14%|          | 128/916 [00:03<00:19, 39.93it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 17% | 160/916 [00:04<00:18, 40.09it/s] |
| Computing ESM2 embeddings: 21% | 192/916 [00:04<00:18, 40.12it/s] |
| Computing ESM2 embeddings: 24% | 224/916 [00:05<00:17, 40.19it/s] |
| Computing ESM2 embeddings: 28% | 256/916 [00:06<00:16, 40.23it/s] |
| Computing ESM2 embeddings: 31% | 288/916 [00:07<00:15, 40.28it/s] |
| Computing ESM2 embeddings: 35% | 320/916 [00:08<00:14, 40.17it/s] |
| Computing ESM2 embeddings: 38% | 352/916 [00:08<00:14, 40.04it/s] |
| Computing ESM2 embeddings: 42% | 384/916 [00:09<00:13, 40.09it/s] |
| Computing ESM2 embeddings: 45% | 416/916 [00:10<00:12, 40.08it/s] |
| Computing ESM2 embeddings: 49% | 448/916 [00:11<00:11, 40.05it/s] |
| Computing ESM2 embeddings: 52% | 480/916 [00:12<00:11, 39.23it/s] |
| Computing ESM2 embeddings: 56% | 512/916 [00:12<00:10, 39.39it/s] |
| Computing ESM2 embeddings: 59% | 544/916 [00:13<00:09, 39.62it/s] |
| Computing ESM2 embeddings: 63% | 576/916 [00:14<00:08, 39.73it/s] |
| Computing ESM2 embeddings: 66% | 608/916 [00:15<00:07, 39.86it/s] |
| Computing ESM2 embeddings: 70% | 640/916 [00:16<00:06, 39.95it/s] |
| Computing ESM2 embeddings: 73% | 672/916 [00:16<00:06, 40.02it/s] |
| Computing ESM2 embeddings: 77% | 704/916 [00:17<00:05, 40.00it/s] |
| Computing ESM2 embeddings: 80% | 736/916 [00:18<00:04, 40.05it/s] |
| Computing ESM2 embeddings: 84% | 768/916 [00:19<00:03, 40.08it/s] |
| Computing ESM2 embeddings: 87% | 800/916 [00:20<00:02, 40.12it/s] |
| Computing ESM2 embeddings: 91% | 832/916 [00:20<00:02, 40.08it/s] |
| Computing ESM2 embeddings: 94% | 864/916 [00:21<00:01, 40.11it/s] |
| Computing ESM2 embeddings: 98% | 896/916 [00:22<00:00, 40.09it/s] |

Computing ESM2 embeddings: 100%| | 916/916 [00:23<00:00, 39.80it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.92it/s]  
 Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.12it/s]  
 Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.70it/s]  
 Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.97it/s]  
 Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 40.06it/s]  
 Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 40.17it/s]  
 Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:16, 40.22it/s]  
 Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:15, 40.25it/s]  
 Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 39.98it/s]  
 Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 39.98it/s]  
 Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:13, 39.97it/s]  
 Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:12, 39.91it/s]  
 Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:11, 40.01it/s]  
 Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.69it/s]  
 Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 38.95it/s]  
 Computing ESM2 embeddings: 57%| | 512/896 [00:12<00:09, 39.32it/s]  
 Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.54it/s]  
 Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.74it/s]  
 Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.89it/s]



```

Computing ESM2 embeddings: 71%|      | 640/896 [00:16<00:06, 40.00it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:16<00:05, 40.02it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:17<00:04, 40.08it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:18<00:03, 40.13it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 40.15it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 40.10it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:20<00:01, 40.13it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 40.15it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.59it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.86it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 39.19it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:20, 39.74it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:19, 39.97it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:18, 40.12it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:04<00:17, 40.15it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 40.23it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 40.26it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:15, 40.30it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:14, 40.10it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:08<00:13, 40.18it/s]

```

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 40.22it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 40.26it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.32it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 39.49it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:12<00:10, 39.72it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.91it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.97it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 40.07it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 40.15it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:16<00:06, 40.20it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 40.17it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 40.23it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 40.27it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:19<00:02, 40.25it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:20<00:02, 39.78it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.79it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.06it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:22<00:00, 39.83it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.89it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:20, 39.63it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:20, 39.92it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:19, 40.05it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:18, 40.13it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:04<00:17, 40.10it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:16, 40.13it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:15, 40.15it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 40.18it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:07<00:14, 40.14it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:08<00:13, 40.16it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:12, 40.19it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:11, 40.23it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 38.73it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 39.08it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:12<00:09, 39.43it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:13<00:08, 39.68it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 39.79it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 39.93it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 40.02it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:16<00:05, 40.10it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:17<00:04, 40.10it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:18<00:03, 40.15it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:19<00:03, 40.19it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 40.22it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:20<00:01, 40.17it/s] |

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 40.20it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.76it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.06it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.33it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.73it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 40.00it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:18, 40.13it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:17, 40.22it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 40.22it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 40.27it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:15, 40.31it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:07<00:14, 40.33it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:13, 40.30it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 40.28it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 40.26it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.05it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 39.31it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:12<00:10, 39.63it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.85it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.95it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 40.07it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 40.15it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:16<00:06, 40.15it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 40.16it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 40.20it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 40.24it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:19<00:02, 40.20it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:20<00:02, 40.23it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 40.26it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.47it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:22<00:00, 39.87it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:25, 33.78it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 37.11it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.48it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.05it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.45it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 39.72it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:16, 39.89it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 39.93it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 40.03it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 40.09it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:13, 40.07it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:12, 40.12it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:11, 40.13it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.30it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.43it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:12<00:09, 39.66it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.83it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.82it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.92it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 40.01it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:16<00:05, 40.06it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 40.03it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:03, 40.09it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 40.13it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 40.07it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:20<00:01, 40.11it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 40.13it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.49it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/580 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/580 [00:00<00:14, 37.96it/s]

Computing ESM2 embeddings: 11%| | 64/580 [00:01<00:13, 39.11it/s]

Computing ESM2 embeddings: 17%| | 96/580 [00:02<00:12, 39.67it/s]

Computing ESM2 embeddings: 22%| | 128/580 [00:03<00:11, 39.91it/s]

Computing ESM2 embeddings: 28%| | 160/580 [00:04<00:10, 39.96it/s]

Computing ESM2 embeddings: 33%| | 192/580 [00:04<00:09, 40.08it/s]

Computing ESM2 embeddings: 39%| | 224/580 [00:05<00:08, 40.14it/s]

Computing ESM2 embeddings: 44%| | 256/580 [00:06<00:08, 40.19it/s]

Computing ESM2 embeddings: 50%| | 288/580 [00:07<00:07, 38.82it/s]

Computing ESM2 embeddings: 55%| | 320/580 [00:08<00:06, 39.14it/s]

Computing ESM2 embeddings: 61%| | 352/580 [00:08<00:05, 39.46it/s]

Computing ESM2 embeddings: 66%| | 384/580 [00:09<00:04, 39.66it/s]

Computing ESM2 embeddings: 72%| | 416/580 [00:10<00:04, 39.84it/s]

Computing ESM2 embeddings: 77%| | 448/580 [00:11<00:03, 39.98it/s]

Computing ESM2 embeddings: 83%| | 480/580 [00:12<00:02, 40.00it/s]

Computing ESM2 embeddings: 88%| | 512/580 [00:12<00:01, 40.07it/s]

Computing ESM2 embeddings: 94%| | 544/580 [00:13<00:00, 40.13it/s]

Computing ESM2 embeddings: 99%| | 576/580 [00:14<00:00, 38.75it/s]

Computing ESM2 embeddings: 100%| | 580/580 [00:14<00:00, 39.42it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/906 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/906 [00:00<00:22, 39.03it/s]

Computing ESM2 embeddings: 7%| | 64/906 [00:01<00:21, 39.81it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/906 [00:02<00:20, 40.00it/s]  |
| Computing ESM2 embeddings: 14% | 128/906 [00:03<00:19, 40.04it/s] |
| Computing ESM2 embeddings: 18% | 160/906 [00:04<00:18, 40.13it/s] |
| Computing ESM2 embeddings: 21% | 192/906 [00:04<00:17, 39.78it/s] |
| Computing ESM2 embeddings: 25% | 224/906 [00:05<00:17, 39.74it/s] |
| Computing ESM2 embeddings: 28% | 256/906 [00:06<00:16, 39.90it/s] |
| Computing ESM2 embeddings: 32% | 288/906 [00:07<00:16, 38.61it/s] |
| Computing ESM2 embeddings: 35% | 320/906 [00:08<00:15, 38.99it/s] |
| Computing ESM2 embeddings: 39% | 352/906 [00:08<00:14, 39.32it/s] |
| Computing ESM2 embeddings: 42% | 384/906 [00:09<00:13, 39.56it/s] |
| Computing ESM2 embeddings: 46% | 416/906 [00:10<00:12, 39.69it/s] |
| Computing ESM2 embeddings: 49% | 448/906 [00:11<00:11, 39.74it/s] |
| Computing ESM2 embeddings: 53% | 480/906 [00:12<00:10, 39.59it/s] |
| Computing ESM2 embeddings: 57% | 512/906 [00:12<00:10, 39.37it/s] |
| Computing ESM2 embeddings: 60% | 544/906 [00:13<00:09, 39.45it/s] |
| Computing ESM2 embeddings: 64% | 576/906 [00:14<00:08, 39.65it/s] |
| Computing ESM2 embeddings: 67% | 608/906 [00:15<00:07, 39.04it/s] |
| Computing ESM2 embeddings: 71% | 640/906 [00:16<00:06, 39.28it/s] |
| Computing ESM2 embeddings: 74% | 672/906 [00:17<00:05, 39.50it/s] |
| Computing ESM2 embeddings: 78% | 704/906 [00:17<00:05, 39.68it/s] |
| Computing ESM2 embeddings: 81% | 736/906 [00:18<00:04, 39.83it/s] |
| Computing ESM2 embeddings: 85% | 768/906 [00:19<00:03, 39.87it/s] |
| Computing ESM2 embeddings: 88% | 800/906 [00:20<00:02, 39.95it/s] |
| Computing ESM2 embeddings: 92% | 832/906 [00:20<00:01, 40.02it/s] |



Computing ESM2 embeddings: 95%| | 864/906 [00:21<00:01, 40.07it/s]  
Computing ESM2 embeddings: 99%| | 896/906 [00:22<00:00, 38.69it/s]  
Computing ESM2 embeddings: 100%| | 906/906 [00:22<00:00, 39.44it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/899 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 4%| | 32/899 [00:00<00:22, 38.31it/s]  
Computing ESM2 embeddings: 7%| | 64/899 [00:01<00:21, 38.98it/s]  
Computing ESM2 embeddings: 11%| | 96/899 [00:02<00:20, 39.52it/s]  
Computing ESM2 embeddings: 14%| | 128/899 [00:03<00:19, 39.80it/s]  
Computing ESM2 embeddings: 18%| | 160/899 [00:04<00:18, 39.95it/s]  
Computing ESM2 embeddings: 21%| | 192/899 [00:04<00:17, 39.98it/s]  
Computing ESM2 embeddings: 25%| | 224/899 [00:05<00:16, 40.07it/s]  
Computing ESM2 embeddings: 28%| | 256/899 [00:06<00:16, 40.08it/s]  
Computing ESM2 embeddings: 32%| | 288/899 [00:07<00:15, 38.61it/s]  
Computing ESM2 embeddings: 36%| | 320/899 [00:08<00:14, 38.94it/s]  
Computing ESM2 embeddings: 39%| | 352/899 [00:08<00:13, 39.31it/s]  
Computing ESM2 embeddings: 43%| | 384/899 [00:09<00:13, 39.56it/s]  
Computing ESM2 embeddings: 46%| | 416/899 [00:10<00:12, 39.70it/s]  
Computing ESM2 embeddings: 50%| | 448/899 [00:11<00:11, 39.84it/s]  
Computing ESM2 embeddings: 53%| | 480/899 [00:12<00:10, 39.95it/s]  
Computing ESM2 embeddings: 57%| | 512/899 [00:12<00:09, 39.95it/s]  
Computing ESM2 embeddings: 61%| | 544/899 [00:13<00:08, 40.01it/s]

Computing ESM2 embeddings: 64%| | 576/899 [00:14<00:08, 38.80it/s]

Computing ESM2 embeddings: 68%| | 608/899 [00:15<00:07, 38.98it/s]

Computing ESM2 embeddings: 71%| | 640/899 [00:16<00:06, 39.20it/s]

Computing ESM2 embeddings: 75%| | 672/899 [00:17<00:05, 39.49it/s]

Computing ESM2 embeddings: 78%| | 704/899 [00:17<00:04, 39.68it/s]

Computing ESM2 embeddings: 82%| | 736/899 [00:18<00:04, 39.76it/s]

Computing ESM2 embeddings: 85%| | 768/899 [00:19<00:03, 39.91it/s]

Computing ESM2 embeddings: 89%| | 800/899 [00:20<00:02, 39.96it/s]

Computing ESM2 embeddings: 93%| | 832/899 [00:21<00:01, 39.94it/s]

Computing ESM2 embeddings: 96%| | 864/899 [00:21<00:00, 38.58it/s]

Computing ESM2 embeddings: 100%| | 899/899 [00:22<00:00, 39.43it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:22, 38.61it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:21, 39.27it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:20, 39.74it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:19, 39.94it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:18, 39.98it/s]

Computing ESM2 embeddings: 21%| | 192/913 [00:04<00:17, 40.06it/s]

Computing ESM2 embeddings: 25%| | 224/913 [00:05<00:17, 40.13it/s]

Computing ESM2 embeddings: 28%| | 256/913 [00:06<00:16, 40.16it/s]

Computing ESM2 embeddings: 32%| | 288/913 [00:07<00:16, 38.83it/s]

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Computing ESM2 embeddings: 35%|          | 320/913 [00:08<00:15, 39.14it/s]
Computing ESM2 embeddings: 39%|          | 352/913 [00:08<00:14, 39.48it/s]
Computing ESM2 embeddings: 42%|          | 384/913 [00:09<00:13, 39.63it/s]
Computing ESM2 embeddings: 46%|          | 416/913 [00:10<00:12, 39.79it/s]
Computing ESM2 embeddings: 49%|          | 448/913 [00:11<00:11, 39.87it/s]
Computing ESM2 embeddings: 53%|          | 480/913 [00:12<00:10, 39.92it/s]
Computing ESM2 embeddings: 56%|          | 512/913 [00:12<00:10, 39.98it/s]
Computing ESM2 embeddings: 60%|          | 544/913 [00:13<00:09, 40.05it/s]
Computing ESM2 embeddings: 63%|          | 576/913 [00:14<00:08, 38.82it/s]
Computing ESM2 embeddings: 67%|          | 608/913 [00:15<00:07, 39.07it/s]
Computing ESM2 embeddings: 70%|          | 640/913 [00:16<00:06, 39.39it/s]
Computing ESM2 embeddings: 74%|          | 672/913 [00:16<00:06, 39.64it/s]
Computing ESM2 embeddings: 77%|          | 704/913 [00:17<00:05, 39.72it/s]
Computing ESM2 embeddings: 81%|          | 736/913 [00:18<00:04, 39.86it/s]
Computing ESM2 embeddings: 84%|          | 768/913 [00:19<00:03, 39.89it/s]
Computing ESM2 embeddings: 88%|          | 800/913 [00:20<00:02, 39.90it/s]
Computing ESM2 embeddings: 91%|          | 832/913 [00:20<00:02, 39.87it/s]
Computing ESM2 embeddings: 95%|          | 864/913 [00:21<00:01, 38.51it/s]
Computing ESM2 embeddings: 98%|          | 896/913 [00:22<00:00, 38.87it/s]

Computing ESM2 embeddings: 100%|         | 913/913 [00:23<00:00, 39.50it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/895 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:22, 38.63it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:21, 39.36it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:20, 39.76it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:19, 39.90it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:18, 39.99it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:04<00:17, 40.09it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:16, 40.07it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:15, 40.13it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:15, 38.81it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:14, 39.13it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:08<00:13, 39.41it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:09<00:12, 39.65it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:10<00:12, 39.82it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:11<00:11, 39.87it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 39.96it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:12<00:09, 40.04it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:13<00:08, 40.08it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:14<00:08, 38.75it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:15<00:07, 39.02it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:16<00:06, 39.30it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:16<00:05, 39.55it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:17<00:04, 39.76it/s] |
| Computing ESM2 embeddings: | 82% | 736/895 [00:18<00:03, 39.85it/s] |
| Computing ESM2 embeddings: | 86% | 768/895 [00:19<00:03, 39.96it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/895 [00:20<00:02, 40.03it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:20<00:01, 40.03it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:21<00:00, 38.69it/s]
Computing ESM2 embeddings: 100%|     | 895/895 [00:22<00:00, 39.53it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/914 [00:00<00:22, 38.71it/s]
Computing ESM2 embeddings: 7%|        | 64/914 [00:01<00:21, 39.31it/s]
Computing ESM2 embeddings: 11%|       | 96/914 [00:02<00:20, 39.76it/s]
Computing ESM2 embeddings: 14%|       | 128/914 [00:03<00:19, 39.98it/s]
Computing ESM2 embeddings: 18%|       | 160/914 [00:04<00:18, 40.10it/s]
Computing ESM2 embeddings: 21%|       | 192/914 [00:04<00:18, 40.10it/s]
Computing ESM2 embeddings: 25%|       | 224/914 [00:05<00:17, 40.15it/s]
Computing ESM2 embeddings: 28%|       | 256/914 [00:06<00:16, 40.19it/s]
Computing ESM2 embeddings: 32%|       | 288/914 [00:07<00:16, 38.83it/s]
Computing ESM2 embeddings: 35%|       | 320/914 [00:08<00:15, 39.14it/s]
Computing ESM2 embeddings: 39%|       | 352/914 [00:08<00:14, 39.21it/s]
Computing ESM2 embeddings: 42%|       | 384/914 [00:09<00:13, 39.40it/s]
Computing ESM2 embeddings: 46%|       | 416/914 [00:10<00:12, 39.65it/s]
Computing ESM2 embeddings: 49%|       | 448/914 [00:11<00:11, 39.77it/s]
Computing ESM2 embeddings: 53%|       | 480/914 [00:12<00:10, 39.90it/s]
Computing ESM2 embeddings: 56%|       | 512/914 [00:12<00:10, 40.00it/s]

```

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 40.02it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.99it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.23it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.46it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:16<00:06, 39.68it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.85it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.90it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.97it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 40.05it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:20<00:02, 40.03it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 38.77it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.05it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.55it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:21, 39.36it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:20, 39.93it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:19, 40.02it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 40.11it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:03<00:18, 40.12it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 40.04it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:16, 40.07it/s]

```

Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:15, 40.11it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 38.70it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 39.05it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:08<00:13, 39.39it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:12, 39.57it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 39.74it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.85it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 39.90it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:12<00:09, 39.97it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:08, 40.05it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 38.61it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.95it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.30it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:16<00:05, 39.50it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 39.69it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 39.83it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 39.88it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 39.97it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:20<00:01, 40.02it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 38.67it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.49it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.23it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:22, 38.56it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.33it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:19, 39.71it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:18, 39.87it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 40.02it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 40.13it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 40.12it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.75it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 39.09it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 39.41it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.69it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.88it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.94it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:10, 40.06it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:12<00:10, 40.12it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 40.12it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 38.92it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.21it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.51it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:16<00:06, 39.75it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:17<00:05, 39.88it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:18<00:04, 39.96it/s] |



```

Computing ESM2 embeddings: 84%|      | 768/914 [00:19<00:03, 40.01it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 40.07it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:20<00:02, 40.05it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:21<00:01, 38.69it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.04it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.48it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/572 [00:00<00:14, 38.05it/s]
Computing ESM2 embeddings: 11%|      | 64/572 [00:01<00:12, 39.11it/s]
Computing ESM2 embeddings: 17%|      | 96/572 [00:02<00:12, 39.45it/s]
Computing ESM2 embeddings: 22%|      | 128/572 [00:03<00:11, 39.75it/s]
Computing ESM2 embeddings: 28%|      | 160/572 [00:04<00:10, 39.90it/s]
Computing ESM2 embeddings: 34%|      | 192/572 [00:04<00:09, 39.97it/s]
Computing ESM2 embeddings: 39%|      | 224/572 [00:05<00:08, 40.05it/s]
Computing ESM2 embeddings: 45%|      | 256/572 [00:06<00:07, 40.12it/s]
Computing ESM2 embeddings: 50%|      | 288/572 [00:07<00:07, 38.67it/s]
Computing ESM2 embeddings: 56%|      | 320/572 [00:08<00:06, 39.03it/s]
Computing ESM2 embeddings: 62%|      | 352/572 [00:08<00:05, 39.39it/s]
Computing ESM2 embeddings: 67%|      | 384/572 [00:09<00:04, 39.58it/s]
Computing ESM2 embeddings: 73%|      | 416/572 [00:10<00:03, 39.78it/s]
Computing ESM2 embeddings: 78%|      | 448/572 [00:11<00:03, 39.91it/s]

```

```

Computing ESM2 embeddings: 84%|          | 480/572 [00:12<00:02, 39.92it/s]
Computing ESM2 embeddings: 90%|          | 512/572 [00:12<00:01, 40.03it/s]
Computing ESM2 embeddings: 95%|          | 544/572 [00:13<00:00, 40.01it/s]
Computing ESM2 embeddings: 100%|         | 572/572 [00:14<00:00, 39.35it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/916 [00:00<?, ?it/s]
Computing ESM2 embeddings: 3%|           | 32/916 [00:00<00:24, 35.43it/s]
Computing ESM2 embeddings: 7%|           | 64/916 [00:01<00:22, 37.92it/s]
Computing ESM2 embeddings: 10%|          | 96/916 [00:02<00:21, 38.84it/s]
Computing ESM2 embeddings: 14%|          | 128/916 [00:03<00:20, 39.39it/s]
Computing ESM2 embeddings: 17%|          | 160/916 [00:04<00:19, 39.70it/s]
Computing ESM2 embeddings: 21%|          | 192/916 [00:04<00:18, 39.79it/s]
Computing ESM2 embeddings: 24%|          | 224/916 [00:05<00:17, 39.94it/s]
Computing ESM2 embeddings: 28%|          | 256/916 [00:06<00:16, 40.04it/s]
Computing ESM2 embeddings: 31%|          | 288/916 [00:07<00:16, 38.52it/s]
Computing ESM2 embeddings: 35%|          | 320/916 [00:08<00:15, 38.67it/s]
Computing ESM2 embeddings: 38%|          | 352/916 [00:09<00:14, 39.04it/s]
Computing ESM2 embeddings: 42%|          | 384/916 [00:09<00:13, 39.32it/s]
Computing ESM2 embeddings: 45%|          | 416/916 [00:10<00:12, 39.58it/s]
Computing ESM2 embeddings: 49%|          | 448/916 [00:11<00:11, 39.78it/s]
Computing ESM2 embeddings: 52%|          | 480/916 [00:12<00:10, 39.86it/s]
Computing ESM2 embeddings: 56%|          | 512/916 [00:12<00:10, 39.96it/s]

```

Computing ESM2 embeddings: 59%| | 544/916 [00:13<00:09, 40.03it/s]

Computing ESM2 embeddings: 63%| | 576/916 [00:14<00:08, 38.37it/s]

Computing ESM2 embeddings: 66%| | 608/916 [00:15<00:07, 38.76it/s]

Computing ESM2 embeddings: 70%| | 640/916 [00:16<00:07, 39.17it/s]

Computing ESM2 embeddings: 73%| | 672/916 [00:17<00:06, 39.41it/s]

Computing ESM2 embeddings: 77%| | 704/916 [00:17<00:05, 39.64it/s]

Computing ESM2 embeddings: 80%| | 736/916 [00:18<00:04, 39.80it/s]

Computing ESM2 embeddings: 84%| | 768/916 [00:19<00:03, 39.86it/s]

Computing ESM2 embeddings: 87%| | 800/916 [00:20<00:02, 39.95it/s]

Computing ESM2 embeddings: 91%| | 832/916 [00:21<00:02, 40.00it/s]

Computing ESM2 embeddings: 94%| | 864/916 [00:21<00:01, 38.74it/s]

Computing ESM2 embeddings: 98%| | 896/916 [00:22<00:00, 39.08it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:23<00:00, 39.32it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:00<00:22, 38.56it/s]

Computing ESM2 embeddings: 7%| | 64/897 [00:01<00:21, 39.26it/s]

Computing ESM2 embeddings: 11%| | 96/897 [00:02<00:20, 39.79it/s]

Computing ESM2 embeddings: 14%| | 128/897 [00:03<00:19, 40.01it/s]

Computing ESM2 embeddings: 18%| | 160/897 [00:04<00:18, 40.06it/s]

Computing ESM2 embeddings: 21%| | 192/897 [00:04<00:17, 40.15it/s]

Computing ESM2 embeddings: 25%| | 224/897 [00:05<00:16, 40.22it/s]

```

Computing ESM2 embeddings: 29%|          | 256/897 [00:06<00:16, 38.69it/s]
Computing ESM2 embeddings: 32%|          | 288/897 [00:07<00:15, 39.06it/s]
Computing ESM2 embeddings: 36%|          | 320/897 [00:08<00:14, 39.44it/s]
Computing ESM2 embeddings: 39%|          | 352/897 [00:08<00:13, 39.66it/s]
Computing ESM2 embeddings: 43%|          | 384/897 [00:09<00:12, 39.85it/s]
Computing ESM2 embeddings: 46%|          | 416/897 [00:10<00:12, 39.99it/s]
Computing ESM2 embeddings: 50%|          | 448/897 [00:11<00:11, 40.03it/s]
Computing ESM2 embeddings: 54%|          | 480/897 [00:12<00:10, 40.11it/s]
Computing ESM2 embeddings: 57%|          | 512/897 [00:12<00:09, 38.86it/s]
Computing ESM2 embeddings: 61%|          | 544/897 [00:13<00:09, 39.11it/s]
Computing ESM2 embeddings: 64%|          | 576/897 [00:14<00:08, 39.45it/s]
Computing ESM2 embeddings: 68%|          | 608/897 [00:15<00:07, 39.71it/s]
Computing ESM2 embeddings: 71%|          | 640/897 [00:16<00:06, 39.83it/s]
Computing ESM2 embeddings: 75%|          | 672/897 [00:16<00:05, 39.98it/s]
Computing ESM2 embeddings: 78%|          | 704/897 [00:17<00:04, 40.06it/s]
Computing ESM2 embeddings: 82%|          | 736/897 [00:18<00:04, 40.06it/s]
Computing ESM2 embeddings: 86%|          | 768/897 [00:19<00:03, 38.82it/s]
Computing ESM2 embeddings: 89%|          | 800/897 [00:20<00:02, 39.13it/s]
Computing ESM2 embeddings: 93%|          | 832/897 [00:21<00:01, 39.39it/s]
Computing ESM2 embeddings: 96%|          | 864/897 [00:21<00:00, 39.66it/s]

```

```

Computing ESM2 embeddings: 100%|         | 897/897 [00:22<00:00, 39.58it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 38.57it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.40it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.83it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:19, 39.95it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:18, 40.09it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:17, 40.17it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 40.14it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 38.73it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.96it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 39.26it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 39.52it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.71it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.78it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.89it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:10, 39.96it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.46it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 38.84it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.22it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.44it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.64it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:16<00:06, 39.78it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:17<00:05, 39.83it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:18<00:04, 39.92it/s] |

```

Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.76it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 39.00it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.33it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:21<00:01, 39.56it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:22<00:00, 39.66it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 39.48it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.59it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.21it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 39.67it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:19, 39.92it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:18, 39.98it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:04<00:17, 40.08it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:16, 40.12it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 39.19it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 39.40it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 39.63it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:08<00:13, 39.74it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:12, 39.88it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 39.97it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.98it/s]

```

```

Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 40.05it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:12<00:09, 40.09it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:09, 38.65it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 38.75it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 39.07it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.33it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:16<00:05, 39.58it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 39.75it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 39.82it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 39.93it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 40.00it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 38.69it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 39.01it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.53it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:22, 39.04it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 39.70it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:20, 39.98it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:19, 40.11it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:18, 40.06it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:04<00:17, 40.13it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:17, 40.18it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:16, 39.24it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:15, 39.45it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:14, 39.70it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:08<00:14, 39.82it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:09<00:13, 39.97it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:10<00:12, 40.07it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:11, 40.09it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:10, 39.97it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:12<00:10, 38.59it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:13<00:09, 38.88it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:14<00:08, 39.25it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:15<00:07, 39.18it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:16<00:06, 39.24it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:16<00:06, 39.50it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:17<00:05, 39.68it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:18<00:04, 39.75it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:19<00:03, 39.87it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:20<00:02, 38.54it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:21<00:02, 38.85it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:21<00:01, 39.23it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:22<00:00, 39.51it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:23<00:00, 39.49it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',



'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:23, 37.24it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.52it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.10it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.58it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.84it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:17, 39.93it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:16, 40.06it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.63it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 38.96it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.37it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 39.63it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:12, 39.77it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 39.92it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 40.04it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 40.02it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:12<00:09, 40.08it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:09, 39.05it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.22it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.52it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.75it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:16<00:05, 39.85it/s] |

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 39.97it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:03, 40.07it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 40.06it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 40.12it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.86it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 39.09it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.50it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.67it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 40.03it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 40.22it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 40.32it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:03<00:18, 40.28it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:17, 40.33it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 40.36it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 38.94it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:15, 39.24it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 39.58it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:14, 39.77it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.96it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 40.08it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 40.12it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 40.20it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:12<00:09, 40.25it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 38.80it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.13it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.49it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.70it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:16<00:06, 39.90it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 40.03it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 40.08it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 40.16it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 40.22it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:20<00:02, 38.79it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 38.84it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.17it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.67it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/572 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/572 [00:00<00:14, 38.47it/s]

Computing ESM2 embeddings: 11%| | 64/572 [00:01<00:12, 39.36it/s]

Computing ESM2 embeddings: 17%| | 96/572 [00:02<00:11, 39.75it/s]

Computing ESM2 embeddings: 22%| | 128/572 [00:03<00:11, 39.99it/s]

Computing ESM2 embeddings: 28%| | 160/572 [00:04<00:10, 40.13it/s]

Computing ESM2 embeddings: 34%| | 192/572 [00:04<00:09, 40.16it/s]

Computing ESM2 embeddings: 39%| | 224/572 [00:05<00:08, 40.24it/s]

Computing ESM2 embeddings: 45%| | 256/572 [00:06<00:08, 38.85it/s]

Computing ESM2 embeddings: 50%| | 288/572 [00:07<00:07, 39.12it/s]

Computing ESM2 embeddings: 56%| | 320/572 [00:08<00:06, 39.48it/s]

Computing ESM2 embeddings: 62%| | 352/572 [00:08<00:05, 39.76it/s]

Computing ESM2 embeddings: 67%| | 384/572 [00:09<00:04, 39.87it/s]

Computing ESM2 embeddings: 73%| | 416/572 [00:10<00:03, 40.02it/s]

Computing ESM2 embeddings: 78%| | 448/572 [00:11<00:03, 40.11it/s]

Computing ESM2 embeddings: 84%| | 480/572 [00:12<00:02, 40.14it/s]

Computing ESM2 embeddings: 90%| | 512/572 [00:12<00:01, 40.20it/s]

Computing ESM2 embeddings: 95%| | 544/572 [00:13<00:00, 39.03it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:14<00:00, 39.59it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:22, 38.39it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:21, 39.25it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:20, 39.68it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:19, 39.92it/s]

Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:18, 40.10it/s]

Computing ESM2 embeddings: 21%| | 192/915 [00:04<00:18, 40.12it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 24% | 224/915 [00:05<00:17, 40.20it/s] |
| Computing ESM2 embeddings: 28% | 256/915 [00:06<00:16, 38.83it/s] |
| Computing ESM2 embeddings: 31% | 288/915 [00:07<00:16, 39.11it/s] |
| Computing ESM2 embeddings: 35% | 320/915 [00:08<00:15, 39.47it/s] |
| Computing ESM2 embeddings: 38% | 352/915 [00:08<00:14, 39.73it/s] |
| Computing ESM2 embeddings: 42% | 384/915 [00:09<00:13, 39.86it/s] |
| Computing ESM2 embeddings: 45% | 416/915 [00:10<00:12, 40.00it/s] |
| Computing ESM2 embeddings: 49% | 448/915 [00:11<00:11, 40.10it/s] |
| Computing ESM2 embeddings: 52% | 480/915 [00:12<00:10, 40.12it/s] |
| Computing ESM2 embeddings: 56% | 512/915 [00:12<00:10, 40.16it/s] |
| Computing ESM2 embeddings: 59% | 544/915 [00:13<00:09, 38.69it/s] |
| Computing ESM2 embeddings: 63% | 576/915 [00:14<00:08, 38.97it/s] |
| Computing ESM2 embeddings: 66% | 608/915 [00:15<00:07, 39.34it/s] |
| Computing ESM2 embeddings: 70% | 640/915 [00:16<00:06, 39.60it/s] |
| Computing ESM2 embeddings: 73% | 672/915 [00:16<00:06, 39.74it/s] |
| Computing ESM2 embeddings: 77% | 704/915 [00:17<00:05, 39.88it/s] |
| Computing ESM2 embeddings: 80% | 736/915 [00:18<00:04, 40.00it/s] |
| Computing ESM2 embeddings: 84% | 768/915 [00:19<00:03, 39.99it/s] |
| Computing ESM2 embeddings: 87% | 800/915 [00:20<00:02, 39.23it/s] |
| Computing ESM2 embeddings: 91% | 832/915 [00:21<00:02, 39.40it/s] |
| Computing ESM2 embeddings: 94% | 864/915 [00:21<00:01, 39.60it/s] |
| Computing ESM2 embeddings: 98% | 896/915 [00:22<00:00, 39.78it/s] |

Computing ESM2 embeddings: 100%| | 915/915 [00:23<00:00, 39.59it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/897 [00:00<00:22, 38.38it/s]
Computing ESM2 embeddings: 7%|          | 64/897 [00:01<00:21, 39.15it/s]
Computing ESM2 embeddings: 11%|         | 96/897 [00:02<00:20, 39.54it/s]
Computing ESM2 embeddings: 14%|         | 128/897 [00:03<00:19, 39.85it/s]
Computing ESM2 embeddings: 18%|         | 160/897 [00:04<00:18, 39.95it/s]
Computing ESM2 embeddings: 21%|         | 192/897 [00:04<00:17, 40.09it/s]
Computing ESM2 embeddings: 25%|         | 224/897 [00:05<00:16, 40.16it/s]
Computing ESM2 embeddings: 29%|         | 256/897 [00:06<00:16, 38.69it/s]
Computing ESM2 embeddings: 32%|         | 288/897 [00:07<00:15, 39.06it/s]
Computing ESM2 embeddings: 36%|         | 320/897 [00:08<00:14, 39.44it/s]
Computing ESM2 embeddings: 39%|         | 352/897 [00:08<00:13, 39.65it/s]
Computing ESM2 embeddings: 43%|         | 384/897 [00:09<00:12, 39.84it/s]
Computing ESM2 embeddings: 46%|         | 416/897 [00:10<00:12, 39.99it/s]
Computing ESM2 embeddings: 50%|         | 448/897 [00:11<00:11, 40.03it/s]
Computing ESM2 embeddings: 54%|         | 480/897 [00:12<00:10, 40.11it/s]
Computing ESM2 embeddings: 57%|         | 512/897 [00:12<00:09, 39.30it/s]
Computing ESM2 embeddings: 61%|         | 544/897 [00:13<00:08, 39.41it/s]
Computing ESM2 embeddings: 64%|         | 576/897 [00:14<00:08, 39.68it/s]
Computing ESM2 embeddings: 68%|         | 608/897 [00:15<00:07, 39.85it/s]
Computing ESM2 embeddings: 71%|         | 640/897 [00:16<00:06, 39.91it/s]
Computing ESM2 embeddings: 75%|         | 672/897 [00:16<00:05, 40.03it/s]
```

```

Computing ESM2 embeddings: 78%|          | 704/897 [00:17<00:04, 39.99it/s]
Computing ESM2 embeddings: 82%|          | 736/897 [00:18<00:04, 39.99it/s]
Computing ESM2 embeddings: 86%|          | 768/897 [00:19<00:03, 38.61it/s]
Computing ESM2 embeddings: 89%|          | 800/897 [00:20<00:02, 38.96it/s]
Computing ESM2 embeddings: 93%|          | 832/897 [00:21<00:01, 39.27it/s]
Computing ESM2 embeddings: 96%|          | 864/897 [00:21<00:00, 39.54it/s]

Computing ESM2 embeddings: 100%|         | 897/897 [00:22<00:00, 39.57it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 37.42it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 38.77it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:20, 39.44it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:19, 39.74it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:18, 39.97it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:04<00:18, 40.11it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:17, 40.13it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 38.69it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 39.06it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 39.39it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:08<00:14, 39.66it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:09<00:13, 39.85it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 39.94it/s]

```

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 40.05it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 40.13it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:12<00:10, 38.95it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.21it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.52it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.69it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.87it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:16<00:06, 40.01it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 40.03it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 40.11it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.14it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.28it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.58it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.80it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.89it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.56it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:21, 39.30it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.15it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.47it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.77it/s]



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:18, 39.87it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:17, 40.00it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:16, 40.10it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 38.66it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 39.00it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 39.35it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:08<00:13, 39.54it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:12, 39.72it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:12, 39.83it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 39.90it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 40.02it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:12<00:09, 39.24it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:13<00:08, 39.38it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 39.61it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 39.81it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 39.89it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:16<00:05, 39.99it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:17<00:04, 40.06it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:18<00:03, 40.04it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 39.31it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 39.46it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:20<00:01, 39.64it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:21<00:00, 39.82it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:22<00:00, 39.67it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/914 [00:00<00:23, 38.32it/s]
Computing ESM2 embeddings: 7%|          | 64/914 [00:01<00:21, 38.77it/s]
Computing ESM2 embeddings: 11%|         | 96/914 [00:02<00:20, 39.35it/s]
Computing ESM2 embeddings: 14%|         | 128/914 [00:03<00:19, 39.68it/s]
Computing ESM2 embeddings: 18%|         | 160/914 [00:04<00:18, 39.80it/s]
Computing ESM2 embeddings: 21%|         | 192/914 [00:04<00:18, 39.94it/s]
Computing ESM2 embeddings: 25%|         | 224/914 [00:05<00:17, 40.04it/s]
Computing ESM2 embeddings: 28%|         | 256/914 [00:06<00:17, 38.54it/s]
Computing ESM2 embeddings: 32%|         | 288/914 [00:07<00:16, 38.92it/s]
Computing ESM2 embeddings: 35%|         | 320/914 [00:08<00:15, 39.30it/s]
Computing ESM2 embeddings: 39%|         | 352/914 [00:08<00:14, 39.50it/s]
Computing ESM2 embeddings: 42%|         | 384/914 [00:09<00:13, 39.70it/s]
Computing ESM2 embeddings: 46%|         | 416/914 [00:10<00:12, 39.87it/s]
Computing ESM2 embeddings: 49%|         | 448/914 [00:11<00:11, 39.91it/s]
Computing ESM2 embeddings: 53%|         | 480/914 [00:12<00:10, 39.99it/s]
Computing ESM2 embeddings: 56%|         | 512/914 [00:13<00:10, 38.60it/s]
Computing ESM2 embeddings: 60%|         | 544/914 [00:13<00:09, 38.56it/s]
Computing ESM2 embeddings: 63%|         | 576/914 [00:14<00:08, 38.62it/s]
Computing ESM2 embeddings: 67%|         | 608/914 [00:15<00:07, 38.68it/s]
Computing ESM2 embeddings: 70%|         | 640/914 [00:16<00:07, 38.96it/s]
```

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 39.31it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:17<00:05, 39.56it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:18<00:04, 39.69it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:19<00:03, 38.41it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 38.80it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.13it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.43it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.64it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.23it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/896 [00:00<00:23, 37.15it/s]
Computing ESM2 embeddings: 7%|        | 64/896 [00:01<00:21, 38.00it/s]
Computing ESM2 embeddings: 11%|       | 96/896 [00:02<00:20, 38.86it/s]
Computing ESM2 embeddings: 14%|       | 128/896 [00:03<00:19, 39.39it/s]
Computing ESM2 embeddings: 18%|       | 160/896 [00:04<00:18, 39.69it/s]
Computing ESM2 embeddings: 21%|       | 192/896 [00:04<00:17, 39.83it/s]
Computing ESM2 embeddings: 25%|       | 224/896 [00:05<00:16, 39.96it/s]
Computing ESM2 embeddings: 29%|       | 256/896 [00:06<00:16, 38.61it/s]
Computing ESM2 embeddings: 32%|       | 288/896 [00:07<00:15, 38.92it/s]
Computing ESM2 embeddings: 36%|       | 320/896 [00:08<00:14, 39.32it/s]
Computing ESM2 embeddings: 39%|       | 352/896 [00:08<00:13, 39.60it/s]

```

```

Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:12, 39.74it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 39.89it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 40.00it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 40.00it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:09, 38.58it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:09, 38.95it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 39.29it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 39.56it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.76it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 39.83it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 39.96it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:03, 40.04it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.78it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 39.11it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.43it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 39.62it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.40it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:22, 38.73it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 39.25it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:20, 39.73it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:19, 39.92it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:18, 39.99it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 40.11it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 40.12it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 38.66it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 39.02it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 39.37it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:08<00:14, 39.58it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 39.73it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 39.83it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.90it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:10, 40.01it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:12<00:10, 39.20it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.29it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.54it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.73it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:06, 39.83it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:16<00:06, 39.93it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 40.00it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 40.01it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 38.64it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 38.99it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:21<00:02, 39.29it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:21<00:01, 39.50it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.68it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.55it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:14, 38.49it/s]

Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:12, 39.35it/s]

Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:11, 39.66it/s]

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:11, 39.88it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:10, 40.02it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:04<00:09, 40.06it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:05<00:08, 40.13it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 38.71it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 39.00it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 39.38it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:08<00:05, 39.66it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:09<00:04, 39.77it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:10<00:03, 39.95it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:11<00:03, 40.02it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 40.02it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:12<00:01, 39.17it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:13<00:00, 39.09it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:14<00:00, 39.02it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/915 [00:00<?, ?it/s]
Computing ESM2 embeddings: 3%|          | 32/915 [00:00<00:22, 39.36it/s]
Computing ESM2 embeddings: 7%|          | 64/915 [00:01<00:21, 39.95it/s]
Computing ESM2 embeddings: 10%|         | 96/915 [00:02<00:20, 40.05it/s]
Computing ESM2 embeddings: 14%|         | 128/915 [00:03<00:19, 40.17it/s]
Computing ESM2 embeddings: 17%|         | 160/915 [00:03<00:18, 40.21it/s]
Computing ESM2 embeddings: 21%|         | 192/915 [00:04<00:18, 40.16it/s]
Computing ESM2 embeddings: 24%|         | 224/915 [00:05<00:17, 40.22it/s]
Computing ESM2 embeddings: 28%|         | 256/915 [00:06<00:16, 38.77it/s]
Computing ESM2 embeddings: 31%|         | 288/915 [00:07<00:16, 39.06it/s]
Computing ESM2 embeddings: 35%|         | 320/915 [00:08<00:15, 39.45it/s]
Computing ESM2 embeddings: 38%|         | 352/915 [00:08<00:14, 39.73it/s]
Computing ESM2 embeddings: 42%|         | 384/915 [00:09<00:13, 39.85it/s]
Computing ESM2 embeddings: 45%|         | 416/915 [00:10<00:12, 39.99it/s]
Computing ESM2 embeddings: 49%|         | 448/915 [00:11<00:11, 40.07it/s]
Computing ESM2 embeddings: 52%|         | 480/915 [00:12<00:11, 38.64it/s]
Computing ESM2 embeddings: 56%|         | 512/915 [00:12<00:10, 38.98it/s]
Computing ESM2 embeddings: 59%|         | 544/915 [00:13<00:09, 39.33it/s]
Computing ESM2 embeddings: 63%|         | 576/915 [00:14<00:08, 39.54it/s]
Computing ESM2 embeddings: 66%|         | 608/915 [00:15<00:07, 39.75it/s]
Computing ESM2 embeddings: 70%|         | 640/915 [00:16<00:06, 39.89it/s]
```

```

Computing ESM2 embeddings: 73%|      | 672/915 [00:16<00:06, 39.93it/s]
Computing ESM2 embeddings: 77%|      | 704/915 [00:17<00:05, 40.01it/s]
Computing ESM2 embeddings: 80%|      | 736/915 [00:18<00:04, 40.06it/s]
Computing ESM2 embeddings: 84%|      | 768/915 [00:19<00:03, 38.46it/s]
Computing ESM2 embeddings: 87%|      | 800/915 [00:20<00:02, 38.84it/s]
Computing ESM2 embeddings: 91%|      | 832/915 [00:21<00:02, 39.23it/s]
Computing ESM2 embeddings: 94%|      | 864/915 [00:21<00:01, 39.47it/s]
Computing ESM2 embeddings: 98%|      | 896/915 [00:22<00:00, 39.70it/s]

Computing ESM2 embeddings: 100%|     | 915/915 [00:23<00:00, 39.57it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/898 [00:00<00:22, 39.24it/s]
Computing ESM2 embeddings: 7%|      | 64/898 [00:01<00:20, 39.81it/s]
Computing ESM2 embeddings: 11%|     | 96/898 [00:02<00:20, 40.04it/s]
Computing ESM2 embeddings: 14%|     | 128/898 [00:03<00:19, 40.10it/s]
Computing ESM2 embeddings: 18%|     | 160/898 [00:03<00:18, 40.20it/s]
Computing ESM2 embeddings: 21%|     | 192/898 [00:04<00:17, 40.27it/s]
Computing ESM2 embeddings: 25%|     | 224/898 [00:05<00:16, 40.22it/s]
Computing ESM2 embeddings: 29%|     | 256/898 [00:06<00:16, 38.88it/s]
Computing ESM2 embeddings: 32%|     | 288/898 [00:07<00:15, 39.21it/s]
Computing ESM2 embeddings: 36%|     | 320/898 [00:08<00:14, 39.51it/s]
Computing ESM2 embeddings: 39%|     | 352/898 [00:08<00:13, 39.75it/s]

```



```

Computing ESM2 embeddings: 43%|          | 384/898 [00:09<00:12, 39.92it/s]
Computing ESM2 embeddings: 46%|          | 416/898 [00:10<00:12, 40.00it/s]
Computing ESM2 embeddings: 50%|          | 448/898 [00:11<00:11, 40.12it/s]
Computing ESM2 embeddings: 53%|          | 480/898 [00:12<00:10, 40.10it/s]
Computing ESM2 embeddings: 57%|          | 512/898 [00:12<00:10, 38.55it/s]
Computing ESM2 embeddings: 61%|          | 544/898 [00:13<00:09, 38.93it/s]
Computing ESM2 embeddings: 64%|          | 576/898 [00:14<00:08, 39.29it/s]
Computing ESM2 embeddings: 68%|          | 608/898 [00:15<00:07, 39.46it/s]
Computing ESM2 embeddings: 71%|          | 640/898 [00:16<00:06, 39.61it/s]
Computing ESM2 embeddings: 75%|          | 672/898 [00:16<00:05, 39.79it/s]
Computing ESM2 embeddings: 78%|          | 704/898 [00:17<00:04, 39.86it/s]
Computing ESM2 embeddings: 82%|          | 736/898 [00:18<00:04, 39.94it/s]
Computing ESM2 embeddings: 86%|          | 768/898 [00:19<00:03, 38.69it/s]
Computing ESM2 embeddings: 89%|          | 800/898 [00:20<00:02, 38.96it/s]
Computing ESM2 embeddings: 93%|          | 832/898 [00:21<00:01, 39.32it/s]
Computing ESM2 embeddings: 96%|          | 864/898 [00:21<00:00, 39.58it/s]

Computing ESM2 embeddings: 100%|         | 898/898 [00:22<00:00, 39.53it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/907 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/907 [00:00<00:22, 38.22it/s]
Computing ESM2 embeddings:  7%|          | 64/907 [00:01<00:22, 36.93it/s]
Computing ESM2 embeddings: 11%|          | 96/907 [00:02<00:21, 38.20it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/907 [00:03<00:20, 38.87it/s] |
| Computing ESM2 embeddings: 18% | 160/907 [00:04<00:18, 39.35it/s] |
| Computing ESM2 embeddings: 21% | 192/907 [00:04<00:18, 39.60it/s] |
| Computing ESM2 embeddings: 25% | 224/907 [00:05<00:17, 39.81it/s] |
| Computing ESM2 embeddings: 28% | 256/907 [00:06<00:16, 38.48it/s] |
| Computing ESM2 embeddings: 32% | 288/907 [00:07<00:15, 38.85it/s] |
| Computing ESM2 embeddings: 35% | 320/907 [00:08<00:14, 39.26it/s] |
| Computing ESM2 embeddings: 39% | 352/907 [00:09<00:14, 39.55it/s] |
| Computing ESM2 embeddings: 42% | 384/907 [00:09<00:13, 39.70it/s] |
| Computing ESM2 embeddings: 46% | 416/907 [00:10<00:12, 39.86it/s] |
| Computing ESM2 embeddings: 49% | 448/907 [00:11<00:11, 39.96it/s] |
| Computing ESM2 embeddings: 53% | 480/907 [00:12<00:10, 39.97it/s] |
| Computing ESM2 embeddings: 56% | 512/907 [00:13<00:10, 38.77it/s] |
| Computing ESM2 embeddings: 60% | 544/907 [00:13<00:09, 39.10it/s] |
| Computing ESM2 embeddings: 64% | 576/907 [00:14<00:08, 39.37it/s] |
| Computing ESM2 embeddings: 67% | 608/907 [00:15<00:07, 39.62it/s] |
| Computing ESM2 embeddings: 71% | 640/907 [00:16<00:06, 39.79it/s] |
| Computing ESM2 embeddings: 74% | 672/907 [00:17<00:05, 39.86it/s] |
| Computing ESM2 embeddings: 78% | 704/907 [00:17<00:05, 39.96it/s] |
| Computing ESM2 embeddings: 81% | 736/907 [00:18<00:04, 40.03it/s] |
| Computing ESM2 embeddings: 85% | 768/907 [00:19<00:03, 38.58it/s] |
| Computing ESM2 embeddings: 88% | 800/907 [00:20<00:02, 38.95it/s] |
| Computing ESM2 embeddings: 92% | 832/907 [00:21<00:01, 39.32it/s] |
| Computing ESM2 embeddings: 95% | 864/907 [00:21<00:01, 39.53it/s] |

Computing ESM2 embeddings: 99%| | 896/907 [00:22<00:00, 39.66it/s]

Computing ESM2 embeddings: 100%| | 907/907 [00:23<00:00, 39.31it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/903 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/903 [00:00<00:22, 38.54it/s]

Computing ESM2 embeddings: 7%| | 64/903 [00:01<00:21, 39.36it/s]

Computing ESM2 embeddings: 11%| | 96/903 [00:02<00:20, 39.75it/s]

Computing ESM2 embeddings: 14%| | 128/903 [00:03<00:19, 39.88it/s]

Computing ESM2 embeddings: 18%| | 160/903 [00:04<00:18, 39.96it/s]

Computing ESM2 embeddings: 21%| | 192/903 [00:04<00:17, 40.06it/s]

Computing ESM2 embeddings: 25%| | 224/903 [00:05<00:16, 40.02it/s]

Computing ESM2 embeddings: 28%| | 256/903 [00:06<00:16, 38.78it/s]

Computing ESM2 embeddings: 32%| | 288/903 [00:07<00:15, 39.09it/s]

Computing ESM2 embeddings: 35%| | 320/903 [00:08<00:14, 39.36it/s]

Computing ESM2 embeddings: 39%| | 352/903 [00:08<00:13, 39.61it/s]

Computing ESM2 embeddings: 43%| | 384/903 [00:09<00:13, 39.78it/s]

Computing ESM2 embeddings: 46%| | 416/903 [00:10<00:12, 39.86it/s]

Computing ESM2 embeddings: 50%| | 448/903 [00:11<00:11, 39.79it/s]

Computing ESM2 embeddings: 53%| | 480/903 [00:12<00:10, 39.91it/s]

Computing ESM2 embeddings: 57%| | 512/903 [00:12<00:10, 38.52it/s]

Computing ESM2 embeddings: 60%| | 544/903 [00:13<00:09, 38.85it/s]

Computing ESM2 embeddings: 64%| | 576/903 [00:14<00:08, 39.23it/s]

Computing ESM2 embeddings: 67%| | 608/903 [00:15<00:07, 39.45it/s]

Computing ESM2 embeddings: 71%| | 640/903 [00:16<00:06, 39.66it/s]

Computing ESM2 embeddings: 74%| | 672/903 [00:16<00:05, 39.81it/s]

Computing ESM2 embeddings: 78%| | 704/903 [00:17<00:04, 39.87it/s]

Computing ESM2 embeddings: 82%| | 736/903 [00:18<00:04, 39.95it/s]

Computing ESM2 embeddings: 85%| | 768/903 [00:19<00:03, 38.63it/s]

Computing ESM2 embeddings: 89%| | 800/903 [00:20<00:02, 38.89it/s]

Computing ESM2 embeddings: 92%| | 832/903 [00:21<00:01, 39.22it/s]

Computing ESM2 embeddings: 96%| | 864/903 [00:21<00:00, 39.48it/s]

Computing ESM2 embeddings: 99%| | 896/903 [00:22<00:00, 39.62it/s]

Computing ESM2 embeddings: 100%| | 903/903 [00:22<00:00, 39.42it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/902 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/902 [00:00<00:23, 37.39it/s]

Computing ESM2 embeddings: 7%| | 64/902 [00:01<00:22, 36.83it/s]

Computing ESM2 embeddings: 11%| | 96/902 [00:02<00:21, 38.33it/s]

Computing ESM2 embeddings: 14%| | 128/902 [00:03<00:19, 39.12it/s]

Computing ESM2 embeddings: 18%| | 160/902 [00:04<00:18, 39.47it/s]

Computing ESM2 embeddings: 21%| | 192/902 [00:04<00:17, 39.78it/s]

Computing ESM2 embeddings: 25%| | 224/902 [00:05<00:17, 39.04it/s]

Computing ESM2 embeddings: 28%| | 256/902 [00:06<00:16, 39.26it/s]

Computing ESM2 embeddings: 32%| | 288/902 [00:07<00:15, 39.58it/s]

Computing ESM2 embeddings: 35%| | 320/902 [00:08<00:14, 39.84it/s]

Computing ESM2 embeddings: 39%| | 352/902 [00:08<00:13, 39.93it/s]

Computing ESM2 embeddings: 43%| | 384/902 [00:09<00:12, 40.04it/s]

Computing ESM2 embeddings: 46%| | 416/902 [00:10<00:12, 40.12it/s]

Computing ESM2 embeddings: 50%| | 448/902 [00:11<00:11, 38.79it/s]

Computing ESM2 embeddings: 53%| | 480/902 [00:12<00:10, 39.11it/s]

Computing ESM2 embeddings: 57%| | 512/902 [00:13<00:09, 39.45it/s]

Computing ESM2 embeddings: 60%| | 544/902 [00:13<00:09, 39.64it/s]

Computing ESM2 embeddings: 64%| | 576/902 [00:14<00:08, 39.84it/s]

Computing ESM2 embeddings: 67%| | 608/902 [00:15<00:07, 39.96it/s]

Computing ESM2 embeddings: 71%| | 640/902 [00:16<00:06, 40.00it/s]

Computing ESM2 embeddings: 75%| | 672/902 [00:17<00:05, 39.26it/s]

Computing ESM2 embeddings: 78%| | 704/902 [00:17<00:05, 39.44it/s]

Computing ESM2 embeddings: 82%| | 736/902 [00:18<00:04, 39.57it/s]

Computing ESM2 embeddings: 85%| | 768/902 [00:19<00:03, 39.76it/s]

Computing ESM2 embeddings: 89%| | 800/902 [00:20<00:02, 39.89it/s]

Computing ESM2 embeddings: 92%| | 832/902 [00:21<00:01, 39.93it/s]

Computing ESM2 embeddings: 96%| | 864/902 [00:21<00:00, 39.99it/s]

Computing ESM2 embeddings: 99%| | 896/902 [00:22<00:00, 39.18it/s]

Computing ESM2 embeddings: 100%| | 902/902 [00:23<00:00, 39.21it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/908 [00:00<?, ?it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/908 [00:00<00:23, 36.67it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/908 [00:01<00:21, 38.46it/s]  |
| Computing ESM2 embeddings: | 11% | 96/908 [00:02<00:20, 39.16it/s]  |
| Computing ESM2 embeddings: | 14% | 128/908 [00:03<00:19, 39.56it/s] |
| Computing ESM2 embeddings: | 18% | 160/908 [00:04<00:18, 39.80it/s] |
| Computing ESM2 embeddings: | 21% | 192/908 [00:04<00:17, 39.88it/s] |
| Computing ESM2 embeddings: | 25% | 224/908 [00:05<00:17, 38.47it/s] |
| Computing ESM2 embeddings: | 28% | 256/908 [00:06<00:16, 38.88it/s] |
| Computing ESM2 embeddings: | 32% | 288/908 [00:07<00:15, 39.23it/s] |
| Computing ESM2 embeddings: | 35% | 320/908 [00:08<00:14, 39.53it/s] |
| Computing ESM2 embeddings: | 39% | 352/908 [00:08<00:13, 39.73it/s] |
| Computing ESM2 embeddings: | 42% | 384/908 [00:09<00:13, 39.81it/s] |
| Computing ESM2 embeddings: | 46% | 416/908 [00:10<00:12, 39.95it/s] |
| Computing ESM2 embeddings: | 49% | 448/908 [00:11<00:11, 39.20it/s] |
| Computing ESM2 embeddings: | 53% | 480/908 [00:12<00:10, 39.29it/s] |
| Computing ESM2 embeddings: | 56% | 512/908 [00:13<00:10, 39.58it/s] |
| Computing ESM2 embeddings: | 60% | 544/908 [00:13<00:09, 39.76it/s] |
| Computing ESM2 embeddings: | 63% | 576/908 [00:14<00:08, 39.81it/s] |
| Computing ESM2 embeddings: | 67% | 608/908 [00:15<00:07, 39.92it/s] |
| Computing ESM2 embeddings: | 70% | 640/908 [00:16<00:06, 39.98it/s] |
| Computing ESM2 embeddings: | 74% | 672/908 [00:16<00:05, 39.96it/s] |
| Computing ESM2 embeddings: | 78% | 704/908 [00:17<00:05, 38.72it/s] |
| Computing ESM2 embeddings: | 81% | 736/908 [00:18<00:04, 39.03it/s] |
| Computing ESM2 embeddings: | 85% | 768/908 [00:19<00:03, 39.30it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/908 [00:20<00:02, 39.49it/s]
Computing ESM2 embeddings: 92%|      | 832/908 [00:21<00:01, 39.65it/s]
Computing ESM2 embeddings: 95%|      | 864/908 [00:21<00:01, 39.75it/s]
Computing ESM2 embeddings: 99%|      | 896/908 [00:22<00:00, 39.87it/s]
Computing ESM2 embeddings: 100%|     | 908/908 [00:23<00:00, 39.43it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/897 [00:00<00:22, 38.00it/s]
Computing ESM2 embeddings:  7%|      | 64/897 [00:01<00:21, 39.36it/s]
Computing ESM2 embeddings: 11%|      | 96/897 [00:02<00:20, 39.82it/s]
Computing ESM2 embeddings: 14%|      | 128/897 [00:03<00:19, 39.94it/s]
Computing ESM2 embeddings: 18%|      | 160/897 [00:04<00:18, 40.04it/s]
Computing ESM2 embeddings: 21%|      | 192/897 [00:04<00:17, 40.18it/s]
Computing ESM2 embeddings: 25%|      | 224/897 [00:05<00:17, 38.62it/s]
Computing ESM2 embeddings: 29%|      | 256/897 [00:06<00:16, 39.02it/s]
Computing ESM2 embeddings: 32%|      | 288/897 [00:07<00:15, 39.41it/s]
Computing ESM2 embeddings: 36%|      | 320/897 [00:08<00:14, 39.63it/s]
Computing ESM2 embeddings: 39%|      | 352/897 [00:08<00:13, 39.83it/s]
Computing ESM2 embeddings: 43%|      | 384/897 [00:09<00:12, 39.97it/s]
Computing ESM2 embeddings: 46%|      | 416/897 [00:10<00:12, 40.01it/s]
Computing ESM2 embeddings: 50%|      | 448/897 [00:11<00:11, 40.08it/s]
Computing ESM2 embeddings: 54%|      | 480/897 [00:12<00:10, 38.76it/s]

```

Computing ESM2 embeddings: 57%| | 512/897 [00:12<00:09, 38.75it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:13<00:09, 39.07it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:14<00:08, 39.28it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:15<00:07, 39.24it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:16<00:06, 39.50it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:17<00:05, 39.70it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:17<00:04, 39.81it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:18<00:04, 38.50it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:19<00:03, 38.88it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:20<00:02, 39.22it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:21<00:01, 39.52it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:21<00:00, 39.74it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:22<00:00, 39.45it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/587 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/587 [00:00<00:14, 38.61it/s]

Computing ESM2 embeddings: 11%| | 64/587 [00:01<00:13, 39.41it/s]

Computing ESM2 embeddings: 16%| | 96/587 [00:02<00:12, 39.78it/s]

Computing ESM2 embeddings: 22%| | 128/587 [00:03<00:11, 40.00it/s]

Computing ESM2 embeddings: 27%| | 160/587 [00:04<00:10, 40.14it/s]

Computing ESM2 embeddings: 33%| | 192/587 [00:04<00:09, 40.14it/s]

Computing ESM2 embeddings: 38%| | 224/587 [00:05<00:09, 38.73it/s]



```

Computing ESM2 embeddings: 44%|          | 256/587 [00:06<00:08, 39.09it/s]
Computing ESM2 embeddings: 49%|          | 288/587 [00:07<00:07, 39.41it/s]
Computing ESM2 embeddings: 55%|          | 320/587 [00:08<00:06, 39.68it/s]
Computing ESM2 embeddings: 60%|          | 352/587 [00:08<00:05, 39.89it/s]
Computing ESM2 embeddings: 65%|          | 384/587 [00:09<00:05, 39.95it/s]
Computing ESM2 embeddings: 71%|          | 416/587 [00:10<00:04, 40.07it/s]
Computing ESM2 embeddings: 76%|          | 448/587 [00:11<00:03, 40.14it/s]
Computing ESM2 embeddings: 82%|          | 480/587 [00:12<00:02, 39.32it/s]
Computing ESM2 embeddings: 87%|          | 512/587 [00:12<00:01, 39.48it/s]
Computing ESM2 embeddings: 93%|          | 544/587 [00:13<00:01, 39.74it/s]
Computing ESM2 embeddings: 98%|          | 576/587 [00:14<00:00, 39.82it/s]
Computing ESM2 embeddings: 100%|         | 587/587 [00:14<00:00, 39.62it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/899 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/899 [00:00<00:22, 37.83it/s]
Computing ESM2 embeddings: 7%|           | 64/899 [00:01<00:21, 38.85it/s]
Computing ESM2 embeddings: 11%|          | 96/899 [00:02<00:20, 39.42it/s]
Computing ESM2 embeddings: 14%|          | 128/899 [00:03<00:19, 39.63it/s]
Computing ESM2 embeddings: 18%|          | 160/899 [00:04<00:18, 39.77it/s]
Computing ESM2 embeddings: 21%|          | 192/899 [00:04<00:17, 39.91it/s]
Computing ESM2 embeddings: 25%|          | 224/899 [00:05<00:17, 39.11it/s]
Computing ESM2 embeddings: 28%|          | 256/899 [00:06<00:16, 38.85it/s]

```

```

Computing ESM2 embeddings: 32%|          | 288/899 [00:07<00:15, 39.12it/s]
Computing ESM2 embeddings: 36%|          | 320/899 [00:08<00:14, 39.42it/s]
Computing ESM2 embeddings: 39%|          | 352/899 [00:08<00:13, 39.58it/s]
Computing ESM2 embeddings: 43%|          | 384/899 [00:09<00:12, 39.76it/s]
Computing ESM2 embeddings: 46%|          | 416/899 [00:10<00:12, 39.92it/s]
Computing ESM2 embeddings: 50%|          | 448/899 [00:11<00:11, 39.91it/s]
Computing ESM2 embeddings: 53%|          | 480/899 [00:12<00:10, 38.60it/s]
Computing ESM2 embeddings: 57%|          | 512/899 [00:13<00:09, 38.95it/s]
Computing ESM2 embeddings: 61%|          | 544/899 [00:13<00:09, 39.28it/s]
Computing ESM2 embeddings: 64%|          | 576/899 [00:14<00:08, 39.57it/s]
Computing ESM2 embeddings: 68%|          | 608/899 [00:15<00:07, 39.78it/s]
Computing ESM2 embeddings: 71%|          | 640/899 [00:16<00:06, 39.89it/s]
Computing ESM2 embeddings: 75%|          | 672/899 [00:17<00:05, 40.00it/s]
Computing ESM2 embeddings: 78%|          | 704/899 [00:17<00:04, 40.08it/s]
Computing ESM2 embeddings: 82%|          | 736/899 [00:18<00:04, 38.71it/s]
Computing ESM2 embeddings: 85%|          | 768/899 [00:19<00:03, 39.03it/s]
Computing ESM2 embeddings: 89%|          | 800/899 [00:20<00:02, 39.39it/s]
Computing ESM2 embeddings: 93%|          | 832/899 [00:21<00:01, 39.59it/s]
Computing ESM2 embeddings: 96%|          | 864/899 [00:21<00:00, 39.80it/s]

Computing ESM2 embeddings: 100%|         | 899/899 [00:22<00:00, 39.44it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          | 0/917 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 3%  | 32/917 [00:00<00:23, 37.25it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/917 [00:01<00:21, 38.81it/s]  |
| Computing ESM2 embeddings: | 10% | 96/917 [00:02<00:20, 39.45it/s]  |
| Computing ESM2 embeddings: | 14% | 128/917 [00:03<00:19, 39.73it/s] |
| Computing ESM2 embeddings: | 17% | 160/917 [00:04<00:18, 39.93it/s] |
| Computing ESM2 embeddings: | 21% | 192/917 [00:04<00:18, 40.04it/s] |
| Computing ESM2 embeddings: | 24% | 224/917 [00:05<00:17, 39.07it/s] |
| Computing ESM2 embeddings: | 28% | 256/917 [00:06<00:16, 39.28it/s] |
| Computing ESM2 embeddings: | 31% | 288/917 [00:07<00:15, 39.55it/s] |
| Computing ESM2 embeddings: | 35% | 320/917 [00:08<00:15, 39.71it/s] |
| Computing ESM2 embeddings: | 38% | 352/917 [00:08<00:14, 39.85it/s] |
| Computing ESM2 embeddings: | 42% | 384/917 [00:09<00:13, 39.95it/s] |
| Computing ESM2 embeddings: | 45% | 416/917 [00:10<00:12, 39.96it/s] |
| Computing ESM2 embeddings: | 49% | 448/917 [00:11<00:11, 40.03it/s] |
| Computing ESM2 embeddings: | 52% | 480/917 [00:12<00:11, 38.68it/s] |
| Computing ESM2 embeddings: | 56% | 512/917 [00:12<00:10, 38.94it/s] |
| Computing ESM2 embeddings: | 59% | 544/917 [00:13<00:09, 39.30it/s] |
| Computing ESM2 embeddings: | 63% | 576/917 [00:14<00:08, 39.57it/s] |
| Computing ESM2 embeddings: | 66% | 608/917 [00:15<00:07, 39.69it/s] |
| Computing ESM2 embeddings: | 70% | 640/917 [00:16<00:06, 39.84it/s] |
| Computing ESM2 embeddings: | 73% | 672/917 [00:16<00:06, 39.93it/s] |
| Computing ESM2 embeddings: | 77% | 704/917 [00:17<00:05, 39.95it/s] |
| Computing ESM2 embeddings: | 80% | 736/917 [00:18<00:04, 39.20it/s] |
| Computing ESM2 embeddings: | 84% | 768/917 [00:19<00:03, 39.36it/s] |

```

Computing ESM2 embeddings: 87%|      | 800/917 [00:20<00:02, 39.55it/s]
Computing ESM2 embeddings: 91%|      | 832/917 [00:21<00:02, 39.74it/s]
Computing ESM2 embeddings: 94%|      | 864/917 [00:21<00:01, 39.87it/s]
Computing ESM2 embeddings: 98%|      | 896/917 [00:22<00:00, 39.91it/s]
Computing ESM2 embeddings: 100%|     | 917/917 [00:23<00:00, 39.53it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/896 [00:00<00:22, 38.14it/s]
Computing ESM2 embeddings: 7%|        | 64/896 [00:01<00:21, 39.23it/s]
Computing ESM2 embeddings: 11%|       | 96/896 [00:02<00:20, 39.76it/s]
Computing ESM2 embeddings: 14%|       | 128/896 [00:03<00:19, 39.96it/s]
Computing ESM2 embeddings: 18%|       | 160/896 [00:04<00:18, 40.02it/s]
Computing ESM2 embeddings: 21%|       | 192/896 [00:04<00:17, 40.10it/s]
Computing ESM2 embeddings: 25%|       | 224/896 [00:05<00:17, 39.30it/s]
Computing ESM2 embeddings: 29%|       | 256/896 [00:06<00:16, 39.44it/s]
Computing ESM2 embeddings: 32%|       | 288/896 [00:07<00:15, 39.73it/s]
Computing ESM2 embeddings: 36%|       | 320/896 [00:08<00:14, 39.91it/s]
Computing ESM2 embeddings: 39%|       | 352/896 [00:08<00:13, 39.98it/s]
Computing ESM2 embeddings: 43%|       | 384/896 [00:09<00:12, 40.08it/s]
Computing ESM2 embeddings: 46%|       | 416/896 [00:10<00:11, 40.11it/s]
Computing ESM2 embeddings: 50%|       | 448/896 [00:11<00:11, 40.05it/s]
Computing ESM2 embeddings: 54%|       | 480/896 [00:12<00:10, 38.69it/s]

```

Computing ESM2 embeddings: 57%| | 512/896 [00:12<00:09, 39.04it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.33it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.58it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.76it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.83it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:16<00:05, 39.95it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 40.02it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 38.65it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.99it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 39.36it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.41it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 39.64it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.50it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.99it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.09it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.53it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 39.71it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:18, 39.88it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.99it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.03it/s]

```

Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:16, 39.29it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:15, 39.58it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:14, 39.72it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:08<00:14, 39.84it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:09<00:13, 39.95it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 39.94it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 40.00it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.76it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:12<00:10, 39.00it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:13<00:09, 39.34it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 39.57it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.69it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:06, 39.84it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:16<00:06, 39.93it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:17<00:05, 39.62it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 39.69it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 39.82it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 39.86it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:20<00:02, 39.94it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:21<00:01, 39.99it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:22<00:00, 39.97it/s]

```

```

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 39.61it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:22, 38.66it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:21, 39.24it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:20, 39.70it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:19, 39.88it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:18, 39.94it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:04<00:17, 40.01it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 38.53it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:16, 38.68it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:15, 39.04it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:14, 39.38it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:08<00:13, 39.27it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:09<00:12, 39.42it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:10<00:12, 39.65it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:11<00:11, 39.75it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 38.95it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:09, 39.16it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:13<00:08, 39.38it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:14<00:08, 39.61it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:15<00:07, 39.79it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:16<00:06, 39.64it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:17<00:05, 39.70it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:17<00:04, 39.82it/s] |

```

Computing ESM2 embeddings: 82%|      | 736/895 [00:18<00:04, 38.45it/s]
Computing ESM2 embeddings: 86%|      | 768/895 [00:19<00:03, 38.57it/s]
Computing ESM2 embeddings: 89%|      | 800/895 [00:20<00:02, 38.91it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:21<00:01, 39.22it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:21<00:00, 39.50it/s]

Computing ESM2 embeddings: 100%|     | 895/895 [00:22<00:00, 39.30it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.16it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:22, 38.35it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:20, 39.13it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:19, 39.51it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 39.60it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:04<00:18, 39.81it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 38.41it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 38.86it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:15, 39.26it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 39.49it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:08<00:14, 39.67it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 39.84it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 39.85it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 39.95it/s]

```



Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 39.21it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 39.33it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.27it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.41it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.59it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.77it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.89it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.90it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 38.61it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.96it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.28it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.56it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.74it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.83it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.35it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.86it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.07it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.66it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.91it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 40.01it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:17, 40.12it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 39.33it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 39.41it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 39.70it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 39.88it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:08<00:13, 39.95it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:12, 40.05it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:11, 40.13it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 39.22it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 39.42it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:12<00:09, 39.65it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:13<00:08, 39.80it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 39.94it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 40.06it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 40.06it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:16<00:05, 38.66it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:17<00:04, 39.01it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:18<00:04, 39.31it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 39.57it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 39.79it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:20<00:01, 39.89it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:21<00:00, 39.98it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:22<00:00, 39.49it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

`'esm.pooler.dense.weight']`

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/570 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 6%| | 32/570 [00:00<00:14, 38.36it/s]  
Computing ESM2 embeddings: 11%| | 64/570 [00:01<00:12, 39.34it/s]  
Computing ESM2 embeddings: 17%| | 96/570 [00:02<00:12, 39.33it/s]  
Computing ESM2 embeddings: 22%| | 128/570 [00:03<00:11, 39.42it/s]  
Computing ESM2 embeddings: 28%| | 160/570 [00:04<00:10, 39.71it/s]  
Computing ESM2 embeddings: 34%| | 192/570 [00:04<00:09, 39.89it/s]  
Computing ESM2 embeddings: 39%| | 224/570 [00:05<00:08, 38.71it/s]  
Computing ESM2 embeddings: 45%| | 256/570 [00:06<00:08, 39.01it/s]  
Computing ESM2 embeddings: 51%| | 288/570 [00:07<00:07, 39.37it/s]  
Computing ESM2 embeddings: 56%| | 320/570 [00:08<00:06, 39.57it/s]  
Computing ESM2 embeddings: 62%| | 352/570 [00:08<00:05, 39.76it/s]  
Computing ESM2 embeddings: 67%| | 384/570 [00:09<00:04, 39.90it/s]  
Computing ESM2 embeddings: 73%| | 416/570 [00:10<00:03, 39.93it/s]  
Computing ESM2 embeddings: 79%| | 448/570 [00:11<00:03, 40.01it/s]  
Computing ESM2 embeddings: 84%| | 480/570 [00:12<00:02, 39.22it/s]  
Computing ESM2 embeddings: 90%| | 512/570 [00:12<00:01, 39.29it/s]  
Computing ESM2 embeddings: 95%| | 544/570 [00:13<00:00, 39.52it/s]

Computing ESM2 embeddings: 100%| | 570/570 [00:14<00:00, 39.47it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/916 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/916 [00:00<00:22, 38.44it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/916 [00:01<00:22, 38.62it/s]  |
| Computing ESM2 embeddings: | 10% | 96/916 [00:02<00:20, 39.23it/s]  |
| Computing ESM2 embeddings: | 14% | 128/916 [00:03<00:19, 39.59it/s] |
| Computing ESM2 embeddings: | 17% | 160/916 [00:04<00:18, 39.87it/s] |
| Computing ESM2 embeddings: | 21% | 192/916 [00:04<00:18, 40.05it/s] |
| Computing ESM2 embeddings: | 24% | 224/916 [00:05<00:17, 38.52it/s] |
| Computing ESM2 embeddings: | 28% | 256/916 [00:06<00:16, 38.93it/s] |
| Computing ESM2 embeddings: | 31% | 288/916 [00:07<00:15, 39.32it/s] |
| Computing ESM2 embeddings: | 35% | 320/916 [00:08<00:15, 39.53it/s] |
| Computing ESM2 embeddings: | 38% | 352/916 [00:08<00:14, 39.74it/s] |
| Computing ESM2 embeddings: | 42% | 384/916 [00:09<00:13, 39.87it/s] |
| Computing ESM2 embeddings: | 45% | 416/916 [00:10<00:12, 39.93it/s] |
| Computing ESM2 embeddings: | 49% | 448/916 [00:11<00:11, 40.00it/s] |
| Computing ESM2 embeddings: | 52% | 480/916 [00:12<00:11, 38.74it/s] |
| Computing ESM2 embeddings: | 56% | 512/916 [00:13<00:10, 38.98it/s] |
| Computing ESM2 embeddings: | 59% | 544/916 [00:13<00:09, 39.34it/s] |
| Computing ESM2 embeddings: | 63% | 576/916 [00:14<00:08, 39.60it/s] |
| Computing ESM2 embeddings: | 66% | 608/916 [00:15<00:07, 39.71it/s] |
| Computing ESM2 embeddings: | 70% | 640/916 [00:16<00:06, 39.85it/s] |
| Computing ESM2 embeddings: | 73% | 672/916 [00:17<00:06, 39.95it/s] |
| Computing ESM2 embeddings: | 77% | 704/916 [00:17<00:05, 39.96it/s] |
| Computing ESM2 embeddings: | 80% | 736/916 [00:18<00:04, 38.74it/s] |

Computing ESM2 embeddings: 84%| | 768/916 [00:19<00:03, 39.04it/s]

Computing ESM2 embeddings: 87%| | 800/916 [00:20<00:02, 39.32it/s]

Computing ESM2 embeddings: 91%| | 832/916 [00:21<00:02, 39.57it/s]

Computing ESM2 embeddings: 94%| | 864/916 [00:21<00:01, 39.75it/s]

Computing ESM2 embeddings: 98%| | 896/916 [00:22<00:00, 39.82it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:23<00:00, 39.49it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/898 [00:00<00:21, 39.56it/s]

Computing ESM2 embeddings: 7%| | 64/898 [00:01<00:20, 39.84it/s]

Computing ESM2 embeddings: 11%| | 96/898 [00:02<00:20, 40.05it/s]

Computing ESM2 embeddings: 14%| | 128/898 [00:03<00:19, 40.14it/s]

Computing ESM2 embeddings: 18%| | 160/898 [00:03<00:18, 40.13it/s]

Computing ESM2 embeddings: 21%| | 192/898 [00:04<00:17, 40.19it/s]

Computing ESM2 embeddings: 25%| | 224/898 [00:05<00:17, 39.30it/s]

Computing ESM2 embeddings: 29%| | 256/898 [00:06<00:16, 39.36it/s]

Computing ESM2 embeddings: 32%| | 288/898 [00:07<00:15, 39.60it/s]

Computing ESM2 embeddings: 36%| | 320/898 [00:08<00:14, 39.78it/s]

Computing ESM2 embeddings: 39%| | 352/898 [00:08<00:13, 39.85it/s]

Computing ESM2 embeddings: 43%| | 384/898 [00:09<00:12, 39.94it/s]

Computing ESM2 embeddings: 46%| | 416/898 [00:10<00:12, 40.00it/s]

Computing ESM2 embeddings: 50%| | 448/898 [00:11<00:11, 39.99it/s]

Computing ESM2 embeddings: 53%| | 480/898 [00:12<00:10, 39.17it/s]

Computing ESM2 embeddings: 57%| | 512/898 [00:12<00:09, 39.36it/s]

Computing ESM2 embeddings: 61%| | 544/898 [00:13<00:08, 39.54it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:14<00:08, 39.72it/s]

Computing ESM2 embeddings: 68%| | 608/898 [00:15<00:07, 39.84it/s]

Computing ESM2 embeddings: 71%| | 640/898 [00:16<00:06, 39.88it/s]

Computing ESM2 embeddings: 75%| | 672/898 [00:16<00:05, 39.98it/s]

Computing ESM2 embeddings: 78%| | 704/898 [00:17<00:04, 40.07it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:18<00:04, 38.69it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:19<00:03, 39.01it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:20<00:02, 39.35it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:20<00:01, 39.53it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:21<00:00, 39.66it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:22<00:00, 39.62it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.90it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 38.96it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.43it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 39.67it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:18, 39.81it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.83it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:17, 38.41it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:16, 38.85it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:15, 39.23it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:15, 39.49it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:08<00:14, 39.70it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:09<00:13, 39.87it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:10<00:12, 39.94it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:11, 40.02it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 38.76it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 39.00it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:13<00:09, 39.37it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:14<00:08, 39.63it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:15<00:07, 39.76it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:16<00:06, 39.90it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 40.01it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:17<00:05, 40.01it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:18<00:04, 39.03it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:19<00:03, 39.26it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:20<00:02, 39.50it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:21<00:02, 39.70it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:21<00:01, 39.87it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:22<00:00, 39.92it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:23<00:00, 39.49it/s] |

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'esm.pooler.dense.weight']

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|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:22, 38.53it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:21, 39.20it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:20, 39.72it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:19, 39.96it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:18, 40.04it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:04<00:17, 40.15it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 38.58it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:16, 38.93it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:15, 39.35it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:14, 39.65it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:08<00:13, 39.78it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:09<00:12, 39.94it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:10<00:11, 40.07it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:11<00:11, 40.07it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 38.79it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:12<00:09, 39.11it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:13<00:08, 39.40it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:14<00:08, 39.65it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:15<00:07, 39.84it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:16<00:06, 39.92it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:16<00:05, 40.03it/s] |



Computing ESM2 embeddings: 79%| | 704/895 [00:17<00:04, 40.12it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:18<00:04, 38.80it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:19<00:03, 39.10it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:20<00:02, 39.45it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:21<00:01, 39.60it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:21<00:00, 39.80it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:22<00:00, 39.54it/s]

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Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.74it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.73it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.92it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 39.28it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.61it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.82it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 38.26it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.44it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.89it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 39.26it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 39.60it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.74it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.80it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.89it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.63it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.86it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.14it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.42it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.57it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.74it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.84it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.87it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 38.62it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.96it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.25it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.52it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.70it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.78it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.28it/s]

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Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.71it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.09it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.71it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.97it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:18, 40.04it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:17, 40.16it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 39.33it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 39.47it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 39.75it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 39.93it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:08<00:13, 40.01it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:12, 40.11it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:11, 40.16it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 40.14it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 38.93it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:12<00:09, 39.20it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:13<00:08, 39.48it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 39.73it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 39.92it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 39.99it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:16<00:05, 40.09it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:17<00:04, 40.16it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:18<00:04, 38.69it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 39.03it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 39.38it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:20<00:01, 39.61it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:21<00:00, 39.81it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:22<00:00, 39.64it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/914 [00:00<00:22, 39.28it/s]
Computing ESM2 embeddings: 7%|          | 64/914 [00:01<00:21, 39.83it/s]
Computing ESM2 embeddings: 11%|         | 96/914 [00:02<00:20, 40.03it/s]
Computing ESM2 embeddings: 14%|         | 128/914 [00:03<00:19, 40.04it/s]
Computing ESM2 embeddings: 18%|         | 160/914 [00:04<00:18, 40.10it/s]
Computing ESM2 embeddings: 21%|         | 192/914 [00:04<00:17, 40.14it/s]
Computing ESM2 embeddings: 25%|         | 224/914 [00:05<00:17, 38.65it/s]
Computing ESM2 embeddings: 28%|         | 256/914 [00:06<00:16, 39.03it/s]
Computing ESM2 embeddings: 32%|         | 288/914 [00:07<00:15, 39.40it/s]
Computing ESM2 embeddings: 35%|         | 320/914 [00:08<00:15, 39.58it/s]
Computing ESM2 embeddings: 39%|         | 352/914 [00:08<00:14, 39.76it/s]
Computing ESM2 embeddings: 42%|         | 384/914 [00:09<00:13, 39.88it/s]
Computing ESM2 embeddings: 46%|         | 416/914 [00:10<00:12, 39.91it/s]
Computing ESM2 embeddings: 49%|         | 448/914 [00:11<00:11, 39.99it/s]
Computing ESM2 embeddings: 53%|         | 480/914 [00:12<00:11, 38.66it/s]
Computing ESM2 embeddings: 56%|         | 512/914 [00:12<00:10, 38.95it/s]
Computing ESM2 embeddings: 60%|         | 544/914 [00:13<00:09, 39.29it/s]
Computing ESM2 embeddings: 63%|         | 576/914 [00:14<00:08, 39.56it/s]
Computing ESM2 embeddings: 67%|         | 608/914 [00:15<00:07, 39.70it/s]
Computing ESM2 embeddings: 70%|         | 640/914 [00:16<00:06, 39.85it/s]
```

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:16<00:06, 39.95it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:17<00:05, 39.96it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:18<00:04, 38.93it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:19<00:03, 39.12it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 39.36it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.59it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:21<00:01, 39.78it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.76it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.52it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/570 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/570 [00:00<00:14, 37.85it/s]
Computing ESM2 embeddings: 11%|      | 64/570 [00:01<00:12, 39.10it/s]
Computing ESM2 embeddings: 17%|      | 96/570 [00:02<00:11, 39.69it/s]
Computing ESM2 embeddings: 22%|      | 128/570 [00:03<00:11, 39.96it/s]
Computing ESM2 embeddings: 28%|      | 160/570 [00:04<00:10, 40.05it/s]
Computing ESM2 embeddings: 34%|      | 192/570 [00:04<00:09, 40.16it/s]
Computing ESM2 embeddings: 39%|      | 224/570 [00:05<00:08, 38.85it/s]
Computing ESM2 embeddings: 45%|      | 256/570 [00:06<00:08, 39.14it/s]
Computing ESM2 embeddings: 51%|      | 288/570 [00:07<00:07, 39.50it/s]
Computing ESM2 embeddings: 56%|      | 320/570 [00:08<00:06, 39.76it/s]
Computing ESM2 embeddings: 62%|      | 352/570 [00:08<00:05, 39.89it/s]

```

Computing ESM2 embeddings: 67%| | 384/570 [00:09<00:04, 40.02it/s]

Computing ESM2 embeddings: 73%| | 416/570 [00:10<00:03, 40.12it/s]

Computing ESM2 embeddings: 79%| | 448/570 [00:11<00:03, 40.15it/s]

Computing ESM2 embeddings: 84%| | 480/570 [00:12<00:02, 38.88it/s]

Computing ESM2 embeddings: 90%| | 512/570 [00:12<00:01, 39.18it/s]

Computing ESM2 embeddings: 95%| | 544/570 [00:13<00:00, 39.47it/s]

Computing ESM2 embeddings: 100%| | 570/570 [00:14<00:00, 39.51it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:22, 38.43it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:21, 39.21it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:20, 39.68it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:19, 39.91it/s]

Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:18, 39.98it/s]

Computing ESM2 embeddings: 21%| | 192/915 [00:04<00:18, 40.08it/s]

Computing ESM2 embeddings: 24%| | 224/915 [00:05<00:17, 38.74it/s]

Computing ESM2 embeddings: 28%| | 256/915 [00:06<00:16, 39.03it/s]

Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:15, 39.40it/s]

Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:15, 39.66it/s]

Computing ESM2 embeddings: 38%| | 352/915 [00:08<00:14, 39.77it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:09<00:13, 39.89it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:10<00:12, 39.99it/s]

```

Computing ESM2 embeddings: 49%|          | 448/915 [00:11<00:11, 40.00it/s]
Computing ESM2 embeddings: 52%|          | 480/915 [00:12<00:11, 38.93it/s]
Computing ESM2 embeddings: 56%|          | 512/915 [00:12<00:10, 39.29it/s]
Computing ESM2 embeddings: 59%|          | 544/915 [00:13<00:09, 39.51it/s]
Computing ESM2 embeddings: 63%|          | 576/915 [00:14<00:08, 39.71it/s]
Computing ESM2 embeddings: 66%|          | 608/915 [00:15<00:07, 39.86it/s]
Computing ESM2 embeddings: 70%|          | 640/915 [00:16<00:06, 39.90it/s]
Computing ESM2 embeddings: 73%|          | 672/915 [00:16<00:06, 39.98it/s]
Computing ESM2 embeddings: 77%|          | 704/915 [00:17<00:05, 38.74it/s]
Computing ESM2 embeddings: 80%|          | 736/915 [00:18<00:04, 38.99it/s]
Computing ESM2 embeddings: 84%|          | 768/915 [00:19<00:03, 39.30it/s]
Computing ESM2 embeddings: 87%|          | 800/915 [00:20<00:02, 39.55it/s]
Computing ESM2 embeddings: 91%|          | 832/915 [00:21<00:02, 39.46it/s]
Computing ESM2 embeddings: 94%|          | 864/915 [00:21<00:01, 39.60it/s]
Computing ESM2 embeddings: 98%|          | 896/915 [00:22<00:00, 39.78it/s]

Computing ESM2 embeddings: 100%|         | 915/915 [00:23<00:00, 39.40it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/898 [00:00<00:22, 38.57it/s]
Computing ESM2 embeddings:  7%|          | 64/898 [00:01<00:21, 39.37it/s]
Computing ESM2 embeddings: 11%|          | 96/898 [00:02<00:20, 39.71it/s]
Computing ESM2 embeddings: 14%|          | 128/898 [00:03<00:19, 39.94it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/898 [00:04<00:18, 40.06it/s] |
| Computing ESM2 embeddings: 21%  | 192/898 [00:04<00:17, 40.07it/s] |
| Computing ESM2 embeddings: 25%  | 224/898 [00:05<00:17, 38.62it/s] |
| Computing ESM2 embeddings: 29%  | 256/898 [00:06<00:16, 39.00it/s] |
| Computing ESM2 embeddings: 32%  | 288/898 [00:07<00:15, 39.32it/s] |
| Computing ESM2 embeddings: 36%  | 320/898 [00:08<00:14, 39.59it/s] |
| Computing ESM2 embeddings: 39%  | 352/898 [00:08<00:13, 39.77it/s] |
| Computing ESM2 embeddings: 43%  | 384/898 [00:09<00:12, 39.86it/s] |
| Computing ESM2 embeddings: 46%  | 416/898 [00:10<00:12, 39.80it/s] |
| Computing ESM2 embeddings: 50%  | 448/898 [00:11<00:11, 39.83it/s] |
| Computing ESM2 embeddings: 53%  | 480/898 [00:12<00:10, 39.87it/s] |
| Computing ESM2 embeddings: 57%  | 512/898 [00:12<00:09, 39.97it/s] |
| Computing ESM2 embeddings: 61%  | 544/898 [00:13<00:08, 39.89it/s] |
| Computing ESM2 embeddings: 64%  | 576/898 [00:14<00:08, 39.91it/s] |
| Computing ESM2 embeddings: 68%  | 608/898 [00:15<00:07, 39.97it/s] |
| Computing ESM2 embeddings: 71%  | 640/898 [00:16<00:06, 38.79it/s] |
| Computing ESM2 embeddings: 75%  | 672/898 [00:17<00:05, 39.01it/s] |
| Computing ESM2 embeddings: 78%  | 704/898 [00:17<00:04, 39.36it/s] |
| Computing ESM2 embeddings: 82%  | 736/898 [00:18<00:04, 39.54it/s] |
| Computing ESM2 embeddings: 86%  | 768/898 [00:19<00:03, 39.64it/s] |
| Computing ESM2 embeddings: 89%  | 800/898 [00:20<00:02, 39.80it/s] |
| Computing ESM2 embeddings: 93%  | 832/898 [00:20<00:01, 39.90it/s] |
| Computing ESM2 embeddings: 96%  | 864/898 [00:21<00:00, 39.92it/s] |
| Computing ESM2 embeddings: 100% | 898/898 [00:22<00:00, 39.45it/s] |



Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.36it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.80it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 40.05it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:19, 40.19it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:03<00:18, 40.16it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:17, 40.22it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.27it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.40it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:15, 39.68it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:14, 39.88it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 39.93it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.98it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 40.07it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 40.06it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 38.85it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:12<00:10, 39.14it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 39.39it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.63it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.81it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.86it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:16<00:06, 39.96it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:17<00:05, 39.21it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:18<00:04, 39.30it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:19<00:03, 39.58it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 39.77it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:20<00:02, 39.83it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:21<00:01, 39.93it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 40.02it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.56it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 37.89it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 39.21it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 39.61it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:19, 39.89it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:18, 40.06it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:17, 39.89it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 38.58it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 38.97it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 39.28it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 39.55it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:08<00:13, 39.73it/s]

```

```

Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:12, 39.78it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 39.89it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.98it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 38.70it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:09, 39.02it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:08, 39.36it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 39.52it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 39.69it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.83it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 39.86it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 38.62it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 38.97it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 39.28it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 39.53it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.67it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 39.74it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.42it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 37.93it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 39.33it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:20, 39.80it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:19, 39.94it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:18, 40.02it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 40.08it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 38.53it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 38.93it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:15, 39.33it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 39.56it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:08<00:14, 39.76it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 39.88it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 39.93it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.98it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 38.52it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 38.80it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.22it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.51it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.65it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:06, 39.81it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:16<00:06, 39.93it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 39.05it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.27it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 39.53it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 39.68it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:21<00:02, 39.83it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:21<00:01, 39.94it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.96it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.40it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 37.66it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:21, 39.09it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:20, 39.65it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:19, 39.93it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:18, 40.00it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:04<00:17, 40.13it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:05<00:17, 38.59it/s]

Computing ESM2 embeddings: 29%| | 256/895 [00:06<00:16, 38.87it/s]

Computing ESM2 embeddings: 32%| | 288/895 [00:07<00:15, 39.27it/s]

Computing ESM2 embeddings: 36%| | 320/895 [00:08<00:14, 39.56it/s]

Computing ESM2 embeddings: 39%| | 352/895 [00:08<00:13, 39.68it/s]

Computing ESM2 embeddings: 43%| | 384/895 [00:09<00:12, 39.84it/s]

Computing ESM2 embeddings: 46%| | 416/895 [00:10<00:12, 39.91it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:11<00:11, 39.94it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:12<00:10, 38.76it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:12<00:09, 39.10it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:13<00:08, 39.36it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:14<00:08, 39.60it/s]

```

Computing ESM2 embeddings: 68%|          | 608/895 [00:15<00:07, 39.77it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:16<00:06, 39.83it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:16<00:05, 39.93it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:17<00:04, 40.00it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:18<00:04, 38.61it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:19<00:03, 38.97it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:20<00:02, 39.32it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:21<00:01, 39.48it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:21<00:00, 39.70it/s]
Computing ESM2 embeddings: 100%|         | 895/895 [00:22<00:00, 39.45it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/914 [00:00<00:22, 39.06it/s]
Computing ESM2 embeddings: 7%|           | 64/914 [00:01<00:21, 39.04it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:20, 39.41it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:19, 39.66it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:18, 39.87it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:04<00:18, 38.54it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:17, 38.92it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:16, 39.30it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:15, 39.57it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:14, 39.69it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:14, 39.86it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.95it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.54it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 38.90it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 39.28it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 39.48it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.69it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.84it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.87it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.87it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.64it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 38.88it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.25it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.53it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.66it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.82it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.84it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 38.95it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.26it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:14, 37.86it/s]

Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:13, 38.86it/s]

Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:12, 39.47it/s]

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:11, 39.77it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:10, 39.87it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:04<00:09, 38.32it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:05<00:08, 38.80it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 39.13it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 39.44it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 39.67it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:08<00:05, 39.77it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:09<00:04, 39.89it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:10<00:03, 39.98it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:11<00:03, 38.77it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 38.79it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 39.08it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:13<00:00, 39.35it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:14<00:00, 39.24it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/916 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/916 [00:00<00:23, 38.02it/s]

Computing ESM2 embeddings: 7%| | 64/916 [00:01<00:21, 39.07it/s]

Computing ESM2 embeddings: 10%| | 96/916 [00:02<00:20, 39.49it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/916 [00:03<00:19, 39.78it/s] |
| Computing ESM2 embeddings: 17% | 160/916 [00:04<00:18, 39.87it/s] |
| Computing ESM2 embeddings: 21% | 192/916 [00:04<00:18, 39.02it/s] |
| Computing ESM2 embeddings: 24% | 224/916 [00:05<00:17, 39.31it/s] |
| Computing ESM2 embeddings: 28% | 256/916 [00:06<00:16, 39.52it/s] |
| Computing ESM2 embeddings: 31% | 288/916 [00:07<00:15, 39.74it/s] |
| Computing ESM2 embeddings: 35% | 320/916 [00:08<00:14, 39.86it/s] |
| Computing ESM2 embeddings: 38% | 352/916 [00:08<00:14, 39.91it/s] |
| Computing ESM2 embeddings: 42% | 384/916 [00:09<00:13, 39.99it/s] |
| Computing ESM2 embeddings: 45% | 416/916 [00:10<00:12, 38.78it/s] |
| Computing ESM2 embeddings: 49% | 448/916 [00:11<00:12, 38.97it/s] |
| Computing ESM2 embeddings: 52% | 480/916 [00:12<00:11, 39.34it/s] |
| Computing ESM2 embeddings: 56% | 512/916 [00:12<00:10, 39.58it/s] |
| Computing ESM2 embeddings: 59% | 544/916 [00:13<00:09, 39.71it/s] |
| Computing ESM2 embeddings: 63% | 576/916 [00:14<00:08, 39.85it/s] |
| Computing ESM2 embeddings: 66% | 608/916 [00:15<00:07, 39.98it/s] |
| Computing ESM2 embeddings: 70% | 640/916 [00:16<00:07, 38.72it/s] |
| Computing ESM2 embeddings: 73% | 672/916 [00:17<00:06, 39.05it/s] |
| Computing ESM2 embeddings: 77% | 704/916 [00:17<00:05, 39.39it/s] |
| Computing ESM2 embeddings: 80% | 736/916 [00:18<00:04, 39.53it/s] |
| Computing ESM2 embeddings: 84% | 768/916 [00:19<00:03, 39.63it/s] |
| Computing ESM2 embeddings: 87% | 800/916 [00:20<00:02, 39.74it/s] |
| Computing ESM2 embeddings: 91% | 832/916 [00:21<00:02, 39.75it/s] |
| Computing ESM2 embeddings: 94% | 864/916 [00:21<00:01, 39.07it/s] |

Computing ESM2 embeddings: 98%| | 896/916 [00:22<00:00, 39.24it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:23<00:00, 39.39it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/898 [00:01<00:28, 30.82it/s]

Computing ESM2 embeddings: 7%| | 64/898 [00:01<00:23, 35.53it/s]

Computing ESM2 embeddings: 11%| | 96/898 [00:02<00:21, 37.45it/s]

Computing ESM2 embeddings: 14%| | 128/898 [00:03<00:20, 38.49it/s]

Computing ESM2 embeddings: 18%| | 160/898 [00:04<00:19, 37.68it/s]

Computing ESM2 embeddings: 21%| | 192/898 [00:05<00:18, 38.28it/s]

Computing ESM2 embeddings: 25%| | 224/898 [00:05<00:17, 38.88it/s]

Computing ESM2 embeddings: 29%| | 256/898 [00:06<00:16, 39.30it/s]

Computing ESM2 embeddings: 32%| | 288/898 [00:07<00:15, 39.52it/s]

Computing ESM2 embeddings: 36%| | 320/898 [00:08<00:14, 39.72it/s]

Computing ESM2 embeddings: 39%| | 352/898 [00:09<00:13, 39.87it/s]

Computing ESM2 embeddings: 43%| | 384/898 [00:09<00:13, 38.66it/s]

Computing ESM2 embeddings: 46%| | 416/898 [00:10<00:12, 39.01it/s]

Computing ESM2 embeddings: 50%| | 448/898 [00:11<00:11, 39.34it/s]

Computing ESM2 embeddings: 53%| | 480/898 [00:12<00:10, 39.53it/s]

Computing ESM2 embeddings: 57%| | 512/898 [00:13<00:09, 39.71it/s]

Computing ESM2 embeddings: 61%| | 544/898 [00:13<00:08, 39.86it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:14<00:08, 39.87it/s]

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Computing ESM2 embeddings: 68%|          | 608/898 [00:15<00:07, 39.97it/s]
Computing ESM2 embeddings: 71%|          | 640/898 [00:16<00:06, 38.63it/s]
Computing ESM2 embeddings: 75%|          | 672/898 [00:17<00:05, 38.87it/s]
Computing ESM2 embeddings: 78%|          | 704/898 [00:18<00:04, 39.25it/s]
Computing ESM2 embeddings: 82%|          | 736/898 [00:18<00:04, 39.53it/s]
Computing ESM2 embeddings: 86%|          | 768/898 [00:19<00:03, 39.65it/s]
Computing ESM2 embeddings: 89%|          | 800/898 [00:20<00:02, 39.82it/s]
Computing ESM2 embeddings: 93%|          | 832/898 [00:21<00:01, 39.96it/s]
Computing ESM2 embeddings: 96%|          | 864/898 [00:22<00:00, 38.68it/s]

Computing ESM2 embeddings: 100%|         | 898/898 [00:23<00:00, 38.94it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:22, 38.70it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 38.79it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 38.94it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:20, 39.28it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 39.50it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:04<00:18, 38.87it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:17, 39.16it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:16, 39.42it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:15, 39.69it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:14, 39.87it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:14, 39.89it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.92it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.98it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 38.35it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.39it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.73it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.10it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.42it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.63it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.74it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.89it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 38.78it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.01it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.35it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.60it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.75it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.84it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.93it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.39it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.51it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/896 [00:01<00:21, 39.33it/s]  |
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:20, 39.63it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:19, 39.81it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:18, 39.95it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:04<00:18, 38.41it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 38.88it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 39.28it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 39.50it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 39.66it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:08<00:13, 39.80it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:12, 39.85it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 38.65it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 38.97it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 39.25it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 39.52it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:13<00:08, 39.70it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 39.78it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 39.89it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 39.96it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 38.58it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:17<00:04, 38.94it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:18<00:04, 39.29it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:19<00:03, 39.49it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 39.68it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.83it/s]

Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 39.86it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.41it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/913 [00:00<00:23, 37.99it/s]

Computing ESM2 embeddings:  7%|      | 64/913 [00:01<00:21, 39.05it/s]

Computing ESM2 embeddings: 11%|      | 96/913 [00:02<00:20, 39.61it/s]

Computing ESM2 embeddings: 14%|      | 128/913 [00:03<00:19, 39.92it/s]

Computing ESM2 embeddings: 18%|      | 160/913 [00:04<00:18, 40.00it/s]

Computing ESM2 embeddings: 21%|      | 192/913 [00:04<00:17, 40.13it/s]

Computing ESM2 embeddings: 25%|      | 224/913 [00:05<00:17, 38.83it/s]

Computing ESM2 embeddings: 28%|      | 256/913 [00:06<00:16, 39.10it/s]

Computing ESM2 embeddings: 32%|      | 288/913 [00:07<00:15, 39.49it/s]

Computing ESM2 embeddings: 35%|      | 320/913 [00:08<00:15, 39.47it/s]

Computing ESM2 embeddings: 39%|      | 352/913 [00:08<00:14, 39.51it/s]

Computing ESM2 embeddings: 42%|      | 384/913 [00:09<00:13, 39.76it/s]

Computing ESM2 embeddings: 46%|      | 416/913 [00:10<00:12, 39.93it/s]

Computing ESM2 embeddings: 49%|      | 448/913 [00:11<00:11, 38.86it/s]

Computing ESM2 embeddings: 53%|      | 480/913 [00:12<00:11, 39.16it/s]

Computing ESM2 embeddings: 56%|      | 512/913 [00:12<00:10, 39.51it/s]

Computing ESM2 embeddings: 60%|      | 544/913 [00:13<00:09, 39.69it/s]

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Computing ESM2 embeddings: 63%|          | 576/913 [00:14<00:08, 39.60it/s]
Computing ESM2 embeddings: 67%|          | 608/913 [00:15<00:07, 39.71it/s]
Computing ESM2 embeddings: 70%|          | 640/913 [00:16<00:06, 39.42it/s]
Computing ESM2 embeddings: 74%|          | 672/913 [00:17<00:06, 39.60it/s]
Computing ESM2 embeddings: 77%|          | 704/913 [00:17<00:05, 39.77it/s]
Computing ESM2 embeddings: 81%|          | 736/913 [00:18<00:04, 39.84it/s]
Computing ESM2 embeddings: 84%|          | 768/913 [00:19<00:03, 39.95it/s]
Computing ESM2 embeddings: 88%|          | 800/913 [00:20<00:02, 40.03it/s]
Computing ESM2 embeddings: 91%|          | 832/913 [00:21<00:02, 40.01it/s]
Computing ESM2 embeddings: 95%|          | 864/913 [00:21<00:01, 38.71it/s]
Computing ESM2 embeddings: 98%|          | 896/913 [00:22<00:00, 39.04it/s]

Computing ESM2 embeddings: 100%|         | 913/913 [00:23<00:00, 39.42it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.22it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.24it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 39.64it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:19, 39.91it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:18, 40.06it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:04<00:17, 39.16it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 39.42it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 39.65it/s]

```

```

Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 39.77it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 39.95it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:08<00:13, 40.05it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:12, 40.07it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.82it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.13it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 39.33it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:12<00:09, 39.61it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:08, 39.81it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 39.89it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 40.03it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.85it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 39.04it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 39.42it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 39.67it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 39.79it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 39.94it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:20<00:01, 40.06it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 38.60it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.46it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]

```



|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 38.78it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.24it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.75it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:19, 39.98it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:18, 40.05it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 39.21it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.44it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.65it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:15, 39.87it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:14, 40.02it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 40.06it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.29it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.47it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.64it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:10, 39.85it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:12<00:10, 40.00it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 40.04it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 38.69it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 38.80it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 39.05it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:16<00:06, 39.43it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:17<00:05, 39.69it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:18<00:04, 39.82it/s] |
| Computing ESM2 embeddings: | 84% | 768/914 [00:19<00:03, 38.53it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 38.93it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.26it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:21<00:01, 39.57it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.78it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.51it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/572 [00:00<00:14, 38.46it/s]
Computing ESM2 embeddings: 11%|      | 64/572 [00:01<00:12, 39.29it/s]
Computing ESM2 embeddings: 17%|      | 96/572 [00:02<00:11, 39.70it/s]
Computing ESM2 embeddings: 22%|      | 128/572 [00:03<00:11, 39.95it/s]
Computing ESM2 embeddings: 28%|      | 160/572 [00:04<00:10, 40.12it/s]
Computing ESM2 embeddings: 34%|      | 192/572 [00:04<00:09, 40.15it/s]
Computing ESM2 embeddings: 39%|      | 224/572 [00:05<00:08, 38.87it/s]
Computing ESM2 embeddings: 45%|      | 256/572 [00:06<00:08, 39.19it/s]
Computing ESM2 embeddings: 50%|      | 288/572 [00:07<00:07, 39.47it/s]
Computing ESM2 embeddings: 56%|      | 320/572 [00:08<00:06, 39.69it/s]
Computing ESM2 embeddings: 62%|      | 352/572 [00:08<00:05, 39.88it/s]
Computing ESM2 embeddings: 67%|      | 384/572 [00:09<00:04, 39.95it/s]
Computing ESM2 embeddings: 73%|      | 416/572 [00:10<00:03, 39.25it/s]
Computing ESM2 embeddings: 78%|      | 448/572 [00:11<00:03, 39.43it/s]
Computing ESM2 embeddings: 84%|      | 480/572 [00:12<00:02, 39.64it/s]

```

Computing ESM2 embeddings: 90%| | 512/572 [00:12<00:01, 39.84it/s]

Computing ESM2 embeddings: 95%| | 544/572 [00:13<00:00, 39.99it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:14<00:00, 39.53it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/904 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/904 [00:00<00:23, 36.60it/s]

Computing ESM2 embeddings: 7%| | 64/904 [00:01<00:21, 38.43it/s]

Computing ESM2 embeddings: 11%| | 96/904 [00:02<00:20, 39.21it/s]

Computing ESM2 embeddings: 14%| | 128/904 [00:03<00:19, 39.50it/s]

Computing ESM2 embeddings: 18%| | 160/904 [00:04<00:18, 39.79it/s]

Computing ESM2 embeddings: 21%| | 192/904 [00:04<00:17, 39.83it/s]

Computing ESM2 embeddings: 25%| | 224/904 [00:05<00:17, 38.61it/s]

Computing ESM2 embeddings: 28%| | 256/904 [00:06<00:16, 38.99it/s]

Computing ESM2 embeddings: 32%| | 288/904 [00:07<00:15, 39.32it/s]

Computing ESM2 embeddings: 35%| | 320/904 [00:08<00:14, 39.62it/s]

Computing ESM2 embeddings: 39%| | 352/904 [00:08<00:13, 39.84it/s]

Computing ESM2 embeddings: 42%| | 384/904 [00:09<00:13, 39.91it/s]

Computing ESM2 embeddings: 46%| | 416/904 [00:10<00:12, 39.21it/s]

Computing ESM2 embeddings: 50%| | 448/904 [00:11<00:11, 39.42it/s]

Computing ESM2 embeddings: 53%| | 480/904 [00:12<00:10, 39.62it/s]

Computing ESM2 embeddings: 57%| | 512/904 [00:12<00:09, 39.82it/s]

Computing ESM2 embeddings: 60%| | 544/904 [00:13<00:09, 39.96it/s]

```

Computing ESM2 embeddings: 64%|          | 576/904 [00:14<00:08, 40.01it/s]
Computing ESM2 embeddings: 67%|          | 608/904 [00:15<00:07, 39.30it/s]
Computing ESM2 embeddings: 71%|          | 640/904 [00:16<00:06, 39.46it/s]
Computing ESM2 embeddings: 74%|          | 672/904 [00:17<00:05, 39.65it/s]
Computing ESM2 embeddings: 78%|          | 704/904 [00:17<00:05, 39.84it/s]
Computing ESM2 embeddings: 81%|          | 736/904 [00:18<00:04, 39.97it/s]
Computing ESM2 embeddings: 85%|          | 768/904 [00:19<00:03, 40.02it/s]
Computing ESM2 embeddings: 88%|          | 800/904 [00:20<00:02, 39.13it/s]
Computing ESM2 embeddings: 92%|          | 832/904 [00:21<00:01, 39.33it/s]
Computing ESM2 embeddings: 96%|          | 864/904 [00:21<00:01, 39.52it/s]
Computing ESM2 embeddings: 99%|          | 896/904 [00:22<00:00, 39.73it/s]

Computing ESM2 embeddings: 100%|         | 904/904 [00:22<00:00, 39.47it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/908 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/908 [00:00<00:23, 36.85it/s]
Computing ESM2 embeddings:  7%|          | 64/908 [00:01<00:21, 38.64it/s]
Computing ESM2 embeddings: 11%|          | 96/908 [00:02<00:20, 39.39it/s]
Computing ESM2 embeddings: 14%|          | 128/908 [00:03<00:19, 39.67it/s]
Computing ESM2 embeddings: 18%|          | 160/908 [00:04<00:18, 39.91it/s]
Computing ESM2 embeddings: 21%|          | 192/908 [00:04<00:18, 38.60it/s]
Computing ESM2 embeddings: 25%|          | 224/908 [00:05<00:17, 38.89it/s]
Computing ESM2 embeddings: 28%|          | 256/908 [00:06<00:16, 39.33it/s]

```

Computing ESM2 embeddings: 32%| | 288/908 [00:07<00:15, 39.63it/s]

Computing ESM2 embeddings: 35%| | 320/908 [00:08<00:14, 39.77it/s]

Computing ESM2 embeddings: 39%| | 352/908 [00:08<00:13, 39.94it/s]

Computing ESM2 embeddings: 42%| | 384/908 [00:09<00:13, 40.06it/s]

Computing ESM2 embeddings: 46%| | 416/908 [00:10<00:12, 38.82it/s]

Computing ESM2 embeddings: 49%| | 448/908 [00:11<00:11, 39.14it/s]

Computing ESM2 embeddings: 53%| | 480/908 [00:12<00:10, 39.49it/s]

Computing ESM2 embeddings: 56%| | 512/908 [00:12<00:09, 39.67it/s]

Computing ESM2 embeddings: 60%| | 544/908 [00:13<00:09, 39.83it/s]

Computing ESM2 embeddings: 63%| | 576/908 [00:14<00:08, 39.93it/s]

Computing ESM2 embeddings: 67%| | 608/908 [00:15<00:07, 38.76it/s]

Computing ESM2 embeddings: 70%| | 640/908 [00:16<00:06, 39.08it/s]

Computing ESM2 embeddings: 74%| | 672/908 [00:17<00:05, 39.40it/s]

Computing ESM2 embeddings: 78%| | 704/908 [00:17<00:05, 39.58it/s]

Computing ESM2 embeddings: 81%| | 736/908 [00:18<00:04, 39.78it/s]

Computing ESM2 embeddings: 85%| | 768/908 [00:19<00:03, 39.89it/s]

Computing ESM2 embeddings: 88%| | 800/908 [00:20<00:02, 38.99it/s]

Computing ESM2 embeddings: 92%| | 832/908 [00:21<00:01, 39.25it/s]

Computing ESM2 embeddings: 95%| | 864/908 [00:21<00:01, 39.52it/s]

Computing ESM2 embeddings: 99%| | 896/908 [00:22<00:00, 39.67it/s]

Computing ESM2 embeddings: 100%| | 908/908 [00:23<00:00, 39.37it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/897 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/897 [00:00<00:22, 38.84it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/897 [00:01<00:21, 39.36it/s]  |
| Computing ESM2 embeddings: | 11% | 96/897 [00:02<00:20, 39.81it/s]  |
| Computing ESM2 embeddings: | 14% | 128/897 [00:03<00:19, 39.99it/s] |
| Computing ESM2 embeddings: | 18% | 160/897 [00:04<00:18, 40.04it/s] |
| Computing ESM2 embeddings: | 21% | 192/897 [00:04<00:18, 38.77it/s] |
| Computing ESM2 embeddings: | 25% | 224/897 [00:05<00:17, 39.16it/s] |
| Computing ESM2 embeddings: | 29% | 256/897 [00:06<00:16, 39.50it/s] |
| Computing ESM2 embeddings: | 32% | 288/897 [00:07<00:15, 39.62it/s] |
| Computing ESM2 embeddings: | 36% | 320/897 [00:08<00:14, 39.83it/s] |
| Computing ESM2 embeddings: | 39% | 352/897 [00:08<00:13, 39.91it/s] |
| Computing ESM2 embeddings: | 43% | 384/897 [00:09<00:13, 39.21it/s] |
| Computing ESM2 embeddings: | 46% | 416/897 [00:10<00:12, 39.43it/s] |
| Computing ESM2 embeddings: | 50% | 448/897 [00:11<00:11, 39.59it/s] |
| Computing ESM2 embeddings: | 54% | 480/897 [00:12<00:10, 39.73it/s] |
| Computing ESM2 embeddings: | 57% | 512/897 [00:12<00:09, 39.67it/s] |
| Computing ESM2 embeddings: | 61% | 544/897 [00:13<00:08, 39.40it/s] |
| Computing ESM2 embeddings: | 64% | 576/897 [00:14<00:08, 38.07it/s] |
| Computing ESM2 embeddings: | 68% | 608/897 [00:15<00:07, 38.53it/s] |
| Computing ESM2 embeddings: | 71% | 640/897 [00:16<00:06, 38.72it/s] |
| Computing ESM2 embeddings: | 75% | 672/897 [00:17<00:05, 39.16it/s] |
| Computing ESM2 embeddings: | 78% | 704/897 [00:17<00:04, 39.44it/s] |
| Computing ESM2 embeddings: | 82% | 736/897 [00:18<00:04, 39.60it/s] |

```

Computing ESM2 embeddings: 86%|          | 768/897 [00:19<00:03, 38.35it/s]
Computing ESM2 embeddings: 89%|          | 800/897 [00:20<00:02, 38.77it/s]
Computing ESM2 embeddings: 93%|          | 832/897 [00:21<00:01, 39.11it/s]
Computing ESM2 embeddings: 96%|          | 864/897 [00:21<00:00, 39.43it/s]
Computing ESM2 embeddings: 100%|         | 897/897 [00:22<00:00, 39.24it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/913 [00:00<00:23, 37.43it/s]
Computing ESM2 embeddings:  7%|          | 64/913 [00:01<00:21, 38.89it/s]
Computing ESM2 embeddings: 11%|          | 96/913 [00:02<00:20, 39.54it/s]
Computing ESM2 embeddings: 14%|          | 128/913 [00:03<00:19, 39.70it/s]
Computing ESM2 embeddings: 18%|          | 160/913 [00:04<00:18, 39.87it/s]
Computing ESM2 embeddings: 21%|          | 192/913 [00:04<00:18, 38.50it/s]
Computing ESM2 embeddings: 25%|          | 224/913 [00:05<00:17, 38.82it/s]
Computing ESM2 embeddings: 28%|          | 256/913 [00:06<00:16, 39.23it/s]
Computing ESM2 embeddings: 32%|          | 288/913 [00:07<00:15, 39.53it/s]
Computing ESM2 embeddings: 35%|          | 320/913 [00:08<00:14, 39.65it/s]
Computing ESM2 embeddings: 39%|          | 352/913 [00:08<00:14, 39.82it/s]
Computing ESM2 embeddings: 42%|          | 384/913 [00:09<00:13, 39.07it/s]
Computing ESM2 embeddings: 46%|          | 416/913 [00:10<00:12, 39.21it/s]
Computing ESM2 embeddings: 49%|          | 448/913 [00:11<00:11, 39.49it/s]
Computing ESM2 embeddings: 53%|          | 480/913 [00:12<00:10, 39.70it/s]

```

Computing ESM2 embeddings: 56%| | 512/913 [00:12<00:10, 39.76it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:13<00:09, 39.90it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:14<00:08, 38.57it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:15<00:07, 38.86it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:16<00:06, 39.24it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:17<00:06, 39.52it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:17<00:05, 39.64it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:18<00:04, 39.81it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:19<00:03, 38.48it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:20<00:02, 38.77it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:21<00:02, 39.23it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:21<00:01, 39.47it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:22<00:00, 39.60it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:23<00:00, 39.27it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:24, 35.42it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 37.75it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.83it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.38it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.58it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:18, 38.36it/s]



```

Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 38.79it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 39.16it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 39.47it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 39.69it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:08<00:13, 39.77it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:12, 39.89it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.59it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 38.87it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 39.21it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:09, 39.50it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:08, 39.62it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 39.77it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.50it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.78it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 39.16it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 39.44it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 39.57it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 39.70it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.60it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 38.90it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 39.26it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.11it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.94it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:22, 38.50it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.29it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:19, 39.58it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:18, 39.79it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 38.15it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 38.61it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.11it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:15, 39.45it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:14, 39.62it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 39.81it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.05it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.18it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.49it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:10, 39.70it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 39.79it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 39.93it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.20it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.26it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.54it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 39.74it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:17<00:05, 39.82it/s] |

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.93it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.56it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.83it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.24it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.52it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.67it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.32it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/887 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/887 [00:00<00:22, 37.61it/s]

Computing ESM2 embeddings: 7%| | 64/887 [00:01<00:21, 38.68it/s]

Computing ESM2 embeddings: 11%| | 96/887 [00:02<00:20, 39.35it/s]

Computing ESM2 embeddings: 14%| | 128/887 [00:03<00:19, 39.64it/s]

Computing ESM2 embeddings: 18%| | 160/887 [00:04<00:18, 39.75it/s]

Computing ESM2 embeddings: 22%| | 192/887 [00:04<00:17, 39.88it/s]

Computing ESM2 embeddings: 25%| | 224/887 [00:05<00:17, 38.54it/s]

Computing ESM2 embeddings: 29%| | 256/887 [00:06<00:16, 38.82it/s]

Computing ESM2 embeddings: 32%| | 288/887 [00:07<00:15, 39.25it/s]

Computing ESM2 embeddings: 36%| | 320/887 [00:08<00:14, 39.56it/s]

Computing ESM2 embeddings: 40%| | 352/887 [00:08<00:13, 39.73it/s]

Computing ESM2 embeddings: 43%| | 384/887 [00:09<00:12, 39.85it/s]

Computing ESM2 embeddings: 47%| | 416/887 [00:10<00:12, 39.16it/s]

```

Computing ESM2 embeddings: 51%|          | 448/887 [00:11<00:11, 39.25it/s]
Computing ESM2 embeddings: 54%|          | 480/887 [00:12<00:10, 39.53it/s]
Computing ESM2 embeddings: 58%|          | 512/887 [00:12<00:09, 39.71it/s]
Computing ESM2 embeddings: 61%|          | 544/887 [00:13<00:08, 39.74it/s]
Computing ESM2 embeddings: 65%|          | 576/887 [00:14<00:07, 39.90it/s]
Computing ESM2 embeddings: 69%|          | 608/887 [00:15<00:07, 38.62it/s]
Computing ESM2 embeddings: 72%|          | 640/887 [00:16<00:06, 38.88it/s]
Computing ESM2 embeddings: 76%|          | 672/887 [00:17<00:05, 39.28it/s]
Computing ESM2 embeddings: 79%|          | 704/887 [00:17<00:04, 39.43it/s]
Computing ESM2 embeddings: 83%|          | 736/887 [00:18<00:03, 39.56it/s]
Computing ESM2 embeddings: 87%|          | 768/887 [00:19<00:02, 39.68it/s]
Computing ESM2 embeddings: 90%|          | 800/887 [00:20<00:02, 38.95it/s]
Computing ESM2 embeddings: 94%|          | 832/887 [00:21<00:01, 39.03it/s]
Computing ESM2 embeddings: 97%|          | 864/887 [00:21<00:00, 39.29it/s]

Computing ESM2 embeddings: 100%|         | 887/887 [00:22<00:00, 38.92it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/599 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|          | 32/599 [00:00<00:14, 38.91it/s]
Computing ESM2 embeddings: 11%|          | 64/599 [00:01<00:13, 39.53it/s]
Computing ESM2 embeddings: 16%|          | 96/599 [00:02<00:12, 39.61it/s]
Computing ESM2 embeddings: 21%|          | 128/599 [00:03<00:11, 39.62it/s]
Computing ESM2 embeddings: 27%|          | 160/599 [00:04<00:11, 39.80it/s]

```

Computing ESM2 embeddings: 32%| | 192/599 [00:04<00:10, 38.51it/s]

Computing ESM2 embeddings: 37%| | 224/599 [00:05<00:09, 38.79it/s]

Computing ESM2 embeddings: 43%| | 256/599 [00:06<00:08, 39.20it/s]

Computing ESM2 embeddings: 48%| | 288/599 [00:07<00:07, 39.48it/s]

Computing ESM2 embeddings: 53%| | 320/599 [00:08<00:07, 39.63it/s]

Computing ESM2 embeddings: 59%| | 352/599 [00:08<00:06, 39.80it/s]

Computing ESM2 embeddings: 64%| | 384/599 [00:09<00:05, 39.03it/s]

Computing ESM2 embeddings: 69%| | 416/599 [00:10<00:04, 39.19it/s]

Computing ESM2 embeddings: 75%| | 448/599 [00:11<00:03, 39.49it/s]

Computing ESM2 embeddings: 80%| | 480/599 [00:12<00:02, 39.69it/s]

Computing ESM2 embeddings: 85%| | 512/599 [00:12<00:02, 39.77it/s]

Computing ESM2 embeddings: 91%| | 544/599 [00:13<00:01, 39.90it/s]

Computing ESM2 embeddings: 96%| | 576/599 [00:14<00:00, 38.57it/s]

Computing ESM2 embeddings: 100%| | 599/599 [00:15<00:00, 39.06it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/898 [00:00<00:22, 37.99it/s]

Computing ESM2 embeddings: 7%| | 64/898 [00:01<00:21, 38.92it/s]

Computing ESM2 embeddings: 11%| | 96/898 [00:02<00:20, 39.30it/s]

Computing ESM2 embeddings: 14%| | 128/898 [00:03<00:19, 39.70it/s]

Computing ESM2 embeddings: 18%| | 160/898 [00:04<00:18, 39.92it/s]

Computing ESM2 embeddings: 21%| | 192/898 [00:04<00:18, 38.55it/s]

Computing ESM2 embeddings: 25%| | 224/898 [00:05<00:17, 38.97it/s]

Computing ESM2 embeddings: 29%| | 256/898 [00:06<00:16, 39.38it/s]

Computing ESM2 embeddings: 32%| | 288/898 [00:07<00:15, 39.58it/s]

Computing ESM2 embeddings: 36%| | 320/898 [00:08<00:14, 39.77it/s]

Computing ESM2 embeddings: 39%| | 352/898 [00:08<00:13, 39.94it/s]

Computing ESM2 embeddings: 43%| | 384/898 [00:09<00:13, 38.75it/s]

Computing ESM2 embeddings: 46%| | 416/898 [00:10<00:12, 39.10it/s]

Computing ESM2 embeddings: 50%| | 448/898 [00:11<00:11, 39.46it/s]

Computing ESM2 embeddings: 53%| | 480/898 [00:12<00:10, 39.63it/s]

Computing ESM2 embeddings: 57%| | 512/898 [00:12<00:09, 39.85it/s]

Computing ESM2 embeddings: 61%| | 544/898 [00:13<00:08, 39.98it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:14<00:08, 39.28it/s]

Computing ESM2 embeddings: 68%| | 608/898 [00:15<00:07, 39.50it/s]

Computing ESM2 embeddings: 71%| | 640/898 [00:16<00:06, 39.66it/s]

Computing ESM2 embeddings: 75%| | 672/898 [00:17<00:05, 39.73it/s]

Computing ESM2 embeddings: 78%| | 704/898 [00:17<00:04, 39.82it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:18<00:04, 39.89it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:19<00:03, 39.91it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:20<00:02, 39.95it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:21<00:01, 38.64it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:21<00:00, 38.86it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:22<00:00, 39.37it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 38.36it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.08it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.65it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:19, 39.94it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:18, 40.00it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 38.54it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 38.95it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.26it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:15, 39.53it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:14, 39.77it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 39.84it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.08it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.30it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.46it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:10, 39.67it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:12<00:10, 39.82it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 39.85it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 38.53it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 38.90it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.17it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 39.43it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:17<00:05, 39.64it/s] |

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.67it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.72it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.15it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.44it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.67it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.85it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.37it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.30it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.24it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.65it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.93it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 40.10it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:18, 38.55it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.99it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 39.40it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 39.64it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 39.85it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:13, 40.02it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 39.11it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 39.34it/s]



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Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.63it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 39.76it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:12<00:09, 39.92it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:08, 40.06it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 38.63it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 39.00it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.38it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 39.59it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 39.80it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 39.95it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.78it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 39.07it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.38it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 39.54it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.38it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 37.17it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:22, 37.06it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 38.40it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:20, 39.12it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 39.45it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/914 [00:05<00:18, 38.28it/s] |
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:17, 38.78it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:16, 39.16it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:15, 39.47it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:14, 39.70it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:08<00:14, 39.79it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:09<00:13, 38.34it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:10<00:12, 38.76it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:11, 39.09it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 39.40it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 39.59it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:13<00:09, 39.69it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:14<00:08, 38.99it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:15<00:07, 39.23it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:16<00:06, 39.41it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 39.65it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:17<00:05, 39.82it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:18<00:04, 39.86it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:19<00:03, 38.61it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:20<00:02, 38.96it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:21<00:02, 39.27it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:22<00:01, 39.54it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:22<00:00, 39.73it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:23<00:00, 39.12it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.68it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.01it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.51it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.85it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 40.03it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.97it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 39.26it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.58it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.73it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.91it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 40.01it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.64it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.98it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.34it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.52it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:12<00:09, 39.72it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:08, 39.87it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.99it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.21it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.51it/s] |

```

Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 39.64it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:17<00:04, 39.81it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:18<00:04, 39.94it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 38.72it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 39.04it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.38it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 39.56it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.44it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 38.25it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 39.02it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:20, 39.63it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:19, 39.90it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:18, 39.97it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:04<00:18, 38.48it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 38.90it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 39.24it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:15, 39.50it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:14, 39.71it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:08<00:14, 39.78it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 38.58it/s]

```

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.95it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.26it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 39.55it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 39.75it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.82it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.59it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.93it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.22it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.51it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.70it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.78it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.06it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.26it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.48it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.69it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.84it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.37it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/828 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/828 [00:00<00:20, 39.02it/s]

Computing ESM2 embeddings: 8%| | 64/828 [00:01<00:19, 39.11it/s]

Computing ESM2 embeddings: 12%| | 96/828 [00:02<00:18, 39.36it/s]

Computing ESM2 embeddings: 15%| | 128/828 [00:03<00:17, 39.74it/s]

Computing ESM2 embeddings: 19%| | 160/828 [00:04<00:16, 39.93it/s]

Computing ESM2 embeddings: 23%| | 192/828 [00:04<00:15, 39.96it/s]

Computing ESM2 embeddings: 27%| | 224/828 [00:05<00:15, 38.91it/s]

Computing ESM2 embeddings: 31%| | 256/828 [00:06<00:14, 39.23it/s]

Computing ESM2 embeddings: 35%| | 288/828 [00:07<00:13, 39.49it/s]

Computing ESM2 embeddings: 39%| | 320/828 [00:08<00:12, 39.70it/s]

Computing ESM2 embeddings: 43%| | 352/828 [00:08<00:11, 39.79it/s]

Computing ESM2 embeddings: 46%| | 384/828 [00:09<00:11, 39.87it/s]

Computing ESM2 embeddings: 50%| | 416/828 [00:10<00:10, 38.75it/s]

Computing ESM2 embeddings: 54%| | 448/828 [00:11<00:09, 39.09it/s]

Computing ESM2 embeddings: 58%| | 480/828 [00:12<00:08, 39.38it/s]

Computing ESM2 embeddings: 62%| | 512/828 [00:12<00:07, 39.60it/s]

Computing ESM2 embeddings: 66%| | 544/828 [00:13<00:07, 39.80it/s]

Computing ESM2 embeddings: 70%| | 576/828 [00:14<00:06, 39.88it/s]

Computing ESM2 embeddings: 73%| | 608/828 [00:15<00:05, 39.15it/s]

Computing ESM2 embeddings: 77%| | 640/828 [00:16<00:04, 39.30it/s]

Computing ESM2 embeddings: 81%| | 672/828 [00:17<00:03, 39.46it/s]

Computing ESM2 embeddings: 85%| | 704/828 [00:17<00:03, 39.70it/s]

Computing ESM2 embeddings: 89%| | 736/828 [00:18<00:02, 39.82it/s]

Computing ESM2 embeddings: 93%| | 768/828 [00:19<00:01, 39.04it/s]

Computing ESM2 embeddings: 97%| | 800/828 [00:20<00:00, 39.24it/s]

Computing ESM2 embeddings: 100%| | 828/828 [00:20<00:00, 39.43it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |      |                                  |
|----------------------------|------|----------------------------------|
| Computing ESM2 embeddings: | 0%   | 0/665 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 5%   | 32/665 [00:00<00:16, 38.37it/s]  |
| Computing ESM2 embeddings: | 10%  | 64/665 [00:01<00:15, 39.26it/s]  |
| Computing ESM2 embeddings: | 14%  | 96/665 [00:02<00:14, 39.72it/s]  |
| Computing ESM2 embeddings: | 19%  | 128/665 [00:03<00:13, 39.85it/s] |
| Computing ESM2 embeddings: | 24%  | 160/665 [00:04<00:12, 40.05it/s] |
| Computing ESM2 embeddings: | 29%  | 192/665 [00:04<00:12, 38.63it/s] |
| Computing ESM2 embeddings: | 34%  | 224/665 [00:05<00:11, 38.97it/s] |
| Computing ESM2 embeddings: | 38%  | 256/665 [00:06<00:10, 39.39it/s] |
| Computing ESM2 embeddings: | 43%  | 288/665 [00:07<00:09, 39.67it/s] |
| Computing ESM2 embeddings: | 48%  | 320/665 [00:08<00:08, 39.80it/s] |
| Computing ESM2 embeddings: | 53%  | 352/665 [00:08<00:07, 39.96it/s] |
| Computing ESM2 embeddings: | 58%  | 384/665 [00:09<00:07, 38.72it/s] |
| Computing ESM2 embeddings: | 63%  | 416/665 [00:10<00:06, 38.98it/s] |
| Computing ESM2 embeddings: | 67%  | 448/665 [00:11<00:05, 39.35it/s] |
| Computing ESM2 embeddings: | 72%  | 480/665 [00:12<00:04, 39.65it/s] |
| Computing ESM2 embeddings: | 77%  | 512/665 [00:12<00:03, 39.74it/s] |
| Computing ESM2 embeddings: | 82%  | 544/665 [00:13<00:03, 39.87it/s] |
| Computing ESM2 embeddings: | 87%  | 576/665 [00:14<00:02, 39.09it/s] |
| Computing ESM2 embeddings: | 91%  | 608/665 [00:15<00:01, 39.22it/s] |
| Computing ESM2 embeddings: | 96%  | 640/665 [00:16<00:00, 39.49it/s] |
| Computing ESM2 embeddings: | 100% | 665/665 [00:17<00:00, 39.03it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/908 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/908 [00:00<00:22, 38.58it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/908 [00:01<00:21, 39.37it/s]  |
| Computing ESM2 embeddings: | 11% | 96/908 [00:02<00:20, 39.83it/s]  |
| Computing ESM2 embeddings: | 14% | 128/908 [00:03<00:19, 39.97it/s] |
| Computing ESM2 embeddings: | 18% | 160/908 [00:04<00:18, 40.12it/s] |
| Computing ESM2 embeddings: | 21% | 192/908 [00:04<00:18, 39.07it/s] |
| Computing ESM2 embeddings: | 25% | 224/908 [00:05<00:17, 39.28it/s] |
| Computing ESM2 embeddings: | 28% | 256/908 [00:06<00:16, 39.61it/s] |
| Computing ESM2 embeddings: | 32% | 288/908 [00:07<00:15, 39.84it/s] |
| Computing ESM2 embeddings: | 35% | 320/908 [00:08<00:14, 39.91it/s] |
| Computing ESM2 embeddings: | 39% | 352/908 [00:08<00:13, 40.03it/s] |
| Computing ESM2 embeddings: | 42% | 384/908 [00:09<00:13, 39.35it/s] |
| Computing ESM2 embeddings: | 46% | 416/908 [00:10<00:12, 39.44it/s] |
| Computing ESM2 embeddings: | 49% | 448/908 [00:11<00:11, 39.73it/s] |
| Computing ESM2 embeddings: | 53% | 480/908 [00:12<00:10, 39.89it/s] |
| Computing ESM2 embeddings: | 56% | 512/908 [00:12<00:09, 39.97it/s] |
| Computing ESM2 embeddings: | 60% | 544/908 [00:13<00:09, 40.08it/s] |
| Computing ESM2 embeddings: | 63% | 576/908 [00:14<00:08, 38.82it/s] |
| Computing ESM2 embeddings: | 67% | 608/908 [00:15<00:07, 39.07it/s] |
| Computing ESM2 embeddings: | 70% | 640/908 [00:16<00:06, 39.44it/s] |



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Computing ESM2 embeddings: 74%|          | 672/908 [00:16<00:05, 39.71it/s]
Computing ESM2 embeddings: 78%|          | 704/908 [00:17<00:05, 39.85it/s]
Computing ESM2 embeddings: 81%|          | 736/908 [00:18<00:04, 39.99it/s]
Computing ESM2 embeddings: 85%|          | 768/908 [00:19<00:03, 38.75it/s]
Computing ESM2 embeddings: 88%|          | 800/908 [00:20<00:02, 39.02it/s]
Computing ESM2 embeddings: 92%|          | 832/908 [00:21<00:01, 39.40it/s]
Computing ESM2 embeddings: 95%|          | 864/908 [00:21<00:01, 39.65it/s]
Computing ESM2 embeddings: 99%|          | 896/908 [00:22<00:00, 39.78it/s]

Computing ESM2 embeddings: 100%|         | 908/908 [00:22<00:00, 39.53it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/899 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/899 [00:00<00:22, 37.98it/s]
Computing ESM2 embeddings: 7%|           | 64/899 [00:01<00:21, 38.95it/s]
Computing ESM2 embeddings: 11%|          | 96/899 [00:02<00:20, 39.50it/s]
Computing ESM2 embeddings: 14%|          | 128/899 [00:03<00:19, 39.85it/s]
Computing ESM2 embeddings: 18%|          | 160/899 [00:04<00:18, 39.97it/s]
Computing ESM2 embeddings: 21%|          | 192/899 [00:04<00:18, 38.54it/s]
Computing ESM2 embeddings: 25%|          | 224/899 [00:05<00:17, 38.75it/s]
Computing ESM2 embeddings: 28%|          | 256/899 [00:06<00:16, 39.05it/s]
Computing ESM2 embeddings: 32%|          | 288/899 [00:07<00:15, 39.46it/s]
Computing ESM2 embeddings: 36%|          | 320/899 [00:08<00:14, 39.74it/s]
Computing ESM2 embeddings: 39%|          | 352/899 [00:08<00:13, 39.88it/s]

```

Computing ESM2 embeddings: 43%| | 384/899 [00:09<00:13, 38.74it/s]

Computing ESM2 embeddings: 46%| | 416/899 [00:10<00:12, 39.10it/s]

Computing ESM2 embeddings: 50%| | 448/899 [00:11<00:11, 39.39it/s]

Computing ESM2 embeddings: 53%| | 480/899 [00:12<00:10, 39.68it/s]

Computing ESM2 embeddings: 57%| | 512/899 [00:12<00:09, 39.87it/s]

Computing ESM2 embeddings: 61%| | 544/899 [00:13<00:08, 39.95it/s]

Computing ESM2 embeddings: 64%| | 576/899 [00:14<00:08, 39.20it/s]

Computing ESM2 embeddings: 68%| | 608/899 [00:15<00:07, 39.40it/s]

Computing ESM2 embeddings: 71%| | 640/899 [00:16<00:06, 39.61it/s]

Computing ESM2 embeddings: 75%| | 672/899 [00:17<00:05, 39.83it/s]

Computing ESM2 embeddings: 78%| | 704/899 [00:17<00:04, 39.93it/s]

Computing ESM2 embeddings: 82%| | 736/899 [00:18<00:04, 40.00it/s]

Computing ESM2 embeddings: 85%| | 768/899 [00:19<00:03, 39.23it/s]

Computing ESM2 embeddings: 89%| | 800/899 [00:20<00:02, 39.35it/s]

Computing ESM2 embeddings: 93%| | 832/899 [00:21<00:01, 39.64it/s]

Computing ESM2 embeddings: 96%| | 864/899 [00:21<00:00, 39.85it/s]

Computing ESM2 embeddings: 100%| | 899/899 [00:22<00:00, 39.46it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.33it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.96it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 40.05it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:19, 40.16it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:03<00:18, 40.18it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 38.62it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 39.03it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 39.43it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:15, 39.64it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:14, 39.84it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:08<00:14, 39.98it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 39.13it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 39.36it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.61it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:10, 39.74it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:12<00:10, 39.85it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.96it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 38.46it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 38.63it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 38.97it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 39.30it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 39.58it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.79it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 38.49it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 38.92it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:21<00:02, 39.25it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:21<00:01, 39.21it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.40it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.37it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 39.06it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:20, 39.86it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.99it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 40.15it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.56it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:18, 38.97it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 39.40it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 39.66it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 39.82it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 40.00it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:14, 38.72it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 39.00it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 39.38it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.65it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.80it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:12<00:09, 39.96it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:09, 38.57it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 38.95it/s]

```

Computing ESM2 embeddings: 68%|      | 608/896 [00:15<00:07, 39.35it/s]
Computing ESM2 embeddings: 71%|      | 640/896 [00:16<00:06, 39.57it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 39.80it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:17<00:04, 39.96it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:18<00:04, 38.68it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 38.74it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 39.06it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.35it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 39.66it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.41it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.71it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 39.37it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:20, 39.86it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:19, 39.99it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:18, 40.04it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:04<00:18, 38.63it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 38.67it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 38.76it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 39.12it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 39.40it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:14, 39.64it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.85it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.94it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 38.67it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 39.05it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 39.01it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.26it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.57it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.75it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.47it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.82it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.24it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.57it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.74it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.89it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.13it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.27it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.53it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.33it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.40it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.48it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.15it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.54it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.82it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.43it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.83it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.27it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.60it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.77it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 39.94it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.74it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 39.03it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.40it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.68it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 39.79it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:08, 39.90it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.22it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.35it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.63it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 39.83it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:17<00:04, 39.91it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:18<00:03, 40.03it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:19<00:03, 39.23it/s] |
| Computing ESM2 embeddings: | 89% | 800/896 [00:20<00:02, 39.40it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.66it/s]

Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 39.80it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.45it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/766 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/766 [00:00<00:19, 37.90it/s]

Computing ESM2 embeddings:  8%|      | 64/766 [00:01<00:17, 39.05it/s]

Computing ESM2 embeddings: 13%|      | 96/766 [00:02<00:16, 39.63it/s]

Computing ESM2 embeddings: 17%|      | 128/766 [00:03<00:16, 39.83it/s]

Computing ESM2 embeddings: 21%|      | 160/766 [00:04<00:15, 40.00it/s]

Computing ESM2 embeddings: 25%|      | 192/766 [00:04<00:14, 38.55it/s]

Computing ESM2 embeddings: 29%|      | 224/766 [00:05<00:13, 38.90it/s]

Computing ESM2 embeddings: 33%|      | 256/766 [00:06<00:12, 39.34it/s]

Computing ESM2 embeddings: 38%|      | 288/766 [00:07<00:12, 39.63it/s]

Computing ESM2 embeddings: 42%|      | 320/766 [00:08<00:11, 39.79it/s]

Computing ESM2 embeddings: 46%|      | 352/766 [00:08<00:10, 39.95it/s]

Computing ESM2 embeddings: 50%|      | 384/766 [00:09<00:09, 39.18it/s]

Computing ESM2 embeddings: 54%|      | 416/766 [00:10<00:08, 39.25it/s]

Computing ESM2 embeddings: 58%|      | 448/766 [00:11<00:08, 39.55it/s]

Computing ESM2 embeddings: 63%|      | 480/766 [00:12<00:07, 39.77it/s]

Computing ESM2 embeddings: 67%|      | 512/766 [00:12<00:06, 39.84it/s]

Computing ESM2 embeddings: 71%|      | 544/766 [00:13<00:05, 39.93it/s]

```



Computing ESM2 embeddings: 75%| | 576/766 [00:14<00:04, 38.83it/s]

Computing ESM2 embeddings: 79%| | 608/766 [00:15<00:04, 39.13it/s]

Computing ESM2 embeddings: 84%| | 640/766 [00:16<00:03, 39.46it/s]

Computing ESM2 embeddings: 88%| | 672/766 [00:17<00:02, 39.63it/s]

Computing ESM2 embeddings: 92%| | 704/766 [00:17<00:01, 39.84it/s]

Computing ESM2 embeddings: 96%| | 736/766 [00:18<00:00, 39.95it/s]

Computing ESM2 embeddings: 100%| | 766/766 [00:19<00:00, 39.05it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/719 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/719 [00:00<00:17, 38.53it/s]

Computing ESM2 embeddings: 9%| | 64/719 [00:01<00:16, 39.25it/s]

Computing ESM2 embeddings: 13%| | 96/719 [00:02<00:15, 39.75it/s]

Computing ESM2 embeddings: 18%| | 128/719 [00:03<00:14, 40.00it/s]

Computing ESM2 embeddings: 22%| | 160/719 [00:04<00:13, 40.06it/s]

Computing ESM2 embeddings: 27%| | 192/719 [00:04<00:13, 39.07it/s]

Computing ESM2 embeddings: 31%| | 224/719 [00:05<00:12, 39.34it/s]

Computing ESM2 embeddings: 36%| | 256/719 [00:06<00:11, 39.57it/s]

Computing ESM2 embeddings: 40%| | 288/719 [00:07<00:10, 39.81it/s]

Computing ESM2 embeddings: 45%| | 320/719 [00:08<00:09, 39.99it/s]

Computing ESM2 embeddings: 49%| | 352/719 [00:08<00:09, 40.04it/s]

Computing ESM2 embeddings: 53%| | 384/719 [00:09<00:08, 39.30it/s]

Computing ESM2 embeddings: 58%| | 416/719 [00:10<00:07, 39.22it/s]

```

Computing ESM2 embeddings: 62%|          | 448/719 [00:11<00:06, 39.38it/s]
Computing ESM2 embeddings: 67%|          | 480/719 [00:12<00:06, 39.65it/s]
Computing ESM2 embeddings: 71%|          | 512/719 [00:12<00:05, 39.79it/s]
Computing ESM2 embeddings: 76%|          | 544/719 [00:13<00:04, 39.96it/s]
Computing ESM2 embeddings: 80%|          | 576/719 [00:14<00:03, 39.17it/s]
Computing ESM2 embeddings: 85%|          | 608/719 [00:15<00:02, 39.28it/s]
Computing ESM2 embeddings: 89%|          | 640/719 [00:16<00:01, 39.57it/s]
Computing ESM2 embeddings: 93%|          | 672/719 [00:16<00:01, 39.78it/s]
Computing ESM2 embeddings: 98%|          | 704/719 [00:17<00:00, 39.86it/s]

Computing ESM2 embeddings: 100%|         | 719/719 [00:18<00:00, 39.60it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/916 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|          | 32/916 [00:00<00:22, 39.39it/s]
Computing ESM2 embeddings:  7%|          | 64/916 [00:01<00:21, 39.96it/s]
Computing ESM2 embeddings: 10%|          | 96/916 [00:02<00:20, 40.19it/s]
Computing ESM2 embeddings: 14%|          | 128/916 [00:03<00:19, 40.13it/s]
Computing ESM2 embeddings: 17%|          | 160/916 [00:03<00:18, 40.14it/s]
Computing ESM2 embeddings: 21%|          | 192/916 [00:04<00:18, 38.63it/s]
Computing ESM2 embeddings: 24%|          | 224/916 [00:05<00:17, 38.99it/s]
Computing ESM2 embeddings: 28%|          | 256/916 [00:06<00:16, 39.36it/s]
Computing ESM2 embeddings: 31%|          | 288/916 [00:07<00:15, 39.65it/s]
Computing ESM2 embeddings: 35%|          | 320/916 [00:08<00:14, 39.78it/s]

```

Computing ESM2 embeddings: 38%| | 352/916 [00:08<00:14, 39.93it/s]

Computing ESM2 embeddings: 42%| | 384/916 [00:09<00:13, 38.63it/s]

Computing ESM2 embeddings: 45%| | 416/916 [00:10<00:12, 38.92it/s]

Computing ESM2 embeddings: 49%| | 448/916 [00:11<00:11, 39.35it/s]

Computing ESM2 embeddings: 52%| | 480/916 [00:12<00:10, 39.66it/s]

Computing ESM2 embeddings: 56%| | 512/916 [00:12<00:10, 39.81it/s]

Computing ESM2 embeddings: 59%| | 544/916 [00:13<00:09, 39.15it/s]

Computing ESM2 embeddings: 63%| | 576/916 [00:14<00:08, 39.36it/s]

Computing ESM2 embeddings: 66%| | 608/916 [00:15<00:07, 39.57it/s]

Computing ESM2 embeddings: 70%| | 640/916 [00:16<00:06, 39.79it/s]

Computing ESM2 embeddings: 73%| | 672/916 [00:16<00:06, 39.97it/s]

Computing ESM2 embeddings: 77%| | 704/916 [00:17<00:05, 40.03it/s]

Computing ESM2 embeddings: 80%| | 736/916 [00:18<00:04, 38.87it/s]

Computing ESM2 embeddings: 84%| | 768/916 [00:19<00:03, 39.10it/s]

Computing ESM2 embeddings: 87%| | 800/916 [00:20<00:02, 39.46it/s]

Computing ESM2 embeddings: 91%| | 832/916 [00:21<00:02, 39.71it/s]

Computing ESM2 embeddings: 94%| | 864/916 [00:21<00:01, 39.82it/s]

Computing ESM2 embeddings: 98%| | 896/916 [00:22<00:00, 39.97it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:23<00:00, 39.44it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/898 [00:00<00:22, 37.92it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/898 [00:01<00:21, 38.55it/s]  |
| Computing ESM2 embeddings: | 11% | 96/898 [00:02<00:20, 38.95it/s]  |
| Computing ESM2 embeddings: | 14% | 128/898 [00:03<00:19, 39.41it/s] |
| Computing ESM2 embeddings: | 18% | 160/898 [00:04<00:18, 39.69it/s] |
| Computing ESM2 embeddings: | 21% | 192/898 [00:04<00:18, 38.26it/s] |
| Computing ESM2 embeddings: | 25% | 224/898 [00:05<00:17, 38.75it/s] |
| Computing ESM2 embeddings: | 29% | 256/898 [00:06<00:16, 39.18it/s] |
| Computing ESM2 embeddings: | 32% | 288/898 [00:07<00:15, 39.43it/s] |
| Computing ESM2 embeddings: | 36% | 320/898 [00:08<00:14, 39.63it/s] |
| Computing ESM2 embeddings: | 39% | 352/898 [00:08<00:13, 39.79it/s] |
| Computing ESM2 embeddings: | 43% | 384/898 [00:09<00:13, 38.53it/s] |
| Computing ESM2 embeddings: | 46% | 416/898 [00:10<00:12, 38.91it/s] |
| Computing ESM2 embeddings: | 50% | 448/898 [00:11<00:11, 39.27it/s] |
| Computing ESM2 embeddings: | 53% | 480/898 [00:12<00:10, 39.49it/s] |
| Computing ESM2 embeddings: | 57% | 512/898 [00:13<00:09, 39.67it/s] |
| Computing ESM2 embeddings: | 61% | 544/898 [00:13<00:08, 39.74it/s] |
| Computing ESM2 embeddings: | 64% | 576/898 [00:14<00:08, 38.47it/s] |
| Computing ESM2 embeddings: | 68% | 608/898 [00:15<00:07, 38.85it/s] |
| Computing ESM2 embeddings: | 71% | 640/898 [00:16<00:06, 39.12it/s] |
| Computing ESM2 embeddings: | 75% | 672/898 [00:17<00:05, 39.41it/s] |
| Computing ESM2 embeddings: | 78% | 704/898 [00:17<00:04, 39.14it/s] |
| Computing ESM2 embeddings: | 82% | 736/898 [00:18<00:04, 37.71it/s] |
| Computing ESM2 embeddings: | 86% | 768/898 [00:19<00:03, 38.30it/s] |
| Computing ESM2 embeddings: | 89% | 800/898 [00:20<00:02, 38.84it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/898 [00:21<00:01, 39.14it/s]

Computing ESM2 embeddings: 96%|      | 864/898 [00:22<00:00, 39.41it/s]

Computing ESM2 embeddings: 100%|     | 898/898 [00:23<00:00, 38.97it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.45it/s]

Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 39.42it/s]

Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:20, 39.75it/s]

Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:19, 40.00it/s]

Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:18, 40.13it/s]

Computing ESM2 embeddings: 21%|      | 192/914 [00:04<00:18, 38.78it/s]

Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 38.83it/s]

Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 39.18it/s]

Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:15, 39.48it/s]

Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:14, 39.74it/s]

Computing ESM2 embeddings: 39%|      | 352/914 [00:08<00:14, 39.92it/s]

Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 38.79it/s]

Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 39.14it/s]

Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 39.38it/s]

Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:10, 39.51it/s]

Computing ESM2 embeddings: 56%|      | 512/914 [00:12<00:10, 39.71it/s]

Computing ESM2 embeddings: 60%|      | 544/914 [00:13<00:09, 39.53it/s]

```

```

Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 38.19it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 38.67it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 39.02it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 39.35it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:17<00:05, 39.60it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 39.70it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.42it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.82it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.12it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 39.43it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:22<00:00, 39.64it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 39.24it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/896 [00:00<00:22, 38.12it/s]
Computing ESM2 embeddings: 7%|           | 64/896 [00:01<00:21, 39.06it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 39.56it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:19, 39.82it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:18, 39.90it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:04<00:18, 38.48it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 38.93it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 39.27it/s]

```

```

Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 39.57it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 39.79it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:08<00:13, 39.87it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:13, 38.50it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.83it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.23it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 39.51it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:09, 39.64it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:08, 39.81it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 38.41it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.72it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.12it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 39.39it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 39.62it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 39.81it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.75it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 39.06it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.40it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 39.53it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.26it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 38.01it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.14it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.66it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:19, 39.86it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.40it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 38.91it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.30it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.62it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:15, 39.84it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:14, 39.91it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 39.21it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.43it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.60it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.80it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:10, 39.89it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:12<00:10, 40.01it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 39.23it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.31it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.59it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.77it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:16<00:06, 39.83it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:17<00:05, 39.97it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:18<00:04, 38.63it/s] |
| Computing ESM2 embeddings: | 84% | 768/914 [00:19<00:03, 38.77it/s] |



```

Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 39.16it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.41it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:21<00:01, 39.59it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.75it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.29it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 37.98it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 39.24it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 39.61it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:19, 39.92it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:18, 40.06it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:18, 39.05it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 39.34it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 39.64it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 39.78it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 39.94it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:08<00:13, 40.01it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:09<00:13, 38.87it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 39.16it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 39.49it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 39.59it/s]

```

Computing ESM2 embeddings: 57%| | 512/896 [00:12<00:09, 39.77it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.88it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.12it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.32it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.50it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:16<00:05, 39.69it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 39.75it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.85it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.63it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.85it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.22it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 39.46it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.43it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/705 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/705 [00:00<00:17, 39.23it/s]

Computing ESM2 embeddings: 9%| | 64/705 [00:01<00:16, 39.77it/s]

Computing ESM2 embeddings: 14%| | 96/705 [00:02<00:15, 39.94it/s]

Computing ESM2 embeddings: 18%| | 128/705 [00:03<00:14, 40.03it/s]

Computing ESM2 embeddings: 23%| | 160/705 [00:04<00:14, 38.29it/s]

Computing ESM2 embeddings: 27%| | 192/705 [00:04<00:13, 38.75it/s]

Computing ESM2 embeddings: 32%| | 224/705 [00:05<00:12, 39.19it/s]

Computing ESM2 embeddings: 36%| | 256/705 [00:06<00:11, 39.43it/s]

Computing ESM2 embeddings: 41%| | 288/705 [00:07<00:10, 39.67it/s]

Computing ESM2 embeddings: 45%| | 320/705 [00:08<00:09, 39.82it/s]

Computing ESM2 embeddings: 50%| | 352/705 [00:08<00:09, 39.01it/s]

Computing ESM2 embeddings: 54%| | 384/705 [00:09<00:08, 39.25it/s]

Computing ESM2 embeddings: 59%| | 416/705 [00:10<00:07, 39.54it/s]

Computing ESM2 embeddings: 64%| | 448/705 [00:11<00:06, 39.65it/s]

Computing ESM2 embeddings: 68%| | 480/705 [00:12<00:05, 39.81it/s]

Computing ESM2 embeddings: 73%| | 512/705 [00:12<00:04, 39.91it/s]

Computing ESM2 embeddings: 77%| | 544/705 [00:13<00:04, 38.66it/s]

Computing ESM2 embeddings: 82%| | 576/705 [00:14<00:03, 38.99it/s]

Computing ESM2 embeddings: 86%| | 608/705 [00:15<00:02, 39.28it/s]

Computing ESM2 embeddings: 91%| | 640/705 [00:16<00:01, 39.53it/s]

Computing ESM2 embeddings: 95%| | 672/705 [00:17<00:00, 39.40it/s]

Computing ESM2 embeddings: 100%| | 705/705 [00:18<00:00, 39.12it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/780 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/780 [00:00<00:19, 38.33it/s]

Computing ESM2 embeddings: 8%| | 64/780 [00:01<00:18, 38.61it/s]

Computing ESM2 embeddings: 12%| | 96/780 [00:02<00:17, 39.10it/s]

Computing ESM2 embeddings: 16%| | 128/780 [00:03<00:16, 39.60it/s]

Computing ESM2 embeddings: 21%| | 160/780 [00:04<00:15, 39.87it/s]

```

Computing ESM2 embeddings: 25%|          | 192/780 [00:04<00:15, 38.34it/s]
Computing ESM2 embeddings: 29%|          | 224/780 [00:05<00:14, 38.73it/s]
Computing ESM2 embeddings: 33%|          | 256/780 [00:06<00:13, 39.12it/s]
Computing ESM2 embeddings: 37%|          | 288/780 [00:07<00:12, 39.48it/s]
Computing ESM2 embeddings: 41%|          | 320/780 [00:08<00:11, 39.74it/s]
Computing ESM2 embeddings: 45%|          | 352/780 [00:08<00:10, 39.87it/s]
Computing ESM2 embeddings: 49%|          | 384/780 [00:09<00:10, 38.67it/s]
Computing ESM2 embeddings: 53%|          | 416/780 [00:10<00:09, 38.96it/s]
Computing ESM2 embeddings: 57%|          | 448/780 [00:11<00:08, 39.36it/s]
Computing ESM2 embeddings: 62%|          | 480/780 [00:12<00:07, 39.64it/s]
Computing ESM2 embeddings: 66%|          | 512/780 [00:13<00:06, 39.78it/s]
Computing ESM2 embeddings: 70%|          | 544/780 [00:13<00:05, 39.93it/s]
Computing ESM2 embeddings: 74%|          | 576/780 [00:14<00:05, 38.61it/s]
Computing ESM2 embeddings: 78%|          | 608/780 [00:15<00:04, 38.97it/s]
Computing ESM2 embeddings: 82%|          | 640/780 [00:16<00:03, 39.31it/s]
Computing ESM2 embeddings: 86%|          | 672/780 [00:17<00:02, 39.60it/s]
Computing ESM2 embeddings: 90%|          | 704/780 [00:17<00:01, 39.80it/s]
Computing ESM2 embeddings: 94%|          | 736/780 [00:18<00:01, 39.90it/s]
Computing ESM2 embeddings: 98%|          | 768/780 [00:19<00:00, 39.34it/s]

Computing ESM2 embeddings: 100%|         | 780/780 [00:19<00:00, 39.26it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/918 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 3%  | 32/918 [00:00<00:22, 38.76it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/918 [00:01<00:21, 39.40it/s]  |
| Computing ESM2 embeddings: | 10% | 96/918 [00:02<00:20, 39.86it/s]  |
| Computing ESM2 embeddings: | 14% | 128/918 [00:03<00:19, 39.93it/s] |
| Computing ESM2 embeddings: | 17% | 160/918 [00:04<00:18, 40.07it/s] |
| Computing ESM2 embeddings: | 21% | 192/918 [00:04<00:18, 39.23it/s] |
| Computing ESM2 embeddings: | 24% | 224/918 [00:05<00:17, 39.40it/s] |
| Computing ESM2 embeddings: | 28% | 256/918 [00:06<00:16, 39.69it/s] |
| Computing ESM2 embeddings: | 31% | 288/918 [00:07<00:15, 39.84it/s] |
| Computing ESM2 embeddings: | 35% | 320/918 [00:08<00:14, 39.96it/s] |
| Computing ESM2 embeddings: | 38% | 352/918 [00:08<00:14, 40.08it/s] |
| Computing ESM2 embeddings: | 42% | 384/918 [00:09<00:13, 39.25it/s] |
| Computing ESM2 embeddings: | 45% | 416/918 [00:10<00:12, 39.44it/s] |
| Computing ESM2 embeddings: | 49% | 448/918 [00:11<00:11, 39.63it/s] |
| Computing ESM2 embeddings: | 52% | 480/918 [00:12<00:10, 39.85it/s] |
| Computing ESM2 embeddings: | 56% | 512/918 [00:12<00:10, 39.95it/s] |
| Computing ESM2 embeddings: | 59% | 544/918 [00:13<00:09, 40.07it/s] |
| Computing ESM2 embeddings: | 63% | 576/918 [00:14<00:08, 38.99it/s] |
| Computing ESM2 embeddings: | 66% | 608/918 [00:15<00:07, 39.24it/s] |
| Computing ESM2 embeddings: | 70% | 640/918 [00:16<00:07, 39.54it/s] |
| Computing ESM2 embeddings: | 73% | 672/918 [00:16<00:06, 39.70it/s] |
| Computing ESM2 embeddings: | 77% | 704/918 [00:17<00:05, 39.85it/s] |
| Computing ESM2 embeddings: | 80% | 736/918 [00:18<00:04, 39.11it/s] |
| Computing ESM2 embeddings: | 84% | 768/918 [00:19<00:03, 39.34it/s] |

```

Computing ESM2 embeddings: 87%|      | 800/918 [00:20<00:02, 39.62it/s]
Computing ESM2 embeddings: 91%|      | 832/918 [00:20<00:02, 39.76it/s]
Computing ESM2 embeddings: 94%|      | 864/918 [00:21<00:01, 39.92it/s]
Computing ESM2 embeddings: 98%|      | 896/918 [00:22<00:00, 40.05it/s]
Computing ESM2 embeddings: 100%|     | 918/918 [00:23<00:00, 39.53it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/898 [00:00<00:22, 39.10it/s]
Computing ESM2 embeddings:  7%|      | 64/898 [00:01<00:20, 39.80it/s]
Computing ESM2 embeddings: 11%|      | 96/898 [00:02<00:20, 40.06it/s]
Computing ESM2 embeddings: 14%|      | 128/898 [00:03<00:19, 40.13it/s]
Computing ESM2 embeddings: 18%|      | 160/898 [00:03<00:18, 40.20it/s]
Computing ESM2 embeddings: 21%|      | 192/898 [00:04<00:18, 39.22it/s]
Computing ESM2 embeddings: 25%|      | 224/898 [00:05<00:17, 39.46it/s]
Computing ESM2 embeddings: 29%|      | 256/898 [00:06<00:16, 39.75it/s]
Computing ESM2 embeddings: 32%|      | 288/898 [00:07<00:15, 39.89it/s]
Computing ESM2 embeddings: 36%|      | 320/898 [00:08<00:14, 40.05it/s]
Computing ESM2 embeddings: 39%|      | 352/898 [00:08<00:13, 40.07it/s]
Computing ESM2 embeddings: 43%|      | 384/898 [00:09<00:13, 39.35it/s]
Computing ESM2 embeddings: 46%|      | 416/898 [00:10<00:12, 39.48it/s]
Computing ESM2 embeddings: 50%|      | 448/898 [00:11<00:11, 39.65it/s]
Computing ESM2 embeddings: 53%|      | 480/898 [00:12<00:10, 39.81it/s]

```

Computing ESM2 embeddings: 57%| | 512/898 [00:12<00:09, 39.87it/s]

Computing ESM2 embeddings: 61%| | 544/898 [00:13<00:08, 40.00it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:14<00:08, 38.87it/s]

Computing ESM2 embeddings: 68%| | 608/898 [00:15<00:07, 39.13it/s]

Computing ESM2 embeddings: 71%| | 640/898 [00:16<00:06, 39.45it/s]

Computing ESM2 embeddings: 75%| | 672/898 [00:16<00:05, 39.56it/s]

Computing ESM2 embeddings: 78%| | 704/898 [00:17<00:04, 39.75it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:18<00:04, 39.87it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:19<00:03, 38.64it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:20<00:02, 39.00it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:21<00:01, 39.25it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:21<00:00, 39.54it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:22<00:00, 39.56it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
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Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.73it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.18it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.72it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 39.99it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.01it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.31it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.57it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.83it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:15, 39.99it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:14, 40.03it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:14, 39.16it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.38it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.65it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.76it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 39.92it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:12<00:10, 39.97it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 38.85it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.14it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.39it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.66it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:16<00:06, 39.77it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.94it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.10it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.30it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.58it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.70it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.86it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.99it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.42it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it



for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.26it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.21it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.70it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.87it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 40.01it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.55it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.98it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.34it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.58it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.81it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 39.90it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.54it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.89it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.21it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.49it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:12<00:09, 39.70it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:08, 39.86it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.12it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.32it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.51it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 39.67it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:17<00:04, 39.80it/s] |

```

Computing ESM2 embeddings: 82%|      | 736/896 [00:18<00:04, 39.85it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 39.18it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 39.27it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.53it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 39.70it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.41it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/913 [00:00<00:22, 38.59it/s]
Computing ESM2 embeddings:  7%|      | 64/913 [00:01<00:21, 39.61it/s]
Computing ESM2 embeddings: 11%|      | 96/913 [00:02<00:20, 39.97it/s]
Computing ESM2 embeddings: 14%|      | 128/913 [00:03<00:19, 40.07it/s]
Computing ESM2 embeddings: 18%|      | 160/913 [00:04<00:18, 40.19it/s]
Computing ESM2 embeddings: 21%|      | 192/913 [00:04<00:18, 38.55it/s]
Computing ESM2 embeddings: 25%|      | 224/913 [00:05<00:17, 39.00it/s]
Computing ESM2 embeddings: 28%|      | 256/913 [00:06<00:16, 39.40it/s]
Computing ESM2 embeddings: 32%|      | 288/913 [00:07<00:15, 39.64it/s]
Computing ESM2 embeddings: 35%|      | 320/913 [00:08<00:14, 39.85it/s]
Computing ESM2 embeddings: 39%|      | 352/913 [00:08<00:14, 39.08it/s]
Computing ESM2 embeddings: 42%|      | 384/913 [00:09<00:13, 39.33it/s]
Computing ESM2 embeddings: 46%|      | 416/913 [00:10<00:12, 39.63it/s]
Computing ESM2 embeddings: 49%|      | 448/913 [00:11<00:11, 39.77it/s]

```

Computing ESM2 embeddings: 53%| | 480/913 [00:12<00:10, 39.94it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:12<00:10, 39.99it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:13<00:09, 38.84it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:14<00:08, 39.13it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:15<00:07, 39.45it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:16<00:06, 39.69it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:16<00:06, 39.85it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:17<00:05, 39.95it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:18<00:04, 39.22it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:19<00:03, 39.39it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:20<00:02, 39.59it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:21<00:02, 39.75it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:21<00:01, 39.84it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:22<00:00, 39.94it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:23<00:00, 39.46it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:21, 39.42it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.34it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.82it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.95it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 40.10it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:18, 38.63it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 39.01it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 39.42it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 39.63it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 39.86it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:08<00:13, 39.95it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:13, 38.72it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:12, 39.08it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 39.36it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 39.64it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:12<00:09, 39.78it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:13<00:08, 39.13it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 39.36it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 39.58it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 39.79it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 39.55it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:17<00:05, 38.14it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:18<00:04, 38.50it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 38.68it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 39.08it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:21<00:01, 39.37it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:22<00:00, 38.28it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:22<00:00, 39.20it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

`'esm.pooler.dense.weight']`

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/642 [00:00<?, ?it/s]
Computing ESM2 embeddings: 5%|          | 32/642 [00:00<00:15, 38.39it/s]
Computing ESM2 embeddings: 10%|         | 64/642 [00:01<00:14, 39.20it/s]
Computing ESM2 embeddings: 15%|         | 96/642 [00:02<00:13, 39.59it/s]
Computing ESM2 embeddings: 20%|         | 128/642 [00:03<00:12, 39.87it/s]
Computing ESM2 embeddings: 25%|         | 160/642 [00:04<00:12, 38.37it/s]
Computing ESM2 embeddings: 30%|         | 192/642 [00:04<00:11, 38.87it/s]
Computing ESM2 embeddings: 35%|         | 224/642 [00:05<00:10, 39.24it/s]
Computing ESM2 embeddings: 40%|         | 256/642 [00:06<00:09, 39.44it/s]
Computing ESM2 embeddings: 45%|         | 288/642 [00:07<00:08, 39.69it/s]
Computing ESM2 embeddings: 50%|         | 320/642 [00:08<00:08, 38.35it/s]
Computing ESM2 embeddings: 55%|         | 352/642 [00:09<00:07, 38.80it/s]
Computing ESM2 embeddings: 60%|         | 384/642 [00:09<00:06, 39.24it/s]
Computing ESM2 embeddings: 65%|         | 416/642 [00:10<00:05, 39.49it/s]
Computing ESM2 embeddings: 70%|         | 448/642 [00:11<00:04, 39.72it/s]
Computing ESM2 embeddings: 75%|         | 480/642 [00:12<00:04, 38.40it/s]
Computing ESM2 embeddings: 80%|         | 512/642 [00:13<00:03, 38.83it/s]
Computing ESM2 embeddings: 85%|         | 544/642 [00:13<00:02, 39.18it/s]
Computing ESM2 embeddings: 90%|         | 576/642 [00:14<00:01, 39.50it/s]
Computing ESM2 embeddings: 95%|         | 608/642 [00:15<00:00, 39.64it/s]
```

```
Computing ESM2 embeddings: 100%|        | 642/642 [00:16<00:00, 39.02it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
```

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/845 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/845 [00:00<00:20, 39.03it/s]  |
| Computing ESM2 embeddings: | 8%  | 64/845 [00:01<00:19, 39.52it/s]  |
| Computing ESM2 embeddings: | 11% | 96/845 [00:02<00:18, 39.83it/s]  |
| Computing ESM2 embeddings: | 15% | 128/845 [00:03<00:17, 40.04it/s] |
| Computing ESM2 embeddings: | 19% | 160/845 [00:04<00:17, 39.91it/s] |
| Computing ESM2 embeddings: | 23% | 192/845 [00:04<00:16, 38.83it/s] |
| Computing ESM2 embeddings: | 27% | 224/845 [00:05<00:15, 39.10it/s] |
| Computing ESM2 embeddings: | 30% | 256/845 [00:06<00:14, 39.45it/s] |
| Computing ESM2 embeddings: | 34% | 288/845 [00:07<00:14, 39.65it/s] |
| Computing ESM2 embeddings: | 38% | 320/845 [00:08<00:13, 39.82it/s] |
| Computing ESM2 embeddings: | 42% | 352/845 [00:08<00:12, 38.66it/s] |
| Computing ESM2 embeddings: | 45% | 384/845 [00:09<00:11, 39.02it/s] |
| Computing ESM2 embeddings: | 49% | 416/845 [00:10<00:10, 39.32it/s] |
| Computing ESM2 embeddings: | 53% | 448/845 [00:11<00:10, 39.59it/s] |
| Computing ESM2 embeddings: | 57% | 480/845 [00:12<00:09, 39.72it/s] |
| Computing ESM2 embeddings: | 61% | 512/845 [00:13<00:08, 38.37it/s] |
| Computing ESM2 embeddings: | 64% | 544/845 [00:13<00:07, 38.68it/s] |
| Computing ESM2 embeddings: | 68% | 576/845 [00:14<00:06, 39.12it/s] |
| Computing ESM2 embeddings: | 72% | 608/845 [00:15<00:06, 39.39it/s] |
| Computing ESM2 embeddings: | 76% | 640/845 [00:16<00:05, 39.63it/s] |
| Computing ESM2 embeddings: | 80% | 672/845 [00:17<00:04, 38.30it/s] |

```

Computing ESM2 embeddings: 83%|      | 704/845 [00:17<00:03, 38.72it/s]
Computing ESM2 embeddings: 87%|      | 736/845 [00:18<00:02, 39.08it/s]
Computing ESM2 embeddings: 91%|      | 768/845 [00:19<00:01, 39.39it/s]
Computing ESM2 embeddings: 95%|      | 800/845 [00:20<00:01, 39.54it/s]
Computing ESM2 embeddings: 98%|      | 832/845 [00:21<00:00, 38.87it/s]

Computing ESM2 embeddings: 100%|     | 845/845 [00:21<00:00, 39.13it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/902 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/902 [00:00<00:22, 38.96it/s]
Computing ESM2 embeddings:  7%|      | 64/902 [00:01<00:21, 39.40it/s]
Computing ESM2 embeddings: 11%|      | 96/902 [00:02<00:20, 39.85it/s]
Computing ESM2 embeddings: 14%|      | 128/902 [00:03<00:19, 39.97it/s]
Computing ESM2 embeddings: 18%|      | 160/902 [00:04<00:18, 40.13it/s]
Computing ESM2 embeddings: 21%|      | 192/902 [00:04<00:18, 38.71it/s]
Computing ESM2 embeddings: 25%|      | 224/902 [00:05<00:17, 39.11it/s]
Computing ESM2 embeddings: 28%|      | 256/902 [00:06<00:16, 39.44it/s]
Computing ESM2 embeddings: 32%|      | 288/902 [00:07<00:15, 39.74it/s]
Computing ESM2 embeddings: 35%|      | 320/902 [00:08<00:14, 39.92it/s]
Computing ESM2 embeddings: 39%|      | 352/902 [00:08<00:14, 38.47it/s]
Computing ESM2 embeddings: 43%|      | 384/902 [00:09<00:13, 38.89it/s]
Computing ESM2 embeddings: 46%|      | 416/902 [00:10<00:12, 39.33it/s]
Computing ESM2 embeddings: 50%|      | 448/902 [00:11<00:11, 39.56it/s]

```

Computing ESM2 embeddings: 53%| | 480/902 [00:12<00:10, 39.80it/s]

Computing ESM2 embeddings: 57%| | 512/902 [00:13<00:10, 38.61it/s]

Computing ESM2 embeddings: 60%| | 544/902 [00:13<00:09, 38.91it/s]

Computing ESM2 embeddings: 64%| | 576/902 [00:14<00:08, 39.32it/s]

Computing ESM2 embeddings: 67%| | 608/902 [00:15<00:07, 39.59it/s]

Computing ESM2 embeddings: 71%| | 640/902 [00:16<00:06, 39.70it/s]

Computing ESM2 embeddings: 75%| | 672/902 [00:17<00:05, 39.87it/s]

Computing ESM2 embeddings: 78%| | 704/902 [00:17<00:05, 39.14it/s]

Computing ESM2 embeddings: 82%| | 736/902 [00:18<00:04, 39.27it/s]

Computing ESM2 embeddings: 85%| | 768/902 [00:19<00:03, 39.54it/s]

Computing ESM2 embeddings: 89%| | 800/902 [00:20<00:02, 39.75it/s]

Computing ESM2 embeddings: 92%| | 832/902 [00:21<00:01, 39.79it/s]

Computing ESM2 embeddings: 96%| | 864/902 [00:21<00:00, 39.81it/s]

Computing ESM2 embeddings: 99%| | 896/902 [00:22<00:00, 39.13it/s]

Computing ESM2 embeddings: 100%| | 902/902 [00:22<00:00, 39.27it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

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Computing ESM2 embeddings: 0%| | 0/912 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/912 [00:00<00:23, 37.91it/s]

Computing ESM2 embeddings: 7%| | 64/912 [00:01<00:21, 38.90it/s]

Computing ESM2 embeddings: 11%| | 96/912 [00:02<00:20, 39.39it/s]

Computing ESM2 embeddings: 14%| | 128/912 [00:03<00:19, 39.60it/s]

Computing ESM2 embeddings: 18%| | 160/912 [00:04<00:19, 38.87it/s]



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/912 [00:04<00:18, 39.18it/s] |
| Computing ESM2 embeddings: 25%  | 224/912 [00:05<00:17, 39.41it/s] |
| Computing ESM2 embeddings: 28%  | 256/912 [00:06<00:16, 39.62it/s] |
| Computing ESM2 embeddings: 32%  | 288/912 [00:07<00:15, 39.78it/s] |
| Computing ESM2 embeddings: 35%  | 320/912 [00:08<00:14, 39.83it/s] |
| Computing ESM2 embeddings: 39%  | 352/912 [00:08<00:14, 39.07it/s] |
| Computing ESM2 embeddings: 42%  | 384/912 [00:09<00:13, 39.29it/s] |
| Computing ESM2 embeddings: 46%  | 416/912 [00:10<00:12, 39.45it/s] |
| Computing ESM2 embeddings: 49%  | 448/912 [00:11<00:11, 39.64it/s] |
| Computing ESM2 embeddings: 53%  | 480/912 [00:12<00:10, 39.79it/s] |
| Computing ESM2 embeddings: 56%  | 512/912 [00:12<00:10, 39.83it/s] |
| Computing ESM2 embeddings: 60%  | 544/912 [00:13<00:09, 39.08it/s] |
| Computing ESM2 embeddings: 63%  | 576/912 [00:14<00:08, 39.29it/s] |
| Computing ESM2 embeddings: 67%  | 608/912 [00:15<00:07, 39.43it/s] |
| Computing ESM2 embeddings: 70%  | 640/912 [00:16<00:06, 39.64it/s] |
| Computing ESM2 embeddings: 74%  | 672/912 [00:17<00:06, 39.78it/s] |
| Computing ESM2 embeddings: 77%  | 704/912 [00:17<00:05, 39.81it/s] |
| Computing ESM2 embeddings: 81%  | 736/912 [00:18<00:04, 38.59it/s] |
| Computing ESM2 embeddings: 84%  | 768/912 [00:19<00:03, 38.94it/s] |
| Computing ESM2 embeddings: 88%  | 800/912 [00:20<00:02, 38.88it/s] |
| Computing ESM2 embeddings: 91%  | 832/912 [00:21<00:02, 39.11it/s] |
| Computing ESM2 embeddings: 95%  | 864/912 [00:21<00:01, 39.40it/s] |
| Computing ESM2 embeddings: 98%  | 896/912 [00:22<00:00, 39.54it/s] |
| Computing ESM2 embeddings: 100% | 912/912 [00:23<00:00, 39.13it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.05it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.90it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.46it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.80it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 38.19it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.74it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 39.20it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.47it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.71it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.90it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 38.75it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 39.07it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 39.00it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.25it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.54it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 39.68it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:09, 38.51it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.91it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.21it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.52it/s] |

```

Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 39.74it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:17<00:04, 39.78it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:18<00:04, 39.02it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 39.26it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 39.47it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.69it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 39.84it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.30it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/914 [00:00<00:23, 38.32it/s]
Computing ESM2 embeddings: 7%|        | 64/914 [00:01<00:21, 38.96it/s]
Computing ESM2 embeddings: 11%|       | 96/914 [00:02<00:20, 39.44it/s]
Computing ESM2 embeddings: 14%|       | 128/914 [00:03<00:19, 39.81it/s]
Computing ESM2 embeddings: 18%|       | 160/914 [00:04<00:19, 38.41it/s]
Computing ESM2 embeddings: 21%|       | 192/914 [00:04<00:18, 38.93it/s]
Computing ESM2 embeddings: 25%|       | 224/914 [00:05<00:17, 39.32it/s]
Computing ESM2 embeddings: 28%|       | 256/914 [00:06<00:16, 39.65it/s]
Computing ESM2 embeddings: 32%|       | 288/914 [00:07<00:15, 39.81it/s]
Computing ESM2 embeddings: 35%|       | 320/914 [00:08<00:15, 38.46it/s]
Computing ESM2 embeddings: 39%|       | 352/914 [00:09<00:14, 38.79it/s]
Computing ESM2 embeddings: 42%|       | 384/914 [00:09<00:13, 38.97it/s]

```

```

Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 39.25it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.49it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:10, 39.73it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.64it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:13<00:09, 38.64it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 39.00it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.38it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.27it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.76it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.96it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 39.12it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 39.42it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.43it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 38.77it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 39.18it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:22<00:00, 39.49it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 39.08it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.49it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.32it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 39.67it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:19, 39.97it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 38.50it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:04<00:18, 38.88it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 39.36it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 39.69it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 39.79it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 39.12it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:08<00:13, 39.37it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:12, 39.60it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 39.82it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 39.88it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 40.02it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 38.86it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:13<00:09, 39.08it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 39.45it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 39.72it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 39.82it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 39.90it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:17<00:05, 38.29it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:18<00:04, 38.64it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:19<00:03, 39.11it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 39.12it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 39.23it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:21<00:00, 39.55it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.13it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.72it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.66it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.82it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 39.30it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.12it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 38.62it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.13it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.42it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:15, 39.66it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:14, 39.83it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.56it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.79it/s]  
 Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.19it/s]  
 Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.42it/s]  
 Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 39.64it/s]  
 Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 39.80it/s]  
 Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 38.38it/s]  
 Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.51it/s]  
 Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.88it/s]

```

Computing ESM2 embeddings: 70%|      | 640/914 [00:16<00:06, 39.18it/s]
Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 39.47it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:17<00:05, 39.66it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:18<00:04, 38.37it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:19<00:03, 38.77it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 39.17it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.38it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.59it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.74it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.97it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/579 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/579 [00:01<00:17, 31.03it/s]
Computing ESM2 embeddings: 11%|      | 64/579 [00:01<00:14, 34.92it/s]
Computing ESM2 embeddings: 17%|      | 96/579 [00:02<00:13, 37.06it/s]
Computing ESM2 embeddings: 22%|      | 128/579 [00:03<00:11, 38.25it/s]
Computing ESM2 embeddings: 28%|      | 160/579 [00:04<00:10, 38.83it/s]
Computing ESM2 embeddings: 33%|      | 192/579 [00:05<00:09, 39.32it/s]
Computing ESM2 embeddings: 39%|      | 224/579 [00:05<00:09, 38.62it/s]
Computing ESM2 embeddings: 44%|      | 256/579 [00:06<00:08, 38.70it/s]
Computing ESM2 embeddings: 50%|      | 288/579 [00:07<00:07, 38.97it/s]
Computing ESM2 embeddings: 55%|      | 320/579 [00:08<00:06, 39.33it/s]

```

Computing ESM2 embeddings: 61%| | 352/579 [00:09<00:05, 39.60it/s]  
 Computing ESM2 embeddings: 66%| | 384/579 [00:09<00:04, 39.70it/s]  
 Computing ESM2 embeddings: 72%| | 416/579 [00:10<00:04, 38.96it/s]  
 Computing ESM2 embeddings: 77%| | 448/579 [00:11<00:03, 39.23it/s]  
 Computing ESM2 embeddings: 83%| | 480/579 [00:12<00:02, 39.43it/s]  
 Computing ESM2 embeddings: 88%| | 512/579 [00:13<00:01, 39.29it/s]  
 Computing ESM2 embeddings: 94%| | 544/579 [00:14<00:00, 39.43it/s]

Computing ESM2 embeddings: 100%| | 579/579 [00:15<00:00, 38.58it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/907 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/907 [00:00<00:23, 37.89it/s]  
 Computing ESM2 embeddings: 7%| | 64/907 [00:01<00:21, 38.94it/s]  
 Computing ESM2 embeddings: 11%| | 96/907 [00:02<00:20, 39.27it/s]  
 Computing ESM2 embeddings: 14%| | 128/907 [00:03<00:19, 39.71it/s]  
 Computing ESM2 embeddings: 18%| | 160/907 [00:04<00:19, 38.40it/s]  
 Computing ESM2 embeddings: 21%| | 192/907 [00:04<00:18, 38.77it/s]  
 Computing ESM2 embeddings: 25%| | 224/907 [00:05<00:17, 39.29it/s]  
 Computing ESM2 embeddings: 28%| | 256/907 [00:06<00:16, 39.63it/s]  
 Computing ESM2 embeddings: 32%| | 288/907 [00:07<00:15, 39.74it/s]  
 Computing ESM2 embeddings: 35%| | 320/907 [00:08<00:14, 39.94it/s]  
 Computing ESM2 embeddings: 39%| | 352/907 [00:08<00:14, 39.27it/s]  
 Computing ESM2 embeddings: 42%| | 384/907 [00:09<00:13, 39.37it/s]



```

Computing ESM2 embeddings: 46%|          | 416/907 [00:10<00:12, 39.66it/s]
Computing ESM2 embeddings: 49%|          | 448/907 [00:11<00:11, 39.90it/s]
Computing ESM2 embeddings: 53%|          | 480/907 [00:12<00:10, 39.91it/s]
Computing ESM2 embeddings: 56%|          | 512/907 [00:12<00:09, 40.05it/s]
Computing ESM2 embeddings: 60%|          | 544/907 [00:13<00:09, 38.77it/s]
Computing ESM2 embeddings: 64%|          | 576/907 [00:14<00:08, 39.01it/s]
Computing ESM2 embeddings: 67%|          | 608/907 [00:15<00:07, 39.39it/s]
Computing ESM2 embeddings: 71%|          | 640/907 [00:16<00:06, 39.68it/s]
Computing ESM2 embeddings: 74%|          | 672/907 [00:17<00:05, 39.77it/s]
Computing ESM2 embeddings: 78%|          | 704/907 [00:17<00:05, 39.93it/s]
Computing ESM2 embeddings: 81%|          | 736/907 [00:18<00:04, 39.04it/s]
Computing ESM2 embeddings: 85%|          | 768/907 [00:19<00:03, 39.22it/s]
Computing ESM2 embeddings: 88%|          | 800/907 [00:20<00:02, 39.51it/s]
Computing ESM2 embeddings: 92%|          | 832/907 [00:21<00:01, 39.76it/s]
Computing ESM2 embeddings: 95%|          | 864/907 [00:21<00:01, 39.84it/s]
Computing ESM2 embeddings: 99%|          | 896/907 [00:22<00:00, 39.98it/s]

Computing ESM2 embeddings: 100%|         | 907/907 [00:23<00:00, 39.40it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/900 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/900 [00:00<00:22, 38.33it/s]
Computing ESM2 embeddings:  7%|          | 64/900 [00:01<00:21, 39.16it/s]
Computing ESM2 embeddings: 11%|          | 96/900 [00:02<00:20, 39.71it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/900 [00:03<00:19, 39.97it/s] |
| Computing ESM2 embeddings: 18% | 160/900 [00:04<00:19, 38.28it/s] |
| Computing ESM2 embeddings: 21% | 192/900 [00:04<00:18, 38.83it/s] |
| Computing ESM2 embeddings: 25% | 224/900 [00:05<00:17, 39.30it/s] |
| Computing ESM2 embeddings: 28% | 256/900 [00:06<00:16, 39.54it/s] |
| Computing ESM2 embeddings: 32% | 288/900 [00:07<00:15, 39.80it/s] |
| Computing ESM2 embeddings: 36% | 320/900 [00:08<00:14, 39.97it/s] |
| Computing ESM2 embeddings: 39% | 352/900 [00:08<00:14, 39.13it/s] |
| Computing ESM2 embeddings: 43% | 384/900 [00:09<00:13, 39.33it/s] |
| Computing ESM2 embeddings: 46% | 416/900 [00:10<00:12, 39.60it/s] |
| Computing ESM2 embeddings: 50% | 448/900 [00:11<00:11, 39.73it/s] |
| Computing ESM2 embeddings: 53% | 480/900 [00:12<00:10, 39.91it/s] |
| Computing ESM2 embeddings: 57% | 512/900 [00:12<00:09, 40.04it/s] |
| Computing ESM2 embeddings: 60% | 544/900 [00:13<00:09, 38.89it/s] |
| Computing ESM2 embeddings: 64% | 576/900 [00:14<00:08, 39.17it/s] |
| Computing ESM2 embeddings: 68% | 608/900 [00:15<00:07, 39.49it/s] |
| Computing ESM2 embeddings: 71% | 640/900 [00:16<00:06, 39.66it/s] |
| Computing ESM2 embeddings: 75% | 672/900 [00:17<00:05, 39.84it/s] |
| Computing ESM2 embeddings: 78% | 704/900 [00:17<00:04, 39.95it/s] |
| Computing ESM2 embeddings: 82% | 736/900 [00:18<00:04, 38.81it/s] |
| Computing ESM2 embeddings: 85% | 768/900 [00:19<00:03, 39.12it/s] |
| Computing ESM2 embeddings: 89% | 800/900 [00:20<00:02, 39.11it/s] |
| Computing ESM2 embeddings: 92% | 832/900 [00:21<00:01, 39.24it/s] |
| Computing ESM2 embeddings: 96% | 864/900 [00:21<00:00, 39.53it/s] |

Computing ESM2 embeddings: 100%| | 896/900 [00:22<00:00, 39.74it/s]

Computing ESM2 embeddings: 100%| | 900/900 [00:22<00:00, 39.35it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.31it/s]  
Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.14it/s]  
Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.69it/s]  
Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 39.90it/s]  
Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.43it/s]  
Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 38.89it/s]  
Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.33it/s]  
Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.65it/s]  
Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:15, 39.79it/s]  
Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:14, 39.96it/s]  
Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:14, 39.12it/s]  
Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.34it/s]  
Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.57it/s]  
Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.72it/s]  
Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 39.88it/s]  
Computing ESM2 embeddings: 56%| | 512/914 [00:12<00:10, 40.02it/s]  
Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 38.70it/s]  
Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.05it/s]

```

Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.42it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:06, 39.61it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 39.83it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:17<00:05, 39.97it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 39.12it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 39.35it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 39.62it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.76it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:21<00:01, 39.91it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:22<00:00, 40.02it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 39.42it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.


Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 39.17it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:20, 39.86it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 40.00it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:19, 40.11it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 38.60it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:04<00:18, 38.97it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 39.34it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 39.65it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 39.80it/s]

```

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 39.95it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:14, 38.73it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 39.08it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 39.44it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.59it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.72it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:12<00:09, 39.88it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:09, 39.00it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.24it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.53it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.68it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 39.82it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 39.12it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 38.85it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 39.14it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 39.39it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.64it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:21<00:00, 39.76it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.29it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.64it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:21, 39.25it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:20, 39.72it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:19, 39.89it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.38it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 38.80it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 39.28it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 39.60it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:15, 39.75it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:14, 39.94it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:08<00:14, 39.19it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 39.31it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 39.62it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.76it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 39.39it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:12<00:10, 39.52it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 38.35it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 38.71it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.18it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:06, 39.50it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 39.67it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 39.85it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.15it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 39.26it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 39.56it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.77it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:21<00:01, 39.85it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.99it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.31it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/896 [00:00<00:21, 39.35it/s]
Computing ESM2 embeddings: 7%|        | 64/896 [00:01<00:20, 39.81it/s]
Computing ESM2 embeddings: 11%|       | 96/896 [00:02<00:19, 40.04it/s]
Computing ESM2 embeddings: 14%|       | 128/896 [00:03<00:19, 40.07it/s]
Computing ESM2 embeddings: 18%|       | 160/896 [00:04<00:19, 38.66it/s]
Computing ESM2 embeddings: 21%|       | 192/896 [00:04<00:18, 38.97it/s]
Computing ESM2 embeddings: 25%|       | 224/896 [00:05<00:17, 39.40it/s]
Computing ESM2 embeddings: 29%|       | 256/896 [00:06<00:16, 39.62it/s]
Computing ESM2 embeddings: 32%|       | 288/896 [00:07<00:15, 39.82it/s]
Computing ESM2 embeddings: 36%|       | 320/896 [00:08<00:14, 39.11it/s]
Computing ESM2 embeddings: 39%|       | 352/896 [00:08<00:13, 39.22it/s]
Computing ESM2 embeddings: 43%|       | 384/896 [00:09<00:12, 39.52it/s]
Computing ESM2 embeddings: 46%|       | 416/896 [00:10<00:12, 39.76it/s]
Computing ESM2 embeddings: 50%|       | 448/896 [00:11<00:11, 39.83it/s]
Computing ESM2 embeddings: 54%|       | 480/896 [00:12<00:10, 38.70it/s]
Computing ESM2 embeddings: 57%|       | 512/896 [00:13<00:09, 39.12it/s]

```

```

Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:08, 39.46it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 39.58it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 39.77it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.91it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.67it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:17<00:04, 39.00it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 39.36it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 39.54it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 39.76it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.89it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:21<00:00, 38.59it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:22<00:00, 39.20it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 38.08it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:22, 38.48it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:20, 39.15it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:19, 39.57it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 38.05it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:04<00:18, 38.63it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:17, 39.05it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:16, 39.41it/s]

```



Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:15, 39.60it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:14, 39.78it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.86it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.05it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.38it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.56it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 39.66it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 39.77it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.04it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.16it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.46it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.68it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.83it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.83it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 38.60it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.92it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.22it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.50it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 39.70it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.79it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.06it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```

Computing ESM2 embeddings: 0%|          | 0/570 [00:00<?, ?it/s]
Computing ESM2 embeddings: 6%|          | 32/570 [00:00<00:14, 36.75it/s]
Computing ESM2 embeddings: 11%|         | 64/570 [00:01<00:13, 38.22it/s]
Computing ESM2 embeddings: 17%|         | 96/570 [00:02<00:12, 39.09it/s]
Computing ESM2 embeddings: 22%|         | 128/570 [00:03<00:11, 39.55it/s]
Computing ESM2 embeddings: 28%|         | 160/570 [00:04<00:10, 38.17it/s]
Computing ESM2 embeddings: 34%|         | 192/570 [00:04<00:09, 38.74it/s]
Computing ESM2 embeddings: 39%|         | 224/570 [00:05<00:08, 39.20it/s]
Computing ESM2 embeddings: 45%|         | 256/570 [00:06<00:07, 39.46it/s]
Computing ESM2 embeddings: 51%|         | 288/570 [00:07<00:07, 39.69it/s]
Computing ESM2 embeddings: 56%|         | 320/570 [00:08<00:06, 39.86it/s]
Computing ESM2 embeddings: 62%|         | 352/570 [00:09<00:05, 38.91it/s]
Computing ESM2 embeddings: 67%|         | 384/570 [00:09<00:04, 39.17it/s]
Computing ESM2 embeddings: 73%|         | 416/570 [00:10<00:03, 39.46it/s]
Computing ESM2 embeddings: 79%|         | 448/570 [00:11<00:03, 39.60it/s]
Computing ESM2 embeddings: 84%|         | 480/570 [00:12<00:02, 39.77it/s]
Computing ESM2 embeddings: 90%|         | 512/570 [00:13<00:01, 39.91it/s]
Computing ESM2 embeddings: 95%|         | 544/570 [00:13<00:00, 39.10it/s]

Computing ESM2 embeddings: 100%|        | 570/570 [00:14<00:00, 39.15it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings: 0%|          | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/914 [00:00<00:23, 38.33it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:21, 39.23it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:20, 39.14it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:19, 39.48it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.19it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 38.78it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 39.23it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 39.61it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:15, 39.78it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:14, 39.65it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:08<00:14, 39.80it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 39.89it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 40.02it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 38.72it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 38.66it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 39.02it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.31it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.58it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.65it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 38.60it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 38.87it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 39.26it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.49it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 39.70it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 39.80it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.09it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:21<00:01, 39.31it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.50it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.31it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/898 [00:00<00:22, 38.78it/s]
Computing ESM2 embeddings:  7%|      | 64/898 [00:01<00:21, 39.54it/s]
Computing ESM2 embeddings: 11%|      | 96/898 [00:02<00:20, 39.79it/s]
Computing ESM2 embeddings: 14%|      | 128/898 [00:03<00:19, 40.03it/s]
Computing ESM2 embeddings: 18%|      | 160/898 [00:04<00:19, 38.58it/s]
Computing ESM2 embeddings: 21%|      | 192/898 [00:04<00:18, 39.00it/s]
Computing ESM2 embeddings: 25%|      | 224/898 [00:05<00:17, 39.35it/s]
Computing ESM2 embeddings: 29%|      | 256/898 [00:06<00:16, 39.66it/s]
Computing ESM2 embeddings: 32%|      | 288/898 [00:07<00:15, 39.80it/s]
Computing ESM2 embeddings: 36%|      | 320/898 [00:08<00:14, 39.98it/s]
Computing ESM2 embeddings: 39%|      | 352/898 [00:08<00:13, 39.27it/s]
Computing ESM2 embeddings: 43%|      | 384/898 [00:09<00:13, 39.37it/s]
Computing ESM2 embeddings: 46%|      | 416/898 [00:10<00:12, 39.67it/s]
Computing ESM2 embeddings: 50%|      | 448/898 [00:11<00:11, 39.87it/s]
Computing ESM2 embeddings: 53%|      | 480/898 [00:12<00:10, 39.58it/s]
Computing ESM2 embeddings: 57%|      | 512/898 [00:12<00:09, 39.66it/s]

```

Computing ESM2 embeddings: 61%| | 544/898 [00:13<00:09, 38.48it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:14<00:08, 38.90it/s]

Computing ESM2 embeddings: 68%| | 608/898 [00:15<00:07, 39.25it/s]

Computing ESM2 embeddings: 71%| | 640/898 [00:16<00:06, 39.56it/s]

Computing ESM2 embeddings: 75%| | 672/898 [00:17<00:05, 39.79it/s]

Computing ESM2 embeddings: 78%| | 704/898 [00:17<00:04, 39.90it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:18<00:04, 39.20it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:19<00:03, 39.40it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:20<00:02, 39.53it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:21<00:01, 39.77it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:21<00:00, 39.94it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:22<00:00, 39.47it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.40it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.13it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.68it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 39.97it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.23it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 38.79it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.26it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.53it/s]

```

Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:15, 39.79it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:14, 39.92it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 38.51it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:09<00:13, 38.93it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 39.28it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.61it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:10, 39.82it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:12<00:10, 39.90it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:13<00:09, 38.69it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 39.04it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.34it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:06, 39.64it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 39.85it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:17<00:05, 38.42it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 38.84it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 39.23it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 39.37it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.61it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 39.03it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:22<00:00, 39.19it/s]

```

```

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 39.20it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:21, 39.34it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:20, 39.76it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:19, 40.05it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 40.10it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 38.63it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.99it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 39.41it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.66it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.80it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.97it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:14, 38.64it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 39.00it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 39.38it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.56it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.77it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:12<00:09, 39.96it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:08, 39.16it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.37it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.67it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.78it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:16<00:05, 39.94it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:17<00:04, 40.06it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:18<00:04, 39.22it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 39.18it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 39.38it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.54it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:21<00:00, 39.76it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.50it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.24it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 36.01it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.60it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:20, 38.45it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.52it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:18, 38.16it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 38.80it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 39.23it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:15, 39.45it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:14, 39.68it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 38.99it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 39.11it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 39.41it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 39.65it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:10, 39.73it/s]

```



Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 39.87it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.11it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.20it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.40it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.61it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.67it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.80it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 38.48it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.75it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.15it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.44it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 39.57it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.75it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.94it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.21it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.14it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.97it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.38it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.05it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:18, 38.55it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 39.08it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 38.46it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 38.83it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 39.22it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 38.24it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:13, 38.70it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 39.16it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 39.41it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 39.66it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 39.83it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 38.50it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 38.85it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 39.23it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 39.38it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 39.63it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:04, 39.79it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:18<00:04, 38.62it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:19<00:03, 38.67it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 38.98it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 39.26it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:22<00:00, 39.51it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.02it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.56it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.97it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 40.03it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:19, 40.17it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.60it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 38.93it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.37it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.68it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:15, 39.80it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:14, 39.96it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 38.66it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 38.90it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.29it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.55it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:10, 39.76it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:12<00:10, 39.92it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 38.79it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.10it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.43it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.61it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 39.79it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:17<00:05, 39.71it/s] |

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.88it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.93it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 40.04it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 40.09it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:21<00:01, 38.91it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.20it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.47it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:13, 39.09it/s]

Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:12, 39.73it/s]

Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:11, 40.04it/s]

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:11, 40.18it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:10, 38.48it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:04<00:09, 38.99it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:05<00:08, 39.42it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:07, 39.63it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 39.86it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 40.03it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:08<00:05, 38.70it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:09<00:04, 39.07it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:10<00:03, 39.41it/s]

```

Computing ESM2 embeddings: 78%|          | 448/571 [00:11<00:03, 39.68it/s]
Computing ESM2 embeddings: 84%|          | 480/571 [00:12<00:02, 38.91it/s]
Computing ESM2 embeddings: 90%|          | 512/571 [00:13<00:01, 39.18it/s]
Computing ESM2 embeddings: 95%|          | 544/571 [00:13<00:00, 39.49it/s]
Computing ESM2 embeddings: 100%|         | 571/571 [00:14<00:00, 39.38it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/897 [00:00<00:22, 38.31it/s]
Computing ESM2 embeddings: 7%|           | 64/897 [00:01<00:21, 39.28it/s]
Computing ESM2 embeddings: 11%|          | 96/897 [00:02<00:20, 39.68it/s]
Computing ESM2 embeddings: 14%|          | 128/897 [00:03<00:19, 39.99it/s]
Computing ESM2 embeddings: 18%|          | 160/897 [00:04<00:19, 38.50it/s]
Computing ESM2 embeddings: 21%|          | 192/897 [00:04<00:18, 38.91it/s]
Computing ESM2 embeddings: 25%|          | 224/897 [00:05<00:17, 39.37it/s]
Computing ESM2 embeddings: 29%|          | 256/897 [00:06<00:16, 39.71it/s]
Computing ESM2 embeddings: 32%|          | 288/897 [00:07<00:15, 39.84it/s]
Computing ESM2 embeddings: 36%|          | 320/897 [00:08<00:14, 40.02it/s]
Computing ESM2 embeddings: 39%|          | 352/897 [00:08<00:13, 39.16it/s]
Computing ESM2 embeddings: 43%|          | 384/897 [00:09<00:13, 39.32it/s]
Computing ESM2 embeddings: 46%|          | 416/897 [00:10<00:12, 39.63it/s]
Computing ESM2 embeddings: 50%|          | 448/897 [00:11<00:11, 39.86it/s]
Computing ESM2 embeddings: 54%|          | 480/897 [00:12<00:10, 40.00it/s]

```

Computing ESM2 embeddings: 57%| | 512/897 [00:12<00:09, 40.06it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:13<00:08, 39.35it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:14<00:08, 39.52it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:15<00:07, 39.68it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:16<00:06, 39.88it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:16<00:05, 40.02it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:17<00:04, 40.06it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:18<00:04, 39.27it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:19<00:03, 39.45it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:20<00:02, 39.65it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:21<00:01, 39.86it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:21<00:00, 39.94it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:22<00:00, 39.46it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/916 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/916 [00:00<00:22, 38.78it/s]

Computing ESM2 embeddings: 7%| | 64/916 [00:01<00:21, 39.74it/s]

Computing ESM2 embeddings: 10%| | 96/916 [00:02<00:20, 39.93it/s]

Computing ESM2 embeddings: 14%| | 128/916 [00:03<00:19, 40.08it/s]

Computing ESM2 embeddings: 17%| | 160/916 [00:04<00:19, 38.54it/s]

Computing ESM2 embeddings: 21%| | 192/916 [00:04<00:18, 39.01it/s]

Computing ESM2 embeddings: 24%| | 224/916 [00:05<00:17, 39.38it/s]

Computing ESM2 embeddings: 28%| | 256/916 [00:06<00:16, 39.70it/s]

Computing ESM2 embeddings: 31%| | 288/916 [00:07<00:16, 38.69it/s]

Computing ESM2 embeddings: 35%| | 320/916 [00:08<00:15, 39.06it/s]

Computing ESM2 embeddings: 38%| | 352/916 [00:08<00:14, 39.38it/s]

Computing ESM2 embeddings: 42%| | 384/916 [00:09<00:13, 39.63it/s]

Computing ESM2 embeddings: 45%| | 416/916 [00:10<00:13, 38.41it/s]

Computing ESM2 embeddings: 49%| | 448/916 [00:11<00:12, 38.86it/s]

Computing ESM2 embeddings: 52%| | 480/916 [00:12<00:11, 39.19it/s]

Computing ESM2 embeddings: 56%| | 512/916 [00:13<00:10, 39.53it/s]

Computing ESM2 embeddings: 59%| | 544/916 [00:13<00:09, 38.39it/s]

Computing ESM2 embeddings: 63%| | 576/916 [00:14<00:08, 38.55it/s]

Computing ESM2 embeddings: 66%| | 608/916 [00:15<00:07, 38.83it/s]

Computing ESM2 embeddings: 70%| | 640/916 [00:16<00:07, 39.28it/s]

Computing ESM2 embeddings: 73%| | 672/916 [00:17<00:06, 38.70it/s]

Computing ESM2 embeddings: 77%| | 704/916 [00:18<00:05, 39.04it/s]

Computing ESM2 embeddings: 80%| | 736/916 [00:18<00:04, 39.34it/s]

Computing ESM2 embeddings: 84%| | 768/916 [00:19<00:03, 39.63it/s]

Computing ESM2 embeddings: 87%| | 800/916 [00:20<00:03, 38.38it/s]

Computing ESM2 embeddings: 91%| | 832/916 [00:21<00:02, 38.59it/s]

Computing ESM2 embeddings: 94%| | 864/916 [00:22<00:01, 38.89it/s]

Computing ESM2 embeddings: 98%| | 896/916 [00:22<00:00, 39.30it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:23<00:00, 38.93it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.82it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.18it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.66it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.93it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 38.53it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 39.00it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 39.39it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.72it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.88it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 40.05it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 38.90it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 39.18it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 39.48it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.74it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.48it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 38.90it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:08, 39.26it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.57it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.87it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.17it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 39.46it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:17<00:04, 39.71it/s] |



```

Computing ESM2 embeddings: 82%|      | 736/896 [00:18<00:04, 38.46it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 38.88it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 39.27it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.58it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 38.41it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.10it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.73it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 39.37it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:20, 39.80it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:19, 39.94it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 38.38it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:04<00:18, 38.83it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 39.31it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 39.59it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 38.33it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 38.73it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 39.17it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 39.46it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 38.54it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 38.90it/s]

```

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 39.32it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 39.55it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 38.92it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.12it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.47it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.63it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.99it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.17it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.49it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.67it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.55it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.82it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 39.23it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.47it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.97it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.76it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.29it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.20it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.51it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.29it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21% | 192/896 [00:04<00:18, 38.75it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 39.27it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 39.55it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 38.42it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 38.79it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:13, 39.24it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:12, 39.50it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 38.36it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 38.76it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 39.21it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 39.49it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 38.42it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 38.78it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 39.22it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 39.45it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 38.42it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:04, 38.77it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:18<00:04, 39.25it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:19<00:03, 39.50it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 38.94it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 39.11it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:22<00:00, 39.46it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 38.98it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.33it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.07it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.58it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:19, 39.88it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.47it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 38.95it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.33it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.65it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.89it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 39.19it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 39.45it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.71it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 38.96it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.24it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:10, 39.47it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 39.73it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 38.50it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 38.89it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.23it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.55it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 38.36it/s] |

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.78it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.14it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.48it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.80it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.11it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 39.39it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.65it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.06it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.95it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.04it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.49it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.78it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.05it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:18, 38.63it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 39.07it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 39.41it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 38.10it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 38.62it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:13, 39.02it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 39.35it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.16it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.66it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 38.99it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 39.34it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.36it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 38.78it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.12it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.43it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.75it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 39.08it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.35it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 39.61it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.90it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.18it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.40it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 38.98it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/589 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/589 [00:00<00:14, 37.94it/s]

Computing ESM2 embeddings: 11%| | 64/589 [00:01<00:13, 39.06it/s]

Computing ESM2 embeddings: 16%| | 96/589 [00:02<00:12, 39.44it/s]

Computing ESM2 embeddings: 22%| | 128/589 [00:03<00:11, 39.81it/s]

Computing ESM2 embeddings: 27%| | 160/589 [00:04<00:11, 38.28it/s]

Computing ESM2 embeddings: 33%| | 192/589 [00:04<00:10, 38.81it/s]

Computing ESM2 embeddings: 38%| | 224/589 [00:05<00:09, 39.23it/s]

Computing ESM2 embeddings: 43%| | 256/589 [00:06<00:08, 39.58it/s]

Computing ESM2 embeddings: 49%| | 288/589 [00:07<00:07, 38.87it/s]

Computing ESM2 embeddings: 54%| | 320/589 [00:08<00:06, 39.16it/s]

Computing ESM2 embeddings: 60%| | 352/589 [00:08<00:06, 39.42it/s]

Computing ESM2 embeddings: 65%| | 384/589 [00:09<00:05, 39.68it/s]

Computing ESM2 embeddings: 71%| | 416/589 [00:10<00:04, 38.97it/s]

Computing ESM2 embeddings: 76%| | 448/589 [00:11<00:03, 39.24it/s]

Computing ESM2 embeddings: 81%| | 480/589 [00:12<00:02, 39.50it/s]

Computing ESM2 embeddings: 87%| | 512/589 [00:13<00:01, 39.74it/s]

Computing ESM2 embeddings: 92%| | 544/589 [00:13<00:01, 38.57it/s]

Computing ESM2 embeddings: 98%| | 576/589 [00:14<00:00, 38.93it/s]

Computing ESM2 embeddings: 100%| | 589/589 [00:15<00:00, 39.06it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/898 [00:00<00:23, 36.36it/s]

Computing ESM2 embeddings: 7%| | 64/898 [00:01<00:21, 38.37it/s]

Computing ESM2 embeddings: 11%| | 96/898 [00:02<00:20, 39.19it/s]

Computing ESM2 embeddings: 14%| | 128/898 [00:03<00:19, 39.64it/s]

Computing ESM2 embeddings: 18%| | 160/898 [00:04<00:19, 38.22it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/898 [00:04<00:18, 38.79it/s] |
| Computing ESM2 embeddings: 25%  | 224/898 [00:05<00:17, 39.23it/s] |
| Computing ESM2 embeddings: 29%  | 256/898 [00:06<00:16, 39.58it/s] |
| Computing ESM2 embeddings: 32%  | 288/898 [00:07<00:15, 38.34it/s] |
| Computing ESM2 embeddings: 36%  | 320/898 [00:08<00:14, 38.84it/s] |
| Computing ESM2 embeddings: 39%  | 352/898 [00:09<00:13, 39.23it/s] |
| Computing ESM2 embeddings: 43%  | 384/898 [00:09<00:12, 39.55it/s] |
| Computing ESM2 embeddings: 46%  | 416/898 [00:10<00:12, 38.13it/s] |
| Computing ESM2 embeddings: 50%  | 448/898 [00:11<00:11, 38.64it/s] |
| Computing ESM2 embeddings: 53%  | 480/898 [00:12<00:10, 39.00it/s] |
| Computing ESM2 embeddings: 57%  | 512/898 [00:13<00:09, 39.38it/s] |
| Computing ESM2 embeddings: 61%  | 544/898 [00:14<00:09, 38.23it/s] |
| Computing ESM2 embeddings: 64%  | 576/898 [00:14<00:08, 38.72it/s] |
| Computing ESM2 embeddings: 68%  | 608/898 [00:15<00:07, 39.11it/s] |
| Computing ESM2 embeddings: 71%  | 640/898 [00:16<00:06, 39.42it/s] |
| Computing ESM2 embeddings: 75%  | 672/898 [00:17<00:05, 38.46it/s] |
| Computing ESM2 embeddings: 78%  | 704/898 [00:18<00:04, 38.85it/s] |
| Computing ESM2 embeddings: 82%  | 736/898 [00:18<00:04, 39.18it/s] |
| Computing ESM2 embeddings: 86%  | 768/898 [00:19<00:03, 39.47it/s] |
| Computing ESM2 embeddings: 89%  | 800/898 [00:20<00:02, 38.73it/s] |
| Computing ESM2 embeddings: 93%  | 832/898 [00:21<00:01, 39.04it/s] |
| Computing ESM2 embeddings: 96%  | 864/898 [00:22<00:00, 39.32it/s] |
| Computing ESM2 embeddings: 100% | 898/898 [00:23<00:00, 38.86it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',



'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/915 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/915 [00:00<00:24, 35.44it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/915 [00:01<00:22, 37.67it/s]  |
| Computing ESM2 embeddings: | 10% | 96/915 [00:02<00:21, 38.82it/s]  |
| Computing ESM2 embeddings: | 14% | 128/915 [00:03<00:20, 39.28it/s] |
| Computing ESM2 embeddings: | 17% | 160/915 [00:04<00:19, 37.99it/s] |
| Computing ESM2 embeddings: | 21% | 192/915 [00:05<00:18, 38.48it/s] |
| Computing ESM2 embeddings: | 24% | 224/915 [00:05<00:17, 39.05it/s] |
| Computing ESM2 embeddings: | 28% | 256/915 [00:06<00:16, 39.35it/s] |
| Computing ESM2 embeddings: | 31% | 288/915 [00:07<00:16, 38.82it/s] |
| Computing ESM2 embeddings: | 35% | 320/915 [00:08<00:15, 39.03it/s] |
| Computing ESM2 embeddings: | 38% | 352/915 [00:09<00:14, 39.39it/s] |
| Computing ESM2 embeddings: | 42% | 384/915 [00:09<00:13, 39.58it/s] |
| Computing ESM2 embeddings: | 45% | 416/915 [00:10<00:12, 38.94it/s] |
| Computing ESM2 embeddings: | 49% | 448/915 [00:11<00:12, 38.91it/s] |
| Computing ESM2 embeddings: | 52% | 480/915 [00:12<00:11, 39.17it/s] |
| Computing ESM2 embeddings: | 56% | 512/915 [00:13<00:10, 39.40it/s] |
| Computing ESM2 embeddings: | 59% | 544/915 [00:14<00:09, 38.36it/s] |
| Computing ESM2 embeddings: | 63% | 576/915 [00:14<00:08, 38.47it/s] |
| Computing ESM2 embeddings: | 66% | 608/915 [00:15<00:07, 38.88it/s] |
| Computing ESM2 embeddings: | 70% | 640/915 [00:16<00:07, 39.19it/s] |
| Computing ESM2 embeddings: | 73% | 672/915 [00:17<00:06, 38.04it/s] |

```

Computing ESM2 embeddings: 77%|          | 704/915 [00:18<00:05, 38.44it/s]
Computing ESM2 embeddings: 80%|          | 736/915 [00:18<00:04, 38.93it/s]
Computing ESM2 embeddings: 84%|          | 768/915 [00:19<00:03, 39.23it/s]
Computing ESM2 embeddings: 87%|          | 800/915 [00:20<00:02, 38.67it/s]
Computing ESM2 embeddings: 91%|          | 832/915 [00:21<00:02, 38.86it/s]
Computing ESM2 embeddings: 94%|          | 864/915 [00:22<00:01, 39.24it/s]
Computing ESM2 embeddings: 98%|          | 896/915 [00:23<00:00, 39.44it/s]

Computing ESM2 embeddings: 100%|         | 915/915 [00:23<00:00, 38.80it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.68it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.51it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 39.89it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:19, 40.01it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 38.44it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:04<00:18, 38.85it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 39.33it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 39.59it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 38.40it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 38.80it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:08<00:13, 39.25it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:12, 39.51it/s]

```

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.39it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.76it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.23it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 39.48it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:09, 38.84it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.09it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.45it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.58it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.95it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.88it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.17it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 39.42it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.31it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.33it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.78it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 38.99it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.08it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 38.01it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.87it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:19, 39.33it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.11it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 38.59it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 39.12it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 39.44it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.19it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 38.59it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 39.07it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 39.37it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 38.37it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 38.71it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 38.80it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 38.99it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 38.51it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 38.81it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.22it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:06, 39.43it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 38.21it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 38.58it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.01it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 39.31it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 38.69it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:21<00:02, 38.93it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:22<00:01, 39.29it/s] |
| Computing ESM2 embeddings: 98% | 896/914 [00:23<00:00, 39.51it/s] |

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.76it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 39.27it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:20, 39.77it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:19, 40.05it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 40.11it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 38.65it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.98it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 39.41it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.63it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 38.46it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 38.83it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 39.26it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:12, 39.55it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.38it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.74it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.12it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 39.41it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:09, 38.83it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.07it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.43it/s] |

```

Computing ESM2 embeddings: 71%|      | 640/896 [00:16<00:06, 39.63it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 38.94it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:17<00:04, 39.11it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:18<00:04, 39.44it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 39.55it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 38.44it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 38.76it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 39.20it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 39.15it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/913 [00:00<00:22, 39.29it/s]
Computing ESM2 embeddings:  7%|      | 64/913 [00:01<00:21, 39.91it/s]
Computing ESM2 embeddings: 11%|      | 96/913 [00:02<00:20, 40.01it/s]
Computing ESM2 embeddings: 14%|      | 128/913 [00:03<00:20, 38.20it/s]
Computing ESM2 embeddings: 18%|      | 160/913 [00:04<00:19, 38.71it/s]
Computing ESM2 embeddings: 21%|      | 192/913 [00:04<00:18, 39.27it/s]
Computing ESM2 embeddings: 25%|      | 224/913 [00:05<00:17, 39.54it/s]
Computing ESM2 embeddings: 28%|      | 256/913 [00:06<00:16, 38.87it/s]
Computing ESM2 embeddings: 32%|      | 288/913 [00:07<00:15, 39.07it/s]
Computing ESM2 embeddings: 35%|      | 320/913 [00:08<00:15, 39.42it/s]
Computing ESM2 embeddings: 39%|      | 352/913 [00:08<00:14, 39.60it/s]

```

Computing ESM2 embeddings: 42%| | 384/913 [00:09<00:13, 38.95it/s]

Computing ESM2 embeddings: 46%| | 416/913 [00:10<00:12, 38.74it/s]

Computing ESM2 embeddings: 49%| | 448/913 [00:11<00:11, 39.07it/s]

Computing ESM2 embeddings: 53%| | 480/913 [00:12<00:11, 39.28it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 38.25it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:13<00:09, 38.58it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:14<00:08, 39.06it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:15<00:07, 39.32it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:16<00:07, 38.25it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:17<00:06, 38.62it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:18<00:05, 39.09it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:18<00:04, 39.31it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:19<00:03, 38.20it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:20<00:02, 38.59it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:21<00:02, 39.07it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:22<00:01, 39.34it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:23<00:00, 38.74it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:23<00:00, 38.86it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.36it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.29it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:20, 39.67it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:19, 39.94it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 38.56it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:04<00:18, 39.02it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 39.38it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 39.69it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 38.65it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 39.02it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:08<00:13, 39.32it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:12, 39.62it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 38.83it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 39.15it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 39.43it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 39.70it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:13<00:09, 38.34it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 38.79it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 39.16it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 39.49it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 38.31it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:04, 38.78it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:18<00:04, 39.16it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:19<00:03, 39.50it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 38.78it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 39.10it/s] |



Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.40it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.17it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/589 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/589 [00:00<00:14, 37.90it/s]

Computing ESM2 embeddings: 11%| | 64/589 [00:01<00:13, 38.94it/s]

Computing ESM2 embeddings: 16%| | 96/589 [00:02<00:12, 39.01it/s]

Computing ESM2 embeddings: 22%| | 128/589 [00:03<00:11, 39.52it/s]

Computing ESM2 embeddings: 27%| | 160/589 [00:04<00:10, 39.75it/s]

Computing ESM2 embeddings: 33%| | 192/589 [00:04<00:09, 39.95it/s]

Computing ESM2 embeddings: 38%| | 224/589 [00:05<00:09, 38.30it/s]

Computing ESM2 embeddings: 43%| | 256/589 [00:06<00:08, 38.80it/s]

Computing ESM2 embeddings: 49%| | 288/589 [00:07<00:07, 39.18it/s]

Computing ESM2 embeddings: 54%| | 320/589 [00:08<00:06, 39.53it/s]

Computing ESM2 embeddings: 60%| | 352/589 [00:09<00:06, 38.28it/s]

Computing ESM2 embeddings: 65%| | 384/589 [00:09<00:05, 38.71it/s]

Computing ESM2 embeddings: 71%| | 416/589 [00:10<00:04, 39.09it/s]

Computing ESM2 embeddings: 76%| | 448/589 [00:11<00:03, 39.42it/s]

Computing ESM2 embeddings: 81%| | 480/589 [00:12<00:02, 38.82it/s]

Computing ESM2 embeddings: 87%| | 512/589 [00:13<00:01, 39.11it/s]

Computing ESM2 embeddings: 92%| | 544/589 [00:13<00:01, 39.39it/s]

Computing ESM2 embeddings: 98%| | 576/589 [00:14<00:00, 39.65it/s]

Computing ESM2 embeddings: 100%| | 589/589 [00:15<00:00, 38.96it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/898 [00:00<00:22, 39.04it/s]  
 Computing ESM2 embeddings: 7%| | 64/898 [00:01<00:20, 39.74it/s]  
 Computing ESM2 embeddings: 11%| | 96/898 [00:02<00:20, 39.88it/s]  
 Computing ESM2 embeddings: 14%| | 128/898 [00:03<00:20, 38.07it/s]  
 Computing ESM2 embeddings: 18%| | 160/898 [00:04<00:19, 38.52it/s]  
 Computing ESM2 embeddings: 21%| | 192/898 [00:04<00:18, 39.08it/s]  
 Computing ESM2 embeddings: 25%| | 224/898 [00:05<00:17, 39.39it/s]  
 Computing ESM2 embeddings: 29%| | 256/898 [00:06<00:16, 38.72it/s]  
 Computing ESM2 embeddings: 32%| | 288/898 [00:07<00:15, 38.96it/s]  
 Computing ESM2 embeddings: 36%| | 320/898 [00:08<00:14, 39.32it/s]  
 Computing ESM2 embeddings: 39%| | 352/898 [00:08<00:13, 39.54it/s]  
 Computing ESM2 embeddings: 43%| | 384/898 [00:09<00:13, 38.31it/s]  
 Computing ESM2 embeddings: 46%| | 416/898 [00:10<00:12, 38.63it/s]  
 Computing ESM2 embeddings: 50%| | 448/898 [00:11<00:11, 39.08it/s]  
 Computing ESM2 embeddings: 53%| | 480/898 [00:12<00:10, 39.35it/s]  
 Computing ESM2 embeddings: 57%| | 512/898 [00:13<00:10, 38.25it/s]  
 Computing ESM2 embeddings: 61%| | 544/898 [00:13<00:09, 38.62it/s]  
 Computing ESM2 embeddings: 64%| | 576/898 [00:14<00:08, 39.09it/s]  
 Computing ESM2 embeddings: 68%| | 608/898 [00:15<00:07, 39.35it/s]

```

Computing ESM2 embeddings: 71%|      | 640/898 [00:16<00:06, 38.76it/s]
Computing ESM2 embeddings: 75%|      | 672/898 [00:17<00:05, 38.61it/s]
Computing ESM2 embeddings: 78%|      | 704/898 [00:18<00:04, 38.95it/s]
Computing ESM2 embeddings: 82%|      | 736/898 [00:18<00:04, 39.25it/s]
Computing ESM2 embeddings: 86%|      | 768/898 [00:19<00:03, 38.08it/s]
Computing ESM2 embeddings: 89%|      | 800/898 [00:20<00:02, 38.47it/s]
Computing ESM2 embeddings: 93%|      | 832/898 [00:21<00:01, 38.95it/s]
Computing ESM2 embeddings: 96%|      | 864/898 [00:22<00:00, 39.25it/s]

Computing ESM2 embeddings: 100%|     | 898/898 [00:23<00:00, 38.76it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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```

Computing ESM2 embeddings:  0%|      |  0/915 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/915 [00:00<00:22, 38.91it/s]
Computing ESM2 embeddings:  7%|      | 64/915 [00:01<00:21, 39.35it/s]
Computing ESM2 embeddings: 10%|      | 96/915 [00:02<00:20, 39.81it/s]
Computing ESM2 embeddings: 14%|      | 128/915 [00:03<00:20, 38.20it/s]
Computing ESM2 embeddings: 17%|      | 160/915 [00:04<00:19, 38.81it/s]
Computing ESM2 embeddings: 21%|      | 192/915 [00:04<00:18, 39.23it/s]
Computing ESM2 embeddings: 24%|      | 224/915 [00:05<00:17, 39.60it/s]
Computing ESM2 embeddings: 28%|      | 256/915 [00:06<00:17, 38.39it/s]
Computing ESM2 embeddings: 31%|      | 288/915 [00:07<00:16, 38.88it/s]
Computing ESM2 embeddings: 35%|      | 320/915 [00:08<00:15, 39.27it/s]
Computing ESM2 embeddings: 38%|      | 352/915 [00:08<00:14, 39.58it/s]

```

Computing ESM2 embeddings: 42%| | 384/915 [00:09<00:13, 38.90it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:10<00:12, 39.19it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:11<00:11, 39.44it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:10, 39.70it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 38.35it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:13<00:09, 38.81it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:14<00:08, 39.14it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:15<00:07, 39.49it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:16<00:07, 38.85it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:17<00:06, 39.07it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:17<00:05, 39.37it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:18<00:04, 39.62it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:19<00:03, 38.32it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:20<00:02, 38.79it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:21<00:02, 39.16it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:22<00:01, 39.47it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:22<00:00, 38.82it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:23<00:00, 39.03it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.10it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.46it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:20, 39.27it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 37.88it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 38.58it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:04<00:18, 39.06it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 39.40it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 38.72it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 39.07it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 39.36it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:13, 39.62it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:13, 38.25it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 38.35it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 38.68it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 39.13it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 38.49it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 38.88it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 39.20it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 39.50it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 38.26it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 38.71it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:04, 39.07it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:18<00:04, 39.40it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:19<00:03, 38.15it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 38.63it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 39.02it/s] |

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.36it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.66it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.30it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.00it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.24it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 37.15it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.00it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:18, 38.73it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.15it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.03it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.48it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 39.01it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 39.31it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.77it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.98it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.33it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 39.54it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.90it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 39.06it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.39it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.56it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.58it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.82it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 39.21it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.45it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.56it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.82it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.19it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 39.41it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.35it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.75it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 38.68it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:21, 39.43it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:20, 39.75it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:20, 38.20it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:18, 38.73it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:04<00:17, 39.28it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:05<00:16, 39.58it/s]

Computing ESM2 embeddings: 29%| | 256/895 [00:06<00:16, 38.50it/s]

Computing ESM2 embeddings: 32%| | 288/895 [00:07<00:15, 38.86it/s]

Computing ESM2 embeddings: 36%| | 320/895 [00:08<00:14, 39.30it/s]

Computing ESM2 embeddings: 39%| | 352/895 [00:08<00:13, 39.54it/s]

Computing ESM2 embeddings: 43%| | 384/895 [00:09<00:13, 38.43it/s]

Computing ESM2 embeddings: 46%| | 416/895 [00:10<00:12, 38.39it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:11<00:11, 38.84it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:12<00:10, 39.20it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 38.15it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 38.59it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:14<00:08, 39.08it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:15<00:07, 39.39it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:16<00:06, 38.27it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:17<00:05, 38.66it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:04, 39.16it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:18<00:04, 39.44it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:19<00:03, 38.88it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:20<00:02, 39.05it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:21<00:01, 39.40it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:22<00:00, 39.60it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:23<00:00, 38.86it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.01it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:23, 36.85it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 38.30it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 37.40it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.25it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:18, 38.79it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 39.23it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 37.98it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.53it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 38.91it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 39.28it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:13, 38.25it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 38.70it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.03it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 39.37it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 38.21it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 38.68it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.04it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.33it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 38.40it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 38.49it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 38.73it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 39.16it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 38.04it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 38.56it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 38.94it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.31it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.30it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.53it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/896 [00:01<00:28, 30.03it/s]
Computing ESM2 embeddings: 7%|        | 64/896 [00:01<00:23, 34.96it/s]
Computing ESM2 embeddings: 11%|       | 96/896 [00:02<00:21, 37.18it/s]
Computing ESM2 embeddings: 14%|       | 128/896 [00:03<00:20, 36.64it/s]
Computing ESM2 embeddings: 18%|       | 160/896 [00:04<00:19, 37.77it/s]
Computing ESM2 embeddings: 21%|       | 192/896 [00:05<00:18, 38.54it/s]
Computing ESM2 embeddings: 25%|       | 224/896 [00:05<00:17, 39.10it/s]
Computing ESM2 embeddings: 29%|       | 256/896 [00:06<00:16, 38.05it/s]
Computing ESM2 embeddings: 32%|       | 288/896 [00:07<00:15, 38.61it/s]
Computing ESM2 embeddings: 36%|       | 320/896 [00:08<00:14, 39.04it/s]
Computing ESM2 embeddings: 39%|       | 352/896 [00:09<00:13, 39.43it/s]
Computing ESM2 embeddings: 43%|       | 384/896 [00:10<00:13, 38.31it/s]
Computing ESM2 embeddings: 46%|       | 416/896 [00:10<00:12, 38.75it/s]
Computing ESM2 embeddings: 50%|       | 448/896 [00:11<00:11, 39.12it/s]
Computing ESM2 embeddings: 54%|       | 480/896 [00:12<00:10, 39.37it/s]
Computing ESM2 embeddings: 57%|       | 512/896 [00:13<00:10, 38.13it/s]

```

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.66it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 39.08it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.42it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.14it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.65it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 39.05it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 39.00it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.68it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.28it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.75it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.18it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.25it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/570 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/570 [00:00<00:14, 38.26it/s]

Computing ESM2 embeddings: 11%| | 64/570 [00:01<00:12, 39.22it/s]

Computing ESM2 embeddings: 17%| | 96/570 [00:02<00:11, 39.59it/s]

Computing ESM2 embeddings: 22%| | 128/570 [00:03<00:11, 38.68it/s]

Computing ESM2 embeddings: 28%| | 160/570 [00:04<00:10, 38.99it/s]

Computing ESM2 embeddings: 34%| | 192/570 [00:04<00:09, 39.39it/s]

Computing ESM2 embeddings: 39%| | 224/570 [00:05<00:08, 39.60it/s]

Computing ESM2 embeddings: 45%| | 256/570 [00:06<00:08, 38.59it/s]

```

Computing ESM2 embeddings: 51%|          | 288/570 [00:07<00:07, 38.86it/s]
Computing ESM2 embeddings: 56%|          | 320/570 [00:08<00:06, 39.26it/s]
Computing ESM2 embeddings: 62%|          | 352/570 [00:08<00:05, 39.50it/s]
Computing ESM2 embeddings: 67%|          | 384/570 [00:09<00:04, 38.24it/s]
Computing ESM2 embeddings: 73%|          | 416/570 [00:10<00:03, 38.62it/s]
Computing ESM2 embeddings: 79%|          | 448/570 [00:11<00:03, 38.71it/s]
Computing ESM2 embeddings: 84%|          | 480/570 [00:12<00:02, 38.93it/s]
Computing ESM2 embeddings: 90%|          | 512/570 [00:13<00:01, 37.99it/s]
Computing ESM2 embeddings: 95%|          | 544/570 [00:14<00:00, 38.47it/s]
Computing ESM2 embeddings: 100%|         | 570/570 [00:14<00:00, 38.83it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/897 [00:00<00:22, 38.51it/s]
Computing ESM2 embeddings: 7%|           | 64/897 [00:01<00:21, 39.11it/s]
Computing ESM2 embeddings: 11%|          | 96/897 [00:02<00:20, 39.66it/s]
Computing ESM2 embeddings: 14%|          | 128/897 [00:03<00:20, 38.11it/s]
Computing ESM2 embeddings: 18%|          | 160/897 [00:04<00:19, 38.75it/s]
Computing ESM2 embeddings: 21%|          | 192/897 [00:04<00:17, 39.20it/s]
Computing ESM2 embeddings: 25%|          | 224/897 [00:05<00:17, 39.55it/s]
Computing ESM2 embeddings: 29%|          | 256/897 [00:06<00:16, 38.16it/s]
Computing ESM2 embeddings: 32%|          | 288/897 [00:07<00:15, 38.69it/s]
Computing ESM2 embeddings: 36%|          | 320/897 [00:08<00:14, 39.08it/s]

```

Computing ESM2 embeddings: 39%| | 352/897 [00:09<00:13, 39.46it/s]

Computing ESM2 embeddings: 43%| | 384/897 [00:09<00:13, 38.31it/s]

Computing ESM2 embeddings: 46%| | 416/897 [00:10<00:12, 38.77it/s]

Computing ESM2 embeddings: 50%| | 448/897 [00:11<00:11, 39.16it/s]

Computing ESM2 embeddings: 54%| | 480/897 [00:12<00:10, 39.13it/s]

Computing ESM2 embeddings: 57%| | 512/897 [00:13<00:10, 37.79it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:14<00:09, 38.39it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:14<00:08, 38.83it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:15<00:07, 39.16it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:16<00:06, 38.63it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:17<00:05, 39.00it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:18<00:04, 39.29it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:18<00:04, 39.58it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:19<00:03, 38.90it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:20<00:02, 39.18it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:21<00:01, 39.43it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:22<00:00, 39.68it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:23<00:00, 38.81it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/916 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/916 [00:00<00:23, 37.71it/s]

Computing ESM2 embeddings: 7%| | 64/916 [00:01<00:22, 38.19it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 10% | 96/916 [00:02<00:21, 38.77it/s]  |
| Computing ESM2 embeddings: 14% | 128/916 [00:03<00:20, 37.74it/s] |
| Computing ESM2 embeddings: 17% | 160/916 [00:04<00:19, 38.38it/s] |
| Computing ESM2 embeddings: 21% | 192/916 [00:04<00:18, 39.01it/s] |
| Computing ESM2 embeddings: 24% | 224/916 [00:05<00:17, 39.35it/s] |
| Computing ESM2 embeddings: 28% | 256/916 [00:06<00:17, 38.78it/s] |
| Computing ESM2 embeddings: 31% | 288/916 [00:07<00:16, 39.03it/s] |
| Computing ESM2 embeddings: 35% | 320/916 [00:08<00:15, 39.42it/s] |
| Computing ESM2 embeddings: 38% | 352/916 [00:09<00:14, 39.62it/s] |
| Computing ESM2 embeddings: 42% | 384/916 [00:09<00:13, 38.99it/s] |
| Computing ESM2 embeddings: 45% | 416/916 [00:10<00:12, 39.14it/s] |
| Computing ESM2 embeddings: 49% | 448/916 [00:11<00:11, 39.19it/s] |
| Computing ESM2 embeddings: 52% | 480/916 [00:12<00:11, 39.28it/s] |
| Computing ESM2 embeddings: 56% | 512/916 [00:13<00:10, 38.70it/s] |
| Computing ESM2 embeddings: 59% | 544/916 [00:13<00:09, 38.74it/s] |
| Computing ESM2 embeddings: 63% | 576/916 [00:14<00:08, 39.11it/s] |
| Computing ESM2 embeddings: 66% | 608/916 [00:15<00:07, 39.37it/s] |
| Computing ESM2 embeddings: 70% | 640/916 [00:16<00:07, 38.32it/s] |
| Computing ESM2 embeddings: 73% | 672/916 [00:17<00:06, 38.68it/s] |
| Computing ESM2 embeddings: 77% | 704/916 [00:18<00:05, 39.14it/s] |
| Computing ESM2 embeddings: 80% | 736/916 [00:18<00:04, 39.41it/s] |
| Computing ESM2 embeddings: 84% | 768/916 [00:19<00:03, 38.80it/s] |
| Computing ESM2 embeddings: 87% | 800/916 [00:20<00:02, 39.02it/s] |
| Computing ESM2 embeddings: 91% | 832/916 [00:21<00:02, 39.37it/s] |

```

Computing ESM2 embeddings: 94%|      | 864/916 [00:22<00:01, 39.55it/s]

Computing ESM2 embeddings: 98%|      | 896/916 [00:23<00:00, 38.28it/s]

Computing ESM2 embeddings: 100%|     | 916/916 [00:23<00:00, 38.86it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|       | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%|       | 32/896 [00:00<00:22, 38.38it/s]

Computing ESM2 embeddings: 7%|       | 64/896 [00:01<00:21, 39.33it/s]

Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 39.70it/s]

Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:19, 38.82it/s]

Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:18, 39.08it/s]

Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:17, 39.49it/s]

Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:16, 39.69it/s]

Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 38.33it/s]

Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 38.71it/s]

Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 39.18it/s]

Computing ESM2 embeddings: 39%|      | 352/896 [00:08<00:13, 39.44it/s]

Computing ESM2 embeddings: 43%|      | 384/896 [00:09<00:13, 38.81it/s]

Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 39.03it/s]

Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 39.39it/s]

Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 39.28it/s]

Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:10, 37.98it/s]

Computing ESM2 embeddings: 61%|      | 544/896 [00:13<00:09, 38.44it/s]

```

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 38.64it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.89it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.48it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.85it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 39.23it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.46it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.38it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.73it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.12it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.33it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.77it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.18it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 38.32it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.18it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 37.82it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.21it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 38.67it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.22it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.29it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.77it/s]



```

Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 39.16it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 39.50it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:09<00:13, 38.37it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 38.53it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 38.86it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 39.28it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.23it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 38.72it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 39.09it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.35it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.67it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.74it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.96it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 39.36it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.15it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.65it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.07it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 39.44it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.34it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.67it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:21, 39.88it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:20, 40.03it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:19, 40.20it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 39.12it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.41it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:17, 39.66it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:16, 39.89it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.04it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.30it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.55it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 39.79it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.50it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.91it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.27it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.27it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.81it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:09, 38.41it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.90it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.31it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.25it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 38.73it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:17<00:04, 39.14it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:18<00:04, 39.48it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:19<00:03, 38.38it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 38.83it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.20it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 39.53it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 38.91it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/913 [00:00<00:23, 37.23it/s]
Computing ESM2 embeddings: 7%|        | 64/913 [00:01<00:22, 37.71it/s]
Computing ESM2 embeddings: 11%|       | 96/913 [00:02<00:21, 38.56it/s]
Computing ESM2 embeddings: 14%|       | 128/913 [00:03<00:20, 38.05it/s]
Computing ESM2 embeddings: 18%|       | 160/913 [00:04<00:19, 38.57it/s]
Computing ESM2 embeddings: 21%|       | 192/913 [00:04<00:18, 39.12it/s]
Computing ESM2 embeddings: 25%|       | 224/913 [00:05<00:17, 39.42it/s]
Computing ESM2 embeddings: 28%|       | 256/913 [00:06<00:16, 38.86it/s]
Computing ESM2 embeddings: 32%|       | 288/913 [00:07<00:15, 39.07it/s]
Computing ESM2 embeddings: 35%|       | 320/913 [00:08<00:15, 39.42it/s]
Computing ESM2 embeddings: 39%|       | 352/913 [00:09<00:14, 39.64it/s]
Computing ESM2 embeddings: 42%|       | 384/913 [00:09<00:13, 39.01it/s]
Computing ESM2 embeddings: 46%|       | 416/913 [00:10<00:12, 39.18it/s]
Computing ESM2 embeddings: 49%|       | 448/913 [00:11<00:11, 39.50it/s]
Computing ESM2 embeddings: 53%|       | 480/913 [00:12<00:10, 39.65it/s]
Computing ESM2 embeddings: 56%|       | 512/913 [00:13<00:10, 38.34it/s]

```

Computing ESM2 embeddings: 60%| | 544/913 [00:13<00:09, 38.69it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:14<00:08, 39.14it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:15<00:07, 39.08it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:16<00:07, 37.88it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:17<00:06, 38.37it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:18<00:05, 38.90it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:18<00:04, 39.21it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:19<00:03, 38.68it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:20<00:02, 38.93it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:21<00:02, 39.31it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:22<00:01, 39.53it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:23<00:00, 38.92it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:23<00:00, 38.79it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.93it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.06it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.46it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 38.71it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 38.98it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 39.41it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:16, 39.60it/s]

```

Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 38.31it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 38.66it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 39.14it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:13, 39.39it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:13, 38.75it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.69it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.02it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 39.27it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:09, 38.65it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:13<00:09, 38.87it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 39.26it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 39.45it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.23it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.57it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:04, 39.03it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 38.92it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 37.26it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 37.90it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 38.56it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 38.96it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.69it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|          | 0/591 [00:00<?, ?it/s]
Computing ESM2 embeddings: 5%|          | 32/591 [00:00<00:14, 38.68it/s]
Computing ESM2 embeddings: 11%|         | 64/591 [00:01<00:13, 38.62it/s]
Computing ESM2 embeddings: 16%|         | 96/591 [00:02<00:12, 39.27it/s]
Computing ESM2 embeddings: 22%|         | 128/591 [00:03<00:12, 37.84it/s]
Computing ESM2 embeddings: 27%|         | 160/591 [00:04<00:11, 38.59it/s]
Computing ESM2 embeddings: 32%|         | 192/591 [00:04<00:10, 39.10it/s]
Computing ESM2 embeddings: 38%|         | 224/591 [00:05<00:09, 39.48it/s]
Computing ESM2 embeddings: 43%|         | 256/591 [00:06<00:08, 38.21it/s]
Computing ESM2 embeddings: 49%|         | 288/591 [00:07<00:07, 38.74it/s]
Computing ESM2 embeddings: 54%|         | 320/591 [00:08<00:06, 39.18it/s]
Computing ESM2 embeddings: 60%|         | 352/591 [00:09<00:06, 39.53it/s]
Computing ESM2 embeddings: 65%|         | 384/591 [00:09<00:05, 38.36it/s]
Computing ESM2 embeddings: 70%|         | 416/591 [00:10<00:04, 38.81it/s]
Computing ESM2 embeddings: 76%|         | 448/591 [00:11<00:03, 39.19it/s]
Computing ESM2 embeddings: 81%|         | 480/591 [00:12<00:02, 39.52it/s]
Computing ESM2 embeddings: 87%|         | 512/591 [00:13<00:02, 38.88it/s]
Computing ESM2 embeddings: 92%|         | 544/591 [00:13<00:01, 39.16it/s]
Computing ESM2 embeddings: 97%|         | 576/591 [00:14<00:00, 39.44it/s]

Computing ESM2 embeddings: 100%|        | 591/591 [00:15<00:00, 39.03it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings: 0%|          | 0/897 [00:00<?, ?it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 4%  | 32/897 [00:00<00:22, 39.31it/s]  |
| Computing ESM2 embeddings: 7%  | 64/897 [00:01<00:20, 39.92it/s]  |
| Computing ESM2 embeddings: 11% | 96/897 [00:02<00:20, 40.04it/s]  |
| Computing ESM2 embeddings: 14% | 128/897 [00:03<00:19, 38.48it/s] |
| Computing ESM2 embeddings: 18% | 160/897 [00:04<00:18, 38.90it/s] |
| Computing ESM2 embeddings: 21% | 192/897 [00:04<00:17, 39.36it/s] |
| Computing ESM2 embeddings: 25% | 224/897 [00:05<00:16, 39.62it/s] |
| Computing ESM2 embeddings: 29% | 256/897 [00:06<00:16, 38.47it/s] |
| Computing ESM2 embeddings: 32% | 288/897 [00:07<00:15, 38.84it/s] |
| Computing ESM2 embeddings: 36% | 320/897 [00:08<00:14, 39.29it/s] |
| Computing ESM2 embeddings: 39% | 352/897 [00:08<00:13, 39.58it/s] |
| Computing ESM2 embeddings: 43% | 384/897 [00:09<00:13, 38.36it/s] |
| Computing ESM2 embeddings: 46% | 416/897 [00:10<00:12, 38.41it/s] |
| Computing ESM2 embeddings: 50% | 448/897 [00:11<00:11, 38.86it/s] |
| Computing ESM2 embeddings: 54% | 480/897 [00:12<00:10, 39.23it/s] |
| Computing ESM2 embeddings: 57% | 512/897 [00:13<00:09, 38.78it/s] |
| Computing ESM2 embeddings: 61% | 544/897 [00:13<00:09, 39.02it/s] |
| Computing ESM2 embeddings: 64% | 576/897 [00:14<00:08, 39.42it/s] |
| Computing ESM2 embeddings: 68% | 608/897 [00:15<00:07, 39.62it/s] |
| Computing ESM2 embeddings: 71% | 640/897 [00:16<00:06, 39.00it/s] |
| Computing ESM2 embeddings: 75% | 672/897 [00:17<00:05, 39.17it/s] |
| Computing ESM2 embeddings: 78% | 704/897 [00:17<00:04, 39.51it/s] |
| Computing ESM2 embeddings: 82% | 736/897 [00:18<00:04, 39.67it/s] |
| Computing ESM2 embeddings: 86% | 768/897 [00:19<00:03, 38.53it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/897 [00:20<00:02, 38.83it/s]
Computing ESM2 embeddings: 93%|      | 832/897 [00:21<00:01, 39.24it/s]
Computing ESM2 embeddings: 96%|      | 864/897 [00:22<00:00, 39.48it/s]
Computing ESM2 embeddings: 100%|     | 897/897 [00:23<00:00, 38.87it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/915 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/915 [00:00<00:23, 37.94it/s]
Computing ESM2 embeddings:  7%|      | 64/915 [00:01<00:21, 38.96it/s]
Computing ESM2 embeddings: 10%|      | 96/915 [00:02<00:20, 39.60it/s]
Computing ESM2 embeddings: 14%|      | 128/915 [00:03<00:20, 38.29it/s]
Computing ESM2 embeddings: 17%|      | 160/915 [00:04<00:19, 38.88it/s]
Computing ESM2 embeddings: 21%|      | 192/915 [00:04<00:18, 39.30it/s]
Computing ESM2 embeddings: 24%|      | 224/915 [00:05<00:17, 39.66it/s]
Computing ESM2 embeddings: 28%|      | 256/915 [00:06<00:17, 38.40it/s]
Computing ESM2 embeddings: 31%|      | 288/915 [00:07<00:16, 38.88it/s]
Computing ESM2 embeddings: 35%|      | 320/915 [00:08<00:15, 39.26it/s]
Computing ESM2 embeddings: 38%|      | 352/915 [00:08<00:14, 39.57it/s]
Computing ESM2 embeddings: 42%|      | 384/915 [00:09<00:13, 38.35it/s]
Computing ESM2 embeddings: 45%|      | 416/915 [00:10<00:12, 38.81it/s]
Computing ESM2 embeddings: 49%|      | 448/915 [00:11<00:11, 39.18it/s]
Computing ESM2 embeddings: 52%|      | 480/915 [00:12<00:11, 39.21it/s]
Computing ESM2 embeddings: 56%|      | 512/915 [00:13<00:10, 37.35it/s]

```



Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 38.09it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:14<00:08, 38.68it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:15<00:07, 39.16it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:16<00:07, 38.06it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:17<00:06, 38.59it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:18<00:05, 39.00it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:18<00:04, 39.39it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:19<00:03, 38.24it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:20<00:02, 38.51it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:21<00:02, 38.82it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:22<00:01, 39.26it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:23<00:00, 38.69it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:23<00:00, 38.70it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:23, 37.13it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:21, 38.42it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:20, 39.29it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:19, 38.49it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:18, 39.02it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:04<00:17, 39.40it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:05<00:16, 39.68it/s]

```

Computing ESM2 embeddings: 29%|          | 256/895 [00:06<00:16, 38.86it/s]
Computing ESM2 embeddings: 32%|          | 288/895 [00:07<00:15, 39.19it/s]
Computing ESM2 embeddings: 36%|          | 320/895 [00:08<00:14, 39.48it/s]
Computing ESM2 embeddings: 39%|          | 352/895 [00:08<00:13, 39.74it/s]
Computing ESM2 embeddings: 43%|          | 384/895 [00:09<00:13, 38.52it/s]
Computing ESM2 embeddings: 46%|          | 416/895 [00:10<00:12, 38.93it/s]
Computing ESM2 embeddings: 50%|          | 448/895 [00:11<00:11, 39.26it/s]
Computing ESM2 embeddings: 54%|          | 480/895 [00:12<00:10, 39.57it/s]
Computing ESM2 embeddings: 57%|          | 512/895 [00:13<00:09, 38.43it/s]
Computing ESM2 embeddings: 61%|          | 544/895 [00:13<00:09, 38.87it/s]
Computing ESM2 embeddings: 64%|          | 576/895 [00:14<00:08, 39.15it/s]
Computing ESM2 embeddings: 68%|          | 608/895 [00:15<00:07, 39.50it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:16<00:06, 38.30it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:17<00:05, 38.76it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:18<00:04, 39.15it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:18<00:04, 39.50it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:19<00:03, 38.34it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:20<00:02, 38.55it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:21<00:01, 38.81it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:22<00:00, 39.25it/s]

```

```

Computing ESM2 embeddings: 100%|         | 895/895 [00:23<00:00, 38.83it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.26it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.83it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.94it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.14it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.65it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 39.21it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.51it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 38.91it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 39.10it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 39.45it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 39.63it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 38.99it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 38.78it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.08it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 39.33it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.28it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 38.65it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.10it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.37it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 38.79it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 39.02it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 39.38it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:18<00:04, 39.58it/s] |

```

Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.39it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.73it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.04it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 39.23it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.08it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.91it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.22it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.20it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 39.62it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 38.24it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:18, 38.75it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:04<00:17, 39.28it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:16, 39.55it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 38.28it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 38.66it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 39.16it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:13, 39.44it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:13, 38.83it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 39.03it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.39it/s]

```

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.59it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 38.34it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:09, 38.69it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 38.82it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.01it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.41it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.51it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.64it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 38.91it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 37.98it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.15it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.62it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.98it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.63it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.02it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.12it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.21it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 37.49it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.37it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 38.97it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:17, 39.40it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:16, 38.76it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:15, 39.13it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:15, 39.44it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 39.73it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:09<00:13, 38.45it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:10<00:12, 38.89it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:11, 39.26it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:10, 39.59it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 38.37it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:13<00:09, 38.83it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:14<00:08, 38.87it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:15<00:07, 39.19it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:16<00:07, 38.28it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 38.72it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 39.04it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:18<00:04, 39.33it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:19<00:03, 38.19it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:20<00:02, 38.43it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:21<00:02, 38.76it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:22<00:01, 38.90it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:23<00:00, 37.58it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:23<00:00, 38.65it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:21, 39.50it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:20, 39.90it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:19, 40.10it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 38.18it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 38.71it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:17, 39.14it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 39.51it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.10it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 38.61it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 38.97it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:13, 39.32it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.13it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.28it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.61it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.05it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.96it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 38.49it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.85it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.21it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.05it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 38.58it/s] |

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.94it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.28it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.36it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.78it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.13it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.45it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.56it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/590 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/590 [00:00<00:15, 36.62it/s]

Computing ESM2 embeddings: 11%| | 64/590 [00:01<00:13, 38.39it/s]

Computing ESM2 embeddings: 16%| | 96/590 [00:02<00:12, 39.19it/s]

Computing ESM2 embeddings: 22%| | 128/590 [00:03<00:11, 38.52it/s]

Computing ESM2 embeddings: 27%| | 160/590 [00:04<00:11, 38.92it/s]

Computing ESM2 embeddings: 33%| | 192/590 [00:04<00:10, 39.38it/s]

Computing ESM2 embeddings: 38%| | 224/590 [00:05<00:09, 39.62it/s]

Computing ESM2 embeddings: 43%| | 256/590 [00:06<00:08, 38.34it/s]

Computing ESM2 embeddings: 49%| | 288/590 [00:07<00:07, 38.72it/s]

Computing ESM2 embeddings: 54%| | 320/590 [00:08<00:06, 39.20it/s]

Computing ESM2 embeddings: 60%| | 352/590 [00:09<00:06, 39.48it/s]

Computing ESM2 embeddings: 65%| | 384/590 [00:09<00:05, 38.89it/s]

Computing ESM2 embeddings: 71%| | 416/590 [00:10<00:04, 39.09it/s]



```

Computing ESM2 embeddings: 76%|      | 448/590 [00:11<00:03, 39.43it/s]
Computing ESM2 embeddings: 81%|      | 480/590 [00:12<00:02, 39.62it/s]
Computing ESM2 embeddings: 87%|      | 512/590 [00:13<00:02, 38.44it/s]
Computing ESM2 embeddings: 92%|      | 544/590 [00:13<00:01, 38.79it/s]
Computing ESM2 embeddings: 98%|      | 576/590 [00:14<00:00, 39.23it/s]

Computing ESM2 embeddings: 100%|     | 590/590 [00:15<00:00, 38.28it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/897 [00:00<00:25, 33.54it/s]
Computing ESM2 embeddings:  7%|      | 64/897 [00:01<00:22, 36.96it/s]
Computing ESM2 embeddings: 11%|      | 96/897 [00:02<00:20, 38.29it/s]
Computing ESM2 embeddings: 14%|      | 128/897 [00:03<00:20, 37.32it/s]
Computing ESM2 embeddings: 18%|      | 160/897 [00:04<00:19, 38.10it/s]
Computing ESM2 embeddings: 21%|      | 192/897 [00:05<00:18, 38.79it/s]
Computing ESM2 embeddings: 25%|      | 224/897 [00:05<00:17, 39.19it/s]
Computing ESM2 embeddings: 29%|      | 256/897 [00:06<00:16, 38.06it/s]
Computing ESM2 embeddings: 32%|      | 288/897 [00:07<00:15, 38.53it/s]
Computing ESM2 embeddings: 36%|      | 320/897 [00:08<00:14, 39.04it/s]
Computing ESM2 embeddings: 39%|      | 352/897 [00:09<00:13, 39.35it/s]
Computing ESM2 embeddings: 43%|      | 384/897 [00:10<00:13, 38.20it/s]
Computing ESM2 embeddings: 46%|      | 416/897 [00:10<00:12, 38.24it/s]
Computing ESM2 embeddings: 50%|      | 448/897 [00:11<00:11, 38.72it/s]

```

Computing ESM2 embeddings: 54%| | 480/897 [00:12<00:10, 39.10it/s]

Computing ESM2 embeddings: 57%| | 512/897 [00:13<00:09, 38.63it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:14<00:09, 38.91it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:14<00:08, 39.30it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:15<00:07, 39.53it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:16<00:06, 38.26it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:17<00:05, 38.56it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:18<00:04, 39.01it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:19<00:04, 39.30it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:19<00:03, 38.75it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:20<00:02, 38.96it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:21<00:01, 39.31it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:22<00:00, 39.15it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:23<00:00, 38.31it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:23, 38.28it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:21, 39.35it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:20, 39.79it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:20, 38.58it/s]

Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:19, 39.06it/s]

Computing ESM2 embeddings: 21%| | 192/915 [00:04<00:18, 39.40it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 24%  | 224/915 [00:05<00:17, 39.65it/s] |
| Computing ESM2 embeddings: 28%  | 256/915 [00:06<00:17, 38.23it/s] |
| Computing ESM2 embeddings: 31%  | 288/915 [00:07<00:16, 38.71it/s] |
| Computing ESM2 embeddings: 35%  | 320/915 [00:08<00:15, 39.07it/s] |
| Computing ESM2 embeddings: 38%  | 352/915 [00:08<00:14, 39.41it/s] |
| Computing ESM2 embeddings: 42%  | 384/915 [00:09<00:13, 38.18it/s] |
| Computing ESM2 embeddings: 45%  | 416/915 [00:10<00:13, 38.30it/s] |
| Computing ESM2 embeddings: 49%  | 448/915 [00:11<00:12, 38.64it/s] |
| Computing ESM2 embeddings: 52%  | 480/915 [00:12<00:11, 39.09it/s] |
| Computing ESM2 embeddings: 56%  | 512/915 [00:13<00:10, 38.47it/s] |
| Computing ESM2 embeddings: 59%  | 544/915 [00:13<00:09, 38.84it/s] |
| Computing ESM2 embeddings: 63%  | 576/915 [00:14<00:08, 39.15it/s] |
| Computing ESM2 embeddings: 66%  | 608/915 [00:15<00:07, 39.44it/s] |
| Computing ESM2 embeddings: 70%  | 640/915 [00:16<00:07, 38.29it/s] |
| Computing ESM2 embeddings: 73%  | 672/915 [00:17<00:06, 38.73it/s] |
| Computing ESM2 embeddings: 77%  | 704/915 [00:18<00:05, 39.08it/s] |
| Computing ESM2 embeddings: 80%  | 736/915 [00:18<00:04, 39.41it/s] |
| Computing ESM2 embeddings: 84%  | 768/915 [00:19<00:03, 38.19it/s] |
| Computing ESM2 embeddings: 87%  | 800/915 [00:20<00:02, 38.67it/s] |
| Computing ESM2 embeddings: 91%  | 832/915 [00:21<00:02, 39.06it/s] |
| Computing ESM2 embeddings: 94%  | 864/915 [00:22<00:01, 39.41it/s] |
| Computing ESM2 embeddings: 98%  | 896/915 [00:23<00:00, 38.33it/s] |
| Computing ESM2 embeddings: 100% | 915/915 [00:23<00:00, 38.77it/s] |

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|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:22, 38.15it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:21, 39.15it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:20, 39.72it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:20, 37.93it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 38.65it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:04<00:17, 39.14it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:16, 39.55it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:16, 38.32it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:15, 38.85it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:14, 39.23it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:13, 39.59it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:09<00:13, 38.36it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:10<00:12, 38.53it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:11<00:11, 38.87it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 39.32it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:09, 38.76it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:13<00:08, 39.12it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:14<00:08, 39.40it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:15<00:07, 39.69it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:16<00:06, 38.53it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:17<00:05, 38.95it/s] |

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:04, 39.28it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:18<00:04, 39.61it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:19<00:03, 38.91it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:20<00:02, 39.19it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:21<00:01, 39.44it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:22<00:00, 39.71it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:22<00:00, 38.92it/s]

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Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.97it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.73it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.91it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.40it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.86it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.33it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.57it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.38it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.57it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 39.03it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:14, 39.35it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.00it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.17it/s]

```

Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.47it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:10, 39.65it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.39it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:13<00:09, 38.73it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 39.17it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.42it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.29it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.64it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.97it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 39.24it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.23it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.57it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.01it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 39.29it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.12it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.85it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 39.03it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:20, 39.80it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 39.96it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 38.39it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:18, 38.85it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:17, 39.34it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:16, 39.59it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 38.85it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 39.11it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 39.46it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:08<00:13, 39.32it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:13, 38.00it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:12, 38.48it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 39.02it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 39.34it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:09, 38.74it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:13<00:09, 38.99it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 39.38it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 39.59it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 39.03it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 39.18it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:17<00:04, 39.52it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:18<00:04, 39.65it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 38.53it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 38.82it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:21<00:01, 39.25it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:22<00:00, 39.15it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:23<00:00, 38.91it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/910 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/910 [00:00<00:22, 38.53it/s]
Computing ESM2 embeddings: 7%|          | 64/910 [00:01<00:21, 39.01it/s]
Computing ESM2 embeddings: 11%|         | 96/910 [00:02<00:20, 39.54it/s]
Computing ESM2 embeddings: 14%|         | 128/910 [00:03<00:20, 38.11it/s]
Computing ESM2 embeddings: 18%|         | 160/910 [00:04<00:19, 38.75it/s]
Computing ESM2 embeddings: 21%|         | 192/910 [00:04<00:18, 39.21it/s]
Computing ESM2 embeddings: 25%|         | 224/910 [00:05<00:17, 39.57it/s]
Computing ESM2 embeddings: 28%|         | 256/910 [00:06<00:17, 38.43it/s]
Computing ESM2 embeddings: 32%|         | 288/910 [00:07<00:15, 38.89it/s]
Computing ESM2 embeddings: 35%|         | 320/910 [00:08<00:15, 39.24it/s]
Computing ESM2 embeddings: 39%|         | 352/910 [00:08<00:14, 39.56it/s]
Computing ESM2 embeddings: 42%|         | 384/910 [00:09<00:13, 38.90it/s]
Computing ESM2 embeddings: 46%|         | 416/910 [00:10<00:12, 38.87it/s]
Computing ESM2 embeddings: 49%|         | 448/910 [00:11<00:11, 39.08it/s]
Computing ESM2 embeddings: 53%|         | 480/910 [00:12<00:10, 39.44it/s]
Computing ESM2 embeddings: 56%|         | 512/910 [00:13<00:10, 38.29it/s]
Computing ESM2 embeddings: 60%|         | 544/910 [00:13<00:09, 38.76it/s]
Computing ESM2 embeddings: 63%|         | 576/910 [00:14<00:08, 39.11it/s]
Computing ESM2 embeddings: 67%|         | 608/910 [00:15<00:07, 39.46it/s]
Computing ESM2 embeddings: 70%|         | 640/910 [00:16<00:07, 38.24it/s]
```



```

Computing ESM2 embeddings: 74%|      | 672/910 [00:17<00:06, 38.69it/s]
Computing ESM2 embeddings: 77%|      | 704/910 [00:18<00:05, 39.10it/s]
Computing ESM2 embeddings: 81%|      | 736/910 [00:18<00:04, 39.45it/s]
Computing ESM2 embeddings: 84%|      | 768/910 [00:19<00:03, 38.75it/s]
Computing ESM2 embeddings: 88%|      | 800/910 [00:20<00:02, 39.09it/s]
Computing ESM2 embeddings: 91%|      | 832/910 [00:21<00:01, 39.35it/s]
Computing ESM2 embeddings: 95%|      | 864/910 [00:22<00:01, 39.63it/s]
Computing ESM2 embeddings: 98%|      | 896/910 [00:23<00:00, 38.41it/s]

Computing ESM2 embeddings: 100%|     | 910/910 [00:23<00:00, 38.90it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/900 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/900 [00:00<00:22, 38.33it/s]
Computing ESM2 embeddings:  7%|      | 64/900 [00:01<00:21, 39.06it/s]
Computing ESM2 embeddings: 11%|      | 96/900 [00:02<00:20, 39.65it/s]
Computing ESM2 embeddings: 14%|      | 128/900 [00:03<00:20, 38.12it/s]
Computing ESM2 embeddings: 18%|      | 160/900 [00:04<00:19, 38.75it/s]
Computing ESM2 embeddings: 21%|      | 192/900 [00:04<00:18, 39.20it/s]
Computing ESM2 embeddings: 25%|      | 224/900 [00:05<00:17, 39.57it/s]
Computing ESM2 embeddings: 28%|      | 256/900 [00:06<00:16, 38.85it/s]
Computing ESM2 embeddings: 32%|      | 288/900 [00:07<00:15, 39.18it/s]
Computing ESM2 embeddings: 36%|      | 320/900 [00:08<00:14, 39.46it/s]
Computing ESM2 embeddings: 39%|      | 352/900 [00:08<00:13, 39.72it/s]

```

```

Computing ESM2 embeddings: 43%|          | 384/900 [00:09<00:13, 38.50it/s]
Computing ESM2 embeddings: 46%|          | 416/900 [00:10<00:12, 38.62it/s]
Computing ESM2 embeddings: 50%|          | 448/900 [00:11<00:11, 38.91it/s]
Computing ESM2 embeddings: 53%|          | 480/900 [00:12<00:10, 39.32it/s]
Computing ESM2 embeddings: 57%|          | 512/900 [00:13<00:10, 38.62it/s]
Computing ESM2 embeddings: 60%|          | 544/900 [00:13<00:09, 39.00it/s]
Computing ESM2 embeddings: 64%|          | 576/900 [00:14<00:08, 39.32it/s]
Computing ESM2 embeddings: 68%|          | 608/900 [00:15<00:07, 39.62it/s]
Computing ESM2 embeddings: 71%|          | 640/900 [00:16<00:06, 38.90it/s]
Computing ESM2 embeddings: 75%|          | 672/900 [00:17<00:05, 39.19it/s]
Computing ESM2 embeddings: 78%|          | 704/900 [00:17<00:04, 39.43it/s]
Computing ESM2 embeddings: 82%|          | 736/900 [00:18<00:04, 39.69it/s]
Computing ESM2 embeddings: 85%|          | 768/900 [00:19<00:03, 39.00it/s]
Computing ESM2 embeddings: 89%|          | 800/900 [00:20<00:02, 39.28it/s]
Computing ESM2 embeddings: 92%|          | 832/900 [00:21<00:01, 39.51it/s]
Computing ESM2 embeddings: 96%|          | 864/900 [00:22<00:00, 39.55it/s]
Computing ESM2 embeddings: 100%|         | 896/900 [00:22<00:00, 37.97it/s]

```

```

Computing ESM2 embeddings: 100%|         | 900/900 [00:23<00:00, 38.88it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']

```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```

Computing ESM2 embeddings:  0%|          |  0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|          | 32/572 [00:00<00:14, 37.76it/s]
Computing ESM2 embeddings: 11%|          | 64/572 [00:01<00:14, 34.73it/s]

```

```

Computing ESM2 embeddings: 17%|          | 96/572 [00:02<00:12, 36.94it/s]
Computing ESM2 embeddings: 22%|          | 128/572 [00:03<00:12, 36.24it/s]
Computing ESM2 embeddings: 28%|          | 160/572 [00:04<00:10, 37.46it/s]
Computing ESM2 embeddings: 34%|          | 192/572 [00:05<00:09, 38.25it/s]
Computing ESM2 embeddings: 39%|          | 224/572 [00:05<00:08, 38.85it/s]
Computing ESM2 embeddings: 45%|          | 256/572 [00:06<00:08, 38.23it/s]
Computing ESM2 embeddings: 50%|          | 288/572 [00:07<00:07, 38.61it/s]
Computing ESM2 embeddings: 56%|          | 320/572 [00:08<00:06, 39.02it/s]
Computing ESM2 embeddings: 62%|          | 352/572 [00:09<00:05, 39.36it/s]
Computing ESM2 embeddings: 67%|          | 384/572 [00:10<00:04, 37.93it/s]
Computing ESM2 embeddings: 73%|          | 416/572 [00:10<00:04, 38.47it/s]
Computing ESM2 embeddings: 78%|          | 448/572 [00:11<00:03, 38.53it/s]
Computing ESM2 embeddings: 84%|          | 480/572 [00:12<00:02, 38.89it/s]
Computing ESM2 embeddings: 90%|          | 512/572 [00:13<00:01, 38.09it/s]
Computing ESM2 embeddings: 95%|          | 544/572 [00:14<00:00, 38.60it/s]

Computing ESM2 embeddings: 100%|         | 572/572 [00:14<00:00, 38.20it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/916 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|          | 32/916 [00:00<00:22, 38.44it/s]
Computing ESM2 embeddings:  7%|          | 64/916 [00:01<00:21, 39.37it/s]
Computing ESM2 embeddings: 10%|          | 96/916 [00:02<00:20, 39.71it/s]
Computing ESM2 embeddings: 14%|          | 128/916 [00:03<00:20, 38.27it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 17% | 160/916 [00:04<00:19, 38.74it/s] |
| Computing ESM2 embeddings: 21% | 192/916 [00:04<00:18, 39.29it/s] |
| Computing ESM2 embeddings: 24% | 224/916 [00:05<00:17, 39.57it/s] |
| Computing ESM2 embeddings: 28% | 256/916 [00:06<00:17, 38.37it/s] |
| Computing ESM2 embeddings: 31% | 288/916 [00:07<00:16, 38.68it/s] |
| Computing ESM2 embeddings: 35% | 320/916 [00:08<00:15, 39.17it/s] |
| Computing ESM2 embeddings: 38% | 352/916 [00:09<00:14, 39.38it/s] |
| Computing ESM2 embeddings: 42% | 384/916 [00:09<00:13, 38.79it/s] |
| Computing ESM2 embeddings: 45% | 416/916 [00:10<00:12, 39.02it/s] |
| Computing ESM2 embeddings: 49% | 448/916 [00:11<00:11, 39.40it/s] |
| Computing ESM2 embeddings: 52% | 480/916 [00:12<00:11, 39.60it/s] |
| Computing ESM2 embeddings: 56% | 512/916 [00:13<00:10, 38.43it/s] |
| Computing ESM2 embeddings: 59% | 544/916 [00:13<00:09, 38.53it/s] |
| Computing ESM2 embeddings: 63% | 576/916 [00:14<00:08, 38.92it/s] |
| Computing ESM2 embeddings: 66% | 608/916 [00:15<00:07, 39.27it/s] |
| Computing ESM2 embeddings: 70% | 640/916 [00:16<00:07, 38.70it/s] |
| Computing ESM2 embeddings: 73% | 672/916 [00:17<00:06, 38.95it/s] |
| Computing ESM2 embeddings: 77% | 704/916 [00:18<00:05, 39.33it/s] |
| Computing ESM2 embeddings: 80% | 736/916 [00:18<00:04, 39.56it/s] |
| Computing ESM2 embeddings: 84% | 768/916 [00:19<00:03, 38.85it/s] |
| Computing ESM2 embeddings: 87% | 800/916 [00:20<00:02, 39.08it/s] |
| Computing ESM2 embeddings: 91% | 832/916 [00:21<00:02, 39.44it/s] |
| Computing ESM2 embeddings: 94% | 864/916 [00:22<00:01, 39.62it/s] |
| Computing ESM2 embeddings: 98% | 896/916 [00:22<00:00, 38.44it/s] |

Computing ESM2 embeddings: 100%| | 916/916 [00:23<00:00, 38.95it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 39.26it/s]  
 Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:20, 39.91it/s]  
 Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 40.00it/s]  
 Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 38.42it/s]  
 Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 38.84it/s]  
 Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 39.32it/s]  
 Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:16, 39.57it/s]  
 Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 38.95it/s]  
 Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 39.15it/s]  
 Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 39.51it/s]  
 Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:13, 39.64it/s]  
 Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 38.61it/s]  
 Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.89it/s]  
 Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.33it/s]  
 Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.55it/s]  
 Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 39.00it/s]  
 Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.16it/s]  
 Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.50it/s]  
 Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.66it/s]

```

Computing ESM2 embeddings: 71%|      | 640/896 [00:16<00:06, 39.03it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 39.16it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:17<00:04, 39.50it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:18<00:04, 39.55it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 38.34it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 38.72it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.03it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 39.32it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:22<00:00, 38.98it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/914 [00:00<00:23, 37.97it/s]
Computing ESM2 embeddings: 7%|      | 64/914 [00:01<00:21, 38.75it/s]
Computing ESM2 embeddings: 11%|     | 96/914 [00:02<00:20, 39.45it/s]
Computing ESM2 embeddings: 14%|     | 128/914 [00:03<00:20, 37.99it/s]
Computing ESM2 embeddings: 18%|     | 160/914 [00:04<00:19, 38.66it/s]
Computing ESM2 embeddings: 21%|     | 192/914 [00:04<00:18, 39.10it/s]
Computing ESM2 embeddings: 25%|     | 224/914 [00:05<00:17, 39.46it/s]
Computing ESM2 embeddings: 28%|     | 256/914 [00:06<00:17, 38.16it/s]
Computing ESM2 embeddings: 32%|     | 288/914 [00:07<00:16, 38.68it/s]
Computing ESM2 embeddings: 35%|     | 320/914 [00:08<00:15, 39.08it/s]
Computing ESM2 embeddings: 39%|     | 352/914 [00:09<00:14, 39.04it/s]

```

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:14, 37.70it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:13, 38.30it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 38.80it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 39.21it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.30it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.77it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.13it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.45it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.17it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.66it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 39.05it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.40it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.14it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.65it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.06it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 39.39it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.17it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.64it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.21it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.01it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:20, 39.60it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 38.15it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:18, 38.76it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:04<00:17, 39.20it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:16, 39.55it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 38.88it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 38.91it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 39.18it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:08<00:13, 39.50it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:13, 38.47it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 38.89it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 39.24it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 39.53it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 38.43it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:13<00:09, 38.85it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 39.20it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 39.51it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 38.79it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 39.10it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:04, 39.37it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:18<00:04, 39.63it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:19<00:03, 38.59it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 38.94it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 39.27it/s] |



Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.56it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.95it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.44it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.78it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.97it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.33it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.79it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.28it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.55it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.41it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.75it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 39.23it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:08<00:14, 39.51it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.88it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.10it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.46it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:10, 39.67it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.91it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.13it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.48it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.66it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.98it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.73it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:17<00:05, 39.07it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.36it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.82it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.04it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.42it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 39.61it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 38.98it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 39.15it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 39.17it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:20, 39.62it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.96it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 38.18it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 38.82it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 39.28it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:16, 39.62it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 38.30it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 38.75it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 39.18it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:13, 39.18it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 37.36it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.09it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.67it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.14it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 38.61it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.96it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.30it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.60it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.86it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 39.16it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 39.44it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.69it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.56it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.92it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.24it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.55it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.78it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.24it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:23, 35.57it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 37.35it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 38.01it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.66it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:18, 39.06it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 39.36it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 38.24it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.65it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 39.11it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 39.35it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:13, 38.20it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 38.55it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.01it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 39.30it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 38.69it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 38.93it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.27it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.47it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 38.29it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 38.65it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 39.08it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.36it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 38.64it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 38.87it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.24it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.45it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.17it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.61it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/572 [00:00<00:15, 34.27it/s]
Computing ESM2 embeddings: 11%|      | 64/572 [00:01<00:13, 37.31it/s]
Computing ESM2 embeddings: 17%|      | 96/572 [00:02<00:12, 38.49it/s]
Computing ESM2 embeddings: 22%|      | 128/572 [00:03<00:11, 37.42it/s]
Computing ESM2 embeddings: 28%|      | 160/572 [00:04<00:10, 38.16it/s]
Computing ESM2 embeddings: 34%|      | 192/572 [00:05<00:09, 38.83it/s]
Computing ESM2 embeddings: 39%|      | 224/572 [00:05<00:08, 39.21it/s]
Computing ESM2 embeddings: 45%|      | 256/572 [00:06<00:08, 38.57it/s]
Computing ESM2 embeddings: 50%|      | 288/572 [00:07<00:07, 38.87it/s]
Computing ESM2 embeddings: 56%|      | 320/572 [00:08<00:06, 39.26it/s]
Computing ESM2 embeddings: 62%|      | 352/572 [00:09<00:05, 39.50it/s]
Computing ESM2 embeddings: 67%|      | 384/572 [00:09<00:04, 38.23it/s]
Computing ESM2 embeddings: 73%|      | 416/572 [00:10<00:04, 38.61it/s]
Computing ESM2 embeddings: 78%|      | 448/572 [00:11<00:03, 39.05it/s]
Computing ESM2 embeddings: 84%|      | 480/572 [00:12<00:02, 39.32it/s]
Computing ESM2 embeddings: 90%|      | 512/572 [00:13<00:01, 38.70it/s]

```

Computing ESM2 embeddings: 95%| | 544/572 [00:14<00:00, 38.93it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:14<00:00, 38.65it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/916 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/916 [00:00<00:26, 32.84it/s]

Computing ESM2 embeddings: 7%| | 64/916 [00:01<00:23, 36.66it/s]

Computing ESM2 embeddings: 10%| | 96/916 [00:02<00:21, 38.21it/s]

Computing ESM2 embeddings: 14%| | 128/916 [00:03<00:21, 37.26it/s]

Computing ESM2 embeddings: 17%| | 160/916 [00:04<00:19, 38.12it/s]

Computing ESM2 embeddings: 21%| | 192/916 [00:05<00:18, 38.72it/s]

Computing ESM2 embeddings: 24%| | 224/916 [00:05<00:17, 39.20it/s]

Computing ESM2 embeddings: 28%| | 256/916 [00:06<00:17, 38.20it/s]

Computing ESM2 embeddings: 31%| | 288/916 [00:07<00:16, 38.70it/s]

Computing ESM2 embeddings: 35%| | 320/916 [00:08<00:15, 39.08it/s]

Computing ESM2 embeddings: 38%| | 352/916 [00:09<00:14, 39.40it/s]

Computing ESM2 embeddings: 42%| | 384/916 [00:10<00:13, 38.13it/s]

Computing ESM2 embeddings: 45%| | 416/916 [00:10<00:12, 38.63it/s]

Computing ESM2 embeddings: 49%| | 448/916 [00:11<00:11, 39.02it/s]

Computing ESM2 embeddings: 52%| | 480/916 [00:12<00:11, 39.36it/s]

Computing ESM2 embeddings: 56%| | 512/916 [00:13<00:10, 38.35it/s]

Computing ESM2 embeddings: 59%| | 544/916 [00:14<00:09, 38.78it/s]

Computing ESM2 embeddings: 63%| | 576/916 [00:14<00:08, 39.15it/s]

```

Computing ESM2 embeddings: 66%|      | 608/916 [00:15<00:07, 39.48it/s]
Computing ESM2 embeddings: 70%|      | 640/916 [00:16<00:07, 38.29it/s]
Computing ESM2 embeddings: 73%|      | 672/916 [00:17<00:06, 38.73it/s]
Computing ESM2 embeddings: 77%|      | 704/916 [00:18<00:05, 39.03it/s]
Computing ESM2 embeddings: 80%|      | 736/916 [00:19<00:04, 39.35it/s]
Computing ESM2 embeddings: 84%|      | 768/916 [00:19<00:03, 38.21it/s]
Computing ESM2 embeddings: 87%|      | 800/916 [00:20<00:03, 38.66it/s]
Computing ESM2 embeddings: 91%|      | 832/916 [00:21<00:02, 39.02it/s]
Computing ESM2 embeddings: 94%|      | 864/916 [00:22<00:01, 39.35it/s]
Computing ESM2 embeddings: 98%|      | 896/916 [00:23<00:00, 38.41it/s]

Computing ESM2 embeddings: 100%|     | 916/916 [00:23<00:00, 38.48it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.05it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 39.13it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 39.69it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 38.19it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 38.52it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:18, 38.46it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 38.90it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 38.07it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 38.62it/s]

```

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 39.08it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:13, 39.46it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 38.27it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.75it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.15it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.49it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 38.31it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.77it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.14it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.48it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.46it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.86it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 39.19it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.49it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.79it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 39.10it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.36it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.62it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.76it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.08it/s]



|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.79it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.96it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.40it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.85it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 39.35it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.54it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 38.37it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.76it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 39.23it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:08<00:14, 39.47it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 38.84it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.06it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.41it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:10, 39.60it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.43it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 38.82it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.24it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.48it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 38.37it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 38.75it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 39.17it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:18<00:04, 39.31it/s] |
| Computing ESM2 embeddings: | 84% | 768/914 [00:19<00:03, 38.71it/s] |
| Computing ESM2 embeddings: | 88% | 800/914 [00:20<00:02, 38.96it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.35it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.56it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 38.40it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.98it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 37.97it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 39.14it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.48it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 38.08it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 38.63it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:17, 39.15it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 38.24it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 38.75it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 39.13it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 39.50it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 38.37it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:09<00:13, 38.81it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 39.14it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 39.47it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 38.40it/s]
Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:09, 38.83it/s]

```

```

Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:08, 39.20it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 39.52it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.39it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.68it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 39.04it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:04, 39.37it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 38.18it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.63it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.98it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.34it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 38.61it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.81it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 38.15it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 39.12it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:20, 39.59it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:20, 38.20it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 38.72it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:04<00:18, 39.27it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:17, 39.56it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 38.29it/s]

```

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.69it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.99it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.68it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.42it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.91it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.32it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.69it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.80it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 39.03it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.39it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.72it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 39.06it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.39it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 39.66it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 38.51it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.92it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.26it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.57it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.30it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.75it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.88it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.77it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.60it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.86it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 38.33it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 38.51it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.99it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 39.33it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.74it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.04it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.45it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 39.65it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.48it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.83it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.01it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.21it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 38.07it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:09, 38.51it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.02it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.38it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.36it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 38.72it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:04, 39.16it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:18<00:04, 39.44it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 38.26it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 38.66it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.14it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 39.44it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 38.76it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.92it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 38.81it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:20, 39.43it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:20, 37.95it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 38.60it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:04<00:18, 38.68it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 39.06it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 38.37it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 38.80it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 39.14it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 39.35it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 38.33it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 38.77it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 39.10it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 39.40it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.62it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.93it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.19it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 39.46it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.13it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.58it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.94it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.29it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.04it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.54it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.96it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 39.30it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.13it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.68it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/590 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/590 [00:00<00:14, 38.74it/s]

Computing ESM2 embeddings: 11%| | 64/590 [00:01<00:13, 39.27it/s]

Computing ESM2 embeddings: 16%| | 96/590 [00:02<00:12, 39.69it/s]

Computing ESM2 embeddings: 22%| | 128/590 [00:03<00:12, 37.96it/s]

Computing ESM2 embeddings: 27%| | 160/590 [00:04<00:11, 38.65it/s]

Computing ESM2 embeddings: 33%| | 192/590 [00:04<00:10, 39.20it/s]

Computing ESM2 embeddings: 38%| | 224/590 [00:05<00:09, 39.53it/s]

Computing ESM2 embeddings: 43%| | 256/590 [00:06<00:08, 38.30it/s]

Computing ESM2 embeddings: 49%| | 288/590 [00:07<00:07, 38.84it/s]

Computing ESM2 embeddings: 54%| | 320/590 [00:08<00:06, 39.22it/s]

Computing ESM2 embeddings: 60%| | 352/590 [00:09<00:06, 39.53it/s]

Computing ESM2 embeddings: 65%| | 384/590 [00:09<00:05, 38.59it/s]

Computing ESM2 embeddings: 71%| | 416/590 [00:10<00:04, 38.93it/s]

Computing ESM2 embeddings: 76%| | 448/590 [00:11<00:03, 39.27it/s]

Computing ESM2 embeddings: 81%| | 480/590 [00:12<00:02, 39.56it/s]

Computing ESM2 embeddings: 87%| | 512/590 [00:13<00:02, 38.36it/s]

Computing ESM2 embeddings: 92%| | 544/590 [00:13<00:01, 38.78it/s]

Computing ESM2 embeddings: 98%| | 576/590 [00:14<00:00, 39.16it/s]

Computing ESM2 embeddings: 100%| | 590/590 [00:15<00:00, 38.87it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:00<00:22, 37.98it/s]

Computing ESM2 embeddings: 7%| | 64/897 [00:01<00:21, 38.90it/s]

Computing ESM2 embeddings: 11%| | 96/897 [00:02<00:20, 39.48it/s]

Computing ESM2 embeddings: 14%| | 128/897 [00:03<00:20, 38.00it/s]

Computing ESM2 embeddings: 18%| | 160/897 [00:04<00:19, 38.65it/s]

Computing ESM2 embeddings: 21%| | 192/897 [00:04<00:18, 39.08it/s]

Computing ESM2 embeddings: 25%| | 224/897 [00:05<00:17, 39.44it/s]



```

Computing ESM2 embeddings: 29%|          | 256/897 [00:06<00:16, 38.17it/s]
Computing ESM2 embeddings: 32%|          | 288/897 [00:07<00:15, 38.67it/s]
Computing ESM2 embeddings: 36%|          | 320/897 [00:08<00:14, 39.06it/s]
Computing ESM2 embeddings: 39%|          | 352/897 [00:09<00:13, 39.40it/s]
Computing ESM2 embeddings: 43%|          | 384/897 [00:09<00:13, 38.18it/s]
Computing ESM2 embeddings: 46%|          | 416/897 [00:10<00:12, 38.66it/s]
Computing ESM2 embeddings: 50%|          | 448/897 [00:11<00:11, 39.04it/s]
Computing ESM2 embeddings: 54%|          | 480/897 [00:12<00:10, 39.37it/s]
Computing ESM2 embeddings: 57%|          | 512/897 [00:13<00:10, 38.13it/s]
Computing ESM2 embeddings: 61%|          | 544/897 [00:14<00:09, 38.62it/s]
Computing ESM2 embeddings: 64%|          | 576/897 [00:14<00:08, 39.01it/s]
Computing ESM2 embeddings: 68%|          | 608/897 [00:15<00:07, 39.34it/s]
Computing ESM2 embeddings: 71%|          | 640/897 [00:16<00:06, 38.34it/s]
Computing ESM2 embeddings: 75%|          | 672/897 [00:17<00:05, 38.42it/s]
Computing ESM2 embeddings: 78%|          | 704/897 [00:18<00:04, 38.71it/s]
Computing ESM2 embeddings: 82%|          | 736/897 [00:18<00:04, 39.12it/s]
Computing ESM2 embeddings: 86%|          | 768/897 [00:19<00:03, 37.94it/s]
Computing ESM2 embeddings: 89%|          | 800/897 [00:20<00:02, 38.44it/s]
Computing ESM2 embeddings: 93%|          | 832/897 [00:21<00:01, 38.86it/s]
Computing ESM2 embeddings: 96%|          | 864/897 [00:22<00:00, 39.20it/s]

```

```

Computing ESM2 embeddings: 100%|         | 897/897 [00:23<00:00, 38.52it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.07it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.82it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 40.00it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.49it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.91it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 39.40it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.67it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 38.35it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.37it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.83it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 39.18it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 38.70it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 38.95it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.35it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:10, 39.58it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.94it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 39.16it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.49it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 39.66it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 39.02it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 39.20it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:17<00:05, 39.50it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:18<00:04, 39.65it/s] |

```

Computing ESM2 embeddings: 84%|      | 768/914 [00:19<00:03, 39.09it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 39.35it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.63it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.74it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 39.13it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.12it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.26it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:22, 37.54it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 38.50it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 37.27it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 38.00it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:18, 38.71it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 39.07it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 37.89it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 38.34it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 38.72it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:13, 39.07it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:09<00:13, 38.56it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 38.80it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 39.20it/s]

```

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.41it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 38.81it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.96it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.32it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.48it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.24it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.19it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.62it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.99it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.04it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.45it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.96it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.24it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.50it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.76it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.20it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.45it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.89it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.43it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.68it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 38.36it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 38.76it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.93it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 39.16it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.66it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.95it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 39.33it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.56it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.42it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.77it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:13<00:09, 39.22it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 39.44it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:08, 38.25it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.63it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 39.12it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 39.40it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 38.84it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 39.05it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.42it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.62it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 39.02it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:22<00:00, 39.20it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.99it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.40it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.11it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 39.70it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 38.13it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 38.78it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:17, 39.17it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:16, 39.54it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.83it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 39.18it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.45it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:08<00:13, 39.73it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.49it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.90it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.26it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 39.58it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 38.53it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:13<00:09, 38.94it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.25it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 39.52it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.31it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 38.46it/s] |

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.80it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.15it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 37.93it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.05it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.23it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.36it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.61it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 35.33it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.82it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.74it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 37.73it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.32it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:18, 38.93it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 39.24it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.12it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.12it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.63it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 39.02it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:13, 37.99it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.43it/s]

```

Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 38.93it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 39.20it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.03it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 38.42it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 38.90it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.21it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.61it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.77it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 39.10it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 39.26it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.66it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.83it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.14it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 39.30it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.59it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.61it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/895 [00:00<00:22, 37.70it/s]
Computing ESM2 embeddings:  7%|          | 64/895 [00:01<00:21, 38.67it/s]
Computing ESM2 embeddings: 11%|          | 96/895 [00:02<00:20, 39.12it/s]
Computing ESM2 embeddings: 14%|          | 128/895 [00:03<00:20, 38.04it/s]

```



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/895 [00:04<00:19, 38.59it/s] |
| Computing ESM2 embeddings: 21%  | 192/895 [00:04<00:17, 39.09it/s] |
| Computing ESM2 embeddings: 25%  | 224/895 [00:05<00:17, 39.24it/s] |
| Computing ESM2 embeddings: 29%  | 256/895 [00:06<00:16, 38.20it/s] |
| Computing ESM2 embeddings: 32%  | 288/895 [00:07<00:15, 38.61it/s] |
| Computing ESM2 embeddings: 36%  | 320/895 [00:08<00:14, 39.06it/s] |
| Computing ESM2 embeddings: 39%  | 352/895 [00:09<00:13, 39.31it/s] |
| Computing ESM2 embeddings: 43%  | 384/895 [00:09<00:13, 38.15it/s] |
| Computing ESM2 embeddings: 46%  | 416/895 [00:10<00:12, 38.54it/s] |
| Computing ESM2 embeddings: 50%  | 448/895 [00:11<00:11, 39.01it/s] |
| Computing ESM2 embeddings: 54%  | 480/895 [00:12<00:10, 39.26it/s] |
| Computing ESM2 embeddings: 57%  | 512/895 [00:13<00:10, 38.21it/s] |
| Computing ESM2 embeddings: 61%  | 544/895 [00:14<00:09, 38.62it/s] |
| Computing ESM2 embeddings: 64%  | 576/895 [00:14<00:08, 39.05it/s] |
| Computing ESM2 embeddings: 68%  | 608/895 [00:15<00:07, 39.31it/s] |
| Computing ESM2 embeddings: 72%  | 640/895 [00:16<00:06, 38.30it/s] |
| Computing ESM2 embeddings: 75%  | 672/895 [00:17<00:05, 38.66it/s] |
| Computing ESM2 embeddings: 79%  | 704/895 [00:18<00:04, 39.08it/s] |
| Computing ESM2 embeddings: 82%  | 736/895 [00:18<00:04, 39.35it/s] |
| Computing ESM2 embeddings: 86%  | 768/895 [00:19<00:03, 38.75it/s] |
| Computing ESM2 embeddings: 89%  | 800/895 [00:20<00:02, 38.97it/s] |
| Computing ESM2 embeddings: 93%  | 832/895 [00:21<00:01, 39.31it/s] |
| Computing ESM2 embeddings: 97%  | 864/895 [00:22<00:00, 39.52it/s] |
| Computing ESM2 embeddings: 100% | 895/895 [00:23<00:00, 38.73it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.07it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:23, 35.61it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.46it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 36.70it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.81it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 38.55it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 39.11it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.48it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 38.92it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 38.97it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:13, 39.26it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 38.17it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.43it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.77it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.41it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 38.68it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:08, 39.11it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 39.39it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.27it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.68it/s] |

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.83it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 39.05it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.09it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.53it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 39.06it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.36it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.73it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.54it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.41it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.28it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.84it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.59it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.15it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.53it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 38.28it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.71it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.93it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 39.32it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.27it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.75it/s]

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Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 39.14it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.46it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.76it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.84it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:13<00:09, 39.06it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 39.41it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 38.76it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 39.10it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 39.40it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 39.67it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 38.51it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.89it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 39.24it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.55it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 38.86it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:22<00:00, 39.13it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.98it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.29it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.23it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 39.66it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 38.34it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:18, 38.82it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:04<00:17, 39.32it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:16, 39.59it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 38.97it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 39.16it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 39.48it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:08<00:13, 39.67it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:13, 39.06it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 38.98it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 39.24it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 39.42it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 38.85it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:13<00:09, 39.07it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 39.42it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 39.62it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 38.43it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 38.41it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:04, 38.83it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:18<00:04, 39.18it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:19<00:03, 38.30it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 38.67it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 39.13it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:22<00:00, 39.40it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.93it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:23, 37.95it/s]  
 Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:21, 39.05it/s]  
 Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:20, 39.64it/s]  
 Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:20, 38.22it/s]  
 Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:19, 38.83it/s]  
 Computing ESM2 embeddings: 21%| | 192/913 [00:04<00:18, 39.23it/s]  
 Computing ESM2 embeddings: 25%| | 224/913 [00:05<00:17, 39.59it/s]  
 Computing ESM2 embeddings: 28%| | 256/913 [00:06<00:17, 38.31it/s]  
 Computing ESM2 embeddings: 32%| | 288/913 [00:07<00:16, 38.50it/s]  
 Computing ESM2 embeddings: 35%| | 320/913 [00:08<00:15, 38.81it/s]  
 Computing ESM2 embeddings: 39%| | 352/913 [00:09<00:14, 39.23it/s]  
 Computing ESM2 embeddings: 42%| | 384/913 [00:09<00:13, 38.03it/s]  
 Computing ESM2 embeddings: 46%| | 416/913 [00:10<00:12, 38.57it/s]  
 Computing ESM2 embeddings: 49%| | 448/913 [00:11<00:11, 39.01it/s]  
 Computing ESM2 embeddings: 53%| | 480/913 [00:12<00:10, 39.38it/s]  
 Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 38.33it/s]  
 Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 38.80it/s]  
 Computing ESM2 embeddings: 63%| | 576/913 [00:14<00:08, 39.15it/s]  
 Computing ESM2 embeddings: 67%| | 608/913 [00:15<00:07, 39.49it/s]

```

Computing ESM2 embeddings: 70%|      | 640/913 [00:16<00:07, 38.78it/s]
Computing ESM2 embeddings: 74%|      | 672/913 [00:17<00:06, 39.08it/s]
Computing ESM2 embeddings: 77%|      | 704/913 [00:18<00:05, 39.37it/s]
Computing ESM2 embeddings: 81%|      | 736/913 [00:18<00:04, 39.65it/s]
Computing ESM2 embeddings: 84%|      | 768/913 [00:19<00:03, 38.94it/s]
Computing ESM2 embeddings: 88%|      | 800/913 [00:20<00:02, 39.22it/s]
Computing ESM2 embeddings: 91%|      | 832/913 [00:21<00:02, 39.47it/s]
Computing ESM2 embeddings: 95%|      | 864/913 [00:22<00:01, 39.72it/s]
Computing ESM2 embeddings: 98%|      | 896/913 [00:23<00:00, 38.42it/s]

Computing ESM2 embeddings: 100%|     | 913/913 [00:23<00:00, 38.86it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/896 [00:00<00:21, 39.39it/s]
Computing ESM2 embeddings: 7%|      | 64/896 [00:01<00:20, 39.71it/s]
Computing ESM2 embeddings: 11%|     | 96/896 [00:02<00:20, 39.97it/s]
Computing ESM2 embeddings: 14%|     | 128/896 [00:03<00:20, 38.39it/s]
Computing ESM2 embeddings: 18%|     | 160/896 [00:04<00:18, 38.93it/s]
Computing ESM2 embeddings: 21%|     | 192/896 [00:04<00:17, 39.31it/s]
Computing ESM2 embeddings: 25%|     | 224/896 [00:05<00:16, 39.64it/s]
Computing ESM2 embeddings: 29%|     | 256/896 [00:06<00:16, 38.87it/s]
Computing ESM2 embeddings: 32%|     | 288/896 [00:07<00:15, 39.20it/s]
Computing ESM2 embeddings: 36%|     | 320/896 [00:08<00:14, 39.47it/s]

```

Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:13, 39.73it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 38.49it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.90it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.25it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.56it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 38.39it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:09, 38.81it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.17it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.47it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.16it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.66it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 39.07it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.42it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.71it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 39.03it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.32it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.60it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 39.01it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.26it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.25it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:20, 39.66it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 38.15it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.69it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 39.26it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 39.54it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 38.88it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.68it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 39.05it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 39.32it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 38.77it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 38.99it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.38it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:10, 39.57it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 39.10it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.37it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.64it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.70it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 38.32it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 38.68it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 39.13it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.35it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 38.83it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 39.03it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:21<00:02, 39.42it/s] |

```

Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.60it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 38.82it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 39.01it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/678 [00:00<?, ?it/s]
Computing ESM2 embeddings: 5%|        | 32/678 [00:00<00:16, 38.03it/s]
Computing ESM2 embeddings: 9%|        | 64/678 [00:01<00:18, 34.05it/s]
Computing ESM2 embeddings: 14%|       | 96/678 [00:02<00:16, 36.09it/s]
Computing ESM2 embeddings: 19%|       | 128/678 [00:03<00:15, 35.83it/s]
Computing ESM2 embeddings: 24%|       | 160/678 [00:04<00:13, 37.07it/s]
Computing ESM2 embeddings: 28%|       | 192/678 [00:05<00:12, 38.04it/s]
Computing ESM2 embeddings: 33%|       | 224/678 [00:05<00:11, 38.65it/s]
Computing ESM2 embeddings: 38%|       | 256/678 [00:06<00:11, 38.24it/s]
Computing ESM2 embeddings: 42%|       | 288/678 [00:07<00:10, 38.59it/s]
Computing ESM2 embeddings: 47%|       | 320/678 [00:08<00:09, 39.07it/s]
Computing ESM2 embeddings: 52%|       | 352/678 [00:09<00:08, 39.33it/s]
Computing ESM2 embeddings: 57%|       | 384/678 [00:10<00:07, 38.22it/s]
Computing ESM2 embeddings: 61%|       | 416/678 [00:10<00:06, 38.59it/s]
Computing ESM2 embeddings: 66%|       | 448/678 [00:11<00:05, 39.06it/s]
Computing ESM2 embeddings: 71%|       | 480/678 [00:12<00:05, 39.25it/s]
Computing ESM2 embeddings: 76%|       | 512/678 [00:13<00:04, 38.18it/s]
Computing ESM2 embeddings: 80%|       | 544/678 [00:14<00:03, 38.20it/s]

```

```

Computing ESM2 embeddings: 85%|      | 576/678 [00:15<00:02, 38.69it/s]
Computing ESM2 embeddings: 90%|      | 608/678 [00:15<00:01, 39.04it/s]
Computing ESM2 embeddings: 94%|      | 640/678 [00:16<00:01, 37.82it/s]
Computing ESM2 embeddings: 99%|      | 672/678 [00:17<00:00, 38.32it/s]
Computing ESM2 embeddings: 100%|     | 678/678 [00:17<00:00, 38.09it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/809 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/809 [00:00<00:22, 33.89it/s]
Computing ESM2 embeddings:  8%|      | 64/809 [00:01<00:20, 37.02it/s]
Computing ESM2 embeddings: 12%|      | 96/809 [00:02<00:18, 38.38it/s]
Computing ESM2 embeddings: 16%|      | 128/809 [00:03<00:18, 37.16it/s]
Computing ESM2 embeddings: 20%|      | 160/809 [00:04<00:17, 38.05it/s]
Computing ESM2 embeddings: 24%|      | 192/809 [00:05<00:15, 38.70it/s]
Computing ESM2 embeddings: 28%|      | 224/809 [00:05<00:14, 39.21it/s]
Computing ESM2 embeddings: 32%|      | 256/809 [00:06<00:14, 38.16it/s]
Computing ESM2 embeddings: 36%|      | 288/809 [00:07<00:13, 38.45it/s]
Computing ESM2 embeddings: 40%|      | 320/809 [00:08<00:12, 38.75it/s]
Computing ESM2 embeddings: 44%|      | 352/809 [00:09<00:11, 39.20it/s]
Computing ESM2 embeddings: 47%|      | 384/809 [00:10<00:11, 38.14it/s]
Computing ESM2 embeddings: 51%|      | 416/809 [00:10<00:10, 38.65it/s]
Computing ESM2 embeddings: 55%|      | 448/809 [00:11<00:09, 38.76it/s]
Computing ESM2 embeddings: 59%|      | 480/809 [00:12<00:08, 39.11it/s]

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Computing ESM2 embeddings: 63%| | 512/809 [00:13<00:07, 38.05it/s]

Computing ESM2 embeddings: 67%| | 544/809 [00:14<00:06, 38.50it/s]

Computing ESM2 embeddings: 71%| | 576/809 [00:14<00:05, 38.94it/s]

Computing ESM2 embeddings: 75%| | 608/809 [00:15<00:05, 39.31it/s]

Computing ESM2 embeddings: 79%| | 640/809 [00:16<00:04, 38.59it/s]

Computing ESM2 embeddings: 83%| | 672/809 [00:17<00:03, 38.95it/s]

Computing ESM2 embeddings: 87%| | 704/809 [00:18<00:02, 39.23it/s]

Computing ESM2 embeddings: 91%| | 736/809 [00:19<00:01, 39.51it/s]

Computing ESM2 embeddings: 95%| | 768/809 [00:19<00:01, 38.33it/s]

Computing ESM2 embeddings: 99%| | 800/809 [00:20<00:00, 38.75it/s]

Computing ESM2 embeddings: 100%| | 809/809 [00:21<00:00, 38.52it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.09it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.10it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.55it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 38.19it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 38.84it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:17, 39.38it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:16, 39.62it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 38.86it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 39.02it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 39.34it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:08<00:13, 39.52it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 38.92it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 39.08it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.41it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.58it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 38.91it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:09, 39.08it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.41it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 39.57it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.31it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.62it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 39.06it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.31it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.67it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.93it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.24it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.46it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.96it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.40it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:21, 39.82it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:20, 40.02it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 38.46it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 39.00it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:04<00:18, 39.36it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 39.67it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 38.39it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.81it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 39.21it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:08<00:14, 39.55it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 38.80it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 39.10it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.36it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:10, 39.60it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 38.80it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:13<00:09, 39.11it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 39.35it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 39.61it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 38.54it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 38.92it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:17<00:05, 39.25it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:18<00:04, 39.51it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 38.21it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 38.66it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.00it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.24it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:22<00:00, 38.04it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.96it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.41it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 39.07it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 39.61it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 38.05it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 38.71it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:18, 39.09it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 39.49it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 38.81it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 39.10it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 39.34it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:08<00:13, 39.41it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:09<00:13, 39.29it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 39.22it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 39.32it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 38.34it/s]
Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:09, 38.68it/s]

```

Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:08, 39.16it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 39.43it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.79it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.01it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 39.38it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:17<00:04, 39.58it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 38.85it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 39.06it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 39.41it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.60it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.39it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:22<00:00, 38.99it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/912 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/912 [00:00<00:22, 39.03it/s]

Computing ESM2 embeddings: 7%| | 64/912 [00:01<00:21, 39.83it/s]

Computing ESM2 embeddings: 11%| | 96/912 [00:02<00:20, 40.00it/s]

Computing ESM2 embeddings: 14%| | 128/912 [00:03<00:20, 38.38it/s]

Computing ESM2 embeddings: 18%| | 160/912 [00:04<00:19, 38.79it/s]

Computing ESM2 embeddings: 21%| | 192/912 [00:04<00:18, 39.29it/s]

Computing ESM2 embeddings: 25%| | 224/912 [00:05<00:17, 39.56it/s]

Computing ESM2 embeddings: 28%| | 256/912 [00:06<00:17, 38.31it/s]



Computing ESM2 embeddings: 32%| | 288/912 [00:07<00:16, 38.72it/s]

Computing ESM2 embeddings: 35%| | 320/912 [00:08<00:15, 39.20it/s]

Computing ESM2 embeddings: 39%| | 352/912 [00:08<00:14, 39.49it/s]

Computing ESM2 embeddings: 42%| | 384/912 [00:09<00:13, 38.91it/s]

Computing ESM2 embeddings: 46%| | 416/912 [00:10<00:12, 39.11it/s]

Computing ESM2 embeddings: 49%| | 448/912 [00:11<00:11, 39.43it/s]

Computing ESM2 embeddings: 53%| | 480/912 [00:12<00:10, 39.63it/s]

Computing ESM2 embeddings: 56%| | 512/912 [00:13<00:10, 38.48it/s]

Computing ESM2 embeddings: 60%| | 544/912 [00:13<00:09, 38.74it/s]

Computing ESM2 embeddings: 63%| | 576/912 [00:14<00:08, 39.11it/s]

Computing ESM2 embeddings: 67%| | 608/912 [00:15<00:07, 39.40it/s]

Computing ESM2 embeddings: 70%| | 640/912 [00:16<00:07, 38.32it/s]

Computing ESM2 embeddings: 74%| | 672/912 [00:17<00:06, 38.67it/s]

Computing ESM2 embeddings: 77%| | 704/912 [00:18<00:05, 39.12it/s]

Computing ESM2 embeddings: 81%| | 736/912 [00:18<00:04, 39.38it/s]

Computing ESM2 embeddings: 84%| | 768/912 [00:19<00:03, 38.82it/s]

Computing ESM2 embeddings: 88%| | 800/912 [00:20<00:02, 39.02it/s]

Computing ESM2 embeddings: 91%| | 832/912 [00:21<00:02, 39.39it/s]

Computing ESM2 embeddings: 95%| | 864/912 [00:22<00:01, 39.59it/s]

Computing ESM2 embeddings: 98%| | 896/912 [00:22<00:00, 38.99it/s]

Computing ESM2 embeddings: 100%| | 912/912 [00:23<00:00, 39.03it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/898 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/898 [00:00<00:22, 38.97it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/898 [00:01<00:20, 39.81it/s]  |
| Computing ESM2 embeddings: | 11% | 96/898 [00:02<00:21, 38.07it/s]  |
| Computing ESM2 embeddings: | 14% | 128/898 [00:03<00:19, 38.81it/s] |
| Computing ESM2 embeddings: | 18% | 160/898 [00:04<00:18, 39.23it/s] |
| Computing ESM2 embeddings: | 21% | 192/898 [00:04<00:18, 38.03it/s] |
| Computing ESM2 embeddings: | 25% | 224/898 [00:05<00:17, 38.25it/s] |
| Computing ESM2 embeddings: | 29% | 256/898 [00:06<00:16, 38.80it/s] |
| Computing ESM2 embeddings: | 32% | 288/898 [00:07<00:15, 39.15it/s] |
| Computing ESM2 embeddings: | 36% | 320/898 [00:08<00:15, 38.08it/s] |
| Computing ESM2 embeddings: | 39% | 352/898 [00:09<00:14, 38.55it/s] |
| Computing ESM2 embeddings: | 43% | 384/898 [00:09<00:13, 39.08it/s] |
| Computing ESM2 embeddings: | 46% | 416/898 [00:10<00:12, 39.41it/s] |
| Computing ESM2 embeddings: | 50% | 448/898 [00:11<00:11, 38.31it/s] |
| Computing ESM2 embeddings: | 53% | 480/898 [00:12<00:10, 38.71it/s] |
| Computing ESM2 embeddings: | 57% | 512/898 [00:13<00:09, 39.14it/s] |
| Computing ESM2 embeddings: | 61% | 544/898 [00:14<00:09, 38.01it/s] |
| Computing ESM2 embeddings: | 64% | 576/898 [00:14<00:08, 38.57it/s] |
| Computing ESM2 embeddings: | 68% | 608/898 [00:15<00:07, 39.00it/s] |
| Computing ESM2 embeddings: | 71% | 640/898 [00:16<00:06, 39.38it/s] |
| Computing ESM2 embeddings: | 75% | 672/898 [00:17<00:05, 38.05it/s] |
| Computing ESM2 embeddings: | 78% | 704/898 [00:18<00:05, 38.59it/s] |
| Computing ESM2 embeddings: | 82% | 736/898 [00:18<00:04, 39.01it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/898 [00:19<00:03, 39.40it/s]
Computing ESM2 embeddings: 89%|      | 800/898 [00:20<00:02, 38.34it/s]
Computing ESM2 embeddings: 93%|      | 832/898 [00:21<00:01, 38.79it/s]
Computing ESM2 embeddings: 96%|      | 864/898 [00:22<00:00, 39.13it/s]
Computing ESM2 embeddings: 100%|     | 898/898 [00:23<00:00, 38.66it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/907 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/907 [00:00<00:23, 37.25it/s]
Computing ESM2 embeddings:  7%|      | 64/907 [00:01<00:21, 38.58it/s]
Computing ESM2 embeddings: 11%|      | 96/907 [00:02<00:20, 39.38it/s]
Computing ESM2 embeddings: 14%|      | 128/907 [00:03<00:20, 37.93it/s]
Computing ESM2 embeddings: 18%|      | 160/907 [00:04<00:19, 38.64it/s]
Computing ESM2 embeddings: 21%|      | 192/907 [00:04<00:18, 39.11it/s]
Computing ESM2 embeddings: 25%|      | 224/907 [00:05<00:17, 39.52it/s]
Computing ESM2 embeddings: 28%|      | 256/907 [00:06<00:16, 38.76it/s]
Computing ESM2 embeddings: 32%|      | 288/907 [00:07<00:15, 39.11it/s]
Computing ESM2 embeddings: 35%|      | 320/907 [00:08<00:14, 39.40it/s]
Computing ESM2 embeddings: 39%|      | 352/907 [00:08<00:13, 39.70it/s]
Computing ESM2 embeddings: 42%|      | 384/907 [00:09<00:13, 38.37it/s]
Computing ESM2 embeddings: 46%|      | 416/907 [00:10<00:12, 38.82it/s]
Computing ESM2 embeddings: 49%|      | 448/907 [00:11<00:11, 39.17it/s]
Computing ESM2 embeddings: 53%|      | 480/907 [00:12<00:10, 39.50it/s]

```

Computing ESM2 embeddings: 56%| | 512/907 [00:13<00:10, 38.12it/s]

Computing ESM2 embeddings: 60%| | 544/907 [00:14<00:09, 38.29it/s]

Computing ESM2 embeddings: 64%| | 576/907 [00:14<00:08, 38.62it/s]

Computing ESM2 embeddings: 67%| | 608/907 [00:15<00:07, 37.70it/s]

Computing ESM2 embeddings: 71%| | 640/907 [00:16<00:06, 38.20it/s]

Computing ESM2 embeddings: 74%| | 672/907 [00:17<00:06, 38.81it/s]

Computing ESM2 embeddings: 78%| | 704/907 [00:18<00:05, 39.16it/s]

Computing ESM2 embeddings: 81%| | 736/907 [00:18<00:04, 38.67it/s]

Computing ESM2 embeddings: 85%| | 768/907 [00:19<00:03, 38.91it/s]

Computing ESM2 embeddings: 88%| | 800/907 [00:20<00:02, 39.30it/s]

Computing ESM2 embeddings: 92%| | 832/907 [00:21<00:01, 39.48it/s]

Computing ESM2 embeddings: 95%| | 864/907 [00:22<00:01, 38.35it/s]

Computing ESM2 embeddings: 99%| | 896/907 [00:23<00:00, 38.66it/s]

Computing ESM2 embeddings: 100%| | 907/907 [00:23<00:00, 38.74it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/627 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/627 [00:00<00:15, 37.43it/s]

Computing ESM2 embeddings: 10%| | 64/627 [00:01<00:14, 38.62it/s]

Computing ESM2 embeddings: 15%| | 96/627 [00:02<00:14, 36.86it/s]

Computing ESM2 embeddings: 20%| | 128/627 [00:03<00:13, 37.84it/s]

Computing ESM2 embeddings: 26%| | 160/627 [00:04<00:12, 38.67it/s]

Computing ESM2 embeddings: 31%| | 192/627 [00:04<00:11, 39.09it/s]

Computing ESM2 embeddings: 36%| | 224/627 [00:05<00:10, 38.00it/s]

Computing ESM2 embeddings: 41%| | 256/627 [00:06<00:09, 38.16it/s]

Computing ESM2 embeddings: 46%| | 288/627 [00:07<00:08, 38.67it/s]

Computing ESM2 embeddings: 51%| | 320/627 [00:08<00:07, 39.04it/s]

Computing ESM2 embeddings: 56%| | 352/627 [00:09<00:07, 38.06it/s]

Computing ESM2 embeddings: 61%| | 384/627 [00:10<00:06, 38.47it/s]

Computing ESM2 embeddings: 66%| | 416/627 [00:10<00:05, 38.98it/s]

Computing ESM2 embeddings: 71%| | 448/627 [00:11<00:04, 39.27it/s]

Computing ESM2 embeddings: 77%| | 480/627 [00:12<00:03, 38.70it/s]

Computing ESM2 embeddings: 82%| | 512/627 [00:13<00:02, 38.90it/s]

Computing ESM2 embeddings: 87%| | 544/627 [00:14<00:02, 39.27it/s]

Computing ESM2 embeddings: 92%| | 576/627 [00:14<00:01, 39.44it/s]

Computing ESM2 embeddings: 97%| | 608/627 [00:15<00:00, 38.16it/s]

Computing ESM2 embeddings: 100%| | 627/627 [00:16<00:00, 38.42it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/848 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/848 [00:00<00:21, 38.14it/s]

Computing ESM2 embeddings: 8%| | 64/848 [00:01<00:23, 33.28it/s]

Computing ESM2 embeddings: 11%| | 96/848 [00:02<00:22, 33.66it/s]

Computing ESM2 embeddings: 15%| | 128/848 [00:03<00:20, 35.89it/s]

Computing ESM2 embeddings: 19%| | 160/848 [00:04<00:18, 37.30it/s]

Computing ESM2 embeddings: 23%| | 192/848 [00:05<00:17, 38.27it/s]

```

Computing ESM2 embeddings: 26%|          | 224/848 [00:06<00:16, 37.31it/s]
Computing ESM2 embeddings: 30%|          | 256/848 [00:06<00:15, 38.03it/s]
Computing ESM2 embeddings: 34%|          | 288/848 [00:07<00:14, 38.62it/s]
Computing ESM2 embeddings: 38%|          | 320/848 [00:08<00:13, 39.13it/s]
Computing ESM2 embeddings: 42%|          | 352/848 [00:09<00:12, 38.53it/s]
Computing ESM2 embeddings: 45%|          | 384/848 [00:10<00:12, 38.63it/s]
Computing ESM2 embeddings: 49%|          | 416/848 [00:11<00:11, 38.92it/s]
Computing ESM2 embeddings: 53%|          | 448/848 [00:11<00:10, 37.99it/s]
Computing ESM2 embeddings: 57%|          | 480/848 [00:12<00:09, 38.17it/s]
Computing ESM2 embeddings: 60%|          | 512/848 [00:13<00:08, 38.39it/s]
Computing ESM2 embeddings: 64%|          | 544/848 [00:14<00:07, 38.76it/s]
Computing ESM2 embeddings: 68%|          | 576/848 [00:15<00:07, 37.88it/s]
Computing ESM2 embeddings: 72%|          | 608/848 [00:16<00:06, 38.36it/s]
Computing ESM2 embeddings: 75%|          | 640/848 [00:16<00:05, 38.89it/s]
Computing ESM2 embeddings: 79%|          | 672/848 [00:17<00:04, 39.22it/s]
Computing ESM2 embeddings: 83%|          | 704/848 [00:18<00:03, 38.34it/s]
Computing ESM2 embeddings: 87%|          | 736/848 [00:19<00:02, 38.68it/s]
Computing ESM2 embeddings: 91%|          | 768/848 [00:20<00:02, 39.10it/s]
Computing ESM2 embeddings: 94%|          | 800/848 [00:20<00:01, 39.03it/s]
Computing ESM2 embeddings: 98%|          | 832/848 [00:21<00:00, 39.01it/s]

Computing ESM2 embeddings: 100%|         | 848/848 [00:22<00:00, 38.13it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 38.05it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 38.66it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.08it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 37.69it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.35it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 39.01it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 37.87it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 38.52it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 39.01it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 39.41it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 38.79it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.14it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 39.43it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 38.32it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 38.68it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 39.16it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 39.43it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 38.40it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 38.77it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.20it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 38.16it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 38.67it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:18<00:04, 39.01it/s] |

```

Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.08it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.50it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.01it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 37.95it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.50it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.67it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/895 [00:00<00:22, 39.19it/s]
Computing ESM2 embeddings:  7%|          | 64/895 [00:01<00:20, 39.81it/s]
Computing ESM2 embeddings: 11%|          | 96/895 [00:02<00:21, 37.86it/s]
Computing ESM2 embeddings: 14%|          | 128/895 [00:03<00:19, 38.72it/s]
Computing ESM2 embeddings: 18%|          | 160/895 [00:04<00:18, 39.17it/s]
Computing ESM2 embeddings: 21%|          | 192/895 [00:04<00:17, 39.50it/s]
Computing ESM2 embeddings: 25%|          | 224/895 [00:05<00:17, 38.39it/s]
Computing ESM2 embeddings: 29%|          | 256/895 [00:06<00:16, 38.89it/s]
Computing ESM2 embeddings: 32%|          | 288/895 [00:07<00:15, 39.22it/s]
Computing ESM2 embeddings: 36%|          | 320/895 [00:08<00:15, 37.99it/s]
Computing ESM2 embeddings: 39%|          | 352/895 [00:09<00:14, 38.16it/s]
Computing ESM2 embeddings: 43%|          | 384/895 [00:09<00:13, 38.65it/s]
Computing ESM2 embeddings: 46%|          | 416/895 [00:10<00:12, 37.66it/s]
Computing ESM2 embeddings: 50%|          | 448/895 [00:11<00:11, 38.26it/s]

```



Computing ESM2 embeddings: 54%| | 480/895 [00:12<00:10, 38.75it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 37.75it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 38.24it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:14<00:08, 38.81it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:15<00:07, 39.16it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:16<00:06, 38.17it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:17<00:05, 38.17it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:04, 38.67it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.82it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:19<00:03, 38.41it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:20<00:02, 38.85it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:21<00:01, 39.23it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:22<00:00, 38.66it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:23<00:00, 38.54it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.03it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 38.96it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.60it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 37.98it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.68it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.15it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:18, 38.14it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:17, 38.60it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:15, 39.13it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:15, 39.44it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 38.41it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:09<00:13, 38.77it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:10<00:12, 39.23it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:12, 38.10it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 38.60it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 39.03it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:13<00:09, 39.37it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:14<00:08, 38.23it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:15<00:07, 38.72it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:16<00:07, 39.13it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 38.18it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 38.61it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:18<00:04, 39.10it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:19<00:03, 39.38it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:20<00:02, 38.18it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:21<00:02, 38.56it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:22<00:01, 39.05it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:23<00:00, 38.42it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:23<00:00, 38.66it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:23, 37.20it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:24, 33.36it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:22, 36.05it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 35.88it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 36.82it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 37.70it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 37.06it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 37.78it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 38.49it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.53it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 38.19it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 38.53it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.75it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.26it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.85it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 38.37it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 38.77it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 39.11it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 38.00it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.49it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 39.00it/s] |

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.51it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.90it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 39.24it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 38.69it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.94it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 39.29it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.01it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.29it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.93it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 40.04it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.51it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.88it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.27it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 38.12it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.67it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 39.10it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.03it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.49it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.00it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:13, 38.00it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 38.56it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.99it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.90it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.34it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.88it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:08, 37.93it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.47it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.92it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.93it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.39it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.93it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:03, 37.99it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.49it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.96it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.07it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.51it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.84it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:20, 39.70it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.89it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 38.21it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:19, 38.64it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:17, 39.22it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 37.96it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 38.59it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 39.04it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 38.57it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:13, 38.86it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:13, 39.31it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:12, 38.28it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 38.82it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 39.19it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:09, 38.77it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 38.99it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 39.38it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 38.16it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 38.68it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 39.07it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:05, 37.96it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 38.41it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 38.95it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 37.93it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:21<00:01, 38.50it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:22<00:00, 38.94it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:23<00:00, 38.46it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/589 [00:00<?, ?it/s]
Computing ESM2 embeddings: 5%|          | 32/589 [00:00<00:15, 36.90it/s]
Computing ESM2 embeddings: 11%|         | 64/589 [00:01<00:13, 38.28it/s]
Computing ESM2 embeddings: 16%|         | 96/589 [00:02<00:12, 39.17it/s]
Computing ESM2 embeddings: 22%|         | 128/589 [00:03<00:12, 37.84it/s]
Computing ESM2 embeddings: 27%|         | 160/589 [00:04<00:11, 38.27it/s]
Computing ESM2 embeddings: 33%|         | 192/589 [00:04<00:10, 38.64it/s]
Computing ESM2 embeddings: 38%|         | 224/589 [00:05<00:09, 37.68it/s]
Computing ESM2 embeddings: 43%|         | 256/589 [00:06<00:08, 38.26it/s]
Computing ESM2 embeddings: 49%|         | 288/589 [00:07<00:07, 38.86it/s]
Computing ESM2 embeddings: 54%|         | 320/589 [00:08<00:07, 37.67it/s]
Computing ESM2 embeddings: 60%|         | 352/589 [00:09<00:06, 38.27it/s]
Computing ESM2 embeddings: 65%|         | 384/589 [00:10<00:05, 38.52it/s]
Computing ESM2 embeddings: 71%|         | 416/589 [00:10<00:04, 37.01it/s]
Computing ESM2 embeddings: 76%|         | 448/589 [00:11<00:03, 37.77it/s]
Computing ESM2 embeddings: 81%|         | 480/589 [00:12<00:02, 38.22it/s]
Computing ESM2 embeddings: 87%|         | 512/589 [00:13<00:02, 36.80it/s]
Computing ESM2 embeddings: 92%|         | 544/589 [00:14<00:01, 37.66it/s]
Computing ESM2 embeddings: 98%|         | 576/589 [00:15<00:00, 38.32it/s]
```

```
Computing ESM2 embeddings: 100%|        | 589/589 [00:15<00:00, 37.76it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
```

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:22, 37.86it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:21, 38.86it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:20, 39.53it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:20, 38.10it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:18, 38.75it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:04<00:17, 39.19it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 38.13it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:16, 38.61it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:15, 39.15it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 38.00it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 38.57it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:09<00:13, 39.00it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:10<00:12, 38.10it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:11<00:11, 38.55it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 39.08it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:10, 37.99it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 38.54it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:14<00:08, 38.98it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:15<00:07, 38.07it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:16<00:06, 38.49it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:17<00:05, 39.02it/s] |



Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:05, 38.01it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 38.14it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:19<00:03, 38.13it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:20<00:02, 36.63it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:21<00:01, 37.49it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:22<00:00, 38.28it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:23<00:00, 38.31it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.72it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.42it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:20, 39.74it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.26it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.78it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 39.31it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 38.61it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.70it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.96it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.02it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.50it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 39.04it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:13, 38.20it/s]

```

Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:12, 38.69it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 39.11it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.07it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 38.53it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 39.04it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:08, 37.87it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.42it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.91it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.07it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 38.51it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 39.01it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:03, 37.92it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 38.48it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 38.93it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 37.98it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.51it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.13it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.17it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 39.61it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 38.08it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:19, 38.63it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:17, 39.19it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 37.99it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 38.57it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 39.01it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 38.09it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 38.55it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:13, 39.06it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:12, 38.20it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 38.70it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 39.09it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 38.13it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 38.55it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 38.96it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 37.99it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 38.53it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 38.97it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:05, 37.94it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 38.36it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 38.92it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 37.90it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:21<00:01, 38.47it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:22<00:00, 38.94it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:23<00:00, 38.44it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 38.47it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.20it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:20, 39.60it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.13it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.80it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 39.25it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 38.13it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 38.59it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:15, 39.14it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.04it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 38.62it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 39.07it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:13, 38.03it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 38.49it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 39.03it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.00it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 38.56it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 39.01it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:08, 38.10it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 38.55it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 39.07it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 38.00it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 38.56it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:19<00:03, 39.00it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 38.07it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 38.49it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 39.02it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 37.93it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.44it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 37.98it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 38.87it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 39.52it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 38.00it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 38.67it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:17, 39.15it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 38.07it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 38.51it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 39.06it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 38.00it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 38.57it/s]

```

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 39.01it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.13it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.56it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.08it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 38.50it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.91it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 38.94it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 37.74it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.24it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.84it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.87it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.45it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.91it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 37.91it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.38it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.92it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.37it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 36.63it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:25, 32.75it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 35.26it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:22, 35.28it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 36.41it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.49it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 36.76it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:17, 37.64it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.28it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.43it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 38.00it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:13, 38.65it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.92it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 38.45it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 38.87it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.87it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 38.30it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:08, 38.74it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.66it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 38.20it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 38.64it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 38.12it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 38.20it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 38.65it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.58it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 38.17it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:22<00:01, 38.64it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.63it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.61it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:23, 37.07it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:22, 37.56it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:20, 38.56it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:20, 37.49it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:19, 38.12it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:18, 38.80it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:05<00:17, 37.53it/s]

Computing ESM2 embeddings: 29%| | 256/895 [00:06<00:16, 37.94it/s]

Computing ESM2 embeddings: 32%| | 288/895 [00:07<00:15, 38.39it/s]

Computing ESM2 embeddings: 36%| | 320/895 [00:08<00:15, 37.72it/s]

Computing ESM2 embeddings: 39%| | 352/895 [00:09<00:14, 38.21it/s]

Computing ESM2 embeddings: 43%| | 384/895 [00:10<00:13, 38.76it/s]

Computing ESM2 embeddings: 46%| | 416/895 [00:10<00:12, 37.84it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:11<00:11, 38.41it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:12<00:10, 38.84it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 37.78it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 38.25it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 38.80it/s]



```

Computing ESM2 embeddings: 68%|      | 608/895 [00:15<00:07, 37.77it/s]
Computing ESM2 embeddings: 72%|      | 640/895 [00:16<00:06, 38.34it/s]
Computing ESM2 embeddings: 75%|      | 672/895 [00:17<00:05, 38.77it/s]
Computing ESM2 embeddings: 79%|      | 704/895 [00:18<00:04, 38.35it/s]
Computing ESM2 embeddings: 82%|      | 736/895 [00:19<00:04, 38.65it/s]
Computing ESM2 embeddings: 86%|      | 768/895 [00:20<00:03, 39.07it/s]
Computing ESM2 embeddings: 89%|      | 800/895 [00:20<00:02, 37.95it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:21<00:01, 38.46it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:22<00:00, 38.87it/s]

Computing ESM2 embeddings: 100%|     | 895/895 [00:23<00:00, 38.13it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/590 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/590 [00:00<00:14, 38.46it/s]
Computing ESM2 embeddings: 11%|      | 64/590 [00:01<00:13, 39.09it/s]
Computing ESM2 embeddings: 16%|      | 96/590 [00:02<00:12, 39.65it/s]
Computing ESM2 embeddings: 22%|      | 128/590 [00:03<00:12, 37.93it/s]
Computing ESM2 embeddings: 27%|      | 160/590 [00:04<00:11, 38.63it/s]
Computing ESM2 embeddings: 33%|      | 192/590 [00:04<00:10, 39.12it/s]
Computing ESM2 embeddings: 38%|      | 224/590 [00:05<00:09, 38.62it/s]
Computing ESM2 embeddings: 43%|      | 256/590 [00:06<00:08, 38.92it/s]
Computing ESM2 embeddings: 49%|      | 288/590 [00:07<00:07, 39.30it/s]
Computing ESM2 embeddings: 54%|      | 320/590 [00:08<00:07, 38.12it/s]

```

Computing ESM2 embeddings: 60%| | 352/590 [00:09<00:06, 38.64it/s]  
 Computing ESM2 embeddings: 65%| | 384/590 [00:09<00:05, 39.08it/s]  
 Computing ESM2 embeddings: 71%| | 416/590 [00:10<00:04, 38.03it/s]  
 Computing ESM2 embeddings: 76%| | 448/590 [00:11<00:03, 38.40it/s]  
 Computing ESM2 embeddings: 81%| | 480/590 [00:12<00:02, 38.95it/s]  
 Computing ESM2 embeddings: 87%| | 512/590 [00:13<00:02, 37.90it/s]  
 Computing ESM2 embeddings: 92%| | 544/590 [00:14<00:01, 38.47it/s]  
 Computing ESM2 embeddings: 98%| | 576/590 [00:14<00:00, 38.92it/s]

Computing ESM2 embeddings: 100%| | 590/590 [00:15<00:00, 38.42it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.27it/s]  
 Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.92it/s]  
 Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 39.53it/s]  
 Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 38.05it/s]  
 Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.67it/s]  
 Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:18, 39.11it/s]  
 Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.53it/s]  
 Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 38.84it/s]  
 Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 39.28it/s]  
 Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 38.00it/s]  
 Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 38.55it/s]

```

Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:13, 38.99it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.06it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 38.52it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 39.00it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.93it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 38.48it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 38.60it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 37.14it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 37.80it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.49it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 37.68it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 38.30it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.72it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 37.84it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 38.34it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 38.88it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.32it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 38.31it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 39.31it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.38it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 38.03it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.59it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.57it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 38.18it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 38.83it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.35it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 38.82it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 39.13it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 38.67it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 38.93it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 39.34it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 38.19it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 38.64it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 39.02it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:14<00:08, 38.00it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 38.46it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 38.95it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.90it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 38.48it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 38.94it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 37.92it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 38.38it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:21<00:02, 38.92it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:22<00:01, 37.90it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.47it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.49it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.27it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.05it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:22, 36.03it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.46it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.29it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.39it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.04it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 38.71it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.69it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 38.35it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 38.83it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 38.39it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.72it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.84it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.42it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 38.11it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.64it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.85it/s]

```

Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.34it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.88it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.34it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:04, 38.76it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 39.10it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 38.60it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.83it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.20it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 37.97it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.32it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:22, 39.03it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 39.78it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.85it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:20, 38.62it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 39.13it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 37.87it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:17, 38.36it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:16, 38.93it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 38.10it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 38.63it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 39.02it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 37.98it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:13, 38.11it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 38.25it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.07it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.85it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.43it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 38.15it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.49it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.96it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.92it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.46it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.87it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 38.37it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.67it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.11it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.54it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.87it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.46it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.29it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.28it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.66it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 38.54it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.08it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 38.03it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.52it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.09it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 38.47it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 38.91it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:13, 39.26it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.22it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.61it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.77it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.18it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.96it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 38.56it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.80it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.32it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.88it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.88it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 38.23it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 38.64it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 37.83it/s] |
| Computing ESM2 embeddings: | 89% | 800/896 [00:20<00:02, 38.35it/s] |



Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.91it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 37.88it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.33it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:22, 38.52it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:21, 39.12it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:21, 37.64it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:20, 38.38it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:19, 39.07it/s]

Computing ESM2 embeddings: 21%| | 192/913 [00:05<00:19, 37.84it/s]

Computing ESM2 embeddings: 25%| | 224/913 [00:05<00:18, 38.25it/s]

Computing ESM2 embeddings: 28%| | 256/913 [00:06<00:16, 38.69it/s]

Computing ESM2 embeddings: 32%| | 288/913 [00:07<00:16, 38.22it/s]

Computing ESM2 embeddings: 35%| | 320/913 [00:08<00:15, 38.64it/s]

Computing ESM2 embeddings: 39%| | 352/913 [00:09<00:14, 39.13it/s]

Computing ESM2 embeddings: 42%| | 384/913 [00:10<00:13, 38.04it/s]

Computing ESM2 embeddings: 46%| | 416/913 [00:10<00:12, 38.32it/s]

Computing ESM2 embeddings: 49%| | 448/913 [00:11<00:12, 38.69it/s]

Computing ESM2 embeddings: 53%| | 480/913 [00:12<00:11, 37.86it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 38.36it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 38.93it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:14<00:08, 38.69it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:15<00:07, 39.16it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:16<00:06, 39.44it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:17<00:06, 38.88it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:18<00:05, 39.06it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:19<00:04, 39.41it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:19<00:03, 38.39it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:20<00:02, 38.84it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:21<00:02, 39.21it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:22<00:01, 38.33it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:23<00:00, 38.70it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:23<00:00, 38.64it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.63it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.13it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.54it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 38.30it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.02it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.86it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.51it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 38.98it/s]

```

Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 38.55it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 38.88it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:13, 39.29it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:13, 38.11it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.62it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.02it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 37.98it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:09, 38.45it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 38.90it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 38.46it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.88it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.23it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.23it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:04, 38.62it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 39.11it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.05it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.60it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.01it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 38.64it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.59it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/572 [00:00<?, ?it/s]

```

Computing ESM2 embeddings: 6%| | 32/572 [00:00<00:14, 37.35it/s]

Computing ESM2 embeddings: 11%| | 64/572 [00:01<00:15, 32.94it/s]

Computing ESM2 embeddings: 17%| | 96/572 [00:02<00:14, 33.66it/s]

Computing ESM2 embeddings: 22%| | 128/572 [00:03<00:12, 35.81it/s]

Computing ESM2 embeddings: 28%| | 160/572 [00:04<00:11, 37.16it/s]

Computing ESM2 embeddings: 34%| | 192/572 [00:05<00:10, 36.69it/s]

Computing ESM2 embeddings: 39%| | 224/572 [00:06<00:09, 37.52it/s]

Computing ESM2 embeddings: 45%| | 256/572 [00:06<00:08, 38.30it/s]

Computing ESM2 embeddings: 50%| | 288/572 [00:07<00:07, 37.31it/s]

Computing ESM2 embeddings: 56%| | 320/572 [00:08<00:06, 38.04it/s]

Computing ESM2 embeddings: 62%| | 352/572 [00:09<00:05, 38.55it/s]

Computing ESM2 embeddings: 67%| | 384/572 [00:10<00:04, 38.06it/s]

Computing ESM2 embeddings: 73%| | 416/572 [00:11<00:04, 38.33it/s]

Computing ESM2 embeddings: 78%| | 448/572 [00:11<00:03, 38.68it/s]

Computing ESM2 embeddings: 84%| | 480/572 [00:12<00:02, 37.66it/s]

Computing ESM2 embeddings: 90%| | 512/572 [00:13<00:01, 38.38it/s]

Computing ESM2 embeddings: 95%| | 544/572 [00:14<00:00, 38.82it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:15<00:00, 37.52it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:22, 38.85it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:21, 39.58it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 10% | 96/915 [00:02<00:21, 38.67it/s]  |
| Computing ESM2 embeddings: 14% | 128/915 [00:03<00:20, 39.03it/s] |
| Computing ESM2 embeddings: 17% | 160/915 [00:04<00:19, 39.49it/s] |
| Computing ESM2 embeddings: 21% | 192/915 [00:04<00:18, 38.26it/s] |
| Computing ESM2 embeddings: 24% | 224/915 [00:05<00:17, 38.77it/s] |
| Computing ESM2 embeddings: 28% | 256/915 [00:06<00:16, 39.17it/s] |
| Computing ESM2 embeddings: 31% | 288/915 [00:07<00:16, 38.17it/s] |
| Computing ESM2 embeddings: 35% | 320/915 [00:08<00:15, 38.58it/s] |
| Computing ESM2 embeddings: 38% | 352/915 [00:09<00:14, 39.11it/s] |
| Computing ESM2 embeddings: 42% | 384/915 [00:09<00:13, 37.99it/s] |
| Computing ESM2 embeddings: 45% | 416/915 [00:10<00:12, 38.49it/s] |
| Computing ESM2 embeddings: 49% | 448/915 [00:11<00:12, 38.54it/s] |
| Computing ESM2 embeddings: 52% | 480/915 [00:12<00:11, 37.34it/s] |
| Computing ESM2 embeddings: 56% | 512/915 [00:13<00:10, 37.95it/s] |
| Computing ESM2 embeddings: 59% | 544/915 [00:14<00:09, 38.62it/s] |
| Computing ESM2 embeddings: 63% | 576/915 [00:14<00:08, 38.15it/s] |
| Computing ESM2 embeddings: 66% | 608/915 [00:15<00:07, 38.65it/s] |
| Computing ESM2 embeddings: 70% | 640/915 [00:16<00:07, 39.04it/s] |
| Computing ESM2 embeddings: 73% | 672/915 [00:17<00:06, 38.06it/s] |
| Computing ESM2 embeddings: 77% | 704/915 [00:18<00:05, 38.49it/s] |
| Computing ESM2 embeddings: 80% | 736/915 [00:19<00:04, 39.01it/s] |
| Computing ESM2 embeddings: 84% | 768/915 [00:19<00:03, 37.95it/s] |
| Computing ESM2 embeddings: 87% | 800/915 [00:20<00:02, 38.50it/s] |
| Computing ESM2 embeddings: 91% | 832/915 [00:21<00:02, 38.90it/s] |

```

Computing ESM2 embeddings: 94%|      | 864/915 [00:22<00:01, 37.88it/s]

Computing ESM2 embeddings: 98%|      | 896/915 [00:23<00:00, 38.33it/s]

Computing ESM2 embeddings: 100%|     | 915/915 [00:23<00:00, 38.49it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|       | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%|       | 32/896 [00:00<00:21, 39.34it/s]

Computing ESM2 embeddings: 7%|       | 64/896 [00:01<00:20, 39.76it/s]

Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 38.74it/s]

Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:19, 39.07it/s]

Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:18, 39.51it/s]

Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:18, 38.39it/s]

Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 38.88it/s]

Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 39.22it/s]

Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 38.19it/s]

Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 38.62it/s]

Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:13, 39.11it/s]

Computing ESM2 embeddings: 43%|      | 384/896 [00:09<00:13, 38.05it/s]

Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 38.59it/s]

Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 39.00it/s]

Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 38.08it/s]

Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:09, 38.53it/s]

Computing ESM2 embeddings: 61%|      | 544/896 [00:14<00:09, 38.65it/s]

```

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 37.37it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.08it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.62it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.73it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 38.23it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.83it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 37.85it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.42it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.85it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.52it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.49it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.45it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.26it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 35.72it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 37.23it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.18it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.32it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 38.00it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.54it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.71it/s]

```

Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 38.29it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 38.76it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:13, 38.25it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 38.57it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.06it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.20it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.68it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 39.00it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:08, 38.01it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 38.37it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.88it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 37.80it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.37it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 38.84it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.81it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.22it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 38.71it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 37.74it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.32it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.13it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/895 [00:00<?, ?it/s]

```



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 4%  | 32/895 [00:00<00:24, 34.83it/s]  |
| Computing ESM2 embeddings: 7%  | 64/895 [00:01<00:22, 37.62it/s]  |
| Computing ESM2 embeddings: 11% | 96/895 [00:02<00:21, 36.97it/s]  |
| Computing ESM2 embeddings: 14% | 128/895 [00:03<00:20, 38.08it/s] |
| Computing ESM2 embeddings: 18% | 160/895 [00:04<00:18, 38.77it/s] |
| Computing ESM2 embeddings: 21% | 192/895 [00:05<00:18, 38.28it/s] |
| Computing ESM2 embeddings: 25% | 224/895 [00:05<00:17, 38.65it/s] |
| Computing ESM2 embeddings: 29% | 256/895 [00:06<00:16, 39.13it/s] |
| Computing ESM2 embeddings: 32% | 288/895 [00:07<00:16, 37.93it/s] |
| Computing ESM2 embeddings: 36% | 320/895 [00:08<00:14, 38.48it/s] |
| Computing ESM2 embeddings: 39% | 352/895 [00:09<00:13, 38.88it/s] |
| Computing ESM2 embeddings: 43% | 384/895 [00:10<00:13, 38.38it/s] |
| Computing ESM2 embeddings: 46% | 416/895 [00:10<00:12, 38.37it/s] |
| Computing ESM2 embeddings: 50% | 448/895 [00:11<00:11, 38.78it/s] |
| Computing ESM2 embeddings: 54% | 480/895 [00:12<00:10, 38.01it/s] |
| Computing ESM2 embeddings: 57% | 512/895 [00:13<00:09, 38.49it/s] |
| Computing ESM2 embeddings: 61% | 544/895 [00:14<00:09, 38.91it/s] |
| Computing ESM2 embeddings: 64% | 576/895 [00:15<00:08, 37.93it/s] |
| Computing ESM2 embeddings: 68% | 608/895 [00:15<00:07, 38.39it/s] |
| Computing ESM2 embeddings: 72% | 640/895 [00:16<00:06, 38.89it/s] |
| Computing ESM2 embeddings: 75% | 672/895 [00:17<00:05, 38.40it/s] |
| Computing ESM2 embeddings: 79% | 704/895 [00:18<00:04, 38.80it/s] |
| Computing ESM2 embeddings: 82% | 736/895 [00:19<00:04, 39.14it/s] |
| Computing ESM2 embeddings: 86% | 768/895 [00:20<00:03, 38.09it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/895 [00:20<00:02, 38.45it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:21<00:01, 38.94it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:22<00:00, 37.84it/s]
Computing ESM2 embeddings: 100%|     | 895/895 [00:23<00:00, 38.34it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/914 [00:00<00:22, 38.78it/s]
Computing ESM2 embeddings: 7%|        | 64/914 [00:01<00:21, 39.31it/s]
Computing ESM2 embeddings: 11%|       | 96/914 [00:02<00:21, 37.60it/s]
Computing ESM2 embeddings: 14%|       | 128/914 [00:03<00:20, 38.04it/s]
Computing ESM2 embeddings: 18%|       | 160/914 [00:04<00:19, 38.69it/s]
Computing ESM2 embeddings: 21%|       | 192/914 [00:05<00:18, 38.30it/s]
Computing ESM2 embeddings: 25%|       | 224/914 [00:05<00:17, 38.82it/s]
Computing ESM2 embeddings: 28%|       | 256/914 [00:06<00:16, 39.23it/s]
Computing ESM2 embeddings: 32%|       | 288/914 [00:07<00:16, 38.17it/s]
Computing ESM2 embeddings: 35%|       | 320/914 [00:08<00:15, 38.60it/s]
Computing ESM2 embeddings: 39%|       | 352/914 [00:09<00:14, 39.10it/s]
Computing ESM2 embeddings: 42%|       | 384/914 [00:09<00:13, 38.57it/s]
Computing ESM2 embeddings: 46%|       | 416/914 [00:10<00:12, 38.97it/s]
Computing ESM2 embeddings: 49%|       | 448/914 [00:11<00:11, 39.30it/s]
Computing ESM2 embeddings: 53%|       | 480/914 [00:12<00:11, 38.10it/s]
Computing ESM2 embeddings: 56%|       | 512/914 [00:13<00:10, 38.53it/s]

```

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 39.01it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.45it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.63it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.88it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.00it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.21it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.62it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 37.88it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.48it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.93it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 37.84it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.33it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.52it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 39.19it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.01it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 36.75it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.83it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.70it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 38.26it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.82it/s]

```

Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 38.85it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 39.19it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 37.90it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 38.34it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:13, 38.76it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.41it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 38.77it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 39.24it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 38.11it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 38.61it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 39.03it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.11it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.51it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 39.03it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 38.03it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 38.56it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.98it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 37.79it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 38.32it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 38.88it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.43it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.22it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.78it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 38.15it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.68it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 39.28it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 38.29it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 38.80it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.22it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.12it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.57it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 39.10it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 37.99it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:13, 38.24it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 38.64it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 38.19it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.15it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 38.66it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 37.73it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 38.38it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 38.86it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 38.50it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 38.84it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:19<00:04, 39.26it/s] |

```

Computing ESM2 embeddings: 84%|      | 768/914 [00:19<00:03, 38.67it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 39.01it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.33it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 38.24it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.54it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.56it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/571 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/571 [00:00<00:14, 37.81it/s]
Computing ESM2 embeddings: 11%|      | 64/571 [00:01<00:13, 38.77it/s]
Computing ESM2 embeddings: 17%|      | 96/571 [00:02<00:12, 37.36it/s]
Computing ESM2 embeddings: 22%|      | 128/571 [00:03<00:11, 38.19it/s]
Computing ESM2 embeddings: 28%|      | 160/571 [00:04<00:10, 38.93it/s]
Computing ESM2 embeddings: 34%|      | 192/571 [00:05<00:10, 37.81it/s]
Computing ESM2 embeddings: 39%|      | 224/571 [00:05<00:09, 38.47it/s]
Computing ESM2 embeddings: 45%|      | 256/571 [00:06<00:08, 38.97it/s]
Computing ESM2 embeddings: 50%|      | 288/571 [00:07<00:07, 38.52it/s]
Computing ESM2 embeddings: 56%|      | 320/571 [00:08<00:06, 38.82it/s]
Computing ESM2 embeddings: 62%|      | 352/571 [00:09<00:05, 39.28it/s]
Computing ESM2 embeddings: 67%|      | 384/571 [00:10<00:04, 38.11it/s]
Computing ESM2 embeddings: 73%|      | 416/571 [00:10<00:04, 38.63it/s]
Computing ESM2 embeddings: 78%|      | 448/571 [00:11<00:03, 39.07it/s]

```

```

Computing ESM2 embeddings: 84%|          | 480/571 [00:12<00:02, 38.60it/s]
Computing ESM2 embeddings: 90%|          | 512/571 [00:13<00:01, 38.89it/s]
Computing ESM2 embeddings: 95%|          | 544/571 [00:14<00:00, 39.00it/s]
Computing ESM2 embeddings: 100%|         | 571/571 [00:14<00:00, 38.35it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/914 [00:01<00:28, 31.16it/s]
Computing ESM2 embeddings: 7%|           | 64/914 [00:01<00:24, 35.24it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:23, 35.11it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 36.70it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 37.79it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 37.53it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:06<00:18, 37.79it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 38.45it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 37.48it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 38.21it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 38.76it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:13, 38.24it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:12, 38.58it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.07it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.35it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.79it/s]

```

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 39.16it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 38.66it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.89it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.28it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.13it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.63it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 39.01it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 38.53it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.83it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.23it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.57it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.92it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.21it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.34it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.90it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.41it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 38.36it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 38.97it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 38.44it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.73it/s]



Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 39.17it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.98it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 38.56it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:13, 38.96it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.99it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.12it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.65it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.54it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 38.19it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.72it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.77it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.26it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.81it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.84it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.44it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.89it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.92it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.38it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.95it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.13it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.37it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 38.22it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.17it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.45it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.40it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 39.00it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.66it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 38.26it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 38.89it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.13it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.66it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 39.04it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:13, 38.15it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 38.56it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.08it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 38.00it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.11it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 38.06it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.13it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:08, 37.81it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 38.53it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 37.67it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 38.32it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:19<00:04, 38.79it/s] |

```

Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.82it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 38.20it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 38.77it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 38.22it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.68it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.33it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.04it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 39.16it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.38it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 37.73it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.81it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 36.89it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 37.39it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 38.07it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.02it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 37.81it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 38.43it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.39it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.93it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 38.57it/s]

```

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.57it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 38.24it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.70it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 38.26it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.58it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.03it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.11it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.61it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 39.00it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 38.10it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.58it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.99it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 37.83it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.09it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:23, 38.05it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:21, 39.02it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:21, 38.17it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:20, 38.51it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:19, 39.10it/s]

Computing ESM2 embeddings: 21%| | 192/913 [00:05<00:18, 38.14it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/913 [00:05<00:17, 38.70it/s] |
| Computing ESM2 embeddings: 28%  | 256/913 [00:06<00:16, 39.09it/s] |
| Computing ESM2 embeddings: 32%  | 288/913 [00:07<00:16, 38.09it/s] |
| Computing ESM2 embeddings: 35%  | 320/913 [00:08<00:15, 38.55it/s] |
| Computing ESM2 embeddings: 39%  | 352/913 [00:09<00:14, 39.04it/s] |
| Computing ESM2 embeddings: 42%  | 384/913 [00:09<00:13, 38.56it/s] |
| Computing ESM2 embeddings: 46%  | 416/913 [00:10<00:12, 39.00it/s] |
| Computing ESM2 embeddings: 49%  | 448/913 [00:11<00:11, 39.29it/s] |
| Computing ESM2 embeddings: 53%  | 480/913 [00:12<00:11, 38.29it/s] |
| Computing ESM2 embeddings: 56%  | 512/913 [00:13<00:10, 38.68it/s] |
| Computing ESM2 embeddings: 60%  | 544/913 [00:14<00:09, 39.13it/s] |
| Computing ESM2 embeddings: 63%  | 576/913 [00:14<00:08, 38.01it/s] |
| Computing ESM2 embeddings: 67%  | 608/913 [00:15<00:07, 38.28it/s] |
| Computing ESM2 embeddings: 70%  | 640/913 [00:16<00:07, 38.67it/s] |
| Computing ESM2 embeddings: 74%  | 672/913 [00:17<00:06, 38.31it/s] |
| Computing ESM2 embeddings: 77%  | 704/913 [00:18<00:05, 38.66it/s] |
| Computing ESM2 embeddings: 81%  | 736/913 [00:19<00:04, 39.11it/s] |
| Computing ESM2 embeddings: 84%  | 768/913 [00:19<00:03, 38.06it/s] |
| Computing ESM2 embeddings: 88%  | 800/913 [00:20<00:02, 38.56it/s] |
| Computing ESM2 embeddings: 91%  | 832/913 [00:21<00:02, 38.98it/s] |
| Computing ESM2 embeddings: 95%  | 864/913 [00:22<00:01, 38.56it/s] |
| Computing ESM2 embeddings: 98%  | 896/913 [00:23<00:00, 38.86it/s] |
| Computing ESM2 embeddings: 100% | 913/913 [00:23<00:00, 38.59it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.91it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.11it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.60it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 38.34it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.03it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 37.86it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.49it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.98it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 38.66it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 39.03it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:13, 39.41it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.77it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 39.10it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.39it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.19it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 38.56it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 39.01it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.49it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.90it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.21it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 38.67it/s] |

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.92it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.31it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.20it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.67it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.07it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.04it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.62it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.80it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.13it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.30it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.29it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.91it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.87it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 38.37it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 38.93it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.85it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.19it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.61it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:13, 38.28it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.64it/s]

```

Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.11it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 37.95it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.48it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 38.93it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 38.41it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 38.68it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:06, 39.14it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.54it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.67it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 38.91it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 37.83it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.29it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 38.85it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 37.88it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.46it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.39it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|          | 32/572 [00:00<00:14, 37.38it/s]
Computing ESM2 embeddings: 11%|          | 64/572 [00:01<00:13, 38.43it/s]
Computing ESM2 embeddings: 17%|          | 96/572 [00:02<00:12, 36.92it/s]
Computing ESM2 embeddings: 22%|          | 128/572 [00:03<00:11, 38.05it/s]

```



Computing ESM2 embeddings: 28%| | 160/572 [00:04<00:10, 38.76it/s]

Computing ESM2 embeddings: 34%| | 192/572 [00:05<00:10, 37.62it/s]

Computing ESM2 embeddings: 39%| | 224/572 [00:05<00:09, 38.20it/s]

Computing ESM2 embeddings: 45%| | 256/572 [00:06<00:08, 38.84it/s]

Computing ESM2 embeddings: 50%| | 288/572 [00:07<00:07, 38.33it/s]

Computing ESM2 embeddings: 56%| | 320/572 [00:08<00:06, 38.76it/s]

Computing ESM2 embeddings: 62%| | 352/572 [00:09<00:05, 39.11it/s]

Computing ESM2 embeddings: 67%| | 384/572 [00:10<00:04, 38.61it/s]

Computing ESM2 embeddings: 73%| | 416/572 [00:10<00:04, 38.86it/s]

Computing ESM2 embeddings: 78%| | 448/572 [00:11<00:03, 39.24it/s]

Computing ESM2 embeddings: 84%| | 480/572 [00:12<00:02, 37.99it/s]

Computing ESM2 embeddings: 90%| | 512/572 [00:13<00:01, 38.54it/s]

Computing ESM2 embeddings: 95%| | 544/572 [00:14<00:00, 38.94it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:15<00:00, 38.11it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.61it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.75it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.55it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 38.32it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.03it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.75it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 38.46it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 38.97it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 38.47it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 38.84it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:13, 39.29it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:13, 38.38it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:12, 38.81it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 39.12it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 38.65it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:09, 38.94it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:08, 39.35it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 38.23it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 38.72it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 39.10it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 38.63it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:04, 38.88it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:18<00:04, 39.30it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 38.21it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 38.71it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:21<00:01, 39.10it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:22<00:00, 38.12it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:23<00:00, 38.53it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 38.70it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.59it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.76it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.59it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 39.12it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 38.58it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 38.89it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.32it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.39it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.85it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 39.17it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 38.56it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 38.51it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 38.90it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 38.21it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.80it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 39.17it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 38.21it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 38.60it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 39.05it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 38.00it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 38.52it/s] |

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.96it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.50it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.79it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.22it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.62it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.96it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.70it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:00<00:22, 37.67it/s]

Computing ESM2 embeddings: 7%| | 64/897 [00:01<00:21, 38.85it/s]

Computing ESM2 embeddings: 11%| | 96/897 [00:02<00:21, 36.50it/s]

Computing ESM2 embeddings: 14%| | 128/897 [00:03<00:20, 37.78it/s]

Computing ESM2 embeddings: 18%| | 160/897 [00:04<00:19, 38.59it/s]

Computing ESM2 embeddings: 21%| | 192/897 [00:05<00:18, 37.71it/s]

Computing ESM2 embeddings: 25%| | 224/897 [00:05<00:17, 38.30it/s]

Computing ESM2 embeddings: 29%| | 256/897 [00:06<00:16, 38.96it/s]

Computing ESM2 embeddings: 32%| | 288/897 [00:07<00:16, 37.91it/s]

Computing ESM2 embeddings: 36%| | 320/897 [00:08<00:14, 38.52it/s]

Computing ESM2 embeddings: 39%| | 352/897 [00:09<00:14, 38.86it/s]

Computing ESM2 embeddings: 43%| | 384/897 [00:10<00:13, 38.43it/s]

Computing ESM2 embeddings: 46%| | 416/897 [00:10<00:12, 38.76it/s]

```

Computing ESM2 embeddings: 50%|          | 448/897 [00:11<00:11, 39.23it/s]
Computing ESM2 embeddings: 54%|          | 480/897 [00:12<00:10, 38.12it/s]
Computing ESM2 embeddings: 57%|          | 512/897 [00:13<00:09, 38.63it/s]
Computing ESM2 embeddings: 61%|          | 544/897 [00:14<00:09, 39.04it/s]
Computing ESM2 embeddings: 64%|          | 576/897 [00:15<00:08, 38.15it/s]
Computing ESM2 embeddings: 68%|          | 608/897 [00:15<00:07, 38.57it/s]
Computing ESM2 embeddings: 71%|          | 640/897 [00:16<00:06, 39.08it/s]
Computing ESM2 embeddings: 75%|          | 672/897 [00:17<00:05, 38.21it/s]
Computing ESM2 embeddings: 78%|          | 704/897 [00:18<00:04, 38.72it/s]
Computing ESM2 embeddings: 82%|          | 736/897 [00:19<00:04, 39.12it/s]
Computing ESM2 embeddings: 86%|          | 768/897 [00:19<00:03, 38.63it/s]
Computing ESM2 embeddings: 89%|          | 800/897 [00:20<00:02, 38.91it/s]
Computing ESM2 embeddings: 93%|          | 832/897 [00:21<00:01, 39.31it/s]
Computing ESM2 embeddings: 96%|          | 864/897 [00:22<00:00, 38.20it/s]

Computing ESM2 embeddings: 100%|         | 897/897 [00:23<00:00, 38.47it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 38.30it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 39.05it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.77it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:20, 38.42it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 39.08it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/914 [00:05<00:19, 37.70it/s] |
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:17, 38.40it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:16, 38.94it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 37.95it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:15, 38.46it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 38.99it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:13, 37.93it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:10<00:12, 38.47it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:11, 38.91it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.95it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 38.45it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 38.99it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:14<00:08, 38.46it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:15<00:07, 38.88it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:16<00:06, 39.22it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 38.77it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 39.01it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:19<00:04, 39.38it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:19<00:03, 38.45it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:20<00:02, 38.88it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:21<00:02, 39.17it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:22<00:01, 38.64it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:23<00:00, 38.92it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:23<00:00, 38.63it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:23, 37.13it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:21, 38.38it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:21, 37.94it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:19, 38.60it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:18, 39.25it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:18, 38.16it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 38.78it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:16, 39.15it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:15, 38.58it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:14, 38.89it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:13, 39.32it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:09<00:13, 38.07it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:10<00:12, 38.61it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:11<00:11, 39.03it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 38.10it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:09, 38.54it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:08, 39.05it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:14<00:08, 38.03it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:15<00:07, 38.58it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:16<00:06, 39.00it/s] |

```

Computing ESM2 embeddings: 75%|      | 672/895 [00:17<00:05, 38.63it/s]
Computing ESM2 embeddings: 79%|      | 704/895 [00:18<00:04, 38.91it/s]
Computing ESM2 embeddings: 82%|      | 736/895 [00:19<00:04, 39.32it/s]
Computing ESM2 embeddings: 86%|      | 768/895 [00:19<00:03, 37.98it/s]
Computing ESM2 embeddings: 89%|      | 800/895 [00:20<00:02, 38.49it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:21<00:01, 38.89it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:22<00:00, 38.00it/s]
Computing ESM2 embeddings: 100%|     | 895/895 [00:23<00:00, 38.54it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/914 [00:00<00:23, 38.26it/s]
Computing ESM2 embeddings: 7%|      | 64/914 [00:01<00:21, 39.20it/s]
Computing ESM2 embeddings: 11%|     | 96/914 [00:02<00:21, 37.77it/s]
Computing ESM2 embeddings: 14%|     | 128/914 [00:03<00:20, 38.58it/s]
Computing ESM2 embeddings: 18%|     | 160/914 [00:04<00:19, 39.11it/s]
Computing ESM2 embeddings: 21%|     | 192/914 [00:04<00:18, 38.60it/s]
Computing ESM2 embeddings: 25%|     | 224/914 [00:05<00:17, 38.86it/s]
Computing ESM2 embeddings: 28%|     | 256/914 [00:06<00:16, 39.30it/s]
Computing ESM2 embeddings: 32%|     | 288/914 [00:07<00:16, 38.38it/s]
Computing ESM2 embeddings: 35%|     | 320/914 [00:08<00:15, 38.82it/s]
Computing ESM2 embeddings: 39%|     | 352/914 [00:09<00:14, 39.20it/s]
Computing ESM2 embeddings: 42%|     | 384/914 [00:09<00:13, 38.75it/s]

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Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 39.08it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.43it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.08it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.58it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 38.89it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 37.91it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 38.35it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.87it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.29it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.72it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:18<00:04, 39.10it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.09it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.50it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 38.96it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 37.85it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.39it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.61it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.78it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.61it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 37.68it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 38.21it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 38.69it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:18, 38.18it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 38.62it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 39.13it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 38.44it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 38.81it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:13, 39.14it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:09<00:13, 38.16it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 38.54it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 39.05it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 38.01it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 38.54it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 38.97it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:14<00:08, 37.96it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 38.40it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 38.89it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 38.12it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:04, 38.63it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 39.00it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:19<00:03, 38.16it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 38.52it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 39.02it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:22<00:00, 38.40it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.57it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/589 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 5%| | 32/589 [00:00<00:14, 39.30it/s]  
 Computing ESM2 embeddings: 11%| | 64/589 [00:01<00:13, 39.84it/s]  
 Computing ESM2 embeddings: 16%| | 96/589 [00:02<00:12, 38.86it/s]  
 Computing ESM2 embeddings: 22%| | 128/589 [00:03<00:11, 38.78it/s]  
 Computing ESM2 embeddings: 27%| | 160/589 [00:04<00:10, 39.17it/s]  
 Computing ESM2 embeddings: 33%| | 192/589 [00:04<00:10, 38.06it/s]  
 Computing ESM2 embeddings: 38%| | 224/589 [00:05<00:09, 38.65it/s]  
 Computing ESM2 embeddings: 43%| | 256/589 [00:06<00:08, 39.08it/s]  
 Computing ESM2 embeddings: 49%| | 288/589 [00:07<00:07, 38.15it/s]  
 Computing ESM2 embeddings: 54%| | 320/589 [00:08<00:06, 38.59it/s]  
 Computing ESM2 embeddings: 60%| | 352/589 [00:09<00:06, 39.06it/s]  
 Computing ESM2 embeddings: 65%| | 384/589 [00:09<00:05, 38.44it/s]  
 Computing ESM2 embeddings: 71%| | 416/589 [00:10<00:04, 38.85it/s]  
 Computing ESM2 embeddings: 76%| | 448/589 [00:11<00:03, 39.19it/s]  
 Computing ESM2 embeddings: 81%| | 480/589 [00:12<00:02, 38.68it/s]  
 Computing ESM2 embeddings: 87%| | 512/589 [00:13<00:01, 38.93it/s]  
 Computing ESM2 embeddings: 92%| | 544/589 [00:14<00:01, 39.03it/s]  
 Computing ESM2 embeddings: 98%| | 576/589 [00:14<00:00, 37.47it/s]  
 Computing ESM2 embeddings: 100%| | 589/589 [00:15<00:00, 38.41it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.03it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.98it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 38.35it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 38.83it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.37it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.01it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.62it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.09it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 38.18it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 38.61it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:13, 39.12it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.00it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.57it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.02it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.07it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 38.50it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 39.04it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.22it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.67it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.08it/s] |

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.15it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 38.29it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.74it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.09it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.64it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.05it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.54it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.58it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.07it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.18it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.47it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.39it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.99it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:18, 38.01it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 38.52it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.08it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.85it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.45it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.93it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.40it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.69it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.16it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.08it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.59it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 39.02it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.01it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.47it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 39.00it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.09it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.60it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 39.05it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.10it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.33it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.79it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 37.71it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.33it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.42it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.06it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.92it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 36.42it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 37.70it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 38.51it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:18, 38.18it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 38.30it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 38.79it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.94it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 38.49it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:13, 38.92it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 38.43it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 38.72it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 39.16it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 38.07it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 38.57it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 38.97it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.93it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 38.35it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 38.87it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 37.93it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:04, 38.45it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 38.86it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.94it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 38.35it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 38.87it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:22<00:00, 37.76it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.29it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.17it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 38.50it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.39it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.11it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.54it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.34it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 38.15it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 38.76it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.36it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.40it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.85it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:13, 38.09it/s]  
 Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.61it/s]  
 Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.06it/s]  
 Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.05it/s]  
 Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.49it/s]  
 Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 39.01it/s]  
 Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 38.01it/s]  
 Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.54it/s]



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Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.99it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.10it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.54it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 39.06it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 38.25it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.72it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.07it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 38.16it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.57it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.36it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:23, 36.86it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 38.09it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 36.59it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 37.65it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 38.54it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:18, 37.57it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 38.32it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 38.82it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 38.33it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 38.63it/s]

```

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:13, 39.09it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 38.45it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.86it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.14it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 37.89it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 38.31it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.86it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 38.27it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.73it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.07it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.17it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.54it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 39.02it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.44it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.84it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.13it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.56it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.45it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:22, 39.15it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:21, 39.88it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/913 [00:02<00:21, 38.26it/s]  |
| Computing ESM2 embeddings: 14% | 128/913 [00:03<00:20, 38.91it/s] |
| Computing ESM2 embeddings: 18% | 160/913 [00:04<00:19, 39.36it/s] |
| Computing ESM2 embeddings: 21% | 192/913 [00:04<00:18, 38.76it/s] |
| Computing ESM2 embeddings: 25% | 224/913 [00:05<00:17, 39.03it/s] |
| Computing ESM2 embeddings: 28% | 256/913 [00:06<00:16, 39.43it/s] |
| Computing ESM2 embeddings: 32% | 288/913 [00:07<00:16, 38.74it/s] |
| Computing ESM2 embeddings: 35% | 320/913 [00:08<00:15, 39.11it/s] |
| Computing ESM2 embeddings: 39% | 352/913 [00:08<00:14, 39.37it/s] |
| Computing ESM2 embeddings: 42% | 384/913 [00:09<00:13, 38.78it/s] |
| Computing ESM2 embeddings: 46% | 416/913 [00:10<00:12, 38.78it/s] |
| Computing ESM2 embeddings: 49% | 448/913 [00:11<00:11, 39.11it/s] |
| Computing ESM2 embeddings: 53% | 480/913 [00:12<00:11, 37.93it/s] |
| Computing ESM2 embeddings: 56% | 512/913 [00:13<00:10, 38.51it/s] |
| Computing ESM2 embeddings: 60% | 544/913 [00:13<00:09, 38.96it/s] |
| Computing ESM2 embeddings: 63% | 576/913 [00:14<00:08, 38.58it/s] |
| Computing ESM2 embeddings: 67% | 608/913 [00:15<00:07, 38.89it/s] |
| Computing ESM2 embeddings: 70% | 640/913 [00:16<00:06, 39.30it/s] |
| Computing ESM2 embeddings: 74% | 672/913 [00:17<00:06, 38.18it/s] |
| Computing ESM2 embeddings: 77% | 704/913 [00:18<00:05, 38.68it/s] |
| Computing ESM2 embeddings: 81% | 736/913 [00:18<00:04, 39.08it/s] |
| Computing ESM2 embeddings: 84% | 768/913 [00:19<00:03, 38.59it/s] |
| Computing ESM2 embeddings: 88% | 800/913 [00:20<00:02, 38.86it/s] |
| Computing ESM2 embeddings: 91% | 832/913 [00:21<00:02, 39.28it/s] |

Computing ESM2 embeddings: 95%| | 864/913 [00:22<00:01, 38.17it/s]  
Computing ESM2 embeddings: 98%| | 896/913 [00:23<00:00, 38.64it/s]  
Computing ESM2 embeddings: 100%| | 913/913 [00:23<00:00, 38.81it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 39.10it/s]  
Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:20, 39.62it/s]  
Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.46it/s]  
Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 39.04it/s]  
Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.42it/s]  
Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:18, 38.76it/s]  
Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 39.02it/s]  
Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 39.43it/s]  
Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 38.30it/s]  
Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 38.79it/s]  
Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:13, 39.15it/s]  
Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 38.74it/s]  
Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.99it/s]  
Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 39.38it/s]  
Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 38.06it/s]  
Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 38.58it/s]  
Computing ESM2 embeddings: 61%| | 544/896 [00:13<00:09, 39.00it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 37.95it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.41it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.97it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.43it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.86it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.20it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.71it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.96it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.35it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.45it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.83it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/588 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/588 [00:00<00:14, 38.28it/s]

Computing ESM2 embeddings: 11%| | 64/588 [00:01<00:13, 39.22it/s]

Computing ESM2 embeddings: 16%| | 96/588 [00:02<00:13, 37.59it/s]

Computing ESM2 embeddings: 22%| | 128/588 [00:03<00:12, 38.14it/s]

Computing ESM2 embeddings: 27%| | 160/588 [00:04<00:11, 38.62it/s]

Computing ESM2 embeddings: 33%| | 192/588 [00:05<00:10, 37.54it/s]

Computing ESM2 embeddings: 38%| | 224/588 [00:05<00:09, 38.16it/s]

Computing ESM2 embeddings: 44%| | 256/588 [00:06<00:08, 38.78it/s]

Computing ESM2 embeddings: 49%| | 288/588 [00:07<00:07, 37.64it/s]

Computing ESM2 embeddings: 54%| | 320/588 [00:08<00:07, 38.27it/s]  
 Computing ESM2 embeddings: 60%| | 352/588 [00:09<00:06, 38.75it/s]  
 Computing ESM2 embeddings: 65%| | 384/588 [00:10<00:05, 37.74it/s]  
 Computing ESM2 embeddings: 71%| | 416/588 [00:10<00:04, 38.20it/s]  
 Computing ESM2 embeddings: 76%| | 448/588 [00:11<00:03, 38.75it/s]  
 Computing ESM2 embeddings: 82%| | 480/588 [00:12<00:02, 37.99it/s]  
 Computing ESM2 embeddings: 87%| | 512/588 [00:13<00:01, 38.23it/s]  
 Computing ESM2 embeddings: 93%| | 544/588 [00:14<00:01, 38.60it/s]  
 Computing ESM2 embeddings: 98%| | 576/588 [00:15<00:00, 37.66it/s]

Computing ESM2 embeddings: 100%| | 588/588 [00:15<00:00, 38.07it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 39.20it/s]  
 Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:20, 39.88it/s]  
 Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.86it/s]  
 Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 38.60it/s]  
 Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.01it/s]  
 Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:18, 38.47it/s]  
 Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.42it/s]  
 Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 38.81it/s]  
 Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 38.02it/s]  
 Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 38.57it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:13, 39.00it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:09<00:13, 37.97it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.44it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.99it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 38.13it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 38.65it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 39.04it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 38.11it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.50it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.02it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.41it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.84it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 39.18it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.07it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.49it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.98it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 37.97it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.54it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:22, 38.46it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:21, 39.09it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/913 [00:02<00:21, 37.80it/s]  |
| Computing ESM2 embeddings: 14% | 128/913 [00:03<00:20, 38.48it/s] |
| Computing ESM2 embeddings: 18% | 160/913 [00:04<00:19, 39.15it/s] |
| Computing ESM2 embeddings: 21% | 192/913 [00:04<00:18, 38.52it/s] |
| Computing ESM2 embeddings: 25% | 224/913 [00:05<00:17, 38.97it/s] |
| Computing ESM2 embeddings: 28% | 256/913 [00:06<00:16, 39.33it/s] |
| Computing ESM2 embeddings: 32% | 288/913 [00:07<00:16, 38.78it/s] |
| Computing ESM2 embeddings: 35% | 320/913 [00:08<00:15, 38.63it/s] |
| Computing ESM2 embeddings: 39% | 352/913 [00:09<00:14, 39.01it/s] |
| Computing ESM2 embeddings: 42% | 384/913 [00:09<00:13, 37.87it/s] |
| Computing ESM2 embeddings: 46% | 416/913 [00:10<00:12, 38.42it/s] |
| Computing ESM2 embeddings: 49% | 448/913 [00:11<00:11, 38.91it/s] |
| Computing ESM2 embeddings: 53% | 480/913 [00:12<00:11, 38.54it/s] |
| Computing ESM2 embeddings: 56% | 512/913 [00:13<00:10, 38.85it/s] |
| Computing ESM2 embeddings: 60% | 544/913 [00:14<00:09, 39.26it/s] |
| Computing ESM2 embeddings: 63% | 576/913 [00:14<00:08, 38.31it/s] |
| Computing ESM2 embeddings: 67% | 608/913 [00:15<00:07, 38.88it/s] |
| Computing ESM2 embeddings: 70% | 640/913 [00:16<00:06, 39.23it/s] |
| Computing ESM2 embeddings: 74% | 672/913 [00:17<00:06, 38.74it/s] |
| Computing ESM2 embeddings: 77% | 704/913 [00:18<00:05, 38.99it/s] |
| Computing ESM2 embeddings: 81% | 736/913 [00:18<00:04, 39.37it/s] |
| Computing ESM2 embeddings: 84% | 768/913 [00:19<00:03, 38.26it/s] |
| Computing ESM2 embeddings: 88% | 800/913 [00:20<00:02, 38.75it/s] |
| Computing ESM2 embeddings: 91% | 832/913 [00:21<00:02, 39.12it/s] |



```

Computing ESM2 embeddings: 95%|      | 864/913 [00:22<00:01, 38.07it/s]

Computing ESM2 embeddings: 98%|      | 896/913 [00:23<00:00, 38.53it/s]

Computing ESM2 embeddings: 100%|     | 913/913 [00:23<00:00, 38.71it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 39.16it/s]

Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:20, 39.66it/s]

Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 38.01it/s]

Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:19, 38.54it/s]

Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:18, 39.16it/s]

Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:18, 37.71it/s]

Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 38.41it/s]

Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 38.89it/s]

Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.97it/s]

Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 38.46it/s]

Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:13, 38.98it/s]

Computing ESM2 embeddings: 43%|      | 384/896 [00:09<00:13, 38.38it/s]

Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 38.81it/s]

Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 39.18it/s]

Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 38.64it/s]

Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:09, 38.89it/s]

Computing ESM2 embeddings: 61%|      | 544/896 [00:14<00:08, 39.30it/s]

```

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 38.09it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.60it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.99it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.14it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.55it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 39.04it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.01it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.54it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.96it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.58it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.54it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 36.41it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.95it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.67it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 37.86it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.58it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.57it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 38.16it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 38.81it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.77it/s]

```

Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 38.07it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 38.49it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:13, 38.21it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 38.59it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.05it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.10it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.61it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 38.99it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:08, 38.51it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 38.78it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:06, 39.18it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.36it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.82it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 39.13it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 38.13it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.50it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.00it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 38.43it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.83it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.38it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          | 0/896 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.95it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.03it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 38.08it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 38.78it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.21it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.64it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.90it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.31it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 38.34it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 38.79it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:13, 39.14it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.09it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.45it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.76it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.38it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 38.10it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 38.63it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 37.83it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 37.99it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.51it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.98it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:04, 38.49it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 38.92it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:19<00:03, 38.22it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 38.59it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.03it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 38.10it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 38.44it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/914 [00:00<00:22, 38.77it/s]
Computing ESM2 embeddings: 7%|        | 64/914 [00:01<00:21, 39.23it/s]
Computing ESM2 embeddings: 11%|       | 96/914 [00:02<00:21, 38.18it/s]
Computing ESM2 embeddings: 14%|       | 128/914 [00:03<00:20, 38.71it/s]
Computing ESM2 embeddings: 18%|       | 160/914 [00:04<00:19, 39.26it/s]
Computing ESM2 embeddings: 21%|       | 192/914 [00:04<00:18, 38.02it/s]
Computing ESM2 embeddings: 25%|       | 224/914 [00:05<00:17, 38.60it/s]
Computing ESM2 embeddings: 28%|       | 256/914 [00:06<00:16, 39.08it/s]
Computing ESM2 embeddings: 32%|       | 288/914 [00:07<00:16, 38.00it/s]
Computing ESM2 embeddings: 35%|       | 320/914 [00:08<00:15, 38.51it/s]
Computing ESM2 embeddings: 39%|       | 352/914 [00:09<00:14, 39.01it/s]
Computing ESM2 embeddings: 42%|       | 384/914 [00:09<00:13, 38.49it/s]
Computing ESM2 embeddings: 46%|       | 416/914 [00:10<00:12, 38.58it/s]
Computing ESM2 embeddings: 49%|       | 448/914 [00:11<00:11, 38.86it/s]
Computing ESM2 embeddings: 53%|       | 480/914 [00:12<00:11, 38.40it/s]
Computing ESM2 embeddings: 56%|       | 512/914 [00:13<00:10, 38.54it/s]

```

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.90it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 37.95it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.51it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.96it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.58it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.85it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 39.23it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.22it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.58it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.76it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.74it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 39.04it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.69it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 39.10it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.56it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.44it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 38.89it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.38it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:04<00:18, 38.08it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.69it/s]

```

Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 39.12it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 38.63it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 38.89it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:13, 39.31it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:09<00:13, 38.34it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.80it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.18it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 38.28it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 38.29it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 38.76it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 38.03it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.55it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.92it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.51it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:04, 38.81it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:18<00:04, 39.25it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.42it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.97it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.31it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 38.76it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.73it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|          | 0/573 [00:00<?, ?it/s]
Computing ESM2 embeddings: 6%|          | 32/573 [00:00<00:14, 38.03it/s]
Computing ESM2 embeddings: 11%|         | 64/573 [00:01<00:13, 36.77it/s]
Computing ESM2 embeddings: 17%|         | 96/573 [00:02<00:12, 36.84it/s]
Computing ESM2 embeddings: 22%|         | 128/573 [00:03<00:11, 38.01it/s]
Computing ESM2 embeddings: 28%|         | 160/573 [00:04<00:10, 38.72it/s]
Computing ESM2 embeddings: 34%|         | 192/573 [00:05<00:10, 37.71it/s]
Computing ESM2 embeddings: 39%|         | 224/573 [00:05<00:09, 38.25it/s]
Computing ESM2 embeddings: 45%|         | 256/573 [00:06<00:08, 38.84it/s]
Computing ESM2 embeddings: 50%|         | 288/573 [00:07<00:07, 38.29it/s]
Computing ESM2 embeddings: 56%|         | 320/573 [00:08<00:06, 38.74it/s]
Computing ESM2 embeddings: 61%|         | 352/573 [00:09<00:05, 39.09it/s]
Computing ESM2 embeddings: 67%|         | 384/573 [00:10<00:04, 38.73it/s]
Computing ESM2 embeddings: 73%|         | 416/573 [00:10<00:04, 38.99it/s]
Computing ESM2 embeddings: 78%|         | 448/573 [00:11<00:03, 39.33it/s]
Computing ESM2 embeddings: 84%|         | 480/573 [00:12<00:02, 38.64it/s]
Computing ESM2 embeddings: 89%|         | 512/573 [00:13<00:01, 38.98it/s]
Computing ESM2 embeddings: 95%|         | 544/573 [00:14<00:00, 39.24it/s]

Computing ESM2 embeddings: 100%|        | 573/573 [00:15<00:00, 37.80it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings: 0%|          | 0/912 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/912 [00:00<00:22, 38.86it/s]

```



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/912 [00:01<00:21, 39.43it/s]  |
| Computing ESM2 embeddings: 11% | 96/912 [00:02<00:21, 38.56it/s]  |
| Computing ESM2 embeddings: 14% | 128/912 [00:03<00:20, 38.96it/s] |
| Computing ESM2 embeddings: 18% | 160/912 [00:04<00:19, 39.39it/s] |
| Computing ESM2 embeddings: 21% | 192/912 [00:04<00:18, 37.96it/s] |
| Computing ESM2 embeddings: 25% | 224/912 [00:05<00:17, 38.58it/s] |
| Computing ESM2 embeddings: 28% | 256/912 [00:06<00:16, 39.04it/s] |
| Computing ESM2 embeddings: 32% | 288/912 [00:07<00:16, 38.11it/s] |
| Computing ESM2 embeddings: 35% | 320/912 [00:08<00:15, 38.58it/s] |
| Computing ESM2 embeddings: 39% | 352/912 [00:09<00:14, 39.06it/s] |
| Computing ESM2 embeddings: 42% | 384/912 [00:09<00:13, 38.00it/s] |
| Computing ESM2 embeddings: 46% | 416/912 [00:10<00:12, 38.54it/s] |
| Computing ESM2 embeddings: 49% | 448/912 [00:11<00:11, 38.98it/s] |
| Computing ESM2 embeddings: 53% | 480/912 [00:12<00:11, 37.92it/s] |
| Computing ESM2 embeddings: 56% | 512/912 [00:13<00:10, 38.35it/s] |
| Computing ESM2 embeddings: 60% | 544/912 [00:14<00:09, 38.92it/s] |
| Computing ESM2 embeddings: 63% | 576/912 [00:14<00:08, 37.96it/s] |
| Computing ESM2 embeddings: 67% | 608/912 [00:15<00:07, 38.54it/s] |
| Computing ESM2 embeddings: 70% | 640/912 [00:16<00:06, 38.94it/s] |
| Computing ESM2 embeddings: 74% | 672/912 [00:17<00:06, 38.55it/s] |
| Computing ESM2 embeddings: 77% | 704/912 [00:18<00:05, 38.77it/s] |
| Computing ESM2 embeddings: 81% | 736/912 [00:19<00:04, 39.22it/s] |
| Computing ESM2 embeddings: 84% | 768/912 [00:19<00:03, 38.57it/s] |
| Computing ESM2 embeddings: 88% | 800/912 [00:20<00:02, 38.96it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/912 [00:21<00:02, 39.25it/s]
Computing ESM2 embeddings: 95%|      | 864/912 [00:22<00:01, 38.75it/s]
Computing ESM2 embeddings: 98%|      | 896/912 [00:23<00:00, 38.99it/s]
Computing ESM2 embeddings: 100%|     | 912/912 [00:23<00:00, 38.70it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/895 [00:00<00:22, 38.46it/s]
Computing ESM2 embeddings: 7%|      | 64/895 [00:01<00:21, 39.39it/s]
Computing ESM2 embeddings: 11%|     | 96/895 [00:02<00:20, 38.36it/s]
Computing ESM2 embeddings: 14%|     | 128/895 [00:03<00:19, 38.95it/s]
Computing ESM2 embeddings: 18%|     | 160/895 [00:04<00:18, 39.37it/s]
Computing ESM2 embeddings: 21%|     | 192/895 [00:04<00:18, 38.68it/s]
Computing ESM2 embeddings: 25%|     | 224/895 [00:05<00:17, 38.99it/s]
Computing ESM2 embeddings: 29%|     | 256/895 [00:06<00:16, 39.39it/s]
Computing ESM2 embeddings: 32%|     | 288/895 [00:07<00:15, 38.20it/s]
Computing ESM2 embeddings: 36%|     | 320/895 [00:08<00:14, 38.72it/s]
Computing ESM2 embeddings: 39%|     | 352/895 [00:09<00:13, 39.12it/s]
Computing ESM2 embeddings: 43%|     | 384/895 [00:09<00:13, 38.19it/s]
Computing ESM2 embeddings: 46%|     | 416/895 [00:10<00:12, 38.21it/s]
Computing ESM2 embeddings: 50%|     | 448/895 [00:11<00:11, 38.70it/s]
Computing ESM2 embeddings: 54%|     | 480/895 [00:12<00:11, 37.59it/s]
Computing ESM2 embeddings: 57%|     | 512/895 [00:13<00:10, 38.22it/s]

```

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 38.70it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:14<00:08, 37.89it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:15<00:07, 38.37it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:16<00:06, 38.86it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:17<00:05, 37.89it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:04, 38.47it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 38.88it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:19<00:03, 38.51it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:20<00:02, 38.81it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:21<00:01, 39.24it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:22<00:00, 38.36it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:23<00:00, 38.58it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.11it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.66it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.75it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 39.10it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.53it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 38.29it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 38.81it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.20it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.28it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.78it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 39.23it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.05it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.52it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 38.95it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.82it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.33it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.88it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 37.85it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.42it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.88it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.05it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.50it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.96it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 37.99it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.54it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.97it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.17it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.71it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.58it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.71it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:20, 39.67it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.81it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 38.60it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.15it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.50it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.84it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.28it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.99it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 38.52it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:13, 38.98it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.09it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.19it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.70it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.73it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 38.34it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 38.83it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 37.85it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.29it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.83it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 38.33it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:04, 38.78it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 39.16it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 38.63it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 38.91it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.30it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 38.61it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 38.58it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.82it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 38.82it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 38.13it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:20, 38.73it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 39.26it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:04<00:18, 38.53it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 38.91it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 39.22it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 38.59it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 38.84it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 39.23it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 38.62it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 38.98it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 39.18it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.93it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.34it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.84it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.33it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.77it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 39.11it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.56it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.77it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.16it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.29it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.72it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.05it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.09it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.47it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.69it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.77it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.05it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.06it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.94it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.73it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.72it/s]

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Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 38.34it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 38.83it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 38.37it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 38.67it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:13, 39.11it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 38.00it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.53it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 38.93it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 37.98it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 38.39it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 38.91it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 38.28it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.72it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.06it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.98it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 38.36it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 38.83it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.29it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.72it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.05it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 38.17it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.46it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```



for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.26it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.77it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.98it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.59it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 39.23it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.88it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 38.51it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.02it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.53it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.82it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 39.17it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 37.92it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 38.48it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 38.96it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 38.52it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.81it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 39.20it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 38.09it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 38.59it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 39.02it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 38.61it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 38.88it/s] |

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.30it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.69it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 39.05it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.33it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.39it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.70it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.69it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:14, 38.43it/s]

Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:12, 39.37it/s]

Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:12, 38.56it/s]

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:11, 38.89it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:10, 39.38it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:04<00:09, 38.49it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:05<00:08, 38.70it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 38.97it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 38.43it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 38.77it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 39.20it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:09<00:04, 38.15it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:10<00:04, 38.67it/s]

```

Computing ESM2 embeddings: 78%|          | 448/571 [00:11<00:03, 39.07it/s]
Computing ESM2 embeddings: 84%|          | 480/571 [00:12<00:02, 38.81it/s]
Computing ESM2 embeddings: 90%|          | 512/571 [00:13<00:01, 38.87it/s]
Computing ESM2 embeddings: 95%|          | 544/571 [00:13<00:00, 39.16it/s]
Computing ESM2 embeddings: 100%|         | 571/571 [00:14<00:00, 38.79it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:27, 32.23it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:23, 36.18it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:22, 35.85it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 37.15it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 38.14it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 37.27it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:06<00:18, 38.03it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 38.44it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 37.51it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 37.83it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 38.48it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 37.73it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:12, 38.32it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:12, 38.62it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 37.76it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.97it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.57it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.91it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 38.14it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.23it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.59it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.01it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.60it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.44it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.94it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.11it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 37.91it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.77it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.63it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.64it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.70it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.54it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 38.41it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.10it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.95it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 38.61it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 39.06it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 38.61it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 38.95it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:13, 39.36it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:09<00:13, 38.78it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:12, 39.08it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 39.39it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 38.40it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:09, 38.44it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 38.89it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 37.86it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 38.46it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 38.89it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 38.52it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:04, 38.79it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:18<00:04, 39.24it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 38.02it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 38.38it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:21<00:01, 38.74it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:22<00:00, 38.31it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:23<00:00, 38.59it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 38.23it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 38.74it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:22, 36.99it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.11it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.88it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:18, 38.42it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 38.79it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.22it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.59it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.99it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 39.34it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 38.27it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 38.66it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.19it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 38.06it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.59it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 39.01it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 38.64it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 38.89it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.31it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 38.67it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 39.02it/s] |

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.35it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.87it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.75it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.09it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.05it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.58it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.70it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.26it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.75it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 36.47it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.75it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.56it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 38.35it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.82it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 39.29it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 38.11it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:14, 38.65it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:13, 39.09it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 38.00it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.45it/s]

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Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 38.99it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 37.90it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:09, 38.43it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 38.90it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 38.52it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.79it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 39.19it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.15it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:04, 38.64it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 39.04it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.67it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.62it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.00it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 38.17it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.51it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:22, 39.41it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 39.77it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 38.35it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:20, 38.94it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 39.44it/s]

```



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/914 [00:04<00:18, 38.08it/s] |
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:17, 38.66it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:16, 39.07it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 38.61it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:15, 38.88it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 39.27it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:09<00:13, 38.17it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:10<00:12, 38.62it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:11, 39.02it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.99it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 38.44it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 38.97it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:14<00:08, 38.44it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:15<00:07, 38.85it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:16<00:06, 39.16it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 38.63it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 38.87it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:18<00:04, 39.27it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:19<00:03, 38.69it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:20<00:02, 39.04it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:21<00:02, 39.32it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:22<00:01, 38.17it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:23<00:00, 38.54it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:23<00:00, 38.75it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.50it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.13it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.95it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 38.60it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.01it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 37.87it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.52it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.01it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.91it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 38.38it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:13, 38.95it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.13it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.64it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.06it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.65it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 38.91it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:08, 39.33it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.11it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.61it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.04it/s] |

```

Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 38.62it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:04, 38.91it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 39.30it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 38.36it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 38.81it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 39.17it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 38.70it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 38.68it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/914 [00:00<00:22, 38.73it/s]
Computing ESM2 embeddings: 7%|        | 64/914 [00:01<00:21, 39.46it/s]
Computing ESM2 embeddings: 11%|       | 96/914 [00:02<00:21, 37.80it/s]
Computing ESM2 embeddings: 14%|       | 128/914 [00:03<00:20, 38.63it/s]
Computing ESM2 embeddings: 18%|       | 160/914 [00:04<00:19, 39.16it/s]
Computing ESM2 embeddings: 21%|       | 192/914 [00:05<00:19, 37.89it/s]
Computing ESM2 embeddings: 25%|       | 224/914 [00:05<00:17, 38.42it/s]
Computing ESM2 embeddings: 28%|       | 256/914 [00:06<00:16, 38.97it/s]
Computing ESM2 embeddings: 32%|       | 288/914 [00:07<00:16, 37.92it/s]
Computing ESM2 embeddings: 35%|       | 320/914 [00:08<00:15, 38.27it/s]
Computing ESM2 embeddings: 39%|       | 352/914 [00:09<00:14, 38.68it/s]
Computing ESM2 embeddings: 42%|       | 384/914 [00:09<00:13, 38.39it/s]

```

```

Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 38.74it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.16it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.33it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.91it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 39.25it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 38.19it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 38.60it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 39.06it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.22it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.70it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 39.10it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.57it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.86it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.27it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 38.17it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.66it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.59it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/571 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|          | 32/571 [00:00<00:14, 37.75it/s]
Computing ESM2 embeddings: 11%|          | 64/571 [00:01<00:12, 39.01it/s]
Computing ESM2 embeddings: 17%|          | 96/571 [00:02<00:12, 37.98it/s]

```

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:11, 38.88it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:10, 39.32it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:04<00:09, 38.62it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:05<00:08, 38.91it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 39.30it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 38.34it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 38.80it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 39.15it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:09<00:04, 38.31it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:10<00:03, 38.81it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:11<00:03, 39.24it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 38.16it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 38.68it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 39.09it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:14<00:00, 38.67it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:23, 37.69it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:21, 38.75it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:21, 37.59it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:20, 38.35it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:19, 39.04it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/913 [00:05<00:18, 38.34it/s] |
| Computing ESM2 embeddings: 25%  | 224/913 [00:05<00:17, 38.96it/s] |
| Computing ESM2 embeddings: 28%  | 256/913 [00:06<00:16, 39.27it/s] |
| Computing ESM2 embeddings: 32%  | 288/913 [00:07<00:16, 38.36it/s] |
| Computing ESM2 embeddings: 35%  | 320/913 [00:08<00:15, 38.81it/s] |
| Computing ESM2 embeddings: 39%  | 352/913 [00:09<00:14, 39.27it/s] |
| Computing ESM2 embeddings: 42%  | 384/913 [00:09<00:13, 38.25it/s] |
| Computing ESM2 embeddings: 46%  | 416/913 [00:10<00:12, 38.41it/s] |
| Computing ESM2 embeddings: 49%  | 448/913 [00:11<00:12, 38.71it/s] |
| Computing ESM2 embeddings: 53%  | 480/913 [00:12<00:11, 37.82it/s] |
| Computing ESM2 embeddings: 56%  | 512/913 [00:13<00:10, 38.30it/s] |
| Computing ESM2 embeddings: 60%  | 544/913 [00:14<00:09, 38.88it/s] |
| Computing ESM2 embeddings: 63%  | 576/913 [00:14<00:08, 37.91it/s] |
| Computing ESM2 embeddings: 67%  | 608/913 [00:15<00:07, 38.48it/s] |
| Computing ESM2 embeddings: 70%  | 640/913 [00:16<00:07, 38.93it/s] |
| Computing ESM2 embeddings: 74%  | 672/913 [00:17<00:06, 38.04it/s] |
| Computing ESM2 embeddings: 77%  | 704/913 [00:18<00:05, 38.46it/s] |
| Computing ESM2 embeddings: 81%  | 736/913 [00:19<00:04, 38.99it/s] |
| Computing ESM2 embeddings: 84%  | 768/913 [00:19<00:03, 38.21it/s] |
| Computing ESM2 embeddings: 88%  | 800/913 [00:20<00:02, 38.69it/s] |
| Computing ESM2 embeddings: 91%  | 832/913 [00:21<00:02, 39.08it/s] |
| Computing ESM2 embeddings: 95%  | 864/913 [00:22<00:01, 38.27it/s] |
| Computing ESM2 embeddings: 98%  | 896/913 [00:23<00:00, 38.72it/s] |
| Computing ESM2 embeddings: 100% | 913/913 [00:23<00:00, 38.57it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.59it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.19it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.75it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 38.48it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.09it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:04<00:18, 38.34it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.57it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.90it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.93it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 38.39it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:13, 38.90it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.44it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.88it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.23it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.86it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 38.94it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:08, 39.23it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 38.55it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.95it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 39.14it/s] |

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.27it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.78it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:18<00:04, 39.22it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.53it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.99it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 39.29it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.17it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.66it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.62it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 38.43it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.48it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 37.75it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.57it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.79it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 38.01it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.58it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.65it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.31it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.82it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:13, 38.62it/s]



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Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 39.00it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.39it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.76it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 39.08it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 39.35it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 38.80it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 39.03it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:06, 39.40it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.37it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.84it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 39.18it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.17it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.56it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.06it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 38.56it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.95it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.62it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.36it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 38.54it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 36.63it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 37.80it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 38.66it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:18, 37.56it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 38.31it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 38.86it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:15, 38.19it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 38.75it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:13, 38.93it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.34it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 38.09it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 38.64it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 38.34it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 38.68it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 38.89it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.46it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 37.94it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 38.40it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 37.63it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 38.17it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 38.74it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 38.42it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 38.87it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 39.21it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:22<00:00, 38.96it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.41it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.38it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.26it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.61it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.54it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.07it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.94it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 38.47it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.04it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.22it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.53it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.87it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.36it/s]  
 Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.73it/s]  
 Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.16it/s]  
 Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.11it/s]  
 Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.61it/s]  
 Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 39.05it/s]  
 Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.63it/s]  
 Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.88it/s]

```

Computing ESM2 embeddings: 70%|      | 640/914 [00:16<00:07, 39.12it/s]
Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 37.37it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 38.12it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 38.34it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.36it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:03, 37.99it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 38.65it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 37.74it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.35it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.38it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/896 [00:00<00:26, 32.62it/s]
Computing ESM2 embeddings: 7%|      | 64/896 [00:01<00:22, 36.52it/s]
Computing ESM2 embeddings: 11%|     | 96/896 [00:02<00:22, 36.02it/s]
Computing ESM2 embeddings: 14%|     | 128/896 [00:03<00:20, 37.40it/s]
Computing ESM2 embeddings: 18%|     | 160/896 [00:04<00:19, 38.26it/s]
Computing ESM2 embeddings: 21%|     | 192/896 [00:05<00:18, 37.48it/s]
Computing ESM2 embeddings: 25%|     | 224/896 [00:05<00:17, 38.11it/s]
Computing ESM2 embeddings: 29%|     | 256/896 [00:06<00:16, 38.73it/s]
Computing ESM2 embeddings: 32%|     | 288/896 [00:07<00:16, 37.92it/s]
Computing ESM2 embeddings: 36%|     | 320/896 [00:08<00:15, 38.35it/s]

```

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 38.45it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.36it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.98it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.61it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 38.38it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 38.89it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:08, 39.20it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 38.34it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.68it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.09it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.05it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.56it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.97it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 38.22it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.57it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.77it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 37.52it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.13it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.93it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.34it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 37.89it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 38.57it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 39.08it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.93it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 38.63it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 39.07it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.99it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 38.44it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 38.94it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:13, 37.89it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 38.46it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:11, 38.89it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.98it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 38.39it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 38.92it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:08, 37.86it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 38.41it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 38.84it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 38.43it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 38.69it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 39.14it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:19<00:03, 38.62it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 39.03it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:21<00:02, 39.28it/s] |

```

Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 38.29it/s]

Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.31it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.54it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/572 [00:00<?, ?it/s]

Computing ESM2 embeddings:  6%|      | 32/572 [00:00<00:13, 38.62it/s]

Computing ESM2 embeddings: 11%|      | 64/572 [00:01<00:12, 39.17it/s]

Computing ESM2 embeddings: 17%|      | 96/572 [00:02<00:12, 37.61it/s]

Computing ESM2 embeddings: 22%|      | 128/572 [00:03<00:11, 37.98it/s]

Computing ESM2 embeddings: 28%|      | 160/572 [00:04<00:10, 38.66it/s]

Computing ESM2 embeddings: 34%|      | 192/572 [00:05<00:10, 37.78it/s]

Computing ESM2 embeddings: 39%|      | 224/572 [00:05<00:09, 38.44it/s]

Computing ESM2 embeddings: 45%|      | 256/572 [00:06<00:08, 38.97it/s]

Computing ESM2 embeddings: 50%|      | 288/572 [00:07<00:07, 38.66it/s]

Computing ESM2 embeddings: 56%|      | 320/572 [00:08<00:06, 38.93it/s]

Computing ESM2 embeddings: 62%|      | 352/572 [00:09<00:05, 39.33it/s]

Computing ESM2 embeddings: 67%|      | 384/572 [00:09<00:04, 38.22it/s]

Computing ESM2 embeddings: 73%|      | 416/572 [00:10<00:04, 38.71it/s]

Computing ESM2 embeddings: 78%|      | 448/572 [00:11<00:03, 39.12it/s]

Computing ESM2 embeddings: 84%|      | 480/572 [00:12<00:02, 38.67it/s]

Computing ESM2 embeddings: 90%|      | 512/572 [00:13<00:01, 38.91it/s]

Computing ESM2 embeddings: 95%|      | 544/572 [00:14<00:00, 39.27it/s]

```

Computing ESM2 embeddings: 100%| | 572/572 [00:14<00:00, 38.48it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/913 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/913 [00:00<00:22, 39.07it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/913 [00:01<00:21, 39.83it/s]  |
| Computing ESM2 embeddings: | 11% | 96/913 [00:02<00:21, 38.13it/s]  |
| Computing ESM2 embeddings: | 14% | 128/913 [00:03<00:20, 38.82it/s] |
| Computing ESM2 embeddings: | 18% | 160/913 [00:04<00:19, 39.29it/s] |
| Computing ESM2 embeddings: | 21% | 192/913 [00:04<00:18, 38.77it/s] |
| Computing ESM2 embeddings: | 25% | 224/913 [00:05<00:17, 38.81it/s] |
| Computing ESM2 embeddings: | 28% | 256/913 [00:06<00:16, 39.15it/s] |
| Computing ESM2 embeddings: | 32% | 288/913 [00:07<00:16, 38.02it/s] |
| Computing ESM2 embeddings: | 35% | 320/913 [00:08<00:15, 38.57it/s] |
| Computing ESM2 embeddings: | 39% | 352/913 [00:09<00:14, 38.99it/s] |
| Computing ESM2 embeddings: | 42% | 384/913 [00:09<00:13, 38.03it/s] |
| Computing ESM2 embeddings: | 46% | 416/913 [00:10<00:13, 38.18it/s] |
| Computing ESM2 embeddings: | 49% | 448/913 [00:11<00:12, 38.57it/s] |
| Computing ESM2 embeddings: | 53% | 480/913 [00:12<00:11, 37.85it/s] |
| Computing ESM2 embeddings: | 56% | 512/913 [00:13<00:10, 38.26it/s] |
| Computing ESM2 embeddings: | 60% | 544/913 [00:14<00:09, 38.62it/s] |
| Computing ESM2 embeddings: | 63% | 576/913 [00:14<00:08, 37.81it/s] |
| Computing ESM2 embeddings: | 67% | 608/913 [00:15<00:07, 38.25it/s] |



```

Computing ESM2 embeddings: 70%|      | 640/913 [00:16<00:07, 38.82it/s]
Computing ESM2 embeddings: 74%|      | 672/913 [00:17<00:06, 38.07it/s]
Computing ESM2 embeddings: 77%|      | 704/913 [00:18<00:05, 38.57it/s]
Computing ESM2 embeddings: 81%|      | 736/913 [00:19<00:04, 38.98it/s]
Computing ESM2 embeddings: 84%|      | 768/913 [00:19<00:03, 38.59it/s]
Computing ESM2 embeddings: 88%|      | 800/913 [00:20<00:02, 38.87it/s]
Computing ESM2 embeddings: 91%|      | 832/913 [00:21<00:02, 39.24it/s]
Computing ESM2 embeddings: 95%|      | 864/913 [00:22<00:01, 38.34it/s]
Computing ESM2 embeddings: 98%|      | 896/913 [00:23<00:00, 38.76it/s]

Computing ESM2 embeddings: 100%|     | 913/913 [00:23<00:00, 38.54it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/914 [00:00<00:22, 38.75it/s]
Computing ESM2 embeddings: 7%|      | 64/914 [00:01<00:21, 39.57it/s]
Computing ESM2 embeddings: 11%|     | 96/914 [00:02<00:21, 37.97it/s]
Computing ESM2 embeddings: 14%|     | 128/914 [00:03<00:20, 38.74it/s]
Computing ESM2 embeddings: 18%|     | 160/914 [00:04<00:19, 39.22it/s]
Computing ESM2 embeddings: 21%|     | 192/914 [00:04<00:18, 38.03it/s]
Computing ESM2 embeddings: 25%|     | 224/914 [00:05<00:17, 38.51it/s]
Computing ESM2 embeddings: 28%|     | 256/914 [00:06<00:16, 39.04it/s]
Computing ESM2 embeddings: 32%|     | 288/914 [00:07<00:16, 38.00it/s]
Computing ESM2 embeddings: 35%|     | 320/914 [00:08<00:15, 38.39it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.83it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.44it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.77it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 39.22it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.29it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.75it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 39.13it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.65it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.91it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.29it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.22it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.69it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:18<00:04, 39.09it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.58it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.81it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 39.24it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.09it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.62it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.64it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:00<00:22, 37.93it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/897 [00:01<00:22, 37.65it/s]  |
| Computing ESM2 embeddings: 11% | 96/897 [00:02<00:21, 36.52it/s]  |
| Computing ESM2 embeddings: 14% | 128/897 [00:03<00:20, 37.77it/s] |
| Computing ESM2 embeddings: 18% | 160/897 [00:04<00:19, 38.51it/s] |
| Computing ESM2 embeddings: 21% | 192/897 [00:05<00:18, 38.32it/s] |
| Computing ESM2 embeddings: 25% | 224/897 [00:05<00:17, 38.79it/s] |
| Computing ESM2 embeddings: 29% | 256/897 [00:06<00:16, 39.22it/s] |
| Computing ESM2 embeddings: 32% | 288/897 [00:07<00:15, 38.29it/s] |
| Computing ESM2 embeddings: 36% | 320/897 [00:08<00:14, 38.78it/s] |
| Computing ESM2 embeddings: 39% | 352/897 [00:09<00:13, 39.08it/s] |
| Computing ESM2 embeddings: 43% | 384/897 [00:09<00:13, 38.53it/s] |
| Computing ESM2 embeddings: 46% | 416/897 [00:10<00:12, 38.81it/s] |
| Computing ESM2 embeddings: 50% | 448/897 [00:11<00:11, 39.21it/s] |
| Computing ESM2 embeddings: 54% | 480/897 [00:12<00:10, 38.17it/s] |
| Computing ESM2 embeddings: 57% | 512/897 [00:13<00:09, 38.61it/s] |
| Computing ESM2 embeddings: 61% | 544/897 [00:14<00:09, 38.98it/s] |
| Computing ESM2 embeddings: 64% | 576/897 [00:14<00:08, 38.48it/s] |
| Computing ESM2 embeddings: 68% | 608/897 [00:15<00:07, 38.71it/s] |
| Computing ESM2 embeddings: 71% | 640/897 [00:16<00:06, 39.10it/s] |
| Computing ESM2 embeddings: 75% | 672/897 [00:17<00:05, 38.23it/s] |
| Computing ESM2 embeddings: 78% | 704/897 [00:18<00:04, 38.67it/s] |
| Computing ESM2 embeddings: 82% | 736/897 [00:19<00:04, 38.67it/s] |
| Computing ESM2 embeddings: 86% | 768/897 [00:19<00:03, 37.61it/s] |
| Computing ESM2 embeddings: 89% | 800/897 [00:20<00:02, 38.11it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/897 [00:21<00:01, 38.69it/s]

Computing ESM2 embeddings: 96%|      | 864/897 [00:22<00:00, 38.24it/s]

Computing ESM2 embeddings: 100%|     | 897/897 [00:23<00:00, 38.38it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.78it/s]

Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 38.91it/s]

Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:22, 36.87it/s]

Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:20, 37.82it/s]

Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 38.68it/s]

Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:18, 38.16it/s]

Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:17, 38.70it/s]

Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 39.08it/s]

Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 38.62it/s]

Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 38.89it/s]

Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 39.31it/s]

Computing ESM2 embeddings: 42%|      | 384/914 [00:09<00:13, 38.56it/s]

Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 38.92it/s]

Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 39.19it/s]

Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 38.02it/s]

Computing ESM2 embeddings: 56%|      | 512/914 [00:13<00:10, 38.41it/s]

Computing ESM2 embeddings: 60%|      | 544/914 [00:14<00:09, 38.90it/s]

```

```

Computing ESM2 embeddings: 63%|          | 576/914 [00:14<00:08, 37.81it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:15<00:07, 38.39it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.85it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 38.47it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.81it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 39.16it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:19<00:03, 38.55it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:20<00:02, 38.91it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 39.22it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 38.15it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 38.53it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:23<00:00, 38.54it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 37.96it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:22, 36.33it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:22, 35.18it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 36.87it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 37.89it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:18, 37.14it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:06<00:17, 37.85it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 38.57it/s]

```

```

Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.62it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 38.27it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 38.75it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 38.46it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.84it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 39.23it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 38.46it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:09, 38.85it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:08, 39.14it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 38.01it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.10it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.58it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.46it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 38.10it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 38.64it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.68it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.82it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 38.37it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 37.70it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.04it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.40it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.80it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.72it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.34it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 39.02it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.84it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 38.48it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 38.96it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.51it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.82it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 39.26it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 38.19it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 38.69it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.08it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 38.70it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.94it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 39.32it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 38.75it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 38.97it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.31it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 38.18it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 38.66it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:18<00:04, 38.99it/s] |
| Computing ESM2 embeddings: | 84% | 768/914 [00:19<00:03, 37.99it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 38.40it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 38.89it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 37.81it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.40it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.59it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/884 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/884 [00:00<00:22, 37.92it/s]
Computing ESM2 embeddings:  7%|      | 64/884 [00:01<00:21, 38.59it/s]
Computing ESM2 embeddings: 11%|      | 96/884 [00:02<00:21, 36.33it/s]
Computing ESM2 embeddings: 14%|      | 128/884 [00:03<00:20, 37.37it/s]
Computing ESM2 embeddings: 18%|      | 160/884 [00:04<00:18, 38.16it/s]
Computing ESM2 embeddings: 22%|      | 192/884 [00:05<00:18, 37.33it/s]
Computing ESM2 embeddings: 25%|      | 224/884 [00:05<00:17, 38.05it/s]
Computing ESM2 embeddings: 29%|      | 256/884 [00:06<00:16, 38.75it/s]
Computing ESM2 embeddings: 33%|      | 288/884 [00:07<00:15, 37.74it/s]
Computing ESM2 embeddings: 36%|      | 320/884 [00:08<00:14, 38.37it/s]
Computing ESM2 embeddings: 40%|      | 352/884 [00:09<00:13, 38.89it/s]
Computing ESM2 embeddings: 43%|      | 384/884 [00:10<00:12, 38.51it/s]
Computing ESM2 embeddings: 47%|      | 416/884 [00:10<00:12, 38.79it/s]
Computing ESM2 embeddings: 51%|      | 448/884 [00:11<00:11, 39.23it/s]
Computing ESM2 embeddings: 54%|      | 480/884 [00:12<00:10, 38.46it/s]

```



Computing ESM2 embeddings: 58%| | 512/884 [00:13<00:09, 38.66it/s]

Computing ESM2 embeddings: 62%| | 544/884 [00:14<00:08, 38.90it/s]

Computing ESM2 embeddings: 65%| | 576/884 [00:15<00:08, 38.09it/s]

Computing ESM2 embeddings: 69%| | 608/884 [00:15<00:07, 38.53it/s]

Computing ESM2 embeddings: 72%| | 640/884 [00:16<00:06, 39.05it/s]

Computing ESM2 embeddings: 76%| | 672/884 [00:17<00:05, 38.20it/s]

Computing ESM2 embeddings: 80%| | 704/884 [00:18<00:04, 38.57it/s]

Computing ESM2 embeddings: 83%| | 736/884 [00:19<00:03, 39.09it/s]

Computing ESM2 embeddings: 87%| | 768/884 [00:20<00:03, 37.98it/s]

Computing ESM2 embeddings: 90%| | 800/884 [00:20<00:02, 38.53it/s]

Computing ESM2 embeddings: 94%| | 832/884 [00:21<00:01, 38.97it/s]

Computing ESM2 embeddings: 98%| | 864/884 [00:22<00:00, 38.55it/s]

Computing ESM2 embeddings: 100%| | 884/884 [00:23<00:00, 38.34it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/601 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/601 [00:00<00:14, 38.25it/s]

Computing ESM2 embeddings: 11%| | 64/601 [00:01<00:14, 36.65it/s]

Computing ESM2 embeddings: 16%| | 96/601 [00:02<00:14, 35.92it/s]

Computing ESM2 embeddings: 21%| | 128/601 [00:03<00:12, 37.08it/s]

Computing ESM2 embeddings: 27%| | 160/601 [00:04<00:11, 37.53it/s]

Computing ESM2 embeddings: 32%| | 192/601 [00:05<00:11, 36.31it/s]

Computing ESM2 embeddings: 37%| | 224/601 [00:06<00:10, 37.29it/s]

```

Computing ESM2 embeddings: 43%|          | 256/601 [00:06<00:09, 38.17it/s]
Computing ESM2 embeddings: 48%|          | 288/601 [00:07<00:08, 37.59it/s]
Computing ESM2 embeddings: 53%|          | 320/601 [00:08<00:07, 38.27it/s]
Computing ESM2 embeddings: 59%|          | 352/601 [00:09<00:06, 38.74it/s]
Computing ESM2 embeddings: 64%|          | 384/601 [00:10<00:05, 37.72it/s]
Computing ESM2 embeddings: 69%|          | 416/601 [00:11<00:04, 38.21it/s]
Computing ESM2 embeddings: 75%|          | 448/601 [00:11<00:03, 38.77it/s]
Computing ESM2 embeddings: 80%|          | 480/601 [00:12<00:03, 38.02it/s]
Computing ESM2 embeddings: 85%|          | 512/601 [00:13<00:02, 38.53it/s]
Computing ESM2 embeddings: 91%|          | 544/601 [00:14<00:01, 38.95it/s]
Computing ESM2 embeddings: 96%|          | 576/601 [00:15<00:00, 38.48it/s]

Computing ESM2 embeddings: 100%|         | 601/601 [00:15<00:00, 37.82it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/895 [00:00<00:22, 38.12it/s]
Computing ESM2 embeddings:  7%|          | 64/895 [00:01<00:21, 39.25it/s]
Computing ESM2 embeddings: 11%|          | 96/895 [00:02<00:21, 37.68it/s]
Computing ESM2 embeddings: 14%|          | 128/895 [00:03<00:19, 38.37it/s]
Computing ESM2 embeddings: 18%|          | 160/895 [00:04<00:18, 39.04it/s]
Computing ESM2 embeddings: 21%|          | 192/895 [00:05<00:18, 37.86it/s]
Computing ESM2 embeddings: 25%|          | 224/895 [00:05<00:17, 38.50it/s]
Computing ESM2 embeddings: 29%|          | 256/895 [00:06<00:16, 38.90it/s]

```

```

Computing ESM2 embeddings: 32%|          | 288/895 [00:07<00:16, 37.92it/s]
Computing ESM2 embeddings: 36%|          | 320/895 [00:08<00:14, 38.37it/s]
Computing ESM2 embeddings: 39%|          | 352/895 [00:09<00:13, 38.89it/s]
Computing ESM2 embeddings: 43%|          | 384/895 [00:09<00:13, 38.43it/s]
Computing ESM2 embeddings: 46%|          | 416/895 [00:10<00:12, 38.86it/s]
Computing ESM2 embeddings: 50%|          | 448/895 [00:11<00:11, 39.19it/s]
Computing ESM2 embeddings: 54%|          | 480/895 [00:12<00:10, 38.09it/s]
Computing ESM2 embeddings: 57%|          | 512/895 [00:13<00:09, 38.48it/s]
Computing ESM2 embeddings: 61%|          | 544/895 [00:14<00:09, 38.99it/s]
Computing ESM2 embeddings: 64%|          | 576/895 [00:14<00:08, 38.17it/s]
Computing ESM2 embeddings: 68%|          | 608/895 [00:15<00:07, 38.69it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:16<00:06, 39.07it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:17<00:05, 38.16it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:18<00:04, 38.51it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:19<00:04, 39.00it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:19<00:03, 37.99it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:20<00:02, 38.55it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:21<00:01, 38.92it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:22<00:00, 38.60it/s]

```

```

Computing ESM2 embeddings: 100%|         | 895/895 [00:23<00:00, 38.54it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']

```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```

Computing ESM2 embeddings:  0%|          | 0/914 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.07it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.82it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 38.00it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.75it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 39.22it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 38.73it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:17, 39.01it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 39.42it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.35it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.89it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 39.24it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:09<00:13, 38.75it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:12, 38.99it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:11, 39.35it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 38.19it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.66it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:13<00:09, 39.05it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:14<00:08, 38.59it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:15<00:07, 38.84it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:06, 39.26it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 38.20it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 38.62it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:18<00:04, 39.07it/s] |
| Computing ESM2 embeddings: | 84% | 768/914 [00:19<00:03, 38.29it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/914 [00:20<00:02, 38.79it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 39.16it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 38.26it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.75it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.73it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/897 [00:00<00:22, 37.68it/s]
Computing ESM2 embeddings:  7%|      | 64/897 [00:01<00:21, 38.88it/s]
Computing ESM2 embeddings: 11%|      | 96/897 [00:02<00:21, 36.89it/s]
Computing ESM2 embeddings: 14%|      | 128/897 [00:03<00:20, 38.01it/s]
Computing ESM2 embeddings: 18%|      | 160/897 [00:04<00:19, 38.72it/s]
Computing ESM2 embeddings: 21%|      | 192/897 [00:05<00:18, 37.63it/s]
Computing ESM2 embeddings: 25%|      | 224/897 [00:05<00:17, 38.24it/s]
Computing ESM2 embeddings: 29%|      | 256/897 [00:06<00:16, 38.83it/s]
Computing ESM2 embeddings: 32%|      | 288/897 [00:07<00:15, 38.31it/s]
Computing ESM2 embeddings: 36%|      | 320/897 [00:08<00:14, 38.73it/s]
Computing ESM2 embeddings: 39%|      | 352/897 [00:09<00:13, 38.99it/s]
Computing ESM2 embeddings: 43%|      | 384/897 [00:10<00:13, 38.06it/s]
Computing ESM2 embeddings: 46%|      | 416/897 [00:10<00:12, 38.47it/s]
Computing ESM2 embeddings: 50%|      | 448/897 [00:11<00:11, 38.95it/s]
Computing ESM2 embeddings: 54%|      | 480/897 [00:12<00:10, 38.43it/s]

```

Computing ESM2 embeddings: 57%| | 512/897 [00:13<00:09, 38.84it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:14<00:09, 39.18it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:14<00:08, 38.77it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:15<00:07, 39.03it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:16<00:06, 39.33it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:17<00:05, 38.20it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:18<00:05, 38.53it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:19<00:04, 38.99it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:19<00:03, 37.79it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:20<00:02, 38.39it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:21<00:01, 38.82it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:22<00:00, 38.36it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:23<00:00, 38.47it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:22, 39.41it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:21, 39.72it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:21, 37.82it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:20, 38.49it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:19, 39.12it/s]

Computing ESM2 embeddings: 21%| | 192/913 [00:04<00:18, 38.10it/s]

Computing ESM2 embeddings: 25%| | 224/913 [00:05<00:17, 38.76it/s]

Computing ESM2 embeddings: 28%| | 256/913 [00:06<00:16, 39.17it/s]

Computing ESM2 embeddings: 32%| | 288/913 [00:07<00:16, 38.66it/s]

Computing ESM2 embeddings: 35%| | 320/913 [00:08<00:15, 39.00it/s]

Computing ESM2 embeddings: 39%| | 352/913 [00:09<00:14, 39.29it/s]

Computing ESM2 embeddings: 42%| | 384/913 [00:09<00:13, 38.77it/s]

Computing ESM2 embeddings: 46%| | 416/913 [00:10<00:12, 38.97it/s]

Computing ESM2 embeddings: 49%| | 448/913 [00:11<00:11, 39.34it/s]

Computing ESM2 embeddings: 53%| | 480/913 [00:12<00:11, 38.17it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 38.64it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 39.02it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:14<00:08, 38.63it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:15<00:07, 38.88it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:16<00:06, 39.23it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:17<00:06, 38.66it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:18<00:05, 38.99it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:18<00:04, 39.28it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:19<00:03, 38.25it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:20<00:02, 38.63it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:21<00:02, 39.09it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:22<00:01, 38.20it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:23<00:00, 38.68it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:23<00:00, 38.75it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:23, 37.46it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.40it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.15it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 38.19it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 38.79it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 38.27it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.59it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.08it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:15, 38.42it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:14, 38.83it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:13, 39.15it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:09<00:13, 38.46it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.86it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 39.14it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.04it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 38.39it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 38.89it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:14<00:08, 37.95it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.48it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.85it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 38.41it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 38.35it/s] |



```

Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 38.76it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:19<00:03, 37.84it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:20<00:02, 38.37it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 38.83it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 37.87it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 38.39it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:25, 34.50it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 36.63it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:23, 35.22it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.78it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 37.96it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 37.35it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 38.13it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 38.67it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.74it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 38.27it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 38.85it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:13, 38.33it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 38.67it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 39.11it/s]

```

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.99it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.51it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.90it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 38.50it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.75it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:06, 39.16it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.55it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.90it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 39.19it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 38.16it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.48it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.95it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.07it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.56it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.22it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/828 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/828 [00:00<00:21, 37.24it/s]

Computing ESM2 embeddings: 8%| | 64/828 [00:01<00:19, 38.63it/s]

Computing ESM2 embeddings: 12%| | 96/828 [00:02<00:19, 37.71it/s]

Computing ESM2 embeddings: 15%| | 128/828 [00:03<00:18, 38.38it/s]

Computing ESM2 embeddings: 19%| | 160/828 [00:04<00:17, 39.01it/s]

```

Computing ESM2 embeddings: 23%|          | 192/828 [00:05<00:16, 37.98it/s]
Computing ESM2 embeddings: 27%|          | 224/828 [00:05<00:15, 38.56it/s]
Computing ESM2 embeddings: 31%|          | 256/828 [00:06<00:14, 38.97it/s]
Computing ESM2 embeddings: 35%|          | 288/828 [00:07<00:14, 37.95it/s]
Computing ESM2 embeddings: 39%|          | 320/828 [00:08<00:13, 38.38it/s]
Computing ESM2 embeddings: 43%|          | 352/828 [00:09<00:12, 38.91it/s]
Computing ESM2 embeddings: 46%|          | 384/828 [00:10<00:11, 37.98it/s]
Computing ESM2 embeddings: 50%|          | 416/828 [00:10<00:10, 38.43it/s]
Computing ESM2 embeddings: 54%|          | 448/828 [00:11<00:09, 38.92it/s]
Computing ESM2 embeddings: 58%|          | 480/828 [00:12<00:09, 37.84it/s]
Computing ESM2 embeddings: 62%|          | 512/828 [00:13<00:08, 38.38it/s]
Computing ESM2 embeddings: 66%|          | 544/828 [00:14<00:07, 38.77it/s]
Computing ESM2 embeddings: 70%|          | 576/828 [00:15<00:06, 37.87it/s]
Computing ESM2 embeddings: 73%|          | 608/828 [00:15<00:05, 38.31it/s]
Computing ESM2 embeddings: 77%|          | 640/828 [00:16<00:04, 38.82it/s]
Computing ESM2 embeddings: 81%|          | 672/828 [00:17<00:04, 37.87it/s]
Computing ESM2 embeddings: 85%|          | 704/828 [00:18<00:03, 38.45it/s]
Computing ESM2 embeddings: 89%|          | 736/828 [00:19<00:02, 38.86it/s]
Computing ESM2 embeddings: 93%|          | 768/828 [00:20<00:01, 37.98it/s]
Computing ESM2 embeddings: 97%|          | 800/828 [00:20<00:00, 38.39it/s]

Computing ESM2 embeddings: 100%|         | 828/828 [00:21<00:00, 38.36it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|          | 0/658 [00:00<?, ?it/s]
Computing ESM2 embeddings: 5%|          | 32/658 [00:00<00:16, 38.71it/s]
Computing ESM2 embeddings: 10%|         | 64/658 [00:01<00:15, 39.16it/s]
Computing ESM2 embeddings: 15%|         | 96/658 [00:02<00:14, 37.77it/s]
Computing ESM2 embeddings: 19%|         | 128/658 [00:03<00:13, 38.59it/s]
Computing ESM2 embeddings: 24%|         | 160/658 [00:04<00:12, 39.21it/s]
Computing ESM2 embeddings: 29%|         | 192/658 [00:05<00:12, 38.07it/s]
Computing ESM2 embeddings: 34%|         | 224/658 [00:05<00:11, 38.70it/s]
Computing ESM2 embeddings: 39%|         | 256/658 [00:06<00:10, 39.13it/s]
Computing ESM2 embeddings: 44%|         | 288/658 [00:07<00:09, 38.20it/s]
Computing ESM2 embeddings: 49%|         | 320/658 [00:08<00:08, 38.56it/s]
Computing ESM2 embeddings: 53%|         | 352/658 [00:09<00:07, 39.08it/s]
Computing ESM2 embeddings: 58%|         | 384/658 [00:09<00:07, 38.06it/s]
Computing ESM2 embeddings: 63%|         | 416/658 [00:10<00:06, 38.60it/s]
Computing ESM2 embeddings: 68%|         | 448/658 [00:11<00:05, 39.04it/s]
Computing ESM2 embeddings: 73%|         | 480/658 [00:12<00:04, 38.16it/s]
Computing ESM2 embeddings: 78%|         | 512/658 [00:13<00:03, 38.62it/s]
Computing ESM2 embeddings: 83%|         | 544/658 [00:14<00:02, 38.78it/s]
Computing ESM2 embeddings: 88%|         | 576/658 [00:15<00:02, 37.33it/s]
Computing ESM2 embeddings: 92%|         | 608/658 [00:15<00:01, 37.97it/s]
Computing ESM2 embeddings: 97%|         | 640/658 [00:16<00:00, 38.62it/s]
Computing ESM2 embeddings: 100%|        | 658/658 [00:17<00:00, 38.33it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.30it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.25it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.65it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 38.51it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:18, 39.07it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 38.16it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.55it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 39.08it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.96it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 38.20it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 38.58it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.65it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.26it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.78it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.00it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:09, 38.47it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 38.99it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.98it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:15<00:07, 38.27it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.63it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.87it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 38.37it/s] |

```

Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 38.92it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 38.02it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.33it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 38.38it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 37.09it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.23it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 38.31it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:21, 38.99it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.52it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:20, 38.28it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 39.02it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:18, 38.05it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:17, 38.66it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:16, 39.07it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 38.13it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 38.21it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 38.72it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:13, 37.96it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:10<00:12, 38.63it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:11, 39.00it/s]

```

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.05it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.46it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.98it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.92it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.50it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.96it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.11it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.53it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 39.05it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 38.04it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.59it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.69it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 37.33it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.98it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.39it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.53it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.11it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.29it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 38.95it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.44it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:04<00:18, 38.20it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 38.75it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 39.14it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:15, 38.20it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:14, 38.60it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 38.81it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 37.26it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:10<00:12, 38.03it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 38.65it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:11, 37.75it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 38.37it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 38.85it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:14<00:08, 38.04it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:15<00:07, 38.51it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:16<00:06, 39.02it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 37.98it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:04, 38.50it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 38.93it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:19<00:03, 38.07it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:20<00:02, 38.50it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:21<00:01, 39.00it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:22<00:00, 37.82it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:23<00:00, 38.43it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',



'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/911 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/911 [00:00<00:22, 38.41it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/911 [00:01<00:21, 39.15it/s]  |
| Computing ESM2 embeddings: | 11% | 96/911 [00:02<00:21, 37.63it/s]  |
| Computing ESM2 embeddings: | 14% | 128/911 [00:03<00:20, 38.40it/s] |
| Computing ESM2 embeddings: | 18% | 160/911 [00:04<00:19, 39.02it/s] |
| Computing ESM2 embeddings: | 21% | 192/911 [00:05<00:18, 38.02it/s] |
| Computing ESM2 embeddings: | 25% | 224/911 [00:05<00:17, 38.62it/s] |
| Computing ESM2 embeddings: | 28% | 256/911 [00:06<00:16, 39.08it/s] |
| Computing ESM2 embeddings: | 32% | 288/911 [00:07<00:16, 38.32it/s] |
| Computing ESM2 embeddings: | 35% | 320/911 [00:08<00:15, 38.54it/s] |
| Computing ESM2 embeddings: | 39% | 352/911 [00:09<00:14, 38.94it/s] |
| Computing ESM2 embeddings: | 42% | 384/911 [00:10<00:13, 37.84it/s] |
| Computing ESM2 embeddings: | 46% | 416/911 [00:10<00:12, 38.42it/s] |
| Computing ESM2 embeddings: | 49% | 448/911 [00:11<00:11, 38.88it/s] |
| Computing ESM2 embeddings: | 53% | 480/911 [00:12<00:11, 38.07it/s] |
| Computing ESM2 embeddings: | 56% | 512/911 [00:13<00:10, 38.47it/s] |
| Computing ESM2 embeddings: | 60% | 544/911 [00:14<00:09, 38.95it/s] |
| Computing ESM2 embeddings: | 63% | 576/911 [00:14<00:08, 38.15it/s] |
| Computing ESM2 embeddings: | 67% | 608/911 [00:15<00:07, 38.74it/s] |
| Computing ESM2 embeddings: | 70% | 640/911 [00:16<00:06, 39.14it/s] |
| Computing ESM2 embeddings: | 74% | 672/911 [00:17<00:06, 38.14it/s] |

Computing ESM2 embeddings: 77%| | 704/911 [00:18<00:05, 38.52it/s]

Computing ESM2 embeddings: 81%| | 736/911 [00:19<00:04, 39.01it/s]

Computing ESM2 embeddings: 84%| | 768/911 [00:19<00:03, 38.03it/s]

Computing ESM2 embeddings: 88%| | 800/911 [00:20<00:02, 38.55it/s]

Computing ESM2 embeddings: 91%| | 832/911 [00:21<00:02, 38.73it/s]

Computing ESM2 embeddings: 95%| | 864/911 [00:22<00:01, 37.40it/s]

Computing ESM2 embeddings: 98%| | 896/911 [00:23<00:00, 38.06it/s]

Computing ESM2 embeddings: 100%| | 911/911 [00:23<00:00, 38.31it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/899 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/899 [00:00<00:23, 37.31it/s]

Computing ESM2 embeddings: 7%| | 64/899 [00:01<00:21, 37.99it/s]

Computing ESM2 embeddings: 11%| | 96/899 [00:02<00:22, 36.31it/s]

Computing ESM2 embeddings: 14%| | 128/899 [00:03<00:20, 37.44it/s]

Computing ESM2 embeddings: 18%| | 160/899 [00:04<00:19, 38.31it/s]

Computing ESM2 embeddings: 21%| | 192/899 [00:05<00:18, 37.28it/s]

Computing ESM2 embeddings: 25%| | 224/899 [00:05<00:17, 38.10it/s]

Computing ESM2 embeddings: 28%| | 256/899 [00:06<00:16, 38.67it/s]

Computing ESM2 embeddings: 32%| | 288/899 [00:07<00:16, 37.72it/s]

Computing ESM2 embeddings: 36%| | 320/899 [00:08<00:15, 38.18it/s]

Computing ESM2 embeddings: 39%| | 352/899 [00:09<00:14, 38.76it/s]

Computing ESM2 embeddings: 43%| | 384/899 [00:10<00:13, 37.87it/s]

Computing ESM2 embeddings: 46%| | 416/899 [00:10<00:12, 38.41it/s]

Computing ESM2 embeddings: 50%| | 448/899 [00:11<00:11, 38.74it/s]

Computing ESM2 embeddings: 53%| | 480/899 [00:12<00:11, 37.60it/s]

Computing ESM2 embeddings: 57%| | 512/899 [00:13<00:10, 37.96it/s]

Computing ESM2 embeddings: 61%| | 544/899 [00:14<00:09, 38.57it/s]

Computing ESM2 embeddings: 64%| | 576/899 [00:15<00:08, 37.60it/s]

Computing ESM2 embeddings: 68%| | 608/899 [00:15<00:07, 38.23it/s]

Computing ESM2 embeddings: 71%| | 640/899 [00:16<00:06, 38.69it/s]

Computing ESM2 embeddings: 75%| | 672/899 [00:17<00:05, 37.86it/s]

Computing ESM2 embeddings: 78%| | 704/899 [00:18<00:05, 38.04it/s]

Computing ESM2 embeddings: 82%| | 736/899 [00:19<00:04, 38.57it/s]

Computing ESM2 embeddings: 85%| | 768/899 [00:20<00:03, 37.52it/s]

Computing ESM2 embeddings: 89%| | 800/899 [00:21<00:02, 37.86it/s]

Computing ESM2 embeddings: 93%| | 832/899 [00:21<00:01, 38.27it/s]

Computing ESM2 embeddings: 96%| | 864/899 [00:22<00:00, 37.50it/s]

Computing ESM2 embeddings: 100%| | 899/899 [00:23<00:00, 37.96it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/906 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/906 [00:00<00:23, 37.05it/s]

Computing ESM2 embeddings: 7%| | 64/906 [00:01<00:21, 38.68it/s]

Computing ESM2 embeddings: 11%| | 96/906 [00:02<00:21, 37.54it/s]

Computing ESM2 embeddings: 14%| | 128/906 [00:03<00:20, 37.55it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18% | 160/906 [00:04<00:19, 37.57it/s] |
| Computing ESM2 embeddings: 21% | 192/906 [00:05<00:19, 36.28it/s] |
| Computing ESM2 embeddings: 25% | 224/906 [00:05<00:18, 37.33it/s] |
| Computing ESM2 embeddings: 28% | 256/906 [00:06<00:17, 38.23it/s] |
| Computing ESM2 embeddings: 32% | 288/906 [00:07<00:16, 37.43it/s] |
| Computing ESM2 embeddings: 35% | 320/906 [00:08<00:15, 37.88it/s] |
| Computing ESM2 embeddings: 39% | 352/906 [00:09<00:14, 38.33it/s] |
| Computing ESM2 embeddings: 42% | 384/906 [00:10<00:13, 37.66it/s] |
| Computing ESM2 embeddings: 46% | 416/906 [00:11<00:12, 38.19it/s] |
| Computing ESM2 embeddings: 49% | 448/906 [00:11<00:11, 38.79it/s] |
| Computing ESM2 embeddings: 53% | 480/906 [00:12<00:11, 37.83it/s] |
| Computing ESM2 embeddings: 57% | 512/906 [00:13<00:10, 38.41it/s] |
| Computing ESM2 embeddings: 60% | 544/906 [00:14<00:09, 38.84it/s] |
| Computing ESM2 embeddings: 64% | 576/906 [00:15<00:08, 38.20it/s] |
| Computing ESM2 embeddings: 67% | 608/906 [00:15<00:07, 38.66it/s] |
| Computing ESM2 embeddings: 71% | 640/906 [00:16<00:06, 39.11it/s] |
| Computing ESM2 embeddings: 74% | 672/906 [00:17<00:06, 38.06it/s] |
| Computing ESM2 embeddings: 78% | 704/906 [00:18<00:05, 38.62it/s] |
| Computing ESM2 embeddings: 81% | 736/906 [00:19<00:04, 38.96it/s] |
| Computing ESM2 embeddings: 85% | 768/906 [00:20<00:03, 37.95it/s] |
| Computing ESM2 embeddings: 88% | 800/906 [00:20<00:02, 38.43it/s] |
| Computing ESM2 embeddings: 92% | 832/906 [00:21<00:01, 38.68it/s] |
| Computing ESM2 embeddings: 95% | 864/906 [00:22<00:01, 37.17it/s] |
| Computing ESM2 embeddings: 99% | 896/906 [00:23<00:00, 37.94it/s] |

Computing ESM2 embeddings: 100%| | 906/906 [00:23<00:00, 38.05it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/779 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 4%| | 32/779 [00:00<00:19, 38.38it/s]  
Computing ESM2 embeddings: 8%| | 64/779 [00:01<00:18, 39.22it/s]  
Computing ESM2 embeddings: 12%| | 96/779 [00:02<00:18, 37.54it/s]  
Computing ESM2 embeddings: 16%| | 128/779 [00:03<00:16, 38.44it/s]  
Computing ESM2 embeddings: 21%| | 160/779 [00:04<00:15, 39.00it/s]  
Computing ESM2 embeddings: 25%| | 192/779 [00:05<00:15, 38.11it/s]  
Computing ESM2 embeddings: 29%| | 224/779 [00:05<00:14, 38.67it/s]  
Computing ESM2 embeddings: 33%| | 256/779 [00:06<00:13, 39.18it/s]  
Computing ESM2 embeddings: 37%| | 288/779 [00:07<00:12, 38.09it/s]  
Computing ESM2 embeddings: 41%| | 320/779 [00:08<00:11, 38.62it/s]  
Computing ESM2 embeddings: 45%| | 352/779 [00:09<00:10, 39.02it/s]  
Computing ESM2 embeddings: 49%| | 384/779 [00:09<00:10, 38.17it/s]  
Computing ESM2 embeddings: 53%| | 416/779 [00:10<00:09, 38.56it/s]  
Computing ESM2 embeddings: 58%| | 448/779 [00:11<00:08, 39.03it/s]  
Computing ESM2 embeddings: 62%| | 480/779 [00:12<00:07, 38.04it/s]  
Computing ESM2 embeddings: 66%| | 512/779 [00:13<00:06, 38.58it/s]  
Computing ESM2 embeddings: 70%| | 544/779 [00:14<00:06, 38.97it/s]  
Computing ESM2 embeddings: 74%| | 576/779 [00:14<00:05, 38.06it/s]  
Computing ESM2 embeddings: 78%| | 608/779 [00:15<00:04, 38.49it/s]

```

Computing ESM2 embeddings: 82%|      | 640/779 [00:16<00:03, 38.97it/s]
Computing ESM2 embeddings: 86%|      | 672/779 [00:17<00:02, 38.04it/s]
Computing ESM2 embeddings: 90%|      | 704/779 [00:18<00:01, 38.56it/s]
Computing ESM2 embeddings: 94%|      | 736/779 [00:19<00:01, 38.98it/s]
Computing ESM2 embeddings: 99%|      | 768/779 [00:19<00:00, 37.96it/s]

Computing ESM2 embeddings: 100%|     | 779/779 [00:20<00:00, 38.35it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/697 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/697 [00:00<00:17, 37.75it/s]
Computing ESM2 embeddings:  9%|      | 64/697 [00:01<00:16, 37.54it/s]
Computing ESM2 embeddings: 14%|      | 96/697 [00:02<00:16, 36.21it/s]
Computing ESM2 embeddings: 18%|      | 128/697 [00:03<00:15, 37.56it/s]
Computing ESM2 embeddings: 23%|      | 160/697 [00:04<00:14, 38.35it/s]
Computing ESM2 embeddings: 28%|      | 192/697 [00:05<00:13, 37.69it/s]
Computing ESM2 embeddings: 32%|      | 224/697 [00:05<00:12, 38.22it/s]
Computing ESM2 embeddings: 37%|      | 256/697 [00:06<00:11, 38.83it/s]
Computing ESM2 embeddings: 41%|      | 288/697 [00:07<00:10, 37.75it/s]
Computing ESM2 embeddings: 46%|      | 320/697 [00:08<00:09, 38.36it/s]
Computing ESM2 embeddings: 51%|      | 352/697 [00:09<00:08, 38.80it/s]
Computing ESM2 embeddings: 55%|      | 384/697 [00:10<00:08, 37.86it/s]
Computing ESM2 embeddings: 60%|      | 416/697 [00:10<00:07, 38.28it/s]
Computing ESM2 embeddings: 64%|      | 448/697 [00:11<00:06, 38.81it/s]

```

Computing ESM2 embeddings: 69%| | 480/697 [00:12<00:05, 37.92it/s]

Computing ESM2 embeddings: 73%| | 512/697 [00:13<00:04, 38.41it/s]

Computing ESM2 embeddings: 78%| | 544/697 [00:14<00:03, 38.83it/s]

Computing ESM2 embeddings: 83%| | 576/697 [00:15<00:03, 37.96it/s]

Computing ESM2 embeddings: 87%| | 608/697 [00:15<00:02, 38.33it/s]

Computing ESM2 embeddings: 92%| | 640/697 [00:16<00:01, 38.79it/s]

Computing ESM2 embeddings: 96%| | 672/697 [00:17<00:00, 37.93it/s]

Computing ESM2 embeddings: 100%| | 697/697 [00:18<00:00, 38.01it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.52it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.88it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.12it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.64it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 39.24it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:04<00:18, 38.15it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:17, 38.70it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:16, 39.13it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.22it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.62it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 39.12it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:09<00:13, 38.11it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:12, 38.47it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:11, 38.86it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.91it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.38it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.94it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:14<00:08, 38.12it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.76it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 39.13it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.15it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.55it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 39.05it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:19<00:03, 37.93it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:02, 38.49it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.93it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.16it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.53it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.58it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.53it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.59it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 36.68it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 37.71it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 38.52it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:18, 37.65it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 38.36it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 38.91it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.96it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:14, 38.52it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:13, 39.05it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 38.12it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:10<00:12, 38.62it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 39.04it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 38.18it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:09, 38.58it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 39.07it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 38.04it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:15<00:07, 38.58it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 38.70it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:06, 37.29it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 37.95it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 38.57it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.72it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:20<00:02, 38.35it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:21<00:01, 38.85it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:22<00:00, 38.03it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.25it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:22, 38.99it/s]  
 Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:21, 39.51it/s]  
 Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:21, 37.79it/s]  
 Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:20, 38.38it/s]  
 Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:19, 39.02it/s]  
 Computing ESM2 embeddings: 21%| | 192/915 [00:05<00:19, 37.94it/s]  
 Computing ESM2 embeddings: 24%| | 224/915 [00:05<00:17, 38.62it/s]  
 Computing ESM2 embeddings: 28%| | 256/915 [00:06<00:16, 39.01it/s]  
 Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:16, 37.89it/s]  
 Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:15, 38.21it/s]  
 Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:14, 38.78it/s]  
 Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 37.85it/s]  
 Computing ESM2 embeddings: 45%| | 416/915 [00:10<00:13, 38.32it/s]  
 Computing ESM2 embeddings: 49%| | 448/915 [00:11<00:12, 38.71it/s]  
 Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 38.04it/s]  
 Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 38.56it/s]  
 Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 38.85it/s]  
 Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:08, 38.03it/s]  
 Computing ESM2 embeddings: 66%| | 608/915 [00:15<00:07, 38.48it/s]

```

Computing ESM2 embeddings: 70%|          | 640/915 [00:16<00:07, 38.96it/s]
Computing ESM2 embeddings: 73%|          | 672/915 [00:17<00:06, 37.99it/s]
Computing ESM2 embeddings: 77%|          | 704/915 [00:18<00:05, 38.51it/s]
Computing ESM2 embeddings: 80%|          | 736/915 [00:19<00:04, 38.91it/s]
Computing ESM2 embeddings: 84%|          | 768/915 [00:20<00:03, 37.94it/s]
Computing ESM2 embeddings: 87%|          | 800/915 [00:20<00:02, 38.38it/s]
Computing ESM2 embeddings: 91%|          | 832/915 [00:21<00:02, 38.90it/s]
Computing ESM2 embeddings: 94%|          | 864/915 [00:22<00:01, 38.07it/s]
Computing ESM2 embeddings: 98%|          | 896/915 [00:23<00:00, 38.64it/s]

Computing ESM2 embeddings: 100%|         | 915/915 [00:23<00:00, 38.45it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/896 [00:00<00:23, 37.35it/s]
Computing ESM2 embeddings: 7%|           | 64/896 [00:01<00:21, 38.70it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 37.40it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 38.14it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:18, 38.86it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:18, 37.99it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 38.36it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 38.69it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.81it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 38.21it/s]

```

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 38.76it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.78it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.39it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 38.62it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 39.05it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 39.30it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.22it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 38.52it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.97it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 37.99it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.50it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.89it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.97it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 38.34it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.84it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 37.80it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.06it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.36it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.60it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.12it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 37.70it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 38.39it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 39.07it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.89it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:17, 38.49it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:16, 38.98it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.09it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 38.52it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 39.04it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:09<00:13, 38.11it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:10<00:12, 38.65it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 38.68it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.32it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 38.05it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 38.70it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:08, 37.80it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:15<00:07, 38.41it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 38.86it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 38.03it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 38.48it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 39.00it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 38.07it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:20<00:02, 38.43it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:21<00:02, 38.75it/s] |

```

Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 37.97it/s]

Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.42it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:23<00:00, 38.33it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 37.89it/s]

Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:22, 37.05it/s]

Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:22, 36.30it/s]

Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 37.47it/s]

Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 38.41it/s]

Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:18, 37.58it/s]

Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 38.29it/s]

Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 38.81it/s]

Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.73it/s]

Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 38.20it/s]

Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 38.75it/s]

Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.75it/s]

Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 38.35it/s]

Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 38.83it/s]

Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 37.96it/s]

Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:10, 38.36it/s]

Computing ESM2 embeddings: 61%|      | 544/896 [00:14<00:09, 38.86it/s]

```

```

Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.94it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.37it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.89it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.77it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 38.36it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 38.81it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.84it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.28it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 38.50it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 36.99it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.02it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/731 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/731 [00:00<00:18, 38.70it/s]
Computing ESM2 embeddings:  9%|          | 64/731 [00:01<00:17, 39.17it/s]
Computing ESM2 embeddings: 13%|          | 96/731 [00:02<00:16, 37.48it/s]
Computing ESM2 embeddings: 18%|          | 128/731 [00:03<00:15, 38.26it/s]
Computing ESM2 embeddings: 22%|          | 160/731 [00:04<00:14, 38.99it/s]
Computing ESM2 embeddings: 26%|          | 192/731 [00:05<00:14, 37.98it/s]
Computing ESM2 embeddings: 31%|          | 224/731 [00:05<00:13, 38.59it/s]
Computing ESM2 embeddings: 35%|          | 256/731 [00:06<00:12, 39.01it/s]
Computing ESM2 embeddings: 39%|          | 288/731 [00:07<00:11, 38.06it/s]

```

Computing ESM2 embeddings: 44%| | 320/731 [00:08<00:10, 38.28it/s]

Computing ESM2 embeddings: 48%| | 352/731 [00:09<00:09, 38.78it/s]

Computing ESM2 embeddings: 53%| | 384/731 [00:10<00:09, 37.72it/s]

Computing ESM2 embeddings: 57%| | 416/731 [00:10<00:08, 38.07it/s]

Computing ESM2 embeddings: 61%| | 448/731 [00:11<00:07, 38.52it/s]

Computing ESM2 embeddings: 66%| | 480/731 [00:12<00:06, 37.75it/s]

Computing ESM2 embeddings: 70%| | 512/731 [00:13<00:05, 38.26it/s]

Computing ESM2 embeddings: 74%| | 544/731 [00:14<00:04, 38.86it/s]

Computing ESM2 embeddings: 79%| | 576/731 [00:15<00:04, 38.02it/s]

Computing ESM2 embeddings: 83%| | 608/731 [00:15<00:03, 38.58it/s]

Computing ESM2 embeddings: 88%| | 640/731 [00:16<00:02, 39.02it/s]

Computing ESM2 embeddings: 92%| | 672/731 [00:17<00:01, 38.09it/s]

Computing ESM2 embeddings: 96%| | 704/731 [00:18<00:00, 38.49it/s]

Computing ESM2 embeddings: 100%| | 731/731 [00:19<00:00, 38.42it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/755 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/755 [00:00<00:18, 38.35it/s]

Computing ESM2 embeddings: 8%| | 64/755 [00:01<00:17, 39.30it/s]

Computing ESM2 embeddings: 13%| | 96/755 [00:02<00:17, 37.64it/s]

Computing ESM2 embeddings: 17%| | 128/755 [00:03<00:16, 38.53it/s]

Computing ESM2 embeddings: 21%| | 160/755 [00:04<00:15, 39.07it/s]

Computing ESM2 embeddings: 25%| | 192/755 [00:05<00:14, 38.02it/s]



Computing ESM2 embeddings: 30%| | 224/755 [00:05<00:13, 38.46it/s]

Computing ESM2 embeddings: 34%| | 256/755 [00:06<00:12, 39.03it/s]

Computing ESM2 embeddings: 38%| | 288/755 [00:07<00:12, 37.81it/s]

Computing ESM2 embeddings: 42%| | 320/755 [00:08<00:11, 38.45it/s]

Computing ESM2 embeddings: 47%| | 352/755 [00:09<00:10, 38.93it/s]

Computing ESM2 embeddings: 51%| | 384/755 [00:10<00:09, 38.05it/s]

Computing ESM2 embeddings: 55%| | 416/755 [00:10<00:08, 38.44it/s]

Computing ESM2 embeddings: 59%| | 448/755 [00:11<00:07, 38.99it/s]

Computing ESM2 embeddings: 64%| | 480/755 [00:12<00:07, 37.96it/s]

Computing ESM2 embeddings: 68%| | 512/755 [00:13<00:06, 38.51it/s]

Computing ESM2 embeddings: 72%| | 544/755 [00:14<00:05, 38.94it/s]

Computing ESM2 embeddings: 76%| | 576/755 [00:15<00:04, 37.93it/s]

Computing ESM2 embeddings: 81%| | 608/755 [00:15<00:03, 38.39it/s]

Computing ESM2 embeddings: 85%| | 640/755 [00:16<00:02, 38.93it/s]

Computing ESM2 embeddings: 89%| | 672/755 [00:17<00:02, 38.04it/s]

Computing ESM2 embeddings: 93%| | 704/755 [00:18<00:01, 38.57it/s]

Computing ESM2 embeddings: 97%| | 736/755 [00:19<00:00, 39.00it/s]

Computing ESM2 embeddings: 100%| | 755/755 [00:19<00:00, 38.34it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:23, 37.19it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:24, 34.81it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/913 [00:02<00:22, 36.70it/s]  |
| Computing ESM2 embeddings: 14% | 128/913 [00:03<00:20, 37.70it/s] |
| Computing ESM2 embeddings: 18% | 160/913 [00:04<00:20, 37.21it/s] |
| Computing ESM2 embeddings: 21% | 192/913 [00:05<00:19, 37.94it/s] |
| Computing ESM2 embeddings: 25% | 224/913 [00:05<00:17, 38.63it/s] |
| Computing ESM2 embeddings: 28% | 256/913 [00:06<00:17, 37.71it/s] |
| Computing ESM2 embeddings: 32% | 288/913 [00:07<00:16, 38.33it/s] |
| Computing ESM2 embeddings: 35% | 320/913 [00:08<00:15, 38.85it/s] |
| Computing ESM2 embeddings: 39% | 352/913 [00:09<00:14, 37.97it/s] |
| Computing ESM2 embeddings: 42% | 384/913 [00:10<00:13, 38.43it/s] |
| Computing ESM2 embeddings: 46% | 416/913 [00:10<00:12, 38.91it/s] |
| Computing ESM2 embeddings: 49% | 448/913 [00:11<00:12, 37.62it/s] |
| Computing ESM2 embeddings: 53% | 480/913 [00:12<00:11, 38.07it/s] |
| Computing ESM2 embeddings: 56% | 512/913 [00:13<00:10, 38.48it/s] |
| Computing ESM2 embeddings: 60% | 544/913 [00:14<00:09, 37.66it/s] |
| Computing ESM2 embeddings: 63% | 576/913 [00:15<00:08, 38.15it/s] |
| Computing ESM2 embeddings: 67% | 608/913 [00:15<00:07, 38.71it/s] |
| Computing ESM2 embeddings: 70% | 640/913 [00:16<00:07, 37.71it/s] |
| Computing ESM2 embeddings: 74% | 672/913 [00:17<00:06, 38.33it/s] |
| Computing ESM2 embeddings: 77% | 704/913 [00:18<00:05, 38.80it/s] |
| Computing ESM2 embeddings: 81% | 736/913 [00:19<00:04, 37.95it/s] |
| Computing ESM2 embeddings: 84% | 768/913 [00:20<00:03, 38.36it/s] |
| Computing ESM2 embeddings: 88% | 800/913 [00:20<00:02, 38.90it/s] |
| Computing ESM2 embeddings: 91% | 832/913 [00:21<00:02, 37.94it/s] |

```

Computing ESM2 embeddings: 95%|      | 864/913 [00:22<00:01, 38.45it/s]

Computing ESM2 embeddings: 98%|      | 896/913 [00:23<00:00, 38.62it/s]

Computing ESM2 embeddings: 100%|     | 913/913 [00:24<00:00, 37.91it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:21, 39.45it/s]

Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 39.62it/s]

Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.97it/s]

Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:19, 38.53it/s]

Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:18, 39.16it/s]

Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:18, 38.07it/s]

Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 38.65it/s]

Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 39.08it/s]

Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 38.07it/s]

Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 38.16it/s]

Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 38.64it/s]

Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.73it/s]

Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 38.30it/s]

Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 38.78it/s]

Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 37.96it/s]

Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:10, 38.37it/s]

Computing ESM2 embeddings: 61%|      | 544/896 [00:14<00:09, 38.89it/s]

```

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.82it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.38it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.83it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.88it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 38.02it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.50it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.59it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.00it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.41it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 37.59it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.29it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:22, 38.61it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:21, 39.13it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:21, 37.63it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:20, 38.18it/s]

Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:19, 38.87it/s]

Computing ESM2 embeddings: 21%| | 192/915 [00:05<00:19, 37.85it/s]

Computing ESM2 embeddings: 24%| | 224/915 [00:05<00:18, 38.37it/s]

Computing ESM2 embeddings: 28%| | 256/915 [00:06<00:16, 38.98it/s]

Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:16, 37.78it/s]

Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:15, 38.38it/s]

Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:14, 38.89it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:13, 38.02it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:10<00:12, 38.46it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:11<00:11, 38.98it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 38.12it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 38.62it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 38.96it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:08, 37.97it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:15<00:07, 38.44it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:16<00:07, 38.96it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:17<00:06, 37.94it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:18<00:05, 38.48it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 38.90it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:03, 38.08it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:20<00:02, 38.46it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:21<00:02, 38.68it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:22<00:01, 37.03it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:23<00:00, 37.82it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:23<00:00, 38.32it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:24, 35.63it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.00it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 36.73it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 37.90it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 38.61it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 37.62it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 38.16it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.47it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 36.96it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.81it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 38.45it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.58it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 38.09it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.36it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 36.84it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.68it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 38.30it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.58it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 38.08it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.67it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.79it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 38.34it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 38.80it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 37.86it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 38.29it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 38.84it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 37.97it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.91it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.94it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:24, 34.04it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:23, 34.81it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.52it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 37.76it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 37.41it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 38.13it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 38.44it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.68it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 38.12it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 38.75it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 37.84it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 38.18it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:12, 38.54it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.63it/s]
Computing ESM2 embeddings: 56%|      | 512/914 [00:13<00:10, 38.14it/s]

```

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.74it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.93it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:07, 38.47it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.88it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.01it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.39it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.89it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.99it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:02, 38.52it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.94it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.05it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.52it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.99it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.63it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.05it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.59it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 38.31it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.01it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.84it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.42it/s]



```

Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 38.87it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.89it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 37.94it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 38.49it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 37.58it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.28it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 38.71it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 37.88it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 38.27it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 38.82it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.92it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.49it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.84it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.33it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:04, 38.63it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 39.05it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 38.16it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.71it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.03it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 37.85it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.30it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|          | 0/674 [00:00<?, ?it/s]
Computing ESM2 embeddings: 5%|          | 32/674 [00:00<00:16, 38.21it/s]
Computing ESM2 embeddings: 9%|          | 64/674 [00:01<00:15, 39.25it/s]
Computing ESM2 embeddings: 14%|         | 96/674 [00:02<00:15, 37.65it/s]
Computing ESM2 embeddings: 19%|         | 128/674 [00:03<00:14, 38.56it/s]
Computing ESM2 embeddings: 24%|         | 160/674 [00:04<00:13, 39.06it/s]
Computing ESM2 embeddings: 28%|         | 192/674 [00:05<00:12, 38.03it/s]
Computing ESM2 embeddings: 33%|         | 224/674 [00:05<00:11, 38.25it/s]
Computing ESM2 embeddings: 38%|         | 256/674 [00:06<00:10, 38.48it/s]
Computing ESM2 embeddings: 43%|         | 288/674 [00:07<00:10, 37.09it/s]
Computing ESM2 embeddings: 47%|         | 320/674 [00:08<00:09, 37.98it/s]
Computing ESM2 embeddings: 52%|         | 352/674 [00:09<00:08, 38.59it/s]
Computing ESM2 embeddings: 57%|         | 384/674 [00:10<00:07, 37.82it/s]
Computing ESM2 embeddings: 62%|         | 416/674 [00:10<00:06, 38.30it/s]
Computing ESM2 embeddings: 66%|         | 448/674 [00:11<00:05, 38.63it/s]
Computing ESM2 embeddings: 71%|         | 480/674 [00:12<00:05, 37.20it/s]
Computing ESM2 embeddings: 76%|         | 512/674 [00:13<00:04, 37.98it/s]
Computing ESM2 embeddings: 81%|         | 544/674 [00:14<00:03, 38.55it/s]
Computing ESM2 embeddings: 85%|         | 576/674 [00:15<00:02, 37.91it/s]
Computing ESM2 embeddings: 90%|         | 608/674 [00:15<00:01, 38.41it/s]
Computing ESM2 embeddings: 95%|         | 640/674 [00:16<00:00, 38.96it/s]

Computing ESM2 embeddings: 100%|        | 674/674 [00:17<00:00, 38.16it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/811 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/811 [00:00<00:20, 38.27it/s]  |
| Computing ESM2 embeddings: | 8%  | 64/811 [00:01<00:20, 36.52it/s]  |
| Computing ESM2 embeddings: | 12% | 96/811 [00:02<00:18, 37.80it/s]  |
| Computing ESM2 embeddings: | 16% | 128/811 [00:03<00:17, 38.74it/s] |
| Computing ESM2 embeddings: | 20% | 160/811 [00:04<00:17, 37.70it/s] |
| Computing ESM2 embeddings: | 24% | 192/811 [00:05<00:16, 38.44it/s] |
| Computing ESM2 embeddings: | 28% | 224/811 [00:05<00:15, 38.96it/s] |
| Computing ESM2 embeddings: | 32% | 256/811 [00:06<00:14, 38.04it/s] |
| Computing ESM2 embeddings: | 36% | 288/811 [00:07<00:13, 38.51it/s] |
| Computing ESM2 embeddings: | 39% | 320/811 [00:08<00:12, 39.07it/s] |
| Computing ESM2 embeddings: | 43% | 352/811 [00:09<00:12, 38.11it/s] |
| Computing ESM2 embeddings: | 47% | 384/811 [00:10<00:11, 38.36it/s] |
| Computing ESM2 embeddings: | 51% | 416/811 [00:10<00:10, 38.65it/s] |
| Computing ESM2 embeddings: | 55% | 448/811 [00:11<00:09, 37.89it/s] |
| Computing ESM2 embeddings: | 59% | 480/811 [00:12<00:08, 38.35it/s] |
| Computing ESM2 embeddings: | 63% | 512/811 [00:13<00:07, 38.93it/s] |
| Computing ESM2 embeddings: | 67% | 544/811 [00:14<00:07, 37.92it/s] |
| Computing ESM2 embeddings: | 71% | 576/811 [00:15<00:06, 38.50it/s] |
| Computing ESM2 embeddings: | 75% | 608/811 [00:15<00:05, 38.93it/s] |
| Computing ESM2 embeddings: | 79% | 640/811 [00:16<00:04, 38.03it/s] |
| Computing ESM2 embeddings: | 83% | 672/811 [00:17<00:03, 38.42it/s] |
| Computing ESM2 embeddings: | 87% | 704/811 [00:18<00:02, 38.94it/s] |

```

Computing ESM2 embeddings: 91%|      | 736/811 [00:19<00:01, 38.01it/s]
Computing ESM2 embeddings: 95%|      | 768/811 [00:20<00:01, 38.57it/s]
Computing ESM2 embeddings: 99%|      | 800/811 [00:20<00:00, 38.98it/s]
Computing ESM2 embeddings: 100%|     | 811/811 [00:21<00:00, 38.29it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:24, 36.60it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:22, 37.64it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.22it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:20, 38.24it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 38.93it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 37.58it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:18, 38.29it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:16, 38.81it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.84it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 38.29it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 38.85it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:13, 37.94it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:12, 38.51it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:11, 38.94it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 38.09it/s]
Computing ESM2 embeddings: 56%|      | 512/914 [00:13<00:10, 38.45it/s]

```

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.96it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 38.05it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:07, 38.57it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.93it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.96it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.34it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.56it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.06it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:03, 37.84it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.43it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 37.63it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.12it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:23<00:00, 38.16it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.43it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.30it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.71it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 38.57it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:18, 39.12it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 38.01it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.47it/s]

```

Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 39.04it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 38.00it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 38.57it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:13, 39.03it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 37.88it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 38.43it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 38.97it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 37.95it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:09, 38.52it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 38.97it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:14<00:08, 38.03it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:15<00:07, 38.46it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.99it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.18it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:04, 38.67it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 39.05it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:19<00:03, 38.21it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:20<00:02, 38.56it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:21<00:01, 39.07it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 38.10it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 38.46it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/913 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/913 [00:00<00:23, 37.98it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/913 [00:01<00:21, 38.94it/s]  |
| Computing ESM2 embeddings: | 11% | 96/913 [00:02<00:21, 37.70it/s]  |
| Computing ESM2 embeddings: | 14% | 128/913 [00:03<00:20, 38.39it/s] |
| Computing ESM2 embeddings: | 18% | 160/913 [00:04<00:19, 39.05it/s] |
| Computing ESM2 embeddings: | 21% | 192/913 [00:05<00:18, 37.98it/s] |
| Computing ESM2 embeddings: | 25% | 224/913 [00:05<00:17, 38.59it/s] |
| Computing ESM2 embeddings: | 28% | 256/913 [00:06<00:16, 38.97it/s] |
| Computing ESM2 embeddings: | 32% | 288/913 [00:07<00:16, 37.98it/s] |
| Computing ESM2 embeddings: | 35% | 320/913 [00:08<00:15, 38.07it/s] |
| Computing ESM2 embeddings: | 39% | 352/913 [00:09<00:14, 38.60it/s] |
| Computing ESM2 embeddings: | 42% | 384/913 [00:10<00:14, 37.76it/s] |
| Computing ESM2 embeddings: | 46% | 416/913 [00:10<00:12, 38.49it/s] |
| Computing ESM2 embeddings: | 49% | 448/913 [00:11<00:11, 38.91it/s] |
| Computing ESM2 embeddings: | 53% | 480/913 [00:12<00:11, 37.97it/s] |
| Computing ESM2 embeddings: | 56% | 512/913 [00:13<00:10, 38.38it/s] |
| Computing ESM2 embeddings: | 60% | 544/913 [00:14<00:09, 38.88it/s] |
| Computing ESM2 embeddings: | 63% | 576/913 [00:15<00:08, 37.93it/s] |
| Computing ESM2 embeddings: | 67% | 608/913 [00:15<00:07, 38.49it/s] |
| Computing ESM2 embeddings: | 70% | 640/913 [00:16<00:07, 38.91it/s] |
| Computing ESM2 embeddings: | 74% | 672/913 [00:17<00:06, 38.19it/s] |
| Computing ESM2 embeddings: | 77% | 704/913 [00:18<00:05, 38.67it/s] |
| Computing ESM2 embeddings: | 81% | 736/913 [00:19<00:04, 39.10it/s] |

```

Computing ESM2 embeddings: 84%|      | 768/913 [00:20<00:03, 37.95it/s]
Computing ESM2 embeddings: 88%|      | 800/913 [00:20<00:02, 38.48it/s]
Computing ESM2 embeddings: 91%|      | 832/913 [00:21<00:02, 38.87it/s]
Computing ESM2 embeddings: 95%|      | 864/913 [00:22<00:01, 38.02it/s]
Computing ESM2 embeddings: 98%|      | 896/913 [00:23<00:00, 38.38it/s]

Computing ESM2 embeddings: 100%|     | 913/913 [00:23<00:00, 38.39it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 39.12it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:20, 39.86it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 38.10it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:19, 38.95it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:18, 39.34it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:04<00:18, 38.14it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 38.58it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 39.13it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 38.15it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:14, 38.68it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:13, 38.90it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:09<00:13, 38.02it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:10<00:12, 38.44it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 38.99it/s]

```



Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 37.95it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:09, 38.50it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.95it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:14<00:08, 38.11it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:15<00:07, 38.52it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 39.03it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.02it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:04, 38.57it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.98it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:19<00:03, 38.15it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:20<00:02, 38.48it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.97it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.07it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 38.44it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/909 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/909 [00:00<00:23, 37.84it/s]

Computing ESM2 embeddings: 7%| | 64/909 [00:01<00:21, 38.78it/s]

Computing ESM2 embeddings: 11%| | 96/909 [00:02<00:21, 37.95it/s]

Computing ESM2 embeddings: 14%| | 128/909 [00:03<00:20, 38.35it/s]

Computing ESM2 embeddings: 18%| | 160/909 [00:04<00:19, 38.95it/s]

Computing ESM2 embeddings: 21%| | 192/909 [00:05<00:19, 37.59it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/909 [00:05<00:17, 38.33it/s] |
| Computing ESM2 embeddings: 28%  | 256/909 [00:06<00:16, 38.86it/s] |
| Computing ESM2 embeddings: 32%  | 288/909 [00:07<00:16, 38.00it/s] |
| Computing ESM2 embeddings: 35%  | 320/909 [00:08<00:15, 38.48it/s] |
| Computing ESM2 embeddings: 39%  | 352/909 [00:09<00:14, 38.76it/s] |
| Computing ESM2 embeddings: 42%  | 384/909 [00:10<00:14, 37.25it/s] |
| Computing ESM2 embeddings: 46%  | 416/909 [00:10<00:12, 38.04it/s] |
| Computing ESM2 embeddings: 49%  | 448/909 [00:11<00:11, 38.62it/s] |
| Computing ESM2 embeddings: 53%  | 480/909 [00:12<00:11, 37.86it/s] |
| Computing ESM2 embeddings: 56%  | 512/909 [00:13<00:10, 38.32it/s] |
| Computing ESM2 embeddings: 60%  | 544/909 [00:14<00:09, 38.90it/s] |
| Computing ESM2 embeddings: 63%  | 576/909 [00:15<00:08, 37.91it/s] |
| Computing ESM2 embeddings: 67%  | 608/909 [00:15<00:07, 38.49it/s] |
| Computing ESM2 embeddings: 70%  | 640/909 [00:16<00:06, 38.94it/s] |
| Computing ESM2 embeddings: 74%  | 672/909 [00:17<00:06, 38.27it/s] |
| Computing ESM2 embeddings: 77%  | 704/909 [00:18<00:05, 38.67it/s] |
| Computing ESM2 embeddings: 81%  | 736/909 [00:19<00:04, 39.10it/s] |
| Computing ESM2 embeddings: 84%  | 768/909 [00:20<00:03, 38.04it/s] |
| Computing ESM2 embeddings: 88%  | 800/909 [00:20<00:02, 38.55it/s] |
| Computing ESM2 embeddings: 92%  | 832/909 [00:21<00:01, 38.97it/s] |
| Computing ESM2 embeddings: 95%  | 864/909 [00:22<00:01, 38.07it/s] |
| Computing ESM2 embeddings: 99%  | 896/909 [00:23<00:00, 38.19it/s] |
| Computing ESM2 embeddings: 100% | 909/909 [00:23<00:00, 38.31it/s] |

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'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/901 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/901 [00:00<00:22, 39.03it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/901 [00:01<00:21, 39.73it/s]  |
| Computing ESM2 embeddings: | 11% | 96/901 [00:02<00:21, 37.81it/s]  |
| Computing ESM2 embeddings: | 14% | 128/901 [00:03<00:20, 38.63it/s] |
| Computing ESM2 embeddings: | 18% | 160/901 [00:04<00:18, 39.13it/s] |
| Computing ESM2 embeddings: | 21% | 192/901 [00:04<00:18, 38.15it/s] |
| Computing ESM2 embeddings: | 25% | 224/901 [00:05<00:17, 38.56it/s] |
| Computing ESM2 embeddings: | 28% | 256/901 [00:06<00:16, 39.09it/s] |
| Computing ESM2 embeddings: | 32% | 288/901 [00:07<00:16, 37.96it/s] |
| Computing ESM2 embeddings: | 36% | 320/901 [00:08<00:15, 38.32it/s] |
| Computing ESM2 embeddings: | 39% | 352/901 [00:09<00:14, 38.67it/s] |
| Computing ESM2 embeddings: | 43% | 384/901 [00:10<00:13, 37.90it/s] |
| Computing ESM2 embeddings: | 46% | 416/901 [00:10<00:12, 38.34it/s] |
| Computing ESM2 embeddings: | 50% | 448/901 [00:11<00:11, 38.91it/s] |
| Computing ESM2 embeddings: | 53% | 480/901 [00:12<00:11, 37.82it/s] |
| Computing ESM2 embeddings: | 57% | 512/901 [00:13<00:10, 38.41it/s] |
| Computing ESM2 embeddings: | 60% | 544/901 [00:14<00:09, 38.84it/s] |
| Computing ESM2 embeddings: | 64% | 576/901 [00:14<00:08, 38.02it/s] |
| Computing ESM2 embeddings: | 67% | 608/901 [00:15<00:07, 38.44it/s] |
| Computing ESM2 embeddings: | 71% | 640/901 [00:16<00:06, 37.53it/s] |
| Computing ESM2 embeddings: | 75% | 672/901 [00:17<00:06, 38.07it/s] |

Computing ESM2 embeddings: 78%| | 704/901 [00:18<00:05, 37.38it/s]

Computing ESM2 embeddings: 82%| | 736/901 [00:19<00:04, 37.97it/s]

Computing ESM2 embeddings: 85%| | 768/901 [00:20<00:03, 37.36it/s]

Computing ESM2 embeddings: 89%| | 800/901 [00:20<00:02, 37.63it/s]

Computing ESM2 embeddings: 92%| | 832/901 [00:21<00:01, 36.36it/s]

Computing ESM2 embeddings: 96%| | 864/901 [00:22<00:00, 37.11it/s]

Computing ESM2 embeddings: 99%| | 896/901 [00:23<00:00, 36.65it/s]

Computing ESM2 embeddings: 100%| | 901/901 [00:23<00:00, 37.88it/s]

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Computing ESM2 embeddings: 0%| | 0/619 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/619 [00:00<00:15, 39.11it/s]

Computing ESM2 embeddings: 10%| | 64/619 [00:01<00:13, 39.79it/s]

Computing ESM2 embeddings: 16%| | 96/619 [00:02<00:13, 37.90it/s]

Computing ESM2 embeddings: 21%| | 128/619 [00:03<00:12, 38.64it/s]

Computing ESM2 embeddings: 26%| | 160/619 [00:04<00:11, 39.13it/s]

Computing ESM2 embeddings: 31%| | 192/619 [00:04<00:11, 38.09it/s]

Computing ESM2 embeddings: 36%| | 224/619 [00:05<00:10, 38.65it/s]

Computing ESM2 embeddings: 41%| | 256/619 [00:06<00:09, 39.08it/s]

Computing ESM2 embeddings: 47%| | 288/619 [00:07<00:08, 38.17it/s]

Computing ESM2 embeddings: 52%| | 320/619 [00:08<00:07, 38.21it/s]

Computing ESM2 embeddings: 57%| | 352/619 [00:09<00:06, 38.70it/s]

Computing ESM2 embeddings: 62%| | 384/619 [00:09<00:06, 37.90it/s]

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Computing ESM2 embeddings: 67%|          | 416/619 [00:10<00:05, 38.56it/s]
Computing ESM2 embeddings: 72%|          | 448/619 [00:11<00:04, 38.99it/s]
Computing ESM2 embeddings: 78%|          | 480/619 [00:12<00:03, 38.11it/s]
Computing ESM2 embeddings: 83%|          | 512/619 [00:13<00:02, 38.48it/s]
Computing ESM2 embeddings: 88%|          | 544/619 [00:14<00:01, 39.00it/s]
Computing ESM2 embeddings: 93%|          | 576/619 [00:14<00:01, 37.97it/s]
Computing ESM2 embeddings: 98%|          | 608/619 [00:15<00:00, 38.51it/s]
Computing ESM2 embeddings: 100%|         | 619/619 [00:16<00:00, 38.43it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
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for predictions and inference.

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Computing ESM2 embeddings: 0%|           | 0/866 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/866 [00:00<00:21, 39.15it/s]
Computing ESM2 embeddings: 7%|           | 64/866 [00:01<00:20, 39.68it/s]
Computing ESM2 embeddings: 11%|          | 96/866 [00:02<00:20, 37.95it/s]
Computing ESM2 embeddings: 15%|          | 128/866 [00:03<00:19, 38.70it/s]
Computing ESM2 embeddings: 18%|          | 160/866 [00:04<00:17, 39.23it/s]
Computing ESM2 embeddings: 22%|          | 192/866 [00:04<00:17, 38.05it/s]
Computing ESM2 embeddings: 26%|          | 224/866 [00:05<00:16, 38.65it/s]
Computing ESM2 embeddings: 30%|          | 256/866 [00:06<00:15, 39.08it/s]
Computing ESM2 embeddings: 33%|          | 288/866 [00:07<00:15, 37.98it/s]
Computing ESM2 embeddings: 37%|          | 320/866 [00:08<00:14, 38.56it/s]
Computing ESM2 embeddings: 41%|          | 352/866 [00:09<00:13, 38.73it/s]
Computing ESM2 embeddings: 44%|          | 384/866 [00:10<00:12, 37.14it/s]

```

```

Computing ESM2 embeddings: 48%|          | 416/866 [00:10<00:11, 38.00it/s]
Computing ESM2 embeddings: 52%|          | 448/866 [00:11<00:10, 38.62it/s]
Computing ESM2 embeddings: 55%|          | 480/866 [00:12<00:10, 37.75it/s]
Computing ESM2 embeddings: 59%|          | 512/866 [00:13<00:09, 38.25it/s]
Computing ESM2 embeddings: 63%|          | 544/866 [00:14<00:08, 37.27it/s]
Computing ESM2 embeddings: 67%|          | 576/866 [00:15<00:07, 37.98it/s]
Computing ESM2 embeddings: 70%|          | 608/866 [00:15<00:06, 37.27it/s]
Computing ESM2 embeddings: 74%|          | 640/866 [00:16<00:05, 38.01it/s]
Computing ESM2 embeddings: 78%|          | 672/866 [00:17<00:05, 37.30it/s]
Computing ESM2 embeddings: 81%|          | 704/866 [00:18<00:04, 37.91it/s]
Computing ESM2 embeddings: 85%|          | 736/866 [00:19<00:03, 37.25it/s]
Computing ESM2 embeddings: 89%|          | 768/866 [00:20<00:02, 37.97it/s]
Computing ESM2 embeddings: 92%|          | 800/866 [00:21<00:01, 37.33it/s]
Computing ESM2 embeddings: 96%|          | 832/866 [00:21<00:00, 37.80it/s]
Computing ESM2 embeddings: 100%|         | 866/866 [00:22<00:00, 37.75it/s]
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for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:23, 36.96it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:02<00:28, 29.36it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:24, 33.03it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:21, 35.40it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:20, 35.48it/s]

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|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:19, 36.36it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:06<00:18, 35.54it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:07<00:17, 36.26it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:08<00:17, 35.50it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:09<00:15, 36.64it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 36.53it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 37.47it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:13, 36.91it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:12<00:11, 37.63it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:13<00:11, 37.09it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:14<00:10, 37.51it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:15<00:09, 36.44it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 36.93it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:08, 35.93it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 36.88it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:18<00:06, 36.45it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:19<00:05, 37.26it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:20<00:04, 36.71it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:21<00:03, 37.45it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 36.98it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 37.54it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:23<00:00, 36.87it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:24<00:00, 36.45it/s] |

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'esm.pooler.dense.weight']

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|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 38.16it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:22, 38.55it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:22, 36.41it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 37.73it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.53it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.73it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 38.27it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 38.82it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.85it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.45it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 38.87it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:13, 37.88it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:13, 38.25it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 38.49it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.13it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.52it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:10, 36.38it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.21it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 36.83it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 37.59it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 36.97it/s] |



Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.64it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 36.99it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.64it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.06it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.72it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 37.16it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.81it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.51it/s]

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Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:23, 36.71it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:24, 34.17it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:23, 34.01it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:21, 35.92it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:19, 37.34it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:19, 36.80it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:06<00:17, 37.73it/s]

Computing ESM2 embeddings: 29%| | 256/895 [00:06<00:16, 38.44it/s]

Computing ESM2 embeddings: 32%| | 288/895 [00:07<00:16, 37.51it/s]

Computing ESM2 embeddings: 36%| | 320/895 [00:08<00:15, 38.06it/s]

Computing ESM2 embeddings: 39%| | 352/895 [00:09<00:14, 38.67it/s]

Computing ESM2 embeddings: 43%| | 384/895 [00:10<00:13, 37.73it/s]

Computing ESM2 embeddings: 46%| | 416/895 [00:11<00:12, 38.32it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:11<00:11, 38.77it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:12<00:10, 37.85it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 38.26it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.42it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.97it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.24it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 37.88it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:17<00:05, 37.20it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:05, 37.81it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.08it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 37.74it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.02it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.77it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.00it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:23<00:00, 37.44it/s]

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Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.50it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.09it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.64it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.37it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 37.45it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:18, 38.11it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.41it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 38.04it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.44it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.76it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 36.59it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.43it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.00it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 37.71it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.21it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.91it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.33it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:08, 37.95it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.27it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 37.91it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.17it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.84it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.25it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.88it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.07it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.77it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:23<00:01, 37.23it/s] |
| Computing ESM2 embeddings: 98% | 896/914 [00:23<00:00, 37.88it/s] |

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.49it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: 4%  | 32/896 [00:00<00:22, 37.82it/s]  |
| Computing ESM2 embeddings: 7%  | 64/896 [00:01<00:21, 38.24it/s]  |
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:21, 36.38it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 37.54it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.00it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:18, 37.85it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:18, 37.19it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 38.02it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.43it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 38.20it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.50it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 38.21it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.39it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 37.90it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:11, 36.53it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.51it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 36.70it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.65it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 36.99it/s] |

```

Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 37.82it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:06, 37.15it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:05, 37.95it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.19it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.97it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.25it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 38.02it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.28it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.45it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 38.09it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 36.31it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:22, 36.05it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 37.29it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 36.81it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 37.64it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 36.92it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 37.66it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.12it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 37.80it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.13it/s]

```

```

Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 37.79it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.19it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:12, 37.76it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 37.01it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 37.73it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 37.15it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:08, 37.80it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.03it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.43it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 36.18it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 37.19it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 36.62it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.50it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 36.94it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.74it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 36.94it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.76it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.10it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/571 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|          | 32/571 [00:00<00:14, 38.07it/s]
Computing ESM2 embeddings: 11%|          | 64/571 [00:01<00:13, 38.46it/s]

```

Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:12, 37.18it/s]

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:11, 38.19it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:11, 37.22it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:05<00:09, 38.08it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:05<00:09, 37.22it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 38.05it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 37.08it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 37.90it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 37.13it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:04, 37.92it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 37.19it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:11<00:03, 37.96it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 37.25it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 37.98it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 37.31it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 37.45it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:24, 35.87it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:22, 38.20it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:21, 37.29it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:20, 38.30it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 17% | 160/915 [00:04<00:20, 37.02it/s] |
| Computing ESM2 embeddings: 21% | 192/915 [00:05<00:19, 37.95it/s] |
| Computing ESM2 embeddings: 24% | 224/915 [00:05<00:18, 37.20it/s] |
| Computing ESM2 embeddings: 28% | 256/915 [00:06<00:17, 37.99it/s] |
| Computing ESM2 embeddings: 31% | 288/915 [00:07<00:16, 37.19it/s] |
| Computing ESM2 embeddings: 35% | 320/915 [00:08<00:15, 38.00it/s] |
| Computing ESM2 embeddings: 38% | 352/915 [00:09<00:15, 37.24it/s] |
| Computing ESM2 embeddings: 42% | 384/915 [00:10<00:13, 38.07it/s] |
| Computing ESM2 embeddings: 45% | 416/915 [00:11<00:13, 37.29it/s] |
| Computing ESM2 embeddings: 49% | 448/915 [00:11<00:12, 37.78it/s] |
| Computing ESM2 embeddings: 52% | 480/915 [00:12<00:11, 36.55it/s] |
| Computing ESM2 embeddings: 56% | 512/915 [00:13<00:10, 37.51it/s] |
| Computing ESM2 embeddings: 59% | 544/915 [00:14<00:10, 36.95it/s] |
| Computing ESM2 embeddings: 63% | 576/915 [00:15<00:08, 37.78it/s] |
| Computing ESM2 embeddings: 66% | 608/915 [00:16<00:08, 37.15it/s] |
| Computing ESM2 embeddings: 70% | 640/915 [00:17<00:07, 37.92it/s] |
| Computing ESM2 embeddings: 73% | 672/915 [00:17<00:06, 37.18it/s] |
| Computing ESM2 embeddings: 77% | 704/915 [00:18<00:05, 37.96it/s] |
| Computing ESM2 embeddings: 80% | 736/915 [00:19<00:04, 37.13it/s] |
| Computing ESM2 embeddings: 84% | 768/915 [00:20<00:03, 37.87it/s] |
| Computing ESM2 embeddings: 87% | 800/915 [00:21<00:03, 37.15it/s] |
| Computing ESM2 embeddings: 91% | 832/915 [00:22<00:02, 37.91it/s] |
| Computing ESM2 embeddings: 94% | 864/915 [00:23<00:01, 37.17it/s] |
| Computing ESM2 embeddings: 98% | 896/915 [00:23<00:00, 37.88it/s] |



Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 37.34it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:23, 36.69it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:22, 36.94it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:22, 35.73it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 37.22it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:20, 36.67it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 37.68it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 37.02it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 37.87it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 36.97it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.83it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.07it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.84it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 36.98it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.29it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 36.08it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 36.86it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 35.88it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 36.97it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 36.42it/s] |

```

Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 37.36it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:18<00:06, 36.88it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:05, 37.70it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.10it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.54it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 36.26it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.24it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 36.74it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.99it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/913 [00:00<00:22, 38.69it/s]
Computing ESM2 embeddings: 7%|      | 64/913 [00:01<00:21, 38.68it/s]
Computing ESM2 embeddings: 11%|     | 96/913 [00:02<00:22, 36.59it/s]
Computing ESM2 embeddings: 14%|     | 128/913 [00:03<00:20, 37.69it/s]
Computing ESM2 embeddings: 18%|     | 160/913 [00:04<00:20, 37.07it/s]
Computing ESM2 embeddings: 21%|     | 192/913 [00:05<00:19, 37.89it/s]
Computing ESM2 embeddings: 25%|     | 224/913 [00:05<00:18, 37.25it/s]
Computing ESM2 embeddings: 28%|     | 256/913 [00:06<00:17, 37.60it/s]
Computing ESM2 embeddings: 32%|     | 288/913 [00:07<00:17, 36.48it/s]
Computing ESM2 embeddings: 35%|     | 320/913 [00:08<00:15, 37.37it/s]
Computing ESM2 embeddings: 39%|     | 352/913 [00:09<00:15, 36.88it/s]

```

```

Computing ESM2 embeddings: 42%|          | 384/913 [00:10<00:14, 37.69it/s]
Computing ESM2 embeddings: 46%|          | 416/913 [00:11<00:13, 37.15it/s]
Computing ESM2 embeddings: 49%|          | 448/913 [00:11<00:12, 37.86it/s]
Computing ESM2 embeddings: 53%|          | 480/913 [00:12<00:11, 37.25it/s]
Computing ESM2 embeddings: 56%|          | 512/913 [00:13<00:10, 37.89it/s]
Computing ESM2 embeddings: 60%|          | 544/913 [00:14<00:09, 37.29it/s]
Computing ESM2 embeddings: 63%|          | 576/913 [00:15<00:08, 37.88it/s]
Computing ESM2 embeddings: 67%|          | 608/913 [00:16<00:08, 37.33it/s]
Computing ESM2 embeddings: 70%|          | 640/913 [00:17<00:07, 37.94it/s]
Computing ESM2 embeddings: 74%|          | 672/913 [00:17<00:06, 37.35it/s]
Computing ESM2 embeddings: 77%|          | 704/913 [00:18<00:05, 37.96it/s]
Computing ESM2 embeddings: 81%|          | 736/913 [00:19<00:04, 37.34it/s]
Computing ESM2 embeddings: 84%|          | 768/913 [00:20<00:03, 38.06it/s]
Computing ESM2 embeddings: 88%|          | 800/913 [00:21<00:03, 37.45it/s]
Computing ESM2 embeddings: 91%|          | 832/913 [00:22<00:02, 38.03it/s]
Computing ESM2 embeddings: 95%|          | 864/913 [00:23<00:01, 37.40it/s]
Computing ESM2 embeddings: 98%|          | 896/913 [00:23<00:00, 37.97it/s]

Computing ESM2 embeddings: 100%|         | 913/913 [00:24<00:00, 37.46it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.41it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.07it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:21, 37.74it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:19, 38.44it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.50it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:18, 38.18it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 37.40it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 38.17it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.30it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 38.08it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.28it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 38.02it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.12it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 37.92it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:11, 37.40it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 38.13it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.61it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 38.25it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.42it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:16<00:06, 38.12it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 37.46it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 37.93it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.05it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.89it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.17it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.69it/s] |

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 36.41it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.55it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.82it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.61it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.15it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 37.53it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.66it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.69it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 36.97it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 37.86it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 36.86it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 37.70it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 36.99it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.82it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.26it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 38.01it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.29it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.92it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.32it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.90it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.17it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.80it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.44it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.13it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.43it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.74it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 36.55it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.36it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 36.73it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.46it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.23it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.02it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:24, 34.31it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:23, 34.29it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 36.19it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:20, 36.03it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.12it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 36.73it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:17, 37.28it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 36.21it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 37.16it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 36.84it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.59it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.10it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:11, 37.81it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.18it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.82it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.25it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.90it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.25it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.92it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:06, 37.12it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.76it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 36.82it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.69it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.05it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.86it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.36it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.07it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.05it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:21, 38.99it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:22, 37.08it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 37.94it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 37.09it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.87it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.17it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 37.85it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.24it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.86it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.05it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.72it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.02it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 37.70it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.00it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.67it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.09it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:08, 37.75it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.05it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.74it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.15it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.77it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.19it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.89it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.21it/s] |



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Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.56it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 36.48it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 37.28it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.25it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/571 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/571 [00:00<00:14, 38.08it/s]
Computing ESM2 embeddings: 11%|      | 64/571 [00:01<00:13, 38.74it/s]
Computing ESM2 embeddings: 17%|      | 96/571 [00:02<00:12, 36.75it/s]
Computing ESM2 embeddings: 22%|      | 128/571 [00:03<00:11, 37.76it/s]
Computing ESM2 embeddings: 28%|      | 160/571 [00:04<00:11, 37.15it/s]
Computing ESM2 embeddings: 34%|      | 192/571 [00:05<00:10, 37.88it/s]
Computing ESM2 embeddings: 39%|      | 224/571 [00:05<00:09, 37.23it/s]
Computing ESM2 embeddings: 45%|      | 256/571 [00:06<00:08, 37.88it/s]
Computing ESM2 embeddings: 50%|      | 288/571 [00:07<00:07, 37.10it/s]
Computing ESM2 embeddings: 56%|      | 320/571 [00:08<00:06, 37.76it/s]
Computing ESM2 embeddings: 62%|      | 352/571 [00:09<00:05, 37.01it/s]
Computing ESM2 embeddings: 67%|      | 384/571 [00:10<00:04, 37.72it/s]
Computing ESM2 embeddings: 73%|      | 416/571 [00:11<00:04, 37.16it/s]
Computing ESM2 embeddings: 78%|      | 448/571 [00:11<00:03, 37.80it/s]
Computing ESM2 embeddings: 84%|      | 480/571 [00:12<00:02, 37.05it/s]
Computing ESM2 embeddings: 90%|      | 512/571 [00:13<00:01, 37.74it/s]

```

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 37.01it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 37.38it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.61it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.47it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.89it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.60it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.26it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:18, 38.15it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 37.31it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.12it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.52it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.22it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.31it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:13, 38.03it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.31it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 37.76it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 36.42it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.38it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:10, 36.77it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.66it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.00it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.80it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.07it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.85it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.23it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.98it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.11it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.88it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.20it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.93it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.52it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.31it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.79it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.34it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 38.27it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.25it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 38.07it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:18, 37.30it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 38.06it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.07it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 37.78it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.20it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.81it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.06it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 37.69it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 36.99it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.37it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 36.28it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.13it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 36.77it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.49it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:06, 36.86it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.54it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 36.92it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.29it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 36.25it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.10it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 36.74it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.23it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.48it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:26, 32.24it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:24, 33.71it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 35.76it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:21, 35.81it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.97it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 36.60it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:17, 37.42it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 36.83it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.63it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 36.89it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.64it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.09it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.73it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 37.19it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.86it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.28it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:08, 37.90it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.07it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.73it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.18it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 37.81it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.12it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.90it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.05it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.85it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.16it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.92it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.02it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/895 [00:00<00:24, 35.67it/s]
Computing ESM2 embeddings:  7%|      | 64/895 [00:01<00:22, 37.57it/s]
Computing ESM2 embeddings: 11%|      | 96/895 [00:02<00:22, 35.96it/s]
Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:20, 37.32it/s]
Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:20, 36.39it/s]
Computing ESM2 embeddings: 21%|      | 192/895 [00:05<00:18, 37.12it/s]
Computing ESM2 embeddings: 25%|      | 224/895 [00:06<00:18, 35.82it/s]
Computing ESM2 embeddings: 29%|      | 256/895 [00:06<00:17, 36.84it/s]
Computing ESM2 embeddings: 32%|      | 288/895 [00:07<00:16, 35.77it/s]
Computing ESM2 embeddings: 36%|      | 320/895 [00:08<00:15, 36.85it/s]
Computing ESM2 embeddings: 39%|      | 352/895 [00:09<00:14, 36.38it/s]
Computing ESM2 embeddings: 43%|      | 384/895 [00:10<00:13, 37.38it/s]
Computing ESM2 embeddings: 46%|      | 416/895 [00:11<00:12, 36.85it/s]
Computing ESM2 embeddings: 50%|      | 448/895 [00:12<00:11, 37.65it/s]
Computing ESM2 embeddings: 54%|      | 480/895 [00:13<00:11, 37.00it/s]
Computing ESM2 embeddings: 57%|      | 512/895 [00:13<00:10, 37.75it/s]

```

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.17it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.94it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.12it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 37.90it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:05, 37.17it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:05, 37.90it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.01it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 37.81it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.12it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.86it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.13it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 37.04it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.34it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.12it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.47it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.20it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.29it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:18, 38.00it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 37.38it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.02it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.35it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 37.74it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 36.55it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.40it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 36.99it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 37.69it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.13it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.81it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.30it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.92it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.32it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.92it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.22it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.89it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.33it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.96it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.27it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.89it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.20it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.87it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.48it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.



|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.78it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.81it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.56it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 38.31it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.41it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 38.09it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 37.43it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.05it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.37it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 38.02it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.24it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.84it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.37it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 37.76it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 36.63it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.44it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 36.98it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.66it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.08it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.82it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:06, 37.24it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 37.65it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 36.52it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.37it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 36.95it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.33it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 36.26it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.34it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.60it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 38.97it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.53it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:20, 38.41it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.37it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:18, 38.18it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:18, 37.37it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 38.13it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.08it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 37.89it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.44it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:13, 38.13it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.33it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:12, 38.02it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.29it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.02it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.28it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.98it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.26it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 37.99it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.24it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.97it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.32it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.88it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 36.89it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.72it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 37.13it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.91it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.54it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:14, 37.16it/s]

Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:13, 36.58it/s]

Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:13, 35.94it/s]

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:11, 37.39it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:11, 37.04it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:05<00:09, 38.04it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:06<00:09, 37.33it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 37.74it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 36.56it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 37.51it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 36.92it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:04, 37.62it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 36.93it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:12<00:03, 37.60it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 36.89it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 37.61it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 36.97it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 37.16it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 38.54it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:21, 39.10it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:21, 37.67it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:19, 38.36it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:19, 37.46it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:18, 38.14it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:05<00:17, 37.35it/s]

Computing ESM2 embeddings: 29%| | 256/895 [00:06<00:16, 38.00it/s]

```

Computing ESM2 embeddings: 32%|          | 288/895 [00:07<00:16, 37.38it/s]
Computing ESM2 embeddings: 36%|          | 320/895 [00:08<00:15, 37.98it/s]
Computing ESM2 embeddings: 39%|          | 352/895 [00:09<00:14, 37.40it/s]
Computing ESM2 embeddings: 43%|          | 384/895 [00:10<00:13, 37.97it/s]
Computing ESM2 embeddings: 46%|          | 416/895 [00:11<00:12, 37.13it/s]
Computing ESM2 embeddings: 50%|          | 448/895 [00:11<00:11, 37.82it/s]
Computing ESM2 embeddings: 54%|          | 480/895 [00:12<00:11, 37.26it/s]
Computing ESM2 embeddings: 57%|          | 512/895 [00:13<00:10, 37.90it/s]
Computing ESM2 embeddings: 61%|          | 544/895 [00:14<00:09, 37.27it/s]
Computing ESM2 embeddings: 64%|          | 576/895 [00:15<00:08, 37.87it/s]
Computing ESM2 embeddings: 68%|          | 608/895 [00:16<00:07, 37.30it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:16<00:06, 37.91it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:17<00:05, 37.80it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:18<00:04, 38.26it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:19<00:04, 37.51it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:20<00:03, 38.07it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:21<00:02, 37.30it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:22<00:01, 37.91it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:22<00:00, 37.27it/s]

Computing ESM2 embeddings: 100%|         | 895/895 [00:23<00:00, 37.65it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 39.01it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:20, 39.78it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.83it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:19, 38.64it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.49it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 38.24it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 37.38it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.15it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.24it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 38.03it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.32it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 38.06it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.35it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.07it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.30it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 38.03it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.28it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.94it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.32it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 37.89it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:06, 37.32it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 37.93it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.09it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 37.50it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 36.47it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.33it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 36.90it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.57it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/914 [00:00<00:23, 37.23it/s]
Computing ESM2 embeddings: 7%|        | 64/914 [00:01<00:23, 36.86it/s]
Computing ESM2 embeddings: 11%|       | 96/914 [00:02<00:22, 37.06it/s]
Computing ESM2 embeddings: 14%|       | 128/914 [00:03<00:20, 37.92it/s]
Computing ESM2 embeddings: 18%|       | 160/914 [00:04<00:20, 37.17it/s]
Computing ESM2 embeddings: 21%|       | 192/914 [00:05<00:19, 37.92it/s]
Computing ESM2 embeddings: 25%|       | 224/914 [00:06<00:18, 37.04it/s]
Computing ESM2 embeddings: 28%|       | 256/914 [00:06<00:17, 37.78it/s]
Computing ESM2 embeddings: 32%|       | 288/914 [00:07<00:16, 37.03it/s]
Computing ESM2 embeddings: 35%|       | 320/914 [00:08<00:15, 37.73it/s]
Computing ESM2 embeddings: 39%|       | 352/914 [00:09<00:15, 37.18it/s]
Computing ESM2 embeddings: 42%|       | 384/914 [00:10<00:14, 37.82it/s]
Computing ESM2 embeddings: 46%|       | 416/914 [00:11<00:13, 37.09it/s]
Computing ESM2 embeddings: 49%|       | 448/914 [00:11<00:12, 37.72it/s]
Computing ESM2 embeddings: 53%|       | 480/914 [00:12<00:11, 36.99it/s]
Computing ESM2 embeddings: 56%|       | 512/914 [00:13<00:10, 37.67it/s]

```

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:10, 36.98it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.66it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.10it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.78it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.65it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.77it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 36.57it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.37it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 36.80it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.53it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 36.86it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.71it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.20it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.87it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.12it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.49it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 38.42it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.59it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 38.32it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 37.44it/s]



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Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 38.17it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.36it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 38.12it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 37.37it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 38.08it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 37.31it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 37.78it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:11, 36.44it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.39it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 36.91it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.76it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.59it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.26it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.47it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 38.14it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.26it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 38.02it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.27it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 38.00it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 37.33it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 37.69it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:24, 35.62it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:22, 37.48it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:22, 36.00it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 37.26it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 36.73it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.59it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.04it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.76it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.42it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 37.99it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 37.37it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:13, 37.97it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.36it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 37.92it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.35it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.93it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.59it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:08, 38.11it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.39it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.93it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 36.92it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 37.38it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:19<00:04, 36.67it/s] |

```

Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.05it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 36.06it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 36.96it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 36.49it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.27it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.07it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/895 [00:00<00:23, 36.35it/s]
Computing ESM2 embeddings:  7%|          | 64/895 [00:01<00:25, 32.75it/s]
Computing ESM2 embeddings: 11%|          | 96/895 [00:02<00:24, 33.29it/s]
Computing ESM2 embeddings: 14%|          | 128/895 [00:03<00:21, 35.38it/s]
Computing ESM2 embeddings: 18%|          | 160/895 [00:04<00:20, 35.71it/s]
Computing ESM2 embeddings: 21%|          | 192/895 [00:05<00:19, 36.88it/s]
Computing ESM2 embeddings: 25%|          | 224/895 [00:06<00:18, 36.72it/s]
Computing ESM2 embeddings: 29%|          | 256/895 [00:07<00:17, 37.50it/s]
Computing ESM2 embeddings: 32%|          | 288/895 [00:07<00:16, 36.79it/s]
Computing ESM2 embeddings: 36%|          | 320/895 [00:08<00:15, 37.24it/s]
Computing ESM2 embeddings: 39%|          | 352/895 [00:09<00:15, 35.93it/s]
Computing ESM2 embeddings: 43%|          | 384/895 [00:10<00:13, 36.89it/s]
Computing ESM2 embeddings: 46%|          | 416/895 [00:11<00:13, 36.61it/s]
Computing ESM2 embeddings: 50%|          | 448/895 [00:12<00:11, 37.36it/s]

```

Computing ESM2 embeddings: 54%| | 480/895 [00:13<00:11, 36.78it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:14<00:10, 37.54it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 36.93it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.55it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 36.89it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 37.56it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:06, 36.89it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:19<00:05, 37.57it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:20<00:04, 37.66it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 38.01it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 36.75it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.47it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.35it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 36.88it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:28, 30.81it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:02<00:26, 32.14it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 35.04it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 35.24it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.62it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.40it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.32it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 36.74it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.38it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.91it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.64it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.13it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.79it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.24it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.83it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.13it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.80it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.75it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.84it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.63it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.38it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.81it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.51it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.88it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.55it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.93it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.62it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.19it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.85it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

`'esm.pooler.dense.weight']`

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/571 [00:00<?, ?it/s]
Computing ESM2 embeddings: 6%|          | 32/571 [00:00<00:14, 36.54it/s]
Computing ESM2 embeddings: 11%|         | 64/571 [00:01<00:13, 38.15it/s]
Computing ESM2 embeddings: 17%|         | 96/571 [00:02<00:13, 36.36it/s]
Computing ESM2 embeddings: 22%|         | 128/571 [00:03<00:11, 37.70it/s]
Computing ESM2 embeddings: 28%|         | 160/571 [00:04<00:11, 37.16it/s]
Computing ESM2 embeddings: 34%|         | 192/571 [00:05<00:09, 38.19it/s]
Computing ESM2 embeddings: 39%|         | 224/571 [00:05<00:09, 37.92it/s]
Computing ESM2 embeddings: 45%|         | 256/571 [00:06<00:08, 38.52it/s]
Computing ESM2 embeddings: 50%|         | 288/571 [00:07<00:07, 37.64it/s]
Computing ESM2 embeddings: 56%|         | 320/571 [00:08<00:06, 38.29it/s]
Computing ESM2 embeddings: 62%|         | 352/571 [00:09<00:05, 37.23it/s]
Computing ESM2 embeddings: 67%|         | 384/571 [00:10<00:04, 38.01it/s]
Computing ESM2 embeddings: 73%|         | 416/571 [00:11<00:04, 37.45it/s]
Computing ESM2 embeddings: 78%|         | 448/571 [00:11<00:03, 38.17it/s]
Computing ESM2 embeddings: 84%|         | 480/571 [00:12<00:02, 37.31it/s]
Computing ESM2 embeddings: 90%|         | 512/571 [00:13<00:01, 38.03it/s]
Computing ESM2 embeddings: 95%|         | 544/571 [00:14<00:00, 37.16it/s]
```

```
Computing ESM2 embeddings: 100%|        | 571/571 [00:15<00:00, 37.55it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/915 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/915 [00:00<00:22, 38.62it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/915 [00:01<00:21, 38.97it/s]  |
| Computing ESM2 embeddings: | 10% | 96/915 [00:02<00:21, 37.97it/s]  |
| Computing ESM2 embeddings: | 14% | 128/915 [00:03<00:20, 38.77it/s] |
| Computing ESM2 embeddings: | 17% | 160/915 [00:04<00:19, 37.78it/s] |
| Computing ESM2 embeddings: | 21% | 192/915 [00:05<00:18, 38.47it/s] |
| Computing ESM2 embeddings: | 24% | 224/915 [00:05<00:18, 37.49it/s] |
| Computing ESM2 embeddings: | 28% | 256/915 [00:06<00:17, 38.07it/s] |
| Computing ESM2 embeddings: | 31% | 288/915 [00:07<00:16, 37.94it/s] |
| Computing ESM2 embeddings: | 35% | 320/915 [00:08<00:15, 38.43it/s] |
| Computing ESM2 embeddings: | 38% | 352/915 [00:09<00:14, 37.55it/s] |
| Computing ESM2 embeddings: | 42% | 384/915 [00:10<00:13, 38.13it/s] |
| Computing ESM2 embeddings: | 45% | 416/915 [00:10<00:13, 37.27it/s] |
| Computing ESM2 embeddings: | 49% | 448/915 [00:11<00:12, 37.75it/s] |
| Computing ESM2 embeddings: | 52% | 480/915 [00:12<00:11, 37.04it/s] |
| Computing ESM2 embeddings: | 56% | 512/915 [00:13<00:10, 37.84it/s] |
| Computing ESM2 embeddings: | 59% | 544/915 [00:14<00:09, 37.45it/s] |
| Computing ESM2 embeddings: | 63% | 576/915 [00:15<00:08, 38.01it/s] |
| Computing ESM2 embeddings: | 66% | 608/915 [00:16<00:08, 37.12it/s] |
| Computing ESM2 embeddings: | 70% | 640/915 [00:16<00:07, 37.71it/s] |
| Computing ESM2 embeddings: | 73% | 672/915 [00:17<00:06, 36.40it/s] |
| Computing ESM2 embeddings: | 77% | 704/915 [00:18<00:05, 37.37it/s] |
| Computing ESM2 embeddings: | 80% | 736/915 [00:19<00:04, 36.60it/s] |

```

Computing ESM2 embeddings: 84%|      | 768/915 [00:20<00:03, 37.51it/s]
Computing ESM2 embeddings: 87%|      | 800/915 [00:21<00:03, 37.05it/s]
Computing ESM2 embeddings: 91%|      | 832/915 [00:22<00:02, 37.90it/s]
Computing ESM2 embeddings: 94%|      | 864/915 [00:22<00:01, 37.01it/s]
Computing ESM2 embeddings: 98%|      | 896/915 [00:23<00:00, 37.82it/s]

Computing ESM2 embeddings: 100%|     | 915/915 [00:24<00:00, 37.57it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 39.05it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 39.02it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 36.94it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 38.02it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.23it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:18, 38.12it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:18, 37.30it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:16, 38.12it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.21it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 38.04it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.83it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 38.45it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.62it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 38.21it/s]

```



Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.64it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 38.34it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.15it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 38.52it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.72it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.24it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:06, 37.31it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.90it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.35it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.93it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.32it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 37.88it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 37.42it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.85it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.18it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 38.97it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.63it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.37it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.48it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.99it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:18, 37.81it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:17, 38.33it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 37.44it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:15, 38.07it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:15, 37.29it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:13, 37.91it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:10<00:13, 37.80it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:12, 38.29it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.56it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 38.10it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 37.93it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:15<00:08, 38.37it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:16<00:08, 37.58it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:16<00:07, 38.13it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 37.30it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 37.88it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:19<00:04, 37.29it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:20<00:03, 37.89it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:21<00:03, 37.31it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:21<00:02, 37.92it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:22<00:01, 37.34it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:23<00:00, 37.93it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:24<00:00, 37.69it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.84it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.09it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.43it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 38.37it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.35it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 38.20it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:18, 37.30it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.06it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.23it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 38.01it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.75it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 38.37it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.47it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.15it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.26it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 38.01it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.79it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 38.39it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.52it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.18it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.63it/s] |

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 38.25it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.50it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 38.18it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.38it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.03it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 37.34it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.83it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:22, 39.24it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:21, 39.91it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:21, 37.99it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:20, 38.71it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:20, 37.46it/s]

Computing ESM2 embeddings: 21%| | 192/913 [00:05<00:18, 38.26it/s]

Computing ESM2 embeddings: 25%| | 224/913 [00:05<00:18, 37.56it/s]

Computing ESM2 embeddings: 28%| | 256/913 [00:06<00:17, 37.99it/s]

Computing ESM2 embeddings: 32%| | 288/913 [00:07<00:17, 36.58it/s]

Computing ESM2 embeddings: 35%| | 320/913 [00:08<00:15, 37.41it/s]

Computing ESM2 embeddings: 39%| | 352/913 [00:09<00:15, 36.88it/s]

Computing ESM2 embeddings: 42%| | 384/913 [00:10<00:14, 37.69it/s]

Computing ESM2 embeddings: 46%| | 416/913 [00:11<00:13, 37.26it/s]

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Computing ESM2 embeddings: 49%|          | 448/913 [00:11<00:12, 37.88it/s]
Computing ESM2 embeddings: 53%|          | 480/913 [00:12<00:11, 37.32it/s]
Computing ESM2 embeddings: 56%|          | 512/913 [00:13<00:10, 37.94it/s]
Computing ESM2 embeddings: 60%|          | 544/913 [00:14<00:09, 37.82it/s]
Computing ESM2 embeddings: 63%|          | 576/913 [00:15<00:08, 38.21it/s]
Computing ESM2 embeddings: 67%|          | 608/913 [00:16<00:08, 37.27it/s]
Computing ESM2 embeddings: 70%|          | 640/913 [00:16<00:07, 37.84it/s]
Computing ESM2 embeddings: 74%|          | 672/913 [00:17<00:06, 37.70it/s]
Computing ESM2 embeddings: 77%|          | 704/913 [00:18<00:05, 38.16it/s]
Computing ESM2 embeddings: 81%|          | 736/913 [00:19<00:04, 37.38it/s]
Computing ESM2 embeddings: 84%|          | 768/913 [00:20<00:03, 37.57it/s]
Computing ESM2 embeddings: 88%|          | 800/913 [00:21<00:03, 37.07it/s]
Computing ESM2 embeddings: 91%|          | 832/913 [00:22<00:02, 37.87it/s]
Computing ESM2 embeddings: 95%|          | 864/913 [00:22<00:01, 37.15it/s]
Computing ESM2 embeddings: 98%|          | 896/913 [00:23<00:00, 37.75it/s]

Computing ESM2 embeddings: 100%|         | 913/913 [00:24<00:00, 37.62it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.66it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.51it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 37.98it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:19, 38.55it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:19, 37.55it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:18, 38.14it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 37.48it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 38.09it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:16, 37.23it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 37.66it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 36.88it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 37.65it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:12, 37.08it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 37.71it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:11, 37.71it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 38.21it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 37.53it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 37.80it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 36.63it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 37.45it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:06, 37.01it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:05, 37.69it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 37.56it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:20<00:03, 38.10it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 37.47it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 38.02it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:22<00:00, 37.68it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:23<00:00, 37.70it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 38.97it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.81it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.78it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.55it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 37.54it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:18, 38.34it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 38.07it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 38.40it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:17, 36.62it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 37.58it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 36.84it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.71it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.07it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 37.87it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.46it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.18it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.34it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:08, 38.09it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.38it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 38.10it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 37.54it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 38.20it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 37.93it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 38.50it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.70it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 38.34it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 37.99it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.53it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.81it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/572 [00:00<00:13, 39.32it/s]
Computing ESM2 embeddings: 11%|      | 64/572 [00:01<00:12, 39.92it/s]
Computing ESM2 embeddings: 17%|      | 96/572 [00:02<00:12, 38.20it/s]
Computing ESM2 embeddings: 22%|      | 128/572 [00:03<00:11, 38.80it/s]
Computing ESM2 embeddings: 28%|      | 160/572 [00:04<00:10, 37.73it/s]
Computing ESM2 embeddings: 34%|      | 192/572 [00:04<00:09, 38.44it/s]
Computing ESM2 embeddings: 39%|      | 224/572 [00:05<00:09, 37.99it/s]
Computing ESM2 embeddings: 45%|      | 256/572 [00:06<00:08, 38.32it/s]
Computing ESM2 embeddings: 50%|      | 288/572 [00:07<00:07, 37.10it/s]
Computing ESM2 embeddings: 56%|      | 320/572 [00:08<00:06, 37.64it/s]
Computing ESM2 embeddings: 62%|      | 352/572 [00:09<00:06, 36.44it/s]

```



Computing ESM2 embeddings: 67%| | 384/572 [00:10<00:05, 37.41it/s]

Computing ESM2 embeddings: 73%| | 416/572 [00:11<00:04, 36.85it/s]

Computing ESM2 embeddings: 78%| | 448/572 [00:11<00:03, 37.73it/s]

Computing ESM2 embeddings: 84%| | 480/572 [00:12<00:02, 37.12it/s]

Computing ESM2 embeddings: 90%| | 512/572 [00:13<00:01, 37.89it/s]

Computing ESM2 embeddings: 95%| | 544/572 [00:14<00:00, 37.70it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:15<00:00, 37.49it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/908 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/908 [00:00<00:23, 38.07it/s]

Computing ESM2 embeddings: 7%| | 64/908 [00:01<00:23, 35.75it/s]

Computing ESM2 embeddings: 11%| | 96/908 [00:02<00:23, 35.21it/s]

Computing ESM2 embeddings: 14%| | 128/908 [00:03<00:21, 36.74it/s]

Computing ESM2 embeddings: 18%| | 160/908 [00:04<00:20, 36.38it/s]

Computing ESM2 embeddings: 21%| | 192/908 [00:05<00:19, 37.22it/s]

Computing ESM2 embeddings: 25%| | 224/908 [00:06<00:18, 37.30it/s]

Computing ESM2 embeddings: 28%| | 256/908 [00:06<00:17, 37.85it/s]

Computing ESM2 embeddings: 32%| | 288/908 [00:07<00:16, 37.05it/s]

Computing ESM2 embeddings: 35%| | 320/908 [00:08<00:15, 37.72it/s]

Computing ESM2 embeddings: 39%| | 352/908 [00:09<00:14, 37.68it/s]

Computing ESM2 embeddings: 42%| | 384/908 [00:10<00:13, 38.17it/s]

Computing ESM2 embeddings: 46%| | 416/908 [00:11<00:12, 38.04it/s]

Computing ESM2 embeddings: 49%| | 448/908 [00:11<00:11, 38.43it/s]

Computing ESM2 embeddings: 53%| | 480/908 [00:12<00:11, 37.82it/s]

Computing ESM2 embeddings: 56%| | 512/908 [00:13<00:10, 38.36it/s]

Computing ESM2 embeddings: 60%| | 544/908 [00:14<00:09, 38.19it/s]

Computing ESM2 embeddings: 63%| | 576/908 [00:15<00:08, 38.64it/s]

Computing ESM2 embeddings: 67%| | 608/908 [00:16<00:07, 37.96it/s]

Computing ESM2 embeddings: 70%| | 640/908 [00:16<00:06, 38.47it/s]

Computing ESM2 embeddings: 74%| | 672/908 [00:17<00:06, 38.28it/s]

Computing ESM2 embeddings: 78%| | 704/908 [00:18<00:05, 38.58it/s]

Computing ESM2 embeddings: 81%| | 736/908 [00:19<00:04, 37.82it/s]

Computing ESM2 embeddings: 85%| | 768/908 [00:20<00:03, 38.36it/s]

Computing ESM2 embeddings: 88%| | 800/908 [00:21<00:02, 37.73it/s]

Computing ESM2 embeddings: 92%| | 832/908 [00:22<00:01, 38.27it/s]

Computing ESM2 embeddings: 95%| | 864/908 [00:22<00:01, 38.10it/s]

Computing ESM2 embeddings: 99%| | 896/908 [00:23<00:00, 38.50it/s]

Computing ESM2 embeddings: 100%| | 908/908 [00:24<00:00, 37.74it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/902 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/902 [00:01<00:27, 31.62it/s]

Computing ESM2 embeddings: 7%| | 64/902 [00:01<00:23, 35.79it/s]

Computing ESM2 embeddings: 11%| | 96/902 [00:02<00:22, 35.86it/s]

Computing ESM2 embeddings: 14%| | 128/902 [00:03<00:20, 37.30it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18% | 160/902 [00:04<00:20, 36.90it/s] |
| Computing ESM2 embeddings: 21% | 192/902 [00:05<00:18, 37.79it/s] |
| Computing ESM2 embeddings: 25% | 224/902 [00:06<00:18, 37.21it/s] |
| Computing ESM2 embeddings: 28% | 256/902 [00:06<00:17, 37.99it/s] |
| Computing ESM2 embeddings: 32% | 288/902 [00:07<00:16, 37.95it/s] |
| Computing ESM2 embeddings: 35% | 320/902 [00:08<00:15, 38.43it/s] |
| Computing ESM2 embeddings: 39% | 352/902 [00:09<00:14, 37.77it/s] |
| Computing ESM2 embeddings: 43% | 384/902 [00:10<00:13, 38.33it/s] |
| Computing ESM2 embeddings: 46% | 416/902 [00:11<00:12, 37.54it/s] |
| Computing ESM2 embeddings: 50% | 448/902 [00:11<00:11, 38.17it/s] |
| Computing ESM2 embeddings: 53% | 480/902 [00:12<00:11, 37.48it/s] |
| Computing ESM2 embeddings: 57% | 512/902 [00:13<00:10, 38.06it/s] |
| Computing ESM2 embeddings: 60% | 544/902 [00:14<00:09, 37.36it/s] |
| Computing ESM2 embeddings: 64% | 576/902 [00:15<00:08, 38.01it/s] |
| Computing ESM2 embeddings: 67% | 608/902 [00:16<00:07, 37.35it/s] |
| Computing ESM2 embeddings: 71% | 640/902 [00:17<00:06, 37.99it/s] |
| Computing ESM2 embeddings: 75% | 672/902 [00:17<00:06, 37.95it/s] |
| Computing ESM2 embeddings: 78% | 704/902 [00:18<00:05, 38.46it/s] |
| Computing ESM2 embeddings: 82% | 736/902 [00:19<00:04, 37.91it/s] |
| Computing ESM2 embeddings: 85% | 768/902 [00:20<00:03, 38.42it/s] |
| Computing ESM2 embeddings: 89% | 800/902 [00:21<00:02, 38.15it/s] |
| Computing ESM2 embeddings: 92% | 832/902 [00:22<00:01, 38.58it/s] |
| Computing ESM2 embeddings: 96% | 864/902 [00:22<00:01, 37.64it/s] |
| Computing ESM2 embeddings: 99% | 896/902 [00:23<00:00, 38.26it/s] |

Computing ESM2 embeddings: 100%| | 902/902 [00:23<00:00, 37.60it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/903 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/903 [00:00<00:23, 37.83it/s]  
 Computing ESM2 embeddings: 7%| | 64/903 [00:01<00:22, 36.90it/s]  
 Computing ESM2 embeddings: 11%| | 96/903 [00:02<00:21, 38.26it/s]  
 Computing ESM2 embeddings: 14%| | 128/903 [00:03<00:20, 37.72it/s]  
 Computing ESM2 embeddings: 18%| | 160/903 [00:04<00:19, 38.47it/s]  
 Computing ESM2 embeddings: 21%| | 192/903 [00:05<00:18, 37.47it/s]  
 Computing ESM2 embeddings: 25%| | 224/903 [00:05<00:17, 38.22it/s]  
 Computing ESM2 embeddings: 28%| | 256/903 [00:06<00:17, 37.51it/s]  
 Computing ESM2 embeddings: 32%| | 288/903 [00:07<00:16, 38.04it/s]  
 Computing ESM2 embeddings: 35%| | 320/903 [00:08<00:15, 37.40it/s]  
 Computing ESM2 embeddings: 39%| | 352/903 [00:09<00:14, 38.01it/s]  
 Computing ESM2 embeddings: 43%| | 384/903 [00:10<00:13, 37.79it/s]  
 Computing ESM2 embeddings: 46%| | 416/903 [00:10<00:12, 38.38it/s]  
 Computing ESM2 embeddings: 50%| | 448/903 [00:11<00:12, 37.65it/s]  
 Computing ESM2 embeddings: 53%| | 480/903 [00:12<00:11, 38.28it/s]  
 Computing ESM2 embeddings: 57%| | 512/903 [00:13<00:10, 37.98it/s]  
 Computing ESM2 embeddings: 60%| | 544/903 [00:14<00:09, 38.52it/s]  
 Computing ESM2 embeddings: 64%| | 576/903 [00:15<00:08, 38.14it/s]  
 Computing ESM2 embeddings: 67%| | 608/903 [00:15<00:07, 38.64it/s]

```

Computing ESM2 embeddings: 71%|          | 640/903 [00:16<00:06, 38.04it/s]
Computing ESM2 embeddings: 74%|          | 672/903 [00:17<00:05, 38.57it/s]
Computing ESM2 embeddings: 78%|          | 704/903 [00:18<00:05, 37.62it/s]
Computing ESM2 embeddings: 82%|          | 736/903 [00:19<00:04, 38.28it/s]
Computing ESM2 embeddings: 85%|          | 768/903 [00:20<00:03, 37.46it/s]
Computing ESM2 embeddings: 89%|          | 800/903 [00:21<00:02, 37.96it/s]
Computing ESM2 embeddings: 92%|          | 832/903 [00:21<00:01, 37.77it/s]
Computing ESM2 embeddings: 96%|          | 864/903 [00:22<00:01, 38.24it/s]
Computing ESM2 embeddings: 99%|          | 896/903 [00:23<00:00, 37.43it/s]

Computing ESM2 embeddings: 100%|         | 903/903 [00:23<00:00, 37.88it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/907 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/907 [00:00<00:22, 38.85it/s]
Computing ESM2 embeddings:  7%|          | 64/907 [00:01<00:21, 39.42it/s]
Computing ESM2 embeddings: 11%|          | 96/907 [00:02<00:21, 37.98it/s]
Computing ESM2 embeddings: 14%|          | 128/907 [00:03<00:20, 38.67it/s]
Computing ESM2 embeddings: 18%|          | 160/907 [00:04<00:19, 37.76it/s]
Computing ESM2 embeddings: 21%|          | 192/907 [00:04<00:18, 38.45it/s]
Computing ESM2 embeddings: 25%|          | 224/907 [00:05<00:17, 38.18it/s]
Computing ESM2 embeddings: 28%|          | 256/907 [00:06<00:16, 38.56it/s]
Computing ESM2 embeddings: 32%|          | 288/907 [00:07<00:16, 38.31it/s]
Computing ESM2 embeddings: 35%|          | 320/907 [00:08<00:15, 38.73it/s]

```

Computing ESM2 embeddings: 39%| | 352/907 [00:09<00:14, 37.91it/s]

Computing ESM2 embeddings: 42%| | 384/907 [00:09<00:13, 38.48it/s]

Computing ESM2 embeddings: 46%| | 416/907 [00:10<00:12, 38.28it/s]

Computing ESM2 embeddings: 49%| | 448/907 [00:11<00:11, 38.73it/s]

Computing ESM2 embeddings: 53%| | 480/907 [00:12<00:11, 38.50it/s]

Computing ESM2 embeddings: 56%| | 512/907 [00:13<00:10, 38.92it/s]

Computing ESM2 embeddings: 60%| | 544/907 [00:14<00:09, 38.59it/s]

Computing ESM2 embeddings: 64%| | 576/907 [00:14<00:08, 39.00it/s]

Computing ESM2 embeddings: 67%| | 608/907 [00:15<00:07, 38.61it/s]

Computing ESM2 embeddings: 71%| | 640/907 [00:16<00:06, 38.99it/s]

Computing ESM2 embeddings: 74%| | 672/907 [00:17<00:06, 38.06it/s]

Computing ESM2 embeddings: 78%| | 704/907 [00:18<00:05, 38.59it/s]

Computing ESM2 embeddings: 81%| | 736/907 [00:19<00:04, 37.86it/s]

Computing ESM2 embeddings: 85%| | 768/907 [00:19<00:03, 38.38it/s]

Computing ESM2 embeddings: 88%| | 800/907 [00:20<00:02, 38.17it/s]

Computing ESM2 embeddings: 92%| | 832/907 [00:21<00:01, 38.51it/s]

Computing ESM2 embeddings: 95%| | 864/907 [00:22<00:01, 38.41it/s]

Computing ESM2 embeddings: 99%| | 896/907 [00:23<00:00, 38.86it/s]

Computing ESM2 embeddings: 100%| | 907/907 [00:23<00:00, 38.47it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/898 [00:00<00:22, 39.09it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/898 [00:01<00:21, 39.60it/s]  |
| Computing ESM2 embeddings: | 11% | 96/898 [00:02<00:21, 38.02it/s]  |
| Computing ESM2 embeddings: | 14% | 128/898 [00:03<00:19, 38.72it/s] |
| Computing ESM2 embeddings: | 18% | 160/898 [00:04<00:19, 37.78it/s] |
| Computing ESM2 embeddings: | 21% | 192/898 [00:04<00:18, 38.47it/s] |
| Computing ESM2 embeddings: | 25% | 224/898 [00:05<00:17, 38.38it/s] |
| Computing ESM2 embeddings: | 29% | 256/898 [00:06<00:16, 38.86it/s] |
| Computing ESM2 embeddings: | 32% | 288/898 [00:07<00:16, 37.96it/s] |
| Computing ESM2 embeddings: | 36% | 320/898 [00:08<00:15, 38.37it/s] |
| Computing ESM2 embeddings: | 39% | 352/898 [00:09<00:14, 37.86it/s] |
| Computing ESM2 embeddings: | 43% | 384/898 [00:10<00:13, 38.40it/s] |
| Computing ESM2 embeddings: | 46% | 416/898 [00:10<00:12, 37.76it/s] |
| Computing ESM2 embeddings: | 50% | 448/898 [00:11<00:11, 38.35it/s] |
| Computing ESM2 embeddings: | 53% | 480/898 [00:12<00:11, 37.77it/s] |
| Computing ESM2 embeddings: | 57% | 512/898 [00:13<00:10, 38.34it/s] |
| Computing ESM2 embeddings: | 61% | 544/898 [00:14<00:09, 37.58it/s] |
| Computing ESM2 embeddings: | 64% | 576/898 [00:15<00:08, 38.26it/s] |
| Computing ESM2 embeddings: | 68% | 608/898 [00:15<00:07, 38.08it/s] |
| Computing ESM2 embeddings: | 71% | 640/898 [00:16<00:06, 38.61it/s] |
| Computing ESM2 embeddings: | 75% | 672/898 [00:17<00:05, 38.39it/s] |
| Computing ESM2 embeddings: | 78% | 704/898 [00:18<00:04, 38.83it/s] |
| Computing ESM2 embeddings: | 82% | 736/898 [00:19<00:04, 37.94it/s] |
| Computing ESM2 embeddings: | 86% | 768/898 [00:20<00:03, 38.48it/s] |
| Computing ESM2 embeddings: | 89% | 800/898 [00:20<00:02, 37.82it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/898 [00:21<00:01, 38.41it/s]

Computing ESM2 embeddings: 96%|      | 864/898 [00:22<00:00, 37.61it/s]

Computing ESM2 embeddings: 100%|     | 898/898 [00:23<00:00, 38.14it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/911 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/911 [00:00<00:22, 39.07it/s]

Computing ESM2 embeddings:  7%|      | 64/911 [00:01<00:21, 39.75it/s]

Computing ESM2 embeddings: 11%|      | 96/911 [00:02<00:21, 37.99it/s]

Computing ESM2 embeddings: 14%|      | 128/911 [00:03<00:20, 38.74it/s]

Computing ESM2 embeddings: 18%|      | 160/911 [00:04<00:19, 37.62it/s]

Computing ESM2 embeddings: 21%|      | 192/911 [00:04<00:18, 38.39it/s]

Computing ESM2 embeddings: 25%|      | 224/911 [00:05<00:18, 37.63it/s]

Computing ESM2 embeddings: 28%|      | 256/911 [00:06<00:17, 38.32it/s]

Computing ESM2 embeddings: 32%|      | 288/911 [00:07<00:16, 37.45it/s]

Computing ESM2 embeddings: 35%|      | 320/911 [00:08<00:15, 37.85it/s]

Computing ESM2 embeddings: 39%|      | 352/911 [00:09<00:15, 37.00it/s]

Computing ESM2 embeddings: 42%|      | 384/911 [00:10<00:13, 37.84it/s]

Computing ESM2 embeddings: 46%|      | 416/911 [00:11<00:13, 37.21it/s]

Computing ESM2 embeddings: 49%|      | 448/911 [00:11<00:12, 37.98it/s]

Computing ESM2 embeddings: 53%|      | 480/911 [00:12<00:11, 37.24it/s]

Computing ESM2 embeddings: 56%|      | 512/911 [00:13<00:10, 37.93it/s]

Computing ESM2 embeddings: 60%|      | 544/911 [00:14<00:09, 37.21it/s]

```



Computing ESM2 embeddings: 63%| | 576/911 [00:15<00:08, 37.95it/s]

Computing ESM2 embeddings: 67%| | 608/911 [00:16<00:08, 37.44it/s]

Computing ESM2 embeddings: 70%| | 640/911 [00:16<00:07, 38.12it/s]

Computing ESM2 embeddings: 74%| | 672/911 [00:17<00:06, 37.39it/s]

Computing ESM2 embeddings: 77%| | 704/911 [00:18<00:05, 38.09it/s]

Computing ESM2 embeddings: 81%| | 736/911 [00:19<00:04, 37.50it/s]

Computing ESM2 embeddings: 84%| | 768/911 [00:20<00:03, 38.17it/s]

Computing ESM2 embeddings: 88%| | 800/911 [00:21<00:02, 37.48it/s]

Computing ESM2 embeddings: 91%| | 832/911 [00:21<00:02, 38.05it/s]

Computing ESM2 embeddings: 95%| | 864/911 [00:22<00:01, 37.81it/s]

Computing ESM2 embeddings: 98%| | 896/911 [00:23<00:00, 38.40it/s]

Computing ESM2 embeddings: 100%| | 911/911 [00:24<00:00, 37.82it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.02it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 38.24it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.04it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 38.11it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.27it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 38.14it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 37.95it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 38.60it/s]

```

Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.99it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:14, 38.43it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 37.24it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 37.84it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:13, 36.67it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 37.62it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:11, 37.50it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 38.17it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.46it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 38.11it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.62it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.00it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:06, 36.99it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 37.51it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 36.64it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.55it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.00it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.82it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.13it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 37.58it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/590 [00:00<?, ?it/s]

```

Computing ESM2 embeddings: 5%| | 32/590 [00:00<00:14, 38.11it/s]

Computing ESM2 embeddings: 11%| | 64/590 [00:01<00:15, 34.19it/s]

Computing ESM2 embeddings: 16%| | 96/590 [00:02<00:14, 34.09it/s]

Computing ESM2 embeddings: 22%| | 128/590 [00:03<00:12, 36.17it/s]

Computing ESM2 embeddings: 27%| | 160/590 [00:04<00:11, 36.43it/s]

Computing ESM2 embeddings: 33%| | 192/590 [00:05<00:10, 37.54it/s]

Computing ESM2 embeddings: 38%| | 224/590 [00:06<00:09, 37.39it/s]

Computing ESM2 embeddings: 43%| | 256/590 [00:06<00:08, 38.16it/s]

Computing ESM2 embeddings: 49%| | 288/590 [00:07<00:07, 37.87it/s]

Computing ESM2 embeddings: 54%| | 320/590 [00:08<00:07, 38.46it/s]

Computing ESM2 embeddings: 60%| | 352/590 [00:09<00:06, 37.82it/s]

Computing ESM2 embeddings: 65%| | 384/590 [00:10<00:05, 38.40it/s]

Computing ESM2 embeddings: 71%| | 416/590 [00:11<00:04, 37.66it/s]

Computing ESM2 embeddings: 76%| | 448/590 [00:11<00:03, 38.00it/s]

Computing ESM2 embeddings: 81%| | 480/590 [00:12<00:03, 36.48it/s]

Computing ESM2 embeddings: 87%| | 512/590 [00:13<00:02, 37.44it/s]

Computing ESM2 embeddings: 92%| | 544/590 [00:14<00:01, 36.69it/s]

Computing ESM2 embeddings: 98%| | 576/590 [00:15<00:00, 37.57it/s]

Computing ESM2 embeddings: 100%| | 590/590 [00:15<00:00, 37.00it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.69it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 37.98it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:22, 36.02it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 37.45it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:20, 36.76it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 37.77it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 36.98it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 37.82it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.15it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.95it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.29it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.93it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.78it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.23it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 37.95it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 38.37it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.50it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 38.04it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.84it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.27it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.94it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 38.33it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.50it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 38.03it/s] |
| Computing ESM2 embeddings: | 89% | 800/896 [00:21<00:02, 37.18it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.80it/s]

Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 37.11it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.61it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.61it/s]

Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 39.44it/s]

Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.79it/s]

Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:20, 38.61it/s]

Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.48it/s]

Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:18, 38.28it/s]

Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:18, 37.62it/s]

Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 38.33it/s]

Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.41it/s]

Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 38.18it/s]

Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 37.93it/s]

Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:13, 38.54it/s]

Computing ESM2 embeddings: 46%|      | 416/914 [00:10<00:13, 37.69it/s]

Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:12, 38.34it/s]

Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 38.02it/s]

Computing ESM2 embeddings: 56%|      | 512/914 [00:13<00:10, 38.59it/s]

Computing ESM2 embeddings: 60%|      | 544/914 [00:14<00:09, 37.96it/s]

```

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 38.54it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:08, 38.19it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.69it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.97it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.54it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.85it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 38.22it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 36.76it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 37.65it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 37.06it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.87it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.93it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.26it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:21, 39.28it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.73it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:19, 38.51it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.76it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 38.45it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:18, 37.31it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:16, 37.74it/s]

```

Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 36.93it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 37.66it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 37.58it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 38.13it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 37.46it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 37.74it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:11, 36.60it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.44it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.42it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 38.02it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.94it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:16<00:06, 38.42it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.15it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:04, 38.53it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.71it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.88it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 36.80it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.54it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 37.07it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 37.70it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.13it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:21, 39.87it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 38.07it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 38.78it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 37.71it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:04<00:18, 38.43it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 38.15it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:16, 38.72it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.62it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 38.30it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.49it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:13, 38.19it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:10<00:13, 37.42it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 38.14it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.26it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 38.02it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.47it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:08, 38.15it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.38it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:16<00:07, 38.07it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 37.87it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 38.46it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:19<00:04, 37.58it/s] |
| Computing ESM2 embeddings: | 84% | 768/914 [00:20<00:03, 38.24it/s] |



```

Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.50it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 38.17it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 37.86it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 38.44it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.92it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/895 [00:00<00:23, 37.38it/s]
Computing ESM2 embeddings:  7%|      | 64/895 [00:02<00:26, 31.12it/s]
Computing ESM2 embeddings: 11%|      | 96/895 [00:02<00:23, 33.89it/s]
Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:22, 33.69it/s]
Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:20, 35.53it/s]
Computing ESM2 embeddings: 21%|      | 192/895 [00:05<00:19, 35.66it/s]
Computing ESM2 embeddings: 25%|      | 224/895 [00:06<00:18, 36.80it/s]
Computing ESM2 embeddings: 29%|      | 256/895 [00:07<00:17, 36.32it/s]
Computing ESM2 embeddings: 32%|      | 288/895 [00:08<00:16, 37.18it/s]
Computing ESM2 embeddings: 36%|      | 320/895 [00:08<00:15, 36.91it/s]
Computing ESM2 embeddings: 39%|      | 352/895 [00:09<00:14, 37.59it/s]
Computing ESM2 embeddings: 43%|      | 384/895 [00:10<00:13, 36.91it/s]
Computing ESM2 embeddings: 46%|      | 416/895 [00:11<00:12, 37.63it/s]
Computing ESM2 embeddings: 50%|      | 448/895 [00:12<00:11, 37.57it/s]
Computing ESM2 embeddings: 54%|      | 480/895 [00:13<00:10, 38.11it/s]

```

Computing ESM2 embeddings: 57%| | 512/895 [00:14<00:10, 37.41it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 38.01it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.74it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 38.21it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 38.00it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:05, 38.36it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:19<00:05, 38.13it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 38.15it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 36.82it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.52it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 36.92it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.57it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 36.88it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.24it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.75it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.00it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.60it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.57it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:18, 38.23it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 37.40it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.04it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.33it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 37.94it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.43it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:13, 37.97it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:10<00:13, 37.89it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 38.35it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.19it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.57it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.79it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 38.30it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:15<00:08, 38.08it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:16<00:07, 38.46it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.24it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.59it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.30it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 38.67it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:20<00:03, 37.84it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:21<00:02, 38.31it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 38.09it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 38.15it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.98it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.02it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 38.81it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.41it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 38.22it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.76it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 38.29it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 37.43it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.18it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.37it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 38.09it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.52it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 38.07it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:10<00:12, 37.92it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 38.37it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.02it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 38.39it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.45it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.99it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.21it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 37.88it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.70it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 38.19it/s] |

```

Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.84it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 38.41it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.45it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:21<00:01, 38.02it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 37.86it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.89it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/590 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/590 [00:00<00:14, 38.86it/s]
Computing ESM2 embeddings: 11%|      | 64/590 [00:01<00:13, 39.56it/s]
Computing ESM2 embeddings: 16%|      | 96/590 [00:02<00:13, 37.92it/s]
Computing ESM2 embeddings: 22%|      | 128/590 [00:03<00:12, 38.49it/s]
Computing ESM2 embeddings: 27%|      | 160/590 [00:04<00:11, 37.60it/s]
Computing ESM2 embeddings: 33%|      | 192/590 [00:05<00:10, 38.22it/s]
Computing ESM2 embeddings: 38%|      | 224/590 [00:05<00:09, 38.31it/s]
Computing ESM2 embeddings: 43%|      | 256/590 [00:06<00:08, 38.60it/s]
Computing ESM2 embeddings: 49%|      | 288/590 [00:07<00:08, 37.50it/s]
Computing ESM2 embeddings: 54%|      | 320/590 [00:08<00:07, 38.11it/s]
Computing ESM2 embeddings: 60%|      | 352/590 [00:09<00:06, 37.93it/s]
Computing ESM2 embeddings: 65%|      | 384/590 [00:10<00:05, 38.41it/s]
Computing ESM2 embeddings: 71%|      | 416/590 [00:10<00:04, 37.66it/s]
Computing ESM2 embeddings: 76%|      | 448/590 [00:11<00:03, 38.20it/s]

```

```

Computing ESM2 embeddings: 81%|          | 480/590 [00:12<00:02, 37.53it/s]
Computing ESM2 embeddings: 87%|          | 512/590 [00:13<00:02, 38.08it/s]
Computing ESM2 embeddings: 92%|          | 544/590 [00:14<00:01, 37.93it/s]
Computing ESM2 embeddings: 98%|          | 576/590 [00:15<00:00, 38.38it/s]
Computing ESM2 embeddings: 100%|         | 590/590 [00:15<00:00, 38.11it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.39it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:21, 39.02it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 37.74it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 38.39it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 37.43it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:18, 38.05it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:18, 37.21it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:16, 37.90it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.79it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 38.29it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 37.51it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 38.09it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:10<00:12, 37.89it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:11, 38.37it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:11, 37.57it/s]

```

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 38.11it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.47it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 38.19it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.39it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:16<00:06, 38.09it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.35it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 38.08it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.47it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 38.16it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.36it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:21<00:01, 38.10it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 37.13it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.79it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 36.65it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:26, 31.85it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:25, 32.58it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 34.85it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:21, 35.08it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.32it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:19, 36.06it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 36.70it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:08<00:17, 36.24it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 37.12it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 36.49it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.34it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 36.91it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.59it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.34it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 37.94it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.51it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 38.01it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.34it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 38.04it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.35it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 38.03it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 37.33it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.95it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.17it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.87it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.17it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.90it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.85it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it



for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.31it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.20it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.67it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 38.38it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.82it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 38.28it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 37.44it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 38.02it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.20it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.85it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.03it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.84it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.12it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 37.87it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.11it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.88it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.20it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.97it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.42it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.07it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.57it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 38.23it/s] |

```

Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.42it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 38.10it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.34it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 38.00it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 36.94it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.59it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 38.21it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 36.41it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:23, 35.37it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.86it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 36.53it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 37.29it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 36.56it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 36.97it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:17, 36.55it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 37.39it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 36.84it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 37.64it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 36.94it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 37.67it/s]

```

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.50it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.00it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.97it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 38.45it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.72it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 38.21it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.61it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 38.10it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.33it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.87it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.64it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 38.19it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.28it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.96it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.23it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.63it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.75it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:23, 34.77it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 36.64it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 36.82it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:18, 37.69it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:06<00:18, 37.02it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 37.89it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:16, 37.16it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 37.87it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 37.10it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 37.72it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:12, 37.06it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:12<00:11, 37.73it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:11, 37.03it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 37.69it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 37.50it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 38.00it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 37.85it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 38.28it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 37.48it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:05, 37.74it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 36.93it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:20<00:03, 37.34it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 36.25it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 37.06it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:23<00:00, 36.37it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:24<00:00, 37.06it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/913 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/913 [00:00<00:22, 39.51it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/913 [00:01<00:22, 37.94it/s]  |
| Computing ESM2 embeddings: | 11% | 96/913 [00:02<00:22, 36.00it/s]  |
| Computing ESM2 embeddings: | 14% | 128/913 [00:03<00:20, 37.48it/s] |
| Computing ESM2 embeddings: | 18% | 160/913 [00:04<00:20, 36.79it/s] |
| Computing ESM2 embeddings: | 21% | 192/913 [00:05<00:19, 37.80it/s] |
| Computing ESM2 embeddings: | 25% | 224/913 [00:06<00:18, 37.01it/s] |
| Computing ESM2 embeddings: | 28% | 256/913 [00:06<00:17, 37.56it/s] |
| Computing ESM2 embeddings: | 32% | 288/913 [00:07<00:17, 36.16it/s] |
| Computing ESM2 embeddings: | 35% | 320/913 [00:08<00:16, 37.04it/s] |
| Computing ESM2 embeddings: | 39% | 352/913 [00:09<00:15, 37.29it/s] |
| Computing ESM2 embeddings: | 42% | 384/913 [00:10<00:13, 38.02it/s] |
| Computing ESM2 embeddings: | 46% | 416/913 [00:11<00:13, 37.49it/s] |
| Computing ESM2 embeddings: | 49% | 448/913 [00:11<00:12, 38.05it/s] |
| Computing ESM2 embeddings: | 53% | 480/913 [00:12<00:11, 37.43it/s] |
| Computing ESM2 embeddings: | 56% | 512/913 [00:13<00:10, 38.00it/s] |
| Computing ESM2 embeddings: | 60% | 544/913 [00:14<00:09, 37.84it/s] |
| Computing ESM2 embeddings: | 63% | 576/913 [00:15<00:08, 38.23it/s] |
| Computing ESM2 embeddings: | 67% | 608/913 [00:16<00:08, 37.52it/s] |
| Computing ESM2 embeddings: | 70% | 640/913 [00:17<00:07, 38.07it/s] |
| Computing ESM2 embeddings: | 74% | 672/913 [00:17<00:06, 37.29it/s] |

Computing ESM2 embeddings: 77%| | 704/913 [00:18<00:05, 37.89it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:19<00:04, 37.10it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:20<00:03, 37.75it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:21<00:03, 37.54it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 38.08it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:22<00:01, 37.88it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:23<00:00, 38.31it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:24<00:00, 37.53it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:21, 39.30it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 36.52it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 38.00it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.20it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.16it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.21it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.03it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 37.37it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 38.15it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 37.36it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.84it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.70it/s]

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Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 37.61it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:12, 36.97it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:11, 37.81it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.09it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.88it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.32it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 38.12it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.27it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.01it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 37.67it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 38.28it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.97it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 38.46it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.50it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 38.07it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 37.59it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|          | 32/572 [00:00<00:14, 36.99it/s]
Computing ESM2 embeddings: 11%|          | 64/572 [00:01<00:13, 38.22it/s]
Computing ESM2 embeddings: 17%|          | 96/572 [00:02<00:13, 36.44it/s]
Computing ESM2 embeddings: 22%|          | 128/572 [00:03<00:11, 37.75it/s]

```

Computing ESM2 embeddings: 28%| | 160/572 [00:04<00:11, 37.16it/s]

Computing ESM2 embeddings: 34%| | 192/572 [00:05<00:10, 37.96it/s]

Computing ESM2 embeddings: 39%| | 224/572 [00:05<00:09, 37.47it/s]

Computing ESM2 embeddings: 45%| | 256/572 [00:06<00:08, 37.92it/s]

Computing ESM2 embeddings: 50%| | 288/572 [00:07<00:07, 36.96it/s]

Computing ESM2 embeddings: 56%| | 320/572 [00:08<00:06, 37.89it/s]

Computing ESM2 embeddings: 62%| | 352/572 [00:09<00:05, 37.69it/s]

Computing ESM2 embeddings: 67%| | 384/572 [00:10<00:04, 38.35it/s]

Computing ESM2 embeddings: 73%| | 416/572 [00:11<00:04, 37.36it/s]

Computing ESM2 embeddings: 78%| | 448/572 [00:11<00:03, 38.10it/s]

Computing ESM2 embeddings: 84%| | 480/572 [00:12<00:02, 37.84it/s]

Computing ESM2 embeddings: 90%| | 512/572 [00:13<00:01, 38.44it/s]

Computing ESM2 embeddings: 95%| | 544/572 [00:14<00:00, 37.47it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:15<00:00, 37.61it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/932 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/932 [00:00<00:23, 37.89it/s]

Computing ESM2 embeddings: 7%| | 64/932 [00:01<00:22, 39.13it/s]

Computing ESM2 embeddings: 10%| | 96/932 [00:02<00:22, 37.57it/s]

Computing ESM2 embeddings: 14%| | 128/932 [00:03<00:20, 38.31it/s]

Computing ESM2 embeddings: 17%| | 160/932 [00:04<00:20, 38.03it/s]

Computing ESM2 embeddings: 21%| | 192/932 [00:05<00:19, 38.19it/s]



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 24%  | 224/932 [00:05<00:19, 37.04it/s] |
| Computing ESM2 embeddings: 27%  | 256/932 [00:06<00:17, 37.82it/s] |
| Computing ESM2 embeddings: 31%  | 288/932 [00:07<00:17, 37.00it/s] |
| Computing ESM2 embeddings: 34%  | 320/932 [00:08<00:16, 37.76it/s] |
| Computing ESM2 embeddings: 38%  | 352/932 [00:09<00:15, 37.19it/s] |
| Computing ESM2 embeddings: 41%  | 384/932 [00:10<00:14, 37.80it/s] |
| Computing ESM2 embeddings: 45%  | 416/932 [00:11<00:13, 37.60it/s] |
| Computing ESM2 embeddings: 48%  | 448/932 [00:11<00:12, 38.20it/s] |
| Computing ESM2 embeddings: 52%  | 480/932 [00:12<00:12, 37.46it/s] |
| Computing ESM2 embeddings: 55%  | 512/932 [00:13<00:11, 38.12it/s] |
| Computing ESM2 embeddings: 58%  | 544/932 [00:14<00:10, 37.16it/s] |
| Computing ESM2 embeddings: 62%  | 576/932 [00:15<00:09, 37.84it/s] |
| Computing ESM2 embeddings: 65%  | 608/932 [00:16<00:08, 37.59it/s] |
| Computing ESM2 embeddings: 69%  | 640/932 [00:16<00:07, 37.81it/s] |
| Computing ESM2 embeddings: 72%  | 672/932 [00:17<00:07, 36.85it/s] |
| Computing ESM2 embeddings: 76%  | 704/932 [00:18<00:06, 37.62it/s] |
| Computing ESM2 embeddings: 79%  | 736/932 [00:19<00:05, 37.09it/s] |
| Computing ESM2 embeddings: 82%  | 768/932 [00:20<00:04, 37.75it/s] |
| Computing ESM2 embeddings: 86%  | 800/932 [00:21<00:03, 37.21it/s] |
| Computing ESM2 embeddings: 89%  | 832/932 [00:22<00:02, 37.91it/s] |
| Computing ESM2 embeddings: 93%  | 864/932 [00:22<00:01, 37.48it/s] |
| Computing ESM2 embeddings: 96%  | 896/932 [00:23<00:00, 38.07it/s] |
| Computing ESM2 embeddings: 100% | 928/932 [00:24<00:00, 37.19it/s] |
| Computing ESM2 embeddings: 100% | 932/932 [00:24<00:00, 37.49it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.10it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:21, 39.08it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.56it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 38.35it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 38.00it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 38.53it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 37.50it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:16, 37.85it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 36.58it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.48it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 36.96it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.75it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.19it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:11, 37.83it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.66it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 38.22it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.96it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 38.45it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.67it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:16<00:06, 38.21it/s] |

```

Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 37.69it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:05, 38.21it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.38it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.99it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.23it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.91it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 37.70it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.80it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.71it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:21, 39.22it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.53it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:20, 38.34it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.61it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:18, 38.26it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:18, 37.94it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 38.07it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:17, 36.81it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 37.62it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 36.90it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 37.68it/s]

```

```

Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.13it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:12, 37.82it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 37.53it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 38.12it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 37.89it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:08, 38.35it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 38.05it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:16<00:07, 38.51it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 37.80it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 38.31it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.37it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.97it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.65it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:21<00:02, 38.20it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 37.40it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 37.70it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.63it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:23, 37.06it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:22, 37.35it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 36.69it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 37.68it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.44it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:18, 38.09it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 37.84it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:16, 38.30it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.30it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 37.92it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.13it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.74it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.19it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:11, 37.85it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:11, 37.30it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.90it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.11it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.76it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 36.97it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:06, 37.68it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:06, 37.17it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 37.83it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.12it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.77it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.03it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.70it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 37.15it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.49it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.51it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.23it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.76it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:20, 38.45it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.46it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:18, 38.15it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 37.62it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 38.23it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.47it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 38.06it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.11it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.81it/s]  
 Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.06it/s]  
 Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 37.79it/s]  
 Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.37it/s]  
 Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 38.02it/s]  
 Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.78it/s]  
 Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 38.28it/s]  
 Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.56it/s]

```

Computing ESM2 embeddings: 70%|      | 640/914 [00:16<00:07, 38.17it/s]
Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 37.91it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 38.39it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 38.13it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 38.53it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.88it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:21<00:02, 38.01it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 37.03it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 37.74it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.76it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/895 [00:00<00:22, 39.14it/s]
Computing ESM2 embeddings:  7%|      | 64/895 [00:01<00:20, 39.70it/s]
Computing ESM2 embeddings: 11%|      | 96/895 [00:02<00:21, 37.84it/s]
Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:20, 38.21it/s]
Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:19, 36.98it/s]
Computing ESM2 embeddings: 21%|      | 192/895 [00:05<00:18, 37.76it/s]
Computing ESM2 embeddings: 25%|      | 224/895 [00:05<00:18, 37.27it/s]
Computing ESM2 embeddings: 29%|      | 256/895 [00:06<00:16, 37.60it/s]
Computing ESM2 embeddings: 32%|      | 288/895 [00:07<00:16, 36.71it/s]
Computing ESM2 embeddings: 36%|      | 320/895 [00:08<00:15, 37.53it/s]

```

Computing ESM2 embeddings: 39%| | 352/895 [00:09<00:14, 37.00it/s]

Computing ESM2 embeddings: 43%| | 384/895 [00:10<00:13, 37.73it/s]

Computing ESM2 embeddings: 46%| | 416/895 [00:11<00:12, 36.98it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:11<00:11, 37.67it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:12<00:11, 36.97it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 37.70it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 36.98it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.66it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.11it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 37.78it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:17<00:05, 37.29it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:05, 37.85it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.08it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 37.80it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.63it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 38.14it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.12it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:23<00:00, 37.50it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.49it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:21, 39.13it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 37.60it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:20, 38.36it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 37.42it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:18, 38.18it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.35it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 37.60it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 36.84it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.77it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.24it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:13, 37.90it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.34it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 37.99it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.51it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 38.11it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.21it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:08, 37.89it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.09it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:16<00:07, 37.83it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.14it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.85it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.21it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.87it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.32it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.71it/s] |

```

Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 36.55it/s]

Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 37.21it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.40it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/572 [00:00<?, ?it/s]

Computing ESM2 embeddings:  6%|      | 32/572 [00:00<00:13, 38.96it/s]

Computing ESM2 embeddings: 11%|      | 64/572 [00:01<00:13, 37.07it/s]

Computing ESM2 embeddings: 17%|      | 96/572 [00:02<00:12, 38.18it/s]

Computing ESM2 embeddings: 22%|      | 128/572 [00:03<00:11, 37.27it/s]

Computing ESM2 embeddings: 28%|      | 160/572 [00:04<00:10, 38.08it/s]

Computing ESM2 embeddings: 34%|      | 192/572 [00:05<00:10, 37.32it/s]

Computing ESM2 embeddings: 39%|      | 224/572 [00:05<00:09, 38.04it/s]

Computing ESM2 embeddings: 45%|      | 256/572 [00:06<00:08, 37.57it/s]

Computing ESM2 embeddings: 50%|      | 288/572 [00:07<00:07, 38.19it/s]

Computing ESM2 embeddings: 56%|      | 320/572 [00:08<00:06, 37.65it/s]

Computing ESM2 embeddings: 62%|      | 352/572 [00:09<00:05, 38.24it/s]

Computing ESM2 embeddings: 67%|      | 384/572 [00:10<00:05, 37.31it/s]

Computing ESM2 embeddings: 73%|      | 416/572 [00:10<00:04, 37.99it/s]

Computing ESM2 embeddings: 78%|      | 448/572 [00:11<00:03, 37.25it/s]

Computing ESM2 embeddings: 84%|      | 480/572 [00:12<00:02, 37.90it/s]

Computing ESM2 embeddings: 90%|      | 512/572 [00:13<00:01, 37.24it/s]

Computing ESM2 embeddings: 95%|      | 544/572 [00:14<00:00, 37.91it/s]

```

Computing ESM2 embeddings: 100%| | 572/572 [00:15<00:00, 37.73it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.99it/s]  
 Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 37.15it/s]  
 Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.25it/s]  
 Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.23it/s]  
 Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 38.10it/s]  
 Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.29it/s]  
 Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 38.04it/s]  
 Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 37.51it/s]  
 Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:15, 38.03it/s]  
 Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.78it/s]  
 Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.61it/s]  
 Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.72it/s]  
 Computing ESM2 embeddings: 46%| | 416/896 [00:10<00:12, 38.44it/s]  
 Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:11, 37.55it/s]  
 Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 38.12it/s]  
 Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.39it/s]  
 Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.78it/s]  
 Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.02it/s]  
 Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.87it/s]

```

Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 37.26it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 37.91it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:05, 36.89it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.65it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.07it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.80it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.14it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 37.86it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.51it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.85it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:22, 36.87it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 38.03it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 37.12it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.97it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 37.00it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 37.76it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 37.57it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 38.16it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 37.26it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.92it/s]

```

```

Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 37.33it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 37.95it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:12, 37.13it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:11, 37.81it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.03it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.82it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.14it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.78it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.06it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.44it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 36.64it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.41it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.36it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.94it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.24it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.88it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 37.40it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/589 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|          | 32/589 [00:00<00:14, 37.87it/s]
Computing ESM2 embeddings: 11%|          | 64/589 [00:01<00:14, 35.42it/s]
Computing ESM2 embeddings: 16%|          | 96/589 [00:02<00:13, 37.22it/s]

```

```

Computing ESM2 embeddings: 22%|          | 128/589 [00:03<00:12, 36.63it/s]
Computing ESM2 embeddings: 27%|          | 160/589 [00:04<00:11, 37.66it/s]
Computing ESM2 embeddings: 33%|          | 192/589 [00:05<00:10, 37.03it/s]
Computing ESM2 embeddings: 38%|          | 224/589 [00:05<00:09, 37.87it/s]
Computing ESM2 embeddings: 43%|          | 256/589 [00:06<00:08, 37.41it/s]
Computing ESM2 embeddings: 49%|          | 288/589 [00:07<00:07, 37.73it/s]
Computing ESM2 embeddings: 54%|          | 320/589 [00:08<00:07, 36.80it/s]
Computing ESM2 embeddings: 60%|          | 352/589 [00:09<00:06, 37.62it/s]
Computing ESM2 embeddings: 65%|          | 384/589 [00:10<00:05, 36.90it/s]
Computing ESM2 embeddings: 71%|          | 416/589 [00:11<00:04, 37.66it/s]
Computing ESM2 embeddings: 76%|          | 448/589 [00:12<00:03, 37.31it/s]
Computing ESM2 embeddings: 81%|          | 480/589 [00:12<00:02, 37.92it/s]
Computing ESM2 embeddings: 87%|          | 512/589 [00:13<00:02, 37.15it/s]
Computing ESM2 embeddings: 92%|          | 544/589 [00:14<00:01, 37.80it/s]
Computing ESM2 embeddings: 98%|          | 576/589 [00:15<00:00, 37.00it/s]

Computing ESM2 embeddings: 100%|         | 589/589 [00:15<00:00, 37.24it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 39.14it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:22, 36.89it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 38.09it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 37.20it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:19, 38.06it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:18, 37.19it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 37.92it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:16, 37.67it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:16, 37.92it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 36.53it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 37.37it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 37.31it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:12, 37.98it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:11, 37.44it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 38.04it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 37.83it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 38.29it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 37.57it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 38.14it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 37.42it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 38.03it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:05, 37.37it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 38.00it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:20<00:03, 37.35it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 37.95it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 37.23it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:22<00:00, 37.92it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:23<00:00, 37.56it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 38.12it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:23, 36.01it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.59it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 36.91it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 37.87it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.16it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 37.94it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.23it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.64it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.62it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 37.44it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.43it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 38.06it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 37.29it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.95it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.77it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 38.26it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.43it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 38.07it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.81it/s] |



```

Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 38.33it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 38.00it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 38.49it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.53it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:02, 38.15it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.43it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:22<00:01, 38.07it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 37.42it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.57it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/898 [00:00<00:24, 35.81it/s]
Computing ESM2 embeddings:  7%|      | 64/898 [00:01<00:24, 34.15it/s]
Computing ESM2 embeddings: 11%|      | 96/898 [00:02<00:22, 34.88it/s]
Computing ESM2 embeddings: 14%|      | 128/898 [00:03<00:22, 34.76it/s]
Computing ESM2 embeddings: 18%|      | 160/898 [00:04<00:20, 36.33it/s]
Computing ESM2 embeddings: 21%|      | 192/898 [00:05<00:19, 36.16it/s]
Computing ESM2 embeddings: 25%|      | 224/898 [00:06<00:18, 37.21it/s]
Computing ESM2 embeddings: 29%|      | 256/898 [00:07<00:17, 36.80it/s]
Computing ESM2 embeddings: 32%|      | 288/898 [00:07<00:16, 37.01it/s]
Computing ESM2 embeddings: 36%|      | 320/898 [00:08<00:16, 36.09it/s]
Computing ESM2 embeddings: 39%|      | 352/898 [00:09<00:14, 37.06it/s]

```

```

Computing ESM2 embeddings: 43%|          | 384/898 [00:10<00:14, 36.55it/s]
Computing ESM2 embeddings: 46%|          | 416/898 [00:11<00:12, 37.40it/s]
Computing ESM2 embeddings: 50%|          | 448/898 [00:12<00:12, 36.79it/s]
Computing ESM2 embeddings: 53%|          | 480/898 [00:13<00:11, 37.59it/s]
Computing ESM2 embeddings: 57%|          | 512/898 [00:14<00:10, 36.90it/s]
Computing ESM2 embeddings: 61%|          | 544/898 [00:14<00:09, 37.68it/s]
Computing ESM2 embeddings: 64%|          | 576/898 [00:15<00:08, 36.62it/s]
Computing ESM2 embeddings: 68%|          | 608/898 [00:16<00:07, 37.36it/s]
Computing ESM2 embeddings: 71%|          | 640/898 [00:17<00:07, 36.79it/s]
Computing ESM2 embeddings: 75%|          | 672/898 [00:18<00:06, 37.54it/s]
Computing ESM2 embeddings: 78%|          | 704/898 [00:19<00:05, 37.13it/s]
Computing ESM2 embeddings: 82%|          | 736/898 [00:19<00:04, 37.77it/s]
Computing ESM2 embeddings: 86%|          | 768/898 [00:20<00:03, 37.01it/s]
Computing ESM2 embeddings: 89%|          | 800/898 [00:21<00:02, 37.74it/s]
Computing ESM2 embeddings: 93%|          | 832/898 [00:22<00:01, 37.15it/s]
Computing ESM2 embeddings: 96%|          | 864/898 [00:23<00:00, 37.82it/s]

Computing ESM2 embeddings: 100%|         | 898/898 [00:24<00:00, 36.81it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:25, 34.39it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:24, 34.12it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:22, 36.36it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 35.92it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 37.16it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.52it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 37.44it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:17, 36.90it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.65it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.13it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 37.83it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.24it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.56it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 36.39it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 37.24it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 36.83it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.58it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.07it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.74it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.27it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.86it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 36.98it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.66it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.23it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.73it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.12it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:23<00:01, 37.79it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.28it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.15it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:24, 35.82it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 34.84it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 36.79it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 36.20it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.32it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.61it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:17, 37.49it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 36.92it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.69it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.98it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.69it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.92it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.63it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.96it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.70it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 36.92it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.62it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.94it/s]

```

Computing ESM2 embeddings: 68%|      | 608/896 [00:16<00:07, 37.39it/s]
Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 36.69it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:18<00:06, 37.14it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:19<00:05, 36.35it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.20it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 36.87it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.59it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.07it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.74it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.06it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:24, 36.53it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:24, 34.96it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:22, 36.88it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.40it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.41it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.81it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.66it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 37.12it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.51it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.42it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.31it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.31it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.97it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.38it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.00it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.13it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.79it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.99it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.78it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.58it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.12it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.44it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.91it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.63it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:02, 38.17it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.33it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.95it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.07it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.27it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.33it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:23, 35.03it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 36.95it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 36.56it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.59it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.97it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:17, 37.83it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 37.61it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.88it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 36.71it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.55it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.04it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.78it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.22it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 37.89it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.48it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 38.08it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.34it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.99it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.28it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.91it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 37.20it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.60it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 36.30it/s] |
| Computing ESM2 embeddings: | 89% | 800/896 [00:21<00:02, 37.23it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 36.77it/s]

Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.56it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.29it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/588 [00:00<?, ?it/s]

Computing ESM2 embeddings:  5%|      | 32/588 [00:00<00:14, 38.94it/s]

Computing ESM2 embeddings: 11%|      | 64/588 [00:01<00:14, 37.04it/s]

Computing ESM2 embeddings: 16%|      | 96/588 [00:02<00:12, 38.12it/s]

Computing ESM2 embeddings: 22%|      | 128/588 [00:03<00:12, 37.23it/s]

Computing ESM2 embeddings: 27%|      | 160/588 [00:04<00:11, 37.71it/s]

Computing ESM2 embeddings: 33%|      | 192/588 [00:05<00:10, 36.23it/s]

Computing ESM2 embeddings: 38%|      | 224/588 [00:06<00:09, 37.25it/s]

Computing ESM2 embeddings: 44%|      | 256/588 [00:06<00:09, 36.87it/s]

Computing ESM2 embeddings: 49%|      | 288/588 [00:07<00:07, 37.68it/s]

Computing ESM2 embeddings: 54%|      | 320/588 [00:08<00:07, 37.05it/s]

Computing ESM2 embeddings: 60%|      | 352/588 [00:09<00:06, 37.79it/s]

Computing ESM2 embeddings: 65%|      | 384/588 [00:10<00:05, 37.16it/s]

Computing ESM2 embeddings: 71%|      | 416/588 [00:11<00:04, 37.87it/s]

Computing ESM2 embeddings: 76%|      | 448/588 [00:11<00:03, 37.44it/s]

Computing ESM2 embeddings: 82%|      | 480/588 [00:12<00:02, 38.07it/s]

Computing ESM2 embeddings: 87%|      | 512/588 [00:13<00:02, 37.47it/s]

Computing ESM2 embeddings: 93%|      | 544/588 [00:14<00:01, 38.09it/s]

```



Computing ESM2 embeddings: 98%| | 576/588 [00:15<00:00, 37.28it/s]

Computing ESM2 embeddings: 100%| | 588/588 [00:15<00:00, 37.23it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.10it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.16it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.09it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 36.52it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.57it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.07it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:17, 37.85it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 37.14it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.87it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 37.33it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 38.00it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.27it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.62it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.49it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.35it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.23it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.92it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.23it/s]

```

Computing ESM2 embeddings: 68%|      | 608/896 [00:16<00:07, 37.92it/s]
Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 37.40it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 37.99it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:05, 37.19it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.90it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.26it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.92it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.33it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.95it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.35it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.80it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:02<00:27, 30.66it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:25, 32.46it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:22, 34.92it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:21, 35.20it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.60it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 36.34it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:07<00:17, 37.32it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:08<00:16, 36.90it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 37.49it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.09it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.79it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.58it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.85it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 36.54it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 37.40it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.29it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.96it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.43it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 38.05it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.77it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 38.22it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.34it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.95it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.65it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 38.18it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.43it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 38.01it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.03it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:00<00:24, 35.69it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/897 [00:01<00:23, 35.42it/s]  |
| Computing ESM2 embeddings: | 11% | 96/897 [00:02<00:21, 37.15it/s]  |
| Computing ESM2 embeddings: | 14% | 128/897 [00:03<00:21, 36.43it/s] |
| Computing ESM2 embeddings: | 18% | 160/897 [00:04<00:19, 37.07it/s] |
| Computing ESM2 embeddings: | 21% | 192/897 [00:05<00:19, 36.13it/s] |
| Computing ESM2 embeddings: | 25% | 224/897 [00:06<00:18, 37.14it/s] |
| Computing ESM2 embeddings: | 29% | 256/897 [00:06<00:17, 36.75it/s] |
| Computing ESM2 embeddings: | 32% | 288/897 [00:07<00:16, 37.55it/s] |
| Computing ESM2 embeddings: | 36% | 320/897 [00:08<00:15, 36.93it/s] |
| Computing ESM2 embeddings: | 39% | 352/897 [00:09<00:14, 37.70it/s] |
| Computing ESM2 embeddings: | 43% | 384/897 [00:10<00:13, 37.07it/s] |
| Computing ESM2 embeddings: | 46% | 416/897 [00:11<00:12, 37.76it/s] |
| Computing ESM2 embeddings: | 50% | 448/897 [00:12<00:12, 37.22it/s] |
| Computing ESM2 embeddings: | 54% | 480/897 [00:12<00:11, 37.86it/s] |
| Computing ESM2 embeddings: | 57% | 512/897 [00:13<00:10, 37.24it/s] |
| Computing ESM2 embeddings: | 61% | 544/897 [00:14<00:09, 37.87it/s] |
| Computing ESM2 embeddings: | 64% | 576/897 [00:15<00:08, 37.28it/s] |
| Computing ESM2 embeddings: | 68% | 608/897 [00:16<00:07, 37.55it/s] |
| Computing ESM2 embeddings: | 71% | 640/897 [00:17<00:06, 36.77it/s] |
| Computing ESM2 embeddings: | 75% | 672/897 [00:18<00:05, 37.56it/s] |
| Computing ESM2 embeddings: | 78% | 704/897 [00:18<00:05, 37.02it/s] |
| Computing ESM2 embeddings: | 82% | 736/897 [00:19<00:04, 37.73it/s] |
| Computing ESM2 embeddings: | 86% | 768/897 [00:20<00:03, 37.01it/s] |
| Computing ESM2 embeddings: | 89% | 800/897 [00:21<00:02, 37.73it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/897 [00:22<00:01, 37.03it/s]

Computing ESM2 embeddings: 96%|      | 864/897 [00:23<00:00, 37.68it/s]

Computing ESM2 embeddings: 100%|     | 897/897 [00:24<00:00, 37.09it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 38.34it/s]

Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 35.71it/s]

Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.36it/s]

Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.76it/s]

Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 37.72it/s]

Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 37.10it/s]

Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:18, 37.92it/s]

Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 37.38it/s]

Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.66it/s]

Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.32it/s]

Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.24it/s]

Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 36.71it/s]

Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.53it/s]

Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 36.98it/s]

Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.69it/s]

Computing ESM2 embeddings: 56%|      | 512/914 [00:13<00:10, 37.13it/s]

Computing ESM2 embeddings: 60%|      | 544/914 [00:14<00:09, 37.85it/s]

```

```

Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.26it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.95it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.19it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 37.85it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 37.15it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.53it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:04, 36.48it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.30it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 36.88it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.58it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.10it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.17it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.08it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:25, 32.28it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:22, 35.04it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:21, 35.14it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:20, 36.45it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:19, 36.08it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:06<00:18, 37.12it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:07<00:17, 36.56it/s]

```

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.42it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.81it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.52it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.79it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.54it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.11it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.78it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.14it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.78it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.16it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.78it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.25it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.86it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 37.09it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.78it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.02it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.64it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.99it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.65it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.91it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/911 [00:00<?, ?it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/911 [00:00<00:22, 39.07it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/911 [00:02<00:30, 28.11it/s]  |
| Computing ESM2 embeddings: | 11% | 96/911 [00:02<00:25, 32.53it/s]  |
| Computing ESM2 embeddings: | 14% | 128/911 [00:03<00:23, 33.73it/s] |
| Computing ESM2 embeddings: | 18% | 160/911 [00:04<00:21, 35.65it/s] |
| Computing ESM2 embeddings: | 21% | 192/911 [00:05<00:20, 35.73it/s] |
| Computing ESM2 embeddings: | 25% | 224/911 [00:06<00:18, 36.92it/s] |
| Computing ESM2 embeddings: | 28% | 256/911 [00:07<00:18, 36.39it/s] |
| Computing ESM2 embeddings: | 32% | 288/911 [00:08<00:16, 37.35it/s] |
| Computing ESM2 embeddings: | 35% | 320/911 [00:08<00:16, 36.90it/s] |
| Computing ESM2 embeddings: | 39% | 352/911 [00:09<00:14, 37.70it/s] |
| Computing ESM2 embeddings: | 42% | 384/911 [00:10<00:14, 37.09it/s] |
| Computing ESM2 embeddings: | 46% | 416/911 [00:11<00:13, 37.52it/s] |
| Computing ESM2 embeddings: | 49% | 448/911 [00:12<00:12, 36.38it/s] |
| Computing ESM2 embeddings: | 53% | 480/911 [00:13<00:11, 37.28it/s] |
| Computing ESM2 embeddings: | 56% | 512/911 [00:14<00:10, 36.86it/s] |
| Computing ESM2 embeddings: | 60% | 544/911 [00:14<00:09, 37.66it/s] |
| Computing ESM2 embeddings: | 63% | 576/911 [00:15<00:09, 37.13it/s] |
| Computing ESM2 embeddings: | 67% | 608/911 [00:16<00:08, 37.85it/s] |
| Computing ESM2 embeddings: | 70% | 640/911 [00:17<00:07, 37.22it/s] |
| Computing ESM2 embeddings: | 74% | 672/911 [00:18<00:06, 37.85it/s] |
| Computing ESM2 embeddings: | 77% | 704/911 [00:19<00:05, 37.12it/s] |
| Computing ESM2 embeddings: | 81% | 736/911 [00:20<00:04, 37.86it/s] |
| Computing ESM2 embeddings: | 84% | 768/911 [00:20<00:03, 37.25it/s] |



```

Computing ESM2 embeddings: 88%|      | 800/911 [00:21<00:02, 38.01it/s]
Computing ESM2 embeddings: 91%|      | 832/911 [00:22<00:02, 37.37it/s]
Computing ESM2 embeddings: 95%|      | 864/911 [00:23<00:01, 37.98it/s]
Computing ESM2 embeddings: 98%|      | 896/911 [00:24<00:00, 37.35it/s]
Computing ESM2 embeddings: 100%|     | 911/911 [00:24<00:00, 36.78it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/898 [00:00<00:22, 39.11it/s]
Computing ESM2 embeddings:  7%|      | 64/898 [00:01<00:22, 37.09it/s]
Computing ESM2 embeddings: 11%|      | 96/898 [00:02<00:21, 38.18it/s]
Computing ESM2 embeddings: 14%|      | 128/898 [00:03<00:20, 37.05it/s]
Computing ESM2 embeddings: 18%|      | 160/898 [00:04<00:19, 37.42it/s]
Computing ESM2 embeddings: 21%|      | 192/898 [00:05<00:19, 36.12it/s]
Computing ESM2 embeddings: 25%|      | 224/898 [00:06<00:18, 37.14it/s]
Computing ESM2 embeddings: 29%|      | 256/898 [00:06<00:17, 36.77it/s]
Computing ESM2 embeddings: 32%|      | 288/898 [00:07<00:16, 37.55it/s]
Computing ESM2 embeddings: 36%|      | 320/898 [00:08<00:15, 37.06it/s]
Computing ESM2 embeddings: 39%|      | 352/898 [00:09<00:14, 37.67it/s]
Computing ESM2 embeddings: 43%|      | 384/898 [00:10<00:13, 37.07it/s]
Computing ESM2 embeddings: 46%|      | 416/898 [00:11<00:12, 37.79it/s]
Computing ESM2 embeddings: 50%|      | 448/898 [00:12<00:12, 36.97it/s]
Computing ESM2 embeddings: 53%|      | 480/898 [00:12<00:11, 37.71it/s]

```

Computing ESM2 embeddings: 57%| | 512/898 [00:13<00:10, 37.11it/s]

Computing ESM2 embeddings: 61%| | 544/898 [00:14<00:09, 37.77it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:15<00:08, 37.09it/s]

Computing ESM2 embeddings: 68%| | 608/898 [00:16<00:07, 37.75it/s]

Computing ESM2 embeddings: 71%| | 640/898 [00:17<00:06, 37.02it/s]

Computing ESM2 embeddings: 75%| | 672/898 [00:17<00:05, 37.68it/s]

Computing ESM2 embeddings: 78%| | 704/898 [00:18<00:05, 37.11it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:19<00:04, 37.48it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:20<00:03, 36.44it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:21<00:02, 37.29it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:22<00:01, 36.74it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:23<00:00, 37.47it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:24<00:00, 37.12it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:14, 37.93it/s]

Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:13, 36.71it/s]

Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:12, 37.95it/s]

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:11, 37.15it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:10, 37.61it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:05<00:10, 36.36it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:06<00:09, 37.36it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 36.89it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 37.69it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 37.11it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 37.83it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:05, 37.05it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 37.75it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:12<00:03, 37.11it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 37.50it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 36.49it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 37.38it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 37.15it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.28it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:02<00:27, 31.11it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 34.58it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 34.94it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.44it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.21it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.23it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.42it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.32it/s]

```

Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:16, 36.84it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 37.64it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 37.05it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.74it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:12<00:12, 36.95it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:13<00:11, 37.63it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 37.02it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 37.75it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.12it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.77it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.37it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 38.00it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 37.10it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.80it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.14it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.82it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.17it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.81it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.10it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 36.92it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.65it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:25, 32.26it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:22, 35.23it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 35.89it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:20, 36.56it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 35.50it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 36.74it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:07<00:17, 36.46it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.25it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 36.93it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.62it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.00it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.72it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.15it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:13<00:10, 37.84it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.24it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.92it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.17it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.54it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:07, 36.38it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:18<00:06, 37.28it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:19<00:05, 36.67it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.19it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 36.08it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 36.94it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 36.33it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.16it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.69it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/915 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/915 [00:00<00:22, 38.63it/s]
Computing ESM2 embeddings:  7%|      | 64/915 [00:01<00:25, 33.69it/s]
Computing ESM2 embeddings: 10%|      | 96/915 [00:02<00:22, 36.07it/s]
Computing ESM2 embeddings: 14%|      | 128/915 [00:03<00:21, 35.81it/s]
Computing ESM2 embeddings: 17%|      | 160/915 [00:04<00:20, 37.06it/s]
Computing ESM2 embeddings: 21%|      | 192/915 [00:05<00:19, 36.44it/s]
Computing ESM2 embeddings: 24%|      | 224/915 [00:06<00:18, 37.29it/s]
Computing ESM2 embeddings: 28%|      | 256/915 [00:07<00:17, 36.76it/s]
Computing ESM2 embeddings: 31%|      | 288/915 [00:07<00:16, 36.99it/s]
Computing ESM2 embeddings: 35%|      | 320/915 [00:08<00:16, 36.05it/s]
Computing ESM2 embeddings: 38%|      | 352/915 [00:09<00:15, 37.05it/s]
Computing ESM2 embeddings: 42%|      | 384/915 [00:10<00:14, 36.58it/s]
Computing ESM2 embeddings: 45%|      | 416/915 [00:11<00:13, 37.38it/s]
Computing ESM2 embeddings: 49%|      | 448/915 [00:12<00:12, 36.71it/s]
Computing ESM2 embeddings: 52%|      | 480/915 [00:13<00:11, 37.51it/s]
Computing ESM2 embeddings: 56%|      | 512/915 [00:13<00:10, 37.01it/s]

```

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 37.64it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 37.01it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.75it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 37.26it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:18<00:06, 37.88it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:19<00:05, 37.05it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 37.24it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:04, 35.94it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 36.91it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 36.30it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:23<00:01, 37.06it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:24<00:00, 36.54it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 36.83it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:00<00:26, 33.25it/s]

Computing ESM2 embeddings: 7%| | 64/897 [00:01<00:24, 33.58it/s]

Computing ESM2 embeddings: 11%| | 96/897 [00:02<00:22, 35.70it/s]

Computing ESM2 embeddings: 14%| | 128/897 [00:03<00:22, 34.90it/s]

Computing ESM2 embeddings: 18%| | 160/897 [00:04<00:20, 36.42it/s]

Computing ESM2 embeddings: 21%| | 192/897 [00:05<00:19, 36.19it/s]

Computing ESM2 embeddings: 25%| | 224/897 [00:06<00:18, 37.17it/s]

```

Computing ESM2 embeddings: 29%|          | 256/897 [00:07<00:17, 36.65it/s]
Computing ESM2 embeddings: 32%|          | 288/897 [00:07<00:16, 37.52it/s]
Computing ESM2 embeddings: 36%|          | 320/897 [00:08<00:15, 36.67it/s]
Computing ESM2 embeddings: 39%|          | 352/897 [00:09<00:14, 37.46it/s]
Computing ESM2 embeddings: 43%|          | 384/897 [00:10<00:13, 37.11it/s]
Computing ESM2 embeddings: 46%|          | 416/897 [00:11<00:12, 37.38it/s]
Computing ESM2 embeddings: 50%|          | 448/897 [00:12<00:12, 36.35it/s]
Computing ESM2 embeddings: 54%|          | 480/897 [00:13<00:11, 37.20it/s]
Computing ESM2 embeddings: 57%|          | 512/897 [00:14<00:10, 36.64it/s]
Computing ESM2 embeddings: 61%|          | 544/897 [00:14<00:09, 37.41it/s]
Computing ESM2 embeddings: 64%|          | 576/897 [00:15<00:08, 36.85it/s]
Computing ESM2 embeddings: 68%|          | 608/897 [00:16<00:07, 37.35it/s]
Computing ESM2 embeddings: 71%|          | 640/897 [00:17<00:07, 36.24it/s]
Computing ESM2 embeddings: 75%|          | 672/897 [00:18<00:06, 37.09it/s]
Computing ESM2 embeddings: 78%|          | 704/897 [00:19<00:05, 36.50it/s]
Computing ESM2 embeddings: 82%|          | 736/897 [00:20<00:04, 37.26it/s]
Computing ESM2 embeddings: 86%|          | 768/897 [00:20<00:03, 36.75it/s]
Computing ESM2 embeddings: 89%|          | 800/897 [00:21<00:02, 37.51it/s]
Computing ESM2 embeddings: 93%|          | 832/897 [00:22<00:01, 36.86it/s]
Computing ESM2 embeddings: 96%|          | 864/897 [00:23<00:00, 37.56it/s]

```

```

Computing ESM2 embeddings: 100%|         | 897/897 [00:24<00:00, 36.68it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```



|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.90it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:24, 35.18it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:22, 37.04it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 36.58it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 37.60it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.97it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.72it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 36.89it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.66it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.95it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.69it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 36.95it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.72it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 36.98it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.68it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 36.90it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.62it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 36.93it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.64it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.00it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 37.40it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 36.38it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:19<00:04, 37.25it/s] |

```

Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 36.81it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.55it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.87it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.61it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.93it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.08it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/895 [00:00<00:22, 37.86it/s]
Computing ESM2 embeddings:  7%|      | 64/895 [00:01<00:23, 35.61it/s]
Computing ESM2 embeddings: 11%|      | 96/895 [00:02<00:21, 37.33it/s]
Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:20, 36.74it/s]
Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:19, 37.72it/s]
Computing ESM2 embeddings: 21%|      | 192/895 [00:05<00:18, 37.03it/s]
Computing ESM2 embeddings: 25%|      | 224/895 [00:05<00:17, 37.93it/s]
Computing ESM2 embeddings: 29%|      | 256/895 [00:06<00:17, 37.19it/s]
Computing ESM2 embeddings: 32%|      | 288/895 [00:07<00:16, 37.61it/s]
Computing ESM2 embeddings: 36%|      | 320/895 [00:08<00:15, 36.49it/s]
Computing ESM2 embeddings: 39%|      | 352/895 [00:09<00:14, 37.38it/s]
Computing ESM2 embeddings: 43%|      | 384/895 [00:10<00:13, 37.00it/s]
Computing ESM2 embeddings: 46%|      | 416/895 [00:11<00:12, 37.77it/s]
Computing ESM2 embeddings: 50%|      | 448/895 [00:12<00:12, 37.06it/s]

```

Computing ESM2 embeddings: 54%| | 480/895 [00:12<00:10, 37.79it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 37.27it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.89it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.18it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.87it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 37.30it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:17<00:05, 38.04it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:05, 37.29it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.84it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 37.09it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.72it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.02it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.48it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 37.17it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.14it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.30it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.36it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 37.28it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.20it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.23it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:18, 37.96it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:17, 37.36it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 37.75it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:16, 36.87it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 37.78it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:14, 37.19it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:11<00:13, 37.88it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:12, 37.40it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.98it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 37.33it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 37.96it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:15<00:09, 37.21it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:16<00:08, 37.91it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:17<00:07, 37.21it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 37.96it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 37.30it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:19<00:04, 37.97it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:20<00:03, 37.32it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:21<00:03, 37.86it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:22<00:02, 37.14it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:22<00:01, 37.92it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:23<00:00, 37.15it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:24<00:00, 37.59it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:13, 39.36it/s]  
Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:13, 37.45it/s]  
Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:12, 38.34it/s]  
Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:11, 37.37it/s]  
Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:10, 38.04it/s]  
Computing ESM2 embeddings: 34%| | 192/571 [00:05<00:10, 37.39it/s]  
Computing ESM2 embeddings: 39%| | 224/571 [00:05<00:09, 38.00it/s]  
Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 37.57it/s]  
Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 37.84it/s]  
Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 36.82it/s]  
Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 37.50it/s]  
Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:05, 36.92it/s]  
Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 37.65it/s]  
Computing ESM2 embeddings: 78%| | 448/571 [00:11<00:03, 37.13it/s]  
Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 37.77it/s]  
Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 37.17it/s]  
Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 37.84it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 37.47it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 38.29it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:24, 35.41it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.25it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 36.49it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 37.31it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:20, 36.07it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.19it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 36.73it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.66it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.97it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.71it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.04it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.76it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 37.28it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.90it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.34it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.95it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.30it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.56it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 36.35it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 37.32it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 36.83it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:19<00:04, 37.56it/s] |

```

Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 36.99it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.65it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.16it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.74it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.08it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.21it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:21, 39.32it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:22, 37.14it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 38.21it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 37.19it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 37.90it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:18, 37.23it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 37.86it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:17, 37.30it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.72it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 36.72it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 37.61it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 37.06it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 37.85it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:12, 37.33it/s]

```

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Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 37.95it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.31it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 38.06it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.35it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.99it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.40it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 38.01it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 37.35it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.95it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.41it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.97it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.37it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 37.95it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 37.46it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/916 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|          | 32/916 [00:00<00:23, 37.87it/s]
Computing ESM2 embeddings:  7%|          | 64/916 [00:01<00:23, 35.73it/s]
Computing ESM2 embeddings: 10%|          | 96/916 [00:02<00:21, 37.53it/s]
Computing ESM2 embeddings: 14%|          | 128/916 [00:03<00:21, 36.73it/s]
Computing ESM2 embeddings: 17%|          | 160/916 [00:04<00:20, 37.67it/s]
Computing ESM2 embeddings: 21%|          | 192/916 [00:05<00:19, 36.87it/s]

```



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 24%  | 224/916 [00:06<00:18, 37.64it/s] |
| Computing ESM2 embeddings: 28%  | 256/916 [00:06<00:17, 37.02it/s] |
| Computing ESM2 embeddings: 31%  | 288/916 [00:07<00:16, 37.49it/s] |
| Computing ESM2 embeddings: 35%  | 320/916 [00:08<00:16, 36.33it/s] |
| Computing ESM2 embeddings: 38%  | 352/916 [00:09<00:15, 37.18it/s] |
| Computing ESM2 embeddings: 42%  | 384/916 [00:10<00:14, 36.82it/s] |
| Computing ESM2 embeddings: 45%  | 416/916 [00:11<00:13, 37.26it/s] |
| Computing ESM2 embeddings: 49%  | 448/916 [00:12<00:12, 36.30it/s] |
| Computing ESM2 embeddings: 52%  | 480/916 [00:12<00:11, 37.20it/s] |
| Computing ESM2 embeddings: 56%  | 512/916 [00:13<00:11, 36.71it/s] |
| Computing ESM2 embeddings: 59%  | 544/916 [00:14<00:09, 37.54it/s] |
| Computing ESM2 embeddings: 63%  | 576/916 [00:15<00:09, 36.96it/s] |
| Computing ESM2 embeddings: 66%  | 608/916 [00:16<00:08, 37.55it/s] |
| Computing ESM2 embeddings: 70%  | 640/916 [00:17<00:07, 36.38it/s] |
| Computing ESM2 embeddings: 73%  | 672/916 [00:18<00:06, 37.08it/s] |
| Computing ESM2 embeddings: 77%  | 704/916 [00:19<00:05, 36.84it/s] |
| Computing ESM2 embeddings: 80%  | 736/916 [00:19<00:04, 37.64it/s] |
| Computing ESM2 embeddings: 84%  | 768/916 [00:20<00:03, 37.23it/s] |
| Computing ESM2 embeddings: 87%  | 800/916 [00:21<00:03, 37.87it/s] |
| Computing ESM2 embeddings: 91%  | 832/916 [00:22<00:02, 37.24it/s] |
| Computing ESM2 embeddings: 94%  | 864/916 [00:23<00:01, 37.89it/s] |
| Computing ESM2 embeddings: 98%  | 896/916 [00:24<00:00, 37.21it/s] |
| Computing ESM2 embeddings: 100% | 916/916 [00:24<00:00, 37.06it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.64it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:24, 33.56it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:22, 36.17it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 36.03it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.16it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.75it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:17, 37.59it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 37.13it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.78it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.26it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 38.00it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.49it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.83it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 36.97it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.80it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.23it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.89it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.24it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.98it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.30it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 38.02it/s] |

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.24it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.99it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.20it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.83it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.22it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.94it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.24it/s]

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Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:27, 32.49it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:25, 33.37it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:22, 36.08it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:21, 35.85it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:20, 36.86it/s]

Computing ESM2 embeddings: 21%| | 192/913 [00:05<00:20, 35.88it/s]

Computing ESM2 embeddings: 25%| | 224/913 [00:06<00:18, 37.09it/s]

Computing ESM2 embeddings: 28%| | 256/913 [00:07<00:17, 36.65it/s]

Computing ESM2 embeddings: 32%| | 288/913 [00:07<00:16, 37.59it/s]

Computing ESM2 embeddings: 35%| | 320/913 [00:08<00:16, 36.96it/s]

Computing ESM2 embeddings: 39%| | 352/913 [00:09<00:14, 37.81it/s]

Computing ESM2 embeddings: 42%| | 384/913 [00:10<00:14, 37.31it/s]

Computing ESM2 embeddings: 46%| | 416/913 [00:11<00:13, 38.12it/s]

Computing ESM2 embeddings: 49%| | 448/913 [00:12<00:12, 37.28it/s]

Computing ESM2 embeddings: 53%| | 480/913 [00:12<00:11, 37.92it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 37.21it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 37.84it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:15<00:09, 37.11it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:16<00:08, 37.60it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:17<00:07, 36.44it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:18<00:06, 37.07it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:19<00:05, 36.16it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:19<00:04, 37.06it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:20<00:03, 36.60it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:21<00:03, 37.39it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 36.91it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:23<00:01, 37.73it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:24<00:00, 36.87it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:24<00:00, 36.82it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:00<00:23, 37.33it/s]

Computing ESM2 embeddings: 7%| | 64/897 [00:01<00:23, 35.42it/s]

Computing ESM2 embeddings: 11%| | 96/897 [00:02<00:21, 37.23it/s]

Computing ESM2 embeddings: 14%| | 128/897 [00:03<00:20, 36.97it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/897 [00:04<00:19, 37.83it/s] |
| Computing ESM2 embeddings: 21%  | 192/897 [00:05<00:18, 37.16it/s] |
| Computing ESM2 embeddings: 25%  | 224/897 [00:05<00:17, 38.02it/s] |
| Computing ESM2 embeddings: 29%  | 256/897 [00:06<00:17, 37.30it/s] |
| Computing ESM2 embeddings: 32%  | 288/897 [00:07<00:16, 37.71it/s] |
| Computing ESM2 embeddings: 36%  | 320/897 [00:08<00:15, 36.60it/s] |
| Computing ESM2 embeddings: 39%  | 352/897 [00:09<00:14, 37.45it/s] |
| Computing ESM2 embeddings: 43%  | 384/897 [00:10<00:13, 37.00it/s] |
| Computing ESM2 embeddings: 46%  | 416/897 [00:11<00:12, 37.84it/s] |
| Computing ESM2 embeddings: 50%  | 448/897 [00:12<00:12, 37.14it/s] |
| Computing ESM2 embeddings: 54%  | 480/897 [00:12<00:11, 37.84it/s] |
| Computing ESM2 embeddings: 57%  | 512/897 [00:13<00:10, 37.23it/s] |
| Computing ESM2 embeddings: 61%  | 544/897 [00:14<00:09, 37.85it/s] |
| Computing ESM2 embeddings: 64%  | 576/897 [00:15<00:08, 37.09it/s] |
| Computing ESM2 embeddings: 68%  | 608/897 [00:16<00:07, 37.82it/s] |
| Computing ESM2 embeddings: 71%  | 640/897 [00:17<00:06, 37.22it/s] |
| Computing ESM2 embeddings: 75%  | 672/897 [00:17<00:05, 37.94it/s] |
| Computing ESM2 embeddings: 78%  | 704/897 [00:18<00:05, 37.25it/s] |
| Computing ESM2 embeddings: 82%  | 736/897 [00:19<00:04, 37.97it/s] |
| Computing ESM2 embeddings: 86%  | 768/897 [00:20<00:03, 37.28it/s] |
| Computing ESM2 embeddings: 89%  | 800/897 [00:21<00:02, 38.01it/s] |
| Computing ESM2 embeddings: 93%  | 832/897 [00:22<00:01, 37.23it/s] |
| Computing ESM2 embeddings: 96%  | 864/897 [00:23<00:00, 37.90it/s] |
| Computing ESM2 embeddings: 100% | 897/897 [00:24<00:00, 37.32it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 38.23it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:23, 35.77it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.39it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 37.36it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.14it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.30it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 38.01it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.28it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.91it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 37.39it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.98it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.41it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.71it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 36.79it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.59it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.07it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.76it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.18it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.84it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.31it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 37.97it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 37.19it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 37.87it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.33it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.96it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.12it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.80it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 37.23it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.48it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/571 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/571 [00:00<00:14, 38.06it/s]
Computing ESM2 embeddings: 11%|      | 64/571 [00:01<00:14, 36.02it/s]
Computing ESM2 embeddings: 17%|      | 96/571 [00:02<00:12, 37.54it/s]
Computing ESM2 embeddings: 22%|      | 128/571 [00:03<00:12, 36.26it/s]
Computing ESM2 embeddings: 28%|      | 160/571 [00:04<00:11, 37.33it/s]
Computing ESM2 embeddings: 34%|      | 192/571 [00:05<00:10, 37.13it/s]
Computing ESM2 embeddings: 39%|      | 224/571 [00:05<00:09, 37.85it/s]
Computing ESM2 embeddings: 45%|      | 256/571 [00:06<00:08, 37.02it/s]
Computing ESM2 embeddings: 50%|      | 288/571 [00:07<00:07, 37.69it/s]
Computing ESM2 embeddings: 56%|      | 320/571 [00:08<00:06, 37.07it/s]
Computing ESM2 embeddings: 62%|      | 352/571 [00:09<00:05, 37.72it/s]

```

Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:05, 37.22it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 37.48it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:12<00:03, 36.51it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 37.28it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 36.85it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 37.53it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 37.07it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 35.98it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 35.39it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 37.17it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 37.27it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 38.04it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.33it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 38.11it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 37.32it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.08it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 37.37it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.12it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.28it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 38.02it/s]



```

Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:12, 37.31it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.04it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 37.32it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 38.03it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.24it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.99it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.30it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 37.92it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 37.19it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.85it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.19it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.84it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.25it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 37.84it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 37.32it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.41it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.


Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:23, 37.30it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:25, 33.03it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:22, 35.66it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:21, 35.94it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:19, 36.83it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:19, 35.97it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:06<00:18, 37.04it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:07<00:17, 36.83it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:16, 37.62it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 37.09it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 37.72it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 37.20it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:12, 37.48it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:12<00:12, 36.38it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:13<00:11, 37.23it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 36.82it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 37.54it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 37.09it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 37.76it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 37.07it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:18<00:05, 37.75it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:19<00:05, 37.14it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 37.81it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:20<00:03, 37.08it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 37.86it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 37.13it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:23<00:00, 37.90it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:24<00:00, 36.97it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.92it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:22, 36.99it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 38.32it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:20, 37.45it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.30it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.30it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 38.12it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.15it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.85it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 37.28it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.92it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.25it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.88it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 37.34it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.65it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 36.56it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.38it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 36.96it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.51it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 36.32it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 37.31it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 36.77it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 37.52it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 36.89it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.75it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.04it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.55it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.42it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.20it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/896 [00:00<00:23, 37.07it/s]
Computing ESM2 embeddings: 7%|      | 64/896 [00:01<00:24, 33.52it/s]
Computing ESM2 embeddings: 11%|     | 96/896 [00:02<00:22, 35.99it/s]
Computing ESM2 embeddings: 14%|     | 128/896 [00:03<00:21, 36.22it/s]
Computing ESM2 embeddings: 18%|     | 160/896 [00:04<00:19, 37.15it/s]
Computing ESM2 embeddings: 21%|     | 192/896 [00:05<00:19, 36.14it/s]
Computing ESM2 embeddings: 25%|     | 224/896 [00:06<00:18, 37.16it/s]
Computing ESM2 embeddings: 29%|     | 256/896 [00:07<00:17, 36.59it/s]
Computing ESM2 embeddings: 32%|     | 288/896 [00:07<00:16, 37.50it/s]
Computing ESM2 embeddings: 36%|     | 320/896 [00:08<00:15, 36.89it/s]
Computing ESM2 embeddings: 39%|     | 352/896 [00:09<00:14, 37.43it/s]

```

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:14, 36.05it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.10it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.67it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.56it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.03it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.79it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.22it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.94it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.25it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.98it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 37.33it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.77it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.70it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.58it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.09it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.90it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.98it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:27, 31.54it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:25, 32.75it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 35.49it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:22, 35.45it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 36.80it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.30it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 37.28it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:17, 36.59it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.45it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.93it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 37.63it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.11it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.73it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.21it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 37.97it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.30it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 38.02it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.23it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.96it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.29it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.65it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 36.54it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.40it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 36.84it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.58it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 36.94it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:23<00:01, 37.64it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.28it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.95it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.95it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:24, 34.20it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 36.40it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 36.24it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 36.90it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 35.90it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 36.95it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:17, 36.68it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.47it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.81it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.57it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.02it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.71it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.20it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.81it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.23it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.82it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.12it/s]

```

Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.75it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.00it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:18<00:05, 37.68it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 37.14it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.72it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.22it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.77it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.19it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.81it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 37.08it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:22, 38.44it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:23, 35.84it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.51it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 37.22it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 37.92it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 36.61it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:06<00:18, 37.47it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 37.04it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 37.78it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 37.16it/s]

```



Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.96it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.24it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.97it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 37.25it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.99it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.21it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.98it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.25it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.99it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.35it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.07it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.31it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.03it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.31it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:02, 38.02it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.37it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.78it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 36.65it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.28it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/572 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/572 [00:00<00:14, 36.54it/s]

Computing ESM2 embeddings: 11%| | 64/572 [00:01<00:14, 34.62it/s]

Computing ESM2 embeddings: 17%| | 96/572 [00:02<00:12, 36.68it/s]

Computing ESM2 embeddings: 22%| | 128/572 [00:03<00:12, 36.21it/s]

Computing ESM2 embeddings: 28%| | 160/572 [00:04<00:11, 37.26it/s]

Computing ESM2 embeddings: 34%| | 192/572 [00:05<00:10, 36.89it/s]

Computing ESM2 embeddings: 39%| | 224/572 [00:06<00:09, 37.63it/s]

Computing ESM2 embeddings: 45%| | 256/572 [00:06<00:08, 36.85it/s]

Computing ESM2 embeddings: 50%| | 288/572 [00:07<00:07, 37.31it/s]

Computing ESM2 embeddings: 56%| | 320/572 [00:08<00:06, 36.27it/s]

Computing ESM2 embeddings: 62%| | 352/572 [00:09<00:05, 37.18it/s]

Computing ESM2 embeddings: 67%| | 384/572 [00:10<00:05, 36.90it/s]

Computing ESM2 embeddings: 73%| | 416/572 [00:11<00:04, 37.61it/s]

Computing ESM2 embeddings: 78%| | 448/572 [00:12<00:03, 37.14it/s]

Computing ESM2 embeddings: 84%| | 480/572 [00:12<00:02, 37.75it/s]

Computing ESM2 embeddings: 90%| | 512/572 [00:13<00:01, 37.24it/s]

Computing ESM2 embeddings: 95%| | 544/572 [00:14<00:00, 37.88it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:15<00:00, 36.88it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.67it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.99it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.58it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 36.71it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.67it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:18, 37.09it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 37.83it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:17, 37.21it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.70it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 36.43it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.36it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.23it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.90it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:11, 37.41it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 38.01it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.60it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 38.39it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.58it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.98it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:06, 36.84it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 37.70it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 37.09it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.89it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.21it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.96it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.24it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 37.74it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.30it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.01it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 35.62it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.26it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.57it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.35it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.27it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.37it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 36.92it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.80it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 37.17it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 38.02it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.37it/s]  
 Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 38.09it/s]  
 Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.51it/s]  
 Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.19it/s]  
 Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.39it/s]  
 Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.99it/s]  
 Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.36it/s]  
 Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.94it/s]

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Computing ESM2 embeddings: 70%|      | 640/914 [00:17<00:07, 37.32it/s]
Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 37.89it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 37.63it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 38.25it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.42it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.80it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.55it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.48it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 37.00it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.40it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/897 [00:00<00:23, 37.13it/s]
Computing ESM2 embeddings: 7%|      | 64/897 [00:02<00:27, 30.11it/s]
Computing ESM2 embeddings: 11%|     | 96/897 [00:02<00:24, 32.82it/s]
Computing ESM2 embeddings: 14%|     | 128/897 [00:03<00:22, 33.67it/s]
Computing ESM2 embeddings: 18%|     | 160/897 [00:04<00:20, 35.23it/s]
Computing ESM2 embeddings: 21%|     | 192/897 [00:05<00:20, 34.91it/s]
Computing ESM2 embeddings: 25%|     | 224/897 [00:06<00:18, 36.25it/s]
Computing ESM2 embeddings: 29%|     | 256/897 [00:07<00:17, 36.16it/s]
Computing ESM2 embeddings: 32%|     | 288/897 [00:08<00:16, 37.13it/s]
Computing ESM2 embeddings: 36%|     | 320/897 [00:09<00:15, 36.59it/s]

```

Computing ESM2 embeddings: 39%| | 352/897 [00:09<00:14, 37.37it/s]

Computing ESM2 embeddings: 43%| | 384/897 [00:10<00:13, 36.88it/s]

Computing ESM2 embeddings: 46%| | 416/897 [00:11<00:12, 37.32it/s]

Computing ESM2 embeddings: 50%| | 448/897 [00:12<00:12, 36.23it/s]

Computing ESM2 embeddings: 54%| | 480/897 [00:13<00:11, 36.85it/s]

Computing ESM2 embeddings: 57%| | 512/897 [00:14<00:10, 36.06it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:15<00:09, 36.99it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:15<00:08, 36.69it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:16<00:07, 37.44it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:17<00:06, 37.05it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:18<00:05, 37.68it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:19<00:05, 37.01it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:20<00:04, 37.67it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:21<00:03, 37.17it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:21<00:02, 37.65it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:22<00:01, 36.48it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:23<00:00, 37.43it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:24<00:00, 36.41it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.38it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.48it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 37.81it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 36.96it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 37.84it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.15it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.55it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:18, 36.40it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.33it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.91it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 37.66it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.14it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.79it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.21it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.84it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.29it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.92it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.16it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.81it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.23it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.88it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.27it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.87it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.28it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.77it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.08it/s] |

```

Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.80it/s]

Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 37.08it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.28it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      |  32/914 [00:00<00:23, 36.88it/s]

Computing ESM2 embeddings:  7%|      |  64/914 [00:01<00:26, 32.54it/s]

Computing ESM2 embeddings: 11%|      |  96/914 [00:02<00:24, 33.51it/s]

Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 35.83it/s]

Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:21, 35.84it/s]

Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 37.13it/s]

Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 36.67it/s]

Computing ESM2 embeddings: 28%|      | 256/914 [00:07<00:17, 37.46it/s]

Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 36.88it/s]

Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 37.76it/s]

Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.20it/s]

Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:13, 37.97it/s]

Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.15it/s]

Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 37.56it/s]

Computing ESM2 embeddings: 53%|      | 480/914 [00:13<00:11, 36.30it/s]

Computing ESM2 embeddings: 56%|      | 512/914 [00:13<00:10, 37.10it/s]

Computing ESM2 embeddings: 60%|      | 544/914 [00:14<00:10, 36.51it/s]

```



```

Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.34it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 36.70it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.59it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 36.89it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 37.71it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.13it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.91it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.20it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.90it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.05it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.83it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 36.91it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:24, 36.65it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:23, 36.12it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.82it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 37.09it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 37.98it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 37.04it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:18, 37.78it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 37.23it/s]

```

```

Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 37.87it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 37.20it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 37.87it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 37.38it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 38.01it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:12, 37.47it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 38.08it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 37.42it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 38.02it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.36it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.98it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.14it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 37.79it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 37.23it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.87it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.33it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.86it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.31it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:22<00:01, 37.91it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 37.38it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.49it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/849 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/849 [00:00<00:24, 32.77it/s]  |
| Computing ESM2 embeddings: | 8%  | 64/849 [00:01<00:23, 34.01it/s]  |
| Computing ESM2 embeddings: | 11% | 96/849 [00:02<00:20, 36.31it/s]  |
| Computing ESM2 embeddings: | 15% | 128/849 [00:03<00:19, 36.23it/s] |
| Computing ESM2 embeddings: | 19% | 160/849 [00:04<00:18, 36.94it/s] |
| Computing ESM2 embeddings: | 23% | 192/849 [00:05<00:18, 36.20it/s] |
| Computing ESM2 embeddings: | 26% | 224/849 [00:06<00:16, 37.25it/s] |
| Computing ESM2 embeddings: | 30% | 256/849 [00:07<00:16, 36.77it/s] |
| Computing ESM2 embeddings: | 34% | 288/849 [00:07<00:14, 37.54it/s] |
| Computing ESM2 embeddings: | 38% | 320/849 [00:08<00:14, 37.34it/s] |
| Computing ESM2 embeddings: | 41% | 352/849 [00:09<00:13, 38.11it/s] |
| Computing ESM2 embeddings: | 45% | 384/849 [00:10<00:12, 37.29it/s] |
| Computing ESM2 embeddings: | 49% | 416/849 [00:11<00:11, 37.52it/s] |
| Computing ESM2 embeddings: | 53% | 448/849 [00:12<00:11, 36.36it/s] |
| Computing ESM2 embeddings: | 57% | 480/849 [00:13<00:09, 37.18it/s] |
| Computing ESM2 embeddings: | 60% | 512/849 [00:13<00:09, 36.70it/s] |
| Computing ESM2 embeddings: | 64% | 544/849 [00:14<00:08, 37.57it/s] |
| Computing ESM2 embeddings: | 68% | 576/849 [00:15<00:07, 37.04it/s] |
| Computing ESM2 embeddings: | 72% | 608/849 [00:16<00:06, 37.85it/s] |
| Computing ESM2 embeddings: | 75% | 640/849 [00:17<00:05, 37.29it/s] |
| Computing ESM2 embeddings: | 79% | 672/849 [00:18<00:04, 37.71it/s] |
| Computing ESM2 embeddings: | 83% | 704/849 [00:19<00:03, 36.51it/s] |
| Computing ESM2 embeddings: | 87% | 736/849 [00:19<00:03, 37.33it/s] |

```

Computing ESM2 embeddings: 90%|      | 768/849 [00:20<00:02, 36.82it/s]
Computing ESM2 embeddings: 94%|      | 800/849 [00:21<00:01, 37.52it/s]
Computing ESM2 embeddings: 98%|      | 832/849 [00:22<00:00, 37.00it/s]
Computing ESM2 embeddings: 100%|     | 849/849 [00:22<00:00, 36.93it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/636 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/636 [00:00<00:16, 37.09it/s]
Computing ESM2 embeddings: 10%|      | 64/636 [00:01<00:15, 36.49it/s]
Computing ESM2 embeddings: 15%|      | 96/636 [00:02<00:14, 37.77it/s]
Computing ESM2 embeddings: 20%|      | 128/636 [00:03<00:13, 37.04it/s]
Computing ESM2 embeddings: 25%|      | 160/636 [00:04<00:12, 37.40it/s]
Computing ESM2 embeddings: 30%|      | 192/636 [00:05<00:12, 36.21it/s]
Computing ESM2 embeddings: 35%|      | 224/636 [00:06<00:11, 37.17it/s]
Computing ESM2 embeddings: 40%|      | 256/636 [00:06<00:10, 36.76it/s]
Computing ESM2 embeddings: 45%|      | 288/636 [00:07<00:09, 37.49it/s]
Computing ESM2 embeddings: 50%|      | 320/636 [00:08<00:08, 37.10it/s]
Computing ESM2 embeddings: 55%|      | 352/636 [00:09<00:07, 37.76it/s]
Computing ESM2 embeddings: 60%|      | 384/636 [00:10<00:06, 37.26it/s]
Computing ESM2 embeddings: 65%|      | 416/636 [00:11<00:05, 37.86it/s]
Computing ESM2 embeddings: 70%|      | 448/636 [00:12<00:05, 37.29it/s]
Computing ESM2 embeddings: 75%|      | 480/636 [00:12<00:04, 37.62it/s]
Computing ESM2 embeddings: 81%|      | 512/636 [00:13<00:03, 36.52it/s]

```

```

Computing ESM2 embeddings: 86%|      | 544/636 [00:14<00:02, 37.43it/s]
Computing ESM2 embeddings: 91%|      | 576/636 [00:15<00:01, 36.88it/s]
Computing ESM2 embeddings: 96%|      | 608/636 [00:16<00:00, 37.39it/s]
Computing ESM2 embeddings: 100%|     | 636/636 [00:17<00:00, 36.88it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.62it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:23, 35.75it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.56it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 36.71it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.82it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:18, 37.10it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 37.97it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 37.40it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 38.11it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 37.39it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 38.12it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.51it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.82it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:12, 36.79it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:11, 37.60it/s]
Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:10, 37.13it/s]

```

```

Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.82it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.01it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.82it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.15it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.66it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 36.41it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.37it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 36.90it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.74it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.28it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.72it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 37.20it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 37.58it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:23, 36.21it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.64it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 36.77it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:20, 37.68it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 37.11it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:05<00:18, 37.82it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 37.23it/s]

```

```

Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 37.86it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:16, 37.11it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 37.83it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 37.26it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.84it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:11<00:12, 37.26it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 37.81it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 37.28it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 38.01it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.33it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.81it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 36.69it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:17<00:06, 37.58it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 37.03it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.85it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.17it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.91it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.33it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 38.05it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:23<00:00, 37.18it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.30it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
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for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/898 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/898 [00:00<00:22, 37.83it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/898 [00:01<00:22, 37.60it/s]  |
| Computing ESM2 embeddings: | 11% | 96/898 [00:02<00:20, 38.32it/s]  |
| Computing ESM2 embeddings: | 14% | 128/898 [00:03<00:20, 37.27it/s] |
| Computing ESM2 embeddings: | 18% | 160/898 [00:04<00:19, 37.62it/s] |
| Computing ESM2 embeddings: | 21% | 192/898 [00:05<00:19, 36.30it/s] |
| Computing ESM2 embeddings: | 25% | 224/898 [00:06<00:18, 37.20it/s] |
| Computing ESM2 embeddings: | 29% | 256/898 [00:06<00:17, 36.71it/s] |
| Computing ESM2 embeddings: | 32% | 288/898 [00:07<00:16, 37.60it/s] |
| Computing ESM2 embeddings: | 36% | 320/898 [00:08<00:15, 36.98it/s] |
| Computing ESM2 embeddings: | 39% | 352/898 [00:09<00:14, 37.80it/s] |
| Computing ESM2 embeddings: | 43% | 384/898 [00:10<00:13, 37.18it/s] |
| Computing ESM2 embeddings: | 46% | 416/898 [00:11<00:12, 37.93it/s] |
| Computing ESM2 embeddings: | 50% | 448/898 [00:12<00:12, 37.15it/s] |
| Computing ESM2 embeddings: | 53% | 480/898 [00:12<00:11, 37.89it/s] |
| Computing ESM2 embeddings: | 57% | 512/898 [00:13<00:10, 37.15it/s] |
| Computing ESM2 embeddings: | 61% | 544/898 [00:14<00:09, 37.87it/s] |
| Computing ESM2 embeddings: | 64% | 576/898 [00:15<00:08, 37.15it/s] |
| Computing ESM2 embeddings: | 68% | 608/898 [00:16<00:07, 37.89it/s] |
| Computing ESM2 embeddings: | 71% | 640/898 [00:17<00:06, 37.17it/s] |
| Computing ESM2 embeddings: | 75% | 672/898 [00:17<00:05, 37.77it/s] |
| Computing ESM2 embeddings: | 78% | 704/898 [00:18<00:05, 37.13it/s] |
| Computing ESM2 embeddings: | 82% | 736/898 [00:19<00:04, 37.73it/s] |



```

Computing ESM2 embeddings: 86%|      | 768/898 [00:20<00:03, 37.01it/s]
Computing ESM2 embeddings: 89%|      | 800/898 [00:21<00:02, 37.65it/s]
Computing ESM2 embeddings: 93%|      | 832/898 [00:22<00:01, 37.02it/s]
Computing ESM2 embeddings: 96%|      | 864/898 [00:23<00:00, 37.74it/s]
Computing ESM2 embeddings: 100%|     | 898/898 [00:24<00:00, 37.20it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
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'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.92it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 36.73it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 38.00it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 37.12it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.67it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.43it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.38it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 37.06it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.81it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 37.48it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 38.05it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 37.48it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 38.11it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:12, 37.45it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 38.05it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.37it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.17it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.39it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 38.11it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.31it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.05it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.48it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 38.15it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.45it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:02, 38.15it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.55it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:22<00:01, 37.97it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 36.68it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.55it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:01<00:34, 24.80it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:02<00:28, 28.72it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:03<00:24, 32.74it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:22, 34.00it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:20, 35.55it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:19, 35.71it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:06<00:18, 36.55it/s]

Computing ESM2 embeddings: 29%| | 256/895 [00:07<00:17, 35.80it/s]

Computing ESM2 embeddings: 32%| | 288/895 [00:08<00:16, 36.87it/s]

Computing ESM2 embeddings: 36%| | 320/895 [00:09<00:15, 36.53it/s]

Computing ESM2 embeddings: 39%| | 352/895 [00:09<00:14, 37.33it/s]

Computing ESM2 embeddings: 43%| | 384/895 [00:10<00:13, 36.96it/s]

Computing ESM2 embeddings: 46%| | 416/895 [00:11<00:12, 37.68it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:12<00:11, 37.41it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:13<00:10, 38.18it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:14<00:10, 37.52it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:15<00:09, 37.89it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:16<00:08, 36.62it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.50it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 36.99it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:05, 37.77it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:19<00:05, 37.15it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:20<00:04, 37.89it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:21<00:03, 37.22it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.84it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.27it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.58it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 36.30it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 38.79it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:22, 36.97it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.71it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 36.23it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 37.30it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.71it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.55it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.01it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.47it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 37.11it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.75it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.22it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.94it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 37.23it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.95it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.17it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.93it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.04it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.82it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.18it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 37.92it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 37.07it/s] |

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.84it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.13it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.87it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.05it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.80it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 36.96it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.28it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/792 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/792 [00:00<00:19, 39.07it/s]

Computing ESM2 embeddings: 8%| | 64/792 [00:01<00:19, 37.22it/s]

Computing ESM2 embeddings: 12%| | 96/792 [00:02<00:18, 38.34it/s]

Computing ESM2 embeddings: 16%| | 128/792 [00:03<00:17, 37.33it/s]

Computing ESM2 embeddings: 20%| | 160/792 [00:04<00:16, 37.64it/s]

Computing ESM2 embeddings: 24%| | 192/792 [00:05<00:16, 36.36it/s]

Computing ESM2 embeddings: 28%| | 224/792 [00:05<00:15, 37.31it/s]

Computing ESM2 embeddings: 32%| | 256/792 [00:06<00:14, 36.91it/s]

Computing ESM2 embeddings: 36%| | 288/792 [00:07<00:13, 37.62it/s]

Computing ESM2 embeddings: 40%| | 320/792 [00:08<00:12, 37.06it/s]

Computing ESM2 embeddings: 44%| | 352/792 [00:09<00:11, 37.47it/s]

Computing ESM2 embeddings: 48%| | 384/792 [00:10<00:11, 36.30it/s]

Computing ESM2 embeddings: 53%| | 416/792 [00:11<00:10, 37.29it/s]

Computing ESM2 embeddings: 57%| | 448/792 [00:12<00:09, 36.79it/s]

Computing ESM2 embeddings: 61%| | 480/792 [00:12<00:08, 37.37it/s]

Computing ESM2 embeddings: 65%| | 512/792 [00:13<00:07, 36.29it/s]

Computing ESM2 embeddings: 69%| | 544/792 [00:14<00:06, 37.27it/s]

Computing ESM2 embeddings: 73%| | 576/792 [00:15<00:05, 36.60it/s]

Computing ESM2 embeddings: 77%| | 608/792 [00:16<00:04, 37.49it/s]

Computing ESM2 embeddings: 81%| | 640/792 [00:17<00:04, 36.95it/s]

Computing ESM2 embeddings: 85%| | 672/792 [00:18<00:03, 37.61it/s]

Computing ESM2 embeddings: 89%| | 704/792 [00:18<00:02, 37.00it/s]

Computing ESM2 embeddings: 93%| | 736/792 [00:19<00:01, 37.65it/s]

Computing ESM2 embeddings: 97%| | 768/792 [00:20<00:00, 37.16it/s]

Computing ESM2 embeddings: 100%| | 792/792 [00:21<00:00, 36.96it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/676 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/676 [00:00<00:16, 38.15it/s]

Computing ESM2 embeddings: 9%| | 64/676 [00:01<00:17, 35.92it/s]

Computing ESM2 embeddings: 14%| | 96/676 [00:02<00:15, 37.51it/s]

Computing ESM2 embeddings: 19%| | 128/676 [00:03<00:14, 37.09it/s]

Computing ESM2 embeddings: 24%| | 160/676 [00:04<00:13, 37.60it/s]

Computing ESM2 embeddings: 28%| | 192/676 [00:05<00:13, 36.36it/s]

Computing ESM2 embeddings: 33%| | 224/676 [00:06<00:12, 37.31it/s]

Computing ESM2 embeddings: 38%| | 256/676 [00:06<00:11, 37.21it/s]

Computing ESM2 embeddings: 43%| | 288/676 [00:07<00:10, 37.85it/s]

Computing ESM2 embeddings: 47%| | 320/676 [00:08<00:09, 37.16it/s]

Computing ESM2 embeddings: 52%| | 352/676 [00:09<00:08, 37.87it/s]

Computing ESM2 embeddings: 57%| | 384/676 [00:10<00:07, 37.24it/s]

Computing ESM2 embeddings: 62%| | 416/676 [00:11<00:06, 38.02it/s]

Computing ESM2 embeddings: 66%| | 448/676 [00:12<00:06, 37.26it/s]

Computing ESM2 embeddings: 71%| | 480/676 [00:12<00:05, 37.81it/s]

Computing ESM2 embeddings: 76%| | 512/676 [00:13<00:04, 36.71it/s]

Computing ESM2 embeddings: 80%| | 544/676 [00:14<00:03, 37.60it/s]

Computing ESM2 embeddings: 85%| | 576/676 [00:15<00:02, 37.07it/s]

Computing ESM2 embeddings: 90%| | 608/676 [00:16<00:01, 37.87it/s]

Computing ESM2 embeddings: 95%| | 640/676 [00:17<00:00, 37.34it/s]

Computing ESM2 embeddings: 99%| | 672/676 [00:17<00:00, 37.98it/s]

Computing ESM2 embeddings: 100%| | 676/676 [00:18<00:00, 37.30it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.44it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.45it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.37it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 37.38it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 37.81it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.42it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:18, 37.39it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:17, 36.73it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 37.68it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:16, 36.98it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 37.82it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:14, 37.19it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:11<00:13, 37.94it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:12, 37.24it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.98it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 37.27it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 37.74it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:15<00:09, 36.65it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:16<00:08, 37.45it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:17<00:07, 37.06it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 37.73it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 37.17it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:19<00:04, 37.92it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:20<00:03, 37.20it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:21<00:03, 37.96it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:22<00:02, 37.26it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:23<00:01, 37.99it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:23<00:00, 37.26it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:24<00:00, 37.47it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',



'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:01<00:30, 28.24it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:02<00:26, 31.06it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:23, 34.12it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:22, 34.13it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:20, 35.82it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 35.92it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 36.91it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:07<00:17, 36.49it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:08<00:16, 37.25it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 36.94it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.64it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.20it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.82it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:11, 37.43it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:13<00:10, 37.94it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:14<00:10, 36.73it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.60it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 36.86it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.68it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.05it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:18<00:05, 37.82it/s] |

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 37.20it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.95it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.35it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 38.02it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.13it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.89it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.65it/s]

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Computing ESM2 embeddings: 0%| | 0/917 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/917 [00:00<00:23, 37.46it/s]

Computing ESM2 embeddings: 7%| | 64/917 [00:01<00:22, 37.44it/s]

Computing ESM2 embeddings: 10%| | 96/917 [00:02<00:21, 38.34it/s]

Computing ESM2 embeddings: 14%| | 128/917 [00:03<00:21, 37.53it/s]

Computing ESM2 embeddings: 17%| | 160/917 [00:04<00:19, 38.26it/s]

Computing ESM2 embeddings: 21%| | 192/917 [00:05<00:19, 37.63it/s]

Computing ESM2 embeddings: 24%| | 224/917 [00:05<00:18, 37.91it/s]

Computing ESM2 embeddings: 28%| | 256/917 [00:06<00:18, 36.72it/s]

Computing ESM2 embeddings: 31%| | 288/917 [00:07<00:16, 37.52it/s]

Computing ESM2 embeddings: 35%| | 320/917 [00:08<00:16, 37.13it/s]

Computing ESM2 embeddings: 38%| | 352/917 [00:09<00:14, 37.76it/s]

Computing ESM2 embeddings: 42%| | 384/917 [00:10<00:14, 37.33it/s]

Computing ESM2 embeddings: 45%| | 416/917 [00:11<00:13, 38.07it/s]

Computing ESM2 embeddings: 49%| | 448/917 [00:11<00:12, 37.47it/s]

Computing ESM2 embeddings: 52%| | 480/917 [00:12<00:11, 38.16it/s]

Computing ESM2 embeddings: 56%| | 512/917 [00:13<00:10, 37.44it/s]

Computing ESM2 embeddings: 59%| | 544/917 [00:14<00:09, 37.90it/s]

Computing ESM2 embeddings: 63%| | 576/917 [00:15<00:09, 36.67it/s]

Computing ESM2 embeddings: 66%| | 608/917 [00:16<00:08, 37.58it/s]

Computing ESM2 embeddings: 70%| | 640/917 [00:17<00:07, 37.05it/s]

Computing ESM2 embeddings: 73%| | 672/917 [00:17<00:06, 37.86it/s]

Computing ESM2 embeddings: 77%| | 704/917 [00:18<00:05, 37.23it/s]

Computing ESM2 embeddings: 80%| | 736/917 [00:19<00:04, 37.99it/s]

Computing ESM2 embeddings: 84%| | 768/917 [00:20<00:03, 37.41it/s]

Computing ESM2 embeddings: 87%| | 800/917 [00:21<00:03, 38.11it/s]

Computing ESM2 embeddings: 91%| | 832/917 [00:22<00:02, 37.40it/s]

Computing ESM2 embeddings: 94%| | 864/917 [00:22<00:01, 37.99it/s]

Computing ESM2 embeddings: 98%| | 896/917 [00:23<00:00, 37.30it/s]

Computing ESM2 embeddings: 100%| | 917/917 [00:24<00:00, 37.52it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 38.11it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:22, 36.59it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:21, 37.67it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:20, 36.80it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/895 [00:04<00:19, 37.88it/s] |
| Computing ESM2 embeddings: 21%  | 192/895 [00:05<00:19, 36.98it/s] |
| Computing ESM2 embeddings: 25%  | 224/895 [00:05<00:17, 37.92it/s] |
| Computing ESM2 embeddings: 29%  | 256/895 [00:06<00:17, 37.19it/s] |
| Computing ESM2 embeddings: 32%  | 288/895 [00:07<00:16, 37.56it/s] |
| Computing ESM2 embeddings: 36%  | 320/895 [00:08<00:15, 36.46it/s] |
| Computing ESM2 embeddings: 39%  | 352/895 [00:09<00:14, 37.44it/s] |
| Computing ESM2 embeddings: 43%  | 384/895 [00:10<00:13, 36.88it/s] |
| Computing ESM2 embeddings: 46%  | 416/895 [00:11<00:12, 37.76it/s] |
| Computing ESM2 embeddings: 50%  | 448/895 [00:12<00:12, 37.12it/s] |
| Computing ESM2 embeddings: 54%  | 480/895 [00:12<00:10, 37.91it/s] |
| Computing ESM2 embeddings: 57%  | 512/895 [00:13<00:10, 37.21it/s] |
| Computing ESM2 embeddings: 61%  | 544/895 [00:14<00:09, 37.98it/s] |
| Computing ESM2 embeddings: 64%  | 576/895 [00:15<00:08, 37.41it/s] |
| Computing ESM2 embeddings: 68%  | 608/895 [00:16<00:07, 37.88it/s] |
| Computing ESM2 embeddings: 72%  | 640/895 [00:17<00:06, 36.76it/s] |
| Computing ESM2 embeddings: 75%  | 672/895 [00:17<00:05, 37.65it/s] |
| Computing ESM2 embeddings: 79%  | 704/895 [00:18<00:05, 37.10it/s] |
| Computing ESM2 embeddings: 82%  | 736/895 [00:19<00:04, 37.91it/s] |
| Computing ESM2 embeddings: 86%  | 768/895 [00:20<00:03, 37.31it/s] |
| Computing ESM2 embeddings: 89%  | 800/895 [00:21<00:02, 38.05it/s] |
| Computing ESM2 embeddings: 93%  | 832/895 [00:22<00:01, 37.34it/s] |
| Computing ESM2 embeddings: 97%  | 864/895 [00:23<00:00, 38.09it/s] |
| Computing ESM2 embeddings: 100% | 895/895 [00:23<00:00, 37.34it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.67it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:22, 37.01it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 38.12it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 37.18it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 37.58it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.40it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.33it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 36.91it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.45it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.11it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 37.19it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.10it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.74it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 37.00it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.43it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:11, 36.44it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.38it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 36.90it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.71it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.01it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:18<00:06, 37.78it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 37.03it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 37.80it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.10it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.87it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.16it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.90it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.27it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.13it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 37.96it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:21, 37.88it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 38.82it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 37.73it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 38.27it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:18, 37.48it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 37.77it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 36.71it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.62it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 37.11it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.51it/s]

```

```

Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:14, 36.42it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 37.24it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:12<00:12, 36.74it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:11, 37.45it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.02it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.67it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.03it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.69it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.00it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.64it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 36.99it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.63it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.14it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.77it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.25it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.84it/s]
Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 37.29it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|          | 0/751 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/751 [00:01<00:23, 31.08it/s]
Computing ESM2 embeddings: 9%|          | 64/751 [00:01<00:20, 33.35it/s]
Computing ESM2 embeddings: 13%|         | 96/751 [00:02<00:18, 35.90it/s]

```

Computing ESM2 embeddings: 17%| | 128/751 [00:03<00:17, 35.72it/s]

Computing ESM2 embeddings: 21%| | 160/751 [00:04<00:15, 36.98it/s]

Computing ESM2 embeddings: 26%| | 192/751 [00:05<00:15, 36.60it/s]

Computing ESM2 embeddings: 30%| | 224/751 [00:06<00:14, 37.49it/s]

Computing ESM2 embeddings: 34%| | 256/751 [00:07<00:13, 36.85it/s]

Computing ESM2 embeddings: 38%| | 288/751 [00:07<00:12, 37.63it/s]

Computing ESM2 embeddings: 43%| | 320/751 [00:08<00:11, 36.80it/s]

Computing ESM2 embeddings: 47%| | 352/751 [00:09<00:10, 37.29it/s]

Computing ESM2 embeddings: 51%| | 384/751 [00:10<00:10, 36.37it/s]

Computing ESM2 embeddings: 55%| | 416/751 [00:11<00:09, 37.21it/s]

Computing ESM2 embeddings: 60%| | 448/751 [00:12<00:08, 36.84it/s]

Computing ESM2 embeddings: 64%| | 480/751 [00:13<00:07, 37.68it/s]

Computing ESM2 embeddings: 68%| | 512/751 [00:13<00:06, 37.08it/s]

Computing ESM2 embeddings: 72%| | 544/751 [00:14<00:05, 37.86it/s]

Computing ESM2 embeddings: 77%| | 576/751 [00:15<00:04, 37.14it/s]

Computing ESM2 embeddings: 81%| | 608/751 [00:16<00:03, 37.90it/s]

Computing ESM2 embeddings: 85%| | 640/751 [00:17<00:02, 37.03it/s]

Computing ESM2 embeddings: 89%| | 672/751 [00:18<00:02, 37.78it/s]

Computing ESM2 embeddings: 94%| | 704/751 [00:19<00:01, 36.95it/s]

Computing ESM2 embeddings: 98%| | 736/751 [00:19<00:00, 37.76it/s]

Computing ESM2 embeddings: 100%| | 751/751 [00:20<00:00, 36.87it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.



|                            |      |                                  |
|----------------------------|------|----------------------------------|
| Computing ESM2 embeddings: | 0%   | 0/735 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%   | 32/735 [00:00<00:18, 37.15it/s]  |
| Computing ESM2 embeddings: | 9%   | 64/735 [00:01<00:18, 35.39it/s]  |
| Computing ESM2 embeddings: | 13%  | 96/735 [00:02<00:17, 37.32it/s]  |
| Computing ESM2 embeddings: | 17%  | 128/735 [00:03<00:16, 36.69it/s] |
| Computing ESM2 embeddings: | 22%  | 160/735 [00:04<00:15, 37.80it/s] |
| Computing ESM2 embeddings: | 26%  | 192/735 [00:05<00:14, 37.23it/s] |
| Computing ESM2 embeddings: | 30%  | 224/735 [00:06<00:13, 37.78it/s] |
| Computing ESM2 embeddings: | 35%  | 256/735 [00:06<00:13, 36.47it/s] |
| Computing ESM2 embeddings: | 39%  | 288/735 [00:07<00:11, 37.45it/s] |
| Computing ESM2 embeddings: | 44%  | 320/735 [00:08<00:11, 37.09it/s] |
| Computing ESM2 embeddings: | 48%  | 352/735 [00:09<00:10, 37.82it/s] |
| Computing ESM2 embeddings: | 52%  | 384/735 [00:10<00:09, 37.21it/s] |
| Computing ESM2 embeddings: | 57%  | 416/735 [00:11<00:08, 37.95it/s] |
| Computing ESM2 embeddings: | 61%  | 448/735 [00:12<00:07, 37.22it/s] |
| Computing ESM2 embeddings: | 65%  | 480/735 [00:12<00:06, 37.96it/s] |
| Computing ESM2 embeddings: | 70%  | 512/735 [00:13<00:05, 37.26it/s] |
| Computing ESM2 embeddings: | 74%  | 544/735 [00:14<00:05, 37.53it/s] |
| Computing ESM2 embeddings: | 78%  | 576/735 [00:15<00:04, 36.50it/s] |
| Computing ESM2 embeddings: | 83%  | 608/735 [00:16<00:03, 37.35it/s] |
| Computing ESM2 embeddings: | 87%  | 640/735 [00:17<00:02, 36.97it/s] |
| Computing ESM2 embeddings: | 91%  | 672/735 [00:18<00:01, 37.64it/s] |
| Computing ESM2 embeddings: | 96%  | 704/735 [00:18<00:00, 37.17it/s] |
| Computing ESM2 embeddings: | 100% | 735/735 [00:19<00:00, 37.20it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 36.99it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:23, 36.38it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.97it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 37.33it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 37.77it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.64it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 37.52it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.04it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.70it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 37.25it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.91it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.34it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.95it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 37.37it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.64it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:11, 36.53it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.31it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 36.91it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.64it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.10it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 37.50it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 36.41it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 37.38it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 36.90it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.41it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.23it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.21it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.78it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.06it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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```

Computing ESM2 embeddings: 0%|      | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/896 [00:00<00:23, 37.29it/s]
Computing ESM2 embeddings: 7%|      | 64/896 [00:01<00:22, 36.22it/s]
Computing ESM2 embeddings: 11%|     | 96/896 [00:02<00:21, 38.00it/s]
Computing ESM2 embeddings: 14%|     | 128/896 [00:03<00:20, 37.14it/s]
Computing ESM2 embeddings: 18%|     | 160/896 [00:04<00:19, 37.81it/s]
Computing ESM2 embeddings: 21%|     | 192/896 [00:05<00:19, 36.46it/s]
Computing ESM2 embeddings: 25%|     | 224/896 [00:06<00:17, 37.52it/s]
Computing ESM2 embeddings: 29%|     | 256/896 [00:06<00:17, 37.01it/s]
Computing ESM2 embeddings: 32%|     | 288/896 [00:07<00:16, 37.87it/s]
Computing ESM2 embeddings: 36%|     | 320/896 [00:08<00:15, 37.16it/s]
Computing ESM2 embeddings: 39%|     | 352/896 [00:09<00:14, 37.82it/s]

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Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 37.18it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 37.94it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:12, 37.27it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 38.02it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.26it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.75it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 36.74it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.54it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.06it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.85it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 37.20it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.96it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.25it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 38.00it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.27it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.74it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 37.29it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/916 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|          | 32/916 [00:00<00:27, 32.30it/s]
Computing ESM2 embeddings:  7%|          | 64/916 [00:01<00:25, 33.12it/s]
Computing ESM2 embeddings: 10%|          | 96/916 [00:02<00:23, 35.32it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/916 [00:03<00:22, 34.87it/s] |
| Computing ESM2 embeddings: 17% | 160/916 [00:04<00:20, 36.40it/s] |
| Computing ESM2 embeddings: 21% | 192/916 [00:05<00:20, 36.08it/s] |
| Computing ESM2 embeddings: 24% | 224/916 [00:06<00:18, 37.13it/s] |
| Computing ESM2 embeddings: 28% | 256/916 [00:07<00:17, 36.73it/s] |
| Computing ESM2 embeddings: 31% | 288/916 [00:07<00:16, 37.24it/s] |
| Computing ESM2 embeddings: 35% | 320/916 [00:08<00:16, 36.04it/s] |
| Computing ESM2 embeddings: 38% | 352/916 [00:09<00:15, 37.02it/s] |
| Computing ESM2 embeddings: 42% | 384/916 [00:10<00:14, 36.76it/s] |
| Computing ESM2 embeddings: 45% | 416/916 [00:11<00:13, 37.52it/s] |
| Computing ESM2 embeddings: 49% | 448/916 [00:12<00:12, 37.25it/s] |
| Computing ESM2 embeddings: 52% | 480/916 [00:13<00:11, 37.95it/s] |
| Computing ESM2 embeddings: 56% | 512/916 [00:13<00:10, 37.31it/s] |
| Computing ESM2 embeddings: 59% | 544/916 [00:14<00:09, 37.90it/s] |
| Computing ESM2 embeddings: 63% | 576/916 [00:15<00:09, 37.31it/s] |
| Computing ESM2 embeddings: 66% | 608/916 [00:16<00:08, 38.04it/s] |
| Computing ESM2 embeddings: 70% | 640/916 [00:17<00:07, 37.29it/s] |
| Computing ESM2 embeddings: 73% | 672/916 [00:18<00:06, 37.70it/s] |
| Computing ESM2 embeddings: 77% | 704/916 [00:19<00:05, 36.48it/s] |
| Computing ESM2 embeddings: 80% | 736/916 [00:19<00:04, 37.40it/s] |
| Computing ESM2 embeddings: 84% | 768/916 [00:20<00:04, 36.92it/s] |
| Computing ESM2 embeddings: 87% | 800/916 [00:21<00:03, 37.62it/s] |
| Computing ESM2 embeddings: 91% | 832/916 [00:22<00:02, 36.98it/s] |
| Computing ESM2 embeddings: 94% | 864/916 [00:23<00:01, 37.64it/s] |

Computing ESM2 embeddings: 98%| | 896/916 [00:24<00:00, 36.91it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:24<00:00, 36.83it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:26, 33.04it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:24, 33.57it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:22, 36.04it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 35.94it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:20, 36.76it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 35.87it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 36.95it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:17, 36.61it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.54it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.94it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.80it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.10it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.88it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.18it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:10, 37.89it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.22it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.89it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.30it/s]

```

Computing ESM2 embeddings: 68%|      | 608/896 [00:16<00:07, 38.01it/s]
Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 37.10it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:18<00:05, 37.74it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:19<00:05, 37.11it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.74it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.20it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.82it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.25it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.87it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.05it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/915 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/915 [00:00<00:22, 38.55it/s]
Computing ESM2 embeddings:  7%|      | 64/915 [00:01<00:23, 36.00it/s]
Computing ESM2 embeddings: 10%|      | 96/915 [00:02<00:21, 37.69it/s]
Computing ESM2 embeddings: 14%|      | 128/915 [00:03<00:21, 36.94it/s]
Computing ESM2 embeddings: 17%|      | 160/915 [00:04<00:19, 37.92it/s]
Computing ESM2 embeddings: 21%|      | 192/915 [00:05<00:19, 37.22it/s]
Computing ESM2 embeddings: 24%|      | 224/915 [00:05<00:18, 38.03it/s]
Computing ESM2 embeddings: 28%|      | 256/915 [00:06<00:17, 37.03it/s]
Computing ESM2 embeddings: 31%|      | 288/915 [00:07<00:16, 37.87it/s]
Computing ESM2 embeddings: 35%|      | 320/915 [00:08<00:16, 37.13it/s]

```

Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:14, 37.82it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 37.19it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 37.82it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:11<00:12, 37.11it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 37.89it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 37.15it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 37.89it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 37.28it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.67it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 36.67it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:17<00:06, 37.55it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:18<00:05, 37.01it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 37.79it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:03, 37.44it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 38.21it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 37.53it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:23<00:01, 38.19it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:23<00:00, 37.42it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 37.48it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.35it/s]



|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:22, 37.29it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:20, 38.45it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 37.42it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.71it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.56it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 37.57it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 37.01it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.86it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.20it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.96it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.22it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.61it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 36.57it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.38it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 36.92it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.74it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.12it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.90it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.24it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.97it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 37.19it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.92it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 37.22it/s] |
| Computing ESM2 embeddings: | 89% | 800/896 [00:21<00:02, 37.66it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 36.50it/s]

Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.19it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.13it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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```

Computing ESM2 embeddings:  0%|      |  0/691 [00:00<?, ?it/s]

Computing ESM2 embeddings:  5%|      | 32/691 [00:00<00:16, 39.02it/s]

Computing ESM2 embeddings:  9%|      | 64/691 [00:01<00:16, 37.24it/s]

Computing ESM2 embeddings: 14%|      | 96/691 [00:02<00:15, 38.33it/s]

Computing ESM2 embeddings: 19%|      | 128/691 [00:03<00:15, 37.20it/s]

Computing ESM2 embeddings: 23%|      | 160/691 [00:04<00:13, 38.00it/s]

Computing ESM2 embeddings: 28%|      | 192/691 [00:05<00:13, 37.31it/s]

Computing ESM2 embeddings: 32%|      | 224/691 [00:05<00:12, 38.09it/s]

Computing ESM2 embeddings: 37%|      | 256/691 [00:06<00:11, 37.30it/s]

Computing ESM2 embeddings: 42%|      | 288/691 [00:07<00:10, 38.07it/s]

Computing ESM2 embeddings: 46%|      | 320/691 [00:08<00:10, 36.87it/s]

Computing ESM2 embeddings: 51%|      | 352/691 [00:09<00:08, 37.79it/s]

Computing ESM2 embeddings: 56%|      | 384/691 [00:10<00:08, 37.13it/s]

Computing ESM2 embeddings: 60%|      | 416/691 [00:11<00:07, 37.52it/s]

Computing ESM2 embeddings: 65%|      | 448/691 [00:11<00:06, 36.53it/s]

Computing ESM2 embeddings: 69%|      | 480/691 [00:12<00:05, 37.27it/s]

Computing ESM2 embeddings: 74%|      | 512/691 [00:13<00:04, 36.98it/s]

Computing ESM2 embeddings: 79%|      | 544/691 [00:14<00:03, 37.61it/s]

```

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Computing ESM2 embeddings: 83%|      | 576/691 [00:15<00:03, 37.18it/s]
Computing ESM2 embeddings: 88%|      | 608/691 [00:16<00:02, 37.66it/s]
Computing ESM2 embeddings: 93%|      | 640/691 [00:17<00:01, 37.20it/s]
Computing ESM2 embeddings: 97%|      | 672/691 [00:17<00:00, 37.77it/s]
Computing ESM2 embeddings: 100%|     | 691/691 [00:18<00:00, 37.36it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/794 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/794 [00:00<00:19, 38.23it/s]
Computing ESM2 embeddings:  8%|      | 64/794 [00:01<00:20, 36.20it/s]
Computing ESM2 embeddings: 12%|      | 96/794 [00:02<00:18, 37.86it/s]
Computing ESM2 embeddings: 16%|      | 128/794 [00:03<00:18, 36.88it/s]
Computing ESM2 embeddings: 20%|      | 160/794 [00:04<00:16, 37.92it/s]
Computing ESM2 embeddings: 24%|      | 192/794 [00:05<00:16, 37.15it/s]
Computing ESM2 embeddings: 28%|      | 224/794 [00:05<00:15, 37.69it/s]
Computing ESM2 embeddings: 32%|      | 256/794 [00:06<00:14, 36.57it/s]
Computing ESM2 embeddings: 36%|      | 288/794 [00:07<00:13, 37.56it/s]
Computing ESM2 embeddings: 40%|      | 320/794 [00:08<00:12, 37.15it/s]
Computing ESM2 embeddings: 44%|      | 352/794 [00:09<00:11, 37.92it/s]
Computing ESM2 embeddings: 48%|      | 384/794 [00:10<00:10, 37.30it/s]
Computing ESM2 embeddings: 52%|      | 416/794 [00:11<00:09, 38.06it/s]
Computing ESM2 embeddings: 56%|      | 448/794 [00:11<00:09, 37.46it/s]
Computing ESM2 embeddings: 60%|      | 480/794 [00:12<00:08, 37.93it/s]

```

```

Computing ESM2 embeddings: 64%|          | 512/794 [00:13<00:07, 36.77it/s]
Computing ESM2 embeddings: 69%|          | 544/794 [00:14<00:06, 37.52it/s]
Computing ESM2 embeddings: 73%|          | 576/794 [00:15<00:05, 37.08it/s]
Computing ESM2 embeddings: 77%|          | 608/794 [00:16<00:04, 37.79it/s]
Computing ESM2 embeddings: 81%|          | 640/794 [00:17<00:04, 37.12it/s]
Computing ESM2 embeddings: 85%|          | 672/794 [00:17<00:03, 37.82it/s]
Computing ESM2 embeddings: 89%|          | 704/794 [00:18<00:02, 37.25it/s]
Computing ESM2 embeddings: 93%|          | 736/794 [00:19<00:01, 37.91it/s]
Computing ESM2 embeddings: 97%|          | 768/794 [00:20<00:00, 37.31it/s]
Computing ESM2 embeddings: 100%|         | 794/794 [00:21<00:00, 37.39it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:22, 38.84it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:23, 36.77it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.55it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 36.22it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:20, 37.11it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 36.50it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:06<00:18, 37.10it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:18, 36.14it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 37.10it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:16, 36.70it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.46it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.98it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.66it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.19it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.83it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.26it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.01it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.26it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 38.01it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.31it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 38.04it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.31it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.93it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.32it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.91it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.29it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.89it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.16it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.21it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.43it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:24, 34.53it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 36.68it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 36.28it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.28it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.86it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 37.18it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:07<00:17, 36.00it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 36.72it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:16, 35.90it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 36.81it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 36.64it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.44it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 36.98it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:13<00:11, 37.66it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.14it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.77it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.29it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.86it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.35it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:18<00:05, 37.61it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:19<00:05, 36.51it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.30it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 36.85it/s] |
| Computing ESM2 embeddings: | 89% | 800/896 [00:21<00:02, 37.56it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.06it/s]

Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.72it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.01it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/916 [00:00<?, ?it/s]

Computing ESM2 embeddings:  3%|      | 32/916 [00:00<00:23, 38.01it/s]

Computing ESM2 embeddings:  7%|      | 64/916 [00:01<00:23, 35.69it/s]

Computing ESM2 embeddings: 10%|      | 96/916 [00:02<00:21, 37.42it/s]

Computing ESM2 embeddings: 14%|      | 128/916 [00:03<00:21, 36.82it/s]

Computing ESM2 embeddings: 17%|      | 160/916 [00:04<00:20, 37.76it/s]

Computing ESM2 embeddings: 21%|      | 192/916 [00:05<00:19, 37.15it/s]

Computing ESM2 embeddings: 24%|      | 224/916 [00:06<00:18, 37.60it/s]

Computing ESM2 embeddings: 28%|      | 256/916 [00:06<00:18, 36.39it/s]

Computing ESM2 embeddings: 31%|      | 288/916 [00:07<00:16, 37.25it/s]

Computing ESM2 embeddings: 35%|      | 320/916 [00:08<00:16, 36.79it/s]

Computing ESM2 embeddings: 38%|      | 352/916 [00:09<00:15, 37.51it/s]

Computing ESM2 embeddings: 42%|      | 384/916 [00:10<00:14, 36.54it/s]

Computing ESM2 embeddings: 45%|      | 416/916 [00:11<00:13, 37.36it/s]

Computing ESM2 embeddings: 49%|      | 448/916 [00:12<00:12, 36.97it/s]

Computing ESM2 embeddings: 52%|      | 480/916 [00:12<00:11, 37.75it/s]

Computing ESM2 embeddings: 56%|      | 512/916 [00:13<00:10, 37.18it/s]

Computing ESM2 embeddings: 59%|      | 544/916 [00:14<00:09, 37.80it/s]

```

Computing ESM2 embeddings: 63%| | 576/916 [00:15<00:09, 37.26it/s]

Computing ESM2 embeddings: 66%| | 608/916 [00:16<00:08, 37.56it/s]

Computing ESM2 embeddings: 70%| | 640/916 [00:17<00:07, 36.51it/s]

Computing ESM2 embeddings: 73%| | 672/916 [00:18<00:06, 37.29it/s]

Computing ESM2 embeddings: 77%| | 704/916 [00:18<00:05, 36.97it/s]

Computing ESM2 embeddings: 80%| | 736/916 [00:19<00:04, 37.70it/s]

Computing ESM2 embeddings: 84%| | 768/916 [00:20<00:03, 37.42it/s]

Computing ESM2 embeddings: 87%| | 800/916 [00:21<00:03, 38.02it/s]

Computing ESM2 embeddings: 91%| | 832/916 [00:22<00:02, 37.48it/s]

Computing ESM2 embeddings: 94%| | 864/916 [00:23<00:01, 38.03it/s]

Computing ESM2 embeddings: 98%| | 896/916 [00:24<00:00, 37.47it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:24<00:00, 37.29it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.23it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 36.50it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.85it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.07it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.88it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.32it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 37.96it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 37.39it/s]



```

Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:15, 38.22it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 37.33it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 38.08it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 37.35it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 38.08it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:12, 37.33it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 38.00it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.32it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 38.05it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.30it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 38.03it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.27it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.97it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 37.30it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.91it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 36.90it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.75it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.26it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:22<00:00, 37.79it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:23<00:00, 37.39it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.92it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:23, 35.51it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.40it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 36.88it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.06it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.24it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 38.09it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.33it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.99it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 37.32it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.65it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 36.52it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.35it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 36.93it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.65it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.18it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.81it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.30it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.99it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.32it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 37.68it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 36.70it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:19<00:04, 37.46it/s] |
| Computing ESM2 embeddings: | 84% | 768/914 [00:20<00:03, 37.07it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.76it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.20it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.84it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 37.25it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.38it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.31it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:23, 35.78it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.39it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 36.94it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.68it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:18, 37.13it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 37.88it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 37.31it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.94it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 37.30it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.90it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.27it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.88it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 37.39it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 38.00it/s]

```

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.53it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.14it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.46it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 38.03it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.36it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.95it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.33it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.91it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.37it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.67it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.56it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.36it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.36it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/632 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/632 [00:00<00:15, 38.16it/s]

Computing ESM2 embeddings: 10%| | 64/632 [00:01<00:15, 36.77it/s]

Computing ESM2 embeddings: 15%| | 96/632 [00:02<00:13, 38.31it/s]

Computing ESM2 embeddings: 20%| | 128/632 [00:03<00:13, 37.41it/s]

Computing ESM2 embeddings: 25%| | 160/632 [00:04<00:12, 38.29it/s]

Computing ESM2 embeddings: 30%| | 192/632 [00:05<00:11, 37.36it/s]

Computing ESM2 embeddings: 35%| | 224/632 [00:05<00:10, 38.15it/s]

Computing ESM2 embeddings: 41%| | 256/632 [00:06<00:10, 37.28it/s]

Computing ESM2 embeddings: 46%| | 288/632 [00:07<00:09, 38.08it/s]

Computing ESM2 embeddings: 51%| | 320/632 [00:08<00:08, 37.34it/s]

Computing ESM2 embeddings: 56%| | 352/632 [00:09<00:07, 38.11it/s]

Computing ESM2 embeddings: 61%| | 384/632 [00:10<00:06, 37.42it/s]

Computing ESM2 embeddings: 66%| | 416/632 [00:10<00:05, 38.14it/s]

Computing ESM2 embeddings: 71%| | 448/632 [00:11<00:04, 37.38it/s]

Computing ESM2 embeddings: 76%| | 480/632 [00:12<00:04, 37.68it/s]

Computing ESM2 embeddings: 81%| | 512/632 [00:13<00:03, 36.88it/s]

Computing ESM2 embeddings: 86%| | 544/632 [00:14<00:02, 37.38it/s]

Computing ESM2 embeddings: 91%| | 576/632 [00:15<00:01, 36.83it/s]

Computing ESM2 embeddings: 96%| | 608/632 [00:16<00:00, 37.60it/s]

Computing ESM2 embeddings: 100%| | 632/632 [00:16<00:00, 37.41it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/853 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/853 [00:00<00:20, 39.23it/s]

Computing ESM2 embeddings: 8%| | 64/853 [00:01<00:21, 37.33it/s]

Computing ESM2 embeddings: 11%| | 96/853 [00:02<00:19, 38.30it/s]

Computing ESM2 embeddings: 15%| | 128/853 [00:03<00:19, 37.38it/s]

Computing ESM2 embeddings: 19%| | 160/853 [00:04<00:18, 37.73it/s]

Computing ESM2 embeddings: 23%| | 192/853 [00:05<00:18, 36.45it/s]

Computing ESM2 embeddings: 26%| | 224/853 [00:05<00:16, 37.40it/s]

Computing ESM2 embeddings: 30%| | 256/853 [00:06<00:16, 36.92it/s]

Computing ESM2 embeddings: 34%| | 288/853 [00:07<00:15, 37.66it/s]

Computing ESM2 embeddings: 38%| | 320/853 [00:08<00:14, 37.09it/s]

Computing ESM2 embeddings: 41%| | 352/853 [00:09<00:13, 37.84it/s]

Computing ESM2 embeddings: 45%| | 384/853 [00:10<00:12, 37.19it/s]

Computing ESM2 embeddings: 49%| | 416/853 [00:11<00:11, 37.65it/s]

Computing ESM2 embeddings: 53%| | 448/853 [00:12<00:11, 36.33it/s]

Computing ESM2 embeddings: 56%| | 480/853 [00:12<00:10, 37.30it/s]

Computing ESM2 embeddings: 60%| | 512/853 [00:13<00:09, 36.78it/s]

Computing ESM2 embeddings: 64%| | 544/853 [00:14<00:08, 37.63it/s]

Computing ESM2 embeddings: 68%| | 576/853 [00:15<00:07, 37.06it/s]

Computing ESM2 embeddings: 71%| | 608/853 [00:16<00:06, 37.78it/s]

Computing ESM2 embeddings: 75%| | 640/853 [00:17<00:05, 37.17it/s]

Computing ESM2 embeddings: 79%| | 672/853 [00:17<00:04, 37.90it/s]

Computing ESM2 embeddings: 83%| | 704/853 [00:18<00:03, 37.26it/s]

Computing ESM2 embeddings: 86%| | 736/853 [00:19<00:03, 37.99it/s]

Computing ESM2 embeddings: 90%| | 768/853 [00:20<00:02, 37.31it/s]

Computing ESM2 embeddings: 94%| | 800/853 [00:21<00:01, 38.02it/s]

Computing ESM2 embeddings: 98%| | 832/853 [00:22<00:00, 37.32it/s]

Computing ESM2 embeddings: 100%| | 853/853 [00:22<00:00, 37.37it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.96it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:25, 32.83it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:22, 35.50it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 35.47it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:20, 36.77it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.54it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:17, 37.46it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:07<00:17, 37.06it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.70it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.11it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.54it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:14, 36.38it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.36it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 36.94it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.74it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.14it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.90it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.24it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.98it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.27it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:18<00:05, 37.69it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:19<00:05, 36.50it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.42it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 36.88it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.73it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.05it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.84it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.99it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/914 [00:00<00:22, 39.36it/s]
Computing ESM2 embeddings: 7%|        | 64/914 [00:01<00:22, 38.08it/s]
Computing ESM2 embeddings: 11%|       | 96/914 [00:02<00:21, 38.79it/s]
Computing ESM2 embeddings: 14%|       | 128/914 [00:03<00:20, 37.53it/s]
Computing ESM2 embeddings: 18%|       | 160/914 [00:04<00:19, 37.84it/s]
Computing ESM2 embeddings: 21%|       | 192/914 [00:05<00:19, 36.61it/s]
Computing ESM2 embeddings: 25%|       | 224/914 [00:05<00:18, 37.50it/s]
Computing ESM2 embeddings: 28%|       | 256/914 [00:06<00:17, 36.98it/s]
Computing ESM2 embeddings: 32%|       | 288/914 [00:07<00:16, 37.71it/s]
Computing ESM2 embeddings: 35%|       | 320/914 [00:08<00:15, 37.32it/s]
Computing ESM2 embeddings: 39%|       | 352/914 [00:09<00:14, 37.84it/s]
Computing ESM2 embeddings: 42%|       | 384/914 [00:10<00:14, 36.59it/s]
Computing ESM2 embeddings: 46%|       | 416/914 [00:11<00:13, 37.44it/s]
Computing ESM2 embeddings: 49%|       | 448/914 [00:11<00:12, 36.93it/s]
Computing ESM2 embeddings: 53%|       | 480/914 [00:12<00:11, 37.63it/s]
Computing ESM2 embeddings: 56%|       | 512/914 [00:13<00:10, 37.17it/s]

```



Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.83it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.26it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.61it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.66it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.55it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 36.97it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.50it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:04, 36.22it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.24it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.82it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.67it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.08it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.26it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/899 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/899 [00:00<00:22, 39.16it/s]

Computing ESM2 embeddings: 7%| | 64/899 [00:01<00:22, 37.54it/s]

Computing ESM2 embeddings: 11%| | 96/899 [00:02<00:20, 38.44it/s]

Computing ESM2 embeddings: 14%| | 128/899 [00:03<00:20, 37.87it/s]

Computing ESM2 embeddings: 18%| | 160/899 [00:04<00:19, 38.43it/s]

Computing ESM2 embeddings: 21%| | 192/899 [00:05<00:18, 37.63it/s]

Computing ESM2 embeddings: 25%| | 224/899 [00:05<00:17, 37.91it/s]

```

Computing ESM2 embeddings: 28%|          | 256/899 [00:06<00:17, 36.69it/s]
Computing ESM2 embeddings: 32%|          | 288/899 [00:07<00:16, 37.51it/s]
Computing ESM2 embeddings: 36%|          | 320/899 [00:08<00:15, 36.96it/s]
Computing ESM2 embeddings: 39%|          | 352/899 [00:09<00:14, 37.74it/s]
Computing ESM2 embeddings: 43%|          | 384/899 [00:10<00:13, 37.17it/s]
Computing ESM2 embeddings: 46%|          | 416/899 [00:11<00:12, 37.82it/s]
Computing ESM2 embeddings: 50%|          | 448/899 [00:11<00:12, 37.22it/s]
Computing ESM2 embeddings: 53%|          | 480/899 [00:12<00:11, 37.84it/s]
Computing ESM2 embeddings: 57%|          | 512/899 [00:13<00:10, 37.27it/s]
Computing ESM2 embeddings: 61%|          | 544/899 [00:14<00:09, 37.93it/s]
Computing ESM2 embeddings: 64%|          | 576/899 [00:15<00:08, 37.31it/s]
Computing ESM2 embeddings: 68%|          | 608/899 [00:16<00:07, 37.74it/s]
Computing ESM2 embeddings: 71%|          | 640/899 [00:17<00:07, 36.53it/s]
Computing ESM2 embeddings: 75%|          | 672/899 [00:17<00:06, 37.38it/s]
Computing ESM2 embeddings: 78%|          | 704/899 [00:18<00:05, 37.04it/s]
Computing ESM2 embeddings: 82%|          | 736/899 [00:19<00:04, 37.71it/s]
Computing ESM2 embeddings: 85%|          | 768/899 [00:20<00:03, 37.22it/s]
Computing ESM2 embeddings: 89%|          | 800/899 [00:21<00:02, 37.87it/s]
Computing ESM2 embeddings: 93%|          | 832/899 [00:22<00:01, 37.29it/s]
Computing ESM2 embeddings: 96%|          | 864/899 [00:23<00:00, 37.90it/s]

Computing ESM2 embeddings: 100%|         | 899/899 [00:24<00:00, 37.41it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.91it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:23, 36.60it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.84it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 37.01it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 37.53it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.43it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.45it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.05it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.73it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 37.20it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.78it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.31it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.64it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 36.52it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.40it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.05it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.70it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.20it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.86it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.31it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 37.87it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 37.25it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:19<00:04, 37.87it/s] |

```

Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.27it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.87it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.16it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.61it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.41it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.28it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/895 [00:01<00:37, 23.10it/s]
Computing ESM2 embeddings:  7%|      | 64/895 [00:02<00:27, 30.50it/s]
Computing ESM2 embeddings: 11%|      | 96/895 [00:03<00:23, 34.13it/s]
Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:21, 34.90it/s]
Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:20, 36.40it/s]
Computing ESM2 embeddings: 21%|      | 192/895 [00:05<00:19, 36.26it/s]
Computing ESM2 embeddings: 25%|      | 224/895 [00:06<00:17, 37.34it/s]
Computing ESM2 embeddings: 29%|      | 256/895 [00:07<00:17, 36.86it/s]
Computing ESM2 embeddings: 32%|      | 288/895 [00:08<00:16, 37.50it/s]
Computing ESM2 embeddings: 36%|      | 320/895 [00:09<00:15, 36.43it/s]
Computing ESM2 embeddings: 39%|      | 352/895 [00:09<00:14, 37.26it/s]
Computing ESM2 embeddings: 43%|      | 384/895 [00:10<00:13, 37.07it/s]
Computing ESM2 embeddings: 46%|      | 416/895 [00:11<00:12, 37.95it/s]
Computing ESM2 embeddings: 50%|      | 448/895 [00:12<00:11, 37.34it/s]

```

Computing ESM2 embeddings: 54%| | 480/895 [00:13<00:10, 37.90it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:14<00:10, 37.17it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.89it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.21it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.88it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 37.32it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:05, 38.03it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:19<00:05, 37.11it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:20<00:04, 37.88it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 37.19it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.97it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.31it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.95it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 36.69it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.14it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.62it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.67it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.90it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 37.89it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.14it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/914 [00:05<00:18, 37.87it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:17, 37.21it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 37.95it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:15, 37.29it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 38.00it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:14, 37.28it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:11<00:13, 37.78it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:11<00:12, 36.99it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.77it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 37.30it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 38.07it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:15<00:09, 37.24it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:16<00:08, 37.86it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:17<00:07, 37.31it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 37.99it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 37.33it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:19<00:04, 37.92it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:20<00:03, 37.34it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:21<00:02, 38.00it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:22<00:02, 37.32it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:22<00:01, 38.05it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:23<00:00, 37.18it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:24<00:00, 37.43it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/574 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 6%| | 32/574 [00:00<00:14, 36.85it/s]  
Computing ESM2 embeddings: 11%| | 64/574 [00:01<00:14, 36.24it/s]  
Computing ESM2 embeddings: 17%| | 96/574 [00:02<00:12, 37.88it/s]  
Computing ESM2 embeddings: 22%| | 128/574 [00:03<00:12, 37.06it/s]  
Computing ESM2 embeddings: 28%| | 160/574 [00:04<00:10, 38.00it/s]  
Computing ESM2 embeddings: 33%| | 192/574 [00:05<00:10, 37.13it/s]  
Computing ESM2 embeddings: 39%| | 224/574 [00:05<00:09, 37.59it/s]  
Computing ESM2 embeddings: 45%| | 256/574 [00:06<00:08, 36.46it/s]  
Computing ESM2 embeddings: 50%| | 288/574 [00:07<00:07, 37.42it/s]  
Computing ESM2 embeddings: 56%| | 320/574 [00:08<00:06, 36.91it/s]  
Computing ESM2 embeddings: 61%| | 352/574 [00:09<00:05, 37.66it/s]  
Computing ESM2 embeddings: 67%| | 384/574 [00:10<00:05, 37.11it/s]  
Computing ESM2 embeddings: 72%| | 416/574 [00:11<00:04, 37.60it/s]  
Computing ESM2 embeddings: 78%| | 448/574 [00:12<00:03, 36.60it/s]  
Computing ESM2 embeddings: 84%| | 480/574 [00:12<00:02, 37.50it/s]  
Computing ESM2 embeddings: 89%| | 512/574 [00:13<00:01, 36.98it/s]  
Computing ESM2 embeddings: 95%| | 544/574 [00:14<00:00, 37.54it/s]

Computing ESM2 embeddings: 100%| | 574/574 [00:15<00:00, 36.98it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/911 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/911 [00:00<00:24, 35.30it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/911 [00:01<00:24, 34.55it/s]  |
| Computing ESM2 embeddings: | 11% | 96/911 [00:02<00:22, 36.65it/s]  |
| Computing ESM2 embeddings: | 14% | 128/911 [00:03<00:21, 36.25it/s] |
| Computing ESM2 embeddings: | 18% | 160/911 [00:04<00:20, 37.14it/s] |
| Computing ESM2 embeddings: | 21% | 192/911 [00:05<00:19, 36.06it/s] |
| Computing ESM2 embeddings: | 25% | 224/911 [00:06<00:18, 37.21it/s] |
| Computing ESM2 embeddings: | 28% | 256/911 [00:07<00:17, 36.74it/s] |
| Computing ESM2 embeddings: | 32% | 288/911 [00:07<00:16, 37.59it/s] |
| Computing ESM2 embeddings: | 35% | 320/911 [00:08<00:16, 36.84it/s] |
| Computing ESM2 embeddings: | 39% | 352/911 [00:09<00:14, 37.44it/s] |
| Computing ESM2 embeddings: | 42% | 384/911 [00:10<00:14, 36.38it/s] |
| Computing ESM2 embeddings: | 46% | 416/911 [00:11<00:13, 37.31it/s] |
| Computing ESM2 embeddings: | 49% | 448/911 [00:12<00:12, 36.67it/s] |
| Computing ESM2 embeddings: | 53% | 480/911 [00:13<00:11, 37.44it/s] |
| Computing ESM2 embeddings: | 56% | 512/911 [00:13<00:10, 36.83it/s] |
| Computing ESM2 embeddings: | 60% | 544/911 [00:14<00:09, 37.67it/s] |
| Computing ESM2 embeddings: | 63% | 576/911 [00:15<00:09, 36.99it/s] |
| Computing ESM2 embeddings: | 67% | 608/911 [00:16<00:08, 37.79it/s] |
| Computing ESM2 embeddings: | 70% | 640/911 [00:17<00:07, 37.14it/s] |
| Computing ESM2 embeddings: | 74% | 672/911 [00:18<00:06, 37.37it/s] |
| Computing ESM2 embeddings: | 77% | 704/911 [00:19<00:05, 36.35it/s] |
| Computing ESM2 embeddings: | 81% | 736/911 [00:19<00:04, 37.27it/s] |



```

Computing ESM2 embeddings: 84%|      | 768/911 [00:20<00:03, 36.83it/s]
Computing ESM2 embeddings: 88%|      | 800/911 [00:21<00:02, 37.67it/s]
Computing ESM2 embeddings: 91%|      | 832/911 [00:22<00:02, 36.91it/s]
Computing ESM2 embeddings: 95%|      | 864/911 [00:23<00:01, 37.73it/s]
Computing ESM2 embeddings: 98%|      | 896/911 [00:24<00:00, 37.09it/s]

Computing ESM2 embeddings: 100%|     | 911/911 [00:24<00:00, 36.89it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.35it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:22, 36.35it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.96it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 37.08it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.96it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:18, 37.16it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 37.83it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 37.08it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.57it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 36.41it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.07it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:14, 36.26it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.34it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:12<00:12, 36.80it/s]

```

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.54it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.08it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.84it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.26it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.89it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.25it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.83it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.24it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.65it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.45it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.39it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.93it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.70it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.15it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:01<00:28, 30.82it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:25, 32.77it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:22, 35.55it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:21, 35.77it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:20, 37.06it/s]

Computing ESM2 embeddings: 21%| | 192/913 [00:05<00:19, 36.67it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/913 [00:06<00:18, 37.19it/s] |
| Computing ESM2 embeddings: 28%  | 256/913 [00:07<00:18, 36.24it/s] |
| Computing ESM2 embeddings: 32%  | 288/913 [00:07<00:16, 37.26it/s] |
| Computing ESM2 embeddings: 35%  | 320/913 [00:08<00:16, 36.76it/s] |
| Computing ESM2 embeddings: 39%  | 352/913 [00:09<00:14, 37.59it/s] |
| Computing ESM2 embeddings: 42%  | 384/913 [00:10<00:14, 37.01it/s] |
| Computing ESM2 embeddings: 46%  | 416/913 [00:11<00:13, 37.71it/s] |
| Computing ESM2 embeddings: 49%  | 448/913 [00:12<00:12, 37.17it/s] |
| Computing ESM2 embeddings: 53%  | 480/913 [00:13<00:11, 37.89it/s] |
| Computing ESM2 embeddings: 56%  | 512/913 [00:13<00:10, 37.27it/s] |
| Computing ESM2 embeddings: 60%  | 544/913 [00:14<00:09, 38.01it/s] |
| Computing ESM2 embeddings: 63%  | 576/913 [00:15<00:09, 37.25it/s] |
| Computing ESM2 embeddings: 67%  | 608/913 [00:16<00:08, 37.90it/s] |
| Computing ESM2 embeddings: 70%  | 640/913 [00:17<00:07, 37.39it/s] |
| Computing ESM2 embeddings: 74%  | 672/913 [00:18<00:06, 37.71it/s] |
| Computing ESM2 embeddings: 77%  | 704/913 [00:19<00:05, 36.59it/s] |
| Computing ESM2 embeddings: 81%  | 736/913 [00:19<00:04, 37.35it/s] |
| Computing ESM2 embeddings: 84%  | 768/913 [00:20<00:03, 36.71it/s] |
| Computing ESM2 embeddings: 88%  | 800/913 [00:21<00:03, 37.61it/s] |
| Computing ESM2 embeddings: 91%  | 832/913 [00:22<00:02, 37.05it/s] |
| Computing ESM2 embeddings: 95%  | 864/913 [00:23<00:01, 37.73it/s] |
| Computing ESM2 embeddings: 98%  | 896/913 [00:24<00:00, 37.15it/s] |
| Computing ESM2 embeddings: 100% | 913/913 [00:24<00:00, 36.89it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/898 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/898 [00:00<00:26, 32.98it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/898 [00:01<00:23, 35.06it/s]  |
| Computing ESM2 embeddings: | 11% | 96/898 [00:02<00:21, 37.15it/s]  |
| Computing ESM2 embeddings: | 14% | 128/898 [00:03<00:21, 36.65it/s] |
| Computing ESM2 embeddings: | 18% | 160/898 [00:04<00:19, 37.31it/s] |
| Computing ESM2 embeddings: | 21% | 192/898 [00:05<00:19, 36.13it/s] |
| Computing ESM2 embeddings: | 25% | 224/898 [00:06<00:18, 37.14it/s] |
| Computing ESM2 embeddings: | 29% | 256/898 [00:07<00:17, 36.65it/s] |
| Computing ESM2 embeddings: | 32% | 288/898 [00:07<00:16, 37.60it/s] |
| Computing ESM2 embeddings: | 36% | 320/898 [00:08<00:15, 36.94it/s] |
| Computing ESM2 embeddings: | 39% | 352/898 [00:09<00:14, 37.71it/s] |
| Computing ESM2 embeddings: | 43% | 384/898 [00:10<00:13, 37.10it/s] |
| Computing ESM2 embeddings: | 46% | 416/898 [00:11<00:12, 37.63it/s] |
| Computing ESM2 embeddings: | 50% | 448/898 [00:12<00:12, 36.63it/s] |
| Computing ESM2 embeddings: | 53% | 480/898 [00:12<00:11, 37.50it/s] |
| Computing ESM2 embeddings: | 57% | 512/898 [00:13<00:10, 36.98it/s] |
| Computing ESM2 embeddings: | 61% | 544/898 [00:14<00:09, 37.68it/s] |
| Computing ESM2 embeddings: | 64% | 576/898 [00:15<00:08, 37.02it/s] |
| Computing ESM2 embeddings: | 68% | 608/898 [00:16<00:07, 37.77it/s] |
| Computing ESM2 embeddings: | 71% | 640/898 [00:17<00:06, 37.06it/s] |
| Computing ESM2 embeddings: | 75% | 672/898 [00:18<00:05, 37.80it/s] |

Computing ESM2 embeddings: 78%| | 704/898 [00:18<00:05, 37.25it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:19<00:04, 37.82it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:20<00:03, 37.07it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:21<00:02, 37.79it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:22<00:01, 36.94it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:23<00:00, 37.64it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:24<00:00, 37.02it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.67it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:26, 32.65it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 35.45it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 35.13it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.72it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.36it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.43it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 37.02it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.87it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 37.38it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.84it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.69it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.63it/s]

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Computing ESM2 embeddings: 49%|          | 448/914 [00:12<00:12, 37.04it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 37.81it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 37.19it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 37.85it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.21it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.95it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.25it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.96it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 37.37it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 38.09it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.39it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.82it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 36.76it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.64it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.16it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.19it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.50it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:23, 35.73it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 37.39it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 36.77it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:19, 37.72it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:18, 37.14it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 37.70it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:17, 36.41it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:16, 37.43it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 36.88it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 37.67it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 37.15it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:12, 37.83it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:12<00:12, 37.21it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 37.88it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 37.30it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 37.71it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 36.54it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 37.49it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 37.13it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 37.82it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:05, 37.25it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 37.93it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:20<00:03, 37.21it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 37.82it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 37.12it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:23<00:00, 37.83it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:24<00:00, 37.32it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.26it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:22, 37.39it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 38.32it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 37.12it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 37.90it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.29it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 38.08it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.44it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.08it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 37.39it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 38.10it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.48it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.80it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 36.65it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.49it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 36.96it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.74it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.17it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.57it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 36.50it/s] |



```

Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 37.40it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 36.92it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 37.43it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 36.52it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.42it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.76it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.64it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 36.99it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.28it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/571 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/571 [00:00<00:14, 37.89it/s]
Computing ESM2 embeddings: 11%|      | 64/571 [00:01<00:15, 32.89it/s]
Computing ESM2 embeddings: 17%|      | 96/571 [00:02<00:13, 35.63it/s]
Computing ESM2 embeddings: 22%|      | 128/571 [00:03<00:12, 35.57it/s]
Computing ESM2 embeddings: 28%|      | 160/571 [00:04<00:11, 36.92it/s]
Computing ESM2 embeddings: 34%|      | 192/571 [00:05<00:10, 36.50it/s]
Computing ESM2 embeddings: 39%|      | 224/571 [00:06<00:09, 37.51it/s]
Computing ESM2 embeddings: 45%|      | 256/571 [00:07<00:08, 36.87it/s]
Computing ESM2 embeddings: 50%|      | 288/571 [00:07<00:07, 37.58it/s]
Computing ESM2 embeddings: 56%|      | 320/571 [00:08<00:06, 36.97it/s]
Computing ESM2 embeddings: 62%|      | 352/571 [00:09<00:05, 37.81it/s]

```

Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:05, 37.12it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 37.84it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:12<00:03, 37.10it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 37.75it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 37.13it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 37.56it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 36.74it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.95it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 34.30it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.73it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.30it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.24it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.75it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.31it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.21it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 36.97it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.39it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.33it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.05it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.90it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.17it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.81it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.09it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.52it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.43it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.37it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.90it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.66it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 37.13it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.88it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.22it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.87it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.19it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.90it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.04it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.05it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 37.72it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:23, 35.47it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:21, 37.21it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:20, 36.66it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/895 [00:04<00:19, 37.73it/s] |
| Computing ESM2 embeddings: 21%  | 192/895 [00:05<00:18, 37.08it/s] |
| Computing ESM2 embeddings: 25%  | 224/895 [00:05<00:17, 37.89it/s] |
| Computing ESM2 embeddings: 29%  | 256/895 [00:06<00:17, 37.19it/s] |
| Computing ESM2 embeddings: 32%  | 288/895 [00:07<00:16, 37.72it/s] |
| Computing ESM2 embeddings: 36%  | 320/895 [00:08<00:15, 36.24it/s] |
| Computing ESM2 embeddings: 39%  | 352/895 [00:09<00:14, 37.28it/s] |
| Computing ESM2 embeddings: 43%  | 384/895 [00:10<00:13, 37.04it/s] |
| Computing ESM2 embeddings: 46%  | 416/895 [00:11<00:12, 37.86it/s] |
| Computing ESM2 embeddings: 50%  | 448/895 [00:12<00:11, 37.31it/s] |
| Computing ESM2 embeddings: 54%  | 480/895 [00:12<00:10, 37.78it/s] |
| Computing ESM2 embeddings: 57%  | 512/895 [00:13<00:10, 36.72it/s] |
| Computing ESM2 embeddings: 61%  | 544/895 [00:14<00:09, 37.58it/s] |
| Computing ESM2 embeddings: 64%  | 576/895 [00:15<00:08, 37.03it/s] |
| Computing ESM2 embeddings: 68%  | 608/895 [00:16<00:07, 37.49it/s] |
| Computing ESM2 embeddings: 72%  | 640/895 [00:17<00:07, 36.40it/s] |
| Computing ESM2 embeddings: 75%  | 672/895 [00:18<00:05, 37.27it/s] |
| Computing ESM2 embeddings: 79%  | 704/895 [00:18<00:05, 36.87it/s] |
| Computing ESM2 embeddings: 82%  | 736/895 [00:19<00:04, 37.68it/s] |
| Computing ESM2 embeddings: 86%  | 768/895 [00:20<00:03, 37.12it/s] |
| Computing ESM2 embeddings: 89%  | 800/895 [00:21<00:02, 38.00it/s] |
| Computing ESM2 embeddings: 93%  | 832/895 [00:22<00:01, 37.35it/s] |
| Computing ESM2 embeddings: 97%  | 864/895 [00:23<00:00, 38.05it/s] |
| Computing ESM2 embeddings: 100% | 895/895 [00:24<00:00, 37.17it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/914 [00:00<00:23, 37.95it/s]
Computing ESM2 embeddings: 7%|          | 64/914 [00:01<00:24, 34.76it/s]
Computing ESM2 embeddings: 11%|         | 96/914 [00:02<00:22, 36.62it/s]
Computing ESM2 embeddings: 14%|         | 128/914 [00:03<00:21, 36.38it/s]
Computing ESM2 embeddings: 18%|         | 160/914 [00:04<00:20, 37.42it/s]
Computing ESM2 embeddings: 21%|         | 192/914 [00:05<00:19, 36.92it/s]
Computing ESM2 embeddings: 25%|         | 224/914 [00:06<00:18, 37.33it/s]
Computing ESM2 embeddings: 28%|         | 256/914 [00:07<00:18, 36.18it/s]
Computing ESM2 embeddings: 32%|         | 288/914 [00:07<00:17, 36.82it/s]
Computing ESM2 embeddings: 35%|         | 320/914 [00:08<00:16, 35.81it/s]
Computing ESM2 embeddings: 39%|         | 352/914 [00:09<00:15, 36.77it/s]
Computing ESM2 embeddings: 42%|         | 384/914 [00:10<00:14, 36.60it/s]
Computing ESM2 embeddings: 46%|         | 416/914 [00:11<00:13, 37.48it/s]
Computing ESM2 embeddings: 49%|         | 448/914 [00:12<00:12, 36.97it/s]
Computing ESM2 embeddings: 53%|         | 480/914 [00:12<00:11, 37.73it/s]
Computing ESM2 embeddings: 56%|         | 512/914 [00:13<00:10, 37.13it/s]
Computing ESM2 embeddings: 60%|         | 544/914 [00:14<00:09, 37.77it/s]
Computing ESM2 embeddings: 63%|         | 576/914 [00:15<00:09, 37.29it/s]
Computing ESM2 embeddings: 67%|         | 608/914 [00:16<00:08, 37.68it/s]
Computing ESM2 embeddings: 70%|         | 640/914 [00:17<00:07, 36.50it/s]
```

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:18<00:06, 37.28it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:19<00:05, 36.92it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 37.63it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.00it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.75it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.06it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.74it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.13it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.96it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/898 [00:00<00:22, 38.09it/s]
Computing ESM2 embeddings: 7%|      | 64/898 [00:01<00:25, 33.10it/s]
Computing ESM2 embeddings: 11%|     | 96/898 [00:02<00:22, 35.53it/s]
Computing ESM2 embeddings: 14%|     | 128/898 [00:03<00:21, 35.54it/s]
Computing ESM2 embeddings: 18%|     | 160/898 [00:04<00:19, 36.92it/s]
Computing ESM2 embeddings: 21%|     | 192/898 [00:05<00:19, 36.29it/s]
Computing ESM2 embeddings: 25%|     | 224/898 [00:06<00:18, 36.87it/s]
Computing ESM2 embeddings: 29%|     | 256/898 [00:07<00:17, 35.89it/s]
Computing ESM2 embeddings: 32%|     | 288/898 [00:07<00:16, 36.93it/s]
Computing ESM2 embeddings: 36%|     | 320/898 [00:08<00:15, 36.37it/s]
Computing ESM2 embeddings: 39%|     | 352/898 [00:09<00:14, 37.28it/s]

```

```

Computing ESM2 embeddings: 43%|          | 384/898 [00:10<00:14, 36.57it/s]
Computing ESM2 embeddings: 46%|          | 416/898 [00:11<00:12, 37.38it/s]
Computing ESM2 embeddings: 50%|          | 448/898 [00:12<00:12, 36.69it/s]
Computing ESM2 embeddings: 53%|          | 480/898 [00:13<00:11, 37.44it/s]
Computing ESM2 embeddings: 57%|          | 512/898 [00:14<00:10, 36.82it/s]
Computing ESM2 embeddings: 61%|          | 544/898 [00:14<00:09, 37.63it/s]
Computing ESM2 embeddings: 64%|          | 576/898 [00:15<00:08, 37.06it/s]
Computing ESM2 embeddings: 68%|          | 608/898 [00:16<00:07, 37.78it/s]
Computing ESM2 embeddings: 71%|          | 640/898 [00:17<00:06, 36.91it/s]
Computing ESM2 embeddings: 75%|          | 672/898 [00:18<00:06, 37.58it/s]
Computing ESM2 embeddings: 78%|          | 704/898 [00:19<00:05, 37.30it/s]
Computing ESM2 embeddings: 82%|          | 736/898 [00:19<00:04, 37.95it/s]
Computing ESM2 embeddings: 86%|          | 768/898 [00:20<00:03, 37.20it/s]
Computing ESM2 embeddings: 89%|          | 800/898 [00:21<00:02, 37.74it/s]
Computing ESM2 embeddings: 93%|          | 832/898 [00:22<00:01, 37.11it/s]
Computing ESM2 embeddings: 96%|          | 864/898 [00:23<00:00, 37.49it/s]

Computing ESM2 embeddings: 100%|         | 898/898 [00:24<00:00, 36.71it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 38.19it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:23, 36.59it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.80it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 36.82it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 37.85it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.29it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.94it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 37.38it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 38.10it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.28it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 38.04it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.28it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 38.07it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 37.25it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.62it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 36.72it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.53it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 36.80it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.65it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.05it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.70it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.20it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.84it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.27it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.95it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.07it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:23<00:01, 37.83it/s] |



Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.01it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.44it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:01<00:33, 25.93it/s]

Computing ESM2 embeddings: 7%| | 64/897 [00:02<00:26, 31.66it/s]

Computing ESM2 embeddings: 11%| | 96/897 [00:02<00:22, 34.87it/s]

Computing ESM2 embeddings: 14%| | 128/897 [00:03<00:21, 35.15it/s]

Computing ESM2 embeddings: 18%| | 160/897 [00:04<00:20, 36.68it/s]

Computing ESM2 embeddings: 21%| | 192/897 [00:05<00:19, 36.35it/s]

Computing ESM2 embeddings: 25%| | 224/897 [00:06<00:18, 37.36it/s]

Computing ESM2 embeddings: 29%| | 256/897 [00:07<00:17, 36.61it/s]

Computing ESM2 embeddings: 32%| | 288/897 [00:08<00:16, 37.39it/s]

Computing ESM2 embeddings: 36%| | 320/897 [00:08<00:15, 37.01it/s]

Computing ESM2 embeddings: 39%| | 352/897 [00:09<00:14, 37.51it/s]

Computing ESM2 embeddings: 43%| | 384/897 [00:10<00:14, 36.41it/s]

Computing ESM2 embeddings: 46%| | 416/897 [00:11<00:12, 37.26it/s]

Computing ESM2 embeddings: 50%| | 448/897 [00:12<00:12, 36.79it/s]

Computing ESM2 embeddings: 54%| | 480/897 [00:13<00:11, 37.54it/s]

Computing ESM2 embeddings: 57%| | 512/897 [00:14<00:10, 36.52it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:14<00:09, 37.38it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:15<00:08, 36.72it/s]

```

Computing ESM2 embeddings: 68%|      | 608/897 [00:16<00:07, 37.53it/s]
Computing ESM2 embeddings: 71%|      | 640/897 [00:17<00:06, 36.87it/s]
Computing ESM2 embeddings: 75%|      | 672/897 [00:18<00:05, 37.53it/s]
Computing ESM2 embeddings: 78%|      | 704/897 [00:19<00:05, 36.94it/s]
Computing ESM2 embeddings: 82%|      | 736/897 [00:20<00:04, 37.25it/s]
Computing ESM2 embeddings: 86%|      | 768/897 [00:21<00:03, 36.02it/s]
Computing ESM2 embeddings: 89%|      | 800/897 [00:21<00:02, 37.01it/s]
Computing ESM2 embeddings: 93%|      | 832/897 [00:22<00:01, 36.66it/s]
Computing ESM2 embeddings: 96%|      | 864/897 [00:23<00:00, 37.51it/s]

Computing ESM2 embeddings: 100%|     | 897/897 [00:24<00:00, 36.49it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/913 [00:00<00:23, 37.63it/s]
Computing ESM2 embeddings:  7%|      | 64/913 [00:01<00:23, 35.55it/s]
Computing ESM2 embeddings: 11%|      | 96/913 [00:02<00:22, 36.76it/s]
Computing ESM2 embeddings: 14%|      | 128/913 [00:03<00:21, 35.77it/s]
Computing ESM2 embeddings: 18%|      | 160/913 [00:04<00:20, 37.12it/s]
Computing ESM2 embeddings: 21%|      | 192/913 [00:05<00:19, 36.64it/s]
Computing ESM2 embeddings: 25%|      | 224/913 [00:06<00:18, 37.50it/s]
Computing ESM2 embeddings: 28%|      | 256/913 [00:06<00:17, 36.90it/s]
Computing ESM2 embeddings: 32%|      | 288/913 [00:07<00:16, 37.38it/s]
Computing ESM2 embeddings: 35%|      | 320/913 [00:08<00:16, 36.20it/s]

```

Computing ESM2 embeddings: 39%| | 352/913 [00:09<00:15, 37.22it/s]

Computing ESM2 embeddings: 42%| | 384/913 [00:10<00:14, 36.80it/s]

Computing ESM2 embeddings: 46%| | 416/913 [00:11<00:13, 37.62it/s]

Computing ESM2 embeddings: 49%| | 448/913 [00:12<00:12, 37.06it/s]

Computing ESM2 embeddings: 53%| | 480/913 [00:12<00:11, 37.83it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 37.16it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 37.87it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:15<00:09, 37.25it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:16<00:08, 37.96it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:17<00:07, 37.28it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:18<00:06, 37.94it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:18<00:05, 37.28it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:19<00:04, 37.61it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:20<00:03, 36.50it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:21<00:03, 37.39it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 36.94it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:23<00:01, 37.63it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:24<00:00, 37.11it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:24<00:00, 37.15it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:14, 36.89it/s]

Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:14, 35.02it/s]

Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:12, 37.10it/s]

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:12, 36.56it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:10, 37.67it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:05<00:10, 37.18it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:06<00:09, 37.97it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 37.27it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 38.05it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 37.51it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 38.18it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:04, 37.43it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 38.14it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:11<00:03, 37.53it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 38.21it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 37.59it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 38.24it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 37.46it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.14it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 36.92it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.73it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:21, 36.43it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.31it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 36.26it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:18, 37.06it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:17, 36.05it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.03it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 36.75it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.68it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.28it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.82it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 36.50it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:11, 37.47it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.02it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.73it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.21it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.96it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:06, 37.29it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:05, 38.01it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 37.34it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 38.07it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.36it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 38.07it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.35it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 38.08it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.23it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.93it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.82it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.01it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 37.02it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 37.95it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.38it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 38.17it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 37.39it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.15it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 37.26it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.99it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.34it/s]  
 Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 38.09it/s]  
 Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 37.42it/s]  
 Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.85it/s]  
 Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 36.63it/s]  
 Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.52it/s]  
 Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.92it/s]  
 Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.65it/s]

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Computing ESM2 embeddings: 70%|      | 640/914 [00:17<00:07, 37.17it/s]
Computing ESM2 embeddings: 74%|      | 672/914 [00:17<00:06, 37.91it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 37.31it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 37.66it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 36.73it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.64it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.15it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.95it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 37.29it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.41it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/896 [00:00<00:23, 37.35it/s]
Computing ESM2 embeddings: 7%|      | 64/896 [00:01<00:23, 35.23it/s]
Computing ESM2 embeddings: 11%|     | 96/896 [00:02<00:21, 37.22it/s]
Computing ESM2 embeddings: 14%|     | 128/896 [00:03<00:20, 36.72it/s]
Computing ESM2 embeddings: 18%|     | 160/896 [00:04<00:19, 37.85it/s]
Computing ESM2 embeddings: 21%|     | 192/896 [00:05<00:18, 37.29it/s]
Computing ESM2 embeddings: 25%|     | 224/896 [00:05<00:17, 38.05it/s]
Computing ESM2 embeddings: 29%|     | 256/896 [00:06<00:17, 37.30it/s]
Computing ESM2 embeddings: 32%|     | 288/896 [00:07<00:15, 38.06it/s]
Computing ESM2 embeddings: 36%|     | 320/896 [00:08<00:15, 37.32it/s]

```

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.63it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.76it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.79it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.22it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 37.91it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.45it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.12it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.35it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 38.07it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.40it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.01it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.30it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.02it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.46it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 38.03it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.44it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:22<00:00, 38.12it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.40it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:23, 37.60it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:23, 36.85it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 10% | 96/915 [00:02<00:21, 38.36it/s]  |
| Computing ESM2 embeddings: 14% | 128/915 [00:03<00:21, 37.13it/s] |
| Computing ESM2 embeddings: 17% | 160/915 [00:04<00:19, 37.92it/s] |
| Computing ESM2 embeddings: 21% | 192/915 [00:05<00:19, 37.20it/s] |
| Computing ESM2 embeddings: 24% | 224/915 [00:05<00:18, 38.03it/s] |
| Computing ESM2 embeddings: 28% | 256/915 [00:06<00:17, 37.54it/s] |
| Computing ESM2 embeddings: 31% | 288/915 [00:07<00:16, 38.30it/s] |
| Computing ESM2 embeddings: 35% | 320/915 [00:08<00:15, 37.42it/s] |
| Computing ESM2 embeddings: 38% | 352/915 [00:09<00:14, 38.00it/s] |
| Computing ESM2 embeddings: 42% | 384/915 [00:10<00:14, 37.33it/s] |
| Computing ESM2 embeddings: 45% | 416/915 [00:11<00:13, 38.04it/s] |
| Computing ESM2 embeddings: 49% | 448/915 [00:11<00:12, 37.33it/s] |
| Computing ESM2 embeddings: 52% | 480/915 [00:12<00:11, 38.06it/s] |
| Computing ESM2 embeddings: 56% | 512/915 [00:13<00:10, 37.36it/s] |
| Computing ESM2 embeddings: 59% | 544/915 [00:14<00:09, 38.06it/s] |
| Computing ESM2 embeddings: 63% | 576/915 [00:15<00:09, 37.46it/s] |
| Computing ESM2 embeddings: 66% | 608/915 [00:16<00:08, 38.04it/s] |
| Computing ESM2 embeddings: 70% | 640/915 [00:16<00:07, 37.43it/s] |
| Computing ESM2 embeddings: 73% | 672/915 [00:17<00:06, 37.81it/s] |
| Computing ESM2 embeddings: 77% | 704/915 [00:18<00:05, 36.70it/s] |
| Computing ESM2 embeddings: 80% | 736/915 [00:19<00:04, 37.49it/s] |
| Computing ESM2 embeddings: 84% | 768/915 [00:20<00:03, 37.10it/s] |
| Computing ESM2 embeddings: 87% | 800/915 [00:21<00:03, 37.86it/s] |
| Computing ESM2 embeddings: 91% | 832/915 [00:22<00:02, 37.36it/s] |

```

Computing ESM2 embeddings: 94%|      | 864/915 [00:22<00:01, 38.03it/s]

Computing ESM2 embeddings: 98%|      | 896/915 [00:23<00:00, 37.35it/s]

Computing ESM2 embeddings: 100%|     | 915/915 [00:24<00:00, 37.51it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%|        | 32/896 [00:00<00:23, 36.80it/s]

Computing ESM2 embeddings: 7%|        | 64/896 [00:01<00:23, 35.82it/s]

Computing ESM2 embeddings: 11%|       | 96/896 [00:02<00:21, 37.71it/s]

Computing ESM2 embeddings: 14%|       | 128/896 [00:03<00:20, 37.12it/s]

Computing ESM2 embeddings: 18%|       | 160/896 [00:04<00:19, 37.92it/s]

Computing ESM2 embeddings: 21%|       | 192/896 [00:05<00:18, 37.24it/s]

Computing ESM2 embeddings: 25%|       | 224/896 [00:05<00:17, 37.78it/s]

Computing ESM2 embeddings: 29%|       | 256/896 [00:06<00:17, 36.67it/s]

Computing ESM2 embeddings: 32%|       | 288/896 [00:07<00:16, 37.48it/s]

Computing ESM2 embeddings: 36%|       | 320/896 [00:08<00:15, 37.08it/s]

Computing ESM2 embeddings: 39%|       | 352/896 [00:09<00:14, 37.88it/s]

Computing ESM2 embeddings: 43%|       | 384/896 [00:10<00:13, 37.30it/s]

Computing ESM2 embeddings: 46%|       | 416/896 [00:11<00:12, 38.05it/s]

Computing ESM2 embeddings: 50%|       | 448/896 [00:11<00:12, 37.33it/s]

Computing ESM2 embeddings: 54%|       | 480/896 [00:12<00:10, 38.05it/s]

Computing ESM2 embeddings: 57%|       | 512/896 [00:13<00:10, 37.45it/s]

Computing ESM2 embeddings: 61%|       | 544/896 [00:14<00:09, 38.23it/s]

```

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.57it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.82it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 36.58it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.53it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.10it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.88it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.25it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 38.03it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.38it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 38.10it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.39it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.48it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 35.53it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.97it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 35.67it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.96it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.69it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.36it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.18it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.29it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.94it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.50it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.34it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.26it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.91it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.64it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.13it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.84it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.21it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.86it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.18it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.64it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.70it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.55it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.05it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.73it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.17it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.85it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.25it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.05it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/895 [00:01<00:27, 31.89it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:24, 33.33it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:22, 35.95it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:21, 35.98it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 36.91it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:19, 36.04it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:06<00:18, 37.14it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:07<00:17, 36.71it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:16, 37.61it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 37.05it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 37.83it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:13, 37.14it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 37.88it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:12<00:12, 37.22it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:13<00:11, 37.53it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:10, 36.36it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.27it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 36.72it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 37.20it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:07, 36.18it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:18<00:06, 37.08it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:19<00:05, 36.60it/s] |
| Computing ESM2 embeddings: | 82% | 736/895 [00:19<00:04, 37.40it/s] |
| Computing ESM2 embeddings: | 86% | 768/895 [00:20<00:03, 36.82it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/895 [00:21<00:02, 37.58it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:22<00:01, 37.00it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:23<00:00, 37.68it/s]
Computing ESM2 embeddings: 100%|     | 895/895 [00:24<00:00, 36.78it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/589 [00:00<?, ?it/s]
Computing ESM2 embeddings: 5%|        | 32/589 [00:01<00:18, 29.48it/s]
Computing ESM2 embeddings: 11%|       | 64/589 [00:01<00:15, 34.38it/s]
Computing ESM2 embeddings: 16%|       | 96/589 [00:02<00:13, 36.76it/s]
Computing ESM2 embeddings: 22%|       | 128/589 [00:03<00:12, 36.41it/s]
Computing ESM2 embeddings: 27%|       | 160/589 [00:04<00:11, 37.53it/s]
Computing ESM2 embeddings: 33%|       | 192/589 [00:05<00:10, 36.83it/s]
Computing ESM2 embeddings: 38%|       | 224/589 [00:06<00:09, 37.39it/s]
Computing ESM2 embeddings: 43%|       | 256/589 [00:07<00:09, 36.33it/s]
Computing ESM2 embeddings: 49%|       | 288/589 [00:07<00:08, 37.33it/s]
Computing ESM2 embeddings: 54%|       | 320/589 [00:08<00:07, 36.86it/s]
Computing ESM2 embeddings: 60%|       | 352/589 [00:09<00:06, 37.71it/s]
Computing ESM2 embeddings: 65%|       | 384/589 [00:10<00:05, 37.07it/s]
Computing ESM2 embeddings: 71%|       | 416/589 [00:11<00:04, 37.80it/s]
Computing ESM2 embeddings: 76%|       | 448/589 [00:12<00:03, 37.23it/s]
Computing ESM2 embeddings: 81%|       | 480/589 [00:12<00:02, 37.84it/s]
Computing ESM2 embeddings: 87%|       | 512/589 [00:13<00:02, 37.25it/s]

```

```

Computing ESM2 embeddings: 92%|      | 544/589 [00:14<00:01, 37.88it/s]

Computing ESM2 embeddings: 98%|      | 576/589 [00:15<00:00, 37.26it/s]

Computing ESM2 embeddings: 100%|     | 589/589 [00:16<00:00, 36.78it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:23, 37.45it/s]

Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:23, 35.85it/s]

Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.67it/s]

Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 36.81it/s]

Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.78it/s]

Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 36.89it/s]

Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 37.74it/s]

Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 37.08it/s]

Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.84it/s]

Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 37.20it/s]

Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.88it/s]

Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.26it/s]

Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.95it/s]

Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:12, 37.33it/s]

Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 37.96it/s]

Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:10, 37.17it/s]

Computing ESM2 embeddings: 61%|      | 544/896 [00:14<00:09, 37.86it/s]

```

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.25it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.91it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.22it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.89it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.06it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.76it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.16it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.87it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.23it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.89it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.36it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:26, 32.79it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:25, 33.17it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 35.80it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 35.87it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.13it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.74it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.62it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 37.11it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.57it/s]



```

Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:16, 36.36it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:15, 37.31it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 36.77it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.37it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:12<00:12, 36.50it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:13<00:11, 37.35it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 36.92it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 37.65it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 36.99it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.46it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 36.16it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.07it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 36.73it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.48it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 36.92it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.63it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.09it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.76it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.31it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 36.83it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          | 0/896 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.00it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:22, 36.80it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.99it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 36.91it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.84it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 37.12it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 37.56it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 36.39it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.36it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 36.88it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.65it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.18it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.85it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.32it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.01it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.29it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.93it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.21it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.85it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.29it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.93it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 37.13it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.80it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 37.18it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.59it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 36.49it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.07it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.15it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
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for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/916 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/916 [00:00<00:25, 35.05it/s]
Computing ESM2 embeddings:  7%|      | 64/916 [00:01<00:24, 35.48it/s]
Computing ESM2 embeddings: 10%|      | 96/916 [00:02<00:22, 37.20it/s]
Computing ESM2 embeddings: 14%|      | 128/916 [00:03<00:21, 36.43it/s]
Computing ESM2 embeddings: 17%|      | 160/916 [00:04<00:20, 37.50it/s]
Computing ESM2 embeddings: 21%|      | 192/916 [00:05<00:19, 36.92it/s]
Computing ESM2 embeddings: 24%|      | 224/916 [00:06<00:18, 37.71it/s]
Computing ESM2 embeddings: 28%|      | 256/916 [00:06<00:17, 37.12it/s]
Computing ESM2 embeddings: 31%|      | 288/916 [00:07<00:16, 37.81it/s]
Computing ESM2 embeddings: 35%|      | 320/916 [00:08<00:15, 37.26it/s]
Computing ESM2 embeddings: 38%|      | 352/916 [00:09<00:14, 37.88it/s]
Computing ESM2 embeddings: 42%|      | 384/916 [00:10<00:14, 37.10it/s]
Computing ESM2 embeddings: 45%|      | 416/916 [00:11<00:13, 37.82it/s]
Computing ESM2 embeddings: 49%|      | 448/916 [00:12<00:12, 37.17it/s]
Computing ESM2 embeddings: 52%|      | 480/916 [00:12<00:11, 37.52it/s]
Computing ESM2 embeddings: 56%|      | 512/916 [00:13<00:11, 36.49it/s]

```

Computing ESM2 embeddings: 59%| | 544/916 [00:14<00:09, 37.34it/s]

Computing ESM2 embeddings: 63%| | 576/916 [00:15<00:09, 37.01it/s]

Computing ESM2 embeddings: 66%| | 608/916 [00:16<00:08, 37.83it/s]

Computing ESM2 embeddings: 70%| | 640/916 [00:17<00:07, 37.18it/s]

Computing ESM2 embeddings: 73%| | 672/916 [00:18<00:06, 37.82it/s]

Computing ESM2 embeddings: 77%| | 704/916 [00:18<00:05, 37.23it/s]

Computing ESM2 embeddings: 80%| | 736/916 [00:19<00:04, 37.99it/s]

Computing ESM2 embeddings: 84%| | 768/916 [00:20<00:03, 37.22it/s]

Computing ESM2 embeddings: 87%| | 800/916 [00:21<00:03, 37.86it/s]

Computing ESM2 embeddings: 91%| | 832/916 [00:22<00:02, 37.06it/s]

Computing ESM2 embeddings: 94%| | 864/916 [00:23<00:01, 37.70it/s]

Computing ESM2 embeddings: 98%| | 896/916 [00:24<00:00, 37.01it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:24<00:00, 37.19it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.72it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:24, 33.89it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:22, 35.78it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 34.98it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:20, 36.48it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.35it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:17, 37.36it/s]

```

Computing ESM2 embeddings: 29%|          | 256/896 [00:07<00:17, 36.85it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.63it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 36.91it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 37.68it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 37.10it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 37.78it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:12<00:12, 37.00it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:11, 37.72it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.17it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.81it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.05it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.71it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.00it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:18<00:05, 37.37it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:19<00:05, 36.15it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 36.80it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 35.79it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 36.84it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 36.58it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.55it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 36.88it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:21, 39.44it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:22, 37.47it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:20, 38.38it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:20, 37.35it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 38.14it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:18, 37.35it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 38.04it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:17, 37.34it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:15, 38.01it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 37.30it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 37.59it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:14, 36.41it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 37.06it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:12<00:12, 36.10it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:11, 37.05it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:10, 36.65it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.42it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 37.03it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 37.74it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:06, 37.24it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:17<00:05, 37.90it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:18<00:05, 37.28it/s] |
| Computing ESM2 embeddings: | 82% | 736/895 [00:19<00:04, 37.92it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/895 [00:20<00:03, 37.25it/s]
Computing ESM2 embeddings: 89%|      | 800/895 [00:21<00:02, 37.60it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:22<00:01, 36.52it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:23<00:00, 37.15it/s]
Computing ESM2 embeddings: 100%|     | 895/895 [00:24<00:00, 37.14it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 38.17it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:02<00:33, 25.76it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:03<00:26, 30.47it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:04<00:24, 31.85it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:22, 34.24it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:20, 34.70it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:19, 36.13it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:07<00:18, 36.08it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:08<00:16, 37.04it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:09<00:16, 36.68it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:10<00:14, 37.48it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 36.87it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.30it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 36.27it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:13<00:11, 37.17it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 36.69it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:15<00:09, 37.49it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:16<00:09, 36.93it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.66it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.95it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.67it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 37.15it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 37.79it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:21<00:03, 37.06it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:22<00:03, 37.46it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:23<00:02, 36.43it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.28it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.87it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:25<00:00, 36.16it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:13, 38.55it/s]

Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:14, 36.04it/s]

Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:12, 37.29it/s]

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:12, 35.93it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:11, 37.10it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:05<00:10, 36.51it/s]



Computing ESM2 embeddings: 39%| | 224/571 [00:06<00:09, 37.11it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:07<00:08, 36.15it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 37.25it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 36.92it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 37.69it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:05, 37.04it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 37.85it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:12<00:03, 37.26it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 37.88it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 37.32it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 37.89it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 37.09it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.37it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:26, 31.73it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 34.59it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 34.76it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.31it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.17it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.17it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 36.77it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.59it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.83it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.56it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.10it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.77it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.19it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.85it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.11it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.81it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.05it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.69it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.01it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.69it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 37.15it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.50it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:04, 36.47it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.32it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.89it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.41it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.31it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.78it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:22, 38.43it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:23, 35.82it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:21, 37.48it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:20, 36.79it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 37.79it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:18, 37.10it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:06<00:17, 37.54it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:17, 36.43it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:16, 37.35it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 36.89it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 37.70it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:13, 37.13it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 37.84it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:12<00:11, 37.25it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 37.91it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:10, 37.28it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.98it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 37.28it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 38.04it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:06, 37.37it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:17<00:05, 37.78it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:18<00:05, 36.57it/s] |
| Computing ESM2 embeddings: | 82% | 736/895 [00:19<00:04, 37.43it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/895 [00:20<00:03, 36.98it/s]
Computing ESM2 embeddings: 89%|      | 800/895 [00:21<00:02, 37.73it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:22<00:01, 37.17it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:23<00:00, 37.85it/s]
Computing ESM2 embeddings: 100%|     | 895/895 [00:23<00:00, 37.30it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/911 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/911 [00:00<00:22, 39.22it/s]
Computing ESM2 embeddings:  7%|      | 64/911 [00:01<00:26, 32.06it/s]
Computing ESM2 embeddings: 11%|      | 96/911 [00:02<00:23, 35.05it/s]
Computing ESM2 embeddings: 14%|      | 128/911 [00:03<00:22, 34.90it/s]
Computing ESM2 embeddings: 18%|      | 160/911 [00:04<00:20, 36.41it/s]
Computing ESM2 embeddings: 21%|      | 192/911 [00:05<00:19, 36.35it/s]
Computing ESM2 embeddings: 25%|      | 224/911 [00:06<00:18, 37.35it/s]
Computing ESM2 embeddings: 28%|      | 256/911 [00:07<00:17, 36.82it/s]
Computing ESM2 embeddings: 32%|      | 288/911 [00:07<00:16, 37.64it/s]
Computing ESM2 embeddings: 35%|      | 320/911 [00:08<00:16, 36.94it/s]
Computing ESM2 embeddings: 39%|      | 352/911 [00:09<00:14, 37.64it/s]
Computing ESM2 embeddings: 42%|      | 384/911 [00:10<00:14, 36.87it/s]
Computing ESM2 embeddings: 46%|      | 416/911 [00:11<00:13, 37.60it/s]
Computing ESM2 embeddings: 49%|      | 448/911 [00:12<00:12, 36.97it/s]
Computing ESM2 embeddings: 53%|      | 480/911 [00:13<00:11, 37.68it/s]

```

Computing ESM2 embeddings: 56%| | 512/911 [00:13<00:10, 37.10it/s]

Computing ESM2 embeddings: 60%| | 544/911 [00:14<00:09, 37.79it/s]

Computing ESM2 embeddings: 63%| | 576/911 [00:15<00:09, 37.16it/s]

Computing ESM2 embeddings: 67%| | 608/911 [00:16<00:08, 37.85it/s]

Computing ESM2 embeddings: 70%| | 640/911 [00:17<00:07, 37.08it/s]

Computing ESM2 embeddings: 74%| | 672/911 [00:18<00:06, 37.79it/s]

Computing ESM2 embeddings: 77%| | 704/911 [00:19<00:05, 37.22it/s]

Computing ESM2 embeddings: 81%| | 736/911 [00:19<00:04, 37.61it/s]

Computing ESM2 embeddings: 84%| | 768/911 [00:20<00:03, 36.36it/s]

Computing ESM2 embeddings: 88%| | 800/911 [00:21<00:02, 37.22it/s]

Computing ESM2 embeddings: 91%| | 832/911 [00:22<00:02, 36.90it/s]

Computing ESM2 embeddings: 95%| | 864/911 [00:23<00:01, 37.63it/s]

Computing ESM2 embeddings: 98%| | 896/911 [00:24<00:00, 36.98it/s]

Computing ESM2 embeddings: 100%| | 911/911 [00:24<00:00, 36.87it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/901 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/901 [00:00<00:22, 39.22it/s]

Computing ESM2 embeddings: 7%| | 64/901 [00:01<00:25, 33.17it/s]

Computing ESM2 embeddings: 11%| | 96/901 [00:02<00:22, 35.68it/s]

Computing ESM2 embeddings: 14%| | 128/901 [00:03<00:21, 35.43it/s]

Computing ESM2 embeddings: 18%| | 160/901 [00:04<00:20, 36.80it/s]

Computing ESM2 embeddings: 21%| | 192/901 [00:05<00:19, 36.35it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/901 [00:06<00:18, 37.33it/s] |
| Computing ESM2 embeddings: 28%  | 256/901 [00:07<00:17, 36.64it/s] |
| Computing ESM2 embeddings: 32%  | 288/901 [00:07<00:16, 37.43it/s] |
| Computing ESM2 embeddings: 36%  | 320/901 [00:08<00:15, 36.82it/s] |
| Computing ESM2 embeddings: 39%  | 352/901 [00:09<00:14, 37.59it/s] |
| Computing ESM2 embeddings: 43%  | 384/901 [00:10<00:13, 37.08it/s] |
| Computing ESM2 embeddings: 46%  | 416/901 [00:11<00:12, 37.72it/s] |
| Computing ESM2 embeddings: 50%  | 448/901 [00:12<00:12, 37.07it/s] |
| Computing ESM2 embeddings: 53%  | 480/901 [00:12<00:11, 37.53it/s] |
| Computing ESM2 embeddings: 57%  | 512/901 [00:13<00:10, 36.39it/s] |
| Computing ESM2 embeddings: 60%  | 544/901 [00:14<00:09, 37.27it/s] |
| Computing ESM2 embeddings: 64%  | 576/901 [00:15<00:08, 36.92it/s] |
| Computing ESM2 embeddings: 67%  | 608/901 [00:16<00:07, 37.79it/s] |
| Computing ESM2 embeddings: 71%  | 640/901 [00:17<00:07, 37.23it/s] |
| Computing ESM2 embeddings: 75%  | 672/901 [00:18<00:06, 37.60it/s] |
| Computing ESM2 embeddings: 78%  | 704/901 [00:19<00:05, 36.50it/s] |
| Computing ESM2 embeddings: 82%  | 736/901 [00:19<00:04, 37.29it/s] |
| Computing ESM2 embeddings: 85%  | 768/901 [00:20<00:03, 36.75it/s] |
| Computing ESM2 embeddings: 89%  | 800/901 [00:21<00:02, 37.54it/s] |
| Computing ESM2 embeddings: 92%  | 832/901 [00:22<00:01, 37.04it/s] |
| Computing ESM2 embeddings: 96%  | 864/901 [00:23<00:00, 37.76it/s] |
| Computing ESM2 embeddings: 99%  | 896/901 [00:24<00:00, 37.19it/s] |
| Computing ESM2 embeddings: 100% | 901/901 [00:24<00:00, 36.86it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/904 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/904 [00:00<00:22, 39.33it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/904 [00:01<00:22, 37.15it/s]  |
| Computing ESM2 embeddings: | 11% | 96/904 [00:02<00:21, 38.27it/s]  |
| Computing ESM2 embeddings: | 14% | 128/904 [00:03<00:20, 37.15it/s] |
| Computing ESM2 embeddings: | 18% | 160/904 [00:04<00:19, 37.98it/s] |
| Computing ESM2 embeddings: | 21% | 192/904 [00:05<00:19, 36.90it/s] |
| Computing ESM2 embeddings: | 25% | 224/904 [00:05<00:18, 37.20it/s] |
| Computing ESM2 embeddings: | 28% | 256/904 [00:06<00:17, 36.04it/s] |
| Computing ESM2 embeddings: | 32% | 288/904 [00:07<00:16, 37.09it/s] |
| Computing ESM2 embeddings: | 35% | 320/904 [00:08<00:16, 36.41it/s] |
| Computing ESM2 embeddings: | 39% | 352/904 [00:09<00:14, 37.25it/s] |
| Computing ESM2 embeddings: | 42% | 384/904 [00:10<00:14, 36.76it/s] |
| Computing ESM2 embeddings: | 46% | 416/904 [00:11<00:12, 37.56it/s] |
| Computing ESM2 embeddings: | 50% | 448/904 [00:12<00:12, 36.83it/s] |
| Computing ESM2 embeddings: | 53% | 480/904 [00:12<00:11, 37.62it/s] |
| Computing ESM2 embeddings: | 57% | 512/904 [00:13<00:10, 36.92it/s] |
| Computing ESM2 embeddings: | 60% | 544/904 [00:14<00:09, 37.62it/s] |
| Computing ESM2 embeddings: | 64% | 576/904 [00:15<00:08, 37.12it/s] |
| Computing ESM2 embeddings: | 67% | 608/904 [00:16<00:07, 37.51it/s] |
| Computing ESM2 embeddings: | 71% | 640/904 [00:17<00:07, 36.43it/s] |
| Computing ESM2 embeddings: | 74% | 672/904 [00:18<00:06, 37.29it/s] |

```

Computing ESM2 embeddings: 78%|      | 704/904 [00:18<00:05, 36.95it/s]
Computing ESM2 embeddings: 81%|      | 736/904 [00:19<00:04, 37.69it/s]
Computing ESM2 embeddings: 85%|      | 768/904 [00:20<00:03, 37.11it/s]
Computing ESM2 embeddings: 88%|      | 800/904 [00:21<00:02, 37.76it/s]
Computing ESM2 embeddings: 92%|      | 832/904 [00:22<00:01, 37.14it/s]
Computing ESM2 embeddings: 96%|      | 864/904 [00:23<00:01, 37.58it/s]
Computing ESM2 embeddings: 99%|      | 896/904 [00:24<00:00, 37.12it/s]
Computing ESM2 embeddings: 100%|     | 904/904 [00:24<00:00, 37.17it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/907 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/907 [00:00<00:22, 39.18it/s]
Computing ESM2 embeddings: 7%|      | 64/907 [00:02<00:27, 30.80it/s]
Computing ESM2 embeddings: 11%|     | 96/907 [00:02<00:23, 34.25it/s]
Computing ESM2 embeddings: 14%|     | 128/907 [00:03<00:22, 34.56it/s]
Computing ESM2 embeddings: 18%|     | 160/907 [00:04<00:20, 35.87it/s]
Computing ESM2 embeddings: 21%|     | 192/907 [00:05<00:20, 35.01it/s]
Computing ESM2 embeddings: 25%|     | 224/907 [00:06<00:18, 36.32it/s]
Computing ESM2 embeddings: 28%|     | 256/907 [00:07<00:17, 36.28it/s]
Computing ESM2 embeddings: 32%|     | 288/907 [00:08<00:16, 37.28it/s]
Computing ESM2 embeddings: 35%|     | 320/907 [00:08<00:15, 36.80it/s]
Computing ESM2 embeddings: 39%|     | 352/907 [00:09<00:14, 37.61it/s]
Computing ESM2 embeddings: 42%|     | 384/907 [00:10<00:14, 37.15it/s]

```



Computing ESM2 embeddings: 46%| | 416/907 [00:11<00:13, 37.76it/s]

Computing ESM2 embeddings: 49%| | 448/907 [00:12<00:12, 36.93it/s]

Computing ESM2 embeddings: 53%| | 480/907 [00:13<00:11, 37.63it/s]

Computing ESM2 embeddings: 56%| | 512/907 [00:14<00:10, 36.98it/s]

Computing ESM2 embeddings: 60%| | 544/907 [00:14<00:09, 37.72it/s]

Computing ESM2 embeddings: 64%| | 576/907 [00:15<00:08, 37.05it/s]

Computing ESM2 embeddings: 67%| | 608/907 [00:16<00:07, 37.41it/s]

Computing ESM2 embeddings: 71%| | 640/907 [00:17<00:07, 36.21it/s]

Computing ESM2 embeddings: 74%| | 672/907 [00:18<00:06, 37.08it/s]

Computing ESM2 embeddings: 78%| | 704/907 [00:19<00:05, 36.79it/s]

Computing ESM2 embeddings: 81%| | 736/907 [00:20<00:04, 37.54it/s]

Computing ESM2 embeddings: 85%| | 768/907 [00:20<00:03, 36.90it/s]

Computing ESM2 embeddings: 88%| | 800/907 [00:21<00:02, 37.61it/s]

Computing ESM2 embeddings: 92%| | 832/907 [00:22<00:02, 37.09it/s]

Computing ESM2 embeddings: 95%| | 864/907 [00:23<00:01, 37.76it/s]

Computing ESM2 embeddings: 99%| | 896/907 [00:24<00:00, 37.21it/s]

Computing ESM2 embeddings: 100%| | 907/907 [00:24<00:00, 36.68it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/898 [00:00<00:22, 37.78it/s]

Computing ESM2 embeddings: 7%| | 64/898 [00:01<00:23, 35.61it/s]

Computing ESM2 embeddings: 11%| | 96/898 [00:02<00:21, 36.92it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/898 [00:03<00:21, 35.57it/s] |
| Computing ESM2 embeddings: 18% | 160/898 [00:04<00:19, 36.94it/s] |
| Computing ESM2 embeddings: 21% | 192/898 [00:05<00:19, 36.57it/s] |
| Computing ESM2 embeddings: 25% | 224/898 [00:06<00:17, 37.49it/s] |
| Computing ESM2 embeddings: 29% | 256/898 [00:06<00:17, 36.97it/s] |
| Computing ESM2 embeddings: 32% | 288/898 [00:07<00:16, 37.49it/s] |
| Computing ESM2 embeddings: 36% | 320/898 [00:08<00:15, 36.20it/s] |
| Computing ESM2 embeddings: 39% | 352/898 [00:09<00:14, 36.94it/s] |
| Computing ESM2 embeddings: 43% | 384/898 [00:10<00:14, 35.98it/s] |
| Computing ESM2 embeddings: 46% | 416/898 [00:11<00:13, 36.95it/s] |
| Computing ESM2 embeddings: 50% | 448/898 [00:12<00:12, 36.59it/s] |
| Computing ESM2 embeddings: 53% | 480/898 [00:13<00:11, 37.44it/s] |
| Computing ESM2 embeddings: 57% | 512/898 [00:13<00:10, 36.85it/s] |
| Computing ESM2 embeddings: 61% | 544/898 [00:14<00:09, 37.57it/s] |
| Computing ESM2 embeddings: 64% | 576/898 [00:15<00:08, 37.06it/s] |
| Computing ESM2 embeddings: 68% | 608/898 [00:16<00:07, 37.77it/s] |
| Computing ESM2 embeddings: 71% | 640/898 [00:17<00:06, 37.16it/s] |
| Computing ESM2 embeddings: 75% | 672/898 [00:18<00:05, 37.80it/s] |
| Computing ESM2 embeddings: 78% | 704/898 [00:19<00:05, 37.05it/s] |
| Computing ESM2 embeddings: 82% | 736/898 [00:19<00:04, 37.78it/s] |
| Computing ESM2 embeddings: 86% | 768/898 [00:20<00:03, 37.19it/s] |
| Computing ESM2 embeddings: 89% | 800/898 [00:21<00:02, 37.85it/s] |
| Computing ESM2 embeddings: 93% | 832/898 [00:22<00:01, 37.25it/s] |
| Computing ESM2 embeddings: 96% | 864/898 [00:23<00:00, 37.87it/s] |

Computing ESM2 embeddings: 100%| | 898/898 [00:24<00:00, 37.00it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/587 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 5%| | 32/587 [00:00<00:14, 38.54it/s]  
 Computing ESM2 embeddings: 11%| | 64/587 [00:01<00:14, 35.64it/s]  
 Computing ESM2 embeddings: 16%| | 96/587 [00:02<00:13, 37.35it/s]  
 Computing ESM2 embeddings: 22%| | 128/587 [00:03<00:12, 36.84it/s]  
 Computing ESM2 embeddings: 27%| | 160/587 [00:04<00:11, 37.83it/s]  
 Computing ESM2 embeddings: 33%| | 192/587 [00:05<00:10, 37.20it/s]  
 Computing ESM2 embeddings: 38%| | 224/587 [00:05<00:09, 37.94it/s]  
 Computing ESM2 embeddings: 44%| | 256/587 [00:06<00:08, 37.26it/s]  
 Computing ESM2 embeddings: 49%| | 288/587 [00:07<00:07, 37.66it/s]  
 Computing ESM2 embeddings: 55%| | 320/587 [00:08<00:07, 36.46it/s]  
 Computing ESM2 embeddings: 60%| | 352/587 [00:09<00:06, 37.27it/s]  
 Computing ESM2 embeddings: 65%| | 384/587 [00:10<00:05, 36.80it/s]  
 Computing ESM2 embeddings: 71%| | 416/587 [00:11<00:04, 37.62it/s]  
 Computing ESM2 embeddings: 76%| | 448/587 [00:12<00:03, 36.79it/s]  
 Computing ESM2 embeddings: 82%| | 480/587 [00:12<00:02, 37.58it/s]  
 Computing ESM2 embeddings: 87%| | 512/587 [00:13<00:02, 37.03it/s]  
 Computing ESM2 embeddings: 93%| | 544/587 [00:14<00:01, 37.76it/s]  
 Computing ESM2 embeddings: 98%| | 576/587 [00:15<00:00, 37.20it/s]  
 Computing ESM2 embeddings: 100%| | 587/587 [00:15<00:00, 37.21it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/896 [00:00<00:22, 39.13it/s]
Computing ESM2 embeddings: 7%|          | 64/896 [00:01<00:22, 37.14it/s]
Computing ESM2 embeddings: 11%|         | 96/896 [00:02<00:21, 37.80it/s]
Computing ESM2 embeddings: 14%|         | 128/896 [00:03<00:21, 36.04it/s]
Computing ESM2 embeddings: 18%|         | 160/896 [00:04<00:19, 37.25it/s]
Computing ESM2 embeddings: 21%|         | 192/896 [00:05<00:19, 36.76it/s]
Computing ESM2 embeddings: 25%|         | 224/896 [00:06<00:17, 37.63it/s]
Computing ESM2 embeddings: 29%|         | 256/896 [00:06<00:17, 37.09it/s]
Computing ESM2 embeddings: 32%|         | 288/896 [00:07<00:16, 37.82it/s]
Computing ESM2 embeddings: 36%|         | 320/896 [00:08<00:15, 36.98it/s]
Computing ESM2 embeddings: 39%|         | 352/896 [00:09<00:14, 37.73it/s]
Computing ESM2 embeddings: 43%|         | 384/896 [00:10<00:13, 37.17it/s]
Computing ESM2 embeddings: 46%|         | 416/896 [00:11<00:12, 37.88it/s]
Computing ESM2 embeddings: 50%|         | 448/896 [00:12<00:12, 37.19it/s]
Computing ESM2 embeddings: 54%|         | 480/896 [00:12<00:10, 37.85it/s]
Computing ESM2 embeddings: 57%|         | 512/896 [00:13<00:10, 37.25it/s]
Computing ESM2 embeddings: 61%|         | 544/896 [00:14<00:09, 37.92it/s]
Computing ESM2 embeddings: 64%|         | 576/896 [00:15<00:08, 37.14it/s]
Computing ESM2 embeddings: 68%|         | 608/896 [00:16<00:07, 37.83it/s]
Computing ESM2 embeddings: 71%|         | 640/896 [00:17<00:06, 37.24it/s]
```

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.65it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 36.68it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.41it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.07it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.83it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.33it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.93it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.26it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.83it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.10it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.42it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 37.35it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 37.78it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.51it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 37.44it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 36.81it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.66it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 37.09it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.95it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.34it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.99it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 37.35it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.99it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.35it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.67it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.58it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.42it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.91it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.66it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.10it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.82it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.26it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.95it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.26it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.93it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.26it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.30it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.60it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.28it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 36.43it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:21, 35.23it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:20, 36.66it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 36.21it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:18, 37.20it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:07<00:17, 36.80it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.39it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 36.20it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.13it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 36.62it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.44it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 37.03it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:13<00:11, 37.74it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.15it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.85it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.23it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.86it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:06, 37.16it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:05, 37.52it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:19<00:05, 36.39it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.23it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 36.74it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.52it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.03it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 37.72it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.92it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:22, 38.80it/s]  
 Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:23, 35.98it/s]  
 Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:21, 37.53it/s]  
 Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:21, 36.81it/s]  
 Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:20, 37.39it/s]  
 Computing ESM2 embeddings: 21%| | 192/915 [00:05<00:19, 36.32it/s]  
 Computing ESM2 embeddings: 24%| | 224/915 [00:06<00:18, 37.15it/s]  
 Computing ESM2 embeddings: 28%| | 256/915 [00:06<00:18, 36.19it/s]  
 Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:16, 37.20it/s]  
 Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:16, 36.72it/s]  
 Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:14, 37.56it/s]  
 Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 37.11it/s]  
 Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 37.79it/s]  
 Computing ESM2 embeddings: 49%| | 448/915 [00:12<00:12, 37.21it/s]  
 Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 37.90it/s]  
 Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 37.28it/s]  
 Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 37.95it/s]  
 Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 37.27it/s]  
 Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.94it/s]



```

Computing ESM2 embeddings: 70%|          | 640/915 [00:17<00:07, 37.27it/s]
Computing ESM2 embeddings: 73%|          | 672/915 [00:18<00:06, 37.66it/s]
Computing ESM2 embeddings: 77%|          | 704/915 [00:18<00:05, 36.37it/s]
Computing ESM2 embeddings: 80%|          | 736/915 [00:19<00:04, 37.26it/s]
Computing ESM2 embeddings: 84%|          | 768/915 [00:20<00:03, 36.79it/s]
Computing ESM2 embeddings: 87%|          | 800/915 [00:21<00:03, 37.58it/s]
Computing ESM2 embeddings: 91%|          | 832/915 [00:22<00:02, 37.09it/s]
Computing ESM2 embeddings: 94%|          | 864/915 [00:23<00:01, 37.82it/s]
Computing ESM2 embeddings: 98%|          | 896/915 [00:24<00:00, 37.19it/s]

Computing ESM2 embeddings: 100%|         | 915/915 [00:24<00:00, 37.22it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/896 [00:00<00:22, 38.42it/s]
Computing ESM2 embeddings: 7%|           | 64/896 [00:01<00:23, 35.85it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 37.44it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 36.90it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 37.46it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:19, 36.29it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:06<00:18, 37.31it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:17, 36.97it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.73it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 37.20it/s]

```

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.64it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.63it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.48it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.82it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.61it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.02it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.48it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.45it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.33it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 36.91it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.67it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.06it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.77it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.26it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.90it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.31it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.96it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.14it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.54it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:25, 33.42it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:22, 35.97it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:22, 35.68it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 36.65it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:20, 35.64it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 36.80it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:18, 36.33it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.00it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.11it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 36.80it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 35.93it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 36.92it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 36.66it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 37.47it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:14<00:10, 37.01it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.73it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.09it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.78it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.08it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.76it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 37.13it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.80it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.09it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.85it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.20it/s] |

```

Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.85it/s]

Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.11it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.83it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/895 [00:00<00:25, 33.76it/s]

Computing ESM2 embeddings:  7%|      | 64/895 [00:01<00:24, 33.97it/s]

Computing ESM2 embeddings: 11%|      | 96/895 [00:02<00:22, 36.28it/s]

Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:21, 35.96it/s]

Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:19, 37.17it/s]

Computing ESM2 embeddings: 21%|      | 192/895 [00:05<00:19, 36.53it/s]

Computing ESM2 embeddings: 25%|      | 224/895 [00:06<00:17, 37.46it/s]

Computing ESM2 embeddings: 29%|      | 256/895 [00:07<00:17, 36.90it/s]

Computing ESM2 embeddings: 32%|      | 288/895 [00:07<00:16, 37.35it/s]

Computing ESM2 embeddings: 36%|      | 320/895 [00:08<00:15, 36.30it/s]

Computing ESM2 embeddings: 39%|      | 352/895 [00:09<00:14, 37.22it/s]

Computing ESM2 embeddings: 43%|      | 384/895 [00:10<00:13, 36.67it/s]

Computing ESM2 embeddings: 46%|      | 416/895 [00:11<00:12, 37.48it/s]

Computing ESM2 embeddings: 50%|      | 448/895 [00:12<00:12, 36.92it/s]

Computing ESM2 embeddings: 54%|      | 480/895 [00:13<00:11, 37.52it/s]

Computing ESM2 embeddings: 57%|      | 512/895 [00:13<00:10, 36.76it/s]

Computing ESM2 embeddings: 61%|      | 544/895 [00:14<00:09, 37.55it/s]

```

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 36.92it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.36it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:07, 36.31it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:05, 37.20it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:19<00:05, 36.69it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.23it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 36.28it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.13it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 36.75it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.52it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 36.75it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/590 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/590 [00:00<00:14, 39.46it/s]

Computing ESM2 embeddings: 11%| | 64/590 [00:01<00:14, 36.56it/s]

Computing ESM2 embeddings: 16%| | 96/590 [00:02<00:13, 37.90it/s]

Computing ESM2 embeddings: 22%| | 128/590 [00:03<00:12, 36.90it/s]

Computing ESM2 embeddings: 27%| | 160/590 [00:04<00:11, 37.84it/s]

Computing ESM2 embeddings: 33%| | 192/590 [00:05<00:10, 37.09it/s]

Computing ESM2 embeddings: 38%| | 224/590 [00:05<00:09, 37.87it/s]

Computing ESM2 embeddings: 43%| | 256/590 [00:06<00:08, 37.13it/s]

Computing ESM2 embeddings: 49%| | 288/590 [00:07<00:08, 37.59it/s]

Computing ESM2 embeddings: 54%| | 320/590 [00:08<00:07, 36.53it/s]

Computing ESM2 embeddings: 60%| | 352/590 [00:09<00:06, 37.33it/s]

Computing ESM2 embeddings: 65%| | 384/590 [00:10<00:05, 36.88it/s]

Computing ESM2 embeddings: 71%| | 416/590 [00:11<00:04, 37.66it/s]

Computing ESM2 embeddings: 76%| | 448/590 [00:12<00:03, 37.12it/s]

Computing ESM2 embeddings: 81%| | 480/590 [00:12<00:02, 37.76it/s]

Computing ESM2 embeddings: 87%| | 512/590 [00:13<00:02, 37.17it/s]

Computing ESM2 embeddings: 92%| | 544/590 [00:14<00:01, 37.84it/s]

Computing ESM2 embeddings: 98%| | 576/590 [00:15<00:00, 37.25it/s]

Computing ESM2 embeddings: 100%| | 590/590 [00:15<00:00, 37.30it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.44it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.80it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.45it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 36.90it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.67it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.45it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:17, 37.46it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 36.97it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.75it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 37.23it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.67it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:14, 36.46it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.36it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.92it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.64it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.05it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.79it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.23it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.93it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.22it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.89it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.30it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.93it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.12it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.81it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.24it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.91it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.31it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.06it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.05it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 37.78it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 36.97it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 37.84it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.12it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.61it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:18, 36.40it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.09it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.12it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.14it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.73it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.55it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.04it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.76it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.17it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.86it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.30it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.92it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.27it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.92it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.34it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.98it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.32it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.75it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 36.29it/s] |



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Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.22it/s]

Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.79it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.22it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%|        | 32/897 [00:00<00:22, 38.19it/s]

Computing ESM2 embeddings: 7%|        | 64/897 [00:01<00:23, 35.73it/s]

Computing ESM2 embeddings: 11%|       | 96/897 [00:02<00:21, 37.44it/s]

Computing ESM2 embeddings: 14%|       | 128/897 [00:03<00:20, 36.77it/s]

Computing ESM2 embeddings: 18%|       | 160/897 [00:04<00:19, 37.75it/s]

Computing ESM2 embeddings: 21%|       | 192/897 [00:05<00:19, 37.09it/s]

Computing ESM2 embeddings: 25%|       | 224/897 [00:06<00:17, 37.57it/s]

Computing ESM2 embeddings: 29%|       | 256/897 [00:06<00:17, 36.35it/s]

Computing ESM2 embeddings: 32%|       | 288/897 [00:07<00:16, 37.28it/s]

Computing ESM2 embeddings: 36%|       | 320/897 [00:08<00:15, 36.76it/s]

Computing ESM2 embeddings: 39%|       | 352/897 [00:09<00:14, 37.54it/s]

Computing ESM2 embeddings: 43%|       | 384/897 [00:10<00:13, 37.04it/s]

Computing ESM2 embeddings: 46%|       | 416/897 [00:11<00:12, 37.78it/s]

Computing ESM2 embeddings: 50%|       | 448/897 [00:12<00:12, 37.22it/s]

Computing ESM2 embeddings: 54%|       | 480/897 [00:12<00:11, 37.85it/s]

Computing ESM2 embeddings: 57%|       | 512/897 [00:13<00:10, 37.25it/s]

Computing ESM2 embeddings: 61%|       | 544/897 [00:14<00:09, 37.60it/s]

```

Computing ESM2 embeddings: 64%| | 576/897 [00:15<00:08, 36.46it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:16<00:07, 37.29it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:17<00:06, 36.87it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:18<00:06, 37.43it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:18<00:05, 36.34it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:19<00:04, 37.26it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:20<00:03, 36.84it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:21<00:02, 37.62it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:22<00:01, 37.12it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:23<00:00, 37.82it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:24<00:00, 37.12it/s]

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Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:22, 38.94it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:24, 34.91it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:22, 36.89it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:21, 36.21it/s]

Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:20, 37.30it/s]

Computing ESM2 embeddings: 21%| | 192/915 [00:05<00:19, 36.93it/s]

Computing ESM2 embeddings: 24%| | 224/915 [00:06<00:18, 37.75it/s]

Computing ESM2 embeddings: 28%| | 256/915 [00:06<00:17, 36.95it/s]

Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:16, 37.73it/s]

Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:16, 36.81it/s]

Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:15, 37.30it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 36.21it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 37.17it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:12<00:12, 36.68it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 37.46it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 36.84it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 37.58it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 37.03it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.40it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 36.28it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:18<00:06, 37.14it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:19<00:05, 36.77it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 37.52it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:03, 36.88it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 37.59it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 37.02it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:23<00:01, 37.67it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:24<00:00, 36.86it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 37.04it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.92it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:23, 35.39it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 36.73it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 35.66it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 36.98it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.46it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 37.11it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:07<00:17, 36.27it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.19it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 36.81it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.59it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 36.98it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.71it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.10it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.80it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.19it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.54it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 36.43it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.31it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 36.75it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:18<00:06, 37.23it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:19<00:05, 36.26it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.17it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 36.62it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.41it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 36.81it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.28it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.77it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 38.04it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:26, 31.83it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:23, 35.04it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:22, 34.99it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 36.47it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:20, 36.07it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.10it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:07<00:17, 36.74it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.37it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.73it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.30it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 36.41it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 36.95it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 36.14it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:13<00:11, 36.85it/s]
Computing ESM2 embeddings: 56%|      | 512/914 [00:14<00:11, 36.03it/s]

```

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:10, 36.96it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.49it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.33it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.91it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.66it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 37.07it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 37.71it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.20it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.50it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.32it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.22it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.63it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.62it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:25, 34.01it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:23, 34.63it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:22, 36.26it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:21, 35.26it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:20, 36.40it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:19, 35.69it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:06<00:18, 36.87it/s]

```

Computing ESM2 embeddings: 29%|          | 256/895 [00:07<00:17, 36.50it/s]
Computing ESM2 embeddings: 32%|          | 288/895 [00:07<00:16, 37.34it/s]
Computing ESM2 embeddings: 36%|          | 320/895 [00:08<00:15, 36.74it/s]
Computing ESM2 embeddings: 39%|          | 352/895 [00:09<00:14, 37.56it/s]
Computing ESM2 embeddings: 43%|          | 384/895 [00:10<00:13, 36.99it/s]
Computing ESM2 embeddings: 46%|          | 416/895 [00:11<00:12, 37.70it/s]
Computing ESM2 embeddings: 50%|          | 448/895 [00:12<00:12, 37.00it/s]
Computing ESM2 embeddings: 54%|          | 480/895 [00:13<00:10, 37.75it/s]
Computing ESM2 embeddings: 57%|          | 512/895 [00:13<00:10, 37.14it/s]
Computing ESM2 embeddings: 61%|          | 544/895 [00:14<00:09, 37.71it/s]
Computing ESM2 embeddings: 64%|          | 576/895 [00:15<00:08, 36.92it/s]
Computing ESM2 embeddings: 68%|          | 608/895 [00:16<00:07, 37.38it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:17<00:07, 36.28it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:18<00:06, 36.94it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:19<00:05, 35.99it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:20<00:04, 36.93it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:20<00:03, 36.48it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:21<00:02, 37.09it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:22<00:01, 36.56it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:23<00:00, 37.40it/s]

```

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Computing ESM2 embeddings: 100%|         | 895/895 [00:24<00:00, 36.68it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|          | 0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings: 6%|          | 32/572 [00:00<00:16, 32.78it/s]
Computing ESM2 embeddings: 11%|         | 64/572 [00:01<00:15, 33.40it/s]
Computing ESM2 embeddings: 17%|         | 96/572 [00:02<00:13, 35.49it/s]
Computing ESM2 embeddings: 22%|         | 128/572 [00:03<00:12, 34.82it/s]
Computing ESM2 embeddings: 28%|         | 160/572 [00:04<00:11, 36.38it/s]
Computing ESM2 embeddings: 34%|         | 192/572 [00:05<00:10, 36.13it/s]
Computing ESM2 embeddings: 39%|         | 224/572 [00:06<00:09, 37.16it/s]
Computing ESM2 embeddings: 45%|         | 256/572 [00:07<00:08, 36.62it/s]
Computing ESM2 embeddings: 50%|         | 288/572 [00:07<00:07, 37.15it/s]
Computing ESM2 embeddings: 56%|         | 320/572 [00:08<00:06, 36.07it/s]
Computing ESM2 embeddings: 62%|         | 352/572 [00:09<00:05, 37.04it/s]
Computing ESM2 embeddings: 67%|         | 384/572 [00:10<00:05, 36.65it/s]
Computing ESM2 embeddings: 73%|         | 416/572 [00:11<00:04, 37.20it/s]
Computing ESM2 embeddings: 78%|         | 448/572 [00:12<00:03, 36.24it/s]
Computing ESM2 embeddings: 84%|         | 480/572 [00:13<00:02, 37.11it/s]
Computing ESM2 embeddings: 90%|         | 512/572 [00:14<00:01, 36.68it/s]
Computing ESM2 embeddings: 95%|         | 544/572 [00:14<00:00, 37.18it/s]

Computing ESM2 embeddings: 100%|        | 572/572 [00:15<00:00, 36.28it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings: 0%|          | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/914 [00:00<00:23, 36.99it/s]

```



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:25, 33.06it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:22, 35.67it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:22, 35.21it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:21, 35.86it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:20, 35.87it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 36.77it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:18, 36.41it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.23it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.57it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.45it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.89it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.65it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.10it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 37.81it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.03it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.50it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 36.43it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.29it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 36.65it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.46it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 36.99it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.65it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.20it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.77it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.18it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.90it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.19it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.86it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:26, 32.42it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:24, 33.36it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:22, 35.85it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:21, 35.70it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 36.97it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 36.45it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:06<00:18, 37.07it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:07<00:17, 36.10it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.01it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 36.57it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.35it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.08it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.76it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:12<00:12, 36.87it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:13<00:11, 37.32it/s]
Computing ESM2 embeddings: 57%|      | 512/896 [00:14<00:10, 36.16it/s]

```

```

Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.10it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 36.53it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.38it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 36.89it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:18<00:05, 37.51it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:19<00:05, 36.39it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:20<00:04, 37.26it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 36.84it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.56it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 36.91it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.61it/s]

```

```

Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 36.70it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/915 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|          | 32/915 [00:00<00:23, 38.30it/s]
Computing ESM2 embeddings:  7%|          | 64/915 [00:01<00:23, 35.88it/s]
Computing ESM2 embeddings: 10%|          | 96/915 [00:02<00:21, 37.48it/s]
Computing ESM2 embeddings: 14%|          | 128/915 [00:03<00:21, 36.79it/s]
Computing ESM2 embeddings: 17%|          | 160/915 [00:04<00:20, 37.74it/s]
Computing ESM2 embeddings: 21%|          | 192/915 [00:05<00:19, 37.04it/s]
Computing ESM2 embeddings: 24%|          | 224/915 [00:06<00:18, 37.57it/s]
Computing ESM2 embeddings: 28%|          | 256/915 [00:06<00:18, 36.43it/s]

```

Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:16, 37.40it/s]

Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:16, 36.87it/s]

Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:14, 37.68it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 37.11it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 37.85it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:12<00:12, 37.17it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 37.57it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:11, 36.33it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 37.25it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 36.87it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.57it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 37.05it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:18<00:06, 37.76it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:18<00:05, 37.03it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 37.56it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:04, 36.57it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 37.23it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 36.09it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:23<00:01, 37.02it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:24<00:00, 36.66it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 37.08it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:01<00:30, 28.30it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:02<00:26, 31.64it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:23, 34.77it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 35.11it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:20, 36.59it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.59it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 37.10it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:07<00:17, 36.16it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:08<00:16, 37.01it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 36.38it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.29it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 36.85it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.62it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.00it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:13<00:11, 37.71it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:14<00:10, 37.10it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.77it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.23it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.89it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.16it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:18<00:05, 37.76it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:19<00:05, 37.03it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:20<00:04, 37.71it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.08it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.75it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.02it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.72it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.74it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 38.11it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 35.75it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.42it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.63it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.63it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 37.16it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:18, 37.94it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 37.30it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.92it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 37.32it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 37.97it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 37.29it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.67it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 36.35it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.26it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 36.67it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.23it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.28it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.22it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.72it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.49it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 36.89it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.61it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.92it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.37it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.33it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.21it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.77it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.06it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:23, 36.88it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:23, 35.64it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:21, 37.34it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:20, 36.72it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:19, 37.67it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:19, 36.89it/s]

```

Computing ESM2 embeddings: 25%|          | 224/895 [00:06<00:17, 37.35it/s]
Computing ESM2 embeddings: 29%|          | 256/895 [00:06<00:17, 36.30it/s]
Computing ESM2 embeddings: 32%|          | 288/895 [00:07<00:16, 37.28it/s]
Computing ESM2 embeddings: 36%|          | 320/895 [00:08<00:15, 36.87it/s]
Computing ESM2 embeddings: 39%|          | 352/895 [00:09<00:14, 37.67it/s]
Computing ESM2 embeddings: 43%|          | 384/895 [00:10<00:13, 37.14it/s]
Computing ESM2 embeddings: 46%|          | 416/895 [00:11<00:12, 37.83it/s]
Computing ESM2 embeddings: 50%|          | 448/895 [00:12<00:11, 37.39it/s]
Computing ESM2 embeddings: 54%|          | 480/895 [00:12<00:10, 37.98it/s]
Computing ESM2 embeddings: 57%|          | 512/895 [00:13<00:10, 37.39it/s]
Computing ESM2 embeddings: 61%|          | 544/895 [00:14<00:09, 38.01it/s]
Computing ESM2 embeddings: 64%|          | 576/895 [00:15<00:08, 37.39it/s]
Computing ESM2 embeddings: 68%|          | 608/895 [00:16<00:07, 37.69it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:17<00:06, 36.51it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:18<00:05, 37.37it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:18<00:05, 36.82it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:19<00:04, 37.60it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:20<00:03, 37.02it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:21<00:02, 37.73it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:22<00:01, 37.18it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:23<00:00, 37.86it/s]

```

```

Computing ESM2 embeddings: 100%|         | 895/895 [00:24<00:00, 37.19it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```



for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 38.58it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:23, 36.26it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.75it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 36.94it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 37.82it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.22it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 37.98it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.28it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 38.00it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 37.19it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.51it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 36.28it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 36.94it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 35.90it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 36.59it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:11, 35.69it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:10, 36.44it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 35.62it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 36.58it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 36.25it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 37.09it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:19<00:05, 36.59it/s] |

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.11it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:04, 36.22it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.14it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.63it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.42it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.96it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.87it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/572 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/572 [00:01<00:18, 29.00it/s]

Computing ESM2 embeddings: 11%| | 64/572 [00:02<00:15, 31.79it/s]

Computing ESM2 embeddings: 17%| | 96/572 [00:02<00:13, 34.90it/s]

Computing ESM2 embeddings: 22%| | 128/572 [00:03<00:12, 35.16it/s]

Computing ESM2 embeddings: 28%| | 160/572 [00:04<00:11, 36.59it/s]

Computing ESM2 embeddings: 34%| | 192/572 [00:05<00:10, 36.17it/s]

Computing ESM2 embeddings: 39%| | 224/572 [00:06<00:09, 37.17it/s]

Computing ESM2 embeddings: 45%| | 256/572 [00:07<00:08, 36.62it/s]

Computing ESM2 embeddings: 50%| | 288/572 [00:07<00:07, 37.48it/s]

Computing ESM2 embeddings: 56%| | 320/572 [00:08<00:06, 36.93it/s]

Computing ESM2 embeddings: 62%| | 352/572 [00:09<00:05, 37.46it/s]

Computing ESM2 embeddings: 67%| | 384/572 [00:10<00:05, 36.89it/s]

Computing ESM2 embeddings: 73%| | 416/572 [00:11<00:04, 37.62it/s]

```

Computing ESM2 embeddings: 78%|          | 448/572 [00:12<00:03, 37.07it/s]
Computing ESM2 embeddings: 84%|          | 480/572 [00:13<00:02, 37.54it/s]
Computing ESM2 embeddings: 90%|          | 512/572 [00:14<00:01, 36.38it/s]
Computing ESM2 embeddings: 95%|          | 544/572 [00:14<00:00, 37.15it/s]
Computing ESM2 embeddings: 100%|         | 572/572 [00:15<00:00, 36.26it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:24, 36.67it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:02<00:27, 30.85it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:23, 34.25it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:22, 34.77it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:20, 36.30it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 36.23it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:06<00:18, 37.25it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:07<00:17, 36.84it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 37.33it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:16, 36.34it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:15, 37.24it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 36.84it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.27it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:12<00:12, 36.11it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:13<00:11, 37.04it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:11, 36.46it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.25it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.78it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.56it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.79it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.57it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.88it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 37.62it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.17it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.83it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.24it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.88it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.15it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.68it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.85it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 36.81it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 38.07it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.01it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.93it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.23it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 37.73it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 36.56it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.18it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.19it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 36.96it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:14, 36.05it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.01it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.74it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.53it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.04it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.76it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.19it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.84it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.47it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 38.17it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.47it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.77it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.49it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.35it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.74it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.52it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.02it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/916 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 3%  | 32/916 [00:00<00:23, 37.42it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/916 [00:01<00:24, 35.47it/s]  |
| Computing ESM2 embeddings: | 10% | 96/916 [00:02<00:22, 37.07it/s]  |
| Computing ESM2 embeddings: | 14% | 128/916 [00:03<00:21, 35.84it/s] |
| Computing ESM2 embeddings: | 17% | 160/916 [00:04<00:20, 37.15it/s] |
| Computing ESM2 embeddings: | 21% | 192/916 [00:05<00:19, 36.78it/s] |
| Computing ESM2 embeddings: | 24% | 224/916 [00:06<00:18, 37.64it/s] |
| Computing ESM2 embeddings: | 28% | 256/916 [00:06<00:17, 36.87it/s] |
| Computing ESM2 embeddings: | 31% | 288/916 [00:07<00:16, 37.46it/s] |
| Computing ESM2 embeddings: | 35% | 320/916 [00:08<00:16, 36.32it/s] |
| Computing ESM2 embeddings: | 38% | 352/916 [00:09<00:15, 37.21it/s] |
| Computing ESM2 embeddings: | 42% | 384/916 [00:10<00:14, 36.90it/s] |
| Computing ESM2 embeddings: | 45% | 416/916 [00:11<00:13, 37.51it/s] |
| Computing ESM2 embeddings: | 49% | 448/916 [00:12<00:12, 36.46it/s] |
| Computing ESM2 embeddings: | 52% | 480/916 [00:12<00:11, 37.36it/s] |
| Computing ESM2 embeddings: | 56% | 512/916 [00:13<00:10, 36.86it/s] |
| Computing ESM2 embeddings: | 59% | 544/916 [00:14<00:09, 37.61it/s] |
| Computing ESM2 embeddings: | 63% | 576/916 [00:15<00:09, 37.15it/s] |
| Computing ESM2 embeddings: | 66% | 608/916 [00:16<00:08, 37.84it/s] |
| Computing ESM2 embeddings: | 70% | 640/916 [00:17<00:07, 37.29it/s] |
| Computing ESM2 embeddings: | 73% | 672/916 [00:18<00:06, 37.64it/s] |
| Computing ESM2 embeddings: | 77% | 704/916 [00:19<00:05, 36.65it/s] |

Computing ESM2 embeddings: 80%| | 736/916 [00:19<00:04, 37.47it/s]

Computing ESM2 embeddings: 84%| | 768/916 [00:20<00:03, 37.07it/s]

Computing ESM2 embeddings: 87%| | 800/916 [00:21<00:03, 37.78it/s]

Computing ESM2 embeddings: 91%| | 832/916 [00:22<00:02, 37.19it/s]

Computing ESM2 embeddings: 94%| | 864/916 [00:23<00:01, 37.62it/s]

Computing ESM2 embeddings: 98%| | 896/916 [00:24<00:00, 36.47it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:24<00:00, 36.96it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.07it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 36.87it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.13it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.01it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.65it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.31it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 37.29it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 36.89it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.39it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.17it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.10it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.70it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.48it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.04it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.74it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.03it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.70it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.95it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.64it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 36.88it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.61it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 36.92it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.62it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.96it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.66it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.14it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.75it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.11it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.12it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 35.19it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.78it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 35.43it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.85it/s]



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/914 [00:05<00:19, 36.46it/s] |
| Computing ESM2 embeddings: 25%  | 224/914 [00:06<00:18, 37.36it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:17, 36.94it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 37.74it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:16, 36.90it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 37.71it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:14, 37.07it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:11<00:13, 37.78it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:12<00:12, 37.19it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.84it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 37.16it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 37.60it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:15<00:09, 36.42it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:16<00:08, 37.31it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:17<00:07, 36.90it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:18<00:06, 37.66it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 37.13it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:19<00:04, 37.80it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:20<00:03, 37.20it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:21<00:03, 37.89it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:22<00:02, 37.22it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:23<00:01, 37.89it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:24<00:00, 37.21it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:24<00:00, 37.06it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|          | 32/913 [00:00<00:23, 37.86it/s]
Computing ESM2 embeddings: 7%|          | 64/913 [00:01<00:23, 35.44it/s]
Computing ESM2 embeddings: 11%|         | 96/913 [00:02<00:21, 37.27it/s]
Computing ESM2 embeddings: 14%|         | 128/913 [00:03<00:21, 36.68it/s]
Computing ESM2 embeddings: 18%|         | 160/913 [00:04<00:19, 37.69it/s]
Computing ESM2 embeddings: 21%|         | 192/913 [00:05<00:19, 37.05it/s]
Computing ESM2 embeddings: 25%|         | 224/913 [00:05<00:18, 37.85it/s]
Computing ESM2 embeddings: 28%|         | 256/913 [00:06<00:17, 37.14it/s]
Computing ESM2 embeddings: 32%|         | 288/913 [00:07<00:16, 37.91it/s]
Computing ESM2 embeddings: 35%|         | 320/913 [00:08<00:15, 37.08it/s]
Computing ESM2 embeddings: 39%|         | 352/913 [00:09<00:14, 37.78it/s]
Computing ESM2 embeddings: 42%|         | 384/913 [00:10<00:14, 37.20it/s]
Computing ESM2 embeddings: 46%|         | 416/913 [00:11<00:13, 37.91it/s]
Computing ESM2 embeddings: 49%|         | 448/913 [00:12<00:12, 37.26it/s]
Computing ESM2 embeddings: 53%|         | 480/913 [00:12<00:11, 37.92it/s]
Computing ESM2 embeddings: 56%|         | 512/913 [00:13<00:10, 37.31it/s]
Computing ESM2 embeddings: 60%|         | 544/913 [00:14<00:09, 37.99it/s]
Computing ESM2 embeddings: 63%|         | 576/913 [00:15<00:09, 37.28it/s]
Computing ESM2 embeddings: 67%|         | 608/913 [00:16<00:08, 37.94it/s]
Computing ESM2 embeddings: 70%|         | 640/913 [00:17<00:07, 37.31it/s]
```

```

Computing ESM2 embeddings: 74%|      | 672/913 [00:17<00:06, 37.96it/s]
Computing ESM2 embeddings: 77%|      | 704/913 [00:18<00:05, 37.25it/s]
Computing ESM2 embeddings: 81%|      | 736/913 [00:19<00:04, 37.92it/s]
Computing ESM2 embeddings: 84%|      | 768/913 [00:20<00:03, 37.06it/s]
Computing ESM2 embeddings: 88%|      | 800/913 [00:21<00:03, 37.49it/s]
Computing ESM2 embeddings: 91%|      | 832/913 [00:22<00:02, 36.49it/s]
Computing ESM2 embeddings: 95%|      | 864/913 [00:23<00:01, 37.12it/s]
Computing ESM2 embeddings: 98%|      | 896/913 [00:24<00:00, 36.20it/s]

Computing ESM2 embeddings: 100%|     | 913/913 [00:24<00:00, 37.19it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/914 [00:00<00:23, 37.96it/s]
Computing ESM2 embeddings: 7%|      | 64/914 [00:01<00:27, 31.42it/s]
Computing ESM2 embeddings: 11%|     | 96/914 [00:02<00:23, 34.38it/s]
Computing ESM2 embeddings: 14%|     | 128/914 [00:03<00:22, 34.54it/s]
Computing ESM2 embeddings: 18%|     | 160/914 [00:04<00:20, 36.19it/s]
Computing ESM2 embeddings: 21%|     | 192/914 [00:05<00:19, 36.16it/s]
Computing ESM2 embeddings: 25%|     | 224/914 [00:06<00:18, 37.16it/s]
Computing ESM2 embeddings: 28%|     | 256/914 [00:07<00:17, 36.70it/s]
Computing ESM2 embeddings: 32%|     | 288/914 [00:07<00:16, 37.54it/s]
Computing ESM2 embeddings: 35%|     | 320/914 [00:08<00:16, 36.91it/s]
Computing ESM2 embeddings: 39%|     | 352/914 [00:09<00:14, 37.75it/s]

```

```

Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 37.10it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.81it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:12<00:12, 37.17it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:13<00:11, 37.86it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 37.10it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 37.81it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.20it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.88it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.27it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.89it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 37.29it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.90it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.16it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.85it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.24it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.87it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.27it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 36.97it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|          | 32/572 [00:00<00:16, 32.09it/s]
Computing ESM2 embeddings: 11%|          | 64/572 [00:01<00:15, 32.72it/s]

```

Computing ESM2 embeddings: 17%| | 96/572 [00:02<00:13, 35.53it/s]

Computing ESM2 embeddings: 22%| | 128/572 [00:03<00:12, 35.67it/s]

Computing ESM2 embeddings: 28%| | 160/572 [00:04<00:11, 36.64it/s]

Computing ESM2 embeddings: 34%| | 192/572 [00:05<00:10, 35.75it/s]

Computing ESM2 embeddings: 39%| | 224/572 [00:06<00:09, 36.88it/s]

Computing ESM2 embeddings: 45%| | 256/572 [00:07<00:08, 36.42it/s]

Computing ESM2 embeddings: 50%| | 288/572 [00:07<00:07, 37.29it/s]

Computing ESM2 embeddings: 56%| | 320/572 [00:08<00:06, 36.73it/s]

Computing ESM2 embeddings: 62%| | 352/572 [00:09<00:05, 37.52it/s]

Computing ESM2 embeddings: 67%| | 384/572 [00:10<00:05, 37.03it/s]

Computing ESM2 embeddings: 73%| | 416/572 [00:11<00:04, 37.75it/s]

Computing ESM2 embeddings: 78%| | 448/572 [00:12<00:03, 37.15it/s]

Computing ESM2 embeddings: 84%| | 480/572 [00:13<00:02, 37.83it/s]

Computing ESM2 embeddings: 90%| | 512/572 [00:13<00:01, 37.27it/s]

Computing ESM2 embeddings: 95%| | 544/572 [00:14<00:00, 37.89it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:15<00:00, 36.61it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:21, 39.41it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 37.41it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:20, 38.41it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.17it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:19, 38.05it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:18, 37.30it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 37.91it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:17, 37.29it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:16, 37.65it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 36.47it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 37.37it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 36.91it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:12, 37.72it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:11<00:12, 37.18it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 37.89it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 37.27it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 37.95it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 37.26it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 37.92it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 37.35it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 37.99it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:05, 37.21it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 37.86it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:20<00:03, 37.36it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 37.98it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 37.15it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:23<00:00, 37.54it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:24<00:00, 37.30it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.91it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:25, 33.50it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:22, 35.62it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:22, 35.13it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 36.55it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.17it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.20it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:07<00:17, 36.76it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.30it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.23it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 37.17it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 36.83it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.65it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 37.08it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:13<00:11, 37.49it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:11, 36.35it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.27it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 36.72it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.49it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 36.79it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:18<00:06, 37.56it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:19<00:05, 37.07it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 37.78it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.18it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.88it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.27it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.90it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.24it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.84it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/897 [00:00<00:22, 38.11it/s]
Computing ESM2 embeddings: 7%|      | 64/897 [00:01<00:22, 37.02it/s]
Computing ESM2 embeddings: 11%|     | 96/897 [00:02<00:20, 38.34it/s]
Computing ESM2 embeddings: 14%|     | 128/897 [00:03<00:20, 37.28it/s]
Computing ESM2 embeddings: 18%|     | 160/897 [00:04<00:19, 38.13it/s]
Computing ESM2 embeddings: 21%|     | 192/897 [00:05<00:18, 37.33it/s]
Computing ESM2 embeddings: 25%|     | 224/897 [00:05<00:17, 38.05it/s]
Computing ESM2 embeddings: 29%|     | 256/897 [00:06<00:17, 37.28it/s]
Computing ESM2 embeddings: 32%|     | 288/897 [00:07<00:16, 37.97it/s]
Computing ESM2 embeddings: 36%|     | 320/897 [00:08<00:15, 37.18it/s]
Computing ESM2 embeddings: 39%|     | 352/897 [00:09<00:14, 37.87it/s]

```



Computing ESM2 embeddings: 43%| | 384/897 [00:10<00:13, 37.25it/s]

Computing ESM2 embeddings: 46%| | 416/897 [00:11<00:12, 37.95it/s]

Computing ESM2 embeddings: 50%| | 448/897 [00:11<00:12, 37.31it/s]

Computing ESM2 embeddings: 54%| | 480/897 [00:12<00:10, 37.96it/s]

Computing ESM2 embeddings: 57%| | 512/897 [00:13<00:10, 37.33it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:14<00:09, 37.96it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:15<00:08, 37.37it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:16<00:07, 38.01it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:17<00:06, 37.26it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:17<00:05, 37.67it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:18<00:05, 36.52it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:19<00:04, 37.15it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:20<00:03, 36.13it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:21<00:02, 37.12it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:22<00:01, 36.71it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:23<00:00, 37.54it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:24<00:00, 37.31it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.23it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.38it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.00it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 37.05it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 37.94it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.21it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.67it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:18, 36.41it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.05it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.10it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.10it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.71it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.56it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.05it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.77it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.17it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.84it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.28it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.64it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 36.74it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.68it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.21it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.87it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.31it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.92it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.32it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:23<00:01, 37.96it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.36it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.30it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.19it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 36.72it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 38.03it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.08it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.96it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:18, 37.10it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 37.85it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 37.23it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.88it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 37.11it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.82it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.22it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.89it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:11<00:12, 37.26it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 37.90it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.26it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.58it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.52it/s]

```

Computing ESM2 embeddings: 68%|      | 608/896 [00:16<00:07, 37.36it/s]
Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 36.83it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 37.57it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:05, 36.79it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.56it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 36.93it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.66it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 36.91it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.43it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.15it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/589 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/589 [00:00<00:14, 38.55it/s]
Computing ESM2 embeddings: 11%|      | 64/589 [00:01<00:14, 35.87it/s]
Computing ESM2 embeddings: 16%|      | 96/589 [00:02<00:13, 37.53it/s]
Computing ESM2 embeddings: 22%|      | 128/589 [00:03<00:12, 36.54it/s]
Computing ESM2 embeddings: 27%|      | 160/589 [00:04<00:11, 37.57it/s]
Computing ESM2 embeddings: 33%|      | 192/589 [00:05<00:10, 36.99it/s]
Computing ESM2 embeddings: 38%|      | 224/589 [00:05<00:09, 37.84it/s]
Computing ESM2 embeddings: 43%|      | 256/589 [00:06<00:08, 37.17it/s]
Computing ESM2 embeddings: 49%|      | 288/589 [00:07<00:07, 37.64it/s]
Computing ESM2 embeddings: 54%|      | 320/589 [00:08<00:07, 36.43it/s]

```

Computing ESM2 embeddings: 60%| | 352/589 [00:09<00:06, 37.35it/s]  
 Computing ESM2 embeddings: 65%| | 384/589 [00:10<00:05, 36.92it/s]  
 Computing ESM2 embeddings: 71%| | 416/589 [00:11<00:04, 37.71it/s]  
 Computing ESM2 embeddings: 76%| | 448/589 [00:12<00:03, 37.12it/s]  
 Computing ESM2 embeddings: 81%| | 480/589 [00:12<00:02, 37.80it/s]  
 Computing ESM2 embeddings: 87%| | 512/589 [00:13<00:02, 37.22it/s]  
 Computing ESM2 embeddings: 92%| | 544/589 [00:14<00:01, 37.91it/s]  
 Computing ESM2 embeddings: 98%| | 576/589 [00:15<00:00, 37.26it/s]

Computing ESM2 embeddings: 100%| | 589/589 [00:15<00:00, 37.26it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:21, 39.52it/s]  
 Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:22, 37.56it/s]  
 Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:20, 38.53it/s]  
 Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:20, 37.16it/s]  
 Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:19, 38.01it/s]  
 Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:18, 37.32it/s]  
 Computing ESM2 embeddings: 25%| | 224/895 [00:05<00:17, 38.02it/s]  
 Computing ESM2 embeddings: 29%| | 256/895 [00:06<00:17, 37.17it/s]  
 Computing ESM2 embeddings: 32%| | 288/895 [00:07<00:16, 37.57it/s]  
 Computing ESM2 embeddings: 36%| | 320/895 [00:08<00:15, 36.44it/s]  
 Computing ESM2 embeddings: 39%| | 352/895 [00:09<00:14, 37.12it/s]

```

Computing ESM2 embeddings: 43%|          | 384/895 [00:10<00:14, 36.15it/s]
Computing ESM2 embeddings: 46%|          | 416/895 [00:11<00:12, 37.12it/s]
Computing ESM2 embeddings: 50%|          | 448/895 [00:12<00:12, 36.72it/s]
Computing ESM2 embeddings: 54%|          | 480/895 [00:12<00:11, 37.55it/s]
Computing ESM2 embeddings: 57%|          | 512/895 [00:13<00:10, 37.03it/s]
Computing ESM2 embeddings: 61%|          | 544/895 [00:14<00:09, 37.76it/s]
Computing ESM2 embeddings: 64%|          | 576/895 [00:15<00:08, 37.18it/s]
Computing ESM2 embeddings: 68%|          | 608/895 [00:16<00:07, 37.82it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:17<00:06, 37.25it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:17<00:05, 37.92it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:18<00:05, 37.27it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:19<00:04, 37.93it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:20<00:03, 37.06it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:21<00:02, 37.54it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:22<00:01, 36.25it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:23<00:00, 37.22it/s]
Computing ESM2 embeddings: 100%|         | 895/895 [00:24<00:00, 37.22it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:22, 39.22it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:22, 37.16it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.88it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 36.31it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 37.41it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.92it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.73it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 37.17it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.92it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.24it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 37.95it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.31it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.92it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 37.22it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.91it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.32it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.97it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.28it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.90it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.30it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.66it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 36.53it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.41it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 36.99it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.71it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.20it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:23<00:01, 37.82it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.19it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.32it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:26, 32.74it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:24, 33.55it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:22, 36.01it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 35.82it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.10it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.41it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 37.32it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:17, 36.71it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.57it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.97it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.41it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:14, 36.19it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:13, 36.84it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.01it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 36.98it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 36.69it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.45it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.87it/s]



```

Computing ESM2 embeddings: 68%|      | 608/896 [00:16<00:07, 37.57it/s]
Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 37.03it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:18<00:05, 37.74it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:19<00:05, 36.96it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.41it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 36.22it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 36.90it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 35.89it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 36.89it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.66it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:01<00:27, 31.52it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:25, 32.76it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:23, 35.51it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:22, 35.68it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 36.93it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.38it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.36it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:07<00:17, 36.92it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.68it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.96it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.70it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.12it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.81it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.16it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.81it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.21it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.84it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.28it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.64it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.45it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.31it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.87it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.60it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.10it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.59it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.61it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.43it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.96it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.92it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.58it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/896 [00:02<00:28, 29.46it/s]  |
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:24, 33.07it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:22, 33.55it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:20, 35.20it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:20, 34.87it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:18, 36.18it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:07<00:17, 36.01it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:08<00:16, 36.73it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:09<00:16, 35.75it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 36.70it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:14, 36.33it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.22it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 36.82it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:13<00:11, 37.50it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:14<00:10, 36.87it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:15<00:09, 37.31it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:16<00:08, 36.17it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.04it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:07, 36.51it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:06, 37.33it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:19<00:05, 36.73it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:20<00:04, 37.26it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:21<00:03, 36.13it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:22<00:02, 37.01it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 36.50it/s]

Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.02it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.10it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings:  3%|      | 32/915 [00:00<00:22, 39.39it/s]

Computing ESM2 embeddings:  7%|      | 64/915 [00:01<00:22, 37.03it/s]

Computing ESM2 embeddings: 10%|      | 96/915 [00:02<00:21, 38.19it/s]

Computing ESM2 embeddings: 14%|      | 128/915 [00:03<00:21, 37.42it/s]

Computing ESM2 embeddings: 17%|      | 160/915 [00:04<00:19, 38.15it/s]

Computing ESM2 embeddings: 21%|      | 192/915 [00:05<00:19, 37.35it/s]

Computing ESM2 embeddings: 24%|      | 224/915 [00:05<00:18, 37.72it/s]

Computing ESM2 embeddings: 28%|      | 256/915 [00:06<00:18, 36.49it/s]

Computing ESM2 embeddings: 31%|      | 288/915 [00:07<00:16, 37.43it/s]

Computing ESM2 embeddings: 35%|      | 320/915 [00:08<00:16, 36.95it/s]

Computing ESM2 embeddings: 38%|      | 352/915 [00:09<00:14, 37.75it/s]

Computing ESM2 embeddings: 42%|      | 384/915 [00:10<00:14, 37.18it/s]

Computing ESM2 embeddings: 45%|      | 416/915 [00:11<00:13, 37.90it/s]

Computing ESM2 embeddings: 49%|      | 448/915 [00:11<00:12, 37.27it/s]

Computing ESM2 embeddings: 52%|      | 480/915 [00:12<00:11, 37.91it/s]

Computing ESM2 embeddings: 56%|      | 512/915 [00:13<00:10, 37.30it/s]

Computing ESM2 embeddings: 59%|      | 544/915 [00:14<00:09, 37.95it/s]

```

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 37.30it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.94it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 37.10it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:17<00:06, 37.83it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:18<00:05, 37.32it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 38.01it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:03, 37.37it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 37.96it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 37.32it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:22<00:01, 37.95it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:23<00:00, 37.34it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 37.49it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/883 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/883 [00:00<00:22, 37.97it/s]

Computing ESM2 embeddings: 7%| | 64/883 [00:01<00:24, 33.21it/s]

Computing ESM2 embeddings: 11%| | 96/883 [00:02<00:22, 35.60it/s]

Computing ESM2 embeddings: 14%| | 128/883 [00:03<00:21, 35.31it/s]

Computing ESM2 embeddings: 18%| | 160/883 [00:04<00:19, 36.79it/s]

Computing ESM2 embeddings: 22%| | 192/883 [00:05<00:18, 36.51it/s]

Computing ESM2 embeddings: 25%| | 224/883 [00:06<00:17, 37.42it/s]

Computing ESM2 embeddings: 29%| | 256/883 [00:07<00:16, 36.89it/s]

```

Computing ESM2 embeddings: 33%|          | 288/883 [00:07<00:15, 37.47it/s]
Computing ESM2 embeddings: 36%|          | 320/883 [00:08<00:15, 36.38it/s]
Computing ESM2 embeddings: 40%|          | 352/883 [00:09<00:14, 37.30it/s]
Computing ESM2 embeddings: 43%|          | 384/883 [00:10<00:13, 36.86it/s]
Computing ESM2 embeddings: 47%|          | 416/883 [00:11<00:12, 37.59it/s]
Computing ESM2 embeddings: 51%|          | 448/883 [00:12<00:11, 36.94it/s]
Computing ESM2 embeddings: 54%|          | 480/883 [00:13<00:10, 37.68it/s]
Computing ESM2 embeddings: 58%|          | 512/883 [00:13<00:10, 37.00it/s]
Computing ESM2 embeddings: 62%|          | 544/883 [00:14<00:09, 37.63it/s]
Computing ESM2 embeddings: 65%|          | 576/883 [00:15<00:08, 36.51it/s]
Computing ESM2 embeddings: 69%|          | 608/883 [00:16<00:07, 37.33it/s]
Computing ESM2 embeddings: 72%|          | 640/883 [00:17<00:06, 36.76it/s]
Computing ESM2 embeddings: 76%|          | 672/883 [00:18<00:05, 37.52it/s]
Computing ESM2 embeddings: 80%|          | 704/883 [00:19<00:04, 36.88it/s]
Computing ESM2 embeddings: 83%|          | 736/883 [00:19<00:03, 37.37it/s]
Computing ESM2 embeddings: 87%|          | 768/883 [00:20<00:03, 36.32it/s]
Computing ESM2 embeddings: 91%|          | 800/883 [00:21<00:02, 37.22it/s]
Computing ESM2 embeddings: 94%|          | 832/883 [00:22<00:01, 36.66it/s]
Computing ESM2 embeddings: 98%|          | 864/883 [00:23<00:00, 37.42it/s]

Computing ESM2 embeddings: 100%|         | 883/883 [00:23<00:00, 36.83it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/602 [00:00<?, ?it/s]

```

Computing ESM2 embeddings: 5%| | 32/602 [00:00<00:16, 33.94it/s]

Computing ESM2 embeddings: 11%| | 64/602 [00:01<00:15, 33.68it/s]

Computing ESM2 embeddings: 16%| | 96/602 [00:02<00:13, 36.17it/s]

Computing ESM2 embeddings: 21%| | 128/602 [00:03<00:13, 36.07it/s]

Computing ESM2 embeddings: 27%| | 160/602 [00:04<00:11, 37.27it/s]

Computing ESM2 embeddings: 32%| | 192/602 [00:05<00:11, 36.71it/s]

Computing ESM2 embeddings: 37%| | 224/602 [00:06<00:10, 37.59it/s]

Computing ESM2 embeddings: 43%| | 256/602 [00:07<00:09, 36.86it/s]

Computing ESM2 embeddings: 48%| | 288/602 [00:07<00:08, 37.67it/s]

Computing ESM2 embeddings: 53%| | 320/602 [00:08<00:07, 37.15it/s]

Computing ESM2 embeddings: 58%| | 352/602 [00:09<00:06, 37.88it/s]

Computing ESM2 embeddings: 64%| | 384/602 [00:10<00:05, 37.23it/s]

Computing ESM2 embeddings: 69%| | 416/602 [00:11<00:04, 37.88it/s]

Computing ESM2 embeddings: 74%| | 448/602 [00:12<00:04, 37.27it/s]

Computing ESM2 embeddings: 80%| | 480/602 [00:12<00:03, 37.88it/s]

Computing ESM2 embeddings: 85%| | 512/602 [00:13<00:02, 37.08it/s]

Computing ESM2 embeddings: 90%| | 544/602 [00:14<00:01, 37.69it/s]

Computing ESM2 embeddings: 96%| | 576/602 [00:15<00:00, 37.04it/s]

Computing ESM2 embeddings: 100%| | 602/602 [00:16<00:00, 36.84it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.84it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:23, 35.60it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.01it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 36.15it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.19it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.77it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:17, 37.67it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 37.14it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.62it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 36.50it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.41it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 36.95it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.74it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:11, 37.38it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 38.01it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.44it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 38.05it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.44it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 38.03it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.15it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.74it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 37.03it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 36.97it/s] |
| Computing ESM2 embeddings: | 86% | 768/896 [00:20<00:03, 35.97it/s] |
| Computing ESM2 embeddings: | 89% | 800/896 [00:21<00:02, 36.88it/s] |



Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.42it/s]  
Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.23it/s]  
Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.02it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.09it/s]  
Computing ESM2 embeddings: 7%| | 64/914 [00:02<00:27, 31.20it/s]  
Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:24, 34.06it/s]  
Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 34.59it/s]  
Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.35it/s]  
Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:20, 36.10it/s]  
Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.00it/s]  
Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.48it/s]  
Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.12it/s]  
Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.09it/s]  
Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.08it/s]  
Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.71it/s]  
Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.51it/s]  
Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.88it/s]  
Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.67it/s]  
Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 37.07it/s]  
Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.78it/s]

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Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 36.98it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.67it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.05it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.57it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 36.96it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:20<00:04, 37.71it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 36.95it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.68it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.07it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.77it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.01it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 36.75it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:21, 39.37it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:22, 37.28it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 38.32it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 37.49it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 38.25it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:18, 37.20it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 37.94it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:17, 37.29it/s]

```

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Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.94it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 37.31it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 37.99it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:13, 37.38it/s]
Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 37.90it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:11<00:12, 36.90it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:11, 37.78it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.09it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.82it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.21it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.89it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.24it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.93it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 37.19it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.53it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 36.65it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.47it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 36.98it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.71it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 37.31it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/913 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/913 [00:00<00:24, 36.34it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/913 [00:02<00:27, 30.88it/s]  |
| Computing ESM2 embeddings: | 11% | 96/913 [00:02<00:24, 33.03it/s]  |
| Computing ESM2 embeddings: | 14% | 128/913 [00:03<00:23, 33.00it/s] |
| Computing ESM2 embeddings: | 18% | 160/913 [00:04<00:21, 35.06it/s] |
| Computing ESM2 embeddings: | 21% | 192/913 [00:05<00:20, 35.16it/s] |
| Computing ESM2 embeddings: | 25% | 224/913 [00:06<00:18, 36.49it/s] |
| Computing ESM2 embeddings: | 28% | 256/913 [00:07<00:18, 36.29it/s] |
| Computing ESM2 embeddings: | 32% | 288/913 [00:08<00:16, 36.93it/s] |
| Computing ESM2 embeddings: | 35% | 320/913 [00:09<00:16, 36.07it/s] |
| Computing ESM2 embeddings: | 39% | 352/913 [00:09<00:15, 37.07it/s] |
| Computing ESM2 embeddings: | 42% | 384/913 [00:10<00:14, 36.57it/s] |
| Computing ESM2 embeddings: | 46% | 416/913 [00:11<00:13, 37.42it/s] |
| Computing ESM2 embeddings: | 49% | 448/913 [00:12<00:12, 36.86it/s] |
| Computing ESM2 embeddings: | 53% | 480/913 [00:13<00:11, 37.58it/s] |
| Computing ESM2 embeddings: | 56% | 512/913 [00:14<00:10, 37.11it/s] |
| Computing ESM2 embeddings: | 60% | 544/913 [00:15<00:09, 37.76it/s] |
| Computing ESM2 embeddings: | 63% | 576/913 [00:15<00:09, 37.07it/s] |
| Computing ESM2 embeddings: | 67% | 608/913 [00:16<00:08, 37.73it/s] |
| Computing ESM2 embeddings: | 70% | 640/913 [00:17<00:07, 37.15it/s] |
| Computing ESM2 embeddings: | 74% | 672/913 [00:18<00:06, 37.71it/s] |
| Computing ESM2 embeddings: | 77% | 704/913 [00:19<00:05, 37.08it/s] |
| Computing ESM2 embeddings: | 81% | 736/913 [00:20<00:04, 37.68it/s] |
| Computing ESM2 embeddings: | 84% | 768/913 [00:21<00:03, 36.80it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/913 [00:21<00:03, 37.58it/s]
Computing ESM2 embeddings: 91%|      | 832/913 [00:22<00:02, 37.03it/s]
Computing ESM2 embeddings: 95%|      | 864/913 [00:23<00:01, 37.71it/s]
Computing ESM2 embeddings: 98%|      | 896/913 [00:24<00:00, 37.11it/s]
Computing ESM2 embeddings: 100%|     | 913/913 [00:24<00:00, 36.58it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.45it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:23, 35.50it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.28it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 36.77it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.71it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 36.88it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:06<00:17, 37.73it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 37.04it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.79it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 37.17it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.86it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.36it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.86it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:12<00:12, 36.60it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:11, 37.45it/s]

```

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 36.94it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.70it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.16it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.83it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.27it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.95it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.34it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.70it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.57it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.38it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.89it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.41it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.10it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:23, 37.91it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:23, 36.90it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:21, 37.67it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:21, 36.30it/s]

Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:20, 37.20it/s]

Computing ESM2 embeddings: 21%| | 192/915 [00:05<00:20, 36.09it/s]

Computing ESM2 embeddings: 24%| | 224/915 [00:06<00:18, 37.18it/s]

Computing ESM2 embeddings: 28%| | 256/915 [00:06<00:17, 36.92it/s]

Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:16, 37.64it/s]

Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:16, 36.91it/s]

Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:14, 37.72it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 37.18it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 37.56it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:12<00:12, 36.41it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 36.98it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:11, 35.94it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:10, 36.96it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 36.61it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.44it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 36.79it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:18<00:06, 37.30it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:19<00:05, 36.02it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 36.96it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:04, 36.56it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 37.35it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 36.83it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:23<00:01, 37.58it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:24<00:00, 36.87it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 36.91it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/826 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/826 [00:00<00:20, 38.27it/s]  |
| Computing ESM2 embeddings: | 8%  | 64/826 [00:01<00:21, 35.68it/s]  |
| Computing ESM2 embeddings: | 12% | 96/826 [00:02<00:19, 37.36it/s]  |
| Computing ESM2 embeddings: | 15% | 128/826 [00:03<00:19, 36.44it/s] |
| Computing ESM2 embeddings: | 19% | 160/826 [00:04<00:17, 37.51it/s] |
| Computing ESM2 embeddings: | 23% | 192/826 [00:05<00:17, 36.77it/s] |
| Computing ESM2 embeddings: | 27% | 224/826 [00:06<00:16, 37.37it/s] |
| Computing ESM2 embeddings: | 31% | 256/826 [00:06<00:15, 36.15it/s] |
| Computing ESM2 embeddings: | 35% | 288/826 [00:07<00:14, 36.85it/s] |
| Computing ESM2 embeddings: | 39% | 320/826 [00:08<00:14, 36.08it/s] |
| Computing ESM2 embeddings: | 43% | 352/826 [00:09<00:12, 37.10it/s] |
| Computing ESM2 embeddings: | 46% | 384/826 [00:10<00:12, 36.76it/s] |
| Computing ESM2 embeddings: | 50% | 416/826 [00:11<00:10, 37.57it/s] |
| Computing ESM2 embeddings: | 54% | 448/826 [00:12<00:10, 37.04it/s] |
| Computing ESM2 embeddings: | 58% | 480/826 [00:12<00:09, 37.76it/s] |
| Computing ESM2 embeddings: | 62% | 512/826 [00:13<00:08, 37.19it/s] |
| Computing ESM2 embeddings: | 66% | 544/826 [00:14<00:07, 37.87it/s] |
| Computing ESM2 embeddings: | 70% | 576/826 [00:15<00:06, 37.23it/s] |
| Computing ESM2 embeddings: | 74% | 608/826 [00:16<00:05, 37.85it/s] |
| Computing ESM2 embeddings: | 77% | 640/826 [00:17<00:04, 37.32it/s] |
| Computing ESM2 embeddings: | 81% | 672/826 [00:18<00:04, 37.99it/s] |
| Computing ESM2 embeddings: | 85% | 704/826 [00:18<00:03, 37.34it/s] |



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Computing ESM2 embeddings: 89%|      | 736/826 [00:19<00:02, 37.67it/s]
Computing ESM2 embeddings: 93%|      | 768/826 [00:20<00:01, 36.52it/s]
Computing ESM2 embeddings: 97%|      | 800/826 [00:21<00:00, 37.12it/s]
Computing ESM2 embeddings: 100%|     | 826/826 [00:22<00:00, 36.95it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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Computing ESM2 embeddings:  0%|      |  0/660 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/660 [00:00<00:16, 37.77it/s]
Computing ESM2 embeddings: 10%|      | 64/660 [00:01<00:16, 36.87it/s]
Computing ESM2 embeddings: 15%|      | 96/660 [00:02<00:14, 37.72it/s]
Computing ESM2 embeddings: 19%|      | 128/660 [00:03<00:14, 36.08it/s]
Computing ESM2 embeddings: 24%|      | 160/660 [00:04<00:13, 36.98it/s]
Computing ESM2 embeddings: 29%|      | 192/660 [00:05<00:12, 36.02it/s]
Computing ESM2 embeddings: 34%|      | 224/660 [00:06<00:11, 36.98it/s]
Computing ESM2 embeddings: 39%|      | 256/660 [00:07<00:11, 36.12it/s]
Computing ESM2 embeddings: 44%|      | 288/660 [00:07<00:10, 36.84it/s]
Computing ESM2 embeddings: 48%|      | 320/660 [00:08<00:09, 36.02it/s]
Computing ESM2 embeddings: 53%|      | 352/660 [00:09<00:08, 37.05it/s]
Computing ESM2 embeddings: 58%|      | 384/660 [00:10<00:07, 36.71it/s]
Computing ESM2 embeddings: 63%|      | 416/660 [00:11<00:06, 37.50it/s]
Computing ESM2 embeddings: 68%|      | 448/660 [00:12<00:05, 37.04it/s]
Computing ESM2 embeddings: 73%|      | 480/660 [00:12<00:04, 37.75it/s]
Computing ESM2 embeddings: 78%|      | 512/660 [00:13<00:03, 37.16it/s]

```

```

Computing ESM2 embeddings: 82%|      | 544/660 [00:14<00:03, 37.63it/s]
Computing ESM2 embeddings: 87%|      | 576/660 [00:15<00:02, 36.26it/s]
Computing ESM2 embeddings: 92%|      | 608/660 [00:16<00:01, 37.15it/s]
Computing ESM2 embeddings: 97%|      | 640/660 [00:17<00:00, 36.77it/s]
Computing ESM2 embeddings: 100%|     | 660/660 [00:17<00:00, 36.78it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.19it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:23, 35.70it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.06it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:21, 35.91it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.13it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 36.49it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:06<00:17, 37.54it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 36.96it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.77it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 37.18it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.89it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.39it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 38.02it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:12<00:11, 37.35it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 37.98it/s]

```

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.42it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.05it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.29it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.96it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.34it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 38.00it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.30it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.94it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.29it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.95it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.36it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.99it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.40it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.60it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.61it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.88it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.93it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 37.91it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.15it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 37.88it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 37.33it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.89it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.47it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.09it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.17it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.14it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.78it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.59it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.11it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.85it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.44it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 38.21it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.32it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.98it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.35it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.96it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.33it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.99it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.30it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.94it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.29it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.40it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:22, 39.07it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:22, 37.14it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:20, 38.21it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:20, 37.07it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 37.95it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:18, 37.22it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 38.00it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:17, 37.24it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:15, 37.98it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 37.28it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 37.98it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:13, 37.34it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 37.99it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:11<00:11, 37.30it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 37.97it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:10, 37.34it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.89it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 37.51it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 38.24it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:06, 37.39it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:17<00:05, 37.77it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:18<00:05, 36.58it/s] |

```

Computing ESM2 embeddings: 82%|      | 736/895 [00:19<00:04, 37.42it/s]
Computing ESM2 embeddings: 86%|      | 768/895 [00:20<00:03, 36.76it/s]
Computing ESM2 embeddings: 89%|      | 800/895 [00:21<00:02, 37.57it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:22<00:01, 37.07it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:23<00:00, 37.43it/s]

Computing ESM2 embeddings: 100%|     | 895/895 [00:23<00:00, 37.32it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/910 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/910 [00:00<00:23, 37.94it/s]
Computing ESM2 embeddings:  7%|      | 64/910 [00:01<00:25, 33.40it/s]
Computing ESM2 embeddings: 11%|      | 96/910 [00:02<00:23, 35.30it/s]
Computing ESM2 embeddings: 14%|      | 128/910 [00:03<00:21, 36.54it/s]
Computing ESM2 embeddings: 18%|      | 160/910 [00:04<00:19, 37.69it/s]
Computing ESM2 embeddings: 21%|      | 192/910 [00:05<00:19, 37.08it/s]
Computing ESM2 embeddings: 25%|      | 224/910 [00:06<00:18, 37.82it/s]
Computing ESM2 embeddings: 28%|      | 256/910 [00:06<00:17, 36.94it/s]
Computing ESM2 embeddings: 32%|      | 288/910 [00:07<00:16, 37.36it/s]
Computing ESM2 embeddings: 35%|      | 320/910 [00:08<00:16, 36.27it/s]
Computing ESM2 embeddings: 39%|      | 352/910 [00:09<00:15, 37.14it/s]
Computing ESM2 embeddings: 42%|      | 384/910 [00:10<00:14, 36.68it/s]
Computing ESM2 embeddings: 46%|      | 416/910 [00:11<00:13, 37.45it/s]
Computing ESM2 embeddings: 49%|      | 448/910 [00:12<00:12, 36.83it/s]

```

```

Computing ESM2 embeddings: 53%|          | 480/910 [00:12<00:11, 37.58it/s]
Computing ESM2 embeddings: 56%|          | 512/910 [00:13<00:10, 37.11it/s]
Computing ESM2 embeddings: 60%|          | 544/910 [00:14<00:09, 37.74it/s]
Computing ESM2 embeddings: 63%|          | 576/910 [00:15<00:09, 37.07it/s]
Computing ESM2 embeddings: 67%|          | 608/910 [00:16<00:08, 37.72it/s]
Computing ESM2 embeddings: 70%|          | 640/910 [00:17<00:07, 36.95it/s]
Computing ESM2 embeddings: 74%|          | 672/910 [00:18<00:06, 37.63it/s]
Computing ESM2 embeddings: 77%|          | 704/910 [00:19<00:05, 37.06it/s]
Computing ESM2 embeddings: 81%|          | 736/910 [00:19<00:04, 37.63it/s]
Computing ESM2 embeddings: 84%|          | 768/910 [00:20<00:03, 37.07it/s]
Computing ESM2 embeddings: 88%|          | 800/910 [00:21<00:02, 37.75it/s]
Computing ESM2 embeddings: 91%|          | 832/910 [00:22<00:02, 37.06it/s]
Computing ESM2 embeddings: 95%|          | 864/910 [00:23<00:01, 37.70it/s]
Computing ESM2 embeddings: 98%|          | 896/910 [00:24<00:00, 37.02it/s]

Computing ESM2 embeddings: 100%|         | 910/910 [00:24<00:00, 36.95it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/900 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/900 [00:00<00:22, 38.14it/s]
Computing ESM2 embeddings:  7%|          | 64/900 [00:01<00:22, 36.85it/s]
Computing ESM2 embeddings: 11%|          | 96/900 [00:02<00:21, 38.06it/s]
Computing ESM2 embeddings: 14%|          | 128/900 [00:03<00:20, 36.95it/s]
Computing ESM2 embeddings: 18%|          | 160/900 [00:04<00:19, 37.88it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/900 [00:05<00:19, 37.07it/s] |
| Computing ESM2 embeddings: 25%  | 224/900 [00:05<00:17, 37.87it/s] |
| Computing ESM2 embeddings: 28%  | 256/900 [00:06<00:17, 37.30it/s] |
| Computing ESM2 embeddings: 32%  | 288/900 [00:07<00:16, 37.96it/s] |
| Computing ESM2 embeddings: 36%  | 320/900 [00:08<00:15, 37.29it/s] |
| Computing ESM2 embeddings: 39%  | 352/900 [00:09<00:14, 37.70it/s] |
| Computing ESM2 embeddings: 43%  | 384/900 [00:10<00:14, 36.67it/s] |
| Computing ESM2 embeddings: 46%  | 416/900 [00:11<00:12, 37.49it/s] |
| Computing ESM2 embeddings: 50%  | 448/900 [00:11<00:12, 37.07it/s] |
| Computing ESM2 embeddings: 53%  | 480/900 [00:12<00:11, 37.74it/s] |
| Computing ESM2 embeddings: 57%  | 512/900 [00:13<00:10, 37.19it/s] |
| Computing ESM2 embeddings: 60%  | 544/900 [00:14<00:09, 37.86it/s] |
| Computing ESM2 embeddings: 64%  | 576/900 [00:15<00:08, 37.26it/s] |
| Computing ESM2 embeddings: 68%  | 608/900 [00:16<00:07, 37.90it/s] |
| Computing ESM2 embeddings: 71%  | 640/900 [00:17<00:07, 37.13it/s] |
| Computing ESM2 embeddings: 75%  | 672/900 [00:17<00:06, 37.82it/s] |
| Computing ESM2 embeddings: 78%  | 704/900 [00:18<00:05, 37.15it/s] |
| Computing ESM2 embeddings: 82%  | 736/900 [00:19<00:04, 37.84it/s] |
| Computing ESM2 embeddings: 85%  | 768/900 [00:20<00:03, 37.15it/s] |
| Computing ESM2 embeddings: 89%  | 800/900 [00:21<00:02, 37.82it/s] |
| Computing ESM2 embeddings: 92%  | 832/900 [00:22<00:01, 37.17it/s] |
| Computing ESM2 embeddings: 96%  | 864/900 [00:23<00:00, 37.79it/s] |
| Computing ESM2 embeddings: 100% | 896/900 [00:23<00:00, 37.08it/s] |
| Computing ESM2 embeddings: 100% | 900/900 [00:24<00:00, 37.32it/s] |



Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/905 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/905 [00:00<00:24, 35.41it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/905 [00:01<00:24, 34.64it/s]  |
| Computing ESM2 embeddings: | 11% | 96/905 [00:02<00:22, 36.70it/s]  |
| Computing ESM2 embeddings: | 14% | 128/905 [00:03<00:21, 36.33it/s] |
| Computing ESM2 embeddings: | 18% | 160/905 [00:04<00:19, 37.43it/s] |
| Computing ESM2 embeddings: | 21% | 192/905 [00:05<00:19, 36.73it/s] |
| Computing ESM2 embeddings: | 25% | 224/905 [00:06<00:18, 37.60it/s] |
| Computing ESM2 embeddings: | 28% | 256/905 [00:06<00:17, 37.02it/s] |
| Computing ESM2 embeddings: | 32% | 288/905 [00:07<00:16, 37.80it/s] |
| Computing ESM2 embeddings: | 35% | 320/905 [00:08<00:15, 37.22it/s] |
| Computing ESM2 embeddings: | 39% | 352/905 [00:09<00:14, 37.63it/s] |
| Computing ESM2 embeddings: | 42% | 384/905 [00:10<00:14, 36.53it/s] |
| Computing ESM2 embeddings: | 46% | 416/905 [00:11<00:13, 37.38it/s] |
| Computing ESM2 embeddings: | 50% | 448/905 [00:12<00:12, 36.96it/s] |
| Computing ESM2 embeddings: | 53% | 480/905 [00:12<00:11, 37.66it/s] |
| Computing ESM2 embeddings: | 57% | 512/905 [00:13<00:10, 37.14it/s] |
| Computing ESM2 embeddings: | 60% | 544/905 [00:14<00:09, 37.83it/s] |
| Computing ESM2 embeddings: | 64% | 576/905 [00:15<00:08, 37.26it/s] |
| Computing ESM2 embeddings: | 67% | 608/905 [00:16<00:07, 37.90it/s] |
| Computing ESM2 embeddings: | 71% | 640/905 [00:17<00:07, 37.28it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/905 [00:18<00:06, 37.86it/s]
Computing ESM2 embeddings: 78%|      | 704/905 [00:18<00:05, 37.07it/s]
Computing ESM2 embeddings: 81%|      | 736/905 [00:19<00:04, 37.44it/s]
Computing ESM2 embeddings: 85%|      | 768/905 [00:20<00:03, 36.27it/s]
Computing ESM2 embeddings: 88%|      | 800/905 [00:21<00:02, 36.95it/s]
Computing ESM2 embeddings: 92%|      | 832/905 [00:22<00:02, 35.98it/s]
Computing ESM2 embeddings: 95%|      | 864/905 [00:23<00:01, 36.98it/s]
Computing ESM2 embeddings: 99%|      | 896/905 [00:24<00:00, 36.73it/s]

Computing ESM2 embeddings: 100%|     | 905/905 [00:24<00:00, 36.89it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/779 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/779 [00:00<00:20, 37.06it/s]
Computing ESM2 embeddings:  8%|      | 64/779 [00:02<00:23, 30.59it/s]
Computing ESM2 embeddings: 12%|      | 96/779 [00:02<00:20, 34.08it/s]
Computing ESM2 embeddings: 16%|      | 128/779 [00:03<00:18, 34.41it/s]
Computing ESM2 embeddings: 21%|      | 160/779 [00:04<00:17, 36.05it/s]
Computing ESM2 embeddings: 25%|      | 192/779 [00:05<00:16, 35.99it/s]
Computing ESM2 embeddings: 29%|      | 224/779 [00:06<00:15, 36.95it/s]
Computing ESM2 embeddings: 33%|      | 256/779 [00:07<00:14, 36.64it/s]
Computing ESM2 embeddings: 37%|      | 288/779 [00:08<00:13, 37.19it/s]
Computing ESM2 embeddings: 41%|      | 320/779 [00:08<00:12, 36.19it/s]
Computing ESM2 embeddings: 45%|      | 352/779 [00:09<00:11, 37.14it/s]

```

Computing ESM2 embeddings: 49%| | 384/779 [00:10<00:10, 36.79it/s]

Computing ESM2 embeddings: 53%| | 416/779 [00:11<00:09, 37.57it/s]

Computing ESM2 embeddings: 58%| | 448/779 [00:12<00:08, 37.01it/s]

Computing ESM2 embeddings: 62%| | 480/779 [00:13<00:07, 37.70it/s]

Computing ESM2 embeddings: 66%| | 512/779 [00:14<00:07, 36.99it/s]

Computing ESM2 embeddings: 70%| | 544/779 [00:14<00:06, 37.67it/s]

Computing ESM2 embeddings: 74%| | 576/779 [00:15<00:05, 37.12it/s]

Computing ESM2 embeddings: 78%| | 608/779 [00:16<00:04, 37.80it/s]

Computing ESM2 embeddings: 82%| | 640/779 [00:17<00:03, 37.11it/s]

Computing ESM2 embeddings: 86%| | 672/779 [00:18<00:02, 37.51it/s]

Computing ESM2 embeddings: 90%| | 704/779 [00:19<00:02, 36.42it/s]

Computing ESM2 embeddings: 94%| | 736/779 [00:20<00:01, 37.27it/s]

Computing ESM2 embeddings: 99%| | 768/779 [00:20<00:00, 36.79it/s]

Computing ESM2 embeddings: 100%| | 779/779 [00:21<00:00, 36.48it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/699 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/699 [00:00<00:18, 36.92it/s]

Computing ESM2 embeddings: 9%| | 64/699 [00:01<00:18, 35.19it/s]

Computing ESM2 embeddings: 14%| | 96/699 [00:02<00:16, 37.10it/s]

Computing ESM2 embeddings: 18%| | 128/699 [00:03<00:15, 36.61it/s]

Computing ESM2 embeddings: 23%| | 160/699 [00:04<00:14, 37.56it/s]

Computing ESM2 embeddings: 27%| | 192/699 [00:05<00:13, 36.79it/s]

Computing ESM2 embeddings: 32%| | 224/699 [00:06<00:12, 37.66it/s]

Computing ESM2 embeddings: 37%| | 256/699 [00:06<00:11, 37.11it/s]

Computing ESM2 embeddings: 41%| | 288/699 [00:07<00:10, 37.84it/s]

Computing ESM2 embeddings: 46%| | 320/699 [00:08<00:10, 37.06it/s]

Computing ESM2 embeddings: 50%| | 352/699 [00:09<00:09, 37.77it/s]

Computing ESM2 embeddings: 55%| | 384/699 [00:10<00:08, 37.00it/s]

Computing ESM2 embeddings: 60%| | 416/699 [00:11<00:07, 37.70it/s]

Computing ESM2 embeddings: 64%| | 448/699 [00:12<00:06, 37.07it/s]

Computing ESM2 embeddings: 69%| | 480/699 [00:12<00:05, 37.53it/s]

Computing ESM2 embeddings: 73%| | 512/699 [00:13<00:05, 36.29it/s]

Computing ESM2 embeddings: 78%| | 544/699 [00:14<00:04, 37.17it/s]

Computing ESM2 embeddings: 82%| | 576/699 [00:15<00:03, 36.85it/s]

Computing ESM2 embeddings: 87%| | 608/699 [00:16<00:02, 37.53it/s]

Computing ESM2 embeddings: 92%| | 640/699 [00:17<00:01, 36.93it/s]

Computing ESM2 embeddings: 96%| | 672/699 [00:18<00:00, 37.62it/s]

Computing ESM2 embeddings: 100%| | 699/699 [00:18<00:00, 37.08it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.00it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.27it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.19it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 37.25it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.02it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.10it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.84it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 37.18it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.61it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.47it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.37it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.87it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.67it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 36.86it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.66it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.19it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.88it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.26it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.89it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.29it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.91it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.29it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.63it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 36.50it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.36it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 36.81it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:23<00:01, 37.61it/s] |
| Computing ESM2 embeddings: 98% | 896/914 [00:23<00:00, 37.04it/s] |

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.26it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.03it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:22, 36.54it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.88it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 36.77it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.69it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 37.06it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 37.78it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 36.95it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.41it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 36.01it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 36.97it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 36.63it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.55it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.05it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.75it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 36.90it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.61it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 36.70it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.47it/s] |

```

Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 36.60it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:18<00:05, 37.38it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:05, 36.61it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.13it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 35.98it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 36.93it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 36.49it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.30it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.85it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 36.92it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:24, 35.13it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:22, 37.06it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.49it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.44it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.65it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.54it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 36.85it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.33it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.33it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.28it/s]

```

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.80it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.56it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.89it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.60it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.10it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.76it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.21it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.87it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.16it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.85it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.17it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.50it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:04, 36.33it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.24it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.78it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.46it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.86it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.01it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.35it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.95it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:21, 37.54it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 36.89it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.83it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 37.05it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 37.85it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:17, 37.24it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.96it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 37.28it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.92it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.29it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.72it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 36.30it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:11, 37.21it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 36.84it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.63it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.05it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.78it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:06, 37.18it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 37.67it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 36.68it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.52it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 36.99it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.73it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.19it/s] |

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.88it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.24it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
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Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:23, 37.90it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:25, 33.99it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:22, 36.16it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:21, 35.93it/s]

Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:20, 37.15it/s]

Computing ESM2 embeddings: 21%| | 192/915 [00:05<00:19, 36.48it/s]

Computing ESM2 embeddings: 24%| | 224/915 [00:06<00:18, 37.42it/s]

Computing ESM2 embeddings: 28%| | 256/915 [00:07<00:17, 36.73it/s]

Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:16, 37.57it/s]

Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:16, 37.02it/s]

Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:15, 37.48it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 36.28it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 37.19it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:12<00:12, 36.82it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 37.56it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 37.07it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 37.77it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 37.20it/s]

```

Computing ESM2 embeddings: 66%|          | 608/915 [00:16<00:08, 37.82it/s]
Computing ESM2 embeddings: 70%|          | 640/915 [00:17<00:07, 37.20it/s]
Computing ESM2 embeddings: 73%|          | 672/915 [00:18<00:06, 37.92it/s]
Computing ESM2 embeddings: 77%|          | 704/915 [00:19<00:05, 37.11it/s]
Computing ESM2 embeddings: 80%|          | 736/915 [00:19<00:04, 37.82it/s]
Computing ESM2 embeddings: 84%|          | 768/915 [00:20<00:03, 37.12it/s]
Computing ESM2 embeddings: 87%|          | 800/915 [00:21<00:03, 37.26it/s]
Computing ESM2 embeddings: 91%|          | 832/915 [00:22<00:02, 36.43it/s]
Computing ESM2 embeddings: 94%|          | 864/915 [00:23<00:01, 37.25it/s]
Computing ESM2 embeddings: 98%|          | 896/915 [00:24<00:00, 36.71it/s]

Computing ESM2 embeddings: 100%|         | 915/915 [00:24<00:00, 37.00it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.38it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:23, 35.82it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 37.48it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:21, 36.52it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 37.59it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:19, 36.94it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:05<00:17, 37.76it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:17, 37.10it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.71it/s]

```

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.94it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.71it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.99it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.72it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.13it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.81it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.00it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.70it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.07it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.75it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.16it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.49it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 36.31it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.21it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.81it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.55it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.00it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.46it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.04it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/729 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/729 [00:00<00:18, 37.54it/s]

```

Computing ESM2 embeddings:  9%|          | 64/729 [00:01<00:18, 35.21it/s]
Computing ESM2 embeddings: 13%|          | 96/729 [00:02<00:17, 37.14it/s]
Computing ESM2 embeddings: 18%|          | 128/729 [00:03<00:16, 36.56it/s]
Computing ESM2 embeddings: 22%|          | 160/729 [00:04<00:15, 37.64it/s]
Computing ESM2 embeddings: 26%|          | 192/729 [00:05<00:14, 36.95it/s]
Computing ESM2 embeddings: 31%|          | 224/729 [00:06<00:13, 37.75it/s]
Computing ESM2 embeddings: 35%|          | 256/729 [00:06<00:12, 37.19it/s]
Computing ESM2 embeddings: 40%|          | 288/729 [00:07<00:11, 37.94it/s]
Computing ESM2 embeddings: 44%|          | 320/729 [00:08<00:10, 37.28it/s]
Computing ESM2 embeddings: 48%|          | 352/729 [00:09<00:09, 37.73it/s]
Computing ESM2 embeddings: 53%|          | 384/729 [00:10<00:09, 36.67it/s]
Computing ESM2 embeddings: 57%|          | 416/729 [00:11<00:08, 37.24it/s]
Computing ESM2 embeddings: 61%|          | 448/729 [00:12<00:07, 36.42it/s]
Computing ESM2 embeddings: 66%|          | 480/729 [00:12<00:06, 37.37it/s]
Computing ESM2 embeddings: 70%|          | 512/729 [00:13<00:05, 36.88it/s]
Computing ESM2 embeddings: 75%|          | 544/729 [00:14<00:04, 37.64it/s]
Computing ESM2 embeddings: 79%|          | 576/729 [00:15<00:04, 37.12it/s]
Computing ESM2 embeddings: 83%|          | 608/729 [00:16<00:03, 37.81it/s]
Computing ESM2 embeddings: 88%|          | 640/729 [00:17<00:02, 37.22it/s]
Computing ESM2 embeddings: 92%|          | 672/729 [00:18<00:01, 37.91it/s]
Computing ESM2 embeddings: 97%|          | 704/729 [00:18<00:00, 37.30it/s]

Computing ESM2 embeddings: 100%|         | 729/729 [00:19<00:00, 37.22it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/756 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/756 [00:00<00:18, 39.28it/s]  |
| Computing ESM2 embeddings: | 8%  | 64/756 [00:01<00:18, 37.40it/s]  |
| Computing ESM2 embeddings: | 13% | 96/756 [00:02<00:17, 38.40it/s]  |
| Computing ESM2 embeddings: | 17% | 128/756 [00:03<00:16, 37.29it/s] |
| Computing ESM2 embeddings: | 21% | 160/756 [00:04<00:15, 38.13it/s] |
| Computing ESM2 embeddings: | 25% | 192/756 [00:05<00:15, 37.33it/s] |
| Computing ESM2 embeddings: | 30% | 224/756 [00:05<00:13, 38.06it/s] |
| Computing ESM2 embeddings: | 34% | 256/756 [00:06<00:13, 37.35it/s] |
| Computing ESM2 embeddings: | 38% | 288/756 [00:07<00:12, 38.04it/s] |
| Computing ESM2 embeddings: | 42% | 320/756 [00:08<00:11, 37.27it/s] |
| Computing ESM2 embeddings: | 47% | 352/756 [00:09<00:10, 37.94it/s] |
| Computing ESM2 embeddings: | 51% | 384/756 [00:10<00:10, 37.13it/s] |
| Computing ESM2 embeddings: | 55% | 416/756 [00:11<00:08, 37.93it/s] |
| Computing ESM2 embeddings: | 59% | 448/756 [00:11<00:08, 37.18it/s] |
| Computing ESM2 embeddings: | 63% | 480/756 [00:12<00:07, 37.84it/s] |
| Computing ESM2 embeddings: | 68% | 512/756 [00:13<00:06, 37.10it/s] |
| Computing ESM2 embeddings: | 72% | 544/756 [00:14<00:05, 37.79it/s] |
| Computing ESM2 embeddings: | 76% | 576/756 [00:15<00:04, 37.19it/s] |
| Computing ESM2 embeddings: | 80% | 608/756 [00:16<00:03, 37.87it/s] |
| Computing ESM2 embeddings: | 85% | 640/756 [00:17<00:03, 37.24it/s] |
| Computing ESM2 embeddings: | 89% | 672/756 [00:17<00:02, 37.87it/s] |
| Computing ESM2 embeddings: | 93% | 704/756 [00:18<00:01, 37.25it/s] |

Computing ESM2 embeddings: 97%| | 736/756 [00:19<00:00, 37.58it/s]

Computing ESM2 embeddings: 100%| | 756/756 [00:20<00:00, 37.37it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.95it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:02<00:28, 29.53it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:24, 33.35it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:23, 34.00it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:21, 35.75it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:20, 35.86it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 36.92it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.36it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:08<00:16, 37.25it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.55it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.40it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.87it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.63it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.98it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.70it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 36.96it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.52it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.96it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.58it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.97it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.65it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.99it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 37.65it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.86it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.59it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.96it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.63it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.95it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.65it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:01<00:27, 31.83it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:25, 32.56it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:22, 35.02it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:22, 34.77it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:20, 36.07it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 35.63it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 36.55it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:17, 35.75it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:08<00:16, 36.84it/s]



Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.44it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.28it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.75it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.52it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.87it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.62it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 36.94it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.67it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.05it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.70it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.08it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.45it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 36.42it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.30it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.87it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.61it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.97it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.67it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.64it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.67it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:23, 35.83it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 37.47it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 36.78it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 37.79it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.92it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.78it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 37.11it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.57it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.36it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.13it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.71it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.54it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.08it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.81it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.24it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.92it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.36it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 38.03it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.38it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.74it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 36.41it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.32it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 36.64it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.41it/s] |

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Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.98it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.47it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.37it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.17it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:25, 34.02it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:24, 34.03it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:22, 36.35it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:21, 36.17it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.28it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 36.65it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:06<00:18, 37.25it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:07<00:17, 36.31it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.25it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 36.76it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.54it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 36.94it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.67it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:12<00:12, 37.00it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:11, 37.68it/s]
Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:10, 36.94it/s]

```

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.46it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.39it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.00it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:07, 36.05it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:06, 36.76it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 35.90it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 36.91it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.51it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.35it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.84it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.26it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.68it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
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Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:22, 38.79it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:23, 36.10it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:22, 37.22it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:21, 36.11it/s]

Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:20, 37.11it/s]

Computing ESM2 embeddings: 21%| | 192/915 [00:05<00:19, 36.25it/s]

Computing ESM2 embeddings: 24%| | 224/915 [00:06<00:18, 37.29it/s]

Computing ESM2 embeddings: 28%| | 256/915 [00:06<00:17, 36.82it/s]

Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:16, 37.37it/s]

Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:16, 36.34it/s]

Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:15, 37.25it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 36.77it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 37.60it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:12<00:12, 37.07it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 37.80it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 37.29it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 37.72it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 36.47it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.17it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 36.12it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:18<00:06, 37.10it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:19<00:05, 36.72it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 37.54it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:03, 37.03it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 37.74it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 37.16it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:23<00:01, 37.81it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:24<00:00, 37.27it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 37.11it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/897 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/897 [00:00<00:22, 39.11it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/897 [00:01<00:22, 37.17it/s]  |
| Computing ESM2 embeddings: | 11% | 96/897 [00:02<00:20, 38.23it/s]  |
| Computing ESM2 embeddings: | 14% | 128/897 [00:03<00:20, 36.99it/s] |
| Computing ESM2 embeddings: | 18% | 160/897 [00:04<00:19, 37.91it/s] |
| Computing ESM2 embeddings: | 21% | 192/897 [00:05<00:18, 37.19it/s] |
| Computing ESM2 embeddings: | 25% | 224/897 [00:05<00:17, 37.94it/s] |
| Computing ESM2 embeddings: | 29% | 256/897 [00:06<00:17, 37.20it/s] |
| Computing ESM2 embeddings: | 32% | 288/897 [00:07<00:16, 37.62it/s] |
| Computing ESM2 embeddings: | 36% | 320/897 [00:08<00:15, 36.55it/s] |
| Computing ESM2 embeddings: | 39% | 352/897 [00:09<00:14, 37.44it/s] |
| Computing ESM2 embeddings: | 43% | 384/897 [00:10<00:13, 36.93it/s] |
| Computing ESM2 embeddings: | 46% | 416/897 [00:11<00:12, 37.67it/s] |
| Computing ESM2 embeddings: | 50% | 448/897 [00:11<00:12, 37.17it/s] |
| Computing ESM2 embeddings: | 54% | 480/897 [00:12<00:11, 37.87it/s] |
| Computing ESM2 embeddings: | 57% | 512/897 [00:13<00:10, 37.23it/s] |
| Computing ESM2 embeddings: | 61% | 544/897 [00:14<00:09, 37.90it/s] |
| Computing ESM2 embeddings: | 64% | 576/897 [00:15<00:08, 37.28it/s] |
| Computing ESM2 embeddings: | 68% | 608/897 [00:16<00:07, 37.88it/s] |
| Computing ESM2 embeddings: | 71% | 640/897 [00:17<00:06, 37.05it/s] |
| Computing ESM2 embeddings: | 75% | 672/897 [00:17<00:05, 37.67it/s] |
| Computing ESM2 embeddings: | 78% | 704/897 [00:18<00:05, 37.06it/s] |
| Computing ESM2 embeddings: | 82% | 736/897 [00:19<00:04, 37.75it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/897 [00:20<00:03, 37.07it/s]
Computing ESM2 embeddings: 89%|      | 800/897 [00:21<00:02, 37.76it/s]
Computing ESM2 embeddings: 93%|      | 832/897 [00:22<00:01, 37.10it/s]
Computing ESM2 embeddings: 96%|      | 864/897 [00:23<00:00, 37.74it/s]
Computing ESM2 embeddings: 100%|     | 897/897 [00:24<00:00, 37.33it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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Computing ESM2 embeddings:  0%|      |  0/669 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/669 [00:00<00:19, 32.99it/s]
Computing ESM2 embeddings: 10%|      | 64/669 [00:01<00:18, 33.45it/s]
Computing ESM2 embeddings: 14%|      | 96/669 [00:02<00:16, 35.63it/s]
Computing ESM2 embeddings: 19%|      | 128/669 [00:03<00:15, 34.70it/s]
Computing ESM2 embeddings: 24%|      | 160/669 [00:04<00:14, 36.27it/s]
Computing ESM2 embeddings: 29%|      | 192/669 [00:05<00:13, 36.00it/s]
Computing ESM2 embeddings: 33%|      | 224/669 [00:06<00:12, 37.05it/s]
Computing ESM2 embeddings: 38%|      | 256/669 [00:07<00:11, 36.71it/s]
Computing ESM2 embeddings: 43%|      | 288/669 [00:07<00:10, 37.23it/s]
Computing ESM2 embeddings: 48%|      | 320/669 [00:08<00:09, 36.18it/s]
Computing ESM2 embeddings: 53%|      | 352/669 [00:09<00:08, 37.10it/s]
Computing ESM2 embeddings: 57%|      | 384/669 [00:10<00:07, 36.79it/s]
Computing ESM2 embeddings: 62%|      | 416/669 [00:11<00:06, 37.32it/s]
Computing ESM2 embeddings: 67%|      | 448/669 [00:12<00:06, 36.24it/s]
Computing ESM2 embeddings: 72%|      | 480/669 [00:13<00:05, 37.16it/s]

```

```

Computing ESM2 embeddings: 77%|          | 512/669 [00:14<00:04, 36.70it/s]
Computing ESM2 embeddings: 81%|          | 544/669 [00:14<00:03, 37.49it/s]
Computing ESM2 embeddings: 86%|          | 576/669 [00:15<00:02, 36.92it/s]
Computing ESM2 embeddings: 91%|          | 608/669 [00:16<00:01, 37.59it/s]
Computing ESM2 embeddings: 96%|          | 640/669 [00:17<00:00, 37.05it/s]

Computing ESM2 embeddings: 100%|         | 669/669 [00:18<00:00, 36.68it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/815 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/815 [00:01<00:24, 31.81it/s]
Computing ESM2 embeddings:  8%|          | 64/815 [00:01<00:22, 32.89it/s]
Computing ESM2 embeddings: 12%|          | 96/815 [00:02<00:20, 35.55it/s]
Computing ESM2 embeddings: 16%|          | 128/815 [00:03<00:19, 35.45it/s]
Computing ESM2 embeddings: 20%|          | 160/815 [00:04<00:17, 36.82it/s]
Computing ESM2 embeddings: 24%|          | 192/815 [00:05<00:17, 36.50it/s]
Computing ESM2 embeddings: 27%|          | 224/815 [00:06<00:15, 37.39it/s]
Computing ESM2 embeddings: 31%|          | 256/815 [00:07<00:15, 36.93it/s]
Computing ESM2 embeddings: 35%|          | 288/815 [00:07<00:13, 37.68it/s]
Computing ESM2 embeddings: 39%|          | 320/815 [00:08<00:13, 37.13it/s]
Computing ESM2 embeddings: 43%|          | 352/815 [00:09<00:12, 37.51it/s]
Computing ESM2 embeddings: 47%|          | 384/815 [00:10<00:11, 36.39it/s]
Computing ESM2 embeddings: 51%|          | 416/815 [00:11<00:10, 37.26it/s]
Computing ESM2 embeddings: 55%|          | 448/815 [00:12<00:09, 36.75it/s]

```



Computing ESM2 embeddings: 59%| | 480/815 [00:13<00:08, 37.53it/s]

Computing ESM2 embeddings: 63%| | 512/815 [00:13<00:08, 36.88it/s]

Computing ESM2 embeddings: 67%| | 544/815 [00:14<00:07, 37.56it/s]

Computing ESM2 embeddings: 71%| | 576/815 [00:15<00:06, 37.07it/s]

Computing ESM2 embeddings: 75%| | 608/815 [00:16<00:05, 37.75it/s]

Computing ESM2 embeddings: 79%| | 640/815 [00:17<00:04, 37.07it/s]

Computing ESM2 embeddings: 82%| | 672/815 [00:18<00:03, 37.71it/s]

Computing ESM2 embeddings: 86%| | 704/815 [00:19<00:03, 36.98it/s]

Computing ESM2 embeddings: 90%| | 736/815 [00:19<00:02, 37.66it/s]

Computing ESM2 embeddings: 94%| | 768/815 [00:20<00:01, 37.11it/s]

Computing ESM2 embeddings: 98%| | 800/815 [00:21<00:00, 37.78it/s]

Computing ESM2 embeddings: 100%| | 815/815 [00:22<00:00, 36.90it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:25, 35.19it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 34.76it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.77it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.44it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.54it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.78it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.33it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.20it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.19it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.70it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.49it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.91it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.66it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.12it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.82it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.28it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.90it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.13it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.80it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.18it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.55it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.69it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.51it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.83it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.57it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.89it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.62it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.97it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.11it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:22, 38.25it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:23, 36.04it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:21, 37.58it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:20, 36.68it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 37.67it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:18, 37.05it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 37.84it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:17, 37.20it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:16, 37.57it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 36.44it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 37.32it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:13, 36.92it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 37.40it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:12<00:12, 36.37it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:11, 37.31it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:10, 36.83it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.63it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 37.08it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 37.82it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:06, 37.18it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:18<00:05, 37.87it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:18<00:05, 37.29it/s] |
| Computing ESM2 embeddings: | 82% | 736/895 [00:19<00:04, 37.93it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/895 [00:20<00:03, 37.27it/s]
Computing ESM2 embeddings: 89%|      | 800/895 [00:21<00:02, 37.95it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:22<00:01, 37.32it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:23<00:00, 37.94it/s]
Computing ESM2 embeddings: 100%|     | 895/895 [00:24<00:00, 37.19it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/910 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/910 [00:00<00:27, 32.39it/s]
Computing ESM2 embeddings:  7%|      | 64/910 [00:02<00:26, 31.47it/s]
Computing ESM2 embeddings: 11%|      | 96/910 [00:02<00:24, 33.67it/s]
Computing ESM2 embeddings: 14%|      | 128/910 [00:03<00:22, 34.08it/s]
Computing ESM2 embeddings: 18%|      | 160/910 [00:04<00:21, 35.65it/s]
Computing ESM2 embeddings: 21%|      | 192/910 [00:05<00:20, 35.74it/s]
Computing ESM2 embeddings: 25%|      | 224/910 [00:06<00:18, 36.81it/s]
Computing ESM2 embeddings: 28%|      | 256/910 [00:07<00:17, 36.42it/s]
Computing ESM2 embeddings: 32%|      | 288/910 [00:08<00:16, 37.06it/s]
Computing ESM2 embeddings: 35%|      | 320/910 [00:09<00:16, 36.21it/s]
Computing ESM2 embeddings: 39%|      | 352/910 [00:09<00:15, 37.13it/s]
Computing ESM2 embeddings: 42%|      | 384/910 [00:10<00:14, 36.49it/s]
Computing ESM2 embeddings: 46%|      | 416/910 [00:11<00:13, 37.34it/s]
Computing ESM2 embeddings: 49%|      | 448/910 [00:12<00:12, 36.80it/s]
Computing ESM2 embeddings: 53%|      | 480/910 [00:13<00:11, 37.53it/s]

```

Computing ESM2 embeddings: 56%| | 512/910 [00:14<00:10, 36.87it/s]

Computing ESM2 embeddings: 60%| | 544/910 [00:14<00:09, 37.60it/s]

Computing ESM2 embeddings: 63%| | 576/910 [00:15<00:09, 36.87it/s]

Computing ESM2 embeddings: 67%| | 608/910 [00:16<00:08, 37.56it/s]

Computing ESM2 embeddings: 70%| | 640/910 [00:17<00:07, 36.86it/s]

Computing ESM2 embeddings: 74%| | 672/910 [00:18<00:06, 37.33it/s]

Computing ESM2 embeddings: 77%| | 704/910 [00:19<00:05, 36.18it/s]

Computing ESM2 embeddings: 81%| | 736/910 [00:20<00:04, 37.12it/s]

Computing ESM2 embeddings: 84%| | 768/910 [00:21<00:03, 36.73it/s]

Computing ESM2 embeddings: 88%| | 800/910 [00:21<00:02, 37.52it/s]

Computing ESM2 embeddings: 91%| | 832/910 [00:22<00:02, 37.01it/s]

Computing ESM2 embeddings: 95%| | 864/910 [00:23<00:01, 37.71it/s]

Computing ESM2 embeddings: 98%| | 896/910 [00:24<00:00, 37.06it/s]

Computing ESM2 embeddings: 100%| | 910/910 [00:24<00:00, 36.51it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/900 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/900 [00:00<00:26, 33.33it/s]

Computing ESM2 embeddings: 7%| | 64/900 [00:01<00:24, 33.77it/s]

Computing ESM2 embeddings: 11%| | 96/900 [00:02<00:22, 36.14it/s]

Computing ESM2 embeddings: 14%| | 128/900 [00:03<00:21, 35.63it/s]

Computing ESM2 embeddings: 18%| | 160/900 [00:04<00:20, 36.95it/s]

Computing ESM2 embeddings: 21%| | 192/900 [00:05<00:19, 36.55it/s]

Computing ESM2 embeddings: 25%| | 224/900 [00:06<00:18, 37.12it/s]

Computing ESM2 embeddings: 28%| | 256/900 [00:07<00:17, 35.81it/s]

Computing ESM2 embeddings: 32%| | 288/900 [00:07<00:16, 36.51it/s]

Computing ESM2 embeddings: 36%| | 320/900 [00:08<00:16, 35.64it/s]

Computing ESM2 embeddings: 39%| | 352/900 [00:09<00:14, 36.74it/s]

Computing ESM2 embeddings: 43%| | 384/900 [00:10<00:14, 36.52it/s]

Computing ESM2 embeddings: 46%| | 416/900 [00:11<00:12, 37.35it/s]

Computing ESM2 embeddings: 50%| | 448/900 [00:12<00:12, 36.89it/s]

Computing ESM2 embeddings: 53%| | 480/900 [00:13<00:11, 37.65it/s]

Computing ESM2 embeddings: 57%| | 512/900 [00:14<00:10, 37.15it/s]

Computing ESM2 embeddings: 60%| | 544/900 [00:14<00:09, 37.83it/s]

Computing ESM2 embeddings: 64%| | 576/900 [00:15<00:08, 37.20it/s]

Computing ESM2 embeddings: 68%| | 608/900 [00:16<00:07, 37.82it/s]

Computing ESM2 embeddings: 71%| | 640/900 [00:17<00:07, 37.08it/s]

Computing ESM2 embeddings: 75%| | 672/900 [00:18<00:06, 37.46it/s]

Computing ESM2 embeddings: 78%| | 704/900 [00:19<00:05, 36.28it/s]

Computing ESM2 embeddings: 82%| | 736/900 [00:20<00:04, 36.85it/s]

Computing ESM2 embeddings: 85%| | 768/900 [00:20<00:03, 35.83it/s]

Computing ESM2 embeddings: 89%| | 800/900 [00:21<00:02, 36.84it/s]

Computing ESM2 embeddings: 92%| | 832/900 [00:22<00:01, 36.66it/s]

Computing ESM2 embeddings: 96%| | 864/900 [00:23<00:00, 37.55it/s]

Computing ESM2 embeddings: 100%| | 896/900 [00:24<00:00, 37.02it/s]

Computing ESM2 embeddings: 100%| | 900/900 [00:24<00:00, 36.64it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/905 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/905 [00:00<00:22, 38.22it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/905 [00:01<00:23, 35.75it/s]  |
| Computing ESM2 embeddings: | 11% | 96/905 [00:02<00:21, 37.40it/s]  |
| Computing ESM2 embeddings: | 14% | 128/905 [00:03<00:21, 36.83it/s] |
| Computing ESM2 embeddings: | 18% | 160/905 [00:04<00:19, 37.80it/s] |
| Computing ESM2 embeddings: | 21% | 192/905 [00:05<00:19, 36.96it/s] |
| Computing ESM2 embeddings: | 25% | 224/905 [00:05<00:18, 37.78it/s] |
| Computing ESM2 embeddings: | 28% | 256/905 [00:06<00:17, 37.14it/s] |
| Computing ESM2 embeddings: | 32% | 288/905 [00:07<00:16, 37.86it/s] |
| Computing ESM2 embeddings: | 35% | 320/905 [00:08<00:15, 37.15it/s] |
| Computing ESM2 embeddings: | 39% | 352/905 [00:09<00:14, 37.61it/s] |
| Computing ESM2 embeddings: | 42% | 384/905 [00:10<00:14, 36.46it/s] |
| Computing ESM2 embeddings: | 46% | 416/905 [00:11<00:13, 37.38it/s] |
| Computing ESM2 embeddings: | 50% | 448/905 [00:12<00:12, 36.95it/s] |
| Computing ESM2 embeddings: | 53% | 480/905 [00:12<00:11, 37.37it/s] |
| Computing ESM2 embeddings: | 57% | 512/905 [00:13<00:10, 36.26it/s] |
| Computing ESM2 embeddings: | 60% | 544/905 [00:14<00:09, 37.21it/s] |
| Computing ESM2 embeddings: | 64% | 576/905 [00:15<00:08, 36.88it/s] |
| Computing ESM2 embeddings: | 67% | 608/905 [00:16<00:07, 37.70it/s] |
| Computing ESM2 embeddings: | 71% | 640/905 [00:17<00:07, 37.12it/s] |
| Computing ESM2 embeddings: | 74% | 672/905 [00:18<00:06, 37.83it/s] |

```

Computing ESM2 embeddings: 78%|      | 704/905 [00:18<00:05, 37.22it/s]
Computing ESM2 embeddings: 81%|      | 736/905 [00:19<00:04, 37.88it/s]
Computing ESM2 embeddings: 85%|      | 768/905 [00:20<00:03, 37.31it/s]
Computing ESM2 embeddings: 88%|      | 800/905 [00:21<00:02, 37.69it/s]
Computing ESM2 embeddings: 92%|      | 832/905 [00:22<00:01, 36.61it/s]
Computing ESM2 embeddings: 95%|      | 864/905 [00:23<00:01, 37.45it/s]
Computing ESM2 embeddings: 99%|      | 896/905 [00:24<00:00, 37.01it/s]

Computing ESM2 embeddings: 100%|     | 905/905 [00:24<00:00, 37.18it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|       | 0/905 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|       | 32/905 [00:00<00:25, 34.91it/s]
Computing ESM2 embeddings: 7%|       | 64/905 [00:01<00:24, 34.83it/s]
Computing ESM2 embeddings: 11%|      | 96/905 [00:02<00:22, 36.16it/s]
Computing ESM2 embeddings: 14%|      | 128/905 [00:03<00:21, 35.54it/s]
Computing ESM2 embeddings: 18%|      | 160/905 [00:04<00:20, 36.86it/s]
Computing ESM2 embeddings: 21%|      | 192/905 [00:05<00:19, 36.38it/s]
Computing ESM2 embeddings: 25%|      | 224/905 [00:06<00:18, 37.30it/s]
Computing ESM2 embeddings: 28%|      | 256/905 [00:07<00:17, 36.68it/s]
Computing ESM2 embeddings: 32%|      | 288/905 [00:07<00:16, 37.50it/s]
Computing ESM2 embeddings: 35%|      | 320/905 [00:08<00:15, 37.01it/s]
Computing ESM2 embeddings: 39%|      | 352/905 [00:09<00:14, 37.74it/s]
Computing ESM2 embeddings: 42%|      | 384/905 [00:10<00:14, 37.01it/s]

```



```

Computing ESM2 embeddings: 46%|          | 416/905 [00:11<00:12, 37.69it/s]
Computing ESM2 embeddings: 50%|          | 448/905 [00:12<00:12, 37.11it/s]
Computing ESM2 embeddings: 53%|          | 480/905 [00:12<00:11, 37.77it/s]
Computing ESM2 embeddings: 57%|          | 512/905 [00:13<00:10, 37.15it/s]
Computing ESM2 embeddings: 60%|          | 544/905 [00:14<00:09, 37.79it/s]
Computing ESM2 embeddings: 64%|          | 576/905 [00:15<00:08, 36.93it/s]
Computing ESM2 embeddings: 67%|          | 608/905 [00:16<00:07, 37.65it/s]
Computing ESM2 embeddings: 71%|          | 640/905 [00:17<00:07, 37.07it/s]
Computing ESM2 embeddings: 74%|          | 672/905 [00:18<00:06, 37.83it/s]
Computing ESM2 embeddings: 78%|          | 704/905 [00:18<00:05, 37.13it/s]
Computing ESM2 embeddings: 81%|          | 736/905 [00:19<00:04, 37.80it/s]
Computing ESM2 embeddings: 85%|          | 768/905 [00:20<00:03, 37.11it/s]
Computing ESM2 embeddings: 88%|          | 800/905 [00:21<00:02, 37.79it/s]
Computing ESM2 embeddings: 92%|          | 832/905 [00:22<00:01, 37.04it/s]
Computing ESM2 embeddings: 95%|          | 864/905 [00:23<00:01, 37.66it/s]
Computing ESM2 embeddings: 99%|          | 896/905 [00:24<00:00, 37.07it/s]

Computing ESM2 embeddings: 100%|         | 905/905 [00:24<00:00, 37.08it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/615 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|          | 32/615 [00:00<00:15, 38.43it/s]
Computing ESM2 embeddings: 10%|          | 64/615 [00:01<00:15, 36.02it/s]
Computing ESM2 embeddings: 16%|          | 96/615 [00:02<00:13, 37.31it/s]

```

Computing ESM2 embeddings: 21%| | 128/615 [00:03<00:13, 36.14it/s]

Computing ESM2 embeddings: 26%| | 160/615 [00:04<00:12, 37.33it/s]

Computing ESM2 embeddings: 31%| | 192/615 [00:05<00:11, 36.82it/s]

Computing ESM2 embeddings: 36%| | 224/615 [00:06<00:10, 37.68it/s]

Computing ESM2 embeddings: 42%| | 256/615 [00:06<00:09, 37.15it/s]

Computing ESM2 embeddings: 47%| | 288/615 [00:07<00:08, 37.86it/s]

Computing ESM2 embeddings: 52%| | 320/615 [00:08<00:07, 37.23it/s]

Computing ESM2 embeddings: 57%| | 352/615 [00:09<00:06, 37.92it/s]

Computing ESM2 embeddings: 62%| | 384/615 [00:10<00:06, 37.26it/s]

Computing ESM2 embeddings: 68%| | 416/615 [00:11<00:05, 37.96it/s]

Computing ESM2 embeddings: 73%| | 448/615 [00:12<00:04, 37.23it/s]

Computing ESM2 embeddings: 78%| | 480/615 [00:12<00:03, 37.91it/s]

Computing ESM2 embeddings: 83%| | 512/615 [00:13<00:02, 37.33it/s]

Computing ESM2 embeddings: 88%| | 544/615 [00:14<00:01, 38.01it/s]

Computing ESM2 embeddings: 94%| | 576/615 [00:15<00:01, 37.33it/s]

Computing ESM2 embeddings: 99%| | 608/615 [00:16<00:00, 37.97it/s]

Computing ESM2 embeddings: 100%| | 615/615 [00:16<00:00, 37.34it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/870 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/870 [00:00<00:22, 37.05it/s]

Computing ESM2 embeddings: 7%| | 64/870 [00:01<00:25, 31.53it/s]

Computing ESM2 embeddings: 11%| | 96/870 [00:02<00:22, 34.03it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 15% | 128/870 [00:03<00:20, 35.51it/s] |
| Computing ESM2 embeddings: 18% | 160/870 [00:04<00:19, 36.98it/s] |
| Computing ESM2 embeddings: 22% | 192/870 [00:05<00:18, 36.51it/s] |
| Computing ESM2 embeddings: 26% | 224/870 [00:06<00:17, 37.05it/s] |
| Computing ESM2 embeddings: 29% | 256/870 [00:07<00:17, 35.89it/s] |
| Computing ESM2 embeddings: 33% | 288/870 [00:07<00:15, 36.92it/s] |
| Computing ESM2 embeddings: 37% | 320/870 [00:08<00:15, 36.50it/s] |
| Computing ESM2 embeddings: 40% | 352/870 [00:09<00:13, 37.33it/s] |
| Computing ESM2 embeddings: 44% | 384/870 [00:10<00:13, 36.73it/s] |
| Computing ESM2 embeddings: 48% | 416/870 [00:11<00:12, 37.02it/s] |
| Computing ESM2 embeddings: 51% | 448/870 [00:12<00:11, 36.34it/s] |
| Computing ESM2 embeddings: 55% | 480/870 [00:13<00:10, 36.96it/s] |
| Computing ESM2 embeddings: 59% | 512/870 [00:14<00:10, 35.75it/s] |
| Computing ESM2 embeddings: 63% | 544/870 [00:14<00:08, 36.79it/s] |
| Computing ESM2 embeddings: 66% | 576/870 [00:15<00:08, 36.55it/s] |
| Computing ESM2 embeddings: 70% | 608/870 [00:16<00:07, 37.36it/s] |
| Computing ESM2 embeddings: 74% | 640/870 [00:17<00:06, 36.76it/s] |
| Computing ESM2 embeddings: 77% | 672/870 [00:18<00:05, 37.52it/s] |
| Computing ESM2 embeddings: 81% | 704/870 [00:19<00:04, 36.85it/s] |
| Computing ESM2 embeddings: 85% | 736/870 [00:20<00:03, 37.60it/s] |
| Computing ESM2 embeddings: 88% | 768/870 [00:20<00:02, 36.97it/s] |
| Computing ESM2 embeddings: 92% | 800/870 [00:21<00:01, 37.65it/s] |
| Computing ESM2 embeddings: 96% | 832/870 [00:22<00:01, 36.99it/s] |
| Computing ESM2 embeddings: 99% | 864/870 [00:23<00:00, 37.66it/s] |

Computing ESM2 embeddings: 100%| | 870/870 [00:23<00:00, 36.63it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: 4%  | 32/896 [00:00<00:22, 38.18it/s]  |
| Computing ESM2 embeddings: 7%  | 64/896 [00:01<00:22, 36.63it/s]  |
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:21, 37.89it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 36.78it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.63it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 37.02it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 37.73it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:17, 37.10it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.80it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 37.06it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.71it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.02it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.73it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 36.98it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:11, 37.66it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.13it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.82it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.04it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.74it/s] |

```

Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 37.20it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 37.84it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:05, 37.26it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.88it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.06it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.71it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.04it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.51it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.13it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/913 [00:00<00:23, 37.10it/s]
Computing ESM2 embeddings: 7%|      | 64/913 [00:01<00:23, 36.75it/s]
Computing ESM2 embeddings: 11%|     | 96/913 [00:02<00:21, 37.79it/s]
Computing ESM2 embeddings: 14%|     | 128/913 [00:03<00:21, 36.58it/s]
Computing ESM2 embeddings: 18%|     | 160/913 [00:04<00:20, 37.63it/s]
Computing ESM2 embeddings: 21%|     | 192/913 [00:05<00:19, 37.04it/s]
Computing ESM2 embeddings: 25%|     | 224/913 [00:05<00:18, 37.81it/s]
Computing ESM2 embeddings: 28%|     | 256/913 [00:06<00:17, 37.18it/s]
Computing ESM2 embeddings: 32%|     | 288/913 [00:07<00:16, 37.72it/s]
Computing ESM2 embeddings: 35%|     | 320/913 [00:08<00:15, 37.07it/s]
Computing ESM2 embeddings: 39%|     | 352/913 [00:09<00:14, 37.80it/s]

```

Computing ESM2 embeddings: 42%| | 384/913 [00:10<00:14, 37.22it/s]

Computing ESM2 embeddings: 46%| | 416/913 [00:11<00:13, 37.91it/s]

Computing ESM2 embeddings: 49%| | 448/913 [00:11<00:12, 37.29it/s]

Computing ESM2 embeddings: 53%| | 480/913 [00:12<00:11, 37.94it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 37.41it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 38.13it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:15<00:08, 37.45it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:16<00:08, 38.02it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:17<00:07, 37.39it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:17<00:06, 37.96it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:18<00:05, 37.14it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:19<00:04, 37.84it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:20<00:03, 37.25it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:21<00:02, 37.92it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 37.30it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:23<00:01, 37.89it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:23<00:00, 37.30it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:24<00:00, 37.49it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.39it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.75it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:21, 37.40it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 36.76it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.67it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 36.89it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 37.69it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:17, 37.08it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.85it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 37.19it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.90it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.29it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.97it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:12, 37.22it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 37.95it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.12it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.82it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.18it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.61it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:07, 36.56it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 37.41it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 36.93it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.68it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.14it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.83it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.16it/s] |

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.79it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.31it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 35.42it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 34.53it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.67it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.18it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.34it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.59it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.46it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 36.91it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.49it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.86it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.60it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.96it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.66it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.14it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.77it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.13it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.80it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.12it/s]



Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.49it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.40it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.26it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.76it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.52it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.95it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.66it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.04it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.66it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.01it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.11it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:21, 39.30it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:02<00:28, 29.08it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:24, 32.07it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:23, 32.27it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:21, 34.31it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:20, 34.21it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 35.42it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:18, 34.84it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:08<00:16, 35.93it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:09<00:16, 35.20it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:10<00:14, 36.40it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:11<00:14, 36.11it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.06it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.67it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.45it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 36.82it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:15<00:09, 37.58it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:16<00:08, 36.96it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.64it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 36.88it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.64it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 37.22it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.89it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:21<00:03, 37.19it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:22<00:02, 37.91it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.20it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.88it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.27it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.66it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:23, 36.33it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 37.57it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 36.77it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 37.73it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.96it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.65it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 37.01it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.48it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.44it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.30it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.74it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.51it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 36.87it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.60it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.10it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.89it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.31it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.90it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.20it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.85it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.02it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.66it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.06it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.69it/s] |

```

Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.97it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.63it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.77it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.27it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/572 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/572 [00:00<00:14, 38.09it/s]
Computing ESM2 embeddings: 11%|      | 64/572 [00:01<00:14, 35.59it/s]
Computing ESM2 embeddings: 17%|      | 96/572 [00:02<00:12, 37.38it/s]
Computing ESM2 embeddings: 22%|      | 128/572 [00:03<00:12, 36.56it/s]
Computing ESM2 embeddings: 28%|      | 160/572 [00:04<00:10, 37.48it/s]
Computing ESM2 embeddings: 34%|      | 192/572 [00:05<00:10, 36.82it/s]
Computing ESM2 embeddings: 39%|      | 224/572 [00:06<00:09, 37.65it/s]
Computing ESM2 embeddings: 45%|      | 256/572 [00:06<00:08, 36.99it/s]
Computing ESM2 embeddings: 50%|      | 288/572 [00:07<00:07, 37.73it/s]
Computing ESM2 embeddings: 56%|      | 320/572 [00:08<00:06, 36.99it/s]
Computing ESM2 embeddings: 62%|      | 352/572 [00:09<00:05, 37.47it/s]
Computing ESM2 embeddings: 67%|      | 384/572 [00:10<00:05, 36.61it/s]
Computing ESM2 embeddings: 73%|      | 416/572 [00:11<00:04, 37.44it/s]
Computing ESM2 embeddings: 78%|      | 448/572 [00:12<00:03, 36.76it/s]
Computing ESM2 embeddings: 84%|      | 480/572 [00:12<00:02, 37.52it/s]
Computing ESM2 embeddings: 90%|      | 512/572 [00:13<00:01, 36.85it/s]

```

Computing ESM2 embeddings: 95%| | 544/572 [00:14<00:00, 37.60it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:15<00:00, 36.94it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 36.49it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:02<00:28, 29.80it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:24, 33.42it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 34.20it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:21, 35.89it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:20, 35.66it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 36.83it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.51it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:08<00:16, 37.37it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.83it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.59it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.87it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.48it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.97it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.63it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 36.89it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.57it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.09it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.74it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.15it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.84it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 37.20it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 37.78it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.13it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.78it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.14it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.63it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.85it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.65it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 37.87it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:23, 35.55it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:21, 36.79it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:21, 35.55it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:19, 36.86it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:19, 36.58it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:06<00:17, 37.52it/s]

Computing ESM2 embeddings: 29%| | 256/895 [00:06<00:17, 36.97it/s]

Computing ESM2 embeddings: 32%| | 288/895 [00:07<00:16, 37.49it/s]

Computing ESM2 embeddings: 36%| | 320/895 [00:08<00:15, 36.38it/s]

Computing ESM2 embeddings: 39%| | 352/895 [00:09<00:14, 37.27it/s]

Computing ESM2 embeddings: 43%| | 384/895 [00:10<00:13, 36.83it/s]

Computing ESM2 embeddings: 46%| | 416/895 [00:11<00:12, 37.57it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:12<00:12, 37.10it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:12<00:10, 37.83it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 37.21it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.90it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.24it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.92it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 37.04it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:05, 37.69it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:05, 37.20it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.87it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 37.16it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.83it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.21it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.84it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 37.12it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.98it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:23, 35.97it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 37.43it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 36.84it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 37.76it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.09it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.88it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 37.21it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.96it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.28it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 37.98it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.29it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.97it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 37.31it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.94it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.42it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 38.04it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.32it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.91it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.26it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.89it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.32it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.95it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.32it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.64it/s] |



```

Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.55it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.35it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 36.84it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.42it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 39.12it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:22, 37.23it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.97it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:21, 36.32it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.43it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 36.95it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 37.75it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 37.10it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.83it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 37.13it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.84it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.28it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.90it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:12, 37.09it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:11, 37.56it/s]
Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:10, 36.27it/s]

```

```

Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.18it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 36.63it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.40it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 36.94it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:17<00:05, 37.69it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 37.20it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.86it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.24it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.59it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 36.44it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.33it/s]

```

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.19it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 37.49it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:24, 35.33it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.19it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 36.77it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:19, 37.77it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 36.88it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:06<00:18, 37.73it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 37.17it/s]

```

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.88it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 37.23it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.93it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.28it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.93it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.27it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.93it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.31it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.86it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.07it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.77it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.24it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.88it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.16it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.85it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.20it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.63it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.43it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.32it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.91it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.27it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.87it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:22, 36.84it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 38.03it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 37.11it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.84it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:18, 37.07it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 37.79it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 37.19it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.72it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.11it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.75it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 36.98it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.63it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 36.97it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.69it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.13it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.77it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.08it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.73it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.06it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.39it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 36.45it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.39it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 36.85it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.55it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.06it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.48it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.11it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 39.21it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:22, 37.38it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 38.40it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 37.27it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 38.08it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 37.29it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:18, 38.02it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 37.31it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.71it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.55it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.45it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 36.95it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.71it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:11<00:12, 37.19it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.87it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.16it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.84it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.34it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.63it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.48it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.38it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 36.93it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.71it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.17it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.88it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.24it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.94it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.31it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.41it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:14, 37.69it/s]

Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:14, 35.53it/s]

Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:12, 36.88it/s]

Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:12, 35.71it/s]

Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:11, 36.95it/s]

Computing ESM2 embeddings: 34%| | 192/571 [00:05<00:10, 36.68it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:06<00:09, 37.59it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 37.05it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 37.52it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 36.53it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 37.53it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:05, 36.98it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 37.48it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:12<00:03, 36.79it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 37.74it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 37.31it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 38.07it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 37.07it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.03it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.83it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 38.07it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 37.13it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 37.94it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.07it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 37.77it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 37.38it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 38.18it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 37.30it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.64it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.71it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.53it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 36.92it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.71it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.14it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.82it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.20it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.86it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.26it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.89it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.30it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.90it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.19it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.79it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.14it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.75it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.03it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.38it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.



|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.97it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:23, 35.47it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 36.85it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 35.60it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 36.96it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.60it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:17, 37.55it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 37.06it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.84it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.11it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.84it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.25it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.93it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.31it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 37.97it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.32it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.98it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.30it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.96it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.40it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.78it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 36.49it/s] |
| Computing ESM2 embeddings: | 82% | 736/896 [00:19<00:04, 37.34it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 36.78it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.57it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.22it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.89it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.20it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/913 [00:00<00:23, 38.07it/s]
Computing ESM2 embeddings:  7%|      | 64/913 [00:01<00:26, 32.17it/s]
Computing ESM2 embeddings: 11%|      | 96/913 [00:02<00:23, 34.96it/s]
Computing ESM2 embeddings: 14%|      | 128/913 [00:03<00:22, 35.26it/s]
Computing ESM2 embeddings: 18%|      | 160/913 [00:04<00:20, 36.69it/s]
Computing ESM2 embeddings: 21%|      | 192/913 [00:05<00:19, 36.33it/s]
Computing ESM2 embeddings: 25%|      | 224/913 [00:06<00:18, 37.33it/s]
Computing ESM2 embeddings: 28%|      | 256/913 [00:07<00:17, 36.80it/s]
Computing ESM2 embeddings: 32%|      | 288/913 [00:07<00:16, 37.67it/s]
Computing ESM2 embeddings: 35%|      | 320/913 [00:08<00:15, 37.11it/s]
Computing ESM2 embeddings: 39%|      | 352/913 [00:09<00:14, 37.83it/s]
Computing ESM2 embeddings: 42%|      | 384/913 [00:10<00:14, 37.02it/s]
Computing ESM2 embeddings: 46%|      | 416/913 [00:11<00:13, 37.75it/s]
Computing ESM2 embeddings: 49%|      | 448/913 [00:12<00:12, 37.12it/s]
Computing ESM2 embeddings: 53%|      | 480/913 [00:12<00:11, 37.83it/s]

```

Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 37.20it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 37.87it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:15<00:09, 37.17it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:16<00:08, 37.83it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:17<00:07, 37.22it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:18<00:06, 37.78it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:19<00:05, 37.10it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:19<00:04, 37.73it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:20<00:03, 37.10it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:21<00:03, 37.52it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 36.58it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:23<00:01, 37.32it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:24<00:00, 36.89it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:24<00:00, 36.94it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.92it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:25, 32.47it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:22, 35.38it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 35.73it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.09it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.63it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:17, 37.59it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:07<00:17, 36.76it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.59it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 37.09it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.77it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.22it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.91it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:11, 37.34it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 37.94it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.08it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.79it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.17it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.85it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:06, 37.21it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:05, 37.89it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 37.09it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.78it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.16it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.81it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.09it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 37.76it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.10it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.21it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:24, 34.28it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:22, 36.12it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:22, 35.29it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 36.67it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.45it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.31it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:07<00:17, 36.67it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.54it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.97it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.72it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.01it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.78it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 37.19it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.87it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.28it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.89it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.29it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.89it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.25it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 37.60it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:19<00:05, 36.54it/s] |

```

Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.37it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 36.97it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.67it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.09it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.78it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.08it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.14it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/897 [00:00<00:22, 38.84it/s]
Computing ESM2 embeddings:  7%|          | 64/897 [00:01<00:22, 36.93it/s]
Computing ESM2 embeddings: 11%|          | 96/897 [00:02<00:21, 38.08it/s]
Computing ESM2 embeddings: 14%|          | 128/897 [00:03<00:20, 36.82it/s]
Computing ESM2 embeddings: 18%|          | 160/897 [00:04<00:19, 37.77it/s]
Computing ESM2 embeddings: 21%|          | 192/897 [00:05<00:19, 37.04it/s]
Computing ESM2 embeddings: 25%|          | 224/897 [00:05<00:17, 37.96it/s]
Computing ESM2 embeddings: 29%|          | 256/897 [00:06<00:17, 37.25it/s]
Computing ESM2 embeddings: 32%|          | 288/897 [00:07<00:16, 37.65it/s]
Computing ESM2 embeddings: 36%|          | 320/897 [00:08<00:15, 36.48it/s]
Computing ESM2 embeddings: 39%|          | 352/897 [00:09<00:14, 37.35it/s]
Computing ESM2 embeddings: 43%|          | 384/897 [00:10<00:13, 36.93it/s]
Computing ESM2 embeddings: 46%|          | 416/897 [00:11<00:12, 37.68it/s]

```

```

Computing ESM2 embeddings: 50%|          | 448/897 [00:12<00:12, 37.14it/s]
Computing ESM2 embeddings: 54%|          | 480/897 [00:12<00:11, 37.85it/s]
Computing ESM2 embeddings: 57%|          | 512/897 [00:13<00:10, 37.19it/s]
Computing ESM2 embeddings: 61%|          | 544/897 [00:14<00:09, 37.62it/s]
Computing ESM2 embeddings: 64%|          | 576/897 [00:15<00:08, 36.49it/s]
Computing ESM2 embeddings: 68%|          | 608/897 [00:16<00:07, 37.06it/s]
Computing ESM2 embeddings: 71%|          | 640/897 [00:17<00:07, 36.04it/s]
Computing ESM2 embeddings: 75%|          | 672/897 [00:18<00:06, 36.99it/s]
Computing ESM2 embeddings: 78%|          | 704/897 [00:18<00:05, 36.48it/s]
Computing ESM2 embeddings: 82%|          | 736/897 [00:19<00:04, 37.29it/s]
Computing ESM2 embeddings: 86%|          | 768/897 [00:20<00:03, 36.88it/s]
Computing ESM2 embeddings: 89%|          | 800/897 [00:21<00:02, 37.60it/s]
Computing ESM2 embeddings: 93%|          | 832/897 [00:22<00:01, 36.96it/s]
Computing ESM2 embeddings: 96%|          | 864/897 [00:23<00:00, 37.66it/s]

Computing ESM2 embeddings: 100%|         | 897/897 [00:24<00:00, 36.99it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 37.83it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:23, 36.72it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.54it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 36.10it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:20, 37.31it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/914 [00:05<00:19, 36.81it/s] |
| Computing ESM2 embeddings: 25%  | 224/914 [00:06<00:18, 37.69it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:17, 36.85it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 37.64it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:16, 37.12it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 37.85it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:14, 37.19it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:11<00:13, 37.89it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:12<00:12, 37.27it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.89it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 37.31it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 37.91it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:15<00:09, 37.17it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:16<00:08, 37.81it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:17<00:07, 37.21it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:17<00:06, 37.83it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 37.25it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:19<00:04, 37.89it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:20<00:03, 37.17it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:21<00:03, 37.81it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:22<00:02, 37.20it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:23<00:01, 37.78it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:23<00:00, 37.15it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:24<00:00, 37.30it/s] |



Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/552 [00:00<?, ?it/s]
Computing ESM2 embeddings: 6%|          | 32/552 [00:00<00:13, 37.46it/s]
Computing ESM2 embeddings: 12%|         | 64/552 [00:02<00:16, 29.68it/s]
Computing ESM2 embeddings: 17%|         | 96/552 [00:02<00:13, 33.01it/s]
Computing ESM2 embeddings: 23%|         | 128/552 [00:03<00:12, 35.13it/s]
Computing ESM2 embeddings: 29%|         | 160/552 [00:04<00:10, 36.74it/s]
Computing ESM2 embeddings: 35%|         | 192/552 [00:05<00:09, 36.39it/s]
Computing ESM2 embeddings: 41%|         | 224/552 [00:06<00:08, 37.35it/s]
Computing ESM2 embeddings: 46%|         | 256/552 [00:07<00:08, 36.86it/s]
Computing ESM2 embeddings: 52%|         | 288/552 [00:07<00:07, 37.35it/s]
Computing ESM2 embeddings: 58%|         | 320/552 [00:08<00:06, 36.27it/s]
Computing ESM2 embeddings: 64%|         | 352/552 [00:09<00:05, 36.95it/s]
Computing ESM2 embeddings: 70%|         | 384/552 [00:10<00:04, 36.01it/s]
Computing ESM2 embeddings: 75%|         | 416/552 [00:11<00:03, 36.98it/s]
Computing ESM2 embeddings: 81%|         | 448/552 [00:12<00:02, 36.58it/s]
Computing ESM2 embeddings: 87%|         | 480/552 [00:13<00:01, 37.41it/s]
Computing ESM2 embeddings: 93%|         | 512/552 [00:14<00:01, 36.71it/s]
Computing ESM2 embeddings: 99%|         | 544/552 [00:14<00:00, 37.50it/s]
```

```
Computing ESM2 embeddings: 100%|        | 552/552 [00:15<00:00, 36.32it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 38.36it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:23, 35.75it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.44it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 36.80it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 37.79it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.11it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 37.90it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.38it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.80it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.48it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 37.43it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 36.97it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.74it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 37.19it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.93it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.34it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.94it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.26it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.87it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.23it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 37.80it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:18<00:05, 37.20it/s] |

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.64it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:04, 36.46it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.34it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.91it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.69it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.06it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.21it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.99it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.31it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 36.96it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 36.47it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.49it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.96it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:17, 37.53it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 36.26it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.24it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.81it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.57it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.17it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.88it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.31it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 37.99it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.43it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 38.06it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.38it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.77it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:07, 36.52it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:06, 37.10it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 36.15it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.13it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.75it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.45it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.93it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.66it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.00it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:30, 29.40it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:02<00:30, 28.30it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:03<00:25, 32.07it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:04<00:24, 32.73it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:21, 34.92it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/914 [00:05<00:20, 34.80it/s] |
| Computing ESM2 embeddings: 25%  | 224/914 [00:06<00:19, 36.01it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:07<00:18, 35.21it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:08<00:17, 36.45it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:09<00:16, 36.17it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:10<00:15, 37.11it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:14, 36.77it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:11<00:13, 37.29it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:12<00:12, 36.29it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:13<00:11, 37.19it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:14<00:10, 36.63it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:15<00:09, 37.44it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:16<00:09, 37.03it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:16<00:08, 37.75it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:17<00:07, 37.15it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:18<00:06, 37.74it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:19<00:05, 37.22it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:20<00:04, 37.87it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:21<00:03, 37.24it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:22<00:03, 37.89it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:22<00:02, 37.15it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:23<00:01, 37.53it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:24<00:00, 36.40it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:25<00:00, 36.16it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 37.94it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:02<00:32, 25.93it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:03<00:26, 30.56it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:04<00:24, 31.85it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:21, 34.25it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:20, 34.87it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 36.21it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:07<00:17, 35.97it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:08<00:16, 37.03it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:09<00:15, 36.61it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:10<00:14, 37.44it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 36.75it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.56it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.05it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:13<00:11, 37.68it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:14<00:10, 37.00it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:15<00:09, 37.66it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:16<00:08, 36.99it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.32it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:07, 36.19it/s] |

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:06, 37.06it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 36.60it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.36it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:21<00:03, 36.97it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:22<00:02, 37.61it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.02it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.66it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.17it/s]

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Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:23, 37.81it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:23, 35.70it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:21, 37.33it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:21, 36.64it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:19, 37.69it/s]

Computing ESM2 embeddings: 21%| | 192/913 [00:05<00:19, 36.88it/s]

Computing ESM2 embeddings: 25%| | 224/913 [00:06<00:18, 37.71it/s]

Computing ESM2 embeddings: 28%| | 256/913 [00:06<00:17, 37.00it/s]

Computing ESM2 embeddings: 32%| | 288/913 [00:07<00:16, 37.76it/s]

Computing ESM2 embeddings: 35%| | 320/913 [00:08<00:15, 37.07it/s]

Computing ESM2 embeddings: 39%| | 352/913 [00:09<00:14, 37.78it/s]

Computing ESM2 embeddings: 42%| | 384/913 [00:10<00:14, 37.04it/s]

Computing ESM2 embeddings: 46%| | 416/913 [00:11<00:13, 37.69it/s]

Computing ESM2 embeddings: 49%| | 448/913 [00:12<00:12, 37.04it/s]

Computing ESM2 embeddings: 53%| | 480/913 [00:12<00:11, 37.66it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 37.04it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 37.64it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:15<00:09, 37.02it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:16<00:08, 37.64it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:17<00:07, 37.10it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:17<00:06, 37.71it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:18<00:05, 37.02it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:19<00:04, 37.71it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:20<00:03, 37.13it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:21<00:02, 37.73it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 37.01it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:23<00:01, 37.66it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:24<00:00, 37.06it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:24<00:00, 37.22it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

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Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.13it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:02<00:27, 30.62it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:23, 33.93it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:22, 34.64it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:20, 36.17it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 36.15it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:18, 36.84it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:07<00:17, 36.29it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:08<00:16, 36.60it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:09<00:16, 35.58it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 36.63it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:14, 36.32it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.17it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 36.78it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:13<00:11, 37.52it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:14<00:10, 36.88it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.62it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 36.94it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.13it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:07, 36.06it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:06, 37.04it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:19<00:05, 36.67it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:20<00:04, 37.50it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:21<00:03, 37.06it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.76it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.12it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 37.78it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.52it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:27, 31.81it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:25, 32.90it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 35.59it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 35.55it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.90it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.40it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.29it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.28it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 36.99it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.61it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.14it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.07it/s]  
 Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.05it/s]  
 Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.63it/s]  
 Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.21it/s]  
 Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:11, 36.25it/s]  
 Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.17it/s]  
 Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.85it/s]  
 Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.62it/s]

```

Computing ESM2 embeddings: 70%|      | 640/914 [00:17<00:07, 37.08it/s]
Computing ESM2 embeddings: 74%|      | 672/914 [00:18<00:06, 37.53it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:19<00:05, 36.39it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:20<00:04, 37.26it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 36.82it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.48it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.88it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.37it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.21it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.63it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/573 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/573 [00:01<00:17, 31.77it/s]
Computing ESM2 embeddings: 11%|      | 64/573 [00:01<00:15, 33.12it/s]
Computing ESM2 embeddings: 17%|      | 96/573 [00:02<00:13, 35.44it/s]
Computing ESM2 embeddings: 22%|      | 128/573 [00:03<00:12, 35.10it/s]
Computing ESM2 embeddings: 28%|      | 160/573 [00:04<00:11, 36.54it/s]
Computing ESM2 embeddings: 34%|      | 192/573 [00:05<00:10, 36.33it/s]
Computing ESM2 embeddings: 39%|      | 224/573 [00:06<00:09, 37.31it/s]
Computing ESM2 embeddings: 45%|      | 256/573 [00:07<00:08, 36.66it/s]
Computing ESM2 embeddings: 50%|      | 288/573 [00:07<00:07, 37.53it/s]
Computing ESM2 embeddings: 56%|      | 320/573 [00:08<00:06, 37.10it/s]

```

Computing ESM2 embeddings: 61%| | 352/573 [00:09<00:05, 37.76it/s]  
 Computing ESM2 embeddings: 67%| | 384/573 [00:10<00:05, 37.15it/s]  
 Computing ESM2 embeddings: 73%| | 416/573 [00:11<00:04, 37.76it/s]  
 Computing ESM2 embeddings: 78%| | 448/573 [00:12<00:03, 37.18it/s]  
 Computing ESM2 embeddings: 84%| | 480/573 [00:13<00:02, 37.79it/s]  
 Computing ESM2 embeddings: 89%| | 512/573 [00:13<00:01, 37.12it/s]  
 Computing ESM2 embeddings: 95%| | 544/573 [00:14<00:00, 37.77it/s]  
 Computing ESM2 embeddings: 100%| | 573/573 [00:15<00:00, 36.78it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:27, 31.64it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 35.20it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.68it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 35.53it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.88it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.61it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.46it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 36.65it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.05it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 35.93it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 36.89it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.45it/s]

```

Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.06it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:12<00:12, 36.02it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:13<00:11, 36.69it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:14<00:11, 35.76it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:10, 36.83it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 36.48it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.32it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 36.92it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.75it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 37.05it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:20<00:04, 37.73it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.15it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.86it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.03it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.69it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.02it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 36.83it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 38.60it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:23, 35.88it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 37.53it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:20, 36.78it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.74it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 36.93it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 37.70it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:17, 37.15it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.88it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 37.27it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.94it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.26it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.95it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:11<00:12, 37.24it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 37.96it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.27it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.69it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 36.49it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.37it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:06, 36.91it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:17<00:05, 37.67it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 37.14it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.86it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.25it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.88it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.33it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 38.00it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.30it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:24, 35.81it/s]  
 Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:24, 34.92it/s]  
 Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:22, 36.99it/s]  
 Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:21, 36.59it/s]  
 Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:19, 37.66it/s]  
 Computing ESM2 embeddings: 21%| | 192/913 [00:05<00:19, 37.04it/s]  
 Computing ESM2 embeddings: 25%| | 224/913 [00:06<00:18, 37.47it/s]  
 Computing ESM2 embeddings: 28%| | 256/913 [00:06<00:18, 36.50it/s]  
 Computing ESM2 embeddings: 32%| | 288/913 [00:07<00:16, 37.53it/s]  
 Computing ESM2 embeddings: 35%| | 320/913 [00:08<00:16, 37.00it/s]  
 Computing ESM2 embeddings: 39%| | 352/913 [00:09<00:14, 37.51it/s]  
 Computing ESM2 embeddings: 42%| | 384/913 [00:10<00:14, 36.42it/s]  
 Computing ESM2 embeddings: 46%| | 416/913 [00:11<00:13, 37.34it/s]  
 Computing ESM2 embeddings: 49%| | 448/913 [00:12<00:12, 37.06it/s]  
 Computing ESM2 embeddings: 53%| | 480/913 [00:12<00:11, 37.82it/s]  
 Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 37.36it/s]  
 Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 38.05it/s]  
 Computing ESM2 embeddings: 63%| | 576/913 [00:15<00:09, 37.37it/s]  
 Computing ESM2 embeddings: 67%| | 608/913 [00:16<00:08, 38.02it/s]

```

Computing ESM2 embeddings: 70%|      | 640/913 [00:17<00:07, 37.15it/s]
Computing ESM2 embeddings: 74%|      | 672/913 [00:18<00:06, 37.84it/s]
Computing ESM2 embeddings: 77%|      | 704/913 [00:18<00:05, 37.23it/s]
Computing ESM2 embeddings: 81%|      | 736/913 [00:19<00:04, 37.92it/s]
Computing ESM2 embeddings: 84%|      | 768/913 [00:20<00:03, 37.24it/s]
Computing ESM2 embeddings: 88%|      | 800/913 [00:21<00:02, 37.87it/s]
Computing ESM2 embeddings: 91%|      | 832/913 [00:22<00:02, 37.30it/s]
Computing ESM2 embeddings: 95%|      | 864/913 [00:23<00:01, 37.95it/s]
Computing ESM2 embeddings: 98%|      | 896/913 [00:24<00:00, 37.32it/s]

Computing ESM2 embeddings: 100%|     | 913/913 [00:24<00:00, 37.25it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/896 [00:00<00:22, 38.22it/s]
Computing ESM2 embeddings: 7%|      | 64/896 [00:01<00:23, 35.88it/s]
Computing ESM2 embeddings: 11%|     | 96/896 [00:02<00:21, 37.62it/s]
Computing ESM2 embeddings: 14%|     | 128/896 [00:03<00:20, 36.84it/s]
Computing ESM2 embeddings: 18%|     | 160/896 [00:04<00:19, 37.82it/s]
Computing ESM2 embeddings: 21%|     | 192/896 [00:05<00:18, 37.21it/s]
Computing ESM2 embeddings: 25%|     | 224/896 [00:05<00:17, 37.98it/s]
Computing ESM2 embeddings: 29%|     | 256/896 [00:06<00:17, 37.33it/s]
Computing ESM2 embeddings: 32%|     | 288/896 [00:07<00:16, 37.76it/s]
Computing ESM2 embeddings: 36%|     | 320/896 [00:08<00:15, 36.73it/s]

```



Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.20it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.68it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.56it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.02it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:10, 37.84it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.15it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.86it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.25it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.58it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:07, 36.45it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:06, 37.33it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 36.89it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.68it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.18it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.84it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.22it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.92it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.22it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.03it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.79it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 37.65it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 36.10it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 37.31it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.72it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 37.62it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 36.85it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.65it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.20it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 37.91it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.42it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 38.06it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:11<00:12, 37.42it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 38.07it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.27it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.90it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.36it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 38.02it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.37it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.96it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 37.32it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.89it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.30it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.62it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 36.38it/s] |

```

Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.28it/s]

Computing ESM2 embeddings: 98%|      | 896/914 [00:23<00:00, 36.81it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.32it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:21, 39.34it/s]

Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:22, 37.45it/s]

Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:20, 38.33it/s]

Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 37.10it/s]

Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.95it/s]

Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:18, 37.20it/s]

Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 37.97it/s]

Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 37.23it/s]

Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.67it/s]

Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 36.45it/s]

Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.38it/s]

Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 36.77it/s]

Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.61it/s]

Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:12, 37.04it/s]

Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:11, 37.80it/s]

Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:10, 37.15it/s]

Computing ESM2 embeddings: 61%|      | 544/896 [00:14<00:09, 37.84it/s]

```

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.21it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.75it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.09it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.77it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.27it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.90it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.10it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.72it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.16it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.82it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.31it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.26it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:26, 31.73it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 34.93it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 35.19it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.58it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.15it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 36.94it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 35.93it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 36.99it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.72it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.23it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.21it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.12it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.85it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.64it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 37.08it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.71it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.18it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.89it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.22it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.87it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 37.28it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.85it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.16it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.76it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.05it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.75it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.06it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.89it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/572 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/572 [00:00<00:13, 39.05it/s]

Computing ESM2 embeddings: 11%| | 64/572 [00:01<00:13, 36.95it/s]

Computing ESM2 embeddings: 17%| | 96/572 [00:02<00:12, 38.15it/s]

Computing ESM2 embeddings: 22%| | 128/572 [00:03<00:11, 37.17it/s]

Computing ESM2 embeddings: 28%| | 160/572 [00:04<00:10, 38.00it/s]

Computing ESM2 embeddings: 34%| | 192/572 [00:05<00:10, 37.28it/s]

Computing ESM2 embeddings: 39%| | 224/572 [00:05<00:09, 37.75it/s]

Computing ESM2 embeddings: 45%| | 256/572 [00:06<00:08, 36.45it/s]

Computing ESM2 embeddings: 50%| | 288/572 [00:07<00:07, 37.43it/s]

Computing ESM2 embeddings: 56%| | 320/572 [00:08<00:06, 37.04it/s]

Computing ESM2 embeddings: 62%| | 352/572 [00:09<00:05, 37.80it/s]

Computing ESM2 embeddings: 67%| | 384/572 [00:10<00:05, 37.23it/s]

Computing ESM2 embeddings: 73%| | 416/572 [00:11<00:04, 37.86it/s]

Computing ESM2 embeddings: 78%| | 448/572 [00:11<00:03, 37.23it/s]

Computing ESM2 embeddings: 84%| | 480/572 [00:12<00:02, 37.92it/s]

Computing ESM2 embeddings: 90%| | 512/572 [00:13<00:01, 37.32it/s]

Computing ESM2 embeddings: 95%| | 544/572 [00:14<00:00, 37.96it/s]

Computing ESM2 embeddings: 100%| | 572/572 [00:15<00:00, 37.37it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/903 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/903 [00:01<00:32, 27.09it/s]

Computing ESM2 embeddings: 7%| | 64/903 [00:02<00:25, 32.85it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/903 [00:02<00:22, 35.34it/s]  |
| Computing ESM2 embeddings: 14% | 128/903 [00:03<00:22, 34.84it/s] |
| Computing ESM2 embeddings: 18% | 160/903 [00:04<00:20, 36.43it/s] |
| Computing ESM2 embeddings: 21% | 192/903 [00:05<00:19, 36.18it/s] |
| Computing ESM2 embeddings: 25% | 224/903 [00:06<00:18, 37.10it/s] |
| Computing ESM2 embeddings: 28% | 256/903 [00:07<00:17, 36.45it/s] |
| Computing ESM2 embeddings: 32% | 288/903 [00:08<00:16, 37.30it/s] |
| Computing ESM2 embeddings: 35% | 320/903 [00:08<00:15, 36.82it/s] |
| Computing ESM2 embeddings: 39% | 352/903 [00:09<00:14, 37.60it/s] |
| Computing ESM2 embeddings: 43% | 384/903 [00:10<00:14, 36.93it/s] |
| Computing ESM2 embeddings: 46% | 416/903 [00:11<00:12, 37.66it/s] |
| Computing ESM2 embeddings: 50% | 448/903 [00:12<00:12, 37.08it/s] |
| Computing ESM2 embeddings: 53% | 480/903 [00:13<00:11, 37.81it/s] |
| Computing ESM2 embeddings: 57% | 512/903 [00:14<00:10, 37.17it/s] |
| Computing ESM2 embeddings: 60% | 544/903 [00:14<00:09, 37.91it/s] |
| Computing ESM2 embeddings: 64% | 576/903 [00:15<00:08, 37.31it/s] |
| Computing ESM2 embeddings: 67% | 608/903 [00:16<00:07, 37.93it/s] |
| Computing ESM2 embeddings: 71% | 640/903 [00:17<00:07, 37.11it/s] |
| Computing ESM2 embeddings: 74% | 672/903 [00:18<00:06, 37.80it/s] |
| Computing ESM2 embeddings: 78% | 704/903 [00:19<00:05, 37.22it/s] |
| Computing ESM2 embeddings: 82% | 736/903 [00:19<00:04, 37.96it/s] |
| Computing ESM2 embeddings: 85% | 768/903 [00:20<00:03, 37.30it/s] |
| Computing ESM2 embeddings: 89% | 800/903 [00:21<00:02, 37.92it/s] |
| Computing ESM2 embeddings: 92% | 832/903 [00:22<00:01, 37.27it/s] |

```

Computing ESM2 embeddings: 96%|      | 864/903 [00:23<00:01, 37.90it/s]

Computing ESM2 embeddings: 99%|      | 896/903 [00:24<00:00, 37.29it/s]

Computing ESM2 embeddings: 100%|     | 903/903 [00:24<00:00, 36.80it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/907 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/907 [00:00<00:22, 38.16it/s]

Computing ESM2 embeddings:  7%|      | 64/907 [00:01<00:23, 35.72it/s]

Computing ESM2 embeddings: 11%|      | 96/907 [00:02<00:21, 36.94it/s]

Computing ESM2 embeddings: 14%|      | 128/907 [00:03<00:21, 35.56it/s]

Computing ESM2 embeddings: 18%|      | 160/907 [00:04<00:20, 36.59it/s]

Computing ESM2 embeddings: 21%|      | 192/907 [00:05<00:20, 35.62it/s]

Computing ESM2 embeddings: 25%|      | 224/907 [00:06<00:18, 36.83it/s]

Computing ESM2 embeddings: 28%|      | 256/907 [00:07<00:17, 36.40it/s]

Computing ESM2 embeddings: 32%|      | 288/907 [00:07<00:16, 37.36it/s]

Computing ESM2 embeddings: 35%|      | 320/907 [00:08<00:15, 36.90it/s]

Computing ESM2 embeddings: 39%|      | 352/907 [00:09<00:14, 37.68it/s]

Computing ESM2 embeddings: 42%|      | 384/907 [00:10<00:14, 37.18it/s]

Computing ESM2 embeddings: 46%|      | 416/907 [00:11<00:12, 37.83it/s]

Computing ESM2 embeddings: 49%|      | 448/907 [00:12<00:12, 37.24it/s]

Computing ESM2 embeddings: 53%|      | 480/907 [00:12<00:11, 37.84it/s]

Computing ESM2 embeddings: 56%|      | 512/907 [00:13<00:10, 37.28it/s]

Computing ESM2 embeddings: 60%|      | 544/907 [00:14<00:09, 37.85it/s]

```



```

Computing ESM2 embeddings: 64%|          | 576/907 [00:15<00:08, 37.05it/s]
Computing ESM2 embeddings: 67%|          | 608/907 [00:16<00:07, 37.73it/s]
Computing ESM2 embeddings: 71%|          | 640/907 [00:17<00:07, 37.15it/s]
Computing ESM2 embeddings: 74%|          | 672/907 [00:18<00:06, 37.80it/s]
Computing ESM2 embeddings: 78%|          | 704/907 [00:18<00:05, 37.13it/s]
Computing ESM2 embeddings: 81%|          | 736/907 [00:19<00:04, 37.76it/s]
Computing ESM2 embeddings: 85%|          | 768/907 [00:20<00:03, 37.24it/s]
Computing ESM2 embeddings: 88%|          | 800/907 [00:21<00:02, 37.87it/s]
Computing ESM2 embeddings: 92%|          | 832/907 [00:22<00:02, 37.25it/s]
Computing ESM2 embeddings: 95%|          | 864/907 [00:23<00:01, 37.85it/s]
Computing ESM2 embeddings: 99%|          | 896/907 [00:24<00:00, 37.34it/s]

Computing ESM2 embeddings: 100%|         | 907/907 [00:24<00:00, 37.21it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/898 [00:00<00:23, 36.54it/s]
Computing ESM2 embeddings:  7%|          | 64/898 [00:01<00:23, 35.71it/s]
Computing ESM2 embeddings: 11%|          | 96/898 [00:02<00:21, 36.91it/s]
Computing ESM2 embeddings: 14%|          | 128/898 [00:03<00:21, 35.83it/s]
Computing ESM2 embeddings: 18%|          | 160/898 [00:04<00:19, 37.05it/s]
Computing ESM2 embeddings: 21%|          | 192/898 [00:05<00:19, 36.71it/s]
Computing ESM2 embeddings: 25%|          | 224/898 [00:06<00:17, 37.60it/s]
Computing ESM2 embeddings: 29%|          | 256/898 [00:06<00:17, 37.04it/s]

```

```

Computing ESM2 embeddings: 32%|          | 288/898 [00:07<00:16, 37.50it/s]
Computing ESM2 embeddings: 36%|          | 320/898 [00:08<00:15, 36.63it/s]
Computing ESM2 embeddings: 39%|          | 352/898 [00:09<00:14, 37.53it/s]
Computing ESM2 embeddings: 43%|          | 384/898 [00:10<00:13, 36.80it/s]
Computing ESM2 embeddings: 46%|          | 416/898 [00:11<00:12, 37.39it/s]
Computing ESM2 embeddings: 50%|          | 448/898 [00:12<00:12, 36.22it/s]
Computing ESM2 embeddings: 53%|          | 480/898 [00:12<00:11, 37.11it/s]
Computing ESM2 embeddings: 57%|          | 512/898 [00:13<00:10, 36.72it/s]
Computing ESM2 embeddings: 61%|          | 544/898 [00:14<00:09, 37.53it/s]
Computing ESM2 embeddings: 64%|          | 576/898 [00:15<00:08, 37.05it/s]
Computing ESM2 embeddings: 68%|          | 608/898 [00:16<00:07, 37.68it/s]
Computing ESM2 embeddings: 71%|          | 640/898 [00:17<00:07, 36.78it/s]
Computing ESM2 embeddings: 75%|          | 672/898 [00:18<00:06, 37.53it/s]
Computing ESM2 embeddings: 78%|          | 704/898 [00:19<00:05, 36.88it/s]
Computing ESM2 embeddings: 82%|          | 736/898 [00:19<00:04, 37.66it/s]
Computing ESM2 embeddings: 86%|          | 768/898 [00:20<00:03, 36.84it/s]
Computing ESM2 embeddings: 89%|          | 800/898 [00:21<00:02, 37.59it/s]
Computing ESM2 embeddings: 93%|          | 832/898 [00:22<00:01, 37.02it/s]
Computing ESM2 embeddings: 96%|          | 864/898 [00:23<00:00, 37.59it/s]

```

```

Computing ESM2 embeddings: 100%|         | 898/898 [00:24<00:00, 36.92it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          | 0/911 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/911 [00:01<00:30, 29.14it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/911 [00:02<00:26, 32.15it/s]  |
| Computing ESM2 embeddings: | 11% | 96/911 [00:02<00:23, 35.22it/s]  |
| Computing ESM2 embeddings: | 14% | 128/911 [00:03<00:22, 35.58it/s] |
| Computing ESM2 embeddings: | 18% | 160/911 [00:04<00:20, 37.02it/s] |
| Computing ESM2 embeddings: | 21% | 192/911 [00:05<00:19, 36.73it/s] |
| Computing ESM2 embeddings: | 25% | 224/911 [00:06<00:18, 37.65it/s] |
| Computing ESM2 embeddings: | 28% | 256/911 [00:07<00:17, 37.11it/s] |
| Computing ESM2 embeddings: | 32% | 288/911 [00:07<00:16, 37.53it/s] |
| Computing ESM2 embeddings: | 35% | 320/911 [00:08<00:16, 36.81it/s] |
| Computing ESM2 embeddings: | 39% | 352/911 [00:09<00:14, 37.61it/s] |
| Computing ESM2 embeddings: | 42% | 384/911 [00:10<00:14, 36.95it/s] |
| Computing ESM2 embeddings: | 46% | 416/911 [00:11<00:13, 37.65it/s] |
| Computing ESM2 embeddings: | 49% | 448/911 [00:12<00:12, 36.91it/s] |
| Computing ESM2 embeddings: | 53% | 480/911 [00:13<00:11, 37.62it/s] |
| Computing ESM2 embeddings: | 56% | 512/911 [00:13<00:10, 36.95it/s] |
| Computing ESM2 embeddings: | 60% | 544/911 [00:14<00:09, 37.52it/s] |
| Computing ESM2 embeddings: | 63% | 576/911 [00:15<00:09, 36.95it/s] |
| Computing ESM2 embeddings: | 67% | 608/911 [00:16<00:08, 37.59it/s] |
| Computing ESM2 embeddings: | 70% | 640/911 [00:17<00:07, 36.94it/s] |
| Computing ESM2 embeddings: | 74% | 672/911 [00:18<00:06, 37.60it/s] |
| Computing ESM2 embeddings: | 77% | 704/911 [00:19<00:05, 36.93it/s] |
| Computing ESM2 embeddings: | 81% | 736/911 [00:19<00:04, 37.33it/s] |
| Computing ESM2 embeddings: | 84% | 768/911 [00:20<00:03, 36.08it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/911 [00:21<00:03, 36.96it/s]
Computing ESM2 embeddings: 91%|      | 832/911 [00:22<00:02, 36.55it/s]
Computing ESM2 embeddings: 95%|      | 864/911 [00:23<00:01, 37.29it/s]
Computing ESM2 embeddings: 98%|      | 896/911 [00:24<00:00, 36.77it/s]
Computing ESM2 embeddings: 100%|     | 911/911 [00:24<00:00, 36.79it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:01<00:29, 29.34it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:02<00:26, 31.29it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:23, 34.51it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:22, 34.72it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:20, 36.02it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 35.22it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:06<00:18, 36.26it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:07<00:18, 35.40it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:08<00:16, 36.43it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:09<00:16, 35.59it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 36.70it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:14, 36.49it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.33it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:12<00:12, 36.89it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:13<00:11, 37.65it/s]

```

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 37.13it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:15<00:09, 37.85it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.27it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.92it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.25it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.68it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 36.52it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.45it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:21<00:03, 36.87it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.64it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.12it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.82it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.48it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:23, 37.09it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:27, 31.49it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:25, 32.52it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:22, 34.98it/s]

Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:21, 35.24it/s]

Computing ESM2 embeddings: 21%| | 192/915 [00:05<00:19, 36.51it/s]

Computing ESM2 embeddings: 24%| | 224/915 [00:06<00:19, 36.20it/s]

Computing ESM2 embeddings: 28%| | 256/915 [00:07<00:17, 37.19it/s]

Computing ESM2 embeddings: 31%| | 288/915 [00:08<00:17, 36.71it/s]

Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:16, 37.16it/s]

Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:15, 36.23it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 37.23it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 36.79it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:12<00:12, 37.29it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:13<00:11, 36.30it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:14<00:10, 37.23it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:10, 36.82it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 37.53it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 36.98it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 37.29it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:18<00:06, 36.32it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:19<00:05, 37.18it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:20<00:04, 36.72it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:21<00:03, 37.45it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 36.96it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 37.65it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:23<00:01, 37.04it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:24<00:00, 37.68it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:25<00:00, 36.60it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.11it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:23, 35.66it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.35it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:20, 36.60it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.64it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 37.04it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:05<00:17, 37.84it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 37.18it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.93it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.22it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 38.05it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.61it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 38.22it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:11<00:12, 37.29it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 37.97it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.32it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.98it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.23it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.89it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.33it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:17<00:05, 37.95it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 37.37it/s] |

```

Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.96it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.17it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.85it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.25it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:22<00:00, 37.88it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:23<00:00, 37.46it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/590 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/590 [00:00<00:14, 38.09it/s]
Computing ESM2 embeddings: 11%|      | 64/590 [00:01<00:14, 35.42it/s]
Computing ESM2 embeddings: 16%|      | 96/590 [00:02<00:13, 37.24it/s]
Computing ESM2 embeddings: 22%|      | 128/590 [00:03<00:12, 36.86it/s]
Computing ESM2 embeddings: 27%|      | 160/590 [00:04<00:11, 37.78it/s]
Computing ESM2 embeddings: 33%|      | 192/590 [00:05<00:10, 36.95it/s]
Computing ESM2 embeddings: 38%|      | 224/590 [00:06<00:09, 37.48it/s]
Computing ESM2 embeddings: 43%|      | 256/590 [00:06<00:09, 36.30it/s]
Computing ESM2 embeddings: 49%|      | 288/590 [00:07<00:08, 37.37it/s]
Computing ESM2 embeddings: 54%|      | 320/590 [00:08<00:07, 36.86it/s]
Computing ESM2 embeddings: 60%|      | 352/590 [00:09<00:06, 37.61it/s]
Computing ESM2 embeddings: 65%|      | 384/590 [00:10<00:05, 37.10it/s]
Computing ESM2 embeddings: 71%|      | 416/590 [00:11<00:04, 37.82it/s]
Computing ESM2 embeddings: 76%|      | 448/590 [00:12<00:03, 37.18it/s]

```



```

Computing ESM2 embeddings: 81%|      | 480/590 [00:12<00:02, 37.86it/s]
Computing ESM2 embeddings: 87%|      | 512/590 [00:13<00:02, 37.27it/s]
Computing ESM2 embeddings: 92%|      | 544/590 [00:14<00:01, 37.96it/s]
Computing ESM2 embeddings: 98%|      | 576/590 [00:15<00:00, 37.30it/s]
Computing ESM2 embeddings: 100%|     | 590/590 [00:15<00:00, 37.24it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.41it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:23, 35.80it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 36.89it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:21, 35.58it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 36.93it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 36.54it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:06<00:17, 37.50it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 36.80it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.31it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 36.29it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.24it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 36.87it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.65it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:12<00:12, 37.22it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:10, 37.92it/s]

```

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.27it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.92it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.31it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.95it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.26it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.96it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.36it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 38.00it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.32it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.88it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.17it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.86it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.23it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 36.72it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 34.99it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.88it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.31it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.41it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.89it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.69it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 37.05it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.48it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.21it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.18it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.76it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.54it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.89it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.65it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.09it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.78it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.15it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.81it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.29it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.91it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.32it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.68it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.52it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.37it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.79it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.58it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.84it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.09it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:01<00:30, 27.96it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:25, 33.10it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:22, 35.87it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:21, 35.78it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 37.10it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:19, 36.74it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:06<00:17, 37.29it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:07<00:17, 36.00it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:16, 36.93it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 36.69it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 37.14it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:14, 36.08it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 36.96it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:12<00:12, 36.49it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:13<00:11, 37.35it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:14<00:10, 36.79it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.35it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 36.25it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 37.17it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:06, 36.69it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:18<00:06, 37.16it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:19<00:05, 36.17it/s] |

```

Computing ESM2 embeddings: 82%|      | 736/895 [00:20<00:04, 37.05it/s]
Computing ESM2 embeddings: 86%|      | 768/895 [00:21<00:03, 36.65it/s]
Computing ESM2 embeddings: 89%|      | 800/895 [00:21<00:02, 37.42it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:22<00:01, 36.92it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:23<00:00, 37.59it/s]

Computing ESM2 embeddings: 100%|     | 895/895 [00:24<00:00, 36.53it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:24, 36.37it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:02<00:27, 30.50it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:24, 33.95it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:22, 34.38it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 36.03it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:20, 35.76it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 36.70it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:07<00:18, 36.36it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:08<00:16, 37.22it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.70it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.45it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 36.89it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.57it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 36.90it/s]

```

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.57it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 36.92it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.64it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.88it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.29it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.15it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.05it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.74it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 37.59it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.05it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.73it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.03it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.67it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.02it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.66it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.43it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 36.62it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.91it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 37.16it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.98it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:18, 37.15it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:17, 37.93it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:17, 37.13it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:16, 37.82it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 37.18it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 37.83it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 37.07it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:12, 37.57it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:12<00:12, 36.45it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:11, 37.30it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 36.90it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 37.64it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 37.10it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 37.77it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 37.20it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 37.83it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:05, 37.19it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 37.76it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:20<00:03, 37.13it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 37.61it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 37.01it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:23<00:00, 37.48it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:24<00:00, 37.14it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:22, 39.14it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:22, 37.00it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 38.25it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 37.19it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:19, 38.02it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.34it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 37.84it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 36.62it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.52it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 37.04it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.81it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.13it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.84it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:11<00:12, 37.23it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.84it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.26it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.63it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 36.33it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.19it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 36.93it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:17<00:06, 37.71it/s] |



Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.03it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.69it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.06it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.80it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.20it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.87it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.30it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.40it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.53it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.89it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 36.96it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 35.74it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.00it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.60it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:17, 37.47it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 36.83it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.37it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.35it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.28it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.84it/s]

```

Computing ESM2 embeddings: 46%|          | 416/896 [00:11<00:12, 37.62it/s]
Computing ESM2 embeddings: 50%|          | 448/896 [00:12<00:12, 37.09it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:11, 37.75it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.16it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.85it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.28it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.77it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 36.59it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:18<00:05, 37.37it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:18<00:05, 37.00it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.48it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 36.29it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.24it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 36.75it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.55it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 37.04it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/590 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|          | 32/590 [00:00<00:15, 35.38it/s]
Computing ESM2 embeddings: 11%|          | 64/590 [00:01<00:15, 34.56it/s]
Computing ESM2 embeddings: 16%|          | 96/590 [00:02<00:13, 36.25it/s]
Computing ESM2 embeddings: 22%|          | 128/590 [00:03<00:13, 35.42it/s]

```

Computing ESM2 embeddings: 27%| | 160/590 [00:04<00:11, 36.71it/s]

Computing ESM2 embeddings: 33%| | 192/590 [00:05<00:10, 36.36it/s]

Computing ESM2 embeddings: 38%| | 224/590 [00:06<00:09, 37.34it/s]

Computing ESM2 embeddings: 43%| | 256/590 [00:07<00:09, 36.77it/s]

Computing ESM2 embeddings: 49%| | 288/590 [00:07<00:08, 37.62it/s]

Computing ESM2 embeddings: 54%| | 320/590 [00:08<00:07, 37.06it/s]

Computing ESM2 embeddings: 60%| | 352/590 [00:09<00:06, 37.69it/s]

Computing ESM2 embeddings: 65%| | 384/590 [00:10<00:05, 37.00it/s]

Computing ESM2 embeddings: 71%| | 416/590 [00:11<00:04, 37.74it/s]

Computing ESM2 embeddings: 76%| | 448/590 [00:12<00:03, 37.20it/s]

Computing ESM2 embeddings: 81%| | 480/590 [00:12<00:02, 37.87it/s]

Computing ESM2 embeddings: 87%| | 512/590 [00:13<00:02, 37.08it/s]

Computing ESM2 embeddings: 92%| | 544/590 [00:14<00:01, 37.77it/s]

Computing ESM2 embeddings: 98%| | 576/590 [00:15<00:00, 36.96it/s]

Computing ESM2 embeddings: 100%| | 590/590 [00:16<00:00, 36.87it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:01<00:30, 28.08it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:02<00:29, 28.05it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:03<00:25, 31.85it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:22, 34.18it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:20, 36.04it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:19, 35.75it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:06<00:18, 36.82it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:07<00:17, 36.30it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:08<00:16, 36.68it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:09<00:16, 35.71it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:10<00:14, 36.80it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:14, 36.40it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:12, 37.33it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:12<00:12, 36.85it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:13<00:11, 37.55it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:14<00:10, 36.91it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:15<00:09, 37.62it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:16<00:08, 37.10it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 37.51it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:07, 36.39it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:18<00:06, 36.99it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:19<00:05, 35.96it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:20<00:04, 36.89it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:21<00:03, 36.44it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:22<00:02, 37.31it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:23<00:01, 36.73it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:23<00:00, 37.46it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:24<00:00, 36.09it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.82it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:23, 35.73it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:21, 37.60it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 36.75it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 37.64it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 37.06it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:05<00:18, 37.82it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.23it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.64it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.47it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 37.31it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 36.77it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.62it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 36.96it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.44it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:11, 36.54it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.32it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 36.78it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.46it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 36.92it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 37.60it/s] |

```

Computing ESM2 embeddings: 77%|      | 704/914 [00:18<00:05, 37.10it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:19<00:04, 37.72it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.09it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.69it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.02it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.67it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.92it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.14it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/895 [00:00<00:21, 39.47it/s]
Computing ESM2 embeddings:  7%|      | 64/895 [00:01<00:22, 37.30it/s]
Computing ESM2 embeddings: 11%|      | 96/895 [00:02<00:21, 37.85it/s]
Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:21, 36.12it/s]
Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:19, 37.21it/s]
Computing ESM2 embeddings: 21%|      | 192/895 [00:05<00:19, 36.78it/s]
Computing ESM2 embeddings: 25%|      | 224/895 [00:06<00:17, 37.37it/s]
Computing ESM2 embeddings: 29%|      | 256/895 [00:06<00:17, 35.98it/s]
Computing ESM2 embeddings: 32%|      | 288/895 [00:07<00:16, 37.03it/s]
Computing ESM2 embeddings: 36%|      | 320/895 [00:08<00:15, 36.74it/s]
Computing ESM2 embeddings: 39%|      | 352/895 [00:09<00:14, 37.45it/s]
Computing ESM2 embeddings: 43%|      | 384/895 [00:10<00:13, 37.07it/s]

```

Computing ESM2 embeddings: 46%| | 416/895 [00:11<00:12, 37.57it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:12<00:12, 36.30it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:12<00:11, 37.25it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 36.89it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.60it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.00it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.41it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:07, 36.36it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:05, 37.28it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:19<00:05, 36.67it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.42it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 36.96it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.67it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.18it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.56it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 36.90it/s]

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You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.17it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:25, 33.61it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 35.69it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 35.85it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 36.98it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.54it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 37.38it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:17, 36.82it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.21it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.12it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.02it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.60it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.37it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 36.94it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 37.61it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.02it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.65it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.01it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.72it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.14it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.80it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 37.00it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.73it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.12it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.51it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 36.28it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:23<00:01, 37.18it/s] |
| Computing ESM2 embeddings: 98% | 896/914 [00:24<00:00, 36.64it/s] |



Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.85it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:01<00:27, 31.04it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:25, 33.20it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:22, 35.60it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 35.65it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 36.95it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.35it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 37.33it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:07<00:17, 36.85it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.64it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.03it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.80it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.21it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.66it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 36.48it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:13<00:11, 37.02it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:14<00:10, 36.01it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.08it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 36.64it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.44it/s] |

```

Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 36.95it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:18<00:05, 37.70it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:19<00:05, 37.12it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.83it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.25it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.90it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.14it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.60it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.75it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/915 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/915 [00:00<00:22, 38.45it/s]
Computing ESM2 embeddings:  7%|      | 64/915 [00:01<00:23, 35.73it/s]
Computing ESM2 embeddings: 10%|      | 96/915 [00:02<00:22, 36.95it/s]
Computing ESM2 embeddings: 14%|      | 128/915 [00:03<00:22, 35.77it/s]
Computing ESM2 embeddings: 17%|      | 160/915 [00:04<00:20, 37.01it/s]
Computing ESM2 embeddings: 21%|      | 192/915 [00:05<00:19, 36.48it/s]
Computing ESM2 embeddings: 24%|      | 224/915 [00:06<00:18, 37.48it/s]
Computing ESM2 embeddings: 28%|      | 256/915 [00:06<00:17, 36.78it/s]
Computing ESM2 embeddings: 31%|      | 288/915 [00:07<00:16, 37.64it/s]
Computing ESM2 embeddings: 35%|      | 320/915 [00:08<00:16, 37.01it/s]
Computing ESM2 embeddings: 38%|      | 352/915 [00:09<00:14, 37.69it/s]

```

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 37.16it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 37.86it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:12<00:12, 37.29it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 37.96it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 37.51it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 37.74it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 36.44it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.35it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 36.87it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:18<00:06, 37.60it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:18<00:05, 36.90it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 37.68it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:03, 36.97it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 37.47it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 36.50it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:23<00:01, 37.37it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:24<00:00, 36.79it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 37.11it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.27it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 37.00it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:21, 37.68it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:21, 36.22it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.37it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 36.86it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:05<00:17, 37.72it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:17, 37.10it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.51it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 36.36it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.28it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 36.75it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.34it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 36.38it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:11, 37.05it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 36.14it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.09it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 36.75it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.48it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:06, 37.06it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:05, 37.47it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:19<00:05, 36.60it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.19it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 36.22it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.11it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 36.75it/s] |

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.56it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.90it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/590 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/590 [00:00<00:14, 38.01it/s]

Computing ESM2 embeddings: 11%| | 64/590 [00:01<00:14, 36.94it/s]

Computing ESM2 embeddings: 16%| | 96/590 [00:02<00:12, 38.11it/s]

Computing ESM2 embeddings: 22%| | 128/590 [00:03<00:12, 37.30it/s]

Computing ESM2 embeddings: 27%| | 160/590 [00:04<00:11, 38.11it/s]

Computing ESM2 embeddings: 33%| | 192/590 [00:05<00:10, 37.22it/s]

Computing ESM2 embeddings: 38%| | 224/590 [00:05<00:09, 37.99it/s]

Computing ESM2 embeddings: 43%| | 256/590 [00:06<00:08, 37.35it/s]

Computing ESM2 embeddings: 49%| | 288/590 [00:07<00:07, 37.99it/s]

Computing ESM2 embeddings: 54%| | 320/590 [00:08<00:07, 37.32it/s]

Computing ESM2 embeddings: 60%| | 352/590 [00:09<00:06, 37.97it/s]

Computing ESM2 embeddings: 65%| | 384/590 [00:10<00:05, 37.32it/s]

Computing ESM2 embeddings: 71%| | 416/590 [00:11<00:04, 38.04it/s]

Computing ESM2 embeddings: 76%| | 448/590 [00:11<00:03, 37.26it/s]

Computing ESM2 embeddings: 81%| | 480/590 [00:12<00:02, 37.74it/s]

Computing ESM2 embeddings: 87%| | 512/590 [00:13<00:02, 36.88it/s]

Computing ESM2 embeddings: 92%| | 544/590 [00:14<00:01, 37.40it/s]

Computing ESM2 embeddings: 98%| | 576/590 [00:15<00:00, 36.25it/s]

Computing ESM2 embeddings: 100%| | 590/590 [00:15<00:00, 37.30it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:30, 28.78it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:02<00:26, 31.60it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 34.70it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 34.92it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.43it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.15it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.21it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 36.72it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.58it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.84it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.60it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.04it/s]  
 Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.69it/s]  
 Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.03it/s]  
 Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.71it/s]  
 Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 37.02it/s]  
 Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.70it/s]  
 Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.93it/s]  
 Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.67it/s]

```

Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.14it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.84it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 37.28it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.89it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.24it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.86it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.10it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.74it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.05it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 36.82it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/896 [00:00<00:26, 32.47it/s]
Computing ESM2 embeddings: 7%|           | 64/896 [00:02<00:29, 27.88it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:03<00:25, 31.18it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:04<00:23, 32.62it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:21, 34.86it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:20, 35.12it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:06<00:18, 36.31it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:07<00:17, 36.11it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:08<00:16, 36.78it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:09<00:16, 35.67it/s]

```

Computing ESM2 embeddings: 39%| | 352/896 [00:10<00:14, 36.37it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:11<00:14, 35.65it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:13, 36.70it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.34it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 36.92it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 35.85it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:15<00:09, 36.73it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:16<00:08, 36.38it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:17<00:07, 37.21it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 36.75it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.47it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 36.92it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.25it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:21<00:03, 36.15it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:22<00:02, 37.02it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:23<00:01, 36.72it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.27it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:25<00:00, 35.83it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:01<00:36, 24.16it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:02<00:27, 31.07it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/913 [00:03<00:25, 32.51it/s]  |
| Computing ESM2 embeddings: 14% | 128/913 [00:03<00:22, 34.95it/s] |
| Computing ESM2 embeddings: 18% | 160/913 [00:04<00:21, 35.25it/s] |
| Computing ESM2 embeddings: 21% | 192/913 [00:05<00:19, 36.60it/s] |
| Computing ESM2 embeddings: 25% | 224/913 [00:06<00:19, 36.18it/s] |
| Computing ESM2 embeddings: 28% | 256/913 [00:07<00:17, 37.20it/s] |
| Computing ESM2 embeddings: 32% | 288/913 [00:08<00:17, 36.56it/s] |
| Computing ESM2 embeddings: 35% | 320/913 [00:09<00:15, 37.38it/s] |
| Computing ESM2 embeddings: 39% | 352/913 [00:09<00:15, 36.92it/s] |
| Computing ESM2 embeddings: 42% | 384/913 [00:10<00:14, 37.70it/s] |
| Computing ESM2 embeddings: 46% | 416/913 [00:11<00:13, 36.96it/s] |
| Computing ESM2 embeddings: 49% | 448/913 [00:12<00:12, 37.68it/s] |
| Computing ESM2 embeddings: 53% | 480/913 [00:13<00:11, 36.92it/s] |
| Computing ESM2 embeddings: 56% | 512/913 [00:14<00:10, 37.39it/s] |
| Computing ESM2 embeddings: 60% | 544/913 [00:15<00:10, 36.28it/s] |
| Computing ESM2 embeddings: 63% | 576/913 [00:15<00:09, 37.07it/s] |
| Computing ESM2 embeddings: 67% | 608/913 [00:16<00:08, 36.69it/s] |
| Computing ESM2 embeddings: 70% | 640/913 [00:17<00:07, 37.43it/s] |
| Computing ESM2 embeddings: 74% | 672/913 [00:18<00:06, 36.79it/s] |
| Computing ESM2 embeddings: 77% | 704/913 [00:19<00:05, 37.47it/s] |
| Computing ESM2 embeddings: 81% | 736/913 [00:20<00:04, 36.93it/s] |
| Computing ESM2 embeddings: 84% | 768/913 [00:21<00:03, 37.59it/s] |
| Computing ESM2 embeddings: 88% | 800/913 [00:21<00:03, 37.13it/s] |
| Computing ESM2 embeddings: 91% | 832/913 [00:22<00:02, 37.73it/s] |

```

Computing ESM2 embeddings: 95%|      | 864/913 [00:23<00:01, 37.08it/s]

Computing ESM2 embeddings: 98%|      | 896/913 [00:24<00:00, 37.71it/s]

Computing ESM2 embeddings: 100%|     | 913/913 [00:25<00:00, 36.42it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 38.81it/s]

Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:22, 36.88it/s]

Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.54it/s]

Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:21, 35.68it/s]

Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 36.87it/s]

Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 36.39it/s]

Computing ESM2 embeddings: 25%|      | 224/896 [00:06<00:18, 37.29it/s]

Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 36.77it/s]

Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:16, 37.16it/s]

Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 36.24it/s]

Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.11it/s]

Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 36.72it/s]

Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 37.56it/s]

Computing ESM2 embeddings: 50%|      | 448/896 [00:12<00:12, 36.89it/s]

Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:11, 37.59it/s]

Computing ESM2 embeddings: 57%|      | 512/896 [00:13<00:10, 36.90it/s]

Computing ESM2 embeddings: 61%|      | 544/896 [00:14<00:09, 37.57it/s]

```

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.97it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.73it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.16it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.83it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.07it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.44it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.37it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.24it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.66it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.46it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.88it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 36.82it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 34.97it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.94it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.42it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.32it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.78it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.25it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.13it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 36.99it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.65it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.40it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.83it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.50it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.03it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.70it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.08it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.66it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.87it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.70it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.05it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.68it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.12it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.74it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.11it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.72it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.16it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.73it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.81it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.98it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 4%  | 32/897 [00:00<00:23, 36.92it/s]  |
| Computing ESM2 embeddings: 7%  | 64/897 [00:01<00:23, 35.10it/s]  |
| Computing ESM2 embeddings: 11% | 96/897 [00:02<00:21, 37.04it/s]  |
| Computing ESM2 embeddings: 14% | 128/897 [00:03<00:20, 36.72it/s] |
| Computing ESM2 embeddings: 18% | 160/897 [00:04<00:19, 37.70it/s] |
| Computing ESM2 embeddings: 21% | 192/897 [00:05<00:19, 37.05it/s] |
| Computing ESM2 embeddings: 25% | 224/897 [00:06<00:17, 37.66it/s] |
| Computing ESM2 embeddings: 29% | 256/897 [00:06<00:17, 37.03it/s] |
| Computing ESM2 embeddings: 32% | 288/897 [00:07<00:16, 37.83it/s] |
| Computing ESM2 embeddings: 36% | 320/897 [00:08<00:15, 37.17it/s] |
| Computing ESM2 embeddings: 39% | 352/897 [00:09<00:14, 37.89it/s] |
| Computing ESM2 embeddings: 43% | 384/897 [00:10<00:13, 37.24it/s] |
| Computing ESM2 embeddings: 46% | 416/897 [00:11<00:12, 37.92it/s] |
| Computing ESM2 embeddings: 50% | 448/897 [00:12<00:12, 37.07it/s] |
| Computing ESM2 embeddings: 54% | 480/897 [00:12<00:11, 37.73it/s] |
| Computing ESM2 embeddings: 57% | 512/897 [00:13<00:10, 37.04it/s] |
| Computing ESM2 embeddings: 61% | 544/897 [00:14<00:09, 37.54it/s] |
| Computing ESM2 embeddings: 64% | 576/897 [00:15<00:08, 36.42it/s] |
| Computing ESM2 embeddings: 68% | 608/897 [00:16<00:07, 37.32it/s] |
| Computing ESM2 embeddings: 71% | 640/897 [00:17<00:06, 36.86it/s] |
| Computing ESM2 embeddings: 75% | 672/897 [00:18<00:05, 37.58it/s] |
| Computing ESM2 embeddings: 78% | 704/897 [00:18<00:05, 37.10it/s] |
| Computing ESM2 embeddings: 82% | 736/897 [00:19<00:04, 37.54it/s] |
| Computing ESM2 embeddings: 86% | 768/897 [00:20<00:03, 36.44it/s] |

```

Computing ESM2 embeddings: 89%|      | 800/897 [00:21<00:02, 37.32it/s]
Computing ESM2 embeddings: 93%|      | 832/897 [00:22<00:01, 36.83it/s]
Computing ESM2 embeddings: 96%|      | 864/897 [00:23<00:00, 37.66it/s]
Computing ESM2 embeddings: 100%|     | 897/897 [00:24<00:00, 37.08it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 38.20it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 35.67it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:22, 37.13it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.51it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.51it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.68it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.60it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 37.03it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.80it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:15, 37.23it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 37.87it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 37.31it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.86it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 37.17it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.78it/s]
Computing ESM2 embeddings: 56%|      | 512/914 [00:13<00:10, 37.22it/s]

```

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.78it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.01it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.68it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.99it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.66it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.08it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.73it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.15it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.82it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.05it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.69it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 37.21it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.28it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 37.87it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:23, 35.53it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:21, 36.80it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:21, 35.71it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:19, 36.99it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:19, 36.71it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:06<00:17, 37.54it/s]

```

Computing ESM2 embeddings: 29%|          | 256/895 [00:06<00:17, 36.78it/s]
Computing ESM2 embeddings: 32%|          | 288/895 [00:07<00:16, 37.56it/s]
Computing ESM2 embeddings: 36%|          | 320/895 [00:08<00:15, 37.05it/s]
Computing ESM2 embeddings: 39%|          | 352/895 [00:09<00:14, 37.74it/s]
Computing ESM2 embeddings: 43%|          | 384/895 [00:10<00:13, 37.30it/s]
Computing ESM2 embeddings: 46%|          | 416/895 [00:11<00:12, 37.98it/s]
Computing ESM2 embeddings: 50%|          | 448/895 [00:12<00:11, 37.48it/s]
Computing ESM2 embeddings: 54%|          | 480/895 [00:12<00:10, 38.04it/s]
Computing ESM2 embeddings: 57%|          | 512/895 [00:13<00:10, 37.33it/s]
Computing ESM2 embeddings: 61%|          | 544/895 [00:14<00:09, 38.00it/s]
Computing ESM2 embeddings: 64%|          | 576/895 [00:15<00:08, 37.23it/s]
Computing ESM2 embeddings: 68%|          | 608/895 [00:16<00:07, 37.91it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:17<00:06, 37.29it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:17<00:05, 37.93it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:18<00:05, 37.45it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:19<00:04, 38.05it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:20<00:03, 37.40it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:21<00:02, 38.01it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:22<00:01, 37.36it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:23<00:00, 37.98it/s]

```

```

Computing ESM2 embeddings: 100%|         | 895/895 [00:24<00:00, 37.27it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```



|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:24, 36.54it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:24, 35.35it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:22, 36.86it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 35.83it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 37.06it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.72it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.52it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 36.66it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.31it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.20it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 37.09it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 36.76it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.56it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 36.92it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.71it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.16it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.61it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 36.33it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.23it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 36.69it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 37.50it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:19<00:05, 36.81it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:19<00:04, 37.60it/s] |

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.88it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.56it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.06it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.43it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.29it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.88it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:01<00:27, 30.86it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:24, 33.50it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:22, 35.96it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 35.91it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.02it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.41it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 37.08it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:17, 36.06it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 36.86it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:16, 35.96it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.00it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.66it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.22it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.33it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.20it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 36.60it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.34it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.67it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.24it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:07, 36.23it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:06, 37.10it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 36.61it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.44it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.97it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.67it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.97it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.44it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.53it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/916 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/916 [00:01<00:32, 27.19it/s]

Computing ESM2 embeddings: 7%| | 64/916 [00:02<00:28, 30.12it/s]

Computing ESM2 embeddings: 10%| | 96/916 [00:02<00:24, 33.78it/s]

Computing ESM2 embeddings: 14%| | 128/916 [00:03<00:22, 34.39it/s]

Computing ESM2 embeddings: 17%| | 160/916 [00:04<00:21, 35.84it/s]

Computing ESM2 embeddings: 21%| | 192/916 [00:05<00:20, 35.14it/s]

Computing ESM2 embeddings: 24%| | 224/916 [00:06<00:18, 36.45it/s]

Computing ESM2 embeddings: 28%| | 256/916 [00:07<00:18, 36.16it/s]

Computing ESM2 embeddings: 31%| | 288/916 [00:08<00:16, 37.10it/s]

Computing ESM2 embeddings: 35%| | 320/916 [00:09<00:16, 36.55it/s]

Computing ESM2 embeddings: 38%| | 352/916 [00:09<00:15, 37.42it/s]

Computing ESM2 embeddings: 42%| | 384/916 [00:10<00:14, 36.92it/s]

Computing ESM2 embeddings: 45%| | 416/916 [00:11<00:13, 37.67it/s]

Computing ESM2 embeddings: 49%| | 448/916 [00:12<00:12, 37.04it/s]

Computing ESM2 embeddings: 52%| | 480/916 [00:13<00:11, 37.72it/s]

Computing ESM2 embeddings: 56%| | 512/916 [00:14<00:10, 36.96it/s]

Computing ESM2 embeddings: 59%| | 544/916 [00:15<00:09, 37.68it/s]

Computing ESM2 embeddings: 63%| | 576/916 [00:15<00:09, 37.09it/s]

Computing ESM2 embeddings: 66%| | 608/916 [00:16<00:08, 37.59it/s]

Computing ESM2 embeddings: 70%| | 640/916 [00:17<00:07, 36.49it/s]

Computing ESM2 embeddings: 73%| | 672/916 [00:18<00:06, 37.35it/s]

Computing ESM2 embeddings: 77%| | 704/916 [00:19<00:05, 36.92it/s]

Computing ESM2 embeddings: 80%| | 736/916 [00:20<00:04, 37.66it/s]

Computing ESM2 embeddings: 84%| | 768/916 [00:21<00:03, 37.05it/s]

Computing ESM2 embeddings: 87%| | 800/916 [00:21<00:03, 37.75it/s]

Computing ESM2 embeddings: 91%| | 832/916 [00:22<00:02, 37.03it/s]

Computing ESM2 embeddings: 94%| | 864/916 [00:23<00:01, 37.78it/s]

Computing ESM2 embeddings: 98%| | 896/916 [00:24<00:00, 37.09it/s]

Computing ESM2 embeddings: 100%| | 916/916 [00:25<00:00, 36.58it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:22, 39.12it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:22, 37.21it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:21, 37.80it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:21, 36.24it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 37.35it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:19, 36.94it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 37.73it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:17, 37.10it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:16, 37.76it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 37.15it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 37.82it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:13, 37.09it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 37.77it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:11<00:12, 37.24it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 37.85it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:10, 37.21it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.85it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 37.12it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 37.54it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:07, 36.31it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:18<00:06, 37.15it/s] |

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:05, 36.81it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.51it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 37.08it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.77it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.18it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.87it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 37.24it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

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Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.66it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:26, 32.59it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 35.38it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 35.28it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.74it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.44it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.45it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 36.98it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.48it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.32it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.24it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.88it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.61it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.10it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.76it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.15it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.78it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.08it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.73it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.07it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.69it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.95it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.62it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.96it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.60it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.99it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.62it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.13it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.98it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/573 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/573 [00:00<00:13, 39.37it/s]

Computing ESM2 embeddings: 11%| | 64/573 [00:01<00:13, 37.45it/s]

Computing ESM2 embeddings: 17%| | 96/573 [00:02<00:12, 38.29it/s]

Computing ESM2 embeddings: 22%| | 128/573 [00:03<00:11, 37.45it/s]

Computing ESM2 embeddings: 28%| | 160/573 [00:04<00:10, 37.91it/s]

Computing ESM2 embeddings: 34%| | 192/573 [00:05<00:10, 36.62it/s]

Computing ESM2 embeddings: 39%| | 224/573 [00:05<00:09, 37.46it/s]

Computing ESM2 embeddings: 45%| | 256/573 [00:06<00:08, 36.91it/s]

Computing ESM2 embeddings: 50%| | 288/573 [00:07<00:07, 37.72it/s]

Computing ESM2 embeddings: 56%| | 320/573 [00:08<00:06, 37.00it/s]

Computing ESM2 embeddings: 61%| | 352/573 [00:09<00:05, 37.79it/s]

Computing ESM2 embeddings: 67%| | 384/573 [00:10<00:05, 37.22it/s]

Computing ESM2 embeddings: 73%| | 416/573 [00:11<00:04, 37.95it/s]

Computing ESM2 embeddings: 78%| | 448/573 [00:11<00:03, 37.27it/s]

Computing ESM2 embeddings: 84%| | 480/573 [00:12<00:02, 37.98it/s]

Computing ESM2 embeddings: 89%| | 512/573 [00:13<00:01, 37.29it/s]

Computing ESM2 embeddings: 95%| | 544/573 [00:14<00:00, 37.94it/s]

Computing ESM2 embeddings: 100%| | 573/573 [00:15<00:00, 37.33it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/904 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/904 [00:00<00:23, 37.75it/s]

Computing ESM2 embeddings: 7%| | 64/904 [00:01<00:24, 34.32it/s]

Computing ESM2 embeddings: 11%| | 96/904 [00:02<00:22, 36.19it/s]

Computing ESM2 embeddings: 14%| | 128/904 [00:03<00:21, 35.97it/s]

Computing ESM2 embeddings: 18%| | 160/904 [00:04<00:20, 37.15it/s]

Computing ESM2 embeddings: 21%| | 192/904 [00:05<00:19, 36.41it/s]



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/904 [00:06<00:18, 37.38it/s] |
| Computing ESM2 embeddings: 28%  | 256/904 [00:07<00:17, 36.69it/s] |
| Computing ESM2 embeddings: 32%  | 288/904 [00:07<00:16, 37.58it/s] |
| Computing ESM2 embeddings: 35%  | 320/904 [00:08<00:15, 36.98it/s] |
| Computing ESM2 embeddings: 39%  | 352/904 [00:09<00:14, 37.50it/s] |
| Computing ESM2 embeddings: 42%  | 384/904 [00:10<00:14, 36.40it/s] |
| Computing ESM2 embeddings: 46%  | 416/904 [00:11<00:13, 37.30it/s] |
| Computing ESM2 embeddings: 50%  | 448/904 [00:12<00:12, 36.83it/s] |
| Computing ESM2 embeddings: 53%  | 480/904 [00:12<00:11, 37.66it/s] |
| Computing ESM2 embeddings: 57%  | 512/904 [00:13<00:10, 37.17it/s] |
| Computing ESM2 embeddings: 60%  | 544/904 [00:14<00:09, 37.66it/s] |
| Computing ESM2 embeddings: 64%  | 576/904 [00:15<00:08, 36.59it/s] |
| Computing ESM2 embeddings: 67%  | 608/904 [00:16<00:07, 37.39it/s] |
| Computing ESM2 embeddings: 71%  | 640/904 [00:17<00:07, 37.00it/s] |
| Computing ESM2 embeddings: 74%  | 672/904 [00:18<00:06, 37.63it/s] |
| Computing ESM2 embeddings: 78%  | 704/904 [00:19<00:05, 36.99it/s] |
| Computing ESM2 embeddings: 81%  | 736/904 [00:19<00:04, 37.36it/s] |
| Computing ESM2 embeddings: 85%  | 768/904 [00:20<00:03, 36.14it/s] |
| Computing ESM2 embeddings: 88%  | 800/904 [00:21<00:02, 37.10it/s] |
| Computing ESM2 embeddings: 92%  | 832/904 [00:22<00:01, 36.57it/s] |
| Computing ESM2 embeddings: 96%  | 864/904 [00:23<00:01, 37.34it/s] |
| Computing ESM2 embeddings: 99%  | 896/904 [00:24<00:00, 36.77it/s] |
| Computing ESM2 embeddings: 100% | 904/904 [00:24<00:00, 36.85it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/905 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/905 [00:00<00:22, 38.33it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/905 [00:01<00:23, 35.49it/s]  |
| Computing ESM2 embeddings: | 11% | 96/905 [00:02<00:21, 37.19it/s]  |
| Computing ESM2 embeddings: | 14% | 128/905 [00:03<00:21, 36.83it/s] |
| Computing ESM2 embeddings: | 18% | 160/905 [00:04<00:19, 37.74it/s] |
| Computing ESM2 embeddings: | 21% | 192/905 [00:05<00:19, 37.25it/s] |
| Computing ESM2 embeddings: | 25% | 224/905 [00:05<00:18, 37.67it/s] |
| Computing ESM2 embeddings: | 28% | 256/905 [00:06<00:17, 36.65it/s] |
| Computing ESM2 embeddings: | 32% | 288/905 [00:07<00:16, 37.52it/s] |
| Computing ESM2 embeddings: | 35% | 320/905 [00:08<00:15, 37.02it/s] |
| Computing ESM2 embeddings: | 39% | 352/905 [00:09<00:14, 37.74it/s] |
| Computing ESM2 embeddings: | 42% | 384/905 [00:10<00:14, 37.18it/s] |
| Computing ESM2 embeddings: | 46% | 416/905 [00:11<00:12, 37.90it/s] |
| Computing ESM2 embeddings: | 50% | 448/905 [00:12<00:12, 37.32it/s] |
| Computing ESM2 embeddings: | 53% | 480/905 [00:12<00:11, 38.02it/s] |
| Computing ESM2 embeddings: | 57% | 512/905 [00:13<00:10, 37.27it/s] |
| Computing ESM2 embeddings: | 60% | 544/905 [00:14<00:09, 37.96it/s] |
| Computing ESM2 embeddings: | 64% | 576/905 [00:15<00:08, 37.28it/s] |
| Computing ESM2 embeddings: | 67% | 608/905 [00:16<00:07, 37.67it/s] |
| Computing ESM2 embeddings: | 71% | 640/905 [00:17<00:07, 36.50it/s] |
| Computing ESM2 embeddings: | 74% | 672/905 [00:17<00:06, 37.37it/s] |

Computing ESM2 embeddings: 78%| | 704/905 [00:18<00:05, 36.94it/s]

Computing ESM2 embeddings: 81%| | 736/905 [00:19<00:04, 37.71it/s]

Computing ESM2 embeddings: 85%| | 768/905 [00:20<00:03, 37.16it/s]

Computing ESM2 embeddings: 88%| | 800/905 [00:21<00:02, 37.85it/s]

Computing ESM2 embeddings: 92%| | 832/905 [00:22<00:01, 37.25it/s]

Computing ESM2 embeddings: 95%| | 864/905 [00:23<00:01, 37.91it/s]

Computing ESM2 embeddings: 99%| | 896/905 [00:23<00:00, 37.27it/s]

Computing ESM2 embeddings: 100%| | 905/905 [00:24<00:00, 37.32it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/900 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/900 [00:00<00:25, 34.46it/s]

Computing ESM2 embeddings: 7%| | 64/900 [00:01<00:24, 34.32it/s]

Computing ESM2 embeddings: 11%| | 96/900 [00:02<00:22, 36.02it/s]

Computing ESM2 embeddings: 14%| | 128/900 [00:03<00:21, 35.41it/s]

Computing ESM2 embeddings: 18%| | 160/900 [00:04<00:20, 36.92it/s]

Computing ESM2 embeddings: 21%| | 192/900 [00:05<00:19, 36.55it/s]

Computing ESM2 embeddings: 25%| | 224/900 [00:06<00:18, 37.41it/s]

Computing ESM2 embeddings: 28%| | 256/900 [00:07<00:17, 36.83it/s]

Computing ESM2 embeddings: 32%| | 288/900 [00:07<00:16, 37.30it/s]

Computing ESM2 embeddings: 36%| | 320/900 [00:08<00:16, 36.08it/s]

Computing ESM2 embeddings: 39%| | 352/900 [00:09<00:14, 37.08it/s]

Computing ESM2 embeddings: 43%| | 384/900 [00:10<00:14, 36.73it/s]

Computing ESM2 embeddings: 46%| | 416/900 [00:11<00:12, 37.49it/s]

Computing ESM2 embeddings: 50%| | 448/900 [00:12<00:12, 37.02it/s]

Computing ESM2 embeddings: 53%| | 480/900 [00:13<00:11, 37.77it/s]

Computing ESM2 embeddings: 57%| | 512/900 [00:13<00:10, 37.18it/s]

Computing ESM2 embeddings: 60%| | 544/900 [00:14<00:09, 37.55it/s]

Computing ESM2 embeddings: 64%| | 576/900 [00:15<00:08, 36.69it/s]

Computing ESM2 embeddings: 68%| | 608/900 [00:16<00:07, 37.46it/s]

Computing ESM2 embeddings: 71%| | 640/900 [00:17<00:07, 36.85it/s]

Computing ESM2 embeddings: 75%| | 672/900 [00:18<00:06, 37.39it/s]

Computing ESM2 embeddings: 78%| | 704/900 [00:19<00:05, 36.47it/s]

Computing ESM2 embeddings: 82%| | 736/900 [00:19<00:04, 37.44it/s]

Computing ESM2 embeddings: 85%| | 768/900 [00:20<00:03, 36.88it/s]

Computing ESM2 embeddings: 89%| | 800/900 [00:21<00:02, 37.36it/s]

Computing ESM2 embeddings: 92%| | 832/900 [00:22<00:01, 36.19it/s]

Computing ESM2 embeddings: 96%| | 864/900 [00:23<00:00, 37.14it/s]

Computing ESM2 embeddings: 100%| | 896/900 [00:24<00:00, 36.74it/s]

Computing ESM2 embeddings: 100%| | 900/900 [00:24<00:00, 36.80it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/910 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/910 [00:00<00:22, 39.36it/s]

Computing ESM2 embeddings: 7%| | 64/910 [00:01<00:22, 37.40it/s]

Computing ESM2 embeddings: 11%| | 96/910 [00:02<00:21, 38.03it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/910 [00:03<00:21, 36.35it/s] |
| Computing ESM2 embeddings: 18% | 160/910 [00:04<00:20, 37.44it/s] |
| Computing ESM2 embeddings: 21% | 192/910 [00:05<00:19, 36.86it/s] |
| Computing ESM2 embeddings: 25% | 224/910 [00:05<00:18, 37.72it/s] |
| Computing ESM2 embeddings: 28% | 256/910 [00:06<00:17, 37.05it/s] |
| Computing ESM2 embeddings: 32% | 288/910 [00:07<00:16, 37.55it/s] |
| Computing ESM2 embeddings: 35% | 320/910 [00:08<00:16, 36.42it/s] |
| Computing ESM2 embeddings: 39% | 352/910 [00:09<00:14, 37.36it/s] |
| Computing ESM2 embeddings: 42% | 384/910 [00:10<00:14, 36.87it/s] |
| Computing ESM2 embeddings: 46% | 416/910 [00:11<00:13, 37.57it/s] |
| Computing ESM2 embeddings: 49% | 448/910 [00:12<00:12, 37.11it/s] |
| Computing ESM2 embeddings: 53% | 480/910 [00:12<00:11, 37.82it/s] |
| Computing ESM2 embeddings: 56% | 512/910 [00:13<00:10, 37.25it/s] |
| Computing ESM2 embeddings: 60% | 544/910 [00:14<00:09, 37.91it/s] |
| Computing ESM2 embeddings: 63% | 576/910 [00:15<00:08, 37.29it/s] |
| Computing ESM2 embeddings: 67% | 608/910 [00:16<00:08, 37.70it/s] |
| Computing ESM2 embeddings: 70% | 640/910 [00:17<00:07, 36.56it/s] |
| Computing ESM2 embeddings: 74% | 672/910 [00:18<00:06, 37.35it/s] |
| Computing ESM2 embeddings: 77% | 704/910 [00:18<00:05, 36.95it/s] |
| Computing ESM2 embeddings: 81% | 736/910 [00:19<00:04, 37.69it/s] |
| Computing ESM2 embeddings: 84% | 768/910 [00:20<00:03, 37.12it/s] |
| Computing ESM2 embeddings: 88% | 800/910 [00:21<00:02, 37.80it/s] |
| Computing ESM2 embeddings: 91% | 832/910 [00:22<00:02, 37.17it/s] |
| Computing ESM2 embeddings: 95% | 864/910 [00:23<00:01, 37.83it/s] |

Computing ESM2 embeddings: 98%| | 896/910 [00:24<00:00, 37.14it/s]

Computing ESM2 embeddings: 100%| | 910/910 [00:24<00:00, 37.20it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:00<00:23, 36.59it/s]

Computing ESM2 embeddings: 7%| | 64/897 [00:02<00:30, 27.07it/s]

Computing ESM2 embeddings: 11%| | 96/897 [00:03<00:25, 31.19it/s]

Computing ESM2 embeddings: 14%| | 128/897 [00:04<00:24, 31.36it/s]

Computing ESM2 embeddings: 18%| | 160/897 [00:05<00:22, 32.54it/s]

Computing ESM2 embeddings: 21%| | 192/897 [00:05<00:21, 32.92it/s]

Computing ESM2 embeddings: 25%| | 224/897 [00:06<00:19, 34.77it/s]

Computing ESM2 embeddings: 29%| | 256/897 [00:07<00:18, 35.13it/s]

Computing ESM2 embeddings: 32%| | 288/897 [00:08<00:16, 36.29it/s]

Computing ESM2 embeddings: 36%| | 320/897 [00:09<00:16, 35.82it/s]

Computing ESM2 embeddings: 39%| | 352/897 [00:10<00:14, 36.86it/s]

Computing ESM2 embeddings: 43%| | 384/897 [00:11<00:14, 36.50it/s]

Computing ESM2 embeddings: 46%| | 416/897 [00:11<00:12, 37.04it/s]

Computing ESM2 embeddings: 50%| | 448/897 [00:12<00:12, 35.99it/s]

Computing ESM2 embeddings: 54%| | 480/897 [00:13<00:11, 36.99it/s]

Computing ESM2 embeddings: 57%| | 512/897 [00:14<00:10, 36.63it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:15<00:09, 37.42it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:16<00:08, 36.92it/s]

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Computing ESM2 embeddings: 68%|          | 608/897 [00:17<00:07, 37.55it/s]
Computing ESM2 embeddings: 71%|          | 640/897 [00:18<00:06, 36.86it/s]
Computing ESM2 embeddings: 75%|          | 672/897 [00:18<00:06, 36.97it/s]
Computing ESM2 embeddings: 78%|          | 704/897 [00:19<00:05, 35.96it/s]
Computing ESM2 embeddings: 82%|          | 736/897 [00:20<00:04, 36.91it/s]
Computing ESM2 embeddings: 86%|          | 768/897 [00:21<00:03, 36.51it/s]
Computing ESM2 embeddings: 89%|          | 800/897 [00:22<00:02, 37.26it/s]
Computing ESM2 embeddings: 93%|          | 832/897 [00:23<00:01, 36.71it/s]
Computing ESM2 embeddings: 96%|          | 864/897 [00:24<00:00, 37.43it/s]

Computing ESM2 embeddings: 100%|         | 897/897 [00:25<00:00, 35.63it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 37.60it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:23, 36.58it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.50it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 36.10it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:20, 37.33it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 36.79it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:06<00:18, 37.66it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 36.84it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 37.58it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:16, 37.12it/s]

```

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.83it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.26it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.83it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.12it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.76it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.10it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.73it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.08it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.72it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.06it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.71it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.02it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.63it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.93it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.33it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.23it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.14it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.82it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.16it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.74it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/896 [00:02<00:27, 30.19it/s]  |
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:23, 33.70it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:22, 34.80it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:20, 36.36it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 36.11it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:18, 36.99it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:07<00:17, 36.60it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.59it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 37.18it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.91it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.47it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 38.13it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:11, 37.45it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:13<00:10, 37.98it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.31it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.85it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.12it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.44it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:07, 36.34it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:06, 37.20it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:19<00:05, 36.63it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.44it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 36.97it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.65it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.05it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.71it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.82it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/591 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/591 [00:00<00:17, 32.30it/s]
Computing ESM2 embeddings: 11%|      | 64/591 [00:01<00:15, 33.16it/s]
Computing ESM2 embeddings: 16%|      | 96/591 [00:02<00:13, 35.67it/s]
Computing ESM2 embeddings: 22%|      | 128/591 [00:03<00:12, 35.86it/s]
Computing ESM2 embeddings: 27%|      | 160/591 [00:04<00:11, 37.16it/s]
Computing ESM2 embeddings: 32%|      | 192/591 [00:05<00:10, 36.44it/s]
Computing ESM2 embeddings: 38%|      | 224/591 [00:06<00:09, 37.44it/s]
Computing ESM2 embeddings: 43%|      | 256/591 [00:07<00:09, 37.00it/s]
Computing ESM2 embeddings: 49%|      | 288/591 [00:07<00:07, 37.90it/s]
Computing ESM2 embeddings: 54%|      | 320/591 [00:08<00:07, 37.38it/s]
Computing ESM2 embeddings: 60%|      | 352/591 [00:09<00:06, 37.82it/s]
Computing ESM2 embeddings: 65%|      | 384/591 [00:10<00:05, 36.55it/s]
Computing ESM2 embeddings: 70%|      | 416/591 [00:11<00:04, 37.35it/s]
Computing ESM2 embeddings: 76%|      | 448/591 [00:12<00:03, 37.02it/s]
Computing ESM2 embeddings: 81%|      | 480/591 [00:13<00:02, 37.53it/s]
Computing ESM2 embeddings: 87%|      | 512/591 [00:13<00:02, 36.32it/s]
Computing ESM2 embeddings: 92%|      | 544/591 [00:14<00:01, 37.25it/s]

```

Computing ESM2 embeddings: 97%| | 576/591 [00:15<00:00, 36.75it/s]

Computing ESM2 embeddings: 100%| | 591/591 [00:16<00:00, 36.75it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 38.97it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:22, 37.13it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:20, 38.29it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:20, 37.06it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:19, 37.66it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:19, 36.50it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:05<00:17, 37.43it/s]

Computing ESM2 embeddings: 29%| | 256/895 [00:06<00:17, 36.76it/s]

Computing ESM2 embeddings: 32%| | 288/895 [00:07<00:16, 37.54it/s]

Computing ESM2 embeddings: 36%| | 320/895 [00:08<00:15, 37.09it/s]

Computing ESM2 embeddings: 39%| | 352/895 [00:09<00:14, 37.73it/s]

Computing ESM2 embeddings: 43%| | 384/895 [00:10<00:13, 37.19it/s]

Computing ESM2 embeddings: 46%| | 416/895 [00:11<00:12, 37.90it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:11<00:12, 37.22it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:12<00:10, 37.84it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 37.20it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.87it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.18it/s]

```

Computing ESM2 embeddings: 68%|          | 608/895 [00:16<00:07, 37.83it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:17<00:06, 37.26it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:17<00:05, 37.65it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:18<00:05, 36.68it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:19<00:04, 37.51it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:20<00:03, 37.05it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:21<00:02, 37.75it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:22<00:01, 37.22it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:23<00:00, 37.88it/s]
Computing ESM2 embeddings: 100%|         | 895/895 [00:23<00:00, 37.34it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/915 [00:00<?, ?it/s]
Computing ESM2 embeddings: 3%|           | 32/915 [00:00<00:22, 39.61it/s]
Computing ESM2 embeddings: 7%|           | 64/915 [00:01<00:22, 37.30it/s]
Computing ESM2 embeddings: 10%|          | 96/915 [00:02<00:21, 38.28it/s]
Computing ESM2 embeddings: 14%|          | 128/915 [00:03<00:21, 37.25it/s]
Computing ESM2 embeddings: 17%|          | 160/915 [00:04<00:19, 38.12it/s]
Computing ESM2 embeddings: 21%|          | 192/915 [00:05<00:19, 37.11it/s]
Computing ESM2 embeddings: 24%|          | 224/915 [00:05<00:18, 37.77it/s]
Computing ESM2 embeddings: 28%|          | 256/915 [00:06<00:17, 37.08it/s]
Computing ESM2 embeddings: 31%|          | 288/915 [00:07<00:16, 37.59it/s]
Computing ESM2 embeddings: 35%|          | 320/915 [00:08<00:16, 36.38it/s]

```

Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:15, 37.23it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 36.81it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 37.56it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:12<00:12, 37.11it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 37.80it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 37.15it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 37.87it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 37.08it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.82it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 37.15it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:17<00:06, 37.87it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:18<00:05, 37.07it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 37.76it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:03, 37.18it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 37.86it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 37.13it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:23<00:01, 37.80it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:23<00:00, 37.22it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 37.41it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 38.30it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:23, 35.68it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:21, 37.27it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:20, 36.69it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 37.71it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:18, 37.10it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 37.81it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:17, 37.21it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:16, 37.53it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 36.28it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 37.26it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:13, 36.85it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 37.57it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:12<00:12, 37.09it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:10, 37.82it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:10, 37.26it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.90it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 37.21it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 37.90it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:06, 37.29it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:17<00:05, 37.97it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:18<00:05, 37.65it/s] |
| Computing ESM2 embeddings: | 82% | 736/895 [00:19<00:04, 38.22it/s] |
| Computing ESM2 embeddings: | 86% | 768/895 [00:20<00:03, 37.53it/s] |
| Computing ESM2 embeddings: | 89% | 800/895 [00:21<00:02, 38.12it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/895 [00:22<00:01, 37.27it/s]

Computing ESM2 embeddings: 97%|      | 864/895 [00:23<00:00, 37.72it/s]

Computing ESM2 embeddings: 100%|     | 895/895 [00:24<00:00, 37.24it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 37.85it/s]

Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:24, 34.07it/s]

Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:22, 36.43it/s]

Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 35.98it/s]

Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.22it/s]

Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.83it/s]

Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.73it/s]

Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 37.04it/s]

Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.36it/s]

Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.23it/s]

Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.10it/s]

Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 36.68it/s]

Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.45it/s]

Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 36.92it/s]

Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.69it/s]

Computing ESM2 embeddings: 56%|      | 512/914 [00:13<00:10, 36.97it/s]

Computing ESM2 embeddings: 60%|      | 544/914 [00:14<00:09, 37.60it/s]

```

```

Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 36.98it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.64it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.08it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.76it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 37.13it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.71it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.09it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.68it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.01it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.68it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.06it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.00it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/898 [00:00<00:23, 37.34it/s]
Computing ESM2 embeddings:  7%|          | 64/898 [00:01<00:26, 32.04it/s]
Computing ESM2 embeddings: 11%|          | 96/898 [00:02<00:24, 32.62it/s]
Computing ESM2 embeddings: 14%|          | 128/898 [00:03<00:22, 34.92it/s]
Computing ESM2 embeddings: 18%|          | 160/898 [00:04<00:21, 35.14it/s]
Computing ESM2 embeddings: 21%|          | 192/898 [00:05<00:19, 36.20it/s]
Computing ESM2 embeddings: 25%|          | 224/898 [00:06<00:19, 35.34it/s]
Computing ESM2 embeddings: 29%|          | 256/898 [00:07<00:17, 36.31it/s]

```



Computing ESM2 embeddings: 32%| | 288/898 [00:08<00:17, 35.60it/s]

Computing ESM2 embeddings: 36%| | 320/898 [00:08<00:15, 36.68it/s]

Computing ESM2 embeddings: 39%| | 352/898 [00:09<00:15, 36.34it/s]

Computing ESM2 embeddings: 43%| | 384/898 [00:10<00:13, 37.26it/s]

Computing ESM2 embeddings: 46%| | 416/898 [00:11<00:13, 36.63it/s]

Computing ESM2 embeddings: 50%| | 448/898 [00:12<00:12, 37.23it/s]

Computing ESM2 embeddings: 53%| | 480/898 [00:13<00:11, 36.16it/s]

Computing ESM2 embeddings: 57%| | 512/898 [00:14<00:10, 37.05it/s]

Computing ESM2 embeddings: 61%| | 544/898 [00:15<00:09, 36.63it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:15<00:08, 37.38it/s]

Computing ESM2 embeddings: 68%| | 608/898 [00:16<00:07, 36.75it/s]

Computing ESM2 embeddings: 71%| | 640/898 [00:17<00:06, 37.46it/s]

Computing ESM2 embeddings: 75%| | 672/898 [00:18<00:06, 36.84it/s]

Computing ESM2 embeddings: 78%| | 704/898 [00:19<00:05, 37.56it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:20<00:04, 36.93it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:21<00:03, 37.51it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:21<00:02, 36.98it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:22<00:01, 37.62it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:23<00:00, 36.94it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:24<00:00, 36.45it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 36.94it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:24, 34.95it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:22, 36.79it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 36.22it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 37.38it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.90it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.72it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:06<00:17, 37.16it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.58it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.43it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 37.26it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 36.89it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.61it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 37.01it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:12<00:11, 37.65it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:13<00:10, 37.06it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.35it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 36.30it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.24it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 36.66it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 37.44it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:19<00:05, 36.94it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:19<00:04, 37.62it/s] |
| Computing ESM2 embeddings: | 84% | 768/914 [00:20<00:03, 37.04it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.49it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.33it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.18it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.75it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.02it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/896 [00:00<00:22, 37.59it/s]
Computing ESM2 embeddings:  7%|      | 64/896 [00:01<00:22, 36.48it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:21, 37.78it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:20, 36.84it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:19, 37.63it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 36.86it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:05<00:17, 37.71it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:06<00:17, 37.39it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:07<00:15, 38.16it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 37.60it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 38.25it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:13, 37.61it/s]
Computing ESM2 embeddings: 46%|      | 416/896 [00:11<00:12, 38.20it/s]
Computing ESM2 embeddings: 50%|      | 448/896 [00:11<00:11, 37.47it/s]
Computing ESM2 embeddings: 54%|      | 480/896 [00:12<00:11, 37.81it/s]

```

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 36.61it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.35it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.81it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.63it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 36.99it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:17<00:05, 37.69it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 36.95it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.64it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.22it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.91it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.23it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.82it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:23<00:00, 37.38it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/591 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/591 [00:00<00:14, 37.67it/s]

Computing ESM2 embeddings: 11%| | 64/591 [00:01<00:14, 36.61it/s]

Computing ESM2 embeddings: 16%| | 96/591 [00:02<00:13, 37.44it/s]

Computing ESM2 embeddings: 22%| | 128/591 [00:03<00:12, 36.04it/s]

Computing ESM2 embeddings: 27%| | 160/591 [00:04<00:11, 37.17it/s]

Computing ESM2 embeddings: 32%| | 192/591 [00:05<00:10, 36.58it/s]

Computing ESM2 embeddings: 38%| | 224/591 [00:06<00:09, 37.46it/s]

```

Computing ESM2 embeddings: 43%|          | 256/591 [00:06<00:09, 36.64it/s]
Computing ESM2 embeddings: 49%|          | 288/591 [00:07<00:08, 37.46it/s]
Computing ESM2 embeddings: 54%|          | 320/591 [00:08<00:07, 36.83it/s]
Computing ESM2 embeddings: 60%|          | 352/591 [00:09<00:06, 37.28it/s]
Computing ESM2 embeddings: 65%|          | 384/591 [00:10<00:05, 36.15it/s]
Computing ESM2 embeddings: 70%|          | 416/591 [00:11<00:04, 37.13it/s]
Computing ESM2 embeddings: 76%|          | 448/591 [00:12<00:03, 36.70it/s]
Computing ESM2 embeddings: 81%|          | 480/591 [00:12<00:02, 37.49it/s]
Computing ESM2 embeddings: 87%|          | 512/591 [00:13<00:02, 36.90it/s]
Computing ESM2 embeddings: 92%|          | 544/591 [00:14<00:01, 37.55it/s]
Computing ESM2 embeddings: 97%|          | 576/591 [00:15<00:00, 36.92it/s]

Computing ESM2 embeddings: 100%|         | 591/591 [00:15<00:00, 36.96it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/895 [00:00<00:24, 35.91it/s]
Computing ESM2 embeddings: 7%|           | 64/895 [00:01<00:23, 34.76it/s]
Computing ESM2 embeddings: 11%|          | 96/895 [00:02<00:22, 36.27it/s]
Computing ESM2 embeddings: 14%|          | 128/895 [00:03<00:21, 35.22it/s]
Computing ESM2 embeddings: 18%|          | 160/895 [00:04<00:20, 36.61it/s]
Computing ESM2 embeddings: 21%|          | 192/895 [00:05<00:19, 36.21it/s]
Computing ESM2 embeddings: 25%|          | 224/895 [00:06<00:18, 37.15it/s]
Computing ESM2 embeddings: 29%|          | 256/895 [00:07<00:17, 36.62it/s]

```

```

Computing ESM2 embeddings: 32%|          | 288/895 [00:07<00:16, 37.09it/s]
Computing ESM2 embeddings: 36%|          | 320/895 [00:08<00:15, 36.05it/s]
Computing ESM2 embeddings: 39%|          | 352/895 [00:09<00:14, 36.97it/s]
Computing ESM2 embeddings: 43%|          | 384/895 [00:10<00:13, 36.57it/s]
Computing ESM2 embeddings: 46%|          | 416/895 [00:11<00:12, 37.37it/s]
Computing ESM2 embeddings: 50%|          | 448/895 [00:12<00:12, 36.76it/s]
Computing ESM2 embeddings: 54%|          | 480/895 [00:13<00:11, 37.55it/s]
Computing ESM2 embeddings: 57%|          | 512/895 [00:13<00:10, 36.94it/s]
Computing ESM2 embeddings: 61%|          | 544/895 [00:14<00:09, 37.64it/s]
Computing ESM2 embeddings: 64%|          | 576/895 [00:15<00:08, 37.04it/s]
Computing ESM2 embeddings: 68%|          | 608/895 [00:16<00:07, 37.74it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:17<00:06, 37.09it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:18<00:05, 37.47it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:19<00:05, 36.23it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:19<00:04, 37.06it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:20<00:03, 36.77it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:21<00:02, 37.60it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:22<00:01, 36.99it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:23<00:00, 37.65it/s]

Computing ESM2 embeddings: 100%|         | 895/895 [00:24<00:00, 36.80it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.93it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:02<00:28, 30.17it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:24, 32.88it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:23, 34.17it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 35.94it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:20, 35.89it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.00it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:07<00:18, 36.43it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:08<00:16, 37.25it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.81it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:14, 37.59it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.08it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.78it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 37.16it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:13<00:11, 37.83it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:14<00:10, 37.31it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.80it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.14it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.80it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.17it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 37.60it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:19<00:05, 36.46it/s] |
| Computing ESM2 embeddings: | 81% | 736/914 [00:20<00:04, 37.33it/s] |
| Computing ESM2 embeddings: | 84% | 768/914 [00:20<00:03, 36.74it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.44it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.89it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.62it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.10it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.70it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/895 [00:00<00:23, 37.43it/s]
Computing ESM2 embeddings:  7%|      | 64/895 [00:01<00:23, 35.40it/s]
Computing ESM2 embeddings: 11%|      | 96/895 [00:02<00:21, 37.18it/s]
Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:20, 36.68it/s]
Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:19, 37.68it/s]
Computing ESM2 embeddings: 21%|      | 192/895 [00:05<00:18, 37.09it/s]
Computing ESM2 embeddings: 25%|      | 224/895 [00:06<00:17, 37.76it/s]
Computing ESM2 embeddings: 29%|      | 256/895 [00:06<00:17, 37.14it/s]
Computing ESM2 embeddings: 32%|      | 288/895 [00:07<00:16, 37.46it/s]
Computing ESM2 embeddings: 36%|      | 320/895 [00:08<00:15, 36.31it/s]
Computing ESM2 embeddings: 39%|      | 352/895 [00:09<00:14, 37.27it/s]
Computing ESM2 embeddings: 43%|      | 384/895 [00:10<00:13, 36.79it/s]
Computing ESM2 embeddings: 46%|      | 416/895 [00:11<00:12, 37.55it/s]
Computing ESM2 embeddings: 50%|      | 448/895 [00:12<00:12, 36.89it/s]
Computing ESM2 embeddings: 54%|      | 480/895 [00:12<00:11, 37.68it/s]

```



Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 37.16it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.84it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.26it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.55it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 36.45it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:05, 37.26it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:18<00:05, 36.79it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.55it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 36.88it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.62it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.05it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.70it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 37.10it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.14it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.06it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.70it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.30it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.34it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.88it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 37.71it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 36.93it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.36it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.27it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.16it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.66it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.51it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.85it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.66it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.02it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.72it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.94it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.68it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.02it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.73it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.14it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.75it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.20it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.84it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.25it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.90it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.26it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.28it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
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for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/898 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/898 [00:00<00:22, 37.84it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/898 [00:01<00:23, 35.70it/s]  |
| Computing ESM2 embeddings: | 11% | 96/898 [00:02<00:21, 37.21it/s]  |
| Computing ESM2 embeddings: | 14% | 128/898 [00:03<00:21, 36.04it/s] |
| Computing ESM2 embeddings: | 18% | 160/898 [00:04<00:19, 37.21it/s] |
| Computing ESM2 embeddings: | 21% | 192/898 [00:05<00:19, 36.62it/s] |
| Computing ESM2 embeddings: | 25% | 224/898 [00:06<00:17, 37.46it/s] |
| Computing ESM2 embeddings: | 29% | 256/898 [00:06<00:17, 36.77it/s] |
| Computing ESM2 embeddings: | 32% | 288/898 [00:07<00:16, 37.56it/s] |
| Computing ESM2 embeddings: | 36% | 320/898 [00:08<00:15, 37.11it/s] |
| Computing ESM2 embeddings: | 39% | 352/898 [00:09<00:14, 37.64it/s] |
| Computing ESM2 embeddings: | 43% | 384/898 [00:10<00:14, 36.65it/s] |
| Computing ESM2 embeddings: | 46% | 416/898 [00:11<00:12, 37.47it/s] |
| Computing ESM2 embeddings: | 50% | 448/898 [00:12<00:12, 36.92it/s] |
| Computing ESM2 embeddings: | 53% | 480/898 [00:12<00:11, 37.62it/s] |
| Computing ESM2 embeddings: | 57% | 512/898 [00:13<00:10, 37.00it/s] |
| Computing ESM2 embeddings: | 61% | 544/898 [00:14<00:09, 37.72it/s] |
| Computing ESM2 embeddings: | 64% | 576/898 [00:15<00:08, 37.10it/s] |
| Computing ESM2 embeddings: | 68% | 608/898 [00:16<00:07, 37.40it/s] |
| Computing ESM2 embeddings: | 71% | 640/898 [00:17<00:07, 36.26it/s] |
| Computing ESM2 embeddings: | 75% | 672/898 [00:18<00:06, 37.21it/s] |
| Computing ESM2 embeddings: | 78% | 704/898 [00:19<00:05, 36.78it/s] |

```

Computing ESM2 embeddings: 82%|      | 736/898 [00:19<00:04, 37.60it/s]
Computing ESM2 embeddings: 86%|      | 768/898 [00:20<00:03, 37.18it/s]
Computing ESM2 embeddings: 89%|      | 800/898 [00:21<00:02, 37.94it/s]
Computing ESM2 embeddings: 93%|      | 832/898 [00:22<00:01, 37.31it/s]
Computing ESM2 embeddings: 96%|      | 864/898 [00:23<00:00, 37.89it/s]

Computing ESM2 embeddings: 100%|     | 898/898 [00:24<00:00, 37.14it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:23, 38.33it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 35.89it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.44it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.89it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:19, 37.84it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 37.02it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:05<00:18, 37.59it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 36.76it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.09it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.39it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.28it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 37.06it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.78it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 37.38it/s]

```

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 38.08it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.53it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.24it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.25it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.81it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.09it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.75it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.00it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.69it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.95it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.70it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.11it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.78it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.13it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.23it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

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Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.66it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:02<00:26, 30.93it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:24, 32.05it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:22, 34.57it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:21, 35.01it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:19, 36.37it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:06<00:18, 36.19it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:07<00:17, 37.09it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:08<00:16, 36.58it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 37.20it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 36.51it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 37.32it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:13, 36.78it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:12<00:11, 37.47it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:13<00:11, 36.86it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:14<00:10, 37.39it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:15<00:09, 36.25it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 37.14it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 36.78it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 37.34it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:18<00:06, 36.68it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:19<00:05, 37.31it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:20<00:04, 36.75it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:21<00:03, 37.52it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 36.87it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 37.62it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:23<00:00, 36.88it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:24<00:00, 36.59it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/591 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 5%| | 32/591 [00:01<00:21, 25.99it/s]  
Computing ESM2 embeddings: 11%| | 64/591 [00:02<00:17, 30.28it/s]  
Computing ESM2 embeddings: 16%| | 96/591 [00:02<00:14, 33.60it/s]  
Computing ESM2 embeddings: 22%| | 128/591 [00:03<00:13, 33.70it/s]  
Computing ESM2 embeddings: 27%| | 160/591 [00:04<00:12, 35.56it/s]  
Computing ESM2 embeddings: 32%| | 192/591 [00:05<00:11, 35.69it/s]  
Computing ESM2 embeddings: 38%| | 224/591 [00:06<00:09, 36.89it/s]  
Computing ESM2 embeddings: 43%| | 256/591 [00:07<00:09, 36.49it/s]  
Computing ESM2 embeddings: 49%| | 288/591 [00:08<00:08, 37.38it/s]  
Computing ESM2 embeddings: 54%| | 320/591 [00:09<00:07, 36.90it/s]  
Computing ESM2 embeddings: 60%| | 352/591 [00:09<00:06, 37.74it/s]  
Computing ESM2 embeddings: 65%| | 384/591 [00:10<00:05, 37.12it/s]  
Computing ESM2 embeddings: 70%| | 416/591 [00:11<00:04, 37.82it/s]  
Computing ESM2 embeddings: 76%| | 448/591 [00:12<00:03, 37.23it/s]  
Computing ESM2 embeddings: 81%| | 480/591 [00:13<00:02, 37.89it/s]  
Computing ESM2 embeddings: 87%| | 512/591 [00:14<00:02, 37.11it/s]  
Computing ESM2 embeddings: 92%| | 544/591 [00:14<00:01, 37.76it/s]  
Computing ESM2 embeddings: 97%| | 576/591 [00:15<00:00, 37.18it/s]

Computing ESM2 embeddings: 100%| | 591/591 [00:16<00:00, 36.07it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/876 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/876 [00:00<00:23, 36.48it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/876 [00:01<00:23, 34.08it/s]  |
| Computing ESM2 embeddings: | 11% | 96/876 [00:02<00:21, 35.97it/s]  |
| Computing ESM2 embeddings: | 15% | 128/876 [00:03<00:21, 35.22it/s] |
| Computing ESM2 embeddings: | 18% | 160/876 [00:04<00:19, 36.28it/s] |
| Computing ESM2 embeddings: | 22% | 192/876 [00:05<00:19, 35.58it/s] |
| Computing ESM2 embeddings: | 26% | 224/876 [00:06<00:17, 36.78it/s] |
| Computing ESM2 embeddings: | 29% | 256/876 [00:07<00:17, 36.26it/s] |
| Computing ESM2 embeddings: | 33% | 288/876 [00:07<00:15, 37.20it/s] |
| Computing ESM2 embeddings: | 37% | 320/876 [00:08<00:15, 36.71it/s] |
| Computing ESM2 embeddings: | 40% | 352/876 [00:09<00:13, 37.53it/s] |
| Computing ESM2 embeddings: | 44% | 384/876 [00:10<00:13, 36.93it/s] |
| Computing ESM2 embeddings: | 47% | 416/876 [00:11<00:12, 37.68it/s] |
| Computing ESM2 embeddings: | 51% | 448/876 [00:12<00:11, 36.93it/s] |
| Computing ESM2 embeddings: | 55% | 480/876 [00:13<00:10, 37.61it/s] |
| Computing ESM2 embeddings: | 58% | 512/876 [00:13<00:09, 36.94it/s] |
| Computing ESM2 embeddings: | 62% | 544/876 [00:14<00:08, 37.63it/s] |
| Computing ESM2 embeddings: | 66% | 576/876 [00:15<00:08, 37.02it/s] |
| Computing ESM2 embeddings: | 69% | 608/876 [00:16<00:07, 37.52it/s] |
| Computing ESM2 embeddings: | 73% | 640/876 [00:17<00:06, 36.14it/s] |
| Computing ESM2 embeddings: | 77% | 672/876 [00:18<00:05, 37.12it/s] |
| Computing ESM2 embeddings: | 80% | 704/876 [00:19<00:04, 36.68it/s] |



```

Computing ESM2 embeddings: 84%|      | 736/876 [00:19<00:03, 37.49it/s]
Computing ESM2 embeddings: 88%|      | 768/876 [00:20<00:02, 36.85it/s]
Computing ESM2 embeddings: 91%|      | 800/876 [00:21<00:02, 37.57it/s]
Computing ESM2 embeddings: 95%|      | 832/876 [00:22<00:01, 36.88it/s]
Computing ESM2 embeddings: 99%|      | 864/876 [00:23<00:00, 37.54it/s]

Computing ESM2 embeddings: 100%|     | 876/876 [00:23<00:00, 36.83it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.39it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 35.62it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.39it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.67it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.60it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.88it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.76it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 37.12it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.62it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.48it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.39it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 37.07it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.48it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 36.25it/s]

```

```

Computing ESM2 embeddings: 53%|          | 480/914 [00:12<00:11, 37.18it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 36.73it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 37.56it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 36.98it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.66it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.05it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.69it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:18<00:05, 36.97it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.45it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:04, 36.44it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.45it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.02it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.65it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.18it/s]

```

```

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.16it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']

```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```

Computing ESM2 embeddings: 0%|           | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/896 [00:00<00:21, 39.33it/s]
Computing ESM2 embeddings: 7%|           | 64/896 [00:01<00:22, 37.26it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:20, 38.23it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 37.11it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 37.68it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:19, 36.40it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:05<00:18, 37.33it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:17, 36.69it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:16, 37.11it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 36.21it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 37.14it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 36.80it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:12, 37.61it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:12<00:12, 37.15it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:10, 37.85it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 37.21it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 37.77it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 37.25it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 37.90it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 37.32it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:17<00:05, 37.95it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:05, 37.11it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 37.80it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:20<00:03, 37.19it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 37.88it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 37.27it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:23<00:00, 37.85it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:24<00:00, 37.25it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/913 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/913 [00:00<00:24, 36.56it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/913 [00:02<00:27, 30.55it/s]  |
| Computing ESM2 embeddings: | 11% | 96/913 [00:02<00:24, 33.98it/s]  |
| Computing ESM2 embeddings: | 14% | 128/913 [00:03<00:22, 34.67it/s] |
| Computing ESM2 embeddings: | 18% | 160/913 [00:04<00:21, 35.73it/s] |
| Computing ESM2 embeddings: | 21% | 192/913 [00:05<00:20, 35.12it/s] |
| Computing ESM2 embeddings: | 25% | 224/913 [00:06<00:18, 36.31it/s] |
| Computing ESM2 embeddings: | 28% | 256/913 [00:07<00:18, 36.08it/s] |
| Computing ESM2 embeddings: | 32% | 288/913 [00:08<00:16, 36.82it/s] |
| Computing ESM2 embeddings: | 35% | 320/913 [00:09<00:16, 35.80it/s] |
| Computing ESM2 embeddings: | 39% | 352/913 [00:09<00:15, 36.88it/s] |
| Computing ESM2 embeddings: | 42% | 384/913 [00:10<00:14, 36.52it/s] |
| Computing ESM2 embeddings: | 46% | 416/913 [00:11<00:13, 37.25it/s] |
| Computing ESM2 embeddings: | 49% | 448/913 [00:12<00:12, 36.76it/s] |
| Computing ESM2 embeddings: | 53% | 480/913 [00:13<00:11, 37.46it/s] |
| Computing ESM2 embeddings: | 56% | 512/913 [00:14<00:10, 36.92it/s] |
| Computing ESM2 embeddings: | 60% | 544/913 [00:14<00:09, 37.62it/s] |
| Computing ESM2 embeddings: | 63% | 576/913 [00:15<00:09, 36.98it/s] |
| Computing ESM2 embeddings: | 67% | 608/913 [00:16<00:08, 37.61it/s] |
| Computing ESM2 embeddings: | 70% | 640/913 [00:17<00:07, 37.13it/s] |
| Computing ESM2 embeddings: | 74% | 672/913 [00:18<00:06, 37.56it/s] |

Computing ESM2 embeddings: 77%| | 704/913 [00:19<00:05, 36.32it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:20<00:04, 37.17it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:21<00:03, 36.74it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:21<00:03, 37.21it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 36.27it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:23<00:01, 37.10it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:24<00:00, 36.65it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:25<00:00, 36.37it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/898 [00:00<00:22, 37.85it/s]

Computing ESM2 embeddings: 7%| | 64/898 [00:01<00:24, 33.80it/s]

Computing ESM2 embeddings: 11%| | 96/898 [00:02<00:22, 36.07it/s]

Computing ESM2 embeddings: 14%| | 128/898 [00:03<00:21, 35.67it/s]

Computing ESM2 embeddings: 18%| | 160/898 [00:04<00:19, 37.13it/s]

Computing ESM2 embeddings: 21%| | 192/898 [00:05<00:19, 36.50it/s]

Computing ESM2 embeddings: 25%| | 224/898 [00:06<00:18, 37.43it/s]

Computing ESM2 embeddings: 29%| | 256/898 [00:06<00:17, 37.00it/s]

Computing ESM2 embeddings: 32%| | 288/898 [00:07<00:16, 37.87it/s]

Computing ESM2 embeddings: 36%| | 320/898 [00:08<00:15, 37.29it/s]

Computing ESM2 embeddings: 39%| | 352/898 [00:09<00:14, 37.49it/s]

Computing ESM2 embeddings: 43%| | 384/898 [00:10<00:14, 36.41it/s]

Computing ESM2 embeddings: 46%| | 416/898 [00:11<00:13, 37.04it/s]

Computing ESM2 embeddings: 50%| | 448/898 [00:12<00:12, 36.02it/s]

Computing ESM2 embeddings: 53%| | 480/898 [00:13<00:11, 36.94it/s]

Computing ESM2 embeddings: 57%| | 512/898 [00:13<00:10, 36.61it/s]

Computing ESM2 embeddings: 61%| | 544/898 [00:14<00:09, 37.44it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:15<00:08, 36.95it/s]

Computing ESM2 embeddings: 68%| | 608/898 [00:16<00:07, 37.58it/s]

Computing ESM2 embeddings: 71%| | 640/898 [00:17<00:06, 36.97it/s]

Computing ESM2 embeddings: 75%| | 672/898 [00:18<00:06, 37.44it/s]

Computing ESM2 embeddings: 78%| | 704/898 [00:19<00:05, 36.32it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:19<00:04, 37.02it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:20<00:03, 36.15it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:21<00:02, 37.04it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:22<00:01, 36.48it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:23<00:00, 37.37it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:24<00:00, 36.67it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.58it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 36.14it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.33it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 35.93it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 37.21it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.71it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 37.65it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:17, 37.02it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.80it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:15, 37.20it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 37.89it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.28it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.93it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.26it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.98it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.26it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.95it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.25it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.85it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.16it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:17<00:06, 37.53it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 36.44it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.38it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 36.92it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.68it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.06it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:23<00:01, 37.72it/s] |
| Computing ESM2 embeddings: 98% | 896/914 [00:24<00:00, 37.08it/s] |

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.33it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/897 [00:01<00:31, 27.17it/s]  
 Computing ESM2 embeddings: 7%| | 64/897 [00:02<00:27, 29.79it/s]  
 Computing ESM2 embeddings: 11%| | 96/897 [00:03<00:24, 33.33it/s]  
 Computing ESM2 embeddings: 14%| | 128/897 [00:03<00:23, 33.27it/s]  
 Computing ESM2 embeddings: 18%| | 160/897 [00:04<00:20, 35.23it/s]  
 Computing ESM2 embeddings: 21%| | 192/897 [00:05<00:19, 35.34it/s]  
 Computing ESM2 embeddings: 25%| | 224/897 [00:06<00:18, 36.62it/s]  
 Computing ESM2 embeddings: 29%| | 256/897 [00:07<00:17, 36.22it/s]  
 Computing ESM2 embeddings: 32%| | 288/897 [00:08<00:16, 37.15it/s]  
 Computing ESM2 embeddings: 36%| | 320/897 [00:09<00:15, 36.63it/s]  
 Computing ESM2 embeddings: 39%| | 352/897 [00:09<00:14, 37.38it/s]  
 Computing ESM2 embeddings: 43%| | 384/897 [00:10<00:13, 36.77it/s]  
 Computing ESM2 embeddings: 46%| | 416/897 [00:11<00:12, 37.31it/s]  
 Computing ESM2 embeddings: 50%| | 448/897 [00:12<00:12, 36.28it/s]  
 Computing ESM2 embeddings: 54%| | 480/897 [00:13<00:11, 37.21it/s]  
 Computing ESM2 embeddings: 57%| | 512/897 [00:14<00:10, 36.80it/s]  
 Computing ESM2 embeddings: 61%| | 544/897 [00:15<00:09, 37.53it/s]  
 Computing ESM2 embeddings: 64%| | 576/897 [00:16<00:08, 36.98it/s]  
 Computing ESM2 embeddings: 68%| | 608/897 [00:16<00:07, 37.72it/s]



```

Computing ESM2 embeddings: 71%|      | 640/897 [00:17<00:06, 37.01it/s]
Computing ESM2 embeddings: 75%|      | 672/897 [00:18<00:05, 37.74it/s]
Computing ESM2 embeddings: 78%|      | 704/897 [00:19<00:05, 36.95it/s]
Computing ESM2 embeddings: 82%|      | 736/897 [00:20<00:04, 37.70it/s]
Computing ESM2 embeddings: 86%|      | 768/897 [00:21<00:03, 37.10it/s]
Computing ESM2 embeddings: 89%|      | 800/897 [00:21<00:02, 37.79it/s]
Computing ESM2 embeddings: 93%|      | 832/897 [00:22<00:01, 37.17it/s]
Computing ESM2 embeddings: 96%|      | 864/897 [00:23<00:00, 37.88it/s]

Computing ESM2 embeddings: 100%|     | 897/897 [00:24<00:00, 36.31it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/589 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/589 [00:00<00:14, 38.04it/s]
Computing ESM2 embeddings: 11%|      | 64/589 [00:01<00:14, 36.49it/s]
Computing ESM2 embeddings: 16%|      | 96/589 [00:02<00:13, 37.82it/s]
Computing ESM2 embeddings: 22%|      | 128/589 [00:03<00:12, 36.97it/s]
Computing ESM2 embeddings: 27%|      | 160/589 [00:04<00:11, 37.77it/s]
Computing ESM2 embeddings: 33%|      | 192/589 [00:05<00:10, 36.86it/s]
Computing ESM2 embeddings: 38%|      | 224/589 [00:05<00:09, 37.73it/s]
Computing ESM2 embeddings: 43%|      | 256/589 [00:06<00:09, 36.99it/s]
Computing ESM2 embeddings: 49%|      | 288/589 [00:07<00:07, 37.76it/s]
Computing ESM2 embeddings: 54%|      | 320/589 [00:08<00:07, 36.96it/s]
Computing ESM2 embeddings: 60%|      | 352/589 [00:09<00:06, 37.63it/s]

```

```

Computing ESM2 embeddings: 65%|          | 384/589 [00:10<00:05, 36.64it/s]
Computing ESM2 embeddings: 71%|          | 416/589 [00:11<00:04, 37.06it/s]
Computing ESM2 embeddings: 76%|          | 448/589 [00:12<00:03, 35.92it/s]
Computing ESM2 embeddings: 81%|          | 480/589 [00:12<00:02, 36.89it/s]
Computing ESM2 embeddings: 87%|          | 512/589 [00:13<00:02, 36.49it/s]
Computing ESM2 embeddings: 92%|          | 544/589 [00:14<00:01, 37.23it/s]
Computing ESM2 embeddings: 98%|          | 576/589 [00:15<00:00, 36.69it/s]

Computing ESM2 embeddings: 100%|         | 589/589 [00:16<00:00, 36.79it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:01<00:29, 29.51it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:24, 34.31it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 36.50it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:21, 36.18it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 37.28it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:19, 36.66it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:06<00:17, 37.45it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:07<00:17, 36.96it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.72it/s]
Computing ESM2 embeddings: 36%|          | 320/896 [00:08<00:15, 37.16it/s]
Computing ESM2 embeddings: 39%|          | 352/896 [00:09<00:14, 37.62it/s]
Computing ESM2 embeddings: 43%|          | 384/896 [00:10<00:14, 36.45it/s]

```

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.22it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.76it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.51it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.02it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.74it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.20it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.75it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.06it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.75it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 37.21it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.85it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.01it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.64it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.00it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.58it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.95it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 36.26it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:26, 31.88it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 34.80it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 34.80it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:21, 35.48it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:20, 34.97it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 36.35it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:18, 36.08it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:08<00:16, 37.05it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.29it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 36.89it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.45it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.37it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 36.84it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 37.51it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:14<00:10, 36.97it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.47it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 36.35it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.14it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 36.68it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.43it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 36.82it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:20<00:04, 37.56it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:21<00:03, 36.94it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.67it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.04it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:23<00:01, 37.46it/s] |
| Computing ESM2 embeddings: 98% | 896/914 [00:24<00:00, 36.31it/s] |

Computing ESM2 embeddings: 100%| | 914/914 [00:25<00:00, 36.44it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: 4%  | 32/896 [00:00<00:23, 36.83it/s]  |
| Computing ESM2 embeddings: 7%  | 64/896 [00:01<00:25, 32.57it/s]  |
| Computing ESM2 embeddings: 11% | 96/896 [00:02<00:22, 34.85it/s]  |
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:22, 34.64it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:20, 36.19it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 35.96it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:18, 36.95it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:07<00:17, 36.38it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.10it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 36.52it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.43it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 36.78it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.59it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 36.99it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:13<00:11, 37.70it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:14<00:10, 37.01it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.64it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.12it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.74it/s] |

```

Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:06, 37.08it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:18<00:05, 37.77it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:19<00:05, 37.19it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.54it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 36.34it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.09it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 36.33it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 36.84it/s]
Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 36.56it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/914 [00:01<00:30, 28.73it/s]
Computing ESM2 embeddings: 7%|           | 64/914 [00:02<00:27, 31.33it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:23, 34.55it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:22, 34.88it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:20, 36.01it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:20, 35.33it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:06<00:18, 36.53it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:07<00:18, 36.17it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:08<00:16, 37.11it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:16, 36.71it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:15, 37.18it/s]

```

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.25it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.18it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.65it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.51it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 36.94it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.57it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.79it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.47it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.94it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.66it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.90it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 37.65it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:21<00:03, 37.06it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.41it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.28it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.16it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.63it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:25<00:00, 36.50it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/898 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/898 [00:00<00:22, 37.88it/s]

Computing ESM2 embeddings: 7%| | 64/898 [00:01<00:22, 36.56it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/898 [00:02<00:21, 37.86it/s]  |
| Computing ESM2 embeddings: 14% | 128/898 [00:03<00:20, 37.17it/s] |
| Computing ESM2 embeddings: 18% | 160/898 [00:04<00:19, 38.14it/s] |
| Computing ESM2 embeddings: 21% | 192/898 [00:05<00:18, 37.26it/s] |
| Computing ESM2 embeddings: 25% | 224/898 [00:05<00:17, 37.93it/s] |
| Computing ESM2 embeddings: 29% | 256/898 [00:06<00:17, 37.32it/s] |
| Computing ESM2 embeddings: 32% | 288/898 [00:07<00:16, 37.91it/s] |
| Computing ESM2 embeddings: 36% | 320/898 [00:08<00:15, 36.75it/s] |
| Computing ESM2 embeddings: 39% | 352/898 [00:09<00:14, 37.44it/s] |
| Computing ESM2 embeddings: 43% | 384/898 [00:10<00:14, 36.33it/s] |
| Computing ESM2 embeddings: 46% | 416/898 [00:11<00:12, 37.15it/s] |
| Computing ESM2 embeddings: 50% | 448/898 [00:12<00:12, 37.00it/s] |
| Computing ESM2 embeddings: 53% | 480/898 [00:12<00:11, 37.91it/s] |
| Computing ESM2 embeddings: 57% | 512/898 [00:13<00:10, 37.34it/s] |
| Computing ESM2 embeddings: 61% | 544/898 [00:14<00:09, 38.05it/s] |
| Computing ESM2 embeddings: 64% | 576/898 [00:15<00:08, 37.19it/s] |
| Computing ESM2 embeddings: 68% | 608/898 [00:16<00:07, 37.85it/s] |
| Computing ESM2 embeddings: 71% | 640/898 [00:17<00:06, 37.25it/s] |
| Computing ESM2 embeddings: 75% | 672/898 [00:17<00:05, 37.91it/s] |
| Computing ESM2 embeddings: 78% | 704/898 [00:18<00:05, 37.05it/s] |
| Computing ESM2 embeddings: 82% | 736/898 [00:19<00:04, 37.78it/s] |
| Computing ESM2 embeddings: 86% | 768/898 [00:20<00:03, 37.16it/s] |
| Computing ESM2 embeddings: 89% | 800/898 [00:21<00:02, 37.85it/s] |
| Computing ESM2 embeddings: 93% | 832/898 [00:22<00:01, 37.21it/s] |



Computing ESM2 embeddings: 96%| | 864/898 [00:23<00:00, 37.88it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:24<00:00, 37.31it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.40it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 35.73it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.47it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.98it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:19, 37.84it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 37.04it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:05<00:18, 37.88it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 37.20it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.90it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:15, 37.32it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.85it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.08it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.80it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:11<00:12, 37.17it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.86it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.23it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.93it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.27it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.92it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.08it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:17<00:06, 37.77it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.08it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.74it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.18it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.55it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.44it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.33it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:23<00:00, 36.95it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.36it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:00<00:22, 38.22it/s]

Computing ESM2 embeddings: 7%| | 64/897 [00:01<00:23, 35.35it/s]

Computing ESM2 embeddings: 11%| | 96/897 [00:02<00:21, 37.07it/s]

Computing ESM2 embeddings: 14%| | 128/897 [00:03<00:20, 36.71it/s]

Computing ESM2 embeddings: 18%| | 160/897 [00:04<00:19, 37.78it/s]

Computing ESM2 embeddings: 21%| | 192/897 [00:05<00:19, 37.01it/s]

Computing ESM2 embeddings: 25%| | 224/897 [00:06<00:17, 37.76it/s]

Computing ESM2 embeddings: 29%| | 256/897 [00:06<00:17, 37.30it/s]

Computing ESM2 embeddings: 32%| | 288/897 [00:07<00:15, 38.07it/s]

Computing ESM2 embeddings: 36%| | 320/897 [00:08<00:15, 37.46it/s]

Computing ESM2 embeddings: 39%| | 352/897 [00:09<00:14, 38.08it/s]

Computing ESM2 embeddings: 43%| | 384/897 [00:10<00:13, 37.40it/s]

Computing ESM2 embeddings: 46%| | 416/897 [00:11<00:12, 37.97it/s]

Computing ESM2 embeddings: 50%| | 448/897 [00:11<00:12, 37.14it/s]

Computing ESM2 embeddings: 54%| | 480/897 [00:12<00:11, 37.61it/s]

Computing ESM2 embeddings: 57%| | 512/897 [00:13<00:10, 36.40it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:14<00:09, 37.33it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:15<00:08, 36.88it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:16<00:07, 37.64it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:17<00:06, 37.16it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:17<00:05, 37.93it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:18<00:05, 37.35it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:19<00:04, 37.96it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:20<00:03, 37.37it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:21<00:02, 38.02it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:22<00:01, 37.31it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:23<00:00, 37.54it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:24<00:00, 37.22it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/590 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/590 [00:00<00:14, 38.96it/s]

Computing ESM2 embeddings: 11%| | 64/590 [00:01<00:14, 37.35it/s]

Computing ESM2 embeddings: 16%| | 96/590 [00:02<00:12, 38.57it/s]

Computing ESM2 embeddings: 22%| | 128/590 [00:03<00:12, 37.65it/s]

Computing ESM2 embeddings: 27%| | 160/590 [00:04<00:11, 38.38it/s]

Computing ESM2 embeddings: 33%| | 192/590 [00:05<00:10, 37.25it/s]

Computing ESM2 embeddings: 38%| | 224/590 [00:05<00:09, 38.04it/s]

Computing ESM2 embeddings: 43%| | 256/590 [00:06<00:08, 37.29it/s]

Computing ESM2 embeddings: 49%| | 288/590 [00:07<00:07, 37.98it/s]

Computing ESM2 embeddings: 54%| | 320/590 [00:08<00:07, 37.29it/s]

Computing ESM2 embeddings: 60%| | 352/590 [00:09<00:06, 37.59it/s]

Computing ESM2 embeddings: 65%| | 384/590 [00:10<00:05, 36.45it/s]

Computing ESM2 embeddings: 71%| | 416/590 [00:11<00:04, 37.39it/s]

Computing ESM2 embeddings: 76%| | 448/590 [00:11<00:03, 36.97it/s]

Computing ESM2 embeddings: 81%| | 480/590 [00:12<00:02, 37.78it/s]

Computing ESM2 embeddings: 87%| | 512/590 [00:13<00:02, 37.16it/s]

Computing ESM2 embeddings: 92%| | 544/590 [00:14<00:01, 37.89it/s]

Computing ESM2 embeddings: 98%| | 576/590 [00:15<00:00, 37.27it/s]

Computing ESM2 embeddings: 100%| | 590/590 [00:15<00:00, 37.53it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 39.14it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:22, 37.41it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/895 [00:02<00:20, 38.20it/s]  |
| Computing ESM2 embeddings: 14% | 128/895 [00:03<00:20, 37.24it/s] |
| Computing ESM2 embeddings: 18% | 160/895 [00:04<00:19, 37.99it/s] |
| Computing ESM2 embeddings: 21% | 192/895 [00:05<00:18, 37.07it/s] |
| Computing ESM2 embeddings: 25% | 224/895 [00:05<00:17, 37.77it/s] |
| Computing ESM2 embeddings: 29% | 256/895 [00:06<00:17, 37.18it/s] |
| Computing ESM2 embeddings: 32% | 288/895 [00:07<00:16, 37.89it/s] |
| Computing ESM2 embeddings: 36% | 320/895 [00:08<00:15, 37.08it/s] |
| Computing ESM2 embeddings: 39% | 352/895 [00:09<00:14, 37.85it/s] |
| Computing ESM2 embeddings: 43% | 384/895 [00:10<00:13, 37.15it/s] |
| Computing ESM2 embeddings: 46% | 416/895 [00:11<00:12, 37.50it/s] |
| Computing ESM2 embeddings: 50% | 448/895 [00:12<00:12, 36.35it/s] |
| Computing ESM2 embeddings: 54% | 480/895 [00:12<00:11, 37.25it/s] |
| Computing ESM2 embeddings: 57% | 512/895 [00:13<00:10, 36.80it/s] |
| Computing ESM2 embeddings: 61% | 544/895 [00:14<00:09, 37.52it/s] |
| Computing ESM2 embeddings: 64% | 576/895 [00:15<00:08, 37.07it/s] |
| Computing ESM2 embeddings: 68% | 608/895 [00:16<00:07, 37.71it/s] |
| Computing ESM2 embeddings: 72% | 640/895 [00:17<00:06, 37.01it/s] |
| Computing ESM2 embeddings: 75% | 672/895 [00:17<00:05, 37.68it/s] |
| Computing ESM2 embeddings: 79% | 704/895 [00:18<00:05, 36.94it/s] |
| Computing ESM2 embeddings: 82% | 736/895 [00:19<00:04, 37.60it/s] |
| Computing ESM2 embeddings: 86% | 768/895 [00:20<00:03, 36.98it/s] |
| Computing ESM2 embeddings: 89% | 800/895 [00:21<00:02, 37.62it/s] |
| Computing ESM2 embeddings: 93% | 832/895 [00:22<00:01, 37.00it/s] |

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.65it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 37.19it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 38.06it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:25, 33.29it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 35.28it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 34.81it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.24it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.11it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.08it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.46it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.29it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.71it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.47it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.85it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.59it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.92it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.62it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.03it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.74it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.11it/s]

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Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.90it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 36.96it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.67it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 37.13it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.73it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.06it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.67it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.19it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.89it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.23it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 36.90it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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Computing ESM2 embeddings: 0%|           | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/896 [00:00<00:22, 38.01it/s]
Computing ESM2 embeddings: 7%|           | 64/896 [00:01<00:25, 32.52it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:22, 35.01it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:21, 35.17it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:20, 36.63it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:19, 36.15it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:06<00:18, 36.83it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:07<00:17, 36.35it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.27it/s]

```

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.82it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.62it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.89it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.55it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.00it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.32it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 36.54it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.13it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.11it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.01it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:07, 36.55it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.42it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 36.91it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.64it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.00it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.35it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.38it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.22it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.66it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.61it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:22, 37.08it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:21, 38.17it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 37.16it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:19, 38.03it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 37.23it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:05<00:18, 37.64it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:06<00:18, 36.43it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.38it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.82it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 37.62it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.99it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.64it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.10it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.78it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.14it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.78it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.20it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.52it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 36.30it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.10it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:18<00:05, 36.62it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.38it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 36.83it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.58it/s] |

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Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.89it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.61it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.96it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 37.23it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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```

Computing ESM2 embeddings:  0%|      |  0/898 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/898 [00:00<00:22, 38.84it/s]
Computing ESM2 embeddings:  7%|      | 64/898 [00:01<00:22, 37.02it/s]
Computing ESM2 embeddings: 11%|      | 96/898 [00:02<00:21, 37.54it/s]
Computing ESM2 embeddings: 14%|      | 128/898 [00:03<00:21, 36.13it/s]
Computing ESM2 embeddings: 18%|      | 160/898 [00:04<00:19, 37.24it/s]
Computing ESM2 embeddings: 21%|      | 192/898 [00:05<00:19, 36.64it/s]
Computing ESM2 embeddings: 25%|      | 224/898 [00:06<00:18, 37.44it/s]
Computing ESM2 embeddings: 29%|      | 256/898 [00:06<00:17, 36.91it/s]
Computing ESM2 embeddings: 32%|      | 288/898 [00:07<00:16, 37.37it/s]
Computing ESM2 embeddings: 36%|      | 320/898 [00:08<00:16, 36.12it/s]
Computing ESM2 embeddings: 39%|      | 352/898 [00:09<00:14, 37.10it/s]
Computing ESM2 embeddings: 43%|      | 384/898 [00:10<00:13, 36.74it/s]
Computing ESM2 embeddings: 46%|      | 416/898 [00:11<00:12, 37.59it/s]
Computing ESM2 embeddings: 50%|      | 448/898 [00:12<00:12, 37.03it/s]
Computing ESM2 embeddings: 53%|      | 480/898 [00:12<00:11, 37.75it/s]
Computing ESM2 embeddings: 57%|      | 512/898 [00:13<00:10, 37.04it/s]

```

Computing ESM2 embeddings: 61%| | 544/898 [00:14<00:09, 37.71it/s]

Computing ESM2 embeddings: 64%| | 576/898 [00:15<00:08, 37.15it/s]

Computing ESM2 embeddings: 68%| | 608/898 [00:16<00:07, 37.83it/s]

Computing ESM2 embeddings: 71%| | 640/898 [00:17<00:06, 37.26it/s]

Computing ESM2 embeddings: 75%| | 672/898 [00:18<00:05, 37.92it/s]

Computing ESM2 embeddings: 78%| | 704/898 [00:18<00:05, 37.07it/s]

Computing ESM2 embeddings: 82%| | 736/898 [00:19<00:04, 37.74it/s]

Computing ESM2 embeddings: 86%| | 768/898 [00:20<00:03, 37.07it/s]

Computing ESM2 embeddings: 89%| | 800/898 [00:21<00:02, 37.78it/s]

Computing ESM2 embeddings: 93%| | 832/898 [00:22<00:01, 37.09it/s]

Computing ESM2 embeddings: 96%| | 864/898 [00:23<00:00, 37.76it/s]

Computing ESM2 embeddings: 100%| | 898/898 [00:24<00:00, 37.13it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 36.94it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:25, 33.20it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 35.89it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 35.77it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.02it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.40it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.28it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 36.73it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.37it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.05it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.02it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.57it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.39it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.86it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.62it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.04it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.78it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.15it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.53it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.31it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.24it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.82it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.61it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.92it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.65it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.10it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.77it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.10it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.93it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/897 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/897 [00:00<00:22, 39.30it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/897 [00:01<00:22, 37.04it/s]  |
| Computing ESM2 embeddings: | 11% | 96/897 [00:02<00:21, 37.98it/s]  |
| Computing ESM2 embeddings: | 14% | 128/897 [00:03<00:20, 37.18it/s] |
| Computing ESM2 embeddings: | 18% | 160/897 [00:04<00:19, 38.06it/s] |
| Computing ESM2 embeddings: | 21% | 192/897 [00:05<00:18, 37.28it/s] |
| Computing ESM2 embeddings: | 25% | 224/897 [00:05<00:17, 37.96it/s] |
| Computing ESM2 embeddings: | 29% | 256/897 [00:06<00:17, 37.13it/s] |
| Computing ESM2 embeddings: | 32% | 288/897 [00:07<00:16, 37.84it/s] |
| Computing ESM2 embeddings: | 36% | 320/897 [00:08<00:15, 37.25it/s] |
| Computing ESM2 embeddings: | 39% | 352/897 [00:09<00:14, 37.90it/s] |
| Computing ESM2 embeddings: | 43% | 384/897 [00:10<00:13, 37.34it/s] |
| Computing ESM2 embeddings: | 46% | 416/897 [00:11<00:12, 37.91it/s] |
| Computing ESM2 embeddings: | 50% | 448/897 [00:11<00:12, 37.31it/s] |
| Computing ESM2 embeddings: | 54% | 480/897 [00:12<00:10, 38.02it/s] |
| Computing ESM2 embeddings: | 57% | 512/897 [00:13<00:10, 37.32it/s] |
| Computing ESM2 embeddings: | 61% | 544/897 [00:14<00:09, 38.00it/s] |
| Computing ESM2 embeddings: | 64% | 576/897 [00:15<00:08, 37.32it/s] |
| Computing ESM2 embeddings: | 68% | 608/897 [00:16<00:07, 37.64it/s] |
| Computing ESM2 embeddings: | 71% | 640/897 [00:17<00:07, 36.51it/s] |
| Computing ESM2 embeddings: | 75% | 672/897 [00:17<00:06, 37.43it/s] |
| Computing ESM2 embeddings: | 78% | 704/897 [00:18<00:05, 36.91it/s] |
| Computing ESM2 embeddings: | 82% | 736/897 [00:19<00:04, 37.69it/s] |

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Computing ESM2 embeddings: 86%|      | 768/897 [00:20<00:03, 37.14it/s]
Computing ESM2 embeddings: 89%|      | 800/897 [00:21<00:02, 37.89it/s]
Computing ESM2 embeddings: 93%|      | 832/897 [00:22<00:01, 37.24it/s]
Computing ESM2 embeddings: 96%|      | 864/897 [00:23<00:00, 37.87it/s]
Computing ESM2 embeddings: 100%|     | 897/897 [00:23<00:00, 37.45it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
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'esm.pooler.dense.weight']
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for predictions and inference.

```

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Computing ESM2 embeddings:  0%|      |  0/571 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|      | 32/571 [00:00<00:14, 37.98it/s]
Computing ESM2 embeddings: 11%|      | 64/571 [00:01<00:14, 35.65it/s]
Computing ESM2 embeddings: 17%|      | 96/571 [00:02<00:12, 37.30it/s]
Computing ESM2 embeddings: 22%|      | 128/571 [00:03<00:11, 37.05it/s]
Computing ESM2 embeddings: 28%|      | 160/571 [00:04<00:10, 38.12it/s]
Computing ESM2 embeddings: 34%|      | 192/571 [00:05<00:10, 37.16it/s]
Computing ESM2 embeddings: 39%|      | 224/571 [00:05<00:09, 37.85it/s]
Computing ESM2 embeddings: 45%|      | 256/571 [00:06<00:08, 37.22it/s]
Computing ESM2 embeddings: 50%|      | 288/571 [00:07<00:07, 37.87it/s]
Computing ESM2 embeddings: 56%|      | 320/571 [00:08<00:06, 37.06it/s]
Computing ESM2 embeddings: 62%|      | 352/571 [00:09<00:05, 37.58it/s]
Computing ESM2 embeddings: 67%|      | 384/571 [00:10<00:05, 36.23it/s]
Computing ESM2 embeddings: 73%|      | 416/571 [00:11<00:04, 37.11it/s]
Computing ESM2 embeddings: 78%|      | 448/571 [00:12<00:03, 36.79it/s]
Computing ESM2 embeddings: 84%|      | 480/571 [00:12<00:02, 37.50it/s]

```

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 37.06it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 37.71it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 37.15it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:24, 36.49it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 34.44it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.72it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.36it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.07it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:20, 36.04it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 36.98it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 36.69it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.59it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.87it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.64it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.90it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.62it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.06it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.73it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.11it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.85it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.20it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.66it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.59it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.35it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.79it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.51it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.93it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.38it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.32it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.25it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.68it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.90it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 37.13it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:02<00:28, 29.04it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:03<00:25, 30.98it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:22, 33.74it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:21, 34.42it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 35.67it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:19, 34.74it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:17, 36.14it/s]



Computing ESM2 embeddings: 32%| | 288/896 [00:08<00:16, 35.90it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:09<00:15, 36.66it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:10<00:15, 35.64it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.76it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:13, 36.33it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.13it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 36.64it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 37.47it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:15<00:09, 36.85it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:16<00:08, 37.55it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.09it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 37.82it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:06, 37.13it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 37.74it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.03it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:21<00:03, 37.64it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:22<00:02, 37.02it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.62it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 36.97it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.22it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 4%  | 32/914 [00:00<00:25, 35.09it/s]  |
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:24, 34.26it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:22, 36.49it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:21, 36.22it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 36.92it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:20, 35.94it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 36.81it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:18, 35.86it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:17, 36.68it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 35.63it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 36.78it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.38it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.29it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 36.77it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 37.57it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:14<00:10, 37.06it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.80it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.16it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.76it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.12it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.77it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 37.17it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.53it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:04, 36.37it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.27it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 36.85it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.59it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.05it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.82it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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```

Computing ESM2 embeddings:  0%|      |  0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/897 [00:00<00:23, 36.80it/s]
Computing ESM2 embeddings:  7%|      | 64/897 [00:02<00:27, 29.90it/s]
Computing ESM2 embeddings: 11%|      | 96/897 [00:02<00:24, 33.30it/s]
Computing ESM2 embeddings: 14%|      | 128/897 [00:03<00:22, 33.60it/s]
Computing ESM2 embeddings: 18%|      | 160/897 [00:04<00:20, 35.31it/s]
Computing ESM2 embeddings: 21%|      | 192/897 [00:05<00:20, 34.74it/s]
Computing ESM2 embeddings: 25%|      | 224/897 [00:06<00:18, 36.17it/s]
Computing ESM2 embeddings: 29%|      | 256/897 [00:07<00:17, 35.93it/s]
Computing ESM2 embeddings: 32%|      | 288/897 [00:08<00:16, 36.76it/s]
Computing ESM2 embeddings: 36%|      | 320/897 [00:09<00:16, 35.80it/s]
Computing ESM2 embeddings: 39%|      | 352/897 [00:09<00:14, 36.81it/s]
Computing ESM2 embeddings: 43%|      | 384/897 [00:10<00:14, 36.44it/s]
Computing ESM2 embeddings: 46%|      | 416/897 [00:11<00:12, 37.23it/s]
Computing ESM2 embeddings: 50%|      | 448/897 [00:12<00:12, 36.85it/s]
Computing ESM2 embeddings: 54%|      | 480/897 [00:13<00:11, 37.62it/s]

```

Computing ESM2 embeddings: 57%| | 512/897 [00:14<00:10, 37.03it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:15<00:09, 37.74it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:15<00:08, 36.86it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:16<00:07, 37.65it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:17<00:06, 37.03it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:18<00:05, 37.79it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:19<00:05, 36.98it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:20<00:04, 37.71it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:21<00:03, 37.15it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:21<00:02, 37.84it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:22<00:01, 37.26it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:23<00:00, 37.93it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:24<00:00, 36.43it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/915 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/915 [00:00<00:22, 38.55it/s]

Computing ESM2 embeddings: 7%| | 64/915 [00:01<00:23, 36.17it/s]

Computing ESM2 embeddings: 10%| | 96/915 [00:02<00:21, 37.61it/s]

Computing ESM2 embeddings: 14%| | 128/915 [00:03<00:21, 36.80it/s]

Computing ESM2 embeddings: 17%| | 160/915 [00:04<00:20, 37.49it/s]

Computing ESM2 embeddings: 21%| | 192/915 [00:05<00:19, 36.25it/s]

Computing ESM2 embeddings: 24%| | 224/915 [00:06<00:18, 37.27it/s]

Computing ESM2 embeddings: 28%| | 256/915 [00:06<00:17, 36.79it/s]

Computing ESM2 embeddings: 31%| | 288/915 [00:07<00:16, 37.62it/s]

Computing ESM2 embeddings: 35%| | 320/915 [00:08<00:16, 36.90it/s]

Computing ESM2 embeddings: 38%| | 352/915 [00:09<00:15, 37.40it/s]

Computing ESM2 embeddings: 42%| | 384/915 [00:10<00:14, 36.34it/s]

Computing ESM2 embeddings: 45%| | 416/915 [00:11<00:13, 37.07it/s]

Computing ESM2 embeddings: 49%| | 448/915 [00:12<00:12, 36.22it/s]

Computing ESM2 embeddings: 52%| | 480/915 [00:12<00:11, 37.16it/s]

Computing ESM2 embeddings: 56%| | 512/915 [00:13<00:10, 36.81it/s]

Computing ESM2 embeddings: 59%| | 544/915 [00:14<00:09, 37.50it/s]

Computing ESM2 embeddings: 63%| | 576/915 [00:15<00:09, 37.04it/s]

Computing ESM2 embeddings: 66%| | 608/915 [00:16<00:08, 37.57it/s]

Computing ESM2 embeddings: 70%| | 640/915 [00:17<00:07, 36.41it/s]

Computing ESM2 embeddings: 73%| | 672/915 [00:18<00:06, 37.31it/s]

Computing ESM2 embeddings: 77%| | 704/915 [00:19<00:05, 36.70it/s]

Computing ESM2 embeddings: 80%| | 736/915 [00:19<00:04, 37.49it/s]

Computing ESM2 embeddings: 84%| | 768/915 [00:20<00:03, 36.90it/s]

Computing ESM2 embeddings: 87%| | 800/915 [00:21<00:03, 37.74it/s]

Computing ESM2 embeddings: 91%| | 832/915 [00:22<00:02, 37.22it/s]

Computing ESM2 embeddings: 94%| | 864/915 [00:23<00:01, 37.79it/s]

Computing ESM2 embeddings: 98%| | 896/915 [00:24<00:00, 37.13it/s]

Computing ESM2 embeddings: 100%| | 915/915 [00:24<00:00, 37.09it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:22, 38.37it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:23, 35.36it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:21, 37.15it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:20, 36.56it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 37.14it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:19, 35.90it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:06<00:18, 36.97it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:17, 36.52it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:16, 37.19it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 36.17it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 36.90it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:14, 35.83it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 36.89it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:12<00:12, 36.56it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:13<00:11, 37.15it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:14<00:10, 36.21it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.08it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 36.65it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 37.46it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:06, 36.95it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:18<00:05, 37.66it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:19<00:05, 37.12it/s] |

```

Computing ESM2 embeddings: 82%|      | 736/895 [00:19<00:04, 37.77it/s]
Computing ESM2 embeddings: 86%|      | 768/895 [00:20<00:03, 36.95it/s]
Computing ESM2 embeddings: 89%|      | 800/895 [00:21<00:02, 37.39it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:22<00:01, 36.26it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:23<00:00, 37.18it/s]

Computing ESM2 embeddings: 100%|     | 895/895 [00:24<00:00, 36.82it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/916 [00:00<?, ?it/s]
Computing ESM2 embeddings:  3%|      | 32/916 [00:00<00:22, 38.60it/s]
Computing ESM2 embeddings:  7%|      | 64/916 [00:01<00:23, 36.04it/s]
Computing ESM2 embeddings: 10%|      | 96/916 [00:02<00:21, 37.55it/s]
Computing ESM2 embeddings: 14%|      | 128/916 [00:03<00:21, 36.89it/s]
Computing ESM2 embeddings: 17%|      | 160/916 [00:04<00:20, 37.50it/s]
Computing ESM2 embeddings: 21%|      | 192/916 [00:05<00:20, 36.08it/s]
Computing ESM2 embeddings: 24%|      | 224/916 [00:06<00:18, 37.11it/s]
Computing ESM2 embeddings: 28%|      | 256/916 [00:06<00:17, 36.70it/s]
Computing ESM2 embeddings: 31%|      | 288/916 [00:07<00:16, 37.45it/s]
Computing ESM2 embeddings: 35%|      | 320/916 [00:08<00:16, 36.84it/s]
Computing ESM2 embeddings: 38%|      | 352/916 [00:09<00:14, 37.66it/s]
Computing ESM2 embeddings: 42%|      | 384/916 [00:10<00:14, 37.09it/s]
Computing ESM2 embeddings: 45%|      | 416/916 [00:11<00:13, 37.66it/s]
Computing ESM2 embeddings: 49%|      | 448/916 [00:12<00:12, 36.42it/s]

```

```

Computing ESM2 embeddings: 52%|          | 480/916 [00:12<00:11, 37.22it/s]
Computing ESM2 embeddings: 56%|          | 512/916 [00:13<00:10, 36.85it/s]
Computing ESM2 embeddings: 59%|          | 544/916 [00:14<00:09, 37.61it/s]
Computing ESM2 embeddings: 63%|          | 576/916 [00:15<00:09, 37.09it/s]
Computing ESM2 embeddings: 66%|          | 608/916 [00:16<00:08, 37.77it/s]
Computing ESM2 embeddings: 70%|          | 640/916 [00:17<00:07, 37.28it/s]
Computing ESM2 embeddings: 73%|          | 672/916 [00:18<00:06, 37.92it/s]
Computing ESM2 embeddings: 77%|          | 704/916 [00:18<00:05, 37.30it/s]
Computing ESM2 embeddings: 80%|          | 736/916 [00:19<00:04, 37.70it/s]
Computing ESM2 embeddings: 84%|          | 768/916 [00:20<00:04, 36.54it/s]
Computing ESM2 embeddings: 87%|          | 800/916 [00:21<00:03, 37.33it/s]
Computing ESM2 embeddings: 91%|          | 832/916 [00:22<00:02, 36.92it/s]
Computing ESM2 embeddings: 94%|          | 864/916 [00:23<00:01, 37.66it/s]
Computing ESM2 embeddings: 98%|          | 896/916 [00:24<00:00, 37.12it/s]

```

```

Computing ESM2 embeddings: 100%|         | 916/916 [00:24<00:00, 37.08it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']

```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```

Computing ESM2 embeddings:  0%|          |  0/571 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|          | 32/571 [00:00<00:14, 36.61it/s]
Computing ESM2 embeddings: 11%|          | 64/571 [00:01<00:15, 32.72it/s]
Computing ESM2 embeddings: 17%|          | 96/571 [00:02<00:13, 35.55it/s]
Computing ESM2 embeddings: 22%|          | 128/571 [00:03<00:12, 35.46it/s]
Computing ESM2 embeddings: 28%|          | 160/571 [00:04<00:11, 36.48it/s]

```



Computing ESM2 embeddings: 34%| | 192/571 [00:05<00:10, 35.95it/s]

Computing ESM2 embeddings: 39%| | 224/571 [00:06<00:09, 37.03it/s]

Computing ESM2 embeddings: 45%| | 256/571 [00:07<00:08, 36.65it/s]

Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 37.45it/s]

Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 36.82it/s]

Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 37.64it/s]

Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:05, 36.85it/s]

Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 37.59it/s]

Computing ESM2 embeddings: 78%| | 448/571 [00:12<00:03, 37.07it/s]

Computing ESM2 embeddings: 84%| | 480/571 [00:13<00:02, 37.79it/s]

Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 37.15it/s]

Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 37.59it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 36.58it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 36.95it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 34.73it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.99it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.52it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.31it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:20, 36.08it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.18it/s]

```

Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 36.81it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:16, 37.60it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:16, 37.05it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 37.62it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 36.35it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.03it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:12<00:12, 36.08it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:13<00:11, 37.04it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:13<00:10, 36.67it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 37.53it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.01it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.75it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.22it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.90it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 37.23it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.90it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.12it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.81it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.21it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.91it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.34it/s]

```

```

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.08it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it

```

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:23, 37.01it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:23, 35.21it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 36.99it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 36.44it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 37.43it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.74it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:17, 37.68it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 37.02it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.76it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.21it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.96it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.26it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.93it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.13it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:11, 37.77it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.17it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.88it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.07it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 37.80it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.11it/s] |
| Computing ESM2 embeddings: | 75% | 672/896 [00:18<00:05, 37.53it/s] |
| Computing ESM2 embeddings: | 79% | 704/896 [00:18<00:05, 36.33it/s] |

```

Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.26it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 36.76it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.58it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 36.86it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.39it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.03it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.89it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:22, 37.06it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.80it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.36it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.45it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.89it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.41it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:18, 36.28it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.32it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.90it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 37.72it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 37.09it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.86it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 37.24it/s]

```

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.91it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.28it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.98it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.32it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.67it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.47it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.40it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 36.91it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.60it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.06it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.80it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.04it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.73it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.20it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.31it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/897 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/897 [00:01<00:28, 30.60it/s]

Computing ESM2 embeddings: 7%| | 64/897 [00:01<00:25, 32.47it/s]

Computing ESM2 embeddings: 11%| | 96/897 [00:02<00:22, 35.01it/s]

Computing ESM2 embeddings: 14%| | 128/897 [00:03<00:22, 34.69it/s]

Computing ESM2 embeddings: 18%| | 160/897 [00:04<00:20, 36.28it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/897 [00:05<00:19, 36.18it/s] |
| Computing ESM2 embeddings: 25%  | 224/897 [00:06<00:18, 36.86it/s] |
| Computing ESM2 embeddings: 29%  | 256/897 [00:07<00:17, 35.85it/s] |
| Computing ESM2 embeddings: 32%  | 288/897 [00:08<00:16, 36.87it/s] |
| Computing ESM2 embeddings: 36%  | 320/897 [00:08<00:15, 36.56it/s] |
| Computing ESM2 embeddings: 39%  | 352/897 [00:09<00:14, 37.51it/s] |
| Computing ESM2 embeddings: 43%  | 384/897 [00:10<00:13, 36.92it/s] |
| Computing ESM2 embeddings: 46%  | 416/897 [00:11<00:12, 37.69it/s] |
| Computing ESM2 embeddings: 50%  | 448/897 [00:12<00:12, 37.13it/s] |
| Computing ESM2 embeddings: 54%  | 480/897 [00:13<00:11, 37.76it/s] |
| Computing ESM2 embeddings: 57%  | 512/897 [00:14<00:10, 37.07it/s] |
| Computing ESM2 embeddings: 61%  | 544/897 [00:14<00:09, 37.75it/s] |
| Computing ESM2 embeddings: 64%  | 576/897 [00:15<00:08, 36.90it/s] |
| Computing ESM2 embeddings: 68%  | 608/897 [00:16<00:07, 37.65it/s] |
| Computing ESM2 embeddings: 71%  | 640/897 [00:17<00:06, 37.16it/s] |
| Computing ESM2 embeddings: 75%  | 672/897 [00:18<00:05, 37.75it/s] |
| Computing ESM2 embeddings: 78%  | 704/897 [00:19<00:05, 37.10it/s] |
| Computing ESM2 embeddings: 82%  | 736/897 [00:20<00:04, 37.71it/s] |
| Computing ESM2 embeddings: 86%  | 768/897 [00:20<00:03, 37.04it/s] |
| Computing ESM2 embeddings: 89%  | 800/897 [00:21<00:02, 37.75it/s] |
| Computing ESM2 embeddings: 93%  | 832/897 [00:22<00:01, 37.16it/s] |
| Computing ESM2 embeddings: 96%  | 864/897 [00:23<00:00, 37.77it/s] |
| Computing ESM2 embeddings: 100% | 897/897 [00:24<00:00, 36.77it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.63it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:02<00:27, 30.87it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:24, 33.90it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:22, 35.53it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 36.94it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.46it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 37.16it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:07<00:18, 35.99it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.04it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.68it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 37.22it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 36.27it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.20it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 36.76it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:13<00:11, 37.34it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:14<00:11, 36.22it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:10, 36.74it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 35.74it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 36.57it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 35.77it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 36.84it/s] |

```

Computing ESM2 embeddings: 77%|      | 704/914 [00:19<00:05, 36.39it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:20<00:04, 37.27it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:21<00:03, 36.88it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.61it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.06it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.78it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.22it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.61it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|       | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|       | 32/896 [00:01<00:28, 30.61it/s]
Computing ESM2 embeddings: 7%|       | 64/896 [00:01<00:24, 33.97it/s]
Computing ESM2 embeddings: 11%|      | 96/896 [00:02<00:22, 36.08it/s]
Computing ESM2 embeddings: 14%|      | 128/896 [00:03<00:21, 35.30it/s]
Computing ESM2 embeddings: 18%|      | 160/896 [00:04<00:20, 36.45it/s]
Computing ESM2 embeddings: 21%|      | 192/896 [00:05<00:19, 35.60it/s]
Computing ESM2 embeddings: 25%|      | 224/896 [00:06<00:18, 36.49it/s]
Computing ESM2 embeddings: 29%|      | 256/896 [00:07<00:17, 35.68it/s]
Computing ESM2 embeddings: 32%|      | 288/896 [00:08<00:16, 36.85it/s]
Computing ESM2 embeddings: 36%|      | 320/896 [00:08<00:15, 36.57it/s]
Computing ESM2 embeddings: 39%|      | 352/896 [00:09<00:14, 37.22it/s]
Computing ESM2 embeddings: 43%|      | 384/896 [00:10<00:14, 36.16it/s]

```



Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.13it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.76it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.42it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 36.81it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.58it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.91it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.62it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 36.88it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.61it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 36.91it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.64it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.05it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.56it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.49it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.14it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.49it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/916 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/916 [00:00<00:23, 37.86it/s]

Computing ESM2 embeddings: 7%| | 64/916 [00:01<00:23, 35.86it/s]

Computing ESM2 embeddings: 10%| | 96/916 [00:02<00:21, 37.36it/s]

Computing ESM2 embeddings: 14%| | 128/916 [00:03<00:21, 36.47it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 17% | 160/916 [00:04<00:20, 37.57it/s] |
| Computing ESM2 embeddings: 21% | 192/916 [00:05<00:19, 36.98it/s] |
| Computing ESM2 embeddings: 24% | 224/916 [00:05<00:18, 37.82it/s] |
| Computing ESM2 embeddings: 28% | 256/916 [00:06<00:17, 37.20it/s] |
| Computing ESM2 embeddings: 31% | 288/916 [00:07<00:16, 37.95it/s] |
| Computing ESM2 embeddings: 35% | 320/916 [00:08<00:16, 37.23it/s] |
| Computing ESM2 embeddings: 38% | 352/916 [00:09<00:14, 37.93it/s] |
| Computing ESM2 embeddings: 42% | 384/916 [00:10<00:14, 37.07it/s] |
| Computing ESM2 embeddings: 45% | 416/916 [00:11<00:13, 37.59it/s] |
| Computing ESM2 embeddings: 49% | 448/916 [00:12<00:12, 36.46it/s] |
| Computing ESM2 embeddings: 52% | 480/916 [00:12<00:11, 37.31it/s] |
| Computing ESM2 embeddings: 56% | 512/916 [00:13<00:10, 36.83it/s] |
| Computing ESM2 embeddings: 59% | 544/916 [00:14<00:09, 37.63it/s] |
| Computing ESM2 embeddings: 63% | 576/916 [00:15<00:09, 37.13it/s] |
| Computing ESM2 embeddings: 66% | 608/916 [00:16<00:08, 37.80it/s] |
| Computing ESM2 embeddings: 70% | 640/916 [00:17<00:07, 37.04it/s] |
| Computing ESM2 embeddings: 73% | 672/916 [00:18<00:06, 37.70it/s] |
| Computing ESM2 embeddings: 77% | 704/916 [00:18<00:05, 36.98it/s] |
| Computing ESM2 embeddings: 80% | 736/916 [00:19<00:04, 37.61it/s] |
| Computing ESM2 embeddings: 84% | 768/916 [00:20<00:03, 37.10it/s] |
| Computing ESM2 embeddings: 87% | 800/916 [00:21<00:03, 37.75it/s] |
| Computing ESM2 embeddings: 91% | 832/916 [00:22<00:02, 37.09it/s] |
| Computing ESM2 embeddings: 94% | 864/916 [00:23<00:01, 37.72it/s] |
| Computing ESM2 embeddings: 98% | 896/916 [00:24<00:00, 36.99it/s] |

Computing ESM2 embeddings: 100%| | 916/916 [00:24<00:00, 37.27it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/571 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 6%| | 32/571 [00:00<00:14, 38.12it/s]  
 Computing ESM2 embeddings: 11%| | 64/571 [00:01<00:14, 35.78it/s]  
 Computing ESM2 embeddings: 17%| | 96/571 [00:02<00:12, 37.23it/s]  
 Computing ESM2 embeddings: 22%| | 128/571 [00:03<00:12, 36.07it/s]  
 Computing ESM2 embeddings: 28%| | 160/571 [00:04<00:11, 37.00it/s]  
 Computing ESM2 embeddings: 34%| | 192/571 [00:05<00:10, 35.93it/s]  
 Computing ESM2 embeddings: 39%| | 224/571 [00:06<00:09, 36.95it/s]  
 Computing ESM2 embeddings: 45%| | 256/571 [00:06<00:08, 36.65it/s]  
 Computing ESM2 embeddings: 50%| | 288/571 [00:07<00:07, 37.43it/s]  
 Computing ESM2 embeddings: 56%| | 320/571 [00:08<00:06, 37.03it/s]  
 Computing ESM2 embeddings: 62%| | 352/571 [00:09<00:05, 37.43it/s]  
 Computing ESM2 embeddings: 67%| | 384/571 [00:10<00:05, 36.22it/s]  
 Computing ESM2 embeddings: 73%| | 416/571 [00:11<00:04, 37.22it/s]  
 Computing ESM2 embeddings: 78%| | 448/571 [00:12<00:03, 36.89it/s]  
 Computing ESM2 embeddings: 84%| | 480/571 [00:12<00:02, 37.76it/s]  
 Computing ESM2 embeddings: 90%| | 512/571 [00:13<00:01, 37.17it/s]  
 Computing ESM2 embeddings: 95%| | 544/571 [00:14<00:00, 37.60it/s]

Computing ESM2 embeddings: 100%| | 571/571 [00:15<00:00, 36.77it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.98it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:24, 34.34it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:22, 36.07it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:22, 35.32it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 36.36it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:20, 35.41it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 36.44it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:07<00:18, 35.46it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:17, 36.61it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.38it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 37.31it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 36.87it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 37.67it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 36.84it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:13<00:11, 37.56it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:14<00:10, 36.93it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.63it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 37.00it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.74it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.14it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 37.79it/s] |

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 37.03it/s]  
Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.70it/s]  
Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.03it/s]  
Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.77it/s]  
Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.10it/s]  
Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.56it/s]  
Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.29it/s]  
Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.78it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.14it/s]  
Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.08it/s]  
Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.74it/s]  
Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.35it/s]  
Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.51it/s]  
Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.94it/s]  
Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.39it/s]  
Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:18, 36.30it/s]  
Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.24it/s]  
Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.81it/s]  
Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.64it/s]  
Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.90it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.67it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.16it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.74it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:11, 36.54it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.46it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.90it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.48it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.41it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.34it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 36.81it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.39it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:04, 36.31it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.03it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.04it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.08it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.66it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.03it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.68it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 36.52it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.50it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:21, 36.17it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.43it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 36.88it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:17, 37.36it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:17, 36.29it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.27it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 36.88it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.65it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 37.12it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.75it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 37.23it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 37.93it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.30it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 38.00it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 37.24it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.65it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:07, 36.50it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:05, 37.42it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 36.90it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.68it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.11it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.89it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.39it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 38.05it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.21it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:28, 31.47it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:02<00:29, 28.48it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:03<00:25, 31.98it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:04<00:24, 32.50it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:22, 34.27it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:21, 34.15it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:19, 35.01it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:19, 34.61it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:08<00:17, 35.91it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:09<00:16, 35.73it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:10<00:15, 36.80it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:11<00:14, 36.44it/s]  
 Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.21it/s]  
 Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.65it/s]  
 Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.29it/s]  
 Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 36.74it/s]  
 Computing ESM2 embeddings: 60%| | 544/914 [00:15<00:09, 37.58it/s]  
 Computing ESM2 embeddings: 63%| | 576/914 [00:16<00:09, 36.93it/s]  
 Computing ESM2 embeddings: 67%| | 608/914 [00:17<00:08, 37.59it/s]



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Computing ESM2 embeddings: 70%|      | 640/914 [00:17<00:07, 36.93it/s]
Computing ESM2 embeddings: 74%|      | 672/914 [00:18<00:06, 37.44it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:19<00:05, 36.20it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:20<00:04, 36.89it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:21<00:04, 35.96it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:22<00:03, 36.96it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:23<00:02, 36.52it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:24<00:01, 37.12it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.10it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:25<00:00, 35.87it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/897 [00:01<00:31, 27.32it/s]
Computing ESM2 embeddings:  7%|      | 64/897 [00:02<00:26, 30.93it/s]
Computing ESM2 embeddings: 11%|      | 96/897 [00:02<00:23, 34.02it/s]
Computing ESM2 embeddings: 14%|      | 128/897 [00:03<00:22, 34.11it/s]
Computing ESM2 embeddings: 18%|      | 160/897 [00:04<00:20, 35.72it/s]
Computing ESM2 embeddings: 21%|      | 192/897 [00:05<00:19, 35.66it/s]
Computing ESM2 embeddings: 25%|      | 224/897 [00:06<00:18, 36.75it/s]
Computing ESM2 embeddings: 29%|      | 256/897 [00:07<00:17, 36.22it/s]
Computing ESM2 embeddings: 32%|      | 288/897 [00:08<00:16, 37.13it/s]
Computing ESM2 embeddings: 36%|      | 320/897 [00:09<00:15, 36.76it/s]

```

Computing ESM2 embeddings: 39%| | 352/897 [00:09<00:14, 37.59it/s]

Computing ESM2 embeddings: 43%| | 384/897 [00:10<00:13, 37.05it/s]

Computing ESM2 embeddings: 46%| | 416/897 [00:11<00:12, 37.79it/s]

Computing ESM2 embeddings: 50%| | 448/897 [00:12<00:12, 37.04it/s]

Computing ESM2 embeddings: 54%| | 480/897 [00:13<00:11, 37.79it/s]

Computing ESM2 embeddings: 57%| | 512/897 [00:14<00:10, 37.10it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:14<00:09, 37.79it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:15<00:08, 37.10it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:16<00:07, 37.81it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:17<00:06, 37.20it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:18<00:05, 37.82it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:19<00:05, 37.08it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:20<00:04, 37.76it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:20<00:03, 37.08it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:21<00:02, 37.52it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:22<00:01, 36.33it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:23<00:00, 37.26it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:24<00:00, 36.51it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.28it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 35.30it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:22, 36.71it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:22, 35.71it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 37.04it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.64it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 37.29it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:18, 36.21it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.22it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.70it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 37.58it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.96it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.76it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.18it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:12<00:11, 37.89it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 37.25it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.71it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 36.56it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.18it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 36.14it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.10it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 36.72it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.51it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:03, 37.07it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 37.82it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:22<00:02, 37.15it/s] |

```

Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.85it/s]

Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.28it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.96it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/872 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/872 [00:00<00:22, 37.36it/s]

Computing ESM2 embeddings:  7%|      | 64/872 [00:01<00:23, 34.05it/s]

Computing ESM2 embeddings: 11%|      | 96/872 [00:02<00:21, 35.35it/s]

Computing ESM2 embeddings: 15%|      | 128/872 [00:03<00:20, 36.34it/s]

Computing ESM2 embeddings: 18%|      | 160/872 [00:04<00:18, 37.61it/s]

Computing ESM2 embeddings: 22%|      | 192/872 [00:05<00:18, 37.04it/s]

Computing ESM2 embeddings: 26%|      | 224/872 [00:06<00:17, 37.74it/s]

Computing ESM2 embeddings: 29%|      | 256/872 [00:06<00:16, 37.02it/s]

Computing ESM2 embeddings: 33%|      | 288/872 [00:07<00:15, 37.57it/s]

Computing ESM2 embeddings: 37%|      | 320/872 [00:08<00:15, 36.38it/s]

Computing ESM2 embeddings: 40%|      | 352/872 [00:09<00:13, 37.32it/s]

Computing ESM2 embeddings: 44%|      | 384/872 [00:10<00:13, 36.80it/s]

Computing ESM2 embeddings: 48%|      | 416/872 [00:11<00:12, 37.37it/s]

Computing ESM2 embeddings: 51%|      | 448/872 [00:12<00:11, 36.18it/s]

Computing ESM2 embeddings: 55%|      | 480/872 [00:13<00:10, 37.08it/s]

Computing ESM2 embeddings: 59%|      | 512/872 [00:13<00:09, 36.61it/s]

Computing ESM2 embeddings: 62%|      | 544/872 [00:14<00:08, 37.47it/s]

```

Computing ESM2 embeddings: 66%| | 576/872 [00:15<00:08, 36.93it/s]

Computing ESM2 embeddings: 70%| | 608/872 [00:16<00:07, 37.70it/s]

Computing ESM2 embeddings: 73%| | 640/872 [00:17<00:06, 37.03it/s]

Computing ESM2 embeddings: 77%| | 672/872 [00:18<00:05, 37.68it/s]

Computing ESM2 embeddings: 81%| | 704/872 [00:19<00:04, 37.25it/s]

Computing ESM2 embeddings: 84%| | 736/872 [00:19<00:03, 37.91it/s]

Computing ESM2 embeddings: 88%| | 768/872 [00:20<00:02, 37.29it/s]

Computing ESM2 embeddings: 92%| | 800/872 [00:21<00:01, 37.94it/s]

Computing ESM2 embeddings: 95%| | 832/872 [00:22<00:01, 37.27it/s]

Computing ESM2 embeddings: 99%| | 864/872 [00:23<00:00, 37.88it/s]

Computing ESM2 embeddings: 100%| | 872/872 [00:23<00:00, 37.08it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/615 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/615 [00:00<00:14, 39.11it/s]

Computing ESM2 embeddings: 10%| | 64/615 [00:01<00:14, 37.12it/s]

Computing ESM2 embeddings: 16%| | 96/615 [00:02<00:13, 38.05it/s]

Computing ESM2 embeddings: 21%| | 128/615 [00:03<00:13, 36.96it/s]

Computing ESM2 embeddings: 26%| | 160/615 [00:04<00:12, 37.63it/s]

Computing ESM2 embeddings: 31%| | 192/615 [00:05<00:11, 36.29it/s]

Computing ESM2 embeddings: 36%| | 224/615 [00:06<00:10, 37.34it/s]

Computing ESM2 embeddings: 42%| | 256/615 [00:06<00:09, 36.78it/s]

Computing ESM2 embeddings: 47%| | 288/615 [00:07<00:08, 37.59it/s]

Computing ESM2 embeddings: 52%| | 320/615 [00:08<00:07, 36.88it/s]  
 Computing ESM2 embeddings: 57%| | 352/615 [00:09<00:07, 37.37it/s]  
 Computing ESM2 embeddings: 62%| | 384/615 [00:10<00:06, 35.95it/s]  
 Computing ESM2 embeddings: 68%| | 416/615 [00:11<00:05, 36.83it/s]  
 Computing ESM2 embeddings: 73%| | 448/615 [00:12<00:04, 36.55it/s]  
 Computing ESM2 embeddings: 78%| | 480/615 [00:12<00:03, 37.37it/s]  
 Computing ESM2 embeddings: 83%| | 512/615 [00:13<00:02, 36.90it/s]  
 Computing ESM2 embeddings: 88%| | 544/615 [00:14<00:01, 37.63it/s]  
 Computing ESM2 embeddings: 94%| | 576/615 [00:15<00:01, 37.05it/s]  
 Computing ESM2 embeddings: 99%| | 608/615 [00:16<00:00, 37.69it/s]

Computing ESM2 embeddings: 100%| | 615/615 [00:16<00:00, 37.03it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/877 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/877 [00:00<00:24, 33.88it/s]  
 Computing ESM2 embeddings: 7%| | 64/877 [00:02<00:26, 30.61it/s]  
 Computing ESM2 embeddings: 11%| | 96/877 [00:02<00:22, 34.08it/s]  
 Computing ESM2 embeddings: 15%| | 128/877 [00:03<00:21, 34.89it/s]  
 Computing ESM2 embeddings: 18%| | 160/877 [00:04<00:19, 36.54it/s]  
 Computing ESM2 embeddings: 22%| | 192/877 [00:05<00:18, 36.28it/s]  
 Computing ESM2 embeddings: 26%| | 224/877 [00:06<00:17, 37.21it/s]  
 Computing ESM2 embeddings: 29%| | 256/877 [00:07<00:16, 36.73it/s]  
 Computing ESM2 embeddings: 33%| | 288/877 [00:07<00:15, 37.59it/s]

Computing ESM2 embeddings: 36%| | 320/877 [00:08<00:15, 36.97it/s]

Computing ESM2 embeddings: 40%| | 352/877 [00:09<00:13, 37.59it/s]

Computing ESM2 embeddings: 44%| | 384/877 [00:10<00:13, 36.96it/s]

Computing ESM2 embeddings: 47%| | 416/877 [00:11<00:12, 37.40it/s]

Computing ESM2 embeddings: 51%| | 448/877 [00:12<00:11, 36.31it/s]

Computing ESM2 embeddings: 55%| | 480/877 [00:13<00:10, 37.22it/s]

Computing ESM2 embeddings: 58%| | 512/877 [00:14<00:09, 36.81it/s]

Computing ESM2 embeddings: 62%| | 544/877 [00:14<00:08, 37.64it/s]

Computing ESM2 embeddings: 66%| | 576/877 [00:15<00:08, 37.13it/s]

Computing ESM2 embeddings: 69%| | 608/877 [00:16<00:07, 37.79it/s]

Computing ESM2 embeddings: 73%| | 640/877 [00:17<00:06, 37.05it/s]

Computing ESM2 embeddings: 77%| | 672/877 [00:18<00:05, 37.62it/s]

Computing ESM2 embeddings: 80%| | 704/877 [00:19<00:04, 37.10it/s]

Computing ESM2 embeddings: 84%| | 736/877 [00:20<00:03, 37.62it/s]

Computing ESM2 embeddings: 88%| | 768/877 [00:20<00:02, 37.10it/s]

Computing ESM2 embeddings: 91%| | 800/877 [00:21<00:02, 37.76it/s]

Computing ESM2 embeddings: 95%| | 832/877 [00:22<00:01, 37.19it/s]

Computing ESM2 embeddings: 99%| | 864/877 [00:23<00:00, 37.83it/s]

Computing ESM2 embeddings: 100%| | 877/877 [00:23<00:00, 36.84it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:27, 32.12it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/914 [00:01<00:25, 32.83it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:02<00:22, 35.66it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:22, 35.67it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 37.05it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:19, 36.59it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 37.56it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:17, 36.91it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.71it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.98it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:14, 37.79it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 37.12it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.58it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 36.33it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 37.22it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:13<00:10, 36.66it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:09, 37.45it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 36.92it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.63it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.03it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.63it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 36.89it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:19<00:04, 37.33it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:20<00:04, 36.15it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:21<00:03, 36.83it/s] |



```

Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 35.76it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 36.75it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 36.47it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.76it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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Computing ESM2 embeddings: 0%|        | 0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|        | 32/896 [00:01<00:27, 31.80it/s]
Computing ESM2 embeddings: 7%|        | 64/896 [00:01<00:25, 32.99it/s]
Computing ESM2 embeddings: 11%|       | 96/896 [00:02<00:22, 35.72it/s]
Computing ESM2 embeddings: 14%|       | 128/896 [00:03<00:21, 35.81it/s]
Computing ESM2 embeddings: 18%|       | 160/896 [00:04<00:19, 37.09it/s]
Computing ESM2 embeddings: 21%|       | 192/896 [00:05<00:19, 36.64it/s]
Computing ESM2 embeddings: 25%|       | 224/896 [00:06<00:17, 37.57it/s]
Computing ESM2 embeddings: 29%|       | 256/896 [00:07<00:17, 36.99it/s]
Computing ESM2 embeddings: 32%|       | 288/896 [00:07<00:16, 37.52it/s]
Computing ESM2 embeddings: 36%|       | 320/896 [00:08<00:15, 36.39it/s]
Computing ESM2 embeddings: 39%|       | 352/896 [00:09<00:14, 37.21it/s]
Computing ESM2 embeddings: 43%|       | 384/896 [00:10<00:13, 36.64it/s]
Computing ESM2 embeddings: 46%|       | 416/896 [00:11<00:12, 37.28it/s]
Computing ESM2 embeddings: 50%|       | 448/896 [00:12<00:12, 36.01it/s]
Computing ESM2 embeddings: 54%|       | 480/896 [00:13<00:11, 36.83it/s]
Computing ESM2 embeddings: 57%|       | 512/896 [00:14<00:10, 36.21it/s]

```

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.08it/s]  
 Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.65it/s]  
 Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.41it/s]  
 Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 36.97it/s]  
 Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.64it/s]  
 Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 37.00it/s]  
 Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.36it/s]  
 Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 36.15it/s]  
 Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 36.88it/s]  
 Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.03it/s]  
 Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.01it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.56it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:27, 31.57it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:02<00:27, 30.44it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:24, 34.04it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 34.79it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 36.12it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:20, 35.40it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 36.53it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 35.75it/s]

```

Computing ESM2 embeddings: 32%|          | 288/914 [00:08<00:16, 36.84it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:09<00:16, 36.51it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:14, 37.50it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 36.97it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.74it/s]
Computing ESM2 embeddings: 49%|          | 448/914 [00:12<00:12, 37.10it/s]
Computing ESM2 embeddings: 53%|          | 480/914 [00:13<00:11, 37.84it/s]
Computing ESM2 embeddings: 56%|          | 512/914 [00:14<00:10, 37.21it/s]
Computing ESM2 embeddings: 60%|          | 544/914 [00:14<00:09, 37.73it/s]
Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.15it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.84it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 37.21it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.61it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 36.49it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:20<00:04, 37.36it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 36.95it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.73it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.11it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.82it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.25it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 36.65it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/895 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/895 [00:00<00:23, 37.47it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/895 [00:02<00:28, 29.57it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:24, 32.80it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:21, 35.00it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:20, 36.67it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:19, 36.14it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:06<00:18, 37.13it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:07<00:17, 36.66it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:16, 37.44it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 36.97it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 37.69it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:13, 37.01it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 37.40it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:12<00:12, 36.13it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:13<00:11, 37.07it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:14<00:10, 36.57it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.35it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 36.69it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 37.50it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:06, 37.03it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:18<00:05, 37.75it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:19<00:05, 37.07it/s] |
| Computing ESM2 embeddings: | 82% | 736/895 [00:20<00:04, 37.76it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/895 [00:20<00:03, 37.04it/s]
Computing ESM2 embeddings: 89%|      | 800/895 [00:21<00:02, 37.80it/s]
Computing ESM2 embeddings: 93%|      | 832/895 [00:22<00:01, 37.19it/s]
Computing ESM2 embeddings: 97%|      | 864/895 [00:23<00:00, 37.88it/s]
Computing ESM2 embeddings: 100%|     | 895/895 [00:24<00:00, 36.69it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.99it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:22, 36.97it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.75it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.20it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.39it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.74it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.28it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:18, 36.11it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.01it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.62it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.40it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 36.93it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.72it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 37.15it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.87it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.21it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.91it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.18it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.57it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.33it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.26it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 36.63it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.47it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.93it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.70it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.12it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.80it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.27it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.20it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/573 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/573 [00:00<00:14, 38.47it/s]

Computing ESM2 embeddings: 11%| | 64/573 [00:01<00:13, 37.12it/s]

Computing ESM2 embeddings: 17%| | 96/573 [00:02<00:12, 37.89it/s]

Computing ESM2 embeddings: 22%| | 128/573 [00:03<00:12, 36.19it/s]

Computing ESM2 embeddings: 28%| | 160/573 [00:04<00:11, 37.30it/s]

Computing ESM2 embeddings: 34%| | 192/573 [00:05<00:10, 36.91it/s]

Computing ESM2 embeddings: 39%| | 224/573 [00:06<00:09, 37.43it/s]

Computing ESM2 embeddings: 45%| | 256/573 [00:06<00:08, 36.16it/s]

Computing ESM2 embeddings: 50%| | 288/573 [00:07<00:07, 37.22it/s]

Computing ESM2 embeddings: 56%| | 320/573 [00:08<00:06, 36.71it/s]

Computing ESM2 embeddings: 61%| | 352/573 [00:09<00:05, 37.40it/s]

Computing ESM2 embeddings: 67%| | 384/573 [00:10<00:05, 36.87it/s]

Computing ESM2 embeddings: 73%| | 416/573 [00:11<00:04, 37.39it/s]

Computing ESM2 embeddings: 78%| | 448/573 [00:12<00:03, 36.88it/s]

Computing ESM2 embeddings: 84%| | 480/573 [00:12<00:02, 37.63it/s]

Computing ESM2 embeddings: 89%| | 512/573 [00:13<00:01, 37.14it/s]

Computing ESM2 embeddings: 95%| | 544/573 [00:14<00:00, 37.81it/s]

Computing ESM2 embeddings: 100%| | 573/573 [00:15<00:00, 37.08it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:26, 32.64it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:25, 33.28it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:23, 35.50it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:22, 34.83it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:20, 36.31it/s]

Computing ESM2 embeddings: 21%| | 192/913 [00:05<00:19, 36.06it/s]

Computing ESM2 embeddings: 25%| | 224/913 [00:06<00:18, 36.81it/s]

Computing ESM2 embeddings: 28%| | 256/913 [00:07<00:18, 35.92it/s]

Computing ESM2 embeddings: 32%| | 288/913 [00:08<00:16, 36.77it/s]

Computing ESM2 embeddings: 35%| | 320/913 [00:08<00:16, 35.91it/s]

Computing ESM2 embeddings: 39%| | 352/913 [00:09<00:15, 36.97it/s]

Computing ESM2 embeddings: 42%| | 384/913 [00:10<00:14, 36.55it/s]

Computing ESM2 embeddings: 46%| | 416/913 [00:11<00:13, 37.43it/s]

Computing ESM2 embeddings: 49%| | 448/913 [00:12<00:12, 36.95it/s]

Computing ESM2 embeddings: 53%| | 480/913 [00:13<00:11, 37.70it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:14<00:10, 37.15it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 37.77it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:15<00:09, 37.05it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:16<00:08, 37.48it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:17<00:07, 36.28it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:18<00:06, 37.12it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:19<00:05, 36.77it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:20<00:04, 37.55it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:20<00:03, 37.03it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:21<00:02, 37.75it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 37.18it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:23<00:01, 37.76it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:24<00:00, 37.13it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:24<00:00, 36.75it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.



|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/897 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/897 [00:00<00:24, 35.54it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/897 [00:01<00:24, 34.71it/s]  |
| Computing ESM2 embeddings: | 11% | 96/897 [00:02<00:22, 36.34it/s]  |
| Computing ESM2 embeddings: | 14% | 128/897 [00:03<00:21, 35.61it/s] |
| Computing ESM2 embeddings: | 18% | 160/897 [00:04<00:20, 36.70it/s] |
| Computing ESM2 embeddings: | 21% | 192/897 [00:05<00:19, 35.74it/s] |
| Computing ESM2 embeddings: | 25% | 224/897 [00:06<00:18, 36.63it/s] |
| Computing ESM2 embeddings: | 29% | 256/897 [00:07<00:17, 35.81it/s] |
| Computing ESM2 embeddings: | 32% | 288/897 [00:07<00:16, 36.87it/s] |
| Computing ESM2 embeddings: | 36% | 320/897 [00:08<00:15, 36.57it/s] |
| Computing ESM2 embeddings: | 39% | 352/897 [00:09<00:14, 37.16it/s] |
| Computing ESM2 embeddings: | 43% | 384/897 [00:10<00:14, 36.10it/s] |
| Computing ESM2 embeddings: | 46% | 416/897 [00:11<00:13, 36.85it/s] |
| Computing ESM2 embeddings: | 50% | 448/897 [00:12<00:12, 35.91it/s] |
| Computing ESM2 embeddings: | 54% | 480/897 [00:13<00:11, 36.95it/s] |
| Computing ESM2 embeddings: | 57% | 512/897 [00:14<00:10, 36.57it/s] |
| Computing ESM2 embeddings: | 61% | 544/897 [00:14<00:09, 37.45it/s] |
| Computing ESM2 embeddings: | 64% | 576/897 [00:15<00:08, 36.76it/s] |
| Computing ESM2 embeddings: | 68% | 608/897 [00:16<00:07, 37.57it/s] |
| Computing ESM2 embeddings: | 71% | 640/897 [00:17<00:06, 37.02it/s] |
| Computing ESM2 embeddings: | 75% | 672/897 [00:18<00:05, 37.77it/s] |
| Computing ESM2 embeddings: | 78% | 704/897 [00:19<00:05, 37.07it/s] |
| Computing ESM2 embeddings: | 82% | 736/897 [00:20<00:04, 37.80it/s] |

```

Computing ESM2 embeddings: 86%|      | 768/897 [00:20<00:03, 37.18it/s]
Computing ESM2 embeddings: 89%|      | 800/897 [00:21<00:02, 37.87it/s]
Computing ESM2 embeddings: 93%|      | 832/897 [00:22<00:01, 37.18it/s]
Computing ESM2 embeddings: 96%|      | 864/897 [00:23<00:00, 37.88it/s]
Computing ESM2 embeddings: 100%|     | 897/897 [00:24<00:00, 36.73it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:22, 38.46it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 36.62it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.44it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 35.87it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.02it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.63it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.21it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:18, 36.18it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.22it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.80it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 37.64it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 37.13it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.76it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 37.19it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.92it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.28it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.93it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.34it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.77it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.11it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.78it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.06it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.55it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:04, 36.37it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.30it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.85it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.64it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.23it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.21it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:01<00:30, 28.27it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:02<00:32, 25.99it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:03<00:26, 30.32it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:04<00:23, 33.17it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:21, 34.97it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:20, 35.07it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:18, 35.88it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:07<00:18, 35.21it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:08<00:16, 36.45it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:09<00:15, 36.20it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:10<00:14, 37.14it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:11<00:13, 36.66it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.47it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 37.01it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:13<00:11, 37.71it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:14<00:10, 37.18it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:15<00:09, 37.85it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:16<00:08, 37.24it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.61it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:07, 36.49it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:05, 37.37it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:19<00:05, 36.79it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:20<00:04, 37.53it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:21<00:03, 36.97it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:22<00:02, 37.68it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:23<00:01, 37.02it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 37.74it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.11it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/913 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/913 [00:00<00:22, 38.61it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/913 [00:01<00:23, 35.89it/s]  |
| Computing ESM2 embeddings: | 11% | 96/913 [00:02<00:21, 37.41it/s]  |
| Computing ESM2 embeddings: | 14% | 128/913 [00:03<00:21, 36.82it/s] |
| Computing ESM2 embeddings: | 18% | 160/913 [00:04<00:20, 37.40it/s] |
| Computing ESM2 embeddings: | 21% | 192/913 [00:05<00:19, 36.18it/s] |
| Computing ESM2 embeddings: | 25% | 224/913 [00:06<00:18, 37.09it/s] |
| Computing ESM2 embeddings: | 28% | 256/913 [00:06<00:17, 36.51it/s] |
| Computing ESM2 embeddings: | 32% | 288/913 [00:07<00:16, 37.12it/s] |
| Computing ESM2 embeddings: | 35% | 320/913 [00:08<00:16, 35.84it/s] |
| Computing ESM2 embeddings: | 39% | 352/913 [00:09<00:15, 36.95it/s] |
| Computing ESM2 embeddings: | 42% | 384/913 [00:10<00:14, 36.60it/s] |
| Computing ESM2 embeddings: | 46% | 416/913 [00:11<00:13, 37.47it/s] |
| Computing ESM2 embeddings: | 49% | 448/913 [00:12<00:12, 36.95it/s] |
| Computing ESM2 embeddings: | 53% | 480/913 [00:12<00:11, 37.45it/s] |
| Computing ESM2 embeddings: | 56% | 512/913 [00:13<00:11, 36.23it/s] |
| Computing ESM2 embeddings: | 60% | 544/913 [00:14<00:09, 37.07it/s] |
| Computing ESM2 embeddings: | 63% | 576/913 [00:15<00:09, 36.60it/s] |
| Computing ESM2 embeddings: | 67% | 608/913 [00:16<00:08, 37.44it/s] |
| Computing ESM2 embeddings: | 70% | 640/913 [00:17<00:07, 36.83it/s] |
| Computing ESM2 embeddings: | 74% | 672/913 [00:18<00:06, 37.63it/s] |
| Computing ESM2 embeddings: | 77% | 704/913 [00:19<00:05, 36.96it/s] |

Computing ESM2 embeddings: 81%| | 736/913 [00:19<00:04, 37.70it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:20<00:03, 37.10it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:21<00:02, 37.79it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 37.19it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:23<00:01, 37.65it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:24<00:00, 36.48it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:24<00:00, 36.90it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.30it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:24, 33.61it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:22, 35.65it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:22, 34.78it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:20, 36.34it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.17it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:19, 35.22it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:18, 35.05it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:08<00:16, 36.34it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.17it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.15it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.59it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.40it/s]

```

Computing ESM2 embeddings: 50%|          | 448/896 [00:12<00:12, 36.95it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:13<00:11, 37.66it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:14<00:10, 36.96it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.50it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 36.32it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.03it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:07, 36.28it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:18<00:06, 37.13it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:19<00:05, 36.71it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:20<00:04, 37.53it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:21<00:03, 36.98it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.73it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.13it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.63it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 36.52it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 37.19it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:24, 35.17it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:22, 37.11it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 36.32it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:20, 37.45it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/914 [00:05<00:19, 36.86it/s] |
| Computing ESM2 embeddings: 25%  | 224/914 [00:06<00:18, 37.70it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:06<00:17, 37.15it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 37.80it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:16, 37.07it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 37.82it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:14, 37.15it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:11<00:13, 37.46it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:12<00:12, 36.41it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.34it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 36.73it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 37.52it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:15<00:09, 36.88it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:16<00:08, 37.66it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:17<00:07, 36.91it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:18<00:06, 37.68it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 37.05it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:19<00:04, 37.75it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:20<00:03, 37.04it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:21<00:03, 37.75it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:22<00:02, 37.13it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:23<00:01, 37.83it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:24<00:00, 36.94it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:24<00:00, 37.12it/s] |



Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
Computing ESM2 embeddings: 0%|          | 0/574 [00:00<?, ?it/s]
Computing ESM2 embeddings: 6%|          | 32/574 [00:00<00:14, 36.86it/s]
Computing ESM2 embeddings: 11%|         | 64/574 [00:02<00:17, 29.04it/s]
Computing ESM2 embeddings: 17%|         | 96/574 [00:02<00:14, 33.07it/s]
Computing ESM2 embeddings: 22%|         | 128/574 [00:03<00:13, 34.10it/s]
Computing ESM2 embeddings: 28%|         | 160/574 [00:04<00:11, 35.64it/s]
Computing ESM2 embeddings: 33%|         | 192/574 [00:05<00:10, 35.04it/s]
Computing ESM2 embeddings: 39%|         | 224/574 [00:06<00:09, 36.16it/s]
Computing ESM2 embeddings: 45%|         | 256/574 [00:07<00:08, 35.88it/s]
Computing ESM2 embeddings: 50%|         | 288/574 [00:08<00:07, 36.89it/s]
Computing ESM2 embeddings: 56%|         | 320/574 [00:09<00:06, 36.55it/s]
Computing ESM2 embeddings: 61%|         | 352/574 [00:09<00:05, 37.17it/s]
Computing ESM2 embeddings: 67%|         | 384/574 [00:10<00:05, 36.14it/s]
Computing ESM2 embeddings: 72%|         | 416/574 [00:11<00:04, 37.13it/s]
Computing ESM2 embeddings: 78%|         | 448/574 [00:12<00:03, 36.75it/s]
Computing ESM2 embeddings: 84%|         | 480/574 [00:13<00:02, 37.45it/s]
Computing ESM2 embeddings: 89%|         | 512/574 [00:14<00:01, 37.02it/s]
Computing ESM2 embeddings: 95%|         | 544/574 [00:15<00:00, 37.76it/s]
```

```
Computing ESM2 embeddings: 100%|        | 574/574 [00:15<00:00, 36.03it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/913 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/913 [00:00<00:22, 39.25it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/913 [00:01<00:22, 36.98it/s]  |
| Computing ESM2 embeddings: | 11% | 96/913 [00:02<00:21, 37.80it/s]  |
| Computing ESM2 embeddings: | 14% | 128/913 [00:03<00:21, 36.19it/s] |
| Computing ESM2 embeddings: | 18% | 160/913 [00:04<00:20, 37.15it/s] |
| Computing ESM2 embeddings: | 21% | 192/913 [00:05<00:19, 36.59it/s] |
| Computing ESM2 embeddings: | 25% | 224/913 [00:06<00:18, 37.51it/s] |
| Computing ESM2 embeddings: | 28% | 256/913 [00:06<00:17, 36.98it/s] |
| Computing ESM2 embeddings: | 32% | 288/913 [00:07<00:16, 37.76it/s] |
| Computing ESM2 embeddings: | 35% | 320/913 [00:08<00:15, 37.17it/s] |
| Computing ESM2 embeddings: | 39% | 352/913 [00:09<00:14, 37.55it/s] |
| Computing ESM2 embeddings: | 42% | 384/913 [00:10<00:14, 36.33it/s] |
| Computing ESM2 embeddings: | 46% | 416/913 [00:11<00:13, 37.18it/s] |
| Computing ESM2 embeddings: | 49% | 448/913 [00:12<00:12, 36.56it/s] |
| Computing ESM2 embeddings: | 53% | 480/913 [00:12<00:11, 37.14it/s] |
| Computing ESM2 embeddings: | 56% | 512/913 [00:13<00:11, 36.19it/s] |
| Computing ESM2 embeddings: | 60% | 544/913 [00:14<00:10, 36.90it/s] |
| Computing ESM2 embeddings: | 63% | 576/913 [00:15<00:09, 36.04it/s] |
| Computing ESM2 embeddings: | 67% | 608/913 [00:16<00:08, 37.01it/s] |
| Computing ESM2 embeddings: | 70% | 640/913 [00:17<00:07, 36.53it/s] |
| Computing ESM2 embeddings: | 74% | 672/913 [00:18<00:06, 37.40it/s] |
| Computing ESM2 embeddings: | 77% | 704/913 [00:19<00:05, 36.84it/s] |

Computing ESM2 embeddings: 81%| | 736/913 [00:19<00:04, 37.64it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:20<00:03, 37.11it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:21<00:02, 37.80it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 37.20it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:23<00:01, 37.64it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:24<00:00, 36.33it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:24<00:00, 36.98it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 38.28it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.73it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 36.99it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 35.76it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.10it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.77it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:17, 37.34it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 36.30it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.18it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.90it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.68it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.16it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.81it/s]

```

Computing ESM2 embeddings: 50%|          | 448/896 [00:12<00:12, 37.24it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:12<00:10, 37.83it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:13<00:10, 37.26it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.74it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 36.55it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.10it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:07, 36.10it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:18<00:06, 37.10it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:19<00:05, 36.68it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:19<00:04, 37.52it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.05it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.67it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.10it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.84it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 37.07it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 38.22it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:01<00:23, 36.02it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:21, 37.19it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 35.94it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:20, 36.96it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/914 [00:05<00:20, 35.94it/s] |
| Computing ESM2 embeddings: 25%  | 224/914 [00:06<00:18, 36.78it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:07<00:18, 36.02it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 37.12it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:16, 36.78it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 37.62it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:14, 37.21it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:11<00:13, 37.84it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:12<00:12, 37.33it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.91it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 37.33it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 37.88it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:15<00:09, 37.31it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:16<00:08, 37.63it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:17<00:07, 36.57it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:18<00:06, 36.71it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 37.47it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:19<00:04, 36.75it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:20<00:03, 37.58it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:21<00:03, 37.28it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:22<00:02, 38.11it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:23<00:01, 37.52it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:24<00:00, 38.02it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:24<00:00, 37.22it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.65it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:23, 35.95it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.09it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 35.72it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:19, 36.97it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.55it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 37.22it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:07<00:17, 36.10it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 37.19it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 36.73it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:14, 37.59it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.07it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:12, 37.80it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:12, 37.18it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:12<00:10, 37.84it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.20it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 37.58it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 38.07it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 36.61it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.49it/s] |

```

Computing ESM2 embeddings: 75%|      | 672/896 [00:18<00:06, 36.96it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:05, 37.78it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.08it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.85it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.26it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.96it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.25it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.22it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/913 [00:00<00:24, 36.67it/s]
Computing ESM2 embeddings: 7%|      | 64/913 [00:02<00:30, 28.01it/s]
Computing ESM2 embeddings: 11%|     | 96/913 [00:03<00:25, 31.79it/s]
Computing ESM2 embeddings: 14%|     | 128/913 [00:03<00:24, 32.68it/s]
Computing ESM2 embeddings: 18%|     | 160/913 [00:04<00:21, 34.88it/s]
Computing ESM2 embeddings: 21%|     | 192/913 [00:05<00:20, 35.18it/s]
Computing ESM2 embeddings: 25%|     | 224/913 [00:06<00:18, 36.51it/s]
Computing ESM2 embeddings: 28%|     | 256/913 [00:07<00:18, 36.19it/s]
Computing ESM2 embeddings: 32%|     | 288/913 [00:08<00:16, 37.00it/s]
Computing ESM2 embeddings: 35%|     | 320/913 [00:09<00:16, 35.97it/s]
Computing ESM2 embeddings: 39%|     | 352/913 [00:09<00:15, 36.72it/s]
Computing ESM2 embeddings: 42%|     | 384/913 [00:10<00:14, 35.80it/s]

```

Computing ESM2 embeddings: 46%| | 416/913 [00:11<00:13, 36.91it/s]

Computing ESM2 embeddings: 49%| | 448/913 [00:12<00:12, 36.58it/s]

Computing ESM2 embeddings: 53%| | 480/913 [00:13<00:11, 37.47it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:14<00:10, 36.84it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:15<00:09, 37.59it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:16<00:09, 37.01it/s]

Computing ESM2 embeddings: 67%| | 608/913 [00:16<00:08, 37.76it/s]

Computing ESM2 embeddings: 70%| | 640/913 [00:17<00:07, 37.10it/s]

Computing ESM2 embeddings: 74%| | 672/913 [00:18<00:06, 37.59it/s]

Computing ESM2 embeddings: 77%| | 704/913 [00:19<00:05, 36.49it/s]

Computing ESM2 embeddings: 81%| | 736/913 [00:20<00:04, 37.26it/s]

Computing ESM2 embeddings: 84%| | 768/913 [00:21<00:03, 36.83it/s]

Computing ESM2 embeddings: 88%| | 800/913 [00:22<00:03, 37.44it/s]

Computing ESM2 embeddings: 91%| | 832/913 [00:22<00:02, 36.31it/s]

Computing ESM2 embeddings: 95%| | 864/913 [00:23<00:01, 37.26it/s]

Computing ESM2 embeddings: 98%| | 896/913 [00:24<00:00, 36.82it/s]

Computing ESM2 embeddings: 100%| | 913/913 [00:25<00:00, 36.17it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.58it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:23, 35.52it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.05it/s]



|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/896 [00:03<00:21, 35.86it/s] |
| Computing ESM2 embeddings: 18% | 160/896 [00:04<00:19, 37.06it/s] |
| Computing ESM2 embeddings: 21% | 192/896 [00:05<00:19, 36.71it/s] |
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:17, 37.38it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:06<00:17, 36.40it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:07<00:16, 37.37it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 36.78it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.61it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 36.94it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.76it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 37.09it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:12<00:10, 37.86it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:13<00:10, 37.25it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.57it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 36.53it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.44it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:06, 36.93it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:05, 37.72it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:18<00:05, 37.11it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:19<00:04, 37.76it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:20<00:03, 37.31it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.98it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.24it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 37.74it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.05it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
 Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:28, 30.71it/s]  
 Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 34.71it/s]  
 Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 36.95it/s]  
 Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.47it/s]  
 Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.45it/s]  
 Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.89it/s]  
 Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 37.73it/s]  
 Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:17, 37.18it/s]  
 Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.92it/s]  
 Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 37.08it/s]  
 Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.79it/s]  
 Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.10it/s]  
 Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.85it/s]  
 Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.19it/s]  
 Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.81it/s]  
 Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.03it/s]  
 Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.68it/s]  
 Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.05it/s]  
 Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.81it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.18it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.81it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.28it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.85it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.33it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.88it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.23it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.80it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.10it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.24it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/574 [00:00<?, ?it/s]

Computing ESM2 embeddings: 6%| | 32/574 [00:00<00:16, 33.83it/s]

Computing ESM2 embeddings: 11%| | 64/574 [00:01<00:15, 33.88it/s]

Computing ESM2 embeddings: 17%| | 96/574 [00:02<00:13, 36.34it/s]

Computing ESM2 embeddings: 22%| | 128/574 [00:03<00:12, 36.05it/s]

Computing ESM2 embeddings: 28%| | 160/574 [00:04<00:11, 36.75it/s]

Computing ESM2 embeddings: 33%| | 192/574 [00:05<00:10, 35.69it/s]

Computing ESM2 embeddings: 39%| | 224/574 [00:06<00:09, 36.87it/s]

Computing ESM2 embeddings: 45%| | 256/574 [00:07<00:08, 36.57it/s]

Computing ESM2 embeddings: 50%| | 288/574 [00:07<00:07, 37.19it/s]

Computing ESM2 embeddings: 56%| | 320/574 [00:08<00:07, 36.18it/s]

```

Computing ESM2 embeddings: 61%|          | 352/574 [00:09<00:05, 37.11it/s]
Computing ESM2 embeddings: 67%|          | 384/574 [00:10<00:05, 36.68it/s]
Computing ESM2 embeddings: 72%|          | 416/574 [00:11<00:04, 37.53it/s]
Computing ESM2 embeddings: 78%|          | 448/574 [00:12<00:03, 36.97it/s]
Computing ESM2 embeddings: 84%|          | 480/574 [00:13<00:02, 37.72it/s]
Computing ESM2 embeddings: 89%|          | 512/574 [00:13<00:01, 37.08it/s]
Computing ESM2 embeddings: 95%|          | 544/574 [00:14<00:00, 37.84it/s]
Computing ESM2 embeddings: 100%|         | 574/574 [00:15<00:00, 36.70it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/906 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/906 [00:01<00:34, 25.54it/s]
Computing ESM2 embeddings: 7%|           | 64/906 [00:02<00:28, 29.60it/s]
Computing ESM2 embeddings: 11%|          | 96/906 [00:03<00:24, 33.21it/s]
Computing ESM2 embeddings: 14%|          | 128/906 [00:03<00:23, 33.41it/s]
Computing ESM2 embeddings: 18%|          | 160/906 [00:04<00:21, 35.25it/s]
Computing ESM2 embeddings: 21%|          | 192/906 [00:05<00:20, 35.49it/s]
Computing ESM2 embeddings: 25%|          | 224/906 [00:06<00:18, 36.37it/s]
Computing ESM2 embeddings: 28%|          | 256/906 [00:07<00:18, 35.52it/s]
Computing ESM2 embeddings: 32%|          | 288/906 [00:08<00:17, 36.29it/s]
Computing ESM2 embeddings: 35%|          | 320/906 [00:09<00:16, 35.61it/s]
Computing ESM2 embeddings: 39%|          | 352/906 [00:10<00:15, 36.72it/s]
Computing ESM2 embeddings: 42%|          | 384/906 [00:10<00:14, 36.25it/s]

```

```

Computing ESM2 embeddings: 46%|          | 416/906 [00:11<00:13, 37.15it/s]
Computing ESM2 embeddings: 49%|          | 448/906 [00:12<00:12, 36.75it/s]
Computing ESM2 embeddings: 53%|          | 480/906 [00:13<00:11, 37.48it/s]
Computing ESM2 embeddings: 57%|          | 512/906 [00:14<00:10, 36.80it/s]
Computing ESM2 embeddings: 60%|          | 544/906 [00:15<00:09, 37.31it/s]
Computing ESM2 embeddings: 64%|          | 576/906 [00:16<00:09, 35.96it/s]
Computing ESM2 embeddings: 67%|          | 608/906 [00:17<00:08, 36.92it/s]
Computing ESM2 embeddings: 71%|          | 640/906 [00:17<00:07, 36.62it/s]
Computing ESM2 embeddings: 74%|          | 672/906 [00:18<00:06, 37.38it/s]
Computing ESM2 embeddings: 78%|          | 704/906 [00:19<00:05, 36.81it/s]
Computing ESM2 embeddings: 81%|          | 736/906 [00:20<00:04, 37.60it/s]
Computing ESM2 embeddings: 85%|          | 768/906 [00:21<00:03, 37.05it/s]
Computing ESM2 embeddings: 88%|          | 800/906 [00:22<00:02, 37.79it/s]
Computing ESM2 embeddings: 92%|          | 832/906 [00:23<00:01, 37.15it/s]
Computing ESM2 embeddings: 95%|          | 864/906 [00:23<00:01, 37.86it/s]
Computing ESM2 embeddings: 99%|          | 896/906 [00:24<00:00, 37.19it/s]

Computing ESM2 embeddings: 100%|         | 906/906 [00:25<00:00, 36.14it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/903 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/903 [00:00<00:22, 37.98it/s]
Computing ESM2 embeddings:  7%|          | 64/903 [00:01<00:23, 35.67it/s]
Computing ESM2 embeddings: 11%|          | 96/903 [00:02<00:21, 36.95it/s]

```

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/903 [00:03<00:21, 35.84it/s] |
| Computing ESM2 embeddings: 18% | 160/903 [00:04<00:20, 37.13it/s] |
| Computing ESM2 embeddings: 21% | 192/903 [00:05<00:19, 36.54it/s] |
| Computing ESM2 embeddings: 25% | 224/903 [00:06<00:18, 37.23it/s] |
| Computing ESM2 embeddings: 28% | 256/903 [00:07<00:17, 36.16it/s] |
| Computing ESM2 embeddings: 32% | 288/903 [00:07<00:16, 37.09it/s] |
| Computing ESM2 embeddings: 35% | 320/903 [00:08<00:15, 36.70it/s] |
| Computing ESM2 embeddings: 39% | 352/903 [00:09<00:14, 37.46it/s] |
| Computing ESM2 embeddings: 43% | 384/903 [00:10<00:14, 36.85it/s] |
| Computing ESM2 embeddings: 46% | 416/903 [00:11<00:12, 37.58it/s] |
| Computing ESM2 embeddings: 50% | 448/903 [00:12<00:12, 36.88it/s] |
| Computing ESM2 embeddings: 53% | 480/903 [00:12<00:11, 37.56it/s] |
| Computing ESM2 embeddings: 57% | 512/903 [00:13<00:10, 36.89it/s] |
| Computing ESM2 embeddings: 60% | 544/903 [00:14<00:09, 37.42it/s] |
| Computing ESM2 embeddings: 64% | 576/903 [00:15<00:09, 36.13it/s] |
| Computing ESM2 embeddings: 67% | 608/903 [00:16<00:07, 37.05it/s] |
| Computing ESM2 embeddings: 71% | 640/903 [00:17<00:07, 36.53it/s] |
| Computing ESM2 embeddings: 74% | 672/903 [00:18<00:06, 37.31it/s] |
| Computing ESM2 embeddings: 78% | 704/903 [00:19<00:05, 36.72it/s] |
| Computing ESM2 embeddings: 82% | 736/903 [00:19<00:04, 37.29it/s] |
| Computing ESM2 embeddings: 85% | 768/903 [00:20<00:03, 36.14it/s] |
| Computing ESM2 embeddings: 89% | 800/903 [00:21<00:02, 37.10it/s] |
| Computing ESM2 embeddings: 92% | 832/903 [00:22<00:01, 36.69it/s] |
| Computing ESM2 embeddings: 96% | 864/903 [00:23<00:01, 37.42it/s] |

Computing ESM2 embeddings: 99%| | 896/903 [00:24<00:00, 36.82it/s]

Computing ESM2 embeddings: 100%| | 903/903 [00:24<00:00, 36.83it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.80it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 36.22it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.65it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:20, 36.97it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:19, 37.73it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.89it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:05<00:17, 37.63it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:06<00:17, 37.02it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.75it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 37.18it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.56it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:14, 36.38it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.27it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.92it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.71it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.21it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.79it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.16it/s]

```

Computing ESM2 embeddings: 68%|      | 608/896 [00:16<00:07, 37.89it/s]
Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:06, 37.28it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:17<00:05, 37.98it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:18<00:05, 37.32it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 37.74it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 36.50it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 37.19it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 36.31it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.36it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 37.06it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:24, 36.28it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:23, 35.72it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:21, 37.50it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:21, 36.58it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 37.68it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.94it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.84it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:06<00:17, 37.11it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.49it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.30it/s]

```



Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.31it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.81it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.67it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.04it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.71it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.17it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.91it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.26it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.92it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.23it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.83it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.11it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.86it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.16it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.81it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.25it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.90it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.11it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.24it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:24, 35.66it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/895 [00:01<00:21, 37.83it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:21, 37.09it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:20, 38.10it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:19, 37.17it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:18, 37.99it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:05<00:17, 37.43it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:06<00:16, 38.02it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:07<00:16, 37.38it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:08<00:15, 37.67it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 36.50it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:13, 37.40it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 36.87it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:11<00:11, 37.69it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:12<00:11, 37.21it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:13<00:10, 37.94it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:14<00:09, 37.32it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 37.95it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 37.31it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:06, 37.72it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:18<00:06, 36.54it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:18<00:05, 37.38it/s] |
| Computing ESM2 embeddings: | 82% | 736/895 [00:19<00:04, 36.98it/s] |
| Computing ESM2 embeddings: | 86% | 768/895 [00:20<00:03, 37.75it/s] |
| Computing ESM2 embeddings: | 89% | 800/895 [00:21<00:02, 37.14it/s] |

```

Computing ESM2 embeddings: 93%|      | 832/895 [00:22<00:01, 37.85it/s]

Computing ESM2 embeddings: 97%|      | 864/895 [00:23<00:00, 37.28it/s]

Computing ESM2 embeddings: 100%|     | 895/895 [00:23<00:00, 37.44it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:25, 35.17it/s]

Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:24, 34.93it/s]

Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:22, 36.37it/s]

Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:22, 35.48it/s]

Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 36.88it/s]

Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:19, 36.53it/s]

Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 37.20it/s]

Computing ESM2 embeddings: 28%|      | 256/914 [00:07<00:18, 36.15it/s]

Computing ESM2 embeddings: 32%|      | 288/914 [00:07<00:16, 37.19it/s]

Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.77it/s]

Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:14, 37.61it/s]

Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 36.80it/s]

Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.63it/s]

Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 37.30it/s]

Computing ESM2 embeddings: 53%|      | 480/914 [00:12<00:11, 37.98it/s]

Computing ESM2 embeddings: 56%|      | 512/914 [00:13<00:10, 37.26it/s]

Computing ESM2 embeddings: 60%|      | 544/914 [00:14<00:09, 37.90it/s]

```

```

Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 37.11it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 37.50it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 36.43it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.23it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 36.75it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:19<00:04, 37.48it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.01it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.65it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.13it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.83it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.13it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 37.06it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/897 [00:01<00:34, 25.08it/s]
Computing ESM2 embeddings:  7%|          | 64/897 [00:02<00:26, 31.55it/s]
Computing ESM2 embeddings: 11%|          | 96/897 [00:03<00:24, 32.61it/s]
Computing ESM2 embeddings: 14%|          | 128/897 [00:03<00:22, 34.95it/s]
Computing ESM2 embeddings: 18%|          | 160/897 [00:04<00:20, 35.21it/s]
Computing ESM2 embeddings: 21%|          | 192/897 [00:05<00:19, 36.23it/s]
Computing ESM2 embeddings: 25%|          | 224/897 [00:06<00:19, 35.16it/s]
Computing ESM2 embeddings: 29%|          | 256/897 [00:07<00:17, 36.33it/s]

```

Computing ESM2 embeddings: 32%| | 288/897 [00:08<00:16, 36.19it/s]

Computing ESM2 embeddings: 36%| | 320/897 [00:09<00:15, 37.06it/s]

Computing ESM2 embeddings: 39%| | 352/897 [00:09<00:14, 36.54it/s]

Computing ESM2 embeddings: 43%| | 384/897 [00:10<00:13, 37.39it/s]

Computing ESM2 embeddings: 46%| | 416/897 [00:11<00:13, 36.90it/s]

Computing ESM2 embeddings: 50%| | 448/897 [00:12<00:11, 37.64it/s]

Computing ESM2 embeddings: 54%| | 480/897 [00:13<00:11, 36.92it/s]

Computing ESM2 embeddings: 57%| | 512/897 [00:14<00:10, 37.59it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:15<00:09, 36.90it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:15<00:08, 37.65it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:16<00:07, 36.91it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:17<00:06, 37.64it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:18<00:06, 36.94it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:19<00:05, 37.58it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:20<00:04, 36.99it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:21<00:03, 37.64it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:21<00:02, 36.97it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:22<00:01, 37.68it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:23<00:00, 37.09it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:24<00:00, 36.38it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/591 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/591 [00:00<00:16, 33.56it/s]

Computing ESM2 embeddings: 11%| | 64/591 [00:01<00:15, 33.81it/s]

Computing ESM2 embeddings: 16%| | 96/591 [00:02<00:13, 36.17it/s]

Computing ESM2 embeddings: 22%| | 128/591 [00:03<00:12, 35.99it/s]

Computing ESM2 embeddings: 27%| | 160/591 [00:04<00:11, 36.80it/s]

Computing ESM2 embeddings: 32%| | 192/591 [00:05<00:11, 35.73it/s]

Computing ESM2 embeddings: 38%| | 224/591 [00:06<00:09, 36.90it/s]

Computing ESM2 embeddings: 43%| | 256/591 [00:07<00:09, 36.56it/s]

Computing ESM2 embeddings: 49%| | 288/591 [00:07<00:08, 37.45it/s]

Computing ESM2 embeddings: 54%| | 320/591 [00:08<00:07, 36.80it/s]

Computing ESM2 embeddings: 60%| | 352/591 [00:09<00:06, 37.59it/s]

Computing ESM2 embeddings: 65%| | 384/591 [00:10<00:05, 37.04it/s]

Computing ESM2 embeddings: 70%| | 416/591 [00:11<00:04, 37.77it/s]

Computing ESM2 embeddings: 76%| | 448/591 [00:12<00:03, 37.26it/s]

Computing ESM2 embeddings: 81%| | 480/591 [00:13<00:02, 37.93it/s]

Computing ESM2 embeddings: 87%| | 512/591 [00:13<00:02, 37.23it/s]

Computing ESM2 embeddings: 92%| | 544/591 [00:14<00:01, 37.79it/s]

Computing ESM2 embeddings: 97%| | 576/591 [00:15<00:00, 37.06it/s]

Computing ESM2 embeddings: 100%| | 591/591 [00:16<00:00, 36.87it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:01<00:31, 27.10it/s]

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 7%  | 64/895 [00:02<00:26, 30.87it/s]  |
| Computing ESM2 embeddings: | 11% | 96/895 [00:02<00:23, 34.25it/s]  |
| Computing ESM2 embeddings: | 14% | 128/895 [00:03<00:22, 34.78it/s] |
| Computing ESM2 embeddings: | 18% | 160/895 [00:04<00:20, 35.97it/s] |
| Computing ESM2 embeddings: | 21% | 192/895 [00:05<00:19, 35.25it/s] |
| Computing ESM2 embeddings: | 25% | 224/895 [00:06<00:18, 36.42it/s] |
| Computing ESM2 embeddings: | 29% | 256/895 [00:07<00:17, 36.17it/s] |
| Computing ESM2 embeddings: | 32% | 288/895 [00:08<00:16, 36.84it/s] |
| Computing ESM2 embeddings: | 36% | 320/895 [00:09<00:15, 35.94it/s] |
| Computing ESM2 embeddings: | 39% | 352/895 [00:09<00:14, 37.00it/s] |
| Computing ESM2 embeddings: | 43% | 384/895 [00:10<00:13, 36.78it/s] |
| Computing ESM2 embeddings: | 46% | 416/895 [00:11<00:12, 37.60it/s] |
| Computing ESM2 embeddings: | 50% | 448/895 [00:12<00:12, 37.06it/s] |
| Computing ESM2 embeddings: | 54% | 480/895 [00:13<00:10, 37.81it/s] |
| Computing ESM2 embeddings: | 57% | 512/895 [00:14<00:10, 37.12it/s] |
| Computing ESM2 embeddings: | 61% | 544/895 [00:15<00:09, 37.64it/s] |
| Computing ESM2 embeddings: | 64% | 576/895 [00:15<00:08, 36.42it/s] |
| Computing ESM2 embeddings: | 68% | 608/895 [00:16<00:07, 37.26it/s] |
| Computing ESM2 embeddings: | 72% | 640/895 [00:17<00:06, 36.66it/s] |
| Computing ESM2 embeddings: | 75% | 672/895 [00:18<00:05, 37.56it/s] |
| Computing ESM2 embeddings: | 79% | 704/895 [00:19<00:05, 37.04it/s] |
| Computing ESM2 embeddings: | 82% | 736/895 [00:20<00:04, 37.57it/s] |
| Computing ESM2 embeddings: | 86% | 768/895 [00:21<00:03, 36.45it/s] |
| Computing ESM2 embeddings: | 89% | 800/895 [00:21<00:02, 37.32it/s] |

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 36.82it/s]  
Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.53it/s]  
Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 36.44it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at  
facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it  
for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:28, 31.19it/s]  
Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:26, 32.68it/s]  
Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:23, 35.02it/s]  
Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 34.66it/s]  
Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 35.92it/s]  
Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:20, 35.23it/s]  
Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 36.42it/s]  
Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 36.28it/s]  
Computing ESM2 embeddings: 32%| | 288/914 [00:08<00:16, 37.29it/s]  
Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.80it/s]  
Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.63it/s]  
Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.14it/s]  
Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.80it/s]  
Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.15it/s]  
Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.55it/s]  
Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:11, 36.42it/s]  
Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.03it/s]



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Computing ESM2 embeddings: 63%|          | 576/914 [00:15<00:09, 36.00it/s]
Computing ESM2 embeddings: 67%|          | 608/914 [00:16<00:08, 36.99it/s]
Computing ESM2 embeddings: 70%|          | 640/914 [00:17<00:07, 36.61it/s]
Computing ESM2 embeddings: 74%|          | 672/914 [00:18<00:06, 37.43it/s]
Computing ESM2 embeddings: 77%|          | 704/914 [00:19<00:05, 36.84it/s]
Computing ESM2 embeddings: 81%|          | 736/914 [00:20<00:04, 37.62it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:20<00:03, 37.07it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:21<00:03, 37.75it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:22<00:02, 37.19it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:23<00:01, 37.84it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:24<00:00, 37.25it/s]

Computing ESM2 embeddings: 100%|         | 914/914 [00:24<00:00, 36.63it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:22, 37.74it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:26, 31.24it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:23, 34.12it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:22, 33.92it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:20, 35.75it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:19, 35.46it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:06<00:18, 36.38it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:07<00:18, 35.49it/s]

```

Computing ESM2 embeddings: 32%| | 288/896 [00:08<00:16, 36.69it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:09<00:15, 36.28it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.22it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.79it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.57it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.05it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.75it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 37.15it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.85it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 37.20it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.59it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:07, 36.46it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:06, 37.29it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 36.73it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.37it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:21<00:03, 36.83it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.60it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 36.94it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.59it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.57it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 4%  | 32/914 [00:01<00:32, 27.02it/s]  |
| Computing ESM2 embeddings: 7%  | 64/914 [00:02<00:29, 29.30it/s]  |
| Computing ESM2 embeddings: 11% | 96/914 [00:03<00:24, 32.96it/s]  |
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:23, 33.21it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:21, 35.29it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:20, 35.31it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 36.66it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:18, 36.22it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:08<00:16, 36.96it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:09<00:16, 36.01it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.02it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.72it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.56it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 36.95it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 37.71it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:14<00:10, 37.17it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:15<00:09, 37.85it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 37.23it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 37.93it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 37.22it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 37.91it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:05, 37.06it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:20<00:04, 37.77it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:21<00:03, 37.08it/s] |

```

Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.77it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.35it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 38.14it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.42it/s]
Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.58it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/895 [00:01<00:29, 29.20it/s]
Computing ESM2 embeddings:  7%|      | 64/895 [00:02<00:25, 32.05it/s]
Computing ESM2 embeddings: 11%|      | 96/895 [00:02<00:23, 34.72it/s]
Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:22, 34.43it/s]
Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:20, 36.15it/s]
Computing ESM2 embeddings: 21%|      | 192/895 [00:05<00:19, 35.92it/s]
Computing ESM2 embeddings: 25%|      | 224/895 [00:06<00:18, 36.71it/s]
Computing ESM2 embeddings: 29%|      | 256/895 [00:07<00:18, 35.48it/s]
Computing ESM2 embeddings: 32%|      | 288/895 [00:08<00:16, 36.60it/s]
Computing ESM2 embeddings: 36%|      | 320/895 [00:09<00:15, 36.41it/s]
Computing ESM2 embeddings: 39%|      | 352/895 [00:09<00:14, 37.07it/s]
Computing ESM2 embeddings: 43%|      | 384/895 [00:10<00:14, 35.97it/s]
Computing ESM2 embeddings: 46%|      | 416/895 [00:11<00:12, 36.99it/s]
Computing ESM2 embeddings: 50%|      | 448/895 [00:12<00:12, 36.61it/s]
Computing ESM2 embeddings: 54%|      | 480/895 [00:13<00:11, 37.48it/s]

```

Computing ESM2 embeddings: 57%| | 512/895 [00:14<00:10, 37.02it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.79it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.10it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.45it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:07, 36.40it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:06, 36.95it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:19<00:05, 37.79it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:20<00:04, 37.03it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:21<00:03, 37.67it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.14it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.81it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 36.95it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 36.60it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 39.16it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:22, 37.16it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:21, 37.90it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.34it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.01it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:20, 35.96it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 36.90it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:18, 35.98it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.01it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.52it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.45it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.95it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.76it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.18it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.91it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.34it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 38.02it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.39it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 38.02it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.32it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.77it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 36.45it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.36it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.93it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.70it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.09it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.57it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.43it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.14it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/897 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/897 [00:00<00:22, 39.14it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/897 [00:01<00:22, 37.39it/s]  |
| Computing ESM2 embeddings: | 11% | 96/897 [00:02<00:21, 38.01it/s]  |
| Computing ESM2 embeddings: | 14% | 128/897 [00:03<00:21, 36.42it/s] |
| Computing ESM2 embeddings: | 18% | 160/897 [00:04<00:19, 37.44it/s] |
| Computing ESM2 embeddings: | 21% | 192/897 [00:05<00:19, 36.70it/s] |
| Computing ESM2 embeddings: | 25% | 224/897 [00:05<00:17, 37.61it/s] |
| Computing ESM2 embeddings: | 29% | 256/897 [00:06<00:17, 36.98it/s] |
| Computing ESM2 embeddings: | 32% | 288/897 [00:07<00:16, 37.57it/s] |
| Computing ESM2 embeddings: | 36% | 320/897 [00:08<00:15, 36.48it/s] |
| Computing ESM2 embeddings: | 39% | 352/897 [00:09<00:14, 37.42it/s] |
| Computing ESM2 embeddings: | 43% | 384/897 [00:10<00:13, 36.94it/s] |
| Computing ESM2 embeddings: | 46% | 416/897 [00:11<00:12, 37.74it/s] |
| Computing ESM2 embeddings: | 50% | 448/897 [00:12<00:12, 37.01it/s] |
| Computing ESM2 embeddings: | 54% | 480/897 [00:12<00:11, 37.76it/s] |
| Computing ESM2 embeddings: | 57% | 512/897 [00:13<00:10, 37.12it/s] |
| Computing ESM2 embeddings: | 61% | 544/897 [00:14<00:09, 37.88it/s] |
| Computing ESM2 embeddings: | 64% | 576/897 [00:15<00:08, 37.25it/s] |
| Computing ESM2 embeddings: | 68% | 608/897 [00:16<00:07, 37.94it/s] |
| Computing ESM2 embeddings: | 71% | 640/897 [00:17<00:06, 37.48it/s] |
| Computing ESM2 embeddings: | 75% | 672/897 [00:17<00:05, 38.10it/s] |
| Computing ESM2 embeddings: | 78% | 704/897 [00:18<00:05, 37.33it/s] |

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Computing ESM2 embeddings: 82%|      | 736/897 [00:19<00:04, 37.95it/s]
Computing ESM2 embeddings: 86%|      | 768/897 [00:20<00:03, 37.18it/s]
Computing ESM2 embeddings: 89%|      | 800/897 [00:21<00:02, 37.89it/s]
Computing ESM2 embeddings: 93%|      | 832/897 [00:22<00:01, 37.21it/s]
Computing ESM2 embeddings: 96%|      | 864/897 [00:23<00:00, 37.66it/s]

Computing ESM2 embeddings: 100%|     | 897/897 [00:24<00:00, 37.28it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/591 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/591 [00:01<00:19, 29.06it/s]
Computing ESM2 embeddings: 11%|      | 64/591 [00:02<00:16, 32.03it/s]
Computing ESM2 embeddings: 16%|      | 96/591 [00:02<00:14, 34.81it/s]
Computing ESM2 embeddings: 22%|      | 128/591 [00:03<00:13, 34.64it/s]
Computing ESM2 embeddings: 27%|      | 160/591 [00:04<00:11, 36.41it/s]
Computing ESM2 embeddings: 32%|      | 192/591 [00:05<00:11, 36.27it/s]
Computing ESM2 embeddings: 38%|      | 224/591 [00:06<00:09, 37.31it/s]
Computing ESM2 embeddings: 43%|      | 256/591 [00:07<00:09, 36.57it/s]
Computing ESM2 embeddings: 49%|      | 288/591 [00:07<00:08, 37.52it/s]
Computing ESM2 embeddings: 54%|      | 320/591 [00:08<00:07, 36.84it/s]
Computing ESM2 embeddings: 60%|      | 352/591 [00:09<00:06, 37.59it/s]
Computing ESM2 embeddings: 65%|      | 384/591 [00:10<00:05, 36.93it/s]
Computing ESM2 embeddings: 70%|      | 416/591 [00:11<00:04, 37.61it/s]
Computing ESM2 embeddings: 76%|      | 448/591 [00:12<00:03, 36.96it/s]

```



Computing ESM2 embeddings: 81%| | 480/591 [00:13<00:02, 37.64it/s]

Computing ESM2 embeddings: 87%| | 512/591 [00:14<00:02, 36.93it/s]

Computing ESM2 embeddings: 92%| | 544/591 [00:14<00:01, 37.62it/s]

Computing ESM2 embeddings: 97%| | 576/591 [00:15<00:00, 36.96it/s]

Computing ESM2 embeddings: 100%| | 591/591 [00:16<00:00, 36.55it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:01<00:30, 28.52it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:02<00:27, 30.65it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:23, 33.75it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:22, 33.57it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:20, 35.36it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:19, 35.56it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:06<00:18, 36.61it/s]

Computing ESM2 embeddings: 29%| | 256/895 [00:07<00:17, 36.06it/s]

Computing ESM2 embeddings: 32%| | 288/895 [00:08<00:16, 37.12it/s]

Computing ESM2 embeddings: 36%| | 320/895 [00:09<00:15, 36.97it/s]

Computing ESM2 embeddings: 39%| | 352/895 [00:09<00:14, 37.76it/s]

Computing ESM2 embeddings: 43%| | 384/895 [00:10<00:13, 37.43it/s]

Computing ESM2 embeddings: 46%| | 416/895 [00:11<00:12, 38.09it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:12<00:11, 37.26it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:13<00:10, 37.89it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:14<00:10, 37.11it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.73it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.18it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.56it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:06, 36.50it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:05, 37.32it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:19<00:05, 36.78it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:20<00:04, 37.25it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:21<00:03, 36.24it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.18it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 36.82it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.60it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 36.54it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:22, 38.94it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:24, 35.37it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 37.18it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 36.60it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.23it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:20, 35.90it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:18, 36.90it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:06<00:18, 36.44it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:07<00:16, 37.37it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:08<00:16, 36.71it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:14, 37.52it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.85it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.54it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.93it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.67it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.10it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.71it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.01it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.65it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.96it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.60it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.77it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.48it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.87it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.53it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.94it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.68it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.09it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.09it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:26, 33.42it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:25, 33.56it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:23, 35.32it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:22, 34.89it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 36.09it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:20, 35.66it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 36.80it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:07<00:18, 36.49it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:07<00:16, 37.17it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.15it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 36.83it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 35.90it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 36.85it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 36.33it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:13<00:11, 36.92it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:14<00:11, 35.96it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:14<00:09, 37.00it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:09, 36.48it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 36.72it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 35.71it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:18<00:06, 36.45it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:19<00:05, 35.33it/s] |

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 36.27it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:21<00:04, 35.93it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:22<00:03, 36.88it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 36.50it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.09it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.08it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:25<00:00, 36.23it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:01<00:27, 31.47it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:25, 33.10it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:22, 35.41it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:22, 34.88it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:20, 36.26it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.23it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 36.99it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:17, 36.00it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.05it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.64it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.42it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.02it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.71it/s]

```

Computing ESM2 embeddings: 50%|          | 448/896 [00:12<00:12, 37.01it/s]
Computing ESM2 embeddings: 54%|          | 480/896 [00:13<00:11, 37.75it/s]
Computing ESM2 embeddings: 57%|          | 512/896 [00:14<00:10, 37.12it/s]
Computing ESM2 embeddings: 61%|          | 544/896 [00:14<00:09, 37.76it/s]
Computing ESM2 embeddings: 64%|          | 576/896 [00:15<00:08, 37.00it/s]
Computing ESM2 embeddings: 68%|          | 608/896 [00:16<00:07, 37.39it/s]
Computing ESM2 embeddings: 71%|          | 640/896 [00:17<00:07, 36.28it/s]
Computing ESM2 embeddings: 75%|          | 672/896 [00:18<00:06, 37.20it/s]
Computing ESM2 embeddings: 79%|          | 704/896 [00:19<00:05, 36.77it/s]
Computing ESM2 embeddings: 82%|          | 736/896 [00:20<00:04, 37.56it/s]
Computing ESM2 embeddings: 86%|          | 768/896 [00:20<00:03, 37.06it/s]
Computing ESM2 embeddings: 89%|          | 800/896 [00:21<00:02, 37.79it/s]
Computing ESM2 embeddings: 93%|          | 832/896 [00:22<00:01, 37.20it/s]
Computing ESM2 embeddings: 96%|          | 864/896 [00:23<00:00, 37.89it/s]

Computing ESM2 embeddings: 100%|         | 896/896 [00:24<00:00, 36.76it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/913 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/913 [00:00<00:24, 36.53it/s]
Computing ESM2 embeddings:  7%|          | 64/913 [00:02<00:28, 29.29it/s]
Computing ESM2 embeddings: 11%|          | 96/913 [00:02<00:25, 32.50it/s]
Computing ESM2 embeddings: 14%|          | 128/913 [00:03<00:22, 34.55it/s]
Computing ESM2 embeddings: 18%|          | 160/913 [00:04<00:20, 36.12it/s]

```

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/913 [00:05<00:20, 36.01it/s] |
| Computing ESM2 embeddings: 25%  | 224/913 [00:06<00:18, 36.91it/s] |
| Computing ESM2 embeddings: 28%  | 256/913 [00:07<00:17, 36.66it/s] |
| Computing ESM2 embeddings: 32%  | 288/913 [00:08<00:16, 37.62it/s] |
| Computing ESM2 embeddings: 35%  | 320/913 [00:08<00:15, 37.15it/s] |
| Computing ESM2 embeddings: 39%  | 352/913 [00:09<00:14, 37.63it/s] |
| Computing ESM2 embeddings: 42%  | 384/913 [00:10<00:14, 36.41it/s] |
| Computing ESM2 embeddings: 46%  | 416/913 [00:11<00:13, 37.29it/s] |
| Computing ESM2 embeddings: 49%  | 448/913 [00:12<00:12, 36.81it/s] |
| Computing ESM2 embeddings: 53%  | 480/913 [00:13<00:11, 37.55it/s] |
| Computing ESM2 embeddings: 56%  | 512/913 [00:14<00:10, 36.96it/s] |
| Computing ESM2 embeddings: 60%  | 544/913 [00:14<00:09, 37.43it/s] |
| Computing ESM2 embeddings: 63%  | 576/913 [00:15<00:09, 36.26it/s] |
| Computing ESM2 embeddings: 67%  | 608/913 [00:16<00:08, 37.18it/s] |
| Computing ESM2 embeddings: 70%  | 640/913 [00:17<00:07, 36.68it/s] |
| Computing ESM2 embeddings: 74%  | 672/913 [00:18<00:06, 37.46it/s] |
| Computing ESM2 embeddings: 77%  | 704/913 [00:19<00:05, 36.87it/s] |
| Computing ESM2 embeddings: 81%  | 736/913 [00:20<00:04, 37.33it/s] |
| Computing ESM2 embeddings: 84%  | 768/913 [00:21<00:03, 36.28it/s] |
| Computing ESM2 embeddings: 88%  | 800/913 [00:21<00:03, 37.14it/s] |
| Computing ESM2 embeddings: 91%  | 832/913 [00:22<00:02, 36.70it/s] |
| Computing ESM2 embeddings: 95%  | 864/913 [00:23<00:01, 37.50it/s] |
| Computing ESM2 embeddings: 98%  | 896/913 [00:24<00:00, 36.95it/s] |
| Computing ESM2 embeddings: 100% | 913/913 [00:24<00:00, 36.55it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/896 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/896 [00:00<00:22, 38.00it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/896 [00:01<00:23, 35.69it/s]  |
| Computing ESM2 embeddings: | 11% | 96/896 [00:02<00:21, 37.33it/s]  |
| Computing ESM2 embeddings: | 14% | 128/896 [00:03<00:21, 36.54it/s] |
| Computing ESM2 embeddings: | 18% | 160/896 [00:04<00:20, 35.34it/s] |
| Computing ESM2 embeddings: | 21% | 192/896 [00:05<00:19, 36.62it/s] |
| Computing ESM2 embeddings: | 25% | 224/896 [00:06<00:18, 36.33it/s] |
| Computing ESM2 embeddings: | 29% | 256/896 [00:06<00:17, 37.32it/s] |
| Computing ESM2 embeddings: | 32% | 288/896 [00:07<00:16, 36.65it/s] |
| Computing ESM2 embeddings: | 36% | 320/896 [00:08<00:15, 37.23it/s] |
| Computing ESM2 embeddings: | 39% | 352/896 [00:09<00:15, 36.19it/s] |
| Computing ESM2 embeddings: | 43% | 384/896 [00:10<00:13, 37.18it/s] |
| Computing ESM2 embeddings: | 46% | 416/896 [00:11<00:13, 36.77it/s] |
| Computing ESM2 embeddings: | 50% | 448/896 [00:12<00:11, 37.45it/s] |
| Computing ESM2 embeddings: | 54% | 480/896 [00:13<00:11, 37.04it/s] |
| Computing ESM2 embeddings: | 57% | 512/896 [00:13<00:10, 37.71it/s] |
| Computing ESM2 embeddings: | 61% | 544/896 [00:14<00:09, 36.96it/s] |
| Computing ESM2 embeddings: | 64% | 576/896 [00:15<00:08, 37.63it/s] |
| Computing ESM2 embeddings: | 68% | 608/896 [00:16<00:07, 36.95it/s] |
| Computing ESM2 embeddings: | 71% | 640/896 [00:17<00:06, 37.69it/s] |



```

Computing ESM2 embeddings: 75%|      | 672/896 [00:18<00:06, 37.12it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:19<00:05, 37.51it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:19<00:04, 36.31it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:20<00:03, 37.16it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:21<00:02, 36.69it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:22<00:01, 37.48it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 36.95it/s]
Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.96it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|        | 0/915 [00:00<?, ?it/s]
Computing ESM2 embeddings: 3%|        | 32/915 [00:00<00:26, 33.91it/s]
Computing ESM2 embeddings: 7%|        | 64/915 [00:02<00:30, 27.54it/s]
Computing ESM2 embeddings: 10%|       | 96/915 [00:03<00:26, 31.34it/s]
Computing ESM2 embeddings: 14%|       | 128/915 [00:03<00:23, 33.07it/s]
Computing ESM2 embeddings: 17%|       | 160/915 [00:04<00:21, 34.78it/s]
Computing ESM2 embeddings: 21%|       | 192/915 [00:05<00:20, 34.47it/s]
Computing ESM2 embeddings: 24%|       | 224/915 [00:06<00:19, 35.65it/s]
Computing ESM2 embeddings: 28%|       | 256/915 [00:07<00:18, 35.04it/s]
Computing ESM2 embeddings: 31%|       | 288/915 [00:08<00:17, 36.04it/s]
Computing ESM2 embeddings: 35%|       | 320/915 [00:09<00:16, 35.36it/s]
Computing ESM2 embeddings: 38%|       | 352/915 [00:10<00:15, 36.46it/s]
Computing ESM2 embeddings: 42%|       | 384/915 [00:11<00:14, 36.32it/s]

```

```

Computing ESM2 embeddings: 45%|          | 416/915 [00:11<00:13, 37.24it/s]
Computing ESM2 embeddings: 49%|          | 448/915 [00:12<00:12, 36.78it/s]
Computing ESM2 embeddings: 52%|          | 480/915 [00:13<00:11, 37.49it/s]
Computing ESM2 embeddings: 56%|          | 512/915 [00:14<00:10, 36.86it/s]
Computing ESM2 embeddings: 59%|          | 544/915 [00:15<00:09, 37.55it/s]
Computing ESM2 embeddings: 63%|          | 576/915 [00:16<00:09, 36.87it/s]
Computing ESM2 embeddings: 66%|          | 608/915 [00:16<00:08, 37.63it/s]
Computing ESM2 embeddings: 70%|          | 640/915 [00:17<00:07, 36.94it/s]
Computing ESM2 embeddings: 73%|          | 672/915 [00:18<00:06, 37.70it/s]
Computing ESM2 embeddings: 77%|          | 704/915 [00:19<00:05, 37.25it/s]
Computing ESM2 embeddings: 80%|          | 736/915 [00:20<00:04, 37.76it/s]
Computing ESM2 embeddings: 84%|          | 768/915 [00:21<00:03, 37.03it/s]
Computing ESM2 embeddings: 87%|          | 800/915 [00:22<00:03, 37.72it/s]
Computing ESM2 embeddings: 91%|          | 832/915 [00:23<00:02, 37.00it/s]
Computing ESM2 embeddings: 94%|          | 864/915 [00:23<00:01, 37.70it/s]
Computing ESM2 embeddings: 98%|          | 896/915 [00:24<00:00, 37.09it/s]

Computing ESM2 embeddings: 100%|         | 915/915 [00:25<00:00, 36.17it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/573 [00:00<?, ?it/s]
Computing ESM2 embeddings:  6%|          | 32/573 [00:01<00:20, 26.97it/s]
Computing ESM2 embeddings: 11%|          | 64/573 [00:02<00:18, 28.13it/s]
Computing ESM2 embeddings: 17%|          | 96/573 [00:03<00:14, 32.01it/s]

```

Computing ESM2 embeddings: 22%| | 128/573 [00:04<00:13, 32.68it/s]

Computing ESM2 embeddings: 28%| | 160/573 [00:04<00:11, 34.88it/s]

Computing ESM2 embeddings: 34%| | 192/573 [00:05<00:10, 35.02it/s]

Computing ESM2 embeddings: 39%| | 224/573 [00:06<00:09, 36.13it/s]

Computing ESM2 embeddings: 45%| | 256/573 [00:07<00:08, 35.46it/s]

Computing ESM2 embeddings: 50%| | 288/573 [00:08<00:07, 36.62it/s]

Computing ESM2 embeddings: 56%| | 320/573 [00:09<00:06, 36.38it/s]

Computing ESM2 embeddings: 61%| | 352/573 [00:10<00:05, 37.28it/s]

Computing ESM2 embeddings: 67%| | 384/573 [00:10<00:05, 36.77it/s]

Computing ESM2 embeddings: 73%| | 416/573 [00:11<00:04, 37.37it/s]

Computing ESM2 embeddings: 78%| | 448/573 [00:12<00:03, 36.54it/s]

Computing ESM2 embeddings: 84%| | 480/573 [00:13<00:02, 37.35it/s]

Computing ESM2 embeddings: 89%| | 512/573 [00:14<00:01, 36.73it/s]

Computing ESM2 embeddings: 95%| | 544/573 [00:15<00:00, 37.29it/s]

Computing ESM2 embeddings: 100%| | 573/573 [00:16<00:00, 35.39it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at  
 facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',  
 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it  
 for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:25, 33.97it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:23, 36.43it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:21, 37.38it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:21, 36.03it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:20, 36.88it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 21%  | 192/913 [00:05<00:20, 35.73it/s] |
| Computing ESM2 embeddings: 25%  | 224/913 [00:06<00:18, 36.57it/s] |
| Computing ESM2 embeddings: 28%  | 256/913 [00:07<00:18, 35.69it/s] |
| Computing ESM2 embeddings: 32%  | 288/913 [00:07<00:17, 36.76it/s] |
| Computing ESM2 embeddings: 35%  | 320/913 [00:08<00:16, 36.44it/s] |
| Computing ESM2 embeddings: 39%  | 352/913 [00:09<00:15, 37.03it/s] |
| Computing ESM2 embeddings: 42%  | 384/913 [00:10<00:14, 36.06it/s] |
| Computing ESM2 embeddings: 46%  | 416/913 [00:11<00:13, 37.10it/s] |
| Computing ESM2 embeddings: 49%  | 448/913 [00:12<00:12, 36.54it/s] |
| Computing ESM2 embeddings: 53%  | 480/913 [00:13<00:11, 37.41it/s] |
| Computing ESM2 embeddings: 56%  | 512/913 [00:14<00:10, 36.75it/s] |
| Computing ESM2 embeddings: 60%  | 544/913 [00:14<00:09, 37.58it/s] |
| Computing ESM2 embeddings: 63%  | 576/913 [00:15<00:09, 37.04it/s] |
| Computing ESM2 embeddings: 67%  | 608/913 [00:16<00:08, 37.52it/s] |
| Computing ESM2 embeddings: 70%  | 640/913 [00:17<00:07, 36.38it/s] |
| Computing ESM2 embeddings: 74%  | 672/913 [00:18<00:06, 37.29it/s] |
| Computing ESM2 embeddings: 77%  | 704/913 [00:19<00:05, 36.75it/s] |
| Computing ESM2 embeddings: 81%  | 736/913 [00:19<00:04, 37.57it/s] |
| Computing ESM2 embeddings: 84%  | 768/913 [00:20<00:03, 37.01it/s] |
| Computing ESM2 embeddings: 88%  | 800/913 [00:21<00:02, 37.73it/s] |
| Computing ESM2 embeddings: 91%  | 832/913 [00:22<00:02, 37.02it/s] |
| Computing ESM2 embeddings: 95%  | 864/913 [00:23<00:01, 37.76it/s] |
| Computing ESM2 embeddings: 98%  | 896/913 [00:24<00:00, 37.15it/s] |
| Computing ESM2 embeddings: 100% | 913/913 [00:24<00:00, 36.91it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/866 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/866 [00:00<00:21, 38.52it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/866 [00:01<00:22, 35.82it/s]  |
| Computing ESM2 embeddings: | 11% | 96/866 [00:02<00:20, 37.04it/s]  |
| Computing ESM2 embeddings: | 15% | 128/866 [00:03<00:20, 36.00it/s] |
| Computing ESM2 embeddings: | 18% | 160/866 [00:04<00:18, 37.23it/s] |
| Computing ESM2 embeddings: | 22% | 192/866 [00:05<00:18, 36.74it/s] |
| Computing ESM2 embeddings: | 26% | 224/866 [00:06<00:17, 37.29it/s] |
| Computing ESM2 embeddings: | 30% | 256/866 [00:06<00:16, 36.54it/s] |
| Computing ESM2 embeddings: | 33% | 288/866 [00:07<00:15, 37.44it/s] |
| Computing ESM2 embeddings: | 37% | 320/866 [00:08<00:14, 36.75it/s] |
| Computing ESM2 embeddings: | 41% | 352/866 [00:09<00:13, 37.59it/s] |
| Computing ESM2 embeddings: | 44% | 384/866 [00:10<00:13, 36.96it/s] |
| Computing ESM2 embeddings: | 48% | 416/866 [00:11<00:11, 37.68it/s] |
| Computing ESM2 embeddings: | 52% | 448/866 [00:12<00:11, 37.06it/s] |
| Computing ESM2 embeddings: | 55% | 480/866 [00:12<00:10, 37.79it/s] |
| Computing ESM2 embeddings: | 59% | 512/866 [00:13<00:09, 37.19it/s] |
| Computing ESM2 embeddings: | 63% | 544/866 [00:14<00:08, 37.90it/s] |
| Computing ESM2 embeddings: | 67% | 576/866 [00:15<00:07, 37.24it/s] |
| Computing ESM2 embeddings: | 70% | 608/866 [00:16<00:06, 37.70it/s] |
| Computing ESM2 embeddings: | 74% | 640/866 [00:17<00:06, 36.48it/s] |

Computing ESM2 embeddings: 78%| | 672/866 [00:18<00:05, 37.36it/s]

Computing ESM2 embeddings: 81%| | 704/866 [00:18<00:04, 36.82it/s]

Computing ESM2 embeddings: 85%| | 736/866 [00:19<00:03, 37.63it/s]

Computing ESM2 embeddings: 89%| | 768/866 [00:20<00:02, 36.93it/s]

Computing ESM2 embeddings: 92%| | 800/866 [00:21<00:01, 37.61it/s]

Computing ESM2 embeddings: 96%| | 832/866 [00:22<00:00, 37.05it/s]

Computing ESM2 embeddings: 100%| | 866/866 [00:23<00:00, 37.00it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/602 [00:00<?, ?it/s]

Computing ESM2 embeddings: 5%| | 32/602 [00:00<00:15, 37.85it/s]

Computing ESM2 embeddings: 11%| | 64/602 [00:02<00:18, 29.12it/s]

Computing ESM2 embeddings: 16%| | 96/602 [00:02<00:15, 32.32it/s]

Computing ESM2 embeddings: 21%| | 128/602 [00:03<00:14, 33.51it/s]

Computing ESM2 embeddings: 27%| | 160/602 [00:04<00:12, 35.26it/s]

Computing ESM2 embeddings: 32%| | 192/602 [00:05<00:11, 34.78it/s]

Computing ESM2 embeddings: 37%| | 224/602 [00:06<00:10, 36.13it/s]

Computing ESM2 embeddings: 43%| | 256/602 [00:07<00:09, 36.05it/s]

Computing ESM2 embeddings: 48%| | 288/602 [00:08<00:08, 36.82it/s]

Computing ESM2 embeddings: 53%| | 320/602 [00:09<00:07, 35.92it/s]

Computing ESM2 embeddings: 58%| | 352/602 [00:09<00:06, 36.88it/s]

Computing ESM2 embeddings: 64%| | 384/602 [00:10<00:05, 36.44it/s]

Computing ESM2 embeddings: 69%| | 416/602 [00:11<00:05, 37.07it/s]

Computing ESM2 embeddings: 74%| | 448/602 [00:12<00:04, 36.37it/s]

Computing ESM2 embeddings: 80%| | 480/602 [00:13<00:03, 37.32it/s]

Computing ESM2 embeddings: 85%| | 512/602 [00:14<00:02, 36.77it/s]

Computing ESM2 embeddings: 90%| | 544/602 [00:15<00:01, 37.60it/s]

Computing ESM2 embeddings: 96%| | 576/602 [00:15<00:00, 37.10it/s]

Computing ESM2 embeddings: 100%| | 602/602 [00:16<00:00, 35.98it/s]

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:00<00:22, 37.77it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:01<00:23, 35.28it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:21, 36.64it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:21, 35.56it/s]

Computing ESM2 embeddings: 18%| | 160/895 [00:04<00:20, 36.57it/s]

Computing ESM2 embeddings: 21%| | 192/895 [00:05<00:19, 35.60it/s]

Computing ESM2 embeddings: 25%| | 224/895 [00:06<00:18, 36.48it/s]

Computing ESM2 embeddings: 29%| | 256/895 [00:07<00:18, 35.50it/s]

Computing ESM2 embeddings: 32%| | 288/895 [00:07<00:16, 36.70it/s]

Computing ESM2 embeddings: 36%| | 320/895 [00:08<00:15, 36.45it/s]

Computing ESM2 embeddings: 39%| | 352/895 [00:09<00:14, 37.33it/s]

Computing ESM2 embeddings: 43%| | 384/895 [00:10<00:13, 36.83it/s]

Computing ESM2 embeddings: 46%| | 416/895 [00:11<00:12, 37.53it/s]

Computing ESM2 embeddings: 50%| | 448/895 [00:12<00:12, 36.85it/s]

Computing ESM2 embeddings: 54%| | 480/895 [00:13<00:11, 37.64it/s]

Computing ESM2 embeddings: 57%| | 512/895 [00:13<00:10, 37.12it/s]

Computing ESM2 embeddings: 61%| | 544/895 [00:14<00:09, 37.81it/s]

Computing ESM2 embeddings: 64%| | 576/895 [00:15<00:08, 37.25it/s]

Computing ESM2 embeddings: 68%| | 608/895 [00:16<00:07, 37.52it/s]

Computing ESM2 embeddings: 72%| | 640/895 [00:17<00:07, 36.30it/s]

Computing ESM2 embeddings: 75%| | 672/895 [00:18<00:05, 37.23it/s]

Computing ESM2 embeddings: 79%| | 704/895 [00:19<00:05, 36.55it/s]

Computing ESM2 embeddings: 82%| | 736/895 [00:19<00:04, 37.44it/s]

Computing ESM2 embeddings: 86%| | 768/895 [00:20<00:03, 36.81it/s]

Computing ESM2 embeddings: 89%| | 800/895 [00:21<00:02, 37.62it/s]

Computing ESM2 embeddings: 93%| | 832/895 [00:22<00:01, 37.00it/s]

Computing ESM2 embeddings: 97%| | 864/895 [00:23<00:00, 37.70it/s]

Computing ESM2 embeddings: 100%| | 895/895 [00:24<00:00, 36.85it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.39it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:23, 35.81it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 37.15it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:21, 35.86it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:20, 37.17it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:19, 36.64it/s]



|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25%  | 224/914 [00:06<00:18, 37.25it/s] |
| Computing ESM2 embeddings: 28%  | 256/914 [00:07<00:18, 36.20it/s] |
| Computing ESM2 embeddings: 32%  | 288/914 [00:07<00:16, 37.22it/s] |
| Computing ESM2 embeddings: 35%  | 320/914 [00:08<00:16, 36.73it/s] |
| Computing ESM2 embeddings: 39%  | 352/914 [00:09<00:14, 37.59it/s] |
| Computing ESM2 embeddings: 42%  | 384/914 [00:10<00:14, 37.06it/s] |
| Computing ESM2 embeddings: 46%  | 416/914 [00:11<00:13, 37.82it/s] |
| Computing ESM2 embeddings: 49%  | 448/914 [00:12<00:12, 37.25it/s] |
| Computing ESM2 embeddings: 53%  | 480/914 [00:12<00:11, 37.91it/s] |
| Computing ESM2 embeddings: 56%  | 512/914 [00:13<00:10, 37.21it/s] |
| Computing ESM2 embeddings: 60%  | 544/914 [00:14<00:09, 37.86it/s] |
| Computing ESM2 embeddings: 63%  | 576/914 [00:15<00:09, 37.08it/s] |
| Computing ESM2 embeddings: 67%  | 608/914 [00:16<00:08, 37.78it/s] |
| Computing ESM2 embeddings: 70%  | 640/914 [00:17<00:07, 37.18it/s] |
| Computing ESM2 embeddings: 74%  | 672/914 [00:18<00:06, 37.78it/s] |
| Computing ESM2 embeddings: 77%  | 704/914 [00:18<00:05, 37.07it/s] |
| Computing ESM2 embeddings: 81%  | 736/914 [00:19<00:04, 37.82it/s] |
| Computing ESM2 embeddings: 84%  | 768/914 [00:20<00:03, 37.23it/s] |
| Computing ESM2 embeddings: 88%  | 800/914 [00:21<00:03, 37.79it/s] |
| Computing ESM2 embeddings: 91%  | 832/914 [00:22<00:02, 37.08it/s] |
| Computing ESM2 embeddings: 95%  | 864/914 [00:23<00:01, 37.77it/s] |
| Computing ESM2 embeddings: 98%  | 896/914 [00:24<00:00, 37.13it/s] |
| Computing ESM2 embeddings: 100% | 914/914 [00:24<00:00, 37.19it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/897 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/897 [00:00<00:24, 35.52it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/897 [00:01<00:24, 34.68it/s]  |
| Computing ESM2 embeddings: | 11% | 96/897 [00:02<00:21, 36.78it/s]  |
| Computing ESM2 embeddings: | 14% | 128/897 [00:03<00:21, 36.37it/s] |
| Computing ESM2 embeddings: | 18% | 160/897 [00:04<00:19, 37.36it/s] |
| Computing ESM2 embeddings: | 21% | 192/897 [00:05<00:19, 36.68it/s] |
| Computing ESM2 embeddings: | 25% | 224/897 [00:06<00:17, 37.54it/s] |
| Computing ESM2 embeddings: | 29% | 256/897 [00:06<00:17, 36.85it/s] |
| Computing ESM2 embeddings: | 32% | 288/897 [00:07<00:16, 37.68it/s] |
| Computing ESM2 embeddings: | 36% | 320/897 [00:08<00:15, 36.90it/s] |
| Computing ESM2 embeddings: | 39% | 352/897 [00:09<00:14, 37.68it/s] |
| Computing ESM2 embeddings: | 43% | 384/897 [00:10<00:13, 37.06it/s] |
| Computing ESM2 embeddings: | 46% | 416/897 [00:11<00:12, 37.45it/s] |
| Computing ESM2 embeddings: | 50% | 448/897 [00:12<00:12, 36.29it/s] |
| Computing ESM2 embeddings: | 54% | 480/897 [00:12<00:11, 37.20it/s] |
| Computing ESM2 embeddings: | 57% | 512/897 [00:13<00:10, 36.75it/s] |
| Computing ESM2 embeddings: | 61% | 544/897 [00:14<00:09, 37.54it/s] |
| Computing ESM2 embeddings: | 64% | 576/897 [00:15<00:08, 37.05it/s] |
| Computing ESM2 embeddings: | 68% | 608/897 [00:16<00:07, 37.75it/s] |
| Computing ESM2 embeddings: | 71% | 640/897 [00:17<00:06, 37.06it/s] |
| Computing ESM2 embeddings: | 75% | 672/897 [00:18<00:05, 37.78it/s] |

```

Computing ESM2 embeddings: 78%|          | 704/897 [00:18<00:05, 37.13it/s]
Computing ESM2 embeddings: 82%|          | 736/897 [00:19<00:04, 37.79it/s]
Computing ESM2 embeddings: 86%|          | 768/897 [00:20<00:03, 37.01it/s]
Computing ESM2 embeddings: 89%|          | 800/897 [00:21<00:02, 37.42it/s]
Computing ESM2 embeddings: 93%|          | 832/897 [00:22<00:01, 36.23it/s]
Computing ESM2 embeddings: 96%|          | 864/897 [00:23<00:00, 37.14it/s]

Computing ESM2 embeddings: 100%|         | 897/897 [00:24<00:00, 36.91it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|           | 0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|           | 32/914 [00:00<00:24, 35.98it/s]
Computing ESM2 embeddings: 7%|           | 64/914 [00:01<00:22, 37.69it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:22, 36.12it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:21, 37.33it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:20, 36.89it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:19, 37.42it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:06<00:19, 36.29it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:06<00:17, 37.28it/s]
Computing ESM2 embeddings: 32%|          | 288/914 [00:07<00:17, 36.73it/s]
Computing ESM2 embeddings: 35%|          | 320/914 [00:08<00:15, 37.52it/s]
Computing ESM2 embeddings: 39%|          | 352/914 [00:09<00:15, 36.92it/s]
Computing ESM2 embeddings: 42%|          | 384/914 [00:10<00:14, 37.62it/s]
Computing ESM2 embeddings: 46%|          | 416/914 [00:11<00:13, 37.01it/s]

```

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.85it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:12<00:11, 37.10it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:13<00:10, 37.82it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:10, 37.00it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:08, 37.76it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.11it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.81it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.22it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:18<00:05, 37.90it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:19<00:04, 37.19it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 37.88it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.13it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.84it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.19it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.61it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 37.14it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 39.00it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:22, 36.90it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:21, 37.63it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 35.98it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/896 [00:04<00:19, 37.20it/s] |
| Computing ESM2 embeddings: 21%  | 192/896 [00:05<00:19, 36.47it/s] |
| Computing ESM2 embeddings: 25%  | 224/896 [00:06<00:17, 37.44it/s] |
| Computing ESM2 embeddings: 29%  | 256/896 [00:06<00:17, 36.87it/s] |
| Computing ESM2 embeddings: 32%  | 288/896 [00:07<00:16, 37.66it/s] |
| Computing ESM2 embeddings: 36%  | 320/896 [00:08<00:15, 36.98it/s] |
| Computing ESM2 embeddings: 39%  | 352/896 [00:09<00:14, 37.62it/s] |
| Computing ESM2 embeddings: 43%  | 384/896 [00:10<00:13, 37.14it/s] |
| Computing ESM2 embeddings: 46%  | 416/896 [00:11<00:12, 37.49it/s] |
| Computing ESM2 embeddings: 50%  | 448/896 [00:12<00:12, 36.30it/s] |
| Computing ESM2 embeddings: 54%  | 480/896 [00:12<00:11, 37.21it/s] |
| Computing ESM2 embeddings: 57%  | 512/896 [00:13<00:10, 36.79it/s] |
| Computing ESM2 embeddings: 61%  | 544/896 [00:14<00:09, 37.51it/s] |
| Computing ESM2 embeddings: 64%  | 576/896 [00:15<00:08, 36.91it/s] |
| Computing ESM2 embeddings: 68%  | 608/896 [00:16<00:07, 37.65it/s] |
| Computing ESM2 embeddings: 71%  | 640/896 [00:17<00:06, 37.07it/s] |
| Computing ESM2 embeddings: 75%  | 672/896 [00:18<00:05, 37.78it/s] |
| Computing ESM2 embeddings: 79%  | 704/896 [00:18<00:05, 37.12it/s] |
| Computing ESM2 embeddings: 82%  | 736/896 [00:19<00:04, 37.70it/s] |
| Computing ESM2 embeddings: 86%  | 768/896 [00:20<00:03, 37.19it/s] |
| Computing ESM2 embeddings: 89%  | 800/896 [00:21<00:02, 37.45it/s] |
| Computing ESM2 embeddings: 93%  | 832/896 [00:22<00:01, 36.33it/s] |
| Computing ESM2 embeddings: 96%  | 864/896 [00:23<00:00, 37.22it/s] |
| Computing ESM2 embeddings: 100% | 896/896 [00:24<00:00, 37.02it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:01<00:28, 31.44it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:02<00:31, 26.58it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:03<00:28, 29.19it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:04<00:24, 32.53it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:05<00:22, 33.63it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:20, 35.37it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:19, 35.50it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:07<00:17, 36.63it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:08<00:17, 36.38it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:09<00:15, 37.31it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:10<00:15, 36.72it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.26it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 36.20it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 37.18it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:13<00:11, 36.88it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:14<00:10, 37.69it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:15<00:09, 37.03it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:16<00:08, 37.82it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:17<00:08, 37.16it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.88it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:18<00:06, 37.23it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:19<00:05, 37.86it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:20<00:04, 37.24it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:21<00:03, 37.92it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:22<00:03, 37.21it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.85it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 36.94it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.70it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:25<00:00, 36.24it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings: 0%|      | 0/826 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/826 [00:01<00:26, 30.16it/s]
Computing ESM2 embeddings: 8%|      | 64/826 [00:02<00:23, 32.20it/s]
Computing ESM2 embeddings: 12%|     | 96/826 [00:02<00:20, 34.90it/s]
Computing ESM2 embeddings: 15%|     | 128/826 [00:03<00:20, 34.48it/s]
Computing ESM2 embeddings: 19%|     | 160/826 [00:04<00:18, 36.16it/s]
Computing ESM2 embeddings: 23%|     | 192/826 [00:05<00:17, 35.90it/s]
Computing ESM2 embeddings: 27%|     | 224/826 [00:06<00:16, 37.03it/s]
Computing ESM2 embeddings: 31%|     | 256/826 [00:07<00:15, 36.65it/s]
Computing ESM2 embeddings: 35%|     | 288/826 [00:08<00:14, 37.51it/s]
Computing ESM2 embeddings: 39%|     | 320/826 [00:08<00:13, 36.99it/s]
Computing ESM2 embeddings: 43%|     | 352/826 [00:09<00:12, 37.72it/s]

```

```

Computing ESM2 embeddings: 46%|          | 384/826 [00:10<00:11, 37.14it/s]
Computing ESM2 embeddings: 50%|          | 416/826 [00:11<00:10, 37.52it/s]
Computing ESM2 embeddings: 54%|          | 448/826 [00:12<00:10, 36.26it/s]
Computing ESM2 embeddings: 58%|          | 480/826 [00:13<00:09, 36.89it/s]
Computing ESM2 embeddings: 62%|          | 512/826 [00:14<00:08, 36.05it/s]
Computing ESM2 embeddings: 66%|          | 544/826 [00:14<00:07, 36.77it/s]
Computing ESM2 embeddings: 70%|          | 576/826 [00:15<00:06, 35.79it/s]
Computing ESM2 embeddings: 74%|          | 608/826 [00:16<00:05, 36.61it/s]
Computing ESM2 embeddings: 77%|          | 640/826 [00:17<00:05, 35.75it/s]
Computing ESM2 embeddings: 81%|          | 672/826 [00:18<00:04, 36.83it/s]
Computing ESM2 embeddings: 85%|          | 704/826 [00:19<00:03, 36.40it/s]
Computing ESM2 embeddings: 89%|          | 736/826 [00:20<00:02, 37.01it/s]
Computing ESM2 embeddings: 93%|          | 768/826 [00:21<00:01, 35.94it/s]
Computing ESM2 embeddings: 97%|          | 800/826 [00:22<00:00, 36.64it/s]
Computing ESM2 embeddings: 100%|         | 826/826 [00:22<00:00, 36.18it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/660 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|          | 32/660 [00:00<00:16, 37.89it/s]
Computing ESM2 embeddings: 10%|          | 64/660 [00:02<00:19, 30.73it/s]
Computing ESM2 embeddings: 15%|          | 96/660 [00:02<00:16, 33.58it/s]
Computing ESM2 embeddings: 19%|          | 128/660 [00:03<00:15, 35.06it/s]
Computing ESM2 embeddings: 24%|          | 160/660 [00:04<00:13, 36.37it/s]

```



Computing ESM2 embeddings: 29%| | 192/660 [00:05<00:12, 36.06it/s]

Computing ESM2 embeddings: 34%| | 224/660 [00:06<00:11, 37.05it/s]

Computing ESM2 embeddings: 39%| | 256/660 [00:07<00:11, 36.61it/s]

Computing ESM2 embeddings: 44%| | 288/660 [00:07<00:09, 37.49it/s]

Computing ESM2 embeddings: 48%| | 320/660 [00:08<00:09, 36.98it/s]

Computing ESM2 embeddings: 53%| | 352/660 [00:09<00:08, 37.64it/s]

Computing ESM2 embeddings: 58%| | 384/660 [00:10<00:07, 36.92it/s]

Computing ESM2 embeddings: 63%| | 416/660 [00:11<00:06, 37.45it/s]

Computing ESM2 embeddings: 68%| | 448/660 [00:12<00:05, 36.36it/s]

Computing ESM2 embeddings: 73%| | 480/660 [00:13<00:04, 37.26it/s]

Computing ESM2 embeddings: 78%| | 512/660 [00:14<00:04, 36.65it/s]

Computing ESM2 embeddings: 82%| | 544/660 [00:14<00:03, 37.13it/s]

Computing ESM2 embeddings: 87%| | 576/660 [00:15<00:02, 36.05it/s]

Computing ESM2 embeddings: 92%| | 608/660 [00:16<00:01, 37.02it/s]

Computing ESM2 embeddings: 97%| | 640/660 [00:17<00:00, 36.55it/s]

Computing ESM2 embeddings: 100%| | 660/660 [00:18<00:00, 36.45it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/895 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/895 [00:01<00:32, 26.47it/s]

Computing ESM2 embeddings: 7%| | 64/895 [00:02<00:27, 30.70it/s]

Computing ESM2 embeddings: 11%| | 96/895 [00:02<00:23, 34.16it/s]

Computing ESM2 embeddings: 14%| | 128/895 [00:03<00:22, 34.63it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 18%  | 160/895 [00:04<00:20, 35.83it/s] |
| Computing ESM2 embeddings: 21%  | 192/895 [00:05<00:20, 34.93it/s] |
| Computing ESM2 embeddings: 25%  | 224/895 [00:06<00:18, 35.97it/s] |
| Computing ESM2 embeddings: 29%  | 256/895 [00:07<00:17, 35.93it/s] |
| Computing ESM2 embeddings: 32%  | 288/895 [00:08<00:16, 36.79it/s] |
| Computing ESM2 embeddings: 36%  | 320/895 [00:09<00:16, 35.75it/s] |
| Computing ESM2 embeddings: 39%  | 352/895 [00:09<00:14, 36.41it/s] |
| Computing ESM2 embeddings: 43%  | 384/895 [00:10<00:14, 35.64it/s] |
| Computing ESM2 embeddings: 46%  | 416/895 [00:11<00:13, 36.69it/s] |
| Computing ESM2 embeddings: 50%  | 448/895 [00:12<00:12, 36.51it/s] |
| Computing ESM2 embeddings: 54%  | 480/895 [00:13<00:11, 37.38it/s] |
| Computing ESM2 embeddings: 57%  | 512/895 [00:14<00:10, 36.86it/s] |
| Computing ESM2 embeddings: 61%  | 544/895 [00:15<00:09, 37.54it/s] |
| Computing ESM2 embeddings: 64%  | 576/895 [00:16<00:08, 36.96it/s] |
| Computing ESM2 embeddings: 68%  | 608/895 [00:16<00:07, 37.71it/s] |
| Computing ESM2 embeddings: 72%  | 640/895 [00:17<00:06, 37.06it/s] |
| Computing ESM2 embeddings: 75%  | 672/895 [00:18<00:05, 37.47it/s] |
| Computing ESM2 embeddings: 79%  | 704/895 [00:19<00:05, 36.43it/s] |
| Computing ESM2 embeddings: 82%  | 736/895 [00:20<00:04, 36.87it/s] |
| Computing ESM2 embeddings: 86%  | 768/895 [00:21<00:03, 36.39it/s] |
| Computing ESM2 embeddings: 89%  | 800/895 [00:22<00:02, 37.21it/s] |
| Computing ESM2 embeddings: 93%  | 832/895 [00:23<00:01, 36.60it/s] |
| Computing ESM2 embeddings: 97%  | 864/895 [00:23<00:00, 37.38it/s] |
| Computing ESM2 embeddings: 100% | 895/895 [00:24<00:00, 36.09it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:00<00:23, 37.46it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:02<00:28, 30.32it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:24, 33.42it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:21, 35.76it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:21, 35.72it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.67it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:19, 35.28it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:07<00:18, 36.31it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:08<00:17, 35.51it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:16, 36.64it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 36.35it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 37.27it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:13, 36.76it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:12, 37.47it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:13<00:11, 36.90it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:14<00:10, 37.62it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:15<00:10, 36.93it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:15<00:08, 37.72it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:16<00:08, 37.11it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:17<00:07, 37.82it/s] |

```

Computing ESM2 embeddings: 74%|      | 672/914 [00:18<00:06, 37.14it/s]
Computing ESM2 embeddings: 77%|      | 704/914 [00:19<00:05, 37.76it/s]
Computing ESM2 embeddings: 81%|      | 736/914 [00:20<00:04, 37.05it/s]
Computing ESM2 embeddings: 84%|      | 768/914 [00:20<00:03, 37.78it/s]
Computing ESM2 embeddings: 88%|      | 800/914 [00:21<00:03, 37.06it/s]
Computing ESM2 embeddings: 91%|      | 832/914 [00:22<00:02, 37.69it/s]
Computing ESM2 embeddings: 95%|      | 864/914 [00:23<00:01, 37.00it/s]
Computing ESM2 embeddings: 98%|      | 896/914 [00:24<00:00, 37.65it/s]

Computing ESM2 embeddings: 100%|     | 914/914 [00:24<00:00, 36.65it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

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```

Computing ESM2 embeddings: 0%|      | 0/897 [00:00<?, ?it/s]
Computing ESM2 embeddings: 4%|      | 32/897 [00:00<00:26, 32.43it/s]
Computing ESM2 embeddings: 7%|      | 64/897 [00:01<00:25, 33.25it/s]
Computing ESM2 embeddings: 11%|     | 96/897 [00:02<00:22, 35.93it/s]
Computing ESM2 embeddings: 14%|     | 128/897 [00:03<00:21, 35.92it/s]
Computing ESM2 embeddings: 18%|     | 160/897 [00:04<00:19, 37.08it/s]
Computing ESM2 embeddings: 21%|     | 192/897 [00:05<00:19, 36.50it/s]
Computing ESM2 embeddings: 25%|     | 224/897 [00:06<00:18, 37.21it/s]
Computing ESM2 embeddings: 29%|     | 256/897 [00:07<00:17, 36.11it/s]
Computing ESM2 embeddings: 32%|     | 288/897 [00:07<00:16, 36.95it/s]
Computing ESM2 embeddings: 36%|     | 320/897 [00:08<00:15, 36.36it/s]
Computing ESM2 embeddings: 39%|     | 352/897 [00:09<00:14, 36.86it/s]

```

Computing ESM2 embeddings: 43%| | 384/897 [00:10<00:14, 35.90it/s]

Computing ESM2 embeddings: 46%| | 416/897 [00:11<00:13, 36.91it/s]

Computing ESM2 embeddings: 50%| | 448/897 [00:12<00:12, 36.46it/s]

Computing ESM2 embeddings: 54%| | 480/897 [00:13<00:11, 37.31it/s]

Computing ESM2 embeddings: 57%| | 512/897 [00:14<00:10, 36.87it/s]

Computing ESM2 embeddings: 61%| | 544/897 [00:14<00:09, 37.28it/s]

Computing ESM2 embeddings: 64%| | 576/897 [00:15<00:08, 37.98it/s]

Computing ESM2 embeddings: 68%| | 608/897 [00:16<00:07, 37.18it/s]

Computing ESM2 embeddings: 71%| | 640/897 [00:17<00:06, 37.55it/s]

Computing ESM2 embeddings: 75%| | 672/897 [00:18<00:06, 36.25it/s]

Computing ESM2 embeddings: 78%| | 704/897 [00:19<00:05, 37.08it/s]

Computing ESM2 embeddings: 82%| | 736/897 [00:20<00:04, 36.41it/s]

Computing ESM2 embeddings: 86%| | 768/897 [00:20<00:03, 37.29it/s]

Computing ESM2 embeddings: 89%| | 800/897 [00:21<00:02, 36.66it/s]

Computing ESM2 embeddings: 93%| | 832/897 [00:22<00:01, 37.32it/s]

Computing ESM2 embeddings: 96%| | 864/897 [00:23<00:00, 36.79it/s]

Computing ESM2 embeddings: 100%| | 897/897 [00:24<00:00, 36.62it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:00<00:23, 37.65it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:01<00:25, 32.82it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:02<00:22, 35.65it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 14% | 128/914 [00:03<00:22, 35.65it/s] |
| Computing ESM2 embeddings: 18% | 160/914 [00:04<00:20, 36.56it/s] |
| Computing ESM2 embeddings: 21% | 192/914 [00:05<00:20, 35.58it/s] |
| Computing ESM2 embeddings: 25% | 224/914 [00:06<00:18, 36.82it/s] |
| Computing ESM2 embeddings: 28% | 256/914 [00:07<00:18, 36.45it/s] |
| Computing ESM2 embeddings: 32% | 288/914 [00:07<00:16, 37.18it/s] |
| Computing ESM2 embeddings: 35% | 320/914 [00:08<00:16, 36.10it/s] |
| Computing ESM2 embeddings: 39% | 352/914 [00:09<00:15, 37.12it/s] |
| Computing ESM2 embeddings: 42% | 384/914 [00:10<00:14, 36.79it/s] |
| Computing ESM2 embeddings: 46% | 416/914 [00:11<00:13, 37.52it/s] |
| Computing ESM2 embeddings: 49% | 448/914 [00:12<00:12, 37.24it/s] |
| Computing ESM2 embeddings: 53% | 480/914 [00:13<00:11, 36.96it/s] |
| Computing ESM2 embeddings: 56% | 512/914 [00:14<00:11, 35.91it/s] |
| Computing ESM2 embeddings: 60% | 544/914 [00:14<00:10, 35.66it/s] |
| Computing ESM2 embeddings: 63% | 576/914 [00:15<00:09, 35.08it/s] |
| Computing ESM2 embeddings: 67% | 608/914 [00:16<00:08, 34.93it/s] |
| Computing ESM2 embeddings: 70% | 640/914 [00:17<00:07, 34.73it/s] |
| Computing ESM2 embeddings: 74% | 672/914 [00:18<00:06, 34.77it/s] |
| Computing ESM2 embeddings: 77% | 704/914 [00:19<00:06, 34.48it/s] |
| Computing ESM2 embeddings: 81% | 736/914 [00:20<00:05, 34.26it/s] |
| Computing ESM2 embeddings: 84% | 768/914 [00:21<00:04, 34.25it/s] |
| Computing ESM2 embeddings: 88% | 800/914 [00:22<00:03, 34.05it/s] |
| Computing ESM2 embeddings: 91% | 832/914 [00:23<00:02, 34.07it/s] |
| Computing ESM2 embeddings: 95% | 864/914 [00:24<00:01, 33.96it/s] |

Computing ESM2 embeddings: 98%| | 896/914 [00:25<00:00, 34.18it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:25<00:00, 35.37it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:22, 37.88it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:02<00:29, 28.55it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:24, 32.27it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:23, 32.70it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:21, 34.51it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:20, 34.33it/s]

Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 35.47it/s]

Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:18, 34.95it/s]

Computing ESM2 embeddings: 32%| | 288/896 [00:08<00:16, 36.24it/s]

Computing ESM2 embeddings: 36%| | 320/896 [00:09<00:15, 36.07it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:10<00:14, 36.98it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:14, 36.50it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.27it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.72it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.54it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 37.00it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:15<00:09, 37.71it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:16<00:08, 36.97it/s]

```

Computing ESM2 embeddings: 68%|      | 608/896 [00:16<00:07, 37.30it/s]
Computing ESM2 embeddings: 71%|      | 640/896 [00:17<00:07, 36.12it/s]
Computing ESM2 embeddings: 75%|      | 672/896 [00:18<00:06, 37.03it/s]
Computing ESM2 embeddings: 79%|      | 704/896 [00:19<00:05, 36.59it/s]
Computing ESM2 embeddings: 82%|      | 736/896 [00:20<00:04, 37.34it/s]
Computing ESM2 embeddings: 86%|      | 768/896 [00:21<00:03, 36.75it/s]
Computing ESM2 embeddings: 89%|      | 800/896 [00:22<00:02, 37.40it/s]
Computing ESM2 embeddings: 93%|      | 832/896 [00:23<00:01, 36.80it/s]
Computing ESM2 embeddings: 96%|      | 864/896 [00:23<00:00, 37.50it/s]

Computing ESM2 embeddings: 100%|     | 896/896 [00:24<00:00, 36.14it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:00<00:25, 34.41it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:01<00:24, 34.07it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:22, 35.97it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:22, 35.21it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:20, 36.40it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:20, 35.57it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:19, 36.27it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:07<00:18, 35.46it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:08<00:17, 36.65it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:08<00:16, 36.54it/s]

```



Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 37.37it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.72it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.56it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.91it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.62it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 37.13it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:14<00:09, 37.83it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.19it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.63it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.42it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.35it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.85it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 37.50it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:20<00:03, 36.99it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.72it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.16it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.86it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.24it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:24<00:00, 36.83it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/768 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/768 [00:01<00:25, 28.68it/s]

|                                 |                                  |
|---------------------------------|----------------------------------|
| Computing ESM2 embeddings: 8%   | 64/768 [00:02<00:23, 30.40it/s]  |
| Computing ESM2 embeddings: 12%  | 96/768 [00:02<00:19, 33.94it/s]  |
| Computing ESM2 embeddings: 17%  | 128/768 [00:03<00:18, 34.37it/s] |
| Computing ESM2 embeddings: 21%  | 160/768 [00:04<00:16, 35.81it/s] |
| Computing ESM2 embeddings: 25%  | 192/768 [00:05<00:16, 35.05it/s] |
| Computing ESM2 embeddings: 29%  | 224/768 [00:06<00:15, 36.09it/s] |
| Computing ESM2 embeddings: 33%  | 256/768 [00:07<00:14, 35.30it/s] |
| Computing ESM2 embeddings: 38%  | 288/768 [00:08<00:13, 36.44it/s] |
| Computing ESM2 embeddings: 42%  | 320/768 [00:09<00:12, 36.14it/s] |
| Computing ESM2 embeddings: 46%  | 352/768 [00:09<00:11, 36.78it/s] |
| Computing ESM2 embeddings: 50%  | 384/768 [00:10<00:10, 35.87it/s] |
| Computing ESM2 embeddings: 54%  | 416/768 [00:11<00:09, 36.92it/s] |
| Computing ESM2 embeddings: 58%  | 448/768 [00:12<00:08, 36.63it/s] |
| Computing ESM2 embeddings: 62%  | 480/768 [00:13<00:07, 37.45it/s] |
| Computing ESM2 embeddings: 67%  | 512/768 [00:14<00:06, 37.00it/s] |
| Computing ESM2 embeddings: 71%  | 544/768 [00:15<00:05, 37.71it/s] |
| Computing ESM2 embeddings: 75%  | 576/768 [00:16<00:05, 36.98it/s] |
| Computing ESM2 embeddings: 79%  | 608/768 [00:16<00:04, 37.71it/s] |
| Computing ESM2 embeddings: 83%  | 640/768 [00:17<00:03, 37.11it/s] |
| Computing ESM2 embeddings: 88%  | 672/768 [00:18<00:02, 37.80it/s] |
| Computing ESM2 embeddings: 92%  | 704/768 [00:19<00:01, 37.20it/s] |
| Computing ESM2 embeddings: 96%  | 736/768 [00:20<00:00, 37.57it/s] |
| Computing ESM2 embeddings: 100% | 768/768 [00:21<00:00, 36.14it/s] |

Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias',

'esm.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/718 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/718 [00:01<00:22, 30.97it/s]  |
| Computing ESM2 embeddings: | 9%  | 64/718 [00:01<00:20, 32.38it/s]  |
| Computing ESM2 embeddings: | 13% | 96/718 [00:02<00:17, 35.28it/s]  |
| Computing ESM2 embeddings: | 18% | 128/718 [00:03<00:16, 35.47it/s] |
| Computing ESM2 embeddings: | 22% | 160/718 [00:04<00:15, 36.92it/s] |
| Computing ESM2 embeddings: | 27% | 192/718 [00:05<00:14, 36.50it/s] |
| Computing ESM2 embeddings: | 31% | 224/718 [00:06<00:13, 37.48it/s] |
| Computing ESM2 embeddings: | 36% | 256/718 [00:07<00:12, 36.90it/s] |
| Computing ESM2 embeddings: | 40% | 288/718 [00:07<00:11, 37.69it/s] |
| Computing ESM2 embeddings: | 45% | 320/718 [00:08<00:10, 36.97it/s] |
| Computing ESM2 embeddings: | 49% | 352/718 [00:09<00:09, 37.57it/s] |
| Computing ESM2 embeddings: | 53% | 384/718 [00:10<00:09, 36.28it/s] |
| Computing ESM2 embeddings: | 58% | 416/718 [00:11<00:08, 37.08it/s] |
| Computing ESM2 embeddings: | 62% | 448/718 [00:12<00:07, 36.66it/s] |
| Computing ESM2 embeddings: | 67% | 480/718 [00:13<00:06, 37.53it/s] |
| Computing ESM2 embeddings: | 71% | 512/718 [00:13<00:05, 36.99it/s] |
| Computing ESM2 embeddings: | 76% | 544/718 [00:14<00:04, 37.76it/s] |
| Computing ESM2 embeddings: | 80% | 576/718 [00:15<00:03, 37.23it/s] |
| Computing ESM2 embeddings: | 85% | 608/718 [00:16<00:02, 37.92it/s] |
| Computing ESM2 embeddings: | 89% | 640/718 [00:17<00:02, 37.26it/s] |
| Computing ESM2 embeddings: | 94% | 672/718 [00:18<00:01, 37.96it/s] |

Computing ESM2 embeddings: 98%| | 704/718 [00:19<00:00, 37.23it/s]

Computing ESM2 embeddings: 100%| | 718/718 [00:19<00:00, 36.86it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/913 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/913 [00:00<00:27, 32.23it/s]

Computing ESM2 embeddings: 7%| | 64/913 [00:01<00:25, 33.42it/s]

Computing ESM2 embeddings: 11%| | 96/913 [00:02<00:22, 35.66it/s]

Computing ESM2 embeddings: 14%| | 128/913 [00:03<00:22, 35.03it/s]

Computing ESM2 embeddings: 18%| | 160/913 [00:04<00:20, 36.54it/s]

Computing ESM2 embeddings: 21%| | 192/913 [00:05<00:19, 36.21it/s]

Computing ESM2 embeddings: 25%| | 224/913 [00:06<00:18, 36.92it/s]

Computing ESM2 embeddings: 28%| | 256/913 [00:07<00:18, 35.79it/s]

Computing ESM2 embeddings: 32%| | 288/913 [00:07<00:16, 36.93it/s]

Computing ESM2 embeddings: 35%| | 320/913 [00:08<00:16, 36.62it/s]

Computing ESM2 embeddings: 39%| | 352/913 [00:09<00:14, 37.54it/s]

Computing ESM2 embeddings: 42%| | 384/913 [00:10<00:14, 36.99it/s]

Computing ESM2 embeddings: 46%| | 416/913 [00:11<00:13, 37.73it/s]

Computing ESM2 embeddings: 49%| | 448/913 [00:12<00:12, 36.93it/s]

Computing ESM2 embeddings: 53%| | 480/913 [00:13<00:11, 37.70it/s]

Computing ESM2 embeddings: 56%| | 512/913 [00:13<00:10, 37.14it/s]

Computing ESM2 embeddings: 60%| | 544/913 [00:14<00:09, 37.78it/s]

Computing ESM2 embeddings: 63%| | 576/913 [00:15<00:09, 36.97it/s]

```

Computing ESM2 embeddings: 67%|          | 608/913 [00:16<00:08, 37.76it/s]
Computing ESM2 embeddings: 70%|          | 640/913 [00:17<00:07, 37.16it/s]
Computing ESM2 embeddings: 74%|          | 672/913 [00:18<00:06, 37.79it/s]
Computing ESM2 embeddings: 77%|          | 704/913 [00:19<00:05, 37.22it/s]
Computing ESM2 embeddings: 81%|          | 736/913 [00:19<00:04, 37.94it/s]
Computing ESM2 embeddings: 84%|          | 768/913 [00:20<00:03, 37.25it/s]
Computing ESM2 embeddings: 88%|          | 800/913 [00:21<00:02, 37.86it/s]
Computing ESM2 embeddings: 91%|          | 832/913 [00:22<00:02, 37.09it/s]
Computing ESM2 embeddings: 95%|          | 864/913 [00:23<00:01, 37.69it/s]
Computing ESM2 embeddings: 98%|          | 896/913 [00:24<00:00, 37.05it/s]

Computing ESM2 embeddings: 100%|         | 913/913 [00:24<00:00, 36.96it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

Computing ESM2 embeddings:  0%|          |  0/896 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/896 [00:00<00:26, 32.55it/s]
Computing ESM2 embeddings:  7%|          | 64/896 [00:01<00:23, 35.89it/s]
Computing ESM2 embeddings: 11%|          | 96/896 [00:02<00:21, 37.64it/s]
Computing ESM2 embeddings: 14%|          | 128/896 [00:03<00:20, 36.77it/s]
Computing ESM2 embeddings: 18%|          | 160/896 [00:04<00:19, 37.72it/s]
Computing ESM2 embeddings: 21%|          | 192/896 [00:05<00:19, 36.91it/s]
Computing ESM2 embeddings: 25%|          | 224/896 [00:06<00:17, 37.77it/s]
Computing ESM2 embeddings: 29%|          | 256/896 [00:06<00:17, 37.11it/s]
Computing ESM2 embeddings: 32%|          | 288/896 [00:07<00:16, 37.90it/s]

```

Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 37.18it/s]

Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.87it/s]

Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 37.24it/s]

Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.89it/s]

Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 37.11it/s]

Computing ESM2 embeddings: 54%| | 480/896 [00:12<00:11, 37.79it/s]

Computing ESM2 embeddings: 57%| | 512/896 [00:13<00:10, 37.25it/s]

Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 37.53it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 36.49it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 37.34it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:06, 36.92it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:05, 37.64it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:18<00:05, 37.00it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:19<00:04, 37.70it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:20<00:03, 37.07it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.80it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.20it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.89it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 37.22it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/916 [00:00<?, ?it/s]

Computing ESM2 embeddings: 3%| | 32/916 [00:01<00:28, 31.38it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 7%  | 64/916 [00:01<00:25, 32.85it/s]  |
| Computing ESM2 embeddings: 10% | 96/916 [00:02<00:23, 35.33it/s]  |
| Computing ESM2 embeddings: 14% | 128/916 [00:03<00:22, 34.79it/s] |
| Computing ESM2 embeddings: 17% | 160/916 [00:04<00:20, 36.38it/s] |
| Computing ESM2 embeddings: 21% | 192/916 [00:05<00:20, 36.07it/s] |
| Computing ESM2 embeddings: 24% | 224/916 [00:06<00:18, 36.96it/s] |
| Computing ESM2 embeddings: 28% | 256/916 [00:07<00:18, 36.41it/s] |
| Computing ESM2 embeddings: 31% | 288/916 [00:07<00:16, 37.28it/s] |
| Computing ESM2 embeddings: 35% | 320/916 [00:08<00:16, 36.51it/s] |
| Computing ESM2 embeddings: 38% | 352/916 [00:09<00:15, 37.45it/s] |
| Computing ESM2 embeddings: 42% | 384/916 [00:10<00:14, 36.84it/s] |
| Computing ESM2 embeddings: 45% | 416/916 [00:11<00:13, 37.49it/s] |
| Computing ESM2 embeddings: 49% | 448/916 [00:12<00:12, 36.96it/s] |
| Computing ESM2 embeddings: 52% | 480/916 [00:13<00:11, 37.47it/s] |
| Computing ESM2 embeddings: 56% | 512/916 [00:14<00:11, 36.26it/s] |
| Computing ESM2 embeddings: 59% | 544/916 [00:14<00:10, 36.96it/s] |
| Computing ESM2 embeddings: 63% | 576/916 [00:15<00:09, 36.07it/s] |
| Computing ESM2 embeddings: 66% | 608/916 [00:16<00:08, 36.75it/s] |
| Computing ESM2 embeddings: 70% | 640/916 [00:17<00:07, 35.73it/s] |
| Computing ESM2 embeddings: 73% | 672/916 [00:18<00:06, 36.46it/s] |
| Computing ESM2 embeddings: 77% | 704/916 [00:19<00:05, 35.64it/s] |
| Computing ESM2 embeddings: 80% | 736/916 [00:20<00:04, 36.49it/s] |
| Computing ESM2 embeddings: 84% | 768/916 [00:21<00:04, 35.74it/s] |
| Computing ESM2 embeddings: 87% | 800/916 [00:22<00:03, 36.78it/s] |

Computing ESM2 embeddings: 91%| | 832/916 [00:22<00:02, 36.37it/s]  
Computing ESM2 embeddings: 94%| | 864/916 [00:23<00:01, 37.29it/s]  
Computing ESM2 embeddings: 98%| | 896/916 [00:24<00:00, 36.85it/s]  
Computing ESM2 embeddings: 100%| | 916/916 [00:25<00:00, 36.44it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]  
Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:26, 32.32it/s]  
Computing ESM2 embeddings: 7%| | 64/896 [00:01<00:25, 33.25it/s]  
Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:22, 35.58it/s]  
Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 35.06it/s]  
Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:20, 36.51it/s]  
Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 36.23it/s]  
Computing ESM2 embeddings: 25%| | 224/896 [00:06<00:18, 36.95it/s]  
Computing ESM2 embeddings: 29%| | 256/896 [00:07<00:17, 36.01it/s]  
Computing ESM2 embeddings: 32%| | 288/896 [00:07<00:16, 37.02it/s]  
Computing ESM2 embeddings: 36%| | 320/896 [00:08<00:15, 36.64it/s]  
Computing ESM2 embeddings: 39%| | 352/896 [00:09<00:14, 37.43it/s]  
Computing ESM2 embeddings: 43%| | 384/896 [00:10<00:13, 36.80it/s]  
Computing ESM2 embeddings: 46%| | 416/896 [00:11<00:12, 37.34it/s]  
Computing ESM2 embeddings: 50%| | 448/896 [00:12<00:12, 36.19it/s]  
Computing ESM2 embeddings: 54%| | 480/896 [00:13<00:11, 37.00it/s]  
Computing ESM2 embeddings: 57%| | 512/896 [00:14<00:10, 35.97it/s]



Computing ESM2 embeddings: 61%| | 544/896 [00:14<00:09, 36.76it/s]

Computing ESM2 embeddings: 64%| | 576/896 [00:15<00:08, 35.76it/s]

Computing ESM2 embeddings: 68%| | 608/896 [00:16<00:07, 36.79it/s]

Computing ESM2 embeddings: 71%| | 640/896 [00:17<00:07, 36.48it/s]

Computing ESM2 embeddings: 75%| | 672/896 [00:18<00:06, 37.28it/s]

Computing ESM2 embeddings: 79%| | 704/896 [00:19<00:05, 36.86it/s]

Computing ESM2 embeddings: 82%| | 736/896 [00:20<00:04, 37.53it/s]

Computing ESM2 embeddings: 86%| | 768/896 [00:21<00:03, 36.92it/s]

Computing ESM2 embeddings: 89%| | 800/896 [00:21<00:02, 37.65it/s]

Computing ESM2 embeddings: 93%| | 832/896 [00:22<00:01, 37.13it/s]

Computing ESM2 embeddings: 96%| | 864/896 [00:23<00:00, 37.81it/s]

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.55it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/914 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/914 [00:01<00:29, 30.18it/s]

Computing ESM2 embeddings: 7%| | 64/914 [00:02<00:28, 29.40it/s]

Computing ESM2 embeddings: 11%| | 96/914 [00:03<00:24, 32.84it/s]

Computing ESM2 embeddings: 14%| | 128/914 [00:03<00:22, 35.25it/s]

Computing ESM2 embeddings: 18%| | 160/914 [00:04<00:21, 34.65it/s]

Computing ESM2 embeddings: 21%| | 192/914 [00:05<00:20, 36.07it/s]

Computing ESM2 embeddings: 25%| | 224/914 [00:06<00:19, 35.90it/s]

Computing ESM2 embeddings: 28%| | 256/914 [00:07<00:17, 36.88it/s]

Computing ESM2 embeddings: 32%| | 288/914 [00:08<00:17, 36.42it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:09<00:15, 37.36it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 36.72it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 37.44it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 36.75it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 37.45it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 36.82it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 37.61it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:15<00:10, 36.96it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.42it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 36.24it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.14it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 36.83it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 37.62it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 36.96it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:21<00:03, 37.69it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 36.95it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.59it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 36.95it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 37.71it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:25<00:00, 36.40it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/883 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/883 [00:01<00:27, 30.70it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/883 [00:01<00:25, 32.48it/s]  |
| Computing ESM2 embeddings: | 11% | 96/883 [00:02<00:22, 34.97it/s]  |
| Computing ESM2 embeddings: | 14% | 128/883 [00:03<00:21, 34.87it/s] |
| Computing ESM2 embeddings: | 18% | 160/883 [00:04<00:19, 36.47it/s] |
| Computing ESM2 embeddings: | 22% | 192/883 [00:05<00:19, 36.26it/s] |
| Computing ESM2 embeddings: | 25% | 224/883 [00:06<00:17, 37.15it/s] |
| Computing ESM2 embeddings: | 29% | 256/883 [00:07<00:17, 36.50it/s] |
| Computing ESM2 embeddings: | 33% | 288/883 [00:07<00:15, 37.47it/s] |
| Computing ESM2 embeddings: | 36% | 320/883 [00:08<00:15, 36.95it/s] |
| Computing ESM2 embeddings: | 40% | 352/883 [00:09<00:14, 37.73it/s] |
| Computing ESM2 embeddings: | 43% | 384/883 [00:10<00:13, 37.10it/s] |
| Computing ESM2 embeddings: | 47% | 416/883 [00:11<00:12, 37.60it/s] |
| Computing ESM2 embeddings: | 51% | 448/883 [00:12<00:11, 36.43it/s] |
| Computing ESM2 embeddings: | 54% | 480/883 [00:13<00:10, 37.12it/s] |
| Computing ESM2 embeddings: | 58% | 512/883 [00:14<00:10, 36.13it/s] |
| Computing ESM2 embeddings: | 62% | 544/883 [00:14<00:09, 37.11it/s] |
| Computing ESM2 embeddings: | 65% | 576/883 [00:15<00:08, 36.61it/s] |
| Computing ESM2 embeddings: | 69% | 608/883 [00:16<00:07, 37.15it/s] |
| Computing ESM2 embeddings: | 72% | 640/883 [00:17<00:06, 36.33it/s] |
| Computing ESM2 embeddings: | 76% | 672/883 [00:18<00:05, 37.23it/s] |
| Computing ESM2 embeddings: | 80% | 704/883 [00:19<00:04, 36.83it/s] |
| Computing ESM2 embeddings: | 83% | 736/883 [00:20<00:03, 37.63it/s] |

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Computing ESM2 embeddings: 87%|      | 768/883 [00:20<00:03, 37.08it/s]
Computing ESM2 embeddings: 91%|      | 800/883 [00:21<00:02, 37.83it/s]
Computing ESM2 embeddings: 94%|      | 832/883 [00:22<00:01, 37.06it/s]
Computing ESM2 embeddings: 98%|      | 864/883 [00:23<00:00, 37.79it/s]
Computing ESM2 embeddings: 100%|     | 883/883 [00:24<00:00, 36.67it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/603 [00:00<?, ?it/s]
Computing ESM2 embeddings:  5%|      | 32/603 [00:01<00:21, 27.09it/s]
Computing ESM2 embeddings: 11%|      | 64/603 [00:02<00:16, 32.27it/s]
Computing ESM2 embeddings: 16%|      | 96/603 [00:02<00:14, 34.94it/s]
Computing ESM2 embeddings: 21%|      | 128/603 [00:03<00:13, 34.47it/s]
Computing ESM2 embeddings: 27%|      | 160/603 [00:04<00:12, 35.94it/s]
Computing ESM2 embeddings: 32%|      | 192/603 [00:05<00:11, 35.20it/s]
Computing ESM2 embeddings: 37%|      | 224/603 [00:06<00:10, 36.02it/s]
Computing ESM2 embeddings: 42%|      | 256/603 [00:07<00:09, 35.28it/s]
Computing ESM2 embeddings: 48%|      | 288/603 [00:08<00:08, 36.28it/s]
Computing ESM2 embeddings: 53%|      | 320/603 [00:09<00:07, 35.58it/s]
Computing ESM2 embeddings: 58%|      | 352/603 [00:09<00:06, 36.73it/s]
Computing ESM2 embeddings: 64%|      | 384/603 [00:10<00:06, 36.35it/s]
Computing ESM2 embeddings: 69%|      | 416/603 [00:11<00:05, 37.23it/s]
Computing ESM2 embeddings: 74%|      | 448/603 [00:12<00:04, 36.68it/s]
Computing ESM2 embeddings: 80%|      | 480/603 [00:13<00:03, 37.48it/s]

```

```

Computing ESM2 embeddings: 85%|      | 512/603 [00:14<00:02, 36.97it/s]
Computing ESM2 embeddings: 90%|      | 544/603 [00:15<00:01, 37.71it/s]
Computing ESM2 embeddings: 96%|      | 576/603 [00:15<00:00, 36.95it/s]
Computing ESM2 embeddings: 100%|     | 603/603 [00:16<00:00, 35.94it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/895 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/895 [00:00<00:26, 32.73it/s]
Computing ESM2 embeddings:  7%|      | 64/895 [00:01<00:24, 33.30it/s]
Computing ESM2 embeddings: 11%|      | 96/895 [00:02<00:22, 36.01it/s]
Computing ESM2 embeddings: 14%|      | 128/895 [00:03<00:21, 35.86it/s]
Computing ESM2 embeddings: 18%|      | 160/895 [00:04<00:19, 36.81it/s]
Computing ESM2 embeddings: 21%|      | 192/895 [00:05<00:19, 35.55it/s]
Computing ESM2 embeddings: 25%|      | 224/895 [00:06<00:18, 36.80it/s]
Computing ESM2 embeddings: 29%|      | 256/895 [00:07<00:17, 36.50it/s]
Computing ESM2 embeddings: 32%|      | 288/895 [00:07<00:16, 37.17it/s]
Computing ESM2 embeddings: 36%|      | 320/895 [00:08<00:15, 36.02it/s]
Computing ESM2 embeddings: 39%|      | 352/895 [00:09<00:14, 37.02it/s]
Computing ESM2 embeddings: 43%|      | 384/895 [00:10<00:13, 36.63it/s]
Computing ESM2 embeddings: 46%|      | 416/895 [00:11<00:12, 37.53it/s]
Computing ESM2 embeddings: 50%|      | 448/895 [00:12<00:12, 36.96it/s]
Computing ESM2 embeddings: 54%|      | 480/895 [00:13<00:11, 37.63it/s]
Computing ESM2 embeddings: 57%|      | 512/895 [00:14<00:10, 36.95it/s]

```

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Computing ESM2 embeddings: 61%|          | 544/895 [00:14<00:09, 37.71it/s]
Computing ESM2 embeddings: 64%|          | 576/895 [00:15<00:08, 37.09it/s]
Computing ESM2 embeddings: 68%|          | 608/895 [00:16<00:07, 37.83it/s]
Computing ESM2 embeddings: 72%|          | 640/895 [00:17<00:06, 37.19it/s]
Computing ESM2 embeddings: 75%|          | 672/895 [00:18<00:05, 37.57it/s]
Computing ESM2 embeddings: 79%|          | 704/895 [00:19<00:05, 36.31it/s]
Computing ESM2 embeddings: 82%|          | 736/895 [00:20<00:04, 36.94it/s]
Computing ESM2 embeddings: 86%|          | 768/895 [00:20<00:03, 36.10it/s]
Computing ESM2 embeddings: 89%|          | 800/895 [00:21<00:02, 37.10it/s]
Computing ESM2 embeddings: 93%|          | 832/895 [00:22<00:01, 36.71it/s]
Computing ESM2 embeddings: 97%|          | 864/895 [00:23<00:00, 37.45it/s]

```

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Computing ESM2 embeddings: 100%|         | 895/895 [00:24<00:00, 36.65it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|          |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|          | 32/914 [00:00<00:23, 37.46it/s]
Computing ESM2 embeddings:  7%|          | 64/914 [00:02<00:29, 29.04it/s]
Computing ESM2 embeddings: 11%|          | 96/914 [00:02<00:24, 32.93it/s]
Computing ESM2 embeddings: 14%|          | 128/914 [00:03<00:23, 33.91it/s]
Computing ESM2 embeddings: 18%|          | 160/914 [00:04<00:21, 35.40it/s]
Computing ESM2 embeddings: 21%|          | 192/914 [00:05<00:20, 34.87it/s]
Computing ESM2 embeddings: 25%|          | 224/914 [00:06<00:19, 36.26it/s]
Computing ESM2 embeddings: 28%|          | 256/914 [00:07<00:18, 36.13it/s]

```

Computing ESM2 embeddings: 32%| | 288/914 [00:08<00:16, 36.87it/s]

Computing ESM2 embeddings: 35%| | 320/914 [00:09<00:16, 35.93it/s]

Computing ESM2 embeddings: 39%| | 352/914 [00:09<00:15, 36.92it/s]

Computing ESM2 embeddings: 42%| | 384/914 [00:10<00:14, 36.66it/s]

Computing ESM2 embeddings: 46%| | 416/914 [00:11<00:13, 37.48it/s]

Computing ESM2 embeddings: 49%| | 448/914 [00:12<00:12, 36.90it/s]

Computing ESM2 embeddings: 53%| | 480/914 [00:13<00:11, 37.63it/s]

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 36.86it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:15<00:09, 37.54it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 37.05it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.51it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 36.37it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.31it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.86it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 37.52it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:21<00:03, 36.91it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:21<00:03, 37.61it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:22<00:02, 37.01it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 37.74it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.88it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:25<00:00, 36.39it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/898 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/898 [00:00<00:22, 37.99it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/898 [00:01<00:26, 32.02it/s]  |
| Computing ESM2 embeddings: | 11% | 96/898 [00:02<00:24, 33.21it/s]  |
| Computing ESM2 embeddings: | 14% | 128/898 [00:03<00:21, 35.58it/s] |
| Computing ESM2 embeddings: | 18% | 160/898 [00:04<00:20, 35.53it/s] |
| Computing ESM2 embeddings: | 21% | 192/898 [00:05<00:19, 36.81it/s] |
| Computing ESM2 embeddings: | 25% | 224/898 [00:06<00:18, 36.27it/s] |
| Computing ESM2 embeddings: | 29% | 256/898 [00:07<00:17, 37.01it/s] |
| Computing ESM2 embeddings: | 32% | 288/898 [00:08<00:16, 36.03it/s] |
| Computing ESM2 embeddings: | 36% | 320/898 [00:08<00:15, 36.81it/s] |
| Computing ESM2 embeddings: | 39% | 352/898 [00:09<00:15, 35.90it/s] |
| Computing ESM2 embeddings: | 43% | 384/898 [00:10<00:13, 36.95it/s] |
| Computing ESM2 embeddings: | 46% | 416/898 [00:11<00:13, 36.65it/s] |
| Computing ESM2 embeddings: | 50% | 448/898 [00:12<00:12, 37.48it/s] |
| Computing ESM2 embeddings: | 53% | 480/898 [00:13<00:11, 37.04it/s] |
| Computing ESM2 embeddings: | 57% | 512/898 [00:14<00:10, 37.77it/s] |
| Computing ESM2 embeddings: | 61% | 544/898 [00:14<00:09, 37.15it/s] |
| Computing ESM2 embeddings: | 64% | 576/898 [00:15<00:08, 37.53it/s] |
| Computing ESM2 embeddings: | 68% | 608/898 [00:16<00:07, 36.43it/s] |
| Computing ESM2 embeddings: | 71% | 640/898 [00:17<00:06, 37.29it/s] |
| Computing ESM2 embeddings: | 75% | 672/898 [00:18<00:06, 36.69it/s] |
| Computing ESM2 embeddings: | 78% | 704/898 [00:19<00:05, 37.44it/s] |
| Computing ESM2 embeddings: | 82% | 736/898 [00:20<00:04, 36.79it/s] |



```

Computing ESM2 embeddings: 86%|      | 768/898 [00:20<00:03, 37.59it/s]
Computing ESM2 embeddings: 89%|      | 800/898 [00:21<00:02, 36.92it/s]
Computing ESM2 embeddings: 93%|      | 832/898 [00:22<00:01, 37.42it/s]
Computing ESM2 embeddings: 96%|      | 864/898 [00:23<00:00, 36.29it/s]
Computing ESM2 embeddings: 100%|     | 898/898 [00:24<00:00, 36.54it/s]
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.

```

```

Computing ESM2 embeddings:  0%|      |  0/914 [00:00<?, ?it/s]
Computing ESM2 embeddings:  4%|      | 32/914 [00:01<00:32, 27.54it/s]
Computing ESM2 embeddings:  7%|      | 64/914 [00:02<00:27, 30.96it/s]
Computing ESM2 embeddings: 11%|      | 96/914 [00:02<00:23, 34.18it/s]
Computing ESM2 embeddings: 14%|      | 128/914 [00:03<00:23, 33.86it/s]
Computing ESM2 embeddings: 18%|      | 160/914 [00:04<00:21, 35.80it/s]
Computing ESM2 embeddings: 21%|      | 192/914 [00:05<00:20, 35.66it/s]
Computing ESM2 embeddings: 25%|      | 224/914 [00:06<00:18, 36.48it/s]
Computing ESM2 embeddings: 28%|      | 256/914 [00:07<00:18, 35.53it/s]
Computing ESM2 embeddings: 32%|      | 288/914 [00:08<00:17, 36.74it/s]
Computing ESM2 embeddings: 35%|      | 320/914 [00:09<00:16, 36.41it/s]
Computing ESM2 embeddings: 39%|      | 352/914 [00:09<00:15, 37.11it/s]
Computing ESM2 embeddings: 42%|      | 384/914 [00:10<00:14, 36.12it/s]
Computing ESM2 embeddings: 46%|      | 416/914 [00:11<00:13, 37.07it/s]
Computing ESM2 embeddings: 49%|      | 448/914 [00:12<00:12, 36.62it/s]
Computing ESM2 embeddings: 53%|      | 480/914 [00:13<00:11, 37.49it/s]

```

Computing ESM2 embeddings: 56%| | 512/914 [00:14<00:10, 36.98it/s]

Computing ESM2 embeddings: 60%| | 544/914 [00:15<00:09, 37.69it/s]

Computing ESM2 embeddings: 63%| | 576/914 [00:15<00:09, 36.88it/s]

Computing ESM2 embeddings: 67%| | 608/914 [00:16<00:08, 37.66it/s]

Computing ESM2 embeddings: 70%| | 640/914 [00:17<00:07, 37.03it/s]

Computing ESM2 embeddings: 74%| | 672/914 [00:18<00:06, 37.55it/s]

Computing ESM2 embeddings: 77%| | 704/914 [00:19<00:05, 36.28it/s]

Computing ESM2 embeddings: 81%| | 736/914 [00:20<00:04, 36.99it/s]

Computing ESM2 embeddings: 84%| | 768/914 [00:21<00:04, 36.01it/s]

Computing ESM2 embeddings: 88%| | 800/914 [00:22<00:03, 36.74it/s]

Computing ESM2 embeddings: 91%| | 832/914 [00:23<00:02, 35.73it/s]

Computing ESM2 embeddings: 95%| | 864/914 [00:23<00:01, 36.63it/s]

Computing ESM2 embeddings: 98%| | 896/914 [00:24<00:00, 36.17it/s]

Computing ESM2 embeddings: 100%| | 914/914 [00:25<00:00, 36.13it/s]  
 Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
 You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Computing ESM2 embeddings: 0%| | 0/896 [00:00<?, ?it/s]

Computing ESM2 embeddings: 4%| | 32/896 [00:00<00:23, 36.45it/s]

Computing ESM2 embeddings: 7%| | 64/896 [00:02<00:26, 30.85it/s]

Computing ESM2 embeddings: 11%| | 96/896 [00:02<00:23, 33.69it/s]

Computing ESM2 embeddings: 14%| | 128/896 [00:03<00:21, 35.12it/s]

Computing ESM2 embeddings: 18%| | 160/896 [00:04<00:20, 36.29it/s]

Computing ESM2 embeddings: 21%| | 192/896 [00:05<00:19, 35.37it/s]

|                                |                                  |
|--------------------------------|----------------------------------|
| Computing ESM2 embeddings: 25% | 224/896 [00:06<00:18, 36.35it/s] |
| Computing ESM2 embeddings: 29% | 256/896 [00:07<00:17, 35.56it/s] |
| Computing ESM2 embeddings: 32% | 288/896 [00:08<00:16, 36.68it/s] |
| Computing ESM2 embeddings: 36% | 320/896 [00:08<00:15, 36.39it/s] |
| Computing ESM2 embeddings: 39% | 352/896 [00:09<00:14, 37.32it/s] |
| Computing ESM2 embeddings: 43% | 384/896 [00:10<00:13, 36.80it/s] |
| Computing ESM2 embeddings: 46% | 416/896 [00:11<00:12, 37.56it/s] |
| Computing ESM2 embeddings: 50% | 448/896 [00:12<00:12, 36.91it/s] |
| Computing ESM2 embeddings: 54% | 480/896 [00:13<00:11, 37.59it/s] |
| Computing ESM2 embeddings: 57% | 512/896 [00:14<00:10, 36.69it/s] |
| Computing ESM2 embeddings: 61% | 544/896 [00:14<00:09, 37.45it/s] |
| Computing ESM2 embeddings: 64% | 576/896 [00:15<00:08, 36.77it/s] |
| Computing ESM2 embeddings: 68% | 608/896 [00:16<00:07, 37.12it/s] |
| Computing ESM2 embeddings: 71% | 640/896 [00:17<00:07, 36.00it/s] |
| Computing ESM2 embeddings: 75% | 672/896 [00:18<00:06, 37.01it/s] |
| Computing ESM2 embeddings: 79% | 704/896 [00:19<00:05, 36.65it/s] |
| Computing ESM2 embeddings: 82% | 736/896 [00:20<00:04, 37.41it/s] |
| Computing ESM2 embeddings: 86% | 768/896 [00:21<00:03, 36.86it/s] |
| Computing ESM2 embeddings: 89% | 800/896 [00:21<00:02, 37.51it/s] |
| Computing ESM2 embeddings: 93% | 832/896 [00:22<00:01, 37.04it/s] |
| Computing ESM2 embeddings: 96% | 864/896 [00:23<00:00, 37.80it/s] |

Computing ESM2 embeddings: 100%| | 896/896 [00:24<00:00, 36.51it/s]  
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it

for predictions and inference.

|                            |     |                                  |
|----------------------------|-----|----------------------------------|
| Computing ESM2 embeddings: | 0%  | 0/914 [00:00<?, ?it/s]           |
| Computing ESM2 embeddings: | 4%  | 32/914 [00:01<00:28, 31.40it/s]  |
| Computing ESM2 embeddings: | 7%  | 64/914 [00:01<00:25, 32.74it/s]  |
| Computing ESM2 embeddings: | 11% | 96/914 [00:02<00:23, 35.28it/s]  |
| Computing ESM2 embeddings: | 14% | 128/914 [00:03<00:22, 34.75it/s] |
| Computing ESM2 embeddings: | 18% | 160/914 [00:04<00:20, 36.40it/s] |
| Computing ESM2 embeddings: | 21% | 192/914 [00:05<00:19, 36.19it/s] |
| Computing ESM2 embeddings: | 25% | 224/914 [00:06<00:18, 36.96it/s] |
| Computing ESM2 embeddings: | 28% | 256/914 [00:07<00:18, 36.08it/s] |
| Computing ESM2 embeddings: | 32% | 288/914 [00:08<00:17, 36.48it/s] |
| Computing ESM2 embeddings: | 35% | 320/914 [00:08<00:15, 37.22it/s] |
| Computing ESM2 embeddings: | 39% | 352/914 [00:09<00:15, 36.74it/s] |
| Computing ESM2 embeddings: | 42% | 384/914 [00:10<00:14, 35.79it/s] |
| Computing ESM2 embeddings: | 46% | 416/914 [00:11<00:14, 35.20it/s] |
| Computing ESM2 embeddings: | 49% | 448/914 [00:12<00:13, 34.99it/s] |
| Computing ESM2 embeddings: | 53% | 480/914 [00:13<00:12, 34.57it/s] |
| Computing ESM2 embeddings: | 56% | 512/914 [00:14<00:11, 34.67it/s] |
| Computing ESM2 embeddings: | 60% | 544/914 [00:15<00:10, 34.26it/s] |
| Computing ESM2 embeddings: | 63% | 576/914 [00:16<00:09, 34.51it/s] |
| Computing ESM2 embeddings: | 67% | 608/914 [00:17<00:08, 34.47it/s] |
| Computing ESM2 embeddings: | 70% | 640/914 [00:18<00:07, 34.38it/s] |
| Computing ESM2 embeddings: | 74% | 672/914 [00:19<00:06, 34.75it/s] |
| Computing ESM2 embeddings: | 77% | 704/914 [00:20<00:06, 34.49it/s] |

```

Computing ESM2 embeddings: 81%|          | 736/914 [00:20<00:05, 34.25it/s]
Computing ESM2 embeddings: 84%|          | 768/914 [00:21<00:04, 34.09it/s]
Computing ESM2 embeddings: 88%|          | 800/914 [00:22<00:03, 33.96it/s]
Computing ESM2 embeddings: 91%|          | 832/914 [00:23<00:02, 33.88it/s]
Computing ESM2 embeddings: 95%|          | 864/914 [00:24<00:01, 33.74it/s]
Computing ESM2 embeddings: 98%|          | 896/914 [00:25<00:00, 33.74it/s]
Computing ESM2 embeddings: 100%|         | 914/914 [00:26<00:00, 34.84it/s]

```

```
[89]: variants_dataset.features
```

```
[89]: {'mutations': Sequence(feature=Value(dtype='string', id=None), length=-1,
id=None),
      'hash': Value(dtype='string', id=None),
      'variant': Value(dtype='string', id=None),
      'embedding': Sequence(feature=Value(dtype='float32', id=None), length=-1,
id=None)}
```

```
[90]: # drop duplicates due to choosing wild type as a variant
# the has column tracks unique variants
# get the first index of each unique variant
hash_set = set()

indexes = []
for index, hash_ in enumerate(variants_dataset['hash']):
    if hash_ in hash_set:
        continue
    hash_set.add(hash_)
    indexes.append(index)
variants_dataset = variants_dataset.select(indexes)
```

```
[91]: variants_dataset.save_to_disk('./data/hf_variants_dataset')
```

```

Saving the dataset (0/5 shards): 0%|          | 0/838413 [00:00<?, ? examples/
s]

```

## 1.5 4. Plate 2: Exploitation

### 1.5.1 4.1 Train Final model on all SSM data

```
[44]: ssm_dataset = datasets.load_from_disk('./data/hf_ssm_dataset')
      ssm_dataset.set_format('numpy', columns=['embedding', 'activity'])
      X_ssm = ssm_dataset['embedding']
      y_ssm = ssm_dataset['activity']
```

```
[45]: with open('./data/best_params.json', 'r') as f:
      best_params = json.load(f)
```

```
[46]: final_model = model_no_embedder.set_params(**best_params)
      final_model.fit(X_ssm, y_ssm)
      final_model.score(X_ssm, y_ssm)
```

```
/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/compose/_target.py:206: UserWarning: The provided functions or
transformer are not strictly inverse of each other. If you are sure you want to
proceed regardless, set 'check_inverse=False'
```

```
warnings.warn(
/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/sklearn/compose/_target.py:206: UserWarning: The provided functions or
transformer are not strictly inverse of each other. If you are sure you want to
proceed regardless, set 'check_inverse=False'
warnings.warn(
```

```
[46]: 0.8426079750061035
```

### 1.5.2 4.2 Run prediction for all 2 and 3 order mutants

```
[47]: variants_dataset = datasets.load_from_disk('./data/hf_variants_dataset')
      variants_dataset.set_format('numpy', columns=['embedding'])
```

```
[48]: # create a batched HF mapping function
      def model_prediction_mapping(examples):
          embeddings = examples['embedding']
          preds, stds = predict_ensemble(final_model, embeddings)
          examples.update({'activity_pred': preds, 'activity_pred_std': stds})
          return examples
```

```
[49]: variants_dataset = variants_dataset.map(model_prediction_mapping, batched=True,
      ↪ batch_size=10000)
```

```
Map: 0%|          | 0/838413 [00:00<?, ? examples/s]
```

### 1.5.3 4.3 Greedy Selection

#### 4.3.1 Sort by prediction mean

```
[50]: # sort variants dataset by predicted activity
sorted_greedy_index = np.array(list(reversed(np.
    ↪argsort(variants_dataset['activity_pred']))))
```

```
[51]: variants_dataset_sorted = variants_dataset.select(sorted_greedy_index)
```

```
[52]: variants_dataset_sorted[0]['activity_pred']
```

```
[52]: 4.5584173
```

**4.3.2 Iteratively add variants greedily, observing mutation counts** Including mutations from plate 1, we track how many times a particular mutation has been seen and cap it out to avoid saturating the plate with the same mutation over and over. Because of this, the model **must try to make the best variant possible without using the highest performing mutations too many times**.

Add the following exception: each top 12 mutant can be used once, in a variant without any other mtop 12 mutations.

```
[62]: plate1_dataset = datasets.load_from_disk('./data/hf_plate1_dataset')
plate1_mutations = plate1_dataset['mutations']
```

```
[63]: for mutations in plate1_mutations:
    add_observed_mutations(mutations)
```

```
[64]: count_top_12_used = {
    k: 0 for k in top_12_ssm_data['mutation_string']
}
```

```
[65]: count_top_12_used
```

```
[65]: {'A399G': 0,
      'V83G': 0,
      'G556R': 0,
      'A420S': 0,
      'A399E': 0,
      'T561S': 0,
      'S569R': 0,
      'D403N': 0,
      'T561K': 0,
      'A570K': 0,
      'S425A': 0,
      'S407Q': 0}
```

Let's actually do the loop. It cannot be parallelized because we have to check the current set of mutations at each evaluation. The dataset is sorted so we do not need to check predictions.

```
[57]: variants_dataset_sorted.set_format('numpy', columns=['mutations'])
```

```

[66]: selected_greedy_indexes = []

i = 0
bar = tqdm(len(variants_dataset_sorted))
class Completed(Exception): pass
try:
    for i, row in enumerate(variants_dataset_sorted):
        mutations = row['mutations']
        viable = True
        mutation_count_overwrite = False # for top 12 mutations

        # CHECK IF WE HAVE FILLED THE ALLOWABLE TOP 12
        n_top_12_in_variant = 0
        for mutation in mutations:
            if mutation in count_top_12_used:
                n_top_12_in_variant += 1
        if n_top_12_in_variant == 1:
            # here we have a variant with one of the top 12 in it
            # check if the one top 12 in this variant is already selected
            for mutation in mutations:
                if mutation in count_top_12_used:
                    if count_top_12_used[mutation] > 0:
                        viable = False
                        break
            if viable:
                # here we have a variant with one of the top 12 in it
                # and that top 12 hasn't been selected yet
                # for this plate
                mutation_count_overwrite = True
                for mutation in mutations:
                    if mutation in count_top_12_used:
                        count_top_12_used[mutation] += 1

            # check if we already have too many of the same mutation
            # observed in this variant
            if not mutation_count_overwrite:
                for mutation in mutations:
                    if MUTATION_COUNTS_OBSERVED.get(mutation, 0) >=
↳ MAX_MUTATION_COUNT:
                        viable = False
                        break
            else:
                pass

        if viable:
            selected_greedy_indexes.append(i)
            add_observed_mutations(mutations)

```



```

        i += 1
        bar.update(1)

        if len(selected_greedy_indexes) == 96:
            raise Completed

except Completed:
    pass

```

Oit [00:00, ?it/s]

```

[67]: greedy_dataset = variants_dataset_sorted.select(selected_greedy_indexes)
def apply_type(example):
    example['type'] = 'Greedy'
    return example
greedy_dataset = greedy_dataset.map(apply_type)

```

Map: 0%| | 0/96 [00:00<?, ? examples/s]

```

[68]: greedy_dataset.save_to_disk('./data/hf_greedy_dataset')

```

Saving the dataset (0/1 shards): 0%| | 0/96 [00:00<?, ? examples/s]

```

[69]: plate2_df = greedy_dataset.to_pandas()
plate2_df = plate2_df[['hash', 'variant', 'mutations']]
plate2_df['type'] = 'Greedy'
plate2_df['mutations'] = plate2_df['mutations'].apply(lambda x: '|'.join(x))
plate2_df.to_csv('./data/plate2.csv', index=False)

```

```

[71]: count_top_12_used

```

```

[71]: {'A399G': 1,
      'V83G': 1,
      'G556R': 1,
      'A420S': 1,
      'A399E': 1,
      'T561S': 1,
      'S569R': 1,
      'D403N': 1,
      'T561K': 1,
      'A570K': 1,
      'S425A': 1,
      'S407Q': 1}

```

```

[73]: plate2_df['count_top_12_used'] = plate2_df['mutations'].apply(lambda x: sum([m_
    ↪in count_top_12_used for m in x.split('|')]))
plate2_df['count_top_12_used'].value_counts()

```

```
[73]: 0    84
      1    12
      Name: count_top_12_used, dtype: int64
```

## 1.6 5. Plate 3: Exploration

### 1.6.1 5.1 UCB selection

Repeat what was done for Plate 2 but with UCB

```
[74]: sorted_ucb_index = np.array(list(reversed(np.
    ↪argsort(variants_dataset['activity_pred'] +
    ↪2*variants_dataset['activity_pred_std']))))
      variants_dataset_sorted = variants_dataset.select(sorted_ucb_index)
```

```
[75]: variants_dataset_sorted.set_format('numpy', columns=['mutations'])
```

```
[76]: selected_ucb_indexes = []

      i = 0
      bar = tqdm(len(variants_dataset_sorted))
      try:
          for i, row in enumerate(variants_dataset_sorted):
              mutations = row['mutations']
              viable = True
              # check if we already have too many of the same mutation
              # observed in this variant
              for mutation in mutations:
                  if MUTATION_COUNTS_OBSERVED.get(mutation, 0) >= MAX_MUTATION_COUNT:
                      viable = False
                      break
              if viable:
                  selected_ucb_indexes.append(i)
                  add_observed_mutations(mutations)

              i += 1
              bar.update(1)

              if len(selected_ucb_indexes) == 96:
                  raise Completed

      except Completed:
          pass
```

0it [00:00, ?it/s]

```
[77]: ucb_dataset = variants_dataset_sorted.select(selected_ucb_indexes)
      def apply_type(example):
          example['type'] = 'UCB'
```

```

    return example
ucb_dataset = ucb_dataset.map(apply_type)

```

Map: 0%| | 0/96 [00:00<?, ? examples/s]

```
[78]: np.max(ucb_dataset['activity_pred_std'])
```

```
[78]: 1.2422927618026733
```

```
[79]: ucb_dataset.save_to_disk('./data/hf_ucb_dataset')
```

Saving the dataset (0/1 shards): 0%| | 0/96 [00:00<?, ? examples/s]

```
[80]: ucb_df = ucb_dataset.to_pandas()
ucb_df = ucb_df[['hash', 'variant', 'mutations']]
ucb_df['type'] = 'UCB'

```

```
[81]: ucb_df['mutations'] = ucb_df['mutations'].apply(lambda x: '|'.join(x))
```

```
[82]: ucb_df.to_csv('./data/plate3.csv', index=False)
```

```
[83]: len(ucb_df)
```

```
[83]: 96
```

```
[84]: del variants_dataset_sorted
```

## 1.7 6. Combine plates into a uniform format

### 1.7.1 6.1 Load previous data

```
[184]: plate1_df = pd.read_csv('./data/plate1.csv')
plate2_df = pd.read_csv('./data/plate2.csv')
plate3_df = pd.read_csv('./data/plate3.csv')

```

```
[185]: plate1_df['plate'] = 1
plate2_df['plate'] = 2
plate3_df['plate'] = 3
combined_df = pd.concat([plate1_df, plate2_df, plate3_df], ignore_index=True)

```

### 1.7.2 6.2 Add columns

```
[186]: # parse mutations back into a list
combined_df['mutations_string'] = combined_df['mutations']
combined_df['mutations'] = combined_df['mutations'].apply(lambda x: x.
    ↪split('|'))

```

```
[187]: combined_df['id'] = combined_df.apply(lambda x:
    ↪f'{x["type"]}_{x["mutations_string"]}', axis=1)

```

```
[188]: combined_df['has_V83'] = combined_df['mutations_string'].apply(lambda x: 'V83' if
    ↪in x)
```

```
[189]: combined_df['has_V83G'] = combined_df['mutations_string'].apply(lambda x:
    ↪'V83G' if x)
```

```
[190]: # get the model predictions
X_seq = ap.ProteinSequences.from_list(combined_df['variant'].tolist())

embedder = ap.ESM2Embedding(
    model_checkpoint='esm2_t12_35M_UR50D',
    metadata_folder='esm2_embedder',
    pool=True,
    device='mps'
)
embedder.fit([])
X_embeddings = embedder.transform(X_seq)
y_pred, y_std = predict_ensemble(final_model, X_embeddings)

combined_df['activity_pred'] = y_pred
combined_df['activity_pred_std'] = y_std
```

/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-packages/huggingface\_hub/file\_download.py:1132: FutureWarning: `resume\_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force\_download=True`.

warnings.warn(
Some weights of EsmModel were not initialized from the model checkpoint at facebook/esm2\_t12\_35M\_UR50D and are newly initialized: ['esm.pooler.dense.bias', 'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
[191]: # also get additive activity for Plate 2 and 3
ssm_data = pd.read_csv('./data/ssm_data_clean.csv').
    ↪set_index('mutation_string')['activity']
def get_additive_score(mutations):
    activities = []
    for mutation in mutations:
        try:
            activities.append(ssm_data[mutation] - 1.0)
        except KeyError:
            return np.nan
    return sum(activities) + 1.0
```

```
[192]: combined_df['additive_activity'] = combined_df.apply(lambda x:
    ↪get_additive_score(x['mutations']), axis=1)
```

Count the number of top 12 mutants in the variant, add a column for the plate number.

```
[193]: ssm_data = pd.read_csv('./data/ssm_data_clean.csv').sort_values(by='activity',
↪ascending=False).reset_index(drop=True).head(12)
ssm_data
```

```
[193]:
```

|    | original_aa | position | mutation | activity | position_on_wt_0_indexed \ |
|----|-------------|----------|----------|----------|----------------------------|
| 0  | A           | 399      | G        | 7.223088 | 388                        |
| 1  | V           | 83       | G        | 2.749185 | 72                         |
| 2  | G           | 556      | R        | 2.208630 | 545                        |
| 3  | A           | 420      | S        | 2.001098 | 409                        |
| 4  | A           | 399      | E        | 1.842972 | 388                        |
| 5  | T           | 561      | S        | 1.764891 | 550                        |
| 6  | S           | 569      | R        | 1.702720 | 558                        |
| 7  | D           | 403      | N        | 1.672030 | 392                        |
| 8  | T           | 561      | K        | 1.632204 | 550                        |
| 9  | A           | 570      | K        | 1.631296 | 559                        |
| 10 | S           | 425      | A        | 1.604971 | 414                        |
| 11 | S           | 407      | Q        | 1.567455 | 396                        |

|    | mutation_string | variant \                                        |
|----|-----------------|--------------------------------------------------|
| 0  | A399G           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 1  | V83G            | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 2  | G556R           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 3  | A420S           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 4  | A399E           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 5  | T561S           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 6  | S569R           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 7  | D403N           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 8  | T561K           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 9  | A570K           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 10 | S425A           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |
| 11 | S407Q           | MAQLTGEQIIARSLKTQGIDKIFGVVGIPVTGIASACQAEGLYTG... |

|    | hash                             |
|----|----------------------------------|
| 0  | 8d5215793ebe5403ed894b7a698082e7 |
| 1  | f94de5cb4860607215a486a4da80dc31 |
| 2  | 6621d218f38d4052e097a83e8f52a119 |
| 3  | ee23d3055f543cc64ae33ac5b2177db0 |
| 4  | 08c92251d61c194bf85c86d2fd28b141 |
| 5  | f74ef3fe4ceb040018154e70f9176083 |
| 6  | 3e0f9211df3e4452a76c1213ffcffa6b |
| 7  | 22f9a6097834a5d25bf164c19d2c7b03 |
| 8  | bb6772b6ea313411519db9033f0df93a |
| 9  | f9e23acb8e41b3eb6eb4dd557977caf3 |
| 10 | e3d6256df4e1c5473872f1e939db99c5 |
| 11 | 2454ca1a68bcee8fbe6b6c4ae44b1fdd |

```
[194]: combined_df['n_mutations'] = combined_df['mutations'].apply(len)

[196]: combined_df['n_top_12'] = combined_df['mutations'].apply(lambda x: sum([m in
↳ ssm_data['mutation_string'].values for m in x]))

[ ]: # do a final check that no mutations occur outside the expected positions
wt_array = np.array(list(wt))
for variant in combined_df['variant']:
    variant_array = np.array(list(variant))
    variable_positions = np.where(variant_array != wt_array)[0]
    if not np.all(np.isin(variable_positions,
↳ putative_mutatable_positions_wt_offset_zero_index)):
        print('Mutation outside expected positions!')
        print(variant)
        print(variable_positions)
        print(putative_mutatable_positions_wt_offset_zero_index)

[233]: # lowercase all but mutated residues
combined_df['variant'] = combined_df.apply(lambda x: ''.join([aa.lower() if i
↳ not in putative_mutatable_positions_wt_offset_zero_index else aa for i, aa
↳ in enumerate(x['variant'])]), axis=1)

[237]: # order columns to make sense
combined_df = combined_df[['id', 'plate', 'type', 'mutations_string',
↳ 'has_V83', 'has_V83G', 'n_mutations', 'n_top_12',
↳ 'variant', 'activity_pred', 'activity_pred_std',
↳ 'additive_activity']]
combined_df = combined_df.sort_values(by=['plate', 'type'])
combined_df.to_csv('./data/round1_variants.csv', index=False)

[238]: combined_df['n_mutations'].value_counts()

[238]: 3    236
      2    46
      4     1
      5     1
      6     1
      7     1
      8     1
      9     1
      Name: n_mutations, dtype: int64

[202]:
```

## 1.8 7. Analyze chosen data

### 1.8.1 7.1 Visualize data space explored

Look at PC's of embeddings vs types of data chosen

#### 7.1.1 First embed full data space

```
[203]: varaints_dataset = datasets.load_from_disk('./data/hf_varaints_dataset')
ssm_dataset = datasets.load_from_disk('./data/hf_ssm_dataset')
combined_dataset = datasets.concatenate_datasets([varaints_dataset,
↪ssm_dataset])
combined_dataset.set_format('numpy', columns=['embedding', 'activity'])

X = combined_dataset['embedding']
pca = PCA(n_components=0.95)
scaler = StandardScaler()
pca_pipe = Pipeline([
    ('scaler', scaler),
    ('pca', pca)
])
X_pca = pca_pipe.fit_transform(X)

joblib.dump(pca_pipe, './data/all_data_pca.pkl')
joblib.dump(X_pca, './data/all_data_pca_embeddings.pkl')
```

```
[203]: ['./data/all_data_pca_embeddings.pkl']
```

#### 7.1.2 Plot embedding of selected variants

```
[204]: ssm_dataset = datasets.load_from_disk('./data/hf_ssm_dataset')
plate1_dataset = datasets.load_from_disk('./data/hf_plate1_dataset')
plate2_dataset = datasets.load_from_disk('./data/hf_greedy_dataset')
plate3_dataset = datasets.load_from_disk('./data/hf_ucb_dataset')
```

```
[205]: def apply_type(example):
        example['type'] = 'SSM'
        return example
ssm_dataset = ssm_dataset.map(apply_type)
```

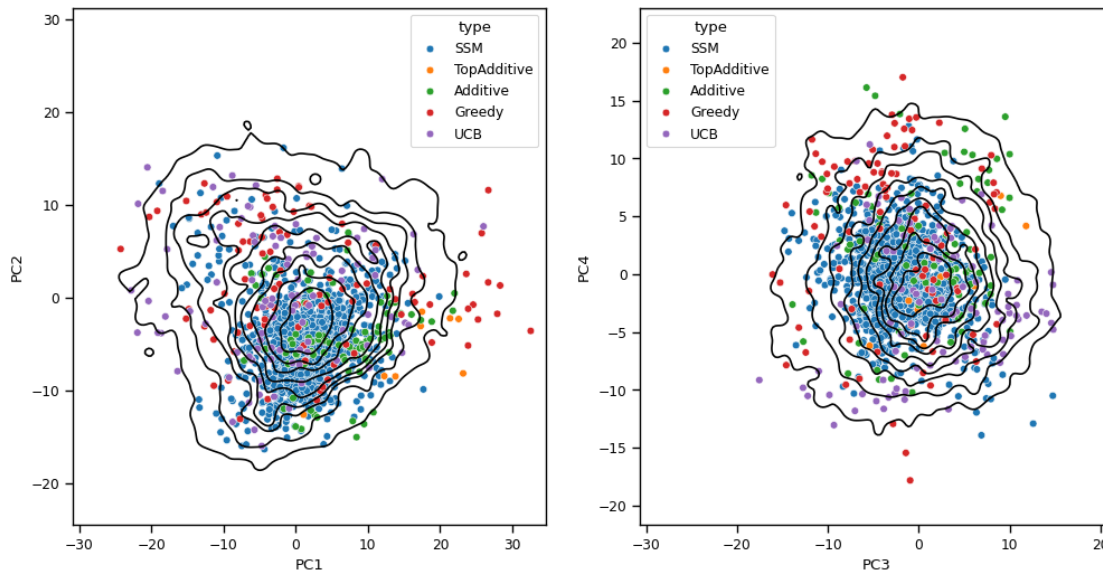
```
[206]: combined_dataset = datasets.concatenate_datasets([ssm_dataset, plate1_dataset,
↪plate2_dataset, plate3_dataset])
combined_dataset.set_format('numpy', columns=['embedding'])
```

```
[207]: chosen_pca = pca_pipe.transform(combined_dataset['embedding'])
```

```
[208]: # select a sample of all PCS
all_pcs_ = X_pca[np.random.choice(X_pca.shape[0], 10000, replace=False)]
```

```
[209]: chosen_df = pd.DataFrame(
        {'type': combined_dataset['type']}
    )
    for i in range(5):
        chosen_df[f'PC{i+1}'] = chosen_pca[:, i]

[210]: fig, ax = plt.subplots(1,2, figsize=(12, 6))
    sns.scatterplot(x='PC1', y='PC2', hue='type', data=chosen_df, ax=ax[0])
    sns.scatterplot(x='PC3', y='PC4', hue='type', data=chosen_df, ax=ax[1])
    sns.kdeplot(x=all_pcs[:, 0], y=all_pcs[:, 1], ax=ax[0], fill=False,
        ↪levels=10, bw_adjust=0.5, color='black')
    sns.kdeplot(x=all_pcs[:, 2], y=all_pcs[:, 3], ax=ax[1], fill=False,
        ↪levels=10, bw_adjust=0.5, color='black')
    plt.savefig('./data/rd1_pca_design_space.png', dpi=300, bbox_inches='tight')
```



## 1.8.2 7.2 View predicted activity vs additive activity for plate 1

```
[211]: combined_df = pd.read_csv('./data/round1_variants.csv')

[212]: plate1_df = combined_df[combined_df['plate'] == 1]

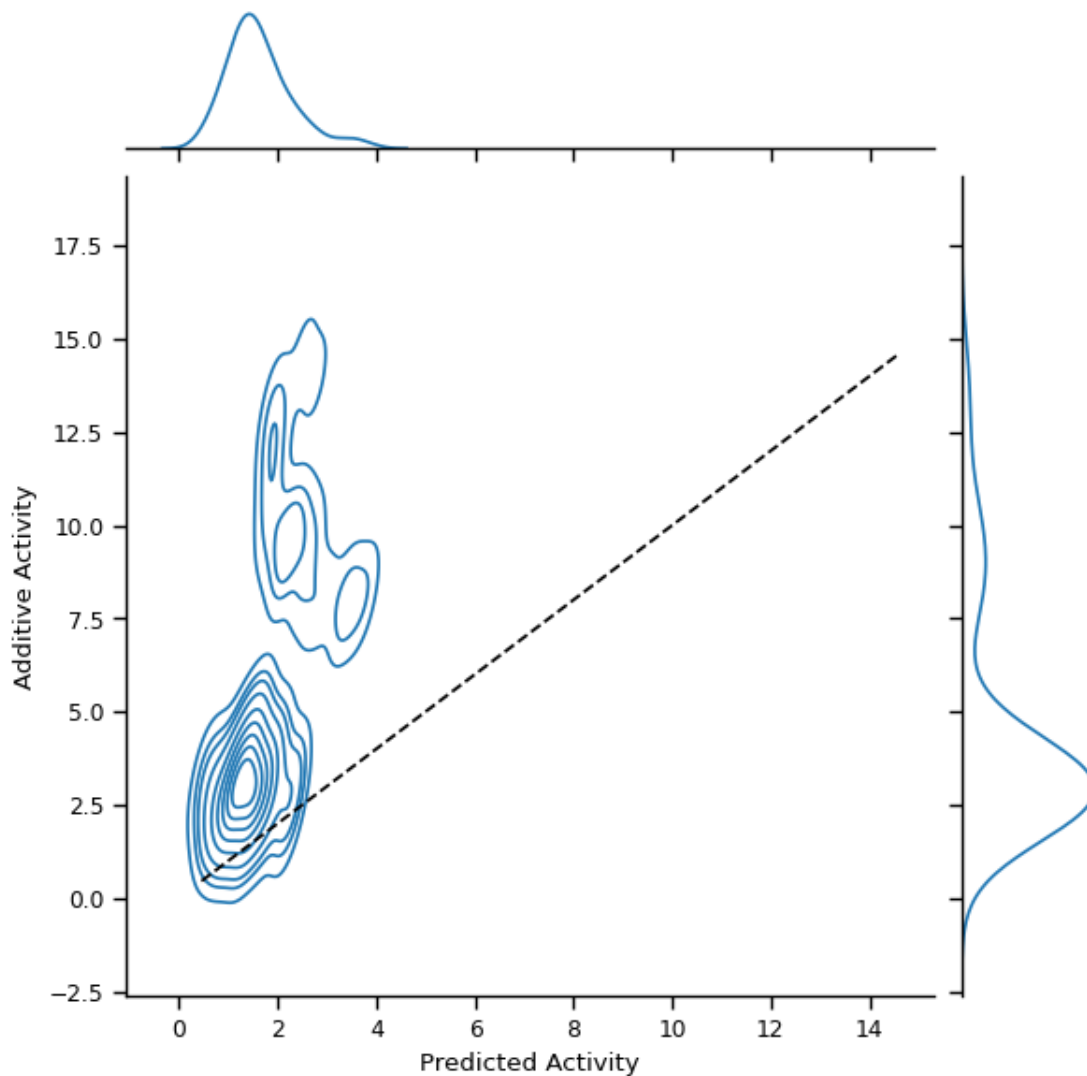
[213]: g = sns.jointplot(data=plate1_df, x='activity_pred', y='additive_activity',
    ↪kind='kde', bw_adjust=0.7)
    ax = g.ax_joint
    # plot the diagonal
    min_min = min(plate1_df['activity_pred'].min(), plate1_df['additive_activity'].
    ↪min())
```



```

max_max = max(plate1_df['activity_pred'].max(), plate1_df['additive_activity'].
    ↪max())
ax.plot([min_min, max_max], [min_min, max_max], color='black', linestyle='--')
ax.set_xlabel('Predicted Activity')
ax.set_ylabel('Additive Activity')
plt.savefig('./data/rd1_plate1_pred_vs_additive.png', dpi=300,
    ↪bbox_inches='tight')

```



### 1.8.3 7.3 Predicted activity for each plate

```

[214]: from matplotlib.lines import Line2D

```

```
[215]: # add a plates worth of random mutations at order 3
AA_LIST = list('ACDEFGHIKLMNPQRSTVWY')
random_variants = []
for i in range(96):
    positions = np.random.
    ↪choice(putative_mutatable_positions_wt_offset_zero_index, 3, replace=False)
    mutations = []
    for position in positions:
        mutation = f'{wt[position]}{position}{np.random.choice(AA_LIST)}'
        mutations.append(mutation)
    variant = mutate_wt(wt, mutations, one_indexed=False, offset=0)
    random_variants.append(variant)
```

```
[216]: random_embeddings = embedder.transform(ap.ProteinSequences.
    ↪from_list(random_variants))
```

```
/Users/ekomp/miniconda3/envs/aidep/lib/python3.9/site-
packages/huggingface_hub/file_download.py:1132: FutureWarning: `resume_download`
is deprecated and will be removed in version 1.0.0. Downloads always resume when
possible. If you want to force a new download, use `force_download=True`.
warnings.warn(
Some weights of EsmModel were not initialized from the model checkpoint at
facebook/esm2_t12_35M_UR50D and are newly initialized: ['esm.pooler.dense.bias',
'esm.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
Computing ESM2 embeddings: 100%|          | 96/96 [00:02<00:00, 37.33it/s]
```

```
[217]: random_predictions = predict_ensemble(final_model, random_embeddings)
```

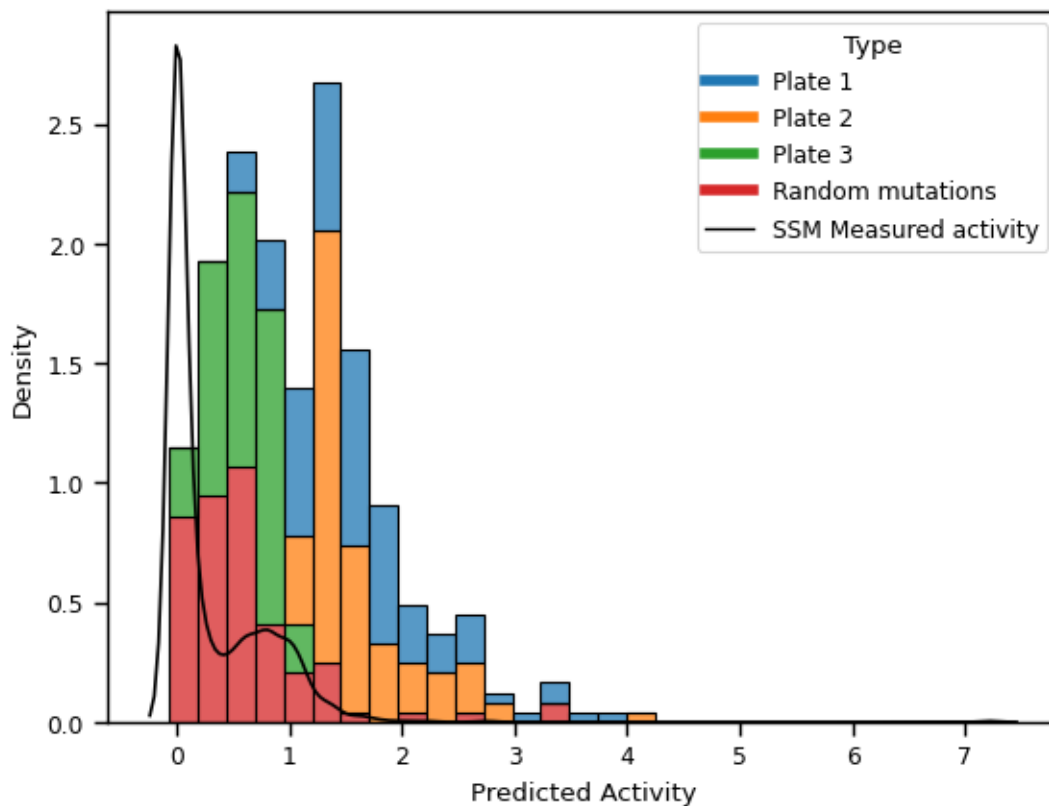
```
[218]: combined_df_ = combined_df.copy()
# add random variants
random_df = pd.DataFrame({
    'type': 'Random',
    'variant': random_variants,
    'activity_pred': random_predictions[0],
    'activity_pred_std': random_predictions[1],
    'plate': 'Random'
})
combined_df_ = pd.concat([combined_df_, random_df], ignore_index=True)
```

```
[219]: g = sns.histplot(
    data=combined_df_, x='activity_pred', hue='plate',
    common_bins=True, common_norm=False, multiple='stack',
    stat='density', palette=sns.color_palette()[:4])
ax = plt.gca()
sns.kdeplot(x=ssm_dataset['activity'], color='black', label='SSM', fill=False,
    ↪bw_adjust=0.7, ax=ax, )
```

```

ax.set_xlabel('Predicted Activity')
ax.legend(title='Type', handles=[
    Line2D([0], [0], color=sns.color_palette()[0], lw=4),
    Line2D([0], [0], color=sns.color_palette()[1], lw=4),
    Line2D([0], [0], color=sns.color_palette()[2], lw=4),
    Line2D([0], [0], color=sns.color_palette()[3], lw=4),
    Line2D([0], [0], color='black', lw=1)
], labels=['Plate 1', 'Plate 2', 'Plate 3', 'Random mutations', 'SSM Measured_
↳activity'])
plt.savefig('./data/rd1_plate_activity_distributions.png', dpi=300,
↳bbox_inches='tight')

```



### 7.3.1 Compute some statistics on likely performance of variants according to model predictions

```

[220]: combined_df['norm_dist'] = combined_df.apply(lambda x: stats.
↳norm(x['activity_pred'], x['activity_pred_std']), axis=1)

```

```

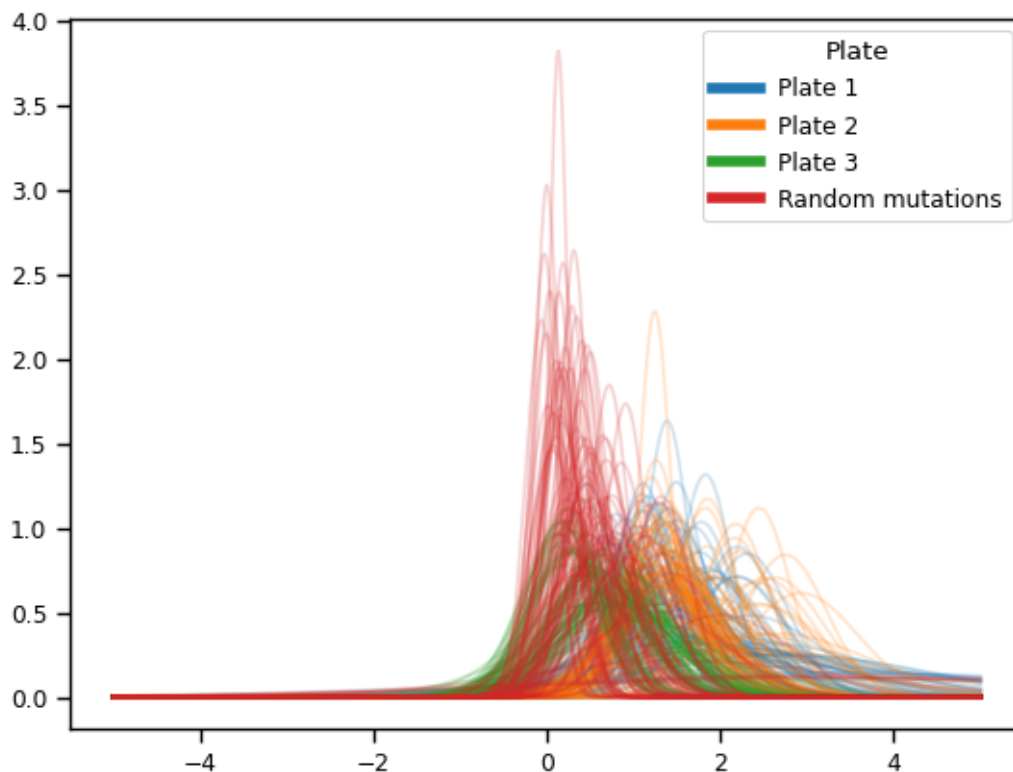
[221]: combined_df['gt_50pc_better_than_wt'] = combined_df['norm_dist'].apply(lambda
↳x: x.cdf(1.0) < 0.5)

```

```
[222]: combined_df['gt_95pc_better_than_0.0'] = combined_df['norm_dist'].
        ↪apply(lambda x: x.cdf(0.0) < 0.05)

[223]: combined_df['p_gt_wt'] = combined_df['norm_dist'].apply(lambda x: 1 - x.cdf(1.
        ↪0))
combined_df['p_gt_0.0'] = combined_df['norm_dist'].apply(lambda x: 1 - x.
        ↪cdf(0.0))

[224]: x = np.linspace(-5, 5, 1000)
colors = dict(zip(combined_df['plate'].unique(), sns.color_palette()[:4]))
for i, row in combined_df.iterrows():
    c = colors[row['plate']]
    plt.plot(x, row['norm_dist'].pdf(x), alpha=0.2, color=c)
plt.legend(title='Plate', handles=[
    Line2D([0], [0], color=sns.color_palette()[0], lw=4),
    Line2D([0], [0], color=sns.color_palette()[1], lw=4),
    Line2D([0], [0], color=sns.color_palette()[2], lw=4),
    Line2D([0], [0], color=sns.color_palette()[3], lw=4),
], labels=['Plate 1', 'Plate 2', 'Plate 3', 'Random mutations'])
ax.set_xlabel('Predicted Activity')
ax.set_ylabel('Density')
plt.savefig('./data/rd1_plate_activity_distributions_pdf.png', dpi=300,
        ↪bbox_inches='tight')
```



```
[225]: combined_df['gt_95pc_better_than_0.0'].groupby(combined_df['plate']).mean()
```

```
[225]: plate
      1      0.666667
      2      0.947917
      3      0.072917
      Random 0.385417
      Name: gt_95pc_better_than_0.0, dtype: float64
```

```
[226]: combined_df['gt_50pc_better_than_wt'].groupby(combined_df['plate']).mean()
```

```
[226]: plate
      1      0.875000
      2      1.000000
      3      0.020833
      Random 0.156250
      Name: gt_50pc_better_than_wt, dtype: float64
```

```
[227]: combined_df['p_gt_wt'].groupby(combined_df['plate']).mean()
```

```
[227]: plate
      1      0.722929
      2      0.802953
      3      0.239066
      Random 0.182478
      Name: p_gt_wt, dtype: float64
```

```
[228]: combined_df['p_gt_0.0'].groupby(combined_df['plate']).mean()
```

```
[228]: plate
      1      0.960400
      2      0.990268
      3      0.823157
      Random 0.867712
      Name: p_gt_0.0, dtype: float64
```

#### 1.8.4 7.4 Mutation space explored

```
[229]: combined_df['mutations'] = combined_df['mutations_string'].apply(lambda x: x.
      ↪split('|'))
```

```
[230]: mutations_tested_count = {}
      for mutations in combined_df['mutations']:
          for mutation in mutations:
              mutations_tested_count[mutation] = mutations_tested_count.get(mutation, 0)
              ↪ + 1
```

```
[231]: mutations_tested_count
```

```
[231]: {'A399E': 23,  
        'T561K': 23,  
        'S407Q': 23,  
        'S425A': 24,  
        'A570K': 23,  
        'D403N': 23,  
        'S569R': 23,  
        'T561S': 23,  
        'A420S': 23,  
        'G556R': 23,  
        'A399G': 22,  
        'V83G': 23,  
        'A420T': 4,  
        'A570G': 4,  
        'T561R': 4,  
        'S569K': 7,  
        'A570R': 7,  
        'A570Q': 5,  
        'A399S': 4,  
        'A399D': 4,  
        'G556N': 4,  
        'T561Q': 4,  
        'G556Q': 4,  
        'G556S': 4,  
        'A570S': 4,  
        'A570N': 4,  
        'T561G': 4,  
        'T561N': 4,  
        'S569Q': 4,  
        'S407E': 4,  
        'T561L': 4,  
        'G556K': 4,  
        'A570F': 4,  
        'T561W': 4,  
        'D403E': 4,  
        'S569T': 4,  
        'S425T': 4,  
        'S407A': 4,  
        'S569G': 4,  
        'A570E': 4,  
        'G556T': 4,  
        'V83A': 4,  
        'S407T': 4,  
        'A399Q': 4,  
        'S569A': 4,
```

'G556P': 4,  
'S569P': 4,  
'S569N': 4,  
'G556D': 4,  
'S407K': 4,  
'T561M': 4,  
'A570M': 4,  
'G556A': 4,  
'S569E': 4,  
'S407R': 4,  
'T561I': 4,  
'A570T': 4,  
'T561P': 4,  
'A570P': 4,  
'S569D': 4,  
'T561E': 4,  
'G556E': 4,  
'S569V': 4,  
'A570V': 4,  
'T561D': 4,  
'A399C': 4,  
'V83S': 4,  
'T561F': 4,  
'A570D': 4,  
'A570I': 4,  
'T561V': 4,  
'S569I': 4,  
'G556V': 4,  
'D403A': 4,  
'S407N': 4,  
'G556L': 4,  
'A570Y': 4,  
'S569M': 4,  
'A420V': 2,  
'S407M': 4,  
'A399N': 4,  
'A570W': 4,  
'A399K': 4,  
'A399M': 4,  
'A420Q': 4,  
'S407D': 4,  
'A420E': 4,  
'D403M': 4,  
'G556H': 4,  
'S425G': 4,  
'S407H': 4,  
'A570L': 4,

'A570H': 4,  
'D403K': 4,  
'V83T': 4,  
'D403Y': 4,  
'D403R': 4,  
'V83I': 4,  
'D403S': 4,  
'A420P': 4,  
'D403G': 4,  
'G556F': 4,  
'D403Q': 4,  
'S425V': 4,  
'G556M': 4,  
'S569C': 4,  
'A399W': 4,  
'D403V': 4,  
'A570C': 4,  
'V83C': 4,  
'V83P': 4,  
'G556Y': 4,  
'S569Y': 4,  
'G556I': 4,  
'S407V': 4,  
'V83R': 4,  
'T561Y': 4,  
'S569W': 4,  
'V83K': 4,  
'S425L': 3,  
'S407F': 3,  
'A420R': 4,  
'S569L': 4,  
'S425I': 4,  
'S425M': 4,  
'T561H': 4,  
'S425W': 1,  
'T561A': 4,  
'D403T': 1,  
'A399R': 4,  
'G556C': 4,  
'S407I': 4,  
'V83D': 4,  
'T561C': 4,  
'V83M': 4,  
'G556W': 4,  
'S569F': 4,  
'D403F': 4,  
'D403I': 4,



```

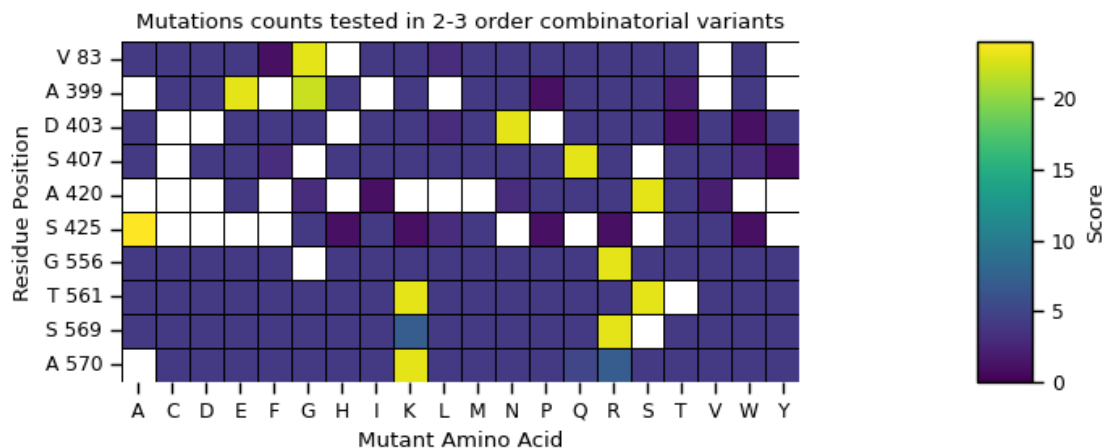
'V83W': 4,
'V83Q': 4,
'A420G': 3,
'S569H': 4,
'S407L': 4,
'D403W': 1,
'V83E': 4,
'S407W': 3,
'S425R': 1,
'S407P': 4,
'V83N': 4,
'V83L': 3,
'D403L': 3,
'A399H': 4,
'A399T': 2,
'A420I': 1,
'V83F': 1,
'S425H': 1,
'S407Y': 1,
'A420N': 3,
'A399P': 1,
'S425P': 1,
'S425K': 1}

```

```

[232]: fig, _, _ = plot_mutation_heatmap(
    list(mutations_tested_count.keys()),
    list(mutations_tested_count.values()),
    zero_minimum=True,
    all_positions=False
)
axes = fig.get_axes()
axes[0].set_title('Mutations counts tested in 2-3 order combinatorial variants')
plt.savefig('./data/rd1_mutation_counts.png', dpi=300, bbox_inches='tight')

```



[ ]: